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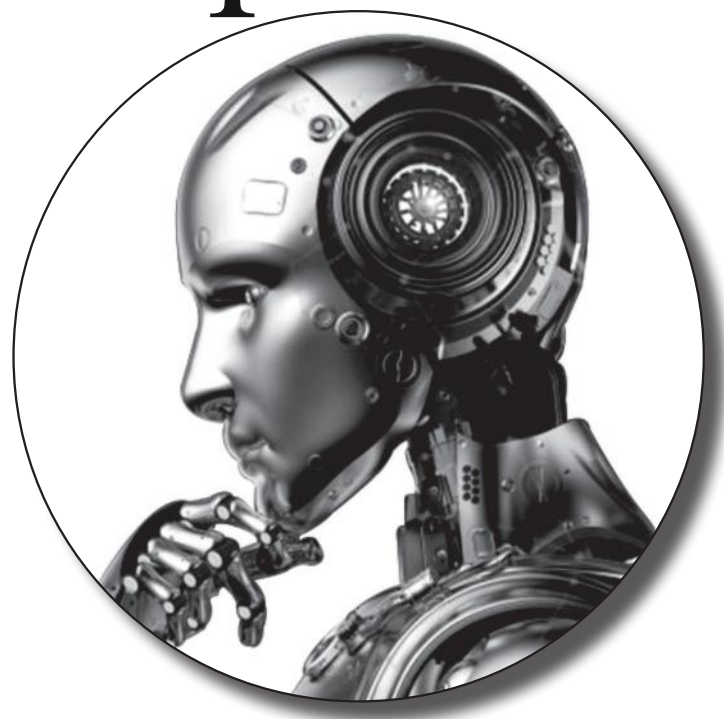
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Chapterwise & Topicwise
Solved Papers



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Opening/ Closing Rank for TOP NITs & List of NITs in India

Opening/ Closing Rank for Top NITS

College Name	OR/CR	CSE	ECE	ME	EE/EEE
NIT Trichy	ORF	2060	5325	4154	5708
	CR	5317	8011	12970	10353
NIT Rourkela	OR	2253	8571	11662	4084
	CR	9420	12009	20304	19168
NIT Surathkal	OR	960	3378	6315	3456
	CR	3181	5608	11788	6801
NIT Warangal	OR	978	2919	4340	5270
	CR	2341	2919	10209	8152
NIT Calicut	OR	2201	8023	10629	9703
	CR	10222	14769	20480	18966
NIT Kurukshetra	OR	2268	8320	11195	9454
	CR	6170	12067	18115	16273
NIT Durgapur	OR	5611	12509	14511	13595
	CR	12095	16098	22753	19325
MNIT Allahabad	OR	1449	3600	5884	5879
	CR	4051	7128	11145	8790
NIT Silchar	OR	8699	17899	21851	32579
	CR	23882	40841	49215	56958
MNIT Jaipur	OR	1148	3881	9277	4119
	CR	3831	7868	11426	9179

List of NITs in India

After IITs, NITs form the second layer of topmost engineering Institutes in India

Rank Of (Amongst NITs)	Name	State	NIRF Score	NIRF Ranking
1	NIT Trichy	Tamil Nadu	61.62	10
2	NIT Rourkela	Odisha	57.75	16
3	NIT Karnataka	Karnataka	55.25	21
4	NIT Warangal	Telangana	53.21	26
5	NIT Calicut	Kerala	52.69	28
6	V-NIT	Maharashtra	51.27	31
7	NIT Kurukshetra	Haryana	47.58	41
7	MN-NIT	Uttar Pradesh	47.49	42
8	NIT Durgapur	West Bengal	46.47	46
9	NIT Silchar	Assam	45.61	51
10	M-NIT	Rajasthan	45.20	53
11	SV-NIT	Gujarat	41.88	58
12	NIT Hamirpur	Himachal Pradesh	41.48	60
13	MA-NIT	Madhya Pradesh	40.98	62
14	NITIE	Maharashtra	40.48	66
15	NIT Meghalaya	Meghalaya	40.32	67
16	NIT Agartala	Tripura	39.53	70
17	NIT Raipur	Chattisgarh	39.09	74
18	NIT Goa	Goa	37.06	87

FAQs - Frequently Asked Questions

QUESTION: Which Colleges other than IITs accept JEE Advanced scores?

ANSWER:

1. Institute of Science (IISc), Bangalore
2. Indian Institute of Petroleum and Energy (IIPE), Visakhapatnam
3. Indian Institutes of Science Education and Research (IISER), Bhopal
4. Indian Institutes of Science Education and Research (IISER), Mohali
5. Indian Institutes of Science Education and Research (IISER), Kolkata
6. Indian Institutes of Science Education and Research (IISER), Pune
7. Indian Institutes of Science Education and Research (IISER), Thiruvananthapuram
8. Indian Institute of Space Science and Technology (IIST), Thiruvananthapuram
9. Rajiv Gandhi Institute of Petroleum Technology (RGIPT), Rae Bareilly

QUESTION: If I am absent in one of the papers (Paper 1, Paper 2), will my result be declared?

ANSWER: NO. You will be considered absent in JEE (Advanced)-2018 and the result will not be prepared/declared. It is compulsory to appear in both the papers for result preparation.

QUESTION: Do I have to choose my question paper language at the time of JEE (Advanced)-2018 registration?

ANSWER: NO. There is no need to indicate question paper language at the time of JEE (Advanced)-2018 registration. Candidates will have the option to choose their preferred language (English or Hindi), as the default language for viewing the questions, at the start of the Computer Based Test (CBT) examination of JEE (Advanced)-2018.

QUESTION: Can I change the language (from English to Hindi and vice versa) of viewing the questions during the CBT of JEE (Advanced)-2018?

ANSWER: Questions will be displayed on the screen of the Candidate in the chosen default language (English or Hindi). Further, the candidate can also switch/toggle between English or Hindi languages, as the viewing language of any question, anytime during the entire period of the examination. The candidate will also be having the option of changing default question viewing language anytime during the examination.

QUESTION: Will I be given rough sheets for my calculations during the CBT of JEE (Advanced)-2018?

ANSWER: Yes, you will be given "Scribble Pad" (containing blank sheets, for rough work) at the start of every paper of JEE (Advanced)-2018. You can do all your calculations inside this "Scribble Pad". Candidates MUST submit their signed Scribble Pads

at the end of each paper of the examination, given to them at the start of the paper.

QUESTION: During examination can I change my answers?

ANSWER: Candidate will have the option to change previously saved answer of any question, anytime during the entire duration of the test.

QUESTION: How can I change a previously saved answer during the CBT of JEE (Advanced)-2018?

ANSWER: To change the answer of a question that has already been answered and saved, first select the corresponding question from the Question Palette, then click on "Clear Response" to clear the previously entered answer and subsequently follow the procedure for answering that type of question.

QUESTION: Will I be given a printout/hard copy of the questions papers along with my responses to questions in Paper-I and Paper-II after the completion of the respective papers?

ANSWER: No.

QUESTION: How will I be getting a copy of the questions papers and my responses to questions in Paper-I and Paper-II?

ANSWER: The responses of all the candidates who have appeared for both Paper 1 and Paper 2, recorded during the exam, along with the questions of each paper, will be electronically mailed to their registered email ids, by Friday, May 25, 2018, 10:00 IST.

QUESTION: Suppose two candidates have same JEE (Advanced)-2018 aggregate marks. Will the two candidates be given the same rank?

ANSWER: If the aggregate marks scored by two or more candidates are the same, then the following tie-break policy will be used for awarding ranks: Step 1: Candidates having higher positive marks will be awarded higher rank. If the tie breaking criterion at Step 1 fails to break the tie, then the following criterion at Step 2 will be followed. Step 2: Higher rank will be assigned to the candidate who has obtained higher marks in Mathematics. If this does not break the tie, higher rank will be assigned to the candidate who has obtained higher marks in Physics. If there is a tie even after this, candidates will be assigned the same rank.

QUESTION: I have read in newspapers that for the academic year 2018-2019, supernumerary seats for female candidates would be there in IITs. Does this mean that the non-females will get reduced number of seats in IITs in 2018?

ANSWER: A decision has been taken at the level of the IIT Council to, inter alia, improve the gender balance in the undergraduate programs at the IITs from the current (approximately) 8% to 14% in 2018-19 by creating supernumerary seats specifically for female candidates, without any reduction in the number of seats that was made available to non-female candidates in the previous academic year (i.e. academic year 2017-2018).

Physical World, Units and Measurements



TOPIC 1 Unit of Physical Quantities



- The density of a material in SI unit is 128 kg m^{-3} . In certain units in which the unit of length is 25 cm and the unit of mass is 50 g, the numerical value of density of the material is: **[10 Jan. 2019 I]**
(a) 40 (b) 16 (c) 640 (d) 410
- A metal sample carrying a current along X-axis with density J_x is subjected to a magnetic field B_z (along z-axis). The electric field E_y developed along Y-axis is directly proportional to J_x as well as B_z . The constant of proportionality has SI unit **[Online April 25, 2013]**

- (a) $\frac{m^2}{A}$ (b) $\frac{m^3}{As}$ (c) $\frac{m^2}{As}$ (d) $\frac{As}{m^3}$

TOPIC 2 Dimensions of Physical Quantities



- The quantities $x = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$, $y = \frac{E}{B}$ and $z = \frac{1}{CR}$ are defined where C -capacitance, R -Resistance, l -length, E -Electric field, B -magnetic field and ϵ_0, μ_0 , - free space permittivity and permeability respectively. Then : **[Sep. 05, 2020 (II)]**
(a) x, y and z have the same dimension.
(b) Only x and z have the same dimension.
(c) Only x and y have the same dimension.
(d) Only y and z have the same dimension.
- Dimensional formula for thermal conductivity is (here K denotes the temperature) : **[Sep. 04, 2020 (I)]**
(a) $MLT^{-2}K$ (b) $MLT^{-2}K^{-2}$
(c) $MLT^{-3}K$ (d) $MLT^{-3}K^{-1}$
- A quantity x is given by (IFv^2/WL^4) in terms of moment of inertia I , force F , velocity v , work W and Length L . The dimensional formula for x is same as that of : **[Sep. 04, 2020 (II)]**
(a) planck's constant (b) force constant
(c) energy density (d) coefficient of viscosity

- Amount of solar energy received on the earth's surface per unit area per unit time is defined a solar constant. Dimension of solar constant is : **[Sep. 03, 2020 (II)]**
(a) ML^2T^{-2} (b) ML^0T^{-3}
(c) $M^2L^0T^{-1}$ (d) MLT^{-2}
- If speed V , area A and force F are chosen as fundamental units, then the dimension of Young's modulus will be : **[Sep. 02, 2020 (I)]**
(a) FA^2V^{-1} (b) FA^2V^{-3}
(c) FA^2V^{-2} (d) $FA^{-1}V^0$
- If momentum (P), area (A) and time (T) are taken to be the fundamental quantities then the dimensional formula for energy is : **[Sep. 02, 2020 (II)]**
(a) $[P^2AT^{-2}]$ (b) $[PA^{-1}T^{-2}]$
(c) $[PA^{1/2}T^{-1}]$ (d) $[P^{1/2}AT^{-1}]$
- Which of the following combinations has the dimension of electrical resistance (ϵ_0 is the permittivity of vacuum and μ_0 is the permeability of vacuum)? **[12 April 2019 I]**

- (a) $\sqrt{\frac{\mu_0}{\epsilon_0}}$ (b) $\frac{\mu_0}{\epsilon_0}$ (c) $\sqrt{\frac{\epsilon_0}{\mu_0}}$ (d) $\frac{\epsilon_0}{\mu_0}$

- In the formula $X = 5YZ^2$, X and Z have dimensions of capacitance and magnetic field, respectively. What are the dimensions of Y in SI units ? **[10 April 2019 II]**
(a) $[M^{-3}L^{-2}T^8A^4]$ (b) $[M^{-1}L^{-2}T^4A^2]$
(c) $[M^{-2}L^0T^{-4}A^{-2}]$ (d) $[M^{-2}L^{-2}T^6A^3]$
- In SI units, the dimensions of $\sqrt{\frac{\epsilon_0}{\mu_0}}$ is: **[8 April 2019 I]**
(a) $A^{-1}TML^3$ (b) $AT^2M^{-1}L^{-1}$
(c) $AT^{-3}ML^{3/2}$ (d) $A^2T^3M^{-1}L^{-2}$
- Let l, r, c and v represent inductance, resistance, capacitance and voltage, respectively. The dimension of $\frac{l}{rcv}$ in SI units will be : **[12 Jan. 2019 II]**
(a) $[LA^{-2}]$ (b) $[A^{-1}]$
(c) $[LTA]$ (d) $[LT^2]$

13. The force of interaction between two atoms is given by

$$F = \alpha\beta \exp\left(-\frac{x^2}{\alpha kT}\right); \text{ where } x \text{ is the distance, } k \text{ is the}$$

Boltzmann constant and T is temperature and α and β are two constants. The dimensions of β is: [11 Jan. 2019 I]

- (a) $M^0L^2T^{-4}$ (b) M^2LT^{-4}
(c) MLT^{-2} (d) $M^2L^2T^{-2}$
14. If speed (V), acceleration (A) and force (F) are considered as fundamental units, the dimension of Young's modulus will be : [11 Jan. 2019 II]
- (a) $V^{-2}A^2F^{-2}$ (b) $V^{-2}A^2F^2$
(c) $V^{-4}A^{-2}F$ (d) $V^{-4}A^2F$

15. A quantity f is given by $f = \frac{\sqrt{hc^5}}{G}$ where c is speed of light, G universal gravitational constant and h is the Planck's constant. Dimension of f is that of: [9 Jan. 2019 I]

- (a) area (b) energy
(c) momentum (d) volume

16. Expression for time in terms of G (universal gravitational constant), h (Planck's constant) and c (speed of light) is proportional to: [9 Jan. 2019 II]

- (a) $\sqrt{\frac{hc^5}{G}}$ (b) $\sqrt{\frac{c^3}{Gh}}$
(c) $\sqrt{\frac{Gh}{c^5}}$ (d) $\sqrt{\frac{Gh}{c^3}}$

17. The dimensions of stopping potential V_0 in photoelectric effect in units of Planck's constant ' h ', speed of light ' c ' and Gravitational constant ' G ' and ampere A is:

[8 Jan. 2019 I]

- (a) $h^{1/3}G^{2/3}c^{1/3}A^{-1}$ (b) $h^{2/3}c^{5/3}G^{1/3}A^{-1}$
(c) $h^{-2/3}e^{-1/3}G^{4/3}A^{-1}$ (d) $h^2G^{3/2}C^{1/3}A^{-1}$

18. The dimensions of $\frac{B^2}{2\mu_0}$, where B is magnetic field and μ_0 is the magnetic permeability of vacuum, is:

[8 Jan. 2019 II]

- (a) MLT^{-2} (b) ML^2T^{-1}
(c) ML^2T^{-2} (d) $ML^{-1}T^{-2}$

19. The characteristic distance at which quantum gravitational effects are significant, the Planck length, can be determined from a suitable combination of the fundamental physical constants G , h and c . Which of the following correctly gives the Planck length? [Online April 15, 2018]

- (a) G^2hc (b) $\left(\frac{Gh}{c^3}\right)^{\frac{1}{2}}$ (c) $G^{\frac{1}{2}}h^2c$ (d) Gh^2c^3

20. Time (T), velocity (C) and angular momentum (h) are chosen as fundamental quantities instead of mass, length and time. In terms of these, the dimensions of mass would be : [Online April 8, 2017]

- (a) $[M] = [T^{-1}C^{-2}h]$ (b) $[M] = [T^{-1}C^2h]$
(c) $[M] = [T^{-1}C^{-2}h^{-1}]$ (d) $[M] = [TC^{-2}h]$

21. A, B, C and D are four different physical quantities having different dimensions. None of them is dimensionless. But we know that the equation $AD = C \ln(BD)$ holds true. Then which of the combination is not a meaningful quantity?

[Online April 10, 2016]

- (a) $\frac{C}{BD} - \frac{AD^2}{C}$ (b) $A^2 - B^2C^2$
(c) $\frac{A}{B} - C$ (d) $\frac{(A-C)}{D}$

22. In the following 'I' refers to current and other symbols have their usual meaning, Choose the option that corresponds to the dimensions of electrical conductivity :

[Online April 9, 2016]

- (a) $M^{-1}L^{-3}T^3I$ (b) $M^{-1}L^{-3}T^3I^2$
(c) $M^{-1}L^3T^3I$ (d) $ML^{-3}T^{-3}I^2$

23. If electronic charge e , electron mass m , speed of light in vacuum c and Planck's constant h are taken as fundamental quantities, the permeability of vacuum μ_0 can be expressed in units of : [Online April 11, 2015]

- (a) $\left(\frac{h}{me^2}\right)$ (b) $\left(\frac{hc}{me^2}\right)$
(c) $\left(\frac{h}{ce^2}\right)$ (d) $\left(\frac{mc^2}{he^2}\right)$

24. If the capacitance of a nanocapacitor is measured in terms of a unit ' u ' made by combining the electric charge ' e ', Bohr radius ' a_0 ', Planck's constant ' h ' and speed of light ' c ' then: [Online April 10, 2015]

- (a) $u = \frac{e^2h}{a_0}$ (b) $u = \frac{hc}{e^2a_0}$
(c) $u = \frac{e^2c}{ha_0}$ (d) $u = \frac{e^2a_0}{hc}$

25. From the following combinations of physical constants (expressed through their usual symbols) the only combination, that would have the same value in different systems of units, is: [Online April 12, 2014]

- (a) $\frac{ch}{2\pi\epsilon_0^2}$
(b) $\frac{e^2}{2\pi\epsilon_0 Gm_e^2}$ (m_e = mass of electron)
(c) $\frac{\mu_0\epsilon_0}{c^2} \frac{G}{he^2}$
(d) $\frac{2\pi\sqrt{\mu_0\epsilon_0}}{ce^2} \frac{h}{G}$

26. In terms of resistance R and time T , the dimensions of ratio $\frac{\mu}{\epsilon}$ of the permeability μ and permittivity ϵ is:

[Online April 11, 2014]

- (a) $[RT^{-2}]$ (b) $[R^2T^{-1}]$ (c) $[R^2]$ (d) $[R^2T^2]$
27. Let $[\epsilon_0]$ denote the dimensional formula of the permittivity of vacuum. If M = mass, L = length, T = time and A = electric current, then:

[2013]

- (a) $\epsilon_0 = [M^{-1} L^{-3} T^2 A]$ (b) $\epsilon_0 = [M^{-1} L^{-3} T^4 A^2]$
- (c) $\epsilon_0 = [M^1 L^2 T^1 A^2]$ (d) $\epsilon_0 = [M^1 L^2 T^1 A]$
28. If the time period t of the oscillation of a drop of liquid of density d , radius r , vibrating under surface tension s is given by the formula $t = \sqrt{r^{2b} s^c d^{a/2}}$. It is observed that the

time period is directly proportional to $\sqrt{\frac{d}{s}}$. The value of b should therefore be :

[Online April 23, 2013]

- (a) $\frac{3}{4}$ (b) $\sqrt{3}$
- (c) $\frac{3}{2}$ (d) $\frac{2}{3}$
29. The dimensions of angular momentum, latent heat and capacitance are, respectively. [Online April 22, 2013]
- (a) $ML^2T^1A^2, L^2T^{-2}, M^{-1}L^{-2}T^{-2}$
- (b) $ML^2T^{-2}, L^2T^2, M^{-1}L^{-2}T^4A^2$
- (c) $ML^2T^{-1}, L^2T^{-2}, ML^2TA^2$
- (d) $ML^2T^{-1}, L^2T^{-2}, M^{-1}L^{-2}T^4A^2$
30. Given that K = energy, V = velocity, T = time. If they are chosen as the fundamental units, then what is dimensional formula for surface tension? [Online May 7, 2012]
- (a) $[KV^{-2}T^{-2}]$ (b) $[K^2V^2T^{-2}]$
- (c) $[K^2V^{-2}T^{-2}]$ (d) $[KV^2T^2]$
31. The dimensions of magnetic field in M, L, T and C (coulomb) is given as [2008]
- (a) $[MLT^{-1}C^{-1}]$ (b) $[MT^2C^{-2}]$
- (c) $[MT^{-1}C^{-1}]$ (d) $[MT^{-2}C^{-1}]$
32. Which of the following units denotes the dimension $\frac{ML^2}{Q^2}$, where Q denotes the electric charge? [2006]
- (a) Wb/m^2 (b) Henry (H)
- (c) H/m^2 (d) Weber (Wb)
33. Out of the following pair, which one does NOT have identical dimensions? [2005]
- (a) Impulse and momentum
- (b) Angular momentum and planck's constant
- (c) Work and torque
- (d) Moment of inertia and moment of a force
34. Which one of the following represents the correct dimensions of the coefficient of viscosity? [2004]

- (a) $[ML^{-1}T^{-1}]$ (b) $[MLT^{-1}]$
- (c) $[ML^{-1}T^{-2}]$ (d) $[ML^{-2}T^{-2}]$

35. Dimensions of $\frac{1}{\mu_0 \epsilon_0}$, where symbols have their usual meaning, are [2003]

- (a) $[L^{-1}T]$ (b) $[L^{-2}T^2]$
- (c) $[L^2T^{-2}]$ (d) $[LT^{-1}]$

36. The physical quantities not having same dimensions are [2003]

- (a) torque and work
- (b) momentum and planck's constant
- (c) stress and young's modulus

- (d) speed and $(\mu_0 \epsilon_0)^{-1/2}$

37. Identify the pair whose dimensions are equal [2002]

- (a) torque and work (b) stress and energy
- (c) force and stress (d) force and work

TOPIC 3 Errors in Measurements



38. A screw gauge has 50 divisions on its circular scale. The circular scale is 4 units ahead of the pitch scale marking, prior to use. Upon one complete rotation of the circular scale, a displacement of 0.5 mm is noticed on the pitch scale. The nature of zero error involved, and the least count of the screw gauge, are respectively :

[Sep. 06, 2020 (I)]

- (a) Negative, $2 \mu m$ (b) Positive, $10 \mu m$
- (c) Positive, $0.1 mm$ (d) Positive, $0.1 \mu m$

39. The density of a solid metal sphere is determined by measuring its mass and its diameter. The maximum error in the density of the sphere is $\left(\frac{x}{100}\right)\%$. If the relative errors in measuring the mass and the diameter are 6.0% and 1.5% respectively, the value of x is _____.

[NA Sep. 06, 2020 (I)]

40. A student measuring the diameter of a pencil of circular cross-section with the help of a vernier scale records the following four readings 5.50 mm, 5.55 mm, 5.45 mm, 5.65 mm. The average of these four reading is 5.5375 mm and the standard deviation of the data is 0.07395 mm. The average diameter of the pencil should therefore be recorded as :

[Sep. 06, 2020 (II)]

- (a) $(5.5375 \pm 0.0739) mm$ (b) $(5.5375 \pm 0.0740) mm$
- (c) $(5.538 \pm 0.074) mm$ (d) $(5.54 \pm 0.07) mm$

41. A physical quantity z depends on four observables a, b, c and d , as $z = \frac{a^2 b^3}{\sqrt{c} d^3}$. The percentages of error in the measurement of a, b, c and d are 2%, 1.5%, 4% and 2.5% respectively. The percentage of error in z is :

[Sep. 05, 2020 (I)]

- (a) 12.25% (b) 16.5%
(c) 13.5% (d) 14.5%
42. Using screw gauge of pitch 0.1 cm and 50 divisions on its circular scale, the thickness of an object is measured. It should correctly be recorded as : [Sep. 03, 2020 (I)]
(a) 2.121 cm (b) 2.124 cm
(c) 2.125 cm (d) 2.123 cm
43. The least count of the main scale of a vernier callipers is 1 mm. Its vernier scale is divided into 10 divisions and coincide with 9 divisions of the main scale. When jaws are touching each other, the 7th division of vernier scale coincides with a division of main scale and the zero of vernier scale is lying right side of the zero of main scale. When this vernier is used to measure length of a cylinder the zero of the vernier scale between 3.1 cm and 3.2 cm and 4th VSD coincides with a main scale division. The length of the cylinder is : (VSD is vernier scale division) [Sep. 02, 2020 (I)]
(a) 3.2 cm (b) 3.21 cm
(c) 3.07 cm (d) 2.99 cm
44. If the screw on a screw-gauge is given six rotations, it moves by 3 mm on the main scale. If there are 50 divisions on the circular scale the least count of the screw gauge is: [9 Jan. 2020 I]
(a) 0.001 cm (b) 0.02 mm
(c) 0.01 cm (d) 0.001 mm
45. For the four sets of three measured physical quantities as given below. Which of the following options is correct? [9 Jan. 2020 II]
(A) $A_1 = 24.36, B_1 = 0.0724, C_1 = 256.2$
(B) $A_2 = 24.44, B_2 = 16.082, C_2 = 240.2$
(C) $A_3 = 25.2, B_3 = 19.2812, C_3 = 236.183$
(D) $A_4 = 25, B_4 = 236.191, C_4 = 19.5$
(a) $A_4 + B_4 + C_4 < A_1 + B_1 + C_1 < A_3 + B_3 + C_3 < A_2 + B_2 + C_2$
(b) $A_1 + B_1 + C_1 = A_2 + B_2 + C_2 = A_3 + B_3 + C_3 = A_4 + B_4 + C_4$
(c) $A_4 + B_4 + C_4 < A_1 + B_1 + C_1 = A_2 + B_2 + C_2 = A_3 + B_3 + C_3$
(d) $A_1 + B_1 + C_1 < A_3 + B_3 + C_3 < A_2 + B_2 + C_2 < A_4 + B_4 + C_4$
46. A simple pendulum is being used to determine the value of gravitational acceleration g at a certain place. The length of the pendulum is 25.0 cm and a stop watch with 1 s resolution measures the time taken for 40 oscillations to be 50 s. The accuracy in g is: [8 Jan. 2020 II]
(a) 5.40% (b) 3.40%
(c) 4.40% (d) 2.40%
47. In the density measurement of a cube, the mass and edge length are measured as (10.00 ± 0.10) kg and (0.10 ± 0.01) m, respectively. The error in the measurement of density is: [9 April 2019 I]
(a) 0.01 kg/m³ (b) 0.10 kg/m³
(c) 0.013 kg/m³ (d) 0.07 kg/m³
48. The area of a square is 5.29 cm². The area of 7 such squares taking into account the significant figures is: [9 April 2019 II]
(a) 37 cm² (b) 37.030 cm²
(c) 37.03 cm² (d) 37.0 cm²
49. In a simple pendulum experiment for determination of acceleration due to gravity (g), time taken for 20 oscillations is measured by using a watch of 1 second least count. The mean value of time taken comes out to be 30 s. The length of pendulum is measured by using a meter scale of least count 1 mm and the value obtained is 55.0 cm. The percentage error in the determination of g is close to : [8 April 2019 II]
(a) 0.7% (b) 0.2% (c) 3.5% (d) 6.8%
50. The least count of the main scale of a screw gauge is 1 mm. The minimum number of divisions on its circular scale required to measure 5 μ m diameter of a wire is: [12 Jan. 2019 I]
(a) 50 (b) 200 (c) 100 (d) 500
51. The diameter and height of a cylinder are measured by a meter scale to be 12.6 ± 0.1 cm and 34.2 ± 0.1 cm, respectively. What will be the value of its volume in appropriate significant figures? [10 Jan. 2019 II]
(a) 4264 ± 81 cm³ (b) 4264.4 ± 81.0 cm³
(c) 4260 ± 80 cm³ (d) 4300 ± 80 cm³
52. The pitch and the number of divisions, on the circular scale for a given screw gauge are 0.5 mm and 100 respectively. When the screw gauge is fully tightened without any object, the zero of its circular scale lies 3 division below the mean line. The readings of the main scale and the circular scale, for a thin sheet, are 5.5 mm and 48 respectively, the thickness of the sheet is: [9 Jan. 2019 II]
(a) 5.755 mm (b) 5.950 mm
(c) 5.725 mm (d) 5.740 mm
53. The density of a material in the shape of a cube is determined by measuring three sides of the cube and its mass. If the relative errors in measuring the mass and length are respectively 1.5% and 1%, the maximum error in determining the density is: [2018]
(a) 2.5% (b) 3.5% (c) 4.5% (d) 6%
54. The percentage errors in quantities P, Q, R and S are 0.5%, 1%, 3% and 1.5% respectively in the measurement of a physical quantity $A = \frac{P^3 Q^2}{\sqrt{RS}}$. The maximum percentage error in the value of A will be [Online April 16, 2018]
(a) 8.5% (b) 6.0%
(c) 7.5% (d) 6.5%
55. The relative uncertainty in the period of a satellite orbiting around the earth is 10^{-2} . If the relative uncertainty in the radius of the orbit is negligible, the relative uncertainty in the mass of the earth is [Online April 16, 2018]
(a) 3×10^{-2} (b) 10^{-2}
(c) 2×10^{-2} (d) 6×10^{-2}

56. The relative error in the determination of the surface area of a sphere is α . Then the relative error in the determination of its volume is **[Online April 15, 2018]**

(a) $\frac{2}{3}\alpha$ (b) $\frac{2}{3}\alpha$ (c) $\frac{3}{2}\alpha$ (d) α

57. In a screw gauge, 5 complete rotations of the screw cause it to move a linear distance of 0.25 cm. There are 100 circular scale divisions. The thickness of a wire measured by this screw gauge gives a reading of 4 main scale divisions and 30 circular scale divisions. Assuming negligible zero error, the thickness of the wire is: **[Online April 15, 2018]**

(a) 0.0430 cm (b) 0.3150 cm
(c) 0.4300 cm (d) 0.2150 cm

58. The following observations were taken for determining surface tension T of water by capillary method :

Diameter of capillary, $D = 1.25 \times 10^{-2}$ m

rise of water, $h = 1.45 \times 10^{-2}$ m

Using $g = 9.80$ m/s² and the simplified relation

$T = \frac{r h g}{2} \times 10^3$ N/m, the possible error in surface tension is closest to : **[2017]**

(a) 2.4 % (b) 10 % (c) 0.15 % (d) 1.5 %

59. A physical quantity P is described by the relation $P = a^{1/2} b^2 c^3 d^{-4}$

If the relative errors in the measurement of a , b , c and d respectively, are 2%, 1%, 3% and 5%, then the relative error in P will be : **[Online April 9, 2017]**

(a) 8 % (b) 12 % (c) 32 % (d) 25 %

60. A screw gauge with a pitch of 0.5 mm and a circular scale with 50 divisions is used to measure the thickness of a thin sheet of Aluminium. Before starting the measurement, it is found that when the two jaws of the screw gauge are brought in contact, the 45th division coincides with the main scale line and the zero of the main scale is barely visible. What is the thickness of the sheet if the main scale reading is 0.5 mm and the 25th division coincides with the main scale line? **[2016]**

(a) 0.70 mm (b) 0.50 mm
(c) 0.75 mm (d) 0.80 mm

61. A student measures the time period of 100 oscillations of a simple pendulum four times. The data set is 90 s, 91 s, 95 s, and 92 s. If the minimum division in the measuring clock is 1 s, then the reported mean time should be: **[2016]**

(a) 92 ± 1.8 s (b) 92 ± 3 s
(c) 92 ± 1.5 s (d) 92 ± 5.0 s

62. The period of oscillation of a simple pendulum is

$T = 2\pi \sqrt{\frac{L}{g}}$. Measured value of L is 20.0 cm known to 1 mm accuracy and time for 100 oscillations of the pendulum is found to be 90 s using a wrist watch of 1 s resolution. The accuracy in the determination of g is : **[2015]**

(a) 1 % (b) 5 % (c) 2 % (d) 3 %

63. Diameter of a steel ball is measured using a Vernier callipers which has divisions of 0.1 cm on its main scale (MS) and 10 divisions of its vernier scale (VS) match 9 divisions on the main scale. Three such measurements for a ball are given as: **[Online April 10, 2015]**

S.No.	MS(cm)	VS divisions
1.	0.5	8
2.	0.5	4
3.	0.5	6

If the zero error is -0.03 cm, then mean corrected diameter is:

(a) 0.52 cm (b) 0.59 cm
(c) 0.56 cm (d) 0.53 cm

64. The current voltage relation of a diode is given by

$I = (e^{1000V/T} - 1)$ mA, where the applied voltage V is in volts and the temperature T is in degree kelvin. If a student makes an error measuring ± 0.01 V while measuring the current of 5 mA at 300 K, what will be the error in the value of current in mA? **[2014]**

(a) 0.2 mA (b) 0.02 mA (c) 0.5 mA (d) 0.05 mA

65. A student measured the length of a rod and wrote it as 3.50 cm. Which instrument did he use to measure it? **[2014]**

- (a) A meter scale.
(b) A vernier calliper where the 10 divisions in vernier scale matches with 9 division in main scale and main scale has 10 divisions in 1 cm.
(c) A screw gauge having 100 divisions in the circular scale and pitch as 1 mm.
(d) A screw gauge having 50 divisions in the circular scale and pitch as 1 mm.

66. Match List - I (Event) with List-II (Order of the time interval for happening of the event) and select the correct option from the options given below the lists:

[Online April 19, 2014]

List - I		List - II	
(1)	Rotation period of earth	(i)	10^5 s
(2)	Revolution period of earth	(ii)	10^7 s
(3)	Period of light wave	(iii)	10^{-15} s
(4)	Period of sound wave	(iv)	10^{-3} s

- (a) (1)-(i), (2)-(ii), (3)-(iii), (4)-(iv)
(b) (1)-(ii), (2)-(i), (3)-(iv), (4)-(iii)
(c) (1)-(i), (2)-(ii), (3)-(iv), (4)-(iii)
(d) (1)-(ii), (2)-(i), (3)-(iii), (4)-(iv)

67. In the experiment of calibration of voltmeter, a standard cell of e.m.f. 1.1 volt is balanced against 440 cm of potential wire. The potential difference across the ends of resistance is found to balance against 220 cm of the wire. The corresponding reading of voltmeter is 0.5 volt. The error in the reading of voltmeter will be: **[Online April 12, 2014]**
 (a) -0.15 volt (b) 0.15 volt
 (c) 0.5 volt (d) -0.05 volt
68. An experiment is performed to obtain the value of acceleration due to gravity g by using a simple pendulum of length L . In this experiment time for 100 oscillations is measured by using a watch of 1 second least count and the value is 90.0 seconds. The length L is measured by using a meter scale of least count 1 mm and the value is 20.0 cm. The error in the determination of g would be:
[Online April 9, 2014]
 (a) 1.7% (b) 2.7% (c) 4.4% (d) 2.27%
69. Resistance of a given wire is obtained by measuring the current flowing in it and the voltage difference applied across it. If the percentage errors in the measurement of the current and the voltage difference are 3% each, then error in the value of resistance of the wire is **[2012]**
 (a) 6% (b) zero (c) 1% (d) 3%
70. A spectrometer gives the following reading when used to measure the angle of a prism.
 Main scale reading : 58.5 degree
 Vernier scale reading : 09 divisions
 Given that 1 division on main scale corresponds to 0.5 degree. Total divisions on the Vernier scale is 30 and match with 29 divisions of the main scale. The angle of the prism from the above data is **[2012]**
 (a) 58.59 degree (b) 58.77 degree
 (c) 58.65 degree (d) 59 degree
71. N divisions on the main scale of a vernier calliper coincide with $(N+1)$ divisions of the vernier scale. If each division of main scale is ' a ' units, then the least count of the instrument is **[Online May 19, 2012]**
 (a) a (b) $\frac{a}{N}$
 (c) $\frac{N}{N+1} \times a$ (d) $\frac{a}{N+1}$
72. A student measured the diameter of a wire using a screw gauge with the least count 0.001 cm and listed the measurements. The measured value should be recorded as **[Online May 12, 2012]**
 (a) 5.3200 cm (b) 5.3 cm
 (c) 5.32 cm (d) 5.320 cm
73. A screw gauge gives the following reading when used to measure the diameter of a wire.
 Main scale reading : 0 mm
 Circular scale reading : 52 divisions
 Given that 1 mm on main scale corresponds to 100 divisions of the circular scale. The diameter of wire from the above data is **[2011]**
 (a) 0.052 cm (b) 0.026 cm
 (c) 0.005 cm (d) 0.52 cm
74. The respective number of significant figures for the numbers 23.023, 0.0003 and 2.1×10^{-3} are **[2010]**
 (a) 5, 1, 2 (b) 5, 1, 5
 (c) 5, 5, 2 (d) 4, 4, 2
75. In an experiment the angles are required to be measured using an instrument, 29 divisions of the main scale exactly coincide with the 30 divisions of the vernier scale. If the smallest division of the main scale is half- a degree ($= 0.5^\circ$), then the least count of the instrument is: **[2009]**
 (a) half minute (b) one degree
 (c) half degree (d) one minute
76. A body of mass $m = 3.513$ kg is moving along the x-axis with a speed of 5.00 ms^{-1} . The magnitude of its momentum is recorded as **[2008]**
 (a) 17.6 kg ms^{-1} (b) $17.565 \text{ kg ms}^{-1}$
 (c) 17.56 kg ms^{-1} (d) 17.57 kg ms^{-1}
77. Two full turns of the circular scale of a screw gauge cover a distance of 1 mm on its main scale. The total number of divisions on the circular scale is 50. Further, it is found that the screw gauge has a zero error of -0.03 mm. While measuring the diameter of a thin wire, a student notes the main scale reading of 3 mm and the number of circular scale divisions in line with the main scale as 35. The diameter of the wire is **[2008]**
 (a) 3.32 mm (b) 3.73 mm
 (c) 3.67 mm (d) 3.38 mm



Hints & Solutions



1. (a) Density of material in SI unit,

$$= \frac{128 \text{ kg}}{\text{m}^3}$$

Density of material in new system

$$= \frac{128(50 \text{ g})(20)}{(25 \text{ cm})^3 (4)^3} = \frac{128}{64} (20) = 40 \text{ units}$$

2. (b) According to question

$$E_y \propto J_x B_z$$

$\therefore$ Constant of proportionality

$$K = \frac{E_y}{B_z J_x} = \frac{C}{J_x} = \frac{\text{m}^3}{\text{As}}$$

$$[\text{As } \frac{E}{B} = C \text{ (speed of light) and } J = \frac{I}{\text{Area}}]$$

3. (a) We know that

$$\text{Speed of light, } c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = x$$

$$\text{Also, } c = \frac{E}{B} = y$$

$$\text{Time constant, } \tau = Rc = t$$

$$\therefore z = \frac{l}{Rc} = \frac{l}{t} = \text{Speed}$$

Thus, x, y, z will have the same dimension of speed.

4. (d) From formula, $\frac{dQ}{dt} = kA \frac{dT}{dx}$

$$\Rightarrow k = \frac{\left(\frac{dQ}{dt}\right)}{A \left(\frac{dT}{dx}\right)}$$

$$[k] = \frac{[\text{ML}^2 \text{T}^{-3}]}{[\text{L}^2][\text{KL}^{-1}]} = [\text{MLT}^{-3} \text{K}^{-1}]$$

5. (c) Dimension of Force $F = \text{ML}^1 \text{T}^{-2}$

$$\text{Dimension of velocity } V = \text{L}^1 \text{T}^{-1}$$

$$\text{Dimension of work} = \text{ML}^2 \text{T}^{-2}$$

$$\text{Dimension of length} = \text{L}$$

$$\text{Moment of inertia} = \text{ML}^2$$

$$\therefore x = \frac{IFV^2}{WL^4}$$

$$= \frac{(\text{M}^1 \text{L}^2)(\text{M}^1 \text{L}^1 \text{T}^{-2})(\text{L}^1 \text{T}^{-2})^2}{(\text{M}^1 \text{L}^2 \text{T}^{-2})(\text{L}^4)}$$

$$= \frac{\text{M}^1 \text{L}^{-2} \text{T}^{-2}}{\text{L}^3} = \text{M}^1 \text{L}^{-1} \text{T}^{-2} = \text{Energy density}$$

6. (b) Solar constant = $\frac{\text{Energy}}{\text{Time Area}}$

$$\text{Dimension of Energy, } E = \text{ML}^2 \text{T}^{-2}$$

$$\text{Dimension of Time} = \text{T}$$

$$\text{Dimension of Area} = \text{L}^2$$

$\therefore$ Dimension of Solar constant

$$= \frac{\text{M}^1 \text{L}^2 \text{T}^{-2}}{\text{TL}^2} = \text{M}^1 \text{L}^0 \text{T}^{-3}$$

7. (d) Young's modulus, $Y = \frac{\text{stress}}{\text{strain}}$

$$\Rightarrow Y = \frac{F}{A} \bigg/ \frac{\Delta \ell}{\ell_0} = F A^{-1} V^0$$

8. (c) Energy, $E \propto A^a T^b P^c$

$$\text{or, } E = k A^a T^b P^c \quad \dots(i)$$

where k is a dimensionless constant and a, b and c are the exponents.

$$\text{Dimension of momentum, } P = \text{M}^1 \text{L}^1 \text{T}^{-1}$$

$$\text{Dimension of area, } A = \text{L}^2$$

$$\text{Dimension of time, } T = \text{T}^1$$

Putting these value in equation (i), we get

$$\text{M}^1 \text{L}^2 \text{T}^{-2} = \text{M}^c \text{L}^{2a+c} \text{T}^{b-c}$$

by comparison

$$c = 1$$

$$2a + c = 2$$

$$b - c = -2$$

$$c = 1, a = 1/2, b = -1$$

$$\therefore E = A^{1/2} T^{-1} P^1$$

9. (a) $\sqrt{\frac{\mu_0}{\epsilon_0}} = \sqrt{\frac{\mu_0^2}{\epsilon_0 \mu_0}} = \mu_0 c \quad \left(\frac{1}{\sqrt{\mu_0 \epsilon_0}} = c \right)$

$$\mu_0 c \rightarrow \text{MLT}^{-2} \text{A}^{-2} \times \text{LT}^{-1}$$

$$\text{ML}^2 \text{T}^{-3} \text{A}^{-2}$$

Dimensions of resistance

10. (a) $X = 5YZ^2$

$$\Rightarrow Y \propto \frac{X}{Z^2} \quad \dots(i)$$

$$X = \text{Capacitance} = \frac{Q}{V} = \frac{Q^2}{W} = \frac{[A^2 T^2]}{[ML^2 T^{-2}]}$$

$$X = [M^{-1} L^{-2} T^4 A^2]$$

$$Z = B = \frac{F}{IL} \quad [\because F = ILB]$$

$$Z = [MT^{-2} A^{-1}]$$

$$Y = \frac{[M^{-1} L^{-2} T^4 A^2]}{[MT^{-2} A^{-1}]^2}$$

$$Y = [M^{-3} L^{-2} T^8 A^4] \quad (\text{Using (i)})$$

$$11. (d) \left[\sqrt{\frac{\epsilon_0}{\mu_0}} \right] = \sqrt{\frac{\epsilon_0^2}{\mu_0 \epsilon_0}} = \left[\frac{\epsilon_0}{\sqrt{\mu_0 \epsilon_0}} \right] = \epsilon_0 C [LT^{-1}] \times [\epsilon_0]$$

$$\left[\because \frac{1}{\sqrt{\mu_0 \epsilon_0}} = C \right]$$

$$\therefore F = \frac{q^2}{4\pi\epsilon_0 r^2}$$

$$\Rightarrow [\epsilon_0] = \frac{[AT]^2}{[MLT^{-2}] \times [L^2]} = [A^2 M^{-1} L^{-3} T^4]$$

$$\therefore \left[\sqrt{\frac{\epsilon_0}{\mu_0}} \right] = [LT^{-1}] \times [A^2 M^{-1} L^{-3} T^4]$$

$$= [M^{-1} L^{-2} T^3 A^2]$$

$$12. (b) \text{ As we know,}$$

$$\left[\frac{\ell}{r} \right] = [T] \text{ and } [cv] = [AT]$$

$$\therefore \left[\frac{\ell}{rcv} \right] = \left[\frac{T}{AT} \right] = [A^{-1}]$$

$$13. (b) \text{ Force of interaction between two atoms,}$$

$$F = \alpha \beta e^{\left(\frac{-x^2}{\alpha k T} \right)}$$

Since exponential terms are dimensionless

$$\therefore \left[\frac{x^2}{\alpha k T} \right] = M^0 L^0 T^0$$

$$\Rightarrow \frac{L^2}{[\alpha] ML^2 T^{-2}} = M^0 L^0 T^0$$

$$\Rightarrow [\alpha] = M^{-1} T^2$$

$$[F] = [\alpha] [\beta]$$

$$MLT^{-2} = M^{-1} T^2 [\beta]$$

$$\Rightarrow [\beta] = M^2 LT^{-4}$$

$$14. (d) \text{ Let } [Y] = [V]^a [F]^b [A]^c$$

$$[ML^{-1} T^{-2}] = [LT^{-1}]^a [MLT^{-2}]^b [LT^{-2}]^c$$

$$[ML^{-1} T^{-2}] = [M^b L^{a+b+c} T^{-a-2b-2c}]$$

Comparing power both side of similar terms we get,

$$b = 1, a + b + c = -1, -a - 2b - 2c = -2$$

solving above equations we get:

$$a = -4, b = 1, c = 2$$

$$\text{so } [Y] = [V^{-4} F A^2] = [V^{-4} A^2 F]$$

$$15. (b) \text{ Dimension of } [h] = [ML^2 T^{-1}]$$

$$[C] = [LT^{-1}]$$

$$[G] = [M^{-1} L^3 T^{-2}]$$

Hence dimension of

$$\left[\sqrt{\frac{hC^5}{G}} \right] = \frac{[ML^2 T^{-1}] \cdot [L^5 T^{-5}]}{[M^{-1} L^3 T^{-2}]}$$

$$= [ML^2 T^{-2}] = \text{energy}$$

$$16. (c) \text{ Let } t \propto G^x h^y C^z$$

$$\text{Dimensions of } G = [M^{-1} L^3 T^{-2}],$$

$$h = [ML^2 T^{-1}] \text{ and } C = [LT^{-1}]$$

$$[T] = [M^{-1} L^3 T^{-2}]^x [ML^2 T^{-1}]^y [LT^{-1}]^z$$

$$[M^0 L^0 T^1] = [M^{-x+y} L^{3x+2y+z} T^{-2x-y-z}]$$

By comparing the powers of M, L, T both the sides

$$-x + y = 0 \Rightarrow x = y$$

$$3x + 2y + z = 0 \Rightarrow 5x + z = 0 \quad \dots (i)$$

$$-2x - y - z = 1 \Rightarrow 3x + z = -1 \quad \dots (ii)$$

Solving eqns. (i) and (ii),

$$x = y = \frac{1}{2}, z = -\frac{5}{2} \therefore t \propto \sqrt{\frac{Gh}{C^5}}$$

$$17. (\text{None})$$

$$\text{Stopping potential } (V_0) \propto h^x I^y G^z C^r$$

$$\text{Here, } h = \text{Planck's constant} = [ML^2 T^{-1}]$$

$$I = \text{current} = [A]$$

$$G = \text{Gravitational constant} = [M^{-1} L^3 T^{-2}]$$

$$\text{and } c = \text{speed of light} = [LT^{-1}]$$

$$V_0 = \text{potential} = [ML^2 T^{-3} A^{-1}]$$

$$\therefore [ML^2 T^{-3} A^{-1}] = [ML^2 T^{-1}]^x [A]^y [M^{-1} L^3 T^{-2}]^z [LT^{-1}]^r$$

$$M^{x-z}, L^{2x+3z+r}, T^{-x-2z-r}, A^y$$

Comparing dimension of M, L, T, A, we get

$$y = -1, x = 0, z = -1, r = 5$$

$$\therefore V_0 \propto h^0 I^{-1} G^{-1} C^5$$

$$18. (d) \text{ The quantity } \frac{B^2}{2\mu_0} \text{ is the energy density of magnetic field.}$$

$$\Rightarrow \left[\frac{B^2}{2\mu_0} \right] = \frac{\text{Energy}}{\text{Volume}} = \frac{\text{Force} \times \text{displacement}}{(\text{displacement})^3}$$

$$= \left[\frac{ML^2 T^{-2}}{L^3} \right] = ML^{-1} T^{-2}$$

19. (b) Planck length is a unit of length, $l_p = 1.616229 \times 10^{-35} \text{ m}$

$$l_p = \sqrt{\frac{hG}{c^3}}$$

20. (a) Let mass, related as $M \propto T^x C^y h^z$

$$M^1 L^0 T^0 = (T)^x (L^1 T^{-1})^y (M^1 L^2 T^{-1})^z$$

$$M^1 L^0 T^0 = M^z L^{y+2z} T^{x-y-z}$$

$$z = 1$$

$$y + 2z = 0 \quad x - y - z = 0$$

$$y = -2 \quad x + 2 - 1 = 0$$

$$x = -1$$

$$M = [T^{-1} C^{-2} h^1]$$

21. (d) Dimension of A $\neq$ dimension of (C)

Hence A – C is not possible.

22. (b) We know that resistivity

$$\rho = \frac{RA}{\ell}$$

$$\text{Conductivity} = \frac{1}{\text{resistivity}} = \frac{\ell}{RA}$$

$$= \frac{\ell I}{VA} \quad (\because V = RI)$$

$$= \frac{[L][I]}{[ML^2 T^{-2}][I][T]} \quad \because V = \frac{W}{q} = \frac{W}{it}$$

$$= [M^{-1} L^{-3} T^3][I^2] = [M^{-1} L^{-3} T^3 I^2]$$

23. (c) Let μ_0 related with e , m , c and h as follows.

$$\mu_0 = k e^a m^b c^c h^d$$

$$[MLT^{-2} A^{-2}] = [AT]^a [M]^b [LT^{-1}]^c [ML^2 T^{-1}]^d$$

$$= [M^{b+d} L^{c+2d} T^{a-c-d} A^a]$$

On comparing both sides we get

$$a = -2 \quad \dots (i)$$

$$b + d = 1 \quad \dots (ii)$$

$$c + 2d = 1 \quad \dots (iii)$$

$$a - c - d = -2 \quad \dots (iv)$$

By equation (i), (ii), (iii) & (iv) we get,

$$a = -2, b = 0, c = -1, d = 1$$

$$\therefore [\mu_0] = \left[\frac{h}{ce^2} \right]$$

24. (d) Let unit 'u' related with e , a_0 , h and c as follows.

$$[u] = [e]^a [a_0]^b [h]^c [C]^d$$

Using dimensional method,

$$[M^{-1} L^{-2} T^{+4} A^{+2}] = [A^1 T^1]^a [L]^b [ML^2 T^{-1}]^c [LT^{-1}]^d$$

$$[M^{-1} L^{-2} T^{+4} A^{+2}] = [M^c L^{b+2c+d} T^{a-c-d} A^a]$$

$$a = 2, b = 1, c = -1, d = -1$$

$$\therefore u = \frac{e^2 a_0}{hc}$$

25. (b) The dimensional formulae of

$$e = [M^0 L^0 T^1 A^1]$$

$$\epsilon_0 = [M^{-1} L^3 T^4 A^2]$$

$$G = [M^{-1} L^3 T^{-2}] \text{ and } m_e = [M^1 L^0 T^0]$$

$$\text{Now, } \frac{e^2}{2\pi\epsilon_0 G m_e^2}$$

$$= \frac{[M^0 L^0 T^1 A^1]^2}{2\pi [M^{-1} L^3 T^4 A^2] [M^{-1} L^3 T^{-2}] [M^1 L^0 T^0]^2}$$

$$= \frac{[T^2 A^2]}{2\pi [M^{-1-1+2} L^{-3+3} T^{4-2} A^2]}$$

$$= \frac{[T^2 A^2]}{2\pi [M^0 L^0 T^2 A^2]} = \frac{1}{2\pi}$$

$$\therefore \frac{1}{2\pi} \text{ is dimensionless thus the combination } \frac{e^2}{2\pi\epsilon_0 G m_e^2}$$

would have the same value in different systems of units.

26. (c) Dimensions of $\mu = [MLT^{-2} A^{-2}]$

$$\text{Dimensions of } \epsilon = [M^{-1} L^{-3} T^4 A^2]$$

$$\text{Dimensions of } R = [ML^2 T^{-3} A^{-2}]$$

$$\therefore \frac{\text{Dimensions of } \mu}{\text{Dimensions of } \epsilon} = \frac{[MLT^{-2} A^{-2}]}{[M^{-1} L^{-3} T^4 A^2]}$$

$$= [M^2 L^4 T^{-6} A^{-4}] = [R^2]$$

27. (b) As we know, $F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{R^2}$

$$\Rightarrow \epsilon_0 = \frac{q_1 q_2}{4\pi F R^2}$$

$$\text{Hence, } \epsilon_0 = \frac{C^2}{N \cdot m^2} = \frac{[AT]^2}{[MLT^{-2}][L^2]} = [M^{-1} L^{-3} T^4 A^2]$$

28. (c)

29. (d) Angular momentum = $m \times v \times r = ML^2 T^{-1}$

$$\text{Latent heat } L = \frac{Q}{m} = \frac{ML^2 T^{-2}}{M} = L^2 T^{-2}$$

$$\text{Capacitance } C = \frac{\text{Charge}}{\text{P.d.}} = M^{-1} L^{-2} T^4 A^2$$

30. (a) Surface tension, $T = \frac{F}{\ell} = \frac{F}{\ell} \cdot \frac{\ell}{\ell} \cdot \frac{T^2}{T^2}$

$$(\text{As, } F \cdot \ell = K (\text{energy}); \frac{T^2}{\ell^2} = V^{-2})$$

$$\text{Therefore, surface tension} = [KV^{-2} T^{-2}]$$

31. (c) Magnitude of Lorentz formula $F = qvB \sin \theta$

$$B = \frac{F}{qv} = \frac{MLT^{-2}}{C \times LT^{-1}} = [MT^{-1} C^{-1}]$$

32. (b) Mutual inductance $= \frac{\phi}{I} = \frac{BA}{I}$
 $[\text{Henry}] = \frac{[MT^{-1}Q^{-1}L^2]}{[QT^{-1}]} = ML^2Q^{-2}$

33. (d) Moment of Inertia, $I = MR^2$
 $[I] = [ML^2]$

Moment of force, $\vec{\tau} = \vec{r} \times \vec{F}$

$\vec{\tau} = [L][MLT^{-2}] = [ML^2T^{-2}]$

34. (a) According to, Stokes law,

$F = 6\pi\eta r v \Rightarrow \eta = \frac{F}{6\pi r v}$

$\eta = \frac{[MLT^{-2}]}{[L][LT^{-1}]} \Rightarrow \eta = [ML^{-1}T^{-1}]$

35. (c) As we know, the velocity of light in free space is given by

$c = \frac{1}{\sqrt{\mu_o \epsilon_o}} \therefore \frac{1}{\mu_o \epsilon_o} = c^2 = Z_0^2 T^2$

$\frac{1}{\mu_o \epsilon_o} = C^2 [m/s]^2$

$= [LT^{-1}]^2$

$= [M^0 L^2 T^{-2}]$

36. (b) Momentum, $= mv = [MLT^{-1}]$

Planck's constant,

$h = \frac{E}{\nu} = \frac{[ML^2T^{-2}]}{[T^{-1}]} = [ML^2T^{-1}]$

37. (a) Work $W = \vec{F} \cdot \vec{s} = Fs \cos \theta$

$\therefore \vec{A} \cdot \vec{B} = AB \cos \theta$

$= [MLT^{-2}][L] = [ML^2T^{-2}];$

Torque, $\vec{\tau} = \vec{r} \times \vec{F} \Rightarrow \tau = rF \sin \theta$

$\therefore \vec{A} \times \vec{B} = AB \sin \theta$

$= [L][MLT^{-2}] = [ML^2T^{-2}]$

38. (b) Given : No. of division on circular scale of screw gauge = 50

Pitch = 0.5 mm

Least count of screw gauge

$= \frac{\text{Pitch}}{\text{No. of division on circular scale}}$

$= \frac{0.5}{50} \text{ mm} = 1 \times 10^{-5} \text{ m} = 10 \mu\text{m}$

And nature of zero error is positive.

39. (1050)

Density, $\rho = \frac{M}{V} = \frac{M}{\frac{4}{3}\pi\left(\frac{D}{2}\right)^3} \Rightarrow \rho = \frac{6}{\pi} MD^{-3}$

$\therefore \% \left(\frac{\Delta \rho}{\rho} \right) = \frac{\Delta m}{m} + 3 \left(\frac{\Delta D}{D} \right) = 6 + 3 \times 1.5 = 10.5\%$

$\% \left(\frac{\Delta \rho}{\rho} \right) = \frac{1050}{100} \% = \left(\frac{x}{100} \right) \%$

$\therefore x = 1050.00$

40. (d) Average diameter, $d_{av} = 5.5375 \text{ mm}$

Deviation of data, $\Delta d = 0.07395 \text{ mm}$

As the measured data are upto two digits after decimal, therefore answer should be in two digits after decimal.

$\therefore d = (5.54 \pm 0.07) \text{ mm}$

41. (d) Given : $Z = \frac{a^2 b^{2/3}}{\sqrt{cd^3}}$

Percentage error in Z,

$= \frac{\Delta Z}{Z} = \frac{2\Delta a}{a} + \frac{2}{3} \frac{\Delta b}{b} + \frac{1}{2} \frac{\Delta c}{c} + \frac{3\Delta d}{d}$

$= 2 \times 2 + \frac{2}{3} \times 1.5 + \frac{1}{2} \times 4 + 3 \times 2.5 = 14.5\%.$

42. (a) Thickness = M.S. Reading + Circular Scale Reading (L.C.)

Here LC = $\frac{\text{Pitch}}{\text{Circular scale division}} = \frac{0.1}{50} = 0.002 \text{ cm per division}$

So, correct measurement is measurement of integral multiple of L.C.

43. (c) L.C. of vernier callipers = 1 MSD – 1 VSD

$= \left(1 - \frac{9}{10} \right) \times 1 = 0.1 \text{ mm} = 0.01 \text{ cm}$

Here 7th division of vernier scale coincides with a division of main scale and the zero of vernier scale is lying right side of the zero of main scale.

Zero error = $7 \times 0.1 = 0.7 \text{ mm} = 0.07 \text{ cm}.$

Length of the cylinder = measured value – zero error

$= (3.1 + 4 \times 0.01) - 0.07 = 3.07 \text{ cm}.$

44. (d) When screw on a screw-gauge is given six rotations, it moves by 3mm on the main scale

$\therefore \text{Pitch} = \frac{3}{6} = 0.5 \text{ mm}$

$\therefore \text{Least count L.C.} = \frac{\text{Pitch}}{\text{CSD}} = \frac{0.5 \text{ mm}}{50}$

$= \frac{1}{100} \text{ mm} = 0.01 \text{ mm} = 0.001 \text{ cm}$

45. (None)

$D_1 = A_1 + B_1 + C_1 = 24.36 + 0.0724 + 256.2 = 280.6$

$D_2 = A_2 + B_2 + C_2 = 24.44 + 16.082 + 240.2 = 280.7$

$D_3 = A_3 + B_3 + C_3 = 25.2 + 19.2812 + 236.183 = 280.7$

$$D_4 = A_4 + B_4 + C_4 = 25 + 236.191 + 19.5 = 281$$

None of the option matches.

46. (c) Given, Length of simple pendulum, $l = 25.0$ cm
Time of 40 oscillation, $T = 50$ s
Time period of pendulum

$$T = 2\pi\sqrt{\frac{l}{g}}$$

$$\Rightarrow T^2 = \frac{4\pi^2 l}{g} \Rightarrow g = \frac{4\pi^2 l}{T^2}$$

$$\Rightarrow \text{Fractional error in } g = \frac{\Delta g}{g} = \frac{\Delta l}{l} + \frac{2\Delta T}{T}$$

$$\Rightarrow \frac{\Delta g}{g} = \left(\frac{0.1}{25.0}\right) + 2\left(\frac{1}{50}\right) = 0.044$$

$$\therefore \text{Percentage error in } g = \frac{\Delta g}{g} \times 100 = 4.4\%$$

47. (Bonus) $\delta = \frac{M}{V} = \frac{M}{l^3} = Ml^{-3}$

$$\frac{\Delta \delta}{\delta} = \frac{\Delta M}{M} + 3 \frac{\Delta l}{l} = \frac{0.10}{10.00} + 3\left(\frac{0.01}{0.10}\right) = 0.31 \text{ kg/m}^3$$

48. (d) $A = 7 \times 5.29 = 37.03 \text{ cm}^2$
The result should have three significant figures, so
 $A = 37.0 \text{ cm}^2$

49. (d) We have

$$T = 2\pi\sqrt{\frac{l}{g}} \text{ or } g = 4\pi^2 \frac{l}{T^2}$$

$$\frac{\Delta g}{g} \times 100 = \frac{\Delta R}{Q} \times 100 + 2 \frac{\Delta T}{T} \times 100$$

$$= \frac{0.1}{55} \times 100 + 2\left(\frac{1}{30}\right) \times 100$$

$$= 0.18 + 6.67 = 6.8\%$$

50. (b) Least count of main scale of screw gauge = 1 mm
Least count of screw gauge

$$= \frac{\text{Pitch}}{\text{Number of division on circular scale}}$$

$$5 \times 10^{-6} = \frac{10^{-3}}{N}$$

$$\Rightarrow N = 200$$

51. (c)

52. (c) Least count of screw gauge,

$$LC = \frac{\text{Pitch}}{\text{No. of division}}$$

$$= 0.5 \times 10^{-3} = 0.5 \times 10^{-2} \text{ mm} + \text{ve error} = 3 \times 0.5 \times 10^{-2} \text{ mm}$$

$$= 1.5 \times 10^{-2} \text{ mm} = 0.015 \text{ mm}$$

$$\text{Reading} = \text{MSR} + \text{CSR} - (\text{+ve error})$$

$$= 5.5 \text{ mm} + (48 \times 0.5 \times 10^{-2}) - 0.015$$

$$= 5.5 + 0.24 - 0.015 = 5.725 \text{ mm}$$

53. (c) $= 1.5\% + 3(1\%) = 4.5\%$

54. (d) Maximum percentage error in A

$$= 3(\% \text{ error in } P) + 2(\% \text{ error in } Q)$$

$$+ \frac{1}{2}(\% \text{ error in } R) + 1(\% \text{ error in } S)$$

$$= 3 \times 0.5 + 2 \times 1 + \frac{1}{2} \times 3 + 1 \times 1.5$$

$$= 1.5 + 2 + 1.5 + 1.5 = 6.5\%$$

55. (c) From Kepler's law, time period of a satellite,

$$T = 2\pi\sqrt{\frac{r^3}{Gm}} \quad T^2 = \frac{4\pi^2}{GM} r^3$$

Relative uncertainty in the mass of the earth

$$\left|\frac{\Delta M}{M}\right| = 2 \frac{\Delta T}{T} = 2 \times 10^{-2} \quad (\because 4\pi \text{ \& } G \text{ constant and}$$

relative uncertainty in radius $\frac{\Delta r}{r}$ negligible)

56. (c) Relative error in Surface area, $\frac{\Delta s}{s} = 2 \times \frac{\Delta r}{r} = \alpha$ and

$$\text{relative error in volume, } \frac{\Delta v}{v} = 3 \times \frac{\Delta r}{r}$$

$\therefore$ Relative error in volume w.r.t. relative error in area,

$$\frac{\Delta v}{v} = \frac{3}{2} \alpha$$

57. (d) Least count = $\frac{\text{Value of 1 part on main scale}}{\text{Number of parts on vernier scale}}$

$$= \frac{0.25}{5 \times 100} \text{ cm} = 5 \times 10^{-4} \text{ cm}$$

$$\text{Reading} = 4 \times 0.05 \text{ cm} + 30 \times 5 \times 10^{-4} \text{ cm}$$

$$= (0.2 + 0.0150) \text{ cm} = 0.2150 \text{ cm (Thickness of wire)}$$

58. (d) Surface tension, $T = \frac{r h g}{2} \times 10^3$

Relative error in surface tension,

$$\frac{\Delta T}{T} = \frac{\Delta r}{r} + \frac{\Delta h}{h} + 0 \quad (\because g, 2 \text{ \& } 10^3 \text{ are constant})$$

Percentage error

$$100 \times \frac{\Delta T}{T} = \left(\frac{10^{-2} \times 0.01}{1.25 \times 10^{-2}} + \frac{10^{-2} \times 0.01}{1.45 \times 10^{-2}} \right) 100$$

$$= (0.8 + 0.689)$$

$$= (1.489) = 1.489\% \approx 1.5\%$$

59. (c) Given, $P = a^{1/2} b^2 c^2 d^{-4}$,

Maximum relative error,

$$\frac{\Delta P}{P} = \frac{1}{2} \frac{\Delta a}{a} + 2 \frac{\Delta b}{b} + 3 \frac{\Delta c}{c} + 4 \frac{\Delta d}{d}$$

$$= \frac{1}{2} \times 2 + 2 \times 1 + 3 \times 3 + 4 \times 5 = 32\%$$

60. (d) L.C. = $\frac{0.5}{50} = 0.01 \text{ mm}$

$$\text{Zero error} = 5 \times 0.01 = 0.05 \text{ mm (Negative)}$$

$$\text{Reading} = (0.5 + 25 \times 0.01) + 0.05 = 0.80 \text{ mm}$$

$$61. (c) \Delta T = \frac{|\Delta T_1| + |\Delta T_2| + |\Delta T_3| + |\Delta T_4|}{4}$$

$$= \frac{2+1+3+0}{4} = 1.5$$

As the resolution of measuring clock is 1.5 therefore the mean time should be 92 ± 1.5

$$62. (d) \text{ As, } g = 4\pi^2 \frac{L}{T^2}$$

$$\text{So, } \frac{\Delta g}{g} \times 100 = \frac{\Delta L}{L} \times 100 + 2 \frac{\Delta T}{T} \times 100$$

$$= \frac{0.1}{20} \times 100 + 2 \times \frac{1}{90} \times 100 = 2.72 \approx 3\%$$

$$63. (b) \text{ Least count} = \frac{0.1}{10} = 0.01 \text{ cm}$$

$$d_1 = 0.5 + 8 \times 0.01 + 0.03 = 0.61 \text{ cm}$$

$$d_2 = 0.5 + 4 \times 0.01 + 0.03 = 0.57 \text{ cm}$$

$$d_3 = 0.5 + 6 \times 0.01 + 0.03 = 0.59 \text{ cm}$$

$$\text{Mean diameter} = \frac{0.61 + 0.57 + 0.59}{3}$$

$$= 0.59 \text{ cm}$$

64. (a) The current voltage relation of diode is

$$I = (e^{1000 V/T} - 1) \text{ mA (given)}$$

$$\text{When, } I = 5 \text{ mA, } e^{1000 V/T} = 6 \text{ mA}$$

$$\text{Also, } dI = (e^{1000 V/T}) \times \frac{1000}{T}$$

$$\text{Error} = \pm 0.01 \text{ (By exponential function)}$$

$$= (6 \text{ mA}) \times \frac{1000}{300} \times (0.01) = 0.2 \text{ mA}$$

65. (b) Measured length of rod = 3.50 cm

For Vernier Scale with 1 Main Scale Division = 1 mm

9 Main Scale Division = 10 Vernier Scale Division,

Least count = 1 MSD - 1 VSD = 0.1 mm

66. (a) Rotation period of earth is about 24 hrs $\approx 10^5$ s

Revolution period of earth is about 365 days $\approx 10^7$ s

Speed of light wave $C = 3 \times 10^8$ m/s

Wavelength of visible light of spectrum

$$\lambda = 4000 - 7800 \text{ \AA}$$

$$C = f\lambda \left(\text{and } T = \frac{1}{f} \right)$$

Therefore period of light wave is 10^{-15} s (approx)

67. (d) In a voltmeter

$$V \propto l$$

$$V = kl$$

Now, it is given $E = 1.1$ volt for $l_1 = 440$ cm

and $V = 0.5$ volt for $l_2 = 220$ cm

Let the error in reading of voltmeter be ΔV then,

$$1.1 = 400 K \text{ and } (0.5 - \Delta V) = 220 K.$$

$$\Rightarrow \frac{1.1}{440} = \frac{0.5 - \Delta V}{220}$$

$$\therefore \Delta V = -0.05 \text{ volt}$$

68. (b) According to the question.

$$t = (90 \pm 1) \text{ or, } \frac{\Delta t}{t} = \frac{1}{90}$$

$$l = (20 \pm 0.1) \text{ or, } \frac{\Delta l}{l} = \frac{0.1}{20}$$

$$\frac{\Delta g}{g} \% = ?$$

As we know,

$$t = 2\pi \sqrt{\frac{l}{g}} \Rightarrow g = \frac{4\pi^2 l}{t^2}$$

$$\text{or, } \frac{\Delta g}{g} = \pm \left(\frac{\Delta l}{l} + 2 \frac{\Delta t}{t} \right) = \left(\frac{0.1}{20} + 2 \times \frac{1}{90} \right) = 0.027$$

$$\therefore \frac{\Delta g}{g} \% = 2.7\%$$

69. (a) According to ohm's law, $V = IR$

$$R = \frac{V}{I}$$

$$\therefore \text{Percentage error} = \frac{\text{Absolute error}}{\text{Measurement}} \times 10^2$$

$$\text{where, } \frac{\Delta V}{V} \times 100 = \frac{\Delta I}{I} \times 100 = 3\%$$

$$\text{then, } \frac{\Delta R}{R} \times 100 = \frac{\Delta V}{V} \times 10^2 + \frac{\Delta I}{I} \times 10^2$$

$$= 3\% + 3\% = 6\%$$

70. (c) $\therefore$ Reading of Vernier = Main scale reading

+ Vernier scale reading $\times$ least count.

Main scale reading = 58.5

Vernier scale reading = 09 division

least count of Vernier = $0.5^\circ/30$

$$\text{Thus, } R = 58.5^\circ + 9 \times \frac{0.5^\circ}{30}$$

$$R = 58.65^\circ$$

71. (d) No. of divisions on main scale = N

No. of divisions on vernier scale = $N + 1$

size of main scale division = a

Let size of vernier scale division be b

then we have

$$aN = b(N + 1) \Rightarrow b = \frac{aN}{N + 1}$$

$$\text{Least count is } a - b = a - \frac{aN}{N + 1}$$

$$= a \left[\frac{N + 1 - N}{N + 1} \right] = \frac{a}{N + 1}$$

72. (d) The least count (L.C.) of a screw guage is the smallest length which can be measured accurately with it.

$$\text{As least count is } 0.001 \text{ cm} = \frac{1}{1000} \text{ cm}$$

Hence measured value should be recorded upto 3 decimal places i.e., 5.320 cm

73. (a) Least count, $L.C. = \frac{1}{100} \text{ mm}$

Diameter of wire = $MSR + CSR \times L.C.$

$\therefore 1 \text{ mm} = 0.1 \text{ cm}$

$= 0 + \frac{1}{100} \times 52 = 0.52 \text{ mm} = 0.052 \text{ cm}$

74. (a) Number of significant figures in 23.023 = 5

Number of significant figures in 0.0003 = 1

Number of significant figures in $2.1 \times 10^{-3} = 2$

So, the radiation belongs to X-rays part of the spectrum.

75. (d) 30 Divisions of V.S. coincide with 29 divisions of M.S.

$\therefore 1 \text{ V.S.D} = \frac{29}{30} \text{ MSD}$

$L.C. = 1 \text{ MSD} - 1 \text{ VSD}$

$= 1 \text{ MSD} - \frac{29}{30} \text{ MSD}$

$= \frac{1}{30} \text{ MSD}$

$= \frac{1}{30} \times 0.5^\circ = 1 \text{ minute.}$

76. (a) Momentum, $p = m \times v$

Given, mass of a body = 3.513 kg speed of body

$= (3.513) \times (5.00) = 17.565 \text{ kg m/s}$

$= 17.6$ (Rounding off to get three significant figures)

77. (d) Least count of screw gauge = 0.01 mm

$\therefore \frac{0.5}{50} \text{ mm}$

Reading = $[M.S.R. + C.S.R. \times L.C.] - (\text{zero error})$

$= [3 + 35 \times 0.01] - (-0.03) = 3.38 \text{ mm}$

2

Motion in a Straight Line



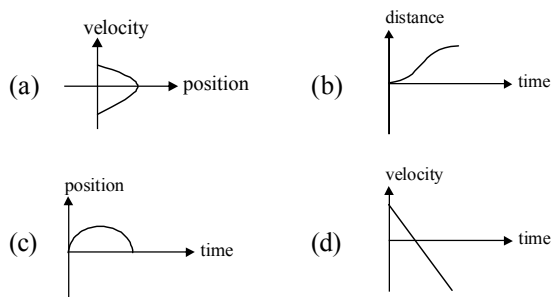
TOPIC 1 Distance, Displacement & Uniform Motion



1. A particle is moving with speed $v = b\sqrt{x}$ along positive x -axis. Calculate the speed of the particle at time $t = \tau$ (assume that the particle is at origin at $t = 0$). [12 Apr. 2019 II]

(a) $\frac{b^2\tau}{4}$ (b) $\frac{b^2\tau}{2}$ (c) $b^2\tau$ (d) $\frac{b^2\tau}{\sqrt{2}}$

2. All the graphs below are intended to represent the same motion. One of them does it incorrectly. Pick it up. [2018]



3. A car covers the first half of the distance between two places at 40 km/h and other half at 60 km/h. The average speed of the car is [Online May 7, 2012]

(a) 40 km/h (b) 45 km/h
(c) 48 km/h (d) 60 km/h

4. The velocity of a particle is $v = v_0 + gt + ft^2$. If its position is $x = 0$ at $t = 0$, then its displacement after unit time ($t = 1$) is [2007]

(a) $v_0 + g/2 + f$ (b) $v_0 + 2g + 3f$
(c) $v_0 + g/2 + f/3$ (d) $v_0 + g + f$

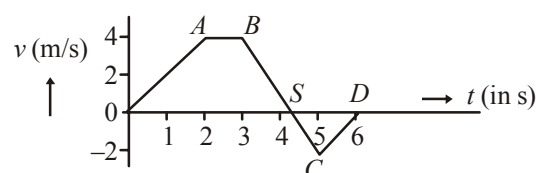
5. A particle located at $x = 0$ at time $t = 0$, starts moving along with the positive x -direction with a velocity ' v ' that varies as $v = \alpha\sqrt{x}$. The displacement of the particle varies with time as [2006]

(a) t^2 (b) t (c) $t^{1/2}$ (d) t^3

TOPIC 2 Non-uniform Motion

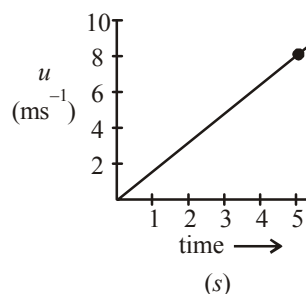


6. The velocity (v) and time (t) graph of a body in a straight line motion is shown in the figure. The point S is at 4.333 seconds. The total distance covered by the body in 6 s is: [05 Sep. 2020 (II)]



(a) $\frac{37}{3}$ m (b) 12 m (c) 11 m (d) $\frac{49}{4}$ m

7. The speed versus time graph for a particle is shown in the figure. The distance travelled (in m) by the particle during the time interval $t = 0$ to $t = 5$ s will be [NA 4 Sep. 2020 (II)]



8. The distance x covered by a particle in one dimensional motion varies with time t as $x^2 = at^2 + 2bt + c$. If the acceleration of the particle depends on x as x^{-n} , where n is an integer, the value of n is [NA 9 Jan 2020 I]

9. A bullet of mass 20g has an initial speed of 1 ms^{-1} , just before it starts penetrating a mud wall of thickness 20 cm. If the wall offers a mean resistance of $2.5 \times 10^{-2} \text{ N}$, the speed of the bullet after emerging from the other side of the wall is close to : [10 Apr. 2019 II]

(a) 0.1 ms^{-1} (b) 0.7 ms^{-1}
(c) 0.3 ms^{-1} (d) 0.4 ms^{-1}

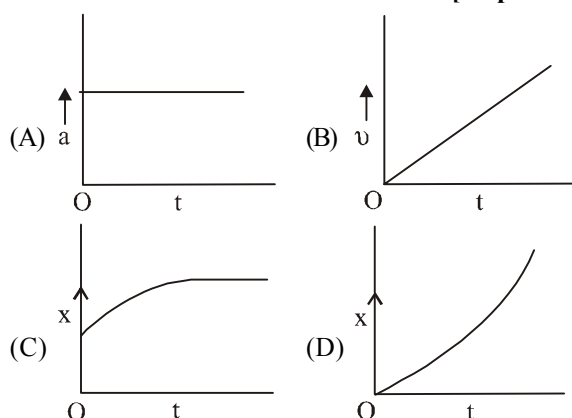
10. The position of a particle as a function of time t , is given by
 $x(t) = at + bt^2 - ct^3$
 where, a , b and c are constants. When the particle attains zero acceleration, then its velocity will be:

[9 Apr. 2019 II]

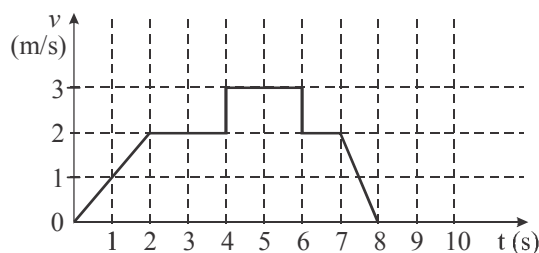
- (a) $a + \frac{b^2}{4c}$ (b) $a + \frac{b^2}{3c}$
 (c) $a + \frac{b^2}{c}$ (d) $a + \frac{b^2}{2c}$

11. A particle starts from origin O from rest and moves with a uniform acceleration along the positive x -axis. Identify all figures that correctly represents the motion qualitatively (a = acceleration, v = velocity, x = displacement, t = time)

[8 Apr. 2019 II]



- (a) (B), (C) (b) (A)
 (c) (A), (B), (C) (d) (A), (B), (D)
12. A particle starts from the origin at time $t = 0$ and moves along the positive x -axis. The graph of velocity with respect to time is shown in figure. What is the position of the particle at time $t = 5$ s? [10 Jan. 2019 II]



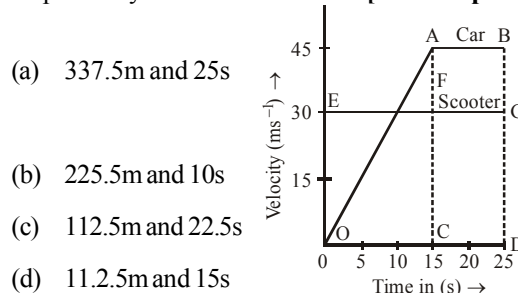
- (a) 10 m (b) 6 m
 (c) 3 m (d) 9 m
13. In a car race on straight road, car A takes a time t less than car B at the finish and passes finishing point with a speed ' v ' more than of car B. Both the cars start from rest and travel with constant acceleration a_1 and a_2 respectively. Then ' v ' is equal to: [9 Jan. 2019 II]

- (a) $\frac{2a_1 a_2}{a_1 + a_2} t$ (b) $\sqrt{2a_1 a_2} t$
 (c) $\sqrt{a_1 a_2} t$ (d) $\frac{a_1 + a_2}{2} t$

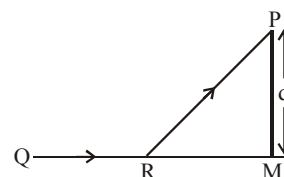
14. An automobile, travelling at 40 km/h, can be stopped at a distance of 40 m by applying brakes. If the same automobile is travelling at 80 km/h, the minimum stopping distance, in metres, is (assume no skidding) [Online April 15, 2018]

(a) 75m (b) 160m (c) 100m (d) 150m

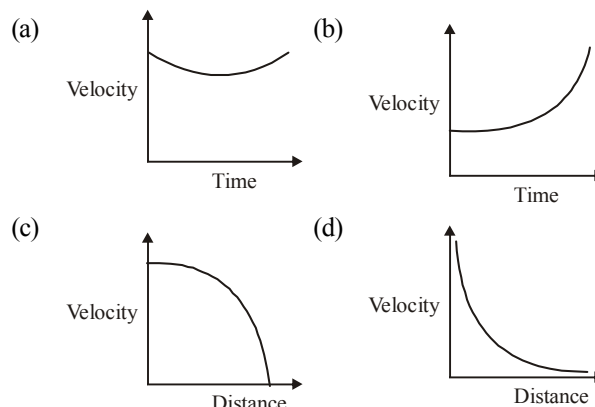
15. The velocity-time graphs of a car and a scooter are shown in the figure. (i) the difference between the distance travelled by the car and the scooter in 15 s and (ii) the time at which the car will catch up with the scooter are, respectively [Online April 15, 2018]



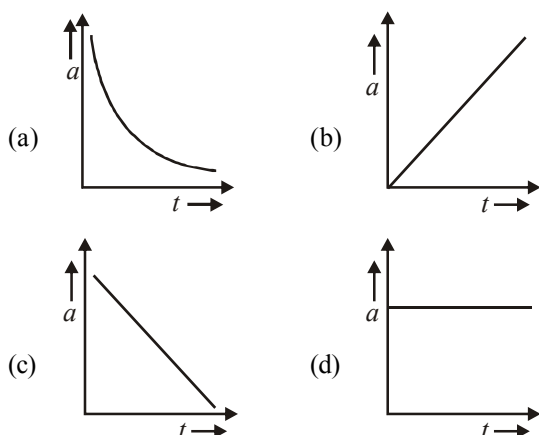
- (a) 337.5m and 25s
 (b) 225.5m and 10s
 (c) 112.5m and 22.5s
 (d) 11.2.5m and 15s
16. A man in a car at location Q on a straight highway is moving with speed v . He decides to reach a point P in a field at a distance d from highway (point M) as shown in the figure. Speed of the car in the field is half to that on the highway. What should be the distance RM, so that the time taken to reach P is minimum? [Online April 15, 2018]



- (a) $\frac{d}{\sqrt{3}}$ (b) $\frac{d}{2}$ (c) $\frac{d}{\sqrt{2}}$ (d) d
17. Which graph corresponds to an object moving with a constant negative acceleration and a positive velocity? [Online April 8, 2017]

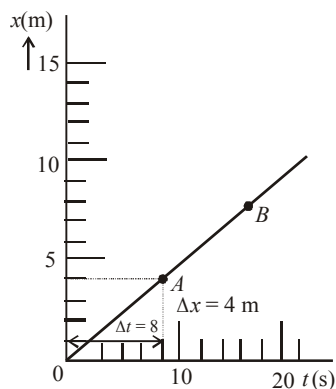


18. The distance travelled by a body moving along a line in time t is proportional to t^3 . The acceleration-time (a , t) graph for the motion of the body will be [Online May 12, 2012]

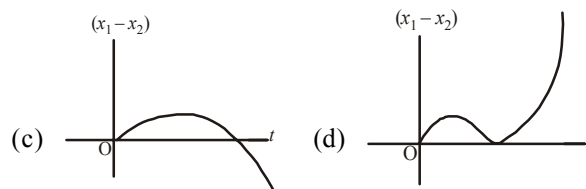
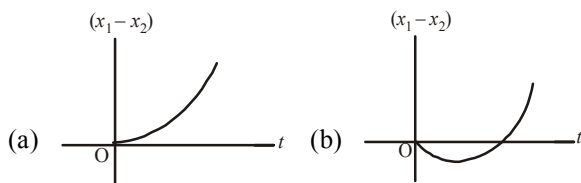


19. The graph of an object's motion (along the x-axis) is shown in the figure. The instantaneous velocity of the object at points A and B are v_A and v_B respectively. Then

[Online May 7, 2012]



- (a) $v_A = v_B = 0.5 \text{ m/s}$ (b) $v_A = 0.5 \text{ m/s} < v_B$
 (c) $v_A = 0.5 \text{ m/s} > v_B$ (d) $v_A = v_B = 2 \text{ m/s}$
20. An object, moving with a speed of 6.25 m/s , is decelerated at a rate given by $\frac{dv}{dt} = -2.5\sqrt{v}$ where v is the instantaneous speed. The time taken by the object, to come to rest, would be: [2011]
 (a) 2 s (b) 4 s (c) 8 s (d) 1 s
21. A body is at rest at $x = 0$. At $t = 0$, it starts moving in the positive x-direction with a constant acceleration. At the same instant another body passes through $x = 0$ moving in the positive x-direction with a constant speed. The position of the first body is given by $x_1(t)$ after time ' t '; and that of the second body by $x_2(t)$ after the same time interval. Which of the following graphs correctly describes $(x_1 - x_2)$ as a function of time ' t '? [2008]



22. A car, starting from rest, accelerates at the rate f through a distance S , then continues at constant speed for time t and then decelerates at the rate $\frac{f}{2}$ to come to rest. If the total distance traversed is $15S$, then [2005]
 (a) $S = \frac{1}{6}ft^2$ (b) $S = ft$
 (c) $S = \frac{1}{4}ft^2$ (d) $S = \frac{1}{72}ft^2$
23. A particle is moving eastwards with a velocity of 5 ms^{-1} . In 10 seconds the velocity changes to 5 ms^{-1} northwards. The average acceleration in this time is [2005]
 (a) $\frac{1}{2} \text{ ms}^{-2}$ towards north
 (b) $\frac{1}{\sqrt{2}} \text{ ms}^{-2}$ towards north - east
 (c) $\frac{1}{\sqrt{2}} \text{ ms}^{-2}$ towards north - west
 (d) zero
24. The relation between time t and distance x is $t = ax^2 + bx$ where a and b are constants. The acceleration is [2005]
 (a) $2bv^3$ (b) $-2abv^2$ (c) $2av^2$ (d) $-2av^3$
25. An automobile travelling with a speed of 60 km/h , can brake to stop within a distance of 20 m . If the car is going twice as fast i.e., 120 km/h , the stopping distance will be [2004]
 (a) 60 m (b) 40 m (c) 20 m (d) 80 m
26. A car, moving with a speed of 50 km/hr , can be stopped by brakes after at least 6 m . If the same car is moving at a speed of 100 km/hr , the minimum stopping distance is [2003]
 (a) 12 m (b) 18 m (c) 24 m (d) 6 m
27. If a body loses half of its velocity on penetrating 3 cm in a wooden block, then how much will it penetrate more before coming to rest? [2002]
 (a) 1 cm (b) 2 cm (c) 3 cm (d) 4 cm
28. Speeds of two identical cars are u and $4u$ at the specific instant. The ratio of the respective distances in which the two cars are stopped from that instant is [2002]
 (a) $1 : 1$ (b) $1 : 4$ (c) $1 : 8$ (d) $1 : 16$

TOPIC 3 Relative Velocity



29. Train A and train B are running on parallel tracks in the opposite directions with speeds of 36 km/hour and 72 km/hour , respectively. A person is walking in train A in the direction opposite to its motion with a speed of 1.8

km/hour. Speed (in ms^{-1}) of this person as observed from train B will be close to : (take the distance between the tracks as negligible) [2 Sep. 2020 (I)]

- (a) 29.5 ms^{-1} (b) 28.5 ms^{-1}
(c) 31.5 ms^{-1} (d) 30.5 ms^{-1}

30. A passenger train of length 60 m travels at a speed of 80 km/hr. Another freight train of length 120 m travels at a speed of 30 km/h. The ratio of times taken by the passenger train to completely cross the freight train when: (i) they are moving in same direction, and (ii) in the opposite directions is: [12 Jan. 2019 II]

- (a) $\frac{11}{5}$ (b) $\frac{5}{2}$ (c) $\frac{3}{2}$ (d) $\frac{25}{11}$

31. A person standing on an open ground hears the sound of a jet aeroplane, coming from north at an angle 60° with ground level. But he finds the aeroplane right vertically above his position. If v is the speed of sound, speed of the plane is: [12 Jan. 2019 II]

- (a) $\frac{\sqrt{3}}{2}v$ (b) $\frac{2v}{\sqrt{3}}$ (c) v (d) $\frac{v}{2}$

32. A car is standing 200 m behind a bus, which is also at rest. The two start moving at the same instant but with different forward accelerations. The bus has acceleration 2 m/s^2 and the car has acceleration 4 m/s^2 . The car will catch up with the bus after a time of:

[Online April 9, 2017]

- (a) $\sqrt{110} \text{ s}$ (b) $\sqrt{120} \text{ s}$
(c) $10\sqrt{2} \text{ s}$ (d) 15 s

33. A person climbs up a stalled escalator in 60 s. If standing on the same but escalator running with constant velocity he takes 40 s. How much time is taken by the person to walk up the moving escalator? [Online April 12, 2014]

- (a) 37 s (b) 27 s (c) 24 s (d) 45 s

34. A goods train accelerating uniformly on a straight railway track, approaches an electric pole standing on the side of track. Its engine passes the pole with velocity u and the guard's room passes with velocity v . The middle wagon of the train passes the pole with a velocity.

[Online May 19, 2012]

- (a) $\frac{u+v}{2}$ (b) $\frac{1}{2}\sqrt{u^2+v^2}$
(c) $\sqrt{uv}$ (d) $\sqrt{\left(\frac{u^2+v^2}{2}\right)}$

TOPIC 4 Motion Under Gravity



35. A helicopter rises from rest on the ground vertically upwards with a constant acceleration g . A food packet is dropped from the helicopter when it is at a height h . The time taken by the packet to reach the ground is close to [g is the acceleration due to gravity]: [5 Sep. 2020 (I)]

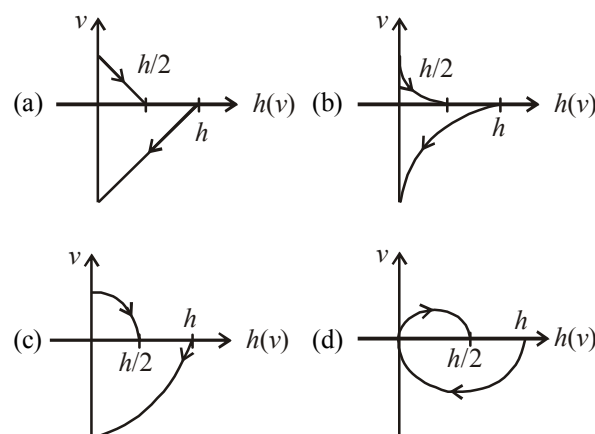
(a) $t = \frac{2}{3}\sqrt{\left(\frac{h}{g}\right)}$ (b) $t = 1.8\sqrt{\frac{h}{g}}$

(c) $t = 3.4\sqrt{\left(\frac{h}{g}\right)}$ (d) $t = \sqrt{\frac{2h}{3g}}$

36. A Tennis ball is released from a height h and after freely falling on a wooden floor it rebounds and reaches height $\frac{h}{2}$. The velocity versus height of the ball during its motion may be represented graphically by:

(graph are drawn schematically and on not to scale)

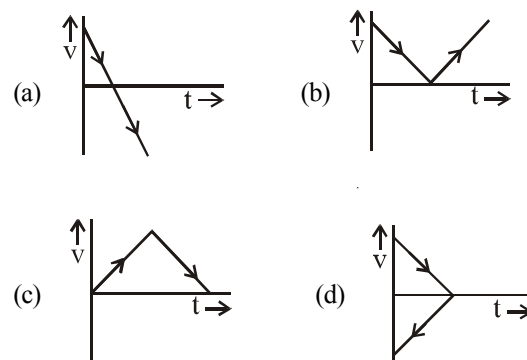
[4 Sep. 2020 (I)]



37. A ball is dropped from the top of a 100 m high tower on a planet. In the last $\frac{1}{2} \text{ s}$ before hitting the ground, it covers a distance of 19 m. Acceleration due to gravity (in ms^{-2}) near the surface on that planet is _____.

[NA 8 Jan. 2020 II]

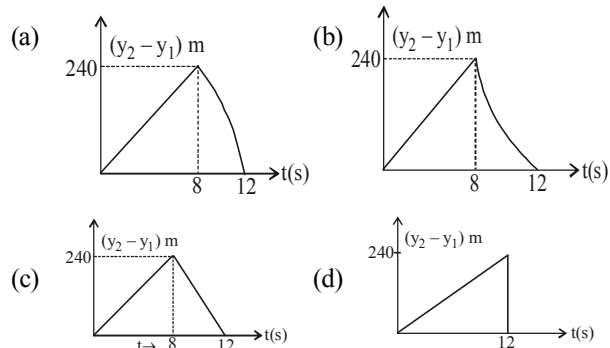
38. A body is thrown vertically upwards. Which one of the following graphs correctly represent the velocity vs time? [2017]



39. Two stones are thrown up simultaneously from the edge of a cliff 240 m high with initial speed of 10 m/s and 40 m/s respectively. Which of the following graph best represents the time variation of relative position of the second stone with respect to the first?

(Assume stones do not rebound after hitting the ground and neglect air resistance, take $g = 10 \text{ m/s}^2$) [2015]

(The figures are schematic and not drawn to scale)

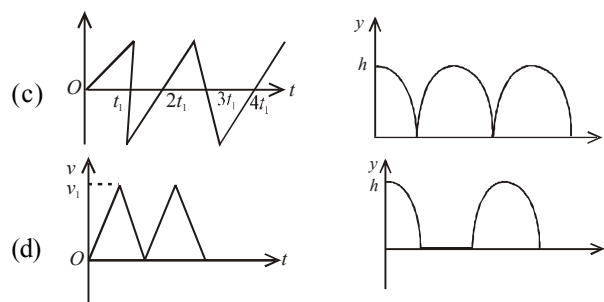
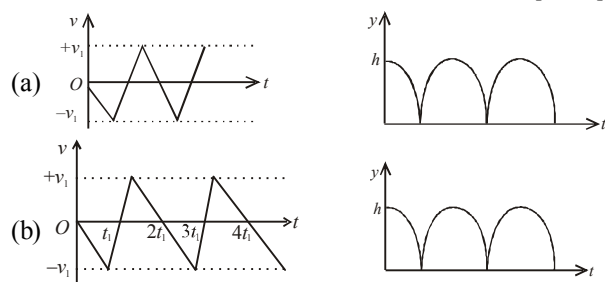


40. From a tower of height H , a particle is thrown vertically upwards with a speed u . The time taken by the particle, to hit the ground, is n times that taken by it to reach the highest point of its path. The relation between H , u and n is: [2014]

(a) $2gH = n^2 u^2$ (b) $gH = (n-2)^2 u^2$
 (c) $2gH = nu^2 (n-2)$ (d) $gH = (n-2)u^2$

41. Consider a rubber ball freely falling from a height $h = 4.9 \text{ m}$ onto a horizontal elastic plate. Assume that the duration of collision is negligible and the collision with the plate is totally elastic.

Then the velocity as a function of time and the height as a function of time will be : [2009]



42. A parachutist after bailing out falls 50 m without friction. When parachute opens, it decelerates at 2 m/s^2 . He reaches the ground with a speed of 3 m/s . At what height, did he bail out? [2005]

(a) 182 m (b) 91 m
 (c) 111 m (d) 293 m

43. A ball is released from the top of a tower of height h meters. It takes T seconds to reach the ground. What is the position

of the ball at $\frac{T}{3}$ second

[2004]

(a) $\frac{8h}{9}$ meters from the ground
 (b) $\frac{7h}{9}$ meters from the ground
 (c) $\frac{h}{9}$ meters from the ground
 (d) $\frac{17h}{18}$ meters from the ground

44. From a building two balls A and B are thrown such that A is thrown upwards and B downwards (both vertically). If v_A and v_B are their respective velocities on reaching the ground, then [2002]

(a) $v_B > v_A$
 (b) $v_A = v_B$
 (c) $v_A > v_B$
 (d) their velocities depend on their masses.



Hints & Solutions



1. (b) Given, $v = b\sqrt{x}$

$$\text{or } \frac{dx}{dt} = b x^{1/2}$$

$$\text{or } \int_0^x x^{-1/2} dx = \int_0^t b dt$$

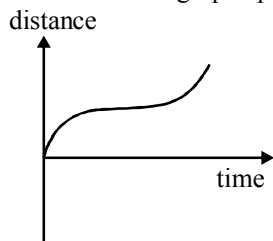
$$\text{or } \frac{x^{1/2}}{1/2} = 6t \quad \text{or } x = \frac{b^2 t^2}{4}$$

Differentiating w. r. t. time, we get

$$\frac{dx}{dt} = \frac{b^2 \times 2t}{4} \quad (t = \tau)$$

$$\text{or } v = \frac{b^2 \tau}{2}$$

2. (b) Graphs in option (c) position-time and option (a) velocity-position are corresponding to velocity-time graph option (d) and its distance-time graph is as given below. Hence distance-time graph option (b) is incorrect.



3. (c) Average speed = $\frac{\text{Total distance travelled}}{\text{Total time taken}} = \frac{x}{T}$

$$= \frac{x}{\frac{x}{2 \times 40} + \frac{x}{2 \times 60}} = 48 \text{ km/h}$$

4. (c) We know that, $v = \frac{dx}{dt}$

$$\Rightarrow dx = v dt$$

$$\text{Integrating, } \int_0^x dx = \int_0^t v dt$$

$$\text{or } x = \int_0^t (v_0 + gt + ft^2) dt = \left[v_0 t + \frac{gt^2}{2} + \frac{ft^3}{3} \right]_0^t$$

$$\text{or, } x = v_0 t + \frac{gt^2}{2} + \frac{ft^3}{3}$$

$$\text{At } t = 1, \quad x = v_0 + \frac{g}{2} + \frac{f}{3}$$

5. (a) $v = \alpha\sqrt{x}$,

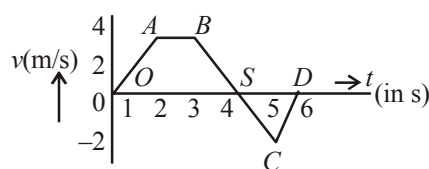
$$\Rightarrow \frac{dx}{dt} = \alpha\sqrt{x} \Rightarrow \frac{dx}{\sqrt{x}} = \alpha dt$$

Integrating both sides,

$$\int_0^x \frac{dx}{\sqrt{x}} = \alpha \int_0^t dt; \left[\frac{2\sqrt{x}}{1} \right]_0^x = \alpha [t]_0^t$$

$$\Rightarrow 2\sqrt{x} = \alpha t \Rightarrow x = \frac{\alpha^2}{4} t^2$$

6. (a)



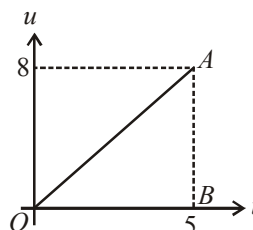
$$OS = 4 + \frac{1}{3} = \frac{13}{3}$$

$$SD = 2 - \frac{1}{3} = \frac{5}{3}$$

Distance covered by the body = area of $v-t$ graph
= ar (OABS) + ar (SCD)

$$= \frac{1}{2} \left(\frac{13}{3} + 1 \right) \times 4 + \frac{1}{2} \times \frac{5}{3} \times 2 = \frac{32}{3} + \frac{5}{3} = \frac{37}{3} \text{ m}$$

7. (20)



Distance travelled = Area of speed-time graph

$$= \frac{1}{2} \times 5 \times 8 = 20 \text{ m}$$

8. (3) Distance X varies with time t as $x^2 = at^2 + 2bt + c$

$$\Rightarrow 2x \frac{dx}{dt} = 2at + 2b$$

$$\Rightarrow x \frac{dx}{dt} = at + b \Rightarrow \frac{dx}{dt} = \frac{(at+b)}{x}$$

$$\Rightarrow x \frac{d^2x}{dt^2} + \left(\frac{dx}{dt} \right)^2 = a$$

$$\Rightarrow \frac{d^2x}{dt^2} = \frac{a - \left(\frac{dx}{dt} \right)^2}{x} = \frac{a - \left(\frac{at+b}{x} \right)^2}{x}$$

$$= \frac{ax^2 - (at+b)^2}{x^3} = \frac{ac - b^2}{x^3}$$

$$\Rightarrow a \propto x^{-3} \quad \text{Hence, } n = 3$$

9. (b) From the third equation of motion

$$v^2 - u^2 = 2aS$$

But, $a = \frac{F}{m}$

$$\therefore v^2 = u^2 - 2\left(\frac{F}{m}\right)S$$

$$\Rightarrow v^2 = (1)^2 - (2)\left[\frac{2.5 \times 10^{-2}}{20 \times 10^{-3}}\right] \frac{20}{100}$$

$$\Rightarrow v^2 = 1 - \frac{1}{2}$$

$$\Rightarrow v = \frac{1}{\sqrt{2}} \text{ m/s} = 0.7 \text{ m/s}$$

10. (b) $x = at + bt^2 - ct^3$

Velocity, $v = \frac{dx}{dt} = \frac{d}{dt}(at + bt^2 + ct^3)$

$$= a + 2bt - 3ct^2$$

Acceleration, $\frac{dv}{dt} = \frac{d}{dt}(a + 2bt - 3ct^2)$

or $0 = 2b - 3c \times 2t \quad \therefore t = \left(\frac{b}{3c}\right)$

and $v = a + 2b\left(\frac{b}{3c}\right) - 3c\left(\frac{b}{3c}\right)^2 = \left(a + \frac{b^2}{3c}\right)$

11. (d) For constant acceleration, there is straight line

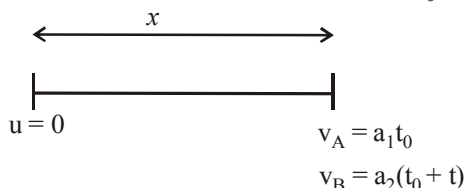
parallel to t -axis on $\vec{a}-t$.

Inclined straight line on $\vec{v}-t$, and parabola on $\vec{x}-t$.

12. (d) Position of the particle,
S = area under graph (time $t = 0$ to 5s)

$$= \frac{1}{2} \times 2 \times 2 + 2 \times 2 + 3 \times 1 = 9 \text{ m}$$

13. (c) Let time taken by A to reach finishing point is t_0
 $\therefore$ Time taken by B to reach finishing point = $t_0 + t$



$$v_A - v_B = v$$

$$\Rightarrow v = a_1 t_0 - a_2(t_0 + t) = (a_1 - a_2)t_0 - a_2 t \quad \dots(i)$$

$$x_B = x_A = \frac{1}{2} a_1 t_0^2 = \frac{1}{2} a_2 (t_0 + t)^2$$

$$\Rightarrow \sqrt{a_1} t_0 = \sqrt{a_2} (t_0 + t)$$

$$\Rightarrow (\sqrt{a_1} - \sqrt{a_2}) t_0 = \sqrt{a_2} t$$

$$\Rightarrow t_0 = \frac{\sqrt{a_2} t}{\sqrt{a_1} - \sqrt{a_2}}$$

Putting this value of t_0 in equation (i)

$$v = (a_1 - a_2) \frac{\sqrt{a_2} t}{\sqrt{a_1} - \sqrt{a_2}} - a_2 t$$

$$= (\sqrt{a_1} + \sqrt{a_2}) \sqrt{a_2} t - a_2 t = \sqrt{a_1 a_2} t + a_2 t - a_2 t$$

$$\text{or, } v = \sqrt{a_1 a_2} t$$

14. (b) According to question, $u_1 = 40 \text{ km/h}$, $v_1 = 0$ and $s_1 = 40 \text{ m}$
using $v^2 - u^2 = 2as$; $0^2 - 40^2 = 2a \times 40 \quad \dots(i)$

Again, $0^2 - 80^2 = 2as \quad \dots(ii)$

From eqn. (i) and (ii)

Stopping distance, $s = 160 \text{ m}$

15. (c) Using equation, $a = \frac{v-u}{t}$ and $S = ut + \frac{1}{2} at^2$

Distance travelled by car in 15 sec = $\frac{1}{2} \frac{(45)}{15} (15)^2$

$$= \frac{675}{2} \text{ m}$$

Distance travelled by scooter in 15 seconds = $30 \times 15 = 450$
($\therefore$ distance = speed $\times$ time)

Difference between distance travelled by car and scooter in 15 sec, $450 - 337.5 = 112.5 \text{ m}$

Let car catches scooter in time t ;

$$\frac{675}{2} + 45(t - 15) = 30t$$

$$337.5 + 45t - 675 = 30t \Rightarrow 15t = 337.5$$

$$\Rightarrow t = 22.5 \text{ sec}$$

16. (a) Let the car turn off the highway at a distance 'x' from the point M. So, $RM = x$

And if speed of car in field is v , then time taken by the car to cover the distance $QR = QM - x$ on the highway,

$$t_1 = \frac{QM - x}{2v} \quad \dots(i)$$

Time taken to travel the distance 'RP' in the field

$$t_2 = \frac{\sqrt{d^2 + x^2}}{v} \quad \dots(ii)$$

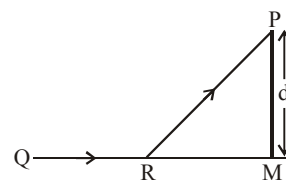
Total time elapsed to move the car from Q to P

$$t = t_1 + t_2 = \frac{QM - x}{2v} + \frac{\sqrt{d^2 + x^2}}{v}$$

For 't' to be minimum $\frac{dt}{dx} = 0$

$$\frac{1}{v} \left[-\frac{1}{2} + \frac{x}{\sqrt{d^2 + x^2}} \right] = 0$$

$$\text{or } x = \frac{d}{\sqrt{2^2 - 1}} = \frac{d}{\sqrt{3}}$$



17. (c) According to question, object is moving with constant negative acceleration i.e., $a = -\text{constant (C)}$

$$\frac{v dv}{dx} = -C$$

$$v dv = -C dx$$

$$\frac{v^2}{2} = -Cx + k \quad x = -\frac{v^2}{2C} + \frac{k}{C}$$

Hence, graph (3) represents correctly.

18. (b) Distance along a line i.e., displacement (s) $= t^3$ ($\because s \propto t^3$ given)

By double differentiation of displacement, we get acceleration.

$$V = \frac{ds}{dt} = \frac{dt^3}{dt} = 3t^2 \quad \text{and} \quad a = \frac{dv}{dt} = \frac{d3t^2}{dt} = 6t$$

$$a = 6t \text{ or } a \propto t$$

Hence graph (b) is correct.

19. (a) Instantaneous velocity $v = \frac{\Delta x}{\Delta t}$

$$\text{From graph, } v_A = \frac{\Delta x_A}{\Delta t_A} = \frac{4m}{8s} = 0.5 \text{ m/s}$$

$$\text{and } v_B = \frac{\Delta x_B}{\Delta t_B} = \frac{8m}{16s} = 0.5 \text{ m/s}$$

$$\text{i.e., } v_A = v_B = 0.5 \text{ m/s}$$

20. (a) Given, $\frac{dv}{dt} = -2.5\sqrt{v}$

$$\Rightarrow \frac{dv}{\sqrt{v}} = -2.5 dt$$

Integrating,

$$\int_{6.25}^0 v^{-1/2} dv = -2.5 \int_0^t dt$$

$$\Rightarrow \left[\frac{v^{+1/2}}{(1/2)} \right]_{6.25}^0 = -2.5 [t]_0^t$$

$$\Rightarrow -2(6.25)^{1/2} = -2.5t$$

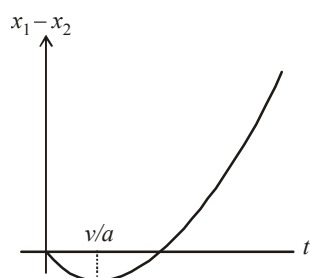
$$\Rightarrow -2 \times 2.5 = -2.5t$$

$$\Rightarrow t = 25$$

21. (b) For the body starting from rest, distance travelled (x_1) is given by

$$x_1 = 0 + \frac{1}{2} at^2$$

$$\Rightarrow x_1 = \frac{1}{2} at^2$$



For the body moving with constant speed

$$x_2 = vt$$

$$\therefore x_1 - x_2 = \frac{1}{2} at^2 - vt$$

$$\text{at } t = 0, x_1 - x_2 = 0$$

This equation is of parabola.

For $t < \frac{v}{a}$; the slope is negative

For $t = \frac{v}{a}$; the slope is zero

For $t > \frac{v}{a}$; the slope is positive

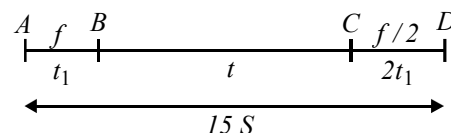
These characteristics are represented by graph (b).

22. (d) Let car starts from A from rest and moves up to point B with acceleration f .

$$\text{Distance, } AB = S = \frac{1}{2} ft_1^2$$

$$\text{Distance, } BC = (ft_1)t$$

$$\text{Distance, } CD = \frac{u^2}{2a} = \frac{(ft_1)^2}{2(f/2)} = ft_1^2 = 2S$$



$$\text{Total distance, } AD = AB + BC + CD = 15S$$

$$AD = S + BC + 2S$$

$$\Rightarrow S + ft_1t + 2S = 15S$$

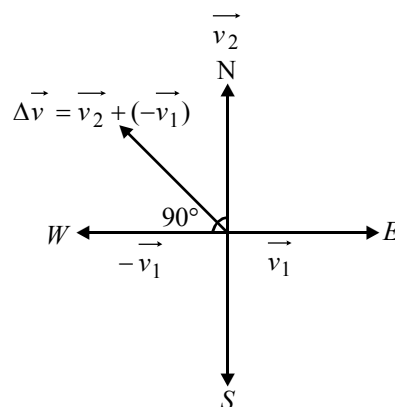
$$\Rightarrow ft_1t = 12S \quad \dots\dots\dots (i)$$

$$\frac{1}{2} ft_1^2 = S \quad \dots\dots\dots (ii)$$

Dividing (i) by (ii), we get $t_1 = \frac{t}{6}$

$$\Rightarrow S = \frac{1}{2} f \left(\frac{t}{6} \right)^2 = \frac{ft^2}{72}$$

23. (c)



Initial velocity, $\vec{v}_1 = 5\hat{i}$,

Final velocity, $\vec{v}_2 = 5\hat{j}$,

Change in velocity $\Delta \vec{v} = (\vec{v}_2 - \vec{v}_1)$

$$= \sqrt{v_1^2 + v_2^2 + 2v_1v_2 \cos 90}$$

$$= \sqrt{5^2 + 5^2 + 0} = 5\sqrt{2} \text{ m/s}$$

[As $|v_1| = |v_2| = 5 \text{ m/s}$]

$$\text{Avg. acceleration} = \frac{\Delta \vec{v}}{t}$$

$$= \frac{5\sqrt{2}}{10} = \frac{1}{\sqrt{2}} \text{ m/s}^2$$

$$\tan \theta = \frac{5}{-5} = -1$$

which means θ is in the second quadrant.
(towards north-west)

24. (d) Given, $t = ax^2 + bx$;

Diff. with respect to time (t)

$$\frac{d}{dt}(t) = a \frac{d}{dt}(x^2) + b \frac{dx}{dt} = a \cdot 2x \frac{dx}{dt} + b \cdot v.$$

$$\Rightarrow 1 = 2axv + bv = v(2ax + b) \quad (v = \text{velocity})$$

$$2ax + b = \frac{1}{v}$$

Again differentiating, we get

$$2a \frac{dx}{dt} + 0 = -\frac{1}{v^2} \frac{dv}{dt}$$

$$\Rightarrow a = \frac{dv}{dt} = -2av^3 \quad \left(\because \frac{dx}{dt} = v \right)$$

25. (d) In first case speed,

$$u = 60 \times \frac{5}{18} \text{ m/s} = \frac{50}{3} \text{ m/s}$$

$$d = 20 \text{ m},$$

Let retardation be a then

$$(0)^2 - u^2 = -2ad$$

$$\text{or } u^2 = 2ad \quad \dots(i)$$

$$\text{In second case speed, } u' = 120 \times \frac{5}{18}$$

$$= \frac{100}{3} \text{ m/s}$$

$$\text{and } (0)^2 - u'^2 = -2ad'$$

$$\text{or } u'^2 = 2ad' \quad \dots(ii)$$

(ii) divided by (i) gives,

$$4 = \frac{d'}{d} \Rightarrow d' = 4 \times 20 = 80 \text{ m}$$

26. (c) **Fir first case :** Initial velocity,

$$u = 50 \times \frac{5}{18} \text{ m/s},$$

$$v = 0, s = 6 \text{ m}, a = a$$

$$\text{Using, } v^2 - u^2 = 2as$$

$$\Rightarrow 0^2 - \left(50 \times \frac{5}{18} \right)^2 = 2 \times a \times 6$$

$$\Rightarrow -\left(50 \times \frac{5}{18} \right)^2 = 2 \times a \times 6$$

$$a = -\frac{250 \times 250}{324 \times 2 \times 6} \approx -16 \text{ ms}^{-2}.$$

Case-2 : Initial velocity, $u = 100 \text{ km/hr}$

$$= 100 \times \frac{5}{18} \text{ m/sec}$$

$$v = 0, s = s, a = a$$

$$\text{As } v^2 - u^2 = 2as$$

$$\Rightarrow 0^2 - \left(100 \times \frac{5}{18} \right)^2 = 2as$$

$$\Rightarrow -\left(100 \times \frac{5}{18} \right)^2 = 2 \times (-16) \times s$$

$$s = \frac{500 \times 500}{324 \times 32} = 24 \text{ m}$$

27. (a) In first case

$$u_1 = u; v_1 = \frac{u}{2}, s_1 = 3 \text{ cm}, a_1 = ?$$

$$\text{Using, } v_1^2 - u_1^2 = 2a_1s_1 \quad \dots(i)$$

$$\left(\frac{u}{2} \right)^2 - u^2 = 2 \times a \times 3$$

$$\Rightarrow a = \frac{-u^2}{8}$$

In second case: Assuming the same retardation

$$u_2 = u/2; v_2 = 0; s_2 = ?; a_2 = \frac{-u^2}{8}$$

$$v_2^2 - u_2^2 = 2a_2s_2 \quad \dots(ii)$$

$$\therefore 0 - \frac{u^2}{4} = 2 \left(\frac{-u^2}{8} \right) \times s_2$$

$$\Rightarrow s_2 = 1 \text{ cm}$$

28. (d) **For first car**

$$u_1 = u, v_1 = 0, a_1 = -a, s_1 = s_1$$

$$\text{As } v_1^2 - u_1^2 = 2a_1s_1$$

$$\Rightarrow -u^2 = -2as_1$$

$$\Rightarrow u^2 = 2as_1$$

$$\Rightarrow s_1 = \frac{u^2}{2a} \quad \dots(i)$$

For second car

$$u_2 = 4u, v_1 = 0, a_2 = -a, s_2 = s_2$$

$$\therefore v_2^2 - u_2^2 = 2a_2s_2$$

$$\Rightarrow -(4u)^2 = 2(-a)s_2$$

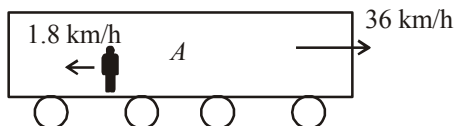
$$\Rightarrow 16u^2 = 2as_2$$

$$\Rightarrow s_2 = \frac{8u^2}{a} \quad \dots(ii)$$

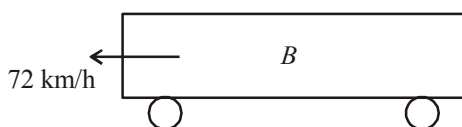
Dividing (i) and (ii),

$$\frac{s_1}{s_2} = \frac{u^2}{2a} \cdot \frac{a}{8u^2} = \frac{1}{16}$$

29. (a) According to question, train A and B are running on parallel tracks in the opposite direction.



$$V_A = 36 \text{ km/h} = 10 \text{ m/s}$$

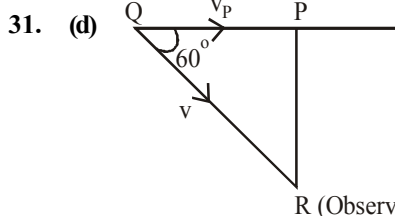


$$V_B = -72 \text{ km/h} = -20 \text{ m/s}$$

$$V_{MA} = -1.8 \text{ km/h} = -0.5 \text{ m/s}$$

$$\begin{aligned} V_{\text{man}, B} &= V_{\text{man}, A} + V_{A, B} \\ &= V_{\text{man}, A} + V_A - V_B = -0.5 + 10 - (-20) \\ &= -0.5 + 30 = 29.5 \text{ m/s.} \end{aligned}$$

30. (a)



Distance, $PQ = v_p \times t$ (Distance = speed $\times$ time)

Distance, $QR = V \cdot t$

$$\cos 60^\circ = \frac{PQ}{QR}$$

$$\frac{1}{2} = \frac{v_p \times t}{V \cdot t} \Rightarrow v_p = \frac{V}{2}$$

32. (c) $\longrightarrow 4 \text{ m/sec}^2$ $\longrightarrow 2 \text{ m/sec}^2$
-

Given, $u_c = u_b = 0$, $a_c = 4 \text{ m/s}^2$, $a_b = 2 \text{ m/s}^2$
hence relative acceleration, $a_{CB} = 2 \text{ m/sec}^2$

Now, we know, $s = ut + \frac{1}{2}at^2$

$$200 = \frac{1}{2} \times 2t^2 \quad \therefore u = 0$$

Hence, the car will catch up with the bus after time

$$t = 10\sqrt{2} \text{ second}$$

33. (c) Person's speed walking only is $\frac{1}{60}$ "escalator" second

Standing the escalator without walking the speed is

$$\frac{1}{40} \text{ "escalator" second}$$

Walking with the escalator going, the speed add.

$$\text{So, the person's speed is } \frac{1}{60} + \frac{1}{40} = \frac{15}{120} \text{ "escalator" second}$$

$$\text{So, the time to go up the escalator } t = \frac{120}{15} = 8 \text{ second.}$$

34. (d) Let 'S' be the distance between two ends 'a' be the constant acceleration

As we know $v^2 - u^2 = 2aS$

$$\text{or, } aS = \frac{v^2 - u^2}{2}$$

Let v be velocity at mid point.

$$\text{Therefore, } v_c^2 - u^2 = 2a \frac{S}{2}$$

$$v_c^2 = u^2 + aS$$

$$v_c^2 = u^2 + \frac{v^2 - u^2}{2}$$

$$v_c = \sqrt{\frac{u^2 + v^2}{2}}$$

35. (c) For upward motion of helicopter,

$$v^2 = u^2 + 2gh \Rightarrow v^2 = 0 + 2gh \Rightarrow v = \sqrt{2gh}$$

Now, packet will start moving under gravity.

Let 't' be the time taken by the food packet to reach the ground.

$$s = ut + \frac{1}{2}at^2$$

$$\Rightarrow -h = \sqrt{2gh}t - \frac{1}{2}gt^2 \Rightarrow \frac{1}{2}gt^2 - \sqrt{2gh}t - h = 0$$

$$\text{or, } t = \frac{\sqrt{2gh} \pm \sqrt{2gh + 4 \times \frac{g}{2} \times h}}{2 \times \frac{g}{2}}$$

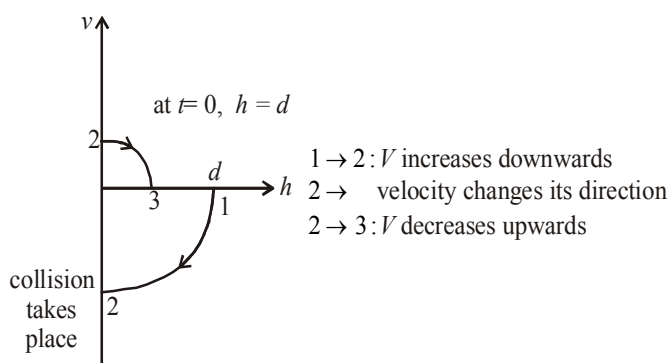
$$\text{or, } t = \sqrt{\frac{2gh}{g}}(1 + \sqrt{2}) \Rightarrow t = \sqrt{\frac{2h}{g}}(1 + \sqrt{2})$$

$$\text{or, } t = 3.4 \sqrt{\frac{h}{g}}$$

36. (c) For uniformly accelerated/ deaccelerated motion :

$$v^2 = u^2 \pm 2gh$$

As equation is quadratic, so, v - h graph will be a parabola



Initially velocity is downwards (–ve) and then after collision it reverses its direction with lesser magnitude, i.e. velocity is upwards (+ve).

Note that time $t = 0$ corresponds to the point on the graph where $h = d$.

Next time collision takes place at 3.

37. (08.00) Let the ball takes time t to reach the ground

$$\text{Using, } S = ut + \frac{1}{2}gt^2$$

$$\Rightarrow S = 0 \times t + \frac{1}{2}gt^2$$

$$\Rightarrow 200 = gt^2 \quad [\because 2S = 100m]$$

$$\Rightarrow t = \sqrt{\frac{200}{g}} \quad \dots (i)$$

In last $\frac{1}{2}$ s, body travels a distance of 19 m, so in $\left(t - \frac{1}{2}\right)$

distance travelled = 81

$$\text{Now, } \frac{1}{2}g\left(t - \frac{1}{2}\right)^2 = 81$$

$$\therefore g\left(t - \frac{1}{2}\right)^2 = 81 \times 2$$

$$\Rightarrow \left(t - \frac{1}{2}\right) = \sqrt{\frac{81 \times 2}{g}}$$

$$\therefore \frac{1}{2} = \frac{1}{\sqrt{g}}(\sqrt{200} - \sqrt{81 \times 2}) \quad \text{using (i)}$$

$$\Rightarrow \sqrt{g} = 2(10\sqrt{2} - 9\sqrt{2})$$

$$\Rightarrow \sqrt{g} = 2\sqrt{2}$$

$$\therefore g = 8 \text{ m/s}^2$$

38. (a) For a body thrown vertically upwards acceleration remains constant ($a = -g$) and velocity at any time t is given by $V = u - gt$

During rise velocity decreases linearly and during fall velocity increases linearly and direction is opposite to each other.

Hence graph (a) correctly depicts velocity versus time.

39. (b) $y_1 = 10t - 5t^2$; $y_2 = 40t - 5t^2$

$$\text{for } y_1 = -240\text{m, } t = 8\text{s}$$

$$\therefore y_2 - y_1 = 30t \text{ for } t \leq 8\text{s.}$$

$$\text{for } t > 8\text{s,}$$

$$y_2 - y_1 = 240 - 40t - \frac{1}{2}gt^2$$

40. (c) Speed on reaching ground

$$v = \sqrt{u^2 + 2gh}$$

$$\text{Now, } v = u + at$$

$$\Rightarrow \sqrt{u^2 + 2gh} = -u + gt$$

$$\text{Time taken to reach highest point is } t = \frac{u}{g},$$

$$\Rightarrow t = \frac{u + \sqrt{u^2 + 2gH}}{g} = \frac{nu}{g}$$

$$(\text{from question})$$

$$\Rightarrow 2gH = n(n-2)u^2$$

41. (b) For downward motion $v = -gt$

The velocity of the rubber ball increases in downward direction and we get a straight line between v and t with a negative slope.

$$\text{Also applying } y - y_0 = ut + \frac{1}{2}at^2$$

$$\text{We get } y - h = -\frac{1}{2}gt^2 \Rightarrow y = h - \frac{1}{2}gt^2$$

The graph between y and t is a parabola with $y = h$ at $t = 0$. As time increases y decreases.

For upward motion.

The ball suffer elastic collision with the horizontal elastic plate therefore the direction of velocity is reversed and the magnitude remains the same.

Here $v = u - gt$ where u is the velocity just after collision. As t increases, v decreases. We get a straight line between v and t with negative slope.

$$\text{Also } y = ut - \frac{1}{2}gt^2$$

All these characteristics are represented by graph (b).

42. (d) Initial velocity of parachute after bailing out,

$$u = \sqrt{2gh}$$

$$u = \sqrt{2 \times 9.8 \times 50} = 14\sqrt{5}$$

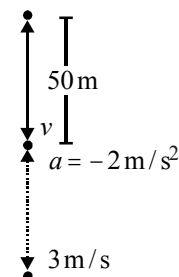
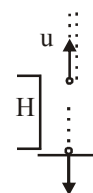
The velocity at ground,

$$v = 3\text{m/s}$$

$$S = \frac{v^2 - u^2}{2 \times 2} = \frac{3^2 - 980}{4} \approx 243 \text{ m}$$

Initially he has fallen 50 m.

$\therefore$ Total height from where he bailed out = $243 + 50 = 293 \text{ m}$



43. (a) We have $s = ut + \frac{1}{2}gt^2$,

$$\Rightarrow h = 0 \times T + \frac{1}{2}gT^2$$

$$\Rightarrow h = \frac{1}{2}gT^2$$

Vertical distance moved in time $\frac{T}{3}$ is

$$h' = \frac{1}{2}g\left(\frac{T}{3}\right)^2 \Rightarrow h' = \frac{1}{2} \times \frac{gT^2}{9} = \frac{h}{9}$$

$$\therefore \text{Position of ball from ground} = h - \frac{h}{9} = \frac{8h}{9}$$

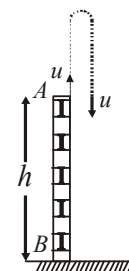
44. (b) Ball A is thrown upwards with velocity u

from the building. During its downward journey when it comes back to the point of throw, its speed is equal to the speed of throw

(u). So, for the journey of both the balls from point A to B .

We can apply $v^2 - u^2 = 2gh$.

As u , g , h are same for both the balls, $v_A = v_B$



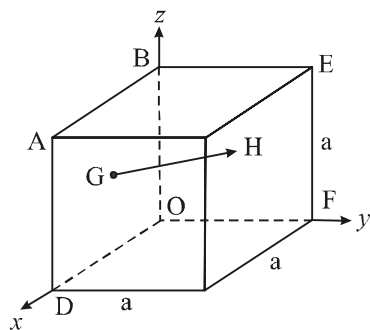
Motion in a Plane



TOPIC 1 Vectors



1. A force $\vec{F} = (\hat{i} + 2\hat{j} + 3\hat{k})$ N acts at a point $(4\hat{i} + 3\hat{j} - \hat{k})$ m. Then the magnitude of torque about the point $(\hat{i} + 2\hat{j} + \hat{k})$ m will be $\sqrt{x}$ N-m. The value of x is _____.
[NA Sep. 05, 2020 (I)]
2. The sum of two forces $\vec{P}$ and $\vec{Q}$ is $\vec{R}$ such that $|\vec{R}| = |\vec{P}|$. The angle θ (in degrees) that the resultant of $2\vec{P}$ and $\vec{Q}$ will make with $\vec{Q}$ is _____.
[NA 7 Jan. 2020 II]
3. Let $|\vec{A}_1| = 3$, $|\vec{A}_2| = 5$ and $|\vec{A}_1 + \vec{A}_2| = 5$. The value of $(2\vec{A}_1 + 3\vec{A}_2) \cdot (3\vec{A}_1 - 2\vec{A}_2)$ is : [8 April 2020 II]
(a) -106.5 (b) -99.5
(c) -112.5 (d) -118.5
4. In the cube of side 'a' shown in the figure, the vector from the central point of the face ABOD to the central point of the face BEFO will be: [10 Jan. 2019 I]

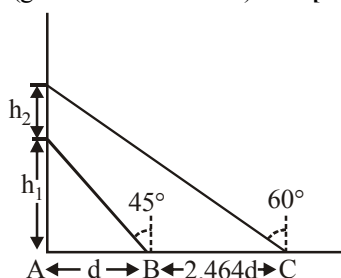


- (a) $\frac{1}{2}a(\hat{k} - \hat{i})$
- (b) $\frac{1}{2}a(\hat{i} - \hat{k})$
- (c) $\frac{1}{2}a(\hat{j} - \hat{i})$
- (d) $\frac{1}{2}a(\hat{j} - \hat{k})$

5. Two forces P and Q, of magnitude 2F and 3F, respectively, are at an angle θ with each other. If the force Q is doubled, then their resultant also gets doubled. Then, the angle θ is: [10 Jan. 2019 II]
(a) 120° (b) 60°
(c) 90° (d) 30°
6. Two vectors $\vec{A}$ and $\vec{B}$ have equal magnitudes. The magnitude of $(\vec{A} + \vec{B})$ is 'n' times the magnitude of $(\vec{A} - \vec{B})$. The angle between $\vec{A}$ and $\vec{B}$ is: [10 Jan. 2019 II]
(a) $\cos^{-1} \left[\frac{n^2 - 1}{n^2 + 1} \right]$ (b) $\cos^{-1} \left[\frac{n - 1}{n + 1} \right]$
(c) $\sin^{-1} \left[\frac{n^2 - 1}{n^2 + 1} \right]$ (d) $\sin^{-1} \left[\frac{n - 1}{n + 1} \right]$
7. Let $\vec{A} = (\hat{i} + \hat{j})$ and $\vec{B} = (\hat{i} - \hat{j})$. The magnitude of a coplanar vector $\vec{C}$ such that $\vec{A} \cdot \vec{C} = \vec{B} \cdot \vec{C} = \vec{A} \cdot \vec{B}$ is given by [Online April 16, 2018]
(a) $\sqrt{\frac{5}{9}}$ (b) $\sqrt{\frac{10}{9}}$
(c) $\sqrt{\frac{20}{9}}$ (d) $\sqrt{\frac{9}{12}}$
8. A vector $\vec{A}$ is rotated by a small angle $\Delta\theta$ radian ($\Delta\theta \ll 1$) to get a new vector $\vec{B}$. In that case $|\vec{B} - \vec{A}|$ is : [Online April 11, 2015]
(a) $|\vec{A}| \Delta\theta$ (b) $|\vec{B}| \Delta\theta - |\vec{A}|$
(c) $|\vec{A}| \left(1 - \frac{\Delta\theta^2}{2} \right)$ (d) 0
9. If $\vec{A} \times \vec{B} = \vec{B} \times \vec{A}$, then the angle between A and B is [2004]
(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$
(c) π (d) $\frac{\pi}{4}$

TOPIC 2 Motion in a Plane with Constant Acceleration


10. A balloon is moving up in air vertically above a point A on the ground. When it is at a height h_1 , a girl standing at a distance d (point B) from A (see figure) sees it at an angle 45° with respect to the vertical. When the balloon climbs up a further height h_2 , it is seen at an angle 60° with respect to the vertical if the girl moves further by a distance $2.464d$ (point C). Then the height h_2 is (given $\tan 30^\circ = 0.5774$): [Sep. 05, 2020 (I)]



- (a) $1.464d$ (b) $0.732d$
(c) $0.464d$ (d) d
11. Starting from the origin at time $t = 0$, with initial velocity $5\hat{j} \text{ ms}^{-1}$, a particle moves in the x - y plane with a constant acceleration of $(10\hat{i} + 4\hat{j}) \text{ ms}^{-2}$. At time t , its coordinates are $(20 \text{ m}, y_0 \text{ m})$. The values of t and y_0 are, respectively: [Sep. 04, 2020 (I)]
- (a) 2 s and 18 m (b) 4 s and 52 m
(c) 2 s and 24 m (d) 5 s and 25 m
12. The position vector of a particle changes with time according to the relation $\vec{r}(t) = 15t^2\hat{i} + (4 - 20t^2)\hat{j}$. What is the magnitude of the acceleration at $t = 1$? [9 April 2019 II]
- (a) 40 (b) 25 (c) 100 (d) 50
13. A particle moves from the point $(2.0\hat{i} + 4.0\hat{j}) \text{ m}$, at $t = 0$, with an initial velocity $(5.0\hat{i} + 4.0\hat{j}) \text{ ms}^{-1}$. It is acted upon by a constant force which produces a constant acceleration $(4.0\hat{i} + 4.0\hat{j}) \text{ ms}^{-2}$. What is the distance of the particle from the origin at time 2s? [11 Jan. 2019 II]
- (a) 15m (b) $20\sqrt{2}\text{m}$
(c) 5m (d) $10\sqrt{2}\text{m}$
14. A particle is moving with a velocity $\vec{v} = K(y\hat{i} + x\hat{j})$, where K is a constant. The general equation for its path is: [9 Jan. 2019 I]
- (a) $y = x^2 + \text{constant}$ (b) $y^2 = x + \text{constant}$
(c) $y^2 = x^2 + \text{constant}$ (d) $xy = \text{constant}$

15. A particle starts from the origin at $t = 0$ with an initial velocity of $3.0\hat{i} \text{ m/s}$ and moves in the x - y plane with a constant acceleration $(6.0\hat{i} + 4.0\hat{j}) \text{ m/s}^2$. The x -coordinate of the particle at the instant when its y -coordinate is 32 m is D meters. The value of D is: [9 Jan. 2020 II]

(a) 32 (b) 50 (c) 60 (d) 40

16. A particle is moving along the x -axis with its coordinate with time ' t ' given by $x(t) = 10 + 8t - 3t^2$. Another particle is moving along the y -axis with its coordinate as a function of time given by $y(t) = 5 - 8t^3$. At $t = 1 \text{ s}$, the speed of the second particle as measured in the frame of the first particle is given as $\sqrt{v}$. Then v (in m/s) is: [NA 8 Jan. 2020 I]
17. A particle moves such that its position vector $\vec{r}(t) = \cos \omega t \hat{i} + \sin \omega t \hat{j}$ where ω is a constant and t is time. Then which of the following statements is true for the velocity $\vec{v}(t)$ and acceleration $\vec{a}(t)$ of the particle: [8 Jan. 2020 II]
- (a) $\vec{v}$ is perpendicular to $\vec{r}$ and $\vec{a}$ is directed away from the origin
(b) $\vec{v}$ and $\vec{a}$ both are perpendicular to $\vec{r}$
(c) $\vec{v}$ and $\vec{a}$ both are parallel to $\vec{r}$
(d) $\vec{v}$ is perpendicular to $\vec{r}$ and $\vec{a}$ is directed towards the origin

18. A particle is moving with velocity $\vec{v} = k(y\hat{i} + x\hat{j})$, where k is a constant. The general equation for its path is: [2010]

(a) $y = x^2 + \text{constant}$ (b) $y^2 = x + \text{constant}$
(c) $xy = \text{constant}$ (d) $y^2 = x^2 + \text{constant}$

19. A particle has an initial velocity of $3\hat{i} + 4\hat{j}$ and an acceleration of $0.4\hat{i} + 0.3\hat{j}$. Its speed after 10 s is: [2009]

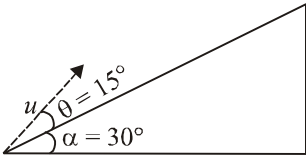
(a) $7\sqrt{2}$ units (b) 7 units
(c) 8.5 units (d) 10 units

20. The co-ordinates of a moving particle at any time ' t ' are given by $x = \alpha t^3$ and $y = \beta t^3$. The speed of the particle at time ' t ' is given by: [2003]

(a) $3t\sqrt{\alpha^2 + \beta^2}$ (b) $3t^2\sqrt{\alpha^2 + \beta^2}$
(c) $t^2\sqrt{\alpha^2 + \beta^2}$ (d) $\sqrt{\alpha^2 + \beta^2}$

TOPIC 3 Projectile Motion


21. A particle of mass m is projected with a speed u from the ground at an angle $\theta = \frac{\pi}{3}$ w.r.t. horizontal (x -axis). When it has reached its maximum height, it collides completely inelastically with another particle of the same mass and velocity $u\hat{i}$. The horizontal distance covered by the combined mass before reaching the ground is: [9 Jan. 2020 II]

- (a) $\frac{3\sqrt{3}}{8} \frac{u^2}{g}$ (b) $\frac{3\sqrt{2}}{4} \frac{u^2}{g}$
 (c) $\frac{5}{8} \frac{u^2}{g}$ (d) $2\sqrt{2} \frac{u^2}{g}$
22. The trajectory of a projectile near the surface of the earth is given as $y = 2x - 9x^2$. If it were launched at an angle θ_0 with speed v_0 then ($g = 10 \text{ ms}^{-2}$): [12 April 2019 I]
 (a) $\theta_0 = \sin^{-1} \frac{1}{\sqrt{5}}$ and $v_0 = \frac{5}{3} \text{ ms}^{-1}$
 (b) $\theta_0 = \cos^{-1} \left(\frac{2}{\sqrt{5}} \right)$ and $v_0 = \frac{3}{5} \text{ ms}^{-1}$
 (c) $\theta_0 = \cos^{-1} \left(\frac{1}{\sqrt{5}} \right)$ and $v_0 = \frac{9}{3} \text{ ms}^{-1}$
 (d) $\theta_0 = \sin^{-1} \left(\frac{2}{\sqrt{5}} \right)$ and $v_0 = \frac{3}{5} \text{ ms}^{-1}$
23. A shell is fired from a fixed artillery gun with an initial speed u such that it hits the target on the ground at a distance R from it. If t_1 and t_2 are the values of the time taken by it to hit the target in two possible ways, the product $t_1 t_2$ is : [12 April 2019 I]
 (a) $R/4g$ (b) R/g (c) $R/2g$ (d) $2R/g$
24. Two particles are projected from the same point with the same speed u such that they have the same range R , but different maximum heights, h_1 and h_2 . Which of the following is correct ? [12 April 2019 II]
 (a) $R^2 = 4 h_1 h_2$ (b) $R^2 = 16 h_1 h_2$
 (c) $R^2 = 2 h_1 h_2$ (d) $R^2 = h_1 h_2$
25. A plane is inclined at an angle $\alpha = 30^\circ$ with respect to the horizontal. A particle is projected with a speed $u = 2 \text{ ms}^{-1}$, from the base of the plane, as shown in figure. The distance from the base, at which the particle hits the plane is close to : (Take $g = 10 \text{ ms}^{-2}$) [10 April 2019 II]
- 
- (a) 20 cm (b) 18 cm (c) 26 cm (d) 14 cm
26. A body is projected at $t = 0$ with a velocity 10 ms^{-1} at an angle of 60° with the horizontal. The radius of curvature of its trajectory at $t = 1 \text{ s}$ is R . Neglecting air resistance and taking acceleration due to gravity $g = 10 \text{ ms}^{-2}$, the value of R is: [11 Jan. 2019 I]
 (a) 10.3 m (b) 2.8 m
 (c) 2.5 m (d) 5.1 m
27. Two guns A and B can fire bullets at speeds 1 km/s and 2 km/s respectively. From a point on a horizontal ground, they are fired in all possible directions. The ratio of maximum areas covered by the bullets fired by the two guns, on the ground is: [10 Jan. 2019 I]
 (a) 1 : 16 (b) 1 : 2 (c) 1 : 4 (d) 1 : 8
28. The initial speed of a bullet fired from a rifle is 630 m/s. The rifle is fired at the centre of a target 700 m away at the same level as the target. How far above the centre of the target ? [Online April 11, 2014]
 (a) 1.0 m (b) 4.2 m (c) 6.1 m (d) 9.8 m
29. The position of a projectile launched from the origin at $t = 0$ is given by $\vec{r} = (40\hat{i} + 50\hat{j}) \text{ m}$ at $t = 2 \text{ s}$. If the projectile was launched at an angle θ from the horizontal, then θ is (take $g = 10 \text{ ms}^{-2}$) [Online April 9, 2014]
 (a) $\tan^{-1} \frac{2}{3}$ (b) $\tan^{-1} \frac{3}{2}$
 (c) $\tan^{-1} \frac{7}{4}$ (d) $\tan^{-1} \frac{4}{5}$
30. A projectile is given an initial velocity of $(\hat{i} + 2\hat{j}) \text{ m/s}$, where $\hat{i}$ is along the ground and $\hat{j}$ is along the vertical. If $g = 10 \text{ m/s}^2$, the equation of its trajectory is : [2013]
 (a) $y = x - 5x^2$ (b) $y = 2x - 5x^2$
 (c) $4y = 2x - 5x^2$ (d) $4y = 2x - 25x^2$
31. The maximum range of a bullet fired from a toy pistol mounted on a car at rest is $R_0 = 40 \text{ m}$. What will be the acute angle of inclination of the pistol for maximum range when the car is moving in the direction of firing with uniform velocity $v = 20 \text{ m/s}$, on a horizontal surface ? ($g = 10 \text{ m/s}^2$) [Online April 25, 2013]
 (a) 30° (b) 60° (c) 75° (d) 45°
32. A ball projected from ground at an angle of 45° just clears a wall in front. If point of projection is 4 m from the foot of wall and ball strikes the ground at a distance of 6 m on the other side of the wall, the height of the wall is : [Online April 22, 2013]
 (a) 4.4 m (b) 2.4 m (c) 3.6 m (d) 1.6 m
33. A boy can throw a stone up to a maximum height of 10 m. The maximum horizontal distance that the boy can throw the same stone up to will be [2012]
 (a) $20\sqrt{2} \text{ m}$ (b) 10 m
 (c) $10\sqrt{2} \text{ m}$ (d) 20 m
34. A water fountain on the ground sprinkles water all around it. If the speed of water coming out of the fountain is v , the total area around the fountain that gets wet is: [2011]
 (a) $\pi \frac{v^4}{g^2}$ (b) $\frac{\pi}{2} \frac{v^4}{g^2}$ (c) $\pi \frac{v^2}{g^2}$ (d) $\pi \frac{v^2}{g}$

35. A projectile can have the same range ' R ' for two angles of projection. If ' T_1 ' and ' T_2 ' to be time of flights in the two cases, then the product of the two time of flights is directly proportional to. [2004]

(a) R (b) $\frac{1}{R}$ (c) $\frac{1}{R^2}$ (d) R^2

36. A ball is thrown from a point with a speed ' v_0 ' at an elevation angle of θ . From the same point and at the same instant, a person starts running with a constant speed $\frac{v_0}{2}$

to catch the ball. Will the person be able to catch the ball? If yes, what should be the angle of projection θ ? [2004]

(a) No (b) Yes, 30°
(c) Yes, 60° (d) Yes, 45°

37. A boy playing on the roof of a 10 m high building throws a ball with a speed of 10 m/s at an angle of 30° with the horizontal. How far from the throwing point will the ball be at the height of 10 m from the ground? [2003]

[$g = 10 \text{ m/s}^2$, $\sin 30^\circ = \frac{1}{2}$, $\cos 30^\circ = \frac{\sqrt{3}}{2}$]

(a) 5.20m (b) 4.33m (c) 2.60m (d) 8.66m

TOPIC 4 Relative Velocity in Two Dimensions & Uniform Circular Motion



38. A clock has a continuously moving second's hand of 0.1 m length. The average acceleration of the tip of the hand (in units of ms^{-2}) is of the order of: [Sep. 06, 2020 (I)]

(a) 10^{-3} (b) 10^{-4}
(c) 10^{-2} (d) 10^{-1}

39. When a car is at rest, its driver sees raindrops falling on it vertically. When driving the car with speed v , he sees that raindrops are coming at an angle 60° from the horizontal. On further increasing the speed of the car to $(1 + \beta)v$, this angle changes to 45° . The value of β is close to:

[Sep. 06, 2020 (II)]

(a) 0.50 (b) 0.41
(c) 0.37 (d) 0.73

40. The stream of a river is flowing with a speed of 2 km/h. A swimmer can swim at a speed of 4 km/h. What should be the direction of the swimmer with respect to the flow of the river to cross the river straight? [9 April 2019 I]

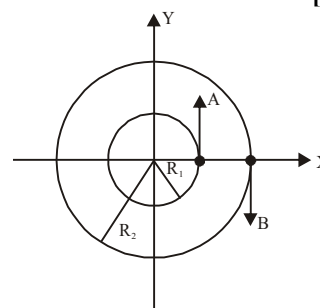
(a) 90° (b) 150°
(c) 120° (d) 60°

41. Ship A is sailing towards north-east with velocity km/hr where points east and north. Ship B is at a distance of 80 km east and 150 km north of Ship A and is sailing towards west at 10 km/hr. A will be at minimum distance from B in:

[8 April 2019 I]

(a) 4.2 hrs. (b) 2.6 hrs.
(c) 3.2 hrs. (d) 2.2 hrs.

42. Two particles A, B are moving on two concentric circles of radii R_1 and R_2 with equal angular speed ω . At $t = 0$, their positions and direction of motion are shown in the figure: [12 Jan. 2019 II]



The relative velocity $\vec{v}_A - \vec{v}_B$ and $t = \frac{\pi}{2\omega}$ is given by:

(a) $\omega(R_1 + R_2)\hat{i}$ (b) $-\omega(R_1 + R_2)\hat{i}$
(c) $\omega(R_2 - R_1)\hat{i}$ (d) $\omega(R_1 - R_2)\hat{i}$

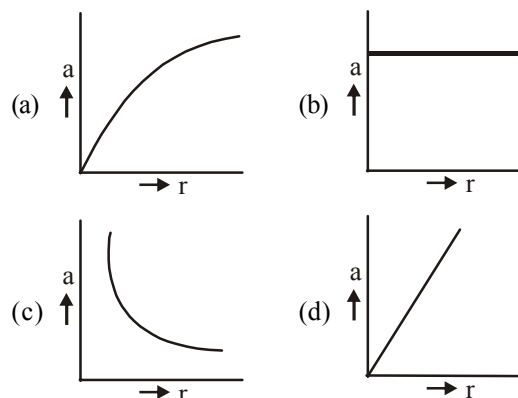
43. A particle is moving along a circular path with a constant speed of 10 ms^{-1} . What is the magnitude of the change in velocity of the particle, when it moves through an angle of 60° around the centre of the circle?

[Online April 10, 2015]

(a) $10\sqrt{3} \text{ m/s}$ (b) zero
(c) $10\sqrt{2} \text{ m/s}$ (d) 10 m/s

44. If a body moving in circular path maintains constant speed of 10 ms^{-1} , then which of the following correctly describes relation between acceleration and radius?

[Online April 10, 2015]





Hints & Solutions



1. (195)

Given : $\vec{F} = (\hat{i} + 2\hat{j} + 3\hat{k}) \text{ N}$

And, $\vec{r} = [(4\hat{i} + 3\hat{j} - \hat{k}) - (\hat{i} + 2\hat{j} + \hat{k})] = 3\hat{i} + \hat{j} - 2\hat{k}$

Torque, $\tau = \vec{r} \times \vec{F} = (3\hat{i} + \hat{j} - 2\hat{k}) \times (\hat{i} + 2\hat{j} + 3\hat{k})$

$$\tau = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & -2 \\ 1 & 2 & 3 \end{vmatrix} = 7\hat{i} - 11\hat{j} + 5\hat{k}$$

Magnitude of torque, $|\vec{\tau}| = \sqrt{195}$.

2. (90)

Given,

$$|\vec{R}| = |\vec{P}| \Rightarrow |\vec{P} + \vec{Q}| = |\vec{P}|$$

$$P^2 + Q^2 + 2PQ \cos \theta = P^2$$

$$\Rightarrow Q + 2P \cos \theta = 0$$

$$\Rightarrow \cos \theta = -\frac{Q}{2P}$$

$$\tan \alpha = \frac{2P \sin \theta}{Q + 2P \cos \theta} = \infty (\because 2P \cos \theta + Q = 0)$$

$$\Rightarrow \alpha = 90^\circ$$

3. (d) Using,

$$R^2 = A_1^2 + A_2^2 + 2A_1A_2 \cos \theta$$

$$5^2 = 3^2 + 5^2 + 2 \times 3 \times 5 \cos \theta$$

$$\text{or } \cos \theta = -0.3$$

$$\left(2\vec{A}_1 + 3\vec{A}_2 \right) \cdot \left(3\vec{A}_1 - 2\vec{A}_2 \right) = 2A_1 \times 3A_1$$

$$\begin{aligned} &+ (3A_2)(3A_1) \cos \theta - (2A_1)(2A_2) \cos \theta - 3A_2 \times 2A_1 \\ &= 6A_1^2 + 9A_1A_2 \cos \theta - 4A_1A_2 \cos \theta - 6A_2^2 \\ &= 6A_1^2 - 6A_2^2 + 5A_1A_2 \cos \theta \\ &= 6 \times 3^2 - 6 \times 5^2 + 5 \times 3 \times 5 (-0.3) \\ &= -118.5 \end{aligned}$$

4. (c) From figure,

$$\vec{r}_G = \frac{a}{2}\hat{i} + \frac{a}{2}\hat{k}$$

$$\vec{r}_H = \frac{a}{2}\hat{j} + \frac{a}{2}\hat{k}$$

$$\therefore \vec{r}_H - \vec{r}_G = \left(\frac{a}{2}\hat{j} + \frac{a}{2}\hat{k} \right) - \left(\frac{a}{2}\hat{i} + \frac{a}{2}\hat{k} \right) = \frac{a}{2}(\hat{j} - \hat{i})$$

5. (a) Using, $R^2 = P^2 + Q^2 + 2PQ \cos \theta$

$$4F^2 + 9F^2 + 12F^2 \cos \theta = R^2$$

When forces Q is doubled,

$$4F^2 + 36F^2 + 24F^2 \cos \theta = 4R^2$$

$$4F^2 + 36F^2 + 24F^2 \cos \theta$$

$$= 4(13F^2 + 12F^2 \cos \theta) = 52F^2 + 48F^2 \cos \theta$$

$$\therefore \cos \theta = -\frac{12F^2}{24F^2} = -\frac{1}{2} \Rightarrow \theta = 120^\circ$$

6. (a) Let magnitude of two vectors $\vec{A}$ and $\vec{B} = a$

$$|\vec{A} + \vec{B}| = \sqrt{a^2 + a^2 + 2a^2 \cos \theta} \text{ and}$$

$$|\vec{A} - \vec{B}| = \sqrt{a^2 + a^2 - 2a^2 [\cos(180^\circ - \theta)]}$$

$$= \sqrt{a^2 + a^2 - 2a^2 \cos \theta}$$

and according to question,

$$|\vec{A} + \vec{B}| = n |\vec{A} - \vec{B}|$$

$$\text{or, } \frac{a^2 + a^2 + 2a^2 \cos \theta}{a^2 + a^2 - 2a^2 \cos \theta} = n^2$$

$$\Rightarrow \frac{1 + 1 + 2 \cos \theta}{1 + 1 - 2 \cos \theta} n^2 = \frac{(1 + \cos \theta)}{(1 - \cos \theta)} = n^2$$

using componendo and dividendo theorem, we get

$$\theta = \cos^{-1} \left(\frac{n^2 - 1}{n^2 + 1} \right)$$

7. (a) If $\vec{C} = a\hat{i} + b\hat{j}$ then $\vec{A} \cdot \vec{C} = \vec{A} \cdot \vec{B}$

$$a + b = 1 \quad \dots (i)$$

$$\vec{B} \cdot \vec{C} = \vec{A} \cdot \vec{B}$$

$$2a - b = 1 \quad \dots (ii)$$

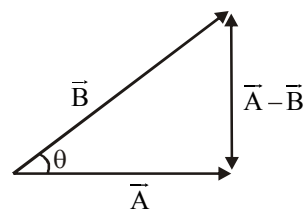
Solving equation (i) and (ii) we get

$$a = \frac{1}{3}, b = \frac{2}{3}$$

$$\therefore \text{Magnitude of coplanar vector, } |\vec{C}| = \sqrt{\frac{1}{9} + \frac{4}{9}} = \sqrt{\frac{5}{9}}$$

8. (a) Arc length = radius $\times$ angle

$$\text{So, } |\vec{B} - \vec{A}| = |\vec{A}| \Delta \theta$$



9. (c) $\vec{A} \times \vec{B} - \vec{B} \times \vec{A} = 0 \Rightarrow \vec{A} \times \vec{B} + \vec{A} \times \vec{B} = 0$

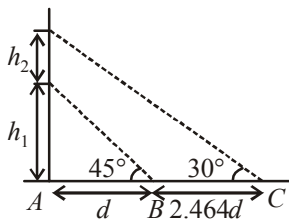
$$\therefore \vec{A} \times \vec{B} = 0$$

Angle between them is $0, \pi$, or 2π

from the given options, $\theta = \pi$

10. (d) From figure/ trigonometry,

$$\frac{h_1}{d} = \tan 45^\circ \quad \therefore h_1 = d$$



$$\text{And, } \frac{h_1 + h_2}{d + 2.464d} = \tan 30^\circ$$

$$\Rightarrow (h_1 + h_2) \times \sqrt{3} = 3.46d$$

$$\Rightarrow (h_1 + h_2) = \frac{3.46d}{\sqrt{3}} \Rightarrow d + h_2 = \frac{3.46d}{\sqrt{3}}$$

$$\therefore h_2 = d$$

11. (a) Given: $\vec{u} = 5\hat{j}$ m/s

Acceleration, $\vec{a} = 10\hat{i} + 4\hat{j}$ and
final coordinate $(20, y_0)$ in time t .

$$S_x = u_x t + \frac{1}{2} a_x t^2 \quad [\because u_x = 0]$$

$$\Rightarrow 20 = 0 + \frac{1}{2} \times 10 \times t^2 \Rightarrow t = 2 \text{ s}$$

$$S_y = u_y t + \frac{1}{2} a_y t^2$$

$$y_0 = 5 \times 2 + \frac{1}{2} \times 4 \times 2^2 = 18 \text{ m}$$

12. (d) $\vec{r} = 15t^2\hat{i} + (4 - 20t^2)\hat{j}$

$$\vec{v} = \frac{d\vec{r}}{dt} = 30t\hat{i} - 40t\hat{j}$$

$$\text{Acceleration, } \vec{a} = \frac{d\vec{v}}{dt} = 30\hat{i} - 40\hat{j}$$

$$\therefore a = \sqrt{30^2 + 40^2} = 50 \text{ m/s}^2$$

13. (b) As $\vec{S} = \vec{u}t + \frac{1}{2}\vec{a}t^2$

$$\vec{S} = (5\hat{i} + 4\hat{j})2 + \frac{1}{2}(4\hat{i} + 4\hat{j})4$$

$$= 10\hat{i} + 8\hat{j} + 8\hat{i} + 8\hat{j}$$

$$\vec{r}_f - \vec{r}_i = 18\hat{i} + 16\hat{j}$$

[as $\vec{S}$ = change in position = $\vec{r}_f - \vec{r}_i$]

$$\vec{r}_f = 20\hat{i} + 20\hat{j}$$

$$|\vec{r}_f| = 20\sqrt{2}$$

14. (c) From given equation,

$$\vec{V} = K(\hat{y}\hat{i} + \hat{x}\hat{j})$$

$$\frac{dx}{dt} = ky \text{ and } \frac{dy}{dt} = kx$$

$$\frac{dy}{dt} = \frac{x}{y} = \frac{dy}{dx} \Rightarrow ydy = xdx$$

Integrating both side

$$y^2 = x^2 + c$$

15. (c) Using $S = ut + \frac{1}{2}at^2$

$$y = u_y t + \frac{1}{2} a_y t^2 \text{ (along y Axis)}$$

$$\Rightarrow 32 = 0 \times t + \frac{1}{2}(4)t^2$$

$$\Rightarrow \frac{1}{2} \times 4 \times t^2 = 32$$

$$\Rightarrow t = 4 \text{ s}$$

$$S_x = u_x t + \frac{1}{2} a_x t^2 \quad \text{(Along x Axis)}$$

$$\Rightarrow x = 3 \times 4 + \frac{1}{2} \times 6 \times 4^2 = 60$$

16. (580)

For particle 'A'

$$X_A = -3t^2 + 8t + 10$$

$$\vec{V}_A = (8 - 6t)\hat{i}$$

$$\vec{a}_A = -6\hat{i}$$

At $t = 1$ sec

$$\vec{V}_A = (8 - 6t)\hat{i} = 2\hat{i} \text{ and } \vec{v}_B = -24\hat{j}$$

$$\therefore \vec{V}_{B/A} = -\vec{v}_A + \vec{v}_B = -2\hat{i} - 24\hat{j}$$

$$\therefore \text{Speed of B w.r.t. A, } \sqrt{v} = \sqrt{2^2 + 24^2}$$

$$= \sqrt{4 + 576} = \sqrt{580}$$

$$\therefore v = 580 \text{ (m/s)}$$

For particle 'B'

$$Y_B = 5 - 8t^3$$

$$\vec{V}_B = -24t^2\hat{j}$$

$$\vec{a}_B = -48t\hat{j}$$

17. (d) Given, Position vector,

$$\vec{r} = \cos \omega t \hat{i} + \sin \omega t \hat{j}$$

$$\text{Velocity, } \vec{v} = \frac{d\vec{r}}{dt} = \omega(-\sin \omega t \hat{i} + \cos \omega t \hat{j})$$

Acceleration,

$$\vec{a} = \frac{d\vec{v}}{dt} = -\omega^2(\cos \omega t \hat{i} + \sin \omega t \hat{j})$$

$$\vec{a} = -\omega^2 \vec{r}$$

$$\therefore \vec{a} \text{ is antiparallel to } \vec{r}$$

$$\text{Also } \vec{v} \cdot \vec{r} = 0 \therefore \vec{v} \perp \vec{r}$$

Thus, the particle is performing uniform circular motion.

18. (d)
- $v = k(yi + xj)$

$$v = kyi + kxj$$

$$\frac{dx}{dt} = ky, \quad \frac{dy}{dt} = kx$$

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$\frac{dy}{dx} = \frac{kx}{ky}$$

$$ydy = xdx \quad \dots(i)$$

Integrating equation (i)

$$\int ydy = \int x \cdot dx$$

$$y^2 = x^2 + c$$

19. (a) Given
- $\vec{u} = 3\hat{i} + 4\hat{j}$
- ,
- $\vec{a} = 0.4\hat{i} + 0.3\hat{j}$
- ,
- $t = 10\text{s}$
-
- From 1st equation of motion.

$$a = \frac{v - u}{t}$$

$$\therefore v = at + u$$

$$\Rightarrow v = (0.4\hat{i} + 0.3\hat{j}) \times 10 + (3\hat{i} + 4\hat{j})$$

$$\Rightarrow 4\hat{i} + 3\hat{j} + 3\hat{i} + 4\hat{j}$$

$$\Rightarrow v = 7\hat{i} + 7\hat{j}$$

$$\Rightarrow |\vec{v}| = \sqrt{7^2 + 7^2} = 7\sqrt{2} \text{ unit.}$$

20. (b) Coordinates of moving particle at time 't' are
-
- $x = \alpha t^3$
- and
- $y = \beta t^3$

$$v_x = \frac{dx}{dt} = 3\alpha t^2 \text{ and } v_y = \frac{dy}{dt} = 3\beta t^2$$

$$\therefore v = \sqrt{v_x^2 + v_y^2} = \sqrt{9\alpha^2 t^4 + 9\beta^2 t^4}$$

$$= 3t^2 \sqrt{\alpha^2 + \beta^2}$$

21. (a) Using principle of conservation of linear momentum
-
- for horizontal motion, we have
-
- $2mv_x = mu + mu \cos 60^\circ$

$$v_x = \frac{3u}{4}$$

For vertical motion

$$h = 0 + \frac{1}{2}gT^2 \Rightarrow T = \sqrt{\frac{2h}{g}}$$

Let R is the horizontal distance travelled by the body.

$$R = v_x T + \frac{1}{2}(0)(T)^2 \text{ (For horizontal motion)}$$

$$R = v_x T = \frac{3u}{4} \times \sqrt{\frac{2h}{g}}$$

$$\Rightarrow R = \frac{3\sqrt{3}u^2}{8g}$$

22. (c) Given,
- $y = 2x - 9x^2$

On comparing with,

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta},$$

We have,

$$\tan \theta = 2 \text{ or } \cos \theta = \frac{1}{\sqrt{5}}$$

$$\text{and } \frac{g}{2u^2 \cos^2 \theta} = 9 \text{ or } \frac{10}{2u^2 (1/\sqrt{5})^2} = 9$$

$$\therefore u = 5/3 \text{ m/s}$$

23. (d) R will be same for
- θ
- and
- $90^\circ - \theta$
- .

Time of flights:

$$t_1 = \frac{2u \sin \theta}{g} \text{ and}$$

$$t_2 = \frac{2u \sin(90^\circ - \theta)}{g} = \frac{2u \cos \theta}{g}$$

$$\text{Now, } t_1 t_2 = \left(\frac{2u \sin \theta}{g} \right) \left(\frac{2u \cos \theta}{g} \right)$$

$$= \frac{2}{g} \left(\frac{u^2 \sin 2\theta}{g} \right) = \frac{2R}{g}$$

24. (b) For same range, the angle of projections are:
-
- θ
- and
- $90^\circ - \theta$
- . So,

$$h_1 = \frac{u^2 \sin^2 \theta}{2g} \text{ and}$$

$$h_2 = \frac{u^2 \sin^2(90^\circ - \theta)}{2g} = \frac{u^2 \cos^2 \theta}{2g}$$

$$\text{Also, } R = \frac{u^2 \sin 2\theta}{g}$$

$$h_1 h_2 = \frac{u^2 \sin^2 \theta}{2g} \times \frac{u^2 \cos^2 \theta}{2g}$$

$$= \frac{u^2}{16} \frac{u^2 (2 \sin \theta \cos \theta)^2}{g^2}$$

$$= \frac{R^2}{16}$$

$$\text{or } R^2 = 16 h_1 h_2$$

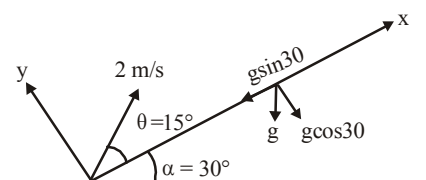
25. (a) On an inclined plane, time of flight (T) is given by

$$T = \frac{2u \sin \theta}{g \cos \alpha}$$

Substituting the values, we get

$$T = \frac{(2)(2 \sin 15^\circ)}{g \cos 30^\circ} = \frac{4 \sin 15^\circ}{10 \cos 30^\circ}$$

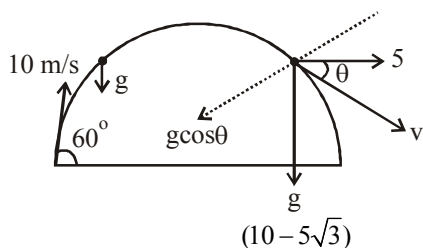
$$\text{Distance, } S = (2 \cos 15^\circ)T - \frac{1}{2}g \sin 30^\circ (T)^2$$



$$= (2 \cos 15^\circ) \frac{4}{10} \frac{\sin 15^\circ}{10 \cos 30^\circ} - \left(\frac{1}{2} \times 10 \sin 30^\circ \right) \frac{16 \sin^2 15^\circ}{100 \cos^2 30^\circ}$$

$$= \frac{16\sqrt{3} - 16}{60} \approx 0.1952 \text{ m} \approx 20 \text{ cm}$$

26. (b)



Horizontal component of velocity

$$v_x = 10 \cos 60^\circ = 5 \text{ m/s}$$

vertical component of velocity

$$v_y = 10 \sin 60^\circ = 5\sqrt{3} \text{ m/s}$$

 After $t = 1$ sec.

 Horizontal component of velocity $v_x = 5 \text{ m/s}$

Vertical component of velocity

$$v_y = |(5\sqrt{3} - 10)| \text{ m/s} = 10 - 5\sqrt{3}$$

 Centripetal, acceleration $a_n = \frac{v^2}{R}$

$$\Rightarrow R = \frac{v_x^2 + v_y^2}{a_n} = \frac{25 + 100 + 75 - 100\sqrt{3}}{10 \cos \theta} \quad \dots(i)$$

From figure (using (i))

$$\tan \theta = \frac{10 - 5\sqrt{3}}{5} = 2 - \sqrt{3} \Rightarrow \theta = 15^\circ$$

$$R = \frac{100(2 - \sqrt{3})}{10 \cos 15^\circ} = 2.8 \text{ m}$$

 27. (a) As we know, range $R = \frac{u^2 \sin 2\theta}{g}$

 and, area $A = \pi R^2$

$$\therefore A \propto R^2 \text{ or, } A \propto u^4$$

$$\therefore \frac{A_1}{A_2} = \frac{u_1^4}{u_2^4} = \left[\frac{1}{2} \right]^4 = \frac{1}{16}$$

 28. (c) Let ' t ' be the time taken by the bullet to hit the target.

$$\therefore 700 \text{ m} = 630 \text{ ms}^{-1} t$$

$$\Rightarrow t = \frac{700 \text{ m}}{630 \text{ ms}^{-1}} = \frac{10}{9} \text{ sec}$$

For vertical motion,

 Here, $u = 0$

$$\therefore h = \frac{1}{2} g t^2$$

$$= \frac{1}{2} \times 10 \times \left(\frac{10}{9} \right)^2$$

$$= \frac{500}{81} \text{ m} = 6.1 \text{ m}$$

Therefore, the rifle must be aimed 6.1 m above the centre of the target to hit the target.

29. (c) From question,

Horizontal velocity (initial),

$$u_x = \frac{40}{2} = 20 \text{ m/s}$$

 Vertical velocity (initial), $50 = u_y t + \frac{1}{2} g t^2$

$$\Rightarrow u_y \times 2 + \frac{1}{2} (-10) \times 4$$

$$\text{or, } 50 = 2u_y - 20$$

$$\text{or, } u_y = \frac{70}{2} = 35 \text{ m/s}$$

$$\therefore \tan \theta = \frac{u_y}{u_x} = \frac{35}{20} = \frac{7}{4}$$

$$\Rightarrow \text{Angle } \theta = \tan^{-1} \frac{7}{4}$$

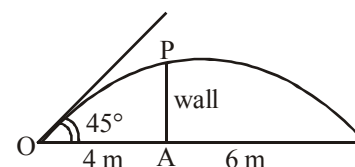
 30. (b) From equation, $\vec{v} = \hat{i} + 2\hat{j}$

$$\Rightarrow x = t \quad \dots(i)$$

$$y = 2t - \frac{1}{2} (10t^2) \quad \dots(ii)$$

 From (i) and (ii), $y = 2x - 5x^2$

31. (b)



32. (b)

 As ball is projected at an angle 45° to the horizontal therefore Range = 4H

$$\text{or } 10 = 4H \Rightarrow H = \frac{10}{4} = 2.5 \text{ m}$$

$$(\because \text{Range} = 4 \text{ m} + 6 \text{ m} = 10 \text{ m})$$

$$\text{Maximum height, } H = \frac{u^2 \sin^2 \theta}{2g}$$

$$\therefore u^2 = \frac{H \times 2g}{\sin^2 \theta} = \frac{2.5 \times 2 \times 10}{\left(\frac{1}{\sqrt{2}} \right)^2} = 100$$

$$\text{or, } u = \sqrt{100} = 10 \text{ ms}^{-1}$$

Height of wall PA

$$= OA \tan \theta - \frac{1}{2} \frac{g(OA)^2}{u^2 \cos^2 \theta}$$

$$= 4 - \frac{1}{2} \times \frac{10 \times 16}{10 \times 10 \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}} = 2.4 \text{ m}$$

33. (d) $R = \frac{u^2 \sin^2 \theta}{g}$, $H = \frac{u^2 \sin^2 \theta}{2g}$

$H_{\max}$ at $2\theta = 90^\circ$

$H_{\max} = \frac{u^2}{2g}$

$\frac{u^2}{2g} = 10 \Rightarrow u^2 = 10g \times 2$

$R = \frac{u^2 \sin 2\theta}{g} \Rightarrow R_{\max} = \frac{u^2}{g}$

$R_{\max} = \frac{10 \times g \times 2}{g} = 20 \text{ metre}$

34. (a) Let, total area around fountain

$A = \pi R_{\max}^2$... (i)

Where $R_{\max} = \frac{v^2 \sin 2\theta}{g} = \frac{v^2 \sin 90^\circ}{g} = \frac{v^2}{g}$... (ii)

From equation (i) and (ii)

$A = \pi \frac{v^4}{g^2}$

35. (a) A projectile have same range for two angle
Let one angle be θ , then other is $90^\circ - \theta$

$T_1 = \frac{2u \sin \theta}{g}$, $T_2 = \frac{2u \cos \theta}{g}$

then, $T_1 T_2 = \frac{4u^2 \sin \theta \cos \theta}{g} = 2R$

$(\because R = \frac{u^2 \sin^2 \theta}{g})$

Thus, it is proportional to R . (Range)

36. (c) Yes, Man will catch the ball, if the horizontal component of velocity becomes equal to the constant speed of man.

$\frac{v_o}{2} = v_o \cos \theta$

or $\theta = 60^\circ$

37. (d) Horizontal range is required

$R = \frac{u^2 \sin 2\theta}{g} = \frac{(10)^2 \sin(2 \times 30^\circ)}{10} = 5\sqrt{3} = 8.66 \text{ m}$

38. (a) Here, $R = 0.1 \text{ m}$

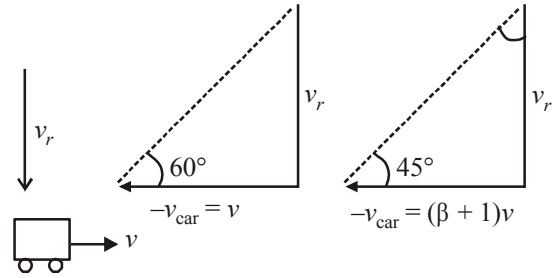
$\omega = \frac{2\pi}{T} = \frac{2\pi}{60} = 0.105 \text{ rad/s}$

Acceleration of the tip of the clock second's hand,

$a = \omega^2 R = (0.105)^2 (0.1) = 0.0011 = 1.1 \times 10^{-3} \text{ m/s}^2$

Hence, average acceleration is of the order of 10^{-3} .

39. (d) The given situation is shown in the diagram. Here v_r be the velocity of rain drop.



When car is moving with speed v ,

$\tan 60^\circ = \frac{v_r}{v}$... (i)

When car is moving with speed $(1 + \beta)v$,

$\tan 45^\circ = \frac{v_r}{(\beta + 1)v}$... (ii)

Dividing (i) by (ii) we get,

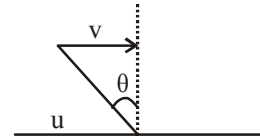
$\sqrt{3}v = (\beta + 1)v \Rightarrow \beta = \sqrt{3} - 1 = 0.732$.

40. (c) $\sin \theta = \frac{u}{v} = \frac{2}{4} = \frac{1}{2}$

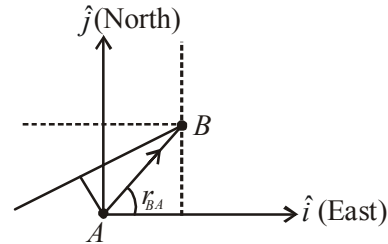
or $\theta = 30^\circ$

with respect to flow,

$= 90^\circ + 30^\circ = 120^\circ$



41. (b)



$\vec{v}_A = 30\hat{i} + 50\hat{j} \text{ km/hr}$

$\vec{v}_B = (-10\hat{i}) \text{ km/hr}$

$r_{BA} = (80\hat{i} + 150\hat{j}) \text{ km}$

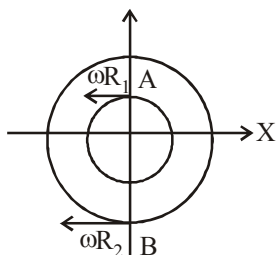
$\vec{v}_{BA} = \vec{v}_B - \vec{v}_A = -10\hat{i} - 30\hat{i} - 50\hat{j} = -40\hat{i} - 50\hat{j}$

$t_{\text{minimum}} = \frac{|(\vec{r}_{BA})(\vec{v}_{BA})|}{|(\vec{v}_{BA})|^2}$

$= \frac{|(80\hat{i} + 150\hat{j})(-40\hat{i} - 50\hat{j})|}{(10\sqrt{41})^2}$

$\therefore t = \frac{10700}{10\sqrt{41} \times 10\sqrt{41}} = \frac{107}{41} = 2.6 \text{ hrs.}$

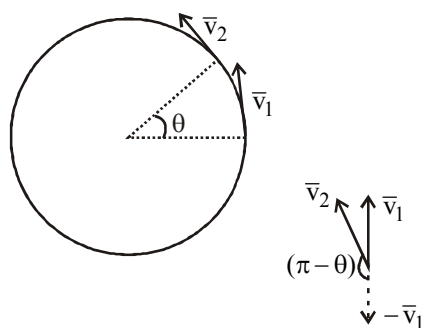
42. (c) From, $\theta = \omega t = \omega \frac{\pi}{2\omega} = \frac{\pi}{2}$
So, both have completed quarter circle



Relative velocity,

$$v_A - v_B = \omega R_1 (-\hat{i}) - \omega R_2 (-\hat{j}) = \omega(R_2 - R_1)\hat{j}$$

43. (d)



Change in velocity,

$$\begin{aligned} |\Delta \vec{v}| &= \sqrt{v_1^2 + v_2^2 + 2v_1 v_2 \cos(\pi - \theta)} \\ &= 2v \sin \frac{\theta}{2} \quad (\because |\vec{v}_1| = |\vec{v}_2| = v) \\ &= (2 \times 10) \times \sin(30^\circ) = 2 \times 10 \times \frac{1}{2} \\ &= 10 \text{ m/s} \end{aligned}$$

44. (c) Speed, $V = \text{constant}$ (from question)
Centripetal acceleration,

$$a = \frac{V^2}{r}$$

$$ra = \text{constant}$$

Hence graph (c) correctly describes relation between acceleration and radius.

Laws of Motion



TOPIC 1 1st, 2nd & 3rd Laws of Motion



1. A particle moving in the xy plane experiences a velocity dependent force $\vec{F} = k(v_y \hat{i} + v_x \hat{j})$, where v_x and v_y are x and y components of its velocity $\vec{v}$. If $\vec{a}$ is the acceleration of the particle, then which of the following statements is true for the particle? [Sep. 06, 2020 (II)]

- (a) quantity $\vec{v} \times \vec{a}$ is constant in time
 (b) $\vec{F}$ arises due to a magnetic field
 (c) kinetic energy of particle is constant in time
 (d) quantity $\vec{v} \cdot \vec{a}$ is constant in time

2. A spaceship in space sweeps stationary interplanetary dust.

As a result, its mass increases at a rate $\frac{dM(t)}{dt} = bv^2(t)$, where $v(t)$ is its instantaneous velocity. The instantaneous acceleration of the satellite is : [Sep. 05, 2020 (II)]

- (a) $-bv^3(t)$ (b) $-\frac{bv^3}{M(t)}$
 (c) $-\frac{2bv^3}{M(t)}$ (d) $-\frac{bv^3}{2M(t)}$

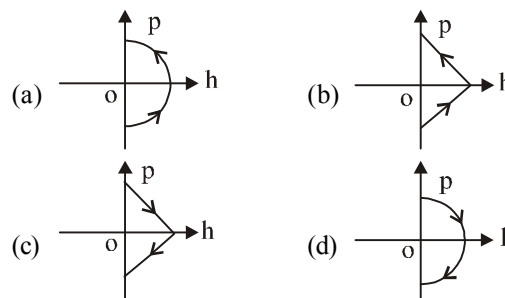
3. A small ball of mass m is thrown upward with velocity u from the ground. The ball experiences a resistive force mkv^2 where v is its speed. The maximum height attained by the ball is : [Sep. 04, 2020 (II)]

- (a) $\frac{1}{2k} \tan^{-1} \frac{ku^2}{g}$ (b) $\frac{1}{k} \ln \left(1 + \frac{ku^2}{2g} \right)$
 (c) $\frac{1}{k} \tan^{-1} \frac{ku^2}{2g}$ (d) $\frac{1}{2k} \ln \left(1 + \frac{ku^2}{g} \right)$

4. A ball is thrown upward with an initial velocity V_0 from the surface of the earth. The motion of the ball is affected by a drag force equal to mvv^2 (where m is mass of the ball, v is its instantaneous velocity and γ is a constant). Time taken by the ball to rise to its zenith is : [10 April 2019 I]

- (a) $\frac{1}{\sqrt{\gamma g}} \tan^{-1} \left(\sqrt{\frac{\gamma}{g}} V_0 \right)$ (b) $\frac{1}{\sqrt{\gamma g}} \sin^{-1} \left(\sqrt{\frac{\gamma}{g}} V_0 \right)$
 (c) $\frac{1}{\sqrt{\gamma g}} \ln \left(1 + \sqrt{\frac{\gamma}{g}} V_0 \right)$ (d) $\frac{1}{\sqrt{2\gamma g}} \tan^{-1} \left(\sqrt{\frac{2\gamma}{g}} V_0 \right)$

5. A ball is thrown vertically up (taken as $+z$ -axis) from the ground. The correct momentum-height (p - h) diagram is: [9 April 2019 I]



6. A particle of mass m is moving in a straight line with momentum p . Starting at time $t = 0$, a force $F = kt$ acts in the same direction on the moving particle during time interval T so that its momentum changes from p to $3p$. Here k is a constant. The value of T is : [11 Jan. 2019 II]

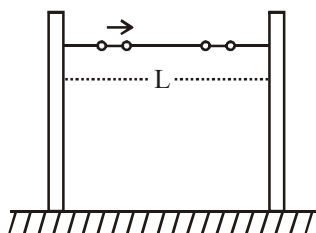
- (a) $2\sqrt{\frac{k}{p}}$ (b) $2\sqrt{\frac{p}{k}}$
 (c) $\sqrt{\frac{2k}{p}}$ (d) $\sqrt{\frac{2p}{k}}$

7. A particle of mass m is acted upon by a force F given by the empirical law $F = \frac{R}{t^2} v(t)$. If this law is to be tested experimentally by observing the motion starting from rest, the best way is to plot : [Online April 10, 2016]

- (a) $\log v(t)$ against $\frac{1}{t}$ (b) $v(t)$ against t^2
 (c) $\log v(t)$ against $\frac{1}{t^2}$ (d) $\log v(t)$ against t

8. A large number (n) of identical beads, each of mass m and radius r are strung on a thin smooth rigid horizontal rod of length L ($L \gg r$) and are at rest at random positions. The rod is mounted between two rigid supports (see figure). If one of the beads is now given a speed v , the average force experienced by each support after a long time is (assume all collisions are elastic):

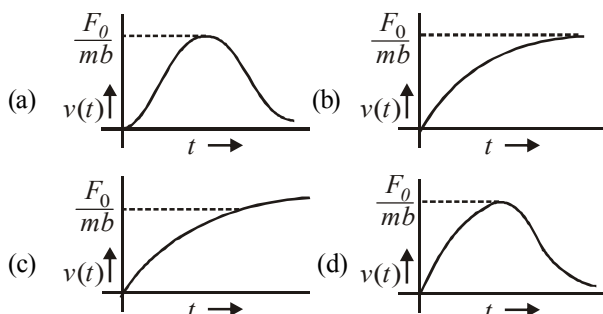
[Online April 11, 2015]



- (a) $\frac{mv^2}{2(L-nr)}$ (b) $\frac{mv^2}{L-2nr}$
 (c) $\frac{mv^2}{L-nr}$ (d) zero
9. A body of mass 5 kg under the action of constant force $\vec{F} = F_x \hat{i} + F_y \hat{j}$ has velocity at $t = 0$ s as $\vec{v} = (6\hat{i} - 2\hat{j})$ m/s and at $t = 10$ s as $\vec{v} = +6\hat{j}$ m/s. The force $\vec{F}$ is:

[Online April 11, 2014]

- (a) $(-3\hat{j} + 4\hat{j})$ N (b) $(-\frac{3}{5}\hat{i} + \frac{4}{5}\hat{j})$ N
 (c) $(3\hat{i} - 4\hat{j})$ N (d) $(\frac{3}{5}\hat{i} - \frac{4}{5}\hat{j})$ N
10. A particle of mass m is at rest at the origin at time $t = 0$. It is subjected to a force $F(t) = F_0 e^{-bt}$ in the x direction. Its speed $v(t)$ is depicted by which of the following curves? [2012]



11. This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.

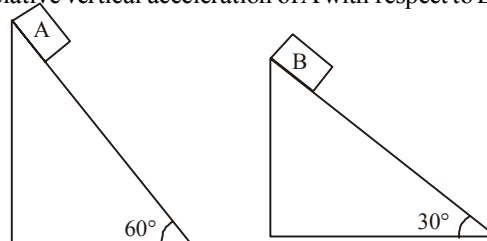
Statement 1: If you push on a cart being pulled by a horse so that it does not move, the cart pushes you back with an equal and opposite force.

Statement 2: The cart does not move because the force described in statement 1 cancel each other.

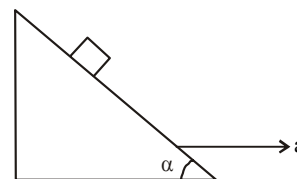
[Online May 26, 2012]

- (a) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.
 (b) Statement 1 is false, Statement 2 is true.
 (c) Statement 1 is true, Statement 2 is false.
 (d) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.

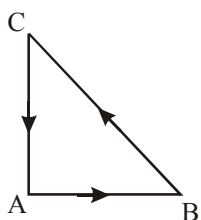
12. Two fixed frictionless inclined planes making an angle 30° and 60° with the vertical are shown in the figure. Two blocks A and B are placed on the two planes. What is the relative vertical acceleration of A with respect to B? [2010]



- (a) 4.9 ms^{-2} in horizontal direction
 (b) 9.8 ms^{-2} in vertical direction
 (c) Zero
 (d) 4.9 ms^{-2} in vertical direction
13. A ball of mass 0.2 kg is thrown vertically upwards by applying a force by hand. If the hand moves 0.2 m while applying the force and the ball goes upto 2 m height further, find the magnitude of the force. (Consider $g = 10 \text{ m/s}^2$). [2006]
 (a) 4 N (b) 16 N (c) 20 N (d) 22 N
14. A player caught a cricket ball of mass 150 g moving at a rate of 20 m/s. If the catching process is completed in 0.1 s, the force of the blow exerted by the ball on the hand of the player is equal to [2006]
 (a) 150 N (b) 3 N (c) 30 N (d) 300 N
15. A particle of mass 0.3 kg subject to a force $F = -kx$ with $k = 15 \text{ N/m}$. What will be its initial acceleration if it is released from a point 20 cm away from the origin? [2005]
 (a) 15 m/s^2 (b) 3 m/s^2 (c) 10 m/s^2 (d) 5 m/s^2
16. A block is kept on a frictionless inclined surface with angle of inclination ' α '. The incline is given an acceleration ' a ' to keep the block stationary. Then a is equal to [2005]



- (a) $g \operatorname{cosec} \alpha$ (b) $g / \tan \alpha$
 (c) $g \tan \alpha$ (d) g
17. A rocket with a lift-off mass 3.5×10^4 kg is blasted upwards with an initial acceleration of 10 m/s^2 . Then the initial thrust of the blast is [2003]
 (a) $3.5 \times 10^5 \text{ N}$ (b) $7.0 \times 10^5 \text{ N}$
 (c) $14.0 \times 10^5 \text{ N}$ (d) $1.75 \times 10^5 \text{ N}$
18. Three forces start acting simultaneously on a particle moving with velocity, $\vec{v}$. These forces are represented in magnitude and direction by the three sides of a triangle ABC. The particle will now move with velocity [2003]



- (a) less than $\vec{v}$
 (b) greater than $\vec{v}$
 (c) $|\vec{v}|$ in the direction of the largest force BC
 (d) $\vec{v}$, remaining unchanged
19. A solid sphere, a hollow sphere and a ring are released from top of an inclined plane (frictionless) so that they slide down the plane. Then maximum acceleration down the plane is for (no rolling) [2002]
- (a) solid sphere (b) hollow sphere
 (c) ring (d) all same

TOPIC 2

**Motion of Connected Bodies,
 Pulley & Equilibrium of
 Forces**


20. A mass of 10 kg is suspended by a rope of length 4 m, from the ceiling. A force F is applied horizontally at the mid-point of the rope such that the top half of the rope makes an angle of 45° with the vertical. Then F equals: (Take $g = 10 \text{ ms}^{-2}$ and the rope to be massless) [7 Jan. 2020 II]
- (a) 100 N (b) 90 N
 (c) 70 N (d) 75 N
21. An elevator in a building can carry a maximum of 10 persons, with the average mass of each person being 68 kg. The mass of the elevator itself is 920 kg and it moves with a constant speed of 3 m/s. The frictional force opposing the motion is 6000 N. If the elevator is moving up with its full capacity, the power delivered by the motor to the elevator ($g = 10 \text{ m/s}^2$) must be at least: [7 Jan. 2020 II]
- (a) 56300 W (b) 62360 W
 (c) 48000 W (d) 66000 W
22. A mass of 10 kg is suspended vertically by a rope from the roof. When a horizontal force is applied on the rope at some point, the rope deviated at an angle of 45° at the roof point. If the suspended mass is at equilibrium, the magnitude of the force applied is ($g = 10 \text{ ms}^{-2}$) [9 Jan. 2019 II]
- (a) 200 N (b) 140 N
 (c) 70 N (d) 100 N
23. A mass ' m ' is supported by a massless string wound around a uniform hollow cylinder of mass m and radius R . If the string does not slip on the cylinder, with what acceleration will the mass fall or release? [2014]

(a) $\frac{2g}{3}$

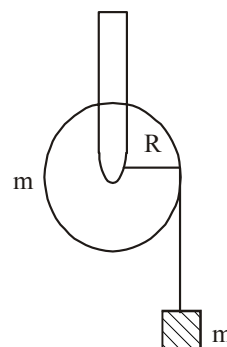
(b) $\frac{g}{2}$

(c) $\frac{5g}{6}$

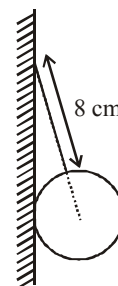
(d) g

24. Two blocks of mass $M_1 = 20 \text{ kg}$ and $M_2 = 12 \text{ kg}$ are connected by a metal rod of mass 8 kg. The system is pulled vertically up by applying a force of 480 N as shown. The tension at the mid-point of the rod is : [Online April 22, 2013]

- (a) 144 N
 (b) 96 N
 (c) 240 N
 (d) 192 N



25. A uniform sphere of weight W and radius 5 cm is being held by a string as shown in the figure. The tension in the string will be : [Online April 9, 2013]



- (a) $12\frac{W}{5}$ (b) $5\frac{W}{12}$ (c) $13\frac{W}{5}$ (d) $13\frac{W}{12}$
26. A spring is compressed between two blocks of masses m_1 and m_2 placed on a horizontal frictionless surface as shown in the figure. When the blocks are released, they have initial velocity of v_1 and v_2 as shown. The blocks travel distances x_1 and x_2 respectively before coming to rest.

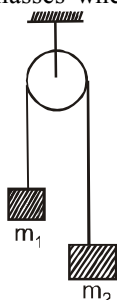
The ratio $\left(\frac{x_1}{x_2}\right)$ is [Online May 12, 2012]

- (a) $\frac{m_2}{m_1}$ (b) $\frac{m_1}{m_2}$ (c) $\sqrt{\frac{m_2}{m_1}}$ (d) $\sqrt{\frac{m_1}{m_2}}$

27. A block of mass m is connected to another block of mass M by a spring (massless) of spring constant k . The blocks are kept on a smooth horizontal plane. Initially the blocks are at rest and the spring is unstretched. Then a constant force F starts acting on the block of mass M to pull it. Find the force of the block of mass m . [2007]

- (a) $\frac{MF}{(m+M)}$ (b) $\frac{mF}{M}$
(c) $\frac{(M+m)F}{m}$ (d) $\frac{mF}{(m+M)}$

28. Two masses $m_1 = 5\text{ g}$ and $m_2 = 4.8\text{ kg}$ tied to a string are hanging over a light frictionless pulley. What is the acceleration of the masses when left free to move? ($g = 9.8\text{ m/s}^2$) [2004]



- (a) 5 m/s^2 (b) 9.8 m/s^2
(c) 0.2 m/s^2 (d) 4.8 m/s^2
29. A spring balance is attached to the ceiling of a lift. A man hangs his bag on the spring and the spring reads 49 N , when the lift is stationary. If the lift moves downward with an acceleration of 5 m/s^2 , the reading of the spring balance will be [2003]
- (a) 24 N (b) 74 N (c) 15 N (d) 49 N
30. A block of mass M is pulled along a horizontal frictionless surface by a rope of mass m . If a force P is applied at the free end of the rope, the force exerted by the rope on the block is [2003]

- (a) $\frac{Pm}{M+m}$ (b) $\frac{Pm}{M-m}$ (c) P (d) $\frac{PM}{M+m}$

31. A light spring balance hangs from the hook of the other light spring balance and a block of mass $M\text{ kg}$ hangs from the former one. Then the true statement about the scale reading is [2003]

- (a) both the scales read $M\text{ kg}$ each
(b) the scale of the lower one reads $M\text{ kg}$ and of the upper one zero
(c) the reading of the two scales can be anything but the sum of the reading will be $M\text{ kg}$
(d) both the scales read $M/2\text{ kg}$ each

32. A lift is moving down with acceleration a . A man in the lift drops a ball inside the lift. The acceleration of the ball as observed by the man in the lift and a man standing stationary on the ground are respectively [2002]

- (a) g, g (b) $g-a, g-a$
(c) $g-a, g$ (d) a, g

33. When forces F_1, F_2, F_3 are acting on a particle of mass m such that F_2 and F_3 are mutually perpendicular, then the particle remains stationary. If the force F_1 is now removed then the acceleration of the particle is [2002]

- (a) F_1/m (b) F_2F_3/mF_1
(c) $(F_2 - F_3)/m$ (d) F_2/m

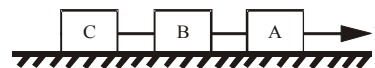
34. Two forces are such that the sum of their magnitudes is 18 N and their resultant is 12 N which is perpendicular to the smaller force. Then the magnitudes of the forces are [2002]

- (a) $12\text{ N}, 6\text{ N}$ (b) $13\text{ N}, 5\text{ N}$
(c) $10\text{ N}, 8\text{ N}$ (d) $16\text{ N}, 2\text{ N}$

35. A light string passing over a smooth light pulley connects two blocks of masses m_1 and m_2 (vertically). If the acceleration of the system is $g/8$, then the ratio of the masses is [2002]

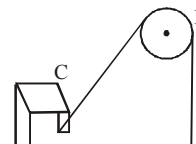
- (a) $8:1$ (b) $9:7$ (c) $4:3$ (d) $5:3$

36. Three identical blocks of masses $m = 2\text{ kg}$ are drawn by a force $F = 10.2\text{ N}$ with an acceleration of 0.6 ms^{-2} on a frictionless surface, then what is the tension (in N) in the string between the blocks B and C ? [2002]



- (a) 9.2 (b) 3.4 (c) 4 (d) 9.8

37. One end of a massless rope, which passes over a massless and frictionless pulley P is tied to a hook C while the other end is free. Maximum tension that the rope can bear is 360 N . With what value of maximum safe acceleration (in ms^{-2}) can a man of 60 kg climb on the rope? [2002]



- (a) 16 (b) 6 (c) 4 (d) 8

TOPIC 3 Friction



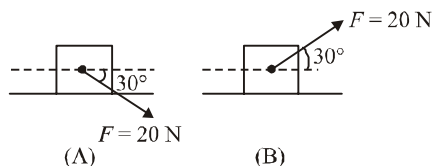
38. An insect is at the bottom of a hemispherical ditch of radius 1 m . It crawls up the ditch but starts slipping after it is at height h from the bottom. If the coefficient of friction between the ground and the insect is 0.75 , then h is : ($g = 10\text{ ms}^{-2}$) [Sep. 06, 2020 (I)]

- (a) 0.20 m (b) 0.45 m
(c) 0.60 m (d) 0.80 m

39. A block starts moving up an inclined plane of inclination 30° with an initial velocity of v_0 . It comes back to its initial position with velocity $\frac{v_0}{2}$. The value of the coefficient of kinetic friction between the block and the inclined plane is close to $\frac{I}{1000}$. The nearest integer to I is _____. [NA Sep. 03, 2020 (II)]

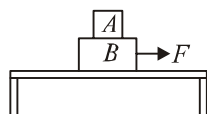
40. A block of mass 5 kg is (i) pushed in case (A) and (ii) pulled in case (B), by a force $F=20$ N, making an angle of 30° with the horizontal, as shown in the figures. The coefficient of friction between the block and floor is $\mu = 0.2$. The difference between the accelerations of the block, in case (B) and case (A) will be : ($g=10 \text{ ms}^{-2}$)

[12 April 2019 II]



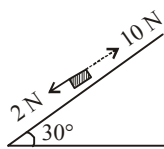
- (a) 0.4 ms^{-2} (b) 3.2 ms^{-2}
(c) 0.8 ms^{-2} (d) 0 ms^{-2}
41. Two blocks A and B masses $m_A=1$ kg and $m_B=3$ kg are kept on the table as shown in figure. The coefficient of friction between A and B is 0.2 and between B and the surface of the table is also 0.2. The maximum force F that can be applied on B horizontally, so that the block A does not slide over the block B is : [Take $g = 10 \text{ m/s}^2$]

[10 April 2019 II]



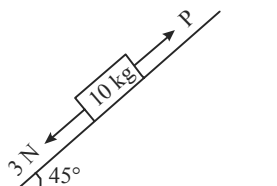
- (a) 8 N (b) 16 N (c) 40 N (d) 12 N
42. A block kept on a rough inclined plane, as shown in the figure, remains at rest upto a maximum force 2 N down the inclined plane. The maximum external force up the inclined plane that does not move the block is 10 N. The coefficient of static friction between the block and the plane is : [Take $g = 10 \text{ m/s}^2$]

[12 Jan. 2019 II]



- (a) $\frac{\sqrt{3}}{2}$ (b) $\frac{\sqrt{3}}{4}$
(c) $\frac{1}{2}$ (d) $\frac{2}{3}$
43. A block of mass 10 kg is kept on a rough inclined plane as shown in the figure. A force of 3 N is applied on the block. The coefficient of static friction between the plane and the block is 0.6. What should be the minimum value of force P , such that the block does not move downward? (take $g = 10 \text{ ms}^{-2}$)

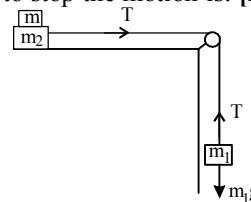
[9 Jan. 2019 I]



- (a) 32 N (b) 18 N (c) 23 N (d) 25 N

44. Two masses $m_1 = 5$ kg and $m_2 = 10$ kg, connected by an inextensible string over a frictionless pulley, are moving as shown in the figure. The coefficient of friction of horizontal surface is 0.15. The minimum weight m that should be put on top of m_2 to stop the motion is: [2018]

- (a) 18.3 kg
(b) 27.3 kg
(c) 43.3 kg
(d) 10.3 kg



45. A given object takes n times more time to slide down a 45° rough inclined plane as it takes to slide down a perfectly smooth 45° incline. The coefficient of kinetic friction between the object and the incline is :

[Online April 15, 2018]

- (a) $\sqrt{1 - \frac{1}{n^2}}$ (b) $1 - \frac{1}{n^2}$
(c) $\frac{1}{2 - n^2}$ (d) $\sqrt{\frac{1}{1 - n^2}}$

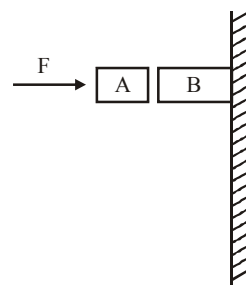
46. A body of mass 2 kg slides down with an acceleration of 3 m/s^2 on a rough inclined plane having a slope of 30° . The external force required to take the same body up the plane with the same acceleration will be: ($g = 10 \text{ m/s}^2$)

[Online April 15, 2018]

- (a) 4 N (b) 14 N (c) 6 N (d) 20 N
47. A rocket is fired vertically from the earth with an acceleration of $2g$, where g is the gravitational acceleration. On an inclined plane inside the rocket, making an angle θ with the horizontal, a point object of mass m is kept. The minimum coefficient of friction $\mu_{\min}$ between the mass and the inclined surface such that the mass does not move is :

[Online April 9, 2016]

- (a) $\tan 2\theta$ (b) $\tan \theta$
(c) $3 \tan \theta$ (d) $2 \tan \theta$
48. Given in the figure are two blocks A and B of weight 20 N and 100 N, respectively. These are being pressed against a wall by a force F as shown. If the coefficient of friction between the blocks is 0.1 and between block B and the wall is 0.15, the frictional force applied by the wall on block B is: [2015]

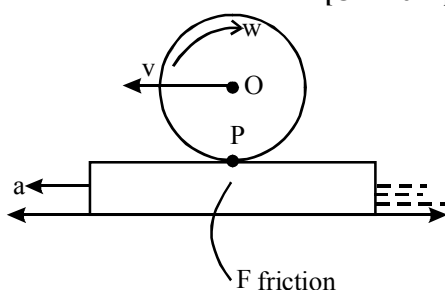


- (a) 120 N (b) 150 N
(c) 100 N (d) 80 N

49. A block of mass $m = 10 \text{ kg}$ rests on a horizontal table. The coefficient of friction between the block and the table is 0.05. When hit by a bullet of mass 50 g moving with speed v , that gets embedded in it, the block moves and comes to stop after moving a distance of 2 m on the table. If a freely falling object were to acquire speed $\frac{v}{10}$ after being dropped from height H , then neglecting energy losses and taking $g = 10 \text{ ms}^{-2}$, the value of H is close to:

[Online April 10, 2015]

- (a) 0.05 km (b) 0.02 km
(c) 0.03 km (d) 0.04 km
50. A block of mass m is placed on a surface with a vertical cross section given by $y = \frac{x^3}{6}$. If the coefficient of friction is 0.5, the maximum height above the ground at which the block can be placed without slipping is: [2014]
- (a) $\frac{1}{6} \text{ m}$ (b) $\frac{2}{3} \text{ m}$ (c) $\frac{1}{3} \text{ m}$ (d) $\frac{1}{2} \text{ m}$
51. Consider a cylinder of mass M resting on a rough horizontal rug that is pulled out from under it with acceleration ' a ' perpendicular to the axis of the cylinder. What is F_{friction} at point P? It is assumed that the cylinder does not slip. [Online April 19, 2014]



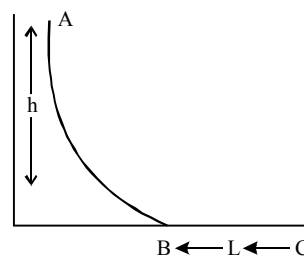
- (a) Mg (b) Ma (c) $\frac{Ma}{2}$ (d) $\frac{Ma}{3}$
52. A heavy box is to be dragged along a rough horizontal floor. To do so, person A pushes it at an angle 30° from the horizontal and requires a minimum force F_A , while person B pulls the box at an angle 60° from the horizontal and needs minimum force F_B . If the coefficient of friction

between the box and the floor is $\frac{\sqrt{3}}{5}$, the ratio $\frac{F_A}{F_B}$ is

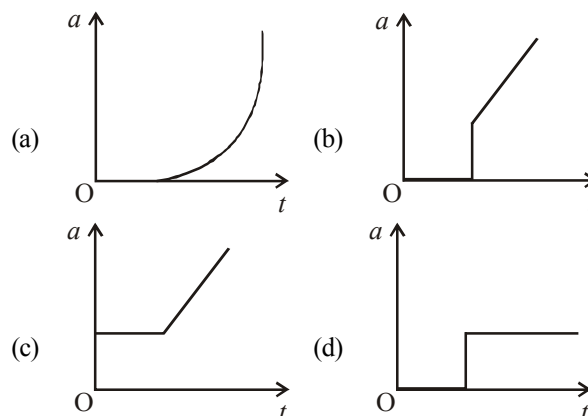
[Online April 19, 2014]

- (a) $\sqrt{3}$ (b) $\frac{5}{\sqrt{3}}$
(c) $\sqrt{\frac{3}{2}}$ (d) $\frac{2}{\sqrt{3}}$
53. A small ball of mass m starts at a point A with speed v_0 and moves along a frictionless track AB as shown. The track BC has coefficient of friction μ . The ball comes to stop at C after travelling a distance L which is:

[Online April 11, 2014]



- (a) $\frac{2h}{\mu} + \frac{v_0^2}{2\mu g}$ (b) $\frac{h}{\mu} + \frac{v_0^2}{2\mu g}$
(c) $\frac{h}{2\mu} + \frac{v_0^2}{\mu g}$ (d) $\frac{h}{2\mu} + \frac{v_0^2}{2\mu g}$
54. A block A of mass 4 kg is placed on another block B of mass 5 kg, and the block B rests on a smooth horizontal table. If the minimum force that can be applied on A so that both the blocks move together is 12 N, the maximum force that can be applied to B for the blocks to move together will be: [Online April 9, 2014]
- (a) 30 N (b) 25 N (c) 27 N (d) 48 N
55. A block is placed on a rough horizontal plane. A time dependent horizontal force $F = kt$ acts on the block, where k is a positive constant. The acceleration - time graph of the block is : [Online April 25, 2013]



56. A body starts from rest on a long inclined plane of slope 45° . The coefficient of friction between the body and the plane varies as $\mu = 0.3x$, where x is distance travelled down the plane. The body will have maximum speed (for $g = 10 \text{ m/s}^2$) when $x =$ [Online April 22, 2013]
- (a) 9.8 m (b) 27 m (c) 12 m (d) 3.33 m
57. A block of weight W rests on a horizontal floor with coefficient of static friction μ . It is desired to make the block move by applying minimum amount of force. The angle θ from the horizontal at which the force should be applied and magnitude of the force F are respectively. [Online May 19, 2012]

- (a) $\theta = \tan^{-1}(\mu), F = \frac{\mu W}{\sqrt{1+\mu^2}}$
(b) $\theta = \tan^{-1}\left(\frac{1}{\mu}\right), F = \frac{\mu W}{\sqrt{1+\mu^2}}$

(c) $\theta = 0, F = \mu W$

(d) $\theta = \tan^{-1} \left(\frac{\mu}{1+\mu} \right), F = \frac{\mu W}{1+\mu}$

58. An insect crawls up a hemispherical surface very slowly. The coefficient of friction between the insect and the surface is $1/3$. If the line joining the centre of the hemispherical surface to the insect makes an angle α with the vertical, the maximum possible value of α so that the insect does not slip is given by [Online May 12, 2012]



- (a) $\cot \alpha = 3$ (b) $\sec \alpha = 3$
(c) $\operatorname{cosec} \alpha = 3$ (d) $\cos \alpha = 3$

59. The minimum force required to start pushing a body up rough (frictional coefficient μ) inclined plane is F_1 while the minimum force needed to prevent it from sliding down is F_2 . If the inclined plane makes an angle θ from the horizontal such that $\tan \theta = 2\mu$ then the ratio $\frac{F_1}{F_2}$ is [2011 RS]
- (a) 1 (b) 2
(c) 3 (d) 4

60. If a spring of stiffness ' k ' is cut into parts ' A ' and ' B ' of length $\ell_A : \ell_B = 2 : 3$, then the stiffness of spring ' A ' is given by [2011 RS]

- (a) $\frac{3k}{5}$ (b) $\frac{2k}{5}$
(c) k (d) $\frac{5k}{2}$

61. A smooth block is released at rest on a 45° incline and then slides a distance ' d '. The time taken to slide is ' n ' times as much to slide on rough incline than on a smooth incline. The coefficient of friction is [2005]

- (a) $\mu_k = \sqrt{1 - \frac{1}{n^2}}$ (b) $\mu_k = 1 - \frac{1}{n^2}$
(c) $\mu_s = \sqrt{1 - \frac{1}{n^2}}$ (d) $\mu_s = 1 - \frac{1}{n^2}$

62. The upper half of an inclined plane with inclination ϕ is perfectly smooth while the lower half is rough. A body starting from rest at the top will again come to rest at the bottom if the coefficient of friction for the lower half is given by [2005]

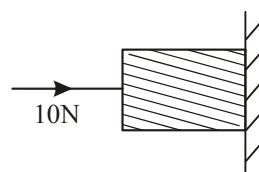
- (a) $2 \cos \phi$ (b) $2 \sin \phi$ (c) $\tan \phi$ (d) $2 \tan \phi$

63. Consider a car moving on a straight road with a speed of 100 m/s . The distance at which car can be stopped is [$\mu_k = 0.5$] [2005]
- (a) 1000 m (b) 800 m (c) 400 m (d) 100 m

64. A block rests on a rough inclined plane making an angle of 30° with the horizontal. The coefficient of static friction between the block and the plane is 0.8 . If the frictional force on the block is 10 N , the mass of the block (in kg) is (take $g = 10 \text{ m/s}^2$) [2004]

- (a) 1.6 (b) 4.0 (c) 2.0 (d) 2.5

65. A horizontal force of 10 N is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and the wall is 0.2 . The weight of the block is [2003]



- (a) 20 N (b) 50 N (c) 100 N (d) 2 N

66. A marble block of mass 2 kg lying on ice when given a velocity of 6 m/s is stopped by friction in 10 s . Then the coefficient of friction is [2003]
- (a) 0.02 (b) 0.03 (c) 0.04 (d) 0.06

TOPIC 4 Circular Motion, Banking of Road



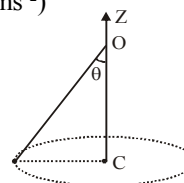
67. A disc rotates about its axis of symmetry in a horizontal plane at a steady rate of 3.5 revolutions per second. A coin placed at a distance of 1.25 cm from the axis of rotation remains at rest on the disc. The coefficient of friction between the coin and the disc is ($g = 10 \text{ m/s}^2$) [Online April 15, 2018]

- (a) 0.5 (b) 0.7 (c) 0.3 (d) 0.6

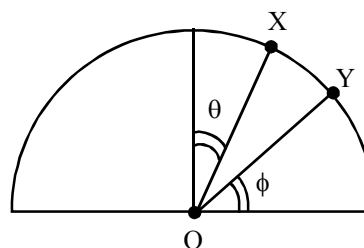
68. A conical pendulum of length 1 m makes an angle $\theta = 45^\circ$ w.r.t. Z-axis and moves in a circle in the XY plane. The radius of the circle is 0.4 m and its centre is vertically below O. The speed of the pendulum, in its circular path, will be : [Online April 9, 2017]

(Take $g = 10 \text{ ms}^{-2}$)

- (a) 0.4 m/s
(b) 4 m/s
(c) 0.2 m/s
(d) 2 m/s



69. A particle is released on a vertical smooth semicircular track from point X so that OX makes angle θ from the vertical (see figure). The normal reaction of the track on the particle vanishes at point Y where OY makes angle ϕ with the horizontal. Then: [Online April 19, 2014]



(a) $\sin \phi = \cos \phi$ (b) $\sin \phi = \frac{1}{2} \cos \theta$

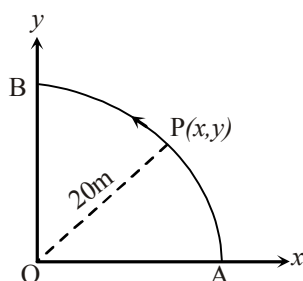
(c) $\sin \phi = \frac{2}{3} \cos \theta$ (d) $\sin \phi = \frac{3}{4} \cos \theta$

70. A body of mass ' m ' is tied to one end of a spring and whirled round in a horizontal plane with a constant angular velocity. The elongation in the spring is 1 cm. If the angular velocity is doubled, the elongation in the spring is 5 cm. The original length of the spring is :

[Online April 23, 2013]

- (a) 15 cm (b) 12 cm
(c) 16 cm (d) 10 cm

71. A point P moves in counter-clockwise direction on a circular path as shown in the figure. The movement of 'P' is such that it sweeps out a length $s = t^3 + 5$, where s is in metres and t is in seconds. The radius of the path is 20 m. The acceleration of 'P' when $t = 2$ s is nearly. [2010]



- (a) 13m/s^2 (b) 12m/s^2
(c) 7.2m/s^2 (d) 14m/s^2

72. For a particle in uniform circular motion, the acceleration $\vec{a}$ at a point P(R, θ) on the circle of radius R is (Here θ is measured from the x-axis) [2010]

(a) $-\frac{v^2}{R} \cos \theta \hat{i} + \frac{v^2}{R} \sin \theta \hat{j}$

(b) $-\frac{v^2}{R} \sin \theta \hat{i} + \frac{v^2}{R} \cos \theta \hat{j}$

(c) $-\frac{v^2}{R} \cos \theta \hat{i} - \frac{v^2}{R} \sin \theta \hat{j}$

(d) $\frac{v^2}{R} \hat{i} + \frac{v^2}{R} \hat{j}$

73. An annular ring with inner and outer radii R_1 and R_2 is rolling without slipping with a uniform angular speed. The ratio of the forces experienced by the two particles situated

on the inner and outer parts of the ring, $\frac{F_1}{F_2}$ is [2005]

(a) $\left(\frac{R_1}{R_2}\right)^2$ (b) $\frac{R_2}{R_1}$

(c) $\frac{R_1}{R_2}$ (d) 1

74. Which of the following statements is FALSE for a particle moving in a circle with a constant angular speed ? [2004]

- (a) The acceleration vector points to the centre of the circle
(b) The acceleration vector is tangent to the circle
(c) The velocity vector is tangent to the circle
(d) The velocity and acceleration vectors are perpendicular to each other.

75. The minimum velocity (in ms^{-1}) with which a car driver must traverse a flat curve of radius 150 m and coefficient of friction 0.6 to avoid skidding is [2002]

- (a) 60 (b) 30
(c) 15 (d) 25



Hints & Solutions



1. (a) Given

$$\vec{F} = k(v_y \hat{i} + v_x \hat{j})$$

$$\therefore F_x = kv_y \hat{i}, F_y = kv_x \hat{j}$$

$$\frac{mdv_x}{dt} = kv_y \Rightarrow \frac{dv_x}{dt} = \frac{k}{m} v_y$$

$$\text{Similarly, } \frac{dv_y}{dt} = \frac{k}{m} v_x$$

$$\frac{dv_y}{dv_x} = \frac{v_x}{v_y} \Rightarrow \int v_y dv_y = \int v_x dv_x$$

$$v_y^2 = v_x^2 + C$$

$$v_y^2 - v_x^2 = \text{constant}$$

$$\vec{v} \times \vec{a} = (v_x \hat{i} + v_y \hat{j}) \times \frac{k}{m} (v_y \hat{i} + v_x \hat{j})$$

$$= (v_x^2 \hat{k} - v_y^2 \hat{k}) \frac{k}{m} = (v_x^2 - v_y^2) \frac{k}{m} \hat{k} = \text{constant}$$

2. (b) From the Newton's second law,

$$F = \frac{dp}{dt} = \frac{d(mv)}{dt} = v \left(\frac{dm}{dt} \right) \quad \dots(i)$$

$$\text{We have given, } \frac{dM(t)}{dt} = bv^2(t) \quad \dots(ii)$$

Thrust on the satellite,

$$F = -v \left(\frac{dm}{dt} \right) = -v(bv^2) = -bv^3 \quad [\text{Using (i) and (ii)}]$$

$$\Rightarrow F = M(t)a = -bv^3 \Rightarrow a = \frac{-bv^3}{M(t)}$$

3. (d)

$$\begin{array}{c} \uparrow v=0 \\ \uparrow u \\ \downarrow H \\ \downarrow (g+kv^2) = a \text{ (acceleration)} \end{array}$$

$$\vec{F} = mkv^2 - mg \quad (\because mg \text{ and } mkv^2 \text{ act opposite to each other})$$

$$\vec{a} = \frac{\vec{F}}{m} = -[kv^2 + g]$$

$$\Rightarrow v \cdot \frac{dv}{dh} = -[kv^2 + g] \quad \left(\because a = v \frac{dv}{dh} \right)$$

$$\Rightarrow \int_u^0 \frac{v \cdot dv}{kv^2 + g} = \int_0^h dh$$

$$\Rightarrow \frac{1}{2k} \ln[kv^2 + g]_u^0 = -h$$

$$\Rightarrow \frac{1}{2k} \ln \left[\frac{ku^2 + g}{g} \right] = h$$

4. (a) Net acceleration

$$\frac{dv}{dt} = a = -(g + \gamma v^2)$$

Let time t required to rise to its zenith ($v = 0$) so,

$$\int_{v_0}^0 \frac{-dv}{g + \gamma v^2} = \int_0^t dt \quad [\text{for } H_{\max}, v = 0]$$

$$\therefore t = \frac{1}{\sqrt{\gamma g}} \tan^{-1} \left(\frac{\sqrt{\gamma} v_0}{\sqrt{g}} \right)$$

5. (d) $v^2 = u^2 - 2gh$

$$\text{or } v = \sqrt{u^2 - 2gh}$$

$$\text{Momentum, } P = mv = m\sqrt{u^2 - 2gh}$$

$$\text{At } h = 0, P = mu \text{ and at } h = \frac{u^2}{2g}, P = 0$$

upward direction is positive and downward direction is negative.

6. (b) From Newton's second law

$$\frac{dp}{dt} = F = kt$$

Integrating both sides we get,

$$\int_p^{3p} dp = \int_0^T kt dt \Rightarrow [p]_p^{3p} = k \left[\frac{t^2}{2} \right]_0^T$$

$$\Rightarrow 2p = \frac{kT^2}{2} \Rightarrow T = 2\sqrt{\frac{p}{k}}$$

7. (a) From $F = \frac{R}{t^2} v(t) \Rightarrow m \frac{dv}{dt} = \frac{R}{t^2} v(t)$

$$\text{Integrating both sides } \int \frac{dv}{v} = \int \frac{R dt}{mt^2}$$

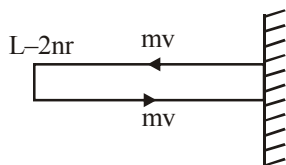
$$\ln v = -\frac{R}{mt}$$

$$\therefore \ln v \propto \frac{1}{t}$$

8. (b) Space between the supports for motion of beads is $L - 2nr$

Average force experienced by each support,

$$F = \frac{2mV}{2(L - 2nr)} = \frac{mV^2}{L - 2nr}$$



9. (a) From question,
Mass of body, $m = 5 \text{ kg}$
Velocity at $t = 0$,

$$u = (6\hat{i} - 2\hat{j}) \text{ m/s}$$

Velocity at $t = 10 \text{ s}$,

$$v = +6\hat{j} \text{ m/s}$$

Force, $F = ?$

$$\text{Acceleration, } a = \frac{v-u}{t}$$

$$= \frac{6\hat{j} - (6\hat{i} - 2\hat{j})}{10} = \frac{-3\hat{i} + 4\hat{j}}{5} \text{ m/s}^2$$

Force, $F = ma$

$$= 5 \times \frac{(-3\hat{i} + 4\hat{j})}{5} = (-3\hat{i} + 4\hat{j}) \text{ N}$$

10. (c) Given that $F(t) = F_0 e^{-bt}$

$$\Rightarrow m \frac{dv}{dt} = F_0 e^{-bt}$$

$$\frac{dv}{dt} = \frac{F_0}{m} e^{-bt}$$

$$\int_0^v dv = \frac{F_0}{m} \int_0^t e^{-bt} dt$$

$$v = \frac{F_0}{m} \left[\frac{e^{-bt}}{-b} \right]_0^t = \frac{F_0}{mb} \left[-\left(e^{-bt} - e^{-0} \right) \right]$$

$$\Rightarrow v = \frac{F_0}{mb} \left[1 - e^{-bt} \right]$$

11. (a) According to Newton's third law of motion i.e. every action is associated with equal and opposite reaction.

12. (d) $mg \sin \theta = ma$

$$\therefore a = g \sin \theta$$

$\therefore$ Vertical component of acceleration

$$= g \sin^2 \theta$$

$\therefore$ Relative vertical acceleration of A with respect to B is

$$g(\sin^2 60^\circ - \sin^2 30^\circ)$$

$$= g \left(\frac{3}{4} - \frac{1}{4} \right) = \frac{g}{2} = 4.9 \text{ m/s}^2$$

in vertical direction

13. (d) For the motion of ball, just after the throwing

$$v = 0, s = 2 \text{ m}, a = -g = -10 \text{ m/s}^2$$

$$v^2 - u^2 = 2as \text{ for upward journey}$$

$$\Rightarrow -u^2 = 2(-10) \times 2 \Rightarrow u^2 = 40$$

When the ball is in the hands of the thrower

$$u = 0, v = \sqrt{40} \text{ m/s}^{-1}$$

$$s = 0.2 \text{ m}$$

$$v^2 - u^2 = 2as$$

$$\Rightarrow 40 - 0 = 2(a) 0.2 \Rightarrow a = 100 \text{ m/s}^2$$

$$\therefore F = ma = 0.2 \times 100 = 20 \text{ N}$$

$$\Rightarrow N - mg = 20 \Rightarrow N = 20 + 2 = 22 \text{ N}$$

Note :

$$W_{\text{hand}} + W_{\text{gravity}} = \Delta K$$

$$\Rightarrow F(0.2) + (0.2)(10)(2.2) = 0 \Rightarrow F = 22 \text{ N}$$

14. (c) Given, mass of cricket ball, $m = 150 \text{ g} = 0.15 \text{ kg}$

Initial velocity, $u = 20 \text{ m/s}$

Force,

$$F = \frac{m(v-u)}{t} = \frac{0.15(0-20)}{0.1} = 30 \text{ N}$$

15. (c) Mass (m) = 0.3 kg

Force, $F = m \cdot a = -kx$

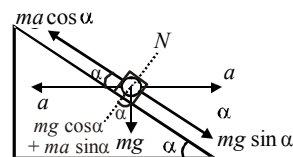
$$\Rightarrow ma = -15x$$

$$\Rightarrow 0.3a = -15x$$

$$\Rightarrow a = -\frac{15}{0.3}x = -\frac{150}{3}x = -50x$$

$$a = -50 \times 0.2 = 10 \text{ m/s}^2$$

16. (c) When the incline is given an acceleration a towards the right, the block receives a reaction ma towards left.



For block to remain stationary, Net force along the incline should be zero.

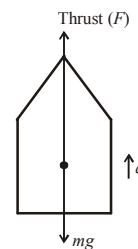
$$mg \sin \alpha = ma \cos \alpha \Rightarrow a = g \tan \alpha$$

17. (b) In the absence of air resistance, if the rocket moves up with an acceleration a , then thrust

$$F = mg + ma$$

$$\therefore F = m(g + a) = 3.5 \times 10^4 (10 + 10)$$

$$= 7 \times 10^5 \text{ N}$$

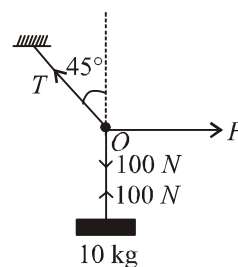


18. (d) Resultant force is zero, as three forces are represented by the sides of a triangle taken in the same order. From Newton's second law, $\vec{F}_{\text{net}} = m\vec{a}$.

Therefore, acceleration is also zero i.e., velocity remains unchanged.

19. (d) This is a case of sliding (if plane is friction less) and therefore the acceleration of all the bodies is same.

20. (a) From the free body diagram



$$T \cos 45^\circ = 100 \text{ N} \quad \dots(i)$$

$$T \sin 45^\circ = F \quad \dots(ii)$$

On dividing (i) by (ii) we get

$$\frac{T \cos 45^\circ}{T \sin 45^\circ} = \frac{100}{F}$$

$$\Rightarrow F = 100 \text{ N}$$

21. (d) Net force on the elevator = force on elevator + frictional force

$$\Rightarrow F = (10m + M)g + f$$

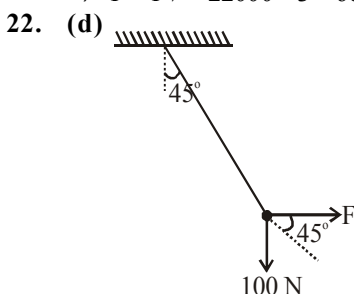
where, m = mass of person, M = mass of elevator,

f = frictional force

$$\Rightarrow F = (10 \times 68 + 920) \times 9.8 + 600$$

$$\Rightarrow F = 22000 \text{ N}$$

$$\Rightarrow P = FV = 22000 \times 3 = 66000 \text{ W}$$

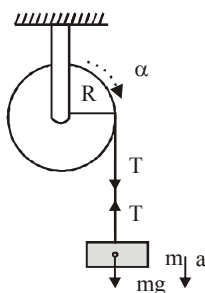


At equilibrium,

$$\tan 45^\circ = \frac{mg}{F} = \frac{100}{F}$$

$$\therefore F = 100 \text{ N}$$

23. (b) From figure,



$$\text{Acceleration } a = R\alpha$$

$$\text{and } mg - T = ma$$

From equation (i) and (ii)

$$T \times R = mR^2\alpha = mR^2\left(\frac{a}{R}\right)$$

$$\text{or } T = ma$$

$$\Rightarrow mg - ma = ma$$

$$\Rightarrow a = \frac{g}{2}$$

24. (d) Acceleration produced in upward direction

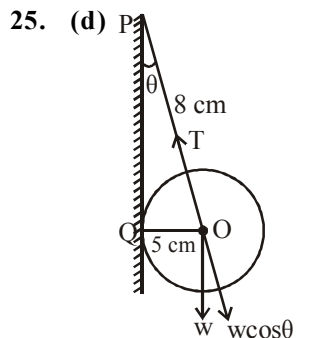
$$a = \frac{F}{M_1 + M_2 + \text{Mass of metal rod}}$$

$$= \frac{480}{20 + 12 + 8} = 12 \text{ ms}^{-2}$$

Tension at the mid point

$$T = \left(M_2 + \frac{\text{Mass of rod}}{2} \right) a$$

$$= (12 + 4) \times 12 = 192 \text{ N}$$



$$PQ = \sqrt{OP^2 + OQ^2}$$

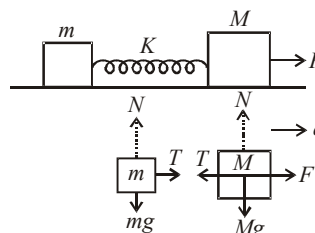
$$= \sqrt{13^2 + 5^2} = 12$$

Tension in the string

$$T = w \cos \theta = \frac{13}{12} W$$

26. (a)

27. (d) Writing free body-diagrams for m & M ,



$$\text{we get } T = ma \text{ and } F - T = Ma$$

where T is force due to spring

$$\Rightarrow F - ma = Ma \text{ or, } F = Ma + ma$$

$\therefore$ Acceleration of the system

$$a = \frac{F}{M + m}$$

Now, force acting on the block of mass m is

$$ma = m \left(\frac{F}{M + m} \right) = \frac{mF}{m + M}$$

If a is the acceleration along the inclined plane, then

28. (c) Here, $m_1 = 5 \text{ kg}$ and $m_2 = 4.8 \text{ kg}$.

If a is the acceleration of the masses,

$$m_1 a = m_1 g - T \quad \dots(i)$$

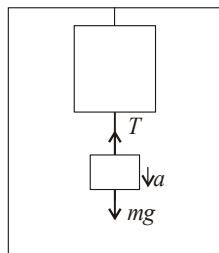
$$m_2 a = T - m_2 g \quad \dots(ii)$$

Solving (i) and (ii) we get

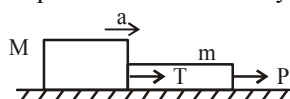
$$a = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) g$$

$$\Rightarrow a = \frac{(5 - 4.8) \times 9.8}{(5 + 4.8)} \text{ m/s}^2 = 0.2 \text{ m/s}^2$$

29. (a) When lift is stationary, $W_1 = mg$... (i)
When the lift descends with acceleration, a
 $W_2 = m(g - a)$
 $W_2 = \frac{49}{10}(10 - 5) = 24.5 \text{ N}$



30. (d) Taking the rope and the block as a system



we get $P = (m + M)a$

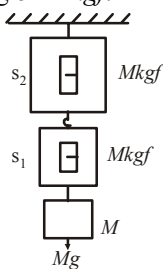
$$\therefore \text{Acceleration produced, } a = \frac{P}{m + M}$$

Taking the block as a system,

Force on the block, $F = Ma$

$$\therefore F = \frac{MP}{m + M}$$

31. (a) The Earth exerts a pulling force Mg . The block in turn exerts a reaction force Mg on the spring of spring balance S_1 which therefore shows a reading of $M \text{ kgf}$. As both the springs are massless. Therefore, it exerts a force of Mg on the spring of spring balance S_2 which shows the reading of $M \text{ kgf}$.



32. (c) **Case - I:** For the man standing in the lift, the acceleration of the ball

$$\vec{a}_{bm} = \vec{a}_b - \vec{a}_m \Rightarrow a_{bm} = g - a$$

Case - II: The man standing on the ground, the acceleration of the ball

$$\vec{a}_{bm} = \vec{a}_b - \vec{a}_m \Rightarrow a_{bm} = g - 0 = g$$

33. (a) When forces F_1 , F_2 and F_3 are acting on the particle, it remains in equilibrium. Force F_2 and F_3 are perpendicular to each other,

$$F_1 = F_2 + F_3$$

$$\therefore F_1 = \sqrt{F_2^2 + F_3^2}$$

The force F_1 is now removed, so, resultant of F_2 and F_3 will now make the particle move with force equal to F_1 .

$$\text{Then, acceleration, } a = \frac{F_1}{m}$$

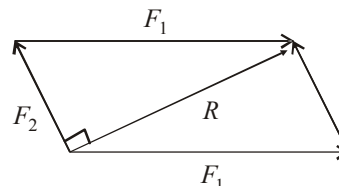
34. (b) Let the two forces be F_1 and F_2 and let $F_2 < F_1$. R is the resultant force.
Given $F_1 + F_2 = 18$... (i)

$$\text{From the figure } F_2^2 + R^2 = F_1^2$$

$$F_1^2 - F_2^2 = R^2$$

$$\therefore F_1^2 - F_2^2 = 144$$
 ... (ii)

Only option (b) follows equation (i) and (ii).

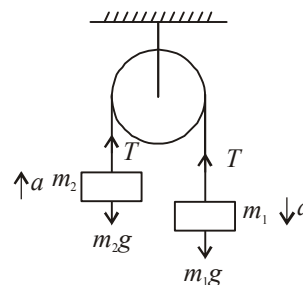


35. (b) For mass m_1

$$m_1 g - T = m_1 a \quad \dots (i)$$

For mass m_2

$$T - m_2 g = m_2 a \quad \dots (ii)$$



Adding the equations we get

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2}$$

$$\text{Here } a = \frac{g}{8}$$

$$\therefore \frac{1}{8} = \frac{\frac{m_1}{m_2} - 1}{\frac{m_1}{m_2} + 1} \Rightarrow \frac{m_1}{m_2} + 1 = 8 \frac{m_1}{m_2} - 8 \Rightarrow \frac{m_1}{m_2} = \frac{9}{7}$$

36. (b) Force = mass $\times$ acceleration

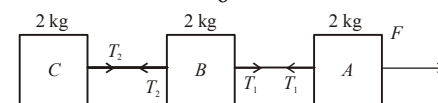
$$\therefore F = (m + m + m) \times a$$

$$F = 3m \times a$$

$$a = \frac{F}{3m}$$

$$\therefore a = \frac{10.2}{6} \text{ m/s}^2$$

$$\therefore T_2 = ma = 2 \times \frac{10.2}{6} = 3.4 \text{ N}$$



37. (c) Tension, $T = 360 \text{ N}$

Mass of a man $m = 60 \text{ kg}$

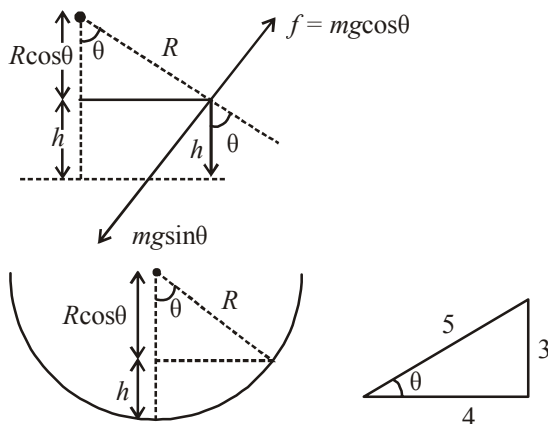
$$mg - T = ma$$

$$\therefore a = g - \frac{T}{m}$$

$$= 10 - \frac{360}{60} = 4 \text{ m/s}^2$$

38. (a) For balancing, $mg \sin \theta = f = \mu mg \cos \theta$

$$\Rightarrow \tan \theta = \mu = \frac{3}{4} = 0.75$$



$$h = R - R \cos \theta = R - R \left(\frac{4}{5} \right) = \frac{R}{5}$$

$$\therefore h = \frac{R}{5} = 0.2 \text{ m} \quad [\because \text{radius, } R = 1 \text{ m}]$$

39. (346)

Acceleration of block while moving up an inclined plane,

$$a_1 = g \sin \theta + \mu g \cos \theta$$

$$\Rightarrow a_1 = g \sin 30^\circ + \mu g \cos 30^\circ$$

$$= \frac{g}{2} + \frac{\mu g \sqrt{3}}{2} \quad \dots(i) \quad (\because \theta = 30^\circ)$$

Using $v^2 - u^2 = 2a(s)$

$$\Rightarrow v_0^2 - 0^2 = 2a_1(s) \quad (\because u = 0)$$

$$\Rightarrow v_0^2 - 2a_1(s) = 0$$

$$\Rightarrow s = \frac{v_0^2}{2a_1} \quad \dots(ii)$$

Acceleration while moving down an inclined plane

$$a_2 = g \sin \theta - \mu g \cos \theta$$

$$\Rightarrow a_2 = g \sin 30^\circ - \mu g \cos 30^\circ$$

$$\Rightarrow a_2 = \frac{g}{2} - \frac{\mu \sqrt{3}}{2} g \quad \dots(iii)$$

Using again $v^2 - u^2 = 2as$ for downward motion

$$\Rightarrow \left(\frac{v_0}{2} \right)^2 = 2a_2(s) \Rightarrow s = \frac{v_0^2}{4a_2} \quad \dots(iv)$$

Equating equation (ii) and (iv)

$$\frac{v_0^2}{a_1} = \frac{v_0^2}{4a_2} \Rightarrow a_1 = 4a_2$$

$$\Rightarrow \frac{g}{2} + \frac{\mu g \sqrt{3}}{2} = 4 \left(\frac{g}{2} - \frac{\mu \sqrt{3}}{2} \right)$$

$$\Rightarrow 5 + 5\sqrt{3}\mu = 4(5 - 5\sqrt{3}\mu) \quad (\text{Substituting, } g = 10 \text{ m/s}^2)$$

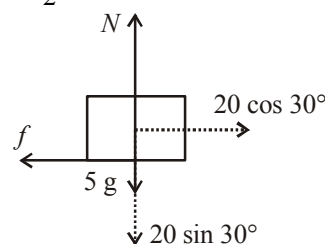
$$\Rightarrow 5 + 5\sqrt{3}\mu = 20 - 20\sqrt{3}\mu \Rightarrow 25\sqrt{3}\mu = 15$$

$$\Rightarrow \mu = \frac{\sqrt{3}}{5} = 0.346 = \frac{346}{1000}$$

$$\text{So, } \frac{I}{1000} = \frac{346}{1000}$$

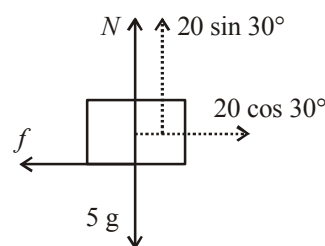
40. (c) A : $N = 5g + 20 \sin 30^\circ$

$$= 50 + 20 \times \frac{1}{2} = 60 \text{ N}$$



$$\text{Acceleration, } a_1 = \frac{F - f}{m} = \frac{20 \cos 30^\circ - \mu N}{5}$$

$$= \left[\frac{20 \times \frac{\sqrt{3}}{2} - 0.2 \times 60}{5} \right] = 1.06 \text{ m/s}^2$$



$$\text{B : } N = 5g - 20 \sin 30^\circ$$

$$= 50 - 20 \times \frac{1}{2} = 40 \text{ N}$$

$$a_2 = \frac{F - f}{m} = \left[\frac{20 \cos 30^\circ - 0.2 \times 40}{5} \right] = 1.86 \text{ m/s}^2$$

$$\text{Now } a_2 - a_1 = 1.86 - 1.06 = 0.8 \text{ m/s}^2$$

41. (b) Taking (A + B) as system

$$F - \mu(M + m)g$$

$$= (M + m)a$$

$$\Rightarrow a = \frac{F - \mu(M + m)g}{(M + m)}$$

$$a = \frac{F - (0.2)4 \times 10}{4} = \left(\frac{F - 8}{4} \right) \quad \dots(i)$$

But, $a_{\max} = \mu g = 0.2 \times 10 = 2$

$$\therefore \frac{F-8}{4} = 2$$

$$\Rightarrow F = 16 \text{ N}$$

42. (a) From figure, $2 + mg \sin 30^\circ = \mu mg \cos 30^\circ$ and

$$10 = mg \sin 30^\circ + \mu mg \cos 30^\circ$$

$$= 2\mu mg \cos 30^\circ - 2$$

$$\Rightarrow 6 = \mu mg \cos 30^\circ \text{ and}$$

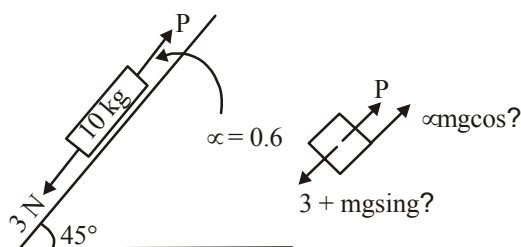
$$4 = mg \cos 30^\circ$$

By dividing above two

$$\Rightarrow \frac{3}{2} = \mu \times \sqrt{3}$$

$$\therefore \text{Coefficient of friction, } \mu = \frac{\sqrt{3}}{2}$$

43. (a)



$$mg \sin 45^\circ = \frac{100}{\sqrt{2}} = 50\sqrt{2}$$

$$[\because m=10\text{kg}, g=9.8\text{m/s}^2]$$

$$\mu mg \cos \theta = 0.6 \times mg \times \frac{1}{\sqrt{2}} = 0.6 \times 50\sqrt{2}$$

$$3 + mg \sin \theta = P + \mu mg \cos \theta$$

$$3 + 50\sqrt{2} = P + 30\sqrt{2}$$

$$\therefore P = 31.28 = 32 \text{ N}$$

44. (b) Given : $m_1 = 5\text{kg}$; $m_2 = 10\text{kg}$; $\mu = 0.15$

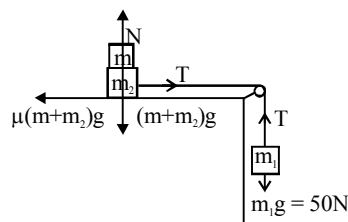
$$\text{FBD for } m_1, m_1 g - T = m_1 a$$

$$\Rightarrow 50 - T = 5 \times a$$

$$\text{and, } T - 0.15(m+10)g = (10+m)a$$

$$\text{For rest } a = 0$$

$$\text{or, } 50 = 0.15(m+10)10$$



$$\Rightarrow 5 = \frac{3}{20}(m+10)$$

$$\frac{100}{3} = m+10 \therefore m = 23.3\text{kg; close to option (b)}$$

45. (b) The coefficients of kinetic friction between the object and the incline

$$\mu = \tan \theta \left(1 - \frac{1}{n^2} \right) \Rightarrow \mu = 1 - \frac{1}{n^2} \quad (\because \theta = 45^\circ)$$

46. (d) Equation of motion when the mass slides down

$$Mg \sin \theta - f = Ma$$

$$\Rightarrow 10 - f = 6 \quad (M = 2 \text{ kg}, a = 3 \text{ m/s}^2, \theta = 30^\circ \text{ given})$$

$$\therefore f = 4 \text{ N}$$

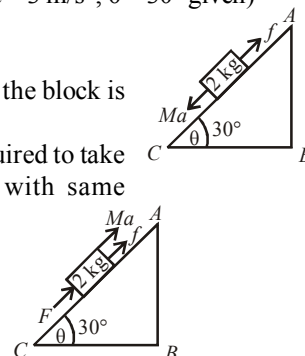
Equation of motion when the block is pushed up

Let the external force required to take the block up the plane with same acceleration be F

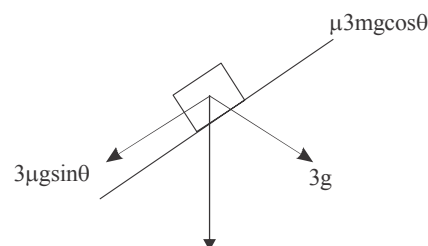
$$F - Mg \sin \theta - f = Ma$$

$$\Rightarrow F - 10 - 4 = 6$$

$$F = 20 \text{ N}$$



47. (b) Let μ be the minimum coefficient of friction

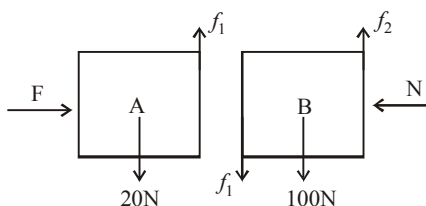


At equilibrium, mass does not move so,

$$3mg \sin \theta = \mu 3mg \cos \theta$$

$$\therefore \mu_{\min} = \tan \theta$$

48. (a)

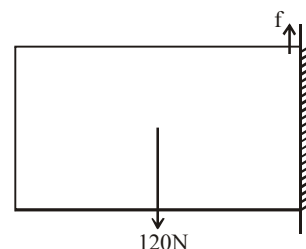


Assuming both the blocks are stationary

$$N = F$$

$$f_1 = 20 \text{ N}$$

$$f_2 = 100 + 20 = 120 \text{ N}$$



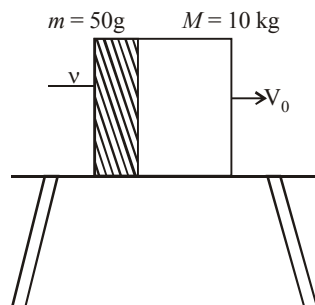
Considering the two blocks as one system and due to equilibrium $f = 120 \text{ N}$

49. (d)
- $f = \mu(M + m)g$

$$a = \frac{f}{M + m} = \frac{\mu(M + m)g}{(M + m)} = \mu g$$

$$= 0.05 \times 10 = 0.5 \text{ ms}^{-2}$$

$$V_0 = \frac{\text{Initial momentum}}{(M + m)} = \frac{0.05V}{10.05}$$



$$v^2 - u^2 = 2as$$

$$0 - u^2 = 2as$$

$$u^2 = 2as$$

$$\left(\frac{0.05v}{10.05}\right)^2 = 2 \times 0.5 \times 2$$

Solving we get $v = 201\sqrt{2}$

Object falling from height H .

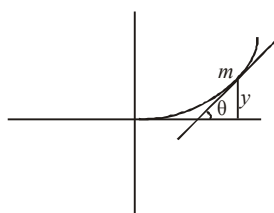
$$\frac{V}{10} = \sqrt{2gH}$$

$$\frac{201\sqrt{2}}{10} = \sqrt{2 \times 10 \times H}$$

$$H = 40 \text{ m} = 0.04 \text{ km}$$

50. (a) At limiting equilibrium,

$$\mu = \tan \theta$$



$$\tan \theta = \mu = \frac{dy}{dx} = \frac{x^2}{2} \quad (\text{from question})$$

$\therefore$ Coefficient of friction $\mu = 0.5$

$$\therefore 0.5 = \frac{x^2}{2}$$

$$\Rightarrow x = \pm 1$$

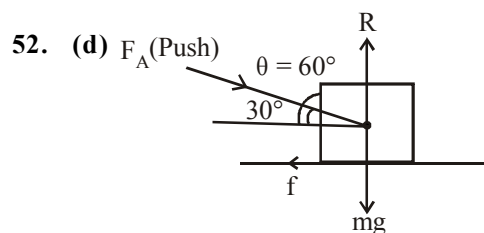
$$\text{Now, } y = \frac{x^3}{6} = \frac{1}{6}m$$

51. (d) Force of friction at point P,

$$F_{\text{friction}} = \frac{1}{3}Ma \sin \theta$$

$$= \frac{1}{3}Ma \sin 90^\circ \quad [\text{here } \theta = 90^\circ]$$

$$= \frac{Ma}{3}$$



$$F_A = \frac{\mu mg}{\sin \theta - \mu \cos \theta}$$

Similarly,

$$F_B = \frac{\mu mg}{\sin \theta + \mu \cos \theta}$$

$$\therefore \frac{F_A}{F_B} = \frac{\frac{\mu mg}{\sin \theta - \mu \cos \theta}}{\frac{\mu mg}{\sin \theta + \mu \cos \theta}}$$

$$= \frac{\mu mg}{\sin 60^\circ - \frac{\sqrt{3}}{5} \cos 60^\circ} \quad \left[\mu = \frac{\sqrt{3}}{5} \text{ given} \right]$$

$$= \frac{\mu mg}{\sin 30^\circ + \frac{\sqrt{3}}{5} \cos 30^\circ}$$

$$= \frac{\mu mg}{\sin 60^\circ - \frac{\sqrt{3}}{5} \cos 60^\circ}$$

$$= \frac{\frac{1}{2} + \frac{\sqrt{3}}{5} \times \frac{\sqrt{3}}{2}}{\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{5} \times \frac{1}{2}}$$

$$= \frac{\frac{1}{2} \left(1 + \frac{3}{5}\right)}{\frac{\sqrt{3}}{5} \left(1 - \frac{1}{5}\right)} = \frac{\frac{1}{2} \times \frac{8}{5}}{\frac{\sqrt{3} \times 4}{10}}$$

$$= \frac{\frac{8}{10}}{\frac{\sqrt{3} \times 4}{10}} = \frac{8}{\sqrt{3} \times 4} = \frac{2}{\sqrt{3}}$$

$$= \frac{2}{\sqrt{3}}$$

53. (b) Initial speed at point A,
- $u = v_0$

Speed at point B, $v = ?$

$$v^2 - u^2 = 2gh$$

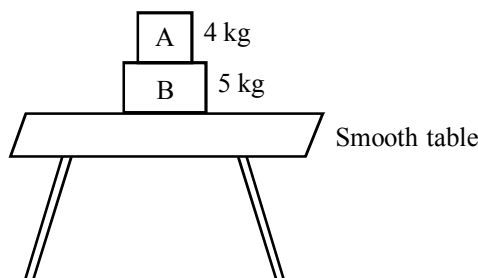
$$v^2 = v_0^2 + 2gh$$

Let ball travels distance ' S ' before coming to rest

$$S = \frac{v^2}{2\mu g} = \frac{v_0^2 + 2gh}{2\mu g}$$

$$= \frac{v_0^2}{2\mu g} + \frac{2gh}{2\mu g} = \frac{h}{\mu} + \frac{v_0^2}{2\mu g}$$

54. (c) Minimum force on A
= frictional force between the surfaces
= 12 N



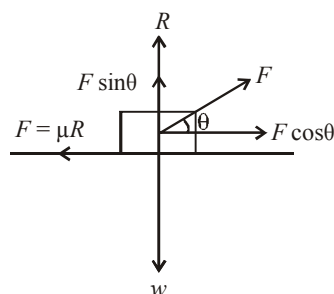
Therefore maximum acceleration

$$a_{\max} = \frac{12 \text{ N}}{4 \text{ kg}} = 3 \text{ m/s}^2$$

Hence maximum force,

$$F_{\max} = \text{total mass} \times a_{\max} \\ = 9 \times 3 = 27 \text{ N}$$

55. (b) Graph (b) correctly depicts the acceleration-time graph of the block.
56. (d) When the body has maximum speed then
 $\mu = 0.3x = \tan 45^\circ$
 $\therefore x = 3.33 \text{ m}$
57. (a) Let the force F is applied at an angle θ with the horizontal.



For horizontal equilibrium,
 $F \cos \theta = \mu R$... (i)

For vertical equilibrium,
 $R + F \sin \theta = mg$
or, $R = mg - F \sin \theta$... (ii)

Substituting this value of R in eq. (i), we get
 $F \cos \theta = \mu (mg - F \sin \theta)$
 $= \mu mg - \mu F \sin \theta$
or, $F (\cos \theta + \mu \sin \theta) = \mu mg$

$$\text{or, } F = \frac{\mu mg}{\cos \theta + \mu \sin \theta} \quad \dots \text{(iii)}$$

For F to be minimum, the denominator $(\cos \theta + \mu \sin \theta)$ should be maximum.

$$\therefore \frac{d}{d\theta} (\cos \theta + \mu \sin \theta) = 0$$

$$\text{or, } -\sin \theta + \mu \cos \theta = 0$$

$$\text{or, } \tan \theta = \mu$$

$$\text{or, } \theta = \tan^{-1}(\mu)$$

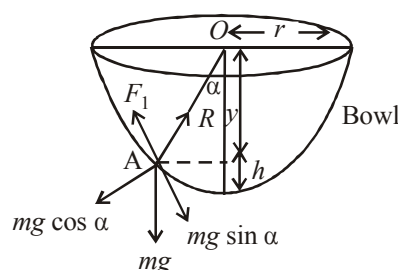
$$\text{Then, } \sin \theta = \frac{\mu}{\sqrt{1+\mu^2}} \text{ and}$$

$$\cos \theta = \frac{1}{\sqrt{1+\mu^2}}$$

Hence, $F_{\min}$

$$= \frac{\mu w}{\frac{1}{\sqrt{1+\mu^2}} + \frac{\mu^2}{\sqrt{1+\mu^2}}} = \frac{\mu w}{\sqrt{1+\mu^2}}$$

58. (a)



The insect crawls up the bowl upto a certain height h only till the component of its weight along the bowl is balanced by limiting frictional force.

For limiting condition at point A

$$R = mg \cos \alpha \quad \dots \text{(i)}$$

$$F_1 = mg \sin \alpha \quad \dots \text{(ii)}$$

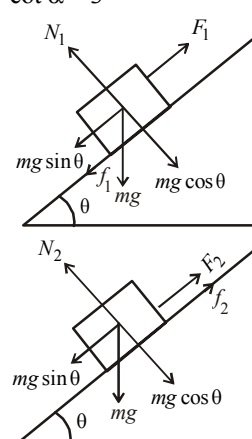
Dividing eq. (ii) by (i)

$$\tan \alpha = \frac{1}{\cot \alpha} = \frac{F_1}{R} = \mu [As F_1 = \mu R]$$

$$\Rightarrow \tan \alpha = \mu = \frac{1}{3} \left[\because \mu = \frac{1}{3} \text{ (Given)} \right]$$

$$\therefore \cot \alpha = 3$$

59. (c)



When the body slides up the inclined plane, then

$$mg \sin \theta + f_1 = F_1$$

$$\text{or, } F_1 = mg \sin \theta + \mu mg \cos \theta$$

When the body slides down the inclined plane, then

$$mg \sin \theta - f_2 = F_2$$

$$\text{or, } F_2 = mg \sin \theta - \mu mg \cos \theta$$

$$\therefore \frac{F_1}{F_2} = \frac{\sin \theta + \mu \cos \theta}{\sin \theta - \mu \cos \theta}$$

$$\Rightarrow \frac{F_1}{F_2} = \frac{\tan \theta + \mu}{\tan \theta - \mu} = \frac{2\mu + \mu}{2\mu - \mu} = \frac{3\mu}{\mu} = 3$$

60. (d) It is given $\ell_A : \ell_B = 2 : 3$

$$\ell_A = \frac{2\ell}{5}, \quad \ell_B = \left(\frac{3\ell}{5}\right)$$

$\therefore$ We know that $k \propto \frac{1}{\ell}$

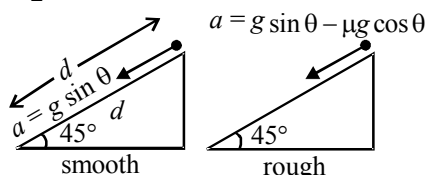
If initial spring constant is k , then

$$k\ell = k_A\ell_A = k_B\ell_B$$

$$k\ell = k_A\left(\frac{2\ell}{5}\right)$$

$$k_A = \frac{5k}{2}$$

61. (b)



On smooth inclined plane, acceleration of the body = $g \sin \theta$. Let d be the distance travelled

$$\therefore d = \frac{1}{2}(g \sin \theta)t_1^2,$$

$$t_1 = \sqrt{\frac{2d}{g \sin \theta}},$$

On rough inclined plane,

$$a = \frac{mg \sin \theta - \mu R}{m}$$

$$\Rightarrow a = \frac{mg \sin \theta - \mu mg \cos \theta}{m}$$

$$\Rightarrow a = g \sin \theta - \mu_k g \cos \theta$$

$$\therefore d = \frac{1}{2}(g \sin \theta - \mu_k g \cos \theta)t_2^2$$

$$t_2 = \sqrt{\frac{2d}{g \sin \theta - \mu_k g \cos \theta}}$$

According to question, $t_2 = nt_1$

$$n\sqrt{\frac{2d}{g \sin \theta}} = \sqrt{\frac{2d}{g \sin \theta - \mu_k g \cos \theta}}$$

Here, μ is coefficient of kinetic friction as the block moves over the inclined plane.

$$\therefore \sin \theta = (\sin \theta - \mu_k \cos \theta)n^2$$

$$\Rightarrow n = \frac{1}{\sqrt{1 - \mu_k}} \Rightarrow n^2 = \frac{1}{1 - \mu_k}$$

$$\Rightarrow \mu_k = 1 - \frac{1}{n^2}$$

62. (d) For first half
acceleration = $g \sin \phi$;
For second half

$$\text{acceleration} = -(g \sin \phi - \mu g \cos \phi)$$

For the block to come to rest at the bottom, acceleration in I half = retardation in II half.

$$g \sin \phi = -(g \sin \phi - \mu g \cos \phi)$$

$$\Rightarrow \mu = 2 \tan \phi$$

NOTE

According to work-energy theorem, $W = \Delta K = 0$

(Since initial and final speeds are zero)

$\therefore$ Workdone by friction + Work done by gravity = 0

$$\text{i.e., } -(\mu mg \cos \phi) \frac{\ell}{2} + mg \ell \sin \phi = 0$$

$$\text{or } \frac{\mu}{2} \cos \phi = \sin \phi \quad \text{or } \mu = 2 \tan \phi$$

63. (a) Given, initial velocity, $u = 100 \text{ m/s}$.

Final velocity, $v = 0$.

Acceleration, $a = \mu_k g = 0.5 \times 10$

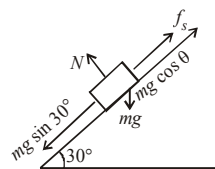
$$v^2 - u^2 = 2as \quad \text{or}$$

$$\Rightarrow 0^2 - u^2 = 2(-\mu_k g)s$$

$$\Rightarrow -100^2 = 2 \times -\frac{1}{2} \times 10 \times s$$

$$\Rightarrow s = 1000 \text{ m}$$

64. (c)



Since the body is at rest on the inclined plane,

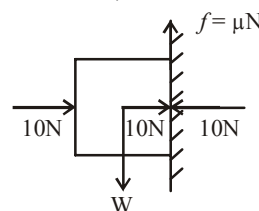
$$mg \sin 30^\circ = \text{Force of friction}$$

$$\Rightarrow m \times 10 \times \sin 30^\circ = 10$$

$$\Rightarrow m \times 5 = 10 \Rightarrow m = 2.0 \text{ kg}$$

65. (d) Horizontal force, $N = 10 \text{ N}$.

Coefficient of friction $\mu = 0.2$.



The block will be stationary so long as

Force of friction = weight of block

$$\therefore \mu N = W$$

$$\Rightarrow 0.2 \times 10 = W$$

$$\Rightarrow W = 2 \text{ N}$$

66. (d) $u = 6 \text{ m/s}$, $v = 0$, $t = 10 \text{ s}$, $a = ?$

$$\text{Acceleration } a = \frac{v - u}{t}$$

$$\Rightarrow a = \frac{0 - 6}{10}$$

$$\Rightarrow a = \frac{-6}{10} = -0.6 \text{ m/s}^2$$



The retardation is due to the frictional force

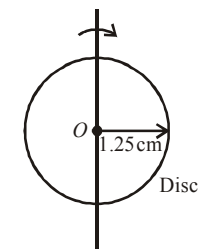
$$\therefore f = ma \Rightarrow \mu N = ma$$

$$\Rightarrow \mu mg = ma \Rightarrow \mu = \frac{ma}{mg}$$

$$\Rightarrow \mu = \frac{a}{g} = \frac{0.6}{10} = 0.06$$

67. (d) Using, $\mu mg = \frac{mv^2}{r} = mr\omega^2$
 $\omega = 2\pi n = 2\pi \times 3.5 = 7\pi \text{ rad/sec}$
 Radius, $r = 1.25 \text{ cm} = 1.25 \times 10^{-2} \text{ m}$
 Coefficient of friction, $\mu = ?$

$$\mu mg = \frac{m(r\omega)^2}{r} (\because v = r\omega)$$



$$\mu mg = mr\omega^2$$

$$\Rightarrow \mu = \frac{r\omega^2}{g} = \frac{1.25 \times 10^{-2} \times \left(7 \times \frac{22}{7}\right)^2}{10}$$

$$= \frac{1.25 \times 10^{-2} \times 22^2}{10} = 0.6$$

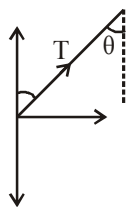
68. (d) Given, $\theta = 45^\circ$, $r = 0.4 \text{ m}$, $g = 10 \text{ m/s}^2$

$$T \sin \theta = \frac{mv^2}{r} \quad \dots\dots (i)$$

$$T \cos \theta = mg \quad \dots\dots (ii)$$

From equation (i) & (ii) we have,

$$\tan \theta = \frac{v^2}{rg}$$



$$v^2 = rg \quad \because \theta = 45^\circ$$

Hence, speed of the pendulum in its circular path,

$$v = \sqrt{rg} = \sqrt{0.4 \times 10} = 2 \text{ m/s}$$

69. (c) 70. (a)

71. (d) $s = t^3 + 5$

$$\Rightarrow \text{velocity, } v = \frac{ds}{dt} = 3t^2$$

$$\text{Tangential acceleration } a_t = \frac{dv}{dt} = 6t$$

$$\text{Radial acceleration } a_c = \frac{v^2}{R} = \frac{9t^4}{R}$$

$$\text{At } t = 2\text{s, } a_t = 6 \times 2 = 12 \text{ m/s}^2$$

$$a_c = \frac{9 \times 16}{20} = 7.2 \text{ m/s}^2$$

$\therefore$ Resultant acceleration

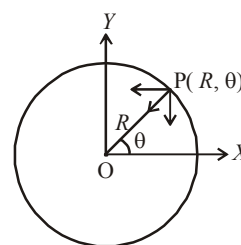
$$= \sqrt{a_t^2 + a_c^2}$$

$$= \sqrt{(12)^2 + (7.2)^2} = \sqrt{144 + 51.84}$$

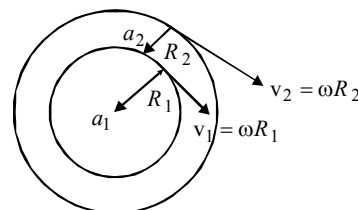
$$= \sqrt{195.84} = 14 \text{ m/s}^2$$

72. (c) Clearly $\vec{a} = a_c \cos \theta (-\hat{i}) + a_c \sin \theta (-\hat{j})$

$$= -\frac{v^2}{R} \cos \theta \hat{i} - \frac{v^2}{R} \sin \theta \hat{j}$$



73. (c)



Let m is the mass of each particle and ω is the angular speed of the annular ring.

$$a_1 = \frac{v_1^2}{R_1} = \frac{\omega^2 R_1^2}{R_1} = \omega^2 R_1$$

$$a_2 = \frac{v_2^2}{R_2} = \omega^2 R_2$$

Taking particle masses equal

$$\frac{F_1}{F_2} = \frac{ma_1}{ma_2} = \frac{mR_1\omega^2}{mR_2\omega^2} = \frac{R_1}{R_2}$$

NOTE :

The force experienced by any particle is only along radial direction.

Force experienced by the particle, $F = m\omega^2 R$

$$\therefore \frac{F_1}{F_2} = \frac{R_1}{R_2}$$

74. (b) Only option (b) is false since acceleration vector is always radial (i.e. towards the center) for uniform circular motion.

75. (b) The maximum velocity of the car is

$$v_{\max} = \sqrt{\mu rg}$$

$$\text{Here } \mu = 0.6, r = 150 \text{ m}, g = 9.8$$

$$v_{\max} = \sqrt{0.6 \times 150 \times 9.8} \simeq 30 \text{ m/s}$$

Work, Energy and Power



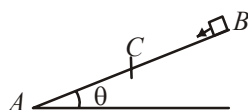
TOPIC 1 Work



1. A person pushes a box on a rough horizontal platform surface. He applies a force of 200 N over a distance of 15 m. Thereafter, he gets progressively tired and his applied force reduces linearly with distance to 100 N. The total distance through which the box has been moved is 30 m. What is the work done by the person during the total movement of the box ? **[4 Sep. 2020 (II)]**

(a) 3280 J (b) 2780 J
(c) 5690 J (d) 5250 J

2.

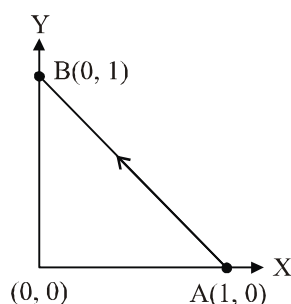


A small block starts slipping down from a point B on an inclined plane AB, which is making an angle θ with the horizontal section BC is smooth and the remaining section CA is rough with a coefficient of friction μ . It is found that the block comes to rest as it reaches the bottom (point A) of the inclined plane. If $BC = 2AC$, the coefficient of friction is given by $\mu = k \tan \theta$. The value of k is _____.

[NA 2 Sep. 2020 (I)]

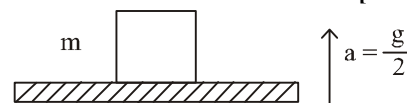
3. Consider a force $\vec{F} = -x\hat{i} + y\hat{j}$. The work done by this force in moving a particle from point A(1, 0) to B(0, 1) along the line segment is: (all quantities are in SI units)

[9 Jan. 2020 I]



(a) 2J (b) $\frac{1}{2}$ J (c) 1J (d) $\frac{3}{2}$ J

4. A block of mass m is kept on a platform which starts from rest with constant acceleration $g/2$ upward, as shown in fig. work done by normal reaction on block in time t is: **[10 Jan. 2019 I]**



(a) $-\frac{m g^2 t^2}{8}$ (b) $\frac{m g^2 t^2}{8}$
(c) 0 (d) $\frac{3m g^2 t^2}{8}$

5. A body of mass starts moving from rest along x-axis so that its velocity varies as $v = a\sqrt{s}$ where a is a constant s and is the distance covered by the body. The total work done by all the forces acting on the body in the first second after the start of the motion is: **[Online April 16, 2018]**

(a) $\frac{1}{8}ma^4t^2$ (b) $4ma^4t^2$
(c) $8ma^4t^2$ (d) $\frac{1}{4}ma^4t^2$

6. When a rubber-band is stretched by a distance x , it exerts restoring force of magnitude $F = ax + bx^2$ where a and b are constants. The work done in stretching the unstretched rubber-band by L is: **[2014]**

(a) $aL^2 + bL^3$ (b) $\frac{1}{2}(aL^2 + bL^3)$
(c) $\frac{aL^2}{2} + \frac{bL^3}{3}$ (d) $\frac{1}{2}\left(\frac{aL^2}{2} + \frac{bL^3}{3}\right)$

7. A uniform chain of length 2 m is kept on a table such that a length of 60 cm hangs freely from the edge of the table. The total mass of the chain is 4 kg. What is the work done in pulling the entire chain on the table? **[2004]**

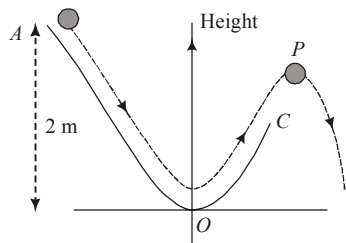
(a) 12 J (b) 3.6 J (c) 7.2 J (d) 1200 J

8. A force $\vec{F} = (5\vec{i} + 3\vec{j} + 2\vec{k})N$ is applied over a particle which displaces it from its origin to the point $\vec{r} = (2\vec{i} - \vec{j})m$. The work done on the particle in joules is [2004]
- (a) +10 (b) +7 (c) -7 (d) +13
9. A spring of spring constant $5 \times 10^3 N/m$ is stretched initially by 5cm from the unstretched position. Then the work required to stretch it further by another 5 cm is [2003]
- (a) 12.50 N-m (b) 18.75 N-m
(c) 25.00 N-m (d) 6.25 N-m
10. A spring of force constant 800 N/m has an extension of 5 cm. The work done in extending it from 5 cm to 15 cm is [2002]
- (a) 16 J (b) 8 J (c) 32 J (d) 24 J

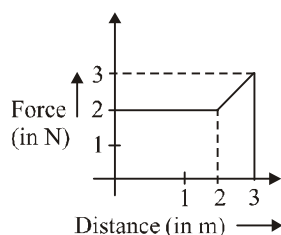
TOPIC 2 Energy



11. A cricket ball of mass 0.15 kg is thrown vertically up by a bowling machine so that it rises to a maximum height of 20 m after leaving the machine. If the part pushing the ball applies a constant force F on the ball and moves horizontally a distance of 0.2 m while launching the ball, the value of F (in N) is ($g = 10 ms^{-2}$) [NA 3 Sep. 2020 (I)]
12. A particle ($m = 1 kg$) slides down a frictionless track (AOC) starting from rest at a point A (height 2 m). After reaching C, the particle continues to move freely in air as a projectile. When it reaching its highest point P (height 1 m), the kinetic energy of the particle (in J) is: (Figure drawn is schematic and not to scale; take $g = 10 ms^{-2}$) [NA 7 Jan. 2020 I]

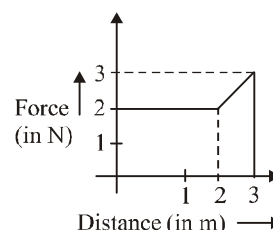


13. A particle moves in one dimension from rest under the influence of a force that varies with the distance travelled by the particle as shown in the figure. The kinetic energy of the particle after it has travelled 3 m is : [7 Jan. 2020 II]



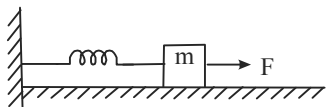
- (a) 4 J (b) 2.5 J
(c) 6.5 J (d) 5 J

14. A spring whose unstretched length is l has a force constant k . The spring is cut into two pieces of unstretched lengths l_1 and l_2 where, $l_1 = nl_2$ and n is an integer. The ratio k_1/k_2 of the corresponding force constants, k_1 and k_2 will be: [12 April 2019 II]
- (a) n (b) $\frac{1}{n^2}$ (c) $\frac{1}{n}$ (d) n^2
15. A body of mass 1 kg falls freely from a height of 100m, on a platform of mass 3 kg which is mounted on a spring having spring constant $k = 1.25 \times 10^6 N/m$. The body sticks to the platform and the spring's maximum compression is found to be x . Given that $g = 10 ms^{-2}$, the value of x will be close to : [11 April 2019 I]
- (a) 40 cm (b) 4 cm (c) 80 cm (d) 8 cm
16. A uniform cable of mass 'M' and length 'L' is placed on a horizontal surface such that its $\left(\frac{1}{n}\right)^{th}$ part is hanging below the edge of the surface. To lift the hanging part of the cable upto the surface, the work done should be: [9 April 2019 I]
- (a) $\frac{MgL}{2n^2}$ (b) $\frac{MgL}{n^2}$ (c) $\frac{2MgL}{n^2}$ (d) $nMgL$
17. A wedge of mass $M = 4m$ lies on a frictionless plane. A particle of mass m approaches the wedge with speed v . There is no friction between the particle and the plane or between the particle and the wedge. The maximum height climbed by the particle on the wedge is given by: [9 April 2019 II]
- (a) $\frac{v^2}{g}$ (b) $\frac{2v^2}{7g}$
(c) $\frac{2v^2}{5g}$ (d) $\frac{v^2}{2g}$
18. A particle moves in one dimension from rest under the influence of a force that varies with the distance travelled by the particle as shown in the figure. The kinetic energy of the particle after it has travelled 3 m is : [8 April 2019 I]



- (a) 4 J (b) 2.5 J
(c) 6.5 J (d) 5 J
19. A particle which is experiencing a force, given by $\vec{F} = 3\vec{i} - 12\vec{j}$, undergoes a displacement of $\vec{d} = 4\vec{i}$. If the particle had a kinetic energy of 3 J at the beginning of the displacement, what is its kinetic energy at the end of the displacement? [10 Jan. 2019 II]
- (a) 9 J (b) 12 J
(c) 10 J (d) 15 J

20. A block of mass m , lying on a smooth horizontal surface, is attached to a spring (of negligible mass) of spring constant k . The other end of the spring is fixed, as shown in the figure. The block is initially at rest in its equilibrium position. If now the block is pulled with a constant force F , the maximum speed of the block is: [9 Jan. 2019 I]



- (a) $\frac{2F}{\sqrt{mk}}$ (b) $\frac{F}{\pi\sqrt{mk}}$ (c) $\frac{\pi F}{\sqrt{mk}}$ (d) $\frac{F}{\sqrt{mk}}$
21. A force acts on a 2 kg object so that its position is given as a function of time as $x = 3t^2 + 5$. What is the work done by this force in first 5 seconds? [9 Jan. 2019 II]

- (a) 850 J (b) 950 J (c) 875 J (d) 900 J
22. A particle is moving in a circular path of radius a under the action of an attractive potential $U = -\frac{k}{2r^2}$. Its total energy is: [2018]

- (a) $-\frac{k}{4a^2}$ (b) $\frac{k}{2a^2}$
(c) zero (d) $-\frac{3}{2}\frac{k}{a^2}$

23. Two particles of the same mass m are moving in circular orbits because of force, given by $F(r) = \frac{-16}{r} - r^3$. The first particle is at a distance $r = 1$, and the second, at $r = 4$. The best estimate for the ratio of kinetic energies of the first and the second particle is closest to [Online April 16, 2018]

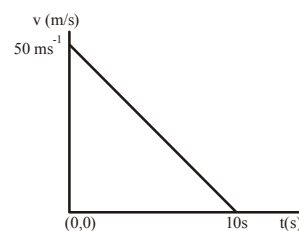
- (a) 10^{-1} (b) 6×10^{-2} (c) 6×10^2 (d) 3×10^{-3}
24. A body of mass $m = 10^{-2}$ kg is moving in a medium and experiences a frictional force $F = -kv^2$. Its initial speed is $v_0 = 10 \text{ ms}^{-1}$. If, after 10 s, its energy is $\frac{1}{8}mv_0^2$, the value of k will be: [2017]

- (a) $10^{-4} \text{ kg m}^{-1}$ (b) $10^{-1} \text{ kg m}^{-1} \text{ s}^{-1}$
(c) $10^{-3} \text{ kg m}^{-1}$ (d) $10^{-3} \text{ kg s}^{-1}$
25. An object is dropped from a height h from the ground. Every time it hits the ground it loses 50% of its kinetic energy. The total distance covered as $t \rightarrow \infty$ is [Online April 8, 2017]

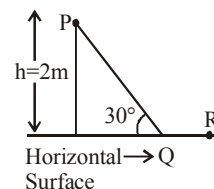
- (a) $3h$ (b) ∞ (c) $\frac{5}{3}h$ (d) $\frac{8}{3}h$

26. A time dependent force $F = 6t$ acts on a particle of mass 1 kg. If the particle starts from rest, the work done by the force during the first 1 second will be [2017]

- (a) 9 J (b) 18 J (c) 4.5 J (d) 22 J
27. Velocity-time graph for a body of mass 10 kg is shown in figure. Work-done on the body in first two seconds of the motion is : [Online April 10, 2016]



- (a) -9300 J (b) 12000 J
(c) -4500 J (d) -12000 J
28. A point particle of mass m , moves along the uniformly rough track PQR as shown in the figure. The coefficient of friction, between the particle and the rough track equals μ . The particle is released, from rest from the point P and it comes to rest at a point R. The energies, lost by the ball, over the parts, PQ and QR, of the track, are equal to each other, and no energy is lost when particle changes direction from PQ to QR. The value of the coefficient of friction μ and the distance $x (= QR)$, are, respectively close to : [2016]



- (a) 0.29 and 3.5 m (b) 0.29 and 6.5 m
(c) 0.2 and 6.5 m (d) 0.2 and 3.5 m
29. A person trying to lose weight by burning fat lifts a mass of 10 kg upto a height of 1 m 1000 times. Assume that the potential energy lost each time he lowers the mass is dissipated. How much fat will he use up considering the work done only when the weight is lifted up? Fat supplies 3.8×10^7 J of energy per kg which is converted to mechanical energy with a 20% efficiency rate. Take $g = 9.8 \text{ ms}^{-2}$: [2016]

- (a) $9.89 \times 10^{-3} \text{ kg}$ (b) $12.89 \times 10^{-3} \text{ kg}$
(c) $2.45 \times 10^{-3} \text{ kg}$ (d) $6.45 \times 10^{-3} \text{ kg}$
30. A particle is moving in a circle of radius r under the action of a force $F = \alpha r^2$ which is directed towards centre of the circle. Total mechanical energy (kinetic energy + potential energy) of the particle is (take potential energy = 0 for $r = 0$): [Online April 11, 2015]

- (a) $\frac{1}{2}\alpha r^3$ (b) $\frac{5}{6}\alpha r^3$ (c) $\frac{4}{3}\alpha r^3$ (d) αr^3
31. A block of mass $m = 0.1$ kg is connected to a spring of unknown spring constant k . It is compressed to a distance x from its equilibrium position and released from rest. After approaching half the distance $\left(\frac{x}{2}\right)$ from equilibrium position, it hits another block and comes to rest momentarily, while the other block moves with a velocity 3 ms^{-1} . The total initial energy of the spring is : [Online April 10, 2015]

- (a) 0.3 J (b) 0.6 J
(c) 0.8 J (d) 1.5 J
32. A bullet loses $\left(\frac{1}{n}\right)^{\text{th}}$ of its velocity passing through one plank. The number of such planks that are required to stop the bullet can be: [Online April 19, 2014]

- (a) $\frac{n^2}{2n-1}$ (b) $\frac{2n^2}{n-1}$ (c) infinite (d) n

33. A spring of unstretched length l has a mass m with one end fixed to a rigid support. Assuming spring to be made of a uniform wire, the kinetic energy possessed by it if its free end is pulled with uniform velocity v is:

[Online April 12, 2014]

- (a) $\frac{1}{2}mv^2$ (b) mv^2 (c) $\frac{1}{3}mv^2$ (d) $\frac{1}{6}mv^2$

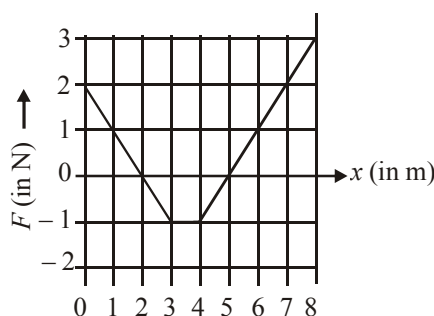
34. Two springs of force constants 300 N/m (Spring A) and 400 N/m (Spring B) are joined together in series. The combination is compressed by 8.75 cm. The ratio of energy stored in A and B is $\frac{E_A}{E_B}$. Then $\frac{E_A}{E_B}$ is equal to:

[Online April 9, 2013]

- (a) $\frac{4}{3}$ (b) $\frac{16}{9}$ (c) $\frac{3}{4}$ (d) $\frac{9}{16}$

35. The force $\vec{F} = F\hat{i}$ on a particle of mass 2 kg, moving along the x -axis is given in the figure as a function of its position x . The particle is moving with a velocity of 5 m/s along the x -axis at $x = 0$. What is the kinetic energy of the particle at $x = 8$ m?

[Online May 26, 2012]



- (a) 34 J (b) 34.5 J (c) 4.5 J (d) 29.4 J
36. A particle gets displaced by $\Delta\vec{r} = (2\hat{i} + 3\hat{j} + 4\hat{k})$ m under the action of a force $\vec{F} = (7\hat{i} + 4\hat{j} + 3\hat{k})$. The change in its kinetic energy is

[Online May 7, 2012]

- (a) 38 J (b) 70 J (c) 52.5 J (d) 126 J
37. At time $t = 0$ a particle starts moving along the x -axis. If its kinetic energy increases uniformly with time ' t ', the net force acting on it must be proportional to [2011 RS]

- (a) constant (b) t
(c) $\frac{1}{\sqrt{t}}$ (d) $\sqrt{t}$

38. The potential energy function for the force between two atoms in a diatomic molecule is approximately given by $U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$, where a and b are constants and x is the distance between the atoms. If the dissociation energy of the molecule is

$$D = [U(x = \infty) - U_{\text{at equilibrium}}], D \text{ is} \quad [2010]$$

- (a) $\frac{b^2}{2a}$ (b) $\frac{b^2}{12a}$ (c) $\frac{b^2}{4a}$ (d) $\frac{b^2}{6a}$

39. An athlete in the olympic games covers a distance of 100 m in 10 s. His kinetic energy can be estimated to be in the range [2008]

- (a) 200 J - 500 J (b) 2×10^5 J - 3×10^5 J
(c) 20,000 J - 50,000 J (d) 2,000 J - 5,000 J

40. A 2 kg block slides on a horizontal floor with a speed of 4 m/s. It strikes a uncompressed spring, and compresses it till the block is motionless. The kinetic friction force is 15 N and spring constant is 10,000 N/m. The spring compresses by [2007]

- (a) 8.5 cm (b) 5.5 cm
(c) 2.5 cm (d) 11.0 cm

41. A particle is projected at 60° to the horizontal with a kinetic energy K . The kinetic energy at the highest point is [2007]

- (a) $K/2$ (b) K
(c) Zero (d) $K/4$

42. A particle of mass 100 g is thrown vertically upwards with a speed of 5 m/s. The work done by the force of gravity during the time the particle goes up is [2006]

- (a) -0.5 J (b) -1.25 J
(c) 1.25 J (d) 0.5 J

43. The potential energy of a 1 kg particle free to move along the x -axis is given by $V(x) = \left(\frac{x^4}{4} - \frac{x^2}{2}\right)$ J.

The total mechanical energy of the particle is 2 J. Then, the maximum speed (in m/s) is [2006]

- (a) $\frac{3}{\sqrt{2}}$ (b) $\sqrt{2}$ (c) $\frac{1}{\sqrt{2}}$ (d) 2

44. A mass of M kg is suspended by a weightless string. The horizontal force that is required to displace it until the string makes an angle of 45° with the initial vertical direction is [2006]

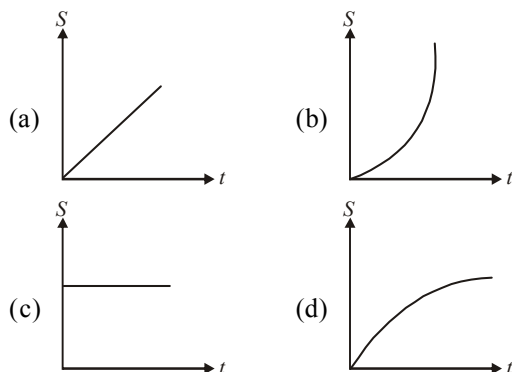
- (a) $Mg(\sqrt{2} + 1)$ (b) $Mg\sqrt{2}$
(c) $\frac{Mg}{\sqrt{2}}$ (d) $Mg(\sqrt{2} - 1)$

45. A spherical ball of mass 20 kg is stationary at the top of a hill of height 100 m. It rolls down a smooth surface to the ground, then climbs up another hill of height 30 m and finally rolls down to a horizontal base at a height of 20 m above the ground. The velocity attained by the ball is [2005]
- (a) 20 m/s (b) 40 m/s
(c) $10\sqrt{30}$ m/s (d) 10 m/s
46. A particle moves in a straight line with retardation proportional to its displacement. Its loss of kinetic energy for any displacement x is proportional to [2004]
- (a) x (b) e^x (c) x^2 (d) $\log_e x$
47. A particle is acted upon by a force of constant magnitude which is always perpendicular to the velocity of the particle, the motion of the particles takes place in a plane. It follows that [2004]
- (a) its kinetic energy is constant
(b) its acceleration is constant
(c) its velocity is constant
(d) it moves in a straight line
48. A wire suspended vertically from one of its ends is stretched by attaching a weight of 200N to the lower end. The weight stretches the wire by 1 mm. Then the elastic energy stored in the wire is [2003]
- (a) 0.2 J (b) 10 J (c) 20 J (d) 0.1 J
49. A ball whose kinetic energy is E , is projected at an angle of 45° to the horizontal. The kinetic energy of the ball at the highest point of its flight will be [2002]
- (a) E (b) $E/\sqrt{2}$ (c) $E/2$ (d) zero
52. A 60 HP electric motor lifts an elevator having a maximum total load capacity of 2000 kg. If the frictional force on the elevator is 4000 N, the speed of the elevator at full load is close to: (1 HP = 746 W, $g = 10 \text{ ms}^{-2}$) [7 Jan. 2020 I]
- (a) 1.7 ms^{-1} (b) 1.9 ms^{-1}
(c) 1.5 ms^{-1} (d) 2.0 ms^{-1}
53. A particle of mass M is moving in a circle of fixed radius R in such a way that its centripetal acceleration at time t is given by $n^2 R t^2$ where n is a constant. The power delivered to the particle by the force acting on it, is : [Online April 10, 2016]
- (a) $\frac{1}{2} M n^2 R^2 t^2$ (b) $M n^2 R^2 t$
(c) $M n R^2 t^2$ (d) $M n R^2 t$
54. A car of weight W is on an inclined road that rises by 100 m over a distance of 1 Km and applies a constant frictional force $\frac{W}{20}$ on the car. While moving uphill on the road at a speed of 10 ms^{-1} , the car needs power P . If it needs power $\frac{P}{2}$ while moving downhill at speed v then value of v is: [Online April 9, 2016]
- (a) 20 ms^{-1} (b) 5 ms^{-1} (c) 15 ms^{-1} (d) 10 ms^{-1}
55. A wind-powered generator converts wind energy into electrical energy. Assume that the generator converts a fixed fraction of the wind energy intercepted by its blades into electrical energy. For wind speed v , the electrical power output will be most likely proportional to [Online April 25, 2013]
- (a) v^4 (b) v^2 (c) v (d) v

TOPIC 3 Power



50. A body of mass 2 kg is driven by an engine delivering a constant power of 1 J/s. The body starts from rest and moves in a straight line. After 9 seconds, the body has moved a distance (in m) _____. [5 Sep. 2020 (II)]
51. A particle is moving unidirectionally on a horizontal plane under the action of a constant power supplying energy source. The displacement (s) - time (t) graph that describes the motion of the particle is (graphs are drawn schematically and are not to scale) : [3 Sep. 2020 (II)]

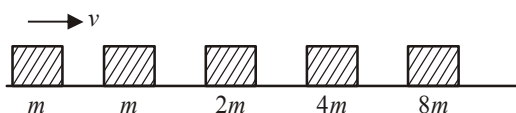


56. A 70 kg man leaps vertically into the air from a crouching position. To take the leap the man pushes the ground with a constant force F to raise himself. The center of gravity rises by 0.5 m before he leaps. After the leap the c.g. rises by another 1 m. The maximum power delivered by the muscles is : (Take $g = 10 \text{ ms}^{-2}$) [Online April 23, 2013]
- (a) 6.26×10^3 Watts at the start
(b) 6.26×10^3 Watts at take off
(c) 6.26×10^4 Watts at the start
(d) 6.26×10^4 Watts at take off
57. A body of mass ' m ', accelerates uniformly from rest to ' v_1 ' in time ' t_1 '. The instantaneous power delivered to the body as a function of time ' t ' is [2004]
- (a) $\frac{mv_1 t^2}{t_1}$ (b) $\frac{mv_1^2 t}{t_1^2}$
(c) $\frac{mv_1 t}{t_1}$ (d) $\frac{mv_1^2 t}{t_1}$
58. A body is moved along a straight line by a machine delivering a constant power. The distance moved by the body in time ' t ' is proportional to [2003]
- (a) $t^{3/4}$ (b) $t^{3/2}$ (c) $t^{1/4}$ (d) $t^{1/2}$

TOPIC 4 Collisions



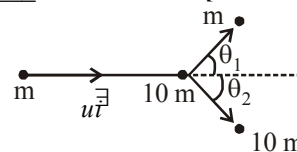
59. Two bodies of the same mass are moving with the same speed, but in different directions in a plane. They have a completely inelastic collision and move together thereafter with a final speed which is half of their initial speed. The angle between the initial velocities of the two bodies (in degree) is _____. [NA 6 Sep. 2020 (I)]
60. Particle A of mass m_1 moving with velocity $(\sqrt{3}\hat{i} + \hat{j}) \text{ ms}^{-1}$ collides with another particle B of mass m_2 which is at rest initially. Let $\vec{v}_1$ and $\vec{v}_2$ be the velocities of particles A and B after collision respectively. If $m_1 = 2m_2$ and after collision $\vec{v}_1 = (\hat{i} + \sqrt{3}\hat{j}) \text{ ms}^{-1}$, the angle between $\vec{v}_1$ and $\vec{v}_2$ is : [6 Sep. 2020 (II)]
- (a) 15° (b) 60°
(c) -45° (d) 105°
61. Blocks of masses $m, 2m, 4m$ and $8m$ are arranged in a line on a frictionless floor. Another block of mass m , moving with speed v along the same line (see figure) collides with mass m in perfectly inelastic manner. All the subsequent collisions are also perfectly inelastic. By the time the last block of mass $8m$ starts moving the total energy loss is $p\%$ of the original energy. Value of ' p ' is close to : [4 Sep. 2020 (I)]



- (a) 77 (b) 94
(c) 37 (d) 87
62. A block of mass 1.9 kg is at rest at the edge of a table, of height 1 m. A bullet of mass 0.1 kg collides with the block and sticks to it. If the velocity of the bullet is 20 m/s in the horizontal direction just before the collision then the kinetic energy just before the combined system strikes the floor, is [Take $g = 10 \text{ m/s}^2$. Assume there is no rotational motion and loss of energy after the collision is negligible.] [3 Sep. 2020 (II)]
- (a) 20 J (b) 21 J (c) 19 J (d) 23 J
63. A particle of mass m with an initial velocity $u\hat{i}$ collides perfectly elastically with a mass $3m$ at rest. It moves with a velocity $v\hat{j}$ after collision, then, v is given by : [2 Sep. 2020 (I)]

- (a) $v = \sqrt{\frac{2}{3}}u$ (b) $v = \frac{u}{\sqrt{3}}$
(c) $v = \frac{u}{\sqrt{2}}$ (d) $v = \frac{1}{\sqrt{6}}u$

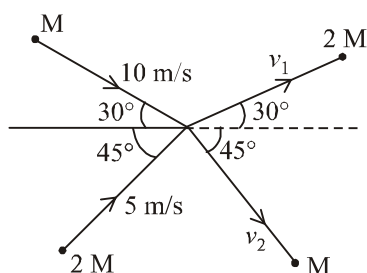
64. A particle of mass m is moving along the x -axis with initial velocity $u\hat{i}$. It collides elastically with a particle of mass $10m$ at rest and then moves with half its initial kinetic energy (see figure). If $\sin \theta_1 = \sqrt{n} \sin \theta_2$, then value of n is _____. [NA 2 Sep. 2020 (II)]



65. Two particles of equal mass m have respective initial velocities $u\hat{i}$ and $u\left(\frac{\hat{i} + \hat{j}}{2}\right)$. They collide completely inelastically. The energy lost in the process is: [9 Jan. 2020 I]
- (a) $\frac{1}{3}mu^2$ (b) $\frac{1}{8}mu^2$
(c) $\frac{3}{4}mu^2$ (d) $\sqrt{\frac{2}{3}}mu^2$
66. A body A , of mass $m = 0.1 \text{ kg}$ has an initial velocity of $3\hat{i} \text{ ms}^{-1}$. It collides elastically with another body, B of the same mass which has an initial velocity of $5\hat{j} \text{ ms}^{-1}$. After collision, A moves with a velocity $\vec{v} = 4(\hat{i} + \hat{j})$. The energy of B after collision is written as $\frac{x}{10} J$. The value of x is _____. [NA 8 Jan. 2020 I]
67. A particle of mass m is dropped from a height h above the ground. At the same time another particle of the same mass is thrown vertically upwards from the ground with a speed of $\sqrt{2gh}$. If they collide head-on completely inelastically, the time taken for the combined mass to reach

the ground, in units of $\sqrt{\frac{h}{g}}$ is: [8 Jan. 2020 II]

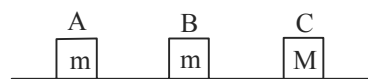
- (a) $\sqrt{\frac{1}{2}}$ (b) $\sqrt{\frac{3}{4}}$
(c) $\frac{1}{2}$ (d) $\sqrt{\frac{3}{2}}$
68. A man (mass = 50 kg) and his son (mass = 20 kg) are standing on a frictionless surface facing each other. The man pushes his son so that he starts moving at a speed of 0.70 ms^{-1} with respect to the man. The speed of the man with respect to the surface is : [12 April 2019 I]
- (a) 0.28 ms^{-1} (b) 0.20 ms^{-1}
(c) 0.47 ms^{-1} (d) 0.14 ms^{-1}
69. Two particles, of masses M and $2M$, moving, as shown, with speeds of 10 m/s and 5 m/s , collide elastically at the origin. After the collision, they move along the indicated directions with speeds v_1 and v_2 , respectively. The values of v_1 and v_2 are nearly : [10 April 2019 I]



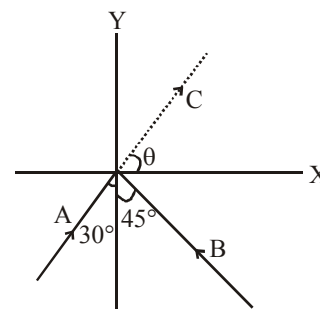
- (a) 6.5 m/s and 6.3 m/s (b) 3.2 m/s and 6.3 m/s
(c) 6.5 m/s and 3.2 m/s (d) 3.2 m/s and 12.6 m/s
70. A body of mass 2 kg makes an elastic collision with a second body at rest and continues to move in the original direction but with one fourth of its original speed. What is the mass of the second body? [9 April 2019 I]
(a) 1.0 kg (b) 1.5 kg (c) 1.8 kg (d) 1.2 kg
71. A particle of mass ' m ' is moving with speed ' $2v$ ' and collides with a mass ' $2m$ ' moving with speed ' v ' in the same direction. After collision, the first mass is stopped completely while the second one splits into two particles each of mass ' m ', which move at angle 45° with respect to the original direction. [9 April 2019 II]
The speed of each of the moving particle will be:
(a) $\sqrt{2} v$ (b) $2\sqrt{2} v$
(c) $v/(2\sqrt{2})$ (d) $v/\sqrt{2}$
72. A body of mass m_1 moving with an unknown velocity of $v_1 \hat{i}$, undergoes a collinear collision with a body of mass m_2 moving with a velocity $v_2 \hat{i}$. After collision, m_1 and m_2 move with velocities of $v_3 \hat{i}$ and $v_4 \hat{i}$, respectively. If $m_2 = 0.5 m_1$ and $v_3 = 0.5 v_1$, then v_4 is : [8 April 2019 II]
(a) $v_4 - \frac{v_2}{2}$ (b) $v_4 - v_2$ (c) $v_4 - \frac{v_2}{4}$ (d) $v_4 + v_2$
73. An alpha-particle of mass m suffers 1-dimensional elastic collision with a nucleus at rest of unknown mass. It is scattered directly backwards losing, 64% of its initial kinetic energy. The mass of the nucleus is : [12 Jan. 2019 II]
(a) $2m$ (b) $3.5m$ (c) $1.5m$ (d) $4m$
74. A piece of wood of mass 0.03 kg is dropped from the top of a 100 m height building. At the same time, a bullet of mass 0.02 kg is fired vertically upward, with a velocity 100 ms^{-1} , from the ground. The bullet gets embedded in the wood. Then the maximum height to which the combined system reaches above the top of the building before falling below is: ($g = 10 \text{ ms}^{-2}$) [10 Jan. 2019 I]
(a) 20m (b) 30m (c) 40m (d) 10m
75. Three blocks A, B and C are lying on a smooth horizontal surface, as shown in the figure. A and B have equal masses, m while C has mass M . Block A is given an initial speed v towards B due to which it collides with B perfectly inelastically. The combined mass collides with

C, also perfectly inelastically $\frac{5}{6}$ th of the initial kinetic energy is lost in whole process. What is value of M/m ?

[9 Jan. 2019 I]



- (a) 5 (b) 2 (c) 4 (d) 3
76. In a collinear collision, a particle with an initial speed v_0 strikes a stationary particle of the same mass. If the final total kinetic energy is 50% greater than the original kinetic energy, the magnitude of the relative velocity between the two particles, after collision, is: [2018]
(a) $\frac{v_0}{4}$ (b) $\sqrt{2}v_0$ (c) $\frac{v_0}{2}$ (d) $\frac{v_0}{\sqrt{2}}$
77. The mass of a hydrogen molecule is $3.32 \times 10^{-27} \text{ kg}$. If 10^{23} hydrogen molecules strike, per second, a fixed wall of area 2 cm^2 at an angle of 45° to the normal, and rebound elastically with a speed of 10^3 m/s , then the pressure on the wall is nearly: [2018]
(a) $2.35 \times 10^3 \text{ N/m}^2$ (b) $4.70 \times 10^3 \text{ N/m}^2$
(c) $2.35 \times 10^2 \text{ N/m}^2$ (d) $4.70 \times 10^2 \text{ N/m}^2$
78. It is found that if a neutron suffers an elastic collinear collision with deuterium at rest, fractional loss of its energy is P_d ; while for its similar collision with carbon nucleus at rest, fractional loss of energy is P_c . The values of P_d and P_c are respectively: [2018]
(a) $(.89, .28)$ (b) $(.28, .89)$ (c) $(0, 0)$ (d) $(0, 1)$
79. A proton of mass m collides elastically with a particle of unknown mass at rest. After the collision, the proton and the unknown particle are seen moving at an angle of 90° with respect to each other. The mass of unknown particle is: [Online April 15, 2018]
(a) $\frac{m}{\sqrt{3}}$ (b) $\frac{m}{2}$ (c) $2m$ (d) m
80. Two particles A and B of equal mass M are moving with the same speed v as shown in the figure. They collide completely inelastically and move as a single particle C. The angle θ that the path of C makes with the X-axis is given by: [Online April 9, 2017]
(a) $\tan \theta = \frac{\sqrt{3} + \sqrt{2}}{1 - \sqrt{2}}$
(b) $\tan \theta = \frac{\sqrt{3} - \sqrt{2}}{1 - \sqrt{2}}$
(c) $\tan \theta = \frac{1 - \sqrt{2}}{\sqrt{2}(1 + \sqrt{3})}$
(d) $\tan \theta = \frac{1 - \sqrt{3}}{1 + \sqrt{2}}$



81. A neutron moving with a speed ' v ' makes a head on collision with a stationary hydrogen atom in ground state. The minimum kinetic energy of the neutron for which inelastic collision will take place is :

[Online April 10, 2016]

(a) 20.4 eV (b) 10.2 eV (c) 12.1 eV (d) 16.8 eV

82. A particle of mass m moving in the x direction with speed $2v$ is hit by another particle of mass $2m$ moving in the y direction with speed v . If the collision is perfectly inelastic, the percentage loss in the energy during the collision is close to :

[2015]

(a) 56% (b) 62% (c) 44% (d) 50%

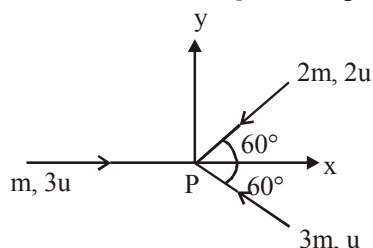
83. A bullet of mass 4 g is fired horizontally with a speed of 300 m/s into 0.8 kg block of wood at rest on a table. If the coefficient of friction between the block and the table is 0.3, how far will the block slide approximately?

[Online April 12, 2014]

(a) 0.19 m (b) 0.379 m (c) 0.569 m (d) 0.758 m

84. Three masses m , $2m$ and $3m$ are moving in x - y plane with speed $3u$, $2u$ and u respectively as shown in figure. The three masses collide at the same point at P and stick together. The velocity of resulting mass will be:

[Online April 12, 2014]



- (a) $\frac{u}{12}(\hat{i} + \sqrt{3}\hat{j})$ (b) $\frac{u}{12}(\hat{i} - \sqrt{3}\hat{j})$
 (c) $\frac{u}{12}(-\hat{i} + \sqrt{3}\hat{j})$ (d) $\frac{u}{12}(-\hat{i} - \sqrt{3}\hat{j})$

85. This question has statement I and statement II. Of the four choices given after the statements, choose the one that best describes the two statements.

Statement - I: A point particle of mass m moving with speed v collides with stationary point particle of mass M . If the maximum energy loss possible is given as

$$f\left(\frac{1}{2}mv^2\right) \text{ then } f = \left(\frac{m}{M+m}\right).$$

Statement - II: Maximum energy loss occurs when the particles get stuck together as a result of the collision.

[2013]

- (a) Statement - I is true, Statement - II is true, Statement - II is the correct explanation of Statement - I.
 (b) Statement - I is true, Statement - II is true, Statement - II is not the correct explanation of Statement - II.
 (c) Statement - I is true, Statement - II is false.
 (d) Statement - I is false, Statement - II is true.

86. A projectile of mass M is fired so that the horizontal range is 4 km. At the highest point the projectile explodes in two parts of masses $M/4$ and $3M/4$ respectively and the heavier part starts falling down vertically with zero initial speed. The horizontal range (distance from point of firing) of the lighter part is :

[Online April 23, 2013]

(a) 16 km (b) 1 km (c) 10 km (d) 2 km

87. A moving particle of mass m , makes a head on elastic collision with another particle of mass $2m$, which is initially at rest. The percentage loss in energy of the colliding particle on collision, is close to

[Online May 19, 2012]

(a) 33% (b) 67% (c) 90% (d) 10%

88. Two bodies A and B of mass m and $2m$ respectively are placed on a smooth floor. They are connected by a spring of negligible mass. A third body C of mass m is placed on the floor. The body C moves with a velocity v_0 along the line joining A and B and collides elastically with A . At a certain time after the collision it is found that the instantaneous velocities of A and B are same and the compression of the spring is x_0 . The spring constant k will be

[Online May 12, 2012]

- (a) $m \frac{v_0^2}{x_0^2}$ (b) $m \frac{v_0}{2x_0}$
 (c) $2m \frac{v_0}{x_0}$ (d) $\frac{2}{3} m \left(\frac{v_0}{x_0}\right)^2$

89. A projectile moving vertically upwards with a velocity of 200 ms^{-1} breaks into two equal parts at a height of 490 m. One part starts moving vertically upwards with a velocity of 400 ms^{-1} . How much time it will take, after the break up with the other part to hit the ground?

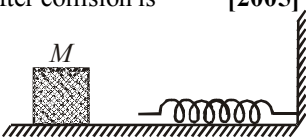
[Online May 12, 2012]

- (a) $2\sqrt{10} \text{ s}$ (b) 5 s
 (c) 10 s (d) $\sqrt{10} \text{ s}$

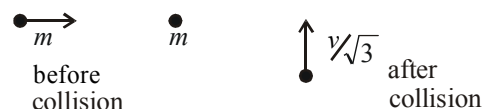
90. **Statement -1:** Two particles moving in the same direction do not lose all their energy in a completely inelastic collision.

Statement -2 : Principle of conservation of momentum holds true for all kinds of collisions. [2010]

- (a) Statement -1 is true, Statement -2 is true ; Statement -2 is the correct explanation of Statement -1.
 (b) Statement -1 is true, Statement -2 is true; Statement -2 is **not** the correct explanation of Statement -1
 (c) Statement -1 is false, Statement -2 is true.
 (d) Statement -1 is true, Statement -2 is false.
 91. A block of mass 0.50 kg is moving with a speed of 2.00 ms^{-1} on a smooth surface. It strikes another mass of 1.00 kg and then they move together as a single body. The energy loss during the collision is [2008]
 (a) 0.16 J (b) 1.00 J (c) 0.67 J (d) 0.34 J

- (a) 0.16 J (b) 1.00 J (c) 0.67 J (d) 0.34 J
92. A bomb of mass 16 kg at rest explodes into two pieces of masses 4 kg and 12 kg. The velocity of the 12 kg mass is 4 ms^{-1} . The kinetic energy of the other mass is [2006]
(a) 144 J (b) 288 J (c) 192 J (d) 96 J
93. The block of mass M moving on the frictionless horizontal surface collides with the spring of spring constant k and compresses it by length L . The maximum momentum of the block after collision is [2005]
- 
- (a) $\frac{kL^2}{2M}$ (b) $\sqrt{Mk} L$ (c) $\frac{ML^2}{k}$ (d) zero
94. A mass 'm' moves with a velocity 'v' and collides inelastically with another identical mass. After collision

the 1st mass moves with velocity $\frac{v}{\sqrt{3}}$ in a direction perpendicular to the initial direction of motion. Find the speed of the 2nd mass after collision. [2005]



- (a) $\sqrt{3}v$ (b) v (c) $\frac{v}{\sqrt{3}}$ (d) $\frac{2}{\sqrt{3}}v$
95. Consider the following two statements : [2003]
A. Linear momentum of a system of particles is zero
B. Kinetic energy of a system of particles is zero.
Then
(a) A does not imply B and B does not imply A
(b) A implies B but B does not imply A
(c) A does not imply B but B implies A
(d) A implies B and B implies A



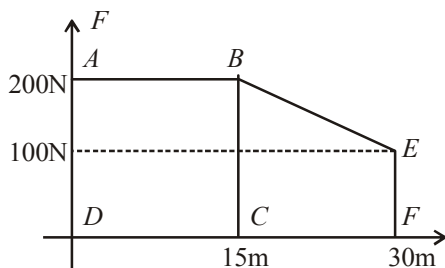
Hints & Solutions



1. (d) The given situation can be drawn graphically as shown in figure.

Work done = Area under F - x graph

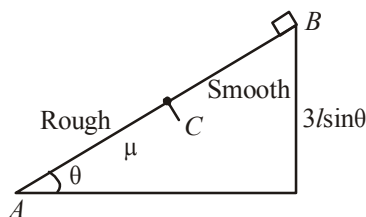
= Area of rectangle $ABCD$ + Area of trapezium $BCFE$



$$W = (200 \times 15) + \frac{1}{2}(100 + 200) \times 15 = 3000 + 2250$$

$$\Rightarrow W = 5250 \text{ J}$$

2. (3) If $AC = l$ then according to question, $BC = 2l$ and $AB = 3l$.



Here, work done by all the forces is zero.

$$W_{\text{friction}} + W_{\text{mg}} = 0$$

$$mg(3l) \sin \theta - \mu mg \cos \theta (l) = 0$$

$$\Rightarrow \mu mg \cos \theta l = 3mgl \sin \theta$$

$$\Rightarrow \mu = 3 \tan \theta = k \tan \theta$$

$$\therefore k = 3$$

3. (c) Work done, $W = \int \vec{F} \cdot d\vec{s}$

$$= (-x\hat{i} + y\hat{j}) \cdot (dx\hat{i} + dy\hat{j})$$

$$\Rightarrow W = -\int_1^0 x dx + \int_0^1 y dy$$

$$= \left(0 + \frac{1}{2}\right) + \frac{1}{2} = 1 \text{ J}$$

4. (d) Here, $N - mg = ma = \frac{3mg}{2} \Rightarrow N = \frac{5mg}{2}$

N = normal reaction

Now, work done by normal reaction ' N ' on

block in time t , $W = \vec{N} \cdot \vec{S} = \left(\frac{5mg}{2}\right) \left(\frac{1}{2}gt^2\right)$

$$\text{or, } W = \frac{3mg^2t^2}{8}$$

5. (a) From question, $v = a\sqrt{s} = \frac{ds}{dt}$

$$\text{or, } 2\sqrt{s} = at \Rightarrow S = \frac{a^2t^2}{4}$$

$$F = m \times \frac{a^2}{2}$$

$$\text{Work done} = \frac{ma^2}{2} \times \frac{a^2t^2}{4} = \frac{1}{8}ma^4t^2$$

6. (c) Work done in stretching the rubber-band by a distance dx is

$$dW = F dx = (ax + bx^2)dx$$

Integrating both sides,

$$W = \int_0^L ax dx + \int_0^L bx^2 dx = \frac{aL^2}{2} + \frac{bL^3}{3}$$

7. (b) Mass of over hanging part of the chain

$$m' = \frac{4}{2} \times (0.6) \text{ kg} = 1.2 \text{ kg}$$

Weight of hanging part of the chain

$$= 1.2 \times 10 = 12 \text{ N}$$

C.M. of hanging part = 0.3 m below the table

$$\text{Workdone in putting the entire chain on the table} = 12 \times 0.30 = 3.6 \text{ J.}$$

8. (b) Given, Force, $\vec{F} = (5\hat{i} + 3\hat{j} + 2\hat{k})$

Displacement, $x = (2\hat{i} - \hat{j})$

Work done,

$$W = \vec{F} \cdot \vec{x} = (5\hat{i} + 3\hat{j} + 2\hat{k}) \cdot (2\hat{i} - \hat{j})$$

$$= 10 - 3 = 7 \text{ joules}$$

9. (b) Spring constant, $k = 5 \times 10^3 \text{ N/m}$

Let x_1 and x_2 be the initial and final stretched position of the spring, then

$$\text{Work done, } W = \frac{1}{2}k(x_2^2 - x_1^2)$$

$$= \frac{1}{2} \times 5 \times 10^3 [(0.1)^2 - (0.05)^2]$$

$$= \frac{5000}{2} \times 0.15 \times 0.05 = 18.75 \text{ Nm}$$

10. (b) Small amount of work done in extending the spring by dx is

$$dW = k x dx$$

$$\therefore W = k \int_{0.05}^{0.15} x dx$$

$$= \frac{800}{2} [(0.15)^2 - (0.05)^2]$$

$$= 400 [(0.15 + 0.05)(0.15 - 0.05)]$$

$$= 400 \times 0.2 \times 0.1 = 8 \text{ J}$$

11. (150.00) From work energy theorem,

$$W = F \cdot s = \Delta KE = \frac{1}{2} mv^2$$

$$\text{Here } V^2 = 2gh$$

$$\therefore F \cdot s = F \times \frac{2}{10} = \frac{1}{2} \times \frac{15}{100} \times 2 \times 10 \times 20$$

$$\therefore F = 150 \text{ N.}$$

12. (10.00) Kinetic energy = change in potential energy of the particle,

$$KE = mg\Delta h$$

$$\text{Given, } m = 1 \text{ kg,}$$

$$\Delta h = h_2 - h_1 = 2 - 1 = 1 \text{ m}$$

$$\therefore KE = 1 \times 10 \times 1 = 10 \text{ J}$$

13. (c) We know area under F-x graph gives the work done by the body

$$\therefore W = \frac{1}{2} \times (3+2) \times (3-2) + 2 \times 2 = 2.5 + 4 = 6.5 \text{ J}$$

Using work energy theorem,

$$\Delta K.E = \text{work done}$$

$$\therefore \Delta K.E = 6.5 \text{ J}$$

14. (c) $l_1 + l_2 = l$ and $l_1 = nl_2$

$$\therefore l_1 = \frac{nl}{n+1} \text{ and } l_2 = \frac{l}{n+1}$$

$$\text{As } k \propto \frac{1}{l},$$

$$\therefore \frac{k_1}{k_2} = \frac{l / (n+1)}{(nl) / (n+1)} = \frac{1}{n}$$

15. (b) Velocity of 1 kg block just before it collides with 3 kg block = $\sqrt{2gh} = \sqrt{2000} \text{ m/s}$

Using principle of conservation of linear momentum just before and just after collision, we get

$$1 \times \sqrt{2000} = 4v \Rightarrow v = \frac{\sqrt{2000}}{4} \text{ m/s}$$

Initial compression of spring

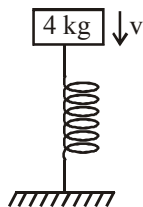
$$1.25 \times 10^6 x_0 = 30 \Rightarrow x_0 \approx 0$$

using work energy theorem,

$$W_g + W_{sp} = \Delta KE$$

$$\Rightarrow 40 \times x + \frac{1}{2} \times 1.25 \times 10^6 (0^2 - x^2)$$

$$= 0 - \frac{1}{2} \times 4 \times v^2$$



solving $x \approx 4 \text{ cm}$

16. (a) $W = u_f - u_i$

$$= 0 - \left(-\frac{mg}{n} \times \frac{L}{2n} \right) = \frac{MgL}{2n^2}$$

17. (c) $mv = (m + M) V'$

$$\text{or } v = \frac{mv}{m + M} = \frac{mv}{m + 4m} = \frac{v}{5}$$

Using conservation of ME, we have

$$\frac{1}{2} mv^2 = \frac{1}{2} (m + 4m) \left(\frac{v}{5} \right)^2 + mgh$$

$$\text{or } h = \frac{2}{5} \frac{v^2}{g}$$

18. (c) We know area under F-x graph gives the work done by the body

$$\therefore W = \frac{1}{2} \times (3+2) \times (3-2) + 2 \times 2$$

$$= 2.5 + 4$$

$$= 6.5 \text{ J}$$

Using work energy theorem,

$$\Delta K.E = \text{work done}$$

$$\therefore \Delta K.E = 6.5 \text{ J}$$

19. (d) Work done = $\vec{F} \cdot \vec{d} = (3\vec{i} - 12\vec{j}) \cdot (4\vec{i}) = 12 \text{ J}$

From work energy theorem,

$$W_{\text{net}} = \Delta K.E. = k_f - k_i$$

$$\Rightarrow 12 = k_f - 3$$

$$\therefore K_f = 15 \text{ J}$$

20. (d) Maximum speed is at mean position or equilibrium

At equilibrium Position

$$F = kx \Rightarrow x = \frac{F}{k}$$

From work-energy theorem,

$$W_F + W_{sp} = \Delta KE$$

$$F(x) - \frac{1}{2} kx^2 = \frac{1}{2} mv^2 - 0$$

$$F\left(\frac{F}{k}\right) - \frac{1}{2} k\left(\frac{F}{k}\right)^2 = \frac{1}{2} mv^2$$

$$\Rightarrow \frac{1}{2} \frac{F^2}{K} = \frac{1}{2} mv^2$$

$$\text{or, } v_{\text{max}} = \frac{F}{\sqrt{mk}}$$

21. (d) Position, $x = 3t^2 + 5$

$$\therefore \text{Velocity, } v = \frac{dx}{dt} \Rightarrow v = \frac{d(3t^2 + 5)}{dt}$$

$$\Rightarrow v = 6t + 0$$

At $t = 0$ $v = 0$
 And, at $t = 5$ sec $v = 30$ m/s
 According to work-energy theorem, $w = \Delta KE$

$$\text{or, } W = \frac{1}{2}mv^2 - 0 = \frac{1}{2}(2)(30)^2 = 900 \text{ J}$$

22. (c) $F = -\frac{\partial u}{\partial r} \hat{r} = \frac{K}{r^3} \hat{r}$

Since particle is moving in circular path

$$F = \frac{mv^2}{r} = \frac{K}{r^3} \Rightarrow mv^2 = \frac{K}{r^2}$$

$$\therefore \text{K.E.} = \frac{1}{2}mv^2 = \frac{K}{2r^2}$$

Total energy = P.E. + K.E.

$$= -\frac{K}{2r^2} + \frac{K}{2r^2} = \text{Zero} \quad (\because \text{P.E.} = -\frac{K}{2r^2} \text{ given})$$

23. (b) As the particles moving in circular orbits, So

$$\frac{mv^2}{r} = \frac{16}{r} + r^2$$

$$\text{Kinetic energy, } KE_0 = \frac{1}{2}mv^2 = \frac{1}{2}[16 + r^4]$$

$$\text{For first particle, } r = 1, K_1 = \frac{1}{2}m(16 + 1)$$

$$\text{Similarly, for second particle, } r = 4, K_2 = \frac{1}{2}m(16 + 256)$$

$$\therefore \frac{K_1}{K_2} = \frac{\frac{16+1}{2}}{\frac{16+256}{2}} = \frac{17}{272} \approx 6 \times 10^{-2}$$

24. (a) Let V_f is the final speed of the body.
 From questions,

$$\frac{1}{2}mV_f^2 = \frac{1}{8}mV_0^2 \Rightarrow V_f = \frac{V_0}{2} = 5 \text{ m/s}$$

$$F = m\left(\frac{dV}{dt}\right) = -kV^2 \therefore (10^{-2})\frac{dV}{dt} = -kV^2$$

$$\int_{10}^5 \frac{dV}{V^2} = -100K \int_0^{10} dt$$

$$\frac{1}{5} - \frac{1}{10} = 100K(10) \quad \text{or, } K = 10^{-4} \text{ kgm}^{-1}$$

25. (a) (K.E.)' = 50% of K.E. after hit i.e.,

$$\frac{1}{2}mv'^2 = \frac{50}{100} \times \frac{1}{2}mv^2 \Rightarrow v' = \frac{v}{\sqrt{2}}$$

$$\text{Coefficient of restitution} = \frac{1}{\sqrt{2}}$$

Now, total distance travelled by object is

$$H = h \left(\frac{1+e^2}{1-e^2} \right) = h \left(\frac{1+\frac{1}{2}}{1-\frac{1}{2}} \right) = 3h$$

26. (c) Using, $F = ma = m \frac{dV}{dt}$

$$6t = 1 \cdot \frac{dV}{dt} \quad [\because m = 1 \text{ kg given}]$$

$$\int_0^v dV = \int 6t dt \quad V = 6 \left[\frac{t^2}{2} \right]_0^1 = 3 \text{ ms}^{-1}$$

$[\because t = 1 \text{ sec given}]$

From work-energy theorem,

$$W = \Delta KE = \frac{1}{2}m(V^2 - u^2) = \frac{1}{2} \times 1 \times 9 = 4.5 \text{ J}$$

27. (c) Acceleration (a) = $\frac{v-4}{t} = \frac{(0-50)}{(10-0)} = -5 \text{ m/s}^2$

$$u = 50 \text{ m/s}$$

$$\therefore v = u + at = 50 - 5t$$

Velocity in first two seconds $t = 2$

$$v_{(at=2)} = 40 \text{ m/s}$$

From work-energy theorem,

$$\Delta \text{K.E.} = W = \frac{1}{2}(40^2 - 50^2) \times 10 = -4500 \text{ J}$$

28. (a) Work done by friction at QR = μmgx

$$\text{In triangle, } \sin 30^\circ = \frac{1}{2} = \frac{2}{PQ}$$

$$\Rightarrow PQ = 4m$$

$$\text{Work done by friction at PQ} = \mu mg \times \cos 30^\circ \times 4$$

$$= \mu mg \times \frac{\sqrt{3}}{2} \times 4 = 2\sqrt{3} \mu mg$$

Since work done by friction on parts PQ and QR are equal,

$$\mu mgx = 2\sqrt{3} \mu mg$$

$$\Rightarrow x = 2\sqrt{3} \approx 3.5m$$

$$\text{Using work energy theorem } mg \sin 30^\circ \times 4 = 2\sqrt{3} \mu mg + \mu mgx$$

$$\Rightarrow 2 = 4\sqrt{3} \mu$$

$$\Rightarrow \mu = 0.29$$

29. (b) $n = \frac{W}{\text{input}} = \frac{mgh \times 1000}{\text{input}} = \frac{10 \times 9.8 \times 1 \times 1000}{\text{input}}$

$$\text{Input} = \frac{98000}{0.2} = 49 \times 10^4 \text{ J}$$

$$\text{Fat used} = \frac{49 \times 10^4}{3.8 \times 10^7} = 12.89 \times 10^{-3} \text{ kg}$$

30. (b) As we know, $dU = F \cdot dr$

$$U = \int_0^r \alpha r^2 dr = \frac{\alpha r^3}{3} \quad \dots(i)$$

$$\text{As, } \frac{mv^2}{r} = \alpha r^2$$

$$m^2 v^2 = m \alpha r^3$$

$$\text{or, } 2m(\text{KE}) = \frac{1}{2} \alpha r^3 \quad \dots(ii)$$

Total energy = Potential energy + kinetic energy

Now, from eqn (i) and (ii)

Total energy = K.E. + P.E.

$$= \frac{\alpha r^3}{3} + \frac{\alpha r^3}{2} = \frac{5}{6} \alpha r^3$$

31. (b) Applying momentum conservation

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$0.1u + m(0) = 0.1(0) + m(3)$$

$$0.1u = 3m$$

$$\frac{1}{2} 0.1u^2 = \frac{1}{2} m(3)^2$$

Solving we get, $u = 3$

$$\frac{1}{2} kx^2 = \frac{1}{2} K \left(\frac{x}{2} \right)^2 + \frac{1}{2} (0.1) 3^2$$

$$\Rightarrow \frac{3}{4} kx^2 = 0.9$$

$$\Rightarrow \frac{3}{2} \times \frac{1}{2} kx^2 = 0.9$$

$$\therefore \frac{1}{2} Kx^2 = 0.6 \text{ J (total initial energy of the spring)}$$

32. (a) Let u be the initial velocity of the bullet of mass m .

After passing through a plank of width x , its velocity decreases to v .

$$\therefore u - v = \frac{4}{n} \text{ or, } v = u - \frac{4}{n} = \frac{u(n-1)}{n}$$

If F be the retarding force applied by each plank, then using work – energy theorem,

$$Fx = \frac{1}{2} mu^2 - \frac{1}{2} mv^2 = \frac{1}{2} mu^2 - \frac{1}{2} mu^2 \frac{(n-1)^2}{n^2}$$

$$= \frac{1}{2} mu^2 \left[\frac{1 - (n-1)^2}{n^2} \right]$$

$$Fx = \frac{1}{2} mu^2 \left(\frac{2n-1}{n^2} \right)$$

Let P be the number of planks required to stop the bullet.

Total distance travelled by the bullet before coming to rest = Px

Using work-energy theorem again,

$$F(Px) = \frac{1}{2} mu^2 - 0$$

$$\text{or, } P(Fx) = P \left[\frac{1}{2} mu^2 \frac{(2n-1)}{n^2} \right] = \frac{1}{2} mu^2$$

$$\therefore P = \frac{n^2}{2n-1}$$

33. (d)

34. (a) Given : $k_A = 300 \text{ N/m}$, $k_B = 400 \text{ N/m}$

Let when the combination of springs is compressed by force F . Spring A is compressed by x . Therefore compression in spring B

$$x_B = (8.75 - x) \text{ cm}$$

$$F = 300 \times x = 400(8.75 - x)$$

Solving we get, $x = 5 \text{ cm}$

$$x_B = 8.75 - 5 = 3.75 \text{ cm}$$

$$\frac{E_A}{E_B} = \frac{\frac{1}{2} k_A (x_A)^2}{\frac{1}{2} k_B (x_B)^2} = \frac{300 \times (5)^2}{400 \times (3.75)^2} = \frac{4}{3}$$

35. (d)

36. (a) According to work-energy theorem, Change in kinetic energy = work done

$$= \vec{F} \cdot \Delta \vec{r} = (7\hat{i} + 4\hat{j} + 3\hat{k}) \cdot (2\hat{i} + 3\hat{j} + 4\hat{k})$$

$$= 14 + 12 + 12 = 38 \text{ J}$$

37. (c) K.E. $\propto t$

$$\text{K.E.} = ct \quad [\text{Here, } c = \text{constant}]$$

$$\Rightarrow \frac{1}{2} mv^2 = ct$$

$$\Rightarrow \frac{(mv)^2}{2m} = ct$$

$$\Rightarrow \frac{p^2}{2m} = ct \quad (\because p = mv)$$

$$\Rightarrow p = \sqrt{2ctm}$$

$$\Rightarrow F = \frac{dp}{dt} = \frac{d(\sqrt{2ctm})}{dt}$$

$$\Rightarrow F = \sqrt{2cm} \times \frac{1}{2\sqrt{t}}$$

$$\Rightarrow F \propto \frac{1}{\sqrt{t}}$$

38. (d) At equilibrium : $F = \frac{-dU(x)}{dx}$

$$\Rightarrow F = \frac{-d}{dx} \left[\frac{a}{x^{12}} - \frac{b}{x^6} \right]$$

$$\Rightarrow F = - \left[\frac{12a}{x^{13}} + \frac{6b}{x^7} \right]$$

$$\Rightarrow \frac{12a}{x^{13}} = \frac{6b}{x^7} \Rightarrow x = \left(\frac{2a}{b} \right)^{\frac{1}{6}}$$

$$\therefore U_{\text{at equilibrium}} = \frac{a}{\left(\frac{2a}{b} \right)^2} - \frac{b}{\left(\frac{2a}{b} \right)} = -\frac{b^2}{4a} \text{ and } U_{(x=\infty)} = 0$$

$$\therefore D = 0 - \left(-\frac{b^2}{4a} \right) = \frac{b^2}{4a}$$

39. (d) The average speed of the athlete

$$v = \frac{5}{t} = \frac{100}{10} = 10 \text{ m/s}$$

$$\therefore \text{K.E.} = \frac{1}{2} mv^2$$

Assuming the mass of athlete to 40 kg his average K.E. would be

$$\text{K.E.} = \frac{1}{2} \times 40 \times (10)^2 = 2000 \text{ J}$$

Assuming mass to 100 kg average kinetic energy

$$\text{K.E.} = \frac{1}{2} \times 100 \times (10)^2 = 5000 \text{ J}$$

40. (b) Suppose the spring gets compressed by x before stopping.
kinetic energy of the block = P.E. stored in the spring + work done against friction.

$$\Rightarrow \frac{1}{2} \times 2 \times (4)^2 = \frac{1}{2} \times 10,000 \times x^2 + 15 \times x$$

$$\Rightarrow 10,000x^2 + 30x - 32 = 0$$

$$\Rightarrow 5000x^2 + 15x - 16 = 0$$

$$\therefore x = \frac{-15 \pm \sqrt{(15)^2 - 4 \times (5000)(-16)}}{2 \times 5000}$$

$$= 0.055 \text{ m} = 5.5 \text{ cm.}$$

41. (d) Let u be the velocity with which the particle is thrown and m be the mass of the particle. Then

$$K = \frac{1}{2} mu^2. \quad \dots (1)$$

At the highest point the velocity is $u \cos 60^\circ$ (only the horizontal component remains, the vertical component being zero at the top-most point). Therefore kinetic energy at the highest point.

$$K' = \frac{1}{2} m(u \cos 60^\circ)^2 = \frac{1}{2} mu^2 \cos^2 60^\circ = \frac{K}{4} \quad [\text{From 1}]$$

42. (b) Given, Mass of the particle, $m = 100 \text{ g}$
Initial speed of the particle, $u = 5 \text{ m/s}$
Final speed of the particle, $v = 0$
Work done by the force of gravity
= Loss in kinetic energy of the body.

$$= \frac{1}{2} m (v^2 - u^2) = \frac{1}{2} \times \frac{100}{1000} (0^2 - 5^2)$$

$$= -1.25 \text{ J}$$

43. (a) Potential energy

$$V(x) = \frac{x^4 - x^2}{4} \text{ joule}$$

For maxima of minima

$$\frac{dV}{dx} = 0 \Rightarrow x^3 - x = 0 \Rightarrow x = \pm 1$$

$$\Rightarrow \text{Min. P.E.} = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4} \text{ J}$$

$$\text{K.E.}_{(\text{max.})} + \text{P.E.}_{(\text{min.})} = 2 \text{ (Given)}$$

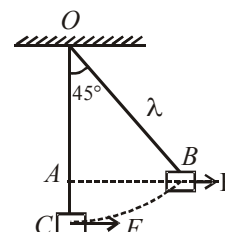
$$\therefore \text{K.E.}_{(\text{max.})} = 2 + \frac{1}{4} = \frac{9}{4}$$

$$\text{K.E.}_{\text{max.}} = \frac{1}{2} mv_{\text{max.}}^2$$

$$\Rightarrow \frac{1}{2} \times 1 \times v_{\text{max.}}^2 = \frac{9}{4} \Rightarrow v_{\text{max.}} = \frac{3}{\sqrt{2}}$$

44. (d) Work done by tension + Work done by force (applied) + Work done by gravitational force = change in kinetic energy

Work done by tension is zero



$$\Rightarrow 0 + F \times AB - Mg \times AC = 0$$

$$\Rightarrow F = Mg \left(\frac{AC}{AB} \right) = Mg \left[\frac{1 - \frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}} \right]$$

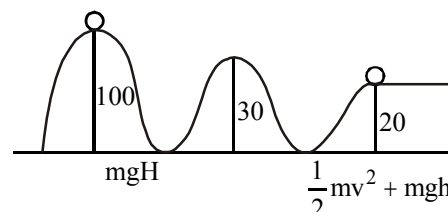
$$[\because AB = l \sin 45^\circ = \frac{l}{\sqrt{2}} \text{ and}]$$

$$AC = OC - OA = l - l \cos 45^\circ = l \left(1 - \frac{1}{\sqrt{2}} \right)$$

where l = length of the string.]

$$\Rightarrow F = Mg(\sqrt{2} - 1)$$

45. (b)



Using conservation of energy,

Total energy at 100 m height

= Total energy at 20m height

$$m (10 \times 100) = m \left(\frac{1}{2} v^2 + 10 \times 20 \right)$$

$$\text{or } \frac{1}{2} v^2 = 800 \text{ or } v = \sqrt{1600} = 40 \text{ m/s}$$

Note :

Loss in potential energy = gain in kinetic energy

$$m \times g \times 80 = \frac{1}{2} mv^2$$

$$10 \times 80 = \frac{1}{2} v^2$$

$$v^2 = 1600 \text{ or } v = 40 \text{ m/s}$$

46. (c) Given : retardation $\propto$ displacement
i.e., $a = -kx$ [Here, k = constant]

$$\text{But } a = v \frac{dv}{dx}$$

$$\therefore \frac{v dv}{dx} = -kx \Rightarrow \int_{v_1}^{v_2} v dv = - \int_0^x kx dx$$

$$\Rightarrow \left[\frac{v^2}{2} \right]_{v_1}^{v_2} = -k \left[\frac{x^2}{2} \right]_0^x$$

$$\Rightarrow (v_2^2 - v_1^2) = -\frac{kx^2}{2}$$

$$\Rightarrow \frac{1}{2}m(v_2^2 - v_1^2) = \frac{1}{2}m \left(\frac{-kx^2}{2} \right)$$

$\therefore$ Loss in kinetic energy, $\therefore \Delta K \propto x^2$

47. (a) Work done by such force is always zero since force is acting in a direction perpendicular to velocity.

$\therefore$ From work-energy theorem $\Delta K = 0$

K remains constant.

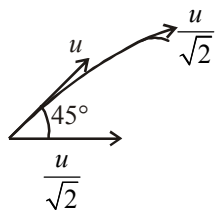
48. (d) The elastic potential energy

$$= \frac{1}{2} \times \text{Force} \times \text{extension}$$

$$= \frac{1}{2} \times F \times x$$

$$= \frac{1}{2} \times 200 \times 0.001 = 0.1 \text{ J}$$

49. (c) Let u be the speed with which the ball of mass m is projected. Then the kinetic energy (E) at the point of projection is



$$E = \frac{1}{2}mu^2 \quad \dots(i)$$

When the ball is at the highest point of its flight, the speed

of the ball is $\frac{u}{\sqrt{2}}$ (Remember that the horizontal component of velocity does not change during a projectile motion).

$\therefore$ The kinetic energy at the highest point

$$= \frac{1}{2}m \left(\frac{u}{\sqrt{2}} \right)^2 = \frac{1}{2} \frac{mu^2}{2} = \frac{E}{2} \quad [\text{From (i)}]$$

50. (18) Given, Mass of the body, $m = 2 \text{ kg}$

Power delivered by engine, $P = 1 \text{ J/s}$

Time, $t = 9 \text{ seconds}$

Power, $P = Fv$

$$\Rightarrow P = mav \quad [\because F = ma]$$

$$\Rightarrow m \frac{dv}{dt} v = P \quad \left(\because a = \frac{dv}{dt} \right)$$

$$\Rightarrow v dv = \frac{P}{m} dt$$

Integrating both sides we get

$$\Rightarrow \int_0^v v dv = \frac{P}{m} \int_0^t dt$$

$$\Rightarrow \frac{v^2}{2} = \frac{Pt}{m} \Rightarrow v = \left(\frac{2Pt}{m} \right)^{1/2}$$

$$\Rightarrow \frac{dx}{dt} = \sqrt{\frac{2P}{m}} t^{1/2} \quad \left(\because v = \frac{dx}{dt} \right)$$

$$\Rightarrow \int_0^x dx = \sqrt{\frac{2P}{m}} \int_0^t t^{1/2} dt$$

$$\therefore \text{Distance, } x = \sqrt{\frac{2P}{m}} \frac{t^{3/2}}{3/2} = \sqrt{\frac{2P}{m}} \times \frac{2}{3} t^{3/2}$$

$$\Rightarrow x = \sqrt{\frac{2 \times 1}{2}} \times \frac{2}{3} \times 9^{3/2} = \frac{2}{3} \times 27 = 18.$$

51. (b) We know that

Power, $P = Fv$

$$\text{But } F = mav = m \frac{dv}{dt} v$$

$$\therefore P = mv \frac{dv}{dt} \Rightarrow P dt = mv dv$$

$$\text{Integrating both sides } \int_0^t P dt = m \int_0^v v dv$$

$$P \cdot t = \frac{1}{2}mv^2 \Rightarrow v = \left(\sqrt{\frac{2P}{m}} \right) t^{1/2}$$

$$\text{Distance, } s = \int_0^t v dt = \sqrt{\frac{2P}{m}} \int_0^t t^{1/2} dt = \sqrt{\frac{2P}{m}} \cdot \frac{t^{3/2}}{3/2}$$

$$\Rightarrow s = \sqrt{\frac{8P}{m}} \cdot t^{3/2} \Rightarrow s \propto t^{3/2}$$

So, graph (b) is correct.

52. (b) Total force required to lift maximum load capacity against frictional force $= 400 \text{ N}$

$$F_{\text{total}} = Mg + \text{friction}$$

$$= 2000 \times 10 + 4000$$

$$= 20,000 + 4000 = 24000 \text{ N}$$

Using power, $P = F \times v$

$$60 \times 746 = 24000 \times v$$

$$\Rightarrow v = 1.86 \text{ m/s} \approx 1.9 \text{ m/s}$$

Hence speed of the elevator at full load is close to 1.9 ms^{-1}

53. (b) Centripetal acceleration $a_c = n^2 R t^2$

$$a_c = \frac{v^2}{R} = n^2 R t^2$$

$$v^2 = n^2 R^2 t^2$$

$$v = n R t$$

$$a_c = \frac{dv}{dt} = n R$$

$$\text{Power} = m a_c v = m n R n R t = M n^2 R^2 t.$$

54. (c) While moving downhill power

$$P = \left(w \sin \theta + \frac{w}{20} \right) 10$$

$$P = \left(\frac{w}{10} + \frac{w}{20} \right) 10 = \frac{3w}{2}$$

$$\frac{P}{2} = \frac{3w}{4} = \left(\frac{w}{10} + \frac{w}{20} \right) V$$

$$\frac{3}{4} = \frac{v}{20} \Rightarrow v = 15 \text{ m/s}$$

$\therefore$ Speed of car while moving downhill $v = 15 \text{ m/s}$.

55. (d) 56. (b)

57. (b) Let a be the acceleration of body

Using, $v = u + at$

$$v_1 = 0 + at_1 \Rightarrow a = \frac{v_1}{t_1}$$

Velocity of the body at instant t ,

$$v = at$$

$$\Rightarrow v = \frac{v_1 t}{t_1}$$

Instantaneous power, $P = \vec{F} \cdot \vec{v} = (m\vec{a}) \cdot \vec{v}$

$$= \left(\frac{mv_1}{t_1} \right) \left(\frac{v_1 t}{t_1} \right) = m \left(\frac{v_1}{t_1} \right)^2 t$$

58. (b) Power, $P = Fv = ma \cdot v$

$$\Rightarrow P = \frac{mdv}{dt} v = c = \text{constant}$$

$$\left(\because F = ma = \frac{mdv}{dt} \right)$$

$$mv_0 v = c dt$$

Integrating both sides, we get

$$m \int_0^v v dv = c \int_0^t dt$$

$$\Rightarrow \frac{1}{2} mv^2 = ct$$

$$\Rightarrow \frac{v^2}{2} = \frac{c \cdot t}{m}$$

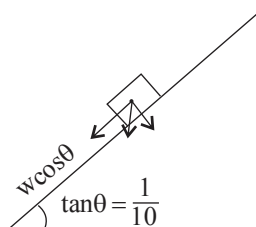
$$\Rightarrow v_2 = \frac{2c \cdot t}{m}$$

$$\Rightarrow v = \sqrt{\frac{2c}{m}} \times t^{1/2}$$

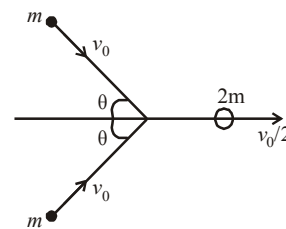
$$\Rightarrow \frac{dx}{dt} = \sqrt{\frac{2c}{m}} \times t^{1/2} \quad \text{where } v = \frac{dx}{dt}$$

$$\Rightarrow \int_e^x dx = \sqrt{\frac{2c}{m}} \times \int_0^t t^{1/2} dt$$

$$\Rightarrow x = \sqrt{\frac{2c}{m}} \times \frac{2t^{3/2}}{3} \Rightarrow x \propto t^{3/2}$$



59. (120)



Momentum conservation along x direction,

$$2mv_0 \cos \theta = 2m \frac{v_0}{2} \Rightarrow \cos \theta = \frac{1}{2} \text{ or } \theta = 60^\circ$$

Hence angle between the initial velocities of the two bodies $= \theta + \theta = 60^\circ + 60^\circ = 120^\circ$.

60. (d) Before collision,

Velocity of particle A , $u_1 = (\sqrt{3}\hat{i} + \hat{j}) \text{ m/s}$

Velocity of particle B , $u_2 = 0$

After collision,

Velocity of particle A , $v_1 = (\hat{i} + \sqrt{3}\hat{j})$

Velocity of particle B , $v_2 = 0$

Using principal of conservation of angular momentum

$$m_1 \vec{u}_1 + m_2 \vec{u}_2 = m_1 \vec{v}_1 + m_2 \vec{v}_2$$

$$\Rightarrow 2m_2(\sqrt{3}\hat{i} + \hat{j}) + m_2 \times 0 = 2m_2(\hat{i} + \sqrt{3}\hat{j}) + m_2 \times \vec{v}_2$$

$$\Rightarrow 2\sqrt{3}\hat{i} + 2\hat{j} = 2\hat{i} + 2\sqrt{3}\hat{j} + \vec{v}_2$$

$$\Rightarrow \vec{v}_2 = (\sqrt{3} - 1)\hat{i} - (\sqrt{3} - 1)\hat{j}$$

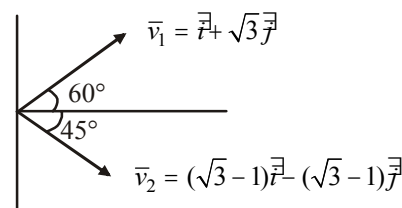
$$\Rightarrow \vec{v}_1 = \hat{i} + \sqrt{3}\hat{j}$$

For angle between $\vec{v}_1$ and $\vec{v}_2$,

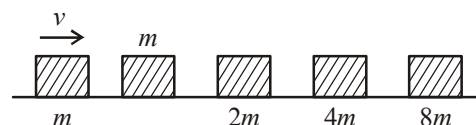
$$\cos \theta = \frac{\vec{v}_1 \cdot \vec{v}_2}{|\vec{v}_1| |\vec{v}_2|} = \frac{2(\sqrt{3} - 1)(1 - \sqrt{3})}{2 \times 2\sqrt{2}(\sqrt{3} - 1)} = \frac{1 - \sqrt{3}}{2\sqrt{2}}$$

$$\Rightarrow \theta = 105^\circ$$

Angle between $\vec{v}_1$ and $\vec{v}_2$ is 105°



61. (b) According to the question, all collisions are perfectly inelastic, so after the final collision, all blocks are moving together.



Let the final velocity be v' , using momentum conservation

$$mv = 16mv' \Rightarrow v' = \frac{v}{16}$$

Now initial energy $E_i = \frac{1}{2}mv^2$

Final energy: $E_f = \frac{1}{2} \times 16m \times \left(\frac{v}{16}\right)^2 = \frac{1}{2} \frac{mv^2}{16}$

Energy loss: $E_i - E_f = \frac{1}{2}mv^2 - \frac{1}{2}m \frac{v^2}{16}$

$$\Rightarrow \frac{1}{2}mv^2 \left[1 - \frac{1}{16}\right] \Rightarrow \frac{1}{2}mv^2 \left[\frac{15}{16}\right]$$

The total energy loss is $P\%$ of the original energy.

$$\therefore \%P = \frac{\text{Energy loss}}{\text{Original energy}} \times 100$$

$$= \frac{\frac{1}{2}mv^2 \left[\frac{15}{16}\right]}{\frac{1}{2}mv^2} \times 100 = 93.75\%$$

Hence, value of P is close to 94.

62. (b) Given,

Mass of block, $m_1 = 1.9$ kg

Mass of bullet, $m_2 = 0.1$ kg

Velocity of bullet, $v_2 = 20$ m/s

Let v be the velocity of the combined system. It is an inelastic collision.

Using conservation of linear momentum

$$m_1 \times 0 + m_2 \times v_2 = (m_1 + m_2)v$$

$$\Rightarrow 0.1 \times 20 = (0.1 + 1.9) \times v$$

$$\Rightarrow v = 1 \text{ m/s}$$

Using work energy theorem

Work done = Change in Kinetic energy

Let K be the Kinetic energy of combined system.

$$(m_1 + m_2)gh$$

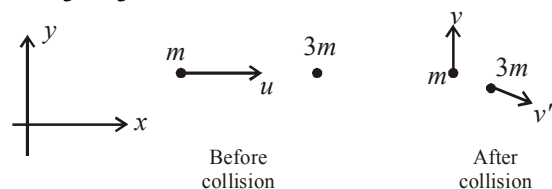
$$= K - \frac{1}{2}(m_1 + m_2)V^2$$

$$\Rightarrow 2 \times g \times 1 = K - \frac{1}{2} \times 2 \times 1^2 \Rightarrow K = 21 \text{ J}$$

63. (c) From conservation of linear momentum

$$mu\hat{i} + 0 = mv\hat{j} + 3m\vec{v}'$$

$$\vec{v}' = \frac{u}{3}\hat{i} - \frac{v}{3}\hat{j}$$



From kinetic energy conservation,

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + \frac{1}{2}(3m)\left[\left(\frac{u}{3}\right)^2 + \left(\frac{v}{3}\right)^2\right]$$

$$\text{or, } mu^2 = mv^2 + \frac{mu^2}{3} + \frac{mv^2}{3}$$

$$\therefore v = \frac{u}{\sqrt{2}}$$

64. (10.00)

From momentum conservation in perpendicular direction of initial motion.

$$mu_1 \sin \theta_1 = 10mv_1 \sin \theta_2 \quad \dots(i)$$

It is given that energy of m reduced by half. If u_1 be velocity of m after collision, then

$$\left(\frac{1}{2}mu^2\right)\frac{1}{2} = \frac{1}{2}mu_1^2$$

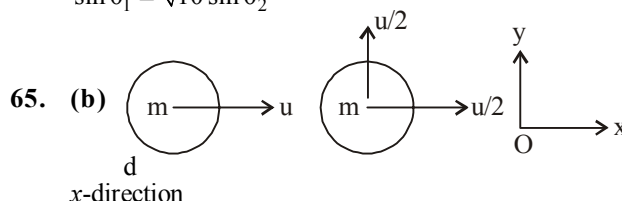
$$\Rightarrow u_1 = \frac{u}{\sqrt{2}}$$

If v_1 be the velocity of mass $10m$ after collision, then

$$\frac{1}{2} \times 10m \times v_1^2 = \frac{1}{2} \frac{u^2}{2} \Rightarrow v_1 = \frac{u}{\sqrt{20}}$$

From equation (i), we have

$$\sin \theta_1 = \sqrt{10} \sin \theta_2$$



x-direction

$$mu + \frac{mu}{2} = 2mv'_x \Rightarrow v'_x = \frac{3u}{4}$$

$$\text{y-direction } 0 + \frac{mu}{2} = 2mv'_y \Rightarrow v'_y = \frac{u}{4}$$

$$K.E._i = \frac{1}{2}mu^2 + \frac{1}{2}m\left[\left(\frac{u}{2}\right)^2 + \left(\frac{u}{2}\right)^2\right]$$

$$= \frac{1}{2}mu^2 + \frac{mu^2}{4} = \frac{3mu^2}{4}$$

$$K.E._f = \frac{1}{2}(2m)(v'_x)^2 + \frac{1}{2}(2m)(v'_y)^2$$

$$= \frac{1}{2}2m\left[\left(\frac{3u}{4}\right)^2 + \left(\frac{u}{4}\right)^2\right] = \frac{5}{8}mu^2$$

$$\therefore \text{Loss in } KE = KE_f - KE_i$$

$$= mu^2 \left(\frac{6}{8} - \frac{5}{8}\right) = \frac{mu^2}{8}$$

66. (a) For elastic collision $KE_i = KE_f$

$$\frac{1}{2}m \times 25 + \frac{1}{2}m \times 9 = \frac{1}{2}m \times 32 + \frac{1}{2}mv_B^2$$

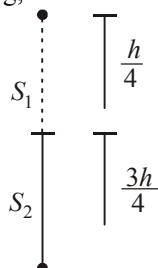
$$34 = 32 + v_B^2 \Rightarrow v_B = \sqrt{2}$$

$$KE_B = \frac{1}{2}mv_B^2 = \frac{1}{2} \times 0.1 \times 2 = 0.1 \text{ J} = \frac{1}{10} \text{ J}$$

$$\therefore x = 1$$

67. (d) Let t be the time taken by the particle dropped from height h to collide with particle thrown upward.

Using,



$$v^2 - u^2 = 2gh$$

$$\Rightarrow v^2 - 0^2 = 2gh$$

$$\Rightarrow v = \sqrt{2gh}$$

Downward distance travelled

$$s_1 = \frac{1}{2}gt^2 = \frac{1}{2}g \cdot \frac{h}{2g} = \frac{h}{4}$$

Distance of collision point from ground

$$s_2 = h - \frac{h}{4} = \frac{3h}{4}$$

Speed of (A) just before collision

$$v_1 = gt = \sqrt{\frac{gh}{2}}$$

And speed of (B) just before collision

$$v_2 = \sqrt{2gh} - \sqrt{\frac{gh}{2}}$$

Using principle of conservation of linear momentum

$$mv_1 + mv_2 = 2mv_f$$

$$\Rightarrow v_f = \frac{m\left(\sqrt{2gh} - \sqrt{\frac{gh}{2}}\right) - m\sqrt{\frac{gh}{2}}}{2m} = 0$$

After collision, time taken (t_1) for combined mass to reach the ground is

$$\Rightarrow \frac{3h}{4} = \frac{1}{2}gt_1^2$$

$$\Rightarrow t_1 = \sqrt{\frac{3h}{2g}}$$

68. (b) $P_i = P_f$
or $0 = 20(0.7 - v) = 50v$
or $v = 0.2 \text{ m/s}$

69. (a) Apply conservation of linear momentum in X and Y direction for the system then
 $M(10\cos 30^\circ) + 2M(5\cos 45^\circ) = 2M(v_1\cos 30^\circ) + M(v_2\cos 45^\circ)$

$$5\sqrt{3} + 5\sqrt{2} = \sqrt{3}v_1 + \frac{v_2}{\sqrt{2}} \quad \dots(1)$$

Also

$$2M(5\sin 45^\circ) - M(10\sin 30^\circ) = 2Mv_1\sin 30^\circ - Mv_2\sin 45^\circ$$

$$5\sqrt{2} - 5 = v_1 - \frac{v_2}{\sqrt{2}} \quad \dots(2)$$

Solving equation (1) and (2)

$$(\sqrt{3} + 1)v_1 = 5\sqrt{3} + 10\sqrt{2} - 5 \Rightarrow v_1 = 6.5 \text{ m/s}$$

$$v_2 = 6.3 \text{ m/s}$$

70. (b) $2u + 0 = 2\left(\frac{u}{4}\right) + mv_2$

$$\text{and } \frac{1}{2} \times 2 \times u^2 + 0 = \frac{1}{2} \times 2 \times \left(\frac{u}{4}\right)^2 + \frac{1}{2}mv_2^2$$

On solving, we get

$$m = 1.5 \text{ kg}$$

71. (b) $m(2v) + 2mv = 0 + 2mv' \cos 45^\circ$

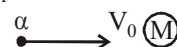
$$\text{or } v' = 2\sqrt{2}v$$

72. (b) $m_1v_1 + m_2v_2 = m_1v_2 + m_2v_1$

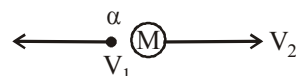
$$\text{or } m_1v_1 + (0.5m_1)v_2 = m_1(0.5v_1) + (0.5m_1)v_4$$

On solving, $v_1 = v_4 - v_2$

73. (d) Using conservation of momentum,
 $mv_0 = mv_2 - mv_1$



After collision



$$\frac{1}{2}mv_1^2 = 0.36 \times \frac{1}{2}mv_0^2$$

$$\Rightarrow v_1 = 0.6v_0$$

The collision is elastic. So,

$$\frac{1}{2}MV_2^2 = 0.64 \times \frac{1}{2}mv_0^2 \quad [\because M = \text{mass of nucleus}]$$

$$\Rightarrow V_2 = \sqrt{\frac{m}{M}} \times 0.8V_0$$

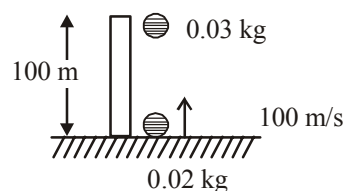
$$mV_0 = \sqrt{mM} \times 0.8V_0 - m \times 0.6V_0$$

$$\Rightarrow 1.6m = 0.8\sqrt{mM}$$

$$\Rightarrow 4m^2 = mM$$

$$\therefore M = 4m$$

74. (c)



Time taken for the particles to collide,

$$t = \frac{d}{V_{\text{rel}}} = \frac{100}{100} = 1 \text{ sec}$$

Speed of wood just before collision = $gt = 10 \text{ m/s}$ and
speed of bullet just before collision = $v - gt$

$$= 100 - 10 = 90 \text{ m/s}$$

$$S = 100 \times 1 - \frac{1}{2} \times 10 \times 1 = 95 \text{ m}$$

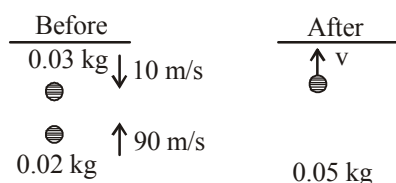
Now, using conservation of linear momentum just before and after the collision

$$-(0.03)(10) + (0.02)(90) = (0.05)v$$

$$\Rightarrow 150 = 5v \quad \therefore v = 30 \text{ m/s}$$

Max. height reached by body

$$h = \frac{30 \times 30}{2 \times 10} = 45 \text{ m}$$



$\therefore$ Height above tower = 40 m

75. (c) Kinetic energy of block A

$$k_1 = \frac{1}{2}mv_0^2$$

$\therefore$ From principle of linear momentum conservation

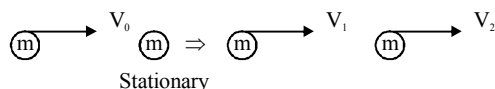
$$mv_0 = (2m + M)v_f \Rightarrow v_f = \frac{mv_0}{2m + M}$$

According to question, of $\frac{5}{6}$ th the initial kinetic energy is lost in whole process.

$$\therefore \frac{k_i}{k_f} = 6 \Rightarrow \frac{\frac{1}{2}mv_0^2}{\frac{1}{2}(2m + M)\left(\frac{mv_0}{2m + M}\right)^2} = 6$$

$$\Rightarrow \frac{2m + M}{m} = 6 \therefore \frac{M}{m} = 4$$

76. (b) Before Collision After Collision



$$\frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 = \frac{3}{2}\left(\frac{1}{2}mv_0^2\right)$$

$$\Rightarrow v_1^2 + v_2^2 = \frac{3}{2}v_0^2 \quad \dots(i)$$

From momentum conservation

$$mv_0 = m(v_1 + v_2) \quad \dots(ii)$$

Squaring both sides,

$$(v_1 + v_2)^2 = v_0^2$$

$$\Rightarrow v_1^2 + v_2^2 + 2v_1v_2 = v_0^2$$

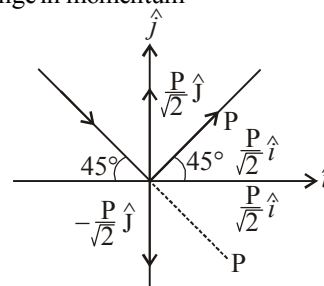
$$2v_1v_2 = -\frac{v_0^2}{2}$$

$$(v_1 - v_2)^2 = v_1^2 + v_2^2 - 2v_1v_2 = \frac{3}{2}v_0^2 + \frac{v_0^2}{2}$$

Solving we get relative velocity between the two particles

$$v_1 - v_2 = \sqrt{2}v_0$$

77. (a) Change in momentum



$$\Delta P = \frac{P}{\sqrt{2}}\hat{j} + \frac{P}{\sqrt{2}}\hat{j} + \frac{P}{\sqrt{2}}\hat{i} - \frac{P}{\sqrt{2}}\hat{i}$$

$$\Delta P = \frac{2P}{\sqrt{2}}\hat{j} = I_H \text{ molecule}$$

$$\Rightarrow I_{\text{wall}} = -\frac{2P}{\sqrt{2}}\hat{j}$$

Pressure, P

$$= \frac{F}{A} = \frac{\sqrt{2}P}{A} n \quad (\because n = \text{no. of particles})$$

$$= \frac{\sqrt{2} \times 3.32 \times 10^{-27} \times 10^3 \times 10^{23}}{2 \times 10^{-4}} = 2.35 \times 10^3 \text{ N/m}^2$$

78. (a) For collision of neutron with deuterium:



Applying conservation of momentum :

$$mv + 0 = mv_1 + 2mv_2 \quad \dots(i)$$

$$v_2 - v_1 = v \quad \dots(ii)$$

$\therefore$ Collision is elastic, $e = 1$

$$\text{From eqn (i) and eqn (ii) } v_1 = -\frac{v}{3}$$

$$P_d = \frac{\frac{1}{2}mv^2 - \frac{1}{2}mv_1^2}{\frac{1}{2}mv^2} = \frac{8}{9} = 0.89$$

Now, For collision of neutron with carbon nucleus



Applying Conservation of momentum

$$mv + 0 = mv_1 + 12mv_2 \quad \dots(iii)$$

$$v = v_2 - v_1 \quad \dots(iv)$$

From eqn (iii) and eqn (iv)

$$v_1 = -\frac{11}{13}v$$

$$P_c = \frac{\frac{1}{2}mv^2 - \frac{1}{2}m\left(\frac{11}{13}v\right)^2}{\frac{1}{2}mv^2} = \frac{48}{169} \approx 0.28$$

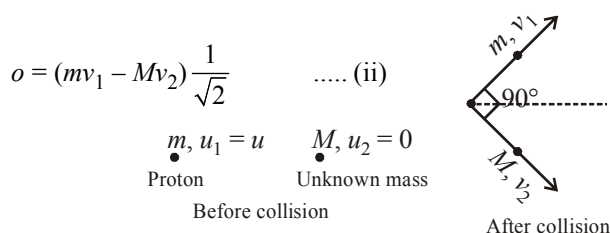
79. (d) Apply principle of conservation of momentum along x-direction,

$$mu = mv_1 \cos 45^\circ + Mv_2 \cos 45^\circ$$

$$mu = \frac{1}{\sqrt{2}}(mv_1 + Mv_2) \quad \dots(i)$$

Along y-direction,

$$0 = mv_1 \sin 45^\circ - Mv_2 \sin 45^\circ$$



$$0 = (mv_1 - Mv_2) \frac{1}{\sqrt{2}} \quad \dots (ii)$$

$$\text{Coefficient of restitution } e = 1 = \frac{v_2 - v_1 \cos 90^\circ}{u \cos 45^\circ}$$

(∴ Collision is elastic)

$$\Rightarrow \frac{v_2}{u} = 1$$

$$\Rightarrow u = \sqrt{2}v_2 \quad \dots (iii)$$

Solving eqs (i), (ii), & (iii), we get mass of unknown particle, $M = m$

80. (a) For particle C, According to law of conservation of linear momentum, vertical component,

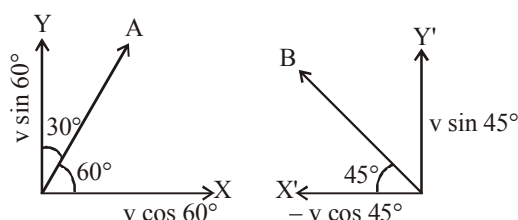
$$2mv' \sin \theta = mv \sin 60^\circ + mv \sin 45^\circ$$

$$2mv' \sin \theta = \frac{mv}{\sqrt{2}} + \frac{mv\sqrt{3}}{2} \quad \dots (i)$$

Horizontal component,

$$2mv' \cos \theta = mv \sin 60^\circ - mv \cos 45^\circ$$

$$2mv' \cos \theta = \frac{mv}{2} + \frac{mv}{\sqrt{2}} \quad \dots (ii)$$



For particle A

For particle B

Dividing eqⁿ (i) by eqⁿ (ii),

$$\tan \theta = \frac{\frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2}}{\frac{1}{2} - \frac{1}{\sqrt{2}}} = \frac{\sqrt{2} + \sqrt{3}}{1 - \sqrt{2}}$$

81. (a) For inelastic collision $v' = \frac{m_1}{(m_1 + m_2)} v$

$$= \frac{1}{(1+1)} v = \frac{v}{2}$$

$n \rightarrow v(H)$ Before

$(n)(H) \rightarrow \frac{v}{2}$ After

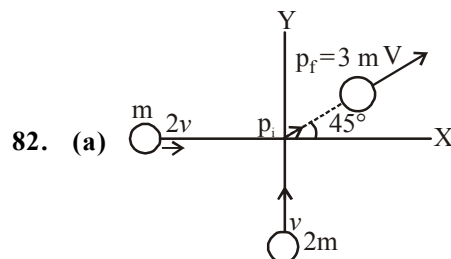
$$\text{Loss in K.E.} = \frac{1}{2}mv^2 - \frac{1}{2}(2m)\left(\frac{v}{2}\right)^2 = \frac{1}{4}mv^2$$

K.E. lost is used to jump from 1st orbit to 2nd orbit

$$\Delta \text{K.E.} = 10.2 \text{ eV}$$

Minimum K.E. of neutron for inelastic collision

$$\frac{1}{2}mv^2 = 2 \times 10.2 = 20.4 \text{ eV}$$



Initial momentum of the system

$$p_i = \sqrt{[m(2V)^2 + 2m(2V)^2]}$$

$$= \sqrt{2}m \times 2V$$

Final momentum of the system = $3mV$

By the law of conservation of momentum

$$2\sqrt{2}mv = 3mV \Rightarrow \frac{2\sqrt{2}v}{3} = V_{\text{combined}}$$

Loss in energy

$$\Delta E = \frac{1}{2}m_1V_1^2 + \frac{1}{2}m_2V_2^2 - \frac{1}{2}(m_1 + m_2)V_{\text{combined}}^2$$

$$\Delta E = 3mv^2 - \frac{4}{3}mv^2 = \frac{5}{3}mv^2 = 55.55\%$$

Percentage loss in energy during the collision $\approx 56\%$

83. (b) Given, $m_1 = 4\text{g}$, $u_1 = 300\text{ m/s}$

$$m_2 = 0.8\text{ kg} = 800\text{ g}$$

$$u_2 = 0\text{ m/s}$$

From law of conservation of momentum,

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

Let the velocity of combined system = $v\text{ m/s}$

then,

$$4 \times 300 + 800 \times 0 = (800 + 4) \times v$$

$$v = \frac{1200}{804} = 1.49\text{ m/s}$$

Now, $\mu = 0.3$ (given)

$$a = \mu g$$

$$a = 0.3 \times 10$$

$$= 3\text{ m/s}^2$$

then, from $v^2 = u^2 + 2as$

$$(1.49)^2 = 0 + 2 \times 3 \times s$$

$$s = \frac{(1.49)^2}{6}$$

$$s = \frac{2.22}{6}$$

$$= 0.379\text{ m}$$

84. (d) From the law of conservation of momentum we know that,

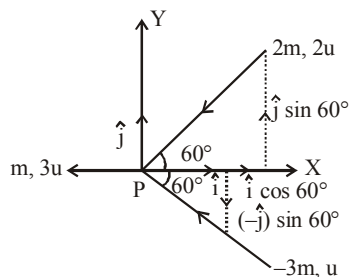
$$m_1u_1 + m_2u_2 + \dots = m_1v_1 + m_2v_2 + \dots$$

Given $m_1 = m$, $m_2 = 2m$ and $m_3 = 3m$

and $u_1 = 3u$, $u_2 = 2u$ and $u_3 = u$

Let the velocity when they stick = v

Then, according to question,



$$\begin{aligned}
 m \times 3u (\hat{i}) + 2m \times 2u (-\hat{i} \cos 60^\circ - \hat{j} \sin 60^\circ) \\
 + 3m \times u (-\hat{i} \cos 60^\circ + \hat{j} \sin 60^\circ) &= (m + 2m + 3m) \vec{v} \\
 \Rightarrow 3mu\hat{i} - 4mu\frac{\hat{i}}{2} - 4mu\left(\frac{\sqrt{3}}{2}\hat{j}\right) - 3mu\frac{\hat{i}}{2} \\
 + 3mu\left(\frac{\sqrt{3}}{2}\hat{j}\right) &= 6m \vec{v} \\
 \Rightarrow mu\hat{i} - \frac{3}{2}mu\hat{i} - \frac{\sqrt{3}}{2}mu\hat{j} &= 6m \vec{v} \\
 \Rightarrow -\frac{1}{2}mu\hat{i} - \frac{\sqrt{3}}{2}mu\hat{j} &= 6m \vec{v} \\
 \Rightarrow \vec{v} &= \frac{u}{12}(-\hat{i} - \sqrt{3}\hat{j})
 \end{aligned}$$

85. (d) Maximum energy loss = $\frac{p^2}{2m} - \frac{p^2}{2(m+M)}$

$$\left[\because \text{K.E.} = \frac{p^2}{2m} = \frac{1}{2}mv^2 \right]$$

$$= \frac{p^2}{2m} \left[\frac{M}{(m+M)} \right] = \frac{1}{2}mv^2 \left\{ \frac{M}{m+M} \right\}$$

Statement II is a case of perfectly inelastic collision.

By comparing the equation given in statement I with above equation, we get

$$f = \left(\frac{M}{m+M} \right) \text{ instead of } \left(\frac{m}{M+m} \right)$$

Hence statement I is wrong and statement II is correct.

86. (c)

87. (c) Fractional decrease in kinetic energy of mass m

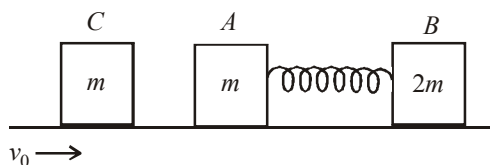
$$= 1 - \left(\frac{m_2 - m_1}{m_2 + m_1} \right)^2 = 1 - \left(\frac{2-1}{2+1} \right)^2$$

$$= 1 - \left(\frac{1}{3} \right)^2 = 1 - \frac{1}{9} = \frac{8}{9}$$

Percentage loss in energy

$$= \frac{8}{9} \times 100 \approx 90\%$$

88. (d)



Initial momentum of the system block (C) = mv_0 . After striking with A, the block C comes to rest and now both block A and B moves with velocity v when compression in spring is x_0 .

By the law of conservation of linear momentum

$$mv_0 = (m + 2m) v \Rightarrow v = \frac{v_0}{3}$$

By the law of conservation of energy

K.E. of block C = K.E. of system + P.E. of system

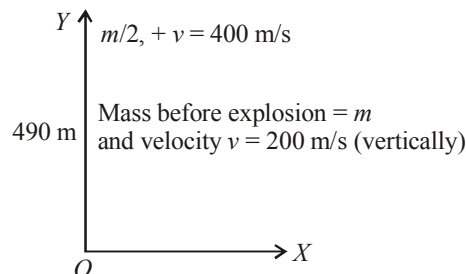
$$\frac{1}{2}mv_0^2 = \frac{1}{2}(3m)\left(\frac{v_0}{3}\right)^2 + \frac{1}{2}kx_0^2$$

$$\Rightarrow \frac{1}{2}mv_0^2 = \frac{1}{6}mv_0^2 + \frac{1}{2}kx_0^2$$

$$\Rightarrow \frac{1}{2}kx_0^2 = \frac{1}{2}mv_0^2 - \frac{1}{6}mv_0^2 = \frac{mv_0^2}{3}$$

$$\therefore k = \frac{2}{3}m\left(\frac{v_0}{x_0}\right)^2$$

89. (c)



Momentum before explosion = Momentum after explosion

$$\begin{aligned}
 m \times 200\hat{j} &= \frac{m}{2} \times 400\hat{j} + \frac{m}{2}v \\
 &= \frac{m}{2}(400\hat{j} + v)
 \end{aligned}$$

$$\Rightarrow 400\hat{j} - 400\hat{j} = v$$

$$\therefore v = 0$$

i.e., the velocity of the other part of the mass, $v = 0$

Let time taken to reach the earth by this part be t

$$\text{Applying formula, } h = ut + \frac{1}{2}gt^2$$

$$490 = 0 + \frac{1}{2} \times 9.8 \times t^2$$

$$\Rightarrow t^2 = \frac{980}{9.8} = 100$$

$$\therefore t = \sqrt{100} = 10 \text{ sec}$$

90. (a) In completely inelastic collision, all initial kinetic energy is not lost but loss in kinetic energy is as large as it can be. Linear momentum remain conserved in all types of collision. Statement -2 explains statement -1 correctly because applying the principle of conservation of momentum, we can get the common velocity and hence the kinetic energy of the combined body.

91. (c) Initial kinetic energy of the system

$$\text{K.E.}_i = \frac{1}{2}mu^2 + \frac{1}{2}M(0)^2$$

$$= \frac{1}{2} \times 0.5 \times 2 \times 2 + 0 = 1 \text{ J}$$

Momentum before collision

= Momentum after collision

$$m_1 u_1 + m_2 u_2 = (m + M) \times v$$

$$\therefore 0.5 \times 2 + 1 \times 0 = (0.5 + 1) \times v \Rightarrow v = \frac{2}{3} \text{ m/s}$$

Final kinetic energy of the system is

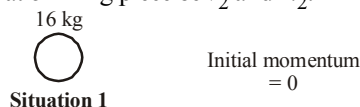
$$\text{K.E}_f = \frac{1}{2} (m + M) v^2$$

$$= \frac{1}{2} (0.5 + 1) \times \frac{2}{3} \times \frac{2}{3} = \frac{1}{3} \text{ J}$$

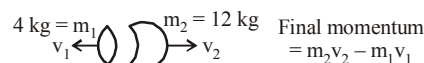
$\therefore$ Energy loss during collision

$$= \left(1 - \frac{1}{3}\right) \text{ J} = 0.67 \text{ J}$$

92. (b) Let the velocity and mass of 4 kg piece be v_1 and m_1 and that of 12 kg piece be v_2 and m_2 .



Situation 1



Situation 2

Applying conservation of linear momentum

$$16 \times 0 = 4 \times v_1 + 12 \times 4$$

$$\Rightarrow v_1 = -\frac{12 \times 4}{4} = -12 \text{ m s}^{-1}$$

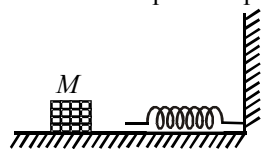
Kinetic energy of 4 kg mass

$$\therefore \text{K.E.} = \frac{1}{2} m_1 v_1^2 = \frac{1}{2} \times 4 \times 144 = 288 \text{ J}$$

93. (b) When the spring gets compressed by length L .
K.E. lost by mass M = P.E. stored in the compressed spring.

$$\frac{1}{2} M v^2 = \frac{1}{2} k L^2$$

$$\Rightarrow v = \sqrt{\frac{k}{M}} \cdot L$$



Momentum of the block, $= M \times v$

$$= M \times \sqrt{\frac{k}{M}} \cdot L = \sqrt{kM} \cdot L$$

94. (d) Considering conservation of momentum along x -direction,

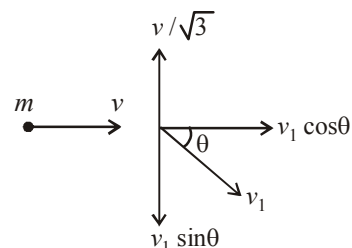
$$mv = mv_1 \cos \theta \quad \dots(1)$$

where v_1 is the velocity of second mass

In y -direction,

$$0 = \frac{mv}{\sqrt{3}} - mv_1 \sin \theta$$

$$\text{or } m_1 v_1 \sin \theta = \frac{mv}{\sqrt{3}} \quad \dots(2)$$



Squaring and adding eqns. (1) and (2) we get

$$v_1^2 = v^2 + \frac{v^2}{3} \Rightarrow v_1 = \frac{2}{\sqrt{3}} v$$

95. (c) Kinetic energy of a system of particle is zero only when the speed of each particles is zero. This implies momentum of each particle is zero, thus linear momentum of the system of particle has to be zero.

Also if linear momentum of the system is zero it does not mean linear momentum of each particle is zero. This is because linear momentum is a vector quantity. In this case the kinetic energy of the system of particles will not be zero.

$\therefore$ A does not imply B but B implies A.

Given, force, $F = 200 \text{ N}$ extension of wire, $x = 1 \text{ mm}$.

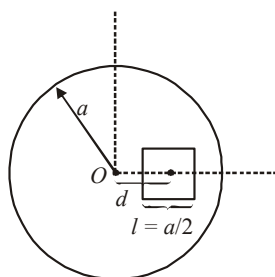
System of Particles and Rotational Motion



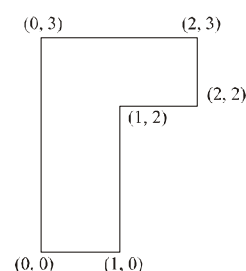
TOPIC 1 Centre of Mass, Centre of Gravity & Principle of Moments



- The centre of mass of a solid hemisphere of radius 8 cm is x cm from the centre of the flat surface. Then value of x is _____. [NA Sep. 06, 2020 (II)]
- A square shaped hole of side $l = \frac{a}{2}$ is carved out at a distance $d = \frac{a}{2}$ from the centre ' O ' of a uniform circular disk of radius a . If the distance of the centre of mass of the remaining portion from O is $-\frac{a}{X}$, value of X (to the nearest integer) is _____. [NA Sep. 02, 2020 (II)]



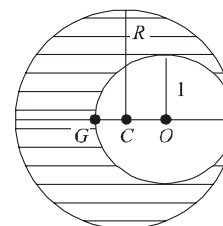
- A rod of length L has non-uniform linear mass density given by $\rho(x) = a + b\left(\frac{x}{L}\right)^2$, where a and b are constants and $0 \leq x \leq L$. The value of x for the centre of mass of the rod is at: [9 Jan. 2020 II]
 - $\frac{3}{2}\left(\frac{a+b}{2a+b}\right)L$
 - $\frac{3}{4}\left(\frac{2a+b}{3a+b}\right)L$
 - $\frac{4}{3}\left(\frac{a+b}{2a+3b}\right)L$
 - $\frac{3}{2}\left(\frac{2a+b}{3a+b}\right)L$
- The coordinates of centre of mass of a uniform flag shaped lamina (thin flat plate) of mass 4 kg. (The coordinates of the same are shown in figure) are: [8 Jan. 2020 I]



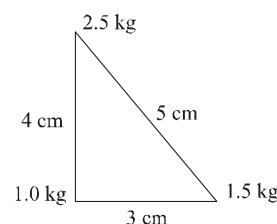
- (1.25 m, 1.50 m)
- (0.75 m, 1.75 m)
- (0.75 m, 0.75 m)
- (1 m, 1.75 m)

- As shown in fig. when a spherical cavity (centred at O) of radius 1 is cut out of a uniform sphere of radius R (centred at C), the centre of mass of remaining (shaded) part of sphere is at G , i.e on the surface of the cavity. R can be determined by the equation: [8 Jan. 2020 II]

- $(R^2 + R + 1)(2 - R) = 1$
- $(R^2 - R - 1)(2 - R) = 1$
- $(R^2 - R + 1)(2 - R) = 1$
- $(R^2 + R - 1)(2 - R) = 1$

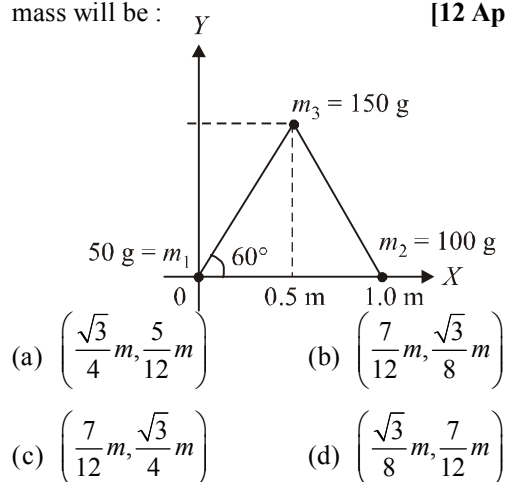


- Three point particles of masses 1.0 kg, 1.5 kg and 2.5 kg are placed at three corners of a right angle triangle of sides 4.0 cm, 3.0 cm and 5.0 cm as shown in the figure. The center of mass of the system is at a point: [7 Jan. 2020 I]

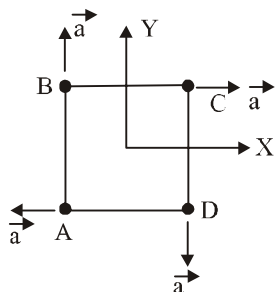


- 0.6 cm right and 2.0 cm above 1 kg mass
- 1.5 cm right and 1.2 cm above 1 kg mass
- 2.0 cm right and 0.9 cm above 1 kg mass
- 0.9 cm right and 2.0 cm above 1 kg mass

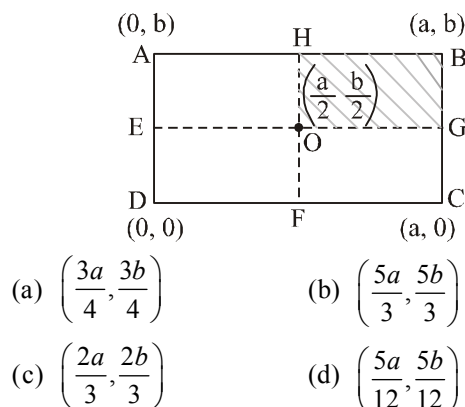
7. Three particles of masses 50 g, 100 g and 150 g are placed at the vertices of an equilateral triangle of side 1 m (as shown in the figure). The (x, y) coordinates of the centre of mass will be : [12 Apr. 2019 II]



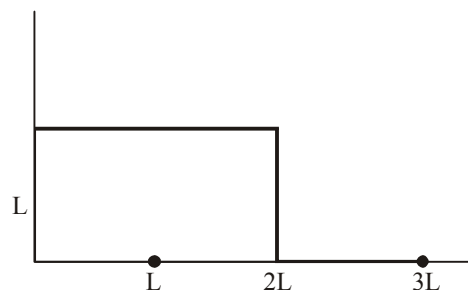
8. Four particles A, B, C and D with masses $m_A = m$, $m_B = 2m$, $m_C = 3m$ and $m_D = 4m$ are at the corners of a square. They have accelerations of equal magnitude with directions as shown. The acceleration of the centre of mass of the particles is : [8 April 2019 I]



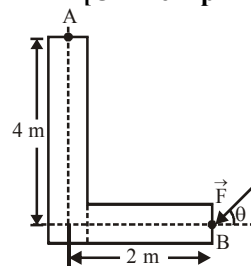
- (a) $\frac{a}{5}(\hat{i} - \hat{j})$ (b) a
- (c) Zero (d) $\frac{a}{5}(\hat{i} + \hat{j})$
9. A uniform rectangular thin sheet ABCD of mass M has length a and breadth b , as shown in the figure. If the shaded portion HBGO is cut-off, the coordinates of the centre of mass of the remaining portion will be : [8 Apr. 2019 II]



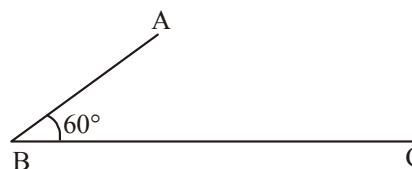
10. The position vector of the centre of mass r_{cm} of an asymmetric uniform bar of negligible area of cross-section as shown in figure is: [12 Jan. 2019 I]



- (a) $\vec{r}_{cm} = \frac{13}{8}L\hat{x} + \frac{5}{8}L\hat{y}$ (b) $\vec{r}_{cm} = \frac{5}{8}L\hat{x} + \frac{13}{8}L\hat{y}$
- (c) $\vec{r}_{cm} = \frac{3}{8}L\hat{x} + \frac{11}{8}L\hat{y}$ (d) $\vec{r}_{cm} = \frac{11}{8}L\hat{x} + \frac{3}{8}L\hat{y}$
11. A force of 40 N acts on a point B at the end of an L-shaped object, as shown in the figure. The angle θ that will produce maximum moment of the force about point A is given by: [Online April 15, 2018]



- (a) $\tan \theta = \frac{1}{4}$
- (b) $\tan \theta = 2$
- (c) $\tan \theta = \frac{1}{2}$
- (d) $\tan \theta = 4$
12. In a physical balance working on the principle of moments, when 5 mg weight is placed on the left pan, the beam becomes horizontal. Both the empty pans of the balance are of equal mass. Which of the following statements is correct? [Online April 8, 2017]
- (a) Left arm is longer than the right arm
- (b) Both the arms are of same length
- (c) Left arm is shorter than the right arm
- (d) Every object that is weighed using this balance appears lighter than its actual weight.
13. In the figure shown ABC is a uniform wire. If centre of mass of wire lies vertically below point A, then $\frac{BC}{AB}$ is close to : [Online April 10, 2016]



- (a) 1.85 (b) 1.5
- (c) 1.37 (d) 3

14. Distance of the centre of mass of a solid uniform cone from its vertex is z_0 . If the radius of its base is R and its height is h then z_0 is equal to : [2015]

(a) $\frac{5h}{8}$ (b) $\frac{3h^2}{8R}$ (c) $\frac{h^2}{4R}$ (d) $\frac{3h}{4}$

15. A uniform thin rod AB of length L has linear mass density $\mu(x) = a + \frac{bx}{L}$, where x is measured from A. If

the CM of the rod lies at a distance of $\left(\frac{7}{12}\right)L$ from A, then a and b are related as : [Online April 11, 2015]

(a) $a = 2b$ (b) $2a = b$
(c) $a = b$ (d) $3a = 2b$

16. A thin bar of length L has a mass per unit length λ , that increases linearly with distance from one end. If its total mass is M and its mass per unit length at the lighter end is λ_0 , then the distance of the centre of mass from the lighter end is: [Online April 11, 2014]

(a) $\frac{L}{2} - \frac{\lambda_0 L^2}{4M}$ (b) $\frac{L}{3} + \frac{\lambda_0 L^2}{8M}$
(c) $\frac{L}{3} + \frac{\lambda_0 L^2}{4M}$ (d) $\frac{2L}{3} - \frac{\lambda_0 L^2}{6M}$

17. A boy of mass 20 kg is standing on a 80 kg free to move long cart. There is negligible friction between cart and ground. Initially, the boy is standing 25 m from a wall. If he walks 10 m on the cart towards the wall, then the final distance of the boy from the wall will be

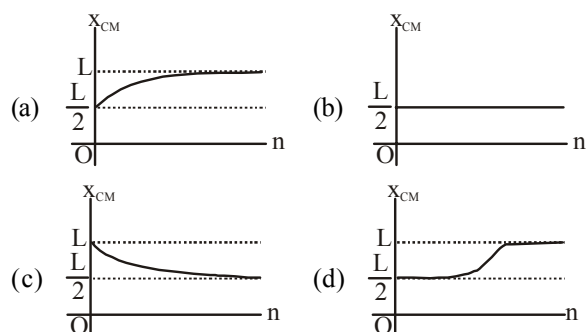
[Online April 23, 2013]

(a) 15m (b) 12.5m (c) 15.5m (d) 17m

18. A thin rod of length ' L ' is lying along the x -axis with its ends at $x = 0$ and $x = L$. Its linear density (mass/length)

varies with x as $k\left(\frac{x}{L}\right)^n$, where n can be zero or any

positive number. If the position x_{CM} of the centre of mass of the rod is plotted against ' n ', which of the following graphs best approximates the dependence of x_{CM} on n ? [2008]



19. A circular disc of radius R is removed from a bigger circular disc of radius $2R$ such that the circumferences of the discs coincide. The centre of mass of the new disc is α/R from the centre of the bigger disc. The value of α is [2007]

(a) $1/4$ (b) $1/3$ (c) $1/2$ (d) $1/6$

20. Consider a two particle system with particles having masses m_1 and m_2 . If the first particle is pushed towards the centre of mass through a distance d , by what distance should the second particle be moved, so as to keep the centre of mass at the same position? [2006]

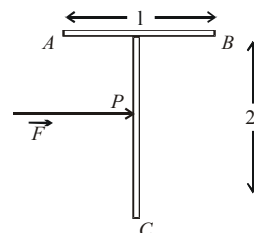
(a) $\frac{m_2}{m_1}d$ (b) $\frac{m_1}{m_1 + m_2}d$
(c) $\frac{m_1}{m_2}d$ (d) d

21. A body A of mass M while falling vertically downwards under gravity breaks into two parts; a body B of mass $\frac{1}{3}M$ and a body C of mass $\frac{2}{3}M$. The centre of mass of

bodies B and C taken together shifts compared to that of body A towards [2005]

- (a) does not shift
(b) depends on height of breaking
(c) body B
(d) body C

22. A 'T' shaped object with dimensions shown in the figure, is lying on a smooth floor. A force ' $\vec{F}$ ' is applied at the point P parallel to AB , such that the object has only the translational motion without rotation. Find the location of P with respect to C . [2005]

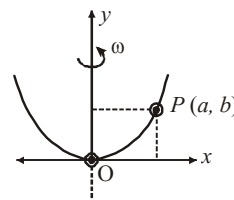


(a) $\frac{3}{2}\ell$ (b) $\frac{2}{3}\ell$ (c) ℓ (d) $\frac{4}{3}\ell$

TOPIC 2 Angular Displacement, Velocity and Acceleration



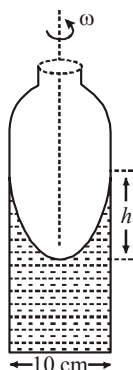
23. A bead of mass m stays at point $P(a, b)$ on a wire bent in the shape of a parabola $y = 4Cx^2$ and rotating with angular speed ω (see figure). The value of ω is (neglect friction): [Sep. 02, 2020 (I)]



(a) $2\sqrt{2gC}$ (b) $2\sqrt{gC}$
(c) $\sqrt{\frac{2gC}{ab}}$ (d) $\sqrt{\frac{2g}{C}}$

24. A cylindrical vessel containing a liquid is rotated about its axis so that the liquid rises at its sides as shown in the figure. The radius of vessel is 5 cm and the angular speed of rotation is ω rad s^{-1} . The difference in the height, h (in cm) of liquid at the centre of vessel and at the side will be :

[Sep. 02, 2020 (I)]



- (a) $\frac{2\omega^2}{25g}$ (b) $\frac{5\omega^2}{2g}$
 (c) $\frac{25\omega^2}{2g}$ (d) $\frac{2\omega^2}{5g}$

25. A spring mass system (mass m , spring constant k and natural length l) rests in equilibrium on a horizontal disc. The free end of the spring is fixed at the centre of the disc. If the disc together with spring mass system, rotates about its axis with an angular velocity ω , ($k \gg m\omega^2$) the relative change in the length of the spring is best given by the option:

[9 Jan. 2020 II]

- (a) $\sqrt{\frac{2}{3}} \left(\frac{m\omega^2}{k} \right)$ (b) $\frac{2m\omega^2}{k}$
 (c) $\frac{m\omega^2}{k}$ (d) $\frac{m\omega^2}{3k}$

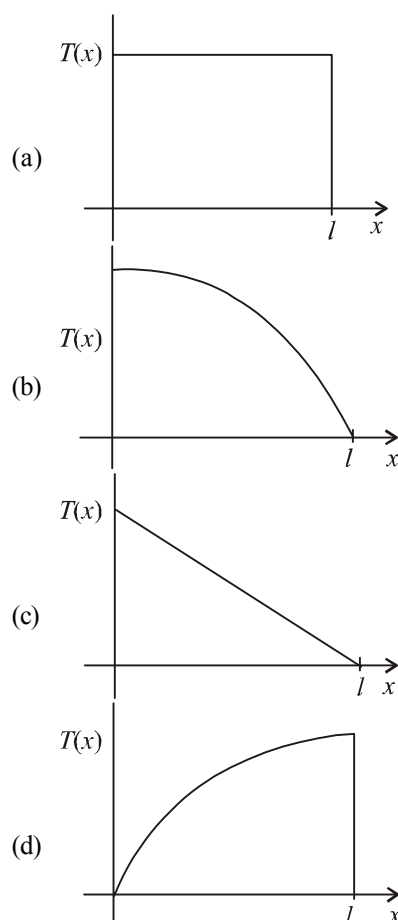
26. A particle of mass m is fixed to one end of a light spring having force constant k and unstretched length l . The other end is fixed. The system is given an angular speed ω about the fixed end of the spring such that it rotates in a circle in gravity free space. Then the stretch in the spring is:

[8 Jan. 2020 I]

- (a) $\frac{ml\omega^2}{k - m\omega^2}$ (b) $\frac{ml\omega^2}{k - m\omega^2}$
 (c) $\frac{ml\omega^2}{k + m\omega^2}$ (d) $\frac{ml\omega^2}{k + m\omega^2}$

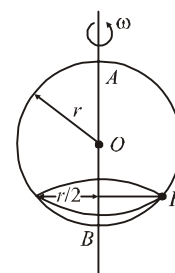
27. A uniform rod of length l is being rotated in a horizontal plane with a constant angular speed about an axis passing through one of its ends. If the tension generated in the rod due to rotation is $T(x)$ at a distance x from the axis, then which of the following graphs depicts it most closely?

[12 Apr. 2019 II]



28. A smooth wire of length $2\pi r$ is bent into a circle and kept in a vertical plane. A bead can slide smoothly on the wire. When the circle is rotating with angular speed ω about the vertical diameter AB, as shown in figure, the bead is at rest with respect to the circular ring at position P as shown. Then the value of ω^2 is equal to :

[12 Apr. 2019 II]



- (a) $\frac{\sqrt{3}g}{2r}$ (b) $2g/(r\sqrt{3})$
 (c) $(g\sqrt{3})/r$ (d) $2g/r$
29. A long cylindrical vessel is half filled with a liquid. When the vessel is rotated about its own vertical axis, the liquid rises up near the wall. If the radius of vessel is 5 cm and its rotational speed is 2 rotations per second, then the difference in the heights between the centre and the sides, in cm, will be :

[12 Jan. 2019 II]

- (a) 2.0 (b) 0.1
 (c) 0.4 (d) 1.2

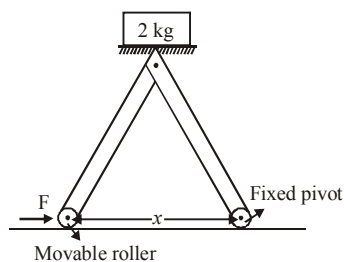
30. A particle is moving with a uniform speed in a circular orbit of radius R in a central force inversely proportional to the n^{th} power of R . If the period of rotation of the particle is T , then: [2018]

(a) $T \propto R^{3/2}$ for any n . (b) $T \propto R^{n/2+1}$

(c) $T \propto R^{(n+1)/2}$ (d) $T \propto R^{n/2}$

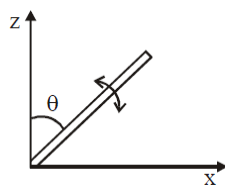
31. The machine as shown has 2 rods of length 1 m connected by a pivot at the top. The end of one rod is connected to the floor by a stationary pivot and the end of the other rod has a roller that rolls along the floor in a slot.

As the roller goes back and forth, a 2 kg weight moves up and down. If the roller is moving towards right at a constant speed, the weight moves up with a : [Online April 9, 2017]



- (a) constant speed
(b) decreasing speed
(c) increasing speed
(d) speed which is $\frac{3}{4}$ th of that of the roller when the weight is 0.4 m above the ground

32. A slender uniform rod of mass M and length ℓ is pivoted at one end so that it can rotate in a vertical plane (see figure). There is negligible friction at the pivot. The free end is held vertically above the pivot and then released. The angular acceleration of the rod when it makes an angle θ with the vertical is [2017]

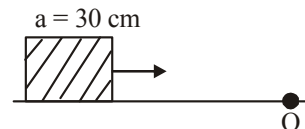


- (a) $\frac{3g}{2\ell} \cos \theta$ (b) $\frac{2g}{3\ell} \cos \theta$
(c) $\frac{3g}{2\ell} \sin \theta$ (d) $\frac{2g}{2\ell} \sin \theta$

33. Concrete mixture is made by mixing cement, stone and sand in a rotating cylindrical drum. If the drum rotates too fast, the ingredients remain stuck to the wall of the drum and proper mixing of ingredients does not take place. The maximum rotational speed of the drum in revolutions per minute (rpm) to ensure proper mixing is close to : (Take the radius of the drum to be 1.25 m and its axle to be horizontal): [Online April 10, 2016]

- (a) 27.0 (b) 0.4 (c) 1.3 (d) 8.0

34. A cubical block of side 30 cm is moving with velocity 2 ms^{-1} on a smooth horizontal surface. The surface has a bump at a point O as shown in figure. The angular velocity (in rad/s) of the block immediately after it hits the bump, is : [Online April 9, 2016]



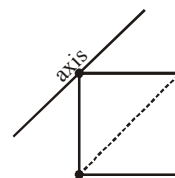
- (a) 13.3 (b) 5.0 (c) 9.4 (d) 6.7
35. Two point masses of mass $m_1 = fM$ and $m_2 = (1-f)M$ ($f < 1$) are in outer space (far from gravitational influence of other objects) at a distance R from each other. They move in circular orbits about their centre of mass with angular velocities ω_1 for m_1 and ω_2 for m_2 . In that case [Online May 19, 2012]

- (a) $(1-f)\omega_1 = f\omega$
(b) $\omega_1 = \omega_2$ and independent of f
(c) $f\omega_1 = (1-f)\omega_2$
(d) $\omega_1 = \omega_2$ and depend on f

TOPIC 3 Torque, Couple and Angular Momentum



36. Four point masses, each of mass m , are fixed at the corners of a square of side l . The square is rotating with angular frequency ω , about an axis passing through one of the corners of the square and parallel to its diagonal, as shown in the figure. The angular momentum of the square about this axis is : [Sep. 06, 2020 (I)]

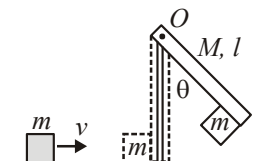


- (a) $ml^2\omega$ (b) $4ml^2\omega$
(c) $3ml^2\omega$ (d) $2ml^2\omega$
37. A thin rod of mass 0.9 kg and length 1 m is suspended, at rest, from one end so that it can freely oscillate in the vertical plane. A particle of mass 0.1 kg moving in a straight line with velocity 80 m/s hits the rod at its bottom most point and sticks to it (see figure). The angular speed (in rad/s) of the rod immediately after the collision will be [NA Sep. 05, 2020 (II)]
38. A person of 80 kg mass is standing on the rim of a circular platform of mass 200 kg rotating about its axis at 5 revolutions per minute (rpm). The person now starts moving towards the centre of the platform. What will be the rotational speed (in rpm) of the platform when the person reaches its centre.

[NA Sep. 03, 2020 (I)]

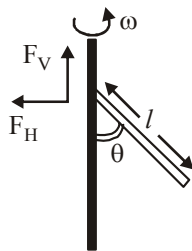
39. A block of mass $m = 1$ kg slides with velocity $v = 6$ m/s on a frictionless horizontal surface and collides with a uniform vertical rod and sticks to it as shown. The rod is pivoted about O and swings as a result of the collision making angle θ before momentarily coming to rest. If the rod has mass $M = 2$ kg, and length $l = 1$ m, the value of θ is approximately: (take $g = 10$ m/s²)

[Sep. 03, 2020 (I)]



- (a) 63° (b) 55°
(c) 69° (d) 49°

40.



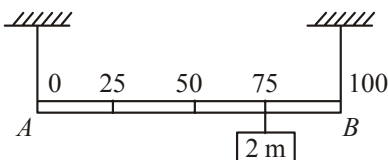
A uniform rod of length ' l ' is pivoted at one of its ends on a vertical shaft of negligible radius. When the shaft rotates at angular speed ω the rod makes an angle θ with it (see figure). To find θ equate the rate of change of angular momentum (direction going into the paper)

$\frac{ml^2}{12} \omega^2 \sin \theta \cos \theta$ about the centre of mass (CM) to the torque provided by the horizontal and vertical forces F_H and F_V about the CM. The value of θ is then such that :

[Sep. 03, 2020 (II)]

- (a) $\cos \theta = \frac{2g}{3l\omega^2}$ (b) $\cos \theta = \frac{g}{2l\omega^2}$
(c) $\cos \theta = \frac{g}{l\omega^2}$ (d) $\cos \theta = \frac{3g}{2l\omega^2}$

41.

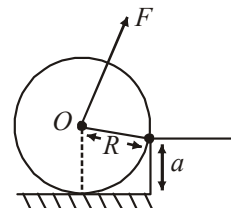


Shown in the figure is rigid and uniform one meter long rod AB held in horizontal position by two strings tied to its ends and attached to the ceiling. The rod is of mass ' m ' and has another weight of mass 2 m hung at a distance of 75 cm from A . The tension in the string at A is :

[Sep. 02, 2020 (I)]

- (a) 0.5 mg (b) 2 mg
(c) 0.75 mg (d) 1 mg

42. A uniform cylinder of mass M and radius R is to be pulled over a step of height a ($a < R$) by applying a force F at its centre ' O ' perpendicular to the plane through the axes of the cylinder on the edge of the step (see figure). The minimum value of F required is : [Sep. 02, 2020 (I)]



- (a) $Mg \sqrt{1 - \left(\frac{R-a}{R}\right)^2}$ (b) $Mg \sqrt{\left(\frac{R}{R-a}\right)^2 - 1}$
(c) $Mg \frac{a}{R}$ (d) $Mg \sqrt{1 - \frac{a^2}{R^2}}$

43. Consider a uniform rod of mass $M = 4m$ and length l pivoted about its centre. A mass m moving with velocity v making angle $\theta = \frac{\pi}{4}$ to the rod's long axis collides with one end of the rod and sticks to it. The angular speed of the rod-mass system just after the collision is:

[8 Jan. 2020 I]

- (a) $\frac{3}{7\sqrt{2}} \frac{v}{l}$ (b) $\frac{3}{7} \frac{v}{l}$
(c) $\frac{3\sqrt{2}}{7} \frac{v}{l}$ (d) $\frac{4}{7} \frac{v}{l}$

44. A particle of mass m is moving along a trajectory given by $x = x_0 + a \cos \omega_1 t$
 $y = y_0 + b \cos \omega_2 t$
The torque, acting on the particle about the origin, at $t = 0$ is:

[10 Apr. 2019 I]

- (a) $m(-x_0 b + y_0 a) \omega_1^2 \hat{k}$ (b) $+m y_0 a \omega_1^2 \hat{k}$
(c) zero (d) $-m(x_0 b \omega_2^2 - y_0 a \omega_1^2) \hat{k}$

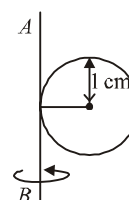
45. The time dependence of the position of a particle of mass $m = 2$ is given by $\vec{r}(t) = 2t \hat{i} - 3t^2 \hat{j}$. Its angular momentum, with respect to the origin, at time $t = 2$ is :

[10 Apr. 2019 II]

- (a) $48(\hat{i} + \hat{j})$ (b) $36\hat{k}$
(c) $-34(\hat{k} - \hat{i})$ (d) $-48\hat{k}$

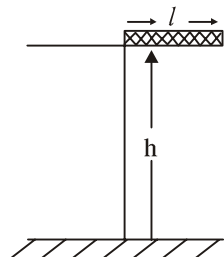
46. A metal coin of mass 5 g and radius 1 cm is fixed to a thin stick AB of negligible mass as shown in the figure. The system is initially at rest. The constant torque, that will make the system rotate about AB at 25 rotations per second in 5 s, is close to :

[10 Apr. 2019 II]



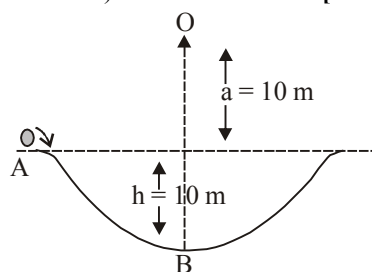
- (a) 4.0×10^{-6} Nm (b) 1.6×10^{-5} Nm
(c) 7.9×10^{-6} Nm (d) 2.0×10^{-5} Nm
47. A rectangular solid box of length 0.3 m is held horizontally, with one of its sides on the edge of a platform of height 5m. When released, it slips off the table in a very short time $\tau = 0.01$ s, remaining essentially horizontal. The angle by which it would rotate when it hits the ground will be (in radians) close to :

[8 Apr. 2019 II]



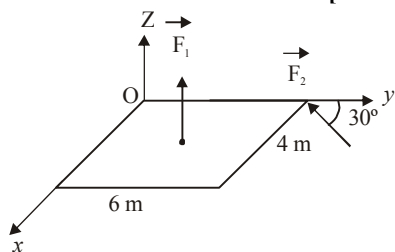
- (a) 0.5 (b) 0.3 (c) 0.02 (d) 0.28
48. A particle of mass 20 g is released with an initial velocity 5 m/s along the curve from the point A, as shown in the figure. The point A is at height h from point B. The particle slides along the frictionless surface. When the particle reaches point B, its angular momentum about O will be : (Take $g = 10 \text{ m/s}^2$)

[12 Jan. 2019 II]



- (a) $2 \text{ kg-m}^2/\text{s}$ (b) $8 \text{ kg-m}^2/\text{s}$
(c) $6 \text{ kg-m}^2/\text{s}$ (d) $3 \text{ kg-m}^2/\text{s}$
49. A slab is subjected to two forces $\vec{F}_1$ and $\vec{F}_2$ of same magnitude F as shown in the figure. Force $\vec{F}_2$ is in XY-plane while force $\vec{F}_1$ acts along z-axis at the point $(2\hat{i} + 3\hat{j})$. The moment of these forces about point O will be :

[11 Jan. 2019 I]



- (a) $(3\hat{i} - 2\hat{j} + 3\hat{k})F$ (b) $(3\hat{i} - 2\hat{j} - 3\hat{k})F$
(c) $(3\hat{i} + 2\hat{j} - 3\hat{k})F$ (d) $(3\hat{i} + 2\hat{j} + 3\hat{k})F$

50. The magnitude of torque on a particle of mass 1 kg is 2.5 Nm about the origin. If the force acting on it is 1 N, and the distance of the particle from the origin is 5m, the angle between the force and the position vector is (in radians):

[11 Jan. 2019 II]

- (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{8}$ (d) $\frac{\pi}{4}$

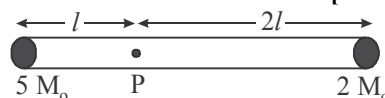
51. To mop-clean a floor, a cleaning machine presses a circular mop of radius R vertically down with a total force F and rotates it with a constant angular speed about its axis. If the force F is distributed uniformly over the mop and if coefficient of friction between the mop and the floor is μ , the torque, applied by the machine on the mop is:

[10 Jan. 2019 I]

- (a) $\mu FR/3$ (b) $\mu FR/6$
(c) $\mu FR/2$ (d) $\frac{2}{3}\mu \mu FR$

52. A rigid massless rod of length $3l$ has two masses attached at each end as shown in the figure. The rod is pivoted at point P on the horizontal axis (see figure). When released from initial horizontal position, its instantaneous angular acceleration will be:

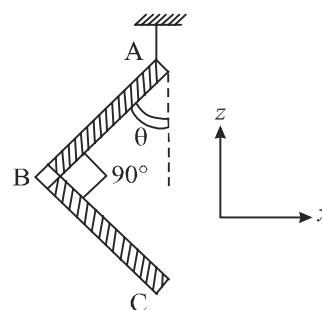
[10 Jan. 2019 II]



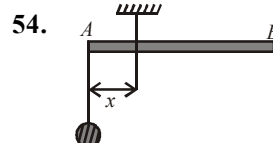
- (a) $\frac{g}{13l}$ (b) $\frac{g}{3l}$
(c) $\frac{g}{2l}$ (d) $\frac{7g}{3l}$

53. An L-shaped object, made of thin rods of uniform mass density, is suspended with a string as shown in figure. If $AB = BC$, and the angle made by AB with downward vertical is θ , then:

[9 Jan. 2019 I]



- (a) $\tan \theta = \frac{1}{2\sqrt{3}}$ (b) $\tan \theta = \frac{1}{2}$
(c) $\tan \theta = \frac{2}{\sqrt{3}}$ (d) $\tan \theta = \frac{1}{3}$



54.

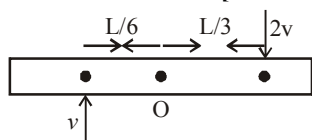
A uniform rod AB is suspended from a point X , at a variable distance from x from A , as shown. To make the rod horizontal, a mass m is suspended from its end A . A set of (m, x) values is recorded. The appropriate variable that give a straight line, when plotted, are:

[Online April 15, 2018]

- (a) $m, \frac{1}{x}$ (b) $m, \frac{1}{x^2}$ (c) m, x (d) m, x^2

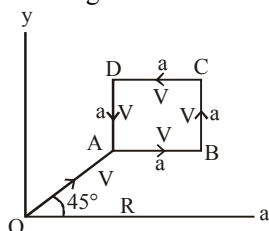
55. A thin uniform bar of length L and mass $8m$ lies on a smooth horizontal table. Two point masses m and $2m$ moving in the same horizontal plane from opposite sides of the bar with speeds $2v$ and v respectively. The masses stick to the bar after collision at a distance $\frac{L}{3}$ and $\frac{L}{6}$ respectively from the centre of the bar. If the bar starts rotating about its center of mass as a result of collision, the angular speed of the bar will be:

[Online April 15, 2018]



- (a) $\frac{v}{6L}$ (b) $\frac{6v}{5L}$ (c) $\frac{3v}{5L}$ (d) $\frac{v}{5L}$
56. A particle of mass m is moving along the side of a square of side ' a ', with a uniform speed v in the x - y plane as shown in the figure:

[2016]



Which of the following statements is false for the angular momentum $\vec{L}$ about the origin?

- (a) $\vec{L} = mv \left[\frac{R}{\sqrt{2}} + a \right] \hat{k}$ when the particle is moving from B to C.
- (b) $\vec{L} = \frac{mv}{\sqrt{2}} R \hat{k}$ when the particle is moving from D to A.
- (c) $\vec{L} = -\frac{mv}{\sqrt{2}} R \hat{k}$ when the particle is moving from A to B.
- (d) $\vec{L} = mv \left[\frac{R}{\sqrt{2}} - a \right] \hat{k}$ when the particle is moving from C to D.
57. A particle of mass 2 kg is on a smooth horizontal table and moves in a circular path of radius 0.6 m . The height of the table from the ground is 0.8 m . If the angular speed of the particle is 12 rad s^{-1} , the magnitude of its angular momentum about a point on the ground right under the centre of the circle is:

[Online April 11, 2015]

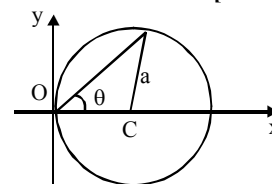
- (a) $14.4 \text{ kg m}^2 \text{ s}^{-1}$ (b) $8.64 \text{ kg m}^2 \text{ s}^{-1}$
(c) $20.16 \text{ kg m}^2 \text{ s}^{-1}$ (d) $11.52 \text{ kg m}^2 \text{ s}^{-1}$

58. A bob of mass m attached to an inextensible string of length l is suspended from a vertical support. The bob rotates in a horizontal circle with an angular speed $\omega \text{ rad/s}$ about the vertical. About the point of suspension: [2014]
- (a) angular momentum is conserved.
(b) angular momentum changes in magnitude but not in direction.
(c) angular momentum changes in direction but not in magnitude.
(d) angular momentum changes both in direction and magnitude.
59. A ball of mass 160 g is thrown up at an angle of 60° to the horizontal at a speed of 10 ms^{-1} . The angular momentum of the ball at the highest point of the trajectory with respect to the point from which the ball is thrown is nearly ($g = 10 \text{ ms}^{-2}$)

[Online April 19, 2014]

- (a) $1.73 \text{ kg m}^2/\text{s}$ (b) $3.0 \text{ kg m}^2/\text{s}$
(c) $3.46 \text{ kg m}^2/\text{s}$ (d) $6.0 \text{ kg m}^2/\text{s}$
60. A particle is moving in a circular path of radius a , with a constant velocity v as shown in the figure. The centre of circle is marked by 'C'. The angular momentum from the origin O can be written as:

[Online April 12, 2014]



- (a) $va(1 + \cos 2\theta)$ (b) $va(1 + \cos \theta)$
(c) $va \cos 2\theta$ (d) va
61. A particle of mass 2 kg is moving such that at time t , its position, in meter, is given by $\vec{r}(t) = 5\hat{i} - 2t^2\hat{j}$. The angular momentum of the particle at $t = 2\text{s}$ about the origin in $\text{kg m}^2 \text{ s}^{-1}$ is:

[Online April 23, 2013]

- (a) $-80\hat{k}$ (b) $(10\hat{i} - 16\hat{j})$
(c) $-40\hat{k}$ (d) $40\hat{k}$
62. A bullet of mass 10 g and speed 500 m/s is fired into a door and gets embedded exactly at the centre of the door. The door is 1.0 m wide and weighs 12 kg . It is hinged at one end and rotates about a vertical axis practically without friction. The angular speed of the door just after the bullet embeds into it will be:

[Online April 9, 2013]

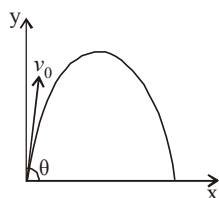
- (a) 6.25 rad/sec (b) 0.625 rad/sec
(c) 3.35 rad/sec (d) 0.335 rad/sec
63. A stone of mass m , tied to the end of a string, is whirled around in a circle on a horizontal frictionless table. The length of the string is reduced gradually keeping the angular momentum of the stone about the centre of the circle constant. Then, the tension in the string is given by $T = Ar^n$, where A is a constant, r is the instantaneous radius of the circle. The value of n is equal to

[Online May 26, 2012]

- (a) -1 (b) -2 (c) -4 (d) -3
64. A thin horizontal circular disc is rotating about a vertical axis passing through its centre. An insect is at rest at a

point near the rim of the disc. The insect now moves along a diameter of the disc to reach its other end. During the journey of the insect, the angular speed of the disc. [2011]

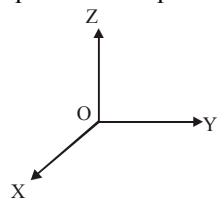
- (a) continuously decreases
(b) continuously increases
(c) first increases and then decreases
(d) remains unchanged
65. A small particle of mass m is projected at an angle θ with the x -axis with an initial velocity v_0 in the x - y plane as shown in the figure. At a time $t < \frac{v_0 \sin \theta}{g}$, the angular momentum of the particle is [2010]



- (a) $-mg v_0 t^2 \cos \theta \hat{j}$ (b) $mg v_0 t \cos \theta \hat{k}$
(c) $-\frac{1}{2} mg v_0 t^2 \cos \theta \hat{k}$ (d) $\frac{1}{2} mg v_0 t^2 \cos \theta \hat{i}$

where $\hat{i}, \hat{j}$ and $\hat{k}$ are unit vectors along x, y and z -axis respectively.

66. Angular momentum of the particle rotating with a central force is constant due to [2007]
(a) constant torque
(b) constant force
(c) constant linear momentum
(d) zero torque
67. A thin circular ring of mass m and radius R is rotating about its axis with a constant angular velocity ω . Two objects each of mass M are attached gently to the opposite ends of a diameter of the ring. The ring now rotates with an angular velocity $\omega' =$ [2006]
(a) $\frac{\omega(m+2M)}{m}$ (b) $\frac{\omega(m-2M)}{(m+2M)}$
(c) $\frac{\omega m}{(m+M)}$ (d) $\frac{\omega m}{(m+2M)}$
68. A force of $-F\hat{k}$ acts on O , the origin of the coordinate system. The torque about the point $(1, -1)$ is [2006]



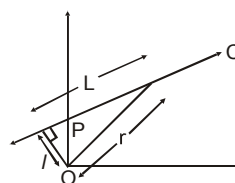
- (a) $F(\hat{i} - \hat{j})$ (b) $-F(\hat{i} + \hat{j})$
(c) $F(\hat{i} + \hat{j})$ (d) $-F(\hat{i} - \hat{j})$
69. A solid sphere is rotating in free space. If the radius of the sphere is increased keeping mass same, which one of the following will not be affected? [2004]

- (a) Angular velocity
(b) Angular momentum
(c) Moment of inertia
(d) Rotational kinetic energy

70. Let $\vec{F}$ be the force acting on a particle having position vector $\vec{r}$, and $\vec{T}$ be the torque of this force about the origin. Then [2003]

- (a) $\vec{r} \cdot \vec{T} = 0$ and $\vec{F} \cdot \vec{T} \neq 0$
(b) $\vec{r} \cdot \vec{T} \neq 0$ and $\vec{F} \cdot \vec{T} = 0$
(c) $\vec{r} \cdot \vec{T} \neq 0$ and $\vec{F} \cdot \vec{T} \neq 0$
(d) $\vec{r} \cdot \vec{T} = 0$ and $\vec{F} \cdot \vec{T} = 0$

71. A particle of mass m moves along line PC with velocity v as shown. What is the angular momentum of the particle about P ? [2002]

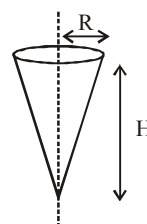


- (a) mvL (b) mvL (c) mvr (d) zero.

TOPIC 4 Moment of Inertia and Rotational K.E.



72. Shown in the figure is a hollow icecream cone (it is open at the top). If its mass is M , radius of its top, R and height, H , then its moment of inertia about its axis is : [Sep. 06, 2020 (I)]



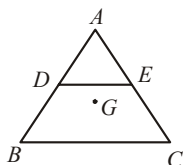
- (a) $\frac{MR^2}{2}$ (b) $\frac{M(R^2 + H^2)}{4}$
(c) $\frac{MH^2}{3}$ (d) $\frac{MR^2}{3}$
73. The linear mass density of a thin rod AB of length L varies from A to B as $\lambda(x) = \lambda_0 \left(1 + \frac{x}{L}\right)$, where x is the distance from A. If M is the mass of the rod then its moment of inertia about an axis passing through A and perpendicular to the rod is : [Sep. 06, 2020 (II)]
(a) $\frac{5}{12} ML^2$ (b) $\frac{7}{18} ML^2$
(c) $\frac{2}{5} ML^2$ (d) $\frac{3}{7} ML^2$
74. A wheel is rotating freely with an angular speed ω on a shaft. The moment of inertia of the wheel is I and the moment of inertia of the shaft is negligible. Another wheel

of moment of inertia $3I$ initially at rest is suddenly coupled to the same shaft. The resultant fractional loss in the kinetic energy of the system is : [Sep. 05, 2020 (I)]

- (a) $\frac{5}{6}$ (b) $\frac{1}{4}$
(c) 0 (d) $\frac{3}{4}$

75. ABC is a plane lamina of the shape of an equilateral triangle. D, E are mid points of AB, AC and G is the centroid of the lamina. Moment of inertia of the lamina about an axis passing through G and perpendicular to the plane ABC is I_0 . If part ADE is removed, the moment of inertia of the remaining part about the same axis is $\frac{NI_0}{16}$ where N is an integer. Value of N is _____.

[NA Sep. 04, 2020 (I)]



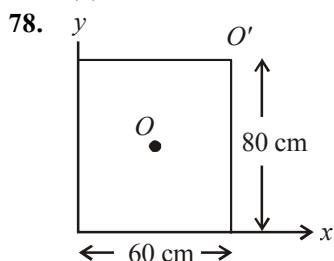
76. A circular disc of mass M and radius R is rotating about its axis with angular speed ω_1 . If another stationary disc having radius $\frac{R}{2}$ and same mass M is dropped co-axially on to the rotating disc. Gradually both discs attain constant angular speed ω_2 . The energy lost in the process is $p\%$ of the initial energy. Value of p is _____.

[NA Sep. 04, 2020 (I)]

77. Consider two uniform discs of the same thickness and different radii $R_1 = R$ and $R_2 = \alpha R$ made of the same material. If the ratio of their moments of inertia I_1 and I_2 , respectively, about their axes is $I_1 : I_2 = 1 : 16$ then the value of α is :

[Sep. 04, 2020 (II)]

- (a) $2\sqrt{2}$ (b) $\sqrt{2}$
(c) 2 (d) 4



For a uniform rectangular sheet shown in the figure, the ratio of moments of inertia about the axes perpendicular to the sheet and passing through O (the centre of mass) and O' (corner point) is :

[Sep. 04, 2020 (II)]

- (a) $2/3$ (b) $1/4$
(c) $1/8$ (d) $1/2$

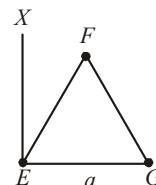
79. Moment of inertia of a cylinder of mass M , length L and radius R about an axis passing through its centre and perpendicular to the axis of the cylinder is

$I = M \left(\frac{R^2}{4} + \frac{L^2}{12} \right)$. If such a cylinder is to be made for a given mass of a material, the ratio L/R for it to have minimum possible I is : [Sep. 03, 2020 (I)]

- (a) $\frac{2}{3}$ (b) $\frac{3}{2}$
(c) $\sqrt{\frac{3}{2}}$ (d) $\sqrt{\frac{2}{3}}$

80. An massless equilateral triangle EFG of side ' a ' (As shown in figure) has three particles of mass m situated at its vertices. The moment of inertia of the system about the line EX perpendicular to EG in the plane of EFG is $\frac{N}{20}ma^2$ where N is an integer. The value of N is _____.

[Sep. 03, 2020 (II)]



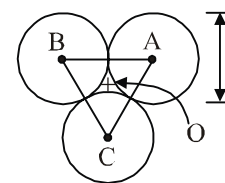
81. Two uniform circular discs are rotating independently in the same direction around their common axis passing through their centres. The moment of inertia and angular velocity of the first disc are 0.1 kg-m^2 and 10 rad s^{-1} respectively while those for the second one are 0.2 kg-m^2 and 5 rad s^{-1} respectively. At some instant they get stuck together and start rotating as a single system about their common axis with some angular speed. The kinetic energy of the combined system is : [Sep. 02, 2020 (II)]

- (a) $\frac{10}{3} \text{ J}$ (b) $\frac{20}{3} \text{ J}$
(c) $\frac{5}{3} \text{ J}$ (d) $\frac{2}{3} \text{ J}$

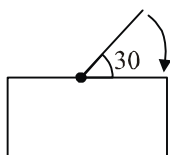
82. Three solid spheres each of mass m and diameter d are stuck together such that the lines connecting the centres form an equilateral triangle of side of length d . The ratio $\frac{I_0}{I_A}$ of moment of inertia I_0 of the system about an axis passing the centroid and about center of any of the spheres I_A and perpendicular to the plane of the triangle is:

[9 Jan. 2020 I]

- (a) $\frac{13}{23}$
(b) $\frac{15}{13}$
(c) $\frac{23}{13}$
(d) $\frac{13}{15}$

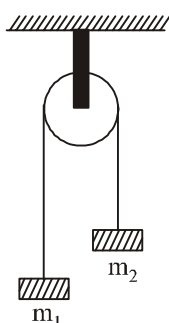


83. One end of a straight uniform 1 m long bar is pivoted on horizontal table. It is released from rest when it makes an angle 30° from the horizontal (see figure). Its angular speed when it hits the table is given as $\sqrt{n}s^{-1}$, where n is an integer. The value of n is _____. [9 Jan. 2020 I]

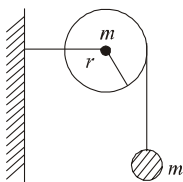


84. A uniformly thick wheel with moment of inertia I and radius R is free to rotate about its centre of mass (see fig). A massless string is wrapped over its rim and two blocks of masses m_1 and m_2 ($m_1 > m_2$) are attached to the ends of the string. The system is released from rest. The angular speed of the wheel when m_1 descends by a distance h is: [9 Jan. 2020 II]

- (a) $\left[\frac{2(m_1 - m_2)gh}{(m_1 + m_2)R^2 + 1} \right]^{1/2}$
 (b) $\left[\frac{2(m_1 + m_2)gh}{(m_1 + m_2)R^2 + 1} \right]^{1/2}$
 (c) $\left[\frac{(m_1 - m_2)}{(m_1 + m_2)R^2 + 1} \right]^{1/2} gh$
 (d) $\left[\frac{m_1 + m_2}{(m_1 + m_2)R^2 + 1} \right]^{1/2} gh$



85. As shown in the figure, a bob of mass m is tied by a massless string whose other end portion is wound on a fly wheel (disc) of radius r and mass m . When released from rest the bob starts falling vertically. When it has covered a distance of h , the angular speed of the wheel will be: [7 Jan. 2020 I]



- (a) $\frac{1}{r} \sqrt{\frac{4gh}{3}}$ (b) $r \sqrt{\frac{3}{2gh}}$
 (c) $\frac{1}{r} \sqrt{\frac{2gh}{3}}$ (d) $r \sqrt{\frac{3}{4gh}}$

86. The radius of gyration of a uniform rod of length l , about an axis passing through a point $\frac{l}{4}$ away from the centre of the rod, and perpendicular to it, is: [7 Jan. 2020 I]

- (a) $\frac{1}{4}l$ (b) $\frac{1}{8}l$ (c) $\sqrt{\frac{7}{48}}l$ (d) $\sqrt{\frac{3}{8}}l$

87. Mass per unit area of a circular disc of radius a depends on the distance r from its centre as $\sigma(r) = A + Br$. The moment of inertia of the disc about the axis, perpendicular to the plane and passing through its centre is:

[7 Jan. 2020 II]

- (a) $2\pi a^4 \left(\frac{A}{4} + \frac{aB}{5} \right)$ (b) $2\pi a^4 \left(\frac{aA}{4} + \frac{B}{5} \right)$
 (c) $\pi a^4 \left(\frac{A}{4} + \frac{aB}{5} \right)$ (d) $2\pi a^4 \left(\frac{A}{4} + \frac{B}{5} \right)$

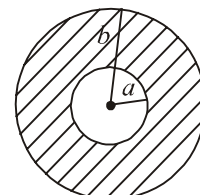
88. A circular disc of radius b has a hole of radius a at its centre (see figure). If the mass per unit area of the disc varies as

$\left(\frac{\sigma_0}{r} \right)$, then the radius of gyration of the disc about its axis

passing through the centre is :

[12 Apr. 2019 I]

- (a) $\sqrt{\frac{a^2 + b^2 + ab}{2}}$ (b) $\frac{a+b}{2}$
 (c) $\sqrt{\frac{a^2 + b^2 + ab}{3}}$ (d) $\frac{a+b}{3}$



89. Two coaxial discs, having moments of inertia I_1 and $\frac{I_1}{2}$, are

rotating with respective angular velocities ω_1 and $\frac{\omega_1}{2}$,

about their common axis. They are brought in contact with each other and thereafter they rotate with a common angular velocity. If E_f and E_i are the final and initial total energies, then $(E_f - E_i)$ is :

[10 Apr. 2019 I]

- (a) $-\frac{I_1 \omega_1^2}{12}$ (b) $\frac{I_1 \omega_1^2}{6}$ (c) $\frac{3}{8} I_1 \omega_1^2$ (d) $-\frac{I_1 \omega_1^2}{24}$

90. A thin disc of mass M and radius R has mass per unit area $\sigma(r) = kr^2$ where r is the distance from its centre. Its moment of inertia about an axis going through its centre of mass and perpendicular to its plane is: [10 Apr. 2019 I]

- (a) $\frac{MR^2}{3}$ (b) $\frac{2MR^2}{3}$
 (c) $\frac{MR^2}{6}$ (d) $\frac{MR^2}{2}$

91. A solid sphere of mass M and radius R is divided into two unequal parts. The first part has a mass of $\frac{7M}{8}$ and is

converted into a uniform disc of radius $2R$. The second part is converted into a uniform solid sphere. Let I_1 be the moment of inertia of the new sphere about its axis. The ratio I_1/I_2 is given by: [10 Apr. 2019 II]

- (a) 185 (b) 140 (c) 285 (d) 65

92. A stationary horizontal disc is free to rotate about its axis. When a torque is applied on it, its kinetic energy as a function of θ , where θ is the angle by which it has rotated, is given as $k\theta^2$. If its moment of inertia is I then the angular acceleration of the disc is: [9 April 2019 I]

(a) $\frac{k}{4I}\theta$ (b) $\frac{k}{I}\theta$ (c) $\frac{k}{2I}\theta$ (d) $\frac{2k}{I}\theta$

93. Moment of inertia of a body about a given axis is 1.5 kg m^2 . Initially the body is at rest. In order to produce a rotational kinetic energy of 1200 J , the angular acceleration of 20 rad/s^2 must be applied about the axis for a duration of: [9 Apr. 2019 II]

(a) 2.5 s (b) 2 s (c) 5 s (d) 3 s

94. A thin smooth rod of length L and mass M is rotating freely with angular speed ω_0 about an axis perpendicular to the rod and passing through its center. Two beads of mass m and negligible size are at the center of the rod initially. The beads are free to slide along the rod. The angular speed of the system, when the beads reach the opposite ends of the rod, will be: [9 Apr. 2019 II]

(a) $\frac{M\omega_0}{M+m}$ (b) $\frac{M\omega_0}{M+3m}$
(c) $\frac{M\omega_0}{M+6m}$ (d) $\frac{M\omega_0}{M+2m}$

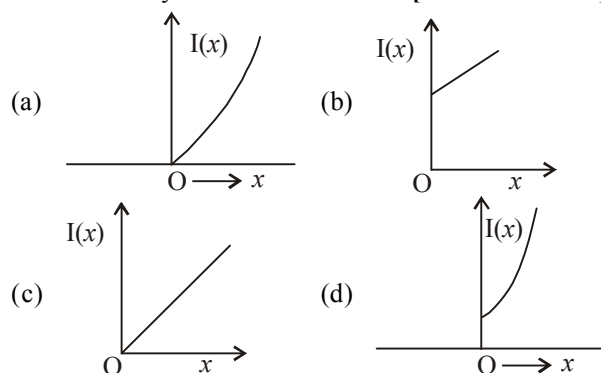
95. A thin circular plate of mass M and radius R has its density varying as $\rho(r) = \rho_0 r$ with ρ_0 as constant and r is the distance from its center. The moment of Inertia of the circular plate about an axis perpendicular to the plate and passing through its edge is $I = a MR^2$. The value of the coefficient a is: [8 April 2019 I]

(a) $\frac{1}{2}$ (b) $\frac{3}{5}$ (c) $\frac{8}{5}$ (d) $\frac{3}{2}$

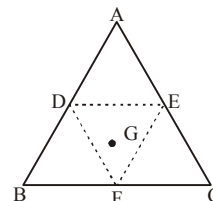
96. Let the moment of inertia of a hollow cylinder of length 30 cm (inner radius 10 cm and outer radius 20 cm), about its axis be I . The radius of a thin cylinder of the same mass such that its moment of inertia about its axis is also I , is: [12 Jan. 2019 I]

(a) 12 cm (b) 16 cm
(c) 14 cm (d) 18 cm

97. The moment of inertia of a solid sphere, about an axis parallel to its diameter and at a distance of x from it, is ' $I(x)$ '. Which one of the graphs represents the variation of $I(x)$ with x correctly? [12 Jan. 2019 II]

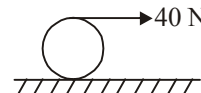


98. An equilateral triangle ABC is cut from a thin solid sheet of wood. (See figure) D , E and F are the mid-points of its sides as shown and G is the centre of the triangle. The moment of inertia of the triangle about an axis passing through G and perpendicular to the plane of the triangle is I_0 . If the smaller triangle DEF is removed from ABC , the moment of inertia of the remaining figure about the same axis is I . Then : [11 Jan. 2019 I]



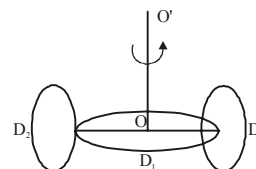
(a) $I = \frac{15}{16}I_0$ (b) $I = \frac{3}{4}I_0$
(c) $I = \frac{9}{16}I_0$ (d) $I = \frac{I_0}{4}$

99. a string is wound around a hollow cylinder of mass 5 kg and radius 0.5 m . If the string is now pulled with a horizontal force of 40 N , and the cylinder is rolling without slipping on a horizontal surface (see figure), then the angular acceleration of the cylinder will be (Neglect the mass and thickness of the string) [11 Jan. 2019 II]



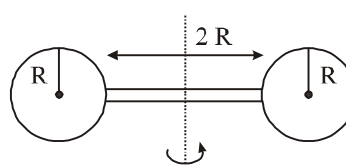
(a) 20 rad/s^2 (b) 16 rad/s^2
(c) 12 rad/s^2 (d) 10 rad/s^2

100. A circular disc D_1 of mass M and radius R has two identical discs D_2 and D_3 of the same mass M and radius R attached rigidly at its opposite ends (see figure). The moment of inertia of the system about the axis OO' , passing through the centre of D_1 , as shown in the figure, will be : [11 Jan. 2019 II]



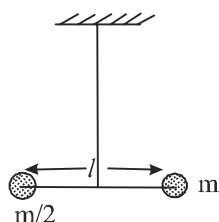
(a) MR^2 (b) $3MR^2$
(c) $\frac{4}{5}MR^2$ (d) $\frac{2}{3}MR^2$

101. Two identical spherical balls of mass M and radius R each are stuck on two ends of a rod of length $2R$ and mass M (see figure). The moment of inertia of the system about the axis passing perpendicularly through the centre of the rod is: [10 Jan. 2019 II]



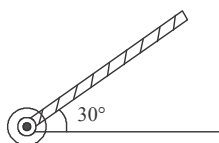
- (a) $\frac{137}{15} MR^2$ (b) $\frac{17}{15} MR^2$
 (c) $\frac{209}{15} MR^2$ (d) $\frac{152}{15} MR^2$

102. Two masses m and $\frac{m}{2}$ are connected at the two ends of a massless rigid rod of length l . The rod is suspended by a thin wire of torsional constant k at the centre of mass of the rod-mass system (see figure). Because of torsional constant k , the restoring torque is $\tau = k\theta$ for angular displacement θ . If the rod is rotated by θ_0 and released, the tension in it when it passes through its mean position will be: [9 Jan. 2019 I]



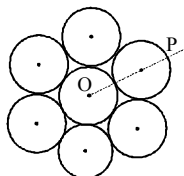
- (a) $\frac{3k\theta_0^2}{l}$ (b) $\frac{2k\theta_0^2}{l}$ (c) $\frac{k\theta_0^2}{l}$ (d) $\frac{k\theta_0^2}{2l}$

103. A rod of length 50 cm is pivoted at one end. It is raised such that it makes an angle of 30° from the horizontal as shown and released from rest. Its angular speed when it passes through the horizontal (in rads^{-1}) will be ($g = 10 \text{ ms}^{-2}$) [9 Jan. 2019 II]



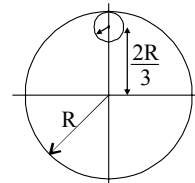
- (a) $\sqrt{\frac{30}{7}}$ (b) $\sqrt{30}$
 (c) $\sqrt{\frac{20}{3}}$ (d) $\sqrt{\frac{30}{2}}$

104. Seven identical circular planar disks, each of mass M and radius R are welded symmetrically as shown. The moment of inertia of the arrangement about the axis normal to the plane and passing through the point P is: [2018]



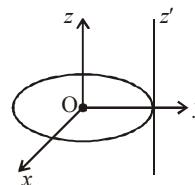
- (a) $\frac{19}{2} MR^2$ (b) $\frac{55}{2} MR^2$ (c) $\frac{73}{2} MR^2$ (d) $\frac{181}{2} MR^2$

105. From a uniform circular disc of radius R and mass $9M$, a small disc of radius $\frac{R}{3}$ is removed as shown in the figure. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through centre of disc is: [2018]



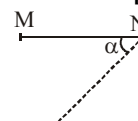
- (a) $4MR^2$ (b) $\frac{40}{9} MR^2$ (c) $10MR^2$ (d) $\frac{37}{9} MR^2$

106. A thin circular disk is in the xy plane as shown in the figure. The ratio of its moment of inertia about z and z' axes will be [Online April 16, 2018]



- (a) 1:2 (b) 1:4 (c) 1:3 (d) 1:5

107. A thin rod MN , free to rotate in the vertical plane about the fixed end N , is held horizontal. When the end M is released the speed of this end, when the rod makes an angle α with the horizontal, will be proportional to: (see figure) [Online April 15, 2018]



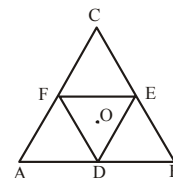
- (a) $\sqrt{\cos \alpha}$ (b) $\cos \alpha$
 (c) $\sin \alpha$ (d) $\sqrt{\sin \alpha}$

108. The moment of inertia of a uniform cylinder of length ℓ and radius R about its perpendicular bisector is I . What is the ratio ℓ/R such that the moment of inertia is minimum? [2017]

- (a) 1 (b) $\frac{3}{\sqrt{2}}$ (c) $\sqrt{\frac{3}{2}}$ (d) $\frac{\sqrt{3}}{2}$

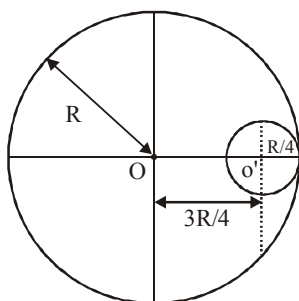
109. Moment of inertia of an equilateral triangular lamina ABC , about the axis passing through its centre O and perpendicular to its plane is I_0 as shown in the figure. A cavity DEF is cut out from the lamina, where D, E, F are the mid points of the sides. Moment of inertia of the remaining part of lamina about the same axis is: [Online April 8, 2017]

- (a) $\frac{7}{8} I_0$ (b) $\frac{15}{16} I_0$
 (c) $\frac{3I_0}{4}$ (d) $\frac{31I_0}{32}$



110. A circular hole of radius $\frac{R}{4}$ is made in a thin uniform disc having mass M and radius R , as shown in figure. The moment of inertia of the remaining portion of the disc about an axis passing through the point O and perpendicular to the plane of the disc is : [Online April 9, 2017]

- (a) $\frac{219MR^2}{256}$
 (b) $\frac{237MR^2}{512}$
 (c) $\frac{19MR^2}{512}$
 (d) $\frac{197MR^2}{256}$



111. From a solid sphere of mass M and radius R a cube of maximum possible volume is cut. Moment of inertia of cube about an axis passing through its center and perpendicular to one of its faces is : [2015]

- (a) $\frac{4MR^2}{9\sqrt{3}\pi}$ (b) $\frac{4MR^2}{3\sqrt{3}\pi}$
 (c) $\frac{MR^2}{32\sqrt{2}\pi}$ (d) $\frac{MR^2}{16\sqrt{2}\pi}$

112. Consider a thin uniform square sheet made of a rigid material. If its side is ' a ' mass m and moment of inertia I about one of its diagonals, then : [Online April 10, 2015]

- (a) $I > \frac{ma^2}{12}$ (b) $\frac{ma^2}{24} < I < \frac{ma^2}{12}$
 (c) $I = \frac{ma^2}{24}$ (d) $I = \frac{ma^2}{12}$

113. A ring of mass M and radius R is rotating about its axis with angular velocity ω . Two identical bodies each of mass m are now gently attached at the two ends of a diameter of the ring. Because of this, the kinetic energy loss will be:

[Online April 25, 2013]

- (a) $\frac{m(M+2m)}{M}\omega^2 R^2$ (b) $\frac{Mm}{(M+m)}\omega^2 R^2$
 (c) $\frac{Mm}{(M+2m)}\omega^2 R^2$ (d) $\frac{(M+m)M}{(M+2m)}\omega^2 R^2$

114. This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.

Statement 1: When moment of inertia I of a body rotating about an axis with angular speed ω increases, its angular momentum L is unchanged but the kinetic energy K increases if there is no torque applied on it.

Statement 2: $L = I\omega$, kinetic energy of rotation $= \frac{1}{2}I\omega^2$ [Online May 12, 2012]

- (a) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.
 (b) Statement 1 is false, Statement 2 is true.

- (c) Statement 1 is true, Statement 2 is true, Statement 2 is correct explanation of the Statement 1.
 (d) Statement 1 is true, Statement 2 is false.

115. A solid sphere having mass m and radius r rolls down an inclined plane. Then its kinetic energy is

[Online May 7, 2012]

- (a) $\frac{5}{7}$ rotational and $\frac{2}{7}$ translational
 (b) $\frac{2}{7}$ rotational and $\frac{5}{7}$ translational
 (c) $\frac{2}{5}$ rotational and $\frac{3}{5}$ translational
 (d) $\frac{1}{2}$ rotational and $\frac{1}{2}$ translational

116. A circular hole of diameter R is cut from a disc of mass M and radius R ; the circumference of the cut passes through the centre of the disc. The moment of inertia of the remaining portion of the disc about an axis perpendicular to the disc and passing through its centre is

[Online May 7, 2012]

- (a) $\left(\frac{15}{32}\right)MR^2$ (b) $\left(\frac{1}{8}\right)MR^2$
 (c) $\left(\frac{3}{8}\right)MR^2$ (d) $\left(\frac{13}{32}\right)MR^2$

117. A mass m hangs with the help of a string wrapped around a pulley on a frictionless bearing. The pulley has mass m and radius R . Assuming pulley to be a perfect uniform circular disc, the acceleration of the mass m , if the string does not slip on the pulley, is: [2011]

- (a) g (b) $\frac{2}{3}g$ (c) $\frac{g}{3}$ (d) $\frac{3}{2}g$

118. A pulley of radius 2 m is rotated about its axis by a force $F = (20t - 5t^2)$ newton (where t is measured in seconds) applied tangentially. If the moment of inertia of the pulley about its axis of rotation is 10 kg-m^2 the number of rotations made by the pulley before its direction of motion is reversed, is: [2011]

- (a) more than 3 but less than 6
 (b) more than 6 but less than 9
 (c) more than 9
 (d) less than 3

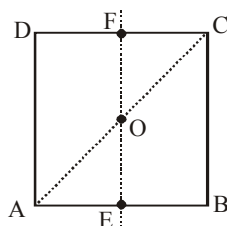
119. A thin uniform rod of length l and mass m is swinging freely about a horizontal axis passing through its end. Its maximum angular speed is ω . Its centre of mass rises to a maximum height of [2009]

- (a) $\frac{l\omega}{6g}$ (b) $\frac{l^2\omega^2}{2g}$
 (c) $\frac{l^2\omega^2}{6g}$ (d) $\frac{l^2\omega^2}{3g}$

120. Consider a uniform square plate of side ' a ' and mass ' M '. The moment of inertia of this plate about an axis perpendicular to its plane and passing through one of its corners is [2008]

- (a) $\frac{5}{6}Ma^2$ (b) $\frac{1}{12}Ma^2$
 (c) $\frac{7}{12}Ma^2$ (d) $\frac{2}{3}Ma^2$

121. For the given uniform square lamina $ABCD$, whose centre is O , [2007]



- (a) $I_{AC} = \sqrt{2} I_{EF}$ (b) $\sqrt{2} I_{AC} = I_{EF}$
 (c) $I_{AD} = 3I_{EF}$ (d) $I_{AC} = I_{EF}$

122. Four point masses, each of value m , are placed at the corners of a square $ABCD$ of side ℓ . The moment of inertia of this system about an axis passing through A and parallel to BD is [2006]

- (a) $2m\ell^2$ (b) $\sqrt{3}m\ell^2$
 (c) $3m\ell^2$ (d) $m\ell^2$

123. The moment of inertia of a uniform semicircular disc of mass M and radius r about a line perpendicular to the plane of the disc through the centre is [2005]

- (a) $\frac{2}{5}Mr^2$ (b) $\frac{1}{4}Mr$
 (c) $\frac{1}{2}Mr^2$ (d) Mr^2

124. One solid sphere A and another hollow sphere B are of same mass and same outer radii. Their moment of inertia about their diameters are respectively I_A and I_B . Such that [2004]

- (a) $I_A < I_B$ (b) $I_A > I_B$
 (c) $I_A = I_B$ (d) $\frac{I_A}{I_B} = \frac{d_A}{d_B}$

where d_A and d_B are their densities.

125. A circular disc X of radius R is made from an iron plate of thickness t , and another disc Y of radius $4R$ is made from an iron plate of thickness $\frac{t}{4}$. Then the relation between the moment of inertia I_X and I_Y is [2003]

- (a) $I_Y = 32I_X$ (b) $I_Y = 16I_X$
 (c) $I_Y = I_X$ (d) $I_Y = 64I_X$

126. A particle performing uniform circular motion has angular frequency is doubled & its kinetic energy halved, then the new angular momentum is [2003]

- (a) $\frac{L}{4}$ (b) $2L$
 (c) $4L$ (d) $\frac{L}{2}$

127. Moment of inertia of a circular wire of mass M and radius R about its diameter is [2002]

- (a) $MR^2/2$ (b) MR^2 (c) $2MR^2$ (d) $MR^2/4$

128. Initial angular velocity of a circular disc of mass M is ω_1 . Then two small spheres of mass m are attached gently to diametrically opposite points on the edge of the disc. What is the final angular velocity of the disc? [2002]

- (a) $\left(\frac{M+m}{M}\right)\omega_1$ (b) $\left(\frac{M+m}{m}\right)\omega_1$
 (c) $\left(\frac{M}{M+4m}\right)\omega_1$ (d) $\left(\frac{M}{M+2m}\right)\omega_1$

TOPIC 5 Rolling Motion

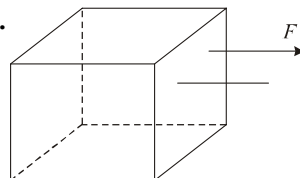


129. A uniform sphere of mass 500 g rolls without slipping on a plane horizontal surface with its centre moving at a speed of 5.00 cm/s. Its kinetic energy is:

[8 Jan. 2020 II]

- (a) $8.75 \times 10^{-4} \text{ J}$ (b) $8.75 \times 10^{-3} \text{ J}$
 (c) $6.25 \times 10^{-4} \text{ J}$ (d) $1.13 \times 10^{-3} \text{ J}$

- 130.



Consider a uniform cubical box of side a on a rough floor that is to be moved by applying minimum possible force F at a point b above its centre of mass (see figure). If the coefficient of friction is $\mu = 0.4$, the maximum possible value

of $100 \times \frac{b}{a}$ for box not to topple before moving is

[NA 7 Jan. 2020 II]

131. A solid sphere and solid cylinder of identical radii approach an incline with the same linear velocity (see figure). Both roll without slipping all throughout. The two climb maximum heights h_{sph} and h_{cyl} on the incline. The ratio

$\frac{h_{sph}}{h_{cyl}}$ is given by :

[8 Apr. 2019 II]



- (a) $\frac{2}{\sqrt{5}}$ (b) 1 (c) $\frac{14}{15}$ (d) $\frac{4}{5}$

132. The following bodies are made to roll up (without slipping) the same inclined plane from a horizontal plane:

- (i) a ring of radius R , (ii) a solid cylinder of radius $\frac{R}{2}$

and (iii) a solid sphere of radius $\frac{R}{4}$. If, in each case, the speed of the center of mass at the bottom of the incline is same, the ratio of the maximum heights they climb is:

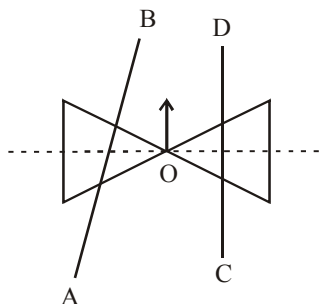
[9 April 2019 I]

- (a) 4 : 3 : 2 (b) 10 : 15 : 7
(c) 14 : 15 : 20 (d) 2 : 3 : 4

133. A homogeneous solid cylindrical roller of radius R and mass M is pulled on a cricket pitch by a horizontal force. Assuming rolling without slipping, angular acceleration of the cylinder is: [10 Jan. 2019 I]

- (a) $\frac{3F}{2mR}$ (b) $\frac{F}{3mR}$
(c) $\frac{F}{2mR}$ (d) $\frac{2F}{3mR}$

134. A roller is made by joining together two cones at their vertices O . It is kept on two rails AB and CD , which are placed asymmetrically (see figure), with its axis perpendicular to CD and its centre O at the centre of line joining AB and Cd (see figure). It is given a light push so that it starts rolling with its centre O moving parallel to CD in the direction shown. As it moves, the roller will tend to: [2016]



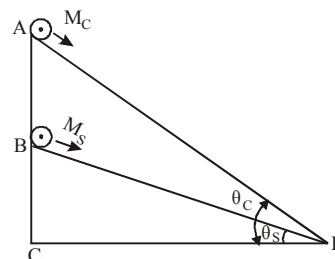
- (a) go straight.
(b) turn left and right alternately.
(c) turn left.
(d) turn right.

135. A uniform solid cylindrical roller of mass ' m ' is being pulled on a horizontal surface with force F parallel to the surface and applied at its centre. If the acceleration of the cylinder is ' a ' and it is rolling without slipping then the value of ' F ' is: [Online April 10, 2015]

- (a) ma (b) $\frac{5}{3}ma$
(c) $\frac{3}{2}ma$ (d) $2ma$

136. A cylinder of mass M_c and sphere of mass M_s are placed at points A and B of two inclines, respectively (See

Figure). If they roll on the incline without slipping such that their accelerations are the same, then the ratio $\frac{\sin \theta_c}{\sin \theta_s}$ is: [Online April 9, 2014]



- (a) $\sqrt{\frac{8}{7}}$ (b) $\sqrt{\frac{15}{14}}$
(c) $\frac{8}{7}$ (d) $\frac{15}{14}$

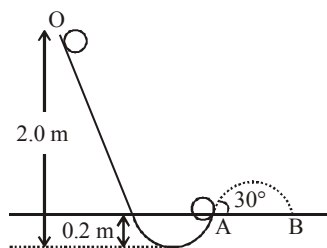
137. A loop of radius r and mass m rotating with an angular velocity ω_0 is placed on a rough horizontal surface. The initial velocity of the centre of the hoop is zero. What will be the velocity of the centre of the hoop when it ceases to slip? [2013]

- (a) $\frac{r\omega_0}{4}$ (b) $\frac{r\omega_0}{3}$
(c) $\frac{r\omega_0}{2}$ (d) $r\omega_0$

138. A tennis ball (treated as hollow spherical shell) starting from O rolls down a hill. At point A the ball becomes air borne leaving at an angle of 30° with the horizontal. The ball strikes the ground at B . What is the value of the distance AB ?

(Moment of inertia of a spherical shell of mass m and radius R about its diameter $= \frac{2}{3}mR^2$)

[Online April 22, 2013]



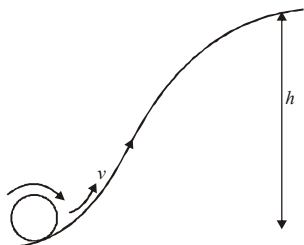
- (a) 1.87 m (b) 2.08 m
(c) 1.57 m (d) 1.77 m

139. A thick-walled hollow sphere has outside radius R_0 . It rolls down an incline without slipping and its speed at the bottom is v_0 . Now the incline is waxed, so that it is practically frictionless and the sphere is observed to slide down (without any rolling). Its speed at the bottom is observed to be $5v_0/4$. The radius of gyration of the hollow sphere about an axis through its centre is [Online May 26, 2012]

- (a) $3R_0/2$ (b) $3R_0/4$
(c) $9R_0/16$ (d) $3R_0$

140. A solid sphere is rolling on a surface as shown in figure, with a translational velocity $v \text{ ms}^{-1}$. If it is to climb the inclined surface continuing to roll without slipping, then minimum velocity for this to happen is

[Online May 12, 2012]



- (a) $\sqrt{2gh}$ (b) $\sqrt{\frac{7}{5}gh}$
 (c) $\sqrt{\frac{7}{2}gh}$ (d) $\sqrt{\frac{10}{7}gh}$

141. A round uniform body of radius R , mass M and moment of inertia I rolls down (without slipping) an inclined plane making an angle θ with the horizontal. Then its acceleration is [2007]

- (a) $\frac{g \sin \theta}{1 - MR^2 / I}$ (b) $\frac{g \sin \theta}{1 + I / MR^2}$
 (c) $\frac{g \sin \theta}{1 + MR^2 / I}$ (d) $\frac{g \sin \theta}{1 - I / MR^2}$

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Hints & Solutions



1. (3) Centre of mass of solid hemisphere of radius R lies at a distance $\frac{3R}{8}$ above the centre of flat side of hemisphere.

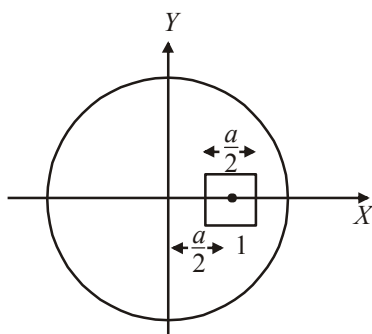
$$\therefore h_{cm} = \frac{3R}{8} = \frac{3 \times 8}{8} = 3 \text{ cm}$$

2. (23.00) Let σ be the mass density of circular disc.

Original mass of the disc, $m_0 = \pi a^2 \sigma$

Removed mass, $m = \frac{a^2}{4} \sigma$

Remaining, mass, $m' = \left(\pi a^2 - \frac{a^2}{4} \right) \sigma = a^2 \left(\frac{4\pi - 1}{4} \right) \sigma$



New position of centre of mass

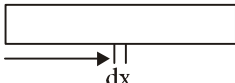
$$X_{CM} = \frac{m_0 x_0 - mx}{m_0 - m} = \frac{\pi a^2 \times 0 - \frac{a^2}{4} \times \frac{a}{2}}{\pi a^2 - \frac{a^2}{4}}$$

$$= \frac{-a^3/8}{\left(\pi - \frac{1}{4}\right)a^2} = \frac{-a}{2(4\pi - 1)} = \frac{-a}{8\pi - 2} = -\frac{a}{23}$$

$\therefore x = 23$

3. (b) Given,

Linear mass density, $\rho(x) = a + b\left(\frac{x}{L}\right)^2$

$$X_{CM} = \frac{\int x dm}{\int dm}$$


$$\int dm = \int_0^L \rho(x) dx$$

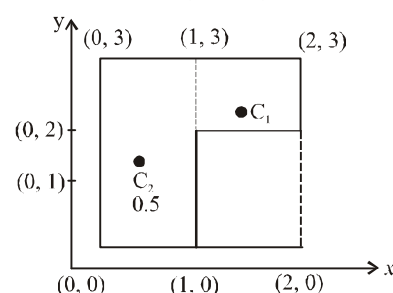
$$= \int_0^L \left[a + b\left(\frac{x}{L}\right)^2 \right] dx = aL + \frac{bL}{3}$$

$$\int_0^L x dm = \int_0^L \left(ax + \frac{bx^3}{L^2} \right) dx = \left(\frac{aL^2}{2} + \frac{bL^2}{4} \right)$$

$$\therefore X_{CM} = \frac{\left(\frac{aL^2}{2} + \frac{bL^2}{4} \right)}{aL + \frac{bL}{3}}$$

$$\Rightarrow X_{CM} = \frac{3L}{4} \left(\frac{2a+b}{3a+b} \right)$$

4. (b)



For given Lamina

$$m_1 = 1, C_1 = (1.5, 2.5)$$

$$m_2 = 3, C_2 = (0.5, 1.5)$$

$$X_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2} = \frac{1.5 + 1.5}{4} = 0.75$$

$$Y_{cm} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2} = \frac{2.5 + 4.5}{4} = 1.75$$

$\therefore$ Coordinate of centre of mass of flag shaped lamina (0.75, 1.75)

5. (a) Mass of sphere = volume of sphere $\times$ density of sphere

$$= \frac{4}{3} \pi R^3 \rho$$

$$\text{Mass of cavity } M_{\text{cavity}} = \frac{4}{3} \pi (1)^3 \rho$$

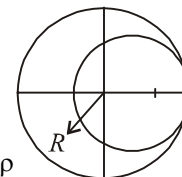
Mass of remaining

$$M_{(\text{Remaining})} = \frac{4}{3} \pi R^3 \rho - \frac{4}{3} \pi (1)^3 \rho$$

Centre of mass of remaining part,

$$X_{COM} = \frac{M_1 r_1 + M_2 r_2}{M_1 + M_2}$$

$$\Rightarrow -(2-R) = \frac{\left[\frac{4}{3} \pi R^3 \rho \right] 0 + \left[\frac{4}{3} \pi (1)^3 (-\rho) \right] [R-1]}{\frac{4}{3} \pi R^3 \rho + \frac{4}{3} \pi (1)^3 (-\rho)}$$

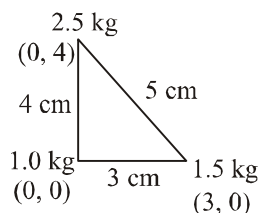


$$\Rightarrow \frac{(R-1)}{(R^3-1)} = 2-R$$

$$\Rightarrow \frac{(R-1)}{(R-1)(R^2+R+1)} = 2-R$$

$$\Rightarrow (R^2+R+1)(2-R) = 1$$

6. (d)



$$X_{cm} = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}$$

$$X_{cm} = \frac{1 \times 0 + 1.5 \times 3 + 2.5 \times 0}{1 + 1.5 + 2.5} = \frac{1.5 \times 3}{5} = 0.9 \text{ cm}$$

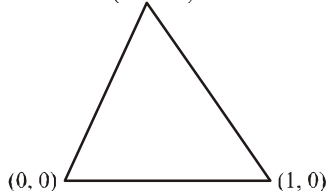
$$Y_{cm} = \frac{m_1y_1 + m_2y_2 + m_3y_3}{m_1 + m_2 + m_3}$$

$$Y_{cm} = \frac{1 \times 0 + 1.5 \times 0 + 2.5 \times 4}{1 + 1.5 + 2.5} = \frac{2.5 \times 4}{5} = 2 \text{ cm}$$

Hence, centre of mass of system is at point (0.9, 2)

$$7. (c) x_{cm} = \frac{50 \times 0 + 100 \times 1 + 150 \times 0.5}{50 + 100 + 150} = \frac{7}{12} \text{ m}$$

$$\left(0.5, \frac{\sqrt{3}}{2}\right)$$



$$y_{cm} = \frac{50 \times 0 + 100 \times 0 + 150 \times \frac{\sqrt{3}}{2}}{50 + 100 + 150} = \frac{\sqrt{3}}{4} \text{ m}$$

8. (a) Acceleration of centre of mass (a_{cm}) is given by

$$\therefore \vec{a}_{cm} = \frac{m_1\vec{a}_1 + m_2\vec{a}_2 + \dots}{m_1 + m_2 + \dots}$$

$$= \frac{(2m)\vec{a}\hat{j} + 3m \times \vec{a}\hat{i} + m\vec{a}(-\hat{i}) + 4m \times \vec{a}(-\hat{j})}{2m + 3m + 4m + m}$$

$$= \frac{2a\hat{i} - 2a\hat{j}}{10} = \frac{a}{5}(\hat{i} - \hat{j})$$

9. (d) With respect to point θ , the CM of the cut-off portion

$$\left(\frac{a}{4}, \frac{b}{4}\right)$$

$$\text{Using, } x_{CM} = \frac{MX - mx}{M - m}$$

$$= \frac{M \times 0 - \frac{M}{4} \times \frac{a}{4}}{M - \frac{M}{4}} = -\frac{a}{12}$$

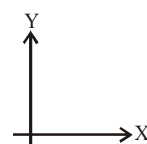
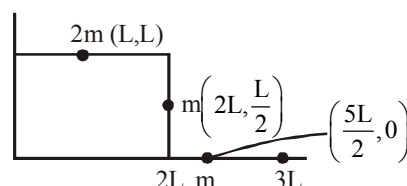
$$\text{and } y_{CM} = -\frac{b}{12}$$

So CM coordinates are

$$x_0 = \frac{a}{2} - \frac{a}{12} = \frac{5a}{12}$$

$$\text{and } y_0 = \frac{b}{2} - \frac{b}{12} = \frac{5b}{12}$$

10. (a)



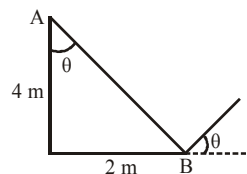
x-coordinate of centre of mass is

$$X_{cm} = \frac{2mL + 2mL + \frac{5mL}{2}}{4m} = \frac{13}{8}L$$

y-coordinate of centre of mass is

$$Y_{cm} = \frac{2m \times L + m \times \left(\frac{L}{2}\right) + m \times 0}{4m} = \frac{5L}{8}$$

11. (c) To produce maximum moment of force line of action of force must be perpendicular to line AB.



$$\therefore \tan \theta = \frac{2}{4} = \frac{1}{2}$$

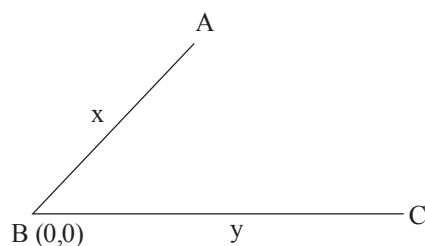
12. (c) According to principle of moments when a system is stable or balance, the anti-clockwise moment is equal to clockwise moment.

i.e., load $\times$ load arm = effort $\times$ effort arm

When 5 mg weight is placed, load arm shifts to left side, hence left arm becomes shorter than right arm.

$$13. (c) \text{ Centre of mass } x_{cm} = \frac{x}{2} \frac{(\rho x) \left(\frac{x}{2}\right) \frac{1}{2} + \rho y^{y/2}}{\rho(x+y)}$$

$$\Rightarrow \frac{1}{2} + \frac{y}{x} = \frac{y^2}{x^2}$$

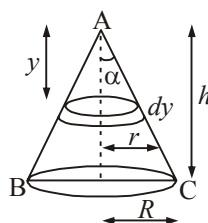


$$\therefore \frac{BC}{AB} \cdot \frac{y}{x} = \frac{1+\sqrt{3}}{2} = 1.37$$

14. (d) Let density of cone = ρ .

$$\text{Centre of mass, } y_{cm} = \frac{\int y dm}{\int dm}$$

$$= \frac{\int_0^h y \pi r^2 dy}{\int_0^h \pi r^2 dy} = \frac{\int_0^h y r^2 dy}{\int_0^h r^2 dy} \quad \dots(i)$$



For a cone, we know that

$$\frac{r}{R} = \frac{y}{h} \quad \therefore r = \frac{y}{h} R$$

$$y_{cm} = \frac{\int_0^h 3y^3 dy}{\int_0^h y^3 dy} = \frac{3 \left[\frac{y^4}{4} \right]_0^h}{\left[\frac{y^4}{4} \right]_0^h} = \frac{3}{4} h$$

15. (b) Centre of mass of the rod is given by:

$$x_{cm} = \frac{\int_0^L (ax + \frac{bx^2}{L}) dx}{\int_0^L (a + \frac{bx}{L}) dx}$$

$$= \frac{\frac{aL^2}{2} + \frac{bL^2}{3}}{aL + \frac{bL}{2}} = \frac{L \left(\frac{a}{2} + \frac{b}{3} \right)}{a + \frac{b}{2}}$$

$$\text{Now } \frac{7L}{12} = \frac{\frac{a}{2} + \frac{b}{3}}{a + \frac{b}{2}}$$

On solving we get, $b = 2a$

16. (c) 17. (d)

18. (a) The linear mass density $\lambda = k \left(\frac{x}{L} \right)^n$

$$\text{Here } \frac{x}{L} \leq 1$$

With increase in the value of n , the centre of mass shift towards the end $x = L$. This is satisfied by only option (a).

$$x_{CM} = \frac{\int_0^L x dm}{\int_0^L dm} = \frac{\int_0^L x (\lambda dx)}{\int_0^L \lambda dx} = \frac{\int_0^L k \left(\frac{x}{L} \right)^n x dx}{\int_0^L k \left(\frac{x}{L} \right)^n dx}$$

$$= \frac{k \left[\frac{x^{n+2}}{(n+2)L^n} \right]_0^L}{k \left[\frac{x^{n+1}}{(n+1)L^n} \right]_0^L} = \frac{L(n+1)}{n+2}$$

$$\text{For } n=0, x_{CM} = \frac{L}{2}; n=1,$$

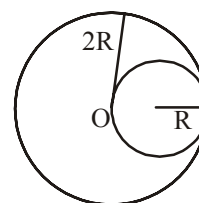
$$x_{CM} = \frac{2L}{3}; n=2, x_{CM} = \frac{3L}{4}; \dots$$

$$\text{For } n \rightarrow \infty, x_{cm} = L$$

Moment of inertia of a square plate about an axis through its centre and perpendicular to its plane is.

19. (b) Let σ be the mass per unit area of the disc.

$$\text{Then the mass of the complete disc} = \sigma(\pi(2R)^2)$$



$$\text{The mass of the removed disc} = \sigma(\pi R^2) = \pi \sigma R^2$$

Let us consider the above situation to be a complete disc of radius $2R$ on which a disc of radius R of negative mass is superimposed. Let O be the origin. Then the above figure can be redrawn keeping in mind the concept of centre of mass as :

$$4\pi\sigma R^2 \quad \xleftarrow{R} \quad \xrightarrow{R} \quad -\pi\sigma R^2$$

$$O \quad \quad \quad O$$

$$x_{c.m} = \frac{(6\pi(2R)^2) \times 0 + (-6(\pi R^2)) R}{4\pi\sigma R^2 - \pi\sigma R^2}$$

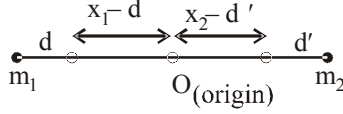
$$\therefore x_{c.m} = \frac{-\pi\sigma R^2 \times R}{3\pi\sigma R^2}$$

$$\therefore x_{c.m} = -\frac{R}{3} = \alpha R \Rightarrow \alpha = \frac{1}{3}$$

20. (c) Initially, m_1 $\xleftarrow{x_1}$ $\xrightarrow{x_2}$ m_2
 $O(\text{origin})$

$$0 = \frac{m_1(-x_1) + m_2x_2}{m_1 + m_2} \Rightarrow m_1x_1 = m_2x_2 \dots (1)$$

Let the particles is displaced through distanced away from centre of mass

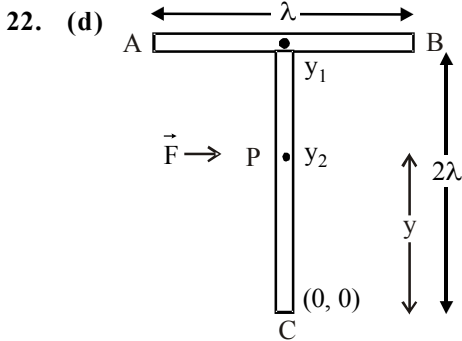


$$\therefore 0 = \frac{m_1(d - x_1) + m_2(x_2 - d')}{m_1 + m_2}$$

$$\Rightarrow 0 = m_1d - m_1x_1 + m_2x_2 - m_2d'$$

$$\Rightarrow d' = \frac{m_1}{m_2}d \quad [\text{From (1).}]$$

21. (a) The centre of mass of bodies B and C taken together does not shift as no external force acts. The centre of mass of the system continues its original path. It is only the internal forces which comes into play while breaking.



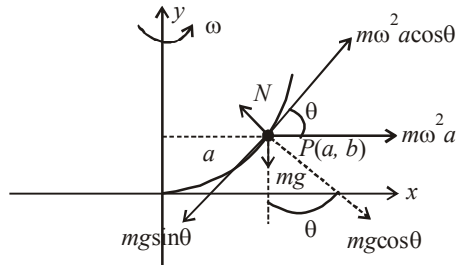
To have translational motion without rotation, the force $\vec{F}$ has to be applied at centre of mass. i.e. the point 'P' has to be at the centre of mass

Taking point C at the origin position, positions of y, and y₂ are $r_1 = 2l$, $r_2 = l$ and $m_1 = m$ and $m_2 = 2m$

$$y = \frac{m_1y_1 + m_2y_2}{m_1 + m_2} = \frac{m \times 2l + 2m \times l}{3m} = \frac{4l}{3}$$

23. (a) $y = 4Cx^2 \Rightarrow \frac{dy}{dx} = \tan \theta = 8Cx$

At P, $\tan \theta = 8Ca$



For steady circular motion

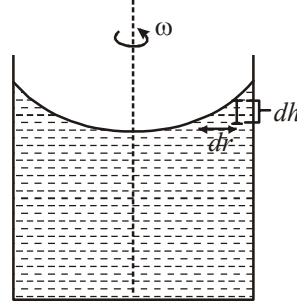
$$m\omega^2 a \cos \theta = mg \sin \theta$$

$$\Rightarrow \omega = \sqrt{\frac{g \tan \theta}{a}}$$

$$\therefore \omega = \sqrt{\frac{g \times 8aC}{a}} = 2\sqrt{2gC}$$

24. (c) Here, $\rho dr \omega^2 r = \rho g dh$

$$\Rightarrow \omega^2 \int_0^R r dr = g \int_0^h dh$$



$$\Rightarrow \frac{\omega^2 R^2}{2} = gh \quad (\text{Given } R = 5 \text{ cm})$$

$$\therefore h = \frac{\omega^2 R^2}{2g} = \frac{25\omega^2}{2g}$$

25. (c) Free body diagram in the frame of disc

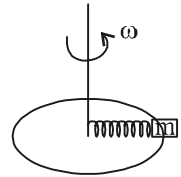
$$\leftarrow kx \quad \boxed{m} \quad \rightarrow m\omega^2(\ell_0 + x)$$

$$\therefore m\omega^2(\ell_0 + x) = kx$$

$$\Rightarrow x = \frac{m\ell_0\omega^2}{k - m\omega^2}$$

For $k \gg m\omega^2$

$$\Rightarrow \frac{x}{\ell_0} = \frac{m\omega^2}{k}$$



26. (b) At elongated position (x),

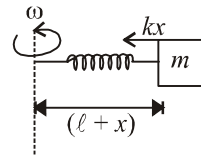
$$F_{\text{radial}} = \frac{mv^2}{r} = mr\omega^2$$

$$\therefore kx = m(\ell + x)\omega^2$$

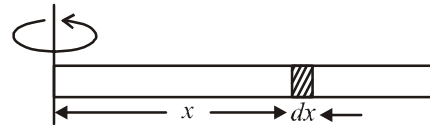
($\because r = \ell + x$ here)

$$kx = m\ell\omega^2 + mx\omega^2$$

$$\therefore x = \frac{m\ell\omega^2}{k - m\omega^2}$$



27. (d) $\int_0^T (-dT) = \int_l^x (dm)\omega^2 x$

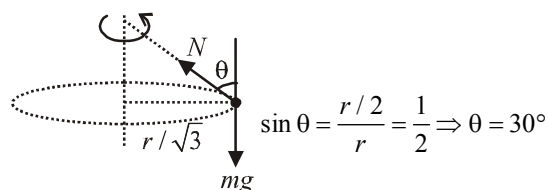


$$-T = \int_l^x \left(\frac{m}{l} dx \right) \omega^2 x$$

$$\text{or } T = \frac{m\omega^2}{l} (l^2 - x^2)$$

It is a parabola between T and x.

28. (b) $N \sin \theta = m\omega^2 (r/2)$... (i)



and $N \cos \theta = mg$... (ii)

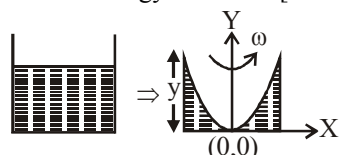
or $\tan \theta = \frac{\omega^2 r}{2g}$

or $\tan 30^\circ = \frac{\omega^2 r}{2g}$

or $\frac{1}{\sqrt{3}} = \frac{\omega^2 r}{2g}$

$$\therefore \omega^2 = \frac{2g}{r\sqrt{3}}$$

29. (a) Using $v^2 = u^2 + 2gy$ [$\therefore u = 0$ at $(0,0)$]
 $v^2 = 2gy$ [$\therefore v = \omega x$]



$$\Rightarrow y = \frac{\omega^2 x^2}{2g} = \frac{(2 \times 2\pi)^2 \times (0.05)^2}{20} \approx 2 \text{ cm}$$

30. (c) $m\omega^2 R = \text{Force} \propto \frac{1}{R^n}$ (Force = $\frac{mv^2}{R}$)

$$\Rightarrow \omega^2 \propto \frac{1}{R^{n+1}} \Rightarrow \omega \propto \frac{1}{R^{\frac{n+1}{2}}}$$

Time period $T = \frac{2\pi}{\omega}$

Time period, $T \propto R^{\frac{n+1}{2}}$

31. (b) decreasing speed

32. (c) Torque at angle θ

$$\tau = Mg \sin \theta \cdot \frac{l}{2}$$

Also $\tau = I\alpha$

$$\therefore I\alpha = Mg \sin \theta \cdot \frac{l}{2}$$

$$\frac{Ml^2}{3} \cdot \alpha = Mg \sin \theta \cdot \frac{l}{2} \quad \left[\because I_{rod} = \frac{Ml^2}{3} \right]$$

$$\Rightarrow \frac{l\alpha}{3} = g \frac{\sin \theta}{2} \quad \therefore \alpha = \frac{3g \sin \theta}{2l}$$

33. (a) For just complete rotation

$$v = \sqrt{Rg} \text{ at top point}$$

The rotational speed of the drum

$$\Rightarrow \omega = \frac{v}{R} = \sqrt{\frac{g}{R}} = \sqrt{\frac{10}{1.25}}$$

The maximum rotational speed of the drum in revolutions per minute

$$\omega(\text{rpm}) = \frac{60}{2\pi} \sqrt{\frac{10}{1.25}} = 27$$

34. (b) Angular momentum, $mvr = I\omega$

Moment of Inertia (I) of cubical block is given by

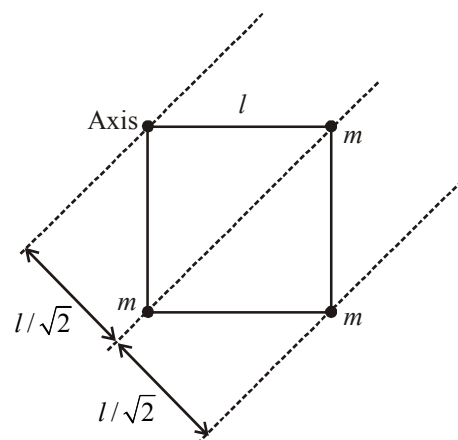
$$I = m \left(\frac{R^2}{6} + \left(\frac{R}{\sqrt{2}} \right)^2 \right) \quad \therefore \omega = \frac{m \cdot 2 \frac{R}{2}}{m \left[\frac{R^2}{6} + \left(\frac{R}{\sqrt{2}} \right)^2 \right]}$$

$$\Rightarrow \omega = \frac{12}{8R} = \frac{3}{2 \times 0.3} = \frac{10}{2} = 5 \text{ rad/s}$$

35. (b) Angular velocity is the angular displacement per unit time i.e., $\omega = \frac{\Delta \theta}{\Delta t}$

 Here $\omega_1 = \omega_2$ and independent of f .

36. (c) Angular momentum, $L = I\omega$

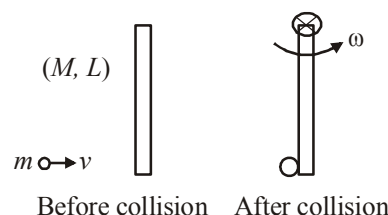


$$I = m(0)^2 + m \left(\frac{l}{\sqrt{2}} \right)^2 \times 2 + m(\sqrt{2}l)^2$$

$$= \frac{2ml^2}{2} + 2ml^2 = 3ml^2$$

Angular momentum $L = I\omega = 3ml^2\omega$

37. (20)



Using principle of conservation of angular momentum we have

$$\vec{L}_i = \vec{L}_f \Rightarrow mvL = I\omega$$

$$\Rightarrow mvL = \left(\frac{ML^2}{3} + mL^2 \right) \omega$$

$$\Rightarrow 0.1 \times 80 \times 1 = \left(\frac{0.9 \times 1^2}{3} + 0.1 \times 1^2 \right) \omega$$

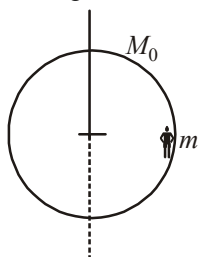
$$\Rightarrow 8 = \left(\frac{3}{10} + \frac{1}{10} \right) \omega \Rightarrow 8 = \frac{4}{10} \omega$$

$$\Rightarrow \omega = 20 \text{ rad/sec.}$$

38. (9.00)

Here $M_0 = 200 \text{ kg}$, $m = 80 \text{ kg}$

Using conservation of angular momentum, $L_i = L_f$



$$I_1 \omega_1 = I_2 \omega_2$$

$$I_1 = (I_M + I_m) = \left(\frac{M_0 R^2}{2} + m R^2 \right)$$

$$I_2 = \frac{1}{2} M_0 R^2 \text{ and } \omega_1 = 5 \text{ rpm}$$

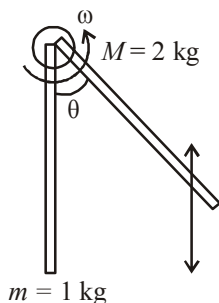
$$\therefore \omega_2 = \left(\frac{M_0 R^2}{2} + m R^2 \right) \times \frac{5}{\frac{M_0 R^2}{2}}$$

$$= \frac{5R^2}{R^2} \times \frac{(80+100)}{100} = 9 \text{ rpm.}$$

39. (a) Using conservation of angular momentum

$$mvl = \left(ml^2 + \frac{2ml^2}{3} \right) \omega \Rightarrow mvl = \frac{5}{3} ml^2 \omega \Rightarrow \omega = \frac{3v}{5l}$$

$$\text{or, } \omega = \frac{3 \times 6}{5 \times 1} = \frac{18}{5} \text{ rad/s}$$



Now using energy conservation, after collision

$$\frac{1}{2} I \omega^2 = 2mg \frac{l}{2} (1 - \cos \theta) + mgl(1 - \cos \theta)$$

$$\Rightarrow \frac{1}{2} \left(\frac{5}{3} ml^2 \right) \frac{9v^2}{25l^2} = 2mgl(1 - \cos \theta)$$

$$\Rightarrow \frac{3}{5 \times 2} mv^2 = 2mgl(1 - \cos \theta)$$

$$\frac{3}{10} \times \frac{36}{2 \times 10} = 1 - \cos \theta \Rightarrow 1 - \frac{27}{50} = \cos \theta$$

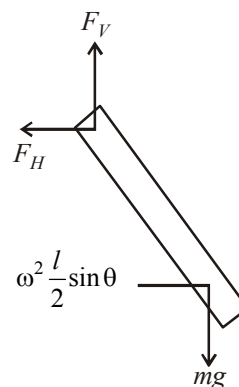
$$\text{or, } \cos \theta = \frac{23}{50} \quad \therefore \theta \approx 63^\circ.$$

40. (d) Vertical force = mg

$$\text{Horizontal force} = \text{Centripetal force} = m\omega^2 \frac{l}{2} \sin \theta$$

$$\text{Torque due to vertical force} = mg \frac{l}{2} \sin \theta$$

$$\text{Torque due to horizontal force} = m\omega^2 \frac{l}{2} \sin \theta \frac{l}{2} \cos \theta$$



Net Torque = Angular momentum

$$mg \frac{l}{2} \sin \theta - m\omega^2 \frac{l}{2} \sin \theta \frac{l}{2} \cos \theta = \frac{ml^2}{12} \omega^2 \sin \theta \cos \theta$$

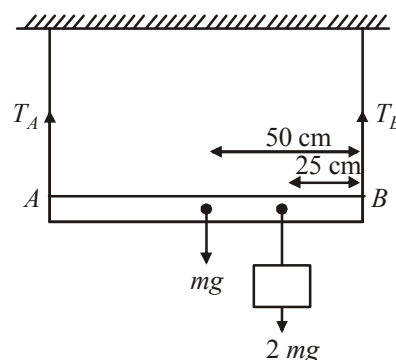
$$\Rightarrow \cos \theta = \frac{3}{2} \frac{g}{\omega^2 l}$$

41. (d) Net torque, τ_{net} about B is zero at equilibrium

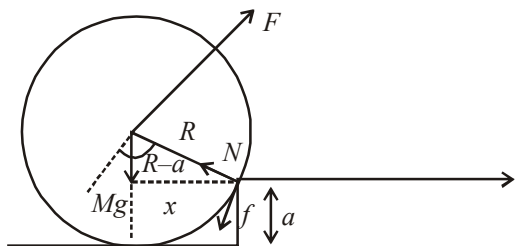
$$\therefore T_A \times 100 - mg \times 50 - 2mg \times 25 = 0$$

$$\Rightarrow T_A \times 100 = 100mg$$

$$\Rightarrow T_A = 1mg \text{ (Tension in the string at A)}$$



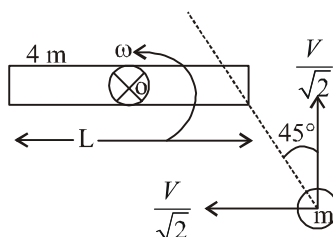
42. (a)


 For step up, $F \times R \geq Mg \times x$

$$x = \sqrt{R^2 - (R-a)^2} \text{ from figure}$$

$$F_{\min} = \frac{Mg}{R} \times \sqrt{R^2 - (R-a)^2} = Mg \sqrt{1 - \left(\frac{R-a}{R}\right)^2}$$

43. (c)



About point O angular momentum

$$L_{\text{initial}} = L_{\text{final}}$$

$$\Rightarrow \frac{mV}{\sqrt{2}} \times \frac{1}{2} = \left[\frac{4mL^2}{12} + \frac{mL^2}{4} \right] \times \omega$$

$$\therefore \omega = \frac{6V}{7\sqrt{2}L} = \frac{3\sqrt{2}V}{7L}$$

 44. (b) Given that, $x = x_0 + a \cos \omega_1 t$
 $y = y_0 + b \sin \omega_2 t$

$$\frac{dx}{dt} = V_x$$

$$\Rightarrow v_x = -a\omega_1 \sin(\omega_1 t), \text{ and } \frac{dy}{dt} = v_y = b\omega_2 \cos(\omega_2 t)$$

$$\frac{dv_x}{dt} = a_x = -a\omega_1^2 \cos(\omega_1 t), \quad \frac{dv_y}{dt} = a_y = -b\omega_2^2 \sin(\omega_2 t)$$

$$\text{At } t = 0, x = x_0 + a, y = y_0$$

$$a_x = -a\omega_1^2, a_y = 0$$

$$\text{Now, } \vec{\tau} = \vec{r} \times \vec{F} = m(\vec{r} \times \vec{a})$$

$$= [(x_0 + a)\hat{i} + y_0\hat{j}] \times m(-a\omega_1^2\hat{i}) = +m y_0 a \omega_1^2 \hat{k}$$

 45. (d) We have given $\vec{r} = 2t\hat{i} - 3t^2\hat{j}$

$$\vec{r}(\text{at } t = 2) = 4\hat{i} - 12\hat{j}$$

$$\text{Velocity, } \vec{v} = \frac{d\vec{r}}{dt} = 2\hat{i} - 6t\hat{j}$$

$$\vec{v}(\text{at } t = 2) = 2\hat{i} - 12\hat{j}$$

$$\vec{L} = mvr \sin \theta \hat{n} = m(\vec{r} \times \vec{v})$$

$$= 2(4\hat{i} - 12\hat{j}) \times (2\hat{i} - 12\hat{j}) = -48\hat{k}$$

46. (d) Angular acceleration,

$$\alpha = \frac{\omega - \omega_0}{t} = \frac{25 \times 2\pi - 0}{5} = 10\pi \text{ rad/s}^2$$

$$\tau = I\alpha$$

$$\Rightarrow \tau = \left(\frac{5}{4} mR^2 \right) \alpha \approx \left(\frac{5}{4} \right) (5 \times 10^{-3}) (10^{-4}) 10\pi$$

$$= 2.0 \times 10^{-5} \text{ Nm}$$

47. (a)

48. (c) According to work-energy theorem

$$mgh = \frac{1}{2}mv_B^2 - \frac{1}{2}mv_A^2$$

$$2gh = v_B^2 - v_A^2$$

$$2 \times 10 \times 10 = v_B^2 - 5^2$$

$$\Rightarrow v_B = 15 \text{ m/s}$$

Angular momentum about O,

$$L_O = mvr$$

$$= 20 \times 10^{-3} \times 20$$

$$L_O = 6 \text{ kg.m}^2/\text{s}$$

 49. (a) Given, $\vec{F}_1 = \frac{F}{2}(-\hat{i}) + \frac{F\sqrt{3}}{2}(-\hat{j})$

$$\vec{r}_1 = 0\hat{i} + 6\hat{j}$$

 Torque due to F_1 force

$$\vec{\tau}_{F_1} = \vec{r}_1 \times \vec{F}_1 = 6\hat{j} \times \left(\frac{F}{2}(-\hat{i}) + \frac{F\sqrt{3}}{2}(-\hat{j}) \right) = 3F(\hat{k})$$

 Torque due to F_2 force

$$\vec{\tau}_{F_2} = (2\hat{i} + 3\hat{j}) \times F\hat{k} = 3F\hat{i} + 2F(-\hat{j})$$

$$\vec{\tau}_{\text{net}} = \vec{\tau}_{F_1} + \vec{\tau}_{F_2} = 3F\hat{i} + 2F(-\hat{j}) + 3F(\hat{k})$$

$$= (3\hat{i} - 2\hat{j} + 3\hat{k})F$$

 50. (a) Torque about the origin = $\vec{\tau} = \vec{r} \times \vec{F}$

$$= r F \sin \theta \Rightarrow 2.5 = 1 \times 5 \sin \theta$$

$$\sin \theta = 0.5 = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{6}$$

51. (d) Consider a strip of radius x and thickness dx,

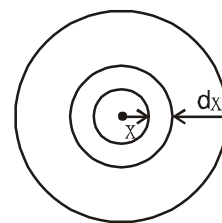
Torque due to friction on this strip

 Net torque = $\sum$ Torque on ring

$$\int d\tau = \int_0^R \frac{\mu F \cdot 2\pi x dx}{\pi R^2}$$

$$\Rightarrow \tau = \frac{2\mu F R^3}{R^2 \cdot 3}$$

$$\tau = \frac{2\mu FR}{3}$$



52. (a) Applying torque equation about point P.

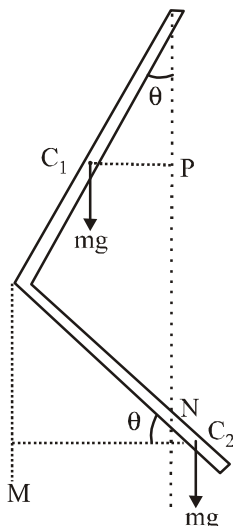
$$\tau = I\alpha = [2M_0(2\ell)^2 + 5M_0\ell^2]\alpha$$

$$\Rightarrow 5M_0 g \ell - 4M_0 g \ell = [2M_0(2\ell)^2 + 5M_0 \ell^2] \alpha$$

$$\Rightarrow M_0 g \ell = (13M_0 g \ell^2) \alpha$$

$$\therefore \alpha = \frac{g}{13\ell}$$

53. (d) Given that, the rod is of uniform mass density and $AB = BC$



Let mass of one rod is m .

Balancing torque about hinge point.

$$mg(C_1P) = mg(C_2N)$$

$$mg\left(\frac{L}{2}\sin\theta\right) = mg\left(\frac{L}{2}\cos\theta - L\sin\theta\right)$$

$$\Rightarrow \frac{3}{2}mgL\sin\theta = \frac{mgL}{2}\cos\theta$$

$$\Rightarrow \frac{\sin\theta}{\cos\theta} = \frac{1}{3} \text{ or, } \tan\theta = \frac{1}{3}$$

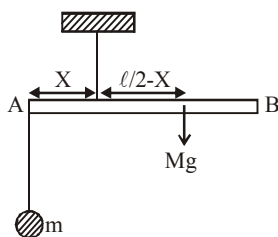
54. (a) Balancing torque w.r.t. point of suspension

$$mgx = Mg\left(\frac{\ell}{2} - x\right)$$

$$\Rightarrow mx = M\frac{\ell}{2} - Mx$$

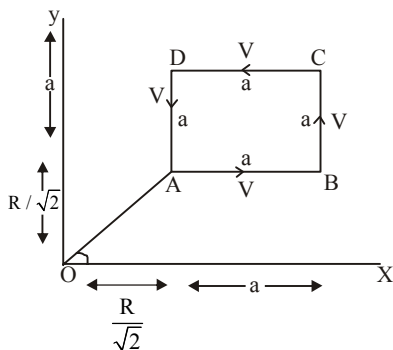
$$m = \left(M\frac{\ell}{2}\right)\frac{1}{x} - M$$

$$y = \alpha\frac{1}{x} - C \quad \text{Straight line equation.}$$



55. (a)

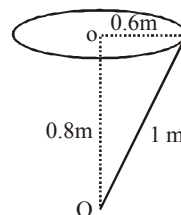
56. (a) We know that $|L| = mvr_{\perp}$



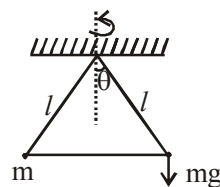
In none of the cases, the perpendicular

distance $r_{\perp}$ is $\left(\frac{R}{\sqrt{2}} + a\right)$

57. (a) Angular momentum,
 $L_0 = mvr \sin 90^\circ$
 $= 2 \times 0.6 \times 12 \times 1 \times 1$
 [As $V = r\omega$, $\sin 90^\circ = 1$]
 So, $L_0 = 14.4 \text{ kgm}^2/\text{s}$



58. (c) Torque working on the bob of mass m is, $\tau = mg \times \ell \sin \theta$. (Direction parallel to plane of rotation of particle)



As τ is perpendicular to $\vec{L}$, direction of L changes but magnitude remains same.

59. (c) Given : $m = 0.160 \text{ kg}$
 $\theta = 60^\circ$
 $v = 10 \text{ m/s}$

$$\text{Angular momentum } L = \vec{r} \times m \vec{v}$$

$$= H m v \cos \theta$$

$$= \frac{v^2 \sin^2 \theta}{2g} \cos \theta \quad \left[H = \frac{v^2 \sin^2 \theta}{2g} \right]$$

$$= \frac{10^2 \times \sin^2 60^\circ \times \cos 60^\circ}{2 \times 10}$$

$$= 3.46 \text{ kg m}^2/\text{s}$$

60. (a)

61. (a) Angular momentum $L = m(v \times r)$

$$= 2 \text{ kg} \left(\frac{dr}{dt} \times r \right) = 2 \text{ kg} (4t \hat{j} \times 5i - 2t^2 \hat{j})$$

$$= 2 \text{ kg} (-20t \hat{k}) = 2 \text{ kg} \times -20 \times 2 \text{ m}^2 \text{ s}^{-1} \hat{k}$$

$$= -80 \hat{k}$$

62. (b) 63. (d)

64. (c) Angular momentum, $L = I\omega \Rightarrow L = mr^2\omega$

As insect moves along a diameter, the effective mass and hence moment of inertia (I) first decreases then increases so from principle of conservation of angular momentum, angular speed ω first increases then decreases.

65. (c) $\vec{L} = m(\vec{r} \times \vec{v})$

$$\vec{L} = m \left[v_0 \cos \theta t \hat{i} + \left(v_0 \sin \theta t - \frac{1}{2} g t^2 \right) \hat{j} \right]$$

$$\begin{aligned}
 & \times [v_0 \cos \theta \hat{i} + (v_0 \sin \theta - gt) \hat{j}] \\
 & = mv_0 \cos \theta \left[-\frac{1}{2}gt \right] \hat{k} \\
 & = -\frac{1}{2}mgv_0 t^2 \cos \theta \hat{k}
 \end{aligned}$$

66. (d) We know Torque $\vec{\tau}_c = \frac{d\vec{L}_c}{dt}$

where $\vec{L}_c$ = Angular momentum about the center of mass of the body. Central forces act along the center of mass. Therefore torque about center of mass is zero.

$$\therefore \tau = \frac{dL}{dt} = 0 \quad \Rightarrow \quad \vec{L}_c = \text{constt.}$$

67. (d) Applying conservation of angular momentum $I'\omega' = I\omega$
 $(mR^2 + 2MR^2)\omega' = mR^2\omega$
 $\Rightarrow (m + 2M)R^2\omega' = mR^2\omega$

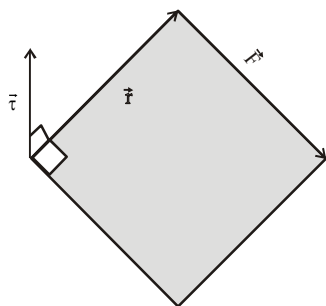
$$\Rightarrow \omega' = \omega \left[\frac{m}{m + 2M} \right]$$

68. (c) Torque $\vec{\tau} = \vec{r} \times \vec{F} = (\hat{i} - \hat{j}) \times (-F\hat{k})$
 $= F[-\hat{i} \times \hat{k} + \hat{j} \times \hat{k}] = F(\hat{j} + \hat{i}) = F(\hat{i} + \hat{j})$

$$\left[\text{Since } \hat{k} \times \hat{i} = \hat{j} \text{ and } \hat{j} \times \hat{k} = \hat{i} \right]$$

69. (b) Angular momentum will remain the same since no external torque act in free space.

70. (d) We know that $\vec{\tau} = \vec{r} \times \vec{F}$



Vector $\vec{\tau}$ is perpendicular to both $\vec{r}$ and $\vec{F}$. We also know that the dot product of two vectors which have an angle of 90° between them is zero.

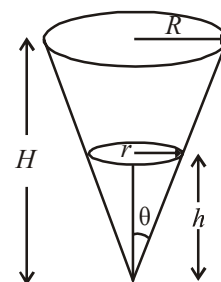
$$\therefore \vec{r} \cdot \vec{\tau} = 0 \text{ and } \vec{F} \cdot \vec{\tau} = 0$$

71. (d) Angular momentum (L)
 $= (\text{linear momentum}) \times (\text{perpendicular distance of the line of action of momentum from the axis of rotation})$
 As the particle moves with velocity V along line PC , the line of motion passes through P .

$$\begin{aligned}
 \therefore L &= mv \times r \\
 &= mv \times 0 \\
 &= 0
 \end{aligned}$$

72. (d) Hollow ice-cream cone can be assume as several parts of discs having different radius, so

$$I = \int dI = \int dm(r^2) \quad \dots(i)$$



From diagram,

$$\frac{r}{h} = \tan \theta = \frac{R}{H} \text{ or } r = \frac{R}{H}h \quad \dots(ii)$$

$$\text{Mass of element, } dm = \rho(\pi r^2)dh \quad \dots(iii)$$

From eq. (i), (ii) and (iii),

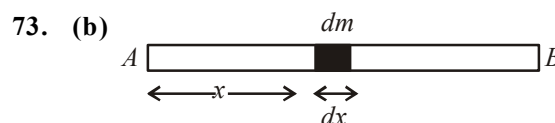
$$\text{Area of element, } dA = 2\pi r dl = 2\pi r \frac{dh}{\cos \theta}$$

$$\text{Mass of element, } dm = \frac{2Mh \tan \theta}{R\sqrt{R^2 + H^2} \cos \theta}$$

(here, $r = h \tan \theta$)

$$\begin{aligned}
 I &= \int dI = \int_0^H dm(r^2) = \int_0^H \rho(\pi r^2)dh \left(\frac{R}{H} \cdot h \right)^2 \\
 &= \int_0^H \rho \left(\pi \left(\frac{R}{H} \cdot h \right)^4 \right) dh
 \end{aligned}$$

$$\text{Solving we get, } I = \frac{MR^2}{2}$$



Mass of the small element of the rod

$$dm = \lambda \cdot dx$$

Moment of inertia of small element,

$$dI = dm \cdot x^2 = \lambda_0 \left(1 + \frac{x}{L} \right) \cdot x^2 dx$$

Moment of inertia of the complete rod can be obtained by integration

$$\begin{aligned}
 I &= \lambda_0 \int_0^L \left(x^2 + \frac{x^3}{L} \right) dx \\
 &= \lambda_0 \left[\frac{x^3}{3} + \frac{x^4}{4L} \right]_0^L = \lambda_0 \left[\frac{L^3}{3} + \frac{L^3}{4} \right]
 \end{aligned}$$

$$\Rightarrow I = \frac{7\lambda_0 L^3}{12} \quad \dots(i)$$

Mass of the thin rod,

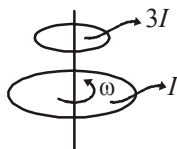
$$M = \int_0^L \lambda dx = \int_0^L \lambda_0 \left(1 + \frac{x}{L}\right) dx = \frac{3\lambda_0 L}{2}$$

$$\therefore \lambda_0 = \frac{2M}{3L}$$

$$\therefore I = \frac{7}{12} \left(\frac{2M}{3L} \right) L^3 \Rightarrow I = \frac{7}{18} ML^2$$

74. (d) By angular momentum conservation, $L_c = L_f$

$$\omega I + 3I \times 0 = 4I \omega' \Rightarrow \omega' = \frac{\omega}{4}$$



$$(KE)_i = \frac{1}{2} I \omega^2$$

$$(KE)_f = \frac{1}{2} (3I + I) \omega'^2$$

$$= \frac{1}{2} \times (4I) \times \left(\frac{\omega}{4} \right)^2 = \frac{I \omega^2}{8}$$

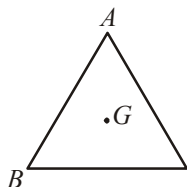
$$\Delta KE = \frac{1}{2} I \omega^2 - \frac{1}{8} I \omega^2 = \frac{3}{8} I \omega^2$$

$$\therefore \text{Fractional loss in K.E.} = \frac{\Delta KE}{KE_i} = \frac{\frac{3}{8} I \omega^2}{\frac{1}{2} I \omega^2} = \frac{3}{4}$$

75. (11) Let mass of triangular lamina = m , and length of side = l , then moment of inertia of lamina about an axis passing through centroid G perpendicular to the plane.

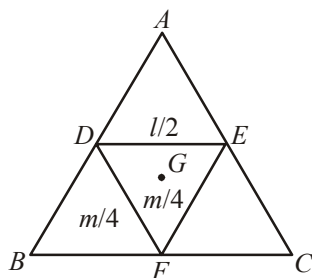
$$I_0 \propto ml^2$$

$$I_0 = kml^2$$



Let moment of inertia of $DEF = I_1$ about G

$$\text{So, } I_1 \propto \left(\frac{m}{4} \right) \left(\frac{l}{2} \right)^2 \propto \frac{ml^2}{16} \text{ or } I_1 = \frac{I_0}{16}$$



$$\text{Let } I_{ADE} = I_{BDF} = I_{EFC} = I_2$$

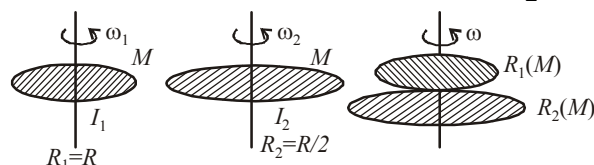
$$\therefore 3I_2 + I_1 = I_0 \Rightarrow 3I_2 + \frac{I_0}{16} = I_0 \Rightarrow I_2 = \frac{5I_0}{16}$$

Hence, moment of inertia of $DECB$ i.e., after removal part ADE

$$= 2I_2 + I_1 = 2 \left(\frac{5I_0}{16} \right) + \left(\frac{I_0}{16} \right) = \frac{11I_0}{16} = \frac{NI_0}{16}$$

Therefore value of $N = 11$.

76. (20) As we know moment of inertia disc, $I_{\text{disc}} = \frac{1}{2} MR^2$



Using angular momentum conservation

$$I_1 \omega_1 + I_2 \omega_2 = (I_1 + I_2) \omega_f$$

$$\frac{MR^2}{2} \times \omega + 0 = \left(\frac{MR^2}{2} + \frac{MR^2}{8} \right) \omega_f \Rightarrow \omega_f = \frac{4}{5} \omega$$

$$\text{Initial K.E., } K_i = \frac{1}{2} I \omega^2 = \frac{1}{2} \left(\frac{MR^2}{2} \right) \omega^2 = \frac{MR^2 \omega^2}{4}$$

$$\text{Final K.E., } K_f = \frac{1}{2} \left(\frac{MR^2}{2} + \frac{MR^2}{8} \right) \frac{16}{25} \omega^2 = \frac{MR^2 \omega^2}{5}$$

Percentage loss in kinetic energy % loss

$$= \frac{\frac{MR^2 \omega^2}{4} - \frac{MR^2 \omega^2}{5}}{\frac{MR^2 \omega^2}{4}} \times 100 = 20\% = P\%$$

Hence, value of $P = 20$.

77. (c) Let p be the density of the discs and t is the thickness of discs.

Moment of inertia of disc is given by

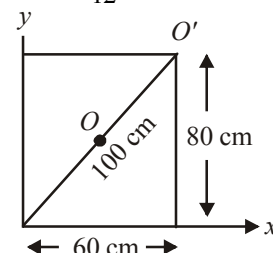
$$I = \frac{MR^2}{2} = \frac{[\rho(\pi R^2)t]R^2}{2}$$

$$I \propto R^4 \quad (\text{As } p \text{ and } t \text{ are same})$$

$$\frac{I_2}{I_1} = \left(\frac{R_2}{R_1} \right)^4 \Rightarrow \frac{16}{1} = \alpha^4 \Rightarrow \alpha = 2$$

78. (b) Moment of inertia of rectangular sheet about an axis passing through O ,

$$I_O = \frac{M}{12} (a^2 + b^2) = \frac{M}{12} [(80)^2 + (60)^2]$$

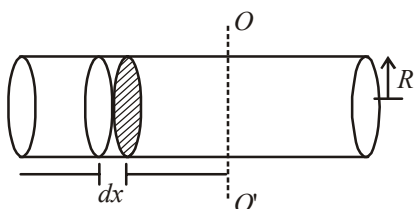


From the parallel axis theorem, moment of inertia about O' ,

$$I_{O'} = I_O + M(50)^2$$

$$\frac{I_O}{I_{O'}} = \frac{\frac{M}{12}(80^2 + 60^2)}{\frac{M}{12}(80^2 + 60^2) + M(50)^2} = \frac{1}{4}$$

79. (c) Let there be a cylinder of mass m length L and radius R . Now, take elementary disc of radius R and thickness dx at a distance of x from axis OO' then moment of inertia about OO' of this element.



$$dI = \frac{dmR^2}{4} + dm x^2$$

$$\Rightarrow I = \int dI = \int \frac{dmR^2}{4} + \int_{n=L/2}^{n=-L/2} \frac{M}{L} dx \times x^2$$

$$\text{Given : } I = \frac{MR^2}{4} + \frac{ML^2}{12}$$

$$\Rightarrow I = \frac{M}{4} \times \frac{V}{\pi L} + \frac{ML^2}{12} \Rightarrow I = \frac{MV}{4\pi L} + \frac{ML^2}{12}$$

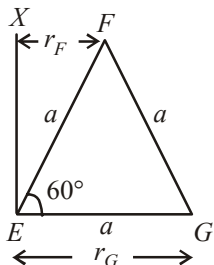
$$\frac{dI}{dL} = -\frac{mV}{4\pi L^2} + \frac{M \times 2L}{12} = 0$$

$$\Rightarrow V = \frac{2}{3} \pi L^3 \Rightarrow \pi R^2 L = \frac{2}{3} \pi L^3$$

$$\therefore \frac{L}{R} = \sqrt{\frac{3}{2}}$$

80. (25)

Moment of inertia of the system about axis XE .



$$I = I_E + I_F + I_G$$

$$\Rightarrow I = m(r_E)^2 + m(r_F)^2 + m(r_G)^2$$

$$\Rightarrow I = m \times 0^2 + m \left(\frac{a}{2} \right)^2 + ma^2 = \frac{5}{4} ma^2 = \frac{25}{20} ma^2$$

$$\therefore N = 25.$$

81. (b) Initial angular momentum = $I_1\omega_1 + I_2\omega_2$

Let ω be angular speed of the combined system.

Final angular momentum = $I_1\omega + I_2\omega$

According to conservation of angular momentum

$$(I_1 + I_2)\omega = I_1\omega_1 + I_2\omega_2$$

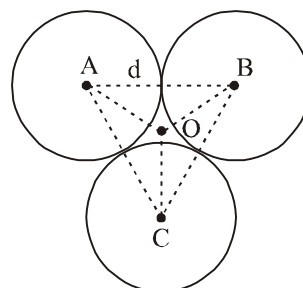
$$\Rightarrow \omega = \frac{I_1\omega_1 + I_2\omega_2}{I_1 + I_2} = \frac{0.1 \times 10 + 0.2 \times 5}{0.1 + 0.2} = \frac{20}{3}$$

Final rotational kinetic energy

$$K_f = \frac{1}{2} I_1 \omega^2 + \frac{1}{2} I_2 \omega^2 = \frac{1}{2} (0.1 + 0.2) \times \left(\frac{20}{3} \right)^2$$

$$\Rightarrow K_f = \frac{20}{3} \text{ J}$$

82. (a) Moment of inertia,



$$I_1 = \frac{2}{5} m \left(\frac{d}{2} \right)^2 + m(AO)^2$$

$$\text{and } AO = \frac{d}{\sqrt{3}}$$

Moment of inertia about 'O'

$$I_0 = 3I_1 = 3 \left[\frac{2}{5} m \left(\frac{d}{2} \right)^2 + m \left(\frac{d}{\sqrt{3}} \right)^2 \right]$$

$$\Rightarrow I_0 = \frac{13}{10} Md^2$$

$$\text{And } I_A = 2 \left[\frac{2}{5} M \left(\frac{d}{2} \right)^2 + Md^2 \right] + \frac{2}{5} M \left(\frac{d}{2} \right)^2$$

$$\Rightarrow I_A = \frac{23}{10} Md^2$$

$$\therefore \frac{I_O}{I_A} = \frac{\frac{13}{10} Md^2}{\frac{23}{10} Md^2} = \frac{13}{23}$$

83. (15) Here, length of bar, $l = 1 \text{ m}$

angle, $\theta = 30^\circ$

$$\Delta PE = \Delta KE \text{ or } mgh = \frac{1}{2} I \omega^2$$

$$\Rightarrow (mg) \frac{l}{2} \sin 30^\circ = \frac{1}{2} \left(\frac{ml^2}{3} \right) \omega^2$$

$$\Rightarrow mg \frac{l}{2} \times \frac{1}{2} = \frac{1}{2} \left(\frac{ml^2}{3} \right) \omega^2$$

$$\Rightarrow \omega = \sqrt{15} \text{ rad/s}$$

84. (a) Using principal of conservation of energy

$$(m_1 - m_2)gh = \frac{1}{2}(m_1 + m_2)v^2 + \frac{1}{2}I\omega^2$$

$$\Rightarrow (m_1 - m_2)gh = \frac{1}{2}(m_1 + m_2)(\omega R)^2 + \frac{1}{2}I\omega^2$$

($\because v = \omega R$)

$$\Rightarrow (m_1 - m_2)gh = \frac{\omega^2}{2} [(m_1 + m_2)R^2 + I]$$

$$\Rightarrow \omega = \sqrt{\frac{2(m_1 - m_2)gh}{(m_1 + m_2)R^2 + I}}$$

85. (a) When the bob covered a distance 'h'

$$\text{Using } mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$= \frac{1}{2}m(\omega r)^2 + \frac{1}{2} \times \frac{mr^2}{2} \times \omega^2 \quad (\because v = \omega r \text{ no slipping})$$

$$\Rightarrow mgh = \frac{3}{4}m\omega^2 r^2$$

$$\Rightarrow \omega = \sqrt{\frac{4gh}{3r^2}} = \frac{1}{r} \sqrt{\frac{4gh}{3}}$$

86. (c) Moment inertia of the rod passing through a point

$\frac{\ell}{4}$ away from the centre of the rod

$$I = I_g + m\ell^2$$

$$\Rightarrow I = \frac{M\ell^2}{12} + M \times \left(\frac{\ell^2}{16} \right) = \frac{7M\ell^2}{48}$$

$$\text{Using } I = MK^2 = \frac{7M\ell^2}{48} \quad (K = \text{radius of gyration})$$

$$\Rightarrow K = \sqrt{\frac{7}{48}}\ell$$

87. (a) Given,

mass per unit area of circular disc, $\sigma = A + Br$

Area of the ring = $2\pi r dr$

Mass of the ring, $dm = \sigma 2\pi r dr$

Moment of inertia,

$$I = \int dm r^2 = \int \sigma 2\pi r dr r^2$$

$$\Rightarrow I = 2\pi \int_0^a (A + Br) r^3 dr = 2\pi \left[\frac{Aa^4}{4} + \frac{Ba^5}{5} \right]$$

$$\Rightarrow I = 2\pi a^4 \left[\frac{A}{4} + \frac{Ba}{5} \right]$$

$$88. (c) I = \int_a^b (dm) r^2$$

$$= \int_a^b \left(\frac{\sigma_0}{r} \times 2\pi r dr \right) r^2 = \frac{2\pi\sigma_0}{3} \left[r^3 \right]_a^b$$

$$= \frac{2\pi\sigma_0}{3} (b^3 - a^3)$$

Mass of the disc,

$$m = \int_a^b \frac{\sigma_0}{r} \times 2\pi r dr = 2\pi\sigma_0 (b - a)$$

Radius of gyration,

$$k = \sqrt{\frac{I}{m}}$$

$$= \sqrt{\frac{(2\pi\sigma_0/3)(b^3 - a^3)}{2\pi\sigma_0(b - a)}} = \sqrt{\frac{a^2 + b^2 + ab}{3}}$$

89. (d) As no external torque is acting so angular momentum should be conserved

$(I_1 + I_2) \omega = I_1 \omega_1 + I_2 \omega_2$ [ω_c = common angular velocity of the system, when discs are in contact]

$$\omega_c = \frac{I_1 \omega_1 + \frac{I_1 \omega_1}{4}}{I_1 + \frac{I_1}{2}} \left(\frac{5}{4} \times \frac{2}{3} \right) \omega_1$$

$$\omega_c = \frac{5\omega_1}{6}$$

$$E_f - E_i = \frac{1}{2}(I_1 + I_2) \omega_c^2 - \frac{1}{2}I_1 \omega_1^2 - \frac{1}{2}I_2 \omega_2^2$$

$$\text{Put } I_2 = I_1/2 \text{ and } \omega_c = \frac{5\omega_1}{6} \Rightarrow 5\omega_1/6$$

We get :

$$E_f - E_i = -\frac{I_1 \omega_1^2}{24}$$

90. (b) As from the question density (σ) = kr^2

$$\text{Mass of disc } M = \int_0^R (kr^2) 2\pi r dr = 2\pi k \frac{R^4}{4} = \frac{\pi k R^4}{2}$$

$$\Rightarrow k = \frac{2M}{\pi R^4} \quad \dots(i)$$

$\therefore$ Moment of inertia about the axis of the disc.

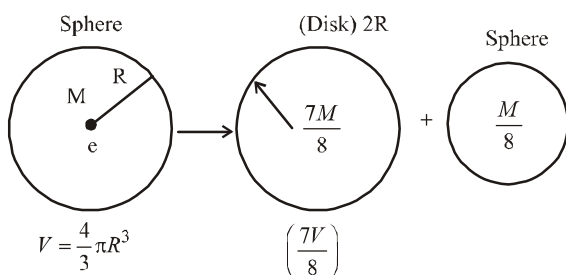
$$I = \int dl = \int (dm) r^2 = \int \sigma dA r^2$$

$$= \int (kr^2) (2\pi r dr) r^2$$

$$= \int_0^R 2\pi k r^5 dr = \frac{\pi k R^6}{3} = \frac{\pi \times \left(\frac{2M}{\pi R^4} \right) \times R^6}{3} = \frac{2}{3} MR^2$$

[putting value of k from eqn(i)]

91. (b)



$$I_1 = \left(\frac{7M}{8}\right)(2R)^2 \frac{1}{2} = \left(\frac{7}{16} \times 4\right)MR^2 = \frac{14}{8}MR^2$$

$$I_2 = \frac{2}{5}\left(\frac{M}{8}\right)r^2$$

$$\Rightarrow I_2 = \frac{2}{5}\left(\frac{M}{8}\right)\left(\frac{R^2}{4}\right) = \frac{MR^2}{80}$$

$$\left[\frac{4}{3}\pi r^3 \rho = \frac{1}{8}\frac{4}{3}\pi R^3 \times \rho\right]$$

$$\Rightarrow r = R/2$$

$$\frac{I_1}{I_2} = \frac{14 \times 80}{8} = 140$$

92. (d) $\frac{1}{2}I\omega^2 = kQ^2$

$$\text{or } \omega = \left(\sqrt{\frac{2k}{I}}\right)Q$$

$$\text{or } \alpha = \frac{d\omega}{dt} = \sqrt{\frac{2k}{I}}\left(\frac{dQ}{dt}\right) = \left(\sqrt{\frac{2k}{I}}\right)\omega$$

$$= \left(\sqrt{\frac{2k}{I}}\right)\left(\sqrt{\frac{2k}{I}}\right)\theta = \frac{2k\theta}{I}$$

93. (b) $\omega = \alpha t = 20t$

$$\text{Given, } \frac{1}{2}I\omega^2 = 1200$$

$$\text{or } \frac{1}{2} \times 1.5 \times (20t)^2 = 1200$$

$$\text{or } t = 2 \text{ s}$$

94. (c) $I_i \omega_i = I_f \omega_f$

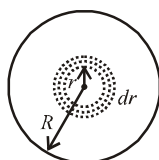
$$\text{or } \left(\frac{ML^2}{12}\right)\omega_0 = \left(\frac{ML^2}{12} + 2m\left(\frac{L}{2}\right)^2\right)\omega_f$$

$$\therefore \omega_f = \left(\frac{M\omega_0}{M + 6m}\right)$$

 95. (c) Taking a circular ring of radius r and thickness dr as a mass element, so total mass,

$$M = \int_0^R \rho_0 r \times 2\pi r dr = \frac{2\pi\rho_0 R^3}{3}$$

$$I_C = \int_0^R \rho_0 r \times 2\pi r dr \times r^2 = \frac{2\pi\rho_0 R^5}{5}$$



Using parallel axis theorem

$$\therefore I = I_C + MR^2 = 2\pi\rho_0 R^5 \left(\frac{1}{3} + \frac{1}{5}\right) = \frac{16\pi\rho_0 R^5}{15}$$

$$= \frac{8}{5} \left[\frac{2}{3}\pi\rho_0 R^3 \right] R^2 = \frac{8}{5}MR^2$$

96. (b)

97. (d) According to parallel axes theorem

$$I = \frac{2}{5}mR^2 + mx^2$$

 Hence graph (d) correctly depicts I vs x .

 98. (a) Let mass of the larger triangle = M

 Side of larger triangle = ℓ

 Moment of inertia of larger triangle = ma^2

 Mass of smaller triangle = $\frac{M}{4}$

 Length of smaller triangle = $\frac{\ell}{2}$

Moment of inertia of removed triangle

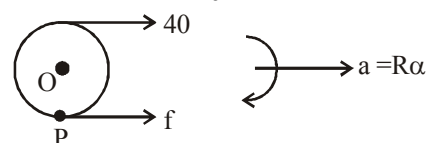
$$= \frac{M}{4} \left(\frac{a}{2}\right)^2$$

$$\therefore \frac{I_{\text{removed}}}{I_{\text{original}}} = \frac{\frac{M}{4} \left(\frac{a}{2}\right)^2}{M \cdot (a)^2}$$

$$I_{\text{removed}} = \frac{I_0}{16}$$

$$\text{So, } I = I_0 - \frac{I_0}{16} = \frac{15I_0}{16}$$

99. (b)



From newton's second law

$$40 + f = m(R\alpha) \quad \dots(i)$$

 Taking torque about O we get

$$40 \times R - f \times R = I\alpha$$

$$40 \times R - f \times R = mR^2 \alpha$$

$$40 - f = mR \alpha$$

Solving equation (i) and (ii)

$$\alpha = \frac{40}{mR} = 16 \text{ rad/s}^2$$

100. (b) Moment of inertia of disc D_1 about $OO' = I_1 = \frac{MR^2}{2}$

 M.O.I of D_2 about OO'

$$= I_2 = \frac{1}{2} \left(\frac{MR^2}{2} \right) + MR^2 = \frac{MR^2}{4} + MR^2$$

 M.O.I of D_3 about OO'

$$= I_3 = \frac{1}{2} \left(\frac{MR^2}{2} \right) + MR^2 = \frac{MR^2}{4} + MR^2$$

so, resultant M.O.I about OO' is $I = I_1 + I_2 + I_3$

$$\Rightarrow I = \frac{MR^2}{2} + 2\left(\frac{MR^2}{4} + MR^2\right)$$

$$= \frac{MR^2}{2} + \frac{MR^2}{2} + 2MR^2 = 3MR^2$$

101. (a) For Ball

using parallel axes theorem, for ball moment of inertia,

$$I_{\text{ball}} = \frac{2}{5}MR^2 + M(2R)^2 = \frac{22}{5}MR^2$$

For two balls $I_{\text{balls}} = 2 \times \frac{22}{5}MR^2$ and,

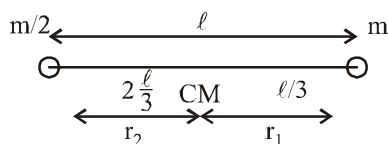
$$I_{\text{rod}} = \frac{M(2R)^2}{12} = \frac{MR^2}{3}$$

$$I_{\text{system}} = I_{\text{balls}} + I_{\text{rod}}$$

$$= \frac{44}{5}MR^2 + \frac{MR^2}{3} = \frac{137}{15}MR^2$$

102. (c) As we know, $\omega = \sqrt{\frac{k}{I}}$

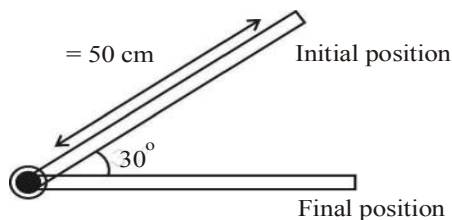
$$\omega = \sqrt{\frac{3k}{m\ell^2}} \left[\because I_{\text{rod}} = \frac{1}{3}m\ell^2 \right]$$



Tension when it passes through the mean position,

$$= m\omega^2 \frac{\ell}{3} = m \frac{3k}{m\ell^2} \theta_0^2 \frac{\ell}{3} = \frac{k\theta_0^2}{\ell}$$

103. (d)



By the law of conservation of energy,

P.E. of rod = Rotational K.E.

$$mg \frac{\ell}{2} \sin \theta = \frac{1}{2} I \omega^2$$

$$\Rightarrow mg \frac{\ell}{2} \sin 30^\circ = \frac{1}{2} \frac{m\ell^2}{3} \omega^2 \Rightarrow mg \frac{\ell}{2} \times \frac{1}{2} = \frac{1}{2} \frac{m\ell^2}{3} \omega^2$$

For complete length of rod,

$$\omega = \sqrt{3g/2(2\ell)} = \sqrt{\frac{30}{2}} \text{ rad s}^{-1}$$

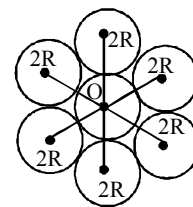
104. (d) Using parallel axes theorem, moment of inertia about 'O'

$$I_o = I_{\text{cm}} + md^2$$

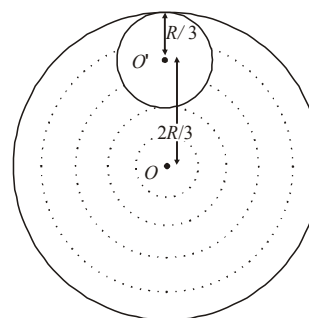
$$= \frac{7MR^2}{2} + 6(M \times (2R)^2) = \frac{55MR^2}{2}$$

Again, moment of inertia about point P, $I_p = I_o + md^2$

$$= \frac{55MR^2}{2} + 7M(3R)^2 = \frac{181}{2}MR^2$$



105. (a) Let σ be the mass per unit area.



The total mass of the disc

$$= \sigma \times \pi R^2 = 9M$$

The mass of the circular disc cut

$$= \sigma \times \pi \left(\frac{R}{3}\right)^2 = \sigma \times \frac{\pi R^2}{9} = M$$

Let us consider the above system as a complete disc of mass $9M$ and a negative mass M super imposed on it.

Moment of inertia (I_1) of the complete disc =

$\frac{1}{2}9MR^2$ about an axis passing through O and perpendicular to the plane of the disc.

M.I. of the cut out portion about an axis passing through O' and perpendicular to the plane of disc

$$= \frac{1}{2} \times M \times \left(\frac{R}{3}\right)^2$$

$\therefore$ M.I. (I_2) of the cut out portion about an axis passing through O and perpendicular to the plane of disc

$$= \left[\frac{1}{2} \times M \times \left(\frac{R}{3}\right)^2 + M \times \left(\frac{2R}{3}\right)^2 \right]$$

[Using perpendicular axis theorem]

$\therefore$ The total M.I. of the system about an axis passing through O and perpendicular to the plane of the disc is

$$I = I_1 + I_2$$

$$= \frac{1}{2}9MR^2 - \left[\frac{1}{2} \times M \times \left(\frac{R}{3}\right)^2 + M \times \left(\frac{2R}{3}\right)^2 \right]$$

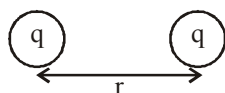
$$= \frac{9MR^2}{2} - \frac{9MR^2}{18} = \frac{(9-1)MR^2}{2} = 4MR^2$$

106. (c) As we know, moment of inertia of a disc about an axis passing through C.G. and perpendicular to its plane,

$$I_z = \frac{mR^2}{2}$$

Moment of inertia of a disc about a tangential axis perpendicular to its own plane,

$$I_z' = \frac{3}{2}mR^2$$



$$\therefore I_z/I_z' = \frac{mR^2}{2} / \frac{3mR^2}{2} = 1/3$$

107. (a) When the rod makes an angle α

Displacement of centre of mass $= \frac{l}{2} \cos \alpha$

$$mg \frac{l}{2} \cos \alpha = \frac{l}{2} I \omega^2$$

$mg \frac{l}{2} \cos \alpha = \frac{ml^2}{6} \omega^2$ ($\because$ M.I. of thin uniform rod about an axis passing through its centre of mass and perpendicular to the rod $I = \frac{ml^2}{12}$)

$$\Rightarrow \omega = \sqrt{\frac{3g \cos \alpha}{l}}$$

Speed of end $= \omega \times l = \sqrt{3g \cos \alpha} l$

i.e., Speed of end, $\omega \propto \sqrt{\cos \alpha}$

108. (c) As we know, moment of inertia of a solid cylinder about an axis which is perpendicular bisector

$$I = \frac{mR^2}{4} + \frac{ml^2}{12}$$

$$I = \frac{m}{4} \left[R^2 + \frac{l^2}{3} \right]$$

$$= \frac{m}{4} \left[\frac{V}{\pi l} + \frac{l^2}{3} \right] \Rightarrow \frac{dl}{dl} = \frac{m}{4} \left[\frac{-V}{\pi l^2} + \frac{2l}{3} \right] = 0$$

$$\frac{V}{\pi l^2} = \frac{2l}{3} \Rightarrow V = \frac{2\pi l^3}{3}$$

$$\pi R^2 l = \frac{2\pi l^3}{3} \Rightarrow \frac{l^2}{R^2} = \frac{3}{2} \text{ or, } \frac{l}{R} = \sqrt{\frac{3}{2}}$$

109. (b) According to theorem of perpendicular axes, moment of inertia of triangle (ABC)

$$I_0 = km^2 \quad \dots (i)$$

$$BC = 1$$

Moment of inertia of a cavity DEF

$$I_{DEF} = K \frac{m}{4} \left(\frac{l}{2} \right)^2$$

$$= \frac{k}{16} ml^2$$

From equation (i),

$$I_{DEF} = \frac{I_0}{16}$$

Moment of inertia of remaining part

$$I_{\text{remain}} = I_0 - \frac{I_0}{16} = \frac{15I_0}{16}$$

110. (b) Moment of Inertia of complete disc about 'O' point

$$I_{\text{total}} = \frac{MR^2}{2}$$

Radius of removed disc $= R/4$

$\therefore$ Mass of removed disc $= M/16$

[As $M \propto R^2$]

M.I of removed disc about its own axis (O')

$$= \frac{1}{2} \frac{M}{16} \left(\frac{R}{4} \right)^2 = \frac{MR^2}{512}$$

M.I of removed disc about O

$$I_{\text{removed disc}} = I_{\text{cm}} + mx^2$$

$$= \frac{MR^2}{512} + \frac{M}{16} \left(\frac{3R}{4} \right)^2 = \frac{19MR^2}{512}$$

M.I of remaining disc

$$I_{\text{remaining}} = \frac{MR^2}{2} - \frac{19}{512} MR^2 = \frac{237}{512} MR^2$$

111. (a) Here $a = \frac{2}{\sqrt{3}} R$

$$\text{Now, } \frac{M}{M'} = \frac{\frac{4}{3} \pi R^3}{a^3}$$

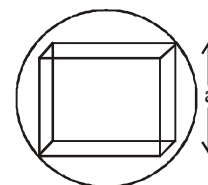
$$= \frac{\frac{4}{3} \pi R^3}{\left(\frac{2}{\sqrt{3}} R \right)^3} = \frac{\sqrt{3}}{2} \pi$$

$$M' = \frac{2M}{\sqrt{3}\pi}$$

Moment of inertia of the cube about the given axis,

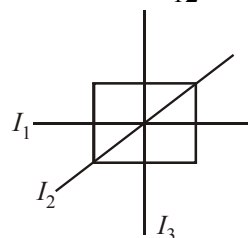
$$I = \frac{M'a^2}{6}$$

$$= \frac{\frac{2M}{\sqrt{3}\pi} \times \left(\frac{2}{\sqrt{3}} R \right)^2}{6} = \frac{4MR^2}{9\sqrt{3}\pi}$$



112. (d) For a thin uniform square sheet

$$I_1 = I_2 = I_3 = \frac{ma^2}{12}$$



113. (c) Kinetic energy_(rotational) $K_R = \frac{1}{2} I \omega^2$

Kinetic energy_(translational) $K_T = \frac{1}{2} M v^2$ ($v = R\omega$)

M.I._(initial) $I_{\text{ring}} = MR^2$; $\omega_{\text{initial}} = \omega$

M.I._(new) $I'_{\text{(system)}} = MR^2 + 2mR^2$

$\omega'_{\text{(system)}} = \frac{M\omega}{M+2m}$

Solving we get loss in K.E.

$= \frac{Mm}{(M+2m)} \omega^2 R^2$

114. (b) As $L = I\omega$ so L increases with increase in ω .
Kinetic energy_(rotational) depends on an angular velocity ' ω ' and moment of inertia of the body I .

i.e., $K.E._{\text{(rotational)}} = \frac{1}{2} I \omega^2$

115. (b) $K.E._{\text{rotational}} = \frac{1}{2} I \omega^2$

$= \frac{1}{2} \cdot \frac{2}{5} \omega r^2 d^2 \left(\because I_{\text{Solid sphere}} = \frac{2}{5} m r^2 \right)$

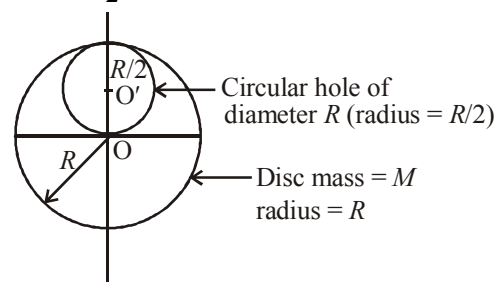
$K.E._{\text{translational}} = \frac{1}{2} m v^2$

$\therefore \frac{K.E._{\text{rotational}}}{K.E._{\text{translational}}} = \frac{2}{5}$

Hence option (b) is correct

116. (d) M.I. of complete disc about its centre O .

$I_{\text{Total}} = \frac{1}{2} MR^2$... (i)



Mass of circular hole (removed)

$= \frac{M}{4}$ (As $M = \pi R^2 t \therefore M \propto R^2$)

M.I. of removed hole about its own axis

$= \frac{1}{2} \left(\frac{M}{4} \right) \left(\frac{R}{2} \right)^2 = \frac{1}{32} MR^2$

M.I. of removed hole about O'

$I_{\text{removed hole}} = I_{\text{cm}} + m x^2$
 $= \frac{MR^2}{32} + \frac{M}{4} \left(\frac{R}{2} \right)^2$
 $= \frac{MR^2}{32} + \frac{MR^2}{16} = \frac{3MR^2}{32}$

M.I. of complete disc can also be written as

$I_{\text{Total}} = I_{\text{removed hole}} + I_{\text{remaining disc}}$

$I_{\text{Total}} = \frac{3MR^2}{32} + I_{\text{remaining disc}}$... (ii)

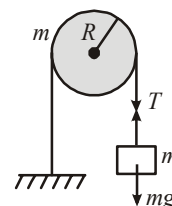
From eq. (i) and (ii),

$\frac{1}{2} MR^2 = \frac{3MR^2}{32} + I_{\text{remaining disc}}$

$\Rightarrow I_{\text{remaining disc}}$

$= \frac{MR^2}{2} - \frac{3MR^2}{32} = \left(\frac{13}{32} \right) MR^2$

117. (b) For translational motion,
 $mg - T = ma$ (1)
 For rotational motion,
 $T.R = I\alpha$



$\Rightarrow T.R = \frac{1}{2} m R^2 \alpha$

Also, acceleration, $a = R\alpha$

$\therefore T = \frac{1}{2} m R \alpha = \frac{1}{2} ma$

Substituting the value of T in equation (1) we get $mg -$

$\frac{1}{2} ma = ma \Rightarrow a = \frac{2}{3} g$

118. (a) Given,

Force, $F = (20t - 5t^2)$

Radius, $r = 2m$

Torque, $T = r f = I\alpha$

$\Rightarrow 2(20t - 5t^2) = 10\alpha$

$\therefore \alpha = 4t - t^2$

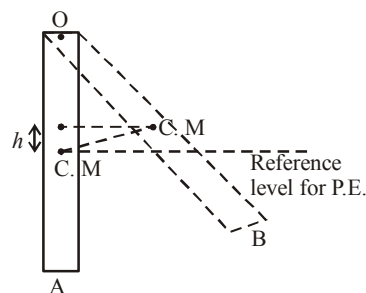
$\Rightarrow \frac{d\omega}{dt} = 4t - t^2 \Rightarrow \int_0^\omega d\omega = \int_0^t (4t - t^2) dt$

$\Rightarrow \omega = 2t^2 - \frac{t^3}{3}$ (as $\omega = 0$ at $t = 0, 6s$)

$\int_0^\theta d\theta = \int_0^6 \left(2t^2 - \frac{t^3}{3} \right) dt$

$\Rightarrow \theta = 36 \text{ rad} \Rightarrow 2\pi n = 36 \Rightarrow n = \frac{36}{2\pi} < 6$

119. (c)



The moment of inertia of the rod about O is $\frac{1}{3}m\ell^2$. The

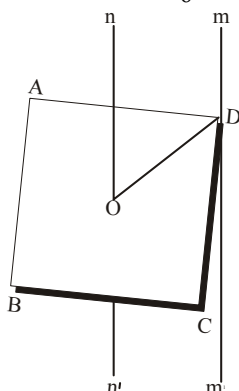
kinetic energy of the rod at position A = $\frac{1}{2}I\omega^2$ where I is the moment of inertia of the rod about O . When the rod is in position B, its angular velocity is zero. In this case, the energy of the rod is mgh where h is the maximum height to which the centre of mass (C.M) rises

Gain in potential energy = Loss in kinetic energy

$$\therefore mgh = \frac{1}{2}I\omega^2 = \frac{1}{2}\left(\frac{1}{3}m\ell^2\right)\omega^2$$

$$\Rightarrow h = \frac{\ell^2\omega^2}{6g}$$

120. (d) $I_{nn'} = \frac{1}{12}M(a^2 + a^2) = \frac{Ma^2}{6}$



Also, $DO = \frac{DB}{2} = \frac{\sqrt{2}a}{2} = \frac{a}{\sqrt{2}}$

By parallel axes theorem, moment of inertia of plate about an axis through one of its corners.

$$I_{mm'} = I_{nn'} + M\left(\frac{a}{\sqrt{2}}\right)^2 = \frac{Ma^2}{6} + \frac{Ma^2}{2}$$

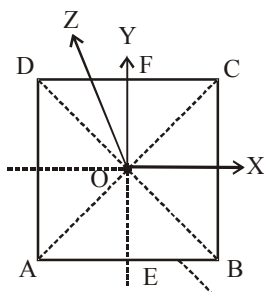
$$= \frac{Ma^2 + 3Ma^2}{6} = \frac{2}{3}Ma^2$$

121. (d) By the theorem of perpendicular axes,

$$I = I_{EF} + I_{GH}$$

Here, I is the moment of inertia of square lamina about an axis through O and perpendicular to its plane.

$$\therefore I_{EF} = I_{GH} \text{ (By Symmetry of Figure)}$$



$$\therefore I_{EF} = \frac{I}{2}$$

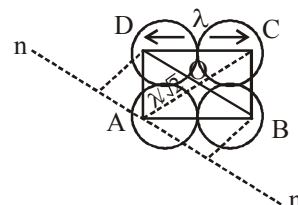
...(i)

Again, by the same theorem $I = I_{AC} + I_{BD} = 2I_{AC}$
 $(\therefore I_{AC} = I_{BD} \text{ by symmetry of the figure})$

$$\therefore I_{AC} = \frac{I}{2} \quad \dots(ii)$$

From (i) and (ii), we get, $I_{EF} = I_{AC}$.

122. (c)



$I_{nn'} = M.I$ due to the point mass at B +
 $M.I$ due to the point mass at D +
 $M.I$ due to the point mass at C.

$$I_{nn'} = m\left(\frac{\ell}{\sqrt{2}}\right)^2$$

$$+ m\left(\frac{\ell}{\sqrt{2}}\right)^2$$

$$+ m(\sqrt{2}\ell)^2$$

$$\Rightarrow I_{nn'} = 2 \times m\left(\frac{\ell}{\sqrt{2}}\right)^2 + m(\sqrt{2}\ell)^2$$

$$= m\ell^2 + 2m\ell^2 = 3m\ell^2$$

123. (c) The disc may be assumed as combination of two semi circular parts. Therefore, circular disc will have twice the mass of semicircular disc.

$$\text{Moment of inertia of disc} = \frac{1}{2}(2m)r^2 = Mr^2$$

Let I be the moment of inertia of the uniform semicircular disc

$$\Rightarrow 2I = Mr^2 \Rightarrow I = \frac{Mr^2}{2}$$

124. (a) The moment of inertia of solid sphere A about its

$$\text{diameter } I_A = \frac{2}{5}MR^2.$$

The moment of inertia of a hollow sphere B about its

$$\text{diameter } I_B = \frac{2}{3}MR^2.$$

$$\therefore I_A < I_B$$

125. (d) We know that density $(d) = \frac{\text{mass}(M)}{\text{volume}(V)}$

$$\therefore M = d \times V = d \times (\pi R^2 \times t).$$

The moment of inertia of a disc is given by $I = \frac{1}{2}MR^2$

$$\therefore I_x = \frac{1}{2}MxR_x^2 = \frac{1}{2}(d \times \pi R^2 \times t)R^2$$

$$= \frac{\pi d}{2} t \times R^4$$

$$I_y = \frac{1}{2} M_y R_y^2 = \frac{1}{2} \left[\pi (4R^2) \left(\frac{1}{4} d \right) \right] \times (4R)^2$$

$$\therefore \frac{I_X}{I_Y} = \frac{t_X R_X^4}{t_Y R_Y^4} = \frac{t \times R^4}{\frac{t}{4} \times (4R)^4} = \frac{1}{64}$$

126. (a) Rotational kinetic energy $= \frac{1}{2} I \omega^2$,

Angular momentum, $L = I \omega \Rightarrow I = \frac{L}{\omega}$

$$\therefore K.E. = \frac{1}{2} \frac{L}{\omega} \times \omega^2 = \frac{1}{2} L \omega$$

$$L' = \frac{2K.E.}{\omega}$$

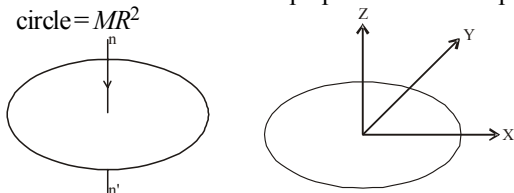
When ω is doubled and K.E is halved.

New angular momentum,

$$L' = \frac{\frac{2K.E.}{\omega}}{2\omega} = \frac{2K.E.}{2\omega}$$

$$\Rightarrow \therefore L' = \frac{L}{4}$$

- 127. (a)** M. I of a circular wire about an axis nn' passing through the centre of the circle and perpendicular to the plane of the circle $= MR^2$



As shown in the figure, X-axis and Y-axis lie along diameter of the ring. Using perpendicular axis theorem

$$I_X + I_Y = I_Z$$

Here, I_X and I_Y are the moment of inertia about the diameter.
 $\Rightarrow 2 I_X = MR^2$ [$\because I_X = I_Y$ (by symmetry) and $I_Z = MR^2$]

$$\therefore I_X = \frac{1}{2} MR^2$$

- 128. (c)** Moment of inertia of circular disc

$$I_1 = \frac{1}{2} MR^2$$

When two small sphere are attached on the edge of the disc, the moment of inertia becomes

$$I_2 = \frac{1}{2} MR^2 + 2mR^2$$

When two small spheres of mass m are attached gently, the external torque, about the axis of rotation, is zero and therefore the angular momentum about the axis of rotation is constant.

$$\therefore I_1 \omega_1 = I_2 \omega_2 \Rightarrow \omega_2 = \frac{I_1}{I_2} \omega_1$$

$$\therefore \omega_2 = \frac{\frac{1}{2} MR^2}{\frac{1}{2} MR^2 + 2mR^2} \times \omega_1 = \frac{M}{M + 4m} \omega_1$$

- 129. (a)** K.E of the sphere = translational K.E + rotational K.E

$$= \frac{1}{2} mv^2 + \frac{1}{2} I \omega^2$$

Where, I = moment of inertia,

ω = Angular, velocity of rotation

m = mass of the sphere

v = linear velocity of centre of mass of sphere

$$\therefore \text{Moment of inertia of sphere } I = \frac{2}{5} mR^2$$

$$\therefore K.E = \frac{1}{2} mv^2 + \frac{1}{2} \times \frac{2}{5} mR^2 \times \omega^2$$

$$\Rightarrow K.E = \frac{1}{2} mv^2 + \frac{1}{2} \times \frac{2}{5} mR^2 \times \left(\frac{v}{R} \right)^2 \left(\because \omega = \frac{v}{R} \right)$$

$$\Rightarrow K.E = \frac{1}{2} \left(\frac{2}{5} mR^2 + mR^2 \right) \left(\frac{v}{R} \right)^2$$

$$\Rightarrow K.E = \frac{1}{2} mR^2 \times \frac{7}{5} \times \frac{v^2}{R^2} = \frac{7}{10} \times \frac{1}{2} \times \frac{25}{10^4}$$

$$\Rightarrow K.E = \frac{35}{4} \times 10^{-4} \text{ joule}$$

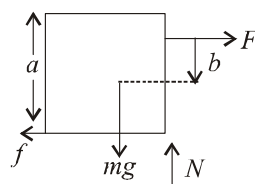
$$\Rightarrow K.E = 8.75 \times 10^{-4} \text{ joule}$$

- 130. (50)** For the box to be slide

$$F = \mu mg = 0.4 mg$$

For no toppling

$$F \left(\frac{a}{2} + b \right) \leq mg \frac{a}{2}$$



$$\Rightarrow 0.4 mg \left(\frac{a}{2} + b \right) \leq mg \frac{a}{2}$$

$$\Rightarrow 0.2 a + 0.4 b \leq 0.5 a$$

$$\Rightarrow \frac{b}{a} \leq \frac{3}{4}$$

i.e. $b \leq 0.75 a$ but this is not possible.

As the maximum value of b can be equal to $0.5a$.

$$\Rightarrow \frac{100b}{a} = 50$$

- 131. (c)** For sphere,

$$\frac{1}{2} mv^2 + I \omega^2 = \frac{1}{2} mgh$$

$$\text{or } \frac{1}{2} mv^2 + \frac{1}{2} \left(\frac{2}{5} mR^2 \right) \frac{v^2}{R^2} = mgh \text{ or } h = \frac{7v^2}{10g}$$

For cylinder

$$\frac{1}{2} mv^2 + \frac{1}{2} \left(\frac{mR^2}{2} \right) \frac{v^2}{R^2} = mgh'$$

$$\text{or } h' = \frac{3v^2}{4g}$$

$$\therefore \frac{h}{h'} = \frac{7v^2/10g}{3v^2/4g} = \frac{14}{15}$$

$$132.(\text{Bonus}) \quad mgh = \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2$$

$$= \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I_{cm}\left(\frac{v_{cm}}{R}\right)^2$$

$$= \frac{1}{2}\left(m + \frac{I_{cm}}{R^2}\right)v_{cm}^2$$

$$\text{For ring: } mgh = \frac{1}{2}\left(m + \frac{mR^2}{R^2}\right)v_{cm}^2$$

$$\therefore h = \frac{v_{cm}^2}{g}$$

$$\text{For solid cylinder, } mgh = \frac{1}{2}\left(m + \frac{mR^2}{2R^2}\right)v_{cm}^2$$

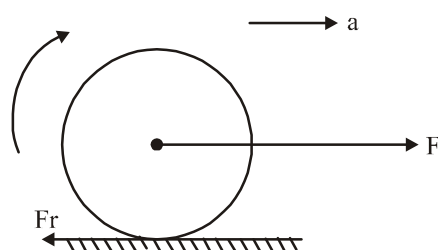
$$\therefore h = \frac{3v_{cm}^2}{4g}$$

$$\text{For sphere, } mgh = \frac{1}{2}\left(m + \frac{2}{5}\frac{mR^2}{R^2}\right)v_{cm}^2$$

$$\therefore h = \frac{7v_{cm}^2}{10g}$$

$$\text{Ratio of heights } 1 : \frac{3}{4} : \frac{7}{10} \Rightarrow 20 : 15 : 14$$

133. (d)



$$F - f_r = ma \quad \dots(i)$$

$$f_r R = I\alpha = \frac{mR^2}{2}\alpha \quad \dots(ii)$$

for pure rolling

$$a = \alpha R \quad \dots(iii)$$

from (1) (2) and (3)

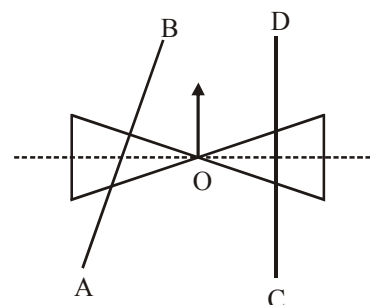
$$F - \frac{mR\alpha}{2} = m\alpha R$$

$$F = \frac{3}{2}mR\alpha$$

$$\alpha = \frac{2F}{3mR}$$

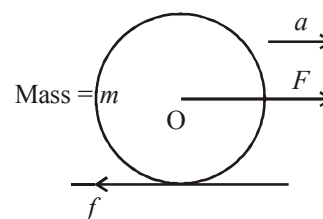
134. (c) As shown in the diagram, the normal reaction of AB on roller will shift towards O.

This will lead to tending of the system of cones to turn left.



135. (c) From figure,

$$ma = F - f \quad \dots(i)$$



And, torque $\tau = I\alpha$

$$\frac{mR^2}{2}\alpha = fR$$

$$\frac{mR^2}{2} \frac{a}{R} = fR \quad \left[\because \alpha = \frac{a}{R} \right]$$

$$\frac{ma}{2} = f \quad \dots(ii)$$

Put this value in equation (i),

$$ma = F - \frac{ma}{2} \text{ or } F = \frac{3ma}{2}$$

136. (d) As we know,

$$\text{Acceleration, } a = \frac{mg \sin \theta}{m + \frac{I}{r^2}}$$

$$\text{For cylinder, } a_c = \frac{M_c g \sin \theta_c}{M_c + \frac{1}{2} \frac{M_c R^2}{R^2}} = \frac{M_c g \sin \theta_c}{M_c + \frac{M_c R^2}{2R^2}}$$

$$\text{or, } a_c = \frac{2}{3} g \sin \theta_c$$

For sphere,

$$a_s = \frac{M_s g \sin \theta_s}{M_s + \frac{I_s}{r^2}} = \frac{M_s g \sin \theta_s}{M_s + \frac{2}{5} \frac{MR^2}{R^2}}$$

$$\text{or, } a_s = \frac{5}{7} g \sin \theta_s$$

$$\text{given, } a_c = a_s$$

$$\text{i.e., } \frac{2}{3} g \sin \theta_c = \frac{5}{7} g \sin \theta_s$$

$$\therefore \frac{\sin \theta_c}{\sin \theta_s} = \frac{\frac{5}{7} g}{\frac{2}{3} g} = \frac{15}{14}$$



From conservation of angular momentum about any fix point on the surface,
 $mr^2\omega_0 = 2mr^2\omega$

$$\Rightarrow \omega = \omega_0/2 \Rightarrow v = \frac{\omega_0 r}{2} \quad [\because v = r\omega]$$

138. (b) Velocity of the tennis ball on the surface of the earth or ground

$$v = \sqrt{\frac{2gh}{1 + \frac{k^2}{R^2}}} \quad (\text{where } k = \text{radius of gyration of spherical shell})$$

$$\text{shell} = \sqrt{\frac{2}{3}} R$$

$$\text{Horizontal range AB} = \frac{v^2 \sin 2\theta}{g}$$

$$= \frac{\left(\sqrt{\frac{2gh}{1 + k^2/R^2}} \right)^2 \sin(2 \times 30^\circ)}{g} = 2.08 \text{ m}$$

139. (b) When body rolls down on inclined plane with velocity V_0 at bottom then body has both rotational and translational kinetic energy.

Therefore, by law of conservation of energy,

$$P.E. = K.E_{\text{trans}} + K.E_{\text{rotational}}$$

$$= \frac{1}{2} m V_0^2 + \frac{1}{2} I \omega^2$$

$$= \frac{1}{2} m V_0^2 + \frac{1}{2} m k^2 \frac{V_0^2}{R_0^2} \quad \dots(i)$$

$$\left[\because I = m k^2, \omega = \frac{V}{R_0} \right]$$

When body is sliding down then body has only translatory motion.

$$\therefore P.E. = K.E_{\text{trans}}$$

$$= \frac{1}{2} m \left(\frac{5}{4} v_0 \right)^2 \quad \dots(ii)$$

Dividing (i) by (ii) we get

$$\frac{P.E.}{P.E.} = \frac{\frac{1}{2} m v_0^2 \left[1 + \frac{K^2}{R_0^2} \right]}{\frac{1}{2} \times \frac{25}{16} \times m V_0^2} = \frac{25}{16} = 1 + \frac{K^2}{R_0^2} \Rightarrow \frac{K^2}{R_0^2} = \frac{9}{16}$$

$$\text{or, } K = \frac{3}{4} R_0.$$

140. (d) Minimum velocity for a body rolling without slipping

$$v = \sqrt{\frac{2gh}{1 + \frac{K^2}{R^2}}}$$

$$\text{For solid sphere, } \frac{K^2}{R^2} = \frac{2}{5}$$

$$\therefore v = \sqrt{\frac{2gh}{1 + \frac{K^2}{R^2}}} = \sqrt{\frac{10}{7} gh}$$

141. (b) Acceleration of the body rolling down an inclined plane is given by.

$$a = \frac{g \sin \theta}{1 + \frac{I}{MR^2}}$$

Gravitation



TOPIC 1 Kepler's Laws of Planetary Motion

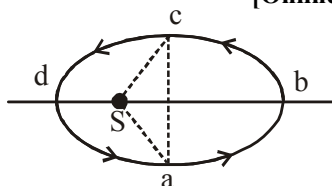


1. If the angular momentum of a planet of mass m , moving around the Sun in a circular orbit is L , about the center of the Sun, its areal velocity is: [9 Jan. 2019 I]

(a) $\frac{L}{m}$ (b) $\frac{4L}{m}$ (c) $\frac{L}{2m}$ (d) $\frac{2L}{m}$

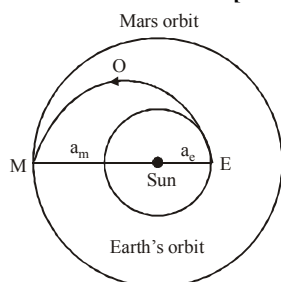
2. Figure shows elliptical path $abcd$ of a planet around the sun S such that the area of triangle csa is $\frac{1}{4}$ the area of the ellipse. (See figure) With db as the semimajor axis, and ca as the semiminor axis. If t_1 is the time taken for planet to go over path abc and t_2 for path taken over cda then:

[Online April 9, 2016]



- (a) $t_1 = 4t_2$ (b) $t_1 = 2t_2$
(c) $t_1 = 3t_2$ (d) $t_1 = t_2$
3. India's Mangalyan was sent to the Mars by launching it into a transfer orbit EOM around the sun. It leaves the earth at E and meets Mars at M . If the semi-major axis of Earth's orbit is $a_e = 1.5 \times 10^{11}$ m, that of Mars orbit $a_m = 2.28 \times 10^{11}$ m, taken Kepler's laws give the estimate of time for Mangalyan to reach Mars from Earth to be close to:

[Online April 9, 2014]



- (a) 500 days (b) 320 days
(c) 260 days (d) 220 days
4. The time period of a satellite of earth is 5 hours. If the separation between the earth and the satellite is increased to 4 times the previous value, the new time period will become [2003]
- (a) 10 hours (b) 80 hours
(c) 40 hours (d) 20 hours

TOPIC 2 Newton's Universal Law of Gravitation



5. A straight rod of length L extends from $x = a$ to $x = L + a$. The gravitational force it exerts on point mass ' m ' at $x = 0$, if the mass per unit length of the rod is $A + Bx^2$, is given by: [12 Jan. 2019 I]

(a) $Gm \left[A \left(\frac{1}{a+L} - \frac{1}{a} \right) - BL \right]$
(b) $Gm \left[A \left(\frac{1}{a} - \frac{1}{a+L} \right) - BL \right]$
(c) $Gm \left[A \left(\frac{1}{a+L} - \frac{1}{a} \right) + BL \right]$
(d) $Gm \left[A \left(\frac{1}{a} - \frac{1}{a+L} \right) + BL \right]$

6. Take the mean distance of the moon and the sun from the earth to be 0.4×10^6 km and 150×10^6 km respectively. Their masses are 8×10^{22} kg and 2×10^{30} kg respectively. The radius of the earth is 6400 km. Let ΔF_1 be the difference in the forces exerted by the moon at the nearest and farthest points on the earth and ΔF_2 be the difference in the force exerted by the sun at the nearest and farthest points on

the earth. Then, the number closest to $\frac{\Delta F_1}{\Delta F_2}$ is:

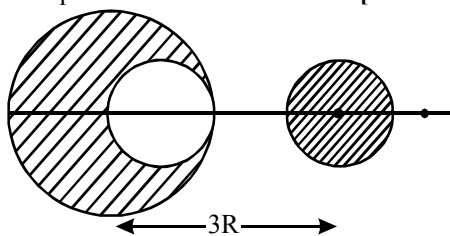
[Online April 15, 2018]

- (a) 2 (b) 6 (c) 10^{-2} (d) 0.6

7. Four particles, each of mass M and equidistant from each other, move along a circle of radius R under the action of their mutual gravitational attraction. The speed of each particle is: [2014]

(a) $\sqrt{\frac{GM}{R}}$ (b) $\sqrt{2\sqrt{2}\frac{GM}{R}}$
 (c) $\sqrt{\frac{GM}{R}(1+2\sqrt{2})}$ (d) $\frac{1}{2}\sqrt{\frac{GM}{R}(1+2\sqrt{2})}$

8. From a sphere of mass M and radius R , a smaller sphere of radius $\frac{R}{2}$ is carved out such that the cavity made in the original sphere is between its centre and the periphery (See figure). For the configuration in the figure where the distance between the centre of the original sphere and the removed sphere is $3R$, the gravitational force between the two sphere is: [Online April 11, 2014]



- (a) $\frac{41 GM^2}{3600 R^2}$ (b) $\frac{41 GM^2}{450 R^2}$ (c) $\frac{59 GM^2}{450 R^2}$ (d) $\frac{GM^2}{225 R^2}$
9. Two particles of equal mass ' m ' go around a circle of radius R under the action of their mutual gravitational attraction. The speed of each particle with respect to their centre of mass is [2011 RS]
- (a) $\sqrt{\frac{Gm}{4R}}$ (b) $\sqrt{\frac{Gm}{3R}}$ (c) $\sqrt{\frac{Gm}{2R}}$ (d) $\sqrt{\frac{Gm}{R}}$
10. Two spherical bodies of mass M and $5M$ & radii R & $2R$ respectively are released in free space with initial separation between their centres equal to $12R$. If they attract each other due to gravitational force only, then the distance covered by the smaller body just before collision is [2003]
- (a) $2.5R$ (b) $4.5R$ (c) $7.5R$ (d) $1.5R$

TOPIC 3 Acceleration due to Gravity



11. The value of acceleration due to gravity is g_1 at a height $h = \frac{R}{2}$ (R = radius of the earth) from the surface of the earth. It is again equal to g_1 at a depth d below the surface of the earth. The ratio $\left(\frac{d}{R}\right)$ equals : [5 Sep. 2020 (I)]
- (a) $\frac{4}{9}$ (b) $\frac{5}{9}$ (c) $\frac{1}{3}$ (d) $\frac{7}{9}$

12. The acceleration due to gravity on the earth's surface at the poles is g and angular velocity of the earth about the axis passing through the pole is ω . An object is weighed at the equator and at a height h above the poles by using a spring balance. If the weights are found to be same, then h is : ($h \ll R$, where R is the radius of the earth)

(a) $\frac{R^2 \omega^2}{2g}$ (b) $\frac{R^2 \omega^2}{g}$ [5 Sep. 2020 (II)]
 (c) $\frac{R^2 \omega^2}{4g}$ (d) $\frac{R^2 \omega^2}{8g}$

13. The height ' h ' at which the weight of a body will be the same as that at the same depth ' h ' from the surface of the earth is (Radius of the earth is R and effect of the rotation of the earth is neglected) : [2 Sep. 2020 (II)]

(a) $\frac{\sqrt{5}}{2}R - R$ (b) $\frac{R}{2}$
 (c) $\frac{\sqrt{5}R - R}{2}$ (d) $\frac{\sqrt{3}R - R}{2}$

14. A box weighs 196 N on a spring balance at the north pole. Its weight recorded on the same balance if it is shifted to the equator is close to (Take $g = 10 \text{ ms}^{-2}$ at the north pole and the radius of the earth = 6400 km): [7 Jan. 2020 II]

(a) 195.66 N (b) 194.32 N
 (c) 194.66 N (d) 195.32 N

15. The ratio of the weights of a body on the Earth's surface to that on the surface of a planet is 9:4. The mass of the planet is $\frac{1}{9}$ th of that of the Earth. If ' R ' is the radius of the

Earth, what is the radius of the planet ? (Take the planets to have the same mass density). [12 April 2019 II]

(a) $\frac{R}{3}$ (b) $\frac{R}{4}$ (c) $\frac{R}{9}$ (d) $\frac{R}{2}$

16. The value of acceleration due to gravity at Earth's surface is 9.8 ms^{-2} . The altitude above its surface at which the acceleration due to gravity decreases to 4.9 ms^{-2} , is close to : (Radius of earth = $6.4 \times 10^6 \text{ m}$) [10 April 2019 I]

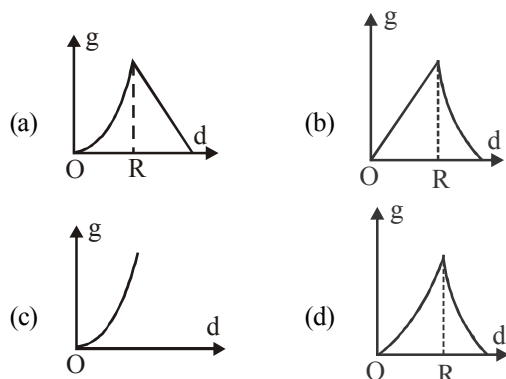
(a) $2.6 \times 10^6 \text{ m}$ (b) $6.4 \times 10^6 \text{ m}$
 (c) $9.0 \times 10^6 \text{ m}$ (d) $1.6 \times 10^6 \text{ m}$

17. Suppose that the angular velocity of rotation of earth is increased. Then, as a consequence.

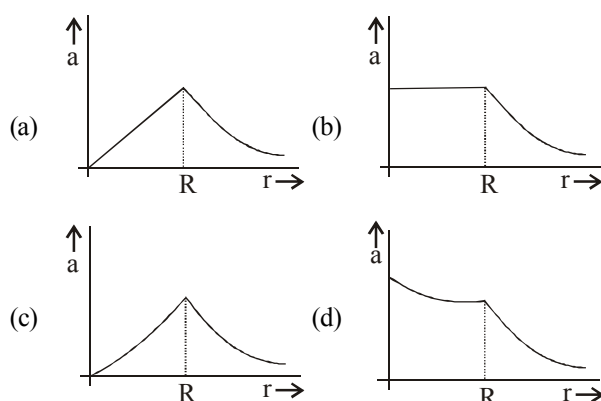
[Online April 16, 2018]

- (a) There will be no change in weight anywhere on the earth
 (b) Weight of the object, everywhere on the earth, will decrease
 (c) Weight of the object, everywhere on the earth, will increase
 (d) Except at poles, weight of the object on the earth will decrease

18. The variation of acceleration due to gravity g with distance d from centre of the earth is best represented by (R = Earth's radius): [2017, Online May 7, 2012]



19. The mass density of a spherical body is given by $\rho(r) = \frac{k}{r}$ for $r \leq R$ and $\rho(r) = 0$ for $r > R$, where r is the distance from the centre. The correct graph that describes qualitatively the acceleration, a , of a test particle as a function of r is : [Online April 9, 2017]



20. If the Earth has no rotational motion, the weight of a person on the equator is W . Determine the speed with which the earth would have to rotate about its axis so that the person

at the equator will weight $\frac{3}{4}W$. Radius of the Earth is

6400 km and $g = 10 \text{ m/s}^2$. [Online April 8, 2017]

- (a) $1.1 \times 10^{-3} \text{ rad/s}$ (b) $0.83 \times 10^{-3} \text{ rad/s}$
(c) $0.63 \times 10^{-3} \text{ rad/s}$ (d) $0.28 \times 10^{-3} \text{ rad/s}$
21. The change in the value of acceleration of earth towards sun, when the moon comes from the position of solar eclipse to the position on the other side of earth in line with sun is:

(mass of the moon = $7.36 \times 10^{22} \text{ kg}$, radius of the moon's orbit = $3.8 \times 10^8 \text{ m}$). [Online April 22, 2013]

- (a) $6.73 \times 10^{-5} \text{ m/s}^2$ (b) $6.73 \times 10^{-3} \text{ m/s}^2$
(c) $6.73 \times 10^{-2} \text{ m/s}^2$ (d) $6.73 \times 10^{-4} \text{ m/s}^2$

22. Assuming the earth to be a sphere of uniform density, the acceleration due to gravity inside the earth at a distance of r from the centre is proportional to [Online May 12, 2012]
(a) r (b) r^{-1} (c) r^2 (d) r^{-2}

23. The height at which the acceleration due to gravity becomes $\frac{g}{9}$ (where g = the acceleration due to gravity on the surface of the earth) in terms of R , the radius of the earth, is [2009]

- (a) $\frac{R}{\sqrt{2}}$ (b) $R/2$ (c) $\sqrt{2}R$ (d) $2R$

24. The change in the value of ' g ' at a height ' h ' above the surface of the earth is the same as at a depth ' d ' below the surface of earth. When both ' d ' and ' h ' are much smaller than the radius of earth, then which one of the following is correct? [2005]

- (a) $d = \frac{3h}{2}$ (b) $d = \frac{h}{2}$
(c) $d = h$ (d) $d = 2h$

25. Average density of the earth [2005]

- (a) is a complex function of g
(b) does not depend on g
(c) is inversely proportional to g
(d) is directly proportional to g

TOPIC 4 Gravitational Field and Potential Energy

26. Two planets have masses M and $16M$ and their radii are a and $2a$, respectively. The separation between the centres of the planets is $10a$. A body of mass m is fired from the surface of the larger planet towards the smaller planet along the line joining their centres. For the body to be able to reach the surface of smaller planet, the minimum firing speed needed is : [16 Sep. 2020 (II)]

- (a) $2\sqrt{\frac{GM}{a}}$ (b) $4\sqrt{\frac{GM}{a}}$
(c) $\sqrt{\frac{GM^2}{ma}}$ (d) $\frac{3}{2}\sqrt{\frac{5GM}{a}}$

27. On the x -axis and at a distance x from the origin, the gravitational field due to a mass distribution is given by

$\frac{Ax}{(x^2 + a^2)^{3/2}}$ in the x -direction. The magnitude of

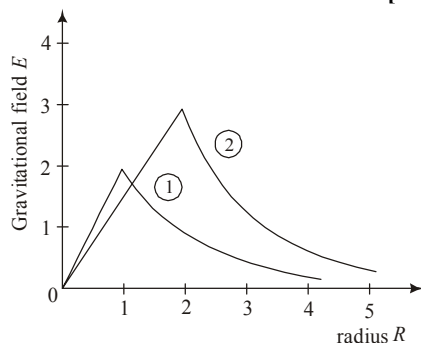
gravitational potential on the x -axis at a distance x , taking its value to be zero at infinity, is : [4 Sep. 2020 (I)]

- (a) $\frac{A}{(x^2 + a^2)^{1/2}}$ (b) $\frac{A}{(x^2 + a^2)^{3/2}}$
(c) $A(x^2 + a^2)^{1/2}$ (d) $A(x^2 + a^2)^{3/2}$

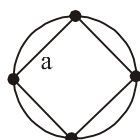
28. The mass density of a planet of radius R varies with the distance r from its centre as $\rho(r) = \rho_0 \left(1 - \frac{r^2}{R^2}\right)$. Then the gravitational field is maximum at : [3 Sep. 2020 (II)]

- (a) $r = \sqrt{\frac{3}{4}}R$ (b) $r = R$
 (c) $r = \frac{1}{\sqrt{3}}R$ (d) $r = \sqrt{\frac{5}{9}}R$

29. Consider two solid spheres of radii $R_1 = 1m$, $R_2 = 2m$ and masses M_1 and M_2 , respectively. The gravitational field due to sphere ① and ② are shown. The value of $\frac{m_1}{m_2}$ is: [8 Jan. 2020 I]



- (a) $\frac{2}{3}$ (b) $\frac{1}{6}$ (c) $\frac{1}{2}$ (d) $\frac{1}{3}$
30. An asteroid is moving directly towards the centre of the earth. When at a distance of $10R$ (R is the radius of the earth) from the earth's centre, it has a speed of 12 km/s. Neglecting the effect of earth's atmosphere, what will be the speed of the asteroid when it hits the surface of the earth (escape velocity from the earth is 11.2 km/s)? Give your answer to the nearest integer in kilometer/s. [NA 8 Jan. 2020 II]
31. A solid sphere of mass ' M ' and radius ' a ' is surrounded by a uniform concentric spherical shell of thickness $2a$ and mass $2M$. The gravitational field at distance ' $3a$ ' from the centre will be: [9 April 2019 I]
- (a) $\frac{2GM}{9a^2}$ (b) $\frac{GM}{9a^2}$ (c) $\frac{GM}{3a^2}$ (d) $\frac{2GM}{3a^2}$
32. Four identical particles of mass M are located at the corners of a square of side ' a '. What should be their speed if each of them revolves under the influence of others' gravitational field in a circular orbit circumscribing the square? [8 April 2019 I]



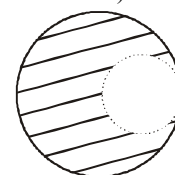
- (a) $1.35 \sqrt{\frac{GM}{a}}$ (b) $1.16 \sqrt{\frac{GM}{a}}$
 (c) $1.21 \sqrt{\frac{GM}{a}}$ (d) $1.41 \sqrt{\frac{GM}{a}}$

33. A test particle is moving in circular orbit in the gravitational field produced by a mass density $\rho(r) = \frac{K}{r^2}$. Identify the correct relation between the radius R of the particle's orbit and its period T : [8 April 2019 II]

- (a) T/R is a constant (b) T^2/R^3 is a constant
 (c) T/R^2 is a constant (d) TR is a constant
34. A body of mass m is moving in a circular orbit of radius R about a planet of mass M . At some instant, it splits into two equal masses. The first mass moves in a circular orbit of radius $\frac{R}{2}$, and the other mass, in a circular orbit of radius $\frac{3R}{2}$. The difference between the final and initial total energies is: [Online April 15, 2018]

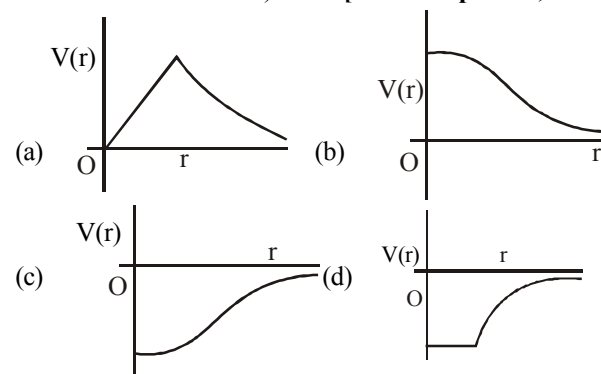
- (a) $-\frac{GMm}{2R}$ (b) $+\frac{GMm}{6R}$ (c) $-\frac{GMm}{6R}$ (d) $\frac{GMm}{2R}$

35. From a solid sphere of mass M and radius R , a spherical portion of radius $R/2$ is removed, as shown in the figure. Taking gravitational potential $V = 0$ at $r = \infty$, the potential at the centre of the cavity thus formed is : [2015]



- (a) $\frac{-2GM}{3R}$ (b) $\frac{-2GM}{R}$ (c) $\frac{-GM}{2R}$ (d) $\frac{-GM}{R}$

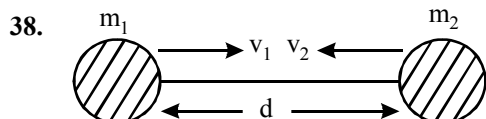
36. Which of the following most closely depicts the correct variation of the gravitational potential $V(r)$ due to a large planet of radius R and uniform mass density? (figures are not drawn to scale) [Online April 11, 2015]



37. The gravitational field in a region is given by $\vec{g} = 5\text{ N/kg}\hat{i} + 12\text{ N/kg}\hat{j}$. The change in the gravitational potential energy of a particle of mass 1 kg when it is taken from the origin to a point (7 m, -3 m) is:

[Online April 19, 2014]

- (a) 71 J (b) $13\sqrt{58}\text{ J}$ (c) -71 J (d) 1 J



Two hypothetical planets of masses m_1 and m_2 are at rest when they are infinite distance apart. Because of the gravitational force they move towards each other along the line joining their centres. What is their speed when their separation is 'd'?

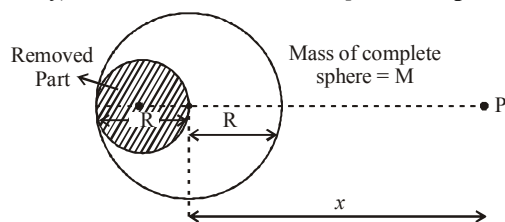
[Online April 12, 2014]

(Speed of m_1 is v_1 and that of m_2 is v_2)

- (a) $v_1 = v_2$
 (b) $v_1 = m_2 \sqrt{\frac{2G}{d(m_1 + m_2)}}$ $v_2 = m_1 \sqrt{\frac{2G}{d(m_1 + m_2)}}$
 (c) $v_1 = m_1 \sqrt{\frac{2G}{d(m_1 + m_2)}}$ $v_2 = m_2 \sqrt{\frac{2G}{d(m_1 + m_2)}}$
 (d) $v_1 = m_2 \sqrt{\frac{2G}{m_1}}$ $v_2 = m_1 \sqrt{\frac{2G}{m_2}}$

39. The gravitational field, due to the 'left over part' of a uniform sphere (from which a part as shown, has been 'removed out'), at a very far off point, P, located as shown, would be (nearly):

[Online April 9, 2013]



- (a) $\frac{5}{6} \frac{GM}{x^2}$ (b) $\frac{8}{9} \frac{GM}{x^2}$ (c) $\frac{7}{8} \frac{GM}{x^2}$ (d) $\frac{6}{7} \frac{GM}{x^2}$

40. The mass of a spaceship is 1000 kg. It is to be launched from the earth's surface out into free space. The value of g and R (radius of earth) are 10 m/s^2 and 6400 km respectively. The required energy for this work will be

[2012]

- (a) $6.4 \times 10^{11}\text{ Joules}$ (b) $6.4 \times 10^8\text{ Joules}$
 (c) $6.4 \times 10^9\text{ Joules}$ (d) $6.4 \times 10^{10}\text{ Joules}$

41. A point particle is held on the axis of a ring of mass m and radius r at a distance r from its centre C . When released, it reaches C under the gravitational attraction of the ring. Its speed at C will be

[Online May 26, 2012]

- (a) $\sqrt{\frac{2Gm}{r}(\sqrt{2}-1)}$ (b) $\sqrt{\frac{Gm}{r}}$
 (c) $\sqrt{\frac{2Gm}{r}\left(1-\frac{1}{\sqrt{2}}\right)}$ (d) $\sqrt{\frac{2Gm}{r}}$

42. Two bodies of masses m and $4m$ are placed at a distance r . The gravitational potential at a point on the line joining them where the gravitational field is zero is:

[2011]

- (a) $-\frac{4Gm}{r}$ (b) $-\frac{6Gm}{r}$ (c) $-\frac{9Gm}{r}$ (d) zero

43. This question contains Statement-1 and Statement-2. Of the four choices given after the statements, choose the one that best describes the two statements.

[2008]

Statement-1 : For a mass M kept at the centre of a cube of side ' a ', the flux of gravitational field passing through its sides $4\pi GM$. and

Statement-2: If the direction of a field due to a point source is radial and its dependence on the distance ' r ' from the

source is given as $\frac{1}{r^2}$, its flux through a closed surface

depends only on the strength of the source enclosed by the surface and not on the size or shape of the surface.

- (a) Statement -1 is false, Statement-2 is true
 (b) Statement -1 is true, Statement-2 is true; Statement -2 is a correct explanation for Statement-1
 (c) Statement -1 is true, Statement-2 is true; Statement -2 is not a correct explanation for Statement-1
 (d) Statement -1 is true, Statement-2 is false

44. A particle of mass 10 g is kept on the surface of a uniform sphere of mass 100 kg and radius 10 cm. Find the work to be done against the gravitational force between them to take the particle far away from the sphere

(you may take $G = 6.67 \times 10^{-11}\text{ Nm}^2/\text{kg}^2$) [2005]

- (a) $3.33 \times 10^{-10}\text{ J}$ (b) $13.34 \times 10^{-10}\text{ J}$
 (c) $6.67 \times 10^{-10}\text{ J}$ (d) $6.67 \times 10^{-9}\text{ J}$

45. If ' g ' is the acceleration due to gravity on the earth's surface, the gain in the potential energy of an object of mass ' m ' raised from the surface of the earth to a height equal to the radius ' R ' of the earth is

[2004]

- (a) $\frac{1}{4}mgR$ (b) $\frac{1}{2}mgR$ (c) $2mgR$ (d) mgR

46. Energy required to move a body of mass m from an orbit of radius $2R$ to $3R$ is

[2002]

- (a) $GMm/12R^2$ (b) $GMm/3R^2$
 (c) $GMm/8R$ (d) $GMm/6R$

TOPIC 5 Motion of Satellites, Escape Speed and Orbital Velocity



47. A satellite is in an elliptical orbit around a planet P. It is observed that the velocity of the satellite when it is farthest from the planet is 6 times less than that when it is closest to the planet. The ratio of distances between the satellite and the planet at closest and farthest points is :

[NA 6 Sep. 2020 (I)]

- (a) 1 : 6 (b) 1 : 3 (c) 1 : 2 (d) 3 : 4
48. A body is moving in a low circular orbit about a planet of mass M and radius R . The radius of the orbit can be taken to be R itself. Then the ratio of the speed of this body in the orbit to the escape velocity from the planet is :

- (a) $\frac{1}{\sqrt{2}}$ (b) 2 [4 Sep. 2020 (II)]

- (c) 1 (d) $\sqrt{2}$

49. A satellite is moving in a low nearly circular orbit around the earth. Its radius is roughly equal to that of the earth's radius R_e . By firing rockets attached to it, its speed is instantaneously increased in the direction of its motion so

that it become $\sqrt{\frac{3}{2}}$ times larger. Due to this the farthest distance from the centre of the earth that the satellite reaches is R . Value of R is : [3 Sep. 2020 (I)]

- (a) $4R_e$ (b) $2.5R_e$ (c) $3R_e$ (d) $2R_e$

50. The mass density of a spherical galaxy varies as $\frac{K}{r}$ over a large distance ' r ' from its centre. In that region, a small star is in a circular orbit of radius R . Then the period of revolution, T depends on R as : [2 Sep. 2020 (I)]

- (a) $T^2 \propto R$ (b) $T^2 \propto R^3$ (c) $T^2 \propto \frac{1}{R^3}$ (d) $T \propto R$

51. A body A of mass m is moving in a circular orbit of radius R about a planet. Another body B of mass $\frac{m}{2}$ collides with

A with a velocity which is half $\left(\frac{\vec{v}}{2}\right)$ the instantaneous

velocity $\vec{v}$ or A. The collision is completely inelastic. Then, the combined body: [9 Jan. 2020 I]

- (a) continues to move in a circular orbit
 (b) Escapes from the Planet's Gravitational field
 (c) Falls vertically downwards towards the planet
 (d) starts moving in an elliptical orbit around the planet
52. The energy required to take a satellite to a height ' h ' above Earth surface (radius of Earth = 6.4×10^3 km) is E_1 and kinetic energy required for the satellite to be in

a circular orbit at this height is E_2 . The value of h for which E_1 and E_2 are equal, is: [9 Jan. 2019 II]

- (a) 1.6×10^3 km (b) 3.2×10^3 km
 (c) 6.4×10^3 km (d) 28×10^4 km

53. Planet A has mass M and radius R . Planet B has half the mass and half the radius of Planet A. If the escape velocities from the Planets A and B are v_A and v_B , respectively, then

$\frac{v_A}{v_B} = \frac{n}{4}$. The value of n is : [9 Jan. 2020 II]

- (a) 4 (b) 1 (c) 2 (d) 3

54. A satellite of mass m is launched vertically upwards with an initial speed u from the surface of the earth. After it reaches height R (R = radius of the earth), it

ejects a rocket of mass $\frac{m}{10}$ so that subsequently the

satellite moves in a circular orbit. The kinetic energy of the rocket is (G is the gravitational constant; M is the mass of the earth): [7 Jan. 2020 I]

- (a) $\frac{m}{20} \left(u^2 + \frac{113}{200} \frac{GM}{R} \right)$ (b) $5m \left(u^2 - \frac{119}{200} \frac{GM}{R} \right)$

- (c) $\frac{3m}{8} \left(u + \sqrt{\frac{5GM}{6R}} \right)^2$ (d) $\frac{m}{20} \left(u - \sqrt{\frac{2GM}{3R}} \right)^2$

55. A spaceship orbits around a planet at a height of 20 km from its surface. Assuming that only gravitational field of the planet acts on the spaceship, what will be the number of complete revolutions made by the spaceship in 24 hours around the planet ? [Given : Mass of Planet = 8×10^{22} kg, Radius of planet = 2×10^6 m, Gravitational constant $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$] [10 April 2019 II]

- (a) 9 (b) 17 (c) 13 (d) 11

56. A rocket has to be launched from earth in such a way that it never returns. If E is the minimum energy delivered by the rocket launcher, what should be the minimum energy that the launcher should have if the same rocket is to be launched from the surface of the moon? Assume that the density of the earth and the moon are equal and that the earth's volume is 64 times the volume of the moon.

[8 April 2019 II]

- (a) $\frac{E}{64}$ (b) $\frac{E}{32}$ (c) $\frac{E}{4}$ (d) $\frac{E}{16}$

57. A satellite of mass M is in a circular orbit of radius R about the centre of the earth. A meteorite of the same mass, falling towards the earth collides with the satellite completely in elastically. The speeds of the satellite and the meteorite are the same, Just before the collision. The subsequent motion of the combined body will be [12 Jan. 2019 I]

- (a) such that it escape to infinity
 (b) In an elliptical orbit
 (c) in the same circular orbit of radius R
 (d) in a circular orbit of a different radius
58. Two satellites, A and B, have masses m and $2m$ respectively. A is in a circular orbit of radius R , and B is in a circular orbit of radius $2R$ around the earth. The ratio of their kinetic energies, T_A/T_B , is : **[12 Jan. 2019 II]**
- (a) $\frac{1}{2}$ (b) 1
 (c) 2 (d) $\sqrt{\frac{1}{2}}$
59. A satellite is revolving in a circular orbit at a height h from the earth surface, such that $h \ll R$ where R is the radius of the earth. Assuming that the effect of earth's atmosphere can be neglected the minimum increase in the speed required so that the satellite could escape from the gravitational field of earth is: **[11 Jan. 2019 I]**
- (a) $\sqrt{2gR}$ (b) $\sqrt{gR}$
 (c) $\sqrt{\frac{gR}{2}}$ (d) $\sqrt{gR}(\sqrt{2}-1)$
60. A satellite is moving with a constant speed v in circular orbit around the earth. An object of mass ' m ' is ejected from the satellite such that it just escapes from the gravitational pull of the earth. At the time of ejection, the kinetic energy of the object is: **[10 Jan. 2019 I]**
- (a) $2m v^2$ (b) $m v^2$
 (c) $\frac{1}{2} m v^2$ (d) $\frac{3}{2} m v^2$
61. Two stars of masses 3×10^{31} kg each, and at distance 2×10^{11} m rotate in a plane about their common centre of mass O. A meteorite passes through O moving perpendicular to the star's rotation plane. In order to escape from the gravitational field of this double star, the minimum speed that meteorite should have at O is: (Take Gravitational constant $G = 66 \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2}$) **[10 Jan. 2019 II]**
- (a) $2.4 \times 10^4 \text{ m/s}$ (b) $1.4 \times 10^5 \text{ m/s}$
 (c) $3.8 \times 10^4 \text{ m/s}$ (d) $2.8 \times 10^5 \text{ m/s}$
62. A satellite is revolving in a circular orbit at a height ' h ' from the earth's surface (radius of earth R ; $h \ll R$). The minimum increase in its orbital velocity required, so that the satellite could escape from the earth's gravitational field, is close to : (Neglect the effect of atmosphere.) **[2016]**
- (a) $\sqrt{gR/2}$ (b) $\sqrt{gR}(\sqrt{2}-1)$
 (c) $\sqrt{2gR}$ (d) $\sqrt{gR}$
63. An astronaut of mass m is working on a satellite orbiting the earth at a distance h from the earth's surface. The radius of the earth is R , while its mass is M . The gravitational pull F_G on the astronaut is : **[Online April 10, 2016]**
- (a) Zero since astronaut feels weightless
 (b) $\frac{GMm}{(R+h)^2} < F_G < \frac{GMm}{R^2}$
 (c) $F_G = \frac{GMm}{(R+h)^2}$
 (d) $0 < F_G < \frac{GMm}{R^2}$
64. A very long (length L) cylindrical galaxy is made of uniformly distributed mass and has radius R ($R < L$). A star outside the galaxy is orbiting the galaxy in a plane perpendicular to the galaxy and passing through its centre. If the time period of star is T and its distance from the galaxy's axis is r , then : **[Online April 10, 2015]**
- (a) $T \propto r$ (b) $T \propto \sqrt{r}$
 (c) $T \propto r^2$ (d) $T^2 \propto r^3$
65. What is the minimum energy required to launch a satellite of mass m from the surface of a planet of mass M and radius R in a circular orbit at an altitude of $2R$? **[2013]**
- (a) $\frac{5GmM}{6R}$ (b) $\frac{2GmM}{3R}$ (c) $\frac{GmM}{2R}$ (d) $\frac{GmM}{2R}$
66. A planet in a distant solar system is 10 times more massive than the earth and its radius is 10 times smaller. Given that the escape velocity from the earth is 11 km s^{-1} , the escape velocity from the surface of the planet would be **[2008]**
- (a) 1.1 km s^{-1} (b) 11 km s^{-1}
 (c) 110 km s^{-1} (d) 0.11 km s^{-1}
67. Suppose the gravitational force varies inversely as the n th power of distance. Then the time period of a planet in circular orbit of radius ' R ' around the sun will be proportional to
- (a) R^n (b) $R^{\left(\frac{n-1}{2}\right)}$ **[2004]**
 (c) $R^{\left(\frac{n+1}{2}\right)}$ (d) $R^{\left(\frac{n-2}{2}\right)}$
68. The time period of an earth satellite in circular orbit is independent of **[2004]**
- (a) both the mass and radius of the orbit
 (b) radius of its orbit
 (c) the mass of the satellite
 (d) neither the mass of the satellite nor the radius of its orbit.

69. A satellite of mass m revolves around the earth of radius R at a height x from its surface. If g is the acceleration due to gravity on the surface of the earth, the orbital speed of the satellite is [2004]
- (a) $\frac{gR^2}{R+x}$ (b) $\frac{gR}{R-x}$ (c) gx (d) $\left(\frac{gR^2}{R+x}\right)^{1/2}$
70. The escape velocity for a body projected vertically upwards from the surface of earth is 11 km/s. If the body is projected at an angle of 45° with the vertical, the escape velocity will be [2003]
- (a) $11\sqrt{2}$ km/s (b) 22 km/s
(c) 11 km/s (d) $\frac{11}{\sqrt{2}}$ km/s
71. The kinetic energy needed to project a body of mass m from the earth surface (radius R) to infinity is [2002]
- (a) $mgR/2$ (b) $2mgR$ (c) mgR (d) $mgR/4$
72. If suddenly the gravitational force of attraction between Earth and a satellite revolving around it becomes zero, then the satellite will [2002]
- (a) continue to move in its orbit with same velocity
(b) move tangentially to the original orbit in the same velocity
(c) become stationary in its orbit
(d) move towards the earth
73. The escape velocity of a body depends upon mass as [2002]
- (a) m^0 (b) m^1 (c) m^2 (d) m^3

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Hints & Solutions



1. (c) Areal velocity; $\frac{dA}{dt}$

$$dA = \frac{1}{2} r^2 d\theta$$

$$\Rightarrow \frac{dA}{dt} = \frac{1}{2} r^2 \frac{d\theta}{dt} = \frac{1}{2} r^2 \omega$$

$$\text{Also, } L = mvr = mr^2\omega$$

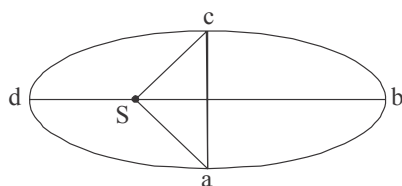
$$\therefore \frac{dA}{dt} = \frac{1}{2} \frac{L}{m}$$

2. (c) Let area of ellipse abcd = x

$$\text{Area of } SabcS = \frac{x}{2} + \frac{x}{4} \quad (\text{i.e., ar of } abca + SabcS)$$

(Area of half ellipse + Area of triangle)

$$= \frac{3x}{4}$$



$$\text{Area of } SadcS = x - \frac{3x}{4} = \frac{x}{4}$$

$$\frac{\text{Area of } SabcS}{\text{Area of } SadcS} = \frac{3x/4}{x/4} = \frac{t_1}{t_2}$$

$$\frac{t_1}{t_2} = 3 \quad \text{or, } t_1 = 3t_2$$

3. (b)

4. (c) According to Kepler's law of periods $T^2 \propto R^3$

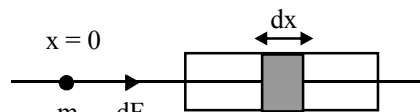
$$\therefore \left(\frac{T_2}{T_1}\right)^2 = \left(\frac{R_2}{R_1}\right)^3$$

$$\Rightarrow T_2 = T_1 \left(\frac{R_2}{R_1}\right)^{3/2} = 5 \times \left[\frac{4R}{R}\right]^{3/2}$$

$$= 5 \times 2^3 = 40 \text{ hours}$$

5. (d) Given $\lambda = (A + Bx^2)$,

Taking small element dm of length dx at a distance x from $x=0$



$$\text{so, } dm = \lambda dx$$

$$dm = (A + Bx^2)dx$$

$$dF = \frac{Gmdm}{x^2}$$

$$\Rightarrow F = \int_a^{a+L} \frac{Gm}{x^2} (A + Bx^2) dx$$

$$= Gm \left[-\frac{A}{x} + Bx \right]_a^{a+L}$$

$$= Gm \left[A \left(\frac{1}{a} - \frac{1}{a+L} \right) + BL \right]$$

6. (a) As we know, Gravitational force of attraction,

$$F = \frac{GMm}{R^2}$$

$$F_1 = \frac{GM_e m}{r_1^2} \quad \text{and} \quad F_2 = \frac{GM_e M_s}{r_2^2}$$

$$\Delta F_1 = \frac{2GM_e m}{r_1^3} \Delta r_1 \quad \text{and} \quad \Delta F_2 = \frac{GM_e M_s}{r_2^3} \Delta r_2$$

$$\frac{\Delta F_1}{\Delta F_2} = \frac{m \Delta r_1}{r_1^3} \frac{r_2^3}{M_s \Delta r_2} = \left(\frac{m}{M_s} \right) \left(\frac{r_2^3}{r_1^3} \right) \left(\frac{\Delta r_1}{\Delta r_2} \right)$$

Using $\Delta r_1 = \Delta r_2 = 2 R_{\text{earth}}$; $m = 8 \times 10^{22} \text{ kg}$;

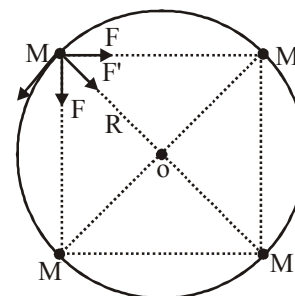
$$M_s = 2 \times 10^{30} \text{ kg}$$

$$r_1 = 0.4 \times 10^6 \text{ km} \quad \text{and} \quad r_2 = 150 \times 10^6 \text{ km}$$

$$\frac{\Delta F_1}{\Delta F_2} = \left(\frac{8 \times 10^{22}}{2 \times 10^{30}} \right) \left(\frac{150 \times 10^6}{0.4 \times 10^6} \right)^3 \times 1 \approx 2$$

7. (d) $2F \cos 45^\circ + F' = \frac{Mv^2}{R}$ (From figure)

$$\text{Where } F = \frac{GM^2}{(\sqrt{2}R)^2} \quad \text{and} \quad F' = \frac{GM^2}{4R^2}$$



$$\Rightarrow \frac{2 \times GM^2}{\sqrt{2}(R\sqrt{2})^2} + \frac{GM^2}{4R^2} = \frac{Mv^2}{R}$$

$$\Rightarrow \frac{GM^2}{R} \left[\frac{1}{4} + \frac{1}{\sqrt{2}} \right] = Mv^2$$

$$\therefore v = \sqrt{\frac{GM}{R} \left(\frac{\sqrt{2} + 4}{4\sqrt{2}} \right)} = \frac{1}{2} \sqrt{\frac{GM}{R} (1 + 2\sqrt{2})}$$

8. (a) Volume of removed sphere

$$V_{\text{remo}} = \frac{4}{3}\pi\left(\frac{R}{2}\right)^3 = \frac{4}{3}\pi R^3\left(\frac{1}{8}\right)$$

Volume of the sphere (remaining)

$$V_{\text{remain}} = \frac{4}{3}\pi R^3 - \frac{4}{3}\pi R^3\left(\frac{1}{8}\right) = \frac{4}{3}\pi R^3\left(\frac{7}{8}\right)$$

Therefore mass of sphere carved and remaining sphere

are at respectively $\frac{1}{8}M$ and $\frac{7}{8}M$.

Therefore, gravitational force between these two sphere,

$$F = \frac{GMm}{r^2} = \frac{G\frac{7M}{8} \times \frac{1}{8}M}{(3R)^2} = \frac{7}{64 \times 9} \frac{GM^2}{R^2}$$

$$\approx \frac{41}{3600} \frac{GM^2}{R^2}$$

9. (a) As two masses revolve about the common centre of mass O .

$\therefore$ Mutual gravitational attraction = centripetal force

$$\frac{Gm^2}{(2R)^2} = m\omega^2 R$$

$$\Rightarrow \frac{Gm}{4R^3} = \omega^2$$

$$\Rightarrow \omega = \sqrt{\frac{Gm}{4R^3}}$$

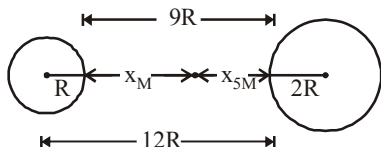
If the velocity of the two particles with respect to the centre of gravity is v then

$$v = \omega R$$

$$v = \sqrt{\frac{Gm}{4R^3}} \times R = \sqrt{\frac{Gm}{4R}}$$

10. (c) We know that

Force = mass $\times$ acceleration.



The gravitational force acting on both the masses is the same.

$$F_1 = F_2$$

$$ma_1 = ma_2$$

$$\Rightarrow \frac{9M}{95M} = \frac{5M}{M} = 5$$

$$\Rightarrow \frac{9M}{95M} = \frac{1}{5}$$

Let t be the time taken for the two masses to collide and x_{5M} , x_M be the distance travelled by the mass $5M$ and M respectively.

For mass $5M$

$$u = 0,$$

$$S = ut + \frac{1}{2}at^2$$

$$\therefore x_{5M} = \frac{1}{2}a_{5M}t^2 \quad \dots(ii)$$

For mass M

$$u = 0, s = x_M, t = t, a = a_M$$

$$\therefore s = ut + \frac{1}{2}at^2$$

$$\Rightarrow x_M = \frac{1}{2}a_M t^2 \quad \dots(iii)$$

Dividing (ii) by (iii)

$$\frac{x_{5M}}{x_M} = \frac{\frac{1}{2}a_{5M}t^2}{\frac{1}{2}a_M t^2} = \frac{a_{5M}}{a_M} = \frac{1}{5} \quad [\text{From (i)}]$$

$$\therefore 5x_{5M} = x_M \quad \dots(iv)$$

From the figure it is clear that

$$x_{5M} + x_M = 9R \quad \dots(v)$$

Where O is the point where the two spheres collide.

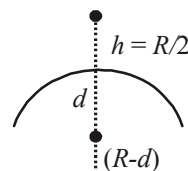
From (iv) and (v)

$$\frac{x_M}{5} + x_M = 9R$$

$$\therefore 6x_M = 45R$$

$$\therefore x_M = \frac{45}{6}R = 7.5R$$

11. (b) According to question, $g_h = g_d = g_1$



$$g_h = \frac{GM}{\left(R + \frac{R}{2}\right)^2} \text{ and } g_d = \frac{GM(R-d)}{R^3}$$

$$\frac{GM}{\left(\frac{3R}{2}\right)^2} = \frac{GM(R-d)}{R^3} \Rightarrow \frac{4}{9} = \frac{(R-d)}{R}$$

$$\Rightarrow 4R = 9R - 9d \Rightarrow 5R = 9d$$

$$\therefore \frac{d}{R} = \frac{5}{9}$$

12. (b) Value of g at equator, $g_A = g - R\omega^2$

Value of g at height h above the pole,

$$g_B = g \cdot \left(1 - \frac{2h}{R}\right)$$

As object is weighed equally at the equator and poles, it means g is same at these places.

$$g_A = g_B$$

$$\Rightarrow g - R\omega^2 = g \left(1 - \frac{2h}{R}\right)$$

$$\Rightarrow R\omega^2 = \frac{2gh}{R} \Rightarrow h = \frac{R^2\omega^2}{2g}$$

13. (c) The acceleration due to gravity at a height h is given by

$$g = \frac{GM}{(R+h)^2}$$

Here, G = gravitation constant

M = mass of earth

The acceleration due to gravity at depth h is

$$g' = \frac{GM}{R^2} \left(1 - \frac{h}{R}\right)$$

Given, $g = g'$

$$\therefore \frac{GM}{(R+h)^2} = \frac{GM}{R^2} \left(1 - \frac{h}{R}\right)$$

$$\therefore R^3 = (R+h)^2(R-h) = (R^2 + h^2 + 2hR)(R-h)$$

$$\Rightarrow R^3 = R^3 + h^2R + 2hR^2 - R^2h - h^3 - 2h^2R$$

$$\Rightarrow h^3 + h^2(2R - R) - R^2h = 0$$

$$\Rightarrow h^3 + h^2R - R^2h = 0$$

$$\Rightarrow h^2 + hR - R^2 = 0$$

$$\Rightarrow h = \frac{-R \pm \sqrt{R^2 + 4(1)R^2}}{2}$$

$$= \frac{-R + \sqrt{5}R}{2} = \frac{(\sqrt{5} - 1)R}{2}$$

14. (d) Weight at pole, $w = mg = 196 \text{ N}$

$$\Rightarrow m = 19.6 \text{ kg}$$

$$\text{Weight at equator, } w' = mg' = m(g - \omega^2 R)$$

$$= 19.6 \left[10 - \left(\frac{2\pi}{24 \times 3600} \right)^2 \times 6400 \times 10^3 \right] \text{ N}$$

$$\left(\because \omega = \frac{2\pi}{T} \right)$$

$$= 19.6 [10 - 0.034] = 195.33 \text{ N}$$

15. (d) $\frac{W_e}{W_p} = \frac{mg_e}{mg_p} = \frac{9}{4}$ or $\frac{g_e}{g_p} = \frac{9}{4}$

$$\text{or } \frac{GM/R^2}{G(M/9)/R_p^2} = \frac{9}{4}$$

$$\therefore R_p = \frac{R}{2}$$

16. (a) Given
Acceleration due to gravity at a height h from earth's surface is

$$g_h = g \left(1 + \frac{h}{R_e}\right)^{-2} \Rightarrow 4.9 = 9.8 \left(1 + \frac{h}{R_e}\right)^{-2}$$

$$\frac{1}{\sqrt{2}} = \left(1 + \frac{h}{R_e}\right) \quad [\text{as } h \ll R_e]$$

$$h = R_e (\sqrt{2} - 1)$$

$$h = 6400 \times 0.414 \text{ km} = 2.6 \times 10^6 \text{ m}$$

17. (d) With rotation of earth or latitude, acceleration due to gravity vary as $g' = g - \omega^2 R \cos^2 \phi$

Where ϕ is latitude, there will be no change in gravity at poles as $\phi = 90^\circ$

At all other points as ω increases g' will decrease hence, weight, $W = mg$ decreases.

18. (b) Variation of acceleration due to gravity, g with distance ' d ' from centre of the earth

$$\text{If } d < R, g = \frac{Gm}{R^2} \cdot d \quad \text{i.e., } g \propto d \text{ (straight line)}$$

$$\text{If } d = R, g_s = \frac{Gm}{R^2}$$

$$\text{If } d > R, g = \frac{Gm}{d^2} \quad \text{i.e., } g \propto \frac{1}{d^2}$$

19. (b) Given that, mass density $\left(\frac{\text{mass}}{\text{volume}}\right)$ of a spherical

$$\text{body } \rho(r) = \frac{k}{r}$$

$$\frac{M}{V} = \frac{k}{r} \text{ for inside } r \leq R$$

$$M = \frac{kv}{r} \quad \dots (i)$$

Inside the surface of sphere Intensity

$$I = \frac{GMr}{R^3} \quad \therefore I = \frac{F}{m}$$

$$g_{\text{inside}} = \frac{GMr}{R^3} \quad \text{or } I = \frac{mg}{m} = g$$

$$= \frac{G}{R^3} \cdot \frac{kv}{r} \cdot r = \text{constant} \quad \text{From eq. (i),}$$

$$\text{Similarly, } g_{\text{out}} = \frac{GM}{r^2}$$

Hence, option (2) is correct graph.

20. (c) We know, $g' = g - \omega^2 R \cos^2 \theta$

$$\frac{3g}{4} = g - \omega^2 R$$

$$\text{Given, } g' = \frac{3}{4}g$$

$$\omega^2 R = \frac{g}{4}$$

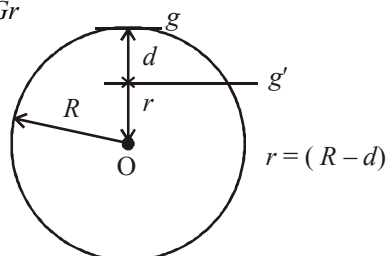
$$\omega = \sqrt{\frac{g}{4R}} = \sqrt{\frac{10}{4 \times 6400 \times 10^3}}$$

$$= \frac{1}{2 \times 8 \times 100} = 0.6 \times 10^{-3} \text{ rad/s}$$

21. (a)

22. (a) Acceleration due to gravity at depth d from the surface of the earth or at a distance r from the centre 'O' of

$$\text{the earth } g' = \frac{4}{3} \pi \rho G r$$

Hence $g' \propto r$ 23. (d) On earth's surface $g = \frac{GM}{R^2}$

At height above earth's surface

$$g_h = \frac{GM}{(R+h)^2}$$

$$\therefore \frac{g_h}{g} = \frac{R^2}{(R+h)^2}$$

$$\Rightarrow \frac{g/9}{g} = \left[\frac{R}{R+h} \right]^2$$

$$\Rightarrow \frac{R}{R+h} = \frac{1}{3}$$

$$\therefore h = 2R$$

24. (d) Value of g with altitude is,

$$g_h = g \left[1 - \frac{2h}{R} \right];$$

Value of g at depth d below earth's surface,

$$g_d = g \left[1 - \frac{d}{R} \right]$$

Equating g_h and g_d , we get $d = 2h$ 25. (d) Value of g on earth's surface,

$$g = \frac{GM}{R^2} = \frac{G\rho \times V}{R^2}$$

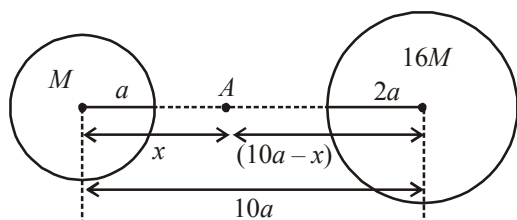
$$\Rightarrow g = \frac{G \times \rho \times \frac{4}{3} \pi R^3}{R^2}$$

$$g = \frac{4}{3} \pi G R \rho \text{ where } \rho \rightarrow \text{average density}$$

$$\rho = \left(\frac{3g}{4\pi GR} \right)$$

 $\Rightarrow \rho$ is directly proportional to g .

26. (d)

Let A be the point where gravitation field of both planets cancel each other i.e. zero.

$$\frac{GM}{x^2} = \frac{G(16M)}{(10a-x)^2}$$

$$\Rightarrow \frac{1}{x} = \frac{4}{(10a-x)} \Rightarrow 4x = 10a - x \Rightarrow x = 2a \quad \dots (i)$$

Using conservation of energy, we have

$$-\frac{GMm}{8a} - \frac{G(16M)m}{2a} + KE = -\frac{GMm}{2a} - \frac{G(16M)m}{8a}$$

$$KE = GMm \left[\frac{1}{8a} + \frac{16}{2a} - \frac{1}{2a} - \frac{16}{8a} \right]$$

$$\Rightarrow KE = GMm \left[\frac{1+64-4-16}{8a} \right]$$

$$\Rightarrow \frac{1}{2}mv^2 = GMm \left[\frac{45}{8a} \right] \Rightarrow v = \sqrt{\frac{90GM}{8a}}$$

$$\Rightarrow v = \frac{3}{2} \sqrt{\frac{5GM}{a}}$$

27. (a) Given : Gravitational field,

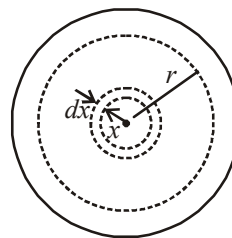
$$E_G = \frac{Ax}{(x^2 + a^2)^{3/2}}, V_\infty = 0$$

$$\int_{V_\infty}^{V_x} dV = - \int_{\infty}^x \vec{E}_G \cdot \vec{d}_x$$

$$\Rightarrow V_x - V_\infty = - \int_{\infty}^x \frac{Ax}{(x^2 + a^2)^{3/2}} dx$$

$$\therefore V_x = \frac{A}{(x^2 + a^2)^{1/2}} - 0 = \frac{A}{(x^2 + a^2)^{1/2}}$$

28. (d)

Mass of small element of planet of radius x and thickness dx .

$$dm = \rho \times 4\pi x^2 dx = \rho_0 \left(1 - \frac{x^2}{R^2} \right) \times 4\pi x^2 dx$$

Mass of the planet

$$M = 4\pi \rho_0 \int_0^R \left(x^2 - \frac{x^4}{R^2} \right) dx$$

$$\Rightarrow M = 4\pi \rho_0 \left[\frac{r^3}{3} - \frac{r^5}{5R^2} \right]$$

Gravitational field,

$$E = \frac{GM}{r^2} = \frac{G}{r^2} \times 4\pi\rho_0 \left(\frac{r^3}{3} - \frac{r^5}{5R^2} \right)$$

$$\Rightarrow E = 4\pi G\rho_0 \left(\frac{r}{3} - \frac{r^3}{5R^2} \right)$$

E is maximum when $\frac{dE}{dr} = 0$

$$\Rightarrow \frac{dE}{dr} = 4\pi G\rho_0 \left(\frac{1}{3} - \frac{3r^2}{5R^2} \right) = 0$$

$$\Rightarrow r = \frac{\sqrt{5}}{3} R$$

29. (b) Gravitation field at the surface

$$E = \frac{Gm}{r^2}$$

$$\therefore E_1 = \frac{Gm_1}{r_1^2} \text{ and } E_2 = \frac{Gm_2}{r_2^2}$$

From the diagram given in question,

$$\frac{E_1}{E_2} = \frac{2}{3} \quad (r_1 = 1m, R_2 = 2m \text{ given})$$

$$\therefore \frac{E_1}{E_2} = \left(\frac{r_2}{r_1} \right)^2 \left(\frac{m_1}{m_2} \right) \Rightarrow \frac{2}{3} = \left(\frac{2}{1} \right)^2 \left(\frac{m_1}{m_2} \right)$$

$$\Rightarrow \left(\frac{m_1}{m_2} \right) = \frac{1}{6}$$

30. (16.00)

Using law of conservation of energy

Total energy at height $10R$ = total energy at earth

$$-\frac{GM_E m}{10R} + \frac{1}{2} m V_0^2 = -\frac{GM_E m}{R} + \frac{1}{2} m V^2$$

$$\left[\because \text{Gravitational potential energy} = -\frac{GMm}{r} \right]$$

$$\Rightarrow \frac{GM_E}{R} \left(1 - \frac{1}{10} \right) + \frac{V_0^2}{2} = \frac{V^2}{2}$$

$$\Rightarrow V^2 = V_0^2 + \frac{9}{5} gR$$

$$\Rightarrow V = \sqrt{V_0^2 + \frac{9}{5} gR} \approx 16 \text{ km/s}$$

[$\because V_0 = 12 \text{ km/s}$ given]

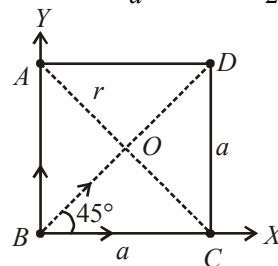
31. (c) $E_g = \frac{GM}{(3a)^2} + \frac{G(2M)}{(3a)^2} = \frac{GM}{3a^2}$

32. (b) $AC = a\sqrt{2} \quad \therefore r = \frac{AC}{2} = \frac{a\sqrt{2}}{2} = \frac{a}{\sqrt{2}}$

Resultant force on the body

$$B = \frac{GM^2}{a^2} \hat{i} + \frac{GM^2}{a^2} \hat{j} + \frac{GM^2}{(a\sqrt{2})^2} (\cos 45^\circ \hat{i} + \sin 45^\circ \hat{j})$$

$$\Rightarrow |F| = \frac{GM^2}{a^2} (\sqrt{2}) + \frac{GM^2}{2a^2}$$



$$\frac{Mv^2}{r} = \text{Resultant force towards centre}$$

$$\therefore \frac{Mv^2}{\left(\frac{a}{\sqrt{2}} \right)} = \frac{GM^2}{a^2} \left(\sqrt{2} + \frac{1}{2} \right)$$

$$\Rightarrow v^2 = \frac{GM}{a} \left(1 + \frac{1}{2\sqrt{2}} \right)$$

$$\Rightarrow v = \sqrt{\frac{GM}{a} \left(1 + \frac{1}{2\sqrt{2}} \right)} = 1.16 \sqrt{\frac{GM}{a}}$$

33. (a) $F = \frac{GMm}{r} = \int a \frac{\rho(dV)m}{r^2}$

$$= mG \int_0^R \frac{k}{r^2} \frac{4\pi r^2 dr}{r^2}$$

$$= -4\pi kGm \left(\frac{1}{r} \right)_0^R$$

$$= -\frac{4\pi kGm}{R}$$

Using Newton's second law, we have

$$\frac{mv_0^2}{R} = \frac{4\pi kGm}{R}$$

or $v_0 = C$ (const.)

$$\text{Time period, } T = \frac{2\pi R}{v_0} = \frac{2\pi R}{C}$$

$$\text{or } \frac{T}{R} = \text{constant.}$$

34. (c) Initial gravitational potential energy, $E_i = -\frac{GMm}{2R}$

Final gravitational potential energy,

$$E_f = -\frac{GMm/2}{2\left(\frac{R}{2}\right)} - \frac{GMm/2}{2\left(\frac{3R}{2}\right)} = -\frac{GMm}{2R} - \frac{GMm}{6R}$$

$$= -\frac{4GMm}{6R} = -\frac{2GMm}{3R}$$

$\therefore$ Difference between initial and final energy,

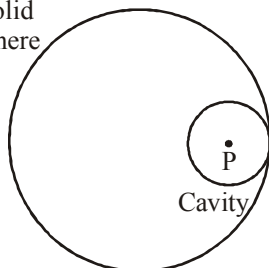
$$E_f - E_i = \frac{GMm}{R} \left(-\frac{2}{3} + \frac{1}{2} \right) = -\frac{GMm}{6R}$$

35. (d) Due to complete solid sphere, potential at point P

$$V_{\text{sphere}} = \frac{-GM}{2R^3} \left[3R^2 - \left(\frac{R}{2} \right)^2 \right]$$

$$= \frac{-GM}{2R^3} \left(\frac{11R^2}{4} \right) = -11 \frac{GM}{8R}$$

Solid
sphere



Due to cavity part potential at point P

$$V_{\text{cavity}} = - \frac{3}{2} \frac{\frac{GM}{R}}{\frac{R}{2}} = - \frac{3GM}{8R}$$

So potential at the centre of cavity

$$= V_{\text{sphere}} - V_{\text{cavity}}$$

$$= - \frac{11GM}{8R} - \left(- \frac{3}{8} \frac{GM}{R} \right) = - \frac{GM}{R}$$

36. (c) As, $V = - \frac{GM}{2R^3} (3R^2 - r^2)$

Graph (c) most closely depicts the correct variation of $v(r)$.

37. (d) Gravitational field, $I = (5\hat{i} + 12\hat{j})$ N/kg

$$I = - \frac{dv}{dr}$$

$$v = - \left[\int_0^x I_x dx + \int_0^y I_y dy \right]$$

$$= - [I_x \cdot x + I_y \cdot y]$$

$$= - [5(7-0) + 12(-3-0)]$$

$$= - [35 + (-36)] = 1 \text{ J/kg}$$

i.e., change in gravitational potential 1 J/kg.

Hence change in gravitational potential energy 1 J

38. (b) We choose reference point, infinity, where total energy of the system is zero.

So, initial energy of the system = 0

$$\text{Final energy} = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 - \frac{Gm_1 m_2}{d}$$

From conservation of energy,

Initial energy = Final energy

$$\therefore 0 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 - \frac{Gm_1 m_2}{d}$$

$$\text{or } \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 = \frac{Gm_1 m_2}{d} \quad \dots(1)$$

By conservation of linear momentum

$$m_1 v_1 + m_2 v_2 = 0 \text{ or } \frac{v_1}{v_2} = - \frac{m_2}{m_1} \Rightarrow v_2 = - \frac{m_1}{m_2} v_1$$

Putting value of v_2 in equation (1), we get

$$m_1 v_1^2 + m_2 \left(- \frac{m_1 v_1}{m_2} \right)^2 = \frac{2Gm_1 m_2}{d}$$

$$\frac{m_1 m_2 v_1^2 + m_1^2 v_1^2}{m_2} = \frac{2Gm_1 m_2}{d}$$

$$v_1 = \sqrt{\frac{2Gm_2^2}{d(m_1 + m_2)}} = m_2 \sqrt{\frac{2G}{d(m_1 + m_2)}}$$

$$\text{Similarly } v_2 = -m_1 \sqrt{\frac{2G}{d(m_1 + m_2)}}$$

39. (c) Let mass of smaller sphere (which has to be removed) is m

$$\text{Radius} = \frac{R}{2} \text{ (from figure)}$$

$$\frac{M}{\frac{4}{3} \pi R^3} = \frac{m}{\frac{4}{3} \pi \left(\frac{R}{2} \right)^3}$$

$$\Rightarrow m = \frac{M}{8}$$

Mass of the left over part of the sphere

$$M' = M - \frac{M}{8} = \frac{7}{8} M$$

Therefore gravitational field due to the left over part of the sphere

$$= \frac{GM'}{x^2} = \frac{7}{8} \frac{GM}{x^2}$$

40. (d) The work done to launch the spaceship

$$W = - \int_R^\infty \vec{F} \cdot \vec{dr} = - \int_R^\infty \frac{GMm}{r^2} dr$$

$$W = + \frac{GMm}{R}$$

... (i)

The force of attraction of the earth on the spaceship, when it was on the earth's surface

$$F = \frac{GMm}{R^2}$$

$$\Rightarrow mg = \frac{GMm}{R^2} \Rightarrow g = \frac{GM}{R^2}$$

... (ii)

Substituting the value of g in (i) we get

$$W = \frac{gR^2 m}{R}$$

$$\Rightarrow W = mgR$$

$$\Rightarrow W = 1000 \times 10 \times 6400 \times 10^3$$

$$= 6.4 \times 10^{10} \text{ Joule}$$

41. (c) Let 'M' be the mass of the particle

Now, $E_{\text{initial}} = E_{\text{final}}$

$$\text{i.e. } \frac{GMm}{\sqrt{2}r} + 0 = \frac{GMm}{r} + \frac{1}{2}MV^2$$

$$\text{or, } \frac{1}{2}MV^2 = \frac{GMm}{r} \left[1 - \frac{1}{\sqrt{2}} \right]$$

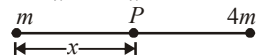
$$\Rightarrow \frac{1}{2}V^2 = \frac{Gm}{r} \left[1 - \frac{1}{\sqrt{2}} \right]$$

$$\text{or, } V = \sqrt{\frac{2Gm}{r} \left(1 - \frac{1}{\sqrt{2}} \right)}$$

42. (c) Let P be the point where gravitational field is zero.

$$\therefore \frac{Gm}{x^2} = \frac{4Gm}{(r-x)^2}$$

$$\Rightarrow \frac{1}{x} = \frac{2}{r-x} \Rightarrow r-x = 2x \Rightarrow x = \frac{r}{3}$$



Gravitational potential at P,

$$V = -\frac{Gm}{\frac{r}{3}} - \frac{4Gm}{\frac{2r}{3}} = -\frac{9Gm}{r}$$

43. (b) Gravitational field, $E = -\frac{GM}{r^2}$

$$\text{Flux, } \phi = \int \vec{E}_g \cdot d\vec{S} = |E| \cdot 4\pi r^2 = -4\pi GM$$

where, M = mass enclosed in the closed surface

This relationship is valid when $|\vec{E}_g| \propto \frac{1}{r^2}$.

44. (c) Initial P.E. $U_i = -\frac{GMm}{R}$

When the particle is far away from the sphere, the P.E. of the system is zero.

$$\therefore U_f = 0$$

$$W = \Delta U = U_f - U_i = 0 - \left[-\frac{GMm}{R} \right]$$

$$W = \frac{6.67 \times 10^{-11} \times 100}{0.1} \times \frac{10}{1000} = 6.67 \times 10^{-10} \text{ J}$$

45. (b) On earth's surface potential energy,

$$U = \frac{GmM}{R}$$

At a height R from the earth's surface, P.E. of system =

$$-\frac{GmM}{2R}$$

$$\therefore \Delta U = \frac{-GmM}{2R} + \frac{GmM}{R};$$

$$\Rightarrow \Delta U = \frac{GmM}{2R}$$

$$\text{Now } \frac{GM}{R^2} = g; \therefore \frac{GM}{R} = gR \quad \therefore \Delta U = \frac{1}{2}mgR$$

46. (d) Gravitational potential energy of mass m in an orbit of radius R

$$u = -\frac{GMm}{R}$$

Energy required = potential energy at 3R – potential energy at 2R

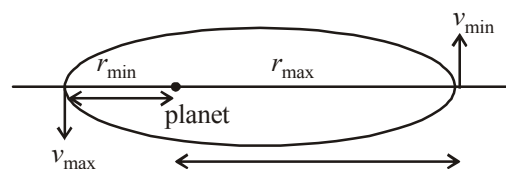
$$= \frac{-GMm}{3R} - \left(\frac{-GMm}{2R} \right)$$

$$= \frac{-GMm}{3R} + \frac{GMm}{2R}$$

$$= \frac{-2GMm + 3GMm}{6R} = \frac{GMm}{6R}$$

47. (a) By angular momentum conservation

$$r_{\min} v_{\max} = m r_{\max} v_{\min}$$



$$\text{Given, } v_{\min} = \frac{v_{\max}}{6}$$

$$\therefore \frac{r_{\min}}{r_{\max}} = \frac{v_{\min}}{v_{\max}} = \frac{1}{6}$$

48. (a) Orbital speed of the body when it revolves very close to the surface of planet

$$V_0 = \sqrt{\frac{GM}{R}} \quad \dots(i)$$

Here, G = gravitational constant

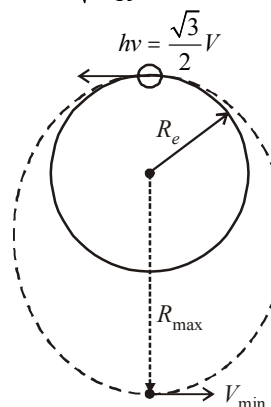
Escape speed from the surface of planet

$$V_e = \sqrt{\frac{2GM}{R}} \quad \dots(ii)$$

Dividing (i) by (ii), we have

$$\frac{V_0}{V_e} = \frac{\sqrt{\frac{GM}{R}}}{\sqrt{\frac{2GM}{R}}} = \frac{1}{\sqrt{2}}$$

49. (c)



Orbital velocity, $V_0 = \sqrt{\frac{GM}{R_e}}$

From energy conversation,

$$-\frac{GMm}{R_e} + \frac{1}{2}m\left(\sqrt{\frac{3}{2}}V\right)^2 = \frac{GMm}{R_{\max}} + \frac{1}{2}mV_{\min}^2 \quad \dots(1)$$

From angular momentum conversation

$$\sqrt{\frac{3}{2}}VR_e = V_{\min}R_{\max} \quad \dots(2)$$

Solving equation (1) and (2) we get,

$$R_{\max} = 3R_e$$

50. (a) According to question, mass density of a spherical galaxy varies as $\frac{k}{r}$.

Mass, $M = \int \rho dV$

$$\Rightarrow M = \int_0^{R_0} \frac{k}{r} 4\pi r^2 dr$$

$$\Rightarrow M = 4\pi k \int_0^{R_0} r dr$$

or, $M = \frac{4\pi k R_0^2}{2} = 2\pi k R^2$

$$F_G = \frac{GMm}{R_0^2} = m\omega_0^2 R (= F_C)$$

$$\Rightarrow \frac{G \frac{4\pi k R^2}{2}}{R^2} = \omega_0^2 R \Rightarrow \omega_0 = \sqrt{\frac{2\pi KG}{R}} \quad \left(\because \omega = \frac{2\pi}{T} \right)$$

$$\therefore T = \frac{2\pi}{\omega_0} = \frac{2\pi\sqrt{R}}{\sqrt{2\pi KG}} = \sqrt{\frac{2\pi R}{KG}} \Rightarrow T^2 = \frac{2\pi R}{KG}$$

$\therefore 2\pi, K$ and G are constants

$$\therefore T^2 \propto R.$$

51. (d) From law of conservation of momentum, $\vec{p}_i = \vec{p}_f$

$$m_1 u_1 + m_2 u_2 = MV_f$$

$$\Rightarrow v_f = \frac{\left(mv + \frac{mv}{4}\right)}{\frac{3m}{2}} = \frac{5v}{6}$$

Clearly, $v_f < v_i \therefore$ Path will be elliptical

52. (b) K.E. of satellite is zero at earth surface and at height h from energy conservation

$$U_{\text{surface}} + E_1 = U_h$$

$$-\frac{GM_em}{R_e} + E_1 = -\frac{GM_em}{(R_e+h)}$$

$$\Rightarrow E_1 = GM_em \left(\frac{1}{R_e} - \frac{1}{R_e+h} \right) \Rightarrow E_1 = \frac{GM_em}{(R_e+h)} \times \frac{h}{R_e}$$

Gravitational attraction

$$F_G = ma_c = \frac{mv^2}{(R_e+h)} = \frac{GM_em}{(R_e+h)^2}$$

$$mv^2 = \frac{GM_em}{(R_e+h)}$$

$$E_2 = \frac{mv^2}{2} = \frac{GM_em}{2(R_e+h)}$$

$$E_1 = E_2$$

$$\text{Clearly, } \frac{h}{R_e} = \frac{1}{2} \Rightarrow h = \frac{R_e}{2} = 3200 \text{ km}$$

53. (a) Escape velocity of the planet A is $V_A = \sqrt{\frac{2GM_A}{R_A}}$

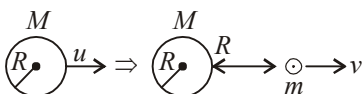
where M_A and R_A be the mass and radius of the planet A .

According to given problem

$$M_B = \frac{M_A}{2}, \quad R_B = \frac{R_A}{2}$$

$$\therefore V_B = \sqrt{\frac{2G \frac{M_A}{2}}{\frac{R_A}{2}}} \therefore \frac{V_A}{V_B} = \sqrt{\frac{\frac{2GM_A}{R_A}}{\frac{2GM_A/2}{R_A/2}}} = \frac{n}{4} = 1$$

$$\Rightarrow n = 4$$

54. (b) 

$$\frac{1}{2}mu^2 + \frac{-GMm}{R} = \frac{1}{2}mv^2 + \frac{-GMm}{2R}$$

$$\Rightarrow \frac{1}{2}m(v^2 - u^2) = \frac{-GMm}{2R}$$

$$\Rightarrow V = \sqrt{V^2 = u^2 - \frac{GM}{R}} \quad \dots(i)$$

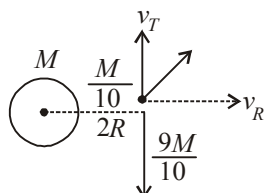
$$v_0 = \sqrt{\frac{GM}{2R}} \therefore v_{\text{rad}} = \frac{m \times v}{\left(\frac{m}{10}\right)} = 10v$$

Ejecting a rocket of mass $\frac{m}{10}$

$$\therefore \frac{9m}{10} \times \sqrt{\frac{GM}{2R}} = \frac{m}{10} \times v_{\tau} \Rightarrow V_{\tau}^2 = 81 \frac{GM}{2R}$$

Kinetic energy of rocket,

$$\begin{aligned} KE_{\text{rocket}} &= \frac{1}{2} \frac{M}{10} (V_T^2 + V_R^2) \\ &= \frac{1}{2} \times \frac{m}{10} \times \left(u^2 - \frac{GM}{R} \right) 100 + 81 \frac{GM}{R} \\ &= \frac{m}{20} \times 100 \left(u^2 - \frac{GM}{R} + \frac{81}{200} \frac{GM}{R} \right) \end{aligned}$$



$$= 5m \left(u^2 - \frac{119}{200} \frac{GM}{R} \right)$$

55. (d) Time period of revolution of satellite,

$$T = \frac{2\pi r}{v}$$

$$v = \sqrt{\frac{GM}{r}}$$

$$\therefore T = 2\pi r \sqrt{\frac{r}{GM}} = 2\pi \sqrt{\frac{r^3}{GM}}$$

Substituting the values, we get

$$T = 2\pi \sqrt{\frac{(202)^3 \times 10^{12}}{6.67 \times 10^{-11} \times 8 \times 10^{22}}} \text{ sec}$$

$$T = 7812.2 \text{ s}$$

$$T \approx 2.17 \text{ hr} \Rightarrow 11 \text{ revolutions.}$$

56. (d) Escape velocity,

$$v_c = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2G\rho V}{R}}$$

$$= \sqrt{\frac{2GS \times 4\pi R^3}{R}} = \sqrt{\frac{8}{3} \pi \rho G R^2}$$

$$\text{For moon, } v_c' = \sqrt{\frac{8}{3} \pi \rho G R_m^2}$$

$$\text{Given, } \frac{4}{3} \pi R^3 = 64 \times \frac{4}{3} \pi R_m^3 \text{ or } R_m = \frac{R}{4}$$

$$\therefore v_e' = \sqrt{\frac{8}{3} \pi \rho G \left(\frac{R}{4} \right)^2} = \frac{v_c}{4}$$

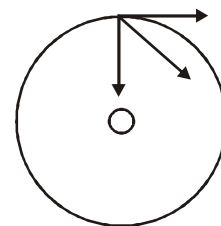
$$\frac{E}{E'} = \frac{\frac{1}{2} m v_e^2}{\frac{1}{2} m v_e'^2} = \frac{v_e^2}{v_e'^2} = \left(\frac{v_e}{v_e'} \right)^2 = 16$$

$$\text{or } E' = \frac{E}{16}$$

57. (b) $m\vec{v}_i + m\vec{v}_j = 2m\vec{v}$

$$\Rightarrow \vec{v} = \frac{v}{2} \hat{i} + \frac{v}{2} \hat{j}$$

$$\begin{aligned} \Rightarrow \vec{v} &= \sqrt{\left(\frac{v}{2} \right)^2 + \left(\frac{v}{2} \right)^2} = \frac{v}{\sqrt{2}} \\ &= \frac{1}{\sqrt{2}} \times \sqrt{\frac{GM}{R}} \end{aligned}$$



58. (b) Orbital, velocity, $v = \sqrt{\frac{GM}{r}}$

Kinetic energy of satellite A,

$$T_A = \frac{1}{2} m_A V_A^2$$

Kinetic energy of satellite B,

$$T_B = \frac{1}{2} m_B V_B^2$$

$$\Rightarrow \frac{T_A}{T_B} = \frac{m \times \frac{GM}{R}}{2m \times \frac{GM}{2R}} = 1$$

59. (d) For a satellite orbiting close to the earth, orbital velocity is given by

$$v_0 = \sqrt{g(R+h)} \approx \sqrt{gR}$$

Escape velocity (v_e) is

$$v_e = \sqrt{2g(R+h)} \approx \sqrt{2gR} \quad [\because h \ll R]$$

$$\Delta v = v_e - v_0 = (\sqrt{2} - 1) \sqrt{gR}$$

60. (b) At height r from center of earth, orbital velocity

$$v = \sqrt{\frac{GM}{r}}$$

By principle of energy conservation

$$KE \text{ of 'm'} + \left(-\frac{GMm}{r} \right) = 0 + 0$$

($\because$ At infinity, PE = KE = 0)

$$\text{or KE of 'm'} = \frac{GMm}{r} = \left(\sqrt{\frac{GM}{r}} \right)^2 m = mv^2$$

61. (d) Let M is mass of star m is mass of meteorite
By energy conservation between 0 and ∞ .

$$-\frac{GMm}{r} + \frac{-GMm}{r} + \frac{1}{2} m V_{\text{esc}}^2 = 0 + 0$$

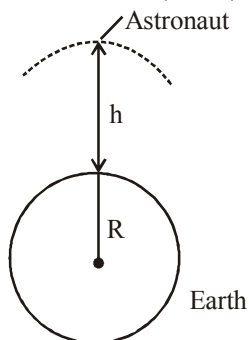
$$\therefore v = \sqrt{\frac{4GM}{r}} = \sqrt{\frac{4 \times 6.67 \times 10^{-11} \times 3 \times 10^{31}}{10^{11}}}$$

$$\approx 2.8 \times 10^5 \text{ m/s}$$

62. (b) For $h \ll R$, the orbital velocity is $\sqrt{gR}$
 Escape velocity = $\sqrt{2gR}$
 $\therefore$ The minimum increase in its orbital velocity
 $= \sqrt{2gR} - \sqrt{gR} = \sqrt{gR} (\sqrt{2} - 1)$

63. (c) According to universal law of Gravitation,

$$\text{Gravitational force } F = \frac{GMm}{(R+h)^2}$$



64. (a) $F = \frac{2GM}{Lr} m$ or, $\frac{mv^2}{r} = \frac{2GM}{Lr} m$
 $mr \left(\frac{2\pi}{T} \right)^2 = \frac{2GMm}{Lr} \left[\because v = r\omega \text{ and } \omega = \frac{2\pi}{T} \right]$

$$\Rightarrow T \propto r$$

65. (a) As we know,

$$\text{Gravitational potential energy} = \frac{-GMm}{r}$$

$$\text{and orbital velocity, } v_0 = \sqrt{GM/R+h}$$

$$E_f = \frac{1}{2}mv_0^2 - \frac{GMm}{3R} = \frac{1}{2}m \frac{GM}{3R} - \frac{GMm}{3R}$$

$$= \frac{GMm}{3R} \left(\frac{1}{2} - 1 \right) = \frac{-GMm}{6R}$$

$$E_i = \frac{-GMm}{R} + K$$

$$E_i = E_f$$

$$\text{Therefore minimum required energy, } K = \frac{5GMm}{6R}$$

66. (c) Escape velocity on earth,

$$v_e = \sqrt{\frac{2GM_e}{R_e}} = 11 \text{ km s}^{-1}$$

$$\therefore \frac{(v_e)_p}{(v_e)_e} = \frac{\sqrt{\frac{2GM_p}{R_p}}}{\sqrt{\frac{2GM_e}{R_e}}} = \sqrt{\frac{M_p}{M_e} \times \frac{R_e}{R_p}}$$

$$= \sqrt{\frac{10M_e}{M_e} \times \frac{R_e}{R_e/10}} = 10$$

$$\therefore (v_e)_p = 10 \times (v_e)_e = 10 \times 11 = 110 \text{ km/s}$$

67. (c) Gravitational force, $F = KR^{-n}$
 This force provides the centripetal force $MR\omega^2$ to the planet at height h above earth's surface.

$$\therefore F = KR^{-n} = MR\omega^2$$

$$\Rightarrow \omega^2 = KR^{-(n+1)}$$

$$\Rightarrow \omega = KR^{\frac{-(n+1)}{2}}$$

$$\frac{2\pi}{T} \propto R^{\frac{-(n+1)}{2}}$$

$$\therefore T \propto R^{\frac{+(n+1)}{2}}$$

68. (c) Time period of satellite is given by

$$T = 2\pi \sqrt{\frac{(R+h)^3}{GM}}$$

Where $R+h$ = radius of orbit of satellite

M = mass of earth.

Time period is independent of mass of satellite.

69. (d) Gravitational force provides the necessary centripetal force.

$\therefore$ Centripetal force on a satellite = Gravitational force

$$\therefore \frac{mv^2}{(R+x)} = \frac{GmM}{(R+x)^2} \text{ also } g = \frac{GM}{R^2}$$

$$\therefore \frac{mv^2}{(R+x)} = m \left(\frac{GM}{R^2} \right) \frac{R^2}{(R+x)^2} \frac{n!}{r!(n-r)!}$$

$$\therefore \frac{mv^2}{(R+x)} = mg \frac{R^2}{(R+x)^2}$$

$$\therefore v^2 = \frac{gR^2}{R+x} \Rightarrow v = \left(\frac{gR^2}{R+x} \right)^{1/2}$$

70. (c) $v_e = \sqrt{2gR}$

Clearly escape velocity does not depend on the angle at which the body is projected.

71. (c) $K.E = \frac{1}{2}mv_e^2$

Here v_e = escape velocity is independent of mass of the body

$$\text{Escape velocity, } v_e = \sqrt{2gR}$$

Substituting value of v_e in above equation we get

$$K.E = \frac{1}{2}m \times 2gR = mgR$$

72. (b) Due to inertia of motion it will move tangentially to the original orbit with the same velocity.

73. (a) Escape velocity, $v_e = \sqrt{2gR} = \sqrt{\frac{2GM}{R}}$

$$\Rightarrow v_e \propto m^0$$

Where M, R are the mass and radius of the planet respectively. Clearly, escape velocity is independent of mass of the body

Mechanical Properties of Solids



TOPIC 1 Hooke's Law & Young's Modulus



1. If the potential energy between two molecules is given by

$U = -\frac{A}{r^6} + \frac{B}{r^{12}}$, then at equilibrium, separation between molecules, and the potential energy are: [Sep. 06, 2020 (I)]

- (a) $\left(\frac{B}{2A}\right)^{\frac{1}{6}}, -\frac{A^2}{2B}$ (b) $\left(\frac{B}{A}\right)^{\frac{1}{6}}, 0$
 (c) $\left(\frac{2B}{A}\right)^{\frac{1}{6}}, -\frac{A^2}{4B}$ (d) $\left(\frac{2B}{A}\right)^{\frac{1}{6}}, -\frac{A^2}{2B}$

2. A body of mass $m = 10$ kg is attached to one end of a wire of length 0.3 m. The maximum angular speed (in rad s^{-1}) with which it can be rotated about its other end in space station is (Breaking stress of wire = $4.8 \times 10^7 \text{ Nm}^{-2}$ and area of cross-section of the wire = 10^{-2} cm^2) is _____.

[9 Jan 2020 I]

3. A uniform cylindrical rod of length L and radius r , is made from a material whose Young's modulus of Elasticity equals Y . When this rod is heated by temperature T and simultaneously subjected to a net longitudinal compressional force F , its length remains unchanged. The coefficient of volume expansion, of the material of the rod, is (nearly) equal to :

[12 April 2019 II]

- (a) $9F/(\pi r^2 Y T)$ (b) $6F/(\pi r^2 Y T)$
 (c) $3F/(\pi r^2 Y T)$ (d) $F/(3\pi r^2 Y T)$

4. In an environment, brass and steel wires of length 1 m each with areas of cross section 1 mm^2 are used. The wires are connected in series and one end of the combined wire is connected to a rigid support and other end is subjected to elongation. The stress required to produce a net elongation of 0.2 mm is,

[Given, the Young's modulus for steel and brass are, respectively, $120 \times 10^9 \text{ N/m}^2$ and $60 \times 10^9 \text{ N/m}^2$]

[10 April 2019 II]

- (a) $1.2 \times 10^6 \text{ N/m}^2$ (b) $4.0 \times 10^6 \text{ N/m}^2$
 (c) $1.8 \times 10^6 \text{ N/m}^2$ (d) $0.2 \times 10^6 \text{ N/m}^2$

5. The elastic limit of brass is 379 MPa. What should be the minimum diameter of a brass rod if it is to support a 400 N load without exceeding its elastic limit?

[10 April 2019 II]

- (a) 1.00 mm (b) 1.16 mm
 (c) 0.90 mm (d) 1.36 mm

6. A steel wire having a radius of 2.0 mm, carrying a load of 4kg, is hanging from a ceiling. Given that $g = 3.1 \text{ Å ms}^{-2}$, what will be the tensile stress that would be developed in the wire?

[9 April 2019 I]

- (a) $6.2 \times 10^6 \text{ Nm}^{-2}$ (b) $5.2 \times 10^6 \text{ Nm}^{-2}$
 (c) $3.1 \times 10^6 \text{ Nm}^{-2}$ (d) $4.8 \times 10^6 \text{ Nm}^{-2}$

7. A steel wire having a radius of 2.0 mm, carrying a load of 4kg, is hanging from a ceiling. Given that $g = 3.1 \text{ Å ms}^{-2}$, what will be the tensile stress that would be developed in the wire?

[8 April 2019 I]

- (a) $6.2 \times 10^6 \text{ Nm}^{-2}$ (b) $5.2 \times 10^6 \text{ Nm}^{-2}$
 (c) $3.1 \times 10^6 \text{ Nm}^{-2}$ (d) $4.8 \times 10^6 \text{ Nm}^{-2}$

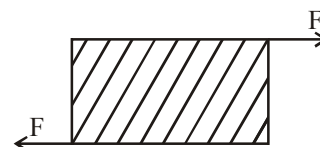
8. Young's moduli of two wires A and B are in the ratio 7 : 4. Wire A is 2 m long and has radius R . Wire B is 1.5 m long and has radius 2 mm. If the two wires stretch by the same length for a given load, then the value of R is close to :

[8 April 2019 II]

- (a) 1.5 mm (b) 1.9 mm (c) 1.7 mm (d) 1.3 mm

9. As shown in the figure, forces of 10^5 N each are applied in opposite directions, on the upper and lower faces of a cube of side 10cm, shifting the upper face parallel to itself by 0.5cm. If the side of another cube of the same material is, 20cm, then under similar conditions as above, the displacement will be:

[Online April 15, 2018]



- (a) 1.00 cm (b) 0.25 cm
 (c) 0.37 cm (d) 0.75 cm

10. A thin 1 m long rod has a radius of 5 mm. A force of $50 \pi kN$ is applied at one end to determine its Young's modulus. Assume that the force is exactly known. If the least count in the measurement of all lengths is 0.01 mm, which of the following statements is false? [Online April 10, 2016]

- (a) The maximum value of Y that can be determined is $2 \times 10^{14} N/m^2$.
 (b) $\frac{\Delta Y}{Y}$ gets minimum contribution from the uncertainty in the length
 (c) $\frac{\Delta Y}{Y}$ gets its maximum contribution from the uncertainty in strain
 (d) The figure of merit is the largest for the length of the rod.

11. A uniformly tapering conical wire is made from a material of Young's modulus Y and has a normal, unextended length L . The radii, at the upper and lower ends of this conical wire, have values R and $3R$, respectively. The upper end of the wire is fixed to a rigid support and a mass M is suspended from its lower end. The equilibrium extended length, of this wire, would equal : [Online April 9, 2016]

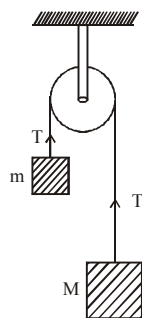
- (a) $L \left(1 + \frac{2}{9} \frac{Mg}{\pi Y R^2} \right)$ (b) $L \left(1 + \frac{1}{9} \frac{Mg}{\pi Y R^2} \right)$
 (c) $L \left(1 + \frac{1}{3} \frac{Mg}{\pi Y R^2} \right)$ (d) $L \left(1 + \frac{2}{3} \frac{Mg}{\pi Y R^2} \right)$

12. The pressure that has to be applied to the ends of a steel wire of length 10 cm to keep its length constant when its temperature is raised by $100^\circ C$ is:

(For steel Young's modulus is $2 \times 10^{11} N/m^2$ and coefficient of thermal expansion is $1.1 \times 10^{-5} K^{-1}$) [2014]

- (a) $2.2 \times 10^8 Pa$ (b) $2.2 \times 10^9 Pa$
 (c) $2.2 \times 10^7 Pa$ (d) $2.2 \times 10^6 Pa$

13. Two blocks of masses m and M are connected by means of a metal wire of cross-sectional area A passing over a frictionless fixed pulley as shown in the figure. The system is then released. If $M = 2m$, then the stress produced in the wire is : [Online April 25, 2013]



- (a) $\frac{2mg}{3A}$ (b) $\frac{4mg}{3A}$ (c) $\frac{mg}{A}$ (d) $\frac{3mg}{4A}$

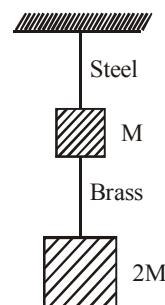
14. A copper wire of length 1.0 m and a steel wire of length 0.5 m having equal cross-sectional areas are joined end to end. The composite wire is stretched by a certain load which stretches the copper wire by 1 mm. If the Young's moduli of copper and steel are respectively $1.0 \times 10^{11} N/m^2$ and $2.0 \times 10^{11} N/m^2$, the total extension of the composite wire is : [Online April 23, 2013]

- (a) 1.75 mm (b) 2.0 mm (c) 1.50 mm (d) 1.25 mm

15. A uniform wire (Young's modulus $2 \times 10^{11} N/m^2$) is subjected to longitudinal tensile stress of $5 \times 10^7 N/m^2$. If the overall volume change in the wire is 0.02%, the fractional decrease in the radius of the wire is close to : [Online April 22, 2013]

- (a) 1.0×10^{-4} (b) 1.5×10^{-4}
 (c) 0.25×10^{-4} (d) 5×10^{-4}

16. If the ratio of lengths, radii and Young's moduli of steel and brass wires in the figure are a , b and c respectively, then the corresponding ratio of increase in their lengths is : [Online April 9, 2013]



- (a) $\frac{3c}{2ab^2}$ (b) $\frac{2a^2c}{b}$ (c) $\frac{3a}{2b^2c}$ (d) $\frac{2ac}{b^2}$

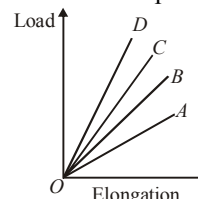
17. A steel wire can sustain 100 kg weight without breaking. If the wire is cut into two equal parts, each part can sustain a weight of [Online May 19, 2012]

- (a) 50 kg (b) 400 kg (c) 100 kg (d) 200 kg

18. A structural steel rod has a radius of 10 mm and length of 1.0 m. A 100 kN force stretches it along its length. Young's modulus of structural steel is $2 \times 10^{11} N/m^2$. The percentage strain is about [Online May 7, 2012]

- (a) 0.16% (b) 0.32% (c) 0.08% (d) 0.24%

19. The load versus elongation graphs for four wires of same length and made of the same material are shown in the figure. The thinnest wire is represented by the line [Online May 7, 2012]



- (a) OA (b) OC (c) OD (d) OB

20. Two wires are made of the same material and have the same volume. However wire 1 has cross-sectional area A and wire 2 has cross-sectional area $3A$. If the length of wire 1 increases by Δx on applying force F , how much force is needed to stretch wire 2 by the same amount? [2009]

- (a) $4F$ (b) $6F$ (c) $9F$ (d) F

21. A wire elongates by l mm when a load W is hanged from it. If the wire goes over a pulley and two weights W each are hung at the two ends, the elongation of the wire will be (in mm)

[2006]

- (a) l (b) $2l$ (c) zero (d) $l/2$

TOPIC 2 Bulk and Rigidity Modulus and Work Done in Stretching a Wire



22. Two steel wires having same length are suspended from a ceiling under the same load. If the ratio of their energy stored per unit volume is $1 : 4$, the ratio of their diameters is:

[9 Jan 2020 II]

- (a) $\sqrt{2} : 1$ (b) $1 : 2$
(c) $2 : 1$ (d) $1 : \sqrt{2}$

23. A boy's catapult is made of rubber cord which is 42 cm long, with 6 mm diameter of cross-section and of negligible mass. The boy keeps a stone weighing 0.02 kg on it and stretches the cord by 20 cm by applying a constant force. When released, the stone flies off with a velocity of 20 ms^{-1} . Neglect the change in the area of cross-section of the cord while stretched. The Young's modulus of rubber is closest to :

[8 April 2019 I]

- (a) 10^6 N/m^2 (b) 10^4 N/m^2
(c) 10^8 N/m^2 (d) 10^3 N/m^2

24. A solid sphere of radius r made of a soft material of bulk modulus K is surrounded by a liquid in a cylindrical container. A massless piston of area a floats on the surface of the liquid, covering entire cross-section of cylindrical container. When a mass m is placed on the surface of the piston to compress the liquid, the fractional decrement in

the radius of the sphere $\left(\frac{dr}{r}\right)$, is: [2018]

- (a) $\frac{Ka}{mg}$ (b) $\frac{Ka}{3mg}$ (c) $\frac{mg}{3Ka}$ (d) $\frac{mg}{Ka}$

25. A bottle has an opening of radius a and length b . A cork of length b and radius $(a + \Delta a)$ where $(\Delta a < a)$ is compressed to fit into the opening completely (see figure). If the bulk modulus of cork is B and frictional coefficient between the bottle and cork is μ then the force needed to push the cork into the bottle is :

[Online April 10, 2016]



- (a) $(\pi\mu B b) a$ (b) $(2\pi\mu B b) \Delta a$
(c) $(\pi\mu B b) \Delta a$ (d) $(4\pi\mu B b) \Delta a$
26. Steel ruptures when a shear of $3.5 \times 10^8 \text{ N m}^{-2}$ is applied. The force needed to punch a 1 cm diameter hole in a steel sheet 0.3 cm thick is nearly: [Online April 12, 2014]
- (a) $1.4 \times 10^4 \text{ N}$ (b) $2.7 \times 10^4 \text{ N}$
(c) $3.3 \times 10^4 \text{ N}$ (d) $1.1 \times 10^4 \text{ N}$
27. The bulk moduli of ethanol, mercury and water are given as 0.9, 25 and 2.2 respectively in units of 10^9 Nm^{-2} . For a given value of pressure, the fractional compression in volume is $\frac{\Delta V}{V}$. Which of the following statements about $\frac{\Delta V}{V}$ for these three liquids is correct? [Online April 11, 2014]
- (a) Ethanol > Water > Mercury
(b) Water > Ethanol > Mercury
(c) Mercury > Ethanol > Water
(d) Ethanol > Mercury > Water
28. In materials like aluminium and copper, the correct order of magnitude of various elastic moduli is: [Online April 9, 2014]
- (a) Young's modulus < shear modulus < bulk modulus.
(b) Bulk modulus < shear modulus < Young's modulus.
(c) Shear modulus < Young's modulus < bulk modulus.
(d) Bulk modulus < Young's modulus < shear modulus.
29. If ' S ' is stress and ' Y ' is young's modulus of material of a wire, the energy stored in the wire per unit volume is [2005]
- (a) $\frac{S^2}{2Y}$ (b) $2S^2Y$ (c) $\frac{S}{2Y}$ (d) $\frac{2Y}{S^2}$
30. A wire fixed at the upper end stretches by length ℓ by applying a force F . The work done in stretching is [2004]
- (a) $2F\ell$ (b) $F\ell$ (c) $\frac{F}{2\ell}$ (d) $\frac{F\ell}{2}$



Hints & Solutions



1. (c) Given: $U = \frac{-A}{r^6} + \frac{B}{r^{12}}$

For equilibrium,

$$F = \frac{dU}{dr} = -(A(-6r^{-7})) + B(-12r^{-13}) = 0$$

$$\Rightarrow 0 = \frac{6A}{r^7} - \frac{12B}{r^{13}} \Rightarrow \frac{6A}{12B} = \frac{1}{r^6}$$

$$\therefore \text{Separation between molecules, } r = \left(\frac{2B}{A}\right)^{1/6}$$

Potential energy,

$$U\left(r = \left(\frac{2B}{A}\right)^{1/6}\right) = -\frac{A}{2B/A} + \frac{B}{4B^2/A^2}$$

$$= \frac{-A^2}{2B} + \frac{A^2}{4B} = \frac{-A^2}{4B}$$

2. (4) Given: Wire length, $l = 0.3$ m

Mass of the body, $m = 10$ kg

Breaking stress, $\sigma = 4.8 \times 10^7 \text{ Nm}^{-2}$

Area of cross-section, $a = 10^{-2} \text{ cm}^2$

Maximum angular speed $\omega = ?$

$$T = Ml\omega^2$$

$$\sigma = \frac{T}{A} = \frac{ml\omega^2}{A}$$

$$\frac{ml\omega^2}{A} \leq 4.8 \times 10^7 \Rightarrow \omega^2 \leq \frac{(4.8 \times 10^7)A}{ml}$$

$$\Rightarrow \omega^2 \leq \frac{(4.8 \times 10^7)(10^{-6})}{10 \times 3} = 16 \Rightarrow \omega_{\max} = 4 \text{ rad/s}$$

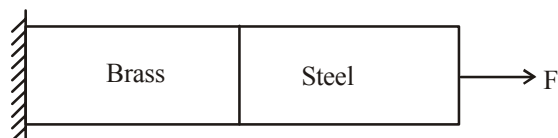
3. (c) $\Delta_{\text{temp}} = \Delta_{\text{force}}$

$$\text{or } L\alpha(\Delta T) = \frac{FL}{AY} \quad \therefore \alpha = \frac{FL}{AYT} = \frac{F}{\pi r^2 YT}$$

Coefficient of volume expansion

$$r = 3\alpha = \frac{3F}{\pi r^2 YT}$$

4. (Bonus)



Young modulus, $Y = \frac{\text{Stress}}{\left(\frac{\Delta l}{L}\right)}$

Let σ be the stress

$$\text{Total elongation } \Delta l_{\text{net}} = \frac{\sigma L_1}{Y_1} + \frac{\sigma L_2}{Y_2}$$

$$\Delta l_{\text{net}} = \sigma \left[\frac{1}{Y_1} + \frac{1}{Y_2} \right] \quad [\because L_1 = L_2 = 1 \text{ m}]$$

$$\sigma = \Delta l \left(\frac{Y_1 Y_2}{Y_1 + Y_2} \right)$$

$$= 0.2 \times 10^{-3} \times \left(\frac{120 \times 60}{180} \right) \times 10^9 = 8 \times 10^6 \frac{\text{N}}{\text{m}^2}$$

5. (b) Stress = $\frac{F}{A} = \frac{400 \times 4}{\pi d^2} = 379 \times 10^6 \text{ N/m}^2$

$$\Rightarrow d^2 = \frac{400 \times 4}{379 \times 10^6 \pi}$$

$$d = 1.15 \text{ mm}$$

6. (c) Given,

Radius of wire, $r = 2$ mm

Mass of the load $m = 4$ kg

$$\text{Stress} = \frac{F}{A} = \frac{mg}{\pi(r)^2}$$

$$= \frac{4 \times 3.1\pi}{\pi \times (2 \times 10^{-3})^2} = 3.1 \times 10^6 \text{ N/m}^2$$

7. (c) Given,

Radius of wire, $r = 2$ mm

Mass of the load $m = 4$ kg

$$\text{Stress} = \frac{F}{A} = \frac{mg}{\pi(r)^2} = \frac{4 \times 3.1\pi}{\pi \times (2 \times 10^{-3})^2} = 3.1 \times 10^6 \text{ N/m}^2$$

8. (c) $\Delta_1 = \Delta_2$

$$\text{or } \frac{Fl_1}{\pi r_1^2 y_1} = \frac{Fl_2}{\pi r_2^2 y_2} \text{ or } \frac{2}{R^2 \times 7} = \frac{1.5}{2^2 \times 4}$$

$$\therefore R = 1.75 \text{ mm}$$

9. (b) For same material the ratio of stress to strain is same
For first cube

$$\text{Stress}_1 = \frac{\text{force}_1}{\text{area}_1} = \frac{10^5}{(0.1^2)}$$

$$\text{Strain}_1 = \frac{\text{change in length}_1}{\text{original length}_1} = \frac{0.5 \times 10^{-2}}{0.1}$$

For second block,

$$\text{stress}_2 = \frac{\text{force}_2}{\text{area}_2} = \frac{10^5}{(0.2)^2}$$

$$\text{strain}_2 = \frac{\text{change in length}_2}{\text{original length}_2} = \frac{x}{0.2}$$

x is the displacement for second block.

$$\text{For same material, } \frac{\text{stress}_1}{\text{strain}_1} = \frac{\text{stress}_2}{\text{strain}_2}$$

$$\text{or, } \frac{\frac{10.5}{(0.1)^2}}{\frac{0.5 \times 10^{-2}}{0.1}} = \frac{\frac{10^5}{(0.2)^2}}{\frac{x}{0.2}}$$

Solving we get, $x = 0.25 \text{ cm}$

10. (a) Young's modulus $Y = \frac{F}{A} \cdot \frac{\Delta \ell}{\ell}$

$$Y = \frac{F \ell}{\pi r^2 \Delta \ell}$$

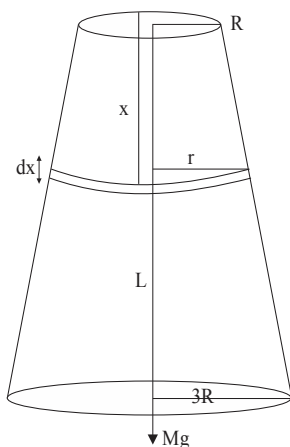
Given, radius $r = 5 \text{ mm}$, force $F = 50 \pi \text{ k N}$,

$$\frac{\ell}{\Delta \ell} = 0.01 \text{ mm}$$

$$\therefore Y = \frac{F}{\pi r^2} \cdot \frac{\ell}{\Delta \ell} = 2 \times 10^{14} \text{ N/m}^2.$$

11. (c) Consider a small element dx of radius r ,

$$r = \frac{2R}{L} x + R$$



At equilibrium change in length of the wire

$$\int_0^L dL = \int \frac{Mg dx}{\pi \left[\frac{2R}{L} x + R \right]^2 y}$$

Taking limit from 0 to L

$$\Delta L = \frac{Mg}{\pi y} \left[-\frac{1}{\left[\frac{2Rx}{L} + R \right]} \times \frac{L}{2R} \right]_0^L = \frac{MgL}{3\pi R^2 y}$$

The equilibrium extended length of wire = $L + \Delta L$

$$= L + \frac{MgL}{3\pi R^2 Y} = L \left(1 + \frac{1}{3} \frac{Mg}{\pi Y R^2} \right)$$

12. (a) Young's modulus $Y = \frac{\text{stress}}{\text{strain}}$

$$\text{stress} = Y \times \text{strain}$$

Stress in steel wire = Applied pressure

$$\text{Pressure} = \text{stress} = Y \times \text{strain}$$

$$\text{Strain} = \frac{\Delta L}{L} = \alpha \Delta T$$

(As length is constant)

$$= 2 \times 10^{11} \times 1.1 \times 10^{-5} \times 100 = 2.2 \times 10^8 \text{ Pa}$$

13. (b) Tension in the wire, $T = \left(\frac{2mM}{m+M} \right) g$

$$\text{Stress} = \frac{\text{Force / Tension}}{\text{Area}} = \frac{2mM}{A(m+M)} g$$

$$= \frac{2(m \times 2m)g}{A(m+2m)} \quad (M = 2m \text{ given})$$

$$= \frac{4m^2}{3mA} g = \frac{4mg}{3A}$$

14. (d) $Y_c \times (\Delta L_c / L_c) = Y_s \times (\Delta L_s / L_s)$

$$\Rightarrow 1 \times 10^{11} \times \left(\frac{1 \times 10^{-3}}{1} \right) = 2 \times 10^{11} \times \left(\frac{\Delta L_s}{0.5} \right)$$

$$\therefore \Delta L_s = \frac{0.5 \times 10^{-3}}{2} = 0.25 \text{ mm}$$

Therefore, total extension of the composite wire

$$= \Delta L_c + \Delta L_s$$

$$= 1 \text{ mm} + 0.25 \text{ mm} = 1.25 \text{ mm}$$

15. (c) Given, $y = 2 \times 10^{11} \text{ Nm}^{-2}$

$$\text{Stress} \left(\frac{F}{A} \right) = 5 \times 10^7 \text{ Nm}^{-2}$$

$$\Delta V = 0.02\% = 2 \times 10^{-4} \text{ m}^3$$

$$\frac{\Delta r}{r} = ?$$

$$\gamma = \frac{\text{stress}}{\text{strain}} \Rightarrow \text{strain} \left(\frac{\Delta \ell}{\ell_0} \right) = \frac{\gamma}{\text{stress}} \quad \dots (i)$$

$$\Delta V = 2\pi r \ell_0 \Delta r - \pi r^2 \Delta \ell \quad \dots (ii)$$

From eqns (i) and (ii) putting the value of $\Delta \ell$, ℓ_0 and ΔV and solving we get

$$\frac{\Delta r}{r} = 0.25 \times 10^{-4}$$

16. (c) According to questions,

$$\frac{\ell_s}{\ell_b} = a, \frac{r_s}{r_b} = b, \frac{y_s}{y_b} = c, \frac{\Delta \ell_s}{\Delta \ell_b} = ?$$

$$\text{As, } y = \frac{F\ell}{A\Delta\ell} \Rightarrow \Delta\ell = \frac{F\ell}{Ay}$$

$$\Delta\ell_s = \frac{3mg\ell_s}{\pi r_s^2 y_s} [\because F_s = (M + 2M)g]$$

$$\Delta\ell_b = \frac{2Mg\ell_b}{\pi r_b^2 y_b} [\because F_b = 2Mg]$$

$$\frac{\Delta\ell_s}{\Delta\ell_b} = \frac{\frac{3Mg\ell_s}{\pi r_s^2 y_s}}{\frac{2Mg\ell_b}{\pi r_b^2 y_b}} = \frac{3a}{2b^2c}$$

17. (c) Breaking force $\propto$ area of cross section of wire
Load hold by wire is independent of length of the wire.

18. (a) Given: $F = 100 \text{ kN} = 10^5 \text{ N}$

$$Y = 2 \times 10^{11} \text{ Nm}^{-2}$$

$$\ell_0 = 1.0 \text{ m}$$

$$\text{radius } r = 10 \text{ mm} = 10^{-2} \text{ m}$$

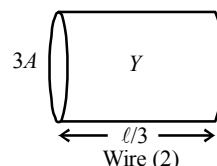
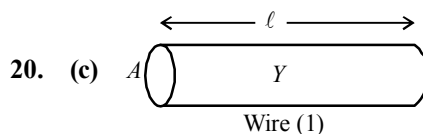
$$\text{From formula, } Y = \frac{\text{Stress}}{\text{Strain}}$$

$$\Rightarrow \text{Strain} = \frac{\text{Stress}}{Y} = \frac{F}{AY}$$

$$= \frac{10^5}{\pi r^2 Y} = \frac{10^5}{3.14 \times 10^{-4} \times 2 \times 10^{11}} = \frac{1}{628}$$

$$\text{Therefore \% strain} = \frac{1}{628} \times 100 = 0.16\%$$

19. (a) From the graph, it is clear that for the same value of load, elongation is maximum for wire OA. Hence OA is the thinnest wire among the four wires.



For wire 1

Length, $L_1 = \ell$

Area, $A_1 = A$

For wire 2

Length, $L_2 = \frac{\ell}{3}$

Area, $A_2 = 3A$

As the wires are made of same material, so they will have same young's modulus.

For wire 1,

$$Y = \frac{F/A}{\Delta x/\ell} \quad \dots (i)$$

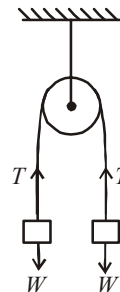
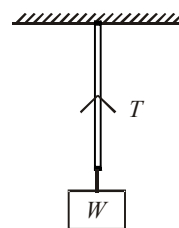
For wire 2,

$$Y = \frac{F'/3A}{\Delta x/(\ell/3)} \quad \dots (ii)$$

From (i) and (ii) we get,

$$\frac{F}{A} \times \frac{\ell}{\Delta x} = \frac{F'}{3A} \times \frac{\ell}{3\Delta x} \Rightarrow F' = 9F$$

21. (a) Case (i)



At equilibrium, $T = W$

$$\text{Young's modulus, } Y = \frac{W/A}{\ell/L} \quad \dots (1)$$

$$\text{Elongation, } \ell = \frac{W}{A} \times \frac{L}{Y}$$

Case (ii) At equilibrium $T = W$

$$\therefore \text{Young's modulus, } Y = \frac{W/A}{\ell/L}$$

$$\Rightarrow Y = \frac{W/A}{\ell/L} \Rightarrow \ell = \frac{W}{A} \times \frac{L}{Y}$$

$\Rightarrow$ Elongation is the same.

22. (a) If force F acts along the length L of the wire of cross-section A , then energy stored in unit volume of wire is given by

$$\text{Energy density} = \frac{1}{2} \text{ stress} \times \text{strain}$$

$$= \frac{1}{2} \times \frac{F}{A} \times \frac{F}{AY} \left(\because \text{stress} = \frac{F}{A} \text{ and strain} = \frac{X}{AY} \right)$$

$$= \frac{1}{2} \frac{F^2}{A^2 Y} = \frac{1}{2} \frac{F^2 \times 16}{(\pi d^2)^2 Y} = \frac{1}{2} \frac{F^2 \times 16}{\pi d^4 Y}$$

If u_1 and u_2 are the densities of two wires, then

$$\frac{u_1}{u_2} = \left(\frac{d_2}{d_1} \right)^4 \Rightarrow \frac{d_1}{d_2} = (4)^{1/4} \Rightarrow \frac{d_1}{d_2} = \sqrt{2} : 1$$

23. (a) When a catapult is stretched up to length l , then the stored energy in it $= \Delta k. E \Rightarrow$

$$\frac{1}{2} \left(\frac{YA}{L} \right) (\Delta l)^2 = \frac{1}{2} mv^2 \Rightarrow y = \frac{mv^2 L}{\Delta (\Delta l)^2}$$

$$m = 0.02 \text{ kg}$$

$$v = 20 \text{ ms}^{-1}$$

$$L = 0.42 \text{ m}$$

$$A = (\pi d^2)/(4)$$

$$d = 6 \times 10^{-3} \text{ m}$$

$$\Delta l = 0.2 \text{ m}$$

$$y = \frac{0.02 \times 400 \times 0.42 \times 4}{\pi \times 36 \times 10^{-6} \times 0.04} = 2.3 \times 10^6 \text{ N/m}^2$$

So, order is 10^6 .

24. (c) Bulk modulus, $K = \frac{\text{volumetric stress}}{\text{volumetric strain}}$

$$K = \frac{mg}{a \left(\frac{dV}{V} \right)}$$

$$\Rightarrow \frac{dV}{V} = \frac{mg}{Ka} \quad \dots(i)$$

$$\text{volume of sphere, } V = \frac{4}{3} \pi R^3$$

$$\text{Fractional change in volume } \frac{dV}{V} = \frac{3dr}{r} \quad \dots(ii)$$

$$\text{Using eq. (i) \& (ii) } \frac{3dr}{r} = \frac{mg}{Ka}$$

$$\therefore \frac{dr}{r} = \frac{mg}{3Ka} \quad (\text{fractional decrement in radius})$$

$$25. (d) \text{ Stress} = \frac{\text{Normal force}}{\text{Area}} = \frac{N}{A} = \frac{N}{(2\pi a)b}$$

Stress = $B \times$ strain

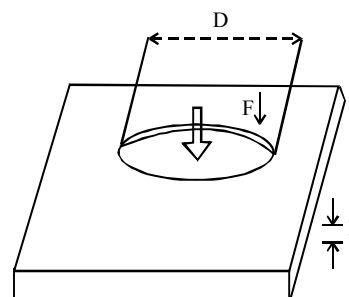
$$\frac{N}{(2\pi a)b} = B \frac{2\pi a \Delta a \times b}{\pi a^2 b}$$

$$\Rightarrow N = B \frac{(2\pi a)^2 \Delta a b^2}{\pi a^2 b}$$

Force needed to push the cork.

$$f = \mu N = \mu 4\pi b \Delta a B = (4\pi \mu B b) \Delta a$$

26. (c)



Shearing strain is created along the side surface of the punched disk. Note that the forces exerted on the disk are exerted along the circumference of the disk, and the total force exerted on its center only.

Let us assume that the shearing stress along the side surface of the disk is uniform, then

$$F = \int_{\text{surface}} dF_{\text{max}} = \int_{\text{surface}} \sigma_{\text{max}} dA = \sigma_{\text{max}} \int_{\text{surface}} dA$$

$$= \int \sigma_{\text{max}} \cdot A = \sigma_{\text{max}} \cdot 2\pi \left(\frac{D}{2} \right) h$$

$$= 3.5 \times 10^8 \times \left(\frac{1}{2} \times 10^{-2} \right) \times 0.3 \times 10^{-2} \times 2\pi$$

$$= 3.297 \times 10^4 \approx 3.3 \times 10^4 \text{ N}$$

$$27. (a) \text{ Compressibility} = \frac{1}{\text{Bulk modulus}}$$

As bulk modulus is least for ethanol (0.9) and maximum for mercury (25) among ethanol, mercury and water. Hence

$$\text{compression in volume } \frac{\Delta V}{V}$$

Ethanol > Water > Mercury

28. (c) Poisson's ratio, $\sigma = \frac{\text{lateral strain } (\beta)}{\text{longitudinal strain } (\alpha)}$

For material like copper, $\sigma = 0.33$

And, $Y = 3k(1 - 2\sigma)$

$$\text{Also, } \frac{9}{Y} = \frac{1}{k} + \frac{3}{\eta}$$

$$Y = 2\eta(1 + \sigma)$$

Hence, $\eta < Y < k$

29. (a) Energy stored in the wire per unit volume,

$$E = \frac{1}{2} \times \text{stress} \times \text{strain} \quad \dots(i)$$

We know that,

$$Y = \frac{\text{stress}}{\text{strain}}$$

$$\Rightarrow \text{strain} = \frac{\text{stress}}{Y}$$

On substituting the expression of strain in equation (i) we get

$$E = \frac{1}{2} \times \text{stress} \times \frac{\text{stress}}{Y} = \frac{1}{2} \cdot \frac{S^2}{Y}$$

30. (d) Let A and L be the area and length of the wire.
Work done by constant force in displacing the wire by a distance ℓ .

= change in potential energy

$$= \frac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume}$$

$$= \frac{1}{2} \times \frac{F}{A} \times \frac{\ell}{L} \times A \times L = \frac{F\ell}{2}$$

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Mechanical Properties of Fluids

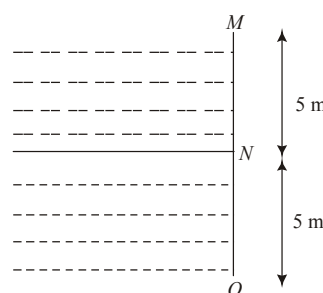


TOPIC 1 Pressure, Density, Pascal's Law and Archimedes' Principle



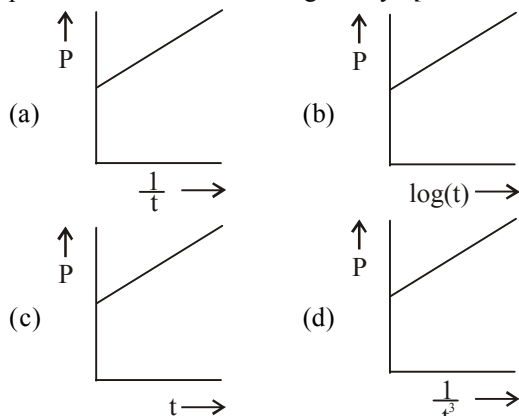
1. A hollow spherical shell at outer radius R floats just submerged under the water surface. The inner radius of the shell is r . If the specific gravity of the shell material is $\frac{27}{8}$ w.r.t water, the value of r is : **[5 Sep. 2020 (I)]**
 (a) $\frac{8}{9}R$ (b) $\frac{4}{9}R$ (c) $\frac{2}{3}R$ (d) $\frac{1}{3}R$
2. An air bubble of radius 1 cm in water has an upward acceleration 9.8 cm s^{-2} . The density of water is 1 gm cm^{-3} and water offers negligible drag force on the bubble. The mass of the bubble is ($g = 980 \text{ cm/s}^2$) **[4 Sep. 2020 (I)]**
 (a) 4.51 gm (b) 3.15 gm (c) 4.15 gm (d) 1.52 gm
3. Two identical cylindrical vessels are kept on the ground and each contain the same liquid of density d . The area of the base of both vessels is S but the height of liquid in one vessel is x_1 and in the other, x_2 . When both cylinders are connected through a pipe of negligible volume very close to the bottom, the liquid flows from one vessel to the other until it comes to equilibrium at a new height. The change in energy of the system in the process is : **[4 Sep. 2020 (II)]**
 (a) $gdS(x_2^2 + x_1^2)$ (b) $gdS(x_2 + x_1)^2$
 (c) $\frac{3}{4}gdS(x_2 - x_1)^2$ (d) $\frac{1}{4}gdS(x_2 - x_1)^2$
4. A leak proof cylinder of length 1 m, made of a metal which has very low coefficient of expansion is floating vertically in water at 0°C such that its height above the water surface is 20 cm. When the temperature of water is increased to 4°C , the height of the cylinder above the water surface becomes 21 cm. The density of water at $T = 4^\circ\text{C}$, relative to the density at $T = 0^\circ\text{C}$ is close to: **[8 Jan 2020 (I)]**
 (a) 1.26 (b) 1.04 (c) 1.01 (d) 1.03
5. Consider a solid sphere of radius R and mass density $\rho(r) = \rho_0 \left(1 - \frac{r^2}{R^2}\right)$, $0 < r \leq R$. The minimum density of a liquid in which it will float is: **[8 Jan 2020 (I)]**
 (a) $\frac{\rho_0}{3}$ (b) $\frac{\rho_0}{5}$ (c) $\frac{2\rho_0}{5}$ (d) $\frac{2\rho_0}{3}$

6.



- Two liquids of densities ρ_1 and ρ_2 ($\rho_2 = 2\rho_1$) are filled up behind a square wall of side 10 m as shown in figure. Each liquid has a height of 5 m. The ratio of the forces due to these liquids exerted on upper part MN to that at the lower part NO is (Assume that the liquids are not mixing): **[8 Jan 2020 (II)]**
 (a) $1/3$ (b) $2/3$ (c) $1/2$ (d) $1/4$
7. A cubical block of side 0.5 m floats on water with 30% of its volume under water. What is the maximum weight that can be put on the block without fully submerging it under water? [Take, density of water = 10^3 kg/m^3] **[10 April 2019 (II)]**
 (a) 46.3 kg (b) 87.5 kg (c) 65.4 kg (d) 30.1 kg
 8. A submarine experiences a pressure of $5.05 \times 10^6 \text{ Pa}$ at depth of d_1 in a sea. When it goes further to a depth of d_2 , it experiences a pressure of $8.08 \times 10^6 \text{ Pa}$. Then $d_2 - d_1$ is approximately (density of water = 10^3 kg/m^3 and acceleration due to gravity = 10 ms^{-2}): **[10 April 2019 (II)]**
 (a) 300m (b) 400m (c) 600m (d) 500m
 9. A wooden block floating in a bucket of water has $\frac{4}{5}$ of its volume submerged. When certain amount of an oil poured into the bucket, it is found that the block is just under the oil surface with half of its volume under water and half in oil. The density of oil relative to that of water is: **[9 April 2019 (II)]**
 (a) 0.5 (b) 0.8 (c) 0.6 (d) 0.7
 10. A load of mass M kg is suspended from a steel wire of length 2 m and radius 1.0 mm in Searle's apparatus experiment. The increase in length produced in the wire is 4.0 mm. Now the load is fully immersed in a liquid of relative density 2. The relative density of the material of load is 8. The new value of increase in length of the steel wire is : **[12 Jan. 2019 (II)]**
 (a) 3.0mm (b) 4.0mm (c) 5.0mm (d) Zero

11. A soap bubble, blown by a mechanical pump at the mouth of a tube, increases in volume, with time, at a constant rate. The graph that correctly depicts the time dependence of pressure inside the bubble is given by: [12 Jan. 2019 (II)]



12. A liquid of density ρ is coming out of a hose pipe of radius a with horizontal speed v and hits a mesh. 50% of the liquid passes through the mesh unaffected. 25% loses all of its momentum and 25% comes back with the same speed. The resultant pressure on the mesh will be: [11 Jan. 2019 (I)]

(a) $\frac{1}{4}\rho v^2$ (b) $\frac{3}{4}\rho v^2$ (c) $\frac{1}{2}\rho v^2$ (d) ρv^2

13. A thin uniform tube is bent into a circle of radius r in the vertical plane. Equal volumes of two immiscible liquids, whose densities are ρ_1 and ρ_2 ($\rho_1 > \rho_2$) fill half the circle. The angle θ between the radius vector passing through the common interface and the vertical is [Online April 15, 2018]

(a) $\theta = \tan^{-1} \left[\frac{\pi}{2} \left(\frac{\rho_1 - \rho_2}{\rho_1 + \rho_2} \right) \right]$

(b) $\theta = \tan^{-1} \frac{\pi}{2} \left(\frac{\rho_1 - \rho_2}{\rho_1 + \rho_2} \right)$

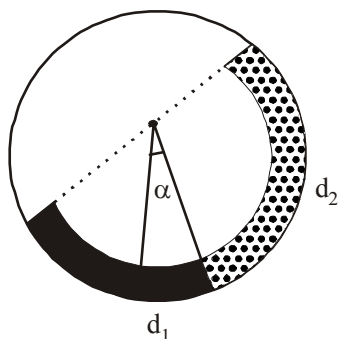
(c) $\theta = \tan^{-1} \pi \left(\frac{\rho_1}{\rho_2} \right)$

(d) None of above

14. There is a circular tube in a vertical plane. Two liquids which do not mix and of densities d_1 and d_2 are filled in the tube. Each liquid subtends 90° angle at centre. Radius joining their interface makes an angle α with vertical. Ratio

$\frac{d_1}{d_2}$ is:

[2014]



(a) $\frac{1 + \sin \alpha}{1 - \sin \alpha}$ (b) $\frac{1 + \cos \alpha}{1 - \cos \alpha}$

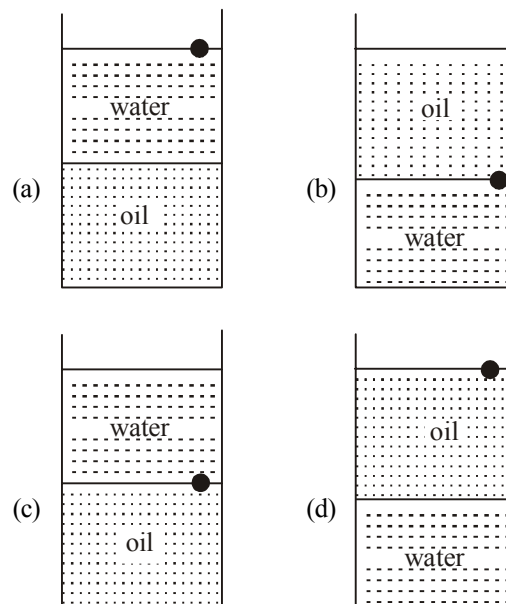
(c) $\frac{1 + \tan \alpha}{1 - \tan \alpha}$ (d) $\frac{1 + \sin \alpha}{1 - \cos \alpha}$

15. A uniform cylinder of length L and mass M having cross-sectional area A is suspended, with its length vertical, from a fixed point by a massless spring such that it is half submerged in a liquid of density σ at equilibrium position. The extension x_0 of the spring when it is in equilibrium is: [2013]

(a) $\frac{Mg}{k}$ (b) $\frac{Mg}{k} \left(1 - \frac{LA\sigma}{M} \right)$

(c) $\frac{Mg}{k} \left(1 - \frac{LA\sigma}{2M} \right)$ (d) $\frac{Mg}{k} \left(1 + \frac{LA\sigma}{M} \right)$

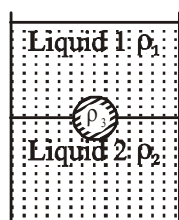
16. A ball is made of a material of density ρ where $\rho_{\text{oil}} < \rho < \rho_{\text{water}}$ with ρ_{oil} and ρ_{water} representing the densities of oil and water, respectively. The oil and water are immiscible. If the above ball is in equilibrium in a mixture of this oil and water, which of the following pictures represents its equilibrium position? [2010]



17. Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle of 30° with each other. When suspended in a liquid of density 0.8 g cm^{-3} , the angle remains the same. If density of the material of the sphere is 1.6 g cm^{-3} , the dielectric constant of the liquid is [2010]

(a) 4 (b) 3 (c) 2 (d) 1

18. A jar is filled with two non-mixing liquids 1 and 2 having densities ρ_1 and ρ_2 respectively. A solid ball, made of a material of density ρ_3 , is dropped in the jar. It comes to equilibrium in the position shown in the figure. Which of the following is true for ρ_1 , ρ_2 and ρ_3 ? [2008]



- (a) $\rho_3 < \rho_1 < \rho_2$ (b) $\rho_1 > \rho_3 > \rho_2$
 (c) $\rho_1 < \rho_2 < \rho_3$ (d) $\rho_1 < \rho_3 < \rho_2$

TOPIC 2 Fluid Flow, Reynold's Number and Bernoulli's Principle

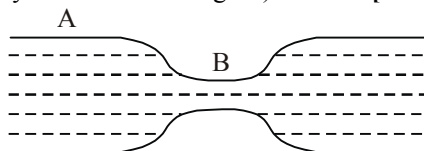


19. A fluid is flowing through a horizontal pipe of varying cross-section, with speed $v \text{ ms}^{-1}$ at a point where the pressure is

P Pascal. At another point where pressure is $\frac{P}{2}$ Pascal its speed is $V \text{ ms}^{-1}$. If the density of the fluid is $\rho \text{ kg m}^{-3}$ and the flow is streamline, then V is equal to: [6 Sep. 2020 (II)]

- (a) $\sqrt{\frac{P}{\rho} + v}$ (b) $\sqrt{\frac{2P}{\rho} + v^2}$
 (c) $\sqrt{\frac{P}{2\rho} + v^2}$ (d) $\sqrt{\frac{P}{\rho} + v^2}$

20. Water flows in a horizontal tube (see figure). The pressure of water changes by 700 Nm^{-2} between A and B where the area of cross section are 40 cm^2 and 20 cm^2 , respectively. Find the rate of flow of water through the tube. (density of water = 1000 kgm^{-3}) [9 Jan. 2020 (I)]



- (a) $3020 \text{ cm}^3/\text{s}$ (b) $2720 \text{ cm}^3/\text{s}$
 (c) $2420 \text{ cm}^3/\text{s}$ (d) $1810 \text{ cm}^3/\text{s}$
21. An ideal fluid flows (laminar flow) through a pipe of non-uniform diameter. The maximum and minimum diameters of the pipes are 6.4 cm and 4.8 cm , respectively. The ratio of the minimum and the maximum velocities of fluid in this pipe is: [7 Jan. 2020 (II)]

- (a) $\frac{9}{16}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{3}{4}$ (d) $\frac{81}{256}$

22. Water from a tap emerges vertically downwards with an initial speed of 1.0 ms^{-1} . The cross-sectional area of the tap is 10^{-4} m^2 . Assume that the pressure is constant throughout the stream of water and that the flow is streamlined. The cross-sectional area of the stream, 0.15 m below the tap would be: [Take $g = 10 \text{ ms}^{-2}$] [10 April 2019 (II)]

- (a) $2 \times 10^{-5} \text{ m}^2$ (b) $5 \times 10^{-5} \text{ m}^2$
 (c) $5 \times 10^{-4} \text{ m}^2$ (d) $1 \times 10^{-5} \text{ m}^2$

23. Water from a pipe is coming at a rate of 100 liters per minute. If the radius of the pipe is 5 cm , the Reynolds number for the flow is of the order of: (density of water = 1000 kg/m^3 , coefficient of viscosity of water = 1 mPa.s) [8 April 2019 I]

- (a) 10^3 (b) 10^4 (c) 10^2 (d) 10^6

24. Water flows into a large tank with flat bottom at the rate of $10^{-4} \text{ m}^3 \text{ s}^{-1}$. Water is also leaking out of a hole of area 1 cm^2 at its bottom. If the height of the water in the tank remains steady, then this height is: [10 Jan. 2019 I]

- (a) 5.1 cm (b) 7 cm (c) 4 cm (d) 9 cm

25. The top of a water tank is open to air and its water level is maintained. It is giving out 0.74 m^3 water per minute through a circular opening of 2 cm radius in its wall. The depth of the centre of the opening from the level of water in the tank is close to: [9 Jan. 2019 (II)]

- (a) 6.0 m (b) 4.8 m (c) 9.6 m (d) 2.9 m

26. When an air bubble of radius r rises from the bottom to the

surface of a lake, its radius becomes $\frac{5r}{4}$. Taking the

atmospheric pressure to be equal to 10 m height of water column, the depth of the lake would approximately be (ignore the surface tension and the effect of temperature):

[Online April 15, 2018]

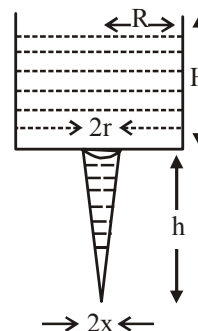
- (a) 10.5 m (b) 8.7 m (c) 11.2 m (d) 9.5 m

27. Two tubes of radii r_1 and r_2 , and lengths l_1 and l_2 , respectively, are connected in series and a liquid flows through each of them in streamline conditions. P_1 and P_2 are pressure

differences across the two tubes. If P_2 is $4P_1$ and l_2 is $\frac{l_1}{4}$, then the radius r_2 will be equal to: [Online April 9, 2017]

- (a) r_1 (b) $2r_1$ (c) $4r_1$ (d) $\frac{r_1}{2}$

28. Consider a water jar of radius R that has water filled up to height H and is kept on a stand of height h (see figure). Through a hole of radius r ($r \ll R$) at its bottom, the water leaks out and the stream of water coming down towards the ground has a shape like a funnel as shown in the figure. If the radius of the cross-section of water stream when it hits the ground is x . Then: [Online April 9, 2016]



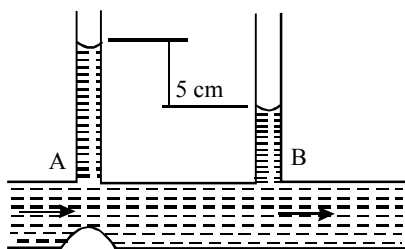
- (a) $x = r \left(\frac{H}{H+h} \right)^{\frac{1}{4}}$ (b) $x = r \left(\frac{H}{H+h} \right)$
 (c) $x = r \left(\frac{H}{H+h} \right)^2$ (d) $x = r \left(\frac{H}{H+h} \right)^{\frac{1}{2}}$

29. If it takes 5 minutes to fill a 15 litre bucket from a water tap

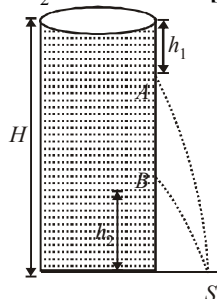
of diameter $\frac{2}{\sqrt{\pi}} \text{ cm}$ then the Reynolds number for the flow is (density of water = 10^3 kg/m^3) and viscosity of water = 10^{-3} Pa.s) close to: [Online April 10, 2015]

- (a) 1100 (b) $11,000$ (c) 550 (d) 5500

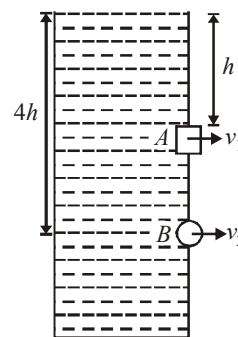
30. An open glass tube is immersed in mercury in such a way that a length of 8 cm extends above the mercury level. The open end of the tube is then closed and sealed and the tube is raised vertically up by additional 46 cm. What will be length of the air column above mercury in the tube now? (Atmospheric pressure = 76 cm of Hg) [2014]
 (a) 16 cm (b) 22 cm (c) 38 cm (d) 6 cm
31. In the diagram shown, the difference in the two tubes of the manometer is 5 cm, the cross section of the tube at A and B is 6 mm² and 10 mm² respectively. The rate at which water flows through the tube is ($g = 10 \text{ ms}^{-2}$) [Online April 19, 2014]



- (a) 7.5 cc/s (b) 8.0 cc/s (c) 10.0 cc/s (d) 12.5 cc/s
32. A cylindrical vessel of cross-section A contains water to a height h . There is a hole in the bottom of radius 'a'. The time in which it will be emptied is: [Online April 12, 2014]
 (a) $\frac{2A}{\pi a^2} \sqrt{\frac{h}{g}}$ (b) $\frac{\sqrt{2}A}{\pi a^2} \sqrt{\frac{h}{g}}$
 (c) $\frac{2\sqrt{2}A}{\pi a^2} \sqrt{\frac{h}{g}}$ (d) $\frac{A}{\sqrt{2}\pi a^2} \sqrt{\frac{h}{g}}$
33. Water is flowing at a speed of 1.5 ms^{-1} through horizontal tube of cross-sectional area 10^{-2} m^2 and you are trying to stop the flow by your palm. Assuming that the water stops immediately after hitting the palm, the minimum force that you must exert should be (density of water = 10^3 kg m^{-3}) [Online April 9, 2014]
 (a) 15 N (b) 22.5 N (c) 33.7 N (d) 45 N
34. Air of density 1.2 kg m^{-3} is blowing across the horizontal wings of an aeroplane in such a way that its speeds above and below the wings are 150 ms^{-1} and 100 ms^{-1} , respectively. The pressure difference between the upper and lower sides of the wings, is : [Online April 22, 2013]
 (a) 60 Nm^{-2} (b) 180 Nm^{-2}
 (c) 7500 Nm^{-2} (d) 12500 Nm^{-2}
35. In a cylindrical water tank, there are two small holes A and B on the wall at a depth of h_1 from the surface of water and at a height of h_2 from the bottom of water tank. Surface of water is at height H from the bottom of water tank. Water coming out from both holes strikes the ground at the same point S. Find the ratio of h_1 and h_2 [Online May 26, 2012]



- (a) Depends on H (b) 1 : 1
 (c) 2 : 2 (d) 1 : 2
36. Water is flowing through a horizontal tube having cross-sectional areas of its two ends being A and A' such that the ratio A/A' is 5. If the pressure difference of water between the two ends is $3 \times 10^5 \text{ N m}^{-2}$, the velocity of water with which it enters the tube will be (neglect gravity effects) [Online May 12, 2012]
 (a) 5 m s^{-1} (b) 10 m s^{-1}
 (c) 25 m s^{-1} (d) $50\sqrt{10} \text{ m s}^{-1}$
37. A square hole of side length ℓ is made at a depth of h and a circular hole of radius r is made at a depth of $4h$ from the surface of water in a water tank kept on a horizontal surface. If $\ell \ll h$, $r \ll h$ and the rate of water flow from the holes is the same, then r is equal to [May 7, 2012]



- (a) $\frac{\ell}{\sqrt{2}\pi}$ (b) $\frac{\ell}{\sqrt{3}\pi}$ (c) $\frac{\ell}{3\pi}$ (d) $\frac{\ell}{2\pi}$
38. Water is flowing continuously from a tap having an internal diameter $8 \times 10^{-3} \text{ m}$. The water velocity as it leaves the tap is 0.4 ms^{-1} . The diameter of the water stream at a distance $2 \times 10^{-1} \text{ m}$ below the tap is close to: [2011]
 (a) $7.5 \times 10^{-3} \text{ m}$ (b) $9.6 \times 10^{-3} \text{ m}$
 (c) $3.6 \times 10^{-3} \text{ m}$ (d) $5.0 \times 10^{-3} \text{ m}$
39. A cylinder of height 20 m is completely filled with water. The velocity of efflux of water (in ms^{-1}) through a small hole on the side wall of the cylinder near its bottom is [2002]
 (a) 10 (b) 20 (c) 25.5 (d) 5

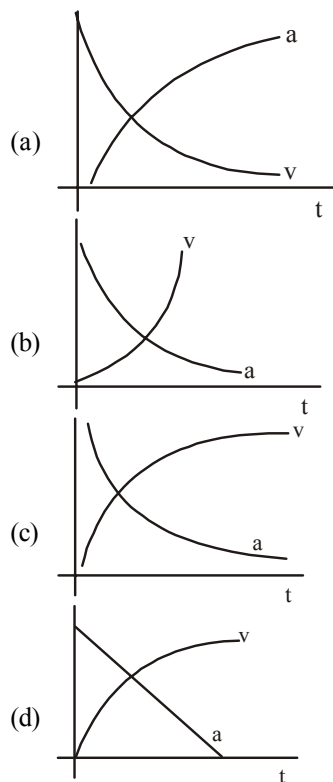
TOPIC 3 Viscosity and Terminal Velocity



40. In an experiment to verify Stokes law, a small spherical ball of radius r and density ρ falls under gravity through a distance h in air before entering a tank of water. If the terminal velocity of the ball inside water is same as its velocity just before entering the water surface, then the value of h is proportional to : (ignore viscosity of air) [5 Sep. 2020 (II)]
 (a) r^4 (b) r (c) r^3 (d) r^2
41. A solid sphere, of radius R acquires a terminal velocity v_1 when falling (due to gravity) through a viscous fluid having a coefficient of viscosity η . The sphere is broken into 27 identical solid spheres. If each of these spheres acquires a terminal velocity, v_2 , when falling through the same fluid, the ratio (v_1/v_2) equals: [12 April 2019 (II)]
 (a) 9 (b) $1/27$ (c) $1/9$ (d) 27

42. Which of the following option correctly describes the variation of the speed v and acceleration ' a ' of a point mass falling vertically in a viscous medium that applies a force $F = -kv$, where ' k ' is a constant, on the body? (Graphs are schematic and not drawn to scale)

[Online April 9, 2016]



43. The velocity of water in a river is 18 km/hr near the surface. If the river is 5 m deep, find the shearing stress between the horizontal layers of water. The co-efficient of viscosity of water = 10^{-2} poise.

[Online April 19, 2014]

- (a) 10^{-1} N/m² (b) 10^{-2} N/m²
(c) 10^{-3} N/m² (d) 10^{-4} N/m²
44. The average mass of rain drops is 3.0×10^{-5} kg and their average terminal velocity is 9 m/s. Calculate the energy transferred by rain to each square metre of the surface at a place which receives 100 cm of rain in a year.

[Online April 11, 2014]

- (a) 3.5×10^5 J (b) 4.05×10^4 J
(c) 3.0×10^5 J (d) 9.0×10^4 J
45. A tank with a small hole at the bottom has been filled with water and kerosene (specific gravity 0.8). The height of water is 3m and that of kerosene 2m. When the hole is opened the velocity of fluid coming out from it is nearly: (take $g = 10$ ms⁻² and density of water = 10^3 kg m⁻³)

[Online April 11, 2014]

- (a) 10.7 ms⁻¹ (b) 9.6 ms⁻¹
(c) 8.5 ms⁻¹ (d) 7.6 ms⁻¹
46. In an experiment, a small steel ball falls through a liquid at a constant speed of 10 cm/s. If the steel ball is pulled upward with a force equal to twice its effective weight, how fast will it move upward?

[Online April 25, 2013]

- (a) 5 cm/s (b) Zero (c) 10 cm/s (d) 20 cm/s

47. The terminal velocity of a small sphere of radius a in a viscous liquid is proportional to [Online May 26, 2012]
(a) a^2 (b) a^3 (c) a (d) a^{-1}

48. If a ball of steel (density $\rho = 7.8$ g cm⁻³) attains a terminal velocity of 10 cm s⁻¹ when falling in water (Coefficient of viscosity $\eta_{\text{water}} = 8.5 \times 10^{-4}$ Pa.s), then, its terminal velocity in glycerine ($\rho = 1.2$ g cm⁻³, $\eta = 13.2$ Pa.s) would be, nearly [2011 RS]

- (a) 6.25×10^{-4} cm s⁻¹ (b) 6.45×10^{-4} cm s⁻¹
(c) 1.5×10^{-5} cm s⁻¹ (d) 1.6×10^{-5} cm s⁻¹

49. A spherical solid ball of volume V is made of a material of density ρ_1 . It is falling through a liquid of density ρ_1 ($\rho_2 < \rho_1$). Assume that the liquid applies a viscous force on the ball that is proportional to the square of its speed v , i.e., $F_{\text{viscous}} = -kv^2$ ($k > 0$). The terminal speed of the ball is [2008]

- (a) $\sqrt{\frac{Vg(\rho_1 - \rho_2)}{k}}$ (b) $\frac{Vg\rho_1}{k}$
(c) $\sqrt{\frac{Vg\rho_1}{k}}$ (d) $\frac{Vg(\rho_1 - \rho_2)}{k}$

50. If the terminal speed of a sphere of gold (density = 19.5 kg/m³) is 0.2 m/s in a viscous liquid (density = 1.5 kg/m³), find the terminal speed of a sphere of silver (density = 10.5 kg/m³) of the same size in the same liquid [2006]

- (a) 0.4 m/s (b) 0.133 m/s
(c) 0.1 m/s (d) 0.2 m/s

51. Spherical balls of radius ' R ' are falling in a viscous fluid of viscosity ' η ' with a velocity ' v '. The retarding viscous force acting on the spherical ball is [2004]

- (a) inversely proportional to both radius ' R ' and velocity ' v '
(b) directly proportional to both radius ' R ' and velocity ' v '
(c) directly proportional to ' R ' but inversely proportional to ' v '
(d) inversely proportional to ' R ' but directly proportional to velocity ' v '

TOPIC 4 Surface Tension, Surface Energy and Capillarity



52. When a long glass capillary tube of radius 0.015 cm is dipped in a liquid, the liquid rises to a height of 15 cm within it. If the contact angle between the liquid and glass is close to 0°, the surface tension of the liquid, in milliNewton m⁻¹, is [$\rho_{\text{(liquid)}} = 900$ kgm⁻³, $g = 10$ ms⁻²] (Give answer in closest integer) _____.

[NA 3 Sep. 2020 (I)]

53. Pressure inside two soap bubbles are 1.01 and 1.02 atmosphere, respectively. The ratio of their volumes is :

[3 Sep. 2020 (I)]

- (a) 4 : 1 (b) 0.8 : 1 (c) 8 : 1 (d) 2 : 1

54. A capillary tube made of glass of radius 0.15 mm is dipped vertically in a beaker filled with methylene iodide (surface tension = 0.05 Nm⁻¹, density = 667 kg m⁻³) which rises

to height h in the tube. It is observed that the two tangents drawn from liquid-glass interfaces (from opp. sides of the capillary) make an angle of 60° with one another. Then h is close to ($g = 10 \text{ ms}^{-2}$). [2 Sep. 2020 (II)]

- (a) 0.049 m (b) 0.087 m
(c) 0.137 m (d) 0.172 m

55. A small spherical droplet of density d is floating exactly half immersed in a liquid of density ρ and surface tension T . The radius of the droplet is (take note that the surface tension applies an upward force on the droplet):

[9 Jan. 2020 (II)]

- (a) $r = \sqrt{\frac{2T}{3(d+\rho)g}}$ (b) $r = \sqrt{\frac{T}{(d-\rho)g}}$
(c) $r = \sqrt{\frac{T}{(d+\rho)g}}$ (d) $r = \sqrt{\frac{3T}{(2d-\rho)g}}$

56. The ratio of surface tensions of mercury and water is given to be 7.5 while the ratio of their densities is 13.6. Their contact angles, with glass, are close to 135° and 0° , respectively. It is observed that mercury gets depressed by an amount h in a capillary tube of radius r_1 , while water rises by the same amount h in a capillary tube of radius r_2 . The ratio, (r_1/r_2) , is then close to :

[10 April 2019 (I)]

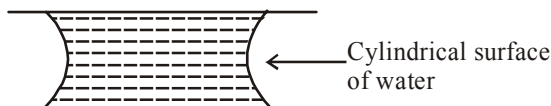
- (a) 4/5 (b) 2/5 (c) 3/5 (d) 2/3

57. If 'M' is the mass of water that rises in a capillary tube of radius 'r', then mass of water which will rise in a capillary tube of radius '2r' is :

[9 April 2019 I]

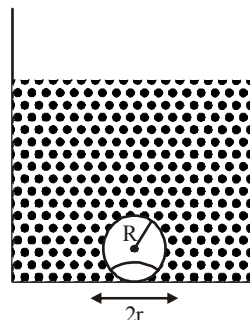
- (a) M (b) $\frac{M}{2}$ (c) 4 M (d) 2 M

58. If two glass plates have water between them and are separated by very small distance (see figure), it is very difficult to pull them apart. It is because the water in between forms cylindrical surface on the side that gives rise to lower pressure in the water in comparison to atmosphere. If the radius of the cylindrical surface is R and surface tension of water is T then the pressure in water between the plates is lower by : [Online April 10, 2015]



- (a) $\frac{2T}{R}$ (b) $\frac{4T}{R}$ (c) $\frac{T}{4R}$ (d) $\frac{T}{R}$

59. On heating water, bubbles being formed at the bottom of the vessel detach and rise. Take the bubbles to be spheres of radius R and making a circular contact of radius r with the bottom of the vessel. If $r \ll R$ and the surface tension of water is T , value of r just before bubbles detach is: (density of water is ρ_w) [2014]



- (a) $R^2 \sqrt{\frac{2\rho_w g}{3T}}$ (b) $R^2 \sqrt{\frac{\rho_w g}{6T}}$
(c) $R^2 \sqrt{\frac{\rho_w g}{T}}$ (d) $R^2 \sqrt{\frac{3\rho_w g}{T}}$

60. A large number of liquid drops each of radius r coalesce to form a single drop of radius R . The energy released in the process is converted into kinetic energy of the big drop so formed. The speed of the big drop is (given, surface tension of liquid T , density ρ)

[Online April 19, 2014, 2012]

- (a) $\sqrt{\frac{T}{\rho} \left(\frac{1}{r} - \frac{1}{R} \right)}$ (b) $\sqrt{\frac{2T}{\rho} \left(\frac{1}{r} - \frac{1}{R} \right)}$
(c) $\sqrt{\frac{4T}{\rho} \left(\frac{1}{r} - \frac{1}{R} \right)}$ (d) $\sqrt{\frac{6T}{\rho} \left(\frac{1}{r} - \frac{1}{R} \right)}$

61. Two soap bubbles coalesce to form a single bubble. If V is the subsequent change in volume of contained air and S change in total surface area, T is the surface tension and P atmospheric pressure, then which of the following relation is correct? [Online April 12, 2014]

- (a) $4PV + 3ST = 0$ (b) $3PV + 4ST = 0$
(c) $2PV + 3ST = 0$ (d) $3PV + 2ST = 0$

62. An air bubble of radius 0.1 cm is in a liquid having surface tension 0.06 N/m and density 10^3 kg/m^3 . The pressure inside the bubble is 1100 Nm^{-2} greater than the atmospheric pressure. At what depth is the bubble below the surface of the liquid? ($g = 9.8 \text{ ms}^{-2}$) [Online April 11, 2014]

- (a) 0.1 m (b) 0.15 m (c) 0.20 m (d) 0.25 m

63. A capillary tube is immersed vertically in water and the height of the water column is x . When this arrangement is taken into a mine of depth d , the height of the water column is y . If R is

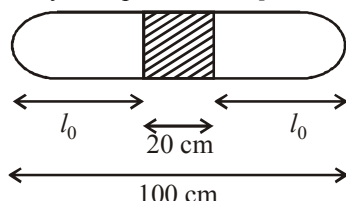
the radius of earth, the ratio $\frac{x}{y}$ is: [Online April 9, 2014]

- (a) $\left(1 - \frac{d}{R} \right)$ (b) $\left(1 - \frac{2d}{R} \right)$
(c) $\left(\frac{R-d}{R+d} \right)$ (d) $\left(\frac{R+d}{R-d} \right)$

64. Wax is coated on the inner wall of a capillary tube and the tube is then dipped in water. Then, compared to the unwaxed capillary, the angle of contact θ and the height h upto which water rises change. These changes are :

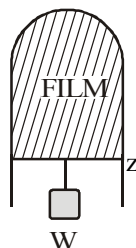
[Online April 23, 2013]

- (a) θ increases and h also increases
 (b) θ decreases and h also decreases
 (c) θ increases and h decreases
 (d) θ decreases and h increases
65. A thin tube sealed at both ends is 100 cm long. It lies horizontally, the middle 20 cm containing mercury and two equal ends containing air at standard atmospheric pressure. If the tube is now turned to a vertical position, by what amount will the mercury be displaced? [Online April 23, 2013]



(Given : cross-section of the tube can be assumed to be uniform)

- (a) 2.95 cm (b) 5.18 cm (c) 8.65 cm (d) 0.0 cm
66. This question has Statement-1 and Statement-2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
- Statement-1:** A capillary is dipped in a liquid and liquid rises to a height h in it. As the temperature of the liquid is raised, the height h increases (if the density of the liquid and the angle of contact remain the same).
- Statement-2:** Surface tension of a liquid decreases with the rise in its temperature. [Online April 9, 2013]
- (a) Statement-1 is true, Statement-2 is true; Statement-2 is **not** the correct explanation for Statement-1.
 (b) Statement-1 is false, Statement-2 is true.
 (c) Statement-1 is true, Statement-2 is false.
 (d) Statement-1 is true, Statement-2 is true; Statement-2 is the correct explanation for Statement-1.
67. A thin liquid film formed between a U-shaped wire and a light slider supports a weight of 1.5×10^{-2} N (see figure). The length of the slider is 30 cm and its weight is negligible. The surface tension of the liquid film is [2012]

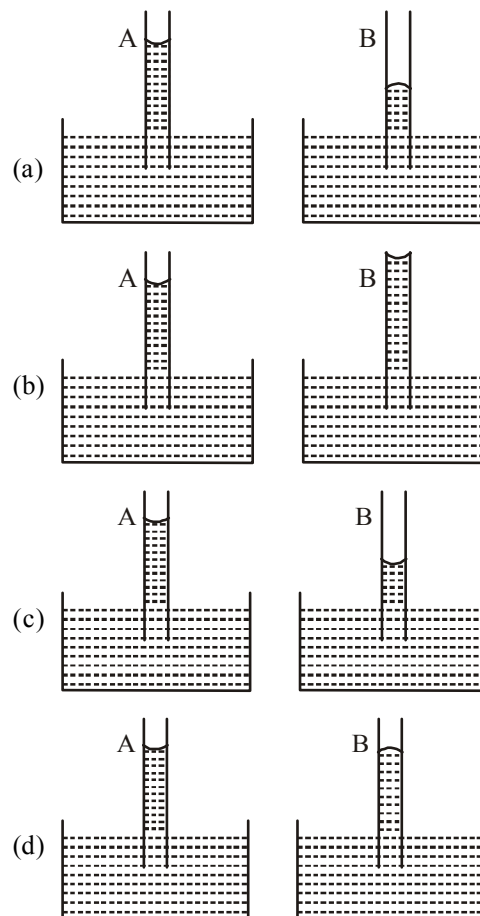


- (a) 0.0125 Nm^{-1} (b) 0.1 Nm^{-1}
 (c) 0.05 Nm^{-1} (d) 0.025 Nm^{-1}
68. Work done in increasing the size of a soap bubble from a radius of 3 cm to 5 cm is nearly (Surface tension of soap solution = 0.03 Nm^{-1}) [2011]
- (a) $0.2 \pi \text{ mJ}$ (b) $2 \pi \text{ mJ}$ (c) $0.4 \pi \text{ mJ}$ (d) $4 \pi \text{ mJ}$

69. Two mercury drops (each of radius ' r ') merge to form bigger drop. The surface energy of the bigger drop, if T is the surface tension, is : [2011 RS]

- (a) $4\pi r^2 T$ (b) $2\pi r^2 T$
 (c) $2^{8/3} \pi r^2 T$ (d) $2^{5/3} \pi r^2 T$

70. A capillary tube (A) is dipped in water. Another identical tube (B) is dipped in a soap-water solution. Which of the following shows the relative nature of the liquid columns in the two tubes? [2008]



71. A 20 cm long capillary tube is dipped in water. The water rises up to 8 cm. If the entire arrangement is put in a freely falling elevator the length of water column in the capillary tube will be [2005]
- (a) 10 cm (b) 8 cm (c) 20 cm (d) 4 cm
72. If two soap bubbles of different radii are connected by a tube [2004]
- (a) air flows from the smaller bubble to the bigger
 (b) air flows from bigger bubble to the smaller bubble till the sizes are interchanged
 (c) air flows from the bigger bubble to the smaller bubble till the sizes become equal
 (d) there is no flow of air.



Hints & Solutions



1. (a) In equilibrium, $mg = F_e$
 $F_B = V\rho_0 g$ and mass = volume $\times$ density

$$\frac{4}{3}\pi(R^3 - r^3)\rho_0 g = \frac{4}{3}\pi R^3 \rho_w g$$

Given, relative density, $\frac{\rho_0}{\rho_w} = \frac{27}{8}$

$$\Rightarrow \left[1 - \left(\frac{r}{R}\right)^3\right] \frac{27}{8} \rho_w = \rho_w$$

$$\Rightarrow 1 - \frac{r^3}{R^3} = \frac{9}{27} \Rightarrow 1 - \frac{1}{3} = \frac{r^3}{R^3} \Rightarrow \frac{2}{3} = \frac{r^3}{R^3}$$

$$\Rightarrow \frac{r}{R} = \left(\frac{2}{3}\right)^{1/3} \Rightarrow 1 - \frac{r^3}{R^3} = \frac{8}{27}$$

$$\Rightarrow \frac{r^3}{R^3} = 1 - \frac{8}{27} = \frac{19}{27}$$

$$\therefore r = 0.89R = \frac{8}{9}R$$

2. (c) Given :

Radius of air bubble = 1 cm,

Upward acceleration of bubble, $a = 9.8 \text{ cm/s}^2$,

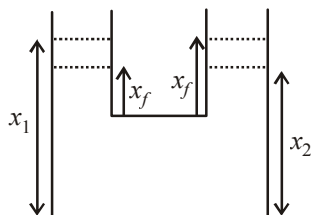
$$\rho_{\text{water}} = 1 \text{ g cm}^{-3}$$

$$\text{Volume } V = \frac{4\pi}{3}r^3 = \frac{4\pi}{3} \times (1)^3 = 4.19 \text{ cm}^3$$

$$F_{\text{buoyant}} - mg = ma \Rightarrow m = \frac{F_{\text{buoyant}}}{g + a}$$

$$\therefore m = \frac{(V\rho_w g)}{g + a} = \frac{V\rho_w}{1 + \frac{a}{g}} = \frac{(4.19) \times 1}{1 + \frac{9.8}{980}} = \frac{4.19}{1.01} = 4.15 \text{ g}$$

3. (d)



Initial potential energy,

$$U_1 = (\rho S x_1) g \cdot \frac{x_1}{2} + (\rho S x_2) g \cdot \frac{x_2}{2}$$

Final potential energy,

$$U_f = (\rho S x_f) g \cdot \frac{x_f}{2} \times 2$$

By volume conservation,

$$Sx_1 + Sx_2 = S(2x_f)$$

$$x_f = \frac{x_1 + x_2}{2}$$

When valve is opened loss in potential energy occur till water level become same.

$$\Delta U = U_i - U_f$$

$$\Delta U = \rho S g \left[\left(\frac{x_1^2}{2} + \frac{x_2^2}{2} \right) - x_f^2 \right]$$

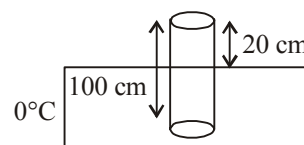
$$= \rho S g \left[\frac{x_1^2}{2} + \frac{x_2^2}{2} - \left(\frac{x_1 + x_2}{2} \right)^2 \right]$$

$$= \frac{\rho S g}{2} \left[\frac{x_1^2}{2} + \frac{x_2^2}{2} - x_1 x_2 \right] = \frac{\rho S g}{4} (x_1 - x_2)^2$$

4. (c) When cylinder is floating in water at 0°C

$$\text{Net thrust} = A(h_2 - h_1)\rho_{0^\circ\text{C}} g$$

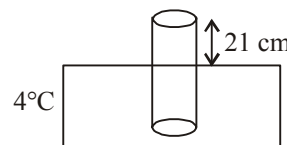
$$= A(100 - 80)\rho_{0^\circ\text{C}} g$$



When cylinder is floating in water at 4°C

$$\text{Net thrust} = A(h_2 - h_1)\rho_{4^\circ\text{C}} g$$

$$= A(100 - 21)\rho_{4^\circ\text{C}} g$$



$$\therefore \frac{\rho_{4^\circ\text{C}}}{\rho_{0^\circ\text{C}}} = \frac{80}{79} = 1.01$$

5. (c) For minimum density of liquid, solid sphere has to float (completely immersed) in the liquid.

$$mg = F_B \text{ (also } V_{\text{immersed}} = V_{\text{total}})$$

$$\text{or } \int \rho dV = \frac{4}{3}\pi R^3 \rho_\ell$$

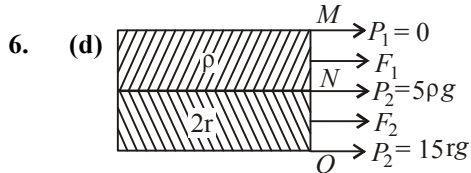
$$\left[\rho(r) = \rho_0 \left(1 - \frac{r^2}{R^2} \right) \right] 0 < r \leq R \text{ given}$$

$$\Rightarrow \int_0^R \rho_0 4\pi \left(1 - \frac{r^2}{R^2} \right) \cdot r^2 dr = \frac{4}{3}\pi R^3 \rho_\ell$$

$$\Rightarrow 4\pi\rho_0 \left[\frac{r^3}{3} - \frac{r^5}{5R^2} \right]_0^R = \frac{4}{3}\pi R^3 \rho_\ell$$

$$\frac{4\pi\rho_0 R^3}{3} \times \frac{2}{5} = \frac{4}{3}\pi R^3 \rho_\ell$$

$$\therefore \rho_\ell = \frac{2\rho_0}{5}$$



Let P_1 , P_2 and P_3 be the pressure at points M , N and O respectively.

Pressure is given by $P = \rho gh$

Now, $P_1 = 0$ ($\because h = 0$)

$$P_2 = \rho g(5)$$

$$P_3 = \rho g(15)$$

$$= 15\rho g$$

$$\text{Force on upper part, } F_1 = \frac{(P_1 + P_2)}{2} A$$

$$\text{Force on lower part, } F_2 = \frac{(P_2 + P_3)}{2} A$$

$$\therefore \frac{F_1}{F_2} = \frac{5\rho g}{20\rho g} = \frac{5}{20} = \frac{1}{4}$$

7. (b) When a body floats then the weight of the body = upthrust

$$\therefore (50)^3 \times \frac{30}{100} \times (1) \times g = M_{\text{cube}} g \quad \dots(i)$$

Let m mass should be placed, then

$$(50)^3 \times (1) \times g = (M_{\text{cube}} + m)g \quad \dots(ii)$$

Subtracting equation (i) from equation (ii), we get

$$\Rightarrow mg = (50)^3 \times g(1 - 0.3) = 125 \times 0.7 \times 10^3 g$$

$$\Rightarrow m = 87.5 \text{ kg}$$

8. (a) $P_1 = P_0 + \rho g d_1$

$$P_2 = P_0 + \rho g d_2$$

$$\Delta P = P_2 - P_1 = \rho g \Delta d$$

$$3.03 \times 10^6 = 10^3 \times 10 \times \Delta d$$

$$\Rightarrow \Delta d \approx 300 \text{ m}$$

9. (c) $Mg = \left(\frac{4V}{5} \right) \rho \omega g$

$$\text{or } \left(\frac{M}{V} \right) = \frac{4\rho\omega}{5} \text{ or } \rho = \frac{4\rho\omega}{5}$$

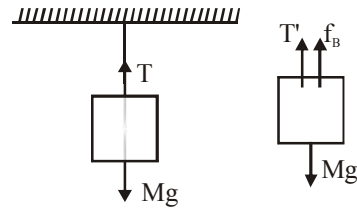
When block floats fully in water and oil, then

$$Mg = F_{b1} + F_{b2}$$

$$(\rho V)g = \left(\frac{V}{2} \right) \rho_{\text{oil}} g + \frac{V}{2} \rho \omega g$$

$$\text{or } \rho_{\text{oil}} = \frac{3}{5} \rho \omega = 0.6\rho \omega$$

10. (a) Using $\frac{F}{A} = Y \cdot \frac{\Delta \ell}{\ell}$



$$\Rightarrow \Delta \ell \propto F$$

$$T = Mg$$

...(i)

$$T = Mg - f_B = Mg - \frac{M}{\rho_b} \cdot \rho_\ell \cdot g$$

$$= \left(1 - \frac{\rho_\ell}{\rho_b} \right) Mg = \left(1 - \frac{2}{8} \right) Mg$$

$$T = \frac{3}{4} Mg$$

From eqn (i)

$$\frac{\Delta \ell'}{\Delta \ell} = \frac{T'}{T} = \frac{3}{4} \quad [\text{Given: } \Delta \ell = 4 \text{ mm}]$$

$$\therefore \Delta \ell' = \frac{3}{4} \cdot \Delta \ell = \frac{3}{4} \times 4 = 3 \text{ mm}$$

11. (d)

$$V = ct \text{ or } \frac{4}{3}\pi r^3 = ct$$

$$\Rightarrow r = kt^{\frac{1}{3}}$$

$$P = P_0 + \frac{4T}{kt^{1/3}}$$

$$P = P_0 + c \left(\frac{1}{t^{1/3}} \right)$$

12. (b) Mass per unit time of the liquid = ρav

Momentum per second carried by liquid

$$= \rho av \times v$$

Net force due to bounced back liquid,

$$F_1 = 2 \times \left[\frac{1}{4} \rho av^2 \right]$$

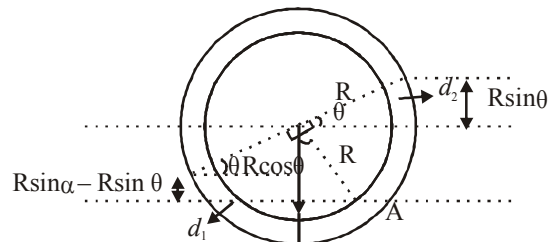
$$\text{Net force due to stopped liquid, } F_2 = \frac{1}{4} \rho av^2$$

Total force,

$$F = F_1 + F_2 = \frac{1}{2} \rho av^2 + \frac{1}{4} \rho av^2 = \frac{3}{4} \rho av^2$$

$$\text{Net pressure} = \frac{3}{4} \rho v^2$$

13. (d) Pressure at interface A must be same from both the sides to be in equilibrium.



$$\therefore (R \cos \theta + R \sin \theta) p_2 g = (R \cos \theta - R \sin \theta) p_1 g$$

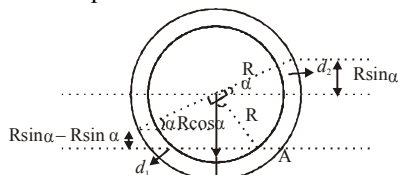
$$\Rightarrow \frac{d_1}{d_2} = \frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta} = \frac{1 + \tan \theta}{1 - \tan \theta}$$

$$\Rightarrow \rho_1 - \rho_1 \tan \theta = \rho_2 + \rho_2 \tan \theta$$

$$\Rightarrow (\rho_1 + \rho_2) \tan \theta = \rho_1 - \rho_2$$

$$\therefore \theta = \tan^{-1} \left(\frac{\rho_1 - \rho_2}{\rho_1 + \rho_2} \right)$$

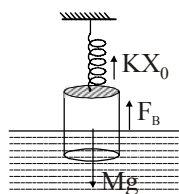
14. (c) Pressure at interface A must be same from both the sides to be in equilibrium.



$$\therefore (R \cos \alpha + R \sin \alpha) d_2 g = (R \cos \alpha - R \sin \alpha) d_1 g$$

$$\Rightarrow \frac{d_1}{d_2} = \frac{\cos \alpha + \sin \alpha}{\cos \alpha - \sin \alpha} = \frac{1 + \tan \alpha}{1 - \tan \alpha}$$

15. (c) From figure, $kx_0 + F_B = Mg$



$$kx_0 + \sigma \frac{L}{2} Ag = Mg$$

$$[\because \text{mass} = \text{density} \times \text{volume}]$$

$$\Rightarrow kx_0 = Mg - \sigma \frac{L}{2} Ag$$

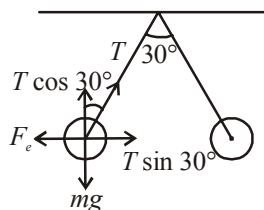
$$\Rightarrow x_0 = \frac{Mg - \frac{\sigma LA g}{2}}{k} = \frac{Mg}{k} \left(1 - \frac{LA \sigma}{2M} \right)$$

Hence, extension of the spring when it is in equilibrium is,

$$x_0 = \frac{Mg}{k} \left(1 - \frac{LA \sigma}{2M} \right)$$

16. (b) Oil will float on water so, (b) or (d) is the correct option. But density of ball is more than that of oil, hence it will sink in oil.

17. (c)



$$F_e = T \sin 30^\circ$$

$$mg = T \cos 30^\circ$$

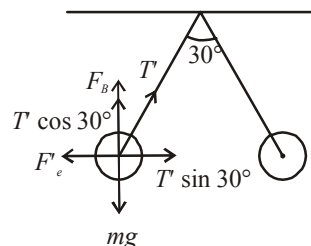
$$\Rightarrow \tan 30^\circ = \frac{F_e}{mg} \quad \dots(1)$$

In liquid,

$$F_e' = T' \sin 30^\circ \quad \dots(A)$$

$$mg = F_B + T' \cos 30^\circ$$

But $F_B = \text{Buoyant force}$



$$= V(d - \rho)g = V(1.6 - 0.8)g = 0.8 Vg$$

$$= 0.8 \frac{m}{d} g = \frac{0.8 mg}{1.6} = \frac{mg}{2}$$

$$\therefore mg = \frac{mg}{2} + T' \cos 30^\circ$$

$$\Rightarrow \frac{mg}{2} = T' \cos 30^\circ \quad \dots(B)$$

$$\text{From (A) and (B), } \tan 30^\circ = \frac{2F_e'}{mg}$$

From (1) and (2)

$$\frac{F_e}{mg} = \frac{2F_e'}{mg} \quad (2)$$

$$\Rightarrow F_e = 2F_e'$$

If K be the dielectric constant, then

$$F_e' = \frac{F_e}{K}$$

$$\therefore F_e = \frac{2F_e}{K} \Rightarrow K = 2$$

18. (d) As liquid 1 floats over liquid 2. The lighter liquid floats over heavier liquid. So, $\rho_1 < \rho_2$

Also $\rho_3 < \rho_2$ because the ball of density ρ_3 does not sink to the bottom of the jar.

Also $\rho_3 > \rho_1$ otherwise the ball would have floated in liquid 1. we conclude that

$$\rho_1 < \rho_3 < \rho_2$$

19. (d) Using Bernoulli's equation

$$P_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2} \rho v_2^2 + \rho g h_2$$

For horizontal pipe, $h_1 = 0$ and $h_2 = 0$ and taking

$$P_1 = P, P_2 = \frac{P}{2}, \text{ we get}$$

$$\Rightarrow P + \frac{1}{2} \rho v^2 = \frac{P}{2} + \frac{1}{2} \rho V^2$$

$$\Rightarrow \frac{P}{2} + \frac{1}{2} \rho v^2 = \frac{1}{2} \rho V^2$$

$$\Rightarrow V = \sqrt{v^2 + \frac{P}{\rho}}$$

20. (b) According to question, area of cross-section at A , $a_A = 40 \text{ cm}^2$ and at B , $a_B = 20 \text{ cm}^2$

Let velocity of liquid flow at A , $= V_A$ and at B , $= V_B$

Using equation of continuity $a_A V_A = a_B V_B$

$$40V_A = 20V_B$$

$$\Rightarrow 2V_A = V_B$$

Now, using Bernoulli's equation

$$P_A + \frac{1}{2}\rho V_A^2 = P_B + \frac{1}{2}\rho V_B^2 \Rightarrow P_A - P_B = \frac{1}{2}\rho(V_B^2 - V_A^2)$$

$$\Rightarrow \Delta P = \frac{1}{2} \times 1000 \left(V_B^2 - \frac{V_B^2}{4} \right) \Rightarrow \Delta P = 500 \times \frac{3V_B^2}{4}$$

$$\Rightarrow V_B = \sqrt{\frac{(\Delta P) \times 4}{1500}} = \sqrt{\frac{(700) \times 4}{1500}} \text{ m/s} = 1.37 \times 10^2 \text{ cm/s}$$

$$\text{Volume flow rate } Q = a_B \times V_B \\ = 20 \times 100 \times V_B = 2732 \text{ cm}^3/\text{s} \approx 2720 \text{ cm}^3/\text{s}$$

21. (a) From the equation of continuity

$$A_1 v_1 = A_2 v_2$$

Here, v_1 and v_2 are the velocities at two ends of pipe.

A_1 and A_2 are the area of pipe at two ends

$$\Rightarrow \frac{v_1}{v_2} = \frac{A_2}{A_1} = \frac{\pi(4.8)^2}{\pi(6.4)^2} = \frac{9}{16}$$

22. (b) Using Bernoulli's equation

$$P + \frac{1}{2}(\rho v_1^2 - \rho v_2^2) + \rho gh = P$$

$$\Rightarrow v_2^2 = v_1^2 + 2gh$$

$$\Rightarrow v_2 = \sqrt{v_1^2 + 2gh}$$

Equation of continuity

$$A_1 v_1 = A_2 v_2$$

$$(1 \text{ cm}^2)(1 \text{ m/s}) = (A_2) \left(\sqrt{(1)^2 + 2 \times 10 \times \frac{15}{100}} \right)$$

$$10^{-4} \times 1 = A_2 \times 2$$

$$\therefore A_2 = \frac{10^{-4}}{2} = 5 \times 10^{-5} \text{ m}^2$$

23. (b) Rate of flow of water (V) = 100 lit/min

$$= \frac{100 \times 10^{-3}}{60} \times \frac{5}{3} \times 10^{-3} \text{ m}^3$$

$$\therefore \text{Velocity of flow of water } (v) = \frac{V}{A} = \frac{5 \times 10^{-3}}{3 \times (5 \times 10^{-2})^2}$$

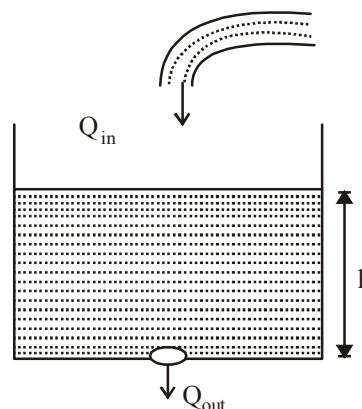
$$= \frac{10}{15\pi} = \frac{2}{3\pi} \text{ m/s} \\ = 0.2 \text{ m/s}$$

$$\therefore \text{Reynold number } (N_R) = \frac{Dvp}{\eta}$$

$$= \frac{(10 \times 10^{-2}) \times \frac{2}{3\pi} \times 1000}{1} = 2 \times 10^4$$

$$\text{Order of } N_R = 10^4$$

24. (a)



Since height of water column is constant therefore, water inflow rate (Q_{in})

= water outflow rate

$$Q_{in} = 10^{-4} \text{ m}^3/\text{s}^{-1}$$

$$Q_{out} = Au = 10^{-4} \times \sqrt{2gh}$$

$$\therefore 10^{-4} = 10^{-4} \times \sqrt{20 \times h}$$

$$\therefore h = \frac{1}{20} \text{ m} = 5 \text{ cm}$$

25. (b) Here, volume flow rate

$$= \frac{0.74}{60} = \pi r^2 v = (\pi \times 4 \times 10^{-4}) \times \sqrt{2gh}$$

$$\Rightarrow \sqrt{2gh} = \frac{74 \times 100}{240\pi} \Rightarrow \sqrt{2gh} = \frac{740}{24\pi}$$

$$\Rightarrow 2gh = \frac{740 \times 740}{24 \times 24 \times 10} (\because \pi^2 = 10)$$

$$\Rightarrow h = \frac{74 \times 74}{2 \times 24 \times 24} \approx 4.8 \text{ m}$$

i.e., The depth of the centre of the opening from the level of water in the tank is close to 4.8 m

26. (d) Using $P_1 V_1 = P_2 V_2$

$$(P_1) \frac{4}{3} \pi r^3 = (P_2) \frac{4}{3} \pi \frac{125r^3}{64}$$

$$\frac{\rho g(10) + \rho gh}{\rho g(10)} = \frac{125}{64}$$

$$640 + 64h = 1250$$

On solving we get $h = 9.5 \text{ m}$

27. (d) The volume of liquid flowing through both the tubes i.e., rate of flow of liquid is same.

Therefore, $V = V_1 = V_2$

$$\text{i.e., } \frac{\pi P_1 r_1^4}{8\eta l_1} = \frac{\pi P_2 r_2^4}{8\eta l_2}$$

$$\text{or } \frac{P_1 r_1^4}{l_1} = \frac{P_2 r_2^4}{l_2}$$

$$\therefore P_2 = 4 P_1 \text{ and } l_2 = l_1/4$$

$$\frac{P_1 r_1^4}{l_1} = \frac{4P_1 r_2^4}{l_1/4} \Rightarrow r_2^4 = \frac{r_1^4}{16}$$

$$r_2 = r_1/2$$

28. (a) According to Bernoulli's Principle,

$$\frac{1}{2} \rho v_1^2 + \rho gh = \frac{1}{2} \rho v_2^2$$

$$v_1^2 + 2gh = v_2^2$$

$$2gH + 2gh = v_2^2 \quad \dots(i)$$

$$a_1 v_1 = a_2 v_2$$

$$\pi r^2 \sqrt{2gh} = \pi x^2 v_2$$

$$\frac{r^2}{x^2} \sqrt{2gh} = v_2$$

Substituting the value of v_2 in equation (i)

$$2gH + 2gh = \frac{r^4}{x^4} 2gh \quad \text{or, } x = r \left[\frac{H}{H+h} \right]^{1/4}$$

29. (d) Given: Diameter of water tap = $\frac{2}{\sqrt{\pi}}$ cm

$$\therefore \text{Radius, } r = \frac{1}{\sqrt{\pi}} \times 10^{-2} \text{ m}$$

$$\frac{dm}{dt} = \rho AV$$

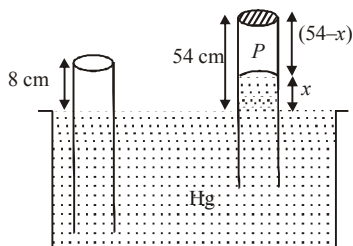
$$\frac{15}{5 \times 60} = 10^3 \times \pi \left(\frac{1}{\sqrt{\pi}} \right)^2 \times 10^{-4} V$$

$$\Rightarrow V = 0.05 \text{ m/s}$$

$$\text{Reynold's number, } R_e = \frac{\rho V r}{\eta}$$

$$= \frac{10^3 \times 0.05 \times \frac{2}{\sqrt{\pi}} \times 10^{-2}}{10^{-3}} \approx 5500$$

30. (a)



Length of the air column above mercury in the tube is,

$$P + x = P_0$$

$$\Rightarrow P = (76 - x)$$

$$\Rightarrow 8 \times A \times 76 = (76 - x) \times A \times (54 - x)$$

$$\therefore x = 38$$

Thus, length of air column

$$= 54 - 38 = 16 \text{ cm.}$$

31. (a) According to Bernoulli's theorem,

$$P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2$$

$$\therefore v_2^2 - v_1^2 = 2gh \quad \dots(1)$$

According to the equation of continuity

$$A_1 v_1 = A_2 v_2 \quad \dots(2)$$

$$\frac{A_1}{A_2} = \frac{6 \text{ mm}^2}{10 \text{ mm}^2}$$

$$\text{From equation (2), } \frac{A_1}{A_2} = \frac{v_2}{v_1} = \frac{6}{10}$$

$$\text{or, } v_2 = \frac{6}{10} v_1$$

Putting this value of v_2 in equation (1)

$$\left(\frac{6}{10} v_1 \right)^2 - (v_1)^2 = 2 \times 10^3 \times 5$$

$$\left[\because g = 10 \text{ m/s}^2 = 10^3 \text{ cm/s}^2 \right]$$

$$\text{and } h = 5 \text{ cm}$$

$$\text{Solving we get } v_1 = \frac{10}{8}$$

Therefore the rate at which water flows through the

$$\text{tube} = A_1 v_1 = A_2 v_2 = \frac{6 \times 10}{8} = 7.5 \text{ cc/s}$$

32. (b) Let the rate of falling water level be $-\frac{dh}{dt}$

Initially at $t = 0; h = h$

$t = t; h = 0$

$$\text{Then, } A \left(-\frac{dh}{dt} \right) = \pi a^2 v$$

$$dt = -\frac{A}{\pi a^2 \sqrt{2gh}} dh$$

$$[\because \text{velocity of efflux of liquid } v = \sqrt{2gh}]$$

Integrating both sides

$$\int_0^t dt = -\frac{A}{\sqrt{2g} \pi a^2} \int_h^0 h^{-1/2} dh$$

$$[t]_0^t = -\frac{A}{\sqrt{2g} \pi a^2} \left[\frac{h^{1/2}}{1/2} \right]_h^0$$

$$t = \frac{\sqrt{2} A}{\pi a^2} \sqrt{h}$$

33. (a) For 1 m length of horizontal tube

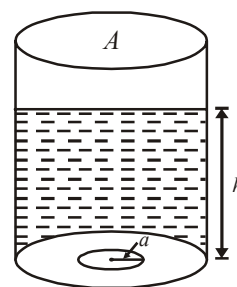
Mass of water $M = \text{density} \times \text{volume}$

$$= 10^3 \times \text{area} \times \text{length}$$

$$= 10^3 \times 10^{-2} \times 1 = 10 \text{ kg}$$

$$\text{Therefore minimum force} = \frac{\Delta p}{\Delta t} \quad (\text{rate of change of momentum})$$

$$= 10 \times 1.5 = 15 \text{ N}$$



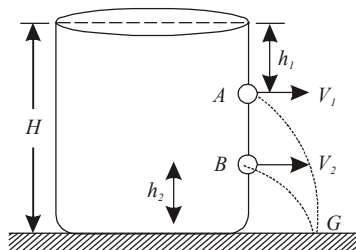
34. (c) Pressure difference

$$P_2 - P_1 = \frac{1}{2} \rho (v_2^2 - v_1^2) = \frac{1}{2} \times 1.2 \left((150)^2 - (100)^2 \right)$$

$$= \frac{1}{2} \times 1.2 (22500 - 10000)$$

$$= 7500 \text{ Nm}^{-2}$$

35. (b)



i.e. $R_1 = R_2 = R$
 or, $v_1 t_1 = v_2 t_2$... (i)

Where v_1 = velocity of efflux at $A = \sqrt{2gh_1}$ and

v_2 = velocity of efflux at $B = \sqrt{2g(H-h_2)}$

t_1 = time of fall water stream through A

$$= \sqrt{\frac{2(H-h_1)}{g}}$$

t_2 = time of fall of the water stream through $B = \sqrt{\frac{2h_2}{g}}$

Putting these values in eqn (i) we get

$$(H-h_1)h_1 = (H-h_2)h_2$$

$$\text{or } [H - (h_1 + h_2)][h_1 - h_2] = 0$$

Here, $H = h_1 + h_2$ is irrelevant because the holes are at

two different heights. Hence $h_1 = h_2$ or, $\frac{h_1}{h_2} = 1$

36. (a) According to Bernoulli's theorem

$$P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2 \quad \dots (i)$$

From question,

$$P_1 - P_2 = 3 \times 10^5, \frac{A_1}{A_2} = 5$$

According to equation of continuity

$$A_1 v_1 = A_2 v_2$$

$$\text{or, } \frac{A_1}{A_2} = \frac{v_2}{v_1} = 5$$

$$\Rightarrow v_2 = 5v_1$$

From equation (i)

$$P_1 - P_2 = \frac{1}{2} \rho (v_2^2 - v_1^2)$$

$$\text{or } 3 \times 10^5 = \frac{1}{2} \times 1000 (5v_1^2 - v_1^2)$$

$$\Rightarrow 600 = 6v_1 \times 4v_1$$

$$\Rightarrow v_1^2 = 25$$

$$\therefore v_1 = 5 \text{ m/s}$$

37. (a) As
- $A_1 v_1 = A_2 v_2$
- (Principle of continuity)

$$\text{or, } \ell^2 \sqrt{2gh} = \pi r^2 \sqrt{2g \times 4h}$$

$$(\text{Efflux velocity} = \sqrt{2gh})$$

$$\therefore r^2 = \frac{\ell^2}{2\pi} \text{ or } r = \sqrt{\frac{\ell^2}{2\pi}} = \frac{\ell}{\sqrt{2\pi}}$$

38. (c) Using Bernoulli's theorem, for horizontal flow

$$P_0 + \frac{1}{2} \rho v_1^2 + \rho gh = P_0 + \frac{1}{2} \rho v_2^2 + 0$$

$$v_2 = \sqrt{v_1^2 + 2gh} = \sqrt{0.16 + 2 \times 10 \times 0.2} = 2.03 \text{ m/s}$$

According to equation of continuity

$$A_2 v_2 = A_1 v_1$$

$$\pi \frac{D_2^2}{4} \times v_2 = \pi \frac{D_1^2}{4} v_1$$

$$\Rightarrow D_2 = D_1 \sqrt{\frac{v_1}{v_2}} = 3.55 \times 10^{-3} \text{ m}$$

39. (b) Given, Height of cylinder,
- $h = 20 \text{ cm}$
- Acceleration due to gravity,
- $g = 10 \text{ ms}^{-2}$
-
- Velocity of efflux

$$v = \sqrt{2gh}$$

Where h is the height of the free surface of liquid from the hole

$$\Rightarrow v = \sqrt{2 \times 10 \times 20} = 20 \text{ m/s}$$

40. (a) Using,
- $v^2 - u^2 = 2gh$

$$\Rightarrow v^2 - 0^2 = 2gh \Rightarrow v = \sqrt{2gh}$$

Terminal velocity,

$$V_T = \frac{2}{9} \frac{r^2 (\rho - \sigma) g}{\eta}$$

After falling through h the velocity should be equal to terminal velocity

$$\therefore \sqrt{2gh} = \frac{2}{9} \frac{r^2 (\rho - \sigma) g}{\eta}$$

$$\Rightarrow 2gh = \frac{4}{81} \frac{r^4 g^2 (\rho - \sigma)^2}{\eta^2}$$

$$\Rightarrow h = \frac{2r^4 g (\rho - \sigma)^2}{81\eta^2} \Rightarrow h \propto r^4$$

41. (a)
- $27 \times \frac{4}{3} \pi r^3 = \frac{4}{3} \pi r^3$

$$\text{or } r = \frac{R}{3}$$

Terminal velocity, $v \propto r^2$

$$\therefore \frac{v_1}{v_2} = \frac{r_1^2}{r_2^2}$$

$$\text{or } v_2 = \left(\frac{r_2}{r_1} \right)^2 v_1 = \left(\frac{R/3}{R} \right)^2 v_1 = \frac{1}{9} v_1$$

$$\text{or } \frac{v_1}{v_2} = 9.$$

42. (c) When a point mass is falling vertically in a viscous medium, the medium or viscous fluid exerts drag force on the body to oppose its motion and at one stage body falling with constant terminal velocity.

43. (b) $\eta = 10^{-2}$ poise

$$v = 18 \text{ km/h} = \frac{18000}{3600} = 5 \text{ m/s}$$

$$l = 5 \text{ m}$$

$$\text{Strain rate} = \frac{v}{l}$$

$$\text{Coefficient of viscosity, } \eta = \frac{\text{shearing stress}}{\text{strain rate}}$$

$$\therefore \text{Shearing stress} = \eta \times \text{strain rate}$$

$$= 10^{-2} \times \frac{5}{5} = 10^{-2} \text{ Nm}^{-2}$$

44. (b) Total volume of rain drops, received 100 cm in a year by area 1 m^2

$$= 1 \text{ m}^2 \times \frac{100}{100} \text{ m} = 1 \text{ m}^3$$

As we know, density of water,
 $d = 10^3 \text{ kg/m}^3$

Therefore, mass of this volume of water
 $M = d \times v = 10^3 \times 1 = 10^3 \text{ kg}$

Average terminal velocity of rain drop
 $v = 9 \text{ m/s}$ (given)

Therefore, energy transferred by rain,

$$E = \frac{1}{2} mv^2$$

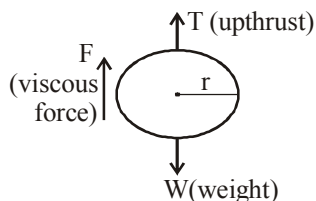
$$= \frac{1}{2} \times 10^3 \times (9)^2$$

$$= \frac{1}{2} \times 10^3 \times 81 = 4.05 \times 10^4 \text{ J}$$

45. (b) According to Toricelli's theorem,
 Velocity of efflux,

$$V_{\text{eff}} = \sqrt{2gh} = \sqrt{2 \times 9.8 \times 5} \approx 9.8 \text{ ms}^{-1}$$

46. (c)



Weight of the body

$$W = mg = \frac{4}{3} \pi r^3 \rho g$$

$$T = \frac{4}{3} \pi r^3 \sigma g$$

$$\text{and } F = 6\pi\eta rv$$

When the body attains terminal velocity net force acting on the body is zero. i.e.,

$$W - T - F = 0$$

$$\text{And terminal velocity } v = \frac{2r^2(\rho - \sigma)g}{9\eta}$$

As in case of upward motion upward force is twice its effective weight, therefore, it will move with same speed 10 cm/s

47. (a) Terminal velocity in a viscous medium is given by:

$$V_T = \frac{2a^2(\rho - \sigma)g}{9\eta}$$

$$\therefore V_T \propto a^2$$

48. (a) When the ball attains terminal velocity
 Weight of the ball = viscous force + buoyant force

$$\therefore V\rho g = 6\pi\eta rv + V\rho_\ell g$$

$$\Rightarrow Vg(\rho - \rho_\ell) = 6\pi\eta rv$$

$$\text{Also } Vg(\rho - \rho'_\ell) = 6\pi\eta'rv'$$

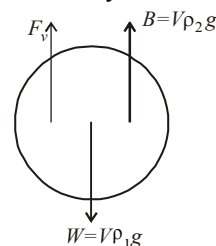
$$\therefore v'\eta' = \frac{(\rho - \rho'_\ell)}{(\rho - \rho_\ell)} \times v\eta$$

$$\Rightarrow v' = \frac{(\rho - \rho'_\ell)}{(\rho - \rho_\ell)} \times \frac{v\eta}{\eta'}$$

$$= \frac{(7.8 - 1.2)}{(7.8 - 1)} \times \frac{10 \times 8.5 \times 10^{-4}}{13.2}$$

$$\therefore v' = 6.25 \times 10^{-4} \text{ cm/s}$$

49. (a) When the ball attains terminal velocity
 Weight of the ball = Buoyant force + Viscous force



$$\therefore V\rho_1 g = V\rho_2 g + kv_t^2 \Rightarrow Vg(\rho_1 - \rho_2) = kv_t^2$$

$$\Rightarrow v_t = \sqrt{\frac{Vg(\rho_1 - \rho_2)}{k}}$$

50. (c) Given,

Density of gold, $\rho_G = 19.5 \text{ kg/m}^3$

Density of silver, $\rho_S = 10.5 \text{ kg/m}^3$

Density of liquid, $\sigma = 1.5 \text{ kg/m}^3$

$$\text{Terminal velocity, } v_T = \frac{2r^2(\rho - \sigma)g}{9\eta}$$

$$\therefore \frac{v_{T_2}}{0.2} = \frac{(10.5 - 1.5)}{(19.5 - 1.5)} \Rightarrow v_{T_2} = 0.2 \times \frac{9}{18}$$

$$\therefore v_{T_2} = 0.1 \text{ m/s}$$

51. (b) From Stoke's law, force of viscosity acting on a spherical body is

$$F = 6\pi\eta rv$$

hence F is directly proportional to radius & velocity.

52. (101)

Given : Radius of capillary tube,

$$r = 0.015 \text{ cm} = 15 \times 10^{-5} \text{ mm}$$

$$h = 15 \text{ cm} = 15 \times 10^{-2} \text{ mm}$$

$$\text{Using, } h = \frac{2T \cos \theta}{\rho g r} \quad [\cos \theta = \cos 0^\circ = 1]$$

Surface tension,

$$T = \frac{r h \rho g}{2} = \frac{15 \times 10^{-5} \times 15 \times 10^{-2} \times 900 \times 10}{2} = 101 \text{ milli newton m}^{-1}$$

53. (c) According to question, pressure inside, 1st soap bubble,

$$\Delta P_1 = P_1 - P_0 = 0.01 = \frac{4T}{R_1} \quad \dots(i)$$

$$\text{And } \Delta P_2 = P_2 - P_0 = 0.02 = \frac{4T}{R_2} \quad \dots(ii)$$

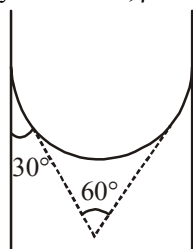
Dividing, equation (ii) by (i),

$$\frac{1}{2} = \frac{R_2}{R_1} \Rightarrow R_1 = 2R_2$$

$$\text{Volume } V = \frac{4}{3} \pi R^3$$

$$\therefore \frac{V_1}{V_2} = \frac{R_1^3}{R_2^3} = \frac{8R_2^3}{R_2^3} = 8$$

54. (b) Given,

Angle of contact $\theta = 30^\circ$ Surface tension, $T = 0.05 \text{ Nm}^{-1}$ Radius of capillary tube, $r = 0.15 \text{ mm} = 0.15 \times 10^{-3} \text{ m}$ Density of methylene iodide, $\rho = 667 \text{ kg m}^{-3}$ 

$$\text{Capillary rise, } h = \frac{2T \cos \theta}{\rho g r}$$

$$= \frac{2 \times 0.05 \times \frac{\sqrt{3}}{2}}{667 \times 10 \times 0.15 \times 10^{-3}} = 0.087 \text{ m}$$

55. (d) For the drops to be in equilibrium upward force on drop = downward force on drop

$$T \cdot 2\pi R = \frac{4}{3} \pi R^3 dg - \frac{2}{3} \pi R^3 \rho g$$

$$\Rightarrow T(2\pi R) = \frac{2}{3} \pi R^3 (2d - \rho)g$$

$$\Rightarrow T = \frac{R^2}{3} (2d - \rho)g \Rightarrow R = \sqrt{\frac{3T}{(2d - \rho)g}}$$

56. (b) As we know that

$$\frac{2T \cos \theta}{r \rho g} = R_h$$

$$\frac{T_{\text{Hg}}}{T_{\text{Water}}} = 7.5$$

$$\frac{\rho_{\text{Hg}}}{\rho_{\text{W}}} = 13.6 \& \frac{\cos \theta_{\text{Hg}}}{\cos \theta_{\text{W}}} = \frac{\cos 135^\circ}{\cos 0^\circ} = \frac{1}{\sqrt{2}}$$

$$\frac{R_{\text{Hg}}}{R_{\text{Water}}} = \left(\frac{T_{\text{Hg}}}{T_{\text{W}}} \right) \left(\frac{\rho_{\text{W}}}{\rho_{\text{Hg}}} \right) \left(\frac{\cos \theta_{\text{Hg}}}{\cos \theta_{\text{W}}} \right)$$

$$= 7.5 \times \frac{1}{13.6} \times \frac{1}{\sqrt{2}} = 0.4 = \frac{2}{5}$$

57. (d) We have, $h = \frac{2T \cos \theta}{r \rho g}$

Mass of the water in the capillary

$$m = \rho V = \rho \times \pi r^2 h = \rho \times \pi r^2 \times \frac{2T \cos \theta}{r \rho g}$$

$$\Rightarrow m \propto r$$

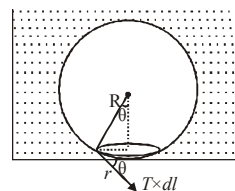
$$\therefore \frac{m_1}{m_2} = \frac{r}{2r}$$

$$\text{or } m_2 = 2m_1 = 2m.$$

58. (d) Here excess pressure, $P_{\text{excess}} = \frac{T}{r_1} + \frac{T}{r_2}$

$$P_{\text{excess}} = \frac{T}{R} \quad \because \begin{cases} r_1 = R \\ r_2 = O \end{cases}$$

59. (a) When the bubble gets detached, Buoyant force = force due to surface tension



Force due to excess pressure = upthrust

$$\text{Access pressure in air bubble} = \frac{2T}{R}$$

$$\frac{2T}{R} (\pi r^2) = \frac{4\pi R^3}{3T} \rho_w g$$

$$\Rightarrow r^2 = \frac{2R^4 \rho_w g}{3T} \Rightarrow r = R^2 \sqrt{\frac{2\rho_w g}{3T}}$$

60. (d) When drops combine to form a single drop of radius R.

$$\text{Then energy released, } E = 4\pi T R^3 \left[\frac{1}{r} - \frac{1}{R} \right]$$

If this energy is converted into kinetic energy then

$$\frac{1}{2} m v^2 = 4\pi R^3 T \left[\frac{1}{r} - \frac{1}{R} \right]$$

$$\frac{1}{2} \times \left[\frac{4}{3} \pi R^3 \rho \right] v^2 = 4\pi R^3 T \left[\frac{1}{r} - \frac{1}{R} \right]$$

$$v^2 = \frac{6T}{\rho} \left[\frac{1}{r} - \frac{1}{R} \right]$$

$$v = \sqrt{\frac{6T}{\rho} \left[\frac{1}{r} - \frac{1}{R} \right]}$$

61. (b)

62. (a) **Given:** Radius of air bubble,

$$r = 0.1 \text{ cm} = 10^{-3} \text{ m}$$

Surface tension of liquid,

$$S = 0.06 \text{ N/m} = 6 \times 10^{-2} \text{ N/m}$$

Density of liquid, $\rho = 10^3 \text{ kg/m}^3$

Excess pressure inside the bubble,

$$p_{\text{excess}} = 1100 \text{ Nm}^{-2}$$

Depth of bubble below the liquid surface,

$$h = ?$$

As we know,

$$p_{\text{Excess}} = h\rho g + \frac{2s}{r}$$

$$\Rightarrow 1100 = h \times 10^3 \times 9.8 + \frac{2 \times 6 \times 10^{-2}}{10^{-3}}$$

$$\Rightarrow 1100 = 9800 h + 120$$

$$\Rightarrow 9800h = 1100 - 120$$

$$\Rightarrow h = \frac{980}{9800} = 0.1 \text{ m}$$

63. (a) Acceleration due to gravity changes with the depth,

$$g' = g \left(1 - \frac{d}{R} \right)$$

Pressure, $P = \rho gh$

$$\text{Hence ratio, } \frac{x}{y} \text{ is } \left(1 - \frac{d}{R} \right)$$

64. (c) Angle of contact θ

$$\cos \theta = \frac{T_{\text{SA}} - T_{\text{SL}}}{T_{\text{LA}}}$$

when water is on a waxy or oily surface

$T_{\text{SA}} < T_{\text{SL}}$ $\cos \theta$ is negative i.e.,

$$90^\circ < \theta < 180^\circ$$

i.e., angle of contact θ increases

And for $\theta > 90^\circ$ liquid level in capillary tube fall.

i.e., h decreases

65. (b)

66. (b) Surface tension of a liquid decreases with the rise in temperature. At the boiling point of liquid, surface tension is zero.

$$\text{Capillary rise } h = \frac{2T \cos \theta}{r \rho g}$$

As surface tension T decreases with rise in temperature hence capillary rise also decreases.

67. (d) Let T is the force due to surface tension per unit length, then

$$F = 2Tl$$

l = length of the slider.

At equilibrium, $F = W$

$$\therefore 2Tl = mg$$

$$\Rightarrow T = \frac{mg}{2l} = \frac{1.5 \times 10^{-2}}{2 \times 30 \times 10^{-2}} = \frac{1.5}{60} = 0.025 \text{ Nm}^{-1}$$

68. (c) Work done = increase in surface area $\times$ surface tension

$$\begin{aligned} \Rightarrow W &= 2T4\pi[(5^2) - (3)^2] \times 10^{-4} \\ &= 2 \times 0.03 \times 4\pi [25 - 9] \times 10^{-4} \text{ J} \\ &= 0.4\pi \times 10^{-3} \text{ J} = 0.4\pi \text{ mJ} \end{aligned}$$

69. (c) As volume remains constant

$\therefore$ Sum of volumes of 2 smaller drops

= Volume of the bigger drop

$$2 \cdot \frac{4}{3} \pi r^3 = \frac{4}{3} \pi R^3 \Rightarrow R = 2^{1/3} r$$

Surface energy = Surface tension $\times$ Surface area = $T \cdot 4\pi R^2$

$$= T4\pi 2^{2/3} r^2 = T \cdot 2^{8/3} \pi r^2.$$

70. (c) In case of water, the meniscus shape is concave

$$\text{upwards. From ascent formula } h = \frac{2\sigma \cos \theta}{r \rho g}$$

The surface tension (σ) of soap solution is less than water. Therefore height of capillary rise for soap solution should be less as compared to water. As in the case of water, the meniscus shape of soap solution is also concave upwards.

71. (c) Water fills the tube entirely in gravityless condition i.e., 20 cm.

72. (a) Let pressure outside be P_0 and r and R be the radius of smaller bubble and bigger bubble respectively.

$$\therefore \text{Pressure } P_1 \text{ For smaller bubble} = P_0 + \frac{2T}{r}$$

$$P_2 \text{ For bigger bubble} = P_0 + \frac{2T}{R} \quad (R > r)$$

$$\therefore P_1 > P_2$$

hence air moves from smaller bubble to bigger bubble.

Thermal Properties of Matter



TOPIC 1 Thermometer & Thermal Expansion



1. Two different wires having lengths L_1 and L_2 , and respective temperature coefficient of linear expansion α_1 and α_2 , are joined end-to-end. Then the effective temperature coefficient of linear expansion is :

[Sep. 05, 2020 (II)]

- (a) $\frac{\alpha_1 L_1 + \alpha_2 L_2}{L_1 + L_2}$ (b) $2\sqrt{\alpha_1 \alpha_2}$
 (c) $\frac{\alpha_1 + \alpha_2}{2}$ (d) $4 \frac{\alpha_1 \alpha_2}{\alpha_1 + \alpha_2} \frac{L_2 L_1}{(L_2 + L_1)^2}$

2. A bakelite beaker has volume capacity of 500 cc at 30°C . When it is partially filled with V_m volume (at 30°C) of mercury, it is found that the unfilled volume of the beaker remains constant as temperature is varied. If $\gamma_{(\text{beaker})} = 6 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$ and $\gamma_{(\text{mercury})} = 1.5 \times 10^{-4} \text{ }^\circ\text{C}^{-1}$, where γ is the coefficient of volume expansion, then V_m (in cc) is close to _____.

[NA Sep. 03, 2020 (I)]

3. When the temperature of a metal wire is increased from 0°C to 10°C , its length increased by 0.02%. The percentage change in its mass density will be closest to :

[Sep. 02, 2020 (II)]

- (a) 0.06 (b) 2.3
 (c) 0.008 (d) 0.8

4. A non-isotropic solid metal cube has coefficients of linear expansion as: $5 \times 10^{-5}/^\circ\text{C}$ along the x -axis and $5 \times 10^{-6}/^\circ\text{C}$ along the y and the z -axis. If the coefficient of volume expansion of the solid is $C \times 10^{-6}/^\circ\text{C}$ then the value of C is _____.

[NA 7 Jan. 2020 I]

5. At 40°C , a brass wire of 1 mm radius is hung from the ceiling. A small mass, M is hung from the free end of the wire. When the wire is cooled down from 40°C to 20°C it regains its original length of 0.2 m. The value of M is close to:

[12 April 2019 I]

(Coefficient of linear expansion and Young's modulus of brass are $10^{-5}/^\circ\text{C}$ and 10^{11} N/m^2 , respectively; $g = 10 \text{ ms}^{-2}$)

- (a) 9 kg (b) 0.5 kg (c) 1.5 kg (d) 0.9 kg

6. Two rods A and B of identical dimensions are at temperature 30°C . If A is heated upto 180°C and B upto

$T^\circ\text{C}$, then the new lengths are the same. If the ratio of the coefficients of linear expansion of A and B is 4 : 3, then the value of T is :

[11 Jan. 2019 II]

- (a) 230°C (b) 270°C
 (c) 200°C (d) 250°C

7. A thermometer graduated according to a linear scale reads a value x_0 when in contact with boiling water, and $x_0/3$ when in contact with ice. What is the temperature of an object in $^\circ\text{C}$, if this thermometer in the contact with the object reads $x_0/2$?

[11 Jan. 2019 II]

- (a) 25 (b) 60 (c) 40 (d) 35

8. A rod, of length L at room temperature and uniform area of cross section A , is made of a metal having coefficient of linear expansion $\alpha/^\circ\text{C}$. It is observed that an external compressive force F , is applied on each of its ends, prevents any change in the length of the rod, when its temperature rises by ΔT K. Young's modulus, Y , for this metal is:

[9 Jan. 2019 I]

- (a) $\frac{F}{A \alpha \Delta T}$ (b) $\frac{F}{A \alpha (\Delta T - 273)}$
 (c) $\frac{F}{2A \alpha \Delta T}$ (d) $\frac{2F}{A \alpha \Delta T}$

9. An external pressure P is applied on a cube at 0°C so that it is equally compressed from all sides. K is the bulk modulus of the material of the cube and α is its coefficient of linear expansion. Suppose we want to bring the cube to its original size by heating. The temperature should be raised by :

[2017]

- (a) $\frac{3\alpha}{PK}$ (b) $3PK\alpha$
 (c) $\frac{P}{3\alpha K}$ (d) $\frac{P}{\alpha K}$

10. A steel rail of length 5 m and area of cross-section 40 cm^2 is prevented from expanding along its length while the temperature rises by 10°C . If coefficient of linear expansion and Young's modulus of steel are $1.2 \times 10^{-5} \text{ K}^{-1}$ and $2 \times 10^{11} \text{ Nm}^{-2}$ respectively, the force developed in the rail is approximately:

[Online April 9, 2017]

- (a) $2 \times 10^7 \text{ N}$ (b) $1 \times 10^5 \text{ N}$
 (c) $2 \times 10^9 \text{ N}$ (d) $3 \times 10^{-5} \text{ N}$

11. A compressive force, F is applied at the two ends of a long thin steel rod. It is heated, simultaneously, such that its temperature increases by ΔT . The net change in its length is zero. Let l be the length of the rod, A its area of cross-section, Y its Young's modulus, and α its coefficient of linear expansion. Then, F is equal to :

[Online April 8, 2017]

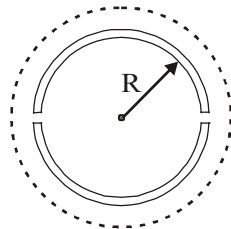
- (a) $l^2 Y \alpha \Delta T$ (b) $l A Y \alpha \Delta T$
 (c) $A Y \alpha \Delta T$ (d) $\frac{AY}{\alpha \Delta T}$
12. The ratio of the coefficient of volume expansion of a glass container to that of a viscous liquid kept inside the container is 1 : 4. What fraction of the inner volume of the container should the liquid occupy so that the volume of the remaining vacant space will be same at all temperatures ?
- [Online April 23, 2013]
- (a) 2 : 5 (b) 1 : 4 (c) 1 : 64 (d) 1 : 8
13. On a linear temperature scale Y , water freezes at $-160^\circ Y$ and boils at $-50^\circ Y$. On this Y scale, a temperature of $340 K$ would be read as : (water freezes at $273 K$ and boils at $373 K$)

[Online April 9, 2013]

- (a) $-73.7^\circ Y$ (b) $-233.7^\circ Y$
 (c) $-86.3^\circ Y$ (d) $-106.3^\circ Y$
14. A wooden wheel of radius R is made of two semicircular part (see figure). The two parts are held together by a ring made of a metal strip of cross sectional area S and length L . L is slightly less than $2\pi R$. To fit the ring on the wheel, it is heated so that its temperature rises by ΔT and it just steps over the wheel. As it cools down to surrounding temperature, it presses the semicircular parts together. If the coefficient of linear expansion of the metal is α , and its Young's modulus is Y , the force that one part of the wheel applies on the other part is :

[2012]

- (a) $2\pi S Y \alpha \Delta T$
 (b) $S Y \alpha \Delta T$
 (c) $\pi S Y \alpha \Delta T$
 (d) $2 S Y \alpha \Delta T$

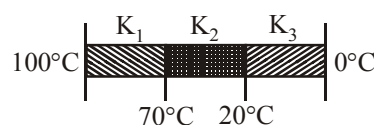


TOPIC 2 Calorimetry and Heat Transfer



15. Three rods of identical cross-section and lengths are made of three different materials of thermal conductivity K_1 , K_2 and K_3 , respectively. They are joined together at their ends to make a long rod (see figure). One end of the long rod is maintained at $100^\circ C$ and the other at $0^\circ C$ (see figure). If the joints of the rod are at $70^\circ C$ and $20^\circ C$ in steady state and there is no loss of energy from the surface of the rod, the correct relationship between K_1 , K_2 and K_3 is :

[Sep. 06, 2020 (II)]



- (a) $K_1 : K_3 = 2 : 3$, $K_1 < K_3 = 2 : 5$
 (b) $K_1 < K_2 < K_3$
 (c) $K_1 : K_2 = 5 : 2$, $K_1 : K_3 = 3 : 5$
 (d) $K_1 > K_2 > K_3$
16. A bullet of mass $5 g$, travelling with a speed of $210 m/s$, strikes a fixed wooden target. One half of its kinetic energy is converted into heat in the bullet while the other half is converted into heat in the wood. The rise of temperature of the bullet if the specific heat of its material is $0.030 cal/(g - ^\circ C)$ ($1 cal = 4.2 \times 10^7 ergs$) close to :

[Sep. 05, 2020 (I)]

- (a) $87.5^\circ C$ (b) $83.3^\circ C$
 (c) $119.2^\circ C$ (d) $38.4^\circ C$
17. The specific heat of water = $4200 J kg^{-1} K^{-1}$ and the latent heat of ice = $3.4 \times 10^5 J kg^{-1}$. $100 g$ of ice at $0^\circ C$ is placed in $200 g$ of water at $25^\circ C$. The amount of ice that will melt as the temperature of water reaches $0^\circ C$ is close to (in grams) :

[Sep. 04, 2020 (I)]

- (a) 61.7 (b) 63.8
 (c) 69.3 (d) 64.6
18. A calorimeter of water equivalent $20 g$ contains $180 g$ of water at $25^\circ C$. ' m ' grams of steam at $100^\circ C$ is mixed in it till the temperature of the mixture is $31^\circ C$. The value of ' m ' is close to (Latent heat of water = $540 cal g^{-1}$, specific heat of water = $1 cal g^{-1} ^\circ C^{-1}$)

[Sep. 03, 2020 (II)]

- (a) 2 (b) 4
 (c) 3.2 (d) 2.6
19. Three containers C_1 , C_2 and C_3 have water at different temperatures. The table below shows the final temperature T when different amounts of water (given in liters) are taken from each container and mixed (assume no loss of heat during the process)

[8 Jan. 2020 II]

C_1	C_2	C_3	T
1 l	2 l	—	$60^\circ C$
—	1 l	2 l	$30^\circ C$
2 l	—	1 l	$60^\circ C$
1 l	1 l	1 l	θ

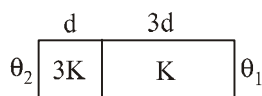
The value of θ (in $^\circ C$ to the nearest integer) is _____.

20. M grams of steam at $100^\circ C$ is mixed with $200 g$ of ice at its melting point in a thermally insulated container. If it produces liquid water at $40^\circ C$ [heat of vaporization of water is $540 cal/g$ and heat of fusion of ice is $80 cal/g$], the value of M is _____
- [NA 7 Jan. 2020 II]
21. When M_1 gram of ice at $-10^\circ C$ (Specific heat = $0.5 cal g^{-1} ^\circ C^{-1}$) is added to M_2 gram of water at $50^\circ C$, finally no ice is left and the water is at $0^\circ C$. The value of latent heat of ice, in $cal g^{-1}$ is :

[12 April 2019 I]

- (a) $\frac{50M_2}{M_1} - 5$ (b) $\frac{5M_1}{M_2} - 50$
 (c) $\frac{50M_2}{M_1}$ (d) $\frac{5M_2}{M_1} - 5$
22. A massless spring ($K = 800 \text{ N/m}$), attached with a mass (500 g) is completely immersed in 1 kg of water. The spring is stretched by 2 cm and released so that it starts vibrating. What would be the order of magnitude of the change in the temperature of water when the vibrations stop completely? (Assume that the water container and spring receive negligible heat and specific heat of mass = 400 J/kg K, specific heat of water = 4184 J/kg K) [9 April 2019 II]
- (a) 10^{-4} K (b) 10^{-5} K (c) 10^{-1} K (d) 10^{-3} K
23. Two materials having coefficients of thermal conductivity '3K' and 'K' and thickness 'd' and '3d', respectively, are joined to form a slab as shown in the figure. The temperatures of the outer surfaces are ' θ_2 ' and ' θ_1 ', respectively, ($\theta_2 > \theta_1$). The temperature at the interface is:

[9 April 2019 II]



- (a) $\frac{\theta_1}{10} + \frac{9\theta_2}{10}$ (b) $\frac{\theta_2 + \theta_1}{2}$
 (c) $\frac{\theta_1}{6} + \frac{5\theta_2}{6}$ (d) $\frac{\theta_1}{3} + \frac{2\theta_2}{3}$
24. A cylinder of radius R is surrounded by a cylindrical shell of inner radius R and outer radius 2R. The thermal conductivity of the material of the inner cylinder is K_1 and that of the outer cylinder is K_2 . Assuming no loss of heat, the effective thermal conductivity of the system for heat flowing along the length of the cylinder is:

[12 Jan. 2019 I]

- (a) $\frac{K_1 + K_2}{2}$ (b) $K_1 + K_2$
 (c) $\frac{2K_1 + 3K_2}{5}$ (d) $\frac{K_1 + 3K_2}{4}$
25. Ice at -20°C is added to 50 g of water at 40°C . When the temperature of the mixture reaches 0°C , it is found that 20 g of ice is still unmelted. The amount of ice added to the water was close to [11 Jan. 2019 I]
- (Specific heat of water = $4.2 \text{ J/g}^\circ\text{C}$
 Specific heat of Ice = $2.1 \text{ J/g}^\circ\text{C}$
 Heat of fusion of water at $0^\circ\text{C} = 334 \text{ J/g}$)
- (a) 50g (b) 100 g
 (c) 60 g (d) 40 g
26. When 100 g of a liquid A at 100°C is added to 50 g of a liquid B at temperature 75°C , the temperature of the mixture becomes 90°C . The temperature of the mixture, if 100 g of liquid A at 100°C is added to 50 g of liquid B at 50°C , will be : [11 Jan. 2019 II]
- (a) 85°C (b) 60°C
 (c) 80°C (d) 70°C

27. A metal ball of mass 0.1 kg is heated upto 500°C and dropped into a vessel of heat capacity 800 JK^{-1} and containing 0.5 kg water. The initial temperature of water and vessel is 30°C . What is the approximate percentage increment in the temperature of the water? [Specific Heat Capacities of water and metal are, respectively, $4200 \text{ Jkg}^{-1}\text{K}^{-1}$ and $400 \text{ Jkg}^{-1}\text{K}^{-1}$] [11 Jan. 2019 II]

- (a) 15% (b) 30%
 (c) 25% (d) 20%
28. A heat source at $T = 10^3 \text{ K}$ is connected to another heat reservoir at $T = 10^2 \text{ K}$ by a copper slab which is 1 m thick. Given that the thermal conductivity of copper is $0.1 \text{ WK}^{-1} \text{ m}^{-1}$, the energy flux through it in the steady state is: [10 Jan. 2019 I]

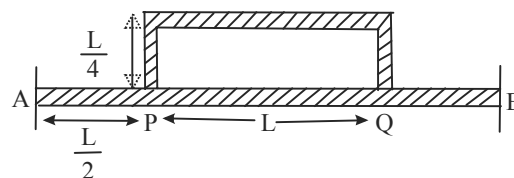
- (a) 90 Wm^{-2} (b) 120 Wm^{-2}
 (c) 65 Wm^{-2} (d) 200 Wm^{-2}
29. An unknown metal of mass 192 g heated to a temperature of 100°C was immersed into a brass calorimeter of mass 128 g containing 240 g of water at a temperature of 8.4°C . Calculate the specific heat of the unknown metal if water temperature stabilizes at 21.5°C . (Specific heat of brass is $394 \text{ J kg}^{-1} \text{ K}^{-1}$) [10 Jan. 2019 II]

- (a) $458 \text{ J kg}^{-1} \text{ K}^{-1}$ (b) $1232 \text{ J kg}^{-1} \text{ K}^{-1}$
 (c) $916 \text{ J kg}^{-1} \text{ K}^{-1}$ (d) $654 \text{ J kg}^{-1} \text{ K}^{-1}$
30. Temperature difference of 120°C is maintained between two ends of a uniform rod AB of length 2L. Another bent

rod PQ, of same cross-section as AB and length $\frac{3L}{2}$, is

connected across AB (See figure). In steady state, temperature difference between P and Q will be close to:

[9 Jan. 2019 I]



- (a) 45°C (b) 75°C (c) 60°C (d) 35°C
31. A copper ball of mass 100 gm is at a temperature T. It is dropped in a copper calorimeter of mass 100 gm, filled with 170 gm of water at room temperature. Subsequently, the temperature of the system is found to be 75°C . T is given by (Given : room temperature = 30°C , specific heat of copper = $0.1 \text{ cal/gm}^\circ\text{C}$) [2017]
- (a) 1250°C (b) 825°C (c) 800°C (d) 885°C
32. In an experiment a sphere of aluminium of mass 0.20 kg is heated upto 150°C . Immediately, it is put into water of volume 150 cc at 27°C kept in a calorimeter of water equivalent to 0.025 kg. Final temperature of the system is 40°C . The specific heat of aluminium is : (take $4.2 \text{ Joule} = 1 \text{ calorie}$) [Online April 8, 2017]
- (a) $378 \text{ J/kg} - ^\circ\text{C}$ (b) $315 \text{ J/kg} - ^\circ\text{C}$
 (c) $476 \text{ J/kg} - ^\circ\text{C}$ (d) $434 \text{ J/kg} - ^\circ\text{C}$

33. An experiment takes 10 minutes to raise the temperature of water in a container from 0°C to 100°C and another 55 minutes to convert it totally into steam by a heater supplying heat at a uniform rate. Neglecting the specific heat of the container and taking specific heat of water to be $1 \text{ cal/g}^{\circ}\text{C}$, the heat of vapourization according to this experiment will come out to be :

[Online April 11, 2015]

- (a) 560 cal/g (b) 550 cal/g
(c) 540 cal/g (d) 530 cal/g
34. Three rods of Copper, Brass and Steel are welded together to form a Y shaped structure. Area of cross - section of each rod = 4 cm^2 . End of copper rod is maintained at 100°C where as ends of brass and steel are kept at 0°C . Lengths of the copper, brass and steel rods are 46, 13 and 12 cms respectively. The rods are thermally insulated from surroundings excepts at ends. Thermal conductivities of copper, brass and steel are 0.92, 0.26 and 0.12 CGS units respectively. Rate of heat flow through copper rod is: [2014]

- (a) 1.2 cal/s (b) 2.4 cal/s
(c) 4.8 cal/s (d) 6.0 cal/s

35. A black coloured solid sphere of radius R and mass M is inside a cavity with vacuum inside. The walls of the cavity are maintained at temperature T_0 . The initial temperature of the sphere is $3T_0$. If the specific heat of the material of the sphere varies as αT^3 per unit mass with the temperature T of the sphere, where α is a constant, then the time taken for the sphere to cool down to temperature $2T_0$ will be (σ is Stefan Boltzmann constant) [Online April 19, 2014]

- (a) $\frac{M\alpha}{4\pi R^2\sigma} \ln\left(\frac{3}{2}\right)$ (b) $\frac{M\alpha}{4\pi R^2\sigma} \ln\left(\frac{16}{3}\right)$
(c) $\frac{M\alpha}{16\pi R^2\sigma} \ln\left(\frac{16}{3}\right)$ (d) $\frac{M\alpha}{16\pi R^2\sigma} \ln\left(\frac{3}{2}\right)$

36. Water of volume 2 L in a closed container is heated with a coil of 1 kW. While water is heated, the container loses energy at a rate of 160 J/s. In how much time will the temperature of water rise from 27°C to 77°C ? (Specific heat of water is 4.2 kJ/kg and that of the container is negligible).

[Online April 9, 2014]

- (a) 8 min 20 s (b) 6 min 2 s
(c) 7 min (d) 14 min
37. Assume that a drop of liquid evaporates by decrease in its surface energy, so that its temperature remains unchanged. What should be the minimum radius of the drop for this to be possible? The surface tension is T , density of liquid is ρ and L is its latent heat of vaporization. [2013]

- (a) $\rho L/T$ (b) $\sqrt{T/\rho L}$ (c) $T/\rho L$ (d) $2T/\rho L$

38. A mass of 50g of water in a closed vessel, with surroundings at a constant temperature takes 2 minutes to cool from 30°C to 25°C . A mass of 100g of another liquid in an identical vessel with identical surroundings takes the same time to cool from 30°C to 25°C . The specific heat of the liquid is :

(The water equivalent of the vessel is 30g.)

[Online April 25, 2013]

- (a) 2.0 kcal/kg (b) 7 kcal/kg
(c) 3 kcal/kg (d) 0.5 kcal/kg
39. 500 g of water and 100 g of ice at 0°C are in a calorimeter whose water equivalent is 40 g. 10 g of steam at 100°C is added to it. Then water in the calorimeter is : (Latent heat of ice = 80 cal/g , Latent heat of steam = 540 cal/g)

[Online April 23, 2013]

- (a) 580 g (b) 590 g (c) 600 g (d) 610 g
40. Given that 1 g of water in liquid phase has volume 1 cm^3 and in vapour phase 1671 cm^3 at atmospheric pressure and the latent heat of vaporization of water is 2256 J/g ; the change in the internal energy in joules for 1 g of water at 373 K when it changes from liquid phase to vapour phase at the same temperature is : [Online April 22, 2013]

- (a) 2256 (b) 167 (c) 2089 (d) 1

41. A large cylindrical rod of length L is made by joining two identical rods of copper and steel of length $\left(\frac{L}{2}\right)$ each.

The rods are completely insulated from the surroundings. If the free end of copper rod is maintained at 100°C and that of steel at 0°C then the temperature of junction is (Thermal conductivity of copper is 9 times that of steel)

[Online May 19, 2012]

- (a) 90°C (b) 50°C (c) 10°C (d) 67°C
42. The heat radiated per unit area in 1 hour by a furnace whose temperature is 3000 K is ($\sigma = 5.7 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$)

[Online May 7, 2012]

- (a) $1.7 \times 10^{10} \text{ J}$ (b) $1.1 \times 10^{12} \text{ J}$
(c) $2.8 \times 10^8 \text{ J}$ (d) $4.6 \times 10^6 \text{ J}$
43. 100g of water is heated from 30°C to 50°C . Ignoring the slight expansion of the water, the change in its internal energy is (specific heat of water is 4184 J/kg/K): [2011]

- (a) 8.4 kJ (b) 84 kJ (c) 2.1 kJ (d) 4.2 kJ

44. The specific heat capacity of a metal at low temperature (T) is given as

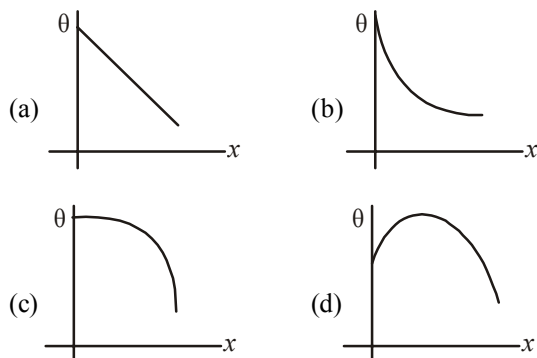
$$C_p (\text{kJ K}^{-1} \text{kg}^{-1}) = 32 \left(\frac{T}{400} \right)^3$$

A 100 gram vessel of this metal is to be cooled from 20°K to 4°K by a special refrigerator operating at room temperature (27°C). The amount of work required to cool the vessel is

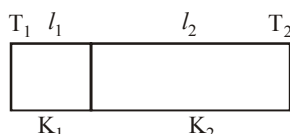
[2011 RS]

- (a) greater than 0.148 kJ
(b) between 0.148 kJ and 0.028 kJ
(c) less than 0.028 kJ
(d) equal to 0.002 kJ

45. A long metallic bar is carrying heat from one of its ends to the other end under steady-state. The variation of temperature θ along the length x of the bar from its hot end is best described by which of the following figures? [2009]



46. One end of a thermally insulated rod is kept at a temperature T_1 and the other at T_2 . The rod is composed of two sections of length l_1 and l_2 and thermal conductivities K_1 and K_2 respectively. The temperature at the interface of the two section is [2007]



- (a) $\frac{(K_1 l_1 T_1 + K_2 l_2 T_2)}{(K_1 l_1 + K_2 l_2)}$ (b) $\frac{(K_2 l_2 T_1 + K_1 l_1 T_2)}{(K_1 l_1 + K_2 l_2)}$
 (c) $\frac{(K_2 l_1 T_1 + K_1 l_2 T_2)}{(K_2 l_1 + K_1 l_2)}$ (d) $\frac{(K_1 l_2 T_1 + K_2 l_1 T_2)}{(K_1 l_2 + K_2 l_1)}$
47. Assuming the Sun to be a spherical body of radius R at a temperature of TK , evaluate the total radiant power incident of Earth at a distance r from the Sun [2006]

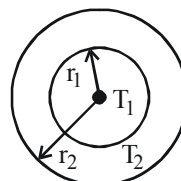
(a) $4\pi r_0^2 R^2 \sigma \frac{T^4}{r^2}$ (b) $\pi r_0^2 R^2 \sigma \frac{T^4}{r^2}$
 (c) $r_0^2 R^2 \sigma \frac{T^4}{4\pi r^2}$ (d) $R^2 \sigma \frac{T^4}{r^2}$

where r_0 is the radius of the Earth and σ is Stefan's constant.

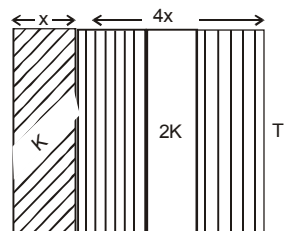
48. Two rigid boxes containing different ideal gases are placed on a table. Box A contains one mole of nitrogen at temperature T_0 , while Box B contains one mole of helium at temperature $\left(\frac{7}{3}\right)T_0$. The boxes are then put into thermal contact with each other, and heat flows between them until the gases reach a common final temperature (ignore the heat capacity of boxes). Then, the final temperature of the gases, T_f in terms of T_0 is [2006]

(a) $T_f = \frac{3}{7}T_0$ (b) $T_f = \frac{7}{3}T_0$
 (c) $T_f = \frac{3}{2}T_0$ (d) $T_f = \frac{5}{2}T_0$

49. The figure shows a system of two concentric spheres of radii r_1 and r_2 are kept at temperatures T_1 and T_2 , respectively. The radial rate of flow of heat in a substance between the two concentric spheres is proportional to [2005]



- (a) $\ln\left(\frac{r_2}{r_1}\right)$ (b) $\frac{(r_2 - r_1)}{(r_1 r_2)}$
 (c) $(r_2 - r_1)$ (d) $\frac{r_1 r_2}{(r_2 - r_1)}$
50. If the temperature of the sun were to increase from T to $2T$ and its radius from R to $2R$, then the ratio of the radiant energy received on earth to what it was previously will be [2004]
- (a) 32 (b) 16 (c) 4 (d) 64
51. The temperature of the two outer surfaces of a composite slab, consisting of two materials having coefficients of thermal conductivity K and $2K$ and thickness x and $4x$, respectively, are T_2 and T_1 ($T_2 > T_1$). The rate of heat transfer through the slab, in a steady state is $\left(\frac{A(T_2 - T_1)K}{x}\right)f$, with f equal to [2004]



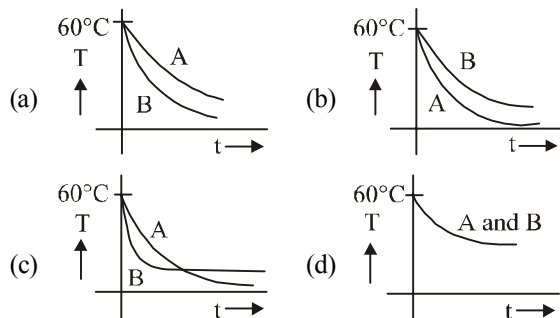
- (a) $\frac{2}{3}$ (b) $\frac{1}{2}$ (c) 1 (d) $\frac{1}{3}$
52. The earth radiates in the infra-red region of the spectrum. The spectrum is correctly given by [2003]
- (a) Rayleigh Jeans law
 (b) Planck's law of radiation
 (c) Stefan's law of radiation
 (d) Wien's law
53. Heat given to a body which raises its temperature by 1°C is [2002]
- (a) water equivalent
 (b) thermal capacity
 (c) specific heat
 (d) temperature gradient

54. Infrared radiation is detected by [2002]
 (a) spectrometer (b) pyrometer
 (c) nanometer (d) photometer
55. Which of the following is more close to a black body? [2002]
 (a) black board paint (b) green leaves
 (c) black holes (d) red roses
56. If mass-energy equivalence is taken into account, when water is cooled to form ice, the mass of water should [2002]
 (a) increase
 (b) remain unchanged
 (c) decrease
 (d) first increase then decrease
57. Two spheres of the same material have radii 1 m and 4 m and temperatures 4000 K and 2000 K respectively. The ratio of the energy radiated per second by the first sphere to that by the second is [2002]
 (a) 1 : 1 (b) 16 : 1
 (c) 4 : 1 (d) 1 : 9.

TOPIC 3 Newton's Law of Cooling

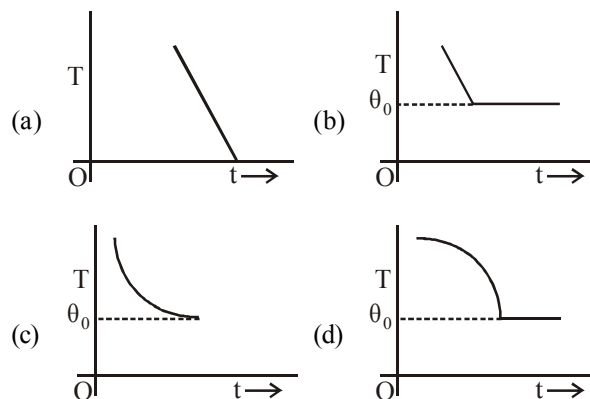


58. A metallic sphere cools from 50°C to 40°C in 300 s. If atmospheric temperature around is 20°C, then the sphere's temperature after the next 5 minutes will be close to : [Sep. 03, 2020 (II)]
 (a) 31°C (b) 33°C (c) 28°C (d) 35°C
59. Two identical beakers A and B contain equal volumes of two different liquids at 60°C each and left to cool down. Liquid in A has density of $8 \times 10^2 \text{ kg/m}^3$ and specific heat of $2000 \text{ J kg}^{-1} \text{ K}^{-1}$ while liquid in B has density of 10^3 kg m^{-3} and specific heat of $4000 \text{ J kg}^{-1} \text{ K}^{-1}$. Which of the following best describes their temperature versus time graph schematically? (assume the emissivity of both the beakers to be the same) [8 April 2019 I]

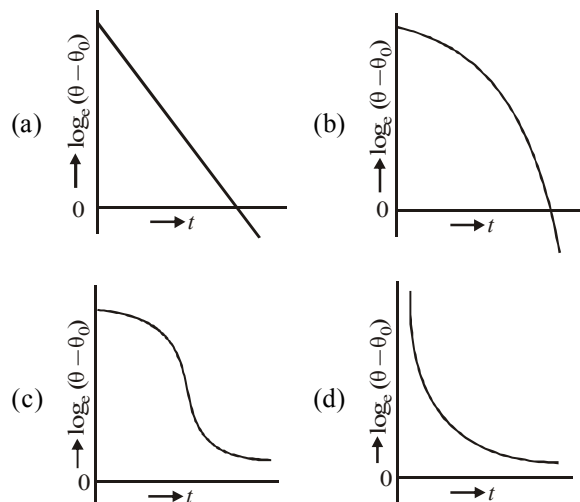


60. A body takes 10 minutes to cool from 60°C to 50°C. The temperature of surroundings is constant at 25°C. Then, the temperature of the body after next 10 minutes will be approximately [Online April 15, 2018]
 (a) 43°C (b) 47°C (c) 41°C (d) 45°C
61. Hot water cools from 60°C to 50°C in the first 10 minutes and to 42°C in the next 10 minutes. The temperature of the surroundings is: [Online April 12, 2014]
 (a) 25°C (b) 10°C (c) 15°C (d) 20°C

62. A hot body, obeying Newton's law of cooling is cooling down from its peak value 80°C to an ambient temperature of 30°C. It takes 5 minutes in cooling down from 80°C to 40°C. How much time will it take to cool down from 62°C to 32°C?
 (Given $\ln 2 = 0.693$, $\ln 5 = 1.609$) [Online April 11, 2014]
 (a) 3.75 minutes (b) 8.6 minutes
 (c) 9.6 minutes (d) 6.5 minutes
63. If a piece of metal is heated to temperature θ and then allowed to cool in a room which is at temperature θ_0 , the graph between the temperature T of the metal and time t will be closest to [2013]



64. A liquid in a beaker has temperature $\theta(t)$ at time t and θ_0 is temperature of surroundings, then according to Newton's law of cooling the correct graph between $\log_e(\theta - \theta_0)$ and t is : [2012]



65. According to Newton's law of cooling, the rate of cooling of a body is proportional to $(\Delta\theta)^n$, where $\Delta\theta$ is the difference of the temperature of the body and the surroundings, and n is equal to [2003]
 (a) two (b) three
 (c) four (d) one



Hints & Solutions



1. (a) Let L'_1 and L'_2 be the lengths of the wire when temperature is changed by $\Delta T^\circ\text{C}$.

At $T^\circ\text{C}$,

$$L_{eq} = L_1 + L_2$$

At $T + \Delta T^\circ\text{C}$

$$L'_{eq} = L'_1 + L'_2$$

$$\therefore L_{eq}(1 + \alpha_{eq}\Delta T) = L_1(1 + \alpha_1\Delta T) + L_2(1 + \alpha_2\Delta T)$$

$$[\because L' = L(1 + \alpha\Delta T)]$$

$$\Rightarrow (L_1 + L_2)(1 + \alpha_{eq}\Delta T) = L_1 + L_2 + L_1\alpha_1\Delta T + L_2\alpha_2\Delta T$$

$$\Rightarrow \alpha_{eq} = \frac{\alpha_1 L_1 + \alpha_2 L_2}{L_1 + L_2}$$

2. (20.00)

Volume capacity of beaker, $V_0 = 500$ cc

$$V_b = V_0 + V_0\gamma_{\text{beaker}}\Delta T$$

When beaker is partially filled with V_m volume of mercury,

$$V_b^1 = V_m + V_m\gamma_m\Delta T$$

$$\text{Unfilled volume } (V_0 - V_m) = (V_b - V_m^1)$$

$$\Rightarrow V_0\gamma_{\text{beaker}} = V_m\gamma_m$$

$$\therefore V_m = \frac{V_0\gamma_{\text{beaker}}}{\gamma_m}$$

$$\text{or, } V_m = \frac{500 \times 6 \times 10^{-6}}{1.5 \times 10^{-4}} = 20 \text{ cc.}$$

3. (a) Change in length of the metal wire (Δl) when its temperature is changed by ΔT is given by

$$\Delta l = l\alpha\Delta T$$

Here, α = Coefficient of linear expansion

Here, $\Delta l = 0.02\%$, $\Delta T = 10^\circ\text{C}$

$$\therefore \alpha = \frac{\Delta l}{l\Delta T} = \frac{0.02}{100 \times 10}$$

$$\Rightarrow \alpha = 2 \times 10^{-5}$$

Volume coefficient of expansion, $\gamma = 3\alpha = 6 \times 10^{-5}$

$$\therefore \rho = \frac{M}{V}$$

$$\frac{\Delta V}{V} \times 100 = \gamma\Delta T = (6 \times 10^{-5} \times 10 \times 100) = 6 \times 10^{-2}$$

Volume increase by 0.06% therefore density decrease by 0.06%.

4. (60.00) Volume, $V = l b h$

$$\therefore \gamma = \frac{\Delta V}{V} = \frac{\Delta l}{l} + \frac{\Delta b}{b} + \frac{\Delta h}{h}$$

(γ = coefficient of volume expansion)

$$\Rightarrow \gamma = 5 \times 10^{-5} + 5 \times 10^{-6} + 5 \times 10^{-6}$$

$$= 60 \times 10^{-6}/^\circ\text{C}$$

$$\therefore \text{Value of } C = 60.00$$

5. (Bonus) $\Delta_{\text{temp}} = \Delta_{\text{load}}$ and $A = \pi r^2 = \pi(10^{-3})^2 = \pi \times 10^{-6}$

$$L \alpha \Delta T = \frac{F L}{A Y}$$

$$\text{or } 0.2 \times 10^{-5} \times 20 = \frac{F \times 0.2}{(\pi \times 10^{-6}) \times 10^{11}}$$

$$\therefore F = 20\pi N \therefore m = \frac{f}{g} = 2\pi = 6.28 \text{ kg}$$

6. (a) Change in length in both rods are same i.e.

$$\Delta l_1 = \Delta l_2$$

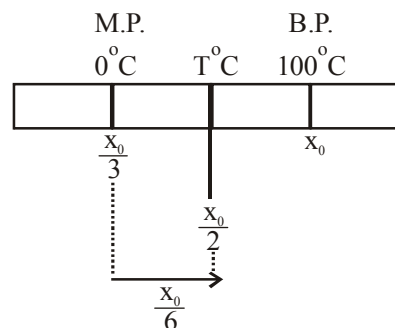
$$l\alpha_1\Delta\theta_1 = l\alpha_2\Delta\theta_2$$

$$\frac{\alpha_1}{\alpha_2} = \frac{\Delta\theta_2}{\Delta\theta_1} \quad \left[\because \frac{\alpha_1}{\alpha_2} = \frac{4}{3} \right]$$

$$\frac{4}{3} = \frac{\theta - 30}{180 - 30}$$

$$\boxed{\theta = 230^\circ\text{C}}$$

7. (a) Let required temperature = $T^\circ\text{C}$



$$\Rightarrow T^\circ\text{C} = \frac{x_0}{2} - \frac{x_0}{3} = \frac{x_0}{6}$$

$$\& \left(x_0 - \frac{x_0}{3} \right) = (100 - 0^\circ\text{C})$$

$$\Rightarrow \frac{2x_0}{3} = 100 \Rightarrow x_0 = \frac{300}{2}$$

$$\Rightarrow T^\circ\text{C} = \frac{x_0}{6} = \frac{150}{6} = 25^\circ\text{C}$$

8. (a) Young's modulus $Y = \frac{\text{stress}}{\text{strain}} = \frac{F/A}{\Delta\ell/\ell}$

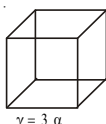
Using, coefficient of linear expansion,

$$\alpha = \frac{\Delta\ell}{\ell\Delta T} \Rightarrow \frac{\Delta\ell}{\ell} = \alpha\Delta T$$

$$\therefore Y = \frac{F}{A(\alpha\Delta T)}$$

9. (c) As we know, Bulk modulus

$$K = \frac{\Delta P}{\left(\frac{-\Delta V}{V}\right)} \Rightarrow \frac{\Delta V}{V} = \frac{P}{K}$$



$$V = V_0(1 + \gamma\Delta T)$$

$$\frac{\Delta V}{V_0} = \gamma\Delta T$$

$$\therefore \frac{P}{K} = \gamma\Delta T \Rightarrow \Delta T = \frac{P}{\gamma K} = \frac{P}{3\alpha K}$$

10. (b) Young's modulus = $\frac{\text{Thermal stress}}{\text{Strain}} = \frac{F/A}{\Delta L/L}$

$$Y = \frac{F}{A \cdot \alpha \Delta \theta} \quad \left(Q \frac{\Delta L}{L} = \alpha \Delta \theta \right)$$

Force developed in the rail $F = YA\alpha\Delta\theta$

$$= 2 \times 10^{11} \times 40 \times 10^{-4} \times 1.2 \times 10^{-5} \times 10$$

$$= 9.6 \times 10^4 = 1 \times 10^5 \text{ N}$$

11. (c) Due to thermal exp., change in length $(\Delta L) = l\alpha\Delta T \dots$
(i)

$$\text{Young's modulus (Y)} = \frac{\text{Normal stress}}{\text{Longitudinal strain}}$$

$$Y = \frac{F/A}{\Delta L/l} \Rightarrow \frac{\Delta L}{l} = \frac{F}{AY}$$

$$\Delta L = \frac{Fl}{AY}$$

$$\text{From eq}^n \text{ (i), } \frac{Fl}{AY} = l\alpha\Delta T$$

$$F = AY\alpha\Delta T$$

12. (b) When there is no change in liquid level in vessel then $\gamma'_{\text{real}} = \gamma'_{\text{vessel}}$

Change in volume in liquid relative to vessel

$$\Delta V_{\text{app}} = V\gamma'_{\text{app}}\Delta\theta = V(\gamma'_{\text{real}} - \gamma'_{\text{vessel}})$$

13. (c) $\frac{\text{Reading on any scale} - \text{LFP}}{\text{UFP} - \text{LFP}}$

= constant for all scales

$$\frac{340 - 273}{373 - 273} = \frac{{}^\circ\text{Y} - (-160)}{-50 - (-160)}$$

$$\Rightarrow \frac{67}{100} = \frac{y + 160}{110}$$

$$\therefore Y = -86.3^\circ\text{Y}$$

14. (d) The Young modulus is given as

$$Y = \frac{\text{stress}}{\text{strain}} = \frac{F/S}{\Delta L/L}$$

Here, $\Delta L = 2\pi R\Delta\theta = 2\pi R$

$$Y = \frac{F}{S2\pi R} \times 2\pi R$$

$$\Rightarrow Y = \frac{FR}{S\Delta R}$$

... (i)

$$\text{The coefficient of linear expansion } \alpha = \frac{\Delta R}{R\Delta T}$$

$$\Rightarrow \frac{\Delta R}{R} = \alpha \Delta T \Rightarrow \frac{R}{\Delta R} = \frac{1}{\alpha \Delta T} \dots \text{ (ii)}$$

From equation (i) and (ii)

$$Y = \frac{F}{S\alpha\Delta T} \Rightarrow F = Y \cdot S \cdot \alpha \Delta T$$

$\therefore$ The ring is pressing the wheel from both sides, Thus

$$F_{\text{net}} = 2F = 2YS\alpha\Delta T$$

15. (a) As the rods are identical, so they have same length (l) and area of cross-section (A). They are connected in series. So, heat current will be same for all rods.

$$\text{Heat current} = \left(\frac{\Delta Q}{\Delta t} \right)_{AB} = \left(\frac{\Delta Q}{\Delta t} \right)_{BC} = \left(\frac{\Delta Q}{\Delta t} \right)_{CD}$$

$$\Rightarrow \frac{(100 - 70)K_1 A}{l} = \frac{(70 - 20)K_2 A}{l} = \frac{(20 - 0)K_3 A}{l}$$

$$\Rightarrow K_1(100 - 70) = K_2(70 - 20) = K_3(20 - 0)$$

$$\Rightarrow K_1(30) = K_2(50) = K_3(20)$$

$$\Rightarrow \frac{K_1}{10} = \frac{K_2}{6} = \frac{K_3}{15}$$

$$\Rightarrow K_1 : K_2 : K_3 = 10 : 6 : 15$$

$$\Rightarrow K_1 : K_3 = 2 : 3.$$

16. (a) According to question, one half of its kinetic energy is converted into heat in the wood.

$$\frac{1}{2}mv^2 \times \frac{1}{2} = ms\Delta T$$

$$\Rightarrow \Delta T = \frac{v^2}{4 \times s} = \frac{210 \times 210}{4 \times 4.2 \times 0.3 \times 1000} = 87.5^\circ\text{C}$$

17. (a) Here ice melts due to water.

Let the amount of ice melts = m_{ice}

$$m_w s_w \Delta\theta = m_{\text{ice}} L_{\text{ice}}$$

$$\therefore m_{ice} = \frac{m_w s_w \Delta \theta}{L_{ice}}$$

$$= \frac{0.2 \times 4200 \times 25}{3.4 \times 10^5} = 0.0617 \text{ kg} = 61.7 \text{ g}$$

18. (a) Heat given by water $= m_w C_w (T_{mix} - T_w)$

$$= 200 \times 1 \times (31 - 25)$$

Heat taken by steam $= m L_{stem} + m C_w (T_s - T_{mix})$

$$= m \times 540 + m(1) \times (100 - 31)$$

$$= m \times 540 + m(1) \times (69)$$

From the principle of calorimetry,

Heat lost = Heat gained

$$\therefore (200)(31 - 25) = m \times 540 + m(1)(69)$$

$$\Rightarrow 1200 = m(609) \Rightarrow m \approx 2.$$

19. (50.00)

Let Q_1, Q_2, Q_3 be the temperatures of container C_1, C_2 and C_3 respectively.

Using principle of calorimetry in container C_1 , we have

$$(\theta_1 - 60) = 2 \text{ ms}(60 - \theta)$$

$$\Rightarrow \theta_1 - 60 = 120 - 2\theta$$

$$\Rightarrow \theta_1 = 180 - 2\theta \quad \dots(i)$$

For container C_2

$$\text{ms}(\theta_2 - 30) = 2\text{ms}(30 - \theta)$$

$$\Rightarrow \theta_2 = 90 - 2\theta \quad \dots(ii)$$

For container C_3

$$2\text{ms}(\theta_1 - 60) = \text{ms}(60 - \theta)$$

$$\Rightarrow 2\theta_1 - 120 = 60 - \theta$$

$$\Rightarrow 2\theta_1 + \theta = 180 \quad \dots(iii)$$

$$\text{Also, } \theta_1 + \theta_2 + \theta_3 = 3\theta \quad \dots(iv)$$

Adding (i), (ii) and (iii)

$$3\theta_1 + 3\theta_2 + 3\theta_3 = 450$$

$$\Rightarrow \theta_1 + \theta_2 + \theta_3 = 150$$

$$\Rightarrow 3\theta = 150 \Rightarrow \theta = 50^\circ\text{C}$$

20. (40) Using the principle of calorimetry

$$M_{ice} L_f + m_{ice} (40 - 0) C_w$$

$$= m_{stream} L_v + m_{stream} (100 - 40) C_w$$

$$\Rightarrow M(540) + M \times 1 \times (100 - 40)$$

$$= 200 \times 80 + 200 \times 1 \times 40$$

$$\Rightarrow 600M = 24000$$

$$\Rightarrow M = 40\text{g}$$

21. (a) $M_1 C_{ice} \times (10) + M_1 L = M_2 C_w (50)$

$$\text{or } M_1 \times C_{ice} (=0.5) \times 10 + M_1 L = M_2 \times 1 \times 50$$

$$\Rightarrow L = \frac{50M_2}{M_1} - 5$$

22. (b) $\frac{1}{2} kx^2 = mC(\Delta T) + m_w C_w \Delta T$

$$\text{or } \frac{1}{2} \times 800 \times 0.02^2 = 0.5 \times 400 \times \Delta T + 1 \times 4184 \times \Delta T$$

$$\therefore \Delta T = 1 \times 10^{-5} \text{ K}$$

23. (a) $H_1 = H_2 \theta_2 \left[\frac{d}{3k} \mid \frac{\theta}{k} \right] \theta_1$

$$\text{or } (3k)A \left(\frac{\theta_2 - \theta}{d} \right) = kA \left(\frac{\theta - \theta_1}{3d} \right)$$

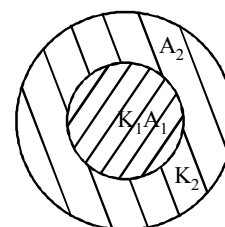
$$\text{or } \theta = \left(\frac{\theta_1 + 9\theta_2}{10} \right)$$

24. (d) Effective thermal conductivity of system

$$K_{eq} = \frac{K_1 A_1 + K_2 A_2}{A_1 + A_2}$$

$$= \frac{K_1 \pi R^2 + K_2 [\pi (2R)^2 - \pi R^2]}{\pi (2R)^2}$$

$$= \frac{K_1 (\pi R^2) + K_2 (3\pi R^2)}{4\pi R^2} = \frac{K_1 + 3K_2}{4}$$



25. (d) Let m gram of ice is added.

From principle of calorimeter

heat gained (by ice) = heat lost (by water)

$$\therefore 20 \times 2.1 \times m + (m - 20) \times 334$$

$$= 50 \times 4.2 \times 40$$

$$376m = 8400 + 6680$$

$$m = 40.1$$

26. (c) Heat loss = Heat gain = $mS\Delta\theta$

$$\text{So, } m_A S_A \Delta\theta_A = m_B S_B \Delta\theta_B$$

$$\Rightarrow 100 \times S_A \times (100 - 90) = 50 \times S_B \times (90 - 75)$$

$$2S_A = 1.5S_B \Rightarrow S_A = \frac{3}{4} S_B$$

$$\text{Now, } 100 \times S_A \times (100 - \theta) = 50 \times S_B \times (\theta - 50)$$

$$2 \times \left(\frac{3}{4} \right) \times (100 - \theta) = (\theta - 50)$$

$$300 - 3\theta = 2\theta - 100$$

$$400 = 5\theta \Rightarrow \theta = 80^\circ\text{C}$$

27. (d) Assume final temperature = $T^\circ\text{C}$

Heat loss = Heat gain = $ms\Delta T$

$$\Rightarrow m_B s_B \Delta T_B = m_w s_w \Delta T_w$$

$$0.1 \times 400 \times (500 - T)$$

$$= 0.5 \times 4200 \times (T - 30) + 800 (T - 30)$$

$$\Rightarrow 40(500 - T) = (T - 30)(2100 + 800)$$

$$\Rightarrow 20000 - 40T = 2900T - 30 \times 2900$$

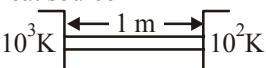
$$\Rightarrow 20000 + 30 \times 2900 = T(2940)$$

$$T = 30.4^\circ\text{C}$$

$$\frac{\Delta T}{T} \times 100 = \frac{6.4}{30} \times 100 = 21\%,$$

so the closest answer is 20%.

Temp. of heat source Temp. of heat reservoir

28. (a) 

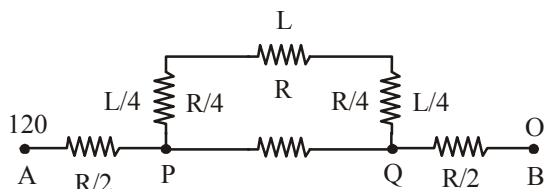
$$\left(\frac{dQ}{dt}\right) = \frac{kA\Delta T}{\ell}$$

$$\text{Energy flux, } \frac{1}{A} \left(\frac{dQ}{dt}\right) = \frac{k\Delta T}{\ell}$$

$$= \frac{(0.1)(900)}{1} = 90 \text{ W/m}^2$$

29. (c) Let specific heat of unknown metal be 's' According to principle of calorimetry, Heat lost = Heat gain $m \times s \Delta \theta = m_1 s_{\text{brass}} (\Delta \theta_1 + m_2 s_{\text{water}} + \Delta \theta_2)$
- $$\Rightarrow 192 \times S \times (100 - 21.5) = 128 \times 394 \times (21.5 - 8.4)$$
- Solving we get, $+240 \times 4200 \times (21.5 - 8.4)$
- $$S = 916 \text{ J kg}^{-1} \text{ K}^{-1}$$

30. (a) $\frac{\Delta T_{AB}}{R_{AB}} = \frac{120}{\frac{8}{5}R} = \frac{120 \times 5}{8R}$



In steady state temperature difference between P and Q,

$$\Delta T_{PQ} = \frac{120 \times 5}{8R} \times \frac{3}{5}R = \frac{360}{8} = 45^\circ\text{C}$$

31. (d) According to principle of calorimetry, Heat lost = Heat gain
- $$100 \times 0.1(T - 75) = 100 \times 0.1 \times 45 + 170 \times 1 \times 45$$
- $$10T - 750 = 450 + 7650 = 8100$$
- $$\Rightarrow T - 75 = 810$$
- $$T = 885^\circ\text{C}$$
32. (d) According to principle of calorimetry, $Q_{\text{given}} = Q_{\text{used}}$
- $$0.2 \times S \times (150 - 40) = 150 \times 1 \times (40 - 27) + 25 \times (40 - 27)$$
- $$0.2 \times S \times 110 = 150 \times 13 + 25 \times 13$$
- Specific heat of aluminium
- $$S = \frac{13 \times 25 \times 7}{0.2 \times 110} = 434 \text{ J/kg}^\circ\text{C}$$
33. (b) As $Pt = mC\Delta T$
- So, $P \times 10 \times 60 = mC 100 \dots(i)$

and $P \times 55 \times 60 = mL \dots(ii)$

Dividing equation (i) by (ii) we get

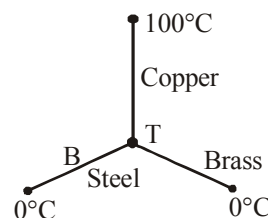
$$\frac{10}{55} = \frac{C \times 100}{L}$$

$$\therefore L = 550 \text{ cal./g.}$$

34. (c) Rate of heat flow is given by,

$$Q = \frac{KA(\theta_1 - \theta_2)}{l}$$

Where, K = coefficient of thermal conductivity
l = length of rod and A = area of cross-section of rod



If the junction temperature is T, then

$$Q_{\text{Copper}} = Q_{\text{Brass}} + Q_{\text{Steel}}$$

$$\frac{0.92 \times 4(100 - T)}{46} = \frac{0.26 \times 4 \times (T - 0)}{13} + \frac{0.12 \times 4 \times (T - 0)}{12}$$

$$\Rightarrow 200 - 2T = 2T + T$$

$$\Rightarrow T = 40^\circ\text{C}$$

$$\therefore Q_{\text{Copper}} = \frac{0.92 \times 4 \times 60}{46} = 4.8 \text{ cal/s}$$

35. (c) In the given problem, fall in temperature of sphere,

$$dT = (3T_0 - 2T_0) = T_0$$

Temperature of surrounding, $T_{\text{surr}} = T_0$

Initial temperature of sphere, $T_{\text{initial}} = 3T_0$

Specific heat of the material of the sphere varies as,

$$c = \alpha T^3 \text{ per unit mass } (\alpha = \text{a constant})$$

Applying formula,

$$\frac{dT}{dt} = \frac{\sigma A}{McJ} (T^4 - T_{\text{surr}}^4)$$

$$\Rightarrow \frac{T_0}{dt} = \frac{\sigma 4\pi R^2}{M\alpha(3T_0)^3 J} [(3T_0)^4 - (T_0)^4]$$

$$\Rightarrow dt = \frac{M\alpha 27T_0^4 J}{\sigma 4\pi R^2 \times 80T_0^4}$$

Solving we get,

Time taken for the sphere to cool down temperature $2T_0$,

$$t = \frac{M\alpha}{16\pi R^2 \sigma} \ln\left(\frac{16}{3}\right)$$

36. (a) From question,

In 1 sec heat gained by water

$$= 1 \text{ KW} - 160 \text{ J/s}$$

$$= 1000 \text{ J/s} - 160 \text{ J/s}$$

$$= 840 \text{ J/s}$$

Total heat required to raise the temperature of water (volume 2L) from 27°C to 77°C

$$= m_{\text{water}} \times \text{sp. ht} \times \Delta\theta$$

$$= 2 \times 10^3 \times 4.2 \times 50 [\because \text{mass} = \text{density} \times \text{volume}]$$

$$\text{And, } 840 \times t = 2 \times 10^3 \times 4.2 \times 50$$

$$\text{or, } t = \frac{2 \times 10^3 \times 4.2 \times 50}{840}$$

$$= 500 \text{ s} = 8 \text{ min } 20 \text{ s}$$

37. (d) When radius is decrease by
- ΔR
- ,

$$4\pi R^2 \Delta R \rho L = 4\pi T [R^2 - (R - \Delta R)^2]$$

$$\Rightarrow \rho R^2 \Delta R L = T [R^2 - R^2 + 2R\Delta R - \Delta R^2]$$

$$\Rightarrow \rho R^2 \Delta R L = T 2R\Delta R \quad [\Delta R \text{ is very small}]$$

$$\Rightarrow R = \frac{2T}{\rho L}$$

38. (d) As the surrounding is identical, vessel is identical time taken to cool both water and liquid (from 30°C to 25°C) is same 2 minutes, therefore

$$\left(\frac{dQ}{dt} \right)_{\text{water}} = \left(\frac{dQ}{dt} \right)_{\text{liquid}}$$

$$\text{or, } \frac{(m_w C_w + W)\Delta T}{t} = \frac{(m_\ell C_\ell + W)\Delta T}{t}$$

(W = water equivalent of the vessel)

$$\text{or, } m_w C_w = m_\ell C_\ell$$

$$\therefore \text{Specific heat of liquid, } C_\ell = \frac{m_w C_w}{m_\ell}$$

$$= \frac{50 \times 1}{100} = 0.5 \text{ kcal / kg}$$

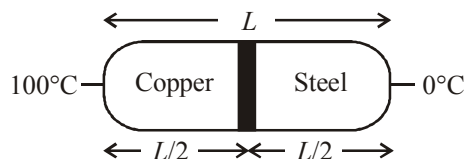
39. (b) As 1g of steam at 100°C melts 8g of ice at 0°C.

10 g of steam will melt 8×10 g of ice at 0°C

$$\text{Water in calorimeter} = 500 + 80 + 10 \text{ g} = 590 \text{ g}$$

40. (c)

41. (a)

Let conductivity of steel $K_{\text{steel}} = k$ then from questionConductivity of copper $K_{\text{copper}} = 9k$

$$\theta_{\text{copper}} = 100^\circ\text{C}$$

$$\theta_{\text{steel}} = 0^\circ\text{C}$$

$$l_{\text{steel}} = l_{\text{copper}} = \frac{L}{2}$$

From formula temperature of junction;

$$\theta = \frac{K_{\text{copper}} \theta_{\text{copper}} l_{\text{steel}} + K_{\text{steel}} \theta_{\text{steel}} l_{\text{copper}}}{K_{\text{copper}} l_{\text{steel}} + K_{\text{steel}} l_{\text{copper}}}$$

$$= \frac{9k \times 100 \times \frac{L}{2} + k \times 0 \times \frac{L}{2}}{9k \times \frac{L}{2} + k \times \frac{L}{2}}$$

$$= \frac{\frac{900}{2} kL}{\frac{10kL}{2}} = 90^\circ\text{C}$$

42. (a) According to Stefan's law

$$E = \sigma T^4$$

Heat radiated per unit area in 1 hour (3600s) is

$$= 5.7 \times 10^{-8} \times (300)^4 \times 3600 = 1.7 \times 10^{10} \text{ J}$$

43. (a)
- $\Delta U = \Delta Q = mc\Delta T$

$$= \frac{100}{1000} \times 4184 (50 - 30) \approx 8.4 \text{ kJ}$$

44. (d) Required work = energy released

$$\text{Here, } Q = \int mc dT$$

$$= \int_{20}^4 0.1 \times 32 \times \left(\frac{T^3}{400^3} \right) dT = \int_{20}^4 \frac{3.2}{64 \times 10^6} T^3 dT$$

$$= 5 \times 10^{-8} \int_{20}^4 T^3 dT = 0.002 \text{ kJ}$$

Therefore, required work = 0.002 kJ

45. (a) Let
- Q
- be the temperature at a distance
- x
- from hot end of bar. Let
- Q
- is the temperature of hot end.

The heat flow rate is given by

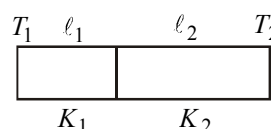
$$\frac{dQ}{dt} = \frac{kA(\theta_1 - \theta)}{x}$$

$$\Rightarrow \theta_1 - \theta = \frac{x}{kA} \frac{dQ}{dt} \quad \Rightarrow \theta = \theta_1 - \frac{x}{kA} \frac{dQ}{dt}$$

Thus, the graph of Q versus x is a straight line with a positive intercept and a negative slope.

The above equation can be graphically represented by option (a).

46. (d) Let
- T
- be the temperature of the interface. In the steady state,
- $Q_1 = Q_2$



$$\therefore \frac{K_1 A (T_1 - T)}{l_1} = \frac{K_2 A (T - T_2)}{l_2}$$

where A is the area of cross-section.

$$\Rightarrow K_1 A(T_1 - T)\ell_2 = K_2 A(T - T_2)\ell_1$$

$$\Rightarrow K_1 T_1 \ell_2 - K_1 T \ell_2 = K_2 T \ell_1 - K_2 T_2 \ell_1$$

$$\Rightarrow (K_2 \ell_1 + K_1 \ell_2)T = K_1 T_1 \ell_2 + K_2 T_2 \ell_1$$

$$\Rightarrow T = \frac{K_1 T_1 \ell_2 + K_2 T_2 \ell_1}{K_2 \ell_1 + K_1 \ell_2}$$

$$= \frac{K_1 \ell_2 T_1 + K_2 \ell_1 T_2}{K_1 \ell_2 + K_2 \ell_1}$$

47. (b) From Stefan's law, total power radiated by Sun, $E = \sigma T^4 \times 4\pi R^2$

The intensity of power Per unit area incident on earth's surface

$$= \frac{\sigma T^4 \times 4\pi R^2}{4\pi r^2}$$

Total power received by Earth

$$E' = \frac{E}{4\pi r^2} \times \text{Cross - Section area of earth facing the}$$

$$\text{sun} = \frac{\sigma T^4 R^2}{r^2} (\pi r_0^2)$$

48. (c) When two gases are mixed together then

Heat lost by He gas = Heat gained by N_2 gas

$$n_1 C_{v1} \Delta T_1 = n_2 C_{v2} \Delta T_2$$

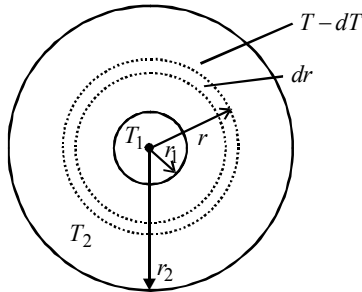
$$\frac{3}{2} R \left[\frac{7}{3} T_0 - T_f \right] = \frac{5}{2} R [T_f - T_0]$$

$$7T_0 - 3T_f = 5T_f - 5T_0$$

$$\Rightarrow 12T_0 = 8T_f \Rightarrow T_f = \frac{12}{8} T_0$$

$$\Rightarrow T_f = \frac{3}{2} T_0 \dots$$

49. (d)



Consider a thin concentric shell of thickness (dr) and of radius (r) and let the temperature of inner and outer surfaces of this shell be T and $(T - dT)$ respectively.

The radial rate of flow of heat through this elementary shell will be

$$\frac{dQ}{dt} = \frac{KA[(T - dT) - T]}{dr} = \frac{-KA dT}{dr}$$

$$= -4\pi K r^2 \frac{dT}{dr} \quad (\because A = 4\pi r^2)$$

Since the area of the surface through which heat will flow is not constant. Integrating both sides between the limits of radii and temperatures of the two shells, we get

$$\left(\frac{dQ}{dt} \right) \int_{r_1}^{r_2} \frac{1}{r^2} dr = -4\pi K \int_{T_1}^{T_2} dT$$

$$\left(\frac{dQ}{dt} \right) \int_{r_1}^{r_2} r^{-2} dr = -4\pi K \int_{T_1}^{T_2} dT$$

$$\frac{dQ}{dt} \left[\frac{1}{r_1} - \frac{1}{r_2} \right] = -4\pi K [T_2 - T_1]$$

$$\text{or } \frac{dQ}{dt} = \frac{-4\pi K r_1 r_2 (T_2 - T_1)}{(r_2 - r_1)}$$

$$\therefore \frac{dQ}{dt} \propto \frac{r_1 r_2}{(r_2 - r_1)}$$

50. (d) From Stefan's law, energy radiated by sun per second

$$E = \sigma A T^4 ;$$

$$\therefore A \propto R^2$$

$$\therefore E \propto R^2 T^4$$

$$\therefore \frac{E_2}{E_1} = \frac{R_2^2 T_2^4}{R_1^2 T_1^4}$$

$$\text{put } R_2 = 2R, R_1 = R ; T_2 = 2T, T_1 = T$$

$$\Rightarrow \frac{E_2}{E_1} = \frac{(2R)^2 (2T)^4}{R^2 T^4} = 64$$

51. (d) The thermal resistance is given by

$$\frac{x}{KA} + \frac{4x}{2KA} = \frac{x}{KA} + \frac{2x}{KA} = \frac{3x}{KA}$$

Amount of heat flow per second,

$$\frac{dQ}{dt} = \frac{\Delta T}{\frac{3x}{KA}} = \frac{(T_2 - T_1)KA}{3x}$$

$$= \frac{1}{3} \left\{ \frac{A(T_2 - T_1)K}{x} \right\} \quad \therefore f = \frac{1}{3}$$

52. (d) Wein's law correctly explains the spectrum
53. (b) Heat required for raising the temperature of a body through 1°C is called its thermal capacity.
54. (b) Pyrometer is used to detect infra-red radiation.
55. (a) Black body is one which absorb all incident radiation. Black board paint is quite approximately equal to black bodies.
56. (c) When water is cooled at 0°C to form ice, energy is released from water in the form of heat. As energy is equivalent to mass, therefore, when water is cooled to ice, its mass decreases.
57. (a) From stefan's law, the energy radiated per second is given by $E = e\sigma T^4 A$

Here, T = temperature of the body

A = surface area of the body

For same material e is same. σ is stefan's constant

Let T_1 and T_2 be the temperature of two spheres. A_1 and A_2 be the area of two spheres.

$$\therefore \frac{E_1}{E_2} = \frac{T_1^4 A_1}{T_2^4 A_2} = \frac{T_1^4 4\pi r_1^2}{T_2^4 4\pi r_2^2}$$

$$= \frac{(4000)^4 \times 1^2}{(2000)^4 \times 4^2} = \frac{1}{1}$$

58. (b) From Newton's Law of cooling,

$$\frac{T_1 - T_2}{t} = K \left[\frac{T_1 + T_2}{2} - T_0 \right]$$

Here, $T_1 = 50^\circ\text{C}$, $T_2 = 40^\circ\text{C}$

and $T_0 = 20^\circ\text{C}$, $t = 600\text{S} = 5 \text{ minutes}$

$$\Rightarrow \frac{50 - 40}{5 \text{ Min}} = K \left(\frac{50 + 40}{2} - 20 \right) \quad \dots(i)$$

Let T be the temperature of sphere after next 5 minutes. Then

$$\frac{40 - T}{5} = K \left(\frac{40 + T}{2} - 20 \right) \quad \dots(ii)$$

Dividing eqn. (ii) by (i), we get

$$\frac{40 - T}{10} = \frac{40 + T - 40}{50 + 40 - 40} = \frac{T}{50}$$

$$\Rightarrow 40 - T = \frac{T}{5} \Rightarrow 200 - 5T = T$$

$$\therefore T = \frac{200}{6} = 33.3^\circ\text{C}$$

59. (b) Rate of Heat loss = $mS \left(\frac{dT}{dt} \right) = e\sigma AT^4$

$$-\frac{dT}{dt} = \frac{e\sigma \times A \times T^4}{\rho \times Vol. \times S} \Rightarrow -\frac{dT}{dt} \propto \frac{1}{\rho S}$$

$$\left(-\frac{dT}{dt} \right)_A = \frac{\rho_B}{\rho_A} \times \frac{S_B}{S_A} = \frac{10^3}{8 \times 10^2} \times \frac{4000}{2000}$$

$$\Rightarrow \left(-\frac{dT}{dt} \right)_A > \left(-\frac{dT}{dt} \right)_B$$

So, A cools down at faster rate.

60. (a) According to Newton's law of cooling,

$$\left(\frac{\theta_1 - \theta_2}{t} \right) = K \left(\frac{\theta_1 + \theta_2}{2} - \theta_0 \right)$$

$$\left(\frac{60 - 50}{10} \right) = K \left(\frac{60 + 50}{2} - 25 \right) \quad \dots(i)$$

$$\text{and, } \left(\frac{50 - \theta}{10} \right) = K \left(\frac{50 + \theta}{2} - 25 \right) \quad \dots(ii)$$

Dividing eq. (i) by (ii),

$$\frac{10}{(50 - \theta)} = \frac{60}{\theta} \Rightarrow \theta = 42.85^\circ\text{C} \cong 43^\circ\text{C}$$

61. (b) By Newton's law of cooling

$$\frac{\theta_1 - \theta_2}{t} = -K \left[\frac{\theta_1 + \theta_2}{2} - \theta_0 \right]$$

where θ_0 is the temperature of surrounding.

Now, hot water cools from 60°C to 50°C in 10 minutes,

$$\frac{60 - 50}{10} = -K \left[\frac{60 + 50}{2} - \theta_0 \right] \quad \dots(i)$$

Again, it cools from 50°C to 42°C in next 10 minutes.

$$\frac{50 - 42}{10} = -K \left[\frac{50 + 42}{2} - \theta_0 \right] \quad \dots(ii)$$

Dividing equations (i) by (ii) we get

$$\frac{1}{0.8} = \frac{55 - \theta_0}{46 - \theta_0}$$

$$\frac{10}{8} = \frac{55 - \theta_0}{46 - \theta_0}$$

$$460 - 10\theta_0 = 440 - 8\theta_0$$

$$2\theta_0 = 20$$

$$\theta_0 = 10^\circ\text{C}$$

62. (b) From Newton's law of cooling,

$$t = \frac{1}{k} \log_e \left(\frac{\theta_2 - \theta_0}{\theta_1 - \theta_0} \right)$$

From question and above equation,

$$5 = \frac{1}{k} \log_e \frac{(40-30)}{(80-30)} \quad \dots(1)$$

$$\text{And, } t = \frac{1}{k} \log_e \frac{(32-30)}{(62-30)} \quad \dots(2)$$

Dividing equation (2) by (1),

$$\frac{t}{5} = \frac{\frac{1}{k} \log_e \frac{(32-30)}{(62-30)}}{\frac{1}{k} \log_e \frac{(40-30)}{(80-30)}}$$

On solving we get, time taken to cool down from 62°C to 32°C, $t = 8.6$ minutes.

63. (c) According to Newton's law of cooling, the temperature goes on decreasing with time non-linearly.

64. (a) According to Newton's law of cooling

$$\frac{d\theta}{dt} = -k(\theta - \theta_0)$$

$$\Rightarrow \frac{d\theta}{(\theta - \theta_0)} = -k dt$$

$$\Rightarrow \int_{\theta_0}^{\theta} \frac{d\theta}{(\theta - \theta_0)} = -k \int_0^t dt$$

$$\Rightarrow \log(\theta - \theta_0) = -kt + c$$

Which represents an equation of straight line.

Thus the option (a) is correct.

65. (d) From Newton's law of cooling $-\frac{dQ}{dt} \propto (\Delta\theta)$

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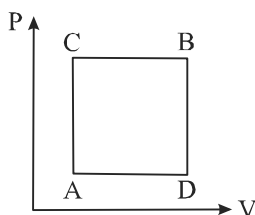
Thermodynamics



TOPIC 1 First Law of Thermodynamics



1. A gas can be taken from A to B via two different processes ACB and ADB.



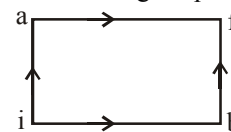
When path ACB is used 60 J of heat flows into the system and 30 J of work is done by the system. If path ADB is used work done by the system is 10 J. The heat flow into the system in path ADB is : **[9 Jan. 2019 I]**

- (a) 40 J (b) 80 J (c) 100 J (d) 20 J
2. 200g water is heated from 40°C to 60°C. Ignoring the slight expansion of water, the change in its internal energy is close to (Given specific heat of water = 4184 J/kgK): **[Online April 9, 2016]**
- (a) 167.4 kJ (b) 8.4 kJ (c) 4.2 kJ (d) 16.7 kJ
3. A gas is compressed from a volume of 2m³ to a volume of 1m³ at a constant pressure of 100 N/m². Then it is heated at constant volume by supplying 150 J of energy. As a result, the internal energy of the gas: **[Online April 19, 2014]**
- (a) increases by 250 J (b) decreases by 250 J
(c) increases by 50 J (d) decreases by 50 J
4. An insulated container of gas has two chambers separated by an insulating partition. One of the chambers has volume V_1 and contains ideal gas at pressure P_1 and temperature T_1 . The other chamber has volume V_2 and contains ideal gas at pressure P_2 and temperature T_2 . If the partition is removed without doing any work on the gas, the final equilibrium temperature of the gas in the container will be **[2008]**

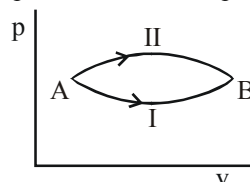
(a) $\frac{T_1 T_2 (P_1 V_1 + P_2 V_2)}{P_1 V_1 T_2 + P_2 V_2 T_1}$ (b) $\frac{P_1 V_1 T_1 + P_2 V_2 T_2}{P_1 V_1 + P_2 V_2}$

(c) $\frac{P_1 V_1 T_2 + P_2 V_2 T_1}{P_1 V_1 + P_2 V_2}$ (d) $\frac{T_1 T_2 (P_1 V_1 + P_2 V_2)}{P_1 V_1 T_1 + P_2 V_2 T_2}$

5. When a system is taken from state i to state f along the path iaf, it is found that $Q=50$ cal and $W=20$ cal. Along the path ibf $Q=36$ cal. W along the path ibf is **[2007]**



- (a) 14 cal (b) 6 cal (c) 16 cal (d) 66 cal
6. A system goes from A to B via two processes I and II as shown in figure. If ΔU_1 and ΔU_2 are the changes in internal energies in the processes I and II respectively, then **[2005]**



- (a) relation between ΔU_1 and ΔU_2 can not be determined
(b) $\Delta U_1 = \Delta U_2$
(c) $\Delta U_2 < \Delta U_1$
(d) $\Delta U_2 > \Delta U_1$
7. Which of the following is incorrect regarding the first law of thermodynamics? **[2005]**
- (a) It is a restatement of the principle of conservation of energy
(b) It is not applicable to any cyclic process
(c) It does not introduce the concept of the entropy
(d) It introduces the concept of the internal energy

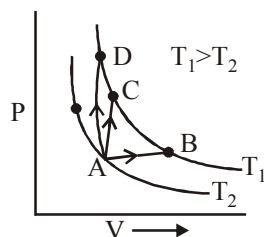
TOPIC 2 Specific Heat Capacity and Thermodynamical Processes



8. Three different processes that can occur in an ideal monoatomic gas are shown in the P vs V diagram. The paths are labelled as A → B, A → C and A → D. The change

in internal energies during these process are taken as E_{AB} , E_{AC} and E_{AD} and the workdone as W_{AB} , W_{AC} and W_{AD} . The correct relation between these parameters are :

[5 Sep. 2020 (I)]



- (a) $E_{AB} = E_{AC} < E_{AD}$, $W_{AB} > 0$, $W_{AC} = 0$, $W_{AD} < 0$
 (b) $E_{AB} = E_{AC} = E_{AD}$, $W_{AB} > 0$, $W_{AC} = 0$, $W_{AD} > 0$
 (c) $E_{AB} < E_{AC} < E_{AD}$, $W_{AB} > 0$, $W_{AC} > W_{AD}$
 (d) $E_{AB} > E_{AC} > E_{AD}$, $W_{AB} < W_{AC} < W_{AD}$

9. In an adiabatic process, the density of a diatomic gas becomes 32 times its initial value. The final pressure of the gas is found to be n times the initial pressure. The value of n is : [5 Sep. 2020 (II)]

- (a) 32 (b) 326 (c) 128 (d) $\frac{1}{32}$

10. Match the thermodynamic processes taking place in a system with the correct conditions. In the table : ΔQ is the heat supplied, ΔW is the work done and ΔU is change in internal energy of the system. [4 Sep. 2020 (II)]

Process	Condition
(I) Adiabatic	(A) $\Delta W = 0$
(II) Isothermal	(B) $\Delta Q = 0$
(III) Isochoric	(C) $\Delta U \neq 0$, $\Delta W \neq 0$, $\Delta Q \neq 0$
(IV) Isobaric	(D) $\Delta U = 0$
(a) (I)-(A), (II)-(B), (III)-(D), (IV)-(D)	
(b) (I)-(B), (II)-(A), (III)-(D), (IV)-(C)	
(c) (I)-(A), (II)-(A), (III)-(B), (IV)-(C)	
(d) (I)-(B), (II)-(D), (III)-(A), (IV)-(C)	

11. A balloon filled with helium (32°C and 1.7 atm.) bursts. Immediately afterwards the expansion of helium can be considered as : [3 Sep. 2020 (I)]

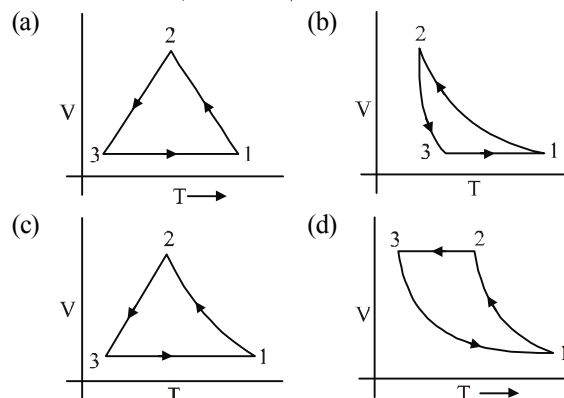
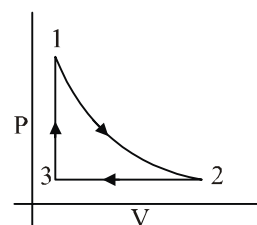
- (a) irreversible isothermal (b) irreversible adiabatic
 (c) reversible adiabatic (d) reversible isothermal

12. An engine takes in 5 mole of air at 20°C and 1 atm, and compresses it adiabatically to $1/10^{\text{th}}$ of the original volume. Assuming air to be a diatomic ideal gas made up of rigid molecules, the change in its internal energy during this process comes out to be $X \text{ kJ}$. The value of X to the nearest integer is _____. [NA 2 Sep. 2020 (I)]

13. Which of the following is an equivalent cyclic process corresponding to the thermodynamic cyclic given in the figure?

where, $1 \rightarrow 2$ is adiabatic.

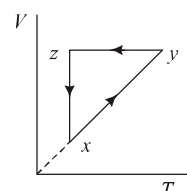
(Graphs are schematic and are not to scale) [9 Jan. 2020 I]



14. Starting at temperature 300 K , one mole of an ideal diatomic gas ($\gamma = 1.4$) is first compressed adiabatically from volume

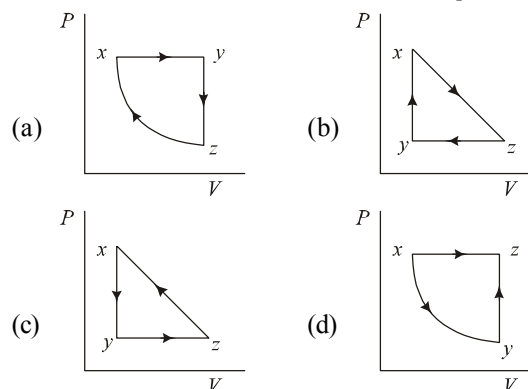
V_1 to $V_2 = \frac{V_1}{16}$. It is then allowed to expand isobarically to volume $2V_2$. If all the processes are the quasi-static then the final temperature of the gas (in $^\circ\text{K}$) is (to the nearest integer) _____. [9 Jan. 2020 II]

15. A thermodynamic cycle $xyzx$ is shown on a V - T diagram.



The P - V diagram that best describes this cycle is: (Diagrams are schematic and not to scale)

[8 Jan. 2020 I]



16. A litre of dry air at STP expands adiabatically to a volume of 3 litres. If $\gamma = 1.40$, the work done by air is: ($3^{1.4} = 4.6555$) [Take air to be an ideal gas]

[7 Jan. 2020 I]

- (a) 60.7 J (b) 90.5 J (c) 100.8 J (d) 48 J

17. Under an adiabatic process, the volume of an ideal gas gets doubled. Consequently the mean collision time between the gas molecule changes from τ_1 to τ_2 . If $\frac{C_p}{C_v} = \gamma$ for this

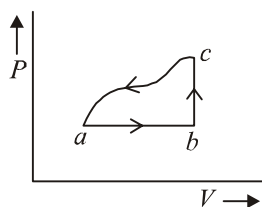
gas then a good estimate for $\frac{\tau_2}{\tau_1}$ is given by:

[7 Jan. 2020 I]

- (a) 2 (b) $\frac{1}{2}$ (c) $\left(\frac{1}{2}\right)^\gamma$ (d) $\left(\frac{1}{2}\right)^{\frac{\gamma+1}{2}}$

18. A sample of an ideal gas is taken through the cyclic process abca as shown in the figure. The change in the internal energy of the gas along the path ca is -180 J, The gas absorbs 250 J of heat along the path ab and 60 J along the path bc. The work done by the gas along the path abc is:

[12 Apr. 2019 I]



- (a) 120 J (b) 130 J (c) 100 J (d) 140 J
19. A cylinder with fixed capacity of 67.2 lit contains helium gas at STP. The amount of heat needed to raise the temperature of the gas by 20°C is: [Given that $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$]

[10 Apr. 2019 I]

- (a) 350 J (b) 374 J (c) 748 J (d) 700 J

20. n moles of an ideal gas with constant volume heat capacity C_v undergo an isobaric expansion by certain volume. The ratio of the work done in the process, to the heat supplied is:

[10 Apr. 2019 I]

- (a) $\frac{nR}{C_v + nR}$ (b) $\frac{nR}{C_v - nR}$
(c) $\frac{4nR}{C_v - nR}$ (d) $\frac{4nR}{C_v + nR}$

21. One mole of an ideal gas passes through a process where

pressure and volume obey the relation $P = P_0 \left[1 - \frac{1}{2} \left(\frac{V_0}{V} \right)^2 \right]$.

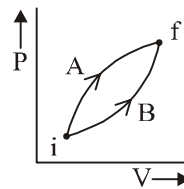
Here P_0 and V_0 are constants. Calculate the change in the temperature of the gas if its volume changes from V_0 to $2V_0$.

[10 Apr. 2019 II]

- (a) $\frac{1}{2} \frac{P_0 V_0}{R}$ (b) $\frac{5}{4} \frac{P_0 V_0}{R}$ (c) $\frac{3}{4} \frac{P_0 V_0}{R}$ (d) $\frac{1}{4} \frac{P_0 V_0}{R}$

22. Following figure shows two processes A and B for a gas. If ΔQ_A and ΔQ_B are the amount of heat absorbed by the system in two cases, and ΔU_A and ΔU_B are changes in internal energies, respectively, then: [9 April 2019 I]

- (a) $\Delta Q_A < \Delta Q_B, \Delta U_A < \Delta U_B$
(b) $\Delta Q_A > \Delta Q_B, \Delta U_A > \Delta U_B$
(c) $\Delta Q_A > \Delta Q_B, \Delta U_A = \Delta U_B$
(d) $\Delta Q_A = \Delta Q_B, \Delta U_A = \Delta U_B$



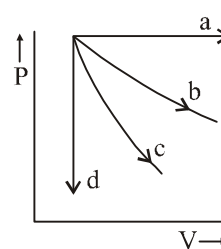
23. A thermally insulated vessel contains 150 g of water at 0°C . Then the air from the vessel is pumped out adiabatically. A fraction of water turns into ice and the rest evaporates at 0°C itself. The mass of evaporated water will be close to: (Latent heat of vaporization of water $= 2.10 \times 10^6 \text{ J kg}^{-1}$ and Latent heat of Fusion of water $= 3.36 \times 10^5 \text{ J kg}^{-1}$)

[8 April 2019 I]

- (a) 150 g (b) 20 g (c) 130 g (d) 35 g

24. The given diagram shows four processes i.e., isochoric, isobaric, isothermal and adiabatic. The correct assignment of the processes, in the same order is given by:

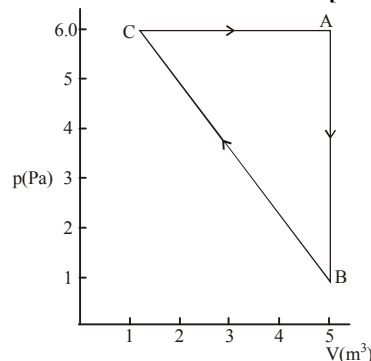
[8 Apr. 2019 II]



- (a) a d b c (b) d a c b (c) a d c b (d) d a b c

25. For the given cyclic process CAB as shown for gas, the work done is:

[12 Jan. 2019 I]



- (a) 30 J (b) 10 J (c) 1 J (d) 5 J

26. A rigid diatomic ideal gas undergoes an adiabatic process at room temperature. The relation between temperature and volume for this process is $TV^x = \text{constant}$, then x is:

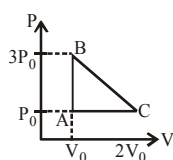
[11 Jan. 2019 I]

- (a) $\frac{3}{5}$ (b) $\frac{2}{5}$ (c) $\frac{2}{3}$ (d) $\frac{5}{3}$

27. Half mole of an ideal monoatomic gas is heated at constant pressure of 1 atm from 20°C to 90°C . Work done by gas is close to: (Gas constant $R = 8.31 \text{ J/mol-K}$) [10 Jan. 2019 II]

- (a) 581 J (b) 291 J (c) 146 J (d) 73 J

28. One mole of an ideal monoatomic gas is taken along the path ABCA as shown in the PV diagram. The maximum temperature attained by the gas along the path BC is given by [Online April 16, 2018]



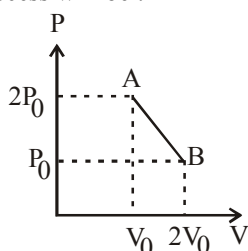
- (a) $\frac{25 P_0 V_0}{8 R}$ (b) $\frac{25 P_0 V_0}{4 R}$ (c) $\frac{25 P_0 V_0}{16 R}$ (d) $\frac{5 P_0 V_0}{8 R}$

29. One mole of an ideal monoatomic gas is compressed isothermally in a rigid vessel to double its pressure at room temperature, 27°C . The work done on the gas will be:

[Online April 15, 2018]

- (a) $300R \ln 6$ (b) $300R$
(c) $300R \ln 7$ (d) $300R \ln 2$

30. 'n' moles of an ideal gas undergoes a process $A \rightarrow B$ as shown in the figure. The maximum temperature of the gas during the process will be: [2016]



- (a) $\frac{9P_0 V_0}{2nR}$ (b) $\frac{9P_0 V_0}{nR}$ (c) $\frac{9P_0 V_0}{4nR}$ (d) $\frac{3P_0 V_0}{2nR}$

31. The ratio of work done by an ideal monoatomic gas to the heat supplied to it in an isobaric process is:

[Online April 9, 2016]

- (a) $\frac{2}{5}$ (b) $\frac{3}{2}$ (c) $\frac{3}{5}$ (d) $\frac{2}{3}$

32. Consider an ideal gas confined in an isolated closed chamber. As the gas undergoes an adiabatic expansion, the average time of collision between molecules increases as V^q , where V is the volume of the gas. The value of q is:

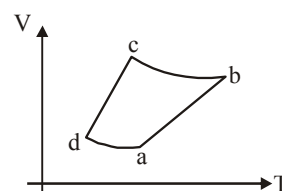
is: $\left(\gamma = \frac{C_p}{C_v} \right)$ [2015]

- (a) $\frac{\gamma+1}{2}$ (b) $\frac{\gamma-1}{2}$ (c) $\frac{3\gamma+5}{6}$ (d) $\frac{3\gamma-5}{6}$

33. Consider a spherical shell of radius R at temperature T . The black body radiation inside it can be considered as an ideal gas of photons with internal energy per unit volume $u = \frac{U}{V} \propto T^4$ and pressure $p = \frac{1}{3} \left(\frac{U}{V} \right)$. If the shell now undergoes an adiabatic expansion the relation between T and R is: [2015]

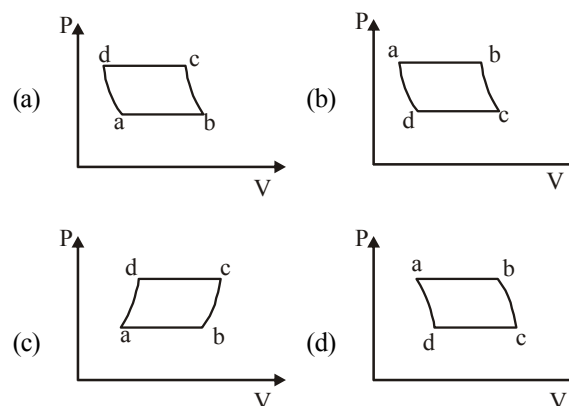
- (a) $T \propto \frac{1}{R}$ (b) $T \propto \frac{1}{R^3}$ (c) $T \propto e^{-R}$ (d) $T \propto e^{-3R}$

34. An ideal gas goes through a reversible cycle $a \rightarrow b \rightarrow c \rightarrow d$ has the $V-T$ diagram shown below. Process $d \rightarrow a$ and $b \rightarrow c$ are adiabatic.



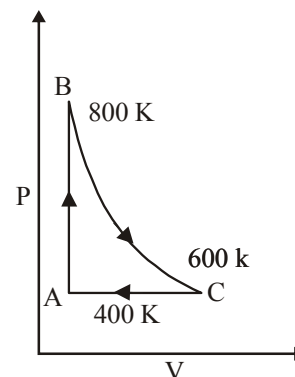
The corresponding $P-V$ diagram for the process is (all figures are schematic and not drawn to scale):

[Online April 10, 2015]



35. One mole of a diatomic ideal gas undergoes a cyclic process ABC as shown in figure. The process BC is adiabatic. The temperatures at A, B and C are 400 K, 800 K and 600 K respectively. Choose the correct statement:

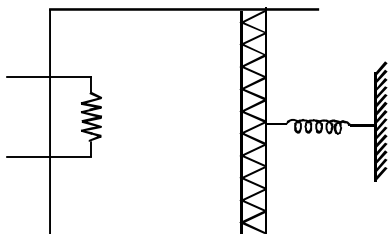
[2014]



- (a) The change in internal energy in whole cyclic process is 250 R.
(b) The change in internal energy in the process CA is 700 R.
(c) The change in internal energy in the process AB is -350 R.
(d) The change in internal energy in the process BC is -500 R.
36. An ideal monoatomic gas is confined in a cylinder by a spring loaded piston of cross section $8.0 \times 10^{-3} \text{ m}^2$. Initially the gas is at 300 K and occupies a volume of $2.4 \times 10^{-3} \text{ m}^3$ and the spring is in its relaxed state as shown in figure. The gas is heated by a small heater until the piston moves out slowly by 0.1 m. The force constant of the spring is 8000 N/m and the atmospheric pressure is $1.0 \times 10^5 \text{ N/m}^2$. The cylinder and

the piston are thermally insulated. The piston and the spring are massless and there is no friction between the piston and the cylinder. The final temperature of the gas will be: (Neglect the heat loss through the lead wires of the heater. The heat capacity of the heater coil is also negligible).

[Online April 11, 2014]



- (a) 300 K (b) 800 K (c) 500 K (d) 1000 K

37. During an adiabatic compression, 830 J of work is done on 2 moles of a diatomic ideal gas to reduce its volume by 50%. The change in its temperature is nearly:

($R = 8.3 \text{ JK}^{-1} \text{ mol}^{-1}$)

[Online April 11, 2014]

- (a) 40 K (b) 33 K (c) 20 K (d) 14 K

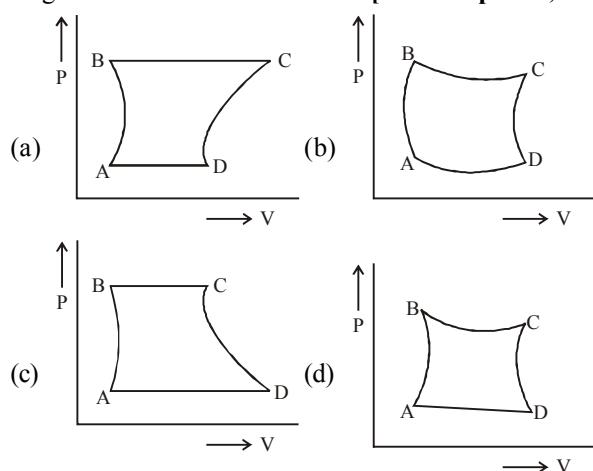
38. The equation of state for a gas is given by $PV = nRT + \alpha V$, where n is the number of moles and α is a positive constant. The initial temperature and pressure of one mole of the gas contained in a cylinder are T_0 and P_0 respectively. The work done by the gas when its temperature doubles isobarically will be:

[Online April 9, 2014]

- (a) $\frac{P_0 T_0 R}{P_0 - \alpha}$ (b) $\frac{P_0 T_0 R}{P_0 + \alpha}$
(c) $P_0 T_0 R \ln 2$ (d) $P_0 T_0 R$

39. A certain amount of gas is taken through a cyclic process (A B C D A) that has two isobars, one isochore and one isothermal. The cycle can be represented on a P-V indicator diagram as:

[Online April 22, 2013]



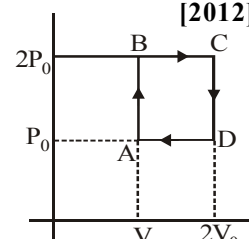
40. An ideal gas at atmospheric pressure is adiabatically compressed so that its density becomes 32 times of its initial value. If the final pressure of gas is 128 atmospheres, the value of ' γ ' of the gas is:

[Online April 22, 2013]

- (a) 1.5 (b) 1.4 (c) 1.3 (d) 1.6

41. Helium gas goes through a cycle ABCDA (consisting of two isochoric and isobaric lines) as shown in figure. The efficiency of this cycle is nearly: (Assume the gas to be close to ideal gas)

[2012]



- (a) 15.4 %
(b) 9.1 %
(c) 10.5 %
(d) 12.5 %

42. An ideal monatomic gas with pressure P , volume V and temperature T is expanded isothermally to a volume $2V$ and a final pressure P_i . If the same gas is expanded adiabatically to a volume $2V$, the final pressure is P_a . The ratio $\frac{P_a}{P_i}$ is

[Online May 26, 2012]

- (a) $2^{-1/3}$ (b) $2^{1/3}$ (c) $2^{2/3}$ (d) $2^{-2/3}$

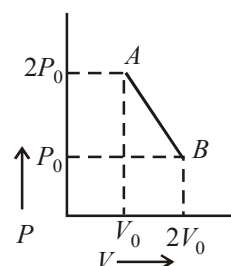
43. The pressure of an ideal gas varies with volume as $P = \alpha V$, where α is a constant. One mole of the gas is allowed to undergo expansion such that its volume becomes ' m ' times its initial volume. The work done by the gas in the process is

[Online May 19, 2012]

- (a) $\frac{\alpha V}{2} (m^2 - 1)$ (b) $\frac{\alpha^2 V^2}{2} (m^2 - 1)$
(c) $\frac{\alpha}{2} (m^2 - 1)$ (d) $\frac{\alpha V^2}{2} (m^2 - 1)$

44. n moles of an ideal gas undergo a process $A \rightarrow B$ as shown in the figure. Maximum temperature of the gas during the process is

[Online May 12, 2012]



- (a) $\frac{9P_0 V_0}{nR}$ (b) $\frac{3P_0 V_0}{2nR}$ (c) $\frac{9P_0 V_0}{2nR}$ (d) $\frac{9P_0 V_0}{4nR}$

45. This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.

Statement 1: In an adiabatic process, change in internal energy of a gas is equal to work done on/by the gas in the process.

Statement 2: The temperature of a gas remains constant in an adiabatic process.

[Online May 7, 2012]

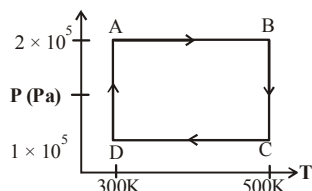
- (a) Statement 1 is true, Statement 2 is true, Statement 2 is a correct explanation of Statement 1.
(b) Statement 1 is true, Statement 2 is false.

- (c) Statement 1 is false, Statement 2 is true.
 (d) Statement 1 is false, Statement 2 is true, Statement 2 is not a correct explanation of Statement 1.
46. A container with insulating walls is divided into equal parts by a partition fitted with a valve. One part is filled with an ideal gas at a pressure P and temperature T , whereas the other part is completely evacuated. If the valve is suddenly opened, the pressure and temperature of the gas will be: [2011 RS]

- (a) $\frac{P}{2}, \frac{T}{2}$ (b) P, T (c) $P, \frac{T}{2}$ (d) $\frac{P}{2}, T$

Directions for questions 47 to 49: Questions are based on the following paragraph.

Two moles of helium gas are taken over the cycle $ABCD$, as shown in the P - T diagram. [2009]



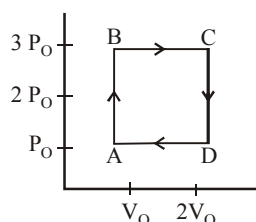
47. Assuming the gas to be ideal the work done on the gas in taking it from A to B is
 (a) 300 R (b) 400 R (c) 500 R (d) 200 R
48. The work done on the gas in taking it from D to A is
 (a) +414 R (b) -690 R (c) +690 R (d) -414 R
49. The net work done on the gas in the cycle $ABCD$ is
 (a) 279 R (b) 1076 R (c) 1904 R (d) zero
50. The work of 146 kJ is performed in order to compress one kilo mole of gas adiabatically and in this process the temperature of the gas increases by 7°C . The gas is [2006] ($R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$)
 (a) diatomic
 (b) triatomic
 (c) a mixture of monoatomic and diatomic
 (d) monoatomic
51. Which of the following parameters does not characterize the thermodynamic state of matter? [2003]
 (a) Temperature (b) Pressure
 (c) Work (d) Volume

TOPIC 3

Carnot Engine, Refrigerators and Second Law of Thermodynamics



52. An engine operates by taking a monatomic ideal gas through the cycle shown in the figure. The percentage efficiency of the engine is close to _____. [NA 6 Sep. 2020 (II)]



53. If minimum possible work is done by a refrigerator in converting 100 grams of water at 0°C to ice, how much heat (in calories) is released to the surroundings at temperature 27°C (Latent heat of ice = 80 Cal/gram) to the nearest integer?

[NA 3 Sep. 2020 (II)]

54. A heat engine is involved with exchange of heat of 1915 J, -40 J, +125 J and $-Q$ J, during one cycle achieving an efficiency of 50.0%. The value of Q is:

[2 Sep. 2020 (II)]

- (a) 640 J (b) 40 J (c) 980 J (d) 400 J

55. A Carnot engine having an efficiency of $\frac{1}{10}$ is being used as a refrigerator. If the work done on the refrigerator is 10 J, the amount of heat absorbed from the reservoir at lower temperature is:
 [8 Jan. 2020 II]

- (a) 99 J (b) 100 J (c) 1 J (d) 90 J

56. A Carnot engine operates between two reservoirs of temperatures 900 K and 300 K. The engine performs 1200 J of work per cycle. The heat energy (in J) delivered by the engine to the low temperature reservoir, in a cycle, is
 [NA 7 Jan. 2020 I]

57. Two ideal Carnot engines operate in cascade (all heat given up by one engine is used by the other engine to produce work) between temperatures, T_1 and T_2 . The temperature of the hot reservoir of the first engine is T_1 and the temperature of the cold reservoir of the second engine is T_2 . T is temperature of the sink of first engine which is also the source for the second engine. How is T related to T_1 and T_2 , if both the engines perform equal amount of work?

[7 Jan. 2020 II]

- (a) $T = \frac{2T_1T_2}{T_1 + T_2}$ (b) $T = \frac{T_1 + T_2}{2}$
 (c) $T = \sqrt{T_1T_2}$ (d) $T = 0$

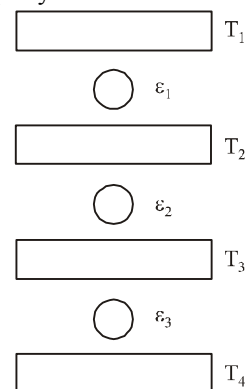
58. A Carnot engine has an efficiency of $1/6$. When the temperature of the sink is reduced by 62°C , its efficiency is doubled. The temperatures of the source and the sink are, respectively.

[12 Apr. 2019 II]

- (a) 62°C , 124°C (b) 99°C , 37°C
 (c) 124°C , 62°C (d) 37°C , 99°C

59. Three Carnot engines operate in series between a heat source at a temperature T_1 and a heat sink at temperature T_4 (see figure). There are two other reservoirs at temperature T_2 and T_3 , as shown, with $T_1 > T_2 > T_3 > T_4$. The three engines are equally efficient if:

[10 Jan. 2019 I]



- (a) $T_2 = (T_1 T_4)^{1/2}$; $T_3 = (T_1^2 T_4)^{1/3}$
 (b) $T_2 = (T_1^2 T_4)^{1/3}$; $T_3 = (T_1 T_4^2)^{1/3}$
 (c) $T_2 = (T_1 T_4^2)^{1/3}$; $T_3 = (T_1^2 T_4)^{1/3}$
 (d) $T_2 = (T_1^3 T_4)^{1/4}$; $T_3 = (T_1 T_4^3)^{1/4}$
60. Two Carnot engines A and B are operated in series. The first one, A receives heat at T_1 ($= 600$ K) and rejects to a reservoir at temperature T_2 . The second engine B receives heat rejected by the first engine and in turn, rejects to a heat reservoir at T_3 ($= 400$ K). Calculate the temperature T_2 if the work outputs of the two engines are equal:

[9 Jan. 2019 II]

- (a) 600 K (b) 400 K (c) 300 K (d) 500 K
61. A Carnot's engine works as a refrigerator between 250 K and 300 K. It receives 500 cal heat from the reservoir at the lower temperature. The amount of work done in each cycle to operate the refrigerator is: [Online April 15, 2018]

(a) 420 J (b) 2100 J (c) 772 J (d) 2520 J

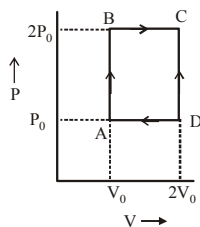
62. Two Carnot engines A and B are operated in series. Engine A receives heat from a reservoir at 600 K and rejects heat to a reservoir at temperature T . Engine B receives heat rejected by engine A and in turn rejects it to a reservoir at 100 K. If the efficiencies of the two engines A and B are represented by η_A and η_B respectively, then what is the value of $\frac{\eta_A}{\eta_B}$

[Online April 15, 2018]

- (a) $\frac{12}{7}$ (b) $\frac{12}{5}$ (c) $\frac{5}{12}$ (d) $\frac{7}{12}$

63. An engine operates by taking n moles of an ideal gas through the cycle ABCDA shown in figure. The thermal efficiency of the engine is: (Take $C_v = 1.5 R$, where R is gas constant) [Online April 8, 2017]

- (a) 0.24
 (b) 0.15
 (c) 0.32
 (d) 0.08



64. A Carnot freezer takes heat from water at 0°C inside it and rejects it to the room at a temperature of 27°C . The latent heat of ice is $336 \times 10^3 \text{ J kg}^{-1}$. If 5 kg of water at 0°C is converted into ice at 0°C by the freezer, then the energy consumed by the freezer is close to:

[Online April 10, 2016]

- (a) $1.51 \times 10^5 \text{ J}$ (b) $1.68 \times 10^6 \text{ J}$
 (c) $1.71 \times 10^7 \text{ J}$ (d) $1.67 \times 10^5 \text{ J}$

65. A solid body of constant heat capacity $1 \text{ J/}^\circ\text{C}$ is being heated by keeping it in contact with reservoirs in two ways: [2015]

- (i) Sequentially keeping in contact with 2 reservoirs such that each reservoir supplies same amount of heat.

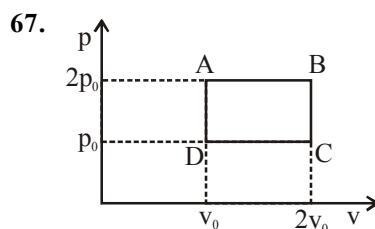
- (ii) Sequentially keeping in contact with 8 reservoirs such that each reservoir supplies same amount of heat.

In both the cases body is brought from initial temperature 100°C to final temperature 200°C . Entropy change of the body in the two cases respectively is:

- (a) $\ln 2, 2\ln 2$ (b) $2\ln 2, 8\ln 2$
 (c) $\ln 2, 4\ln 2$ (d) $\ln 2, \ln 2$

66. A Carnot engine absorbs 1000 J of heat energy from a reservoir at 127°C and rejects 600 J of heat energy during each cycle. The efficiency of engine and temperature of sink will be: [Online April 12, 2014]

- (a) 20% and -43°C (b) 40% and -33°C
 (c) 50% and -20°C (d) 70% and -10°C



The above p-v diagram represents the thermodynamic cycle of an engine, operating with an ideal monatomic gas. The amount of heat, extracted from the source in a single cycle is [2013]

- (a) $p_0 v_0$ (b) $\left(\frac{13}{2}\right) p_0 v_0$
 (c) $\left(\frac{11}{2}\right) p_0 v_0$ (d) $4p_0 v_0$

68. A Carnot engine, whose efficiency is 40%, takes in heat from a source maintained at a temperature of 500 K. It is desired to have an engine of efficiency 60%. Then, the intake temperature for the same exhaust (sink) temperature must be: [2012]

- (a) efficiency of Carnot engine cannot be made larger than 50%
 (b) 1200 K
 (c) 750 K
 (d) 600 K

69. The door of a working refrigerator is left open in a well insulated room. The temperature of air in the room will [Online May 26, 2012]

- (a) decrease
 (b) increase in winters and decrease in summers
 (c) remain the same
 (d) increase

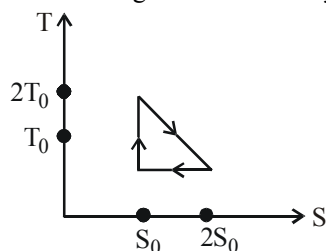
70. This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.

Statement 1: An inventor claims to have constructed an engine that has an efficiency of 30% when operated between the boiling and freezing points of water. This is not possible.

Statement 2: The efficiency of a real engine is always less than the efficiency of a Carnot engine operating between the same two temperatures.

[Online May 19, 2012]

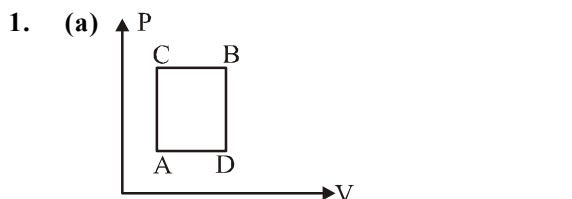
- (a) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.
 (b) Statement 1 is true, Statement 2 is false.
 (c) Statement 1 is false, Statement 2 is true.
 (d) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.
71. A Carnot engine operating between temperatures T_1 and T_2 has efficiency $\frac{1}{6}$. When T_2 is lowered by 62 K its efficiency increases to $\frac{1}{3}$. Then T_1 and T_2 are, respectively: [2011]
 (a) 372 K and 310 K (b) 330 K and 268 K
 (c) 310 K and 248 K (d) 372 K and 310 K
72. A diatomic ideal gas is used in a Carnot engine as the working substance. If during the adiabatic expansion part of the cycle the volume of the gas increases from V to $32V$, the efficiency of the engine is [2010]
 (a) 0.5 (b) 0.75 (c) 0.99 (d) 0.25
73. A Carnot engine, having an efficiency of $\eta = 1/10$ as heat engine, is used as a refrigerator. If the work done on the system is 10 J, the amount of energy absorbed from the reservoir at lower temperature is [2007]
 (a) 100 J (b) 99 J (c) 90 J (d) 1 J
74. The temperature-entropy diagram of a reversible engine cycle is given in the figure. Its efficiency is [2005]



- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{2}{3}$ (d) $\frac{1}{3}$
75. Which of the following statements is **correct** for any thermodynamic system? [2004]
 (a) The change in entropy can never be zero
 (b) Internal energy and entropy are state functions
 (c) The internal energy changes in all processes
 (d) The work done in an adiabatic process is always zero.
76. "Heat cannot by itself flow from a body at lower temperature to a body at higher temperature" is a statement or consequence of [2003]
 (a) second law of thermodynamics
 (b) conservation of momentum
 (c) conservation of mass
 (d) first law of thermodynamics
77. A Carnot engine takes 3×10^6 cal of heat from a reservoir at 627°C , and gives it to a sink at 27°C . The work done by the engine is [2003]
 (a) 4.2×10^6 J (b) 8.4×10^6 J
 (c) 16.8×10^6 J (d) zero
78. Which statement is incorrect? [2002]
 (a) All reversible cycles have same efficiency
 (b) Reversible cycle has more efficiency than an irreversible one
 (c) Carnot cycle is a reversible one
 (d) Carnot cycle has the maximum efficiency in all cycles
79. Even Carnot engine cannot give 100% efficiency because we cannot [2002]
 (a) prevent radiation
 (b) find ideal sources
 (c) reach absolute zero temperature
 (d) eliminate friction



Hints & Solutions



ΔU remains same for both paths ACB and ADB

$$\Delta Q_{ACB} = \Delta W_{ACB} + \Delta U_{ACB}$$

$$\Rightarrow 60 \text{ J} = 30 \text{ J} + \Delta U_{ACB}$$

$$\Rightarrow \Delta U_{ACB} = 30 \text{ J}$$

$$\therefore \Delta U_{ADB} = \Delta U_{ACB} = 30 \text{ J}$$

$$\Delta Q_{ADB} = \Delta U_{ADB} + \Delta W_{ADB}$$

$$= 10 \text{ J} + 30 \text{ J} = 40 \text{ J}$$

2. (d) Volume of water does not change, no work is done on or by the system ($W = 0$)

According to first law of thermodynamics

$$Q = \Delta U + W$$

For Isochoric process $Q = \Delta U$

$$\Delta U = \mu c d T = 2 \times 4184 \times 20 = 16.7 \text{ kJ}$$

3. (a) As we know, $\Delta Q = \Delta u + \Delta w$ (Ist law of thermodynamics)

$$\Rightarrow \Delta Q = \Delta u + P \Delta v$$

$$\text{or } 150 = \Delta u + 100(1 - 2)$$

$$= \Delta u - 100$$

$$\therefore \Delta u = 150 + 100 = 250 \text{ J}$$

Thus the internal energy of the gas increases by 250 J

4. (a) Here $Q = 0$ and $W = 0$. Therefore from first law of thermodynamics $\Delta U = Q + W = 0$

Internal energy of first vessel + Internal energy of second vessel = Internal energy of combined vessel

$$n_1 C_v T_1 + n_2 C_v T_2 = (n_1 + n_2) C_v T$$

$$\therefore T = \frac{n_1 T_1 + n_2 T_2}{n_1 + n_2}$$

For first vessel $n_1 = \frac{P_1 V_1}{RT_1}$ and for second vessel

$$n_2 = \frac{P_2 V_2}{RT_2}$$

$$\therefore T = \frac{\frac{P_1 V_1}{RT_1} \times T_1 + \frac{P_2 V_2}{RT_2} \times T_2}{\frac{P_1 V_1}{RT_1} + \frac{P_2 V_2}{RT_2}}$$

$$= \frac{T_1 T_2 (P_1 V_1 + P_2 V_2)}{P_1 V_1 T_2 + P_2 V_2 T_1}$$

5. (b) For path iaf ,
 $Q_1 = 50 \text{ cal}$, $W_1 = 20 \text{ cal}$
 By first law of thermodynamics,
 $\Delta U = Q_1 - W_1 = 50 - 20 = 30 \text{ cal}$.

For path ibf

$$Q_2 = 36 \text{ cal}$$

$$W_2 = ?$$

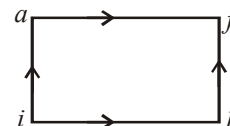
$$\Delta U_{ibf} = Q_2 - W_2$$

Since, the change in internal energy does not depend on the path, therefore $\Delta U_{iaf} = \Delta U_{ibf}$

$$\Delta U_{iaf} = \Delta U_{ibf}$$

$$\Rightarrow 30 = Q_2 - W_2$$

$$\Rightarrow W_2 = 36 - 30 = 6 \text{ cal}$$



6. (b) Change in internal energy is independent of path taken by the process. It only depends on initial and final states *i.e.*,

$$\Delta U_1 = \Delta U_2$$

7. (b, c) First law is applicable to a cyclic process. Concept of entropy is introduced by the second law of thermodynamics.

8. (b) Temperature change ΔT is same for all three processes

$A \rightarrow B$; $A \rightarrow C$ and $A \rightarrow D$

$$\Delta U = n C_v \Delta T = \text{same}$$

$$E_{AB} = E_{AC} = E_{AD}$$

Work done, $W = P \times \Delta V$

$$AB \rightarrow \text{volume is increasing} \Rightarrow W_{AB} > 0$$

$$AD \rightarrow \text{volume is decreasing} \Rightarrow W_{AD} < 0$$

$$AC \rightarrow \text{volume is constant} \Rightarrow W_{AC} = 0$$

9. (c) In adiabatic process

$$PV^\gamma = \text{constant}$$

$$\therefore P \left(\frac{m}{\rho} \right)^\gamma = \text{constant} \quad \left(\because V = \frac{m}{\rho} \right)$$

As mass is constant

$$\therefore P \propto \rho^\gamma$$

If P_i and P_f be the initial and final pressure of the gas and

ρ_i and ρ_f be the initial and final density of the gas. Then

$$\frac{P_f}{P_i} = \left(\frac{\rho_f}{\rho_i} \right)^\gamma = (32)^{7/5}$$

$$\Rightarrow \frac{n P_i}{P_i} = (2^5)^{7/5} = 2^7$$

$$\Rightarrow n = 2^7 = 128$$

10. (d)

(I) **Adiabatic process** : No exchange of heat takes place with surroundings.

$$\Rightarrow \Delta Q = 0$$

(II) **Isothermal process** : Temperature remains constant

$$\therefore \Delta T = 0 \Rightarrow \Delta U = \frac{f}{2} nR\Delta T \Rightarrow \Delta U = 0$$

No change in internal energy [$\Delta U = 0$].

(III) Isochoric process volume remains constant

$$\Delta V = 0 \Rightarrow W = \int P \cdot dV = 0$$

Hence work done is zero.

(IV) In isobaric process pressure remains constant.

$$W = P \cdot \Delta V \neq 0$$

$$\Delta U = \frac{f}{2} nR\Delta T = \frac{f}{2} [P\Delta V] \neq 0$$

$$\therefore \Delta Q = nC_p \Delta T \neq 0$$

11. (b) Bursting of helium balloon is irreversible and in this process $\Delta Q = 0$, so adiabatic.

12. (46)

For adiabatic process, $TV^{\gamma-1} = \text{constant}$

$$\text{or, } T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$T_1 = 20^\circ\text{C} + 273 = 293 \text{ K}, V_2 = \frac{V_1}{10} \text{ and } \gamma = \frac{7}{5}$$

$$T_1 (V_1)^{\gamma-1} = T_2 \left(\frac{V_1}{10}\right)^{\gamma-1}$$

$$\Rightarrow 293 = T_2 \left(\frac{1}{10}\right)^{2/5} \Rightarrow T_2 = 293(10)^{2/5} \approx 736 \text{ K}$$

$$\Delta T = 736 - 293 = 443 \text{ K}$$

During the process, change in internal energy

$$\Delta U = NC_V \Delta T = 5 \times \frac{5}{2} \times 8.3 \times 443 \approx 46 \times 10^3 \text{ J} = X \text{ kJ}$$

$$\therefore X = 46.$$

13. (c) For process 3 $\rightarrow$ 1 volume is constant

$\therefore$ Graph given in option (d) is wrong.

And process 1 $\rightarrow$ 2 is adiabatic $\therefore$ graph in option (1) is wrong

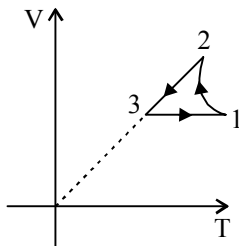
$\therefore v = \text{constant}$

$$P \uparrow, T \uparrow$$

For Process 2 $\rightarrow$ 3 Pressure constant i.e., $P = \text{constant}$

$$\therefore V \downarrow, T \downarrow$$

Hence graph (c) is the correct $V - T$ graph of given $P - V$ graph



14. (1818) For an adiabatic process,
 $TV^{\gamma-1} = \text{constant}$

$$\therefore T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$\Rightarrow T_2 = (300) \times \left(\frac{V_1}{V_2}\right)^{1.4-1}$$

$$\Rightarrow T_2 = 300 \times (16)^{0.4}$$

Ideal gas equation, $PV = nRT$

$$\therefore V = \frac{nRT}{P}$$

$\Rightarrow V = kT$ (since pressure is constant for isobaric process)

So, during isobaric process

$$V_2 = kT_2 \quad \dots(i)$$

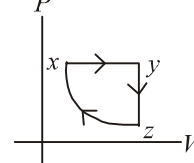
$$2V_2 = kT_f \quad \dots(ii)$$

Dividing (i) by (ii)

$$\frac{1}{2} = \frac{T_2}{T_f}$$

$$T_f = 2T_2 = 300 \times 2 \times (16)^{0.4} = 1818 \text{ K}$$

15. (a) From the corresponding V-T graph given in question,
Process $xy \rightarrow$ Isobaric expansion,
Process $yz \rightarrow$ Isochoric (Pressure decreases)
Process $zx \rightarrow$ Isothermal compression
Therefore, corresponding PV graph is as shown in figure



16. (b) Given, $V_1 = 1$ litre, $P_1 = 1$ atm
 $V_2 = 3$ litre, $\gamma = 1.40$,

$$\text{Using, } PV^\gamma = \text{constant} \Rightarrow P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$\Rightarrow P_2 = P_1 \times \left(\frac{1}{3}\right)^{1.4} = \frac{1}{4.6555} \text{ atm}$$

$$\therefore \text{Work done, } W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$$

$$= \frac{\left(1 \times 1 - \frac{1}{4.6555} \times 3\right) 1.01325 \times 10^5 \times 10^{-3}}{0.4} = 90.1 \text{ J}$$

Closest value of $W = 90.5 \text{ J}$

17. (Bonus) We know that Relaxation time,

$$T \propto \frac{V}{\sqrt{T}} \quad \dots(i)$$

Equation of adiabatic process is

$$TV^{\gamma-1} = \text{constant}$$

$$\Rightarrow T \propto \frac{1}{V^{\gamma-1}}$$

$$\Rightarrow T \propto V^{\frac{1+\gamma}{2}} \quad \text{using (i)}$$

$$\Rightarrow T \propto V^{\frac{1+\gamma}{2}}$$

$$\Rightarrow \frac{T_f}{T_i} = \left(\frac{2V}{V} \right)^{\frac{1+\gamma}{2}} = (2)^{\frac{1+\gamma}{2}}$$

18. (b) $\Delta U_{ac} = -(\Delta U_{ca}) = -(-180) = 180 \text{ J}$

$$Q = 250 + 60 = 310 \text{ J}$$

$$\text{Now } Q = \Delta U + W$$

$$\text{or } 310 = 180 + W$$

$$\text{or } W = 130 \text{ J}$$

19. (c) As the process is isochoric so,

$$Q = nc_v \Delta T = \frac{67.2}{22.4} \times \frac{3R}{2} \times 20 = 90R = 90 \times 8.31 \approx 748 \text{ J}$$

20. (a) At constant volume

$$\text{Work done (W)} = nR\Delta T$$

$$\text{Heat given } Q = C_v \Delta T + nR\Delta T$$

$$\text{So, } \therefore \frac{W}{Q} = \frac{nR\Delta T}{C_v \Delta T + nR\Delta T} = \frac{nR}{C_v + nR}$$

21. (b) We have given,

$$P = P_0 \left[1 - \frac{1}{2} \left(\frac{V_0}{V} \right)^2 \right]$$

$$\text{When } V_1 = V_0$$

$$\Rightarrow P_1 = P_0 \left[1 - \frac{1}{2} \right] = \frac{P_0}{2}$$

$$\text{When } V_2 = 2V_0$$

$$\Rightarrow P_2 = P_0 \left[1 - \frac{1}{2} \left(\frac{1}{4} \right) \right] = \left(\frac{7P_0}{8} \right)$$

$$\Delta T = T_2 - T_1 = \left| \frac{P_1 V_1}{nR} - \frac{P_2 V_2}{nR} \right| \left[\because T = \frac{PV}{nR} \right]$$

$$\Delta T = \left| \left(\frac{1}{nR} \right) (P_1 V_1 - P_2 V_2) \right| = \left(\frac{1}{nR} \right) \left| \left(\frac{P_0 V_0}{2} - \frac{7P_0 V_0}{4} \right) \right|$$

$$= \frac{5P_0 V_0}{4nR} = \frac{5P_0 V_0}{4R} \quad (\because n=1)$$

22. (c) Internal energy depends only on initial and final state

$$\text{So, } \Delta U_A = \Delta U_B$$

$$\text{Also } \Delta Q = \Delta U + W$$

$$\text{As } W_A > W_B \Rightarrow \Delta Q_A > \Delta Q_B$$

23. (b) Suppose amount of water evaporated be M gram.

Then $(150 - M)$ gram water converted into ice.

so, heat consumed in evaporation = Heat released in fusion

$$M \times L_v = (150 - M) \times L_s$$

$$M \times 2.1 \times 10^6 = (150 - M) \times 3.36 \times 10^5$$

$$\Rightarrow M = 20 \text{ g}$$

24. (d) a $\rightarrow$ Isobaric, b $\rightarrow$ Isothermal, c $\rightarrow$ Adiabatic, d $\rightarrow$ Isochoric

25. (b) Total work done by the gas during the cycle is equal to area of triangle ABC.

$$\therefore \Delta W = \frac{1}{2} \times 4 \times 5 = 10 \text{ J}$$

26. (b) Equation of adiabatic change is $TV^{\gamma-1} = \text{constant}$

$$\text{Put } \gamma = \frac{7}{5}, \text{ we get: } \gamma - 1 = \frac{7}{5} - 1$$

$$\therefore x = \frac{2}{5}$$

27. (b) Work done,

$$W = P\Delta V = nR\Delta T = \frac{1}{2} \times 8.31 \times 70 \approx 291 \text{ J}$$

28. (a) Equation of the BC

$$P = P_0 - \frac{2P_0}{V_0} (V - 2V_0)$$

$$\text{using } PV = nRT$$

$$\text{Temperature, } T = \frac{P_0 V - \frac{2P_0 V^2}{V_0} + 4P_0 V}{1 \times R}$$

$$(\because n = 1 \text{ mole given})$$

$$T = \frac{P_0}{R} \left[5V - \frac{2V^2}{V_0} \right]$$

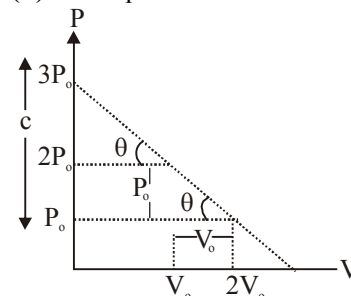
$$\frac{dT}{dV} = 0 \Rightarrow 5 - \frac{4V}{V_0} = 0 \Rightarrow V = \frac{5}{4} V_0$$

$$T = \frac{P_0}{R} \left[5 \times \frac{5V_0}{4} - \frac{2}{V_0} \times \frac{25}{16} V_0^2 \right] = \frac{25}{8} \frac{P_0 V_0}{R}$$

29. (d) Work done on gas = $nRT \ln \left(\frac{P_f}{P_i} \right) = R(300) \ln(2)$

$$= 300 R \ln 2 \quad \left(\because \frac{P_f}{P_i} = 2 \text{ given} \right)$$

30. (c) The equation for the line is



$$P = \frac{-P_0}{V_0} V + 3P_0 \quad [\text{slope} = \frac{-P_0}{V_0}, c = 3P_0]$$

$$PV_0 + P_0 V = 3P_0 V_0$$

$$\text{But } pV = nRT$$

$$\therefore P = \frac{nRT}{V}$$

$$\text{From (i) \& (ii) } \frac{nRT}{V} V_0 + P_0 V = 3P_0 V_0$$

...(i)

...(ii)

From (i) & (ii) $\frac{nRT}{V} V_0 + P_0 V = 3P_0 V_0$
 $\therefore nRT V_0 + P_0 V^2 = 3P_0 V_0 V \quad \dots(iii)$

For temperature to be maximum $\frac{dT}{dV} = 0$

Differentiating e.q. (iii) by 'V' we get

$$nR V_0 \frac{dT}{dV} + P_0 (2V) = 3P_0 V_0$$

$$\therefore nR V_0 \frac{dT}{dV} = 3P_0 V_0 - 2P_0 V$$

$$\frac{dT}{dV} = \frac{3P_0 V_0 - 2P_0 V}{nR V_0} = 0$$

$$V = \frac{3V_0}{2} \quad \therefore P = \frac{3P_0}{2} \quad [\text{From (i)}]$$

$$\therefore T_{\max} = \frac{9P_0 V_0}{4nR} \quad [\text{From (iii)}]$$

8. (a) Efficiency of heat engine is given by

$$\eta = \frac{W}{Q} = 1 - \frac{C_V}{C_P} = \frac{R}{C_P} = \frac{R}{\frac{5R}{2}} = \frac{2}{5}$$

$$(\because C_P - C_V = R)$$

For monoatomic gas $C_P = \frac{5}{2} R$.

9. (a) $\tau = \frac{1}{\sqrt{2\pi d^2 \left(\frac{N}{V}\right) \sqrt{\frac{3RT}{M}}}}$

$$t \propto \frac{V}{\sqrt{T}}$$

$$\text{As, } TV^{\gamma-1} = K$$

$$\text{So, } \tau \propto V^{\gamma+1/2}$$

Therefore, $q = \frac{g+1}{2}$

10. (a) As, $P = \frac{1}{3} \left(\frac{U}{V} \right)$

But $\frac{U}{V} = KT^4$

So, $P = \frac{1}{3} KT^4$

or $\frac{uRT}{V} = \frac{1}{3} KT^4$ [As $PV = uRT$]

$$\frac{4}{3} pR^3 T^3 = \text{constant}$$

Therefore, $T \propto \frac{1}{R}$

11. (b) In VT graph

ab-process : Isobaric, temperature increases.

bc process : Adiabatic, pressure decreases.

cd process : Isobaric, volume decreases.

da process : Adiabatic, pressure increases.

The above processes correctly represented in P - V diagram (b).

12. (d) In cyclic process, change in total internal energy is zero.

$$\Delta U_{\text{cyclic}} = 0$$

$$\Delta U_{BC} = nC_V \Delta T = 1 \times \frac{5R}{2} \Delta T$$

Where, C_V = molar specific heat at constant volume.

For BC, $\Delta T = -200$ K

$$\therefore \Delta U_{BC} = -500R$$

13. (c)

14. (c) Given : work done, $W = 830$ J

No. of moles of gas, $\mu = 2$

For diatomic gas $\gamma = 1.4$

Work done during an adiabatic change

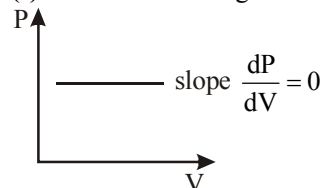
$$W = \frac{\mu R(T_1 - T_2)}{\gamma - 1}$$

$$\Rightarrow 830 = \frac{2 \times 8.3(\Delta T)}{1.4 - 1} = \frac{2 \times 8.3(\Delta T)}{0.4}$$

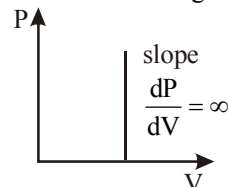
$$\Rightarrow \Delta T = \frac{830 \times 0.4}{2 \times 8.3} = 20 \text{ K}$$

15. (a)

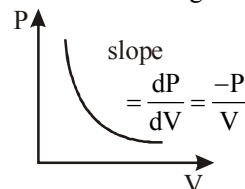
16. (c) P - V indicator diagram for isobaric



P - V indicator diagram for isochoric process



P - V indicator diagram for isothermal process



17. (b) Volume of the gas

$$v = \frac{m}{d} \text{ and}$$

Using $PV^\gamma = \text{constant}$

$$\frac{P'}{P} = \frac{V}{V'} = \left(\frac{d'}{d} \right)^\gamma$$

$$\text{or } 128 = (32)^\gamma$$

$$\therefore \gamma = \frac{7}{5} = 1.4$$

18. (a) The efficiency

$$\eta = \frac{\text{output work}}{\text{heat given to the system}}$$

$$= n \frac{3}{2} R \Delta T = \frac{3}{2} V_0 \Delta P = \frac{3}{2} P_0 V_0$$

$$W_i = \frac{n}{2} (P_0 V_0) + \frac{n}{2} (2P_0 V_0) + 2P_0 V_0$$

Heat given in going B to C = $nC_p \Delta T$

$$= n \left(\frac{5}{2} R \right) \Delta T = \frac{5}{2} (2P_0) \Delta V$$

$$= 5P_0 V_0$$

and W_0 = area under PV diagram $P_0 V_0$

$$\eta = \frac{W}{Q} = \frac{P_0 V_0}{\frac{13}{2} P_0 V_0} = \frac{2}{13}$$

Efficiency in %

$$\eta = \frac{2}{13} \times 100 = \frac{200}{13} \approx 15.4\%$$

42. (d) For isothermal process :

$$PV = P_i \cdot 2V$$

$$P = 2P_i \quad \dots(i)$$

For adiabatic process

$$PV^\gamma = P_a (2V)^\gamma$$

($\because$ for monatomic gas $\gamma = 5/3$)

$$\text{or, } 2P_i V^{\frac{5}{3}} = P_a (2V)^{\frac{5}{3}} \quad [\text{From (i)}]$$

$$\Rightarrow \frac{P_a}{P_i} = \frac{2}{\frac{5}{2^{\frac{2}{3}}}}$$

$$\Rightarrow \frac{P_a}{P_i} = 2^{\frac{2}{3}}$$

43. (d) Given $P = \alpha V$

$$\text{Work done, } w = \int_V^{mV} P dV$$

$$= \int_V^{mV} \alpha V dV = \frac{\alpha V^2}{2} (m^2 - 1)$$

44. (b) Work done during the process $A \rightarrow B$

= Area of trapezium (= area bounded by indicator diagram with V-axis)

$$= \frac{1}{2} (2P_0 + P_0) (2V_0 - V_0) = \frac{3}{2} P_0 V_0$$

Ideal gas eqn : $PV = nRT$

$$\Rightarrow T = \frac{PV}{nR} = \frac{3P_0 V_0}{2nR}$$

45. (b) In an adiabatic process, $\delta H = 0$

And according to first law of thermodynamics

$$\delta H = \delta U + W$$

$$\therefore W = -\delta U$$



It is the free expansion

$\therefore$ So, T remains constant

$$\Rightarrow P_1 V_1 = P_2 V_2$$

$$\Rightarrow P \frac{V}{2} = P_2 (V)$$

$$P_2 = \left(\frac{P}{2} \right)$$

47. (b) The process $A \rightarrow B$ is isobaric.

$\therefore$ work done $W_{AB} = nR(T_2 - T_1)$

$$= 2R(500 - 300) = 400R$$

48. (a) The process $D \rightarrow A$ is isothermal as temperature is constant.

$$\text{Work done, } W_{DA} = 2.303nRT \log_{10} \frac{P_D}{P_A}$$

$$= 2.303 \times 2 R \times 300$$

$$\log_{10} \frac{1 \times 10^5}{2 \times 10^5} = -414R$$

Therefore, work done on the gas is +414 R.

49. (a) The net work in the cycle ABCDA is

$$W = W_{AB} + W_{BC} + W_{CD} + W_{DA}$$

$$= 400R + 2.303nRT \log \frac{P_B}{P_C} + (-400R) - 414R$$

$$= 2.303 \times 2R \times 500 \log \frac{2 \times 10^5}{1 \times 10^5} - 414R$$

$$= 693.2R - 414R = 279.2R$$

50. (a) Work done in adiabatic compression is given by

$$W = \frac{nR\Delta T}{1-\gamma}$$

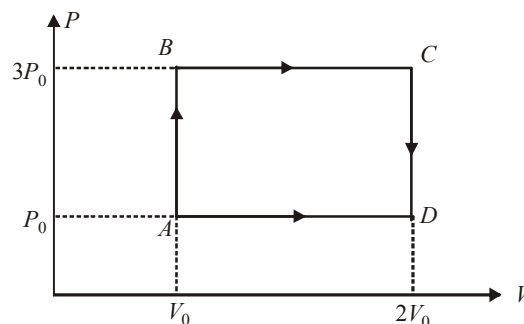
$$\Rightarrow -146000 = \frac{1000 \times 8.3 \times 7}{1-\gamma}$$

$$\text{or } 1-\gamma = -\frac{58.1}{146} \Rightarrow \gamma = 1 + \frac{58.1}{146} = 1.4$$

Hence the gas is diatomic.

51. (c) Work is not a state function. The remaining three parameters are state function.

52. (19)



From the figure,

$$\text{Work, } W = 2P_0V_0$$

$$\text{Heat given, } Q_{\text{in}} = W_{AB} + W_{BC} = n \cdot C_V \Delta T_{AB} + n C_P \Delta T_{BC}$$

$$= n \frac{3R}{2} (T_B - T_A) + \frac{n5R}{2} (T_C - T_B)$$

$$\left(\because C_V = \frac{3R}{2} \text{ and } C_P = \frac{5R}{2} \right)$$

$$= \frac{3}{2} (P_B V_B - P_A V_A) + \frac{5}{2} (P_C V_C - P_B V_B)$$

$$= \frac{3}{2} \times [3P_0V_0 - P_0V_0] + \frac{5}{2} [6P_0V_0 - 3P_0V_0]$$

$$= 3P_0V_0 + \frac{15}{2} P_0V_0 = \frac{21}{2} P_0V_0$$

$$\text{Efficiency, } \eta = \frac{W}{Q_{\text{in}}} = \frac{2P_0V_0}{\frac{21}{2} P_0V_0} = \frac{4}{21}$$

$$\eta\% = \frac{400}{21} \approx 19.$$

53. (8791)

Given,

$$\text{Heat absorbed, } Q_2 = mL = 80 \times 100 = 8000 \text{ Cal}$$

$$\text{Temperature of ice, } T_2 = 273 \text{ K}$$

$$\text{Temperature of surrounding,}$$

$$T_1 = 273 + 27 = 300 \text{ K}$$

$$\text{Efficiency} = \frac{w}{Q_2} = \frac{Q_1 - Q_2}{Q_2} = \frac{T_1 - T_2}{T_2} = \frac{300 - 273}{273}$$

$$\Rightarrow \frac{Q_1 - 8000}{8000} = \frac{27}{273} \Rightarrow Q_1 = 8791 \text{ Cal}$$

54. (c) Efficiency, $\eta = \frac{\text{Work done}}{\text{Heat absorbed}} = \frac{W}{\Sigma Q}$

$$= \frac{Q_1 + Q_2 + Q_3 + Q_4}{Q_1 + Q_3} = 0.5$$

$$\text{Here, } Q_1 = 1915 \text{ J, } Q_2 = -40 \text{ J and } Q_3 = 125 \text{ J}$$

$$\therefore \frac{1915 - 40 + 125 + Q_4}{1915 + 125} = 0.5$$

$$\Rightarrow 1915 - 40 + 125 + Q_4 = 1020$$

$$\Rightarrow Q_4 = 1020 - 2000$$

$$\Rightarrow Q_4 = -Q = -980 \text{ J}$$

$$\Rightarrow Q = 980 \text{ J}$$

55. (d) For carnot refrigerator

$$\text{Efficiency} = \frac{Q_1 - Q_2}{Q_1}$$

Where,

Q_1 = heat lost from surrounding

Q_2 = heat absorbed from reservoir at low temperature.

$$\text{Also, } \frac{Q_1 - Q_2}{Q_1} = \frac{w}{Q_1}$$

$$\Rightarrow \frac{1}{10} = \frac{w}{Q_1}$$

$$\Rightarrow Q_1 = w \times 10 = 100 \text{ J}$$

$$\text{So, } Q_1 - Q_2 = w$$

$$\Rightarrow Q_2 = Q_1 - w$$

$$\Rightarrow 100 - 10 = Q_2 = 90 \text{ J}$$

56. (600.00) Given; $T_1 = 900 \text{ K, } T_2 = 300 \text{ K, } W = 1200 \text{ J}$

$$\text{Using, } 1 - \frac{T_2}{T_1} = \frac{W}{Q_1}$$

$$\Rightarrow 1 - \frac{300}{900} = \frac{1200}{Q_1}$$

$$\Rightarrow \frac{2}{3} = \frac{1200}{Q_1} \Rightarrow Q_1 = 1800$$

Therefore heat energy delivered by the engine to the low temperature reservoir, $Q_2 = Q_1 - W = 1800 - 1200 = 600.00 \text{ J}$

57. (b) Let Q_H = Heat taken by first engine

Q_L = Heat rejected by first engine

Q_2 = Heat rejected by second engine

Work done by 1st engine = work done by 2nd engine

$$W = Q_H - Q_L = Q_L - Q_2 \Rightarrow 2Q_L = Q_H + Q_2$$

$$2 = \frac{\theta_H}{\theta_L} + \frac{\theta_2}{\theta_L}$$

Let T be the temperature of cold reservoir of first engine. Then in carnot engine.

$$\frac{Q_H}{Q_L} = \frac{T_1}{T} \text{ and } \frac{Q_L}{Q_2} = \frac{T}{T_2}$$

$$\Rightarrow 2 = \frac{T_1}{T} + \frac{T_2}{T} \quad \text{using (i)}$$

$$\Rightarrow 2T = T_1 + T_2 \Rightarrow T = \frac{T_1 + T_2}{2}$$

58. (b) Using, $n = 1 - \frac{T_2}{T_1}$

$$n = \frac{1}{6} = 1 - \frac{T_2}{T_1}$$

$$\text{and } \frac{T}{3} = 1 - \frac{T_2 - 62}{T_1}$$

On solving, we get

$$T_1 = 99^\circ\text{C and } T_2 = 37^\circ\text{C}$$

59. (b) According to question, $\eta_1 = \eta_2 = \eta_3$

$$\therefore 1 - \frac{T_2}{T_1} = 1 - \frac{T_3}{T_2} = 1 - \frac{T_4}{T_3}$$

[$\because$ Three engines are equally efficient]

$$\Rightarrow \frac{T_2}{T_1} = \frac{T_3}{T_2} = \frac{T_4}{T_3}$$

$$\Rightarrow T_2 = \sqrt{T_1 T_3} \quad \dots(i)$$

$$T_3 = \sqrt{T_2 T_4} \quad \dots(ii)$$

From (i) and (ii)

$$T_2 = (T_1^2 T_4)^{1/3}$$

$$T_3 = (T_1 T_4^2)^{1/3}$$

$$60. (d) \eta_A = \frac{T_1 - T_2}{T_1} = \frac{W_A}{Q_1}$$

$$\text{and, } \eta_B = \frac{T_2 - T_3}{T_2} = \frac{W_B}{Q_2}$$

According to question,

$$W_A = W_B$$

$$\therefore \frac{Q_1}{Q_2} = \frac{T_1}{T_2} \times \frac{T_2 - T_3}{T_1 - T_2} = \frac{T_1}{T_2}$$

$$\therefore T_2 = \frac{T_1 + T_3}{2}$$

$$= \frac{600 + 400}{2}$$

$$= 500K$$

61. (a) Given: Temperature of cold body, $T_2 = 250$ K
temperature of hot body, $T_1 = 300$ K
Heat received, $Q_2 = 500$ cal work done, $W = ?$

$$\text{Efficiency} = 1 - \frac{T_2}{T_1} = \frac{W}{Q_2 + W} \Rightarrow 1 - \frac{250}{300} = \frac{W}{Q_2 + W}$$

$$W = \frac{Q_2}{5} = \frac{500 \times 4.2}{5} J = 420 J$$

62. (d) Efficiency of engine A, $n_A = \frac{T_1 - T_2}{T_1}$

$$\text{and } n_B = \frac{T_2 - T_3}{T_2}; T_2 = \frac{T_1 + T_3}{2} = 350 K$$

$$\text{or } \frac{n_A}{n_B} = \frac{\frac{600 - 350}{350 - 100}}{\frac{600}{350}} = \frac{7}{12}$$

63. (b) Work-done (W) = $P_0 V_0$

According to principle of calorimetry

$$\text{Heat given} = Q_{AB} = Q_{BC}$$

$$= nC_V dT_{AB} + nC_P dT_{BC}$$

$$= \frac{3}{2}(nRT_B - nRT_A) + \frac{5}{2}(nRT_C - nRT_B)$$

$$= \frac{3}{2}(2P_0 V_0 - P_0 V_0) + \frac{5}{2}(4P_0 V_0 - 2P_0 V_0)$$

$$= \frac{13}{2} P_0 V_0$$

$$\text{Thermal efficiency of engine } (\eta) = \frac{W}{Q_{\text{given}}} = \frac{2}{13} = 0.15$$

64. (d) $\Delta H = mL = 5 \times 336 \times 10^3 = Q_{\text{sink}}$

$$\frac{Q_{\text{sink}}}{Q_{\text{source}}} = \frac{T_{\text{sink}}}{T_{\text{source}}}$$

$$\therefore Q_{\text{source}} = \frac{T_{\text{source}}}{T_{\text{sink}}} \times Q_{\text{sink}}$$

Energy consumed by freezer

$$\therefore W_{\text{output}} = Q_{\text{source}} - Q_{\text{sink}} = Q_{\text{sink}} \left(\frac{T_{\text{source}}}{T_{\text{sink}}} - 1 \right)$$

$$\text{Given: } T_{\text{source}} = 27^\circ\text{C} + 273 = 300K,$$

$$T_{\text{sink}} = 0^\circ\text{C} + 273 = 273 K$$

$$W_{\text{output}} = 5 \times 336 \times 10^3 \left(\frac{300}{273} - 1 \right) = 1.67 \times 10^5 J$$

65. (d) The entropy change of the body in the two cases is same as entropy is a state function.

66. (b) Given: $Q_1 = 1000 J$

$$Q_2 = 600 J$$

$$T_1 = 127^\circ\text{C} = 400 K$$

$$T_2 = ?$$

$$\eta = ?$$

Efficiency of carnot engine,

$$\eta = \frac{W}{Q_1} \times 100\%$$

$$\text{or, } \eta = \frac{Q_2 - Q_1}{Q_1} \times 100\%$$

$$\text{or, } \eta = \frac{1000 - 600}{1000} \times 100\%$$

$$\eta = 40\%$$

$$\text{Now, for carnot cycle } \frac{Q_2}{Q_1} = \frac{T_2}{T_1}$$

$$\frac{600}{1000} = \frac{T_2}{400}$$

$$T_2 = \frac{600 \times 400}{1000}$$

$$= 240 K$$

$$= 240 - 273$$

$$\therefore T_2 = -33^\circ\text{C}$$

67. (b) Heat is extracted from the source in path DA and AB is

$$\Delta Q = \frac{3}{2} R \left(\frac{P_0 V_0}{R} \right) + \frac{5}{2} R \left(\frac{2P_0 V_0}{R} \right)$$

$$\Rightarrow \frac{3}{2} P_0 V_0 + \frac{5}{2} 2P_0 V_0 = \left(\frac{13}{2} \right) P_0 V_0$$

68. (c) The efficiency of the Carnot's heat engine is given as

$$\eta = \left(1 - \frac{T_2}{T_1}\right) \times 100$$

When efficiency is 40%,

$$T_1 = 500 \text{ K}; \eta = 40$$

$$40 = \left(1 - \frac{T_2}{500}\right) \times 100$$

$$\Rightarrow \frac{40}{100} = 1 - \frac{T_2}{500}$$

$$\Rightarrow \frac{T_2}{500} = \frac{60}{100} \Rightarrow T_2 = 300 \text{ K}$$

When efficiency is 60%, then

$$\frac{60}{100} = \left(1 - \frac{300}{T_2}\right) \Rightarrow \frac{300}{T_2} = \frac{40}{100}$$

$$\Rightarrow T_2 = \frac{100 \times 300}{40} \Rightarrow T_2 = 750 \text{ K}$$

69. (d) In a refrigerator, the heat dissipated in the atmosphere is more than that taken from the cooling chamber, therefore the room is heated. If the door of a refrigerator is kept open.
70. (d) According to Carnot's theorem - no heat engine working between two given temperatures of source and sink can be more efficient than a perfectly reversible engine i.e. Carnot engine working between the same two temperatures.

$$\text{Efficiency of Carnot's engine, } \eta = 1 - \frac{T_2}{T_1}$$

where, T_1 = temperature of source

T_2 = temperature of sink

71. (d) Efficiency of engine

$$\eta_1 = 1 - \frac{T_2}{T_1} = \frac{1}{6}$$

$$\Rightarrow \frac{T_2}{T_1} = \frac{5}{6} \quad \dots(i)$$

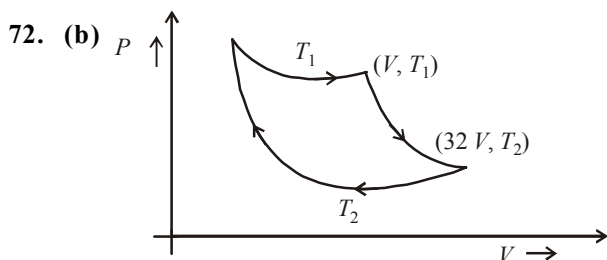
When T_2 is lowered by 62K, then

$$\text{Again, } \eta_2 = 1 - \frac{T_2 - 62}{T_1}$$

$$= 1 - \frac{T_2}{T_1} + \frac{62}{T_1} = \frac{1}{3} \quad \dots(ii)$$

Solving (i) and (ii), we get,

$$T_1 = 372 \text{ K and } T_2 = \frac{5}{6} \times 372 = 310 \text{ K}$$



$$\text{For adiabatic expansion } T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$\Rightarrow T_1 V^{g-1} = T_2 (32V)^{g-1}$$

$$\Rightarrow \frac{T_1}{T_2} = (32)^{\gamma-1}$$

$$\text{For diatomic gas, } \gamma = \frac{7}{5}$$

$$\therefore \gamma - 1 = \frac{2}{5}$$

$$\therefore \frac{T_1}{T_2} = (32)^{\frac{2}{5}} \Rightarrow T_1 = 4T_2$$

$$\text{Now, efficiency} = 1 - \frac{T_2}{T_1}$$

$$= 1 - \frac{T_2}{4T_2} = 1 - \frac{1}{4} = \frac{3}{4} = 0.75.$$

73. (c) The efficiency (η) of a Carnot engine and the coefficient of performance (β) of a refrigerator are related as

$$\beta = \frac{1 - \eta}{\eta}$$

$$\text{Also, } \beta = \frac{Q_2}{W}$$

$$\therefore \beta = \frac{1 - \eta}{\eta} = \frac{Q_2}{W}$$

$$\therefore \beta = \frac{1 - \frac{1}{10}}{\left(\frac{1}{10}\right)} = \frac{Q_2}{W}$$

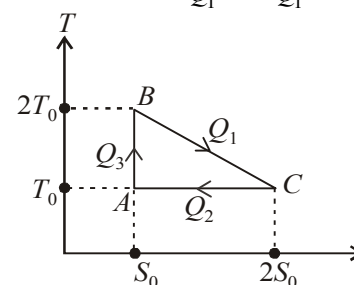
is independent of path taken by the process.

$$\Rightarrow 9 = \frac{Q_2}{10}$$

$$\Rightarrow Q_2 = 90 \text{ J.}$$

74. (d) Q_1 = area under BC = $T_0 S_0 + \frac{1}{2} T_0 S_0$
 Q_2 = area under AC = $T_0 (2S_0 - S_0) = T_0 S_0$
 and $Q_3 = 0$

$$\text{Efficiency, } \eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1}$$



$$= 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_0 S_0}{\frac{3}{2} T_0 S_0} = \frac{1}{3}$$

75. (b) Internal energy and entropy are state function, they are independent of path taken.
76. (a) This is a consequence of second law of thermodynamics
77. (b) Here, $T_1 = 627 + 273 = 900$ K
 $T_2 = 27 + 273 = 300$ K

$$\text{Efficiency, } \eta = 1 - \frac{T_2}{T_1}$$

$$= 1 - \frac{300}{900} = 1 - \frac{1}{3} = \frac{2}{3}$$

$$\text{But } \eta = \frac{W}{Q}$$

$$\therefore \frac{W}{Q} = \frac{2}{3} \Rightarrow W = \frac{2}{3} \times Q = \frac{2}{3} \times 3 \times 10^6$$

$$= 2 \times 10^6 \text{ cal}$$

$$= 2 \times 10^6 \times 4.2 \text{ J} = 8.4 \times 10^6 \text{ J}$$

78. (a) All reversible engines have same efficiencies if they are working for the same temperature of source and sink. If the temperatures are different, the efficiency is different.
79. (c) In Carnot's cycle we assume frictionless piston, absolute insulation and ideal source and sink (reservoirs).

$$\text{The efficiency of carnot's cycle } \eta = 1 - \frac{T_2}{T_1}$$

The efficiency of carnot engine will be 100% when its sink (T_2) is at 0 K.

The temperature of 0 K (absolute zero) cannot be realised in practice so, efficiency is never 100%.

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Kinetic Theory



TOPIC 1 Kinetic Theory of an Ideal Gas and Gas Laws



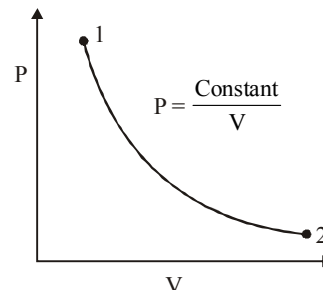
- Initially a gas of diatomic molecules is contained in a cylinder of volume V_1 at a pressure P_1 and temperature 250 K. Assuming that 25% of the molecules get dissociated causing a change in number of moles. The pressure of the resulting gas at temperature 2000 K, when contained in a volume $2V_1$ is given by P_2 . The ratio P_2/P_1 is _____. [NA Sep. 06, 2020 (I)]
- The change in the magnitude of the volume of an ideal gas when a small additional pressure ΔP is applied at a constant temperature, is the same as the change when the temperature is reduced by a small quantity ΔT at constant pressure. The initial temperature and pressure of the gas were 300 K and 2 atm. respectively. If $|\Delta T| = C |\Delta P|$, then value of C in (K/atm.) is _____. [NA Sep. 04, 2020 (II)]
- The number density of molecules of a gas depends on their distance r from the origin as, $n(r) = n_0 e^{-ar^4}$. Then the total number of molecules is proportional to : [12 April 2019 II]
 - $n_0 \alpha^{-3/4}$
 - $\sqrt{n_0} \alpha^{1/2}$
 - $n_0 \alpha^{1/4}$
 - $n_0 \alpha^{-3}$
- A vertical closed cylinder is separated into two parts by a frictionless piston of mass m and of negligible thickness. The piston is free to move along the length of the cylinder. The length of the cylinder above the piston is l_1 , and that below the piston is l_2 , such that $l_1 > l_2$. Each part of the cylinder contains n moles of an ideal gas at equal temperature T . If the piston is stationary, its mass, m , will be given by: (R is universal gas constant and g is the acceleration due to gravity) [12 Jan. 2019 II]

$$(a) \frac{RT}{ng} \left[\frac{l_1 - 3l_2}{l_1 l_2} \right]$$

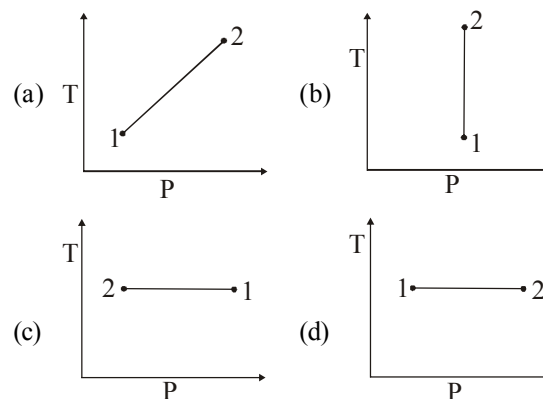
$$(b) \frac{RT}{g} \left[\frac{2l_1 + l_2}{l_1 l_2} \right]$$

$$(c) \frac{nRT}{g} \left[\frac{1}{l_2} + \frac{1}{l_1} \right] \quad (d) \frac{nRT}{g} \left[\frac{l_1 - l_2}{l_1 l_2} \right]$$

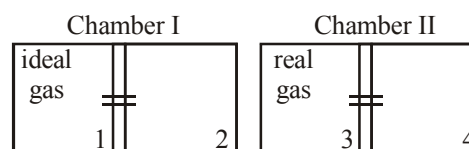
- The temperature of an open room of volume 30 m^3 increases from 17°C to 27°C due to sunshine. The atmospheric pressure in the room remains $1 \times 10^5 \text{ Pa}$. If n_i and n_f are the number of molecules in the room before and after heating, then $n_f - n_i$ will be : [2017]
 - 2.5×10^{25}
 - -2.5×10^{25}
 - -1.61×10^{23}
 - 1.38×10^{23}
- For the P-V diagram given for an ideal gas,



out of the following which one correctly represents the T-P diagram ? [Online April 9, 2017]



7.



There are two identical chambers, completely thermally insulated from surroundings. Both chambers have a partition wall dividing the chambers in two compartments. Compartment 1 is filled with an ideal gas and Compartment 3 is filled with a real gas. Compartments 2 and 4 are vacuum. A small hole (orifice) is made in the partition walls and the gases are allowed to expand in vacuum.

Statement-1: No change in the temperature of the gas takes place when ideal gas expands in vacuum. However, the temperature of real gas goes down (cooling) when it expands in vacuum.

Statement-2: The internal energy of an ideal gas is only kinetic. The internal energy of a real gas is kinetic as well as potential. [Online April 9, 2013]

- (a) Statement-1 is false and Statement-2 is true.
 (b) Statement-1 and Statement-2 both are true. Statement-2 is the correct explanation of Statement-1.
 (c) Statement-1 is true and Statement-2 is false.
 (d) Statement-1 and Statement-2 both are true. Statement-2 is **not** correct explanation of Statement-1.
8. Cooking gas containers are kept in a lorry moving with uniform speed. The temperature of the gas molecules inside will [2002]
 (a) increase
 (b) decrease
 (c) remain same
 (d) decrease for some, while increase for others

TOPIC

2 Speed of Gas, Pressure and Kinetic Energy



9. Number of molecules in a volume of 4 cm^3 of a perfect monoatomic gas at some temperature T and at a pressure of 2 cm of mercury is close to? (Given, mean kinetic energy of a molecule (at T) is $4 \times 10^{-14} \text{ erg}$, $g = 980 \text{ cm/s}^2$, density of mercury = 13.6 g/cm^3) [Sep. 05, 2020 (I)]
 (a) 4.0×10^{18} (b) 4.0×10^{16}
 (c) 5.8×10^{16} (d) 5.8×10^{18}
10. Nitrogen gas is at 300°C temperature. The temperature (in K) at which the rms speed of a H_2 molecule would be equal to the rms speed of a nitrogen molecule, is _____. (Molar mass of N_2 gas 28 g); [NA Sep. 05, 2020 (II)]
11. For a given gas at 1 atm pressure, rms speed of the molecules is 200 m/s at 127°C . At 2 atm pressure and at 227°C , the rms speed of the molecules will be: [9 April 2019 I]
 (a) 100 m/s (b) $80\sqrt{5}$ m/s
 (c) $100\sqrt{5}$ m/s (d) 80 m/s
12. If 10^{22} gas molecules each of mass 10^{-26} kg collide with a surface (perpendicular to it) elastically per second over an area 1 m^2 with a speed 10^4 m/s , the pressure exerted by the gas molecules will be of the order of: [8 April 2019 I]

- (a) 10^4 N/m^2 (b) 10^8 N/m^2
 (c) 10^3 N/m^2 (d) 10^{16} N/m^2

13. The temperature, at which the root mean square velocity of hydrogen molecules equals their escape velocity from the earth, is closest to: [8 April 2019 II]

[Boltzmann Constant $k_B = 1.38 \times 10^{-23} \text{ J/K}$

Avogadro Number $N_A = 6.02 \times 10^{26} / \text{kg}$

Radius of Earth : $6.4 \times 10^6 \text{ m}$

Gravitational acceleration on Earth = 10 ms^{-2}

- (a) 800 K (b) $3 \times 10^5 \text{ K}$
 (c) 10^4 K (d) 650 K
14. A mixture of 2 moles of helium gas (atomic mass = $4u$), and 1 mole of argon gas (atomic mass = $40u$) is kept at 300 K in a container. The ratio of their rms speeds

$\left[\frac{V_{\text{rms}}(\text{helium})}{V_{\text{rms}}(\text{argon})} \right]$ is close to: [9 Jan. 2019 I]

- (a) 3.16 (b) 0.32
 (c) 0.45 (d) 2.24
15. N moles of a diatomic gas in a cylinder are at a temperature T . Heat is supplied to the cylinder such that the temperature remains constant but n moles of the diatomic gas get converted into monoatomic gas. What is the change in the total kinetic energy of the gas? [Online April 9, 2017]

- (a) $\frac{1}{2}nRT$ (b) 0
 (c) $\frac{3}{2}nRT$ (d) $\frac{5}{2}nRT$
16. In an ideal gas at temperature T , the average force that a molecule applies on the walls of a closed container depends on T as T^q . A good estimate for q is: [Online April 10, 2015]

- (a) $\frac{1}{2}$ (b) 2
 (c) 1 (d) $\frac{1}{4}$
17. A gas molecule of mass M at the surface of the Earth has kinetic energy equivalent to 0°C . If it were to go up straight without colliding with any other molecules, how high it would rise? Assume that the height attained is much less than radius of the earth. (k_B is Boltzmann constant). [Online April 19, 2014]

- (a) 0 (b) $\frac{273k_B}{2Mg}$
 (c) $\frac{546k_B}{3Mg}$ (d) $\frac{819k_B}{2Mg}$
18. At room temperature a diatomic gas is found to have an r.m.s. speed of 1930 ms^{-1} . The gas is: [Online April 12, 2014]

- (a) H_2 (b) Cl_2
 (c) O_2 (d) F_2

19. In the isothermal expansion of 10g of gas from volume V to $2V$ the work done by the gas is 575J. What is the root mean square speed of the molecules of the gas at that temperature? [Online April 25, 2013]

(a) 398m/s (b) 520m/s
(c) 499m/s (d) 532m/s

20. A perfect gas at 27°C is heated at constant pressure so as to double its volume. The final temperature of the gas will be, close to [Online May 7, 2012]

(a) 327°C (b) 200°C
(c) 54°C (d) 300°C

21. A thermally insulated vessel contains an ideal gas of molecular mass M and ratio of specific heats γ . It is moving with speed v and it's suddenly brought to rest. Assuming no heat is lost to the surroundings, its temperature increases by: [2011]

(a) $\frac{(\gamma-1)}{2\gamma R} Mv^2 K$ (b) $\frac{\gamma M^2 v}{2R} K$
(c) $\frac{(\gamma-1)}{2R} Mv^2 K$ (d) $\frac{(\gamma-1)}{2(\gamma+1)R} Mv^2 K$

22. Three perfect gases at absolute temperatures T_1 , T_2 and T_3 are mixed. The masses of molecules are m_1 , m_2 and m_3 and the number of molecules are n_1 , n_2 and n_3 respectively. Assuming no loss of energy, the final temperature of the mixture is : [2011]

(a) $\frac{n_1 T_1 + n_2 T_2 + n_3 T_3}{n_1 + n_2 + n_3}$ (b) $\frac{n_1 T_1^2 + n_2 T_2^2 + n_3 T_3^2}{n_1 T_1 + n_2 T_2 + n_3 T_3}$
(c) $\frac{n_1^2 T_1^2 + n_2^2 T_2^2 + n_3^2 T_3^2}{n_1 T_1 + n_2 T_2 + n_3 T_3}$ (d) $\frac{(T_1 + T_2 + T_3)}{3}$

23. One kg of a diatomic gas is at a pressure of $8 \times 10^4 \text{ N/m}^2$. The density of the gas is 4 kg/m^3 . What is the energy of the gas due to its thermal motion? [2009]

(a) $5 \times 10^4 \text{ J}$ (b) $6 \times 10^4 \text{ J}$
(c) $7 \times 10^4 \text{ J}$ (d) $3 \times 10^4 \text{ J}$

24. The speed of sound in oxygen (O_2) at a certain temperature is 460 ms^{-1} . The speed of sound in helium (He) at the same temperature will be (assume both gases to be ideal) [2008]

(a) 1421 ms^{-1} (b) 500 ms^{-1}
(c) 650 ms^{-1} (d) 330 ms^{-1}

25. At what temperature is the r.m.s velocity of a hydrogen molecule equal to that of an oxygen molecule at 47°C ? [2002]

(a) 80 K (b) -73 K
(c) 3 K (d) 20 K

TOPIC 3

Degree of Freedom, Specific Heat Capacity, and Mean Free Path



26. Molecules of an ideal gas are known to have three translational degrees of freedom and two rotational degrees of freedom. The gas is maintained at a temperature of T .

The total internal energy, U of a mole of this gas, and the value of $\gamma \left(= \frac{C_p}{C_v} \right)$ are given, respectively, by:

[Sep. 06, 2020 (I)]

(a) $U = \frac{5}{2} RT$ and $\gamma = \frac{6}{5}$ (b) $U = 5RT$ and $\gamma = \frac{7}{5}$
(c) $U = \frac{5}{2} RT$ and $\gamma = \frac{7}{5}$ (d) $U = 5RT$ and $\gamma = \frac{6}{5}$

27. In a dilute gas at pressure P and temperature T , the mean time between successive collisions of a molecule varies with T is : [Sep. 06, 2020 (II)]

(a) T (b) $\frac{1}{\sqrt{T}}$
(c) $\frac{1}{T}$ (d) $\sqrt{T}$

28. Match the C_p/C_v ratio for ideal gases with different type of molecules : [Sep. 04, 2020 (I)]

Column-I

Molecule Type

Column-II

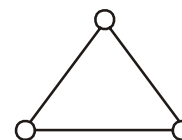
 C_p/C_v

(A) Monatomic (I) 7/5
(B) Diatomic rigid molecules (II) 9/7
(C) Diatomic non-rigid molecules (III) 4/3
(D) Triatomic rigid molecules (IV) 5/3
(a) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)
(b) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)
(c) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)
(d) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

29. A closed vessel contains 0.1 mole of a monatomic ideal gas at 200 K. If 0.05 mole of the same gas at 400 K is added to it, the final equilibrium temperature (in K) of the gas in the vessel will be close to _____.

[NA Sep. 04, 2020 (I)]

- 30.

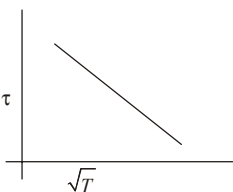


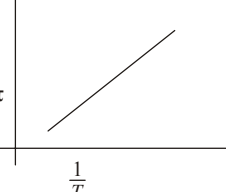
Consider a gas of triatomic molecules. The molecules are assumed to be triangular and made of massless rigid rods whose vertices are occupied by atoms. The internal energy of a mole of the gas at temperature T is :

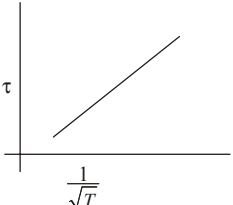
[Sep. 03, 2020 (I)]

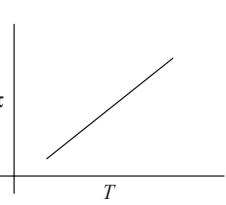
(a) $\frac{5}{2} RT$ (b) $\frac{3}{2} RT$
(c) $\frac{9}{2} RT$ (d) $3RT$

31. To raise the temperature of a certain mass of gas by 50°C at a constant pressure, 160 calories of heat is required. When the same mass of gas is cooled by 100°C at constant

- volume, 240 calories of heat is released. How many degrees of freedom does each molecule of this gas have (assume gas to be ideal)? [Sep. 03, 2020 (II)]
- (a) 5 (b) 6
(c) 3 (d) 7
32. A gas mixture consists of 3 moles of oxygen and 5 moles of argon at temperature T . Assuming the gases to be ideal and the oxygen bond to be rigid, the total internal energy (in units of RT) of the mixture is : [Sep. 02, 2020 (I)]
- (a) 15 (b) 13
(c) 20 (d) 11
33. An ideal gas in a closed container is slowly heated. As its temperature increases, which of the following statements are true? [Sep. 02, 2020 (II)]
- (1) The mean free path of the molecules decreases
(2) The mean collision time between the molecules decreases
(3) The mean free path remains unchanged
(4) The mean collision time remains unchanged
- (a) (2) and (3) (b) (1) and (2)
(c) (3) and (4) (d) (1) and (4)
34. Consider two ideal diatomic gases A and B at some temperature T . Molecules of the gas A are rigid, and have a mass m . Molecules of the gas B have an additional vibrational mode, and have a mass $\frac{m}{4}$. The ratio of the specific heats (C_V^A and C_V^B) of gas A and B, respectively is: [9 Jan 2020 I]
- (a) 7 : 9 (b) 5 : 9
(c) 3 : 5 (d) 5 : 7
35. Two gases-argon (atomic radius 0.07 nm, atomic weight 40) and xenon (atomic radius 0.1 nm, atomic weight 140) have the same number density and are at the same temperature. The ratio of their respective mean free times is closest to: [9 Jan 2020 II]
- (a) 3.67 (b) 1.83
(c) 2.3 (d) 4.67
36. The plot that depicts the behavior of the mean free time τ (time between two successive collisions) for the molecules of an ideal gas, as a function of temperature (T), qualitatively, is: (Graphs are schematic and not drawn to scale) [8 Jan. 2020 I]
- (a) 

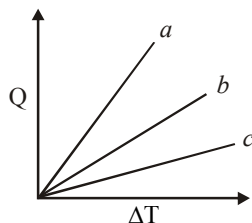
(b) 

(c) 

(d) 
37. Consider a mixture of n moles of helium gas and $2n$ moles of oxygen gas (molecules taken to be rigid) as an ideal gas. Its C_p/C_V value will be: [8 Jan. 2020 II]
- (a) 19/13 (b) 67/45
(c) 40/27 (d) 23/15
38. Two moles of an ideal gas with $\frac{C_p}{C_V} = \frac{5}{3}$ are mixed with 3 moles of another ideal gas with $\frac{C_p}{C_V} = \frac{4}{3}$. The value of $\frac{C_p}{C_V}$ for the mixture is: [7 Jan. 2020 I]
- (a) 1.45 (b) 1.50
(c) 1.47 (d) 1.42
39. Two moles of helium gas is mixed with three moles of hydrogen molecules (taken to be rigid). What is the molar specific heat of mixture at constant volume? ($R = 8.3 \text{ J/mol K}$) [12 April 2019 I]
- (a) 19.7 J/mol L (b) 15.7 J/mol K
(c) 17.4 J/mol K (d) 21.6 J/mol K
40. A diatomic gas with rigid molecules does 10 J of work when expanded at constant pressure. What would be the heat energy absorbed by the gas, in this process? [12 April 2019 II]
- (a) 25 J (b) 35 J
(c) 30 J (d) 40 J
41. A $25 \times 10^{-3} \text{ m}^3$ volume cylinder is filled with 1 mol of O_2 gas at room temperature (300 K). The molecular diameter of O_2 , and its root mean square speed, are found to be 0.3 nm and 200 m/s, respectively. What is the average collision rate (per second) for an O_2 molecule? [10 April 2019 I]
- (a) $\sim 10^{12}$ (b) $\sim 10^{11}$
(c) $\sim 10^{10}$ (d) $\sim 10^{13}$
42. When heat Q is supplied to a diatomic gas of rigid molecules, at constant volume its temperature increases by ΔT . The heat required to produce the same change in temperature, at a constant pressure is : [10 April 2019 II]
- (a) $\frac{2}{3}Q$ (b) $\frac{5}{3}Q$
(c) $\frac{7}{5}Q$ (d) $\frac{3}{2}Q$
43. An HCl molecule has rotational, translational and vibrational motions. If the rms velocity of HCl molecules in its gaseous phase is $\bar{v}$, m is its mass and k_B is Boltzmann constant, then its temperature will be: [9 April 2019 I]
- (a) $\frac{m\bar{v}^2}{6k_B}$ (b) $\frac{m\bar{v}^2}{3k_B}$
(c) $\frac{m\bar{v}^2}{7k_B}$ (d) $\frac{m\bar{v}^2}{5k_B}$

44. The specific heats, C_p and C_v of a gas of diatomic molecules, A, are given (in units of $\text{J mol}^{-1} \text{K}^{-1}$) by 29 and 22, respectively. Another gas of diatomic molecules, B, has the corresponding values 30 and 21. If they are treated as ideal gases, then: [9 April 2019 II]
 (a) A is rigid but B has a vibrational mode.
 (b) A has a vibrational mode but B has none.
 (c) A has one vibrational mode and B has two.
 (d) Both A and B have a vibrational mode each.
45. An ideal gas occupies a volume of 2 m^3 at a pressure of $3 \times 10^6 \text{ Pa}$. The energy of the gas: [12 Jan. 2019 I]
 (a) $9 \times 10^6 \text{ J}$ (b) $6 \times 10^4 \text{ J}$
 (c) 10^8 J (d) $3 \times 10^2 \text{ J}$
46. An ideal gas is enclosed in a cylinder at pressure of 2 atm and temperature, 300 K. The mean time between two successive collisions is $6 \times 10^{-8} \text{ s}$. If the pressure is doubled and temperature is increased to 500 K, the mean time between two successive collisions will be close to: [12 Jan. 2019 II]
 (a) $2 \times 10^{-7} \text{ s}$ (b) $4 \times 10^{-8} \text{ s}$
 (c) $0.5 \times 10^{-8} \text{ s}$ (d) $3 \times 10^{-6} \text{ s}$
47. A gas mixture consists of 3 moles of oxygen and 5 moles of argon at temperature T. Considering only translational and rotational modes, the total internal energy of the system is: [11 Jan. 2019 I]
 (a) 15 RT (b) 12 RT
 (c) 4 RT (d) 20 RT
48. In a process, temperature and volume of one mole of an ideal monoatomic gas are varied according to the relation $VT = K$, where K is a constant. In this process the temperature of the gas is increased by ΔT . The amount of heat absorbed by gas is (R is gas constant): [11 Jan. 2019 II]
 (a) $\frac{1}{2} R \Delta T$ (b) $\frac{1}{2} K R \Delta T$
 (c) $\frac{3}{2} R \Delta T$ (d) $\frac{2K}{3} \Delta T$
49. Two kg of a monoatomic gas is at a pressure of $4 \times 10^4 \text{ N/m}^2$. The density of the gas is 8 kg/m^3 . What is the order of energy of the gas due to its thermal motion? [10 Jan 2019 II]
 (a) 10^3 J (b) 10^5 J
 (c) 10^4 J (d) 10^6 J
50. A 15 g mass of nitrogen gas is enclosed in a vessel at a temperature 27°C . Amount of heat transferred to the gas, so that rms velocity of molecules is doubled, is about: [Take $R = 8.3 \text{ J/K mole}$] [9 Jan. 2019 II]
 (a) 0.9 kJ (b) 6 kJ
 (c) 10 kJ (d) 14 kJ
51. Two moles of an ideal monoatomic gas occupies a volume V at 27°C . The gas expands adiabatically to a volume 2 V. Calculate (1) the final temperature of the gas and (2) change in its internal energy. [2018]
 (a) (1) 189 K (2) 2.7 kJ
 (b) (1) 195 K (2) -2.7 kJ
 (c) (1) 189 K (2) -2.7 kJ
 (d) (1) 195 K (2) 2.7 kJ
52. Two moles of helium are mixed with n with moles of hydrogen. If $\frac{C_p}{C_v} = \frac{3}{2}$ for the mixture, then the value of n is [Online April 16, 2018]
 (a) 3/2 (b) 2
 (c) 1 (d) 3
53. C_p and C_v are specific heats at constant pressure and constant volume respectively. It is observed that $C_p - C_v = a$ for hydrogen gas
 $C_p - C_v = b$ for nitrogen gas
 The correct relation between a and b is: [2017]
 (a) $a = 14 b$ (b) $a = 28 b$
 (c) $a = \frac{1}{14} b$ (d) $a = b$
54. An ideal gas has molecules with 5 degrees of freedom. The ratio of specific heats at constant pressure (C_p) and at constant volume (C_v) is: [Online April 8, 2017]
 (a) 6 (b) $\frac{7}{2}$
 (c) $\frac{5}{2}$ (d) $\frac{7}{5}$
55. An ideal gas undergoes a quasi static, reversible process in which its molar heat capacity C remains constant. If during this process the relation of pressure P and volume V is given by $PV^n = \text{constant}$, then n is given by (Here C_p and C_v are molar specific heat at constant pressure and constant volume, respectively): [2016]
 (a) $n = \frac{C_p - C_v}{C - C_v}$ (b) $n = \frac{C - C_v}{C - C_p}$
 (c) $n = \frac{C_p}{C_v}$ (d) $n = \frac{C - C_p}{C - C_v}$
56. Using equipartition of energy, the specific heat (in $\text{J kg}^{-1} \text{K}^{-1}$) of aluminium at room temperature can be estimated to be (atomic weight of aluminium = 27) [Online April 11, 2015]
 (a) 410 (b) 25
 (c) 1850 (d) 925
57. Modern vacuum pumps can evacuate a vessel down to a pressure of $4.0 \times 10^{-15} \text{ atm}$. at room temperature (300 K). Taking $R = 8.0 \text{ JK}^{-1} \text{ mole}^{-1}$, $1 \text{ atm} = 10^5 \text{ Pa}$ and $N_{\text{Avogadro}} = 6 \times 10^{23} \text{ mole}^{-1}$, the mean distance between molecules of gas in an evacuated vessel will be of the order of: [Online April 9, 2014]
 (a) 0.2 μm (b) 0.2 mm
 (c) 0.2 cm (d) 0.2 nm

58. Figure shows the variation in temperature (ΔT) with the amount of heat supplied (Q) in an isobaric process corresponding to a monoatomic (M), diatomic (D) and a polyatomic (P) gas. The initial state of all the gases are the same and the scales for the two axes coincide. Ignoring vibrational degrees of freedom, the lines a , b and c respectively correspond to : [Online April 9, 2013]



- (a) P, M and D (b) M, D and P
(c) P, D and M (d) D, M and P
59. A given ideal gas with $\gamma = \frac{C_p}{C_v} = 1.5$ at a temperature T . If the gas is compressed adiabatically to one-fourth of its initial volume, the final temperature will be [Online May 12, 2012]
- (a) $2\sqrt{2}T$ (b) $4T$
(c) $2T$ (d) $8T$

60. If C_p and C_v denote the specific heats of nitrogen per unit mass at constant pressure and constant volume respectively, then [2007]

(a) $C_p - C_v = 28R$ (b) $C_p - C_v = R/28$
(c) $C_p - C_v = R/14$ (d) $C_p - C_v = R$

61. A gaseous mixture consists of 16 g of helium and 16 g of oxygen. The ratio $\frac{C_p}{C_v}$ of the mixture is [2005]

(a) 1.62 (b) 1.59
(c) 1.54 (d) 1.4

62. One mole of ideal monatomic gas ($\gamma = 5/3$) is mixed with one mole of diatomic gas ($\gamma = 7/5$). What is γ for the mixture? γ Denotes the ratio of specific heat at constant pressure, to that at constant volume [2004]

(a) 35/23 (b) 23/15
(c) 3/2 (d) 4/3

63. During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its absolute temperature. The ratio C_p/C_v for the gas is [2003]

(a) $\frac{4}{3}$ (b) 2
(c) $\frac{5}{3}$ (d) $\frac{3}{2}$



Hints & Solutions



1. (5) Using ideal gas equation, $PV = nRT$

$$\Rightarrow P_1 V_1 = nR \times 250 \quad [\because T_1 = 250 \text{ K}] \quad \dots(i)$$

$$P_2 (2V_1) = \frac{5n}{4} R \times 2000 \quad [\because T_2 = 2000 \text{ K}] \quad \dots(ii)$$

Dividing eq. (i) by (ii),

$$\frac{P_1}{2P_2} = \frac{4 \times 250}{5 \times 2000} \Rightarrow \frac{P_1}{P_2} = \frac{1}{5}$$

$$\therefore \frac{P_2}{P_1} = 5.$$

2. (150) In first case,
From ideal gas equation

$$PV = nRT$$

$$P\Delta V + V\Delta P = 0 \quad (\text{As temperature is constant})$$

$$\Delta V = -\frac{\Delta P}{P} V \quad \dots(i)$$

In second case, using ideal gas equation again

$$P\Delta V = -nR\Delta T$$

$$\Delta V = -\frac{nR\Delta T}{P} \quad \dots(ii)$$

Equating (i) and (ii), we get

$$\frac{nR\Delta T}{P} = -\frac{\Delta P}{P} V \Rightarrow \Delta T = \Delta P \frac{V}{nR}$$

Comparing the above equation with $|\Delta T| = C |\Delta P|$, we have

$$C = \frac{V}{nR} = \frac{\Delta T}{\Delta P} = \frac{300 \text{ K}}{2 \text{ atm}} = 150 \text{ K/atm}$$

3. (a) $N = \int \rho(dv)$

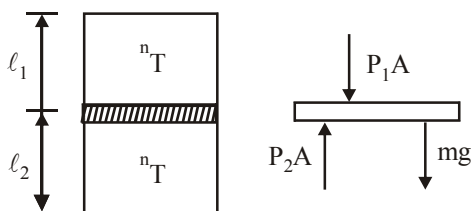
$$= \int_0^r n_0 e^{-\alpha r^4} \times 4\pi r^2 dr = 4\pi n_0 \int_0^r r^2 (e^{-\alpha r^4}) dr$$

$$\propto n_0 a^{-3/4}$$

4. (d) Clearly from figure,

$$P_2 A = P_1 A + mg$$

$$\text{or, } \frac{nRT \cdot A}{A\ell_2} = \frac{nRT \cdot A}{A\ell_1} + mg$$



$$\Rightarrow nRT \left(\frac{1}{\ell_2} - \frac{1}{\ell_1} \right) = mg$$

$$\therefore m = \frac{nRT}{g} \left(\frac{\ell_1 - \ell_2}{\ell_1 \cdot \ell_2} \right)$$

5. (b) Given: Temperature $T_i = 17 + 273 = 290 \text{ K}$

Temperature $T_f = 27 + 273 = 300 \text{ K}$

Atmospheric pressure, $P_0 = 1 \times 10^5 \text{ Pa}$

Volume of room, $V_0 = 30 \text{ m}^3$

Difference in number of molecules, $n_f - n_i = ?$

Using ideal gas equation, $PV = nRT(N_0)$,

$N_0 = \text{Avogadro's number}$

$$\Rightarrow n = \frac{PV}{RT} (N_0)$$

$$\therefore n_f - n_i = \frac{P_0 V_0}{R} \left(\frac{1}{T_f} - \frac{1}{T_i} \right) N_0$$

$$= \frac{1 \times 10^5 \times 30}{8.314} \times 6.023 \times 10^{23} \left(\frac{1}{300} - \frac{1}{290} \right)$$

$$= -2.5 \times 10^{25}$$

6. (c) From P-V graph,

$$P \propto \frac{1}{V}, \quad T = \text{constant and Pressure is increasing from 2}$$

to 1 so option (3) represents correct T-P graph.

7. (a) In ideal gases the molecules are considered as point particles and for point particles, there is no internal excitation, no vibration and no rotation. For an ideal gas the internal energy can only be translational kinetic energy and for real gas both kinetic as well as potential energy.

8. (c) The centre of mass of gas molecules also moves with lorry with uniform speed. As there is no relative motion of gas molecule. So, kinetic energy and hence temperature remain same.

9. (c) Given: $\text{K.E.}_{\text{mean}} = \frac{3}{2} kT = 4 \times 10^{-14}$

$$P = 2 \text{ cm of Hg, } V = 4 \text{ cm}^3$$

$$N = \frac{PV}{KT} = \frac{P \rho g V}{KT} = \frac{2 \times 13.6 \times 980 \times 4}{\frac{8}{3} \times 10^{-14}} \approx 4 \times 10^{18}$$

10. (41) Room mean square speed is given by

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

Here, M = Molar mass of gas molecule

T = temperature of the gas molecule

We have given $v_{N_2} = v_{H_2}$

$$\therefore \sqrt{\frac{3RT_{N_2}}{M_{N_2}}} = \sqrt{\frac{3RT_{H_2}}{M_{H_2}}}$$

$$\Rightarrow \frac{T_{H_2}}{2} = \frac{573}{28} \Rightarrow T_{H_2} = 41 \text{ K}$$

11. (c) $V_{rms} = \sqrt{\frac{3RT}{M}}$

$$\frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}} = \frac{(273+127)}{(273+237)} = \sqrt{\frac{400}{500}} = \sqrt{\frac{4}{5}} = \frac{2}{\sqrt{5}}$$

$$\therefore v_2 = \frac{\sqrt{5}}{2} v_1 = \frac{\sqrt{5}}{2} \times 200 = 100\sqrt{5} \text{ m/s.}$$

12. (Bouns) Rate of change of momentum during collision

$$= \frac{mv - (-mv)}{\Delta t} = \frac{2mv}{\Delta t} N$$

$$\text{so pressure } P = \frac{N \times (2mv)}{\Delta t \times A}$$

$$= \frac{10^{22} \times 2 \times 10^{-26} \times 10^4}{1 \times 1} = 2 \text{ N/m}^2$$

13. (c) $v_{rms} = v_e$

$$\sqrt{\frac{3RT}{M}} = 11.2 \times 10^3$$

$$\text{or } \sqrt{\frac{3kT}{m}} = 11.2 \times 10^3$$

$$\text{or } \sqrt{\frac{3 \times 1.38 \times 10^{-23} T}{2 \times 10^{-3}}} = 11.2 \times 10^3 \quad \therefore T = 10^4 \text{ K}$$

14. (a) Using $\frac{V_{rms}}{V_{2rms}} = \sqrt{\frac{M_2}{M_1}}$

$$\frac{V_{rms}(\text{He})}{V_{rms}(\text{Ar})} = \sqrt{\frac{M_{Ar}}{M_{He}}} = \sqrt{\frac{40}{4}} = 3.16$$

15. (a) Energy associated with N moles of diatomic gas,

$$U_i = N \frac{5}{2} RT$$

Energy associated with n moles of monoatomic gas

$$= n \frac{3}{2} RT$$

Total energy when n moles of diatomic gas converted into

$$\text{monoatomic } (U_f) = 2n \frac{3}{2} RT + (N-n) \frac{5}{2} RT$$

$$= \frac{1}{2} nRT + \frac{5}{2} NRT$$

Now, change in total kinetic energy of the gas

$$\Delta U = Q = \frac{1}{2} nRT$$

16. (c) Pressure, $P = \frac{1}{3} \frac{mN}{V} V_{rms}^2$

$$\text{or, } P = \frac{(mN)T}{V}$$

If the gas mass and temperature are constant then

$$P \propto (V_{rms})^2 \propto T$$

$$\text{So, force } \propto (V_{rms})^2 \propto T$$

$$\text{i.e., Value of } q = 1$$

17. (d) Kinetic energy of each molecule,

$$\text{K.E.} = \frac{3}{2} K_B T$$

In the given problem,

$$\text{Temperature, } T = 0^\circ\text{C} = 273 \text{ K}$$

Height attained by the gas molecule, $h = ?$

$$\text{K.E.} = \frac{3}{2} K_B (273) = \frac{819 K_B}{2}$$

$$\text{K.E.} = \text{P.E.}$$

$$\Rightarrow \frac{819 K_B}{2} = Mgh$$

$$\text{or } h = \frac{819 K_B}{2Mg}$$

18. (a) $\therefore C = \sqrt{\frac{3RT}{M}}$

$$(1930)^2 = \frac{3 \times 8.314 \times 300}{M}$$

$$M = \frac{3 \times 8.314 \times 300}{1930 \times 1930} \approx 2 \times 10^{-3} \text{ kg}$$

The gas is H_2 .

19. (c) $v_{rms} = \sqrt{\frac{3\rho v}{\text{mass of the gas}}}$

20. (a) Given, $V_1 = V$

$$V_2 = 2V$$

$$T_1 = 27^\circ + 273 = 300 \text{ K}$$

$$T_2 = ?$$

From charle's law

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \quad (\because \text{Pressure is constant})$$

$$\text{or, } \frac{V}{300} = \frac{2V}{T_2}$$

$$\therefore T_2 = 600 \text{ K} = 600 - 273 = 327^\circ\text{C}$$

21. (c) As, work done is zero.

So, loss in kinetic energy = heat gain by the gas

$$\frac{1}{2} mv^2 = nC_v \Delta T = n \frac{R}{\gamma - 1} \Delta T$$

$$\frac{1}{2} mv^2 = \frac{m}{M} \frac{R}{\gamma - 1} \Delta T$$

$$\therefore \Delta T = \frac{Mv^2(\gamma - 1)}{2R} K$$

22. (a) Number of moles of first gas = $\frac{n_1}{N_A}$

Number of moles of second gas = $\frac{n_2}{N_A}$

Number of moles of third gas = $\frac{n_3}{N_A}$

If there is no loss of energy then

$$P_1 V_1 + P_2 V_2 + P_3 V_3 = PV$$

$$\frac{n_1}{N_A} RT_1 + \frac{n_2}{N_A} RT_2 + \frac{n_3}{N_A} RT_3$$

$$= \frac{n_1 + n_2 + n_3}{N_A} RT_{\text{mix}}$$

$$T_{\text{mix}} = \frac{n_1 T_1 + n_2 T_2 + n_3 T_3}{n_1 + n_2 + n_3}$$

23. (a) Given, mass = 1 kg

Density = 4 kg m⁻³

$$\text{Volume} = \frac{\text{mass}}{\text{density}} = \frac{1}{4} \text{ m}^3$$

Internal energy of the diatomic gas

$$= \frac{5}{2} PV = \frac{5}{2} \times 8 \times 10^4 \times \frac{1}{4} = 5 \times 10^4 \text{ J}$$

Alternatively:

$$\text{K.E} = \frac{5}{2} nRT = \frac{5}{2} \frac{m}{M} RT = \frac{5}{2} \frac{m}{M} \times \frac{PM}{d} \quad [\because PM = dRT]$$

$$= \frac{5}{2} \frac{mP}{d} = \frac{5}{2} \times \frac{1 \times 8 \times 10^4}{4} = 5 \times 10^4 \text{ J}$$

24. (a) The speed of sound in a gas is given by $v = \sqrt{\frac{\gamma RT}{M}}$

$$\therefore v \propto \sqrt{\frac{\gamma}{M}} \quad [\text{As } R \text{ and } T \text{ is constant}]$$

$$\therefore \frac{v_{\text{O}_2}}{v_{\text{He}}} = \sqrt{\frac{\gamma_{\text{O}_2}}{\gamma_{\text{He}}} \times \frac{M_{\text{He}}}{M_{\text{O}_2}}}$$

$$= \sqrt{\frac{1.4}{32} \times \frac{4}{1.67}} = 0.3237$$

$$\therefore v_{\text{He}} = \frac{v_{\text{O}_2}}{0.3237} = \frac{460}{0.3237} = 1421 \text{ m/s}$$

25. (d) RMS velocity of a gas molecule is given by

$$V_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

Let T be the temperature at which the velocity of hydrogen molecule is equal to the velocity of oxygen molecule.

$$\therefore \sqrt{\frac{3RT}{2}} = \sqrt{\frac{3R \times (273 + 47)}{32}}$$

$$\Rightarrow T = 20\text{K}$$

26. (c) Total degree of freedom $f = 3 + 2 = 5$

$$\text{Total energy, } U = \frac{nfRT}{2} = \frac{5RT}{2}$$

$$\text{And } \gamma = \frac{C_p}{C_v} = 1 + \frac{2}{f} = 1 + \frac{2}{5} = \frac{7}{5}$$

27. (b) Mean free path, $\lambda = \frac{1}{\sqrt{2} \pi n d^2}$

where, d = diameter of the molecule

n = number of molecules per unit volume

But, mean time of collision, $\tau = \frac{\lambda}{v_{\text{rms}}}$

$$\text{But } v_{\text{rms}} = \sqrt{\frac{3kT}{m}}$$

$$\therefore \tau = \frac{\lambda}{\sqrt{\frac{3kT}{m}}} \Rightarrow \tau \propto \frac{1}{\sqrt{T}}$$

28. (c) As we know,

$$\gamma = \frac{C_p}{C_v} = 1 + \frac{2}{f}, \text{ where } f = \text{degree of freedom}$$

(A) Monatomic, $f = 3$

$$\therefore \gamma = 1 + \frac{2}{3} = \frac{5}{3}$$

(B) Diatomic rigid molecules, $f = 5$

$$\therefore \gamma = 1 + \frac{2}{5} = \frac{7}{5}$$

(C) Diatomic non-rigid molecules, $f = 7$

$$\therefore \gamma = 1 + \frac{2}{7} = \frac{9}{7}$$

(D) Triatomic rigid molecules, $f = 6$

$$\therefore \gamma = 1 + \frac{2}{6} = \frac{4}{3}$$

29. (266.67) Here work done on gas and heat supplied to the gas are zero.

Let T be the final equilibrium temperature of the gas in the vessel.

Total internal energy of gases remain same.

$$\text{i.e., } u_1 + u_2 = u'_1 + u'_2$$

$$\text{or, } n_1 C_v \Delta T_1 + n_2 C_v \Delta T_2 = (n_1 + n_2) C_v T$$

$$\Rightarrow (0.1) C_v (200) + (0.05) C_v (400) = (0.15) C_v T$$

$$\therefore T = \frac{800}{3} = 266.67 \text{ K}$$

30. (d) Here degree of freedom, $f = 3 + 3 = 6$ for triatomic non-linear molecule.

Internal energy of a mole of the gas at temperature T ,

$$U = \frac{f}{2} nRT = \frac{6}{2} RT = 3RT$$

31. (b) Let C_p and C_v be the specific heat capacity of the gas at constant pressure and volume.

At constant pressure, heat required

$$\Delta Q_1 = nC_p \Delta T$$

$$\Rightarrow 160 = nC_p \cdot 50 \quad \dots(i)$$

At constant volume, heat required

$$\Delta Q_2 = nC_v \Delta T$$

$$\Rightarrow 240 = nC_v \cdot 100 \quad \dots(ii)$$

Dividing (i) by (ii), we get

$$\frac{160}{240} = \frac{C_p}{C_v} \cdot \frac{50}{100} \Rightarrow \frac{C_p}{C_v} = \frac{4}{3}$$

$$\gamma = \frac{C_p}{C_v} = \frac{4}{3} = 1 + \frac{2}{f} \quad (\text{Here, } f = \text{degree of freedom})$$

$$\Rightarrow f = 6.$$

32. (a) Total energy of the gas mixture,

$$\begin{aligned} E_{\text{mix}} &= \frac{f_1 n_1 RT_1}{2} + \frac{f_2 n_2 RT_2}{2} \\ &= 3 \times \frac{5}{2} RT + \frac{5}{2} \times 3RT = 15RT \end{aligned}$$

33. (a) As we know mean free path

$$\lambda = \frac{1}{\sqrt{2} \left(\frac{N}{V} \right) \pi d^2}$$

Here, N = no. of molecule
 V = volume of container
 d = diameter of molecule

$$\text{But } PV = nRT = nNKT$$

$$\Rightarrow \frac{N}{V} = \frac{P}{KT} = n$$

$$\lambda = \frac{1}{\sqrt{2}} \frac{KT}{\pi d^2 P}$$

For constant volume and hence constant number density

n of gas molecules $\frac{P}{T}$ is constant.

So mean free path remains same.

As temperature increases no. of collision increases so relaxation time decreases.

34. (d) Specific heat of gas at constant volume

$$C_v = \frac{1}{2} fR; f = \text{degree of freedom}$$

For gas A (diatomic)

$f = 5$ (3 translational + 2 rotational)

$$\therefore C_v^A = \frac{5}{2} R$$

For gas B (diatomic) in addition to (3 translational + 2 rotational) 2 vibrational degree of freedom.

$$\therefore C_v^B = \frac{7}{2} R \quad \text{Hence } \frac{C_v^A}{C_v^B} = \frac{\frac{5}{2} R}{\frac{7}{2} R} = \frac{5}{7}$$

35. (Bonus) Mean free path of a gas molecule is given by

$$\lambda = \frac{1}{\sqrt{2} \pi d^2 n}$$

Here, n = number of collisions per unit volume

d = diameter of the molecule

If average speed of molecule is v then

$$\text{Mean free time, } \tau = \frac{\lambda}{v}$$

$$\Rightarrow \tau = \frac{1}{\sqrt{2} \pi n d^2 v} = \frac{1}{\sqrt{2} \pi n d^2} \sqrt{\frac{M}{3RT}}$$

$$\left(\because v = \sqrt{\frac{3RT}{M}} \right)$$

$$\therefore \tau \propto \frac{\sqrt{M}}{d^2} \therefore \frac{\tau_1}{\tau_2} = \frac{\sqrt{M_1}}{d_1^2} \times \frac{d_2^2}{\sqrt{M_2}}$$

$$= \sqrt{\frac{40}{140}} \times \left(\frac{0.1}{0.07} \right)^2 = 1.09$$

36. (c) Relaxation time (τ) $\propto \frac{\text{mean free path}}{\text{speed}} \Rightarrow \tau \propto \frac{1}{v}$

$$\text{and, } v \propto \sqrt{T}$$

$$\therefore \tau \propto \frac{1}{\sqrt{T}}$$

Hence graph between τ v/s $\frac{1}{\sqrt{T}}$ is a straight line which is correctly depicted by graph shown in option (c).

37. (a) Helium is a monoatomic gas and Oxygen is a diatomic gas.

$$\text{For helium, } C_{V1} = \frac{3}{2} R \text{ and } C_{P1} = \frac{5}{2} R$$

$$\text{For oxygen, } C_{V2} = \frac{5}{2} R \text{ and } C_{P2} = \frac{7}{2} R$$

$$\gamma = \frac{N_1 C_{P1} + N_2 C_{P2}}{N_1 C_{V1} + N_2 C_{V2}}$$

$$\Rightarrow \gamma = \frac{n \cdot \frac{5}{2} R + 2n \cdot \frac{7}{2} R}{n \cdot \frac{3}{2} R + 2n \cdot \frac{5}{2} R} = \frac{19nR \times 2}{2(13nR)}$$

$$\therefore \left(\frac{C_P}{C_V} \right)_{\text{mixture}} = \frac{19}{13}$$

38. (d) Using, $\gamma_{\text{mixture}} = \frac{n_1 C_{p1} + n_2 C_{p2}}{n_1 C_{v1} + n_2 C_{v2}}$

$$\Rightarrow \frac{n_1}{\gamma_1 - 1} + \frac{n_2}{\gamma_2 - 1} = \frac{n_1 + n_2}{\gamma_m - 1}$$

$$\Rightarrow \frac{3}{\frac{4}{3} - 1} + \frac{2}{\frac{5}{3} - 1} = \frac{5}{\gamma_m - 1}$$

$$\Rightarrow \frac{9}{1} + \frac{2 \times 3}{2} = \frac{5}{\gamma_m - 1} \Rightarrow \gamma_m - 1 = \frac{5}{12}$$

$$\Rightarrow \gamma_m = \frac{17}{12} = 1.42$$

39. (c) $[C_v]_{\text{min}} = \frac{n_1 [C_{v1}] + n_2 [C_{v2}]}{n_1 + n_2}$

$$= \left[\frac{2 \times \frac{3R}{2} + 3 \times \frac{5R}{2}}{2 + 3} \right]$$

$$= 2.1 R = 2.1 \times 8.3 = 17.4 \text{ J/mol-K}$$

40. (b) $F = \frac{C_v}{C_p} = \frac{1}{r} = \frac{1}{(7/5)} = \frac{5}{7}$

$$\text{or } \frac{W}{Q} = 1 - \frac{5}{7} = \frac{2}{7}$$

$$\text{or } Q = \frac{7}{2} W = \frac{7 \times 10}{2} = 35 \text{ J}$$

41. (c) $V = 25 \times 10^{-3} \text{ m}^3$, $N = 1 \text{ mole of O}_2$

$$T = 300 \text{ K}$$

$$V_{\text{rms}} = 200 \text{ m/s}$$

$$\therefore \lambda = \frac{1}{\sqrt{2N\pi r^2}}$$

$$\text{Average time } \frac{1}{\tau} = \frac{\langle V \rangle}{\lambda} = 200 \cdot N\pi r^2 \cdot \sqrt{2}$$

$$= \frac{\sqrt{2} \times 200 \times 6.023 \times 10^{23}}{25 \times 10^{-3}} \cdot \pi \times 10^{-18} \times 0.09$$

The closest value in the given option is $= 10^{10}$

42. (c) Amount of heat required (Q) to raise the temperature at constant volume

$$Q = nC_v \Delta T \quad \dots (i)$$

Amount of heat required (Q_1) at constant pressure

$$Q_1 = nC_p \Delta T \quad \dots (ii)$$

Dividing equation (ii) by (i), we get

$$\therefore \frac{Q_1}{Q} = \frac{C_p}{C_v}$$

$$\Rightarrow Q_1 = (Q) \left(\frac{7}{5} \right) \left(\because \gamma = \frac{C_p}{C_v} = \frac{7}{5} \right)$$

43. (a) $\frac{1}{2} m \bar{v}^2 = 3k_B T$

$$\text{or } T = \frac{m \bar{v}^2}{6k_B}$$

44. (b) $\gamma_A = \frac{C_p}{C_v} = \frac{29}{22} = 1.32 < 1.4$ (diatomic)

$$\text{and } \gamma_B = \frac{30}{21} = \frac{10}{7} = 1.43 > 1.4$$

Gas A has more than 5-degrees of freedom.

45. (a) Energy of the gas, E

$$= \frac{f}{2} nRT = \frac{f}{2} PV$$

$$= \frac{f}{2} (3 \times 10^6)(2) = f \times 3 \times 10^6$$

Considering gas is monoatomic i.e., $f = 3$

$$\text{Energy, } E = 9 \times 10^6 \text{ J}$$

46. (b) Using, $\tau = \frac{1}{2n\pi d^2 V_{\text{avg}}}$

$$\therefore t \propto \frac{\sqrt{T}}{P} \left[\because n = \frac{\text{no. of molecules}}{\text{Volume}} \right]$$

$$\text{or, } \frac{t_1}{6 \times 10^{-8}} = \frac{\sqrt{500}}{2P} \times \frac{P}{\sqrt{300}} \approx 4 \times 10^{-8}$$

47. (a) $U = \frac{f_1}{2} n_1 RT + \frac{f_2}{2} n_2 RT$

Considering translational and rotational modes, degrees of freedom $f_1 = 5$ and $f_2 = 3$

$$\therefore U = \frac{5}{2} (3RT) + \frac{3}{2} \times 5RT$$

$$U = 15RT$$

48. (a) According to question $VT = K$

we also know that $PV = nRT$

$$\Rightarrow T = \left(\frac{PV}{nR} \right)$$

$$\Rightarrow V \left(\frac{PV}{nR} \right) = K \Rightarrow PV^2 = K$$

$$\therefore C = \frac{R}{1-x} + C_v \text{ (For polytropic process)}$$

$$C = \frac{R}{1-2} + \frac{3R}{2} = \frac{R}{2}$$

$$\therefore \Delta Q = nC \Delta T$$

$$= \frac{R}{2} \times \Delta T \text{ [here, } n = 1 \text{ mole]}$$

49. (c) Thermal energy of N molecule

$$= N \left(\frac{3}{2} kT \right)$$

$$= \frac{N}{N_A} \frac{3}{2} RT = \frac{3}{2} (nRT) = \frac{3}{2} PV$$

$$= \frac{3}{2} P \left(\frac{m}{\rho} \right) = \frac{3}{2} P \left(\frac{2}{8} \right)$$

$$= \frac{3}{2} \times 4 \times 10^4 \times \frac{2}{8} = 1.5 \times 10^4 \text{ J}$$

therefore, order = 10^4 J

50. (c) Heat transferred,

$Q = nC_v \Delta T$ as gas in closed vessel

To double the rms speed, temperature should be 4 times

i.e., $T' = 4T$ as $v_{rms} = \sqrt{3RT/M}$

$$\therefore Q = \frac{15}{28} \times \frac{5 \times R}{2} \times (4T - T)$$

$$\left[\therefore \frac{CP}{CV} = \gamma_{diatomic} = \frac{7}{5} \text{ \& } C_p - C_v = R \right]$$

or, $Q = 10000 \text{ J} = 10 \text{ kJ}$

51. (c) In an adiabatic process

$$TV^{\gamma-1} = \text{Constant} \quad \text{or, } T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

For monoatomic gas $\gamma = \frac{5}{3}$

$$(300)V^{2/3} = T_2(2V)^{2/3} \Rightarrow T_2 = \frac{300}{(2)^{2/3}}$$

$T_2 = 189 \text{ K}$ (final temperature)

Change in internal energy $\Delta U = n \frac{f}{2} R \Delta T$

$$= 2 \left(\frac{3}{2} \right) \left(\frac{25}{3} \right) (-111) = -2.7 \text{ kJ}$$

52. (b) Using formula,

$$\gamma_{\text{mixture}} = \left(\frac{C_p}{C_v} \right)_{\text{mix}} = \frac{\frac{n_1 \gamma_1}{\gamma_1 - 1} + \frac{n_2 \gamma_2}{\gamma_2 - 1}}{\frac{n_1}{\gamma_1 - 1} + \frac{n_2}{\gamma_2 - 1}}$$

Putting the value of $n_1 = 2$, $n_2 = n$, $\left(\frac{C_p}{C_v} \right)_{\text{mix}} = \frac{3}{2}$

$\gamma_1 = \frac{5}{3}$, $\gamma_2 = \frac{7}{5}$ and solving we get, $n = 2$

53. (a) As we know, $C_p - C_v = R$ where C_p and C_v are molar specific heat capacities

$$\text{or, } C_p - C_v = \frac{R}{M}$$

For hydrogen ($M = 2$) $C_p - C_v = a = \frac{R}{2}$

For nitrogen ($M = 28$) $C_p - C_v = b = \frac{R}{28}$

$$\therefore \frac{a}{b} = 14 \quad \text{or, } a = 14b$$

54. (d) The ratio of specific heats at constant pressure (C_p) and constant volume (C_v)

$$\frac{C_p}{C_v} = \gamma = \left(1 + \frac{2}{f} \right)$$

where f is degree of freedom

$$\frac{C_p}{C_v} = \left(1 + \frac{2}{5} \right) = \frac{7}{5}$$

55. (d) For a polytropic process

$$C = C_v + \frac{R}{1-n} \quad \therefore C - C_v = \frac{R}{1-n}$$

$$\therefore 1-n = \frac{R}{C-C_v} \quad \therefore 1 - \frac{R}{C-C_v} = n$$

$$\therefore n = \frac{C - C_v - R}{C - C_v} = \frac{C - C_v - C_p + C_v}{C - C_v}$$

$$= \frac{C - C_p}{C - C_v} \quad (\because C_p - C_v = R)$$

56. (d) Using equipartition of energy, we have

$$\frac{6}{2} KT = mCT$$

$$C = \frac{3 \times 1.38 \times 10^{-23} \times 6.02 \times 10^{23}}{27 \times 10^{-3}}$$

$$\therefore C = 925 \text{ J/kgK}$$

57. (b)

58. (b) On giving same amount of heat at constant pressure, there is no change in temperature for mono, dia and polyatomic.

$$(\Delta Q)_p = \mu C_p \Delta T \left(\mu = \frac{\text{No. of molecules}}{\text{Avogadro's no.}} \right)$$

$$\text{or } \Delta T \propto \frac{1}{\text{no. of molecules}}$$

59. (c) $TV^{\gamma-1} = \text{constant}$

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$\Rightarrow T(V)^{1/2} = T_2 \left(\frac{V}{4} \right)^{1/2}$$

$$\left[\because \gamma = 1.5, T_1 = T, V_1 = V \text{ and } V_2 = \frac{V}{4} \right]$$

$$\therefore T_2 = \left(\frac{4V}{V} \right)^{1/2} T = 2T$$

60. (b) According to Mayer's relationship

$C_p - C_v = R$, as per the question $(C_p - C_v) M = R$

$$\Rightarrow C_p - C_v = R/28$$

Here $M = 28 = \text{mass of 1 unit of } N_2$

61. (a) For mixture of gas specific heat at constant volume

$$C_v = \frac{n_1 C_{v1} + n_2 C_{v2}}{n_1 + n_2}$$

No. of moles of helium,

$$n_1 = \frac{m_{He}}{M_{He}} = \frac{16}{4} = 4$$

Number of moles of oxygen,

$$n_2 = \frac{16}{32} = \frac{1}{2}$$

$$\therefore C_v = \frac{4 \times \frac{3}{2}R + \frac{1}{2} \times \frac{5}{2}R}{\left(4 + \frac{1}{2}\right)} = \frac{6R + \frac{5}{4}R}{\frac{9}{2}}$$

$$= \frac{29R \times 2}{9 \times 4} = \frac{29R}{18} \text{ and}$$

Specific heat at constant pressure

$$C_p = \frac{n_1 C_{p1} + n_2 C_{p2}}{(n_1 + n_2)} = \frac{4 \times \frac{5R}{2} + \frac{1}{2} \times \frac{7R}{2}}{\left(4 + \frac{1}{2}\right)}$$

$$= \frac{10R + \frac{7}{4}R}{\frac{9}{2}} = \frac{47R}{18}$$

$$\therefore \frac{C_p}{C_v} = \frac{47R}{18} \times \frac{18}{29R} = 1.62$$

$$62. \text{ (c) } \gamma_1 = \frac{5}{3} \quad \gamma_2 = \frac{7}{5}$$

$$n_1 = 1, n_2 = 1$$

$$\frac{n_1 + n_2}{\gamma - 1} = \frac{n_1}{\gamma_1 - 1} + \frac{n_2}{\gamma_2 - 1}$$

$$\Rightarrow \frac{1+1}{\gamma-1} = \frac{1}{\frac{5}{3}-1} + \frac{1}{\frac{7}{5}-1} = \frac{3}{2} + \frac{5}{2} = 4$$

$$\therefore \frac{2}{\gamma-1} = 4 \Rightarrow \gamma = \frac{3}{2}$$

$$63. \text{ (d) } P \propto T^3 \Rightarrow PT^{-3} = \text{constant} \quad \dots(i)$$

But for an adiabatic process, the pressure temperature relationship is given by

$$P^{1-\gamma} T^\gamma = \text{constant}$$

$$\Rightarrow PT^{\frac{\gamma}{1-\gamma}} = \text{constt.} \quad \dots(ii)$$

$$\text{From (i) and (ii) } \frac{\gamma}{1-\gamma} = -3 \Rightarrow \gamma = -3 + 3\gamma \Rightarrow \gamma = \frac{3}{2}$$

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Oscillations



TOPIC 1 Displacement, Phase, Velocity and Acceleration in S.H.M.



1. The position co-ordinates of a particle moving in a 3-D coordinate system is given by [9 Jan 2019, II]

$$x = a \cos \omega t$$

$$y = a \sin \omega t$$

$$\text{and } z = a \omega t$$

The speed of the particle is:

- (a) $\sqrt{2} a \omega$ (b) $a \omega$ (c) $\sqrt{3} a \omega$ (d) $2 a \omega$
2. Two simple harmonic motions, as shown, are at right angles. They are combined to form Lissajous figures.
 $x(t) = A \sin(at + \delta)$
 $y(t) = B \sin(bt)$
 Identify the correct match below

[Online April 15, 2018]

- (a) Parameters: $A = B$, $a = 2b$; $\delta = \frac{\pi}{2}$; Curve: Circle
- (b) Parameters: $A = B$, $a = b$; $\delta = \frac{\pi}{2}$; Curve: Line
- (c) Parameters: $A \neq B$, $a = b$; $\delta = \frac{\pi}{2}$; Curve: Ellipse
- (d) Parameters: $A \neq B$, $a = b$; $\delta = 0$; Curve: Parabola
3. The ratio of maximum acceleration to maximum velocity in a simple harmonic motion is 10 s^{-1} . At, $t = 0$ the displacement is 5 m. What is the maximum acceleration? The initial phase is $\frac{\pi}{4}$

[Online April 8, 2017]

- (a) 500 m/s^2 (b) $500 \sqrt{2} \text{ m/s}^2$
- (c) 750 m/s^2 (d) $750 \sqrt{2} \text{ m/s}^2$
4. A particle performs simple harmonic motion with amplitude A . Its speed is trebled at the instant that it is at a distance $\frac{2A}{3}$ from equilibrium position. The new amplitude of the motion is:

[2016]

- (a) $A\sqrt{3}$ (b) $\frac{7A}{3}$ (c) $\frac{A}{3}\sqrt{41}$ (d) $3A$

5. Two particles are performing simple harmonic motion in a straight line about the same equilibrium point. The amplitude and time period for both particles are same and equal to A and T , respectively. At time $t = 0$ one particle has displacement A while the other one has displacement

$$\frac{-A}{2} \text{ and they are moving towards each other. If they cross each other at time } t, \text{ then } t \text{ is:}$$

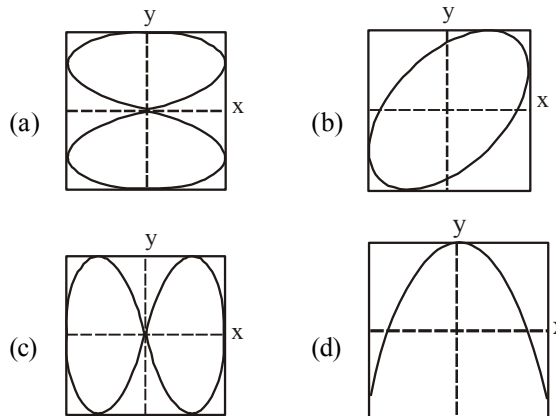
[Online April 9, 2016]

- (a) $\frac{5T}{6}$ (b) $\frac{T}{3}$ (c) $\frac{T}{4}$ (d) $\frac{T}{6}$
6. A simple harmonic oscillator of angular frequency 2 rad s^{-1} is acted upon by an external force $F = \sin t \text{ N}$. If the oscillator is at rest in its equilibrium position at $t = 0$, its position at later times is proportional to:

[Online April 10, 2015]

- (a) $\sin t + \frac{1}{2} \cos 2t$ (b) $\cos t - \frac{1}{2} \sin 2t$
- (c) $\sin t - \frac{1}{2} \sin 2t$ (d) $\sin t + \frac{1}{2} \sin 2t$
7. x and y displacements of a particle are given as $x(t) = a \sin \omega t$ and $y(t) = a \sin 2\omega t$. Its trajectory will look like:

[Online April 10, 2015]



8. A body is in simple harmonic motion with time period half second ($T = 0.5$ s) and amplitude one cm ($A = 1$ cm). Find the average velocity in the interval in which it moves from equilibrium position to half of its amplitude.

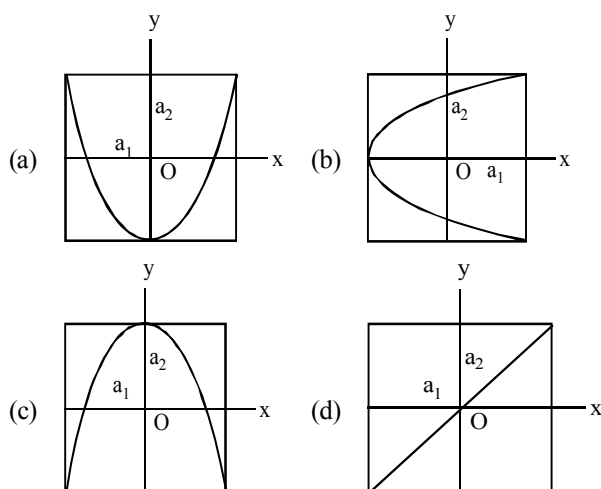
[Online April 19, 2014]

- (a) 4 cm/s (b) 6 cm/s (c) 12 cm/s (d) 16 cm/s
9. Which of the following expressions corresponds to simple harmonic motion along a straight line, where x is the displacement and a, b, c are positive constants?

[Online April 12, 2014]

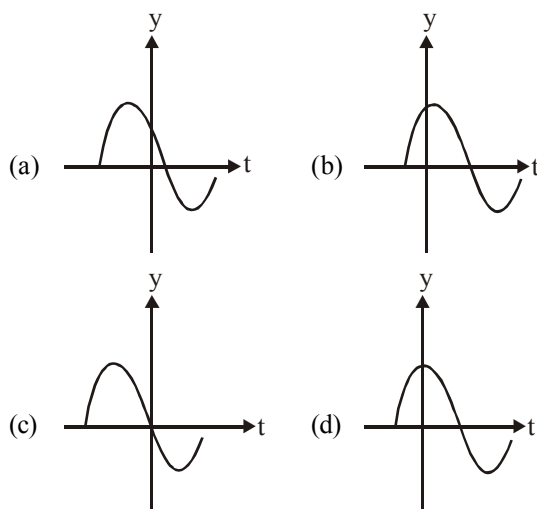
- (a) $a + bx - cx^2$ (b) bx^2
(c) $a - bx + cx^2$ (d) $-bx$
10. A particle which is simultaneously subjected to two perpendicular simple harmonic motions represented by; $x = a_1 \cos \omega t$ and $y = a_2 \cos 2\omega t$ traces a curve given by:

[Online April 9, 2014]



11. The displacement $y(t) = A \sin(\omega t + \phi)$ of a pendulum for $\phi = \frac{2\pi}{3}$ is correctly represented by

[Online May 19, 2012]



12. Two particles are executing simple harmonic motion of the same amplitude A and frequency ω along the x -axis. Their mean position is separated by distance X_0 ($X_0 > A$). If the maximum separation between them is $(X_0 + A)$, the phase difference between their motion is: [2011]

- (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{6}$ (d) $\frac{\pi}{2}$

13. A mass M , attached to a horizontal spring, executes S.H.M. with amplitude A_1 . When the mass M passes through its mean position then a smaller mass m is placed over it and both of them move together with amplitude A_2 . The ratio

of $\left(\frac{A_1}{A_2}\right)$ is: [2011]

- (a) $\frac{M+m}{M}$ (b) $\left(\frac{M}{M+m}\right)^{\frac{1}{2}}$
(c) $\left(\frac{M+m}{M}\right)^{\frac{1}{2}}$ (d) $\frac{M}{M+m}$

14. A point mass oscillates along the x -axis according to the law $x = x_0 \cos(\omega t - \pi/4)$. If the acceleration of the particle is written as $a = A \cos(\omega t + \delta)$, then [2007]

- (a) $A = x_0 \omega^2, \delta = 3\pi/4$ (b) $A = x_0 \omega^2, \delta = -\pi/4$
(c) $A = x_0 \omega^2, \delta = \pi/4$ (d) $A = x_0 \omega^2, \delta = -\pi/4$

15. A coin is placed on a horizontal platform which undergoes vertical simple harmonic motion of angular frequency ω . The amplitude of oscillation is gradually increased. The coin will leave contact with the platform for the first time

- (a) at the mean position of the platform [2006]
(b) for an amplitude of $\frac{g}{\omega^2}$
(c) for an amplitude of $\frac{g^2}{\omega^2}$
(d) at the highest position of the platform

16. The maximum velocity of a particle, executing simple harmonic motion with an amplitude 7 mm, is 4.4 m/s. The period of oscillation is [2006]

- (a) 0.01 s (b) 10 s (c) 0.1 s (d) 100 s

17. The function $\sin^2(\omega t)$ represents [2005]

- (a) a periodic, but not simple harmonic motion with a period $\frac{\pi}{\omega}$
(b) a periodic, but not simple harmonic motion with a period $\frac{2\pi}{\omega}$
(c) a simple harmonic motion with a period $\frac{\pi}{\omega}$
(d) a simple harmonic motion with a period $\frac{2\pi}{\omega}$

18. Two simple harmonic motions are represented by the equations $y_1 = 0.1 \sin \left(100\pi t + \frac{\pi}{3} \right)$ and $y_2 = 0.1 \cos \pi t$.

The phase difference of the velocity of particle 1 with respect to the velocity of particle 2 is [2005]

- (a) $\frac{\pi}{3}$ (b) $-\frac{\pi}{6}$ (c) $\frac{\pi}{6}$ (d) $-\frac{\pi}{3}$
19. Two particles A and B of equal masses are suspended from two massless springs of spring constants k_1 and k_2 , respectively. If the maximum velocities, during oscillation, are equal, the ratio of amplitude of A and B is [2003]

- (a) $\sqrt{\frac{k_1}{k_2}}$ (b) $\frac{k_2}{k_1}$ (c) $\sqrt{\frac{k_2}{k_1}}$ (d) $\frac{k_1}{k_2}$

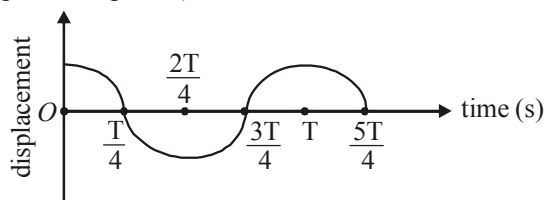
20. The displacement of a particle varies according to the relation $x = 4(\cos \pi t + \sin \pi t)$. The amplitude of the particle is [2003]

- (a) -4 (b) 4 (c) $4\sqrt{2}$ (d) 8

TOPIC 2 Energy in Simple Harmonic Motion



21. The displacement time graph of a particle executing S.H.M. is given in figure : (sketch is schematic and not to scale)



Which of the following statements is/are true for this motion? [Sep. 02, 2020 (II)]

- (1) The force is zero at $t = \frac{3T}{4}$
 (2) The acceleration is maximum at $t = T$
 (3) The speed is maximum at $t = \frac{T}{4}$
 (4) The P.E. is equal to K.E. of the oscillation at $t = \frac{T}{2}$
- (a) (1), (2) and (4) (b) (2), (3) and (4)
 (c) (1), (2) and (3) (d) (1) and (4)
22. A particle undergoing simple harmonic motion has time

dependent displacement given by $x(t) = A \sin \frac{\pi t}{90}$. The

ratio of kinetic to potential energy of this particle at $t = 210$ s will be : [11 Jan 2019, I]

- (a) $\frac{1}{9}$ (b) 1 (c) 2 (d) $\frac{1}{3}$

23. A pendulum is executing simple harmonic motion and its maximum kinetic energy is K_1 . If the length of the pendulum is doubled and it performs simple harmonic motion with the same amplitude as in the first case, its maximum kinetic energy is K_2 . [11 Jan 2019, II]

- (a) $K_2 = 2K_1$ (b) $K_2 = \frac{K_1}{2}$

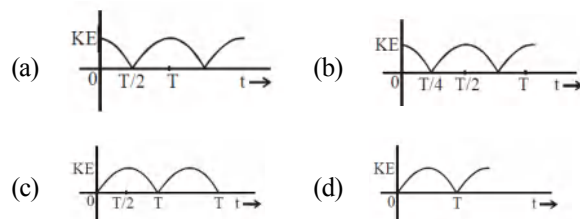
- (c) $K_2 = \frac{K_1}{4}$ (d) $K_2 = K_1$

24. A particle is executing simple harmonic motion (SHM) of amplitude A, along the x-axis, about $x = 0$. When its potential Energy (PE) equals kinetic energy (KE), the position of the particle will be: [9 Jan 2019, II]

- (a) $\frac{A}{2}$ (b) $\frac{A}{2\sqrt{2}}$

- (c) $\frac{A}{\sqrt{2}}$ (d) A

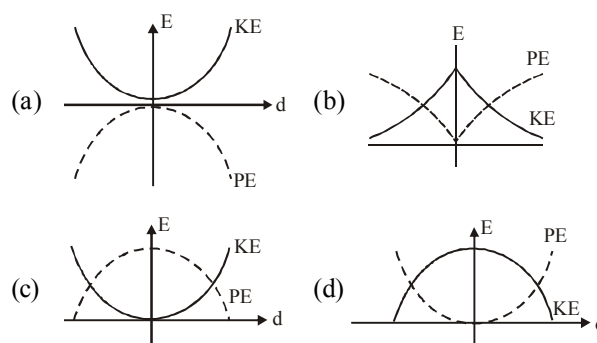
25. A particle is executing simple harmonic motion with a time period T. At time $t = 0$, it is at its position of equilibrium. The kinetic energy-time graph of the particle will look like: [2017]



26. A block of mass 0.1 kg is connected to an elastic spring of spring constant 640 Nm^{-1} and oscillates in a medium of constant $10^{-2} \text{ kg s}^{-1}$. The system dissipates its energy gradually. The time taken for its mechanical energy of vibration to drop to half of its initial value, is closest to : [Online April 9, 2017]

- (a) 2 s (b) 3.5 s (c) 5 s (d) 7 s

27. For a simple pendulum, a graph is plotted between its kinetic energy (KE) and potential energy (PE) against its displacement d. Which one of the following represents these correctly? (graphs are schematic and not drawn to scale) [2015]



28. A pendulum with time period of 1s is losing energy. At certain time its energy is 45 J. If after completing 15 oscillations, its energy has become 15 J, its damping constant (in s^{-1}) is : **[Online April 11, 2015]**

(a) $\frac{1}{2}$ (b) $\frac{1}{30}\ln 3$ (c) 2 (d) $\frac{1}{15}\ln 3$

29. This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.

If two springs S_1 and S_2 of force constants k_1 and k_2 respectively, are stretched by the same force, it is found that more work is done on spring S_1 than on spring S_2 .

Statement 1 : If stretched by the same amount work done on S_1

Statement 2 : $k_1 < k_2$ **[2012]**

- (a) Statement 1 is false, Statement 2 is true.
 (b) Statement 1 is true, Statement 2 is false.
 (c) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation for Statement 1
 (d) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation for Statement 1

30. A particle of mass m executes simple harmonic motion with amplitude a and frequency ν . The average kinetic energy during its motion from the position of equilibrium to the end is **[2007]**

(a) $2\pi^2 ma^2 \nu^2$ (b) $\pi^2 ma^2 \nu^2$
 (c) $\frac{1}{4} ma^2 \nu^2$ (d) $4\pi^2 ma^2 \nu^2$

31. Starting from the origin a body oscillates simple harmonically with a period of 2 s. After what time will its kinetic energy be 75% of the total energy? **[2006]**

(a) $\frac{1}{6}\text{s}$ (b) $\frac{1}{4}\text{s}$ (c) $\frac{1}{3}\text{s}$ (d) $\frac{1}{12}\text{s}$

32. The total energy of a particle, executing simple harmonic motion is **[2004]**

(a) independent of x (b) $\propto x^2$
 (c) $\propto x$ (d) $\propto x^{1/2}$

where x is the displacement from the mean position, hence total energy is independent of x .

33. A body executes simple harmonic motion. The potential energy (P.E), the kinetic energy (K.E) and total energy (T.E) are measured as a function of displacement x . Which of the following statements is true ? **[2003]**

- (a) K.E. is maximum when $x = 0$
 (b) T.E is zero when $x = 0$
 (c) K.E is maximum when x is maximum
 (d) P.E is maximum when $x = 0$

34. In a simple harmonic oscillator, at the mean position **[2002]**

- (a) kinetic energy is minimum, potential energy is maximum
 (b) both kinetic and potential energies are maximum
 (c) kinetic energy is maximum, potential energy is minimum
 (d) both kinetic and potential energies are minimum

TOPIC 3 Time Period, Frequency, Simple Pendulum and Spring Pendulum



35. An object of mass m is suspended at the end of a massless wire of length L and area of cross-section, A . Young modulus of the material of the wire is Y . If the mass is pulled down slightly its frequency of oscillation along the vertical direction is: **[Sep. 06, 2020 (I)]**

(a) $f = \frac{1}{2\pi} \sqrt{\frac{mL}{YA}}$ (b) $f = \frac{1}{2\pi} \sqrt{\frac{YA}{mL}}$
 (c) $f = \frac{1}{2\pi} \sqrt{\frac{mA}{YL}}$ (d) $f = \frac{1}{2\pi} \sqrt{\frac{YL}{mA}}$

36. When a particle of mass m is attached to a vertical spring of spring constant k and released, its motion is described by $y(t) = y_0 \sin^2 \omega t$, where ' y ' is measured from the lower end of unstretched spring. Then ω is : **[Sep. 06, 2020 (II)]**

(a) $\frac{1}{2} \sqrt{\frac{g}{y_0}}$ (b) $\sqrt{\frac{g}{y_0}}$
 (c) $\sqrt{\frac{g}{2y_0}}$ (d) $\sqrt{\frac{2g}{y_0}}$

37. A block of mass m attached to a massless spring is performing oscillatory motion of amplitude ' A ' on a frictionless horizontal plane. If half of the mass of the block breaks off when it is passing through its equilibrium point, the amplitude of oscillation for the remaining system become fA . The value of f is : **[Sep. 03, 2020 (II)]**

(a) $\frac{1}{\sqrt{2}}$ (b) 1 (c) $\frac{1}{2}$ (d) $\sqrt{2}$

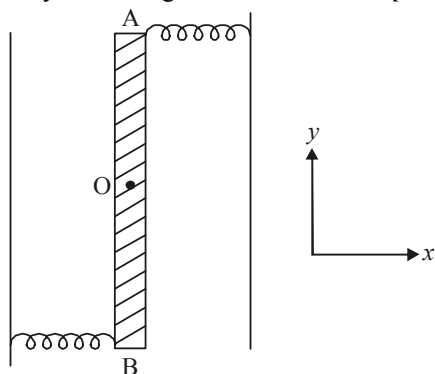
38. A person of mass M is, sitting on a swing of length L and swinging with an angular amplitude θ_0 . If the person stands up when the swing passes through its lowest point, the work done by him, assuming that his centre of mass moves by a distance l ($l \ll L$), is close to : **[12 April 2019, II]**

(a) $mg l (1 - \theta_0^2)$ (b) $mg l (1 + \theta_0^2)$
 (c) $mg l$ (d) $Mg l \left(1 + \frac{\theta_0^2}{2}\right)$

39. A simple pendulum oscillating in air has period T . The bob of the pendulum is completely immersed in a non-viscous liquid. The density of the liquid is $\frac{1}{16}$ th of the material of the bob. If the bob is inside liquid all the time, its period of oscillation in this liquid is : **[9 April 2019 I]**

(a) $2T \sqrt{\frac{1}{10}}$ (b) $2T \sqrt{\frac{1}{14}}$ (c) $4T \sqrt{\frac{1}{15}}$ (d) $4T \sqrt{\frac{1}{14}}$

40. Two light identical springs of spring constant k are attached horizontally at the two ends of a uniform horizontal rod AB of length l and mass m . The rod is pivoted at its centre 'O' and can rotate freely in horizontal plane. The other ends of two springs are fixed to rigid supports as shown in figure. The rod is gently pushed through a small angle and released. The frequency of resulting oscillation is: [12 Jan 2019, I]



- (a) $\frac{1}{2\pi} \sqrt{\frac{3k}{m}}$ (b) $\frac{1}{2\pi} \sqrt{\frac{2k}{m}}$ (c) $\frac{1}{2\pi} \sqrt{\frac{6k}{m}}$ (d) $\frac{1}{2\pi} \sqrt{\frac{k}{m}}$
41. A simple pendulum, made of a string of length l and a bob of mass m , is released from a small angle θ_0 . It strikes a block of mass M , kept on a horizontal surface at its lowest point of oscillations, elastically. It bounces back and goes up to an angle θ_1 . The M is given by: [12 Jan 2019, I]
- (a) $\frac{m}{2} \left(\frac{\theta_0 + \theta_1}{\theta_0 - \theta_1} \right)$ (b) $m \left(\frac{\theta_0 - \theta_1}{\theta_0 + \theta_1} \right)$
- (c) $m \left(\frac{\theta_0 + \theta_1}{\theta_0 - \theta_1} \right)$ (d) $\frac{m}{2} \left(\frac{\theta_0 - \theta_1}{\theta_0 + \theta_1} \right)$
42. A simple harmonic motion is represented by:
 $y = 5(\sin 3\pi t + \sqrt{3} \cos 3\pi t)$ cm
 The amplitude and time period of the motion are: [12 Jan 2019, II]
- (a) 10 cm, $\frac{2}{3}$ s (b) 10 cm, $\frac{3}{2}$ s
- (c) 5 cm, $\frac{3}{2}$ s (d) 5 cm, $\frac{2}{3}$ s
43. A simple pendulum of length 1 m is oscillating with an angular frequency 10 rad/s. The support of the pendulum starts oscillating up and down with a small angular frequency of 1 rad/s and an amplitude of 10^{-2} m. The relative change in the angular frequency of the pendulum is best given by: [11 Jan 2019, II]
- (a) 10^{-3} rad/s (b) 1 rad/s
- (c) 10^{-1} rad/s (d) 10^{-5} rad/s
44. The mass and the diameter of a planet are three times the respective values for the Earth. The period of oscillation of a simple pendulum on the Earth is 2s. The period of oscillation of the same pendulum on the planet would be: [11 Jan 2019, II]

- (a) $\frac{\sqrt{3}}{2}$ s (b) $\frac{2}{\sqrt{3}}$ s
- (c) $\frac{3}{2}$ s (d) $2\sqrt{3}$ s
45. A particle executes simple harmonic motion with an amplitude of 5 cm. When the particle is at 4 cm from the mean position, the magnitude of its velocity in SI units is equal to that of its acceleration. Then, its periodic time in seconds is: [10 Jan 2019, II]
- (a) $\frac{4\pi}{3}$ (b) $\frac{3}{8}\pi$
- (c) $\frac{8\pi}{3}$ (d) $\frac{7}{3}\pi$
46. A cylindrical plastic bottle of negligible mass is filled with 310 ml of water and left floating in a pond with still water. If pressed downward slightly and released, it starts performing simple harmonic motion at angular frequency ω . If the radius of the bottle is 2.5 cm then ω is close to: (density of water = 10^3 kg/m³) [10 Jan 2019, II]
- (a) 3.75 rad s⁻¹ (b) 1.25 rad s⁻¹
- (c) 2.50 rad s⁻¹ (d) 5.00 rad s⁻¹
47. A rod of mass 'M' and length '2L' is suspended at its middle by a wire. It exhibits torsional oscillations; If two masses each of 'm' are attached at distance 'L/2' from its centre on both sides, it reduces the oscillation frequency by 20%. The value of ratio m/M is close to: [9 Jan 2019, II]
- (a) 0.77 (b) 0.57
- (c) 0.37 (d) 0.17
48. A silver atom in a solid oscillates in simple harmonic motion in some direction with a frequency of 10^{12} /sec. What is the force constant of the bonds connecting one atom with the other? (Mole wt. of silver = 108 and Avagadro number = 6.02×10^{23} gm mole⁻¹) [2018]
- (a) 6.4 N/m (b) 7.1 N/m (c) 2.2 N/m (d) 5.5 N/m
49. A particle executes simple harmonic motion and is located at $x = a$, b and c at times t_0 , $2t_0$ and $3t_0$ respectively. The frequency of the oscillation is [Online April 16, 2018]
- (a) $\frac{1}{2\pi t_0} \cos^{-1} \left(\frac{a+b}{2c} \right)$ (b) $\frac{1}{2\pi t_0} \cos^{-1} \left(\frac{a+b}{3c} \right)$
- (c) $\frac{1}{2\pi t_0} \cos^{-1} \left(\frac{2a+3c}{b} \right)$ (d) $\frac{1}{2\pi t_0} \cos^{-1} \left(\frac{a+c}{2b} \right)$
50. In an experiment to determine the period of a simple pendulum of length 1 m, it is attached to different spherical bobs of radii r_1 and r_2 . The two spherical bobs have uniform mass distribution. If the relative difference in the periods, is found to be 5×10^{-4} s, the difference in radii, $|r_1 - r_2|$ is best given by: [Online April 9, 2017]
- (a) 1 cm (b) 0.1 cm (c) 0.5 cm (d) 0.01 cm

51. A 1 kg block attached to a spring vibrates with a frequency of 1 Hz on a frictionless horizontal table. Two springs identical to the original spring are attached in parallel to an 8 kg block placed on the same table. So, the frequency of vibration of the 8 kg block is : **[Online April 8, 2017]**

(a) $\frac{1}{4}$ Hz (b) $\frac{1}{2\sqrt{2}}$ Hz (c) $\frac{1}{2}$ Hz (d) 2 Hz

52. A pendulum clock loses 12 s a day if the temperature is 40°C and gains 4 s a day if the temperature is 20°C . The temperature at which the clock will show correct time, and the co-efficient of linear expansion (α) of the metal of the pendulum shaft are respectively : **[2016]**

(a) 30°C ; $\alpha = 1.85 \times 10^{-3}/^\circ\text{C}$
 (b) 55°C ; $\alpha = 1.85 \times 10^{-2}/^\circ\text{C}$
 (c) 25°C ; $\alpha = 1.85 \times 10^{-5}/^\circ\text{C}$
 (d) 60°C ; $\alpha = 1.85 \times 10^{-4}/^\circ\text{C}$

53. In an engine the piston undergoes vertical simple harmonic motion with amplitude 7 cm. A washer rests on top of the piston and moves with it. The motor speed is slowly increased. The frequency of the piston at which the washer no longer stays in contact with the piston, is close to : **[Online April 10, 2016]**

(a) 0.7 Hz (b) 1.9 Hz (c) 1.2 Hz (d) 0.1 Hz

54. A pendulum made of a uniform wire of cross sectional area A has time period T . When an additional mass M is added to its bob, the time period changes to T_M . If the Young's modulus of the material of the wire is Y then $\frac{1}{Y}$ is equal to:

to:

($g = \text{gravitational acceleration}$) **[2015]**

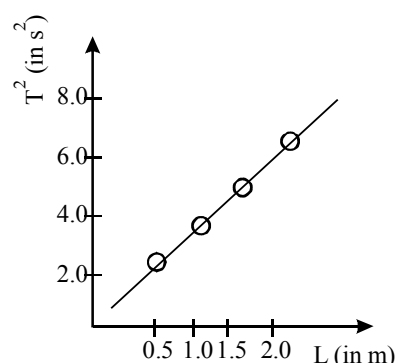
(a) $\left[1 - \left(\frac{T_M}{T}\right)^2\right] \frac{A}{Mg}$ (b) $\left[1 - \left(\frac{T}{T_M}\right)^2\right] \frac{A}{Mg}$
 (c) $\left[\left(\frac{T_M}{T}\right)^2 - 1\right] \frac{A}{Mg}$ (d) $\left[\left(\frac{T_M}{T}\right)^2 - 1\right] \frac{Mg}{A}$

55. A particle moves with simple harmonic motion in a straight line. In first τ s, after starting from rest it travels a distance a , and in next τ s it travels $2a$, in same direction, then:

(a) amplitude of motion is $3a$ **[2014]**
 (b) time period of oscillations is 8τ
 (c) amplitude of motion is $4a$
 (d) time period of oscillations is 6τ

56. In an experiment for determining the gravitational acceleration g of a place with the help of a simple pendulum, the measured time period square is plotted against the string length of the pendulum in the figure.

[Online April 19, 2014]



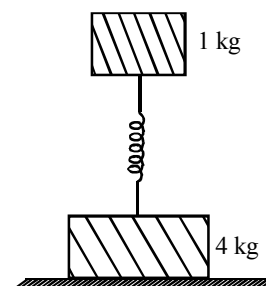
What is the value of g at the place?

(a) 9.81 m/s^2 (b) 9.87 m/s^2
 (c) 9.91 m/s^2 (d) 10.0 m/s^2

57. The amplitude of a simple pendulum, oscillating in air with a small spherical bob, decreases from 10 cm to 8 cm in 40 seconds. Assuming that Stokes law is valid, and ratio of the coefficient of viscosity of air to that of carbon dioxide is 1.3. The time in which amplitude of this pendulum will reduce from 10 cm to 5 cm in carbon dioxide will be close to ($\ln 5 = 1.601$, $\ln 2 = 0.693$). **[Online April 9, 2014]**

(a) 231 s (b) 208 s (c) 161 s (d) 142 s

58. Two bodies of masses 1 kg and 4 kg are connected to a vertical spring, as shown in the figure. The smaller mass executes simple harmonic motion of angular frequency 25 rad/s, and amplitude 1.6 cm while the bigger mass remains stationary on the ground. The maximum force exerted by the system on the floor is (take $g = 10 \text{ ms}^{-2}$) **[Online April 9, 2014]**



(a) 20 N (b) 10 N (c) 60 N (d) 40 N

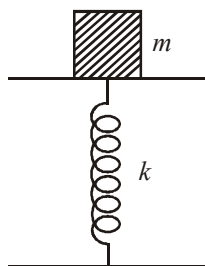
59. An ideal gas enclosed in a vertical cylindrical container supports a freely moving piston of mass M . The piston and the cylinder have equal cross sectional area A . When the piston is in equilibrium, the volume of the gas is V_0 and its pressure is P_0 . The piston is slightly displaced from the equilibrium position and released. Assuming that the system is completely isolated from its surrounding, the piston executes a simple harmonic motion with frequency **[2013]**

(a) $\frac{1}{2\pi} \frac{A\gamma P_0}{V_0 M}$ (b) $\frac{1}{2\pi} \frac{V_0 M P_0}{A^2 \gamma}$
 (c) $\frac{1}{2\pi} \sqrt{\frac{A^2 \gamma P_0}{M V_0}}$ (d) $\frac{1}{2\pi} \sqrt{\frac{M V_0}{A \gamma P_0}}$

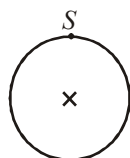
60. A mass $m = 1.0$ kg is put on a flat pan attached to a vertical spring fixed on the ground. The mass of the spring and the pan is negligible. When pressed slightly and released, the mass executes simple harmonic motion. The spring constant is 500 N/m. What is the amplitude A of the motion, so that the mass m tends to get detached from the pan?

(Take $g = 10$ m/s²).

The spring is stiff enough so that it does not get distorted during the motion. [Online April 22, 2013]



- (a) $A > 2.0$ cm (b) $A = 2.0$ cm
(c) $A < 2.0$ cm (d) $A = 1.5$ cm
61. Two simple pendulums of length 1 m and 4 m respectively are both given small displacement in the same direction at the same instant. They will be again in phase after the shorter pendulum has completed number of oscillations equal to : [Online April 9, 2013]
- (a) 2 (b) 7 (c) 5 (d) 3
62. If a simple pendulum has significant amplitude (up to a factor of $1/e$ of original) only in the period between $t = 0$ s to $t = \tau$ s, then τ may be called the average life of the pendulum. When the spherical bob of the pendulum suffers a retardation (due to viscous drag) proportional to its velocity with b as the constant of proportionality, the average life time of the pendulum in second is (assuming damping is small) [2012]
- (a) $\frac{0.693}{b}$ (b) b (c) $\frac{1}{b}$ (d) $\frac{2}{b}$
63. A ring is suspended from a point S on its rim as shown in the figure. When displaced from equilibrium, it oscillates with time period of 1 second. The radius of the ring is (take $g = \pi^2$) [Online May 19, 2012]



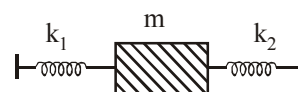
- (a) 0.15 m (b) 1.5 m (c) 1.0 m (d) 0.5 m
64. A wooden cube (density of wood ' d ') of side ' ℓ ' floats in a liquid of density ' ρ ' with its upper and lower surfaces horizontal. If the cube is pushed slightly down and released, it performs simple harmonic motion of period T [2011 RS]

(a) $2\pi\sqrt{\frac{\ell d}{\rho g}}$ (b) $2\pi\sqrt{\frac{\ell \rho}{dg}}$
(c) $2\pi\sqrt{\frac{\ell d}{(\rho - d)g}}$ (d) $2\pi\sqrt{\frac{\ell \rho}{(\rho - d)g}}$

65. If x , v and a denote the displacement, the velocity and the acceleration of a particle executing simple harmonic motion of time period T , then, which of the following does not change with time? [2009]

(a) aT/x (b) $aT + 2\pi v$
(c) aT/v (d) $a^2T^2 + 4\pi^2v^2$

66. Two springs, of force constants k_1 and k_2 are connected to a mass m as shown. The frequency of oscillation of the mass is f . If both k_1 and k_2 are made four times their original values, the frequency of oscillation becomes [2007]



- (a) $2f$ (b) $f/2$ (c) $f/4$ (d) $4f$
67. The displacement of an object attached to a spring and executing simple harmonic motion is given by $x = 2 \times 10^{-2} \cos \pi t$ metre. The time at which the maximum speed first occurs is [2007]

(a) 0.25 s (b) 0.5 s (c) 0.75 s (d) 0.125 s

68. The bob of a simple pendulum is a spherical hollow ball filled with water. A plugged hole near the bottom of the oscillating bob gets suddenly unplugged. During observation, till water is coming out, the time period of oscillation would [2005]

- (a) first decrease and then increase to the original value
(b) first increase and then decrease to the original value
(c) increase towards a saturation value
(d) remain unchanged

69. If a simple harmonic motion is represented by $\frac{d^2x}{dt^2} + \alpha x = 0$, its time period is [2005]

(a) $\frac{2\pi}{\sqrt{\alpha}}$ (b) $\frac{2\pi}{\alpha}$ (c) $2\pi\sqrt{\alpha}$ (d) $2\pi\alpha$

70. The bob of a simple pendulum executes simple harmonic motion in water with a period t , while the period of oscillation of the bob is t_0 in air. Neglecting frictional force of water and given that the density of the bob is $(4/3) \times 1000$ kg/m³. Which relationship between t and t_0 is true? [2004]

(a) $t = 2t_0$ (b) $t = t_0/2$
(c) $t = t_0$ (d) $t = 4t_0$

71. A particle at the end of a spring executes S.H.M with a period t_1 , while the corresponding period for another spring is t_2 . If the period of oscillation with the two springs in series is T then [2004]
- (a) $T^{-1} = t_1^{-1} + t_2^{-1}$ (b) $T^2 = t_1^2 + t_2^2$
 (c) $T = t_1 + t_2$ (d) $T^{-2} = t_1^{-2} + t_2^{-2}$
72. A mass M is suspended from a spring of negligible mass. The spring is pulled a little and then released so that the mass executes SHM of time period T . If the mass is increased by m , the time period becomes $\frac{5T}{3}$. Then the ratio of $\frac{m}{M}$ is [2003]
- (a) $\frac{3}{5}$ (b) $\frac{25}{9}$ (c) $\frac{16}{9}$ (d) $\frac{5}{3}$
73. The length of a simple pendulum executing simple harmonic motion is increased by 21%. The percentage increase in the time period of the pendulum of increased length is [2003]
- (a) 11% (b) 21% (c) 42% (d) 10%
74. If a spring has time period T , and is cut into n equal parts, then the time period of each part will be [2002]
- (a) $T\sqrt{n}$ (b) $T/\sqrt{n}$ (c) nT (d) T
75. A child swinging on a swing in sitting position, stands up, then the time period if the swing will [2002]
- (a) increase
 (b) decrease
 (c) remains same
 (d) increases if the child is long and decreases if the child is short
78. An oscillator of mass M is at rest in its equilibrium position in a potential $V = \frac{1}{2}k(x - X)^2$. A particle of mass m comes from right with speed u and collides completely inelastically with M and sticks to it. This process repeats every time the oscillator crosses its equilibrium position. The amplitude of oscillations after 13 collisions is: ($M = 10, m = 5, u = 1, k = 1$). [Online April 16, 2018]
- (a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{3}}$
 (c) $\frac{2}{3}$ (d) $\sqrt{\frac{3}{5}}$
79. The angular frequency of the damped oscillator is given by, $\omega = \sqrt{\left(\frac{k}{m} - \frac{r^2}{4m^2}\right)}$ where k is the spring constant, m is the mass of the oscillator and r is the damping constant. If the ratio $\frac{r^2}{mk}$ is 8%, the change in time period compared to the undamped oscillator is approximately as follows: [Online April 11, 2014]
- (a) increases by 1% (b) increases by 8%
 (c) decreases by 1% (d) decreases by 8%
80. The amplitude of a damped oscillator decreases to 0.9 times its original magnitude in 5s. In another 10s it will decrease to α times its original magnitude, where α equals [2013]
- (a) 0.7 (b) 0.81 (c) 0.729 (d) 0.6
81. A uniform cylinder of length L and mass M having cross-sectional area A is suspended, with its length vertical, from a fixed point by a massless spring, such that it is half submerged in a liquid of density σ at equilibrium position. When the cylinder is given a downward push and released, it starts oscillating vertically with a small amplitude. The time period T of the oscillations of the cylinder will be : [Online April 25, 2013]

TOPIC 4 Damped, Forced Oscillations and Resonance



76. A damped harmonic oscillator has a frequency of 5 oscillations per second. The amplitude drops to half its value for every 10 oscillations. The time it will take to drop to $\frac{1}{1000}$ of the original amplitude is close to : [8 April 2019, II]
- (a) 50s (b) 100s (c) 20s (d) 10s
77. The displacement of a damped harmonic oscillator is given by $x(t) = e^{-0.1t} \cdot \cos(10\pi t + \phi)$. Here t is in seconds. The time taken for its amplitude of vibration to drop to half of its initial value is close to : [9 Jan 2019, II]
- (a) 4s (b) 7s (c) 13s (d) 27s

- (a) Smaller than $2\pi \left[\frac{M}{(k + A\sigma g)} \right]^{\frac{1}{2}}$
 (b) $2\pi \sqrt{\frac{M}{k}}$
 (c) Larger than $2\pi \left[\frac{M}{(k + A\sigma g)} \right]^{\frac{1}{2}}$
 (d) $2\pi \left[\frac{M}{(k + A\sigma g)} \right]^{\frac{1}{2}}$

82. Bob of a simple pendulum of length l is made of iron. The pendulum is oscillating over a horizontal coil carrying direct current. If the time period of the pendulum is T then : **[Online April 23, 2013]**
- $T < 2\pi\sqrt{\frac{l}{g}}$ and damping is smaller than in air alone.
 - $T = 2\pi\sqrt{\frac{l}{g}}$ and damping is larger than in air alone.
 - $T > 2\pi\sqrt{\frac{l}{g}}$ and damping is smaller than in air alone.
 - $T < 2\pi\sqrt{\frac{l}{g}}$ and damping is larger than in air alone.
83. In forced oscillation of a particle the amplitude is maximum for a frequency ω_1 of the force while the energy is maximum for a frequency ω_2 of the force; then **[2004]**
- $\omega_1 < \omega_2$ when damping is small and $\omega_1 > \omega_2$ when damping is large
 - $\omega_1 > \omega_2$
 - $\omega_1 = \omega_2$
 - $\omega_1 < \omega_2$
84. A particle of mass m is attached to a spring (of spring constant k) and has a natural angular frequency ω_0 . An external force $F(t)$ proportional to $\cos \omega t$ ($\omega \neq \omega_0$) is applied to the oscillator. The displacement of the oscillator will be proportional to **[2004]**
- $\frac{1}{m(\omega_0^2 + \omega^2)}$
 - $\frac{1}{m(\omega_0^2 - \omega^2)}$
 - $\frac{m}{\omega_0^2 - \omega^2}$
 - $\frac{m}{(\omega_0^2 + \omega^2)}$



Hints & Solutions



1. (a) Here, $v_x = -a\omega \sin \omega t$, $v_y = a\omega \cos \omega t$ and $v_z = a\omega$

$$\Rightarrow v = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

$$\Rightarrow v = \sqrt{(-a\omega \sin \omega t)^2 + (a\omega \cos \omega t)^2 + (a\omega)^2}$$

$$v = \sqrt{2}a\omega$$

2. (c) From the two mutually perpendicular S.H.M. 's, the general equation of Lissajous figure,

$$\frac{x^2}{A^2} + \frac{y^2}{B^2} - \frac{2xy}{AB} \cos \delta = \sin^2 \delta$$

$$x = A \sin (at + \delta)$$

$$y = B \sin (bt + r)$$

Clearly $A \neq B$ hence ellipse.

3. (b) Maximum velocity in SHM, $v_{\max} = a\omega$
Maximum acceleration in SHM, $A_{\max} = a\omega^2$
where a and ω are maximum amplitude and angular frequency.

Given that, $\frac{A_{\max}}{v_{\max}} = 10$

i.e., $\omega = 10 \text{ s}^{-1}$

Displacement is given by

$$x = a \sin (\omega t + \pi/4)$$

at $t=0$, $x=5$

$$5 = a \sin \pi/4$$

$$5 = a \sin 45^\circ \Rightarrow a = 5\sqrt{2}$$

Maximum acceleration $A_{\max} = a\omega^2 = 500\sqrt{2} \text{ m/s}^2$

4. (b) We know that $V = \omega \sqrt{A^2 - x^2}$

$$\text{Initially } V = \omega \sqrt{A^2 - \left(\frac{2A}{3}\right)^2}$$

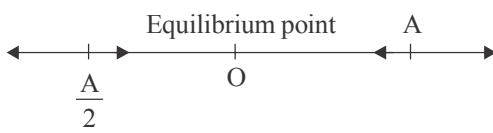
$$\text{Finally } 3V = \omega \sqrt{A'^2 - \left(\frac{2A}{3}\right)^2}$$

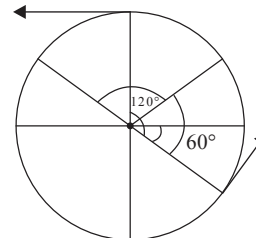
Where A' = final amplitude (Given at $x = \frac{2A}{3}$, velocity to trebled)

On dividing we get

$$\frac{3}{1} = \frac{\sqrt{A'^2 - \left(\frac{2A}{3}\right)^2}}{\sqrt{A^2 - \left(\frac{2A}{3}\right)^2}}$$

$$9 \left[A^2 - \frac{4A^2}{9} \right] = A'^2 - \frac{4A^2}{9} \therefore A' = \frac{7A}{3}$$

5. (d) 
(at time $t=0$)



Angle covered to meet $\theta = 60^\circ = \frac{\pi}{3} \text{ rad}$.

If they cross each other at time t then

$$t = \frac{\theta}{2\pi} = \frac{\pi}{3 \times 2\pi} T = \frac{T}{6}$$

6. (c) As we know,

$$F = ma \Rightarrow a \propto F$$

$$\text{or, } a \propto \sin t$$

$$\Rightarrow \frac{dv}{dt} \propto \sin t$$

$$\Rightarrow \int_0^t dV \propto \int_0^t \sin t \, dt$$

$$V \propto -\cos t + 1$$

$$\int_0^x dx = \int_0^t (-\cos t + 1) \, dt$$

$$x = \sin t - \frac{1}{2} \sin 2t$$

7. (c) At $t=0$, $x(t)=0$; $y(t)=0$

$x(t)$ is a sinusoidal function

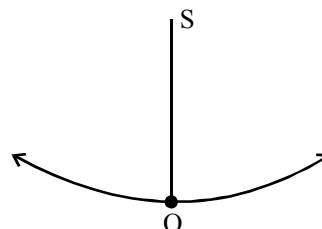
$$\text{At } t = \frac{\pi}{2\omega}; x(t) = a \text{ and } y(t) = 0$$

Hence trajectory of particle will look like as (c).

8. (c) **Given:** Time period, $T = 0.5 \text{ sec}$

Amplitude, $A = 1 \text{ cm}$

Average velocity in the interval in which body moves from equilibrium to half of its amplitude, $v = ?$



Time taken to a displacement $A/2$ where A is the amplitude of oscillation from the mean position 'O' is $\frac{T}{12}$

Therefore, time, $t = \frac{0.5}{12}$ sec

Displacement, $s = \frac{A}{2} = \frac{1}{2}$ cm

$\therefore$ Average velocity, $v = \frac{\frac{A}{2}}{\frac{0.5}{12}} = \frac{\frac{1}{2}}{\frac{0.5}{12}} = 12$ cm/s

9. (d) In linear S.H.M., the restoring force acting on particle should always be proportional to the displacement of the particle and directed towards the equilibrium position.

i.e., $F \propto x$

or $F = -bx$ where b is a positive constant.

10. (b) Two perpendicular S.H.M.s are

$$x = a_1 \cos \omega t \quad \dots(1)$$

$$\text{and } y = a_2 2 \cos \omega t \quad \dots(2)$$

From eqn (1)

$$\frac{x}{a_1} = \cos \omega t$$

and from eqn (2)

$$\frac{y}{a_2} = 2 \cos \omega t$$

$$\therefore y = 2 \frac{a_2}{a_1} x$$

11. (a) Displacement $y(t) = A \sin(\omega t + \phi)$ [Given]

$$\text{For } \phi = \frac{2\pi}{3}$$

$$\text{at } t = 0; y = A \sin \phi = A \sin \frac{2\pi}{3}$$

$$= A \sin 120^\circ = 0.87A \quad [\because \sin 120^\circ = 0.866]$$

Graph (a) depicts $y = 0.87A$ at $t = 0$

12. (a) Let, $x_1 = A \sin \omega t$ and $x_2 = A \sin(\omega t + \phi)$

$$x_2 - x_1 = 2A \cos\left(\omega t + \frac{\phi}{2}\right) \sin \frac{\phi}{2}$$

The above equation is SHM with amplitude $2A \sin \frac{\phi}{2}$

$$\therefore 2A \sin \frac{\phi}{2} = A$$

$$\Rightarrow \sin \frac{\phi}{2} = \frac{1}{2} \Rightarrow \phi = \frac{\pi}{3}$$

13. (c) At mean position, $F_{\text{net}} = 0$

Therefore, by principle of conservation of linear momentum.

$$\therefore Mv_1 = (M+m)v_2$$

$$Mw, a, = (M+m)w_2 a_2$$

$$MA_1 \sqrt{\frac{k}{M}} = (M+m)A_2 \sqrt{\frac{k}{m+M}}$$

$$\therefore \left(V = A_1 \sqrt{\frac{k}{M}} \right)$$

$$\Rightarrow A_1 \sqrt{M} = A_2 \sqrt{M+m}$$

$$\Rightarrow \frac{A_1}{A_2} = \sqrt{\frac{m+M}{M}}$$

14. (a) Given,

Displacement, $x = x_0 \cos(\omega t - \pi/4)$

$$\therefore \text{Velocity, } v = \frac{dx}{dt} = -x_0 \omega \sin\left(\omega t - \frac{\pi}{4}\right)$$

Acceleration,

$$a = \frac{dv}{dt} = -x_0 \omega^2 \cos\left(\omega t - \frac{\pi}{4}\right)$$

$$= x_0 \omega^2 \cos\left[\pi + \left(\omega t - \frac{\pi}{4}\right)\right]$$

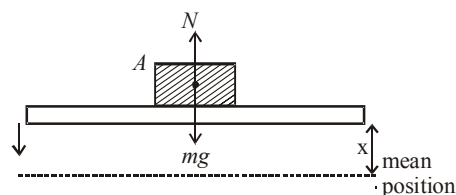
$$= x_0 \omega^2 \cos\left(\omega t + \frac{3\pi}{4}\right) \quad \dots(1)$$

Acceleration, $a = A \cos(\omega t + \delta) \quad \dots(2)$

Comparing the two equations, we get

$$A = x_0 \omega^2 \text{ and } \delta = \frac{3\pi}{4}$$

15. (b) For block A to move in SHM.



$$mg - N = m\omega^2 x$$

where x is the distance from mean position

For block to leave contact $N = 0$

$$\Rightarrow mg = m\omega^2 x \Rightarrow x = \frac{g}{\omega^2}$$

16. (a) Maximum velocity,

$$v_{\text{max}} = a\omega$$

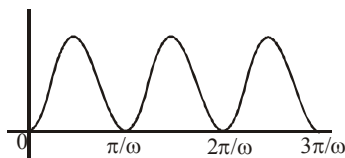
Here, a = amplitude of SHM

ω = angular velocity of SHM

$$v_{\text{max}} = a \times \frac{2\pi}{T} \therefore \left(\because \omega = \frac{2\pi}{T} \right)$$

$$\Rightarrow T = \frac{2\pi a}{v_{\text{max}}} = \frac{2 \times 3.14 \times 7 \times 10^{-3}}{4.4} \approx 0.01 \text{ s}$$

17. (a) Clearly $\sin 2\omega t$ is a periodic function with period $\frac{\pi}{\omega}$



For SHM $\frac{d^2y}{dt^2} \propto -y$

$$y = \sin^2 \omega t = \frac{1 - \cos 2\omega t}{2}$$

$$= \frac{1}{2} - \frac{1}{2} \cos 2\omega t$$

$$v = \frac{dy}{dt} = \frac{1}{2} \times 2\omega \sin 2\omega t = 2\omega \sin \omega t \cos \omega t$$

$$= \omega \sin 2\omega t$$

$$\text{Acceleration, } a = \frac{d^2y}{dt^2} = 2\omega^2 \cos 2\omega t \text{ which is not}$$

proportional to $-y$. Hence, it is not in SHM.

18. (b) Velocity of particle 1,

$$v_1 = \frac{dy_1}{dt} = 0.1 \times 100\pi \cos\left(100\pi t + \frac{\pi}{3}\right)$$

Velocity of particle 2,

$$v_2 = \frac{dy_2}{dt} = -0.1\pi \sin \pi t = 0.1\pi \cos\left(\pi t + \frac{\pi}{2}\right)$$

$\therefore$ Phase difference of velocity of particle 1 with respect to the velocity of particle 2 is

$$= \phi_1 - \phi_2 = \frac{\pi}{3} - \frac{\pi}{2} = \frac{2\pi - 3\pi}{6} = -\frac{\pi}{6}$$

19. (c) Maximum velocity during SHM, $V_{\max} = A\omega$
But $k = m\omega^2$

$$\therefore \omega = \sqrt{\frac{k}{m}}$$

$\therefore$ Maximum velocity of the body in SHM

$$= A\sqrt{\frac{k}{m}}$$

As maximum velocities are equal

$$\therefore A_1\sqrt{\frac{k_1}{m}} = A_2\sqrt{\frac{k_2}{m}}$$

$$\Rightarrow A_1\sqrt{k_1} = A_2\sqrt{k_2} \Rightarrow \frac{A_1}{A_2} = \sqrt{\frac{k_2}{k_1}}$$

20. (c) Displacement, $x = 4(\cos \pi t + \sin \pi t)$

$$= \sqrt{2} \times 4 \left(\frac{\sin \pi t}{\sqrt{2}} + \frac{\cos \pi t}{\sqrt{2}} \right)$$

$$= 4\sqrt{2}(\sin \pi t \cos 45^\circ + \cos \pi t \sin 45^\circ)$$

$$x = 4\sqrt{2} \sin(\pi t + 45^\circ)$$

On comparing it with standard equation $x = A \sin(\omega t + \phi)$

we get $A = 4\sqrt{2}$

21. (c) From graph equation of SHM

$$X = A \cos \omega t$$

- (1) At $\frac{3T}{4}$ particle is at mean position.

$\therefore$ Acceleration = 0, Force = 0

- (2) At T particle again at extreme position so acceleration is maximum.

- (3) At $t = \frac{T}{4}$, particle is at mean position so velocity is maximum.

Acceleration = 0

- (4) When KE = PE

$$\Rightarrow \frac{1}{2}k(A^2 - x^2) = \frac{1}{2}kx^2$$

Here, A = amplitude of SHM

x = displacement from mean position

$$\Rightarrow A^2 = 2x^2 \Rightarrow x = \frac{+A}{\sqrt{2}}$$

$$\Rightarrow \frac{A}{\sqrt{2}} = A \cos \omega t \Rightarrow t = \frac{T}{2}$$

$\therefore x = -A$ which is not possible

$\therefore$ 1, 2 and 3 are correct.

22. (d) Kinetic energy, $K = \frac{1}{2}m\omega^2 A^2 \cos^2 \omega t$

$$\text{Potential energy, } U = \frac{1}{2}m\omega^2 A^2 \sin^2 \omega t$$

$$\frac{K}{U} = \cot^2 \omega t = \cot^2 \frac{\pi}{90} (210) = \frac{1}{3}$$

23. (a) $K = \frac{1}{2}m\omega^2 x^2$

$$\Rightarrow K_{\max} = \frac{1}{2}m\omega^2 A^2$$

$$A = L\theta$$

$$\omega = \sqrt{\frac{g}{L}}$$

$$\Rightarrow K = \frac{1}{2}m \cdot \frac{g}{L} \cdot L^2 \theta^2$$

$$= \frac{1}{2}mgL\theta^2$$

$$\therefore \frac{K_1}{K_2} = \frac{L}{2L} = \frac{1}{2} \Rightarrow K_2 = 2K_1$$

$$\therefore \frac{K_1}{K_2} = \frac{L}{2L} = \frac{1}{2} \Rightarrow K_2 = 2K_1$$

24. (c) Potential energy (U) = $\frac{1}{2}kx^2$

$$\text{Kinetic energy (K)} = \frac{1}{2}kA^2 - \frac{1}{2}kx^2$$

According to the question, U = K

$$\therefore \frac{1}{2}kx^2 = \frac{1}{2}kA^2 - \frac{1}{2}kx^2$$

$$\Rightarrow x^2 = A^2 \text{ or, } x = \pm \frac{A}{\sqrt{2}}$$

25. (b) For a particle executing SHM

At mean position; $t = 0, \omega t = 0, y = 0, V = V_{\max} = a\omega$

$$\therefore K.E. = KE_{\max} = \frac{1}{2}m\omega^2 a^2$$

At extreme position : $t = \frac{T}{4}, \omega t = \frac{\pi}{2}, y = A, V = V_{\min} = 0$

$$\therefore K.E. = KE_{\min} = 0$$

$$\text{Kinetic energy in SHM, } KE = \frac{1}{2}m\omega^2(a^2 - y^2)$$

$$= \frac{1}{2}m\omega^2 a^2 \cos^2 \omega t$$

Hence graph (b) correctly depicts kinetic energy time graph.

26. (b) Since system dissipates its energy gradually, and hence amplitude will also decreases with time according to

$$a = a_0 e^{-bt/m} \dots\dots (i)$$

$\therefore$ Energy of vibration drop to half of its initial value (E_0), as $E \propto a^2 \Rightarrow a \propto \sqrt{E}$

$$a = \frac{a_0}{\sqrt{2}} \Rightarrow \frac{bt}{m} = \frac{10^{-2}t}{0.1} = \frac{t}{10}$$

From eqⁿ (i),

$$\frac{a_0}{\sqrt{2}} = a_0 e^{-t/10}$$

$$\frac{1}{\sqrt{2}} = e^{-t/10} \text{ or } \sqrt{2} = e^{t/10}$$

$$\ln \sqrt{2} = \frac{t}{10} \therefore t = 3.5 \text{ seconds}$$

27. (d) $K.E. = \frac{1}{2}k(A^2 - d^2)$

$$\text{and P.E.} = \frac{1}{2}kd^2$$

At mean position $d = 0$. At extreme positions $d = A$

28. (d) As we know, $E = E_0 e^{-\frac{bt}{m}}$

$$15 = 45 e^{-\frac{bt}{m}}$$

[As no. of oscillations = 15 so $t = 15\text{sec}$]

$$\frac{1}{3} = e^{-\frac{bt}{m}}$$

Taking log on both sides

$$\frac{b}{m} = \frac{1}{15} \ln 3$$

29. (b) Work done, $w = \frac{1}{2}kx^2$

$$\text{Work done by spring } S_1, w_1 = \frac{1}{2}k_1x^2$$

$$\text{Work done by spring } S_2, w_2 = \frac{1}{2}k_2x^2$$

Since $w_1 > w_2$ Thus ($k_1 > k_2$)

30. (b) The kinetic energy of a particle executing S.H.M. at any instant t is given by

$$K = \frac{1}{2}ma^2\omega^2 \sin^2 \omega t$$

where, m = mass of particle

a = amplitude

ω = angular frequency

t = time

The average value of $\sin^2 \omega t$ over a cycle is $\frac{1}{2}$.

$$\therefore K.E. = \frac{1}{2}m\omega^2 a^2 \left(\frac{1}{2} \right) \left(\because \langle \sin^2 \theta \rangle = \frac{1}{2} \right)$$

$$= \frac{1}{4}m\omega^2 a^2 = \frac{1}{4}ma^2 (2\pi\nu)^2 \quad (\because \omega = 2\pi\nu)$$

$$\text{or, } \langle K \rangle = \pi^2 ma^2 \nu^2$$

31. (a) K.E. of a body undergoing SHM is given by,

$$K.E. = \frac{1}{2}ma^2\omega^2 \cos^2 \omega t$$

Here, a = amplitude of SHM

ω = angular velocity of SHM

$$\text{Total energy in S.H.M} = \frac{1}{2}ma^2\omega^2$$

Given K.E. = 75% T.E.

$$\frac{1}{2}ma^2\omega^2 \cos^2 \omega t = \frac{75}{100} \times \frac{1}{2}ma^2\omega^2$$

$$\Rightarrow 0.75 = \cos^2 \omega t \Rightarrow \omega t = \frac{\pi}{6}$$

$$\Rightarrow t = \frac{\pi}{6\omega} \Rightarrow t = \frac{\pi \times 2}{6 \times 2\pi} \Rightarrow t = \frac{1}{6} \text{ s}$$

32. (a) At any instant the total energy in SHM is

$$\frac{1}{2}kA_0^2 = \text{constant,}$$

where A_0 = amplitude

k = spring constant

hence total energy is independent of x .

33. (a) K.E. of simple harmonic motion

$$= \frac{1}{2}m\omega^2(a^2 - x^2)$$

34. (c) The kinetic energy (K. E.) of particle in SHM is given by,

$$\text{K.E} = \frac{1}{2}k(A^2 - x^2);$$

Potential energy of particle in SHM is $U = \frac{1}{2}kx^2$

Where A = amplitude and $k = m\omega^2$

x = displacement from the mean position

At the mean position $x = 0$

$$\therefore \text{K.E.} = \frac{1}{2}kA^2 = \text{Maximum}$$

and $U = 0$

35. (b) An elastic wire can be treated as a spring and its spring constant.

$$k = \frac{YA}{L} \quad \left[\because Y = \frac{F}{A} / \frac{\Delta l}{l_0} \right]$$

Frequency of oscillation,

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{1}{2\pi} \sqrt{\frac{YA}{mL}}$$

36. (c) $y = y_0 \sin^2 \omega t$

$$\Rightarrow y = \frac{y_0}{2}(1 - \cos 2\omega t) \quad \left(\because \sin^2 \omega t = \frac{1 - \cos 2\omega t}{2} \right)$$

$$\Rightarrow y - \frac{y_0}{2} = -\frac{y_0}{2} \cos 2\omega t$$

$$\Rightarrow y = A \cos 2\omega t$$

$$\therefore \text{Amplitude} = \frac{y_0}{2}$$

Angular velocity = 2ω

$$\text{For equilibrium of mass, } \frac{ky_0}{2} = mg \Rightarrow \frac{k}{m} = \frac{2g}{y_0}$$

Also, spring constant $k = m(2\omega)^2$

$$\Rightarrow 2\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{2g}{y_0}} \Rightarrow \omega = \frac{1}{2} \sqrt{\frac{2g}{y_0}} = \sqrt{\frac{g}{2y_0}}$$

37. (a) Potential energy of spring = $\frac{1}{2}kx^2$

Here, x = distance of block from mean position,

k = spring constant

$$\text{At mean position, potential energy} = \frac{1}{2}kA^2$$

At equilibrium position, half of the mass of block breaks off, so its potential energy becomes half.

$$\text{Remaining energy} = \frac{1}{2} \left(\frac{1}{2}kA^2 \right) = \frac{1}{2}kA'^2$$

Here, A' = New distance of block from mean position

$$\Rightarrow A' = \frac{A}{\sqrt{2}}$$

38. (b)

39. (c) $T = 2\pi \sqrt{\frac{l}{g}}$

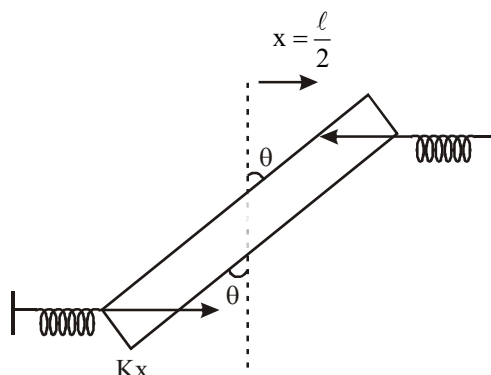
When immersed non viscous liquid

$$a_{mt} = \left(g - \frac{g}{16} \right) = \frac{15g}{16}$$

$$\text{Now } T' = 2\pi \sqrt{\frac{l}{a_{\text{net}}}} = 2\pi \sqrt{\frac{l}{\frac{15g}{16}}} = \frac{4}{\sqrt{15}} T$$

40. (c) Net torque due to spring force:

$$\tau = -2Kx \frac{\ell}{2} \cos \theta$$

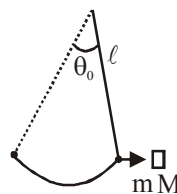


$$\Rightarrow \tau = \left(\frac{K\ell^2}{2} \right) \theta = -C\theta \quad \left[\text{let } C = \frac{K\ell^2}{2} \right]$$

$\Rightarrow$ So, frequency of resulting oscillations

$$f = \frac{1}{2\pi} \sqrt{\frac{C}{I}} = \frac{1}{2\pi} \sqrt{\frac{\frac{K\ell^2}{2}}{\frac{M\ell^2}{12}}} = \frac{1}{2\pi} \sqrt{\frac{6K}{M}}$$

41. (b)



$$\text{Velocity before collision } v = \sqrt{2g\ell(1 - \cos \theta_0)}$$

Velocity after collision

$$v_1 = \sqrt{2g\ell(1 - \cos \theta_1)}$$

Using momentum conservation

$$mv = MV_m - mV_1$$

$$m\sqrt{2g\ell(1 - \cos \theta_0)} = MV_m - m\sqrt{2g\ell(1 - \cos \theta)}$$

$$\Rightarrow m\sqrt{2g\ell} \{ \sqrt{1 - \cos \theta_0} + \sqrt{1 - \cos \theta_1} \} = MV_m$$

$$\text{and } e = 1 = \frac{V_m + \sqrt{2g\ell(1 - \cos \theta_1)}}{\sqrt{2g\ell(1 - \cos \theta_0)}}$$

$$\sqrt{2g\ell}(\sqrt{1-\cos\theta_0}-\sqrt{1-\cos\theta_1})=V_m \quad \dots(i)$$

$$m\sqrt{2g\ell}(\sqrt{1-\cos\theta_0}+\sqrt{1-\cos\theta_1})=MV_M \quad \dots(ii)$$

Dividing (ii) by (i) we get

$$\frac{(\sqrt{1-\cos\theta_0}+\sqrt{1-\cos\theta_1})}{(\sqrt{1-\cos\theta_0}-\sqrt{1-\cos\theta_1})}=\frac{M}{m}$$

By componendo and dividendo rule

$$\frac{m-M}{m+M}=\frac{\sqrt{1-\cos\theta_1}}{\sqrt{1-\cos\theta_0}}=\frac{\sin\left(\frac{\theta_1}{2}\right)}{\sin\left(\frac{\theta_0}{2}\right)}$$

$$\Rightarrow \frac{M}{m}=\frac{\theta_0-\theta_1}{\theta_0+\theta_1} \Rightarrow M=m\frac{\theta_0-\theta_1}{\theta_0+\theta_1}$$

42. (a) Given: $y=5[\sin(3\pi t)+\sqrt{3}\cos(3\pi t)]$

$$\Rightarrow y=10\sin\left(3\pi t+\frac{\pi}{3}\right)$$

$\therefore$ Amplitude = 10 cm

$$\text{Time period, } T=\frac{2\pi}{\omega}=\frac{2\pi}{3\pi}=\frac{2}{3}\text{ s}$$

43. (a) Angular frequency of pendulum $\omega=\sqrt{\frac{g}{\ell}}$

$\therefore$ relative change in angular frequency

$$\frac{\Delta\omega}{\omega}=\frac{1}{2}\frac{\Delta g}{g} \quad [\text{as length remains constant}]$$

$\Delta g=2A\omega_s^2$ [ω_s = angular frequency of support and, A = amplitude]

$$\frac{\Delta\omega}{\omega}=\frac{1}{2}\times\frac{2A\omega_s^2}{g}$$

$$\Delta\omega=\frac{1}{2}\times\frac{2\times 1^2\times 10^{-2}}{10}=10^{-3}\text{ rad/sec.}$$

44. (d) Acceleration due to gravity $g=\frac{GM}{R^2}$

$$\frac{g_p}{g_e}=\frac{M_p}{M_e}\left(\frac{R_e}{R_p}\right)^2=3\left(\frac{1}{3}\right)^2=\frac{1}{3}$$

$$\text{Also } T\propto\frac{1}{\sqrt{g}}\Rightarrow\frac{T_p}{T_e}=\sqrt{\frac{g_e}{g_p}}=\sqrt{3}$$

$$\Rightarrow T_p=2\sqrt{3}\text{ s}$$

45. (c) Velocity, $v=\omega\sqrt{A^2-x^2}$... (i)

acceleration, $a=-\omega^2x$... (ii)

and according to question,

$$|v|=|a|$$

$$\Rightarrow\omega\sqrt{A^2-x^2}=\omega^2x$$

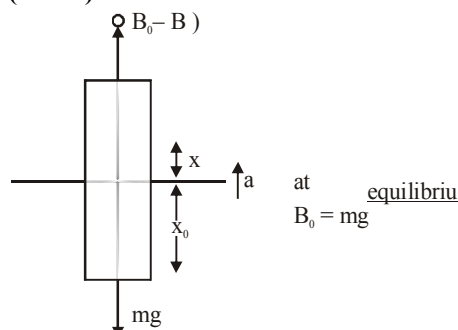
$$\Rightarrow A^2-x^2=\omega^2x^2$$

$$\Rightarrow 5^2-4^2=\omega^2(4^2)$$

$$3=\omega\times 4\Rightarrow\omega=\frac{3}{4}$$

$$\therefore T=2\pi/\omega=\frac{2\pi}{3/4}=\frac{8\pi}{3}$$

46. (Bonus)



Extra buoyant force = $\rho A x g$

$$B_0 + B = mg + ma$$

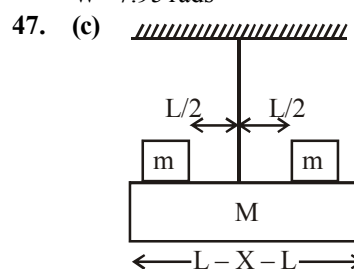
$$\therefore B = ma = \rho A x g = (\pi r^2 \rho g) x$$

$$a = \frac{(\pi r^2 \rho g) x}{m}$$

using, $a = \omega^2 x$

$$\Rightarrow \omega = \sqrt{\frac{\pi r^2 \rho g}{m}}$$

$$\omega \approx 7.95 \text{ rad s}^{-1}$$



$$f_1 = \frac{1}{2\pi} \sqrt{\frac{C}{1}} \quad \dots(i)$$

$$= \frac{1}{2} \sqrt{\frac{3C}{ML^2}}$$

$$f_2 = \frac{1}{2\pi} \sqrt{\frac{C}{L^2 \left(\frac{M}{3} + \frac{M}{2} \right)}} \quad \dots(ii)$$

As frequency reduces by 80%

$$\therefore f_2 = 0.8 f_1 \Rightarrow \frac{f_2}{f_1} = 0.8 \quad \dots(iii)$$

Solving equations (i), (ii) & (iii)

$$\text{Ratio } \frac{m}{M} = 0.37$$

As frequency reduces by 80%

$$\therefore f_2 = 0.8 f_1 \Rightarrow \frac{f_2}{f_1} = 0.8 \quad \dots(\text{iii})$$

Solving equations (i), (ii) & (iii)

$$\text{Ratio } \frac{m}{M} = 0.37$$

48. (b) As we know, frequency in SHM

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = 10^{12}$$

where m = mass of one atom

$$\text{Mass of one atom of silver, } = \frac{108}{(6.02 \times 10^{23})} \times 10^{-3} \text{ kg}$$

$$\frac{1}{2\pi} \sqrt{\frac{k}{108 \times 10^{-3}}} \times 6.02 \times 10^{23} = 10^{12}$$

Solving we get, spring constant,

$$K = 7.1 \text{ N/m}$$

49. (d) Using $y = A \sin \omega t$

$$a = A \sin \omega t_0$$

$$b = A \sin 2\omega t_0$$

$$c = A \sin 3\omega t_0$$

$$a + c = A[\sin \omega t_0 + A \sin 3\omega t_0] = 2A \sin 2\omega t_0 \cos \omega t_0$$

$$\frac{a+c}{b} = 2 \cos \omega t_0$$

$$\Rightarrow \omega = \frac{1}{t_0} \cos^{-1} \left(\frac{a+c}{2b} \right) \Rightarrow f = \frac{1}{2\pi t_0} \cos^{-1} \left(\frac{a+c}{2b} \right)$$

50. (b) As we know, Time-period of simple pendulum,

$$T \propto \sqrt{l}$$

$$\text{differentiating both side, } \frac{\Delta T}{T} = \frac{1}{2} \frac{\Delta l}{l}$$

$$\therefore \text{change in length } \Delta l = r_1 - r_2$$

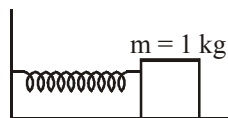
$$5 \times 10^{-4} = \frac{1}{2} \frac{r_1 - r_2}{1}$$

$$r_1 - r_2 = 10 \times 10^{-4}$$

$$10^{-3} \text{ m} = 10^{-1} \text{ cm} = 0.1 \text{ cm}$$

51. (c) Frequency of spring (f) = $\frac{1}{2\pi} \sqrt{\frac{k}{m}} = 1 \text{ Hz}$

$$\Rightarrow 4\pi^2 = \frac{k}{m}$$

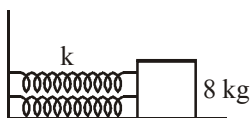


If block of mass $m = 1 \text{ kg}$ is attached then,

$$k = 4\pi^2$$

Now, identical springs are attached in parallel with mass $m = 8 \text{ kg}$. Hence,

$$k_{\text{eq}} = 2k$$



$$F = \frac{1}{2\pi} \sqrt{\frac{k \times 2}{g}} = \frac{1}{2} \text{ Hz}$$

52. (c) Time lost/gained per day = $\frac{1}{2} \propto \Delta \theta \times 86400 \text{ second}$

$$12 = \frac{1}{2} \alpha (40 - \theta) \times 86400 \quad \dots(\text{i})$$

$$4 = \frac{1}{2} \alpha (\theta - 20) \times 86400 \quad \dots(\text{ii})$$

$$\text{On dividing we get, } 3 = \frac{40 - \theta}{\theta - 20}$$

$$3\theta - 60 = 40 - \theta$$

$$4\theta = 100 \Rightarrow \theta = 25^\circ \text{C}$$

53. (b) Washer contact with piston $\Rightarrow N = 0$

Given Amplitude $A = 7 \text{ cm} = 0.07 \text{ m}$.

$$a_{\text{max}} = g = \omega^2 A$$

The frequency of piston

$$f = \frac{\omega}{2\pi} = \sqrt{\frac{g}{A}} \frac{1}{2\pi} = \sqrt{\frac{1000}{7}} \frac{1}{2\pi} = 1.9 \text{ Hz.}$$

54. (c) As we know, time period, $T = 2\pi \sqrt{\frac{\ell}{g}}$

When additional mass M is added then

$$T_M = 2\pi \sqrt{\frac{\ell + \Delta \ell}{g}}$$

$$\frac{T_M}{T} = \sqrt{\frac{\ell + \Delta \ell}{\ell}}$$

$$\Rightarrow \left(\frac{T_M}{T} \right)^2 = \frac{\ell + \Delta \ell}{\ell}$$

$$\text{or, } \left(\frac{T_M}{T} \right)^2 = 1 + \frac{Mg}{AY} \quad \left[\because \Delta \ell = \frac{Mg\ell}{AY} \right]$$

$$\therefore \frac{1}{Y} = \left[\left(\frac{T_M}{T} \right)^2 - 1 \right] \frac{A}{Mg}$$

55. (d) In simple harmonic motion, starting from rest,

At $t = 0$, $x = A$

$$x = A \cos \omega t \quad \dots(\text{i})$$

When $t = \tau$, $x = A - a$

When $t = 2\tau$, $x = A - 3a$

From equation (i)

$$A - a = A \cos \omega \tau \quad \dots(\text{ii})$$

$$A - 3a = A \cos 2\omega \tau \quad \dots(\text{iii})$$

As $\cos 2\omega \tau = 2 \cos^2 \omega \tau - 1 \dots(\text{iv})$

From equation (ii), (iii) and (iv)

$$\frac{A - 3a}{A} = 2 \left(\frac{A - a}{A} \right)^2 - 1$$

$$\Rightarrow \frac{A - 3a}{A} = \frac{2A^2 + 2a^2 - 4Aa - A^2}{A^2}$$

$$\Rightarrow A^2 - 3aA = A^2 + 2a^2 - 4Aa$$

$$\Rightarrow 2a^2 = aA$$

$$\Rightarrow A = 2a$$

$$\Rightarrow \frac{a}{A} = \frac{1}{2}$$

$$\text{Now, } A - a = A \cos \omega \tau$$

$$\Rightarrow \cos \omega \tau = \frac{A - a}{A}$$

$$\Rightarrow \cos \omega \tau = \frac{1}{2} \quad \text{or} \quad \frac{2\pi}{T} \tau = \frac{\pi}{3}$$

$$\Rightarrow T = 6 \tau$$

56. (b) From graph it is clear that when

$$L = 1 \text{ m}, T^2 = 4 \text{ s}^2$$

As we know,

$$T = 2\pi \sqrt{\frac{L}{g}}$$

$$\Rightarrow g = \frac{4\pi^2 L}{T^2}$$

$$= 4 \times \left(\frac{22}{7}\right)^2 \times \frac{1}{4} = \left(\frac{22}{7}\right)^2$$

$$\therefore g = \frac{484}{49} = 9.87 \text{ m/s}^2$$

57. (d) As we know,

$$x = x_0 e^{-bt/2m}$$

From question,

$$8 = 10e^{-\frac{40b}{2m}} \quad \dots(i)$$

$$\text{Similarly, } 5 = 10e^{-\frac{bt}{2m}} \quad \dots(ii)$$

Solving eqns (i) and (ii) we get

$$t \cong 142 \text{ s}$$

58. (c) Mass of bigger body $M = 4 \text{ kg}$

Mass of smaller body $m = 1 \text{ kg}$

Smaller mass ($m = 1 \text{ kg}$) executes S.H.M of angular frequency $\omega = 25 \text{ rad/s}$

Amplitude $x = 1.6 \text{ cm} = 1.6 \times 10^{-2}$

As we know,

$$T = 2\pi \sqrt{\frac{m}{K}}$$

$$\text{or, } \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{K}}$$

$$\text{or, } \frac{1}{25} = \sqrt{\frac{1}{K}} \quad [\because m = 1 \text{ kg; } \omega = 25 \text{ rad/s}]$$

$$\text{or, } K = 625 \text{ Nm}^{-1}$$

The maximum force exerted by the system on the floor

$$= Mg + Kx + mg$$

$$= 4 \times 10 + 625 \times 1.6 \times 10^{-2} + 1 \times 10$$

$$= 40 + 10 + 10$$

$$= 60 \text{ N}$$

59. (c) $\frac{Mg}{A} = P_0$

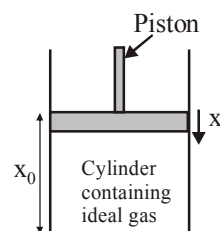
$$P_0 V_0^\gamma = P V^\gamma$$

$$Mg = P_0 A$$

Let piston is displaced by distance x

$$P_0 A x_0^\gamma = P A (x_0 - x)^\gamma$$

$$P = \frac{P_0 x_0^\gamma}{(x_0 - x)^\gamma}$$



$$Mg - \left(\frac{P_0 x_0^\gamma}{(x_0 - x)^\gamma} \right) A = F_{\text{restoring}}$$

$$P_0 A \left(1 - \frac{x_0^\gamma}{(x_0 - x)^\gamma} \right) = F_{\text{restoring}} \quad [x_0 - x \approx x_0]$$

$$F = -\frac{\gamma P_0 A x}{x_0}$$

$\therefore$ Frequency with which piston executes SHM.

$$f = \frac{1}{2\pi} \sqrt{\frac{\gamma P_0 A}{x_0 M}} = \frac{1}{2\pi} \sqrt{\frac{\gamma P_0 A^2}{M V_0}}$$

60. (c) As $F = -kx$

61. (a) Let T_1 and T_2 be the time period of the two pendulums

$$T_1 = 2\pi \sqrt{\frac{l}{g}} \quad \text{and} \quad T_2 = 2\pi \sqrt{\frac{4}{g}}$$

As $\ell_1 < \ell_2$ therefore $T_1 < T_2$

Let at $t = 0$ they start swinging together. Since their time periods are different, the swinging will not be in unison always. Only when number of completed oscillations differ by an integer, the two pendulums will again begin to swing together

Let longer length pendulum complete n oscillation and shorter length pendulum complete $(n + 1)$ oscillation. For unison swinging

$$(n + 1)T_1 = nT_2$$

$$(n + 1) \times 2\pi \sqrt{\frac{l}{g}} = (n) \times 2\pi \sqrt{\frac{4}{g}}$$

$$\Rightarrow n = 1$$

$$\therefore n + 1 = 1 + 1 = 2$$

62. (d) The equation of motion for the pendulum, for damped harmonic motion

$$F = -kx - bv$$

$$\Rightarrow ma + kx + bv = 0$$

$$\Rightarrow m \frac{d^2x}{dt^2} + kx + b \frac{dx}{dt} = 0$$

$$\Rightarrow \frac{d^2x}{dt^2} + \frac{k}{m}x + \frac{b}{m} \frac{dx}{dt} = 0$$

$$\Rightarrow \frac{d^2x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \frac{k}{m}x = 0 \quad \dots (1)$$

Let $x = e^{\lambda t}$ is the solution of the equation (1)

$$\frac{dx}{dt} = \lambda e^{\lambda t} \Rightarrow \frac{d^2x}{dt^2} = \lambda^2 e^{\lambda t}$$

Substituting in the equation (1)

$$\lambda^2 e^{\lambda t} + \frac{b}{m} \lambda e^{\lambda t} + \frac{k}{m} e^{\lambda t} = 0$$

$$\lambda^2 + \frac{b}{m} \lambda + \frac{k}{m} = 0$$

$$\lambda = \frac{-\frac{b}{m} \pm \sqrt{\frac{b^2}{m^2} - 4 \frac{k}{m}}}{2} = \frac{-b \pm \sqrt{b^2 - 4km}}{2m}$$

Solving the equation (1) for x , we have

$$x = e^{\frac{-b}{2m}t}$$

$$\omega = \sqrt{\omega_0^2 - \lambda^2} \text{ where } \omega_0 = \frac{k}{m}, \lambda = \frac{+b}{2}$$

$$\text{The average life} = \frac{1}{\lambda} = \frac{2}{b}$$

63. (a)

64. (a) Let the cube be at a depth x from the equilibrium position.

Force acting on the cube = up thrust on the portion of length x .

$$F = -\rho \ell^2 x g \quad [\because \text{mass density} \times \text{volume}] \quad \dots (i)$$

Clearly $F \propto -x$, Hence it is a SHM.

$$\text{Equation of SHM is } F = -kx \quad \dots (ii)$$

Comparing equation (i) and (ii) we have

$$k = \rho \ell^2 g$$

$$\text{Now, Time period, } T = 2\pi \sqrt{\frac{m}{k}}$$

$$\Rightarrow T = 2\pi \sqrt{\frac{\ell^3 d}{\rho \ell^2 g}}$$

$$= 2\pi \sqrt{\frac{\ell d}{\rho g}}$$

Comparing the above equation with

$$a = -\omega^2 x, \text{ we get}$$

$$\therefore \omega = \sqrt{\frac{\rho g}{d \ell}} \Rightarrow T = 2\pi \sqrt{\frac{\ell d}{\rho g}}$$

65. (a) For an SHM, the acceleration

$$a = -\omega^2 x \text{ where } \omega^2 \text{ is a constant.}$$

$$a = \frac{-4\pi^2 x}{T^2} \Rightarrow \frac{aT}{x} = \frac{-4\pi^2}{T}$$

The time period T is also constant. Therefore, $\frac{aT}{x}$ is a constant.

66. (a) The two springs are in parallel.

$\therefore$ Effective spring constant,

$$k = k_1 + k_2$$

Initial frequency of oscillation is given by

$$v = \frac{1}{2\pi} \sqrt{\frac{k_1 + k_2}{m}} \quad \dots (i)$$

When both k_1 and k_2 are made four times their original values, the new frequency is given by

$$v' = \frac{1}{2\pi} \sqrt{\frac{4k_1 + 4k_2}{m}} \\ = \frac{1}{2\pi} \sqrt{\frac{4(k_1 + k_2)}{m}} = 2 \left(\frac{1}{2\pi} \sqrt{\frac{k_1 + k_2}{m}} \right) = 2v$$

67. (b) Here, Displacement, $x = 2 \times 10^{-2} \cos \pi t$

Velocity is given by

$$v = \frac{dx}{dt} = 2 \times 10^{-2} \pi \sin \pi t$$

For the first time, the when velocity becomes maximum, $\sin \pi t = 1$

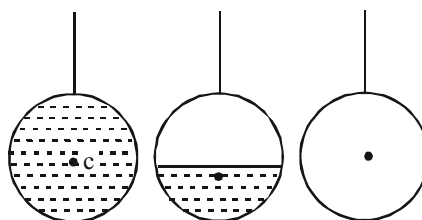
$$\Rightarrow \sin \pi t = \sin \frac{\pi}{2}$$

$$\Rightarrow \pi t = \frac{\pi}{2} \text{ or, } t = \frac{1}{2} = 0.5 \text{ sec.}$$

68. (b) When plugged hole near the bottom of the oscillating bob gets suddenly unplugged, centre of mass of combination of liquid and hollow portion (at position ℓ), first goes down (to $\ell + \Delta \ell$) and when total water is drained out, centre of mass regain its original position (to ℓ),

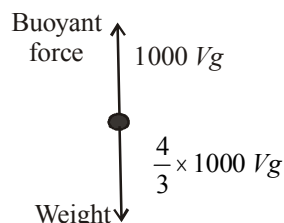
$$\text{Time period, } T = 2\pi \sqrt{\frac{\ell}{g}}$$

$\therefore$ 'T' first increases and then decreases to original value.



70. (a) Time period, $t = 2\pi\sqrt{\frac{\ell}{g_{\text{eff}}}}$;

In air, $t_0 = 2\pi\sqrt{\frac{\ell}{g}}$



Net force $= \left(\frac{4}{3} - 1\right) \times 1000 Vg = \frac{1000}{3} Vg$

$g_{\text{eff}} = \frac{1000 Vg}{3 \times \frac{4}{3} \times 1000 V} = \frac{g}{4}$

$\therefore t = 2\pi\sqrt{\frac{\ell}{g/4}} = 2 \times 2\pi\sqrt{\frac{\ell}{g}}$

$t = 2t_0$

71. (b) Time period for first spring, $t_1 = 2\pi\sqrt{\frac{m}{k_1}}$,

Time period for second spring, $t_2 = 2\pi\sqrt{\frac{m}{k_2}}$

Force constant of the series combination $k_{\text{eff}} = \frac{k_1 k_2}{k_1 + k_2}$

$\therefore$ Time period of oscillation for series combination

$T = 2\pi\sqrt{\frac{m(k_1 + k_2)}{k_1 k_2}}$

$\therefore T = 2\pi\sqrt{\frac{m}{k_2} + \frac{m}{k_1}} = 2\pi\sqrt{\frac{t_2^2}{(2\pi)^2} + \frac{t_1^2}{(2\pi)^2}}$

$\Rightarrow T^2 = t_1^2 + t_2^2$

where x is the displacement from the mean position

72. (c) With mass M , the time period of the spring.

$T = 2\pi\sqrt{\frac{M}{k}}$

With mass $M + m$, the time period becomes,

$T' = 2\pi\sqrt{\frac{M+m}{k}} = \frac{5T}{3}$

$\therefore 2\pi\sqrt{\frac{M+m}{k}} = \frac{5}{3} \times 2\pi\sqrt{\frac{M}{k}}$

$\Rightarrow M + m = \frac{25}{9} \times M$

$\Rightarrow 1 + \frac{m}{M} = \frac{25}{9}$

$\Rightarrow \frac{m}{M} = \frac{25}{9} - 1 = \frac{16}{9}$

73. (d) Time period, $T = 2\pi\sqrt{\frac{\ell}{g}}$

New length, $\ell' = \ell + 21\% \text{ of } \ell$

$\ell' = \ell + 0.21\ell$

$\Rightarrow \ell' = 1.21\ell$

$T' = 2\pi\sqrt{\frac{1.21\ell}{g}}$

% increase in length $= \frac{T' - T}{T} \times 100$

$= \frac{\sqrt{1.21\ell} - \sqrt{\ell}}{\sqrt{\ell}} \times 100 = (\sqrt{1.21} - \sqrt{1}) \times 100$

$= (1.1 - 1) \times 100 = 10\%$

74. (b) Let k be the spring constant of the original spring.

Time period $T = 2\pi\sqrt{\frac{m}{k}}$ where m is the mass of oscillating body.

When the spring is cut into n equal parts, the spring constant of one part becomes nk . Therefore the new time period,

$T' = 2\pi\sqrt{\frac{m}{nk}} = \frac{T}{\sqrt{n}}$

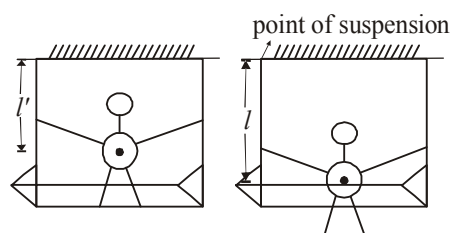
75. (b) The time period $T = 2\pi\sqrt{\frac{\ell}{g}}$ where ℓ = distance

between the point of suspension and the centre of mass of the child.

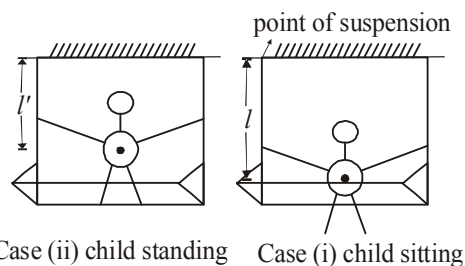
As the child stands up, her centre of mass is raised. The distance between point of suspension and centre of mass decreases i.e. length ℓ decreases.

$\therefore \ell' < \ell$

$\therefore T' < T$ i.e., the period decreases.



Case (ii) child standing Case (i) child sitting



Case (ii) child standing Case (i) child sitting

76. (c) Time of half the amplitude is $= 2s$

Using, $A = A_0 e^{-kt}$

$$\frac{A_0}{2} = A_0 e^{-k \times 2} \quad \dots(i)$$

$$\text{and } \frac{A_0}{1000} = A_0 e^{-kt} \quad \dots(ii)$$

Dividing (i) by (ii) and solving, we get

$$t = 20 \text{ s}$$

77. (b) Amplitude of vibration at time $t = 0$ is given by

$$A = A_0 e^{-0.1 \times 0} = 1 \times A_0 = A_0$$

$$\text{also at } t = t, \text{ if } A = \frac{A_0}{2}$$

$$\Rightarrow \frac{1}{2} = e^{-0.1t}$$

$$t = 10 \ln 2 \approx 7 \text{ s}$$

78. (b) In first collision mu momentum will be imparted to system, in second collision when momentum of $(M + m)$ is in opposite direction mu momentum of particle will make its momentum zero.

$$\text{On 13th collision, } [m] \rightarrow [M + 12]; [M + 13m] \rightarrow V$$

$$mu = (M + 13m)v \Rightarrow v = \frac{mu}{M + 13m} = \frac{u}{15}$$

$$v = \omega A \Rightarrow \frac{u}{15} = \sqrt{\frac{K}{M - 13m}} \times A$$

Putting value of M , m , u and K we get amplitude

$$A = \frac{1}{15} \sqrt{\frac{75}{1}} = \frac{1}{\sqrt{3}}$$

79. (b) The change in time period compared to the undamped oscillator increases by 8%.

$$80. (c) \because A = A_0 e^{-\frac{bt}{2m}}$$

(where, A_0 = maximum amplitude)

According to the questions, after 5 second,

$$0.9A_0 = A_0 e^{-\frac{b(5)}{2m}} \quad \dots (i)$$

After 10 more second,

$$A = A_0 e^{-\frac{b(15)}{2m}} \quad \dots(ii)$$

From eqⁿs (i) and (ii)

$$A = 0.729 A_0$$

$$\therefore \alpha = 0.729$$

81. (a)

82. (d) When the pendulum is oscillating over a current carrying coil, and when the direction of oscillating pendulum bob is opposite to the direction of current. Its instantaneous acceleration increases.

$$\text{Hence time period } T < 2\pi \sqrt{\frac{\ell}{g}}$$

and damping is larger than in air alone due energy dissipation.

83. (c) As energy $\propto (\text{Amplitude})^2$, the maximum for both of them occurs at the same frequency and this is only possible in case of resonance.

In resonance state $\omega_1 = \omega_2$

84. (b) Equation of displacement in forced oscillation is given by

$$y = \frac{F_0}{m(\omega_0^2 - \omega^2)^2}$$

$$= \frac{F_0}{m(\omega_0^2 - \omega^2)}$$

Here damping effect is considered to be zero

$$\therefore x \propto \frac{1}{m(\omega_0^2 - \omega^2)}$$

Waves


**TOPIC 1 Basic of Mechanical Waves,
Progressive and Stationary
Waves**

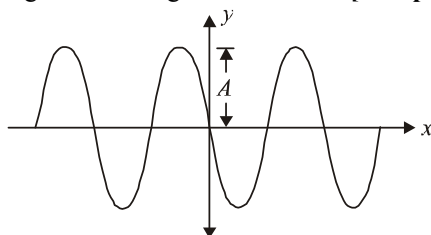

1. Assume that the displacement (s) of air is proportional to the pressure difference (Δp) created by a sound wave. Displacement (s) further depends on the speed of sound (v), density of air (ρ) and the frequency (f). If $\Delta p \sim 10 \text{ Pa}$, $v \sim 300 \text{ m/s}$, $\rho \sim 1 \text{ kg/m}^3$ and $f \sim 1000 \text{ Hz}$, then s will be of the order of (take the multiplicative constant to be 1) **[Sep. 05, 2020 (I)]**

- (a) $\frac{3}{100} \text{ mm}$ (b) 10 mm
(c) $\frac{1}{10} \text{ mm}$ (d) 1 mm

2. For a transverse wave travelling along a straight line, the distance between two peaks (crests) is 5 m , while the distance between one crest and one trough is 1.5 m . The possible wavelengths (in m) of the waves are : **[Sep. 04, 2020 (I)]**

- (a) $1, 3, 5, \dots$ (b) $\frac{1}{1}, \frac{1}{3}, \frac{1}{5}, \dots$
(c) $1, 2, 3, \dots$ (d) $\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \dots$

3. A progressive wave travelling along the positive x -direction is represented by $y(x, t) = A \sin(kx - \omega t + \phi)$. Its snapshot at $t = 0$ is given in the figure. **[12 April 2019 I]**



For this wave, the phase ϕ is :

- (a) $-\frac{\pi}{2}$ (b) π (c) 0 (d) $\frac{\pi}{2}$

4. A small speaker delivers 2 W of audio output. At what distance from the speaker will one detect 120 dB intensity sound ? [Given reference intensity of sound as 10^{-12} W/m^2] **[12 April 2019 II]**

- (a) 40 cm (b) 20 cm (c) 10 cm (d) 30 cm

5. The pressure wave, $P = 0.01 \sin[1000t - 3x] \text{ Nm}^{-2}$, corresponds to the sound produced by a vibrating blade on a day when atmospheric temperature is 0°C . On some other day when temperature is T , the speed of sound produced by the same blade and at the same frequency is found to be 336 ms^{-1} . Approximate value of T is : **[9 April 2019 I]**

- (a) 4°C (b) 11°C (c) 12°C (d) 15°C

6. A travelling harmonic wave is represented by the equation $y(x, t) = 10^{-3} \sin(50t + 2x)$, where x and y are in meter and t is in seconds. Which of the following is a correct statement about the wave? **[12 Jan. 2019 I]**

- (a) The wave is propagating along the negative x -axis with speed 25 ms^{-1} .
(b) The wave is propagating along the positive x -axis with speed 100 ms^{-1} .
(c) The wave is propagating along the positive x -axis with speed 25 ms^{-1} .
(d) The wave is propagating along the negative x -axis with speed 100 ms^{-1} .

7. A transverse wave is represented by

$$y = \frac{10}{\pi} \sin\left(\frac{2\pi}{T}t - \frac{2\pi}{\lambda}x\right)$$

For what value of the wavelength the wave velocity is twice the maximum particle velocity? **[Online April 9, 2014]**

- (a) 40 cm (b) 20 cm (c) 10 cm (d) 60 cm

8. In a transverse wave the distance between a crest and neighbouring trough at the same instant is 4.0 cm and the distance between a crest and trough at the same place is 1.0 cm . The next crest appears at the same place after a time interval of 0.4 s . The maximum speed of the vibrating particles in the medium is : **[Online April 25, 2013]**

- (a) $\frac{3\pi}{2} \text{ cm/s}$ (b) $\frac{5\pi}{2} \text{ cm/s}$
(c) $\frac{\pi}{2} \text{ cm/s}$ (d) $2\pi \text{ cm/s}$

9. When two sound waves travel in the same direction in a medium, the displacements of a particle located at 'x' at time 't' is given by :

$$y_1 = 0.05 \cos (0.50 \pi x - 100 \pi t)$$

$$y_2 = 0.05 \cos (0.46 \pi x - 92 \pi t)$$

where y_1 , y_2 and x are in meters and t in seconds. The speed of sound in the medium is : [Online April 9, 2013]

- (a) 92 m/s (b) 200 m/s (c) 100 m/s (d) 332 m/s
10. The disturbance $y(x, t)$ of a wave propagating in the positive x -direction is given by $y = \frac{1}{1+x^2}$ at time $t = 0$

and by $y = \frac{1}{1+(x-1)^2}$ at $t = 2$ s, where x and y are in meters. The shape of the wave disturbance does not change during the propagation. The velocity of wave in m/s is

[Online May 26, 2012]

- (a) 2.0 (b) 4.0 (c) 0.5 (d) 1.0
11. The transverse displacement $y(x, t)$ of a wave is given by

$$y(x, t) = e^{-\left(ax^2 + bt^2 + 2\sqrt{ab}xt\right)}$$

This represents a:

[2011]

- (a) wave moving in $-x$ direction with speed $\sqrt{\frac{b}{a}}$
- (b) standing wave of frequency $\sqrt{b}$
- (c) standing wave of frequency $\frac{1}{\sqrt{b}}$
- (d) wave moving in $+x$ direction speed $\sqrt{\frac{a}{b}}$
12. A wave travelling along the x -axis is described by the equation $y(x, t) = 0.005 \cos (\alpha x - \beta t)$. If the wavelength and the time period of the wave are 0.08 m and 2.0 s, respectively, then α and β in appropriate units are [2008]
- (a) $\alpha = 25.00 \pi$, $\beta = \pi$ (b) $\alpha = \frac{0.08}{\pi}$, $\beta = \frac{2.0}{\pi}$
- (c) $\alpha = \frac{0.04}{\pi}$, $\beta = \frac{1.0}{\pi}$ (d) $\alpha = 12.50\pi$, $\beta = \frac{\pi}{2.0}$

13. A sound absorber attenuates the sound level by 20 dB. The intensity decreases by a factor of [2007]

(a) 100 (b) 1000 (c) 10000 (d) 10

14. The displacement y of a particle in a medium can be

expressed as, $y = 10^{-6} \sin \left(100t + 20x + \frac{\pi}{4} \right)$ m where t is

in second and x in meter. The speed of the wave is [2004]

- (a) 20 m/s (b) 5 m/s
- (c) 2000 m/s (d) 5π m/s
15. The displacement y of a wave travelling in the x -direction

is given by $y = 10^{-4} \sin \left(600t - 2x + \frac{\pi}{3} \right)$ metres

where x is expressed in metres and t in seconds. The speed of the wave - motion, in ms^{-1} , is [2003]

(a) 300 (b) 600 (c) 1200 (d) 200

16. When temperature increases, the frequency of a tuning fork [2002]

(a) increases

(b) decreases

(c) remains same

(d) increases or decreases depending on the material

TOPIC 2 Vibration of String and Organ Pipe



17. In a resonance tube experiment when the tube is filled with water up to a height of 17.0 cm from bottom, it resonates with a given tuning fork. When the water level is raised the next resonance with the same tuning fork occurs at a height of 24.5 cm. If the velocity of sound in air is 330 m/s, the tuning fork frequency is :

[Sep. 05, 2020 (I)]

(a) 2200 Hz (b) 550 Hz

(c) 1100 Hz (d) 3300 Hz

18. A uniform thin rope of length 12 m and mass 6 kg hangs vertically from a rigid support and a block of mass 2 kg is attached to its free end. A transverse short wave-train of wavelength 6 cm is produced at the lower end of the rope. What is the wavelength of the wavetrain (in cm) when it reaches the top of the rope ? [Sep. 03, 2020 (I)]

(a) 3 (b) 6

(c) 12 (d) 9

19. Two identical strings X and Z made of same material have tension T_X and T_Z in them. If their fundamental frequencies are 450 Hz and 300 Hz, respectively, then the ratio T_X/T_Z is: [Sep. 02, 2020 (I)]

(a) 2.25 (b) 0.44

(c) 1.25 (d) 1.5

20. A wire of density $9 \times 10^{-3} \text{ kg cm}^{-3}$ is stretched between two clamps 1 m apart. The resulting strain in the wire is 4.9×10^{-4} . The lowest frequency of the transverse vibrations in the wire is

(Young's modulus of wire $Y = 9 \times 10^{10} \text{ Nm}^{-2}$), (to the nearest integer), [Sep. 02, 2020 (II)]

21. A one metre long (both ends open) organ pipe is kept in a gas that has double the density of air at STP. Assuming the speed of sound in air at STP is 300 m/s, the frequency difference between the fundamental and second harmonic of this pipe is _____ Hz. [NA 8 Jan. 2020 (I)]

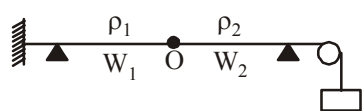
22. A transverse wave travels on a taut steel wire with a velocity of v when tension in it is $2.06 \times 10^4 \text{ N}$. When the tension is changed to T , the velocity changed to $v/2$. The value of T is close to: [8 Jan. 2020 (II)]

(a) $2.50 \times 10^4 \text{ N}$ (b) $5.15 \times 10^3 \text{ N}$

(c) $30.5 \times 10^4 \text{ N}$ (d) $10.2 \times 10^2 \text{ N}$

23. Speed of a transverse wave on a straight wire (mass 6.0 g, length 60 cm and area of cross-section 1.0 mm^2) is 90 ms^{-1} . If the Young's modulus of wire is $16 \times 10^{11} \text{ Nm}^{-2}$ the extension of wire over its natural length is:

[7 Jan. 2020 (I)]

- (a) 0.03 mm (b) 0.02 mm
(c) 0.04 mm (d) 0.01 mm
24. Equation of travelling wave on a stretched string of linear density 5 g/m is $y = 0.03 \sin(450t - 9x)$ where distance and time are measured in SI units. The tension in the string is:
[11 Jan 2019 (I)]
(a) 10 N (b) 7.5 N (c) 12.5 N (d) 5 N
25. A heavy ball of mass M is suspended from the ceiling of a car by a light string of mass m ($m \ll M$). When the car is at rest, the speed of transverse waves in the string is 60 ms^{-1} . When the car has acceleration a , the wave-speed increases to 60.5 ms^{-1} . The value of a , in terms of gravitational acceleration g , is closest to: [9 Jan. 2019 (I)]
(a) $\frac{g}{30}$ (b) $\frac{g}{5}$ (c) $\frac{g}{10}$ (d) $\frac{g}{20}$
26. A wire of length L and mass per unit length $6.0 \times 10^{-3} \text{ kgm}^{-1}$ is put under tension of 540 N. Two consecutive frequencies that it resonates at are: 420 Hz and 490 Hz. Then L in meters is: [9 Jan. 2020 (II)]
(a) 2.1 m (b) 1.1 m
(c) 8.1 m (d) 5.1 m
27. A tuning fork of frequency 480 Hz is used in an experiment for measuring speed of sound (v) in air by resonance tube method. Resonance is observed to occur at two successive lengths of the air column, $l_1 = 30 \text{ cm}$ and $l_2 = 70 \text{ cm}$. Then, v is equal to: [12 April 2019 (II)]
(a) 332 ms^{-1} (b) 384 ms^{-1}
(c) 338 ms^{-1} (d) 379 ms^{-1}
28. A string 2.0 m long and fixed at its ends is driven by a 240 Hz vibrator. The string vibrates in its third harmonic mode. The speed of the wave and its fundamental frequency is:
[9 April 2019 (II)]
(a) 180 m/s, 80 Hz (b) 320 m/s, 80 Hz
(c) 320 m/s, 120 Hz (d) 180 m/s, 120 Hz
29. A string is clamped at both the ends and it is vibrating in its 4th harmonic. The equation of the stationary wave is $Y = 0.3 \sin(0.157x) \cos(200At)$. The length of the string is: (All quantities are in SI units.) [9 April 2019 (I)]
(a) 20m (b) 80m (c) 40m (d) 60m
30. A wire of length $2L$, is made by joining two wires A and B of same length but different radii r and $2r$ and made of the same material. It is vibrating at a frequency such that the joint of the two wires forms a node. If the number of antinodes in wire A is p and that in B is q then the ratio $p : q$ is: [8 April 2019 (I)]
- 
- (a) 3 : 5 (b) 4 : 9 (c) 1 : 2 (d) 1 : 4
31. A closed organ pipe has a fundamental frequency of 1.5 kHz. The number of overtones that can be distinctly heard by a person with this organ pipe will be: (Assume that the highest frequency a person can hear is 20,000 Hz)
[10 Jan. 2019 (I)]
(a) 6 (b) 4
(c) 7 (d) 5
32. A string of length 1 m and mass 5 g is fixed at both ends. The tension in the string is 8.0 N. The string is set into vibration using an external vibrator of frequency 100 Hz. The separation between successive nodes on the string is close to: [10 Jan. 2019 (I)]
(a) 10.0 cm (b) 33.3 cm (c) 16.6 cm (d) 20.0 cm
33. A granite rod of 60 cm length is clamped at its middle point and is set into longitudinal vibrations. The density of granite is $2.7 \times 10^3 \text{ kg/m}^3$ and its Young's modulus is $9.27 \times 10^{10} \text{ Pa}$. What will be the fundamental frequency of the longitudinal vibrations? [2018]
(a) 5 kHz (b) 2.5 kHz (c) 10 kHz (d) 7.5 kHz
34. The end correction of a resonance column is 1 cm. If the shortest length resonating with the tuning fork is 10 cm, the next resonating length should be
[Online April 16, 2018]
(a) 32 cm (b) 40 cm (c) 28 cm (d) 36 cm
35. Two wires W_1 and W_2 have the same radius r and respective densities ρ_1 and ρ_2 such that $\rho_2 = 4\rho_1$. They are joined together at the point O, as shown in the figure. The combination is used as a sonometer wire and kept under tension T . The point O is midway between the two bridges. When a stationary waves is set up in the composite wire, the joint is found to be a node. The ratio of the number of antinodes formed in W_1 to W_2 is :
[Online April 8, 2017]
- 
- (a) 1 : 1 (b) 1 : 2 (c) 1 : 3 (d) 4 : 1
36. A uniform string of length 20 m is suspended from a rigid support. A short wave pulse is introduced at its lowest end. It starts moving up the string. The time taken to reach the supports is : [2016] (take $g = 10 \text{ ms}^{-2}$)
(a) $2\sqrt{2} \text{ s}$ (b) $\sqrt{2} \text{ s}$ (c) $2\pi\sqrt{2} \text{ s}$ (d) 2 s
37. A pipe open at both ends has a fundamental frequency f in air. The pipe is dipped vertically in water so that half of it is in water. The fundamental frequency of the air column is now : [2016]
(a) $2f$ (b) f (c) $\frac{f}{2}$ (d) $\frac{3f}{4}$
38. A pipe of length 85 cm is closed from one end. Find the number of possible natural oscillations of air column in the pipe whose frequencies lie below 1250 Hz. The velocity of sound in air is 340 m/s. [2014]
(a) 12 (b) 8 (c) 6 (d) 4
39. The total length of a sonometer wire between fixed ends is 110 cm. Two bridges are placed to divide the length of wire in ratio 6 : 3 : 2. The tension in the wire is 400 N and the mass per unit length is 0.01 kg/m. What is the minimum common frequency with which three parts can vibrate?
[Online April 19, 2014]

- (a) 1100 Hz (b) 1000 Hz
(c) 166 Hz (d) 100 Hz
40. A sonometer wire of length 1.5 m is made of steel. The tension in it produces an elastic strain of 1%. What is the fundamental frequency of steel if density and elasticity of steel are $7.7 \times 10^3 \text{ kg/m}^3$ and $2.2 \times 10^{11} \text{ N/m}^2$ respectively?
(a) 188.5 Hz (b) 178.2 Hz [2013]
(c) 200.5 Hz (d) 770 Hz
41. A sonometer wire of length 114 cm is fixed at both the ends. Where should the two bridges be placed so as to divide the wire into three segments whose fundamental frequencies are in the ratio 1 : 3 : 4? [Online April 23, 2013]
(a) At 36 cm and 84 cm from one end
(b) At 24 cm and 72 cm from one end
(c) At 48 cm and 96 cm from one end
(d) At 72 cm and 96 cm from one end
42. A cylindrical tube, open at both ends, has a fundamental frequency f in air. The tube is dipped vertically in water so that half of it is in water. The fundamental frequency of the air-column is now : [2012]
(a) f (b) $f/2$ (c) $3f/4$ (d) $2f$
43. An air column in a pipe, which is closed at one end, will be in resonance with a vibrating tuning fork of frequency 264 Hz if the length of the column in cm is (velocity of sound = 330 m/s) [Online May 26, 2012]
(a) 125.00 (b) 93.75 (c) 62.50 (d) 187.50
44. A uniform tube of length 60.5 cm is held vertically with its lower end dipped in water. A sound source of frequency 500 Hz sends sound waves into the tube. When the length of tube above water is 16 cm and again when it is 50 cm, the tube resonates with the source of sound. Two lowest frequencies (in Hz), to which tube will resonate when it is taken out of water, are (approximately). [Online May 19, 2012]
(a) 281, 562 (b) 281, 843
(c) 276, 552 (d) 272, 544
45. The equation of a wave on a string of linear mass density 0.04 kg m^{-1} is given by

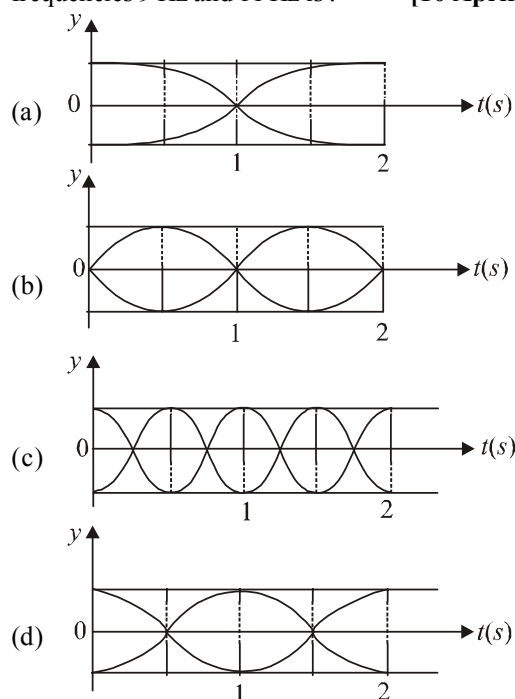
$$y = 0.02(m) \sin \left[2\pi \left(\frac{t}{0.04(s)} - \frac{x}{0.50(m)} \right) \right]$$
The tension in the string is [2010]
(a) 4.0 N (b) 12.5 N (c) 0.5 N (d) 6.25 N
46. While measuring the speed of sound by performing a resonance column experiment, a student gets the first resonance condition at a column length of 18 cm during winter. Repeating the same experiment during summer, she measures the column length to be x cm for the second resonance. Then [2008]
(a) $18 > x$ (b) $x > 54$
(c) $54 > x > 36$ (d) $36 > x > 18$
47. A string is stretched between fixed points separated by 75.0 cm. It is observed to have resonant frequencies of 420 Hz and 315 Hz. There are no other resonant frequencies between these two. Then, the lowest resonant frequency for this string is [2006]

- (a) 105 Hz (b) 1.05 Hz
(c) 1050 Hz (d) 10.5 Hz
48. Tube A has both ends open while tube B has one end closed, otherwise they are identical. The ratio of fundamental frequency of tube A and B is [2002]
(a) 1 : 2 (b) 1 : 4 (c) 2 : 1 (d) 4 : 1
49. A wave $y = a \sin(\omega t - kx)$ on a string meets with another wave producing a node at $x = 0$. Then the equation of the unknown wave is [2002]
(a) $y = a \sin(\omega t + kx)$
(b) $y = -a \sin(\omega t + kx)$
(c) $y = a \sin(\omega t - kx)$
(d) $y = -a \sin(\omega t - kx)$

TOPIC 3 Beats, Interference and Superposition of Waves



50. There harmonic waves having equal frequency ν and same intensity I_0 , have phase angles 0 , $\frac{\pi}{4}$ and $-\frac{\pi}{4}$ respectively. When they are superimposed the intensity of the resultant wave is close to: [9 Jan. 2020 I]
(a) $5.8I_0$ (b) $0.2I_0$ (c) $3I_0$ (d) I_0
51. The correct figure that shows, schematically, the wave pattern produced by superposition of two waves of frequencies 9 Hz and 11 Hz is : [10 April 2019 II]



52. A resonance tube is old and has jagged end. It is still used in the laboratory to determine velocity of sound in air. A tuning fork of frequency 512 Hz produces first resonance when the tube is filled with water to a mark 11 cm below a reference mark, near the open end of the tube. The experiment is repeated with another fork of frequency 256 Hz which produces first resonance when water

reaches a mark 27 cm below the reference mark. The velocity of sound in air, obtained in the experiment, is close to:

[12 Jan. 2019 II]

- (a) 322 ms⁻¹ (b) 341 ms⁻¹
(c) 335 ms⁻¹ (d) 328 ms⁻¹

53. A tuning fork vibrates with frequency 256 Hz and gives one beat per second with the third normal mode of vibration of an open pipe. What is the length of the pipe? (Speed of sound of air is 340 ms⁻¹)

[Online April 15, 2018]

- (a) 190 cm (b) 180 cm
(c) 220 cm (d) 200 cm

54. 5 beats/ second are heard when a tuning fork is sounded with a sonometer wire under tension, when the length of the sonometer wire is either 0.95m or 1m. The frequency of the fork will be:

[Online April 15, 2018]

- (a) 195Hz (b) 251Hz (c) 150Hz (d) 300Hz

55. A standing wave is formed by the superposition of two waves travelling in opposite directions. The transverse displacement is given by

$$y(x, t) = 0.5 \sin \left(\frac{5\pi}{4} x \right) \cos(200\pi t).$$

What is the speed of the travelling wave moving in the positive x direction?

(x and t are in meter and second, respectively.)

[Online April 9, 2017]

- (a) 160 m/s (b) 90 m/s (c) 180 m/s (d) 120 m/s

56. A wave represented by the equation $y_1 = a \cos(kx - \omega t)$ is superimposed with another wave to form a stationary wave such that the point $x = 0$ is node. The equation for the other wave is

[Online May 12, 2012]

- (a) $a \cos(kx - \omega t + \pi)$ (b) $a \cos(kx + \omega t + \pi)$
(c) $a \cos\left(kx + \omega t + \frac{\pi}{2}\right)$ (d) $a \cos\left(kx - \omega t + \frac{\pi}{2}\right)$

57. Following are expressions for four plane simple harmonic waves

[Online May 7, 2012]

- (i) $y_1 = A \cos 2\pi \left(n_1 t + \frac{x}{\lambda_1} \right)$
(ii) $y_2 = A \cos 2\pi \left(n_1 t + \frac{x}{\lambda_1} + \pi \right)$
(iii) $y_3 = A \cos 2\pi \left(n_2 t + \frac{x}{\lambda_2} \right)$
(iv) $y_4 = A \cos 2\pi \left(n_2 t - \frac{x}{\lambda_2} \right)$

The pairs of waves which will produce destructive interference and stationary waves respectively in a medium, are

- (a) (iii, iv), (i, ii) (b) (i, iii), (ii, iv)
(c) (i, iv), (ii, iii) (d) (i, ii), (iii, iv)

58. A travelling wave represented by

$y = A \sin(\omega t - kx)$ is superimposed on another wave represented by $y = A \sin(\omega t + kx)$. The resultant is

- (a) A wave travelling along + x direction [2011 RS]
(b) A wave travelling along - x direction
(c) A standing wave having nodes at

$$x = \frac{n\lambda}{2}, n = 0, 1, 2, \dots$$

- (d) A standing wave having nodes at

$$x = \left(n + \frac{1}{2} \right) \frac{\lambda}{2}; n = 0, 1, 2, \dots$$

59. **Statement - 1** : Two longitudinal waves given by equations : $y_1(x, t) = 2a \sin(\omega t - kx)$ and $y_2(x, t) = a \sin(2\omega t - 2kx)$ will have equal intensity. [2011 RS]

Statement - 2 : Intensity of waves of given frequency in same medium is proportional to square of amplitude only.

- (a) Statement-1 is true, statement-2 is false.
(b) Statement-1 is true, statement-2 is true, statement-2 is the correct explanation of statement-1
(c) Statement-1 is true, statement-2 is true, statement-2 is not the correct explanation of statement-1
(d) Statement-1 is false, statement-2 is true.

60. Three sound waves of equal amplitudes have frequencies $(v-1)$, v , $(v+1)$. They superpose to give beats. The number of beats produced per second will be : [2009]

- (a) 3 (b) 2 (c) 1 (d) 4

61. When two tuning forks (fork 1 and fork 2) are sounded simultaneously, 4 beats per second are heard. Now, some tape is attached on the prong of the fork 2. When the tuning forks are sounded again, 6 beats per second are heard. If the frequency of fork 1 is 200 Hz, then what was the original frequency of fork 2? [2005]

- (a) 202 Hz (b) 200 Hz (c) 204 Hz (d) 196 Hz

62. A tuning fork of known frequency 256 Hz makes 5 beats per second with the vibrating string of a piano. The beat frequency decreases to 2 beats per second when the tension in the piano string is slightly increased. The frequency of the piano string before increasing the tension was [2003]

- (a) (256+2) Hz (b) (256-2) Hz
(c) (256-5) Hz (d) (256+5) Hz

63. A tuning fork arrangement (pair) produces 4 beats/sec with one fork of frequency 288 cps. A little wax is placed on the unknown fork and it then produces 2 beats/sec. The frequency of the unknown fork is [2002]

- (a) 286 cps (b) 292 cps (c) 294 cps (d) 288 cps

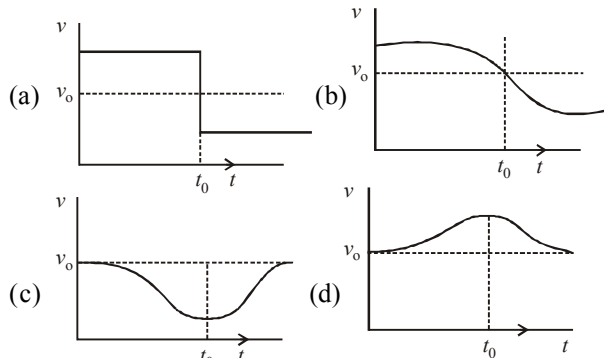
TOPIC 4 Musical Sound and Doppler's Effect



64. A sound source S is moving along a straight track with speed v , and is emitting sound of frequency ν_0 (see figure). An observer is standing at a finite distance, at the point O , from the track. The time variation of frequency heard by the observer is best represented by:

[Sep. 06, 2020 (I)]

(t_0 represents the instant when the distance between the source and observer is minimum)



65. A driver in a car, approaching a vertical wall notices that the frequency of his car horn, has changed from 440 Hz to 480 Hz, when it gets reflected from the wall. If the speed of sound in air is 345 m/s, then the speed of the car is :

[Sep. 05, 2020 (II)]

- (a) 54 km/hr (b) 36 km/hr
(c) 18 km/hr (d) 24 km/hr

66. The driver of a bus approaching a big wall notices that the frequency of his bus's horn changes from 420 Hz to 490 Hz when he hears it after it gets reflected from the wall. Find the speed of the bus if speed of the sound is 330 ms⁻¹.

[Sep. 04, 2020 (II)]

- (a) 91 kmh⁻¹ (b) 81 kmh⁻¹
(c) 61 kmh⁻¹ (d) 71 kmh⁻¹

67. Magnetic materials used for making permanent magnets (P) and magnets in a transformer (T) have different properties of the following, which property best matches for the type of magnet required? [Sep. 02, 2020 (I)]

- (a) T : Large retentivity, small coercivity
(b) P : Small retentivity, large coercivity
(c) T : Large retentivity, large coercivity
(d) P : Large retentivity, large coercivity

68. A stationary observer receives sound from two identical tuning forks, one of which approaches and the other one recedes with the same speed (much less than the speed of sound). The observer hears 2 beats/sec. The oscillation frequency of each tuning fork is $\nu_0 = 1400$ Hz and the velocity of sound in air is 350 m/s. The speed of each tuning fork is close to:

[7 Jan. 2020 I]

- (a) $\frac{1}{2}$ m/s (b) 1 m/s (c) $\frac{1}{4}$ m/s (d) $\frac{1}{8}$ m/s

69. A submarine (A) travelling at 18 km/hr is being chased along the line of its velocity by another submarine (B) travelling at 27 km/hr. B sends a sonar signal of 500 Hz to detect A and receives a reflected sound of frequency ν . The value of ν is close to :

[12 April 2019 I]

(Speed of sound in water = 1500 ms⁻¹)

- (a) 504 Hz (b) 507 Hz
(c) 499 Hz (d) 502 Hz

70. Two sources of sound S_1 and S_2 produce sound waves of same frequency 660 Hz. A listener is moving from source S_1 towards S_2 with a constant speed u m/s and he hears

10 beats/s. The velocity of sound is 330 m/s. Then u equals:

[12 April 2019 II]

- (a) 5.5 m/s (b) 15.0 m/s
(c) 2.5 m/s (d) 10.0 m/s

71. A stationary source emits sound waves of frequency 500 Hz. Two observers moving along a line passing through the source detect sound to be of frequencies 480 Hz and 530 Hz. Their respective speeds are, in ms⁻¹,

(Given speed of sound = 300 m/s) [10 April 2019 I]

- (a) 12, 16 (b) 12, 18 (c) 16, 14 (d) 8, 18

72. A source of sound S is moving with a velocity of 50 m/s towards a stationary observer. The observer measures the frequency of the source as 1000 Hz. What will be the apparent frequency of the source when it is moving away from the observer after crossing him? (Take velocity of sound in air 350 m/s)

[10 April 2019 II]

- (a) 750 Hz (b) 857 Hz (c) 1143 Hz (d) 807 Hz

73. Two cars A and B are moving away from each other in opposite directions. Both the cars are moving with a speed of 20 ms⁻¹ with respect to the ground. If an observer in car A detects a frequency 2000 Hz of the sound coming from car B, what is the natural frequency of the sound source in car B?

(speed of sound in air = 340 ms⁻¹) [9 April 2019 II]

- (a) 2250 Hz (b) 2060 Hz
(c) 2300 Hz (d) 2150 Hz

74. A train moves towards a stationary observer with speed 34 m/s. The train sounds a whistle and its frequency registered by the observer is f_1 . If the speed of the train is reduced to 17 m/s, the frequency registered is f_2 . If speed of sound is 340 m/s, then the ratio f_1/f_2 is:

[10 Jan. 2019 I]

- (a) 18/17 (b) 19/18 (c) 20/19 (d) 21/20

75. A musician using an open flute of length 50 cm produces second harmonic sound waves. A person runs towards the musician from another end of a hall at a speed of 10 km/h. If the wave speed is 330 m/s, the frequency heard by the running person shall be close to:

[9 Jan. 2019 II]

- (a) 666 Hz (b) 753 Hz
(c) 500 Hz (d) 333 Hz

76. Two sitar strings, A and B, playing the note 'Dha' are slightly out of tune and produce beats and frequency 5 Hz. The tension of the string B is slightly increased and the beat frequency is found to decrease by 3 Hz. If the frequency of A is 425 Hz, the original frequency of B is

[Online April 16, 2018]

- (a) 430 Hz (b) 428 Hz (c) 422 Hz (d) 420 Hz

77. A toy-car, blowing its horn, is moving with a steady speed of 5 m/s, away from a wall. An observer, towards whom the toy car is moving, is able to hear 5 beats per second. If the velocity of sound in air is 340 m/s, the frequency of the horn of the toy car is close to :

[Online April 10, 2016]

- (a) 680 Hz (b) 510 Hz (c) 340 Hz (d) 170 Hz

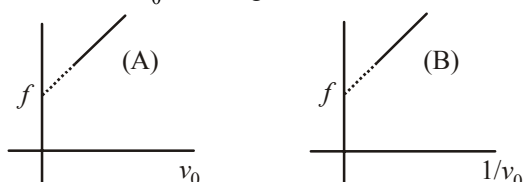
78. Two engines pass each other moving in opposite directions with uniform speed of 30 m/s. One of them is blowing a whistle of frequency 540 Hz. Calculate the

frequency heard by driver of second engine before they pass each other. Speed of sound is 330 m/sec:

[Online April 9, 2016]

- (a) 450 Hz (b) 540 Hz (c) 270 Hz (d) 648 Hz
79. A train is moving on a straight track with speed 20 ms^{-1} . It is blowing its whistle at the frequency of 1000 Hz. The percentage change in the frequency heard by a person standing near the track as the train passes him is (speed of sound = 320 ms^{-1}) close to : [2015]

- (a) 18% (b) 24% (c) 6% (d) 12%
80. A source of sound emits sound waves at frequency f_0 . It is moving towards an observer with fixed speed v_s ($v_s < v$, where v is the speed of sound in air). If the observer were to move towards the source with speed v_0 , one of the following two graphs (A and B) will give the correct variation of the frequency f heard by the observer as v_0 is changed.



The variation of f with v_0 is given correctly by :

[Online April 11, 2015]

- (a) graph A with slope = $\frac{f_0}{(v + v_s)}$
- (b) graph B with slope = $\frac{f_0}{(v - v_s)}$
- (c) graph A with slope = $\frac{f_0}{(v - v_s)}$
- (d) graph B with slope = $\frac{f_0}{(v + v_s)}$
81. A bat moving at 10 ms^{-1} towards a wall sends a sound signal of 8000 Hz towards it. On reflection it hears a sound of frequency f . The value of f in Hz is close to (speed of sound = 320 ms^{-1})
- [Online April 10, 2015]
- (a) 8516 (b) 8258 (c) 8424 (d) 8000
82. A source of sound A emitting waves of frequency 1800 Hz is falling towards ground with a terminal speed v . The observer B on the ground directly beneath the source receives waves of frequency 2150 Hz. The source A receives waves, reflected from ground of frequency nearly: (Speed of sound = 343 m/s)
- [Online April 12, 2014]
- (a) 2150 Hz (b) 2500 Hz
- (c) 1800 Hz (d) 2400 Hz
83. Two factories are sounding their sirens at 800 Hz. A man goes from one factory to other at a speed of 2 m/s. The velocity of sound is 320 m/s. The number of beats heard by the person in one second will be:
- [Online April 11, 2014]

- (a) 2 (b) 4
- (c) 8 (d) 10

84. A and B are two sources generating sound waves. A listener is situated at C. The frequency of the source at A is 500 Hz. A, now, moves towards C with a speed 4 m/s. The number of beats heard at C is 6. When A moves away from C with speed 4 m/s, the number of beats heard at C is 18. The speed of sound is 340 m/s. The frequency of the source at B is : [Online April 22, 2013]



- (a) 500 Hz (b) 506 Hz (c) 512 Hz (d) 494 Hz
85. An engine approaches a hill with a constant speed. When it is at a distance of 0.9 km, it blows a whistle whose echo is heard by the driver after 5 seconds. If the speed of sound in air is 330 m/s, then the speed of the engine is : [Online April 9, 2013]
- (a) 32 m/s (b) 27.5 m/s (c) 60 m/s (d) 30 m/s
86. This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.

Statement 1: Bats emitting ultrasonic waves can detect the location of a prey by hearing the waves reflected from it.

Statement 2: When the source and the detector are moving, the frequency of reflected waves is changed.

[Online May 12, 2012]

- (a) Statement 1 is false, Statement 2 is true.
- (b) Statement 1 is true, Statement 2 is false.
- (c) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.
- (d) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.
87. A motor cycle starts from rest and accelerates along a straight path at 2 m/s^2 . At the starting point of the motor cycle there is a stationary electric siren. How far has the motor cycle gone when the driver hears the frequency of the siren at 94% of its value when the motor cycle was at rest? (Speed of sound = 330 ms^{-1}) [2009]
- (a) 98 m (b) 147 m (c) 196 m (d) 49 m
88. A whistle producing sound waves of frequencies 9500 Hz and above is approaching a stationary person with speed $v \text{ ms}^{-1}$. The velocity of sound in air is 300 ms^{-1} . If the person can hear frequencies upto a maximum of 10,000 Hz, the maximum value of v upto which he can hear whistle is [2006]
- (a) $15\sqrt{2} \text{ ms}^{-1}$ (b) $\frac{15}{\sqrt{2}} \text{ ms}^{-1}$
- (c) 15 ms^{-1} (d) 30 ms^{-1}
89. An observer moves towards a stationary source of sound, with a velocity one-fifth of the velocity of sound. What is the percentage increase in the apparent frequency? [2005]
- (a) 0.5% (b) zero (c) 20% (d) 5%



Hints & Solutions



1. (a) As we know,

$$\text{Pressure amplitude, } \Delta P_0 = aKB = S_0KB = S_0 \times \frac{\omega}{V} \times \rho V^2$$

$$\left[\because K = \frac{\omega}{V}, V = \sqrt{\frac{B}{\rho}} \right]$$

$$\Rightarrow S_0 = \frac{\Delta P_0}{\rho V \omega} \approx \frac{10}{1 \times 300 \times 1000} \text{ m} = \frac{1}{30} \text{ mm} \approx \frac{3}{100} \text{ mm}$$

2. (b) Given : Distance between one crest and one trough = 1.5 m

$$= (2n_1 + 1) \frac{\lambda}{2}$$

$$\text{Distance between two crests} = 5 \text{ m} = n_2 \lambda$$

$$\frac{1.5}{5} = \frac{(2n_1 + 1)}{2n_2} \Rightarrow 3n_2 = 10n_1 + 5$$

Here n_1 and n_2 are integer.

$$\text{If } n_1 = 1, n_2 = 5 \quad \therefore \lambda = 1$$

$$n_1 = 4, n_2 = 15 \quad \therefore \lambda = 1/3$$

$$n_1 = 7, n_2 = 25 \quad \therefore \lambda = 1/5$$

Hence possible wavelengths $\frac{1}{1}, \frac{1}{3}, \frac{1}{5}$ metre.

3. (b) At $t = 0, x = 0, y = 0$

$$\phi = \pi \text{ rad}$$

4. (a) Using, $\beta = 10$

$$\text{or } 120 = 10 \log_{10} \left(\frac{I}{10^{-12}} \right) \quad \dots(i)$$

$$\text{Also } I = \frac{P}{4\pi r^2} = \frac{2}{4\pi r^2} \quad \dots(ii)$$

On solving above equations, we get

$$r = 40 \text{ cm.}$$

5. (a) On comparing with $P = P_0 \sin(\omega t - kx)$, we have $\omega = 1000 \text{ rad/s}$, $K = 3 \text{ m}^{-1}$

$$\therefore v_0 = \frac{\omega}{k} = \frac{1000}{3} = 333.3 \text{ m/s}$$

$$\frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}}$$

$$\text{or } \frac{333.3}{336} = \sqrt{\frac{273+0}{273+t}}$$

$$\therefore t = 4^\circ\text{C}$$

6. (a) Comparing the given equation $y = 10^{-3} \sin(50t + 2x)$ with standard equation, $y = a \sin(\omega t - kx)$
 $\Rightarrow$ wave is moving along -ve x-axis with speed

$$v = \frac{\omega}{K} \Rightarrow v = \frac{50}{2} = 25 \text{ m/sec.}$$

7. (a) Given, amplitude $a = 10 \text{ cm}$
 wave velocity = $2 \times$ maximum particle velocity

$$\text{i.e., } \frac{\omega \lambda}{2\pi} = 2 \frac{a\omega}{\pi}$$

$$\text{or, } \lambda = 4a = 4 \times 10 = 40 \text{ cm}$$

8. (b)

9. (b) Standard equation

$$y(x, t) = A \cos \left(\frac{\omega}{V} x - \omega t \right)$$

From any of the displacement equation

Say y_1

$$\frac{\omega}{V} = 0.50\pi \text{ and } \omega = 100\pi$$

$$\therefore \frac{100\pi}{V} = 0.5\pi$$

$$\therefore V = \frac{100\pi}{0.5\pi} = 200 \text{ m/s}$$

10. (c) The equation of wave at any time is obtained by putting $X = x - vt$

$$y = \frac{1}{1+x^2} = \frac{1}{1+(x-vt)^2} \quad \dots(i)$$

We know at $t = 2 \text{ sec}$,

$$y = \frac{1}{1+(x-1)^2} \quad \dots(ii)$$

On comparing (i) and (ii) we get

$$vt = 1$$

$$V = \frac{1}{t}$$

As $t = 2 \text{ sec}$

$$\therefore V = \frac{1}{2} = 0.5 \text{ m/s.}$$

11. (a) Given

$$y(x, t) = e^{(-ax^2 + bt^2 + 2\sqrt{ab}xt)}$$

$$= e^{-(\sqrt{ax})^2 + (\sqrt{bt})^2 + 2\sqrt{ax} \cdot \sqrt{bt}}$$

$$= e^{-(\sqrt{ax} + \sqrt{bt})^2}$$

$$= e^{-\left(x + \sqrt{\frac{b}{a}}t\right)^2}$$

It is a function of type $y = f(x + vt)$

$\therefore y(x, t)$ represents wave travelling along -ve x direction

$$\Rightarrow \text{Speed of wave} = \frac{\omega}{k} = \sqrt{\frac{b}{a}}$$

12. (a) Given,
Wavelength, $\lambda = 0.08\text{m}$
Time period, $T = 2.05$
 $y(x, t) = 0.005 \cos(\alpha x - \beta t)$ (Given)
Comparing it with the standard equation of wave
 $y(x, t) = a \cos(kx - \omega t)$ we get

$$k = \alpha = \frac{2\pi}{\lambda} \text{ and } \omega = \beta = \frac{2\pi}{T}$$

$$\therefore \alpha = \frac{2\pi}{0.08} = 25\pi \text{ and } \beta = \frac{2\pi}{2} = \pi$$

13. (a) Loudness of sound. $L_1 = 10 \log\left(\frac{I_1}{I_0}\right)$;

$$L_2 = 10 \log\left(\frac{I_2}{I_0}\right)$$

$$\therefore L_1 - L_2 = 10 \log\left(\frac{I_1}{I_0}\right) - 10 \log\left(\frac{I_2}{I_0}\right)$$

$$\text{or, } \Delta L = 10 \log\left(\frac{I_1}{I_0} \times \frac{I_0}{I_2}\right)$$

$$\text{or, } \Delta L = 10 \log\left(\frac{I_1}{I_2}\right)$$

The sound level attenuated by 20 dB ie
 $L_1 - L_2 = 20 \text{ dB}$

$$\text{or, } 20 = 10 \log\left(\frac{I_1}{I_2}\right) \text{ or, } 2 = \log\left(\frac{I_1}{I_2}\right)$$

$$\text{or, } \frac{I_1}{I_2} = 10^2 \text{ or, } I_2 = \frac{I_1}{100}$$

$\Rightarrow$ Intensity decreases by a factor 100.

14. (b) Given, $y = 10^{-6} \sin\left(100t + 20x + \frac{\pi}{4}\right) \text{m}$

Comparing it with standard equation, we get
 $\omega = 100$ and $k = 20$

$$v = \frac{\omega}{k} = \frac{100}{20} = 5 \text{ m/s}$$

15. (a) $y = 10^{-4} \sin\left(600t - 2x + \frac{\pi}{3}\right)$

On comparing with standard equation

$$y = A \sin(\omega t - kx + \phi)$$

we get $\omega = 600$; $k = 2$

Velocity of wave

$$v = \frac{\omega}{k} = \frac{600}{2} = 300 \text{ ms}^{-1}$$

16. (b) The frequency of a tuning fork is given by

$$f = \frac{m^2 k}{4\sqrt{3} \pi \ell^2} \sqrt{\frac{Y}{\rho}}$$

As temperature increases, the length or dimension of the prongs increases and also young's modulus increases therefore f decreases.

17. (a) Here, $l_1 = 17 \text{ cm}$ and $l_2 = 24.5 \text{ cm}$, $V = 330 \text{ m/s}$,
 $f = ?$

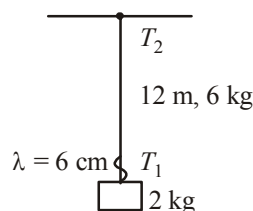
$$\lambda = 2(l_2 - l_1) = 2 \times (24.5 - 17) = 15 \text{ cm}$$

$$\text{Now, from } v = f\lambda \Rightarrow 330 = \lambda \times 15 \times 10^{-2}$$

$$\therefore \lambda = \frac{330}{15} \times 100 = \frac{1100 \times 100}{5} = 2200 \text{ Hz}$$

18. (c) Using, $V = f\lambda$

$$\frac{V_1}{\lambda_1} = \frac{V_2}{\lambda_2} \Rightarrow \lambda_2 = \frac{V_2}{V_1} \lambda_1$$



Again using,

$$n = \frac{V}{\lambda} = \sqrt{\frac{T}{M}} \lambda_2 = \sqrt{\frac{T_2}{T_1}} \lambda_1 \quad T_2 = 8g \text{ (Top)}$$

$$= \sqrt{\frac{8g}{2g}} \lambda_1 = 2\lambda_1 = 12 \text{ cm } T_1 = 2g \text{ (Bottom)}$$

19. (a) Using $f = \frac{1}{2\ell} \sqrt{\frac{T}{\mu}}$,

where, T = tension and $\mu = \frac{\text{mass}}{\text{length}}$

$$f_x = \frac{1}{2\ell} \sqrt{\frac{T_x}{\mu}} \text{ and } f_z = \frac{1}{2\ell} \sqrt{\frac{T_z}{\mu}}$$

$$\frac{f_x}{f_z} = \frac{450}{300} = \sqrt{\frac{T_x}{T_z}}$$

$$\therefore \frac{T_x}{T_z} = \frac{9}{4} = 2.25$$

20. (35.00)

Given,

Density of wire, $\sigma = 9 \times 10^{-3} \text{ kg cm}^{-3}$

Young's modulus of wire, $Y = 9 \times 10^{10} \text{ Nm}^{-2}$

Strain $= 4.9 \times 10^{-4}$

$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{T/A}{\text{Strain}}$$

$$\therefore \frac{T}{A} = Y \times \text{Strain} = 9 \times 10^9 \times 4.9 \times 10^{-4}$$

Also, mass of wire, $m = A\ell\sigma$

$$\text{Mass per unit length, } \mu = \frac{m}{\ell} = A\sigma$$

Fundamental frequency in the string

$$f = \frac{1}{2l} \sqrt{\frac{T}{\mu}} = \frac{1}{2l} \sqrt{\frac{T}{\sigma A}}$$

$$= \frac{1}{2 \times 1} \sqrt{\frac{9 \times 10^9 \times 4.9 \times 10^{-4}}{9 \times 10^3}}$$

$$= \frac{1}{2} \sqrt{49 \times 10^{9-4-3}} = \frac{1}{2} \times 70 = 35 \text{ Hz}$$

21. (106) Given : $V_{\text{air}} = 300 \text{ m/s}$, $\rho_{\text{gas}} = 2 \rho_{\text{air}}$

Using, $V = \sqrt{\frac{B}{\rho}}$

$$\frac{V_{\text{gas}}}{V_{\text{air}}} = \frac{\sqrt{\frac{B}{2\rho_{\text{air}}}}}{\sqrt{\frac{B}{\rho_{\text{air}}}}}$$

$$\Rightarrow V_{\text{gas}} = \frac{V_{\text{air}}}{\sqrt{2}} = \frac{300}{\sqrt{2}} = 150\sqrt{2} \text{ m/s}$$

And $f_{\text{nth harmonic}} = \frac{nv}{2L}$ (in open organ pipe)

($L = 1$ metre given)

$$\therefore f_{2\text{nd harmonic}} - f_{\text{fundamental}} = \frac{2v}{2 \times 1} - \frac{v}{2 \times 1} = \frac{v}{2}$$

$$\therefore f_{2\text{n}} \text{ harmonic} - f_{\text{fundamental}} = \frac{150\sqrt{2}}{2} = \frac{150}{\sqrt{2}} \approx 106 \text{ Hz}$$

22. (b) The velocity of a transverse wave in a stretched wire is given by

$$v = \sqrt{\frac{T}{\mu}}$$

Where,

T = Tension in the wire

μ = linear density of wire

($\therefore V \propto T$)

$$\therefore \frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}}$$

$$\Rightarrow \frac{v}{v} \times 2 = \sqrt{\frac{2.06 \times 10^4}{T_2}}$$

$$\Rightarrow T_2 = \frac{2.06 \times 10^4}{4} = 0.515 \times 10^4 \text{ N}$$

$$\Rightarrow T_2 = 5.15 \times 10^3 \text{ N}$$

23. (a) Given, $l = 60 \text{ cm}$, $m = 6 \text{ g}$, $A = 1 \text{ mm}^2$, $v = 90 \text{ m/s}$ and $Y = 16 \times 10^{11} \text{ Nm}^{-2}$

Using, $v = \sqrt{\frac{T}{m}} \times l \Rightarrow T = \frac{mv^2}{l}$

Again from, $Y = \frac{T}{A} \Delta L / L_0$

$$\Delta L = \frac{Tl}{YA} = \frac{mv^2 \times l}{l(YA)}$$

$$= \frac{6 \times 10^{-3} \times 90^2}{16 \times 10^{11} \times 10^{-6}} = 3 \times 10^{-4} \text{ m}$$

$$= 0.03 \text{ mm}$$

24. (c) We have given,

$$y = 0.03 \sin(450t - 9x)$$

Comparing it with standard equation of wave, we get $\omega = 450 \text{ k} = 9$

$$\therefore v = \frac{\omega}{k} = \frac{450}{9} = 50 \text{ m/s}$$

Velocity of travelling wave on a stretched string is given by

$$v = \sqrt{\frac{T}{\mu}} \Rightarrow \frac{T}{\mu} = 2500$$

μ = linear mass density

$$\Rightarrow T = 2500 \times 5 \times 10^{-3}$$

$$\Rightarrow 12.5 \text{ N}$$

25. (b) Wave speed $V = \sqrt{\frac{T}{\mu}}$

when car is at rest $a = 0$

$$\therefore 60 = \sqrt{\frac{Mg}{\mu}}$$

Similarly when the car is moving with acceleration a ,

$$60.5 = \sqrt{\frac{M(g^2 + a^2)^{1/2}}{\mu}}$$

$$\frac{60.5}{60} = \sqrt{\frac{g^2 + a^2}{g^2}}$$

$$\left(1 + \frac{0.5}{60}\right)^4 = \frac{g^2 + a^2}{g^2} = 1 + \frac{2}{60}$$

$$\Rightarrow g^2 + a^2 = g^2 + g^2 \times \frac{2}{60}$$

$$a = g \sqrt{\frac{2}{60}} = \frac{g}{\sqrt{30}} \quad [\text{which is closest to } g/5]$$

26. (a) Fundamental frequency, $f = 70 \text{ Hz}$.

The fundamental frequency of wire vibrating under tension T is given by

$$f = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

Here, μ = mass per unit length of the wire
 L = length of wire

$$70 = \frac{1}{2L} \sqrt{\frac{540}{6 \times 10^{-3}}}$$

$$\Rightarrow L \approx 2.14 \text{ m}$$

$$27. \text{ (b) } V = f\lambda = f \times 2(\ell_2 - \ell_1) \\ = 480 \times 2(0.70 - 0.30) \\ = 384 \text{ m/s}$$

$$28. \text{ (b) } \frac{3\lambda}{2} = 2 \text{ or } \lambda = \frac{4}{3} \text{ m}$$

$$\text{Velocity, } v = f\lambda = 240 \times \frac{4}{3} = 320 \text{ m/sec}$$

$$\text{Also } f_1 = \frac{240}{3} = 80 \text{ Hz}$$

$$29. \text{ (b) Given, } y = 0.3 \sin(0.157x) \cos(200\pi t) \\ \text{So } k = 0.157 \text{ and } \omega = 200\pi$$

$$\text{or } f = 100 \text{ Hz, } v = \frac{\omega}{k} = \frac{200\pi}{0.157} = 4000 \text{ m/s}$$

$$\text{Now, using } f = \frac{nv}{2l} = \frac{4v}{2l} = \frac{2v}{l}$$

$$\therefore l = \frac{2v}{f} = \frac{2 \times 4000}{100} = 80 \text{ m}$$

$$30. \text{ (c) As there must be node at both ends and at the joint of the wire A and B so}$$

$$\frac{V_A}{V_B} = \sqrt{\frac{u_B}{u_A}} = \frac{r_B}{r_A} = 2 = \frac{\lambda_A}{\lambda_B}$$

$$\Rightarrow \lambda_A = 2\lambda_B$$

$$\Rightarrow \frac{P}{q} = \frac{1}{2}$$

$$31. \text{ (a) If a closed pipe vibration in } N^{\text{th}} \text{ mode then frequency}$$

$$\text{of vibration } n = \frac{(2N-1)v}{4l} = (2N-1)n_1$$

(where n_1 = fundamental frequency of vibration)

$$\text{Hence } 20,000 = (2N-1) \times 1500$$

$$\Rightarrow N = 7.1 \approx 7$$

$$\therefore \text{Number of over tones} = (\text{No. of mode of vibration}) - 1 \\ = 7 - 1 = 6$$

$$32. \text{ (d) Velocity of wave on string}$$

$$V = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{8}{5}} \times 1000 = 40 \text{ m/s}$$

Here, T = tension and μ = mass/length

$$\text{Wavelength of wave } \lambda = \frac{v}{n} = \frac{40}{100} \text{ m}$$

Separation b/w successive nodes,

$$\frac{\lambda}{2} = \frac{40}{2 \times 100} = \frac{20}{100} \text{ m} = 20 \text{ cm}$$

$$33. \text{ (a) In solids, Velocity of wave}$$

$$V = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{9.27 \times 10^{10}}{2.7 \times 10^3}}$$

$$v = 5.85 \times 10^3 \text{ m/sec}$$

Since rod is clamped at middle fundamental wave shape is as follow

$$\frac{\lambda}{2} = L \Rightarrow \lambda = 2L$$

$$\lambda = 1.2 \text{ m } (\because L = 60 \text{ cm} = 0.6 \text{ m (given)})$$

Using $v = f\lambda$

$$\Rightarrow f = \frac{v}{\lambda} = \frac{5.85 \times 10^3}{1.2}$$

$$= 4.88 \times 10^3 \text{ Hz} \approx 5 \text{ KHz}$$

$$34. \text{ (a) For first resonance, } \frac{\lambda}{4} = \ell_1 + e = 11 \text{ cm} \\ (\because \text{end correction } e = 1 \text{ cm given})$$

$$\text{For second resonance, } \frac{3\lambda}{4} = \ell_2 + e$$

$$\Rightarrow \ell_2 = 3 \times 11 - 1 = 32 \text{ cm}$$

$$35. \text{ (b) } n_1 = n_2 \\ T \rightarrow \text{Same} \\ r \rightarrow \text{Same} \\ l \rightarrow \text{Same}$$

Frequency of vibration

$$n = \frac{p}{2l} \sqrt{\frac{T}{\pi r^2 \rho}}$$

As T , r , and l are same for both the wires

$$n_1 = n_2$$

$$\frac{p_1}{\sqrt{\rho_1}} = \frac{p_2}{\sqrt{\rho_2}}$$

$$\Rightarrow \frac{p_1}{p_2} = \frac{1}{2}$$

$$\because \rho_2 = 4\rho_1$$

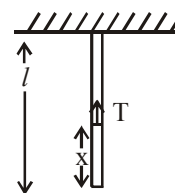
$$36. \text{ (a) We know that velocity in string is given by}$$

$$v = \sqrt{\frac{T}{\mu}} \quad \dots(i)$$

$$\text{where } \mu = \frac{m}{l} = \frac{\text{mass of string}}{\text{length of string}}$$

$$\text{The tension } T = \frac{m}{\ell} \times x \times g \quad \dots(ii)$$

From (1) and (2)



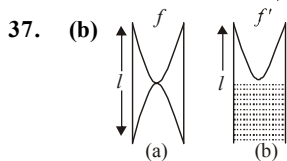
$$\frac{dx}{dt} = \sqrt{gx}$$

$$x^{-1/2} dx = \sqrt{g} dt$$

$$\therefore \int_0^{\ell} x^{-1/2} dx = \sqrt{g} \int_0^{\ell} dt$$

$$\Rightarrow 2\sqrt{\ell}$$

$$= \sqrt{g} \times t \quad \therefore t = 2\sqrt{\frac{\ell}{g}} = 2\sqrt{\frac{20}{10}} = 2\sqrt{2}$$



The fundamental frequency in case (a) is $f = \frac{v}{2\ell}$

The fundamental frequency in case (b) is

$$f' = \frac{v}{4(\ell/2)} = \frac{v}{2\ell} = f$$

38. (c) Length of pipe = 85 cm = 0.85 m

Frequency of oscillations of air column in closed organ pipe is given by,

$$f = \frac{(2n-1)v}{4L}$$

$$f = \frac{(2n-1)v}{4L} \leq 1250$$

$$\Rightarrow \frac{(2n-1) \times 340}{0.85 \times 4} \leq 1250$$

$$\Rightarrow 2n-1 \leq 12.5 \approx 6$$

39. (b) Total length of sonometer wire, $l = 110 \text{ cm} = 1.1 \text{ m}$
Length of wire is in ratio, 6 : 3 : 2 i.e. 60 cm, 30 cm, 20 cm.

Tension in the wire, $T = 400 \text{ N}$

Mass per unit length, $m = 0.01 \text{ kg}$

Minimum common frequency = ?

As we know,

$$\text{Frequency, } v = \frac{1}{2l} \sqrt{\frac{T}{m}} = \frac{1000}{11} \text{ Hz}$$

$$\text{Similarly, } v_1 = \frac{1000}{6} \text{ Hz}$$

$$v_2 = \frac{1000}{3} \text{ Hz}$$

$$v_3 = \frac{1000}{2} \text{ Hz}$$

Hence common frequency = 1000 Hz

40. (b) Fundamental frequency,

$$f = \frac{v}{2\ell} = \frac{1}{2\ell} \sqrt{\frac{T}{\mu}} = \frac{1}{2\ell} \sqrt{\frac{T}{A\rho}} \quad \left[\because v = \sqrt{\frac{T}{\mu}} \text{ and } \mu = \frac{m}{\ell} \right]$$

$$\text{Also, } Y = \frac{T\ell}{A\Delta\ell} \Rightarrow \frac{T}{A} = \frac{Y\Delta\ell}{\ell}$$

$$\Rightarrow f = \frac{1}{2\ell} \sqrt{\frac{Y\Delta\ell}{\ell\rho}} \quad \dots(i)$$

$$\ell = 1.5 \text{ m, } \frac{\Delta\ell}{\ell} = 0.01,$$

$$\rho = 7.7 \times 10^3 \text{ kg/m}^3 \text{ (given)}$$

$$Y = 2.2 \times 10^{11} \text{ N/m}^2 \text{ (given)}$$

Putting the value of ℓ , $\frac{\Delta\ell}{\ell}$, ρ and Y in eqⁿ. (i) we get,

$$f = \sqrt{\frac{2}{7}} \times \frac{10^3}{3} \text{ or } f \approx 178.2 \text{ Hz}$$

41. (d) Total length of the wire, $L = 114 \text{ cm}$

$$n_1 : n_2 : n_3 = 1 : 3 : 4$$

Let L_1 , L_2 and L_3 be the lengths of the three parts

$$\text{As } n \propto \frac{1}{L}$$

$$\therefore L_1 : L_2 : L_3 = \frac{1}{1} : \frac{1}{3} : \frac{1}{4} = 12 : 4 : 3$$

$$\therefore L_1 = \left(\frac{12}{12+4+3} \times 114 \right) = 72 \text{ cm}$$

$$L_2 = \left(\frac{4}{12+4+3} \times 114 \right) = 24 \text{ cm}$$

$$\text{and } L_3 = \left(\frac{3}{12+4+3} \times 114 \right) = 18 \text{ cm}$$

Hence the bridges should be placed at 72 cm and $72 + 24 = 96 \text{ cm}$ from one end.

42. (a) Initially for open organ pipe, fundamental frequency

$$v_0 = \frac{v}{2l_0} \quad \dots(i)$$

where l_0 is the length of the tube

v = speed of sound

But when it is half dipped in water, it becomes closed organ

pipe of length $\frac{l_0}{2}$.

Fundamental frequency of closed organ pipe

$$v_c = \frac{v}{4l_c} \quad \dots(ii)$$

$$\text{New length, } l_c = \frac{l_0}{2}$$

$$\text{Thus } v_c = \frac{v}{4l_0/2} \Rightarrow v_c = \frac{v}{2l} \quad \dots(iii)$$

From equations (i) and (iii)

$$v_0 = v_c$$

Thus, $v_c = f$ ($\because v_0 = f$ is given)

43. (b) Given : Frequency of tuning fork, $n = 264 \text{ Hz}$

Length of column $L = ?$

For closed organ pipe

$$n = \frac{v}{4l}$$

$$\Rightarrow l = \frac{v}{4n} = \frac{330}{4 \times 264} = 0.3125$$

$$\text{or, } l = 0.3125 \times 100 = 31.25 \text{ cm}$$

In case of closed organ pipe only odd harmonics are possible.

Therefore value of l will be $(2n-1)l$

Hence option (b) i.e. $3 \times 31.25 = 93.75 \text{ cm}$ is correct.

44. (d) Two lowest frequencies to which tube will resonates are 272 Hz and 544 Hz.

$$45. (d) y = 0.02(m) \sin \left[2\pi \left(\frac{t}{0.04(s)} - \frac{x}{0.50(m)} \right) \right]$$

Comparing it with the standard wave equation

$$y = a \sin(\omega t - kx)$$

we get

$$\omega = \frac{2\pi}{0.04} \text{ rad s}^{-1}$$

$$\text{and } k = \frac{2\pi}{0.50}$$

$$\text{Wave velocity, } v = \frac{w}{k}$$

$$\Rightarrow v = \frac{2\pi / 0.04}{2\pi / 0.5} = 12.5 \text{ m/s}$$

Velocity on a string is given by

$$v = \sqrt{\frac{T}{\mu}}$$

$$\therefore T = v^2 \times \mu = (12.5)^2 \times 0.04 = 6.25 \text{ N}$$

46. (b) Fundamental frequency for first resonant length

$$v = \frac{v}{4\ell_1} = \frac{v}{4 \times 18} \text{ (in winter)}$$

Fundamental frequency for second resonant length

$$v' = \frac{3v'}{4\ell_2} = \frac{3v'}{4x} \text{ (in summer)}$$

According to questions,

$$\therefore \frac{v}{4 \times 18} = \frac{3v'}{4 \times x}$$

$$\therefore x = 3 \times 18 \times \frac{v'}{v}$$

$$\therefore x = 54 \times \frac{v'}{v} \text{ cm}$$

$v' > v$ because velocity of light is greater in summer as compared to winter ($v \propto \sqrt{T}$)

$$\therefore x > 54 \text{ cm}$$

47. (a) It is given that 315 Hz and 420 Hz are two resonant frequencies, let these be n^{th} and $(n+1)^{\text{th}}$ harmonies, then

$$\text{we have } \frac{nv}{2\ell} = 315$$

$$\text{and } (n+1) \frac{v}{2\ell} = 420$$

$$\Rightarrow \frac{n+1}{n} = \frac{420}{315}$$

$$\Rightarrow n = 3$$

$$\text{Hence } 3 \times \frac{v}{2\ell} = 315 \Rightarrow \frac{v}{2\ell} = 105 \text{ Hz}$$

The lowest resonant frequency is when $n=1$

Therefore lowest resonant frequency = 105 Hz.

48. (c) The fundamental frequency for tube B closed at one end is given by

$$v_B = \frac{v}{4\ell} \quad \left[\because \ell = \frac{\lambda}{4} \right]$$

Where ℓ = length of the tube and v is the velocity of sound in air.

The fundamental frequency for tube A open with both ends is given by

$$v_A = \frac{v}{2\ell} \quad \left[\because \ell = \frac{\lambda}{2} \right]$$

$$\therefore \frac{v_A}{v_B} = \frac{v}{2\ell} \times \frac{4\ell}{v} = \frac{2}{1}$$

49. (b) To form a node there should be superposition of this wave with the reflected wave. The reflected wave should travel in opposite direction with a phase change of π . The equation of the reflected wave will be

$$y = a \sin(\omega t + kx + \pi)$$

$$\Rightarrow y = -a \sin(\omega t + kx)$$

50. (a)

51. (c) Beat frequency
= difference in frequencies of two waves
= $11 - 9 = 2 \text{ Hz}$

52. (d)

53. (d) According to question, tuning fork gives 1 beat/second with (N) 3^{rd} normal mode. Therefore, organ pipe will have frequency $(256 \pm 1) \text{ Hz}$. In open organ pipe, frequency

$$n = \frac{NV}{2\ell}$$

$$\text{or, } 255 = \frac{3 \times 340}{2 \times \ell} \Rightarrow \ell = 2 \text{ m} = 200 \text{ cm}$$

54. (a) Probable frequencies of tuning fork be $n \pm 5$

$$\text{Frequency of sonometer wire, } n \propto \frac{1}{l}$$

$$\therefore \frac{n+5}{n-5} = \frac{100}{95} \Rightarrow 95(n+5) = 100(n-5)$$

$$\text{or, } 95n + 475 = 100n - 500$$

$$\text{or, } 5n = 975$$

$$\text{or, } n = \frac{975}{5} = 195 \text{ Hz}$$

55. (a) Given, $y(x, t) = 0.5 \sin\left(\frac{5\pi}{4}x\right) \cos(200\pi t)$,

comparing with equation $y(x, t) = 2a \sin kx \cos \omega t$

$$\omega = 200\pi, k = \frac{5\pi}{4}$$

$$\text{speed of travelling wave } v = \frac{\omega}{k} = \frac{200\pi}{5\pi/4} = 160 \text{ m/s}$$

56. (b) Since the point $x=0$ is a node and reflection is taking place from point $x=0$. This means that reflection must be taking place from the fixed end and hence the reflected ray must suffer an additional phase change of π or a path

change of $\frac{\lambda}{2}$.

$$\text{So, if } y_{\text{incident}} = a \cos(kx - \omega t)$$

$$\Rightarrow y_{\text{incident}} = a \cos(-kx - \omega t + \pi)$$

$$= -a \cos(\omega t + kx)$$

Hence equation for the other wave

$$y = a \cos(kx + \omega t + \pi)$$

57. (d) In case of destructive interference

Phase difference $\phi = 180^\circ$ or π

So wave pair (i) and (ii) will produce destructive interference.

Stationary or standing waves will produce by equations (iii) & (iv) as two waves travelling along the same line but in opposite direction.

$$n' = n + x$$

58. (d) $y = A \sin(\omega t - kx) + A \sin(\omega t + kx)$

$$y = 2A \sin \omega t \cos kx$$

This is an equation of standing wave. For position of nodes

$$\cos kx = 0$$

$$\Rightarrow \frac{2\pi}{\lambda} x = (2n+1) \frac{\pi}{2}$$

$$\Rightarrow x = \frac{(2n+1)\lambda}{4}, n = 0, 1, 2, 3, \dots$$

59. (a) Intensity of a wave

$$I = \frac{1}{2} \rho w^2 A^2 v$$

$$\text{Since, } I \propto A^2 \omega^2$$

$$\therefore I_1 \propto (2a)^2 \omega^2$$

$$\text{and } I_2 \propto a^2 (2\omega)^2$$

$$I_1 = I_2$$

In the same medium, ρ and v are same.

Intensity depends on amplitude and frequency.

60. (b) Maximum number of beats

= Maximum frequency - Minimum frequency

$$= (v+1) - (v-1) = 2 \text{ Beats per second}$$

61. (d) Frequency of fork 1, $n_0 = 200$ Hz

No. of beats heard when fork 2 is sounded with fork 1 = $\Delta n = 4$

Now on loading (attaching tape) on unknown fork, the mass of tuning fork increases. So the beat frequency increases (from 4 to 6 in this case) then the frequency of the unknown fork 2 is given by,

$$n = n_0 - \Delta n = 200 - 4 = 196 \text{ Hz}$$

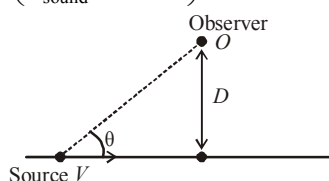
62. (c) It is given that tuning fork of frequency 256 Hz makes 5 beats/second with the vibrating string of a piano. Therefore, possible frequency of the piano are (256 ± 5) Hz. i.e., either 261 Hz or 251 Hz. When the tension in the piano string increases, its frequency will increase. As the original frequency was 261 Hz, the beat frequency should decrease, we can conclude that the frequency of piano string is 251 Hz

63. (b) Frequency of unknown fork = known frequency $\pm$ Beat frequency = $288 + 4$ cps or $288 - 4$ cps i.e. 292 cps or 284 cps. When a little wax is placed on the unknown fork, it produces 2 beats/sec. When a little wax is placed on the unknown fork, its frequency decreases and simultaneously the beat frequency decreases confirming that the frequency of the unknown fork is 292 cps.

Note : Had the frequency of unknown fork been 284 cps, then on placing wax its frequency would have decreased thereby increasing the gap between its frequency and the frequency of known fork. This would produce high beat frequency.

64. (b) Frequency heard by the observer

$$v_{\text{observed}} = \left(\frac{v_{\text{sound}}}{v_{\text{sound}} - v \cos \theta} \right) v_0$$



Initially θ will be less so $\cos \theta$ more.

$\therefore v_{\text{observed}}$ more, then it will decrease.

65. (a) Let f_1 be the frequency heard by wall,

$$f_1 = \left(\frac{v}{v - v_c} \right) f_0$$

Here, v = Velocity of sound,

v_c = Velocity of Car,

f_0 = actual frequency of car horn

Let f_2 be the frequency heard by driver after reflection from wall.

$$f_2 = \left(\frac{v + v_c}{v} \right) f_1 = \left(\frac{v + v_c}{v - v_c} \right) f_0$$

$$\Rightarrow 480 = \left[\frac{345 + v_c}{345 - v_c} \right] 440 \Rightarrow \frac{12}{11} = \frac{345 + v_c}{345 - v_c}$$

$$\Rightarrow v_c = 54 \text{ km/hr}$$

66. (a) From the Doppler's effect of sound, frequency appeared at wall

$$f_w = \frac{330}{330 - v} \cdot f \quad \dots(i)$$

Here, v = speed of bus,

f = actual frequency of source

Frequency heard after reflection from wall (f') is

$$f' = \frac{330 + v}{330} \cdot f_w = \frac{330 + v}{330 - v} \cdot f$$

$$\Rightarrow 490 = \frac{330 + v}{330 - v} \cdot 420$$

$$\Rightarrow v = \frac{330 \times 7}{91} \approx 25.38 \text{ m/s} = 91 \text{ km/s}$$

67. (d) Permanent magnets (P) are made of materials with large retentivity and large coercivity. Transformer cores (T) are made of materials with low retentivity and low coercivity.

68. (c) From Doppler's effect, frequency of sound heard (f_1) when source is approaching

$$f_1 = f_0 \frac{c}{c - v}$$

Here, c = velocity of sound

v = velocity of source

Frequency of sound heard (f_2) when source is receding

$$f_2 = f_0 \frac{c}{c + v}$$

$$\text{Beat frequency} = f_1 - f_2$$

$$\Rightarrow 2 = f_1 - f_2 = f_0 c \left[\frac{1}{c - v} - \frac{1}{c + v} \right]$$

$$= f_0 c \frac{2v}{c^2 \left[1 - \frac{v^2}{c^2} \right]}$$

For $c \gg v$

$$\Rightarrow v = \frac{2c}{2f_0} = \frac{c}{f_0} = \frac{350}{1400} = \frac{1}{4} \text{ m/s}$$

69. (d) $f_1 = f \left(\frac{v - v_o}{v - v_s} \right) = f \left(\frac{1500 - 5}{1500 - 7.5} \right)$

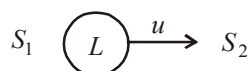
No reflected signal,

$$f_2 = f_1 \left(\frac{v + v_o}{v + v_s} \right) = f_1 \left(\frac{1500 + 7.5}{1500 + 5} \right)$$

$$f_2 = 500 \left(\frac{1500 - 5}{1500 - 7.5} \right) \left(\frac{1500 + 7.5}{1500 + 5} \right)$$

502 Hz

70. (c) $f_1 = f \frac{v - v_0}{v}$ and $f_2 = f \frac{v + v_0}{v}$



But frequency,

$$f_2 - f_1 = f \times \frac{2v_0}{v}$$

$$\text{or } 10 = 660 \times \frac{2u}{330}$$

$$\therefore u = 2.5 \text{ m/s.}$$

71. (b) Frequency of sound source (f_0) = 500 Hz
When observer is moving away from the source

$$\text{Apparent frequency } f_1 = 480 = f_0 \left(\frac{v - v_0''}{v} \right) \dots (i)$$

And when observer is moving towards the source

$$f_2 = 530 = f_0 \left(\frac{v - v_0''}{v} \right) \dots (ii)$$

From equation (i)

$$480 = 500 \left(\frac{300 - v_0''}{300} \right)$$

$$v_0'' = 12 \text{ m/s}$$

From equation (ii)

$$530 = 500 \left(1 + \frac{v_0''}{v} \right)$$

$$\therefore v_0'' = 18 \text{ m/s}$$

72. (a) When source is moving towards a stationary observer,

$$f_{\text{app}} = f_{\text{source}} \left(\frac{V - 0}{V - 50} \right)$$

$$1000 = f_{\text{source}} \left(\frac{350}{300} \right)$$

When source is moving away from observer

$$f' = f_{\text{source}} \left(\frac{350}{350 + 50} \right)$$

$$f' = \frac{1000 \times 300}{350} \times \frac{350}{400}$$

$f' \approx 750 \text{ Hz}$

73. (a) $f' = f \frac{v - v_0}{v + v_s}$

$$\text{or } 2000 = f \frac{340 - 20}{340 + 20}$$

$$\therefore f = 2250 \text{ Hz.}$$

74. (b) According to Doppler's effect, when source is moving but observer at rest

$$f_{\text{app}} = f_0 \left[\frac{V}{V - V_s} \right] \Rightarrow f_1 = f_0 \left[\frac{340}{340 - 34} \right]$$

$$\text{and, } f_2 = f_0 \left[\frac{340}{340 - 17} \right]$$

$$\therefore \frac{f_1}{f_2} = \frac{340 - 17}{340 - 34} = \frac{323}{306} \text{ or, } \frac{f_1}{f_2} = \frac{19}{18}$$

75. (a) Frequency of the sound produced by open flute.

$$f = 2 \left(\frac{v}{2\ell} \right) = \frac{2 \times 330}{2 \times 0.5} = 660 \text{ Hz}$$

$$\text{Velocity of observer, } v_0 = 10 \times \frac{5}{18} = \frac{25}{9} \text{ m/s}$$

As the source is moving towards the observer therefore, according to Doppler's effect.

$\therefore$ Frequency detected by observer,

$$f' = \left[\frac{v + v_0}{v} \right] f = \left[\frac{\frac{25}{9} + 330}{330} \right] 660$$

$$= \frac{2995}{9 \times 330} \times 660 \text{ or, } f' = 665.55 \approx 666 \text{ Hz}$$

76. (d) $n_A = 425 \text{ Hz}$, $n_B = ?$

Beat frequency $x = 5 \text{ Hz}$ which is decreasing ($5 \rightarrow 3$) after increasing the tension of the string B.

Also tension of string B increasing so

$$n_B \uparrow (\because n \propto \sqrt{T})$$

$$\text{Hence } n_A - n_B \uparrow = x \downarrow \longrightarrow \text{correct}$$

$$n_B \uparrow - n_A = x \downarrow \longrightarrow \text{incorrect}$$

$$\therefore n_B = n_A - x = 425 - 5 = 420 \text{ Hz}$$

77. (d) From Doppler's effect

$$f(\text{direct}) = f \left(\frac{340}{340 - 5} \right) = f_1$$

$$f(\text{by wall}) = f \left(\frac{340}{340 + 5} \right) = f_2$$

$$\text{Beats} = (f_1 - f_2)$$

$$5 = f \left(\frac{340}{340 - 5} - \frac{340}{340 + 5} \right)$$

$$\Rightarrow f = 170 \text{ Hz.}$$

78. (d) We know that the apparent frequency

$$f' = \left(\frac{v - v_0}{v - v_s} \right) f \text{ from Doppler's effect}$$

where $v_0 = v_s = 30 \text{ m/s}$, velocity of observer and source
Speed of sound $v = 330 \text{ m/s}$

$$\therefore f' = \frac{330 + 30}{330 - 30} \times 540 = 648 \text{ Hz.}$$

$$\therefore \text{Frequency of whistle } (f) = 540 \text{ Hz.}$$

79. (d) $f_1 = f \left[\frac{v}{v - v_s} \right] = f \times \frac{320}{300} \text{ Hz}$

$$f_2 = f \left[\frac{v}{v + v_s} \right] = f \times \frac{320}{340} \text{ Hz}$$

$$\left(\frac{f_2}{f_1} - 1\right) \times 100 = \left(\frac{300}{340} - 1\right) \times 100 = 12\%$$

80. (c) According to Doppler's effect,

$$\text{Apparent, frequency } f = \left(\frac{V + V_0}{V - V_s}\right) f_0$$

$$\text{Now, } f = \left(\frac{f_0}{V - V_s}\right) V_0 + \frac{V f_0}{V - V_s}$$

$$\text{So, slope} = \frac{f_0}{V - V_s}$$

Hence, option (c) is the correct answer.

81. (a) Reflected frequency of sound reaching bat

$$= \left[\frac{V - (-V_0)}{V - V_s}\right] f = \left[\frac{V + V_0}{V - V_s}\right] f = \frac{V + 10}{V - 10} f$$

$$= \left(\frac{320 + 10}{320 - 10}\right) \times 8000 = 8516 \text{ Hz}$$

82. (b) Given $f_A = 1800 \text{ Hz}$

$$v_t = v$$

$$f_B = 2150 \text{ Hz}$$

Reflected wave frequency received by A, $f_A' = ?$

Applying doppler's effect of sound,

$$f' = \frac{v_s f}{v_s - v_t}$$

$$\text{here, } v_t = v_s \left(1 - \frac{f_A}{f_B}\right)$$

$$= 343 \left(1 - \frac{1800}{2150}\right)$$

$$v_t = 55.8372 \text{ m/s}$$

Now, for the reflected wave,

$$\therefore f_A' = \left(\frac{v_s + v_t}{v_s - v_t}\right) f_A$$

$$= \left(\frac{343 + 55.83}{343 - 55.83}\right) \times 1800$$

$$= 2499.44 \approx 2500 \text{ Hz}$$

83. (d) **Given:** Frequency of sound produced by siren, $f = 800 \text{ Hz}$

Speed of observer, $u = 2 \text{ m/s}$

Velocity of sound, $v = 320 \text{ m/s}$

No. of beats heard per second = ?

No. of extra waves received by the observer per second = $\pm 4\lambda$

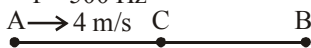
$\therefore$ No. of beats/sec

$$= \frac{2}{\lambda} - \left(-\frac{2}{\lambda}\right) = \frac{4}{\lambda}$$

$$= \frac{2 \times 2}{\frac{320}{800}} \quad \left(\because \lambda = \frac{V}{f}\right)$$

$$= \frac{2 \times 2 \times 800}{320} = 10$$

84. (c) $f = 500 \text{ Hz}$



Listener

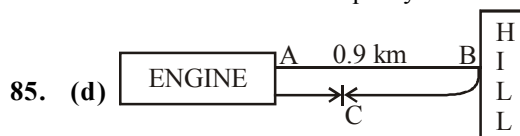
Case 1 : When source is moving towards stationary listener

$$\text{apparent frequency } \eta' = \eta \left(\frac{v}{v - v_s}\right) = 500 \left(\frac{340}{336}\right) = 506 \text{ Hz}$$

Case 2 : When source is moving away from the stationary listener

$$\eta'' = \eta \left(\frac{v}{v + v_s}\right) = 500 \left(\frac{340}{344}\right) = 494 \text{ Hz}$$

In case 1 number of beats heard is 6 and in case 2 number of beats heard is 18 therefore frequency of the source at B = 512 Hz



85. (d)

Let after 5 sec engine at point C

$$t = \frac{AB}{330} + \frac{BC}{330} \quad 5 = \frac{0.9 \times 1000}{330} + \frac{BC}{330}$$

$$\therefore BC = 750 \text{ m}$$

Distance travelled by engine in 5 sec

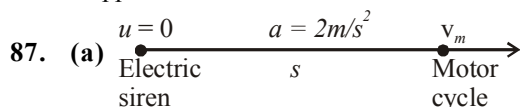
$$= 900 \text{ m} - 750 \text{ m} = 150 \text{ m}$$

Therefore velocity of engine

$$= \frac{150 \text{ m}}{5 \text{ sec}} = 30 \text{ m/s}$$

86. (c) Bats catch the prey by hearing reflected ultrasonic waves.

When the source and the detector (observer) are moving, frequency of reflected waves change. This is according to Doppler's effect.



87. (a)

Electric siren

Motor cycle

Let the motorcycle has travelled a distances, its velocity at that point

$$v_m^2 - u^2 = 2as \quad \therefore v_m^2 = 2 \times 2 \times s$$

$$\therefore v_m = 2\sqrt{s}$$

The observed frequency will be

$$v' = v \left[\frac{v - v_m}{v}\right]$$

$$0.94v = v \left[\frac{330 - 2\sqrt{s}}{330}\right] \Rightarrow s = 98.01 \text{ m}$$

88. (c) Apparent frequency $v' = v \left[\frac{v}{v - v_s}\right]$

$$\Rightarrow 10000 = 9500 \left[\frac{300}{300 - v}\right] \Rightarrow 300 - v = 300 \times 0.95$$

$$\Rightarrow v = 300 - 285 = 15 \text{ ms}^{-1}$$

89. (c) Apparent frequency

$$n' = n \left[\frac{v + v_0}{v}\right] = n \left[\frac{v + \frac{v}{5}}{v}\right] = n \left[\frac{6}{5}\right] \quad \frac{n'}{n} = \frac{6}{5}$$

The percentage increase in apparent

$$\text{frequency } \frac{n' - n}{n} = \frac{6 - 5}{5} \times 100 = 20\%$$

Electric Charges and Fields



TOPIC 1 Electric Charges and Coulomb's Law



1. Three charges $+Q$, q , $+Q$ are placed respectively, at distance, $d/2$ and d from the origin, on the x -axis. If the net force experienced by $+Q$, placed at $x = 0$, is zero, then value of q is: [9 Jan. 2019 I]

(a) $-Q/4$ (b) $+Q/2$ (c) $+Q/4$ (d) $-Q/2$

2. Charge is distributed within a sphere of radius R with a volume charge density $\rho(r) = \frac{A}{r^2} e^{-2r/a}$ where A and a

are constants. If Q is the total charge of this charge distribution, the radius R is: [9 Jan. 2019, II]

(a) $a \log \left(1 - \frac{Q}{2\pi a A} \right)$ (b) $\frac{a}{2} \log \left(\frac{1}{1 - \frac{Q}{2\pi a A}} \right)$

(c) $a \log \left(\frac{1}{1 - \frac{Q}{2\pi a A}} \right)$ (d) $\frac{a}{2} \log \left(1 - \frac{Q}{2\pi a A} \right)$

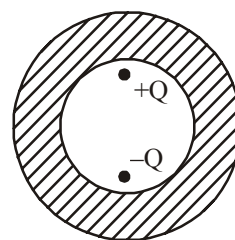
3. Two identical conducting spheres A and B, carry equal charge. They are separated by a distance much larger than their diameter, and the force between them is F . A third identical conducting sphere, C, is uncharged. Sphere C is first touched to A, then to B, and then removed. As a result, the force between A and B would be equal to

[Online April 16, 2018]

(a) $\frac{3F}{4}$ (b) $\frac{F}{2}$ (c) F (d) $\frac{3F}{8}$

4. Shown in the figure are two point charges $+Q$ and $-Q$ inside the cavity of a spherical shell. The charges are kept near the surface of the cavity on opposite sides of the centre of the shell. If σ_1 is the surface charge on the inner surface and Q_1 net charge on it and σ_2 the surface charge on the outer surface and Q_2 net charge on it then :

[Online April 10, 2015]



- (a) $\sigma_1 \neq 0, Q_1 = 0$ (b) $\sigma_1 \neq 0, Q_1 = 0$
 $\sigma_2 = 0, Q_2 = 0$ $\sigma_2 \neq 0, Q_2 = 0$
 (c) $\sigma_1 = 0, Q_1 = 0$ (d) $\sigma_1 \neq 0, Q_1 \neq 0$
 $\sigma_2 = 0, Q_2 = 0$ $\sigma_2 \neq 0, Q_2 \neq 0$

5. Two charges, each equal to q , are kept at $x = -a$ and $x = a$

on the x -axis. A particle of mass m and charge $q_0 = \frac{q}{2}$ is placed at the origin. If charge q_0 is given a small displacement ($y \ll a$) along the y -axis, the net force acting on the particle is proportional to [2013]

- (a) y (b) $-y$ (c) $\frac{1}{y}$ (d) $-\frac{1}{y}$

6. Two balls of same mass and carrying equal charge are hung from a fixed support of length l . At electrostatic equilibrium, assuming that angles made by each thread is small, the separation, x between the balls is proportional to : [Online April 9, 2013]

- (a) l (b) l^2 (c) $l^{2/3}$ (d) $l^{1/3}$

7. Two identical charged spheres suspended from a common point by two massless strings of length l are initially a distance d ($d \ll l$) apart because of their mutual repulsion. The charge begins to leak from both the spheres at a constant rate. As a result charges approach each other with a velocity v . Then as a function of distance x between them, [2011]

- (a) $v \propto x^{-1}$ (b) $v \propto x^{1/2}$ (c) $v \propto x$ (d) $v \propto x^{-1/2}$

8. A charge Q is placed at each of the opposite corners of a square. A charge q is placed at each of the other two corners. If the net electrical force on Q is zero, then Q/q equals: [2009]

- (a) -1 (b) 1 (c) $-\frac{1}{\sqrt{2}}$ (d) $-2\sqrt{2}$

9. If g_E and g_M are the accelerations due to gravity on the surfaces of the earth and the moon respectively and if Millikan's oil drop experiment could be performed on the

two surfaces, one will find the ratio
electronic charge on the moon to be
electronic charge on the earth

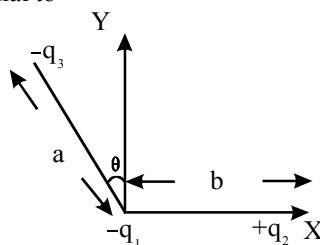
[2007]

- (a) g_M / g_E (b) 1 (c) 0 (d) g_E / g_M

10. Two spherical conductors B and C having equal radii and carrying equal charges on them repel each other with a force F when kept apart at some distance. A third spherical conductor having same radius as that B but **uncharged** is brought in contact with B , then brought in contact with C and finally removed away from both. The new force of repulsion between B and C is [2004]

- (a) $F/8$ (b) $3F/4$ (c) $F/4$ (d) $3F/8$

11. Three charges $-q_1$, $+q_2$ and $-q_3$ are placed as shown in the figure. The x -component of the force on $-q_1$ is proportional to [2003]



- (a) $\frac{q_2}{b^2} - \frac{q_3}{a^2} \cos \theta$ (b) $\frac{q_2}{b^2} + \frac{q_3}{a^2} \sin \theta$
(c) $\frac{q_2}{b^2} + \frac{q_3}{a^2} \cos \theta$ (d) $\frac{q_2}{b^2} - \frac{q_3}{a^2} \sin \theta$

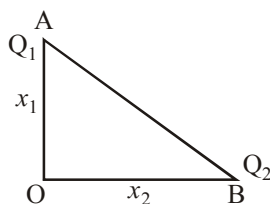
12. If a charge q is placed at the centre of the line joining two equal charges Q such that the system is in equilibrium then the value of q is [2002]

- (a) $Q/2$ (b) $-Q/2$ (c) $Q/4$ (d) $-Q/4$

TOPIC 2 Electric Field and Electric Field Lines



13. Charges Q_1 and Q_2 are at points A and B of a right angle triangle OAB (see figure). The resultant electric field at point O is perpendicular to the hypotenuse, then Q_1/Q_2 is proportional to : [Sep. 06, 2020 (I)]



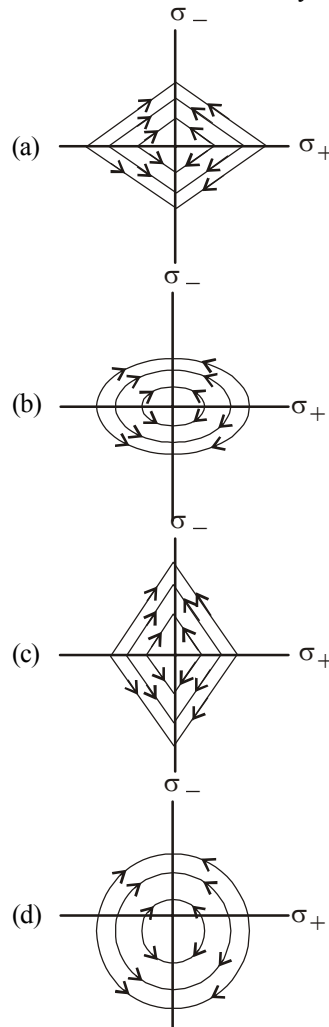
- (a) $\frac{x_1^3}{x_2^3}$ (b) $\frac{x_2}{x_1}$ (c) $\frac{x_1}{x_2}$ (d) $\frac{x_2^2}{x_1^2}$

14. Consider the force F on a charge ' q ' due to a uniformly charged spherical shell of radius R carrying charge Q distributed uniformly over it. Which one of the following statements is true for F , if ' q ' is placed at distance r from the centre of the shell? [Sep. 06, 2020 (II)]

(a) $F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{R^2}$ for $r < R$ (b) $\frac{1}{4\pi\epsilon_0} \frac{Qq}{R^2} > F > 0$ for $r < R$

(c) $F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{R^2}$ for $r > R$ (d) $F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{R^2}$ for all r

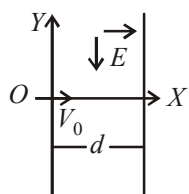
15. Two charged thin infinite plane sheets of uniform surface charge density σ_+ and σ_- , where $|\sigma_+| > |\sigma_-|$, intersect at right angle. Which of the following best represents the electric field lines for this system? [Sep. 04, 2020 (I)]



16. A particle of charge q and mass m is subjected to an electric field $E = E_0(1 - ax^2)$ in the x -direction, where a and E_0 are constants. Initially the particle was at rest at $x = 0$. Other than the initial position the kinetic energy of the particle becomes zero when the distance of the particle from the origin is : [Sep. 04, 2020 (II)]

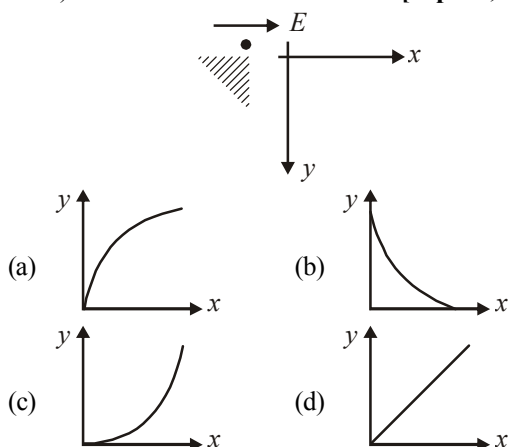
- (a) a (b) $\sqrt{\frac{2}{a}}$ (c) $\sqrt{\frac{3}{a}}$ (d) $\sqrt{\frac{1}{a}}$

17. A charged particle (mass m and charge q) moves along X axis with velocity V_0 . When it passes through the origin it enters a region having uniform electric field $\vec{E} = -E\hat{j}$ which extends upto $x = d$. Equation of path of electron in the region $x > d$ is : [Sep. 02, 2020 (I)]

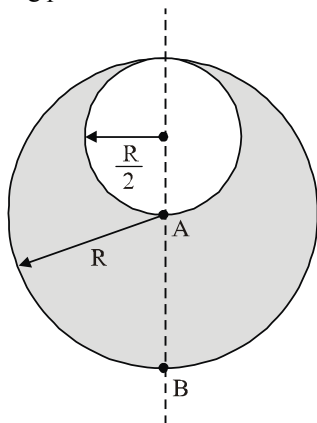


- (a) $y = \frac{qEd}{mV_0^2}(x-d)$ (b) $y = \frac{qEd}{mV_0^2}\left(\frac{d}{2}-x\right)$
 (c) $y = \frac{qEd}{mV_0^2}x$ (d) $y = \frac{qEd^2}{mV_0^2}x$

18. A small point mass carrying some positive charge on it, is released from the edge of a table. There is a uniform electric field in this region in the horizontal direction. Which of the following options then correctly describe the trajectory of the mass? (Curves are drawn schematically and are not to scale). [Sep. 02, 2020 (II)]



19. Consider a sphere of radius R which carries a uniform charge density ρ . If a sphere of radius $\frac{R}{2}$ is carved out of it, as shown, the ratio $\left|\frac{\vec{E}_A}{\vec{E}_B}\right|$ of magnitude of electric field $\vec{E}_A$ and $\vec{E}_B$, respectively, at points A and B due to the remaining portion is: [9 Jan. 2020, I]



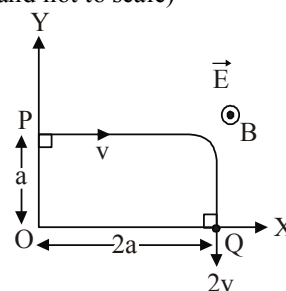
- (a) $\frac{21}{34}$ (b) $\frac{18}{34}$ (c) $\frac{17}{54}$ (d) $\frac{18}{54}$

20. An electric dipole of moment $\vec{p} = (\hat{i} - 3\hat{j} + 2\hat{k}) \times 10^{-29}$ C.m is at the origin $(0, 0, 0)$. The electric field due to this dipole at $\vec{r} = +\hat{i} + 3\hat{j} + 5\hat{k}$

(note that $\vec{r} \cdot \vec{p} = 0$) is parallel to: [9 Jan. 2020, I]

- (a) $(+\hat{i} - 3\hat{j} - 2\hat{k})$ (b) $(-\hat{i} + 3\hat{j} - 2\hat{k})$
 (c) $(+\hat{i} + 3\hat{j} - 2\hat{k})$ (d) $(-\hat{i} - 3\hat{j} + 2\hat{k})$

21. A charged particle of mass ' m ' and charge ' q ' moving under the influence of uniform electric field $E\hat{i}$ and a uniform magnetic field $B\hat{k}$ follows a trajectory from point P to Q as shown in figure. The velocities at P and Q are respectively, $v\hat{i}$ and $-2v\hat{j}$. Then which of the following statements (A, B, C, D) are the correct? (Trajectory shown is schematic and not to scale) [9 Jan. 2020, I]



(A) $E = \frac{3}{4} \left(\frac{mv^2}{qa} \right)$

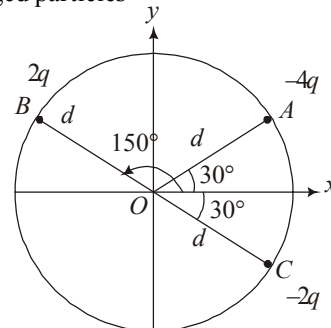
(B) Rate of work done by the electric field at P is $\frac{3}{4} \left(\frac{mv^2}{a} \right)$

(C) Rate of work done by both the fields at Q is zero

(D) The difference between the magnitude of angular momentum of the particle at P and Q is $2mv$.

- (a) (A), (C), (D) (b) (B), (C), (D)
 (c) (A), (B), (C) (d) (A), (B), (C), (D)

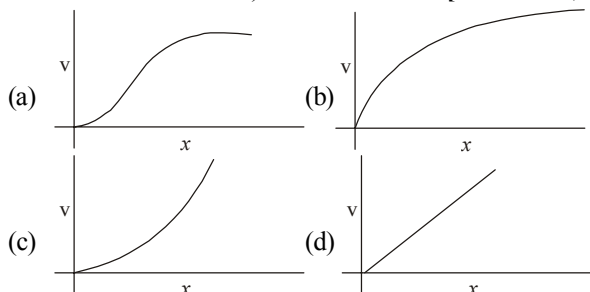
22. Three charged particles



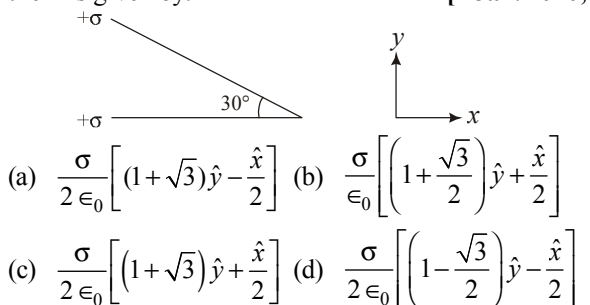
A, B and C with charges $-4q$, $2q$ and $-2q$ are present on the circumference of a circle of radius d . The charged particles A, C and centre O of the circle formed an equilateral triangle as shown in figure. Electric field at O along x-direction is: [8 Jan. 2020, I]

- (a) $\frac{\sqrt{3}q}{\pi \epsilon_0 d^2}$ (b) $\frac{2\sqrt{3}q}{\pi \epsilon_0 d^2}$ (c) $\frac{\sqrt{3}q}{4\pi \epsilon_0 d^2}$ (d) $\frac{3\sqrt{3}q}{4\pi \epsilon_0 d^2}$

23. A particle of mass m and charge q is released from rest in a uniform electric field. If there is no other force on the particle, the dependence of its speed v on the distance x travelled by it is correctly given by (graphs are schematic and not drawn to scale) [8 Jan. 2020, II]



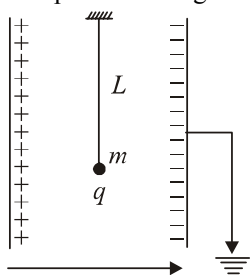
24. Two infinite planes each with uniform surface charge density $+\sigma$ are kept in such a way that the angle between them is 30° . The electric field in the region shown between them is given by: [7 Jan. 2020, I]



25. A particle of mass m and charge q has an initial velocity $\vec{v} = v_0 \hat{j}$. If an electric field $\vec{E} = E_0 \hat{i}$ and magnetic field $\vec{B} = B_0 \hat{i}$ act on the particle, its speed will double after a time: [7 Jan 2020, II]

(a) $\frac{2mv_0}{qE_0}$ (b) $\frac{3mv_0}{qE_0}$ (c) $\frac{\sqrt{3}mv_0}{qE_0}$ (d) $\frac{\sqrt{2}mv_0}{qE_0}$

26. A simple pendulum of length L is placed between the plates of a parallel plate capacitor having electric field E , as shown in figure. Its bob has mass m and charge q . The time period of the pendulum is given by: [10 April 2019, II]



(a) $2\pi \sqrt{\frac{L}{\left(g + \frac{qE}{m} \right)}}$ (b) $2\pi \sqrt{\frac{L}{\sqrt{g^2 - \frac{q^2 E^2}{m^2}}}}$
 (c) $2\pi \sqrt{\frac{L}{\left(g - \frac{qE}{m} \right)}}$ (d) $2\pi \sqrt{\frac{L}{\sqrt{g^2 + \left(\frac{qE}{m} \right)^2}}}$

27. Four point charges $-q, +q, +q$ and $-q$ are placed on y -axis at $y = -2d, y = -d, y = +d$ and $y = +2d$, respectively. The magnitude of the electric field E at a point on the x -axis at $x = D$, with $D \gg d$, will behave as: [9 April 2019, II]

(a) $E \propto \frac{1}{D^3}$ (b) $E \propto \frac{1}{D}$ (c) $E \propto \frac{1}{D^4}$ (d) $E \propto \frac{1}{D^2}$

28. The bob of a simple pendulum has mass 2 g and a charge of $5.0 \text{ } \mu\text{C}$. It is at rest in a uniform horizontal electric field of intensity 2000 V/m . At equilibrium, the angle that the pendulum makes with the vertical is: [8 April 2019 I] (take $g = 10 \text{ m/s}^2$)

(a) $\tan^{-1}(2.0)$ (b) $\tan^{-1}(0.2)$
 (c) $\tan^{-1}(5.0)$ (d) $\tan^{-1}(0.5)$

29. For a uniformly charged ring of radius R , the electric field on its axis has the largest magnitude at a distance h from its centre. Then value of h is: [9 Jan. 2019 I]

(a) $\frac{R}{\sqrt{5}}$ (b) $\frac{R}{\sqrt{2}}$ (c) R (d) $R\sqrt{2}$

30. Two point charges $q_1 (\sqrt{10} \text{ } \mu\text{C})$ and $q_2 (-25 \text{ } \mu\text{C})$ are placed on the x -axis at $x = 1 \text{ m}$ and $x = 4 \text{ m}$ respectively. The electric field (in V/m) at a point $y = 3 \text{ m}$ on y -axis is, [9 Jan 2019, II]

[take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2 \text{C}^{-2}$]

(a) $(63\hat{i} - 27\hat{j}) \times 10^2$ (b) $(-63\hat{i} + 27\hat{j}) \times 10^2$
 (c) $(81\hat{i} - 81\hat{j}) \times 10^2$ (d) $(-81\hat{i} + 81\hat{j}) \times 10^2$

31. A body of mass M and charge q is connected to a spring of spring constant k . It is oscillating along x -direction about its equilibrium position, taken to be at $x = 0$, with an amplitude A . An electric field E is applied along the x -direction. Which of the following statements is correct? [Online April 15, 2018]

(a) The total energy of the system is $\frac{1}{2}m\omega^2 A^2 + \frac{1}{2} \frac{q^2 E^2}{k}$
 (b) The new equilibrium position is at a distance: $\frac{2qE}{k}$ from $x=0$
 (c) The new equilibrium position is at a distance: $\frac{qE}{2k}$ from $x=0$
 (d) The total energy of the system is $\frac{1}{2}m\omega^2 A^2 - \frac{1}{2} \frac{q^2 E^2}{k}$

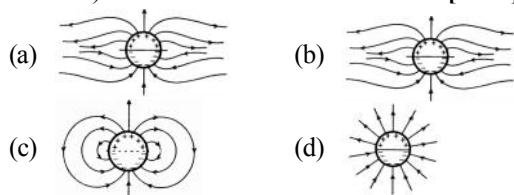
32. A solid ball of radius R has a charge density ρ given by

$\rho = \rho_0 \left(1 - \frac{r}{R} \right)$ for $0 \leq r \leq R$. The electric field outside the ball is:

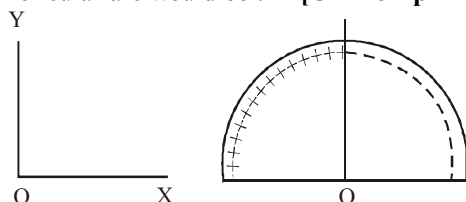
[Online April 15, 2018]

(a) $\frac{\rho_0 R^3}{\epsilon_0 r^2}$ (b) $\frac{4\rho_0 R^3}{3\epsilon_0 r^2}$ (c) $\frac{3\rho_0 R^3}{4\epsilon_0 r^2}$ (d) $\frac{\rho_0 R^3}{12\epsilon_0 r^2}$

33. A long cylindrical shell carries positive surface charge σ in the upper half and negative surface charge $-\sigma$ in the lower half. The electric field lines around the cylinder will look like figure given in : (figures are schematic and not drawn to scale) [2015]

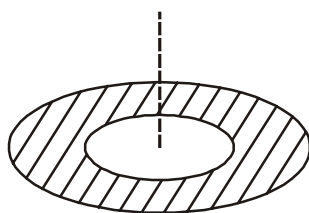


34. A wire of length L ($=20$ cm), is bent into a semicircular arc. If the two equal halves of the arc were each to be uniformly charged with charges $\pm Q$, [$|Q| = 10^3 \epsilon_0$ Coulomb where ϵ_0 is the permittivity (in SI units) of free space] the net electric field at the centre O of the semicircular arc would be : [Online April 11, 2015]



- (a) $(50 \times 10^3 \text{ N/C}) \hat{j}$ (b) $(50 \times 10^3 \text{ N/C}) \hat{i}$
(c) $(25 \times 10^3 \text{ N/C}) \hat{j}$ (d) $(25 \times 10^3 \text{ N/C}) \hat{i}$
35. A thin disc of radius $b = 2a$ has a concentric hole of radius ' a ' in it (see figure). It carries uniform surface charge ' σ ' on it. If the electric field on its axis at height ' h ' ($h \ll a$) from its centre is given as ' Ch ' then value of ' C ' is : [Online April 10, 2015]

- (a) $\frac{\sigma}{4a\epsilon_0}$
(b) $\frac{\sigma}{8a\epsilon_0}$
(c) $\frac{\sigma}{a\epsilon_0}$
(d) $\frac{\sigma}{2a\epsilon_0}$



36. A spherically symmetric charge distribution is characterised by a charge density having the following variations:

$$\rho(r) = \rho_0 \left(1 - \frac{r}{R} \right) \text{ for } r < R$$

$$\rho(r) = 0 \text{ for } r \geq R$$

Where r is the distance from the centre of the charge distribution ρ_0 is a constant. The electric field at an internal point ($r < R$) is: [Online April 12, 2014]

- (a) $\frac{\rho_0}{4\epsilon_0} \left(\frac{r}{3} - \frac{r^2}{4R} \right)$ (b) $\frac{\rho_0}{\epsilon_0} \left(\frac{r}{3} - \frac{r^2}{4R} \right)$
(c) $\frac{\rho_0}{3\epsilon_0} \left(\frac{r}{3} - \frac{r^2}{4R} \right)$ (d) $\frac{\rho_0}{12\epsilon_0} \left(\frac{r}{3} - \frac{r^2}{4R} \right)$

37. The magnitude of the average electric field normally present in the atmosphere just above the surface of the Earth is about 150 N/C , directed inward towards the center of the Earth. This gives the total net surface charge carried by the Earth to be: [Online April 9, 2014]

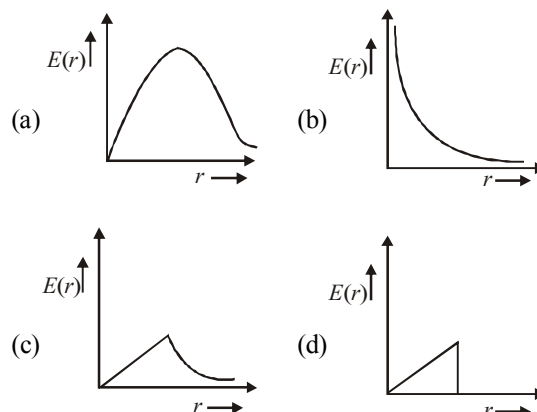
[Given $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N-m}^2$, $R_E = 6.37 \times 10^6 \text{ m}$]

- (a) $+670 \text{ kC}$ (b) -670 kC
(c) -680 kC (d) $+680 \text{ kC}$
38. The surface charge density of a thin charged disc of radius R is σ . The value of the electric field at the centre of the disc is $\frac{\sigma}{2\epsilon_0}$. With respect to the field at the centre, the

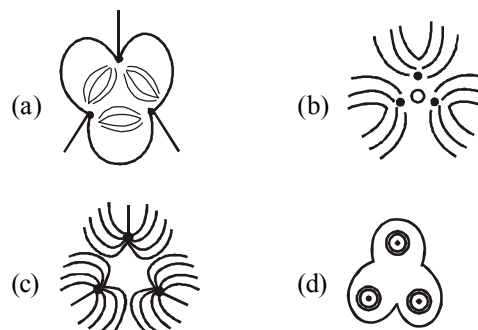
electric field along the axis at a distance R from the centre of the disc : [Online April 25, 2013]

- (a) reduces by 70.7% (b) reduces by 29.3%
(c) reduces by 9.7% (d) reduces by 14.6%
39. A liquid drop having 6 excess electrons is kept stationary under a uniform electric field of 25.5 kV m^{-1} . The density of liquid is $1.26 \times 10^3 \text{ kg m}^{-3}$. The radius of the drop is (neglect buoyancy). [Online April 23, 2013]

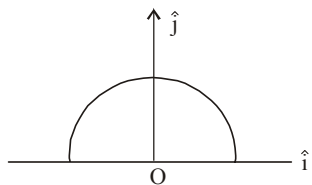
- (a) $4.3 \times 10^{-7} \text{ m}$ (b) $7.8 \times 10^{-7} \text{ m}$
(c) $0.078 \times 10^{-7} \text{ m}$ (d) $3.4 \times 10^{-7} \text{ m}$
40. In a uniformly charged sphere of total charge Q and radius R , the electric field E is plotted as function of distance from the centre, The graph which would correspond to the above will be: [2012]



41. Three positive charges of equal value q are placed at vertices of an equilateral triangle. The resulting lines of force should be sketched as in [Online May 26, 2012]



42. A thin semi-circular ring of radius r has a positive charge q distributed uniformly over it. The net field $\vec{E}$ at the centre O is [2010]



- (a) $\frac{q}{4\pi^2 \epsilon_0 r^2} \hat{j}$ (b) $-\frac{q}{4\pi^2 \epsilon_0 r^2} \hat{j}$
 (c) $-\frac{q}{2\pi^2 \epsilon_0 r^2} \hat{j}$ (d) $\frac{q}{2\pi^2 \epsilon_0 r^2} \hat{j}$

43. Let there be a spherically symmetric charge distribution with charge density varying as $\rho(r) = \rho_0 \left(\frac{5}{4} - \frac{r}{R} \right)$ upto $r = R$, and $\rho(r) = 0$ for $r > R$, where r is the distance from the origin. The electric field at a distance $r (r < R)$ from the origin is given by [2010]

- (a) $\frac{\rho_0 r}{4\epsilon_0} \left(\frac{5}{3} - \frac{r}{R} \right)$ (b) $\frac{4\pi \rho_0 r}{3\epsilon_0} \left(\frac{5}{3} - \frac{r}{R} \right)$
 (c) $\frac{\rho_0 r}{4\epsilon_0} \left(\frac{5}{4} - \frac{r}{R} \right)$ (d) $\frac{\rho_0 r}{3\epsilon_0} \left(\frac{5}{4} - \frac{r}{R} \right)$

44. This question contains Statement-1 and Statement-2. Of the four choices given after the statements, choose the one that best describes the two statements.

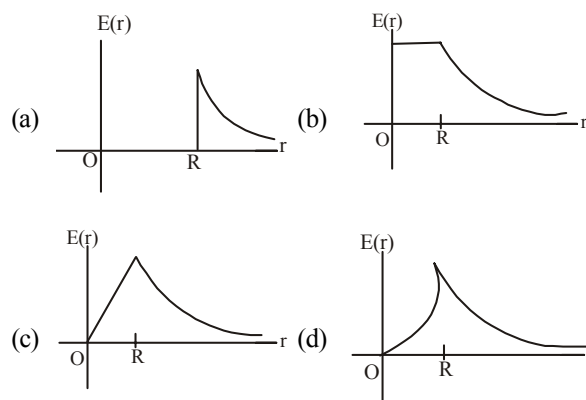
Statement-1 : For a charged particle moving from point P to point Q , the net work done by an electrostatic field on the particle is independent of the path connecting point P to point Q .

Statement-2 : The net work done by a conservative force on an object moving along a closed loop is zero. [2009]

- (a) Statement-1 is true, Statement-2 is true; Statement-2 is the correct explanation of Statement-1.
 (b) Statement-1 is true, Statement-2 is true; Statement-2 is not the correct explanation of Statement-1.
 (c) Statement-1 is false, Statement-2 is true.
 (d) Statement-1 is true, Statement-2 is false.
45. Let $\rho(r) = \frac{Q}{\pi R^4} r$ be the charge density distribution for a solid sphere of radius R and total charge Q . For a point 'P' inside the sphere at distance r_1 from the centre of the sphere, the magnitude of electric field is : [2009]

- (a) $\frac{Q}{4\pi \epsilon_0 r_1^2}$ (b) $\frac{Q r_1^2}{4\pi \epsilon_0 R^4}$
 (c) $\frac{Q r_1^2}{3\pi \epsilon_0 R^4}$ (d) 0

46. A thin spherical shell of radius R has charge Q spread uniformly over its surface. Which of the following graphs most closely represents the electric field $E(r)$ produced by the shell in the range $0 \leq r < \infty$, where r is the distance from the centre of the shell? [2008]



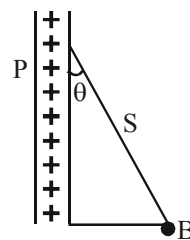
47. Two spherical conductors A and B of radii 1 mm and 2 mm are separated by a distance of 5 cm and are uniformly charged. If the spheres are connected by a conducting wire then in equilibrium condition, the ratio of the magnitude of the electric fields at the surfaces of spheres A and B is [2006]

- (a) 4 : 1 (b) 1 : 2 (c) 2 : 1 (d) 1 : 4

48. Two point charges $+8q$ and $-2q$ are located at $x = 0$ and $x = L$ respectively. The location of a point on the x axis at which the net electric field due to these two point charges is zero is [2005]

- (a) $\frac{L}{4}$ (b) $2L$ (c) $4L$ (d) $8L$

49. A charged ball B hangs from a silk thread S , which makes an angle θ with a large charged conducting sheet P , as shown in the figure. The surface charge density σ of the sheet is proportional to [2005]



- (a) $\cot \theta$ (b) $\cos \theta$ (c) $\tan \theta$ (d) $\sin \theta$

50. Four charges equal to $-Q$ are placed at the four corners of a square and a charge q is at its centre. If the system is in equilibrium the value of q is [2004]

- (a) $-\frac{Q}{2}(1+2\sqrt{2})$ (b) $\frac{Q}{4}(1+2\sqrt{2})$
 (c) $-\frac{Q}{4}(1+2\sqrt{2})$ (d) $\frac{Q}{2}(1+2\sqrt{2})$

51. A charged oil drop is suspended in a uniform field of 3×10^4 v/m so that it neither falls nor rises. The charge on the drop will be (Take the mass of the charge $= 9.9 \times 10^{-15}$ kg and $g = 10$ m/s²) [2004]

- (a) 1.6×10^{-18} C (b) 3.2×10^{-18} C
 (c) 3.3×10^{-18} C (d) 4.8×10^{-18} C

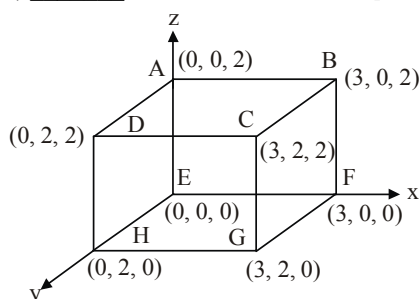
TOPIC 3 Electric Dipole, Electric Flux and Gauss's Law



52. Two identical electric point dipoles have dipole moments $\vec{P}_1 = P\hat{i}$ and $\vec{P}_2 = -P\hat{i}$ and are held on the x axis at distance 'a' from each other. When released, they move along x -axis with the direction of their dipole moments remaining unchanged. If the mass of each dipole is 'm', their speed when they are infinitely far apart is : [Sep. 06, 2020 (II)]

- (a) $\frac{P}{a} \sqrt{\frac{1}{\pi\epsilon_0 m a}}$ (b) $\frac{P}{a} \sqrt{\frac{1}{2\pi\epsilon_0 m a}}$
 (c) $\frac{P}{a} \sqrt{\frac{2}{\pi\epsilon_0 m a}}$ (d) $\frac{P}{a} \sqrt{\frac{2}{2\pi\epsilon_0 m a}}$

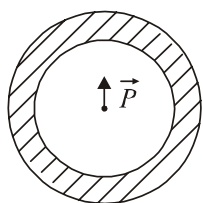
53. An electric field $\vec{E} = 4x\hat{i} - (y^2 + 1)\hat{j}$ N/C passes through the box shown in figure. The flux of the electric field through surfaces ABCD and BCGF are marked as ϕ_1 and ϕ_{11} respectively. The difference between $(\phi_1 - \phi_{11})$ is (in Nm^2/C) _____. [9 Jan 2020, II]



54. In finding the electric field using Gauss law the formula $|\vec{E}| = \frac{q_{enc}}{\epsilon_0 |A|}$ is applicable. In the formula ϵ_0 is permittivity of free space, A is the area of Gaussian surface and q_{enc} is charge enclosed by the Gaussian surface. This equation can be used in which of the following situation? [8 Jan 2020, I]

- (a) Only when the Gaussian surface is an equipotential surface.
 Only when the Gaussian surface is an
 (b) equipotential surface and $|\vec{E}|$ is constant on the surface.
 (c) Only when $|\vec{E}| = \text{constant}$ on the surface.
 (d) For any choice of Gaussian surface.

55. Shown in the figure is a shell made of a conductor. It has inner radius a and outer radius b , and carries charge Q . At its centre is a dipole $\vec{P}$ as shown. In this case : [12 April 2019, I]



- (a) surface charge density on the inner surface is uniform and equal to $\frac{Q/2}{4\pi a^2}$

- (b) electric field outside the shell is the same as that of a point charge at the centre of the shell.
 (c) surface charge density on the outer surface depends on $|\vec{P}|$
 (d) surface charge density on the inner surface of the shell is zero everywhere.-

56. Let a total charge $2Q$ be distributed in a sphere of radius R , with the charge density given by $\rho(r) = kr$, where r is the distance from the centre. Two charges A and B , of $-Q$ each, are placed on diametrically opposite points, at equal distance, a , from the centre. If A and B do not experience any force, then. [12 April 2019, II]

- (a) $a = 8^{-1/4} R$ (b) $a = \frac{3R}{2^{1/4}}$
 (c) $a = 2^{-1/4} R$ (d) $a = R/\sqrt{3}$

57. An electric dipole is formed by two equal and opposite charges q with separation d . The charges have same mass m . It is kept in a uniform electric field E . If it is slightly rotated from its equilibrium orientation, then its angular frequency ω is : [8 April 2019, II]

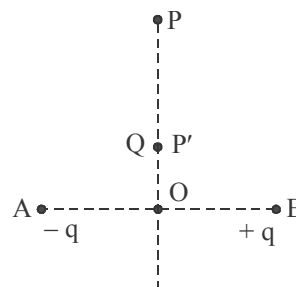
- (a) $\sqrt{\frac{qE}{md}}$ (b) $\sqrt{\frac{2qE}{md}}$ (c) $2\sqrt{\frac{qE}{md}}$ (d) $\sqrt{\frac{qE}{2md}}$

58. An electric field of 1000 V/m is applied to an electric dipole at angle of 45° . The value of electric dipole moment is 10^{-29} C.m . What is the potential energy of the electric dipole? [11 Jan 2019, II]

- (a) $-20 \times 10^{-18} \text{ J}$ (b) $-7 \times 10^{-27} \text{ J}$
 (c) $-10 \times 10^{-29} \text{ J}$ (d) $-9 \times 10^{-20} \text{ J}$

59. Charges $-q$ and $+q$ located at A and B , respectively, constitute an electric dipole. Distance $AB = 2a$, O is the mid point of the dipole and OP is perpendicular to AB . A charge Q is placed at P where $OP = y$ and $y \gg 2a$. The charge Q experiences an electrostatic force F . If Q is now moved along the equatorial line to P' such that $OP' = \left(\frac{y}{3}\right)$, the force on Q will be close to: $\left(\frac{y}{3} \gg 2a\right)$

[10 Jan 2019, II]

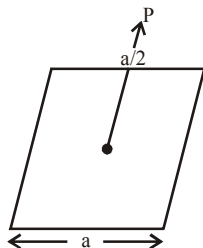


- (a) $3F$ (b) $\frac{F}{3}$ (c) $9F$ (d) $27F$

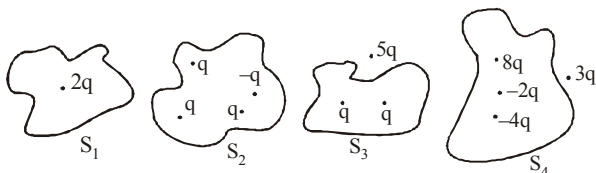
60. A charge Q is placed at a distance $a/2$ above the centre of the square surface of edge a as shown in the figure. The electric flux through the square surface is:

[Online April 15, 2018]

- (a) $\frac{Q}{3\epsilon_0}$
 (b) $\frac{Q}{6\epsilon_0}$
 (c) $\frac{Q}{2\epsilon_0}$
 (d) $\frac{Q}{\epsilon_0}$

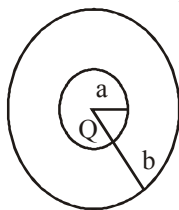


61. An electric dipole has a fixed dipole moment $\vec{p}$, which makes angle θ with respect to x -axis. When subjected to an electric field $\vec{E}_1 = E\hat{i}$, it experiences a torque $\vec{T}_1 = \tau\hat{i}$. When subjected to another electric field $\vec{E}_2 = \sqrt{3}E_1\hat{j}$ it experiences torque $\vec{T}_2 = -\vec{T}_1$. The angle θ is: [2017]
 (a) 60° (b) 90° (c) 30° (d) 45°
62. Four closed surfaces and corresponding charge distributions are shown below. [Online April 9, 2017]



Let the respective electric fluxes through the surfaces be Φ_1, Φ_2, Φ_3 , and Φ_4 . Then :

- (a) $\Phi_1 < \Phi_2 = \Phi_3 > \Phi_4$ (b) $\Phi_1 > \Phi_2 > \Phi_3 > \Phi_4$
 (c) $\Phi_1 = \Phi_2 = \Phi_3 = \Phi_4$ (d) $\Phi_1 > \Phi_3; \Phi_2 < \Phi_4$
63. The region between two concentric spheres of radii 'a' and 'b', respectively (see figure), have volume charge density $\rho = \frac{A}{r}$, where A is a constant and r is the distance from the centre. At the centre of the spheres is a point charge Q . The value of A such that the electric field in the region between the spheres will be constant, is: [2016]



- (a) $\frac{2Q}{\pi(a^2 - b^2)}$ (b) $\frac{2Q}{\pi a^2}$
 (c) $\frac{Q}{2\pi a^2}$ (d) $\frac{Q}{2\pi(b^2 - a^2)}$

64. The electric field in a region of space is given by, $\vec{E} = E_0\hat{i} + 2E_0\hat{j}$ where $E_0 = 100 \text{ N/C}$. The flux of the field

through a circular surface of radius 0.02 m parallel to the Y - Z plane is nearly:

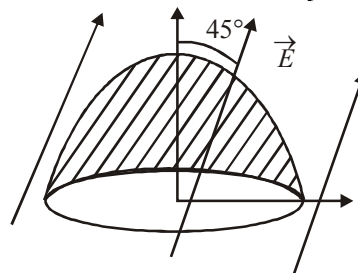
[Online April 19, 2014]

- (a) $0.125 \text{ Nm}^2/\text{C}$ (b) $0.02 \text{ Nm}^2/\text{C}$
 (c) $0.005 \text{ Nm}^2/\text{C}$ (d) $3.14 \text{ Nm}^2/\text{C}$

65. Two point dipoles of dipole moment $\vec{p}_1$ and $\vec{p}_2$ are at a distance x from each other and $\vec{p}_1 \parallel \vec{p}_2$. The force between the dipoles is: [Online April 9, 2013]

- (a) $\frac{1}{4\pi\epsilon_0} \frac{4p_1p_2}{x^4}$ (b) $\frac{1}{4\pi\epsilon_0} \frac{3p_1p_2}{x^3}$
 (c) $\frac{1}{4\pi\epsilon_0} \frac{6p_1p_2}{x^4}$ (d) $\frac{1}{4\pi\epsilon_0} \frac{8p_1p_2}{x^4}$

66. The flat base of a hemisphere of radius a with no charge inside it lies in a horizontal plane. A uniform electric field $\vec{E}$ is applied at an angle $\frac{\pi}{4}$ with the vertical direction. The electric flux through the curved surface of the hemisphere is [Online May 19, 2012]

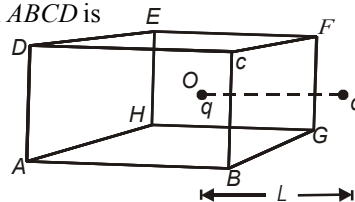


- (a) $\pi a^2 E$ (b) $\frac{\pi a^2 E}{\sqrt{2}}$
 (c) $\frac{\pi a^2 E}{2\sqrt{2}}$ (d) $\frac{(\pi + 2)\pi a^2 E}{(2\sqrt{2})^2}$

67. An electric dipole is placed at an angle of 30° to a non-uniform electric field. The dipole will experience [2006]
 (a) a translational force only in the direction of the field
 (b) a translational force only in a direction normal to the direction of the field
 (c) a torque as well as a translational force
 (d) a torque only
68. If the electric flux entering and leaving an enclosed surface respectively is ϕ_1 and ϕ_2 , the electric charge inside the surface will be [2003]

- (a) $(\phi_2 - \phi_1)\epsilon_0$ (b) $(\phi_1 - \phi_2)/\epsilon_0$
 (c) $(\phi_2 - \phi_1)/\epsilon_0$ (d) $(\phi_1 - \phi_2)\epsilon_0$

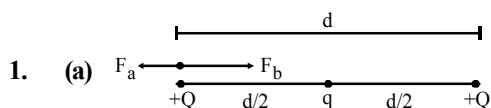
69. A charged particle q is placed at the centre O of cube of length L ($AB C D E F G H$). Another same charge q is placed at a distance L from O . Then the electric flux through $ABCD$ is [2002]



- (a) $q/4 \pi \epsilon_0 L$ (b) zero
 (c) $q/2 \pi \epsilon_0 L$ (d) $q/3 \pi \epsilon_0 L$



Hints & Solutions



Force due to charge +Q,

$$F_a = \frac{KQq}{d^2}$$

Force due to charge q,

$$F_b = \frac{KQq}{\left(\frac{d}{2}\right)^2}$$

For equilibrium,

$$\vec{F}_a + \vec{F}_b = 0$$

$$\Rightarrow \frac{kQq}{d^2} + \frac{kQq}{(d/2)^2} = 0 \quad \therefore q = -\frac{Q}{4}$$

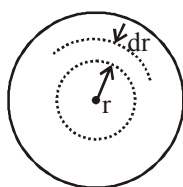
2. (b) $Q = \int \rho dv = \int_0^R \frac{A}{r^2} e^{-2r/a} (4\pi r^2 dr)$

$$= 4\pi A \int_0^R e^{-2r/a} dr = 4\pi A \left[\frac{e^{-2r/a}}{-2/a} \right]_0^R$$

$$= 4\pi A \left(-\frac{a}{2} \right) (e^{-2R/a} - 1)$$

$$Q = 2\pi a A (1 - e^{-2R/a})$$

$$R = \frac{a}{2} \log \left(\frac{1}{1 - \frac{Q}{2\pi a A}} \right)$$



3. (d) Spheres A and B carry equal charge say 'q'

$$\therefore \text{Force between them, } F = \frac{kqq}{r^2}$$

When A and C are touched, charge on both $q_A = q_C = \frac{q}{2}$

Then when B and C are touched, charge on B

$$q_B = \frac{\frac{q}{2} + q}{2} = \frac{3q}{4}$$

Now, the force between charge q_A and q_B

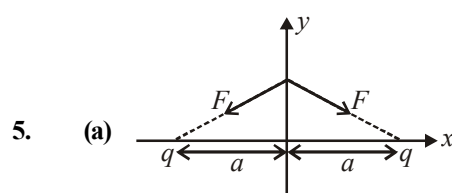
$$F' = \frac{kq_A q_B}{r^2} = \frac{k \times \frac{q}{2} \times \frac{3q}{4}}{r^2} = \frac{3kq^2}{8r^2} = \frac{3}{8} F$$

4. (c) Inside the cavity net charge is zero.

$$\therefore Q_1 = 0 \text{ and } \sigma_1 = 0$$

There is no effect of point charges +Q, -Q and induced charge on inner surface on the outer surface.

$$\therefore Q_2 = 0 \text{ and } \sigma_2 = 0$$



$$\Rightarrow F \sin \theta \leftarrow \quad \rightarrow F \sin \theta$$

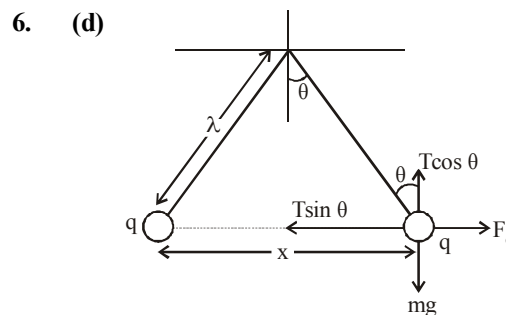
$$2F \cos \theta$$

$$\Rightarrow F_{\text{net}} = 2F \cos \theta$$

$$F_{\text{net}} = \frac{2kq \left(\frac{q}{2} \right)}{\left(\sqrt{y^2 + a^2} \right)^2} \cdot \frac{y}{\sqrt{y^2 + a^2}}$$

$$F_{\text{net}} = \frac{2kq \left(\frac{q}{2} \right) y}{(y^2 + a^2)^{3/2}} \quad (\because y \ll a)$$

$$\Rightarrow \frac{kq^2 y}{a^3} \quad \text{So, } F \propto y$$



In equilibrium, $F_e = T \sin \theta$

$$mg = T \cos \theta$$

$$\tan \theta = \frac{F_e}{mg} = \frac{q^2}{4\pi \epsilon_0 x^2 \times mg}$$

$$\text{also } \tan \theta \approx \sin \theta = \frac{x/2}{l}$$

$$\text{Hence, } \frac{x}{2l} = \frac{q^2}{4\pi \epsilon_0 x^2 \times mg}$$

$$\Rightarrow x^3 = \frac{2q^2 l}{4\pi \epsilon_0 mg}$$

$$\therefore x = \left(\frac{q^2 l}{2\pi \epsilon_0 mg} \right)^{1/3}$$

Therefore $x \propto l^{1/3}$

7. (d) From figure

$$T \cos \theta = mg \quad \dots(i)$$

$$T \sin \theta = F_e \quad \dots(ii)$$

Dividing equation (ii) by (i), we get

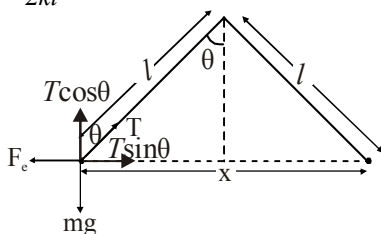
$$\Rightarrow \frac{\sin \theta}{\cos \theta} = \frac{F_e}{mg} \quad \Rightarrow F_e = mg \tan \theta$$

$$\Rightarrow \frac{kq^2}{x^2} = mg \tan \theta \quad \Rightarrow q^2 = \frac{x^2 mg \tan \theta}{k}$$

Since θ is small

$$\therefore \tan \theta \approx \sin \theta = \frac{x}{2l}$$

$$\therefore q^2 = \frac{x^3 mg}{2kl} \Rightarrow q^2 \propto x^{3/2}$$

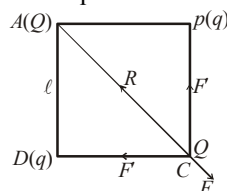


$$\Rightarrow \frac{dq}{dt} \propto \frac{3}{2} \sqrt{x} \frac{dx}{dt} = \frac{3}{2} \sqrt{x} V$$

Since $\frac{dq}{dt} = \text{const.}$

$$\Rightarrow v \propto x^{-1/2} \quad [\because q^2 \propto x^{3/2}]$$

8. (d) Let F be the force between Q and Q . The force between q and Q should be attractive for net force on Q to be zero. Let F' be the force between Q and q . The resultant of F' and F is R . For equilibrium



Net force on Q at C is zero.

$$\therefore \vec{R} + \vec{F} = 0 \Rightarrow \sqrt{2} F' = -F$$

$$\Rightarrow \sqrt{2} \times k \frac{Qq}{\ell^2} = -k \frac{Q^2}{(\sqrt{2} \ell)^2}$$

$$\Rightarrow \frac{Q}{q} = -2\sqrt{2}$$

9. (b) It is obvious that by charge conservation law, electronic charge does not depend on acceleration due to gravity as it is a universal constant. So, electronic charge on earth = electronic charge on moon. $\therefore$ Required ratio = 1.

10. (d)

$$\text{Initial force, } F = K \frac{Q_B Q_C}{x^2}$$

x is distance between the spheres. When third spherical conductor comes in contact with B charge on B is halved

i.e., $\frac{Q}{2}$ and charge on third sphere becomes $\frac{Q}{2}$. Now it is touched to C , charge then equally distributes themselves to make potential same, hence charge on C becomes $\left(Q + \frac{Q}{2}\right) \frac{1}{2} = \frac{3Q}{4}$.

$$\therefore F_{\text{new}} = k \frac{Q'_C Q'_B}{x^2} = k \frac{\left(\frac{3Q}{4}\right) \left(\frac{Q}{2}\right)}{x^2} = k \frac{3Q^2}{8x^2}$$

$$\text{or } F_{\text{new}} = \frac{3}{8} F$$

11. (b) Force applied by charge q_2 on q_1

$$F_{12} = k \frac{q_1 q_2}{b^2}$$

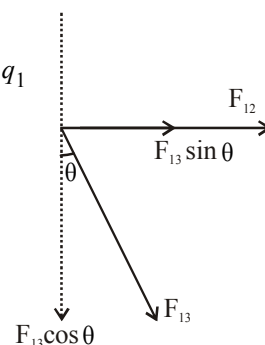
Force applied by charge q_3 on q_1

$$F_{13} = k \frac{q_1 q_3}{a^2}$$

The X-component of net force (F_x) on q_1 is $F_{12} + F_{13} \sin \theta$

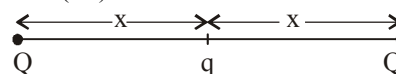
$$\therefore F_x = k \frac{q_1 q_2}{b^2} + k \frac{q_1 q_3}{a^2} \sin \theta$$

$$\therefore F_x \propto \frac{q_2}{b^2} + \frac{q_3}{a^2} \sin \theta$$



12. (d) At equilibrium net force is zero,

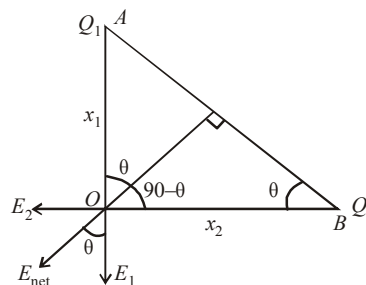
$$\therefore k \frac{Q \times Q}{(2x)^2} + k \frac{Qq}{x^2} = 0$$



$$\Rightarrow q = -\frac{Q}{4}$$

13. (c) Electric field due charge Q_2 , $E_2 = \frac{kQ_2}{x_2^2}$

Electric field due charge Q_1 , $E_1 = \frac{kQ_1}{x_1^2}$



From figure,

$$\tan \theta = \frac{E_2}{E_1} = \frac{x_1}{x_2} \Rightarrow \frac{kQ_2}{x_2^2 \times \frac{kQ_1}{x_1^2}} = \frac{x_1}{x_2}$$

$$\Rightarrow \frac{Q_2 x_1^2}{Q_1 x_2^2} = \frac{x_1}{x_2} \Rightarrow \frac{Q_2}{Q_1} = \frac{x_2}{x_1} \text{ or, } \frac{Q_1}{Q_2} = \frac{x_1}{x_2}.$$

14. (c) For spherical shell

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \quad (\text{if } r \geq R)$$

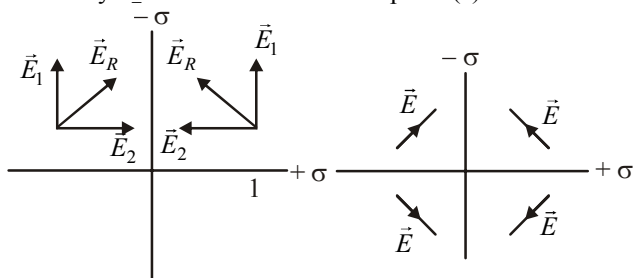
$$= 0 \quad (\text{if } r < R)$$

Force on charge in electric field, $F = qE$

$$\therefore F = 0 \quad (\text{For } r < R)$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2} \quad (\text{For } r > R)$$

15. (c) The electric field produced due to uniformly charged infinite plane is uniform. So option (b) and (d) are wrong. And +ve charge density σ_+ is bigger in magnitude so its field along Y direction will be bigger than field of -ve charge density σ_- in X direction. Hence option (c) is correct.



16. (c) Given,

$$\text{Electric field, } E = E_0(1 - x^2)$$

$$\therefore \text{Force, } F = qE = qE_0(1 - x^2)$$

$$\text{Also, } F = ma = mv \frac{dv}{dx} \quad \left(\because a = v \frac{dv}{dx} \right)$$

$$\therefore mv \frac{dv}{dx} = qE_0(1 - x^2)$$

$$\Rightarrow v dv = \frac{qE_0(1 - x^2)dx}{m}$$

Integrating both sides we get,

$$\Rightarrow \int_0^v v dv = \int_0^x \frac{qE_0(1 - x^2)dx}{m}$$

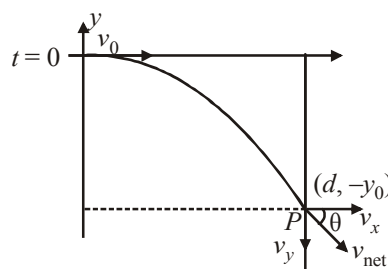
$$\Rightarrow \frac{v^2}{2} = \frac{qE_0}{m} \left(x - \frac{9x^3}{3} \right) = 0$$

$$\Rightarrow x = \sqrt{\frac{3}{a}}$$

17. (b) $F_x = 0$, $a_x = 0$, $(v)_x = \text{constant}$

$$\text{Time taken to reach at 'P' } = \frac{d}{v_0} = t_0 \text{ (let)} \quad \dots(i)$$

$$(\text{Along } -y), y_0 = 0 + \frac{1}{2} \cdot \frac{qE}{m} \cdot t_0^2 \quad \dots(ii)$$



$$\tan \theta = \frac{v_y}{v_x} = \frac{qEt_0}{m \cdot v_0}, \quad \left(t = \frac{d}{v_0} \right)$$

$$\tan \theta = \frac{qEd}{m \cdot v_0^2}, \quad \text{Slope} = \frac{-qEd}{mv_0^2}$$

No electric field $\Rightarrow F_{\text{net}} = 0$, $\vec{v} = \text{const.}$

$$y = mx + c, \quad \begin{cases} m = \frac{qEd}{mv_0^2} \\ (d, -y_0) \end{cases}$$

$$-y_0 = \frac{-qEd}{mv_0^2}, \quad d + c \Rightarrow c = -y_0 + \frac{qEd^2}{mv_0^2}$$

$$y = \frac{-qEd}{mv_0^2} x - y_0 + \frac{qEd^2}{mv_0^2}$$

$$y_0 = \frac{1}{2} \cdot \frac{qE}{m} \left(\frac{d}{v_0} \right)^2 = \frac{1}{2} \frac{qEd^2}{mv_0^2}$$

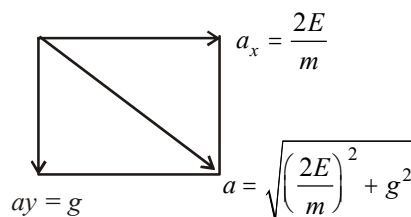
$$y = \frac{-qEdx}{mv_0^2} - \frac{1}{2} \frac{qEd^2}{mv_0^2} + \frac{qEd^2}{mv_0^2}$$

$$y = \frac{-qEd}{mv_0^2} x + \frac{1}{2} \frac{qEd^2}{mv_0^2} \Rightarrow y = \frac{qEd}{mv_0^2} \left(\frac{d}{2} - x \right)$$

18. (d) Net force acting on the particle,

$$\vec{F} = qE\hat{i} + mg\hat{j}$$

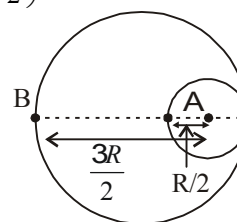
Net acceleration of particle is constant, initial velocity is zero therefore path is straight line.



19. (b) Electric field at A $\left(R' = \frac{R}{2} \right)$

$$E_A \cdot ds = \frac{q}{\epsilon_0}$$

$$\Rightarrow \vec{E}_A = \frac{\rho \times \frac{4}{3} \pi \left(\frac{R}{2} \right)^3}{\epsilon_0 \cdot 4\pi \left(\frac{R}{2} \right)^2}$$



$$\Rightarrow \vec{E}_A = \frac{\sigma(R/2)}{3\epsilon_0} = \left(\frac{\sigma R}{6\epsilon_0} \right)$$

Electric fields at 'B'

$$\vec{E}_B = \frac{k \times \rho \times \frac{4}{3} \pi R^3}{R^2} - \frac{k \times \rho \times \frac{4}{3} \pi \left(\frac{R}{2} \right)^3}{\left(\frac{3R}{2} \right)^2}$$

$$\Rightarrow \vec{E}_B = \frac{\sigma R}{3\epsilon_0} - \left(\frac{1}{4\pi\epsilon_0} \right) \frac{(\sigma)}{\left(\frac{3R}{2} \right)^2} \frac{4\pi}{3} \left(\frac{R}{2} \right)^3$$

$$\Rightarrow \vec{E}_B = \frac{\sigma R}{3\epsilon_0} - \frac{\sigma R}{54\epsilon_0}$$

$$\Rightarrow E_B = \frac{17}{54} \left(\frac{\sigma R}{\epsilon_0} \right)$$

$$\left| \frac{E_A}{E_B} \right| = \frac{1 \times 54}{6 \times 17} = \left(\frac{9}{17} \right) = \frac{9}{17} \times \frac{2}{2} = \frac{18}{34}$$

20. (c) Since $\vec{r} \cdot \vec{p} = 0$

$\vec{E}$ must be antiparallel to $\vec{p}$

$\therefore \hat{E}$ is parallel to $(\hat{i} + 3\hat{j} - 2\hat{k})$

21. (c) (A) By work energy theorem

$$W_{mg} + W_{ele} = \frac{1}{2} m (2v)^2 - \frac{1}{2} m (v)^2$$

$$0 + qE_0 2a = \frac{3}{2} mv^2 \Rightarrow E_0 = \frac{3}{4} \frac{mv^2}{qa}$$

(B) Rate of work done at P = power of electric force

$$= qE_0 V = \frac{3}{4} \frac{mv^3}{a}$$

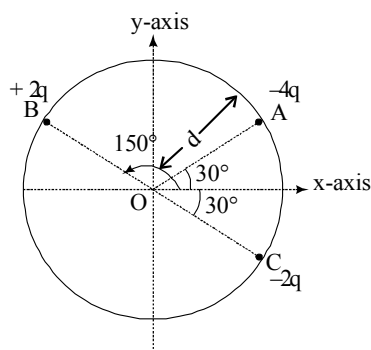
(C) At, Q, $\frac{dw}{dt} = 0$ for both the fields

(D) The difference of magnitude of angular momentum of the particle at P and Q,

$$\Delta \vec{L} = (-m2v2a\hat{k}) - (-mva\hat{k})$$

$$|\Delta \vec{L}| = 3mva$$

22. (a)



Electric field due to charge +2q at centre O

$$\vec{E}_1 = \frac{1}{4\pi\epsilon_0} \times \frac{2q}{d^2} \left[\frac{\sqrt{3}\hat{i} - \hat{j}}{2} \right]$$

Electric field due to charge -2q at centre O

$$\vec{E}_2 = \frac{1}{4\pi\epsilon_0} \times \frac{2q}{d^2} \left[\frac{\sqrt{3}\hat{i} - \hat{j}}{2} \right]$$

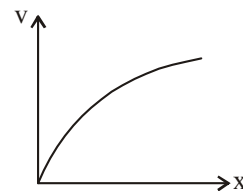
Electric field due to charge -4q at centre O

$$\vec{E}_3 = \frac{1}{4\pi\epsilon_0} \times \frac{4q}{d^2} \left[\frac{\sqrt{3}\hat{i} + \hat{j}}{2} \right]$$

$\therefore$ Net electric field at point O

$$\vec{E}_0 = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 = \frac{\sqrt{3}q}{\pi\epsilon_0 d^2} \hat{i}$$

23. (b)



Using

$$v^2 - u^2 = 2aS \dots (i)$$

Here, $u = 0, s = x$

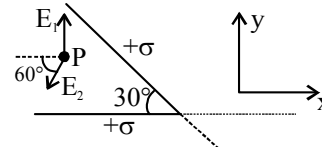
Also, $F_{\text{electric}} = ma$

$$\Rightarrow qE = ma \Rightarrow a = \frac{qE}{m} \Rightarrow a = \frac{qE}{m}$$

Substituting the values in (i) we get

$$v^2 = \frac{2qE}{m} x$$

24. (d)



From figure,

$$\vec{E}_1 = \frac{\sigma}{2\epsilon_0} \hat{y} \quad \text{and} \quad \vec{E}_2 = \frac{\sigma}{2\epsilon_0} (-\cos 60^\circ \hat{x} - \sin 60^\circ \hat{y})$$

$$= \frac{\sigma}{2\epsilon_0} \left(-\frac{1}{2} \hat{x} - \frac{\sqrt{3}}{2} \hat{y} \right)$$

Electric field in the region shown in figure (P)

$$\vec{E}_P = \vec{E}_1 + \vec{E}_2 = \frac{\sigma}{2\epsilon_0} \left[-\frac{1}{2} \hat{x} + \left(1 - \frac{\sqrt{3}}{2} \right) \hat{y} \right]$$

$$\text{or, } \vec{E}_P = \frac{\sigma}{2\epsilon_0} \left[\left(1 - \frac{\sqrt{3}}{2} \right) \hat{y} - \frac{\hat{x}}{2} \right]$$

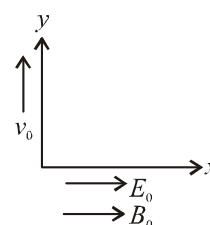
25. (c) In the x direction

$$F_x = qE$$

$$\Rightarrow ma_x = qE$$

$$\Rightarrow a_x = \frac{E_0 q}{m}$$

For speed to be double,



$$v_0^2 + v_x^2 = (2v_0)^2$$

$$\Rightarrow v_x = \sqrt{3} v_0 = a_x t$$

$$\Rightarrow \sqrt{3} v_0 = 0 + \frac{qE_0 t}{m} \Rightarrow t = \frac{\sqrt{3} v_0 m}{E_0 q}$$

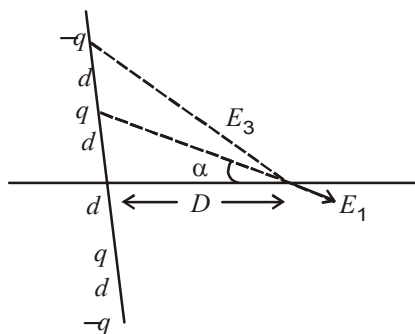
26. (d) Time period of the pendulum (T) is given by

$$T = 2\pi \sqrt{\frac{L}{g_{\text{eff}}}}$$

$$g_{\text{eff}} = \frac{\sqrt{(mg)^2 + (qE)^2}}{m}$$

$$\Rightarrow g_{\text{eff}} = \sqrt{g^2 + \left(\frac{qE}{m}\right)^2} \Rightarrow T = 2\pi \sqrt{\frac{L}{\sqrt{g^2 + \left(\frac{qE}{m}\right)^2}}}$$

27. (d) $\vec{E} = (\vec{E}_1 + \vec{E}_2) + (\vec{E}_3 + \vec{E}_4)$
or $E = 2E \cos \alpha - 2E \cos \beta$



$$= \frac{2kq}{(D^2 + d^2)} \times \frac{D}{\sqrt{D^2 + d^2}} - \frac{2kq}{(D^2 + (2d)^2)} \times \frac{D}{\sqrt{D^2 + (2d)^2}}$$

$$= \frac{2kqD}{(D^2 + d^2)^{3/2}} - \frac{2kqD}{[D^2 + (2d)^2]^{3/2}}$$

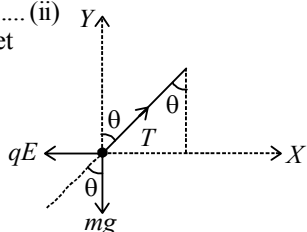
For $d \ll D$

$$E \propto \frac{D}{D^3} \propto \frac{1}{D^2}$$

28. (d) At equilibrium resultant force on bob must be zero, so
 $T \cos \theta = mg$ (i)
 $T \sin \theta = qE$ (ii)
 Solving (i) and (ii) we get

$$\tan \theta = \frac{qE}{mg}$$

$$\tan \theta = \frac{5 \times 10^{-6} \times 2000}{2 \times 10^{-3} \times 10} = \frac{1}{2}$$



[Here, $q = 5 \times 10^{-6} \text{ C}$,
 $E = 2000 \text{ V/m}$, $m = 2 \times 10^{-3} \text{ kg}$]

$$\Rightarrow \tan^{-1}\left(\frac{1}{2}\right)$$

29. (b) Electric field on the axis of a ring of radius R at a distance h from the centre,

$$E = \frac{kQh}{(h^2 + R^2)^{3/2}}$$

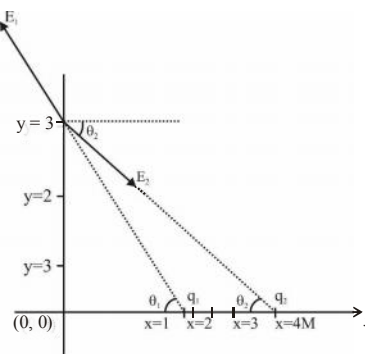
Condition: for maximum electric field $\frac{dE}{dh} = 0$

$$\Rightarrow \frac{d}{dh} \left[\frac{kQh}{(R^2 + h^2)^{3/2}} \right] = 0$$

By using the concept of maxima and minima we get,

$$h = \frac{R}{\sqrt{2}}$$

30. (a)



Let $\vec{E}_1$ and $\vec{E}_2$ are the values of electric field due to charge, q_1 and q_2 respectively

$$\text{magnitude of } E_1 = \frac{1}{4\pi \epsilon_0} \frac{q_1}{r_1^2}$$

$$= \frac{1}{4\pi \epsilon_0} \frac{\sqrt{10} \times 10^{-6}}{(1^2 + 3^2)}$$

$$= (9 \times 10^9) \times \sqrt{10} \times 10^{-7}$$

$$= 9\sqrt{10} \times 10^2$$

$$\therefore \vec{E}_1 = 9\sqrt{10} \times 10^2 [\cos \theta_1 (-\hat{i}) + \sin \theta_1 \hat{j}]$$

$$\Rightarrow E_1 = 9 \times \sqrt{10} \times 10^2 \left[\frac{1}{\sqrt{10}} (-\hat{i}) + \frac{3}{\sqrt{10}} \hat{j} \right]$$

$$\Rightarrow E_1 = 9 \times 10^2 [-\hat{i} + 3\hat{j}] = [-9\hat{i} + 27\hat{j}] 10^2$$

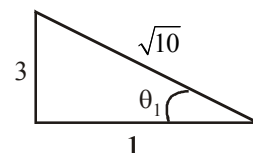
$$\text{Similarly, } E_2 = \frac{1}{4\pi \epsilon_0} \frac{q_2}{r^2}$$

$$E_2 = \frac{9 \times 10^9 \times (25) \times 10^{-6}}{(4^2 + 3^2)} \quad E_2 = 9 \times 10^3 \text{ V/m}$$

$$\therefore \vec{E}_2 = 9 \times 10^3 (\cos \theta_2 \hat{i} - \sin \theta_2 \hat{j}) \because \tan \theta_2 = \frac{3}{4}$$

$$\therefore \vec{E}_2 = 9 \times 10^3 \left(\frac{4}{5} \hat{i} - \frac{3}{5} \hat{j} \right) = (72\hat{i} - 54\hat{j}) \times 10^2$$

$$\therefore \vec{E} = \vec{E}_1 + \vec{E}_2 = (63\hat{i} - 27\hat{j}) \times 10^2 \text{ V/m}$$



31. (a) Equilibrium position will shift to point where resultant force = 0

$$kx_{eq} = qE \Rightarrow x_{eq} = \frac{qE}{k}$$

$$\text{Total energy} = \frac{1}{2}m\omega^2 A^2 + \frac{1}{2}kx_{eq}^2$$

$$\text{Total energy} = \frac{1}{2}m\omega^2 A^2 + \frac{1}{2} \frac{q^2 E^2}{k}$$

32. (d) Charge density, $\rho = \rho_0 \left(1 - \frac{r}{R}\right)$

$$dq = \rho dv$$

$$q_{in} = \int dq = \rho dv$$

$$= \rho_0 \left(1 - \frac{r}{R}\right) 4\pi r^2 dr \quad (\because dv = 4\pi r^2 dr)$$

$$= 4\pi\rho_0 \int_0^R \left(1 - \frac{r}{R}\right) r^2 dr$$

$$= 4\pi\rho_0 \int_0^R r^2 dr - \frac{r^2}{R} dr$$

$$= 4\pi\rho_0 \left[\left[\frac{r^3}{3} \right]_0^R - \left[\frac{r^4}{4R} \right]_0^R \right] = 4\pi\rho_0 \left[\frac{R^3}{3} - \frac{R^4}{4R} \right]$$

$$= 4\pi\rho_0 \left[\frac{R^3}{3} - \frac{R^3}{4} \right] = 4\pi\rho_0 \left[\frac{R^3}{12} \right]$$

$$q = \frac{\pi\rho_0 R^3}{3}$$

$$E \cdot 4\pi r^2 = \left(\frac{\pi\rho_0 R^3}{3\epsilon_0} \right)$$

$$\therefore \text{Electric field outside the ball, } E = \frac{\rho_0 R^3}{12\epsilon_0 r^2}$$

33. (c) Field lines originate perpendicular from positive charge and terminate perpendicular at negative charge. Further this system can be treated as an electric dipole.

34. (d) Given: Length of wire $L = 20$ cm

$$\text{charge } Q = 10^3 \epsilon_0$$

We know, electric field at the centre of the semicircular arc

$$E = \frac{2K\lambda}{r}$$

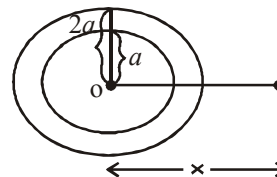
$$\text{or, } E = \frac{2K \left(\frac{2Q}{\pi r} \right)}{r} \left[\text{As } \lambda = \frac{2Q}{\pi r} \right]$$

$$= \frac{4KQ}{\pi r^2} = \frac{4KQ\pi^2}{\pi L^2} = \frac{4\pi KQ}{L^2} = 25 \times 10^3 \text{ N/C}$$

35. (a) Electric field due to complete disc ($R = 2a$) at a distance x and on its axis

$$E_1 = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{x}{\sqrt{R^2 + x^2}} \right] \quad E_1 = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{h}{\sqrt{4a^2 + h^2}} \right]$$

$$= \frac{\sigma}{2\epsilon_0} \left[1 - \frac{h}{2a} \right] \quad \left[\text{here } x=h \text{ and } R=2a \right]$$



Similarly, electric field due to disc ($R = a$)

$$E_2 = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{h}{a} \right)$$

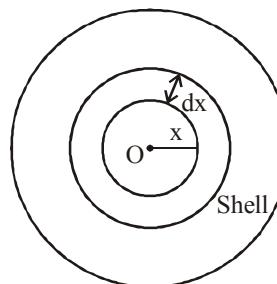
Electric field due to given disc

$$E = E_1 - E_2$$

$$\frac{\sigma}{2\epsilon_0} \left[1 - \frac{h}{2a} \right] - \frac{\sigma}{2\epsilon_0} \left[1 - \frac{h}{a} \right] = \frac{\sigma h}{4\epsilon_0 a}$$

$$\text{Hence, } c = \frac{\sigma}{4a\epsilon_0}$$

36. (b) Let us consider a spherical shell of radius x and thickness dx .



Charge on this shell

$$dq = \rho \cdot 4\pi x^2 dx = \rho_0 \left(1 - \frac{x}{R} \right) \cdot 4\pi x^2 dx$$

$\therefore$ Total charge in the spherical region from centre to r ($r < R$) is

$$q = \int dq = 4\pi\rho_0 \int_0^r \left(1 - \frac{x}{R} \right) x^2 dx$$

$$= 4\pi\rho_0 \left[\frac{x^3}{3} - \frac{x^4}{4R} \right]_0^r = 4\pi\rho_0 \left[\frac{r^3}{3} - \frac{r^4}{4R} \right] = 4\pi\rho_0 r^3 \left[\frac{1}{3} - \frac{r}{4R} \right]$$

$$\therefore \text{Electric field at } r, E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{4\pi\rho_0 r^3 \left[\frac{1}{3} - \frac{r}{4R} \right]}{r^2} = \frac{\rho_0}{\epsilon_0} \left[\frac{r}{3} - \frac{r^2}{4R} \right]$$

37. (c) Given,

 Electric field $E = 150 \text{ N/C}$

 Total surface charge carried by earth $q = ?$

$$\text{or, } q = \epsilon_0 E A$$

$$= \epsilon_0 E \pi r^2$$

$$= 8.85 \times 10^{-12} \times 150 \times (6.37 \times 10^6)^2$$

$$\approx 680 \text{ Kc}$$

As electric field directed inward hence

$$q = -680 \text{ Kc}$$

38. (a) Electric field intensity at the centre of the disc.

$$E = \frac{\sigma}{2\epsilon_0} \text{ (given)}$$

 Electric field along the axis at any distance x from the centre of the disc

$$E' = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{x}{\sqrt{x^2 + R^2}} \right)$$

 From question, $x = R$ (radius of disc)

$$\therefore E' = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{R}{\sqrt{R^2 + R^2}} \right)$$

$$= \frac{\sigma}{2\epsilon_0} \left(\frac{\sqrt{2}R - R}{\sqrt{2}R} \right)$$

$$= \frac{4}{14} E$$

 $\therefore$ % reduction in the value of electric field

$$= \frac{\left(E - \frac{4}{14} E \right) \times 100}{E} = \frac{1000}{14} \% \approx 70.7\%$$

39. (b)
- $F = qE = mg$
- (
- $q = 6e = 6 \times 1.6 \times 10^{-19}$
-)

$$\text{Density (d)} = \frac{\text{mass}}{\text{volume}} = \frac{m}{\frac{4}{3}\pi r^3}$$

$$\text{or } r^3 = \frac{m}{\frac{4}{3}\pi d}$$

 Putting the value of d and m ($= \frac{qE}{g}$) and solving we get $r = 7.8 \times 10^{-7} \text{ m}$

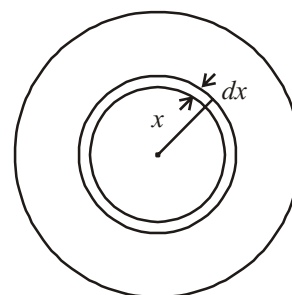
40. (a) Let us consider a spherical shell of radius
- x
- and thickness
- dx
- .

Charge on this shell

$$dq = \rho \cdot 4\pi x^2 dx = \rho_0 \left(\frac{5}{4} - \frac{x}{R} \right) \cdot 4\pi x^2 dx$$

 $\therefore$ Total charge in the spherical region from centre to r ($r < R$) is

$$q = \int dq = 4\pi \rho_0 \int_0^r \left(\frac{5}{4} - \frac{x}{R} \right) x^2 dx$$



$$= 4\pi \rho_0 \left[\frac{5}{4} \cdot \frac{r^3}{3} - \frac{1}{R} \cdot \frac{r^4}{4} \right] = \pi \rho_0 r^3 \left(\frac{5}{3} - \frac{r}{R} \right)$$

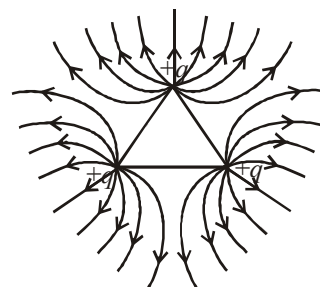
 $\therefore$ Electric field at r ,

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{\pi \rho_0 r^3 \left(\frac{5}{3} - \frac{r}{R} \right)}{r^2} = \frac{\rho_0 r}{4\epsilon_0} \left(\frac{5}{3} - \frac{r}{R} \right)$$

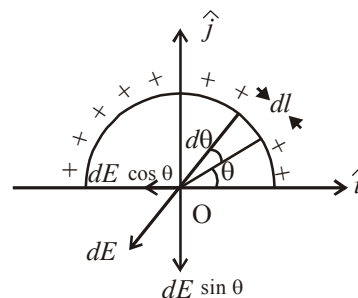
41. (c) Electric lines of force due to a positive charge is spherically symmetric.

All the charges are positive and equal in magnitude. So repulsion takes place. Due to which no lines of force are present inside the equilateral triangle and the resulting lines of force obtained as shown:



42. (c) Let us consider a differential element
- dl
- subtending at angle
- $d\theta$
- at the centre
- O
- as shown in the figure. Linear charge density

$$\lambda = \frac{q}{Qr}$$



$$\text{Charge on the element, } dq = \left(\frac{q}{\pi r} \right) dl$$

$$= \frac{q}{\pi r} (r d\theta)$$

$$(\because dl = r d\theta)$$

$$= \left(\frac{q}{\pi} \right) d\theta$$

Electric field at the center O due to dq is

$$dE = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r^2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{\pi r^2} d\theta$$

Resolving dE into two rectangular component, we find the component $dE \cos \theta$ will be counter balanced by another element on left portion. Hence resultant field at O is the resultant of the component $dE \sin \theta$ only.

$$\therefore E = \int dE \sin \theta = \int_0^\pi \frac{q}{4\pi^2 r^2 \epsilon_0} \sin \theta d\theta$$

$$= \frac{q}{4\pi^2 r^2 \epsilon_0} [-\cos \theta]_0^\pi$$

$$= \frac{q}{4\pi^2 r^2 \epsilon_0} (+1+1) = \frac{q}{2\pi^2 r^2 \epsilon_0}$$

The direction of E is towards negative y -axis.

$$\therefore \vec{E} = -\frac{q}{2\pi^2 r^2 \epsilon_0} \hat{j}$$

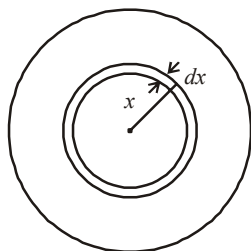
43. (a) Let us consider a spherical shell of radius x and thickness dx .

Due to spherically symmetric charge distribution, the charge on the spherical surface of radius x is

$$dq = dV\rho = 4\pi x^2 dx = \rho_0 \left(\frac{5}{4} - \frac{x}{R} \right) \cdot 4\pi x^2 dx$$

$\therefore$ Total charge in the spherical region from centre to r ($r < R$) is

$$q = \int dq = 4\pi\rho_0 \int_0^r \left(\frac{5}{4} - \frac{x}{R} \right) x^2 dx$$



$$= 4\pi\rho_0 \left[\frac{5}{4} \cdot \frac{r^3}{3} - \frac{1}{R} \cdot \frac{r^4}{4} \right] = \pi\rho_0 r^3 \left(\frac{5}{3} - \frac{r}{R} \right)$$

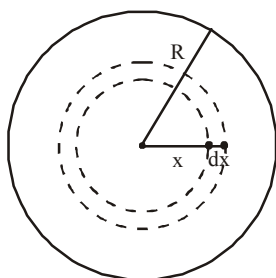
$\therefore$ Electric field intensity at a point on this spherical surface

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{\pi\rho_0 r^3}{r^2} \left(\frac{5}{3} - \frac{r}{R} \right) = \frac{\rho_0 r}{4\epsilon_0} \left(\frac{5}{3} - \frac{r}{R} \right)$$

44. (a)

45. (b)



Let us consider a spherical shell of thickness dx and radius x . The area of this spherical shell $= 4\pi x^2$.

The volume of this spherical shell $= 4\pi x^2 dx$. The charge enclosed within shell

$$dq = \left[\frac{Q}{\pi R^4} \right] [4\pi x^2 dx] = \frac{4Q}{R^4} x^3 dx$$

The charge enclosed in a sphere of radius r_1 can be calculated by

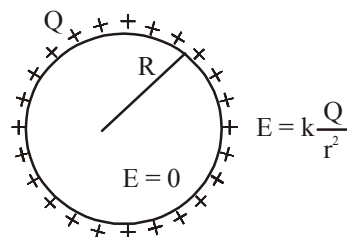
$$Q = \int dq = \frac{4Q}{R^4} \int_0^{r_1} x^3 dx = \frac{4Q}{R^4} \left[\frac{x^4}{4} \right]_0^{r_1} = \frac{Q}{R^4} r_1^4$$

$\therefore$ The electric field at point P inside the sphere at a distance r_1 from the centre of the sphere is

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r_1^2}$$

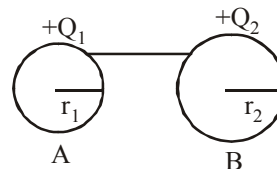
$$\Rightarrow E = \frac{1}{4\pi\epsilon_0} \frac{\left[\frac{Q}{R^4} r_1^4 \right]}{r_1^2} = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^4} r_1^2$$

46. (a) The electric field inside a thin spherical shell of radius R has charge Q spread uniformly over its surface is zero.



Outside the shell the electric field is $E = k \frac{Q}{r^2}$. These characteristics are represented by graph (a).

47. (c)



When the two conducting spheres are connected by a conducting wire, charge will flow from one to other till both acquire same potential.

$\therefore$ After connection, $V_1 = V_2$

$$\Rightarrow k \frac{Q_1}{r_1} = k \frac{Q_2}{r_2} \Rightarrow \frac{Q_1}{r_1} = \frac{Q_2}{r_2}$$

The ratio of electric fields

$$\frac{E_1}{E_2} = \frac{k \frac{Q_1}{r_1^2}}{k \frac{Q_2}{r_2^2}} \Rightarrow \frac{E_1}{E_2} = \frac{Q_1}{r_1^2} \times \frac{r_2^2}{Q_2}$$

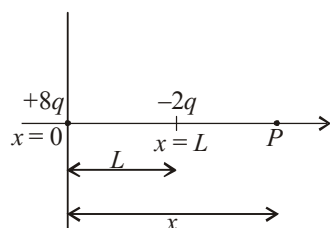
$$\Rightarrow \frac{E_1}{E_2} = \frac{r_1 \times r_2^2}{r_1^2 \times r_2} \Rightarrow \frac{E_1}{E_2} = \frac{r_2}{r_1} = \frac{2}{1}$$

48. (b) At P $\frac{-K2q}{(x-L)^2} + \frac{K8q}{x^2} = 0$

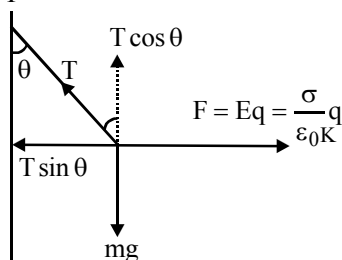
$$\Rightarrow \frac{1}{(x-L)^2} = \frac{4}{x^2}$$

$$\text{or } \frac{1}{x-L} = \frac{2}{x}$$

$$\Rightarrow x = 2x - 2L \text{ or } x = 2L$$



49. (c) P



$$T \sin \theta = qE \quad \dots (i)$$

$$T \cos \theta = mg \quad \dots (ii)$$

Dividing (i) by (ii),

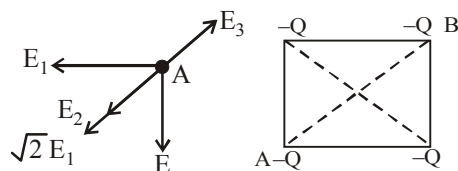
$$\tan \theta = \frac{qE}{mg} = \frac{q}{mg} \left(\frac{\sigma}{\epsilon_0 K} \right) \frac{\sigma q}{\epsilon_0 K \cdot mg}$$

$$\therefore \sigma \propto \tan \theta$$

50. (b) For the system to be equilibrium, net field at A should be zero

$$\sqrt{2} E_1 + E_2 = E_3$$

$$\therefore \frac{kQ \times \sqrt{2}}{a^2} + \frac{kQ}{(\sqrt{2}a)^2} = \frac{kq}{\left(\frac{a}{\sqrt{2}}\right)^2}$$



$$\Rightarrow \frac{Q\sqrt{2}}{1} + \frac{Q}{2} = 2q \Rightarrow q = \frac{Q}{4}(2\sqrt{2} + 1)$$

51. (c) Given, Electric field, $E = 3 \times 10^4$

Mass of the drop, $m = 9.9 \times 10^{-15} \text{ kg}$

At equilibrium, coulomb force on drop balances weight of drop.

$$qE = mg$$

$$\Rightarrow q = \frac{mg}{E} \Rightarrow q = \frac{9.9 \times 10^{-15} \times 10}{3 \times 10^4} = 3.3 \times 10^{-18} \text{ C}$$

52. (b) Let v be the speed of dipole.

Using energy conservation

$$K_i + U_i = K_f + U_f$$

$$\Rightarrow 0 - \frac{2k \cdot p_1}{r^3} p_2 \cos(180^\circ) = \frac{1}{2}mv^2 + \frac{1}{2}mv^2 + 0$$

$$\left(\because \text{Potential energy of interaction between dipole} \right.$$

$$\left. = \frac{-2p_1 p_2 \cos \theta}{4\pi \epsilon_0 r^3} \right)$$

$$\Rightarrow mv^2 = \frac{2kp_1 p_2}{r^3} \Rightarrow v = \sqrt{\frac{2kp_1 p_2}{mr^3}}$$

When $p_1 = p_2 = p$ and $r = a$

$$v = \frac{p}{a} \sqrt{\frac{1}{2\pi \epsilon_0 ma}}$$

53. (-48)

Flux of electric field $\vec{E}$ through any area $\vec{A}$ is defined as

$$\phi = \int \vec{E} \cdot \vec{A} \cos \theta$$

Here, θ = angle between electric field and area vector of a surface

For surface ABCD Angle, $\theta = 90^\circ$

$$\therefore \phi_1 = \int \vec{E} \cdot \vec{A} \cos 90^\circ = 0$$

For surface BCGF $\phi_n = \int \vec{E} \cdot \vec{dA}$

$$\therefore \phi_{11} = [4 \times \hat{i} - (y^2 + 1) \hat{j}] \cdot 4\hat{i} = 16x$$

$$\phi_{11} = 48 \frac{Nm^2}{C}$$

$$\phi_1 - \phi_{11} = -48$$

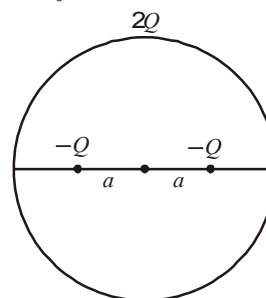
54. (a)

55. (b) Surface charge density depends only due to Q . Also

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$$

$$\text{or } E \times 4\pi r^2 = \frac{Q}{\epsilon_0} \Rightarrow E = \frac{1}{4\pi \epsilon_0} \frac{Q}{r^2}, r \geq R$$

56. (a) $\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$



$$\text{or } E \times 4\pi r^2 = \frac{1}{\epsilon_0} \int S(4\pi r^2) dr$$

$$\text{or } E \times 4\pi r^2 = \frac{1}{\epsilon_0} \int_0^r (kr)(4\pi r^2) dr$$

$$\text{or } E \times 4\pi r^2 = \frac{4\pi k}{\epsilon_0} \left(\frac{r^4}{4} \right)$$

$$\therefore E = \frac{k}{4\epsilon_0} r^2 \quad \dots(i)$$

$$\text{Also } 2Q = \int_0^R (kr)(4\pi r^2) dr = 4\pi k \left[\frac{r^4}{4} \right]_0^R$$

$$Q = \frac{\pi k R^4}{2} \quad \dots(ii)$$

From above equations,

$$E = \frac{Qr^2}{2\pi\epsilon_0 R^4} \quad \dots(iii)$$

According to given condition

$$= EQ \frac{Q^4}{4\pi\epsilon_0 (20)^2} \quad \dots(iv)$$

From equations (iii) and (iv), we have

$$a = 8^{-1/4} R.$$

57. (b) $\tau = -PE \sin \theta$
or $I\alpha = -PE(\theta)$

$$\alpha = \frac{PE}{I}(-\theta)$$

On comparing with
 $\alpha = -\omega^2 \theta$

$$\omega = \sqrt{\frac{PE}{I}} = \sqrt{\frac{qdE}{2m\left(\frac{d}{2}\right)^2}} = \sqrt{\frac{2qE}{md}}$$

58. (b) Potential energy of a dipole is given by

$$U = -\vec{P} \cdot \vec{E}$$

$$= -PE \cos \theta$$

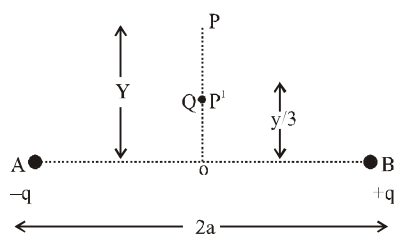
[Where θ = angle between dipole and perpendicular to the field]

$$= -(10^{-29})(10^3) \cos 45^\circ$$

$$= -0.707 \times 10^{-26} \text{ J} = -7 \times 10^{-27} \text{ J}$$

59. (d) Electric field of equatorial plane of dipole

$$= -\frac{K\vec{P}}{r^3}$$



$$\therefore \text{At point P, } = +\frac{KP}{y^3} Q$$

$$\text{At Point P}^1, F^1 = +\frac{KPQ}{(y/3)^3} = 27F.$$

60. (b) When cube is of side a and point charge Q is at the center of the cube then the total electric flux due to this charge will pass evenly through the six faces of the cube. So, the electric flux through one face will be equal to 1/6 of the total electric flux due to this charge.

$$\text{Flux through 6 faces} = \frac{Q}{\epsilon_0}$$

$$\therefore \text{Flux through 1 face, } = \frac{Q}{6\epsilon_0}$$

61. (a) $T = PE \sin \theta$ Torque experienced by the dipole in an electric field, $\vec{T} = \vec{P} \times \vec{E}$

$$\vec{p} = p \cos \theta \hat{i} + p \sin \theta \hat{j}$$

$$\vec{E}_1 = E \hat{i}$$

$$\vec{T}_1 = \vec{p} \times \vec{E}_1 = (p \cos \theta \hat{i} + p \sin \theta \hat{j}) \times E(\hat{i})$$

$$\tau \hat{k} = pE \sin \theta (-\hat{k}) \quad \dots(i)$$

$$\vec{E}_2 = \sqrt{3} E_1 \hat{j}$$

$$\vec{T}_2 = p \cos \theta \hat{i} + p \sin \theta \hat{j} \times \sqrt{3} E_1 \hat{j}$$

$$\tau \hat{k} = \sqrt{3} pE_1 \cos \theta \hat{k} \quad \dots(ii)$$

From eqns. (i) and (ii)

$$pE \sin \theta = \sqrt{3} pE \cos \theta$$

$$\tan \theta = \sqrt{3} \quad \therefore \theta = 60^\circ$$

62. (c) The net flux linked with closed surfaces S_1, S_2, S_3 & S_4 are

$$\text{For surface } S_1, \phi_1 = \frac{1}{\epsilon_0} (2q)$$

$$\text{For surface } S_2, \phi_2 = \frac{1}{\epsilon_0} (q + q + q - q) = \frac{1}{\epsilon_0} 2q$$

$$\text{For surface } S_3, \phi_3 = \frac{1}{\epsilon_0} (q + q) = \frac{1}{\epsilon_0} (2q)$$

$$\text{For surface } S_4, \phi_4 = \frac{1}{\epsilon_0} (8q - 2q - 4q) = \frac{1}{\epsilon_0} (2q)$$

Hence, $\phi_1 = \phi_2 = \phi_3 = \phi_4$ i.e. net electric flux is same for all surfaces.

Keep in mind, the electric field due to a charge outside (S_3 and S_4), the Gaussian surface contributes zero net flux through the surface, because as many lines due to that charge enter the surface as leave it.

63. (c) Applying Gauss's law

$$\oint_S \vec{E} \cdot d\vec{s} = \frac{Q}{\epsilon_0}$$

$$\therefore E \times 4\pi r^2 = \frac{Q + 2\pi A r^2 - 2\pi A a^2}{\epsilon_0}$$

$$\rho = \frac{dr}{dV}$$

$$Q = \rho 4\pi r^2$$

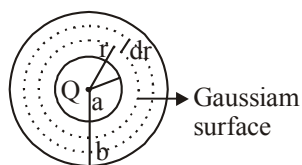
$$Q = \int_a^r \frac{A}{r} 4\pi r^2 dr = 2\pi A[r^2 - a^2]$$

$$E = \frac{1}{4\pi\epsilon_0} \left[\frac{Q - 2\pi Aa^2}{r^2} + 2\pi A \right]$$

For E to be independent of ' r '

$$Q - 2\pi Aa^2 = 0$$

$$\therefore A = \frac{Q}{2\pi a^2}$$



64. (a) $\vec{E} = E_0\hat{i} + 2E_0\hat{j}$

Given, $E_0 = 100 \text{ N/C}$

So, $\vec{E} = 100\hat{i} + 200\hat{j}$

Radius of circular surface = 0.02 m

$$\text{Area} = \pi r^2 = \frac{22}{7} \times 0.02 \times 0.02$$

$$= 1.25 \times 10^{-3} \hat{i} \text{ m}^2 \quad [\text{Loop is parallel to Y-Z plane}]$$

Now, flux (ϕ) = $EA \cos \theta$

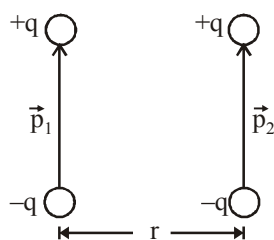
$$= (100\hat{i} + 200\hat{j}) \cdot 1.25 \times 10^{-3} \hat{i} \cos \theta^\circ \quad [\theta = 0^\circ]$$

$$= 125 \times 10^{-3} \text{ Nm}^2/\text{C}$$

$$= 0.125 \text{ Nm}^2/\text{C}$$

65. (c) Force of interaction

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{6p_1p_2}{r^4}$$



66. (b) We know that,

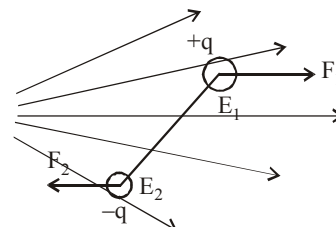
$$\phi = \oint E \cdot dS = E \oint dS \cos 45^\circ$$

In case of hemisphere

$$\phi_{\text{curved}} = \phi_{\text{circular}}$$

$$\text{Therefore, } \phi_{\text{curved}} = E\pi a^2 \cdot \frac{1}{\sqrt{2}} = \frac{E\pi a^2}{\sqrt{2}}$$

67. (c)



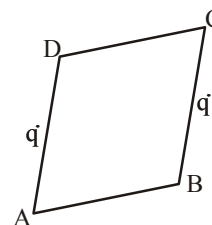
As the dipole is placed in non-uniform field, so the force acting on the dipole will not cancel each other. This will result in a force as well as torque.

68. (a) The electric flux ϕ_1 entering an enclosed surface is taken as negative and the electric flux ϕ_2 leaving the surface is taken as positive, by convention. Therefore the net flux leaving the enclosed surface, $\phi = \phi_2 - \phi_1$
According to Gauss theorem

$$\phi = \frac{q}{\epsilon_0} \Rightarrow q = \epsilon_0 \phi = \epsilon_0 (\phi_2 - \phi_1)$$

69. (None) Electric flux due to charge placed outside is zero. But for the charge inside the cube, flux due to each face is

$$\frac{1}{6} \left[\frac{q}{\epsilon_0} \right] \text{ which is not in option.}$$



Electrostatic Potential and Capacitance



TOPIC 1 Electrostatic Potential and Equipotential Surfaces



1. Ten charges are placed on the circumference of a circle of radius R with constant angular separation between successive charges. Alternate charges 1, 3, 5, 7, 9 have charge $(+q)$ each, while 2, 4, 6, 8, 10 have charge $(-q)$ each. The potential V and the electric field E at the centre of the circle are respectively :

(Take $V = 0$ at infinity)

[Sep. 05, 2020 (II)]

(a) $V = \frac{10q}{4\pi\epsilon_0 R}$; $E = 0$

(b) $V = 0$; $E = \frac{10q}{4\pi\epsilon_0 R^2}$

(c) $V = 0$; $E = 0$

(d) $V = \frac{10q}{4\pi\epsilon_0 R}$; $E = \frac{10q}{4\pi\epsilon_0 R^2}$

2. Two isolated conducting spheres S_1 and S_2 of radius $\frac{2}{3}R$ and $\frac{1}{3}R$ have $12 \mu\text{C}$ and $-3 \mu\text{C}$ charges, respectively, and are at a large distance from each other. They are now connected by a conducting wire. A long time after this is done the charges on S_1 and S_2 are respectively :

[Sep. 03, 2020 (I)]

(a) $4.5 \mu\text{C}$ on both (b) $+4.5 \mu\text{C}$ and $-4.5 \mu\text{C}$

(c) $3 \mu\text{C}$ and $6 \mu\text{C}$ (d) $6 \mu\text{C}$ and $3 \mu\text{C}$

3. Concentric metallic hollow spheres of radii R and $4R$ hold charges Q_1 and Q_2 respectively. Given that surface charge densities of the concentric spheres are equal, the potential difference $V(R) - V(4R)$ is :

[Sep. 03, 2020 (II)]

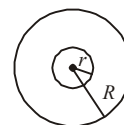
(a) $\frac{3Q_1}{16\pi\epsilon_0 R}$

(b) $\frac{3Q_2}{4\pi\epsilon_0 R}$

(c) $\frac{Q_2}{4\pi\epsilon_0 R}$

(d) $\frac{3Q_1}{4\pi\epsilon_0 R}$

4. A charge Q is distributed over two concentric conducting thin spherical shells radii r and R ($R > r$). If the surface charge densities on the two shells are equal, the electric potential at the common centre is : [Sep. 02, 2020 (II)]



(a) $\frac{1}{4\pi\epsilon_0} \frac{(R+r)}{2(R^2+r^2)} Q$

(b) $\frac{1}{4\pi\epsilon_0} \frac{(2R+r)}{(R^2+r^2)} Q$

(c) $\frac{1}{4\pi\epsilon_0} \frac{(R+2r)Q}{2(R^2+r^2)}$

(d) $\frac{1}{4\pi\epsilon_0} \frac{(R+r)}{(R^2+r^2)} Q$

5. A point dipole $= \vec{p} - p_o \hat{x}$ kept at the origin. The potential and electric field due to this dipole on the y -axis at a distance d are, respectively : (Take $V = 0$ at infinity)

[12 April 2019 I]

(a) $\frac{|\vec{p}|}{4\pi\epsilon_0 d^2}, \frac{\vec{p}}{4\pi\epsilon_0 d^3}$

(b) $0, \frac{-\vec{p}}{4\pi\epsilon_0 d^3}$

(c) $0, \frac{|\vec{p}|}{4\pi\epsilon_0 d^3}$

(d) $\frac{|\vec{p}|}{4\pi\epsilon_0 d^2}, \frac{-\vec{p}}{4\pi\epsilon_0 d^3}$

6. A uniformly charged ring of radius $3a$ and total charge q is placed in xy -plane centred at origin. A point charge q is moving towards the ring along the z -axis and has speed v at $z = 4a$. The minimum value of v such that it crosses the origin is :

[10 April 2019 I]

(a) $\sqrt{\frac{2}{m} \left(\frac{4}{15} \frac{q^2}{4\pi\epsilon_0 a} \right)^{1/2}}$

(b) $\sqrt{\frac{2}{m} \left(\frac{1}{5} \frac{q^2}{4\pi\epsilon_0 a} \right)^{1/2}}$

(c) $\sqrt{\frac{2}{m} \left(\frac{2}{15} \frac{q^2}{4\pi\epsilon_0 a} \right)^{1/2}}$

(d) $\sqrt{\frac{2}{m} \left(\frac{1}{15} \frac{q^2}{4\pi\epsilon_0 a} \right)^{1/2}}$

7. A solid conducting sphere, having a charge Q , is surrounded by an uncharged conducting hollow spherical shell. Let the potential difference between the surface of the solid sphere and that of the outer surface of the hollow shell be V . If the shell is now given a charge of $-4Q$, the new potential difference between the same two surfaces is :

[8 April 2019 I]

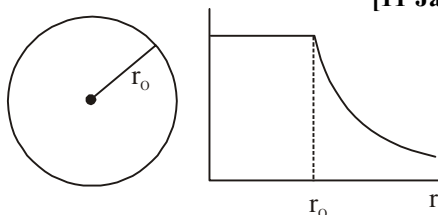
(a) $-2V$

(b) $2V$

(c) $4V$

(d) V

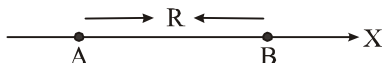
8. The electric field in a region is given by $\vec{E} = (Ax + B)\hat{i}$, where E is in NC^{-1} and x is in metres. The values of constants are $A = 20$ SI unit and $B = 10$ SI unit. If the potential at $x = 1$ is V_1 and that at $x = -5$ is V_2 , then $V_1 - V_2$ is: [8 Jan. 2019 II]
 (a) 320 V (b) -48 V (c) 180 V (d) -520 V
9. The given graph shows variation (with distance r from centre) of: [11 Jan. 2019 I]



- (a) Electric field of a uniformly charged sphere
 (b) Potential of a uniformly charged spherical shell
 (c) Potential of a uniformly charged sphere
 (d) Electric field of a uniformly charged spherical shell
10. A charge Q is distributed over three concentric spherical shells of radii a, b, c ($a < b < c$) such that their surface charge densities are equal to one another. The total potential at a point at distance r from their common centre, where $r < a$, would be:

[10 Jan. 2019 I]

- (a) $\frac{Q}{12\pi\epsilon_0} \frac{ab + bc + ca}{abc}$ (b) $\frac{Q(a^2 + b^2 + c^2)}{4\pi\epsilon_0(a^3 + b^3 + c^3)}$
 (c) $\frac{Q}{4\pi\epsilon_0(a + b + c)}$ (d) $\frac{Q(a + b + c)}{4\pi\epsilon_0(a^2 + b^2 + c^2)}$
11. Two electric dipoles, A, B with respective dipole moments $\vec{d}_A = -4qa\hat{i}$ and $\vec{d}_B = -2qa\hat{i}$ are placed on the x -axis with a separation R , as shown in the figure



The distance from A at which both of them produce the same potential is:

[10 Jan. 2019 I]

- (a) $\frac{R}{\sqrt{2} + 1}$ (b) $\frac{\sqrt{2}R}{\sqrt{2} + 1}$
 (c) $\frac{R}{\sqrt{2} - 1}$ (d) $\frac{\sqrt{2}R}{\sqrt{2} - 1}$
12. Consider two charged metallic spheres S_1 and S_2 of radii R_1 and R_2 , respectively. The electric fields E_1 (on S_1) and E_2 (on S_2) on their surfaces are such that $E_1/E_2 = R_1/R_2$. Then the ratio V_1 (on S_1)/ V_2 (on S_2) of the electrostatic potentials on each sphere is: [8 Jan. 2019 II]
 (a) R_1/R_2 (b) $(R_1/R_2)^2$
 (c) (R_2/R_1) (d) $\left(\frac{R_1}{R_2}\right)^3$
13. Three concentric metal shells A, B and C of respective radii a, b and c ($a < b < c$) have surface charge densities $+\sigma, -\sigma$ and $+\sigma$ respectively. The potential of shell B is: [2018]

(a) $\frac{\sigma}{\epsilon_0} \left[\frac{a^2 - b^2}{a} + c \right]$ (b) $\frac{\sigma}{\epsilon_0} \left[\frac{a^2 - b^2}{b} + c \right]$
 (c) $\frac{\sigma}{\epsilon_0} \left[\frac{b^2 - c^2}{b} + a \right]$ (d) $\frac{\sigma}{\epsilon_0} \left[\frac{b^2 - c^2}{c} + a \right]$

14. There is a uniform electrostatic field in a region. The potential at various points on a small sphere centred at P, in the region, is found to vary between in the limits 589.0 V to 589.8 V. What is the potential at a point on the sphere whose radius vector makes an angle of 60° with the direction of the field? [Online April 8, 2017]
 (a) 589.5 V (b) 589.2 V (c) 589.4 V (d) 589.6 V
15. Within a spherical charge distribution of charge density $\rho(r)$, N equipotential surfaces of potential $V_0, V_0 + \Delta V, V_0 + 2\Delta V, \dots, V_0 + N\Delta V$ ($\Delta V > 0$), are drawn and have increasing radii $r_0, r_1, r_2, \dots, r_N$, respectively. If the difference in the radii of the surfaces is constant for all values of V_0 and ΔV then: [Online April 10, 2016]
 (a) $\rho(r) = \text{constant}$ (b) $\rho(r) \propto \frac{1}{r^2}$
 (c) $\rho(r) \propto \frac{1}{r}$ (d) $\rho(r) \propto r$
16. The potential (in volts) of a charge distribution is given by $V(z) = 30 - 5z^2$ for $|z| \leq 1$ m
 $V(z) = 35 - 10|z|$ for $|z| \geq 1$ m.
 $V(z)$ does not depend on x and y . If this potential is generated by a constant charge per unit volume ρ_0 (in units of ϵ_0) which is spread over a certain region, then choose the correct statement. [Online April 9, 2016]
 (a) $\rho_0 = 20 \epsilon_0$ in the entire region
 (b) $\rho_0 = 10 \epsilon_0$ for $|z| \leq 1$ m and $\rho_0 = 0$ elsewhere
 (c) $\rho_0 = 20 \epsilon_0$ for $|z| \leq 1$ m and $\rho_0 = 0$ elsewhere
 (d) $\rho_0 = 40 \epsilon_0$ in the entire region
17. A uniformly charged solid sphere of radius R has potential V_0 (measured with respect to ∞) on its surface. For this sphere the equipotential surfaces with potentials $\frac{3V_0}{2}, \frac{5V_0}{4}, \frac{3V_0}{4}$ and $\frac{V_0}{4}$ have radius R_1, R_2, R_3 and R_4 respectively. Then [2015]
 (a) $R_1 = 0$ and $R_2 < (R_4 - R_3)$
 (b) $2R = R_4$
 (c) $R_1 = 0$ and $R_2 > (R_4 - R_3)$
 (d) $R_1 \neq 0$ and $(R_2 - R_1) > (R_4 - R_3)$
18. An electric field $\vec{E} = (25\hat{i} + 30\hat{j})\text{NC}^{-1}$ exists in a region of space. If the potential at the origin is taken to be zero then the potential at $x = 2$ m, $y = 2$ m is: [Online April 11, 2015]
 (a) -110 J (b) -140 J (c) -120 J (d) -130 J
19. Assume that an electric field $\vec{E} = 30x^2\hat{i}$ exists in space. Then the potential difference $V_A - V_0$, where V_0 is the potential at the origin and V_A the potential at $x = 2$ m is: [2014]
 (a) 120 J/C (b) -120 J/C
 (c) -80 J/C (d) 80 J/C

20. Consider a finite insulated, uncharged conductor placed near a finite positively charged conductor. The uncharged body must have a potential : **[Online April 23, 2013]**
- less than the charged conductor and more than at infinity.
 - more than the charged conductor and less than at infinity.
 - more than the charged conductor and more than at infinity.
 - less than the charged conductor and less than at infinity.

21. Two small equal point charges of magnitude q are suspended from a common point on the ceiling by insulating mass less strings of equal lengths. They come to equilibrium with each string making angle θ from the vertical. If the mass of each charge is m , then the electrostatic potential at the centre of line joining them will

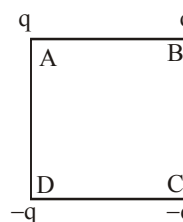
be $\left(\frac{1}{4\pi\epsilon_0} = k \right)$. **[Online April 22, 2013]**

- $2\sqrt{kmg \tan \theta}$
 - $\sqrt{kmg \tan \theta}$
 - $4\sqrt{kmg / \tan \theta}$
 - $\sqrt{kmg / \tan \theta}$
22. A point charge of magnitude $+1 \mu\text{C}$ is fixed at $(0, 0, 0)$. An isolated uncharged spherical conductor, is fixed with its center at $(4, 0, 0)$. The potential and the induced electric field at the centre of the sphere is : **[Online April 22, 2013]**
- $1.8 \times 10^5 \text{ V}$ and $-5.625 \times 10^6 \text{ V/m}$
 - 0 V and 0 V/m
 - $2.25 \times 10^5 \text{ V}$ and $-5.625 \times 10^6 \text{ V/m}$
 - $2.25 \times 10^5 \text{ V}$ and 0 V/m
23. A charge of total amount Q is distributed over two concentric hollow spheres of radii r and R ($R > r$) such that the surface charge densities on the two spheres are equal. The electric potential at the common centre is

[Online May 19, 2012]

- $\frac{1}{4\pi\epsilon_0} \frac{(R-r)Q}{(R^2+r^2)}$
 - $\frac{1}{4\pi\epsilon_0} \frac{(R+r)Q}{2(R^2+r^2)}$
 - $\frac{1}{4\pi\epsilon_0} \frac{(R+r)Q}{(R^2+r^2)}$
 - $\frac{1}{4\pi\epsilon_0} \frac{(R-r)Q}{2(R^2+r^2)}$
24. The electric potential $V(x)$ in a region around the origin is given by $V(x) = 4x^2$ volts. The electric charge enclosed in a cube of 1 m side with its centre at the origin is (in coulomb) **[Online May 7, 2012]**
- $8\epsilon_0$
 - $-4\epsilon_0$
 - 0
 - $-8\epsilon_0$
25. The electrostatic potential inside a charged spherical ball is given by $\phi = ar^2 + b$ where r is the distance from the centre and a, b are constants. Then the charge density inside the ball is: **[2011]**
- $-6a\epsilon_0 r$
 - $-24\pi a\epsilon_0$
 - $-6a\epsilon_0$
 - $-24\pi a\epsilon_0 r$

26. An electric charge $10^{-3} \mu\text{C}$ is placed at the origin $(0, 0)$ of $X-Y$ co-ordinate system. Two points A and B are situated at $(\sqrt{2}, \sqrt{2})$ and $(2, 0)$ respectively. The potential difference between the points A and B will be **[2007]**
- 4.5 volts
 - 9 volts
 - Zero
 - 2 volt
27. Charges are placed on the vertices of a square as shown. Let $\vec{E}$ be the electric field and V the potential at the centre. If the charges on A and B are interchanged with those on D and C respectively, then **[2007]**



- $\vec{E}$ changes, V remains unchanged
 - $\vec{E}$ remains unchanged, V changes
 - both $\vec{E}$ and V change
 - $\vec{E}$ and V remain unchanged
28. The potential at a point x (measured in μm) due to some charges situated on the x -axis is given by $V(x) = 20/(x^2 - 4)$ volt. The electric field E at $x = 4 \mu\text{m}$ is given by **[2007]**
- $(10/9) \text{ volt}/\mu\text{m}$ and in the $+ve x$ direction
 - $(5/3) \text{ volt}/\mu\text{m}$ and in the $-ve x$ direction
 - $(5/3) \text{ volt}/\mu\text{m}$ and in the $+ve x$ direction
 - $(10/9) \text{ volt}/\mu\text{m}$ and in the $-ve x$ direction
29. Two thin wire rings each having a radius R are placed at a distance d apart with their axes coinciding. The charges on the two rings are $+q$ and $-q$. The potential difference between the centres of the two rings is **[2005]**
- $\frac{q}{2\pi\epsilon_0} \left[\frac{1}{R} - \frac{1}{\sqrt{R^2 + d^2}} \right]$
 - $\frac{qR}{4\pi\epsilon_0 d^2}$
 - $\frac{q}{4\pi\epsilon_0} \left[\frac{1}{R} - \frac{1}{\sqrt{R^2 + d^2}} \right]$
 - zero
30. A thin spherical conducting shell of radius R has a charge q . Another charge Q is placed at the centre of the shell.

The electrostatic potential at a point P , a distance $\frac{R}{2}$ from the centre of the shell is **[2003]**

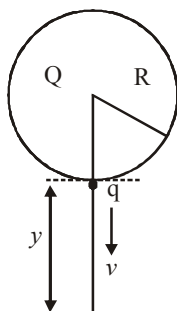
- $\frac{2Q}{4\pi\epsilon_0 R}$
- $\frac{2Q}{4\pi\epsilon_0 R} - \frac{2q}{4\pi\epsilon_0 R}$
- $\frac{2Q}{4\pi\epsilon_0 R} + \frac{q}{4\pi\epsilon_0 R}$
- $\frac{(q+Q)2}{4\pi\epsilon_0 R}$

TOPIC 2 Electric Potential Energy and Work Done in Carrying a Charge



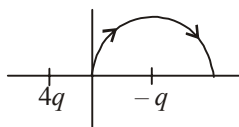
31. A solid sphere of radius R carries a charge $Q + q$ distributed uniformly over its volume. A very small point like piece of it of mass m gets detached from the bottom of the sphere and falls down vertically under gravity. This piece carries charge q . If it acquires a speed v when it has fallen through a vertical height y (see figure), then : (assume the remaining portion to be spherical).

[Sep. 05, 2020 (I)]



- (a) $v^2 = y \left[\frac{qQ}{4\pi\epsilon_0 R^2 y m} + g \right]$
 (b) $v^2 = y \left[\frac{qQ}{4\pi\epsilon_0 R(R+y)m} + g \right]$
 (c) $v^2 = 2y \left[\frac{QqR}{4\pi\epsilon_0 (R+y)^3 m} + g \right]$
 (d) $v^2 = 2y \left[\frac{qQ}{4\pi\epsilon_0 R(R+y)m} + g \right]$
32. A two point charges $4q$ and $-q$ are fixed on the x -axis at $x = -\frac{d}{2}$ and $x = \frac{d}{2}$, respectively. If a third point charge ' q ' is taken from the origin to $x = d$ along the semicircle as shown in the figure, the energy of the charge will :

[Sep. 04, 2020 (I)]



- (a) increase by $\frac{3q^2}{4\pi\epsilon_0 d}$
 (b) increase by $\frac{2q^2}{3\pi\epsilon_0 d}$
 (c) decrease by $\frac{q^2}{4\pi\epsilon_0 d}$
 (d) decrease by $\frac{4q^2}{3\pi\epsilon_0 d}$

33. Hydrogen ion and singly ionized helium atom are accelerated, from rest, through the same potential difference. The ratio of final speeds of hydrogen and helium ions is close to : [Sep. 03, 2020 (II)]

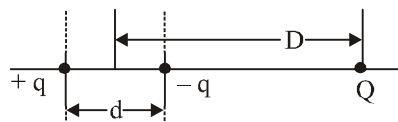
- (a) 1 : 2 (b) 10 : 7
 (c) 2 : 1 (d) 5 : 7

34. In free space, a particle A of charge $1 \mu\text{C}$ is held fixed at a point P. Another particle B of the same charge and mass $4 \mu\text{g}$ is kept at a distance of 1 mm from P. If B is released, then its velocity at a distance of 9 mm from P is :

$\left[\text{Take } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-2} \right]$ [10 April 2019 II]

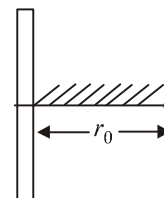
- (a) 1.0 m/s (b) 3.0×10^4 m/s
 (c) 2.0×10^3 m/s (d) 1.5×10^2 m/s

35. A system of three charges are placed as shown in the figure:

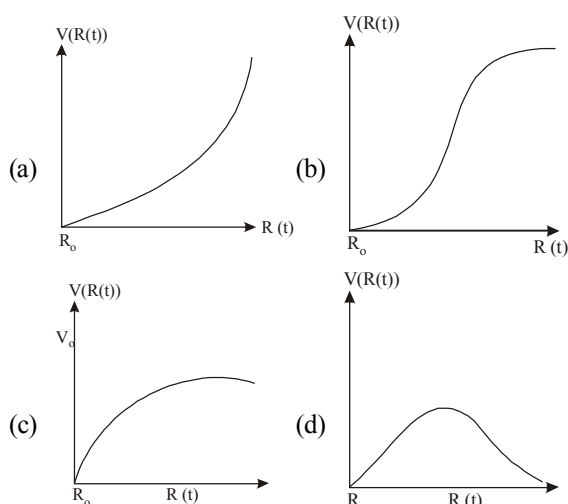


If $D \gg d$, the potential energy of the system is best given by [9 April 2019 I]

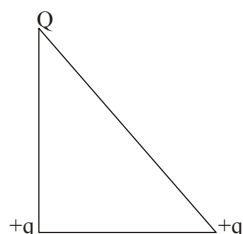
- (a) $\frac{1}{4\pi\epsilon_0} \left[\frac{-q^2}{d} - \frac{qQd}{2D^2} \right]$ (b) $\frac{1}{4\pi\epsilon_0} \left[\frac{-q^2}{d} + \frac{2qQd}{D^2} \right]$
 (c) $\frac{1}{4\pi\epsilon_0} \left[+\frac{q^2}{d} + \frac{qQd}{D^2} \right]$ (d) $\frac{1}{4\pi\epsilon_0} \left[-\frac{q^2}{d} - \frac{qQd}{D^2} \right]$
36. A positive point charge is released from rest at a distance r_0 from a positive line charge with uniform density. The speed (v) of the point charge, as a function of instantaneous distance r from line charge, is proportional to : [8 April 2019 II]



- (a) $v \propto e^{+r/r_0}$ (b) $v \propto \sqrt{\ln\left(\frac{r}{r_0}\right)}$
 (c) $v \propto \ln\left(\frac{r}{r_0}\right)$ (d) $v \propto \left(\frac{r}{r_0}\right)$
37. There is a uniform spherically symmetric surface charge density at a distance R_0 from the origin. The charge distribution is initially at rest and starts expanding because of mutual repulsion. The figure that represents best the speed V ($R(t)$) of the distribution as a function of its instantaneous radius $R(t)$ is: [12 Jan. 2019 I]



38. Three charges Q , $+q$ and $+q$ are placed at the vertices of a right-angle isosceles triangle as shown below. The net electrostatic energy of the configuration is zero, if the value of Q is : [11 Jan. 2019 I]



- (a) $+q$ (b) $\frac{-\sqrt{2}q}{\sqrt{2}+1}$
 (c) $\frac{-q}{1+\sqrt{2}}$ (d) $-2q$
39. Four equal point charges Q each are placed in the xy plane at $(0, 2)$, $(4, 2)$, $(4, -2)$ and $(0, -2)$. The work required to put a fifth charge Q at the origin of the coordinate system will be: [10 Jan. 2019 II]

- (a) $\frac{Q^2}{4\pi\epsilon_0} \left(1 + \frac{1}{\sqrt{3}}\right)$ (b) $\frac{Q^2}{4\pi\epsilon_0} \left(1 + \frac{1}{\sqrt{5}}\right)$
 (c) $\frac{Q^2}{2\sqrt{2}\pi\epsilon_0}$ (d) $\frac{Q^2}{4\pi\epsilon_0}$

40. **Statement 1** : No work is required to be done to move a test charge between any two points on an equipotential surface.

Statement 2 : Electric lines of force at the equipotential surfaces are mutually perpendicular to each other.

[Online April 25, 2013]

- (a) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.
 (b) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.
 (c) Statement 1 is true, Statement 2 is false.
 (d) Statement 1 is false, Statement 2 is true.

41. An insulating solid sphere of radius R has a uniformly positive charge density ρ . As a result of this uniform charge distribution there is a finite value of electric potential at the centre of the sphere, at the surface of the sphere and also at a point outside the sphere. The electric potential at infinite is zero. [2012]

Statement -1 When a charge q is taken from the centre to the surface of the sphere its potential energy changes by $\frac{q\rho}{3\epsilon_0}$.

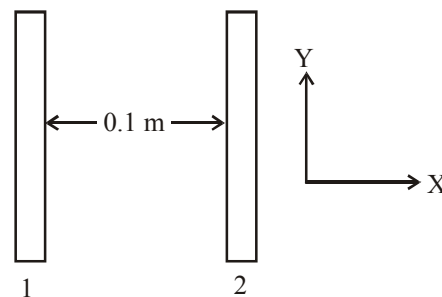
Statement -2 The electric field at a distance r ($r < R$) from the centre of the sphere is $\frac{\rho r}{3\epsilon_0}$.

- (a) Statement 1 is true, Statement 2 is true; Statement 2 is not the correct explanation of statement 1.
 (b) Statement 1 is true Statement 2 is false.
 (c) Statement 1 is false Statement 2 is true.
 (d) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1
42. Two positive charges of magnitude ' q ' are placed, at the ends of a side (side 1) of a square of side ' $2a$ '. Two negative charges of the same magnitude are kept at the other corners. Starting from rest, if a charge Q moves from the middle of side 1 to the centre of square, its kinetic energy at the centre of square is [2011 RS]

- (a) zero (b) $\frac{1}{4\pi\epsilon_0} \frac{2qQ}{a} \left(1 + \frac{1}{\sqrt{5}}\right)$
 (c) $\frac{1}{4\pi\epsilon_0} \frac{2qQ}{a} \left(1 - \frac{2}{\sqrt{5}}\right)$ (d) $\frac{1}{4\pi\epsilon_0} \frac{2qQ}{a} \left(1 - \frac{1}{\sqrt{5}}\right)$

43. Two points P and Q are maintained at the potentials of 10 V and -4 V, respectively. The work done in moving 100 electrons from P to Q is: [2009]
 (a) 9.60×10^{-17} J (b) -2.24×10^{-16} J
 (c) 2.24×10^{-16} J (d) -9.60×10^{-17} J

44. Two insulating plates are both uniformly charged in such a way that the potential difference between them is $V_2 - V_1 = 20$ V. (i.e., plate 2 is at a higher potential). The plates are separated by $d = 0.1$ m and can be treated as infinitely large. An electron is released from rest on the inner surface of plate 1. What is its speed when it hits plate 2? ($e = 1.6 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg) [2006]



- (a) 2.65×10^6 m/s (b) 7.02×10^{12} m/s
 (c) 1.87×10^6 m/s (d) 32×10^{-19} m/s

45. A charged particle ' q ' is shot towards another charged particle ' Q ' which is fixed, with a speed ' v '. It approaches ' Q ' upto a closest distance r and then returns. If q were given a speed of ' $2v$ ' the closest distances of approach would be [2004]
 (a) $r/2$ (b) $2r$ (c) r (d) $r/4$
46. On moving a charge of 20 coulomb by 2 cm, 2 J of work is done, then the potential difference between the points is [2002]
 (a) 0.1 V (b) 8 V (c) 2 V (d) 0.5 V

TOPIC 3 Capacitors, Grouping of Capacitors and Energy Stored in a Capacitor



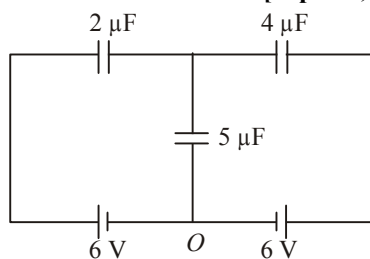
47. Two capacitors of capacitances C and $2C$ are charged to potential differences V and $2V$, respectively. These are then connected in parallel in such a manner that the positive terminal of one is connected to the negative terminal of the other. The final energy of this configuration is :

[Sep. 05, 2020 (I)]

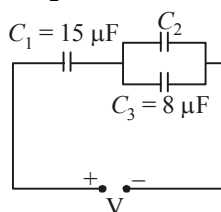
- (a) $\frac{25}{6}CV^2$ (b) $\frac{3}{2}CV^2$
 (c) zero (d) $\frac{9}{2}CV^2$

48. In the circuit shown, charge on the $5 \mu\text{F}$ capacitor is :

[Sep. 05, 2020 (II)]



- (a) $18.00 \mu\text{C}$ (b) $10.90 \mu\text{C}$
 (c) $16.36 \mu\text{C}$ (d) $5.45 \mu\text{C}$
49. A capacitor C is fully charged with voltage V_0 . After disconnecting the voltage source, it is connected in parallel with another uncharged capacitor of capacitance $\frac{C}{2}$. The energy loss in the process after the charge is distributed between the two capacitors is : [Sep. 04, 2020 (II)]
- (a) $\frac{1}{2}CV_0^2$ (b) $\frac{1}{3}CV_0^2$
 (c) $\frac{1}{4}CV_0^2$ (d) $\frac{1}{6}CV_0^2$
50. In the circuit shown in the figure, the total charge is $750 \mu\text{C}$ and the voltage across capacitor C_2 is 20 V. Then the charge on capacitor C_2 is : [Sep. 03, 2020 (I)]



- (a) $450 \mu\text{C}$ (b) $590 \mu\text{C}$
 (c) $160 \mu\text{C}$ (d) $650 \mu\text{C}$

51. A $5 \mu\text{F}$ capacitor is charged fully by a 220 V supply. It is then disconnected from the supply and is connected in series to another uncharged $2.5 \mu\text{F}$ capacitor. If the energy

change during the charge redistribution is $\frac{X}{100}$ J then value

of X to the nearest integer is _____.

[NA Sep. 02, 2020 (I)]

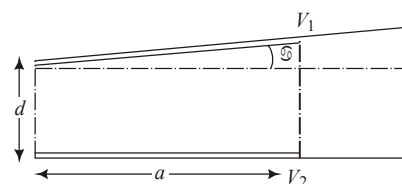
52. A $10 \mu\text{F}$ capacitor is fully charged to a potential difference of 50 V. After removing the source voltage it is connected to an uncharged capacitor in parallel. Now the potential difference across them becomes 20 V. The capacitance of the second capacitor is : [Sep. 02, 2020 (II)]

- (a) $15 \mu\text{F}$ (b) $30 \mu\text{F}$
 (c) $20 \mu\text{F}$ (d) $10 \mu\text{F}$

53. Effective capacitance of parallel combination of two capacitors C_1 and C_2 is $10 \mu\text{F}$. When these capacitors are individually connected to a voltage source of 1 V, the energy stored in the capacitor C_2 is 4 times that of C_1 . If these capacitors are connected in series, their effective capacitance will be: [8 Jan. 2020 I]

- (a) $4.2 \mu\text{F}$ (b) $3.2 \mu\text{F}$ (c) $1.6 \mu\text{F}$ (d) $8.4 \mu\text{F}$

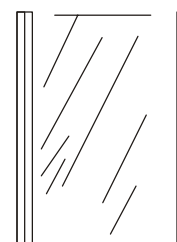
54. A capacitor is made of two square plates each of side ' a ' making a very small angle α between them, as shown in figure. The capacitance will be close to: [8 Jan. 2020 II]



- (a) $\frac{\epsilon_0 a^2}{d} \left(1 - \frac{\alpha a}{2d}\right)$ (b) $\frac{\epsilon_0 a^2}{d} \left(1 - \frac{\alpha a}{4d}\right)$
 (c) $\frac{\epsilon_0 a^2}{d} \left(1 + \frac{\alpha a}{d}\right)$ (d) $\frac{\epsilon_0 a^2}{d} \left(1 - \frac{3\alpha a}{2d}\right)$

55. A parallel plate capacitor has plates of area A separated by distance ' d ' between them. It is filled with a dielectric which has a dielectric constant that varies as $k(x) = K(1 + \alpha x)$ where ' x ' is the distance measured from one of the plates. If $(\alpha d) \ll 1$, the total capacitance of the system is best given by the expression: [7 Jan. 2020 I]

- (a) $\frac{AK\epsilon_0}{d} \left(1 + \frac{\alpha d}{2}\right)$
 (b) $\frac{A\epsilon_0 K}{d} \left(1 + \left(\frac{\alpha d}{2}\right)^2\right)$
 (c) $\frac{A\epsilon_0 K}{d} \left(1 + \frac{\alpha^2 d^2}{2}\right)$
 (d) $\frac{AK\epsilon_0}{d} (1 + \alpha d)$



56. A 60 pF capacitor is fully charged by a 20 V supply. It is then disconnected from the supply and is connected to another uncharged 60 pF capacitor in parallel. The electrostatic energy that is lost in this process by the time the charge is redistributed between them is (in nJ)

[NA 7 Jan. 2020 II]

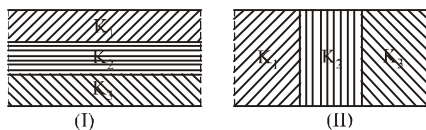
57. The parallel combination of two air filled parallel plate capacitors of capacitance C and nC is connected to a battery of voltage, V . When the capacitors are fully charged, the battery is removed and after that a dielectric material of dielectric constant K is placed between the two plates of the first capacitor. The new potential difference of the combined system is: [9 April 2020 II]

- (a) $\frac{nV}{K+n}$ (b) V
(c) $\frac{V}{K+n}$ (d) $\frac{(n+1)V}{(K+n)}$

58. Two identical parallel plate capacitors, of capacitance C each, have plates of area A , separated by a distance d . The space between the plates of the two capacitors, is filled with three dielectrics, of equal thickness and dielectric constants K_1 , K_2 and K_3 . The first capacitor is filled as shown in Fig. I, and the second one is filled as shown in Fig. II.

If these two modified capacitors are charged by the same potential V , the ratio of the energy stored in the two, would be (E_1 refers to capacitors (I) and E_2 to capacitors (II) :

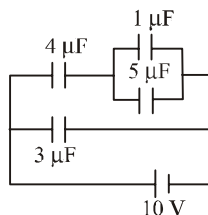
[12 April 2019 I]



- (a) $\frac{E_1}{E_2} = \frac{K_1 K_2 K_3}{(K_1 + K_2 + K_3)(K_2 K_3 + K_3 K_1 + K_1 K_2)}$
(b) $\frac{E_1}{E_2} = \frac{(K_1 + K_2 + K_3)(K_2 K_3 + K_3 K_1 + K_1 K_2)}{K_1 K_2 K_3}$
(c) $\frac{E_1}{E_2} = \frac{9 K_1 K_2 K_3}{(K_1 + K_2 + K_3)(K_2 K_3 + K_3 K_1 + K_1 K_2)}$
(d) $\frac{E_1}{E_2} = \frac{(K_1 + K_2 + K_3)(K_2 K_3 + K_3 K_1 + K_1 K_2)}{9 K_1 K_2 K_3}$

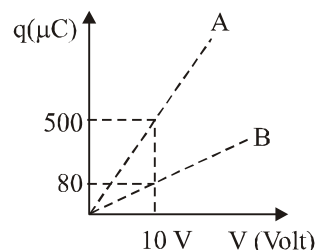
59. In the given circuit, the charge on $4 \mu\text{F}$ capacitor will be :

[12 April 2019 II]

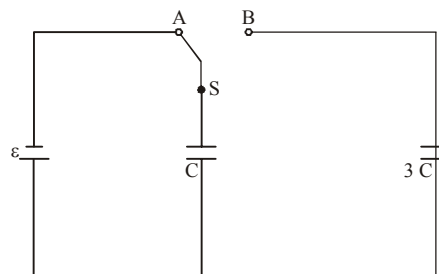


- (a) $5.4 \mu\text{C}$ (b) $9.6 \mu\text{C}$ (c) $13.4 \mu\text{C}$ (d) $24 \mu\text{C}$

60. Figure shows charge (q) versus voltage (V) graph for series and parallel combination of two given capacitors. The capacitances are : [10 April 2019 I]



- (a) $40 \mu\text{F}$ and $10 \mu\text{F}$ (b) $60 \mu\text{F}$ and $40 \mu\text{F}$
(c) $50 \mu\text{F}$ and $30 \mu\text{F}$ (d) $20 \mu\text{F}$ and $30 \mu\text{F}$
61. A capacitor with capacitance $5 \mu\text{F}$ is charged to $5 \mu\text{C}$. If the plates are pulled apart to reduce the capacitance to $2 \frac{1}{4} \text{ F}$, how much work is done? [9 April 2019 I]
- (a) $6.25 \times 10^{-6} \text{ J}$ (b) $3.75 \times 10^{-6} \text{ J}$
(c) $2.16 \times 10^{-6} \text{ J}$ (d) $2.55 \times 10^{-6} \text{ J}$
62. Voltage rating of a parallel plate capacitor is 500 V . Its dielectric can withstand a maximum electric field of 10^6 V/m . The plate area is 10^{-4} m^2 . What is the dielectric constant if the capacitance is 15 pF ? [8 April 2019 I]
- (given $\epsilon_0 = 8.86 \times 10^{-12} \text{ C}^2/\text{Nm}^2$)
- (a) 3.8 (b) 8.5 (c) 4.5 (d) 6.2
63. A parallel plate capacitor has $1 \mu\text{F}$ capacitance. One of its two plates is given $+2 \mu\text{C}$ charge and the other plate, $+4 \mu\text{C}$ charge. The potential difference developed across the capacitor is : [8 April 2019 II]
- (a) 3 V (b) 1 V (c) 5 V (d) 2 V
64. In the figure shown, after the switch 'S' is turned from position 'A' to position 'B', the energy dissipated in the circuit in terms of capacitance 'C' and total charge 'Q' is: [12 Jan. 2019 I]

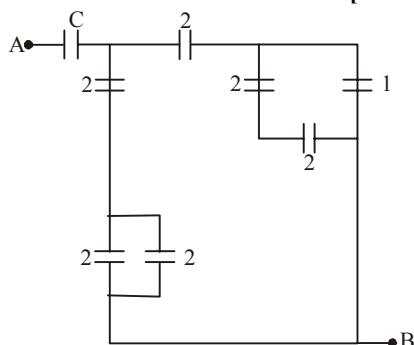


- (a) $\frac{1}{8} \frac{Q^2}{C}$ (b) $\frac{3}{8} \frac{Q^2}{C}$ (c) $\frac{5}{8} \frac{Q^2}{C}$ (d) $\frac{3}{4} \frac{Q^2}{C}$
65. A parallel plate capacitor with plates of area 1 m^2 each, are at a separation of 0.1 m . If the electric field between the plates is 100 N/C , the magnitude of charge on each plate is :

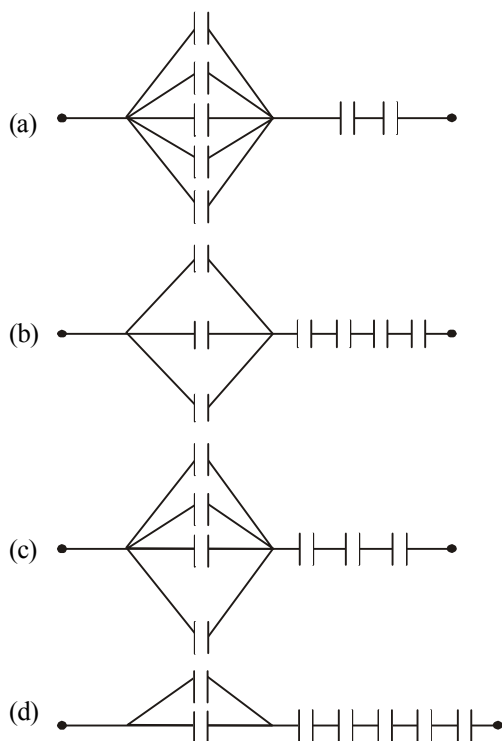
(Take $\epsilon_0 = 8.85 \times 10^{-12} \frac{\text{C}^2}{\text{N}-\text{M}^2}$) [12 Jan. 2019 II]

- (a) $7.85 \times 10^{-10} \text{ C}$ (b) $6.85 \times 10^{-10} \text{ C}$
(c) $8.85 \times 10^{-10} \text{ C}$ (d) $9.85 \times 10^{-10} \text{ C}$

66. In the circuit shown, find C if the effective capacitance of the whole circuit is to be $0.5 \mu\text{F}$. All values in the circuit are in μF . [12 Jan. 2019 II]



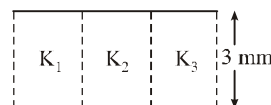
- (a) $\frac{7}{11} \mu\text{F}$ (b) $\frac{6}{5} \mu\text{F}$ (c) $4 \mu\text{F}$ (d) $\frac{7}{10} \mu\text{F}$
67. In the figure shown below, the charge on the left plate of the $10 \mu\text{F}$ capacitor is $-30 \mu\text{C}$. The charge on the right plate of the $6 \mu\text{F}$ capacitor is : [11 Jan. 2019 I]
-
- (a) $-12 \mu\text{C}$ (b) $+12 \mu\text{C}$
(c) $-18 \mu\text{C}$ (d) $+18 \mu\text{C}$
68. Seven capacitors, each of capacitance $2 \mu\text{F}$, are to be connected in a configuration to obtain an effective capacitance of $\left(\frac{6}{13}\right) \mu\text{F}$. Which of the combinations, shown in figures below, will achieve the desired value? [11 Jan. 2019 II]



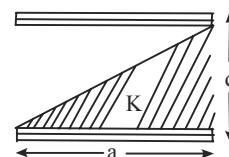
69. A parallel plate capacitor having capacitance 12 pF is charged by a battery to a potential difference of 10 V between its plates. The charging battery is now disconnected and a porcelain slab of dielectric constant 6.5 is slipped between the plates. The work done by the capacitor on the slab is: [10 Jan. 2019 II]

- (a) 692 pJ (b) 508 pJ
(c) 560 pJ (d) 600 pJ

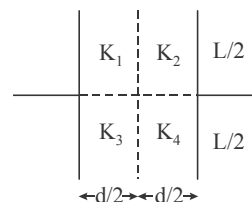
70. A parallel plate capacitor is of area 6 cm^2 and a separation 3 mm . The gap is filled with three dielectric materials of equal thickness (see figure) with dielectric constants $K_1 = 10$, $K_2 = 12$ and $K_3 = 1$ (4) The dielectric constant of a material which when fully inserted in above capacitor, gives same capacitance would be: [10 Jan. 2019 I]



- (a) 4 (b) 14 (c) 12 (d) 36
71. A parallel plate capacitor is made of two square plates of side 'a', separated by a distance d ($d \ll a$). The lower triangular portion is filled with a dielectric of dielectric constant K , as shown in the figure. Capacitance of this capacitor is: [9 Jan. 2019 I]



- (a) $\frac{K \epsilon_0 a^2}{2d(K+1)}$ (b) $\frac{K \epsilon_0 a^2}{d(K-1)} \ln K$
(c) $\frac{K \epsilon_0 a^2}{d} \ln K$ (d) $\frac{1}{2} \frac{K \epsilon_0 a^2}{d}$
72. A parallel plate capacitor with square plates is filled with four dielectrics of dielectric constants K_1 , K_2 , K_3 , K_4 arranged as shown in the figure. The effective dielectric constant K will be: [9 Jan. 2019 II]



- (a) $K = \frac{(K_1 + K_3)(K_2 + K_4)}{K_1 + K_2 + K_3 + K_4}$
(b) $K = \frac{(K_1 + K_2)(K_3 + K_4)}{2(K_1 + K_2 + K_3 + K_4)}$

$$(c) K = \frac{(K_1 + K_2)(K_3 + K_4)}{K_1 + K_2 + K_3 + K_4}$$

$$(d) K = \frac{(K_1 + K_4)(K_2 + K_3)}{2(K_1 + K_2 + K_3 + K_4)}$$

73. A parallel plate capacitor of capacitance 90 pF is connected to a battery of emf 20V. If a dielectric material of dielectric constant $k = \frac{5}{3}$ is inserted between the plates, the magnitude of the induced charge will be: [2018]

(a) 1.2 nC (b) 0.3 nC (c) 2.4 nC (d) 0.9 nC

74. In the following circuit, the switch S is closed at $t = 0$. The charge on the capacitor C_1 as a function of time will be

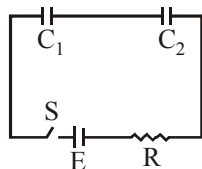
given by $\left(C_{eq} = \frac{C_1 C_2}{C_1 + C_2} \right)$. [Online April 16, 2018]

(a) $C_{eq} E [1 - \exp(-t/RC_{eq})]$

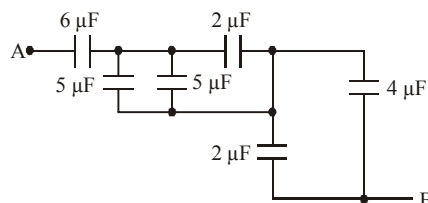
(b) $C_1 E [1 - \exp(-tR/C_1)]$

(c) $C_2 E [1 - \exp(-t/RC_2)]$

(d) $C_{eq} E \exp(-t/RC_{eq})$



75. The equivalent capacitance between A and B in the circuit given below is:



[Online April 15, 2018]

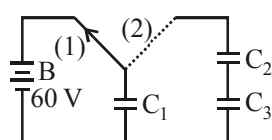
(a) 4.9 μF (b) 3.6 μF (c) 5.4 μF (d) 2.4 μF

76. A parallel plate capacitor with area 200 cm² and separation between the plates 1.5 cm, is connected across a battery of emf V. If the force of attraction between the plates is 25×10^{-6} N, the value of V is approximately: [Online April 15, 2018]

$$\left(\epsilon_0 = 8.85 \times 10^{-12} \frac{C^2}{N \cdot m^2} \right)$$

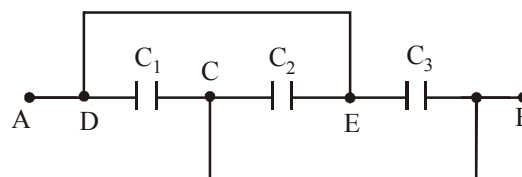
(a) 150V (b) 100V (c) 250V (d) 300V

77. A capacitor C_1 is charged up to a voltage $V = 60$ V by connecting it to battery B through switch (1). Now C_1 is disconnected from battery and connected to a circuit consisting of two uncharged capacitors $C_2 = 3.0 \mu F$ and $C_3 = 6.0 \mu F$ through a switch (2) as shown in the figure. The sum of final charges on C_2 and C_3 is: [Online April 15, 2018]



(a) 36 μC (b) 20 μC (c) 54 μC (d) 40 μC

78. A capacitance of $2 \mu F$ is required in an electrical circuit across a potential difference of 1.0 kV. A large number of $1 \mu F$ capacitors are available which can withstand a potential difference of not more than 300 V. The minimum number of capacitors required to achieve this is [2017]
- (a) 24 (b) 32 (c) 2 (d) 16
79. A combination of parallel plate capacitors is maintained at a certain potential difference.



When a 3 mm thick slab is introduced between all the plates, in order to maintain the same potential difference, the distance between the plates is increased by 2.4 mm. Find the dielectric constant of the slab.

[Online April 9, 2017]

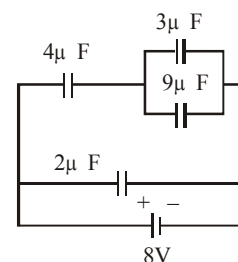
(a) 3 (b) 4 (c) 5 (d) 6

80. The energy stored in the electric field produced by a metal sphere is 4.5 J. If the sphere contains $4 \mu C$ charge, its radius will be: [Take : $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$]

[Online April 8, 2017]

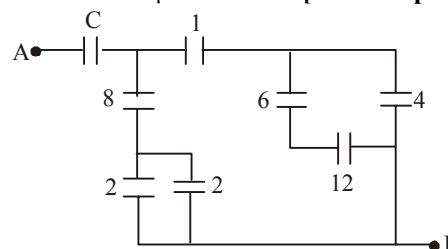
(a) 20 mm (b) 32 mm (c) 28 mm (d) 16 mm

81. A combination of capacitors is set up as shown in the figure. The magnitude of the electric field, due to a point charge Q (having a charge equal to the sum of the charges on the $4 \mu F$ and $9 \mu F$ capacitors), at a point distance 30 m from it, would equal : [2016]



(a) 420 N/C (b) 480 N/C
(c) 240 N/C (d) 360 N/C

82. Figure shows a network of capacitors where the numbers indicates capacitances in micro Farad. The value of capacitance C if the equivalent capacitance between point A and B is to be $1 \mu F$ is : [Online April 10, 2016]

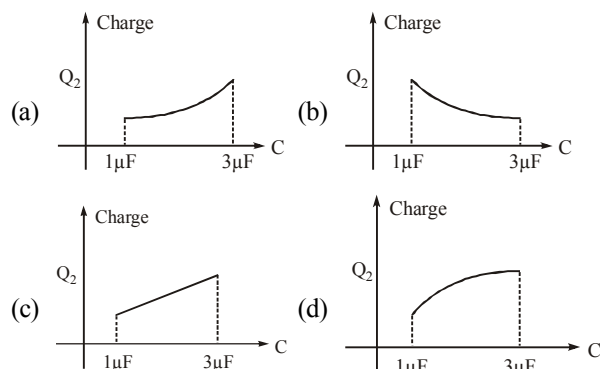
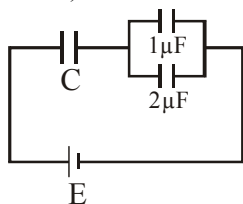


- (a) $\frac{32}{23} \mu\text{F}$ (b) $\frac{31}{23} \mu\text{F}$ (c) $\frac{33}{23} \mu\text{F}$ (d) $\frac{34}{23} \mu\text{F}$

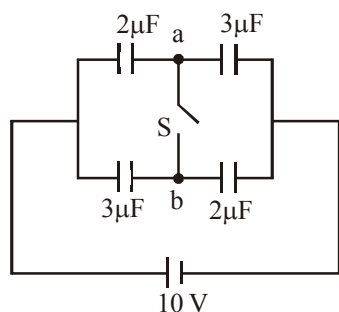
83. Three capacitors each of $4 \mu\text{F}$ are to be connected in such a way that the effective capacitance is $6 \mu\text{F}$. This can be done by connecting them : [Online April 9, 2016]

- (a) all in series
(b) all in parallel
(c) two in parallel and one in series
(d) two in series and one in parallel

84. In the given circuit, charge Q_2 on the $2 \mu\text{F}$ capacitor changes as C is varied from $1 \mu\text{F}$ to $3 \mu\text{F}$. Q_2 as a function of ' C ' is given properly by: (figures are drawn schematically and are not to scale) [2015]



85. In figure a system of four capacitors connected across a 10 V battery is shown. Charge that will flow from switch S when it is closed is : [Online April 11, 2015]



- (a) $5 \mu\text{C}$ from b to a (b) $20 \mu\text{C}$ from a to b
(c) zero (d) $5 \mu\text{C}$ from a to b

86. A parallel plate capacitor is made of two circular plates separated by a distance 5 mm and with a dielectric of dielectric constant 2.2 between them. When the electric field in the dielectric is $3 \times 10^4 \text{ V/m}$ the charge density of the positive plate will be close to: [2014]

- (a) $6 \times 10^{-7} \text{ C/m}^2$ (b) $3 \times 10^{-7} \text{ C/m}^2$
(c) $3 \times 10^4 \text{ C/m}^2$ (d) $6 \times 10^4 \text{ C/m}^2$

87. The gap between the plates of a parallel plate capacitor of area A and distance between plates d , is filled with a dielectric whose permittivity varies linearly from ϵ_1 at one plate to ϵ_2 at the other. The capacitance of capacitor is:

[Online April 19, 2014]

- (a) $\epsilon_0 (\epsilon_1 + \epsilon_2) A / d$
(b) $\epsilon_0 (\epsilon_2 + \epsilon_1) A / 2d$
(c) $\epsilon_0 A / [d \ln(\epsilon_2 / \epsilon_1)]$
(d) $\epsilon_0 (\epsilon_2 - \epsilon_1) A / [d \ln(\epsilon_2 / \epsilon_1)]$

88. The space between the plates of a parallel plate capacitor is filled with a 'dielectric' whose 'dielectric constant' varies with distance as per the relation:

$$K(x) = K_0 + \lambda x \quad (\lambda = \text{a constant})$$

The capacitance C , of the capacitor, would be related to its vacuum capacitance C_0 for the relation :

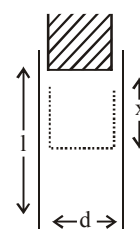
[Online April 12, 2014]

- (a) $C = \frac{\lambda d}{\ln(1 + K_0 \lambda d)} C_0$ (b) $C = \frac{\lambda}{d \ln(1 + K_0 \lambda d)} C_0$
(c) $C = \frac{\lambda d}{\ln(1 + \lambda d / K_0)} C_0$ (d) $C = \frac{\lambda}{d \ln(1 + K_0 / \lambda d)} C_0$

89. A parallel plate capacitor is made of two plates of length l , width w and separated by distance d . A dielectric slab (dielectric constant K) that fits exactly between the plates is held near the edge of the plates. It is pulled into the

capacitor by a force $F = -\frac{\partial U}{\partial x}$ where U is the energy of the capacitor when dielectric is inside the capacitor up to distance x (See figure). If the charge on the capacitor is Q then the force on the dielectric when it is near the edge is:

[Online April 11, 2014]



- (a) $\frac{Q^2 d}{2\omega l^2 \epsilon_0} K$ (b) $\frac{Q^2 \omega}{2d l^2 \epsilon_0} (K - 1)$
(c) $\frac{Q^2 d}{2w l^2 \epsilon_0} (K - 1)$ (d) $\frac{Q^2 w}{2d l^2 \epsilon_0} K$

90. Three capacitors, each of $3 \mu\text{F}$, are provided. These cannot be combined to provide the resultant capacitance of:

[Online April 9, 2014]

- (a) $1 \mu\text{F}$ (b) $2 \mu\text{F}$ (c) $4.5 \mu\text{F}$ (d) $6 \mu\text{F}$

91. A parallel plate capacitor having a separation between the plates d , plate area A and material with dielectric constant K has capacitance C_0 . Now one-third of the material is replaced by another material with dielectric constant $2K$, so that effectively there are two capacitors one with area

$\frac{1}{3}$ A, dielectric constant $2K$ and another with area $\frac{2}{3}$ A and dielectric constant K . If the capacitance of this new capacitor is C then $\frac{C}{C_0}$ is [Online April 25, 2013]

- (a) 1 (b) $\frac{4}{3}$ (c) $\frac{2}{3}$ (d) $\frac{1}{3}$

92. To establish an instantaneous current of 2 A through a 1 μ F capacitor; the potential difference across the capacitor plates should be changed at the rate of:

[Online April 22, 2013]

- (a) 2×10^4 V/s (b) 4×10^6 V/s
(c) 2×10^6 V/s (d) 4×10^4 V/s

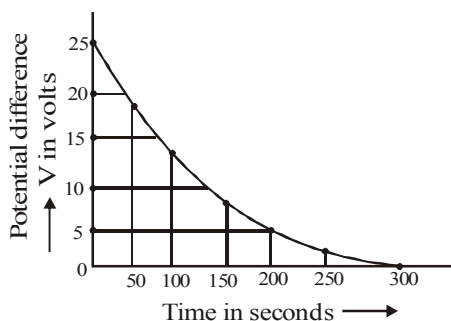
93. A uniform electric field $\vec{E}$ exists between the plates of a charged condenser. A charged particle enters the space between the plates and perpendicular to $\vec{E}$. The path of the particle between the plates is a:

[Online April 9, 2013]

- (a) straight line (b) hyperbola
(c) parabola (d) circle

94. The figure shows an experimental plot discharging of a capacitor in an RC circuit. The time constant τ of this circuit lies between:

[2012]



- (a) 150 sec and 200 sec (b) 0 sec and 50 sec
(c) 50 sec and 100 sec (d) 100 sec and 150 sec

95. The capacitor of an oscillatory circuit is enclosed in a container. When the container is evacuated, the resonance frequency of the circuit is 10 kHz. When the container is filled with a gas, the resonance frequency changes by 50 Hz. The dielectric constant of the gas is

[Online May 26, 2012]

- (a) 1.001 (b) 2.001 (c) 1.01 (d) 3.01

96. **Statement 1:** It is not possible to make a sphere of capacity 1 farad using a conducting material.

Statement 2: It is possible for earth as its radius is 6.4×10^6 m.

[Online May 26, 2012]

- (a) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.
(b) Statement 1 is false, Statement 2 is true.
(c) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.
(d) Statement 1 is true, Statement 2 is false.

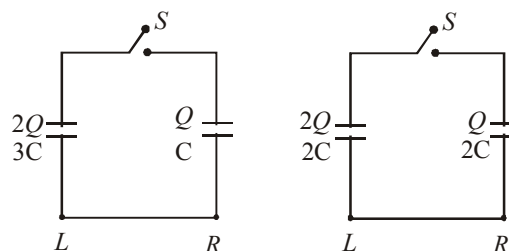
97. A series combination of n_1 capacitors, each of capacity C_1 is charged by source of potential difference 4 V. When another parallel combination of n_2 capacitors each of capacity C_2 is charged by a source of potential difference V , it has the same total energy stored in it as the first combination has. The value of C_2 in terms of C_1 is then

[Online May 12, 2012]

- (a) $16 \frac{n_2}{n_1} C_1$ (b) $\frac{2 C_1}{n_1 n_2}$
(c) $2 \frac{n_2}{n_1} C_1$ (d) $\frac{16 C_1}{n_1 n_2}$

98. Two circuits (a) and (b) have charged capacitors of capacitance C , $2C$ and $3C$ with open switches. Charges on each of the capacitor are as shown in the figures. On closing the switches

[Online May 7, 2012]



Circuit (a)

Circuit (b)

- (a) No charge flows in (a) but charge flows from R to L in (b)
(b) Charges flow from L to R in both (a) and (b)
(c) Charges flow from R to L in (a) and from L to R in (b)
(d) No charge flows in (a) but charge flows from L to R in (b)
99. Let C be the capacitance of a capacitor discharging through a resistor R . Suppose t_1 is the time taken for the energy stored in the capacitor to reduce to half its initial value and t_2 is the time taken for the charge to reduce to one-fourth its initial value. Then the ratio t_1/t_2 will be [2010]

- (a) 1 (b) $\frac{1}{2}$ (c) $\frac{1}{4}$ (d) 2

100. A parallel plate capacitor with air between the plates has capacitance of 9 pF. The separation between its plates is 'd'. The space between the plates is now filled with two dielectrics. One of the dielectrics has dielectric constant

$k_1 = 3$ and thickness $\frac{d}{3}$ while the other one has dielectric

constant $k_2 = 6$ and thickness $\frac{2d}{3}$. Capacitance of the capacitor is now [2008]

- (a) 1.8 pF (b) 45 pF (c) 40.5 pF (d) 20.25 pF

101. A parallel plate condenser with a dielectric of dielectric constant K between the plates has a capacity C and is charged to a potential V volt. The dielectric slab is slowly removed from between the plates and then reinserted. The net work done by the system in this process is [2007]

- (a) zero (b) $\frac{1}{2}(K-1) CV^2$
- (c) $\frac{CV^2(K-1)}{K}$ (d) $(K-1) CV^2$
- 102.** A parallel plate capacitor is made by stacking n equally spaced plates connected alternatively. If the capacitance between any two adjacent plates is ' C ' then the resultant capacitance is [2005]
- (a) $(n+1)C$ (b) $(n-1)C$
- (c) nC (d) C
- 103.** A fully charged capacitor has a capacitance ' C '. It is discharged through a small coil of resistance wire embedded in a thermally insulated block of specific heat capacity ' s ' and mass ' m '. If the temperature of the block is raised by ' ΔT ', the potential difference ' V ' across the capacitance is [2005]
- (a) $\frac{mC\Delta T}{s}$ (b) $\sqrt{\frac{2mC\Delta T}{s}}$
- (c) $\sqrt{\frac{2ms\Delta T}{C}}$ (d) $\frac{ms\Delta T}{C}$
- 104.** A sheet of aluminium foil of negligible thickness is introduced between the plates of a capacitor. The capacitance of the capacitor [2003]
- (a) decreases (b) remains unchanged
- (c) becomes infinite (d) increases
- 105.** The work done in placing a charge of 8×10^{-18} coulomb on a condenser of capacity 100 micro-farad is [2003]
- (a) 16×10^{-32} joule (b) 3.1×10^{-26} joule
- (c) 4×10^{-10} joule (d) 32×10^{-32} joule
- 106.** If there are n capacitors in parallel connected to V volt source, then the energy stored is equal to [2002]
- (a) CV (b) $\frac{1}{2}nCV^2$ (c) CV^2 (d) $\frac{1}{2n}CV^2$
- 107.** Capacitance (in F) of a spherical conductor with radius 1 m is [2002]
- (a) 1.1×10^{-10} (b) 10^{-6}
- (c) 9×10^{-9} (d) 10^{-3}



Hints & Solutions

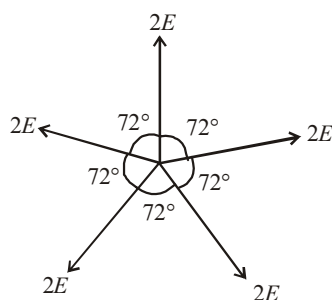


1. (c) Potential at the centre, $V_C = \frac{KQ_{\text{net}}}{R}$

$$\because Q_{\text{net}} = 0$$

$$\therefore V_C = 0$$

Let E be electric field produced by each charge at the centre, then resultant electric field will be $E_C = 0$, since equal electric field vectors are acting at equal angle so their resultant is equal to zero.



2. (d) Total charge $Q_1 + Q_2 = Q'_1 + Q'_2$
 $= 12\mu\text{C} - 3\mu\text{C} = 9\mu\text{C}$

Two isolated conducting spheres S_1 and S_2 are now connected by a conducting wire.

$$\therefore V_1 = V_2 = \frac{KQ'_1}{\frac{2}{3}R} = \frac{KQ'_2}{\frac{R}{3}} = 12 - 3 = 9\mu\text{C}$$

$$Q'_1 = 2Q'_2 \Rightarrow 2Q'_2 + Q'_2 = 9\mu\text{C}$$

$$\therefore Q'_1 = 6\mu\text{C} \text{ and } Q'_2 = 3\mu\text{C}$$

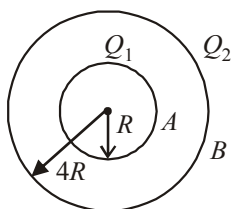
3. (a) We have given two metallic hollow spheres of radii R and $4R$ having charges Q_1 and Q_2 respectively.

Potential on the surface of inner sphere (at A)

$$V_A = \frac{kQ_1}{R} + \frac{kQ_2}{4R}$$

Potential on the surface of outer sphere (at B)

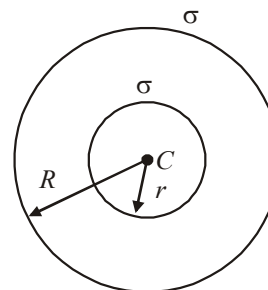
$$V_B = \frac{kQ_1}{4R} + \frac{kQ_2}{4R} \quad \left(\text{Here, } k = \frac{1}{4\pi\epsilon_0} \right)$$



Potential difference,

$$\Delta V = V_A - V_B = \frac{3}{4} \cdot \frac{kQ_1}{R} = \frac{3}{16\pi\epsilon_0} \cdot \frac{Q_1}{R}$$

4. (d) Let σ be the surface charge density of the shells.



Charge on the inner shell, $Q_1 = \sigma 4\pi r^2$

Charge on the outer shell, $Q_2 = \sigma 4\pi R^2$

$$\therefore \text{Total charge, } Q = \sigma 4\pi(r^2 + R^2)$$

$$\Rightarrow \sigma = \frac{Q}{4\pi(r^2 + R^2)}$$

Potential at the common centre,

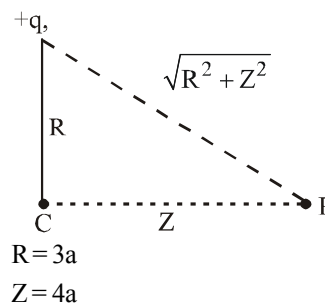
$$\begin{aligned} V_C &= \frac{KQ_1}{r} + \frac{KQ_2}{R} \quad \left(\text{where } K = \frac{1}{4\pi\epsilon_0} \right) \\ &= \frac{K\sigma 4\pi r^2}{r} + \frac{K\sigma 4\pi R^2}{R} \\ &= K\sigma 4\pi(r + R) \\ &= \frac{KQ 4\pi(r + R)}{4\pi(r^2 + R^2)} \\ &= \frac{1}{4\pi\epsilon_0} \frac{(r + R)Q}{(r^2 + R^2)} \end{aligned}$$

5. (b) The electric potential at the bisector is zero and electric field is antiparallel to the dipole moment.

$$\therefore V = 0 \text{ and } \vec{E} = \frac{1}{4\pi\epsilon_0} \left(\frac{-\vec{P}}{d^3} \right)$$

6. (c) Potential at any point of the charged ring

$$V_p = \frac{Kq}{\sqrt{R^2 + Z^2}}$$



$$\ell = \sqrt{R^2 + Z^2} = 5a$$

The minimum velocity (v_0) should just sufficient to reach the point charge at the center, therefore

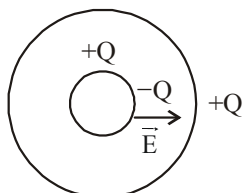
$$\frac{1}{2}mv_0^2 = q[V_C - V_P]$$

$$= q \left[\frac{Kq}{3a} - \frac{Kq}{5a} \right]$$

$$v_0^2 = \frac{4Kq^2}{15ma} = \frac{4}{15} \frac{1}{4\pi\epsilon_0} \frac{q^2}{ma}$$

$$\Rightarrow v_0 = \sqrt{\frac{2}{m} \left(\frac{2q^2}{15 \times 4\pi\epsilon_0 a} \right)^{\frac{1}{2}}}$$

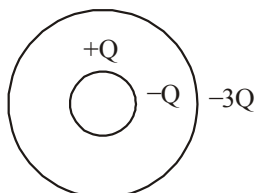
7. (d) When charge Q is on inner solid conducting sphere



Electric field between spherical surface

$$\vec{E} = \frac{KQ}{r^2} \text{ So } \int \vec{E} \cdot d\vec{r} = V \text{ given}$$

Now when a charge $-4Q$ is given to hollow shell



Electric field between surface remain unchanged.

$$\vec{E} = \frac{KQ}{r^2}$$

as, field inside the hollow spherical shell = 0

$\therefore$ Potential difference between them remain unchanged

$$\text{i.e. } \int \vec{E} \cdot d\vec{r} = V$$

8. (c) Given, $\vec{E} = (Ax + B)\hat{i}$

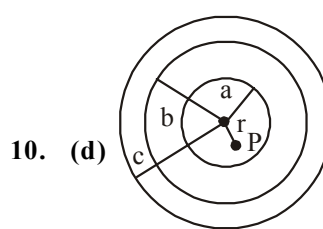
$$\text{or } E = 20x + 10$$

Using $V = \int E dx$, we have

$$V_2 - V_1 = \int_{-5}^1 (20x + 10) dx = -180 \text{ V}$$

$$\text{or } V_1 - V_2 = 180 \text{ V}$$

9. (b) Electric potential is constant inside a charged spherical shell.



10. (d)

$$\text{Potential at point P, } V = \frac{kQ_a}{a} + \frac{kQ_b}{b} + \frac{kQ_c}{c}$$

Since surface charge densities are equal to one another

$$\text{i.e., } \sigma_a = \sigma_b = \sigma_c$$

$$\therefore Q_a : Q_b : Q_c :: a^2 : b^2 : c^2$$

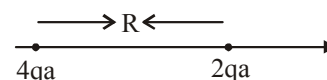
$$\therefore Q_a = \left[\frac{a^2}{a^2 + b^2 + c^2} \right] Q$$

$$Q_b = \left[\frac{b^2}{a^2 + b^2 + c^2} \right] Q$$

$$Q_c = \left[\frac{c^2}{a^2 + b^2 + c^2} \right] Q$$

$$\therefore V = \frac{Q}{4\pi\epsilon_0} \left[\frac{(a + b + c)}{a^2 + b^2 + c^2} \right]$$

11. (d) Let at a distance 'x' from point B, both the dipoles produce same potential



$$\therefore \frac{4qa}{(R+x)} = \frac{2qa}{(x^2)}$$

$$\Rightarrow \sqrt{2x} = R + x \Rightarrow x = \frac{R}{\sqrt{2}-1}$$

Therefore distance from A at which both of them produce the same potential

$$= \frac{R}{\sqrt{2}-1} + R = \frac{\sqrt{2}R}{\sqrt{2}-1}$$

12. (b) Electric field at a point outside the sphere is given by

$$E = \frac{1Q}{4\pi\epsilon_0 r^2} \text{ But } \rho = \frac{Q}{\frac{4}{3}\pi R^3}$$

$$\therefore E = \frac{\rho R^3}{3\epsilon_0 r^2}$$

At surface $r = R$

$$\therefore E = \frac{\rho R^3}{3\epsilon_0}$$

Let ρ_1 and ρ_2 are the charge densities of two sphere.

$$E_1 = \frac{\rho_1 R_1}{3\epsilon_0} \text{ and } E_2 = \frac{\rho_2 R_2}{3\epsilon_0}$$

$$\therefore \frac{E_1}{E_2} = \frac{\rho_1 R_1}{\rho_2 R_2} = \frac{R_1}{R_2}$$

This gives $\rho_1 = \rho_2 = \rho$

Potential at a point outside the sphere

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{r} = \frac{\rho R^3}{3\epsilon_0 r} \left(\because \rho = \frac{Q}{\frac{4}{3}\pi R^3} \right)$$

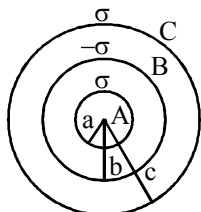
At surface, $r = R$

$$V = \frac{\rho R^2}{3\epsilon_0} \text{ so, } V_1 = \frac{\rho R_1^2}{3\epsilon_0} \text{ and } V_2 = \frac{\rho R_2^2}{3\epsilon_0}$$

$$\therefore \frac{V_1}{V_2} = \left(\frac{R_1}{R_2} \right)^2$$

13. (b) Potential outside the shell, $V_{\text{outside}} = \frac{KQ}{r}$
where r is distance of point from the centre of shell

Potential inside the shell, $V_{\text{inside}} = \frac{KQ}{R}$
where 'R' is radius of the shell



$$V_B = \frac{Kq_A}{r_b} + \frac{Kq_B}{r_b} + \frac{Kq_C}{r_c}$$

$$V_B = \frac{1}{4\pi\epsilon_0} \left[\frac{\sigma 4\pi a^2}{b} - \frac{\sigma 4\pi b^2}{b} + \frac{\sigma 4\pi c^2}{c} \right]$$

$$V_B = \frac{\sigma}{\epsilon_0} \left[\frac{a^2 - b^2}{b} + c \right]$$

14. (c) Potential gradient is given by,
 $\Delta V = E \cdot d$
 $0.8 = E d (\max)$
 $\Delta V = E d \cos \theta = 0.8 \times \cos 60 = 0.4$
Hence, maximum potential at a point on the sphere
 $= 589.4 \text{ V}$

15. (c) As we know electric field, $E = \frac{-dv}{dr}$

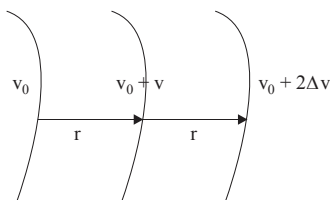
$E = \text{constant} \therefore dv$ and dr same

$$E = \frac{K\phi}{r^2} = c$$

$$\Rightarrow \phi \propto r^2$$

$$\therefore \phi = \int_0^r \rho 4\pi r^2 dr$$

$$\Rightarrow \rho \propto \frac{1}{r}$$



16. (b) $\Sigma_1 = \frac{-dv}{dr} = 10 |z|$

$$\Sigma_2 = \frac{-dv}{dr} = 10 \quad (\text{constant : } E)$$

$\therefore$ The source is an infinity large non conducting thick plate of thickness 2 m.

$$\therefore 10Z \cdot 10A = \frac{\rho \cdot A \propto Z}{\epsilon_0}$$

$$r_0 = 10e_0 \text{ for } |z| \leq 1 \text{ m.}$$

17. (a) We know, $V_0 = \frac{Kq}{R} = V_{\text{surface}}$

$$\text{Now, } V_i = \frac{Kq}{2R^3} (3R^2 - r^2) \quad [\text{For } r < R]$$

At the centre of sphere $r = 0$. Here

$$V = \frac{3}{2} V_0$$

$$\text{Now, } \frac{5}{4} \frac{Kq}{R} = \frac{Kq}{2R^3} (3R^2 - r^2)$$

$$\Rightarrow R_2 = \frac{R}{\sqrt{2}}$$

$$\frac{3}{4} \frac{Kq}{R} = \frac{Kq}{R^3}$$

$$\frac{1}{4} \frac{Kq}{R} = \frac{Kq}{R_4}$$

$$R_4 = 4R$$

Also, $R_1 = 0$ and $R_2 < (R_4 - R_3)$

18. (a) As we know, $E = -\frac{dv}{dx}$

Potential at the point $x = 2\text{m}$, $y = 2\text{m}$ is given by :

$$\int_0^V dV = - \int_0^{2,2} (25dx + 30dy)$$

on solving we get,

$$V = -110 \text{ volt.}$$

19. (c) Potential difference between any two points in an electric field is given by,

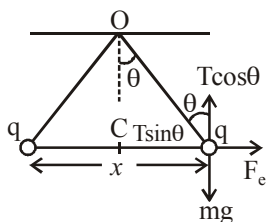
$$dV = -\vec{E} \cdot \vec{dx}$$

$$\int_{V_O}^{V_A} dV = - \int_0^2 30x^2 dx$$

$$V_A - V_O = -[10x^3]_0^2 = -80 \text{ J/C}$$

20. (a) The potential of uncharged body is less than that of the charged conductor and more than at infinity.

21. (c)

In equilibrium, $F_e = T \sin \theta$ $mg = T \cos \theta$

$$\tan \theta = \frac{F_e}{mg} = \frac{q^2}{4\pi \epsilon_0 x^2 \times mg}$$

$$\therefore x = \sqrt{\frac{q^2}{4\pi \epsilon_0 \tan \theta \times mg}}$$

Electric potential at the centre of the line

$$V = \frac{kq}{x/2} + \frac{kq}{x/2} = 4\sqrt{kmg / \tan \theta}$$

22. (c) $q = 1 \mu\text{C} = 1 \times 10^{-6} \text{C}$ $r = 4 \text{ cm} = 4 \times 10^{-2} \text{ m}$

$$\text{Potential } V = \frac{kq}{r} = \frac{9 \times 10^9 \times 10^{-6}}{4 \times 10^{-2}} = 2.25 \times 10^5 \text{ V.}$$

$$\text{Induced electric field } E = -\frac{kq}{r^2}$$

$$= \frac{9 \times 10^9 \times 1 \times 10^{-6}}{16 \times 10^{-4}} = -5.625 \times 10^6 \text{ V/m}$$

23. (c) Let q_1 and q_2 be charge on two spheres of radius ' r ' and ' R ' respectivelyAs, $q_1 + q_2 = Q$ and $\sigma_1 = \sigma_2$ [Surface charge density are equal]

$$\therefore \frac{q_1}{r\pi r^2} = \frac{q_2}{4\pi R^2}$$

$$\text{So, } q_1 = \frac{QR^2}{R^2 + r^2} \text{ and } q_2 = \frac{Qr^2}{R^2 + r^2}$$

$$\text{Now, potential, } V = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1}{r} + \frac{q_2}{R} \right]$$

$$= \frac{1}{4\pi\epsilon_0} \left[\frac{Qr}{R^2 + r^2} + \frac{QR}{R^2 + r^2} \right]$$

$$= \frac{Q(R+r)}{R^2 + r^2} \cdot \frac{1}{4\pi\epsilon_0}$$

24. (c) Charges reside only on the outer surface of a conductor with cavity.

25. (c) Electric field

$$E = -\frac{d\phi}{dr} = -2ar \quad \dots(i)$$

By Gauss's theorem

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \quad \dots(ii)$$

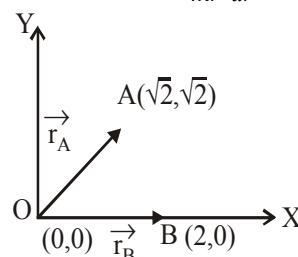
From (i) and (ii),

$$Q = -8\pi\epsilon_0 ar^3$$

$$\Rightarrow dq = -24\pi\epsilon_0 ar^2 dr$$

$$\text{Charge density, } \rho = \frac{dq}{4\pi r^2 dr} = -6\epsilon_0 a$$

26. (c)

The distance of point $A(\sqrt{2}, \sqrt{2})$ from the origin,

$$r_A = \sqrt{(\sqrt{2})^2 + (\sqrt{2})^2} = \sqrt{4} = 2 \text{ units.}$$

The distance of point $B(2, 0)$ from the origin,

$$r_B = \sqrt{(2)^2 + (0)^2} = 2 \text{ units.}$$

Now, potential at A , due to charge $q = 10^{-3} \mu\text{C}$

$$V_A = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{r_A}$$

$$\text{Potential at } B, \text{ due to charge } Q = 10^{-3} \text{ QC } V_B = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{r_B}$$

 $\therefore$ Potential difference between the points A and B is given by

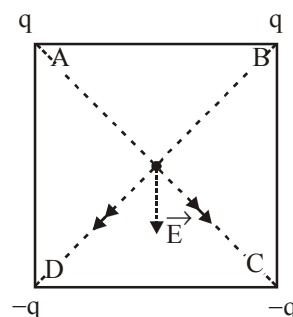
$$V_A - V_B = \frac{1}{4\pi\epsilon_0} \cdot \frac{10^{-3}}{r_A} - \frac{1}{4\pi\epsilon_0} \cdot \frac{10^{-3}}{r_B}$$

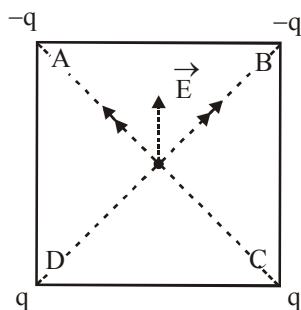
$$= \frac{10^{-3}}{4\pi\epsilon_0} \left(\frac{1}{r_A} - \frac{1}{r_B} \right) = \frac{10^{-3}}{4\pi\epsilon_0} \left(\frac{1}{2} - \frac{1}{2} \right)$$

$$= \frac{Q}{4\pi\epsilon_0} \times 0 = 0.$$

27. (a) As shown in the figure, the resultant electric fields before and after interchanging the charges will have the same magnitude, but opposite directions.

As potential is a scalar quantity, So the potential will be same in both cases.





28. (a) Given, potential $V(x) = \frac{20}{x^2 - 4}$ volt

$$\text{Electric field } E = -\frac{dV}{dx} = -\frac{d}{dx} \left(\frac{20}{x^2 - 4} \right)$$

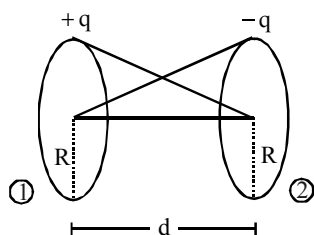
$$\Rightarrow E = +\frac{40x}{(x^2 - 4)^2}$$

At $x = 4 \mu\text{m}$,

$$E = +\frac{40 \times 4}{(4^2 - 4)^2} = +\frac{160}{144} = +\frac{10}{9} \text{ volt}/\mu\text{m}.$$

Positive sign indicates that $\vec{E}$ is in +ve x -direction.

29. (a)



Potential at the center of ring of charge $+q$ = potential due to itself + potential due to other ring of charge $-q$.

$$\Rightarrow V_1 = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{R} - \frac{q}{\sqrt{R^2 + d^2}} \right]$$

Potential at the centre of ring of charge $-q$ = potential due to itself + potential due to other ring of charge $+q$.

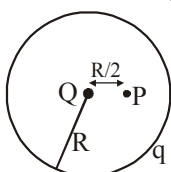
$$\Rightarrow V_2 = \frac{1}{4\pi\epsilon_0} \left[\frac{-q}{R} + \frac{q}{\sqrt{R^2 + d^2}} \right]$$

$$\Delta V = V_1 - V_2$$

$$= \frac{1}{4\pi\epsilon_0} \left[\frac{q}{R} + \frac{q}{R} - \frac{q}{\sqrt{R^2 + d^2}} - \frac{q}{\sqrt{R^2 + d^2}} \right]$$

$$= \frac{1}{2\pi\epsilon_0} \left[\frac{q}{R} - \frac{q}{\sqrt{R^2 + d^2}} \right]$$

30. (c) Electric potential due to charge Q at point P is



$$V_1 = \frac{1}{4\pi\epsilon_0} \frac{Q}{R/2} = \frac{1}{4\pi\epsilon_0} \frac{2Q}{R}$$

Electric potential due to charge q inside the shell is

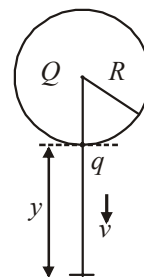
$$V_2 = \frac{1}{4\pi\epsilon_0} \frac{q}{R}$$

$\therefore$ The net electric potential at point P is

$$V = V_1 + V_2 = \frac{1}{4\pi\epsilon_0} \frac{2Q}{R} + \frac{1}{4\pi\epsilon_0} \frac{q}{R}$$

31. (d) By using energy conservation,

$$\Delta KE + (\Delta PE)_{\text{Electro}} + (\Delta PE)_{\text{gravitational}} = 0$$



$$\frac{1}{2} m V^2 + \left(k \frac{Qq}{R+y} - k \frac{Qq}{R} \right) + (-mgy) = 0$$

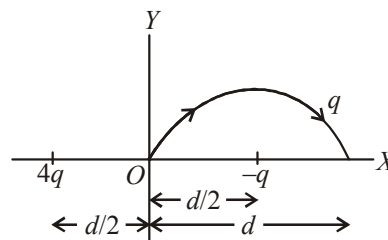
$$\Rightarrow \frac{1}{2} m V^2 = mgy + kQq \left(\frac{1}{R} - \frac{1}{R+y} \right)$$

$$\Rightarrow V^2 = 2gy + \frac{2kQq}{m} \frac{y}{R(R+y)}$$

$$\text{or, } V^2 = 2y \left[\frac{qQ}{4\pi\epsilon_0 R(R+y)m} + g \right]$$

32. (d) Change in potential energy, $\Delta u = q(V_f - V_i)$

Potential of $-q$ is same as initial and final point of the path.



$$\Delta u = q \left(\frac{k4q}{3d/2} - \frac{k4q}{d/2} \right) = -\frac{4q^2}{3\pi\epsilon_0 d}$$

–ve sign shows the energy of the charge is decreasing.

33. (c) According to work energy theorem, gain in kinetic energy is equal to work done in displacement of charge.

$$\therefore \frac{1}{2} m v^2 = q \Delta V$$

Here, ΔV = potential difference between two positions of charge q .

For same q and ΔV .

$$v \propto \frac{1}{\sqrt{m}}$$

Mass of hydrogen ion $m_H = 1$

Mass of helium ion $m_{He} = 4$

$$\therefore \frac{v_H}{v_{He}} = \sqrt{\frac{4}{1}} = 2:1.$$

34. (c) Using conservation of energy

$$U_i = U_f + \frac{1}{2}mv^2$$

$$\frac{kq_1q_2}{r_1} = \frac{kq_1q_2}{r_2} + \frac{1}{2}mv^2$$

$$\Rightarrow \frac{1}{2}mv^2 = kq_1q_2 \left[\frac{1}{r_1} - \frac{1}{r_2} \right]$$

$$v^2 = \frac{2kq_1q_2}{m} \left[\frac{1}{r_1} - \frac{1}{r_2} \right]$$

$$= \frac{2 \times 9 \times 10^9 \times 10^{-12}}{4 \times 10^{-6} \times 10^{-3}} \left[1 - \frac{1}{9} \right] = 4 \times 10^6$$

$$v = 2 \times 10^3 \text{ m/s}$$

35. (d)
$$U = \frac{1}{4\pi\epsilon_0} \left[\frac{q(-q)}{d} + \frac{qQ}{\left(D + \frac{d}{2}\right)} + \frac{(-q)Q}{\left(D - \frac{d}{2}\right)} \right]$$

$$= \frac{1}{4\pi\epsilon_0} \left[-\frac{q^2}{d} - \frac{qQd}{D^2} \right], \text{ Ignoring } \frac{d^2}{4}$$

36. (b) Using, $[K + U]_i = [K + U]_f$

or $0 + Vq = mv^2 + v'q$

or $mv^2 = (V - V')q$

$$= -q \int_{r_0}^r E dr = q \int_{r_0}^r \frac{\lambda}{2\pi\epsilon_0 r} dr = \frac{\lambda q}{2\pi\epsilon_0} \left(\ln \frac{r}{r_0} \right)$$

$$\Rightarrow v \propto \sqrt{\ln \frac{r}{r_0}}$$

37. (c) Total energy of charge distribution is constant at any instant t.

$$U_f + K_f = U_i + K_i$$

$$\text{i.e., } \frac{1}{2}mV^2 + \frac{KQ^2}{2R} = 0 + \frac{KQ^2}{2R_0}$$

$$\therefore \frac{1}{2}mV^2 = \frac{KQ^2}{2R_0} - \frac{KQ^2}{2R}$$

$$\therefore V = \sqrt{\frac{2}{m} \frac{KQ^2}{2} \left(\frac{1}{R_0} - \frac{1}{R} \right)}$$

$$\therefore V = \sqrt{\frac{KQ^2}{m} \left(\frac{1}{R_0} - \frac{1}{R} \right)} = C \sqrt{\frac{1}{R_0} - \frac{1}{R}}$$

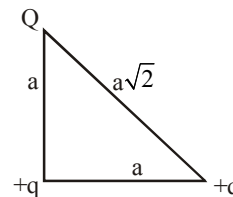
Also the slope of $V - R$ curve will go on decreasing.

38. (b) Net electrostatic energy for the system

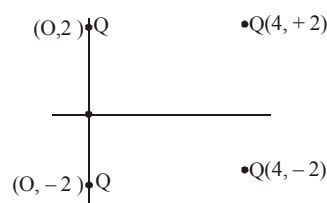
$$U = K \left[\frac{q^2}{a} + \frac{Qq}{a} + \frac{Qq}{a\sqrt{2}} \right] = 0$$

$$\Rightarrow q = -Q \left[1 + \frac{1}{\sqrt{2}} \right]$$

$$\Rightarrow Q = \frac{-q\sqrt{2}}{\sqrt{2}+1}$$



39. (b)



Potential at origin

$$V = \frac{KQ}{2} + \frac{KQ}{2} + \frac{KQ}{\sqrt{20}} + \frac{KQ}{\sqrt{20}}$$

$$\text{and potential at } \infty = 0 = KQ \left(1 + \frac{1}{\sqrt{5}} \right)$$

$\therefore$ Work required to put a fifth charge Q at origin $W =$

$$VQ = \frac{Q^2}{4\pi\epsilon_0} \left(1 + \frac{1}{\sqrt{5}} \right)$$

40. (c) The work done in moving a charge along an equipotential surface is always zero.

The direction of electric field is perpendicular to the equipotential surface or lines.

41. (c) The potential energy at the centre of the sphere

$$U_c = \frac{3}{2} \frac{KQq}{R}$$

The potential energy at the surface of the sphere

$$U_s = \frac{KqQ}{R}$$

Now change in the energy

$$\Delta U = U_c - U_s = \frac{KQq}{R} \left[\frac{3}{2} - 1 \right] = \frac{KQq}{2R}$$

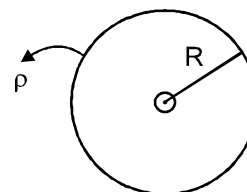
$$\text{Where } Q = \rho V = \rho \cdot \frac{4}{3} \pi R^3$$

$$\Delta U = \frac{2K}{3} \frac{\pi R^3 \rho q}{R}$$

$$\Delta U = \frac{2}{3} \times \frac{1}{4\pi\epsilon_0} \frac{\pi R^3 \rho q}{R}$$

$$\Delta U = \frac{R^2 \rho q}{6 \epsilon_0}$$

Using Gauss's law



$$\int \vec{E} \cdot d\vec{A} = \frac{q_{en}}{E_0} = \frac{\beta \times \frac{4}{3} \pi R^3}{E_0}$$

$$\Rightarrow \int E dA (\cos \theta) = \frac{\beta \times 4 \pi R^3}{3 E_0}$$

$$\Rightarrow E(4 \pi R^2) = \beta \times \frac{4}{3} \pi R^3 \times \frac{1}{E_0}$$

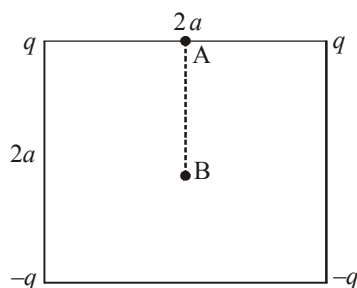
$$\Rightarrow E = \frac{\beta r}{3 E_0} \quad (r < R)$$

42. (d) Initial potential of the charge,

$$V_A = \frac{2kq}{a} - \frac{2kq}{a\sqrt{5}}$$

$$\Rightarrow V_A = \frac{1}{4\pi\epsilon_0} \frac{2q}{a} \left(1 - \frac{1}{\sqrt{5}}\right)$$

(Here potential due to each $q = \frac{kq}{a}$ and potential due to each $-q = \frac{-kq}{a\sqrt{5}}$)



Final potential of the charge

$$V_B = 0$$

($\because$ Point B is equidistant from all the four charges)

$\therefore$ Using work energy theorem,

$$(W_{AB})_{\text{electric}} = Q(V_A - V_B)$$

$$= \frac{2qQ}{4\pi\epsilon_0 a} \left[1 - \frac{1}{\sqrt{5}}\right]$$

$$= \left(\frac{1}{4\pi\epsilon_0}\right) \frac{2Qq}{a} \left[1 - \frac{1}{\sqrt{5}}\right]$$

43. (c) Work done, $W_{PQ} = q(V_Q - V_P)$

$$= (-100 \times 1.6 \times 10^{-19})(-4 - 10)$$

$$= +2.24 \times 10^{-16} \text{ J}$$

44. (a) Gain in kinetic energy = work done by potential difference

$$eV = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{2eV}{m}}$$

$$= \sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 20}{9.1 \times 10^{-31}}} = 2.65 \times 10^6 \text{ m/s}$$

45. (d) $\frac{1}{2}mv^2 = \frac{kQq}{r}$

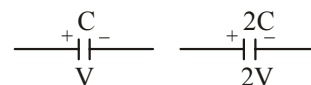
$$\Rightarrow \frac{1}{2}m(2v)^2 = \frac{kqQ}{r'} \Rightarrow r' = \frac{r}{4}$$

46. (a) By using

$$W = q(V_B - V_A)$$

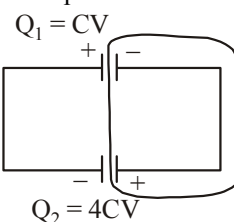
$$\therefore V_B - V_A = \frac{2J}{20C} = 0.1 \text{ J/C} = 0.1 \text{ V}$$

47. (b) When capacitors C and $2C$ capacitance are charged to V and $2V$ respectively.



$$Q_1 = CV \quad Q_2 = 2C \times 2V = 4CV$$

When connected in parallel



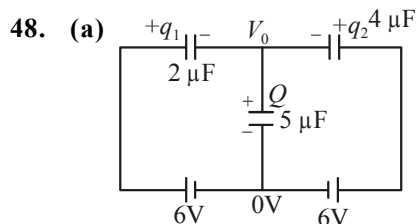
By conservation of charge

$$4CV - CV = (C + 2C)V_{\text{common}}$$

$$V_{\text{common}} = \frac{3CV}{3C} = V$$

Therefore final energy of this configuration,

$$U_f = \left(\frac{1}{2}CV^2 + \frac{1}{2} \times 2CV^2\right) = \frac{3}{2}CV^2$$



Let q_1 and q_2 be the charge on the capacitors of $2\mu\text{F}$ and $4\mu\text{F}$. Then charge on capacitor of $5\mu\text{F}$

$$Q = q_1 + q_2$$

$$\Rightarrow 5V_0 = 2(6 - V_0) + 4(6 - V_0)$$

$$\Rightarrow 5V_0 = 12 - 2V_0 + 24 - 4V_0$$

$$\Rightarrow 11V_0 = 36 \Rightarrow V_0 = \frac{36}{11} \text{ V}$$

$$\Rightarrow Q = 5V_0 = \frac{180}{11} \mu\text{C}$$

49. (d) When two capacitors with capacitance C_1 and C_2 at potential V_1 and V_2 connected to each other by wire, charge begins to flow from higher to lower potential till they acquire common potential. Here, some loss of energy takes place which is given by.

$$\text{Heat loss, } H = \frac{C_1 C_2}{2(C_1 + C_2)} (V_1 - V_2)^2$$

In the equation, put $V_2 = 0$, $V_1 = V_0$

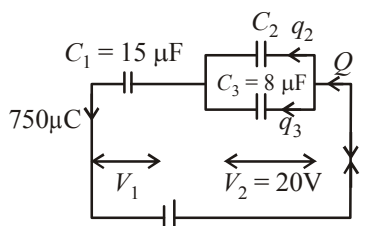
$$C_1 = C, C_2 = \frac{C}{2}$$

$$\text{Loss of heat} = \frac{C \times \frac{C}{2}}{2\left(C + \frac{C}{2}\right)} (V_0 - 0)^2 = \frac{C}{6} V_0^2$$

$$H = \frac{1}{6} C V_0^2$$

50. (b) According to question,

$$Q = 750 \mu C = q_2 + q_3$$



Capacitors C_2 and C_3 are in parallel hence,
Voltage across $C_2 =$ voltage across $C_3 = 20$ V
Change on capacitor C_3 ,

$$q_3 = C_3 \times V_3 = 8 \times 20 = 160 \mu C$$

$$\therefore q_2 = 750 \mu C - 160 \mu C = 590 \mu C$$

51. (4)

Given, $C_1 = 5 \mu F$ and $V_1 = 220$ Volt

When capacitor C_1 fully charged it is disconnected from the supply and connected to uncharged capacitor C_2 .

$C_2 = 2.5 \mu F$, $V_2 = 0$

Energy change during the charge redistribution,

$$\begin{aligned} \Delta U &= U_i - U_f = \frac{1}{2} \frac{C_1 C_2}{C_1 + C_2} (V_1 - V_2)^2 \\ &= \frac{1}{2} \times \frac{5 \times 2.5}{(5 + 2.5)} (220 - 0)^2 \mu J \\ &= \frac{5}{2 \times 3} \times 22 \times 22 \times 100 \times 10^{-6} J \\ &= \frac{5 \times 11 \times 22}{3} \times 10^{-4} J = \frac{55 \times 22}{3} \times 10^{-4} J \\ &= \frac{1210}{3} \times 10^{-4} J = \frac{1210}{3} \times 10^{-3} J \approx 4 \times 10^{-2} J \end{aligned}$$

According to questions, $\frac{x}{100} = 4 \times 10^{-2}$

$$\therefore x = 4$$

52. (a) Given,

Capacitance of capacitor, $C_1 = 10 \mu F$

Potential difference before removing the source voltage,
 $V_1 = 50$ V

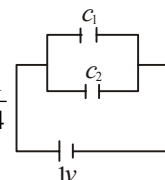
If C_2 be the capacitance of uncharged capacitor, then common potential is

$$V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$$

$$\Rightarrow 20 = \frac{10 \times 50 + 0}{20 + C} \Rightarrow C = 15 \mu F$$

53. (c) In parallel combination, $C_{eq} = C_1 + C_2 = 10 \mu F$
When connected across 1 V battery, then

$$\frac{U_1}{U_2} = \frac{\left(\frac{1}{2} C_1 V^2\right)}{\left(\frac{1}{2} C_2 V^2\right)} = \frac{1}{4} \Rightarrow \frac{C_1}{C_2} = \frac{1}{4}$$

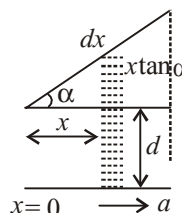


$$\therefore C_2 = 8 \mu F \text{ and } C_1 = 2 \mu F$$

Now C_1 and C_2 are connected in series combination,

$$\therefore C_{equivalent} = \frac{C_1 C_2}{C_1 + C_2} = \frac{2 \times 8}{2 + 8} = \frac{16}{10} = 1.6 \mu F$$

54. (a)



Consider an infinitesimal strip of capacitor of thickness dx at a distance x as shown.

Capacitance of parallel plate capacitor of area A is given

$$\text{by } C = \frac{\epsilon_0 A}{t}$$

[Here $t =$ separation between plates]

So, capacitance of thickness dx will be

$$\therefore dC = \frac{\epsilon_0 a dx}{d + x \tan \alpha}$$

Total capacitance of system can be obtained by integrating with limits $x = 0$ to $x = a$

$$\therefore C_{eq} = \int dC = a \epsilon_0 \int_{x=0}^{x=a} \frac{dx}{x \tan \alpha + d}$$

[By Binomial expansion]

$$\Rightarrow C_{eq} = \frac{a \epsilon_0}{d} \int_0^a \left(1 - \frac{x \tan \alpha}{d}\right) dx = \frac{a \epsilon_0}{d} \left(x - \frac{x^2 \tan \alpha}{2d}\right)_0^a$$

$$\Rightarrow C_{eq} = \frac{a^2 \epsilon_0}{d} = \left(1 - \frac{a \tan \alpha}{2d}\right) = \frac{\epsilon_0 a^2}{d} \left(1 - \frac{\alpha a}{2d}\right)$$

55. (a) Given, $K(x) = K(1 + \alpha x)$

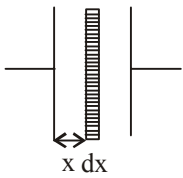
$$\text{Capacitance of element, } C_{el} = \frac{K \epsilon_0 A}{dx}$$

$$\Rightarrow C_{el} = \frac{\epsilon_0 K(1+\alpha x)A}{dx}$$

$$\therefore \int d\left(\frac{1}{C}\right) = \frac{1}{C_{el}} = \int_0^d \left(\frac{dx}{\epsilon_0 KA(1+\alpha x)} \right)$$

$$\Rightarrow \frac{1}{C} = \frac{1}{\epsilon_0 KA\alpha} [\ln(1+\alpha x)]_0^d$$

$$\Rightarrow \frac{1}{C} = \frac{1}{\epsilon_0 KA\alpha} \ln(1+\alpha d) [\alpha d \ll 1]$$

$$= \frac{1}{\epsilon_0 KA\alpha} \left[\alpha d - \frac{\alpha^2 d^2}{2} \right]$$


$$= \frac{1}{\epsilon_0 KA} \left[1 - \frac{\alpha d}{2} \right]$$

$$\therefore C = \frac{\epsilon_0 KA}{d \left(1 - \frac{\alpha d}{2} \right)} \Rightarrow C = \frac{\epsilon_0 KA}{d} \left(1 + \frac{\alpha d}{2} \right)$$

56. (6) In the first condition, electrostatic energy is

$$U_i = \frac{1}{2} CV_0^2 = \frac{1}{2} \times 60 \times 10^{-12} \times 400 = 12 \times 10^{-9} J$$

In the second condition $U_F = \frac{1}{2} C' V'^2$

$$U_f = \frac{1}{2} 2C \left(\frac{V_0}{2} \right)^2 \quad \left(\because C' = 2C, V' = \frac{V_0}{2} \right)$$

$$= \frac{1}{4} \times 60 \times 10^{-12} \times (20)^2 = 6 \times 10^{-9} J$$

$$\text{Energy lost} = U_i - U_f = 12 \times 10^{-9} J - 6 \times 10^{-9} J = 6 nJ$$

57. (d) $V\phi = \frac{CV + (nC)V}{kC + nC}$

$$\frac{(n+1)V}{k+n}$$

58. (c) $\frac{1}{C_1} = \frac{d/3}{k_1 \epsilon_0 A} + \frac{d/3}{k_2 \epsilon_0 A} + \frac{d/3}{k_3 \epsilon_0 A}$

$$\text{or } C_1 = \frac{3k_1 k_2 k_3 \epsilon_0 A}{d(k_1 k_2 + k_2 k_3 + k_3 k_1)}$$

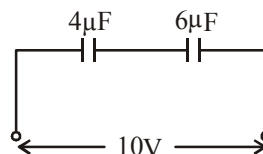
$$= \frac{k_1 \epsilon_0 (A/3)}{d} + \frac{k_2 \epsilon_0 (A/3)}{d} + \frac{k_3 \epsilon_0 (A/3)}{d}$$

$$= \frac{(k_1 + k_2 + k_3) \epsilon_0 A}{3d}$$

$$\frac{U_1}{U_2} = \frac{\frac{1}{2} C_1 V^2}{\frac{1}{2} C_2 V^2}$$

$$= \frac{E_1}{E_2} = \frac{9k_1 k_2 k_3}{(k_1 + k_2 + k_3)(k_1 k_2 + k_2 k_3 + k_3 k_1)}$$

59. (d) $V_1 + V_2 = 10$



$$\text{and } 4V_1 = 6V_2$$

On solving above equations, we get

$$V_1 = 6 V$$

Charge on $4 \mu F$,

$$q = CV_1 = 4 \times 6 = 24 \mu C.$$

60. (a) Equivalent capacitance in series combination (C') is given by

$$\frac{1}{C'} = \frac{1}{C_1} + \frac{1}{C_2} \Rightarrow C' = \frac{C_1 C_2}{C_1 + C_2}$$

For parallel combination equivalent capacitance

$$C'' = C_1 + C_2$$

For parallel combination

$$q = 10(C_1 + C_2)$$

$$q_1 = 500 \mu C$$

$$500 = 10(C_1 + C_2)$$

$$C_1 + C_2 = 50 \mu F$$

....(i)

For Series Combination—

$$q_2 = 10 \frac{C_1 C_2}{(C_1 + C_2)}$$

$$80 = 10 \frac{C_1 C_2}{50} \text{ From equation } \dots(ii)$$

$$C_1 C_2 = 400 \dots(iii)$$

From equation (i) and (ii)

$$C_1 = 10 \mu F \quad C_2 = 40 \mu F$$

61. (b) $\omega = \omega_f - v_i = \frac{q}{2} \left(\frac{1}{C_f} - \frac{1}{C_i} \right)$

$$= \frac{(5 \times 10)^2}{2} \left(\frac{1}{2} - \frac{1}{5} \right) \times 10^6$$

$$= 3.75 \times 10^{-6} J$$

62. (b) Capacitance of a capacitor with a dielectric of dielectric constant k is given by

$$C = \frac{k \epsilon_0 A}{d}$$

$$\therefore E = \frac{V}{d} \quad \therefore C = \frac{k \epsilon_0 A E}{V}$$

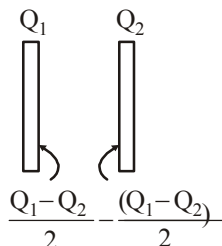
$$15 \times 10^{-12} = \frac{k \times 8.86 \times 10^{-12} \times 10^{-4} \times 10^6}{500}$$

$$k = 8.5$$

63. (b) $V = \frac{Q}{C}$

$$= \left(\frac{Q_1 - Q_2}{2C} \right)$$

$$= \left(\frac{4 - 2}{2 \times 1} \right) = 1 \text{ V}$$



64. (b) Energy stored in the system initially

$$U_i = \frac{1}{2} CE^2$$

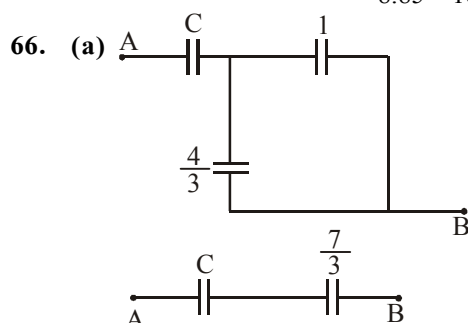
$$U_f = \frac{1}{2} \frac{Q^2}{C_{eq}} = \frac{(CE)^2}{2 \times 4C} = \frac{1}{2} \frac{CE^2}{4}$$

[As $Q = CE$, and $C_{eq} = 4C$]

$$\Delta U = \frac{1}{2} CE^2 \times \frac{3}{4} = \frac{3}{8} CE^2 = \frac{3}{8} \frac{Q^2}{C}$$

65. (c) $E = \frac{\sigma}{\epsilon_0} = \frac{Q}{A\epsilon_0}$

$$\therefore Q = \epsilon_0 \cdot E \cdot A = 8.85 \times 10^{-12} \times 100 \times 1 = 8.85 \times 10^{-10} \text{ C}$$



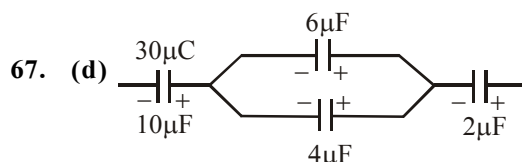
For series combination

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$\Rightarrow \frac{\frac{7C}{3}}{\frac{7}{3} + C} = \frac{1}{2}$$

$$\Rightarrow 14C = 7 + 3C$$

$$\Rightarrow C = \frac{7}{11} \mu\text{F}$$



As given in the figure, $6\mu\text{F}$ and $4\mu\text{F}$ are in parallel. Now using charge conservation

$$\text{Charge on } 6\mu\text{F capacitor} = \frac{6}{6+4} \times 30 = 18\mu\text{C}$$

Since charge is asked on right plate therefore is $+18\mu\text{C}$

68. (b) As required equivalent capacitance should be

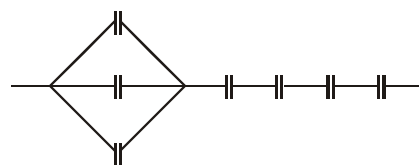
$$C_{eq} = \frac{6}{13} \mu\text{F}$$

Therefore three capacitors must be in parallel and 4 must be in series with it.

$$\frac{1}{C_{eq}} = \left[\frac{1}{3C} \right] + \left[\frac{1}{C} + \frac{1}{C} + \frac{1}{C} \right]$$

$$C_{eq} = \frac{3C}{13} = \frac{6}{13} \mu\text{F} \quad [\text{as } C = 2\mu\text{F}]$$

So, desired combination will be as below:



69. (b) $W = -\Delta u$

$$= (-1) \left[\frac{(c\epsilon)^2}{2kc} - \frac{(c\epsilon)^2}{2c} \right]$$

$$= \frac{\epsilon^2 c}{2} \frac{k-1}{k}$$

$$= 508 \text{ J}$$

70. (c) Let dielectric constant of material used be K .

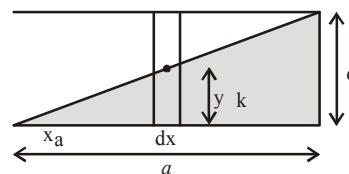
$$\frac{k_1 \epsilon_0 A_1}{d} + \frac{k_2 \epsilon_0 A_2}{d} + \frac{k_3 \epsilon_0 A_3}{d} = \frac{k \epsilon_0 A}{d}$$

$$\text{or } \frac{10 \epsilon_0 A/3}{d} + \frac{12 \epsilon_0 A/3}{d} + \frac{14 \epsilon_0 A/3}{d} = \frac{K \epsilon_0 A}{d}$$

$$\frac{\epsilon_0 A}{d} \left(\frac{10}{3} + \frac{12}{3} + \frac{14}{3} \right) = \frac{K \epsilon_0 A}{d}$$

$$\therefore K = 12$$

71. (b)



$$\text{From figure, } \frac{y}{x} = \frac{d}{a} \Rightarrow y = \frac{d}{a} x$$

$$dy = \frac{d}{a} (dx) \Rightarrow \frac{1}{dc} = \frac{y}{K\epsilon_0 a dx} + \frac{(d-y)}{\epsilon_0 a dx}$$

$$\frac{1}{dc} = \frac{y}{\epsilon_0 a bx} \left(\frac{y}{k} + d - y \right)$$

$$\int dc = \int \frac{\epsilon_0 a dx}{\frac{y}{k} + d - y}$$

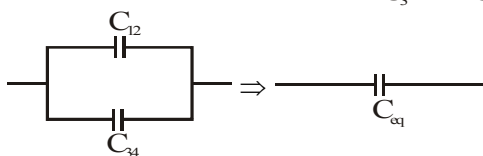
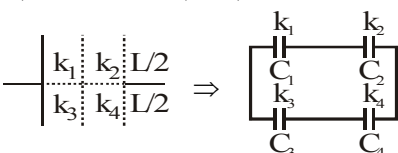
$$\text{or, } c = \epsilon_0 a \cdot \frac{a}{d} \int_0^d \frac{dy}{d + y \left(\frac{1}{k} - 1 \right)}$$

$$= \frac{\epsilon_0 a^2}{\left(\frac{1}{k}-1\right)d} \left[\ln \left(d + y \left(\frac{1}{k}-1 \right) \right) \right]_0^d$$

$$= \frac{k \epsilon_0 a^2}{(1-k)d} \ln \left(\frac{d + d \left(\frac{1}{k}-1 \right)}{d} \right)$$

$$= \frac{k \epsilon_0 a^2}{(1-k)d} \ln \left(\frac{1}{k} \right) = \frac{k \epsilon_0 a^2 \ln k}{(k-1)d}$$

72. (Bonus)



$$C_{12} = \frac{C_1 C_2}{C_1 + C_2} = \frac{\frac{k_1 \epsilon_0 \frac{L}{2} \times L}{d/2} \cdot \frac{k_2 \epsilon_0 \frac{L}{2} \times L}{d/2}}{\left[\frac{k_1 \epsilon_0 \frac{L}{2} \times L}{d/2} + \frac{k_2 \epsilon_0 \frac{L}{2} \times L}{d/2} \right]}$$

$$C_{12} = \frac{k_1 k_2 \epsilon_0 L^2}{(k_1 + k_2) d}$$

in the same way we get,

$$C_{34} = \frac{k_3 k_4 \epsilon_0 L^2}{(k_3 + k_4) d}$$

$$\therefore C_{eq} = C_{12} + C_{34} = \left[\frac{k_1 k_2}{k_1 + k_2} + \frac{k_3 k_4}{k_3 + k_4} \right] \frac{\epsilon_0 L^2}{d} \quad \dots (i)$$

$$\text{Now if } k_{eq} = K, C_{eq} = \frac{K \epsilon_0 L^2}{d} \quad \dots (ii)$$

on comparing equation (i) to equation (ii), we get

$$k_{eq} = \frac{k_1 k_2 (k_3 + k_4) + k_3 k_4 (k_1 + k_2)}{(k_1 + k_2)(k_3 + k_4)}$$

This does not match with any of the options so this must be a bonus.

73. (a) Charge on Capacitor, $Q_i = CV$

After inserting dielectric of dielectric constant

$$= K Q_f = (kC) V$$

Induced charges on dielectric

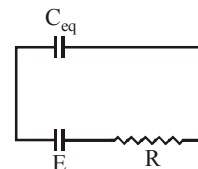
$$Q_{ind} = Q_f - Q_i = KCV - CV$$

$$(K-1)CV = \left(\frac{5}{3} - 1 \right) \times 90 \text{ pF} \times 2V = 1.2 \text{ nc}$$

74. (a) During charging charge on the capacitor increases with time. Charge on the capacitor C_1 as a function of time, $Q = Q_0(1 - e^{-t/RC})$

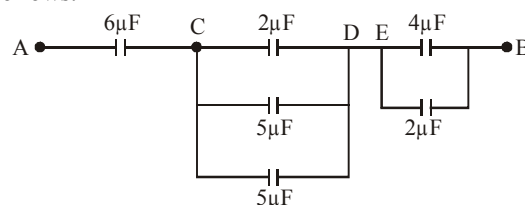
$$Q = C_{eq} E [1 - e^{-t/RC_{eq}}]$$

$$(\because Q_0 = C_{eq} E)$$



Both capacitor will have charge as they are connected in series

75. (d) The simplified circuit of the circuit given in question as follows:

The equivalent capacitance between C & D capacitors of $2\mu\text{F}$, $5\mu\text{F}$ and $5\mu\text{F}$ are in parallel.

$$\therefore C_{CD} = 2 + 5 + 5 = 12 \mu\text{F} \quad (\because \text{In parallel grouping } C_{eq} = C_1 + C_2 + \dots + C_n)$$

Similarly equivalent capacitance between E & B $C_{EB} = 4 + 2 = 6\mu\text{F}$

Now equivalent capacitance between A & B

$$\frac{1}{C_{eq}} = \frac{1}{6} + \frac{1}{12} + \frac{1}{6} = \frac{5}{12}$$

$$\Rightarrow C_{eq} = \frac{12}{5} = 2.4 \mu\text{F} \quad (\because \text{In series grouping,})$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_n}$$

76. (c) Given area of Parallel plate capacitor, $A = 200 \text{ cm}^2$ Separation between the plates, $d = 1.5 \text{ cm}$ Force of attraction between the plates, $F = 25 \times 10^{-6} \text{ N}$
 $F = QE$

$$F = \frac{Q^2}{2A \epsilon_0} \quad (E \text{ due to parallel plate} = \frac{\sigma}{2\epsilon_0} = \frac{Q}{2A \epsilon_0})$$

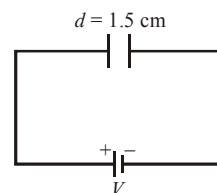
$$\text{But } Q = CV = \frac{\epsilon_0 A(V)}{d}$$

$$\therefore F = \frac{(\epsilon_0 AV)^2}{d^2 \times 2A \epsilon_0}$$

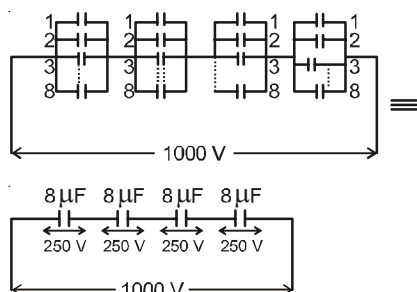
$$= \frac{(\epsilon_0 A)^2 \times V^2}{d^2 \times 2 \times (A \epsilon_0)} = \frac{(\epsilon_0 A) \times V^2}{d^2 \times 2}$$

$$\text{or, } 25 \times 10^{-6} = \frac{(8.85 \times 10^{-12}) \times (200 \times 10^{-4}) \times V^2}{2.25 \times 10^{-4} \times 2}$$

$$\Rightarrow V = \sqrt{\frac{25 \times 10^{-6} \times 2.25 \times 10^{-4} \times 2}{8.85 \times 10^{-12} \times 200 \times 10^{-4}}} \approx 250 \text{ V}$$



77. (a) The sum of final charges on C_2 and C_3 is $36 \mu\text{C}$.
78. (b) To get a capacitance of $2 \mu\text{F}$ arrangement of capacitors of capacitance $1 \mu\text{F}$ as shown in figure 8 capacitors of $1 \mu\text{F}$ in parallel with four such branches in series i.e., 32 such capacitors are required.



$$\frac{1}{C_{eq}} = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} \quad \therefore C_{eq} = 2 \mu\text{F}$$

79. (c) Before introducing a slab capacitance of plates

$$C_1 = \frac{\epsilon_0 A}{3}$$

If a slab of dielectric constant K is introduced between plates then

$$C = \frac{K\epsilon_0 A}{d} \text{ then } C'_1 = \frac{\epsilon_0 A}{2.4}$$

C_1 and C'_1 are in series hence,

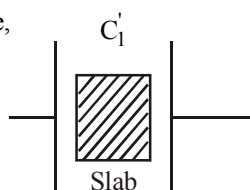
$$\frac{\epsilon_0 A}{3} = \frac{k \frac{\epsilon_0 A}{3} \cdot \frac{\epsilon_0 A}{2.4}}{k \frac{\epsilon_0 A}{3} + \frac{\epsilon_0 A}{2.4}}$$

$$3k = 2.4k + 3$$

$$0.6k = 3$$

Hence, the dielectric constant of slab is given by,

$$k = \frac{30}{6} = 5$$



80. (d) Energy of sphere = $\frac{Q^2}{2C}$

$$4.5 = \frac{16 \times 10^{-12}}{2C}$$

$$C = \frac{16 \times 10^{-12}}{9} = 4\pi\epsilon_0 R$$

(capacity of spherical conductor)

$$R = \frac{16 \times 10^{-12}}{9} \times \frac{1}{4\pi\epsilon_0} \quad \therefore \frac{1}{4\pi\epsilon_0} = 9 \times 10^9$$

$$= 9 \times 10^9 \times \frac{16}{9} \times 10^{-12} = 16 \text{ mm}$$

81. (a)

$$\text{Charge on } C_1 \text{ is } q_1 = \left[\left(\frac{12}{4+12} \right) \times 8 \right] \times 4 = 24 \mu\text{C}$$

$$\text{The voltage across } C_p \text{ is } V_p = \frac{4}{4+12} \times 8 = 2\text{V}$$

$\therefore$ Voltage across $9 \mu\text{F}$ is also 2V

$\therefore$ Charge on $9 \mu\text{F}$ capacitor = $9 \times 2 = 18 \mu\text{C}$

$\therefore$ Total charge on $4 \mu\text{F}$ and $9 \mu\text{F}$ = $42 \mu\text{C}$

$$\therefore E = \frac{KQ}{r^2} = 9 \times 10^9 \times \frac{42 \times 10^{-6}}{30 \times 30} = 420 \text{ NC}^{-1}$$

82. (a) Capacitors $2 \mu\text{F}$ and $2 \mu\text{F}$ are parallel, their equivalent = $4 \mu\text{F}$

$6 \mu\text{F}$ and $12 \mu\text{F}$ are in series, their equivalent = $4 \mu\text{F}$

Now $4 \mu\text{F}$ (2 and $2 \mu\text{F}$) and $8 \mu\text{F}$ in series = $\frac{3}{8} \mu\text{F}$

And $4 \mu\text{F}$ (12 & $6 \mu\text{F}$) and $4 \mu\text{F}$ in parallel = $4 + 4 = 8 \mu\text{F}$

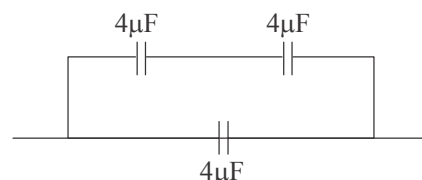
$8 \mu\text{F}$ in series with $1 \mu\text{F} = \frac{1}{8} + 1 \Rightarrow \frac{8}{9} \mu\text{F}$

$$\text{Now } C_{eq} = \frac{8}{9} + \frac{8}{3} = \frac{32}{9}$$

$$C_{eq} \text{ of circuit} = \frac{32}{9}$$

$$\text{With } C - \frac{1}{C_{eq}} = \frac{1}{C} + \frac{9}{32} = 1 \Rightarrow C = \frac{32}{23}$$

83. (d) To get effective capacitance of $6 \mu\text{F}$ two capacitors of $4 \mu\text{F}$ each connected in series and one of $4 \mu\text{F}$ capacitor in parallel with them.



Two capacitances in series

$$\therefore \frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

1 capacitor in parallel

$$\therefore C_{eq} = C_3 + C = 4 + 2 = 6 \mu\text{F}$$

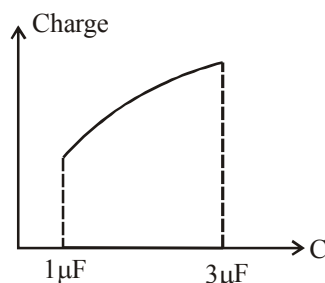
84. (d)

$$\text{From figure, } Q_2 = \frac{2}{2+1} Q = \frac{2}{3} Q$$

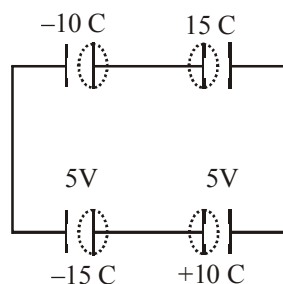
$$Q = E \left(\frac{C \times 3}{C+3} \right)$$

$$\therefore Q_2 = \frac{2}{3} \left(\frac{3CE}{C+3} \right) = \frac{2CE}{C+3}$$

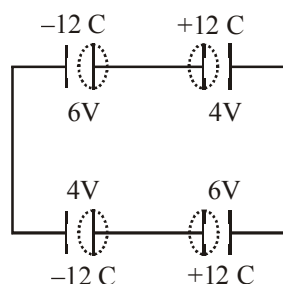
Therefore graph d correctly depicts.



85. (a) when switch is closed



When switch is open



Charge of $5\mu\text{C}$ flows from b to a

86. (a) Electric field in presence of dielectric between the two plates of a parallel plate capacitor is given by,

$$E = \frac{\sigma}{K\epsilon_0}$$

Then, charge density

$$\begin{aligned} \sigma &= K\epsilon_0 E \\ &= 2.2 \times 8.85 \times 10^{-12} \times 3 \times 10^4 \\ &\approx 6 \times 10^{-7} \text{ C/m}^2 \end{aligned}$$

87. (d)

88. (c) The value of dielectric constant is given as,

$$K = K_0 + \lambda x$$

$$\text{And, } V = \int_0^d E dr \Rightarrow V = \int_0^d \frac{\sigma}{K} dx$$

$$= \sigma \int_0^d \frac{1}{(K_0 + \lambda x)} dx = \frac{\sigma}{\lambda} [\ln(K_0 + \lambda d) - \ln K_0]$$

$$= \frac{\sigma}{\lambda} \ln \left(1 + \frac{\lambda d}{K_0} \right)$$

Now it is given that capacitance of vacuum = C_0 .

$$\text{Thus, } C = \frac{Q}{V}$$

$$= \frac{\sigma \cdot s}{V} \quad (\text{Let surface area of plates} = s)$$

$$= \frac{\sigma \cdot s}{\frac{\sigma}{\lambda} \ln \left(1 + \frac{\lambda d}{K_0} \right)}$$

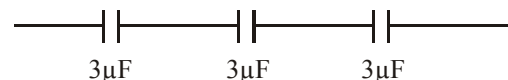
$$= s \lambda \cdot \frac{d}{\ln \left(1 + \frac{\lambda d}{K_0} \right)} \quad (\because \text{in vacuum } \epsilon_0 = 1)$$

$$C = \frac{\lambda d}{\ln \left(1 + \frac{\lambda d}{K_0} \right)} \cdot C_0 \quad \left(\text{here, } C_0 = \frac{s}{d} \right)$$

89. (c)

90. (d) Possible combination of capacitors

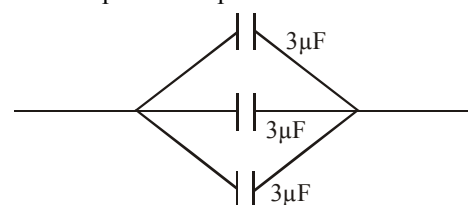
- (i) Three capacitors in series combination



$$\frac{1}{C_{eq}} = \frac{1}{3} + \frac{1}{3} + \frac{1}{3}$$

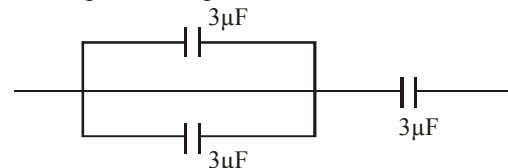
$$\therefore \frac{1}{C_{eq}} = 1\mu\text{F}$$

- (ii) Three capacitors in parallel combination



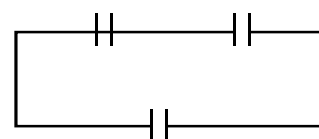
$$C_{eq} = 3 + 3 + 3 = 9\mu\text{F}$$

- (iii) Two capacitors in parallel and one is in series



$$C_{eq} = 2\mu\text{F}$$

- (iv) Two capacitors in series and one is in parallel



$$C_{eq} = 4.5\mu\text{F}$$

91. (b) $C_0 = \frac{k \epsilon_0 A}{d}$

$$C = \frac{k \epsilon_0 2}{3d} + \frac{2k \epsilon_0 A}{3d} = \frac{4k \epsilon_0 A}{3d}$$

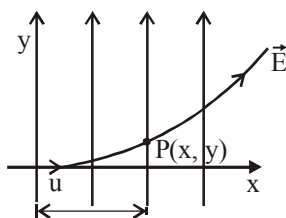
$$\therefore \frac{C}{C_0} = \frac{\frac{4k \epsilon_0 A}{3d}}{\frac{k \epsilon_0 A}{d}} = \frac{4}{3}$$

92. (c) As, $C = \frac{Q}{V} = \frac{It}{V}$
 $\Rightarrow \frac{V}{t} = \frac{I}{C} = \frac{2}{1 \times 10^{-6}}$
 $= 2 \times 10^6 \text{ V/s}$

93. (c) When charged particle enters perpendicularly in an electric field, it describes a parabolic path

$$y = \frac{1}{2} \left(\frac{QE}{m} \right) \left(\frac{x}{4} \right)^2$$

This is the equation of parabola.



94. (d) The discharging of a capacitor is given as

$$q = q_0 \exp[-t/RC]$$

$RC = \text{time constant} = \tau$

$$q = q_0 e^{-t/\tau}$$

If e is the capacitance of the capacitor

$$q = CV \text{ and } q = CV_0$$

$$\text{Thus, } CV = CV_0 e^{-t/\tau}$$

$$V = V_0 e^{-t/\tau} \dots (i)$$

From the graph (given in the problem

when $t = 0.5$, $V = 25$ i.e.,

$V_0 = 25$ volt.

and when $t = 200$, $V = 5$ volt

Thus equation (i) becomes

$$5 = 25 e^{-200/\tau}$$

$$\Rightarrow 1/5 = e^{-200/\tau}$$

Taking $\log_e$ on both sides

$$\log_e \frac{1}{5} = -200/\tau \Rightarrow -\frac{200}{\tau} = \log_e 5$$

$$\tau = \frac{200}{\log_e 5}$$

$$\text{or } \tau = \frac{200}{\log_e \left(\frac{10}{2} \right)} = \frac{200}{\log_e 10 - \log_e 2}$$

$$\tau = \frac{200}{2.302 - 0.693} = \frac{200}{1.609} = 124.300$$

Which lies between 100 s and 150 s

95. (c) The dielectric constant of the gas is 1.01

96. (d) Capacitance of sphere is given by :

$$C = 4\pi \epsilon_0 r$$

If, $C = 1\text{F}$ then radius of sphere needed:

$$r = \frac{C}{4\pi \epsilon_0} = \frac{1}{4\pi \times 8.85 \times 10^{-12}}$$

$$\text{or, } r = \frac{10^{12}}{4\pi \times 8.85} = 9 \times 10^9 \text{ m}$$

$9 \times 10^9 \text{ m}$ is very large, it is not possible to obtain such a large sphere. Infact earth has radius $6.4 \times 10^6 \text{ m}$ only and capacitance of earth is $711 \mu\text{F}$.

97. (d) Equivalent capacitance of n_2 number of capacitors each of capacitance C_2 in parallel $= n_2 C_2$

Equivalent capacitance of n_1 number of capacitors each of capacitances C_1 in series.

$$\text{Capacitance of each is } C_1 = \frac{C_1}{n_1}$$

According to question, total energy stored in both the combinations are same

$$\text{i.e., } \frac{1}{2} \left(\frac{C_1}{n_1} \right) (4V)^2 = \frac{1}{2} (n_2 C_2) V^2$$

$$\therefore C_2 = \frac{16C_1}{n_1 n_2}$$

98. (c) Charge (or current) always flows from higher potential to lower potential.

$$\text{Potential} = \frac{\text{Charge}}{\text{Capacitance}}$$

99. (c) Initial energy of capacitor, $E_1 = \frac{q_1^2}{2C}$

Final energy of capacitor,

$$E_2 = \frac{1}{2} E_1 = \frac{q_1^2}{4C} = \left(\frac{q_1}{\sqrt{2}} \right)^2 \frac{1}{2C}$$

$\therefore t_1 = \text{time for the charge to reduce to } \frac{1}{\sqrt{2}} \text{ of its initial value}$

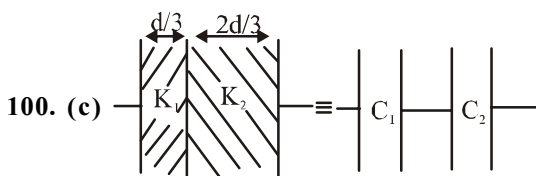
and $t_2 = \text{time for the charge to reduce to } \frac{1}{4} \text{ of its initial value}$

$$\text{We have, } q_2 = q_1 e^{-t/CR} \Rightarrow \ln \left(\frac{q_2}{q_1} \right) = -\frac{t}{CR}$$

$$\therefore \ln \left(\frac{1}{\sqrt{2}} \right) = \frac{-t_1}{CR} \dots (1)$$

$$\text{and } \ln \left(\frac{1}{4} \right) = \frac{-t_2}{CR} \dots (2)$$

$$\text{By (1) and (2), } \frac{t_1}{t_2} = \frac{\ln \left(\frac{1}{\sqrt{2}} \right)}{\ln \left(\frac{1}{4} \right)} = \frac{1}{2} \frac{\ln \left(\frac{1}{2} \right)}{2 \ln \left(\frac{1}{2} \right)} = \frac{1}{4}$$



The capacitance with air between the plates

$$C = \frac{\epsilon_0 A}{d} = 9 \text{ pF}$$

On introducing two dielectric between the plates, the given capacitance is equal to two capacitances connected in series where

$$C_1 = \frac{k_1 \epsilon_0 A}{d/3} = \frac{3 \epsilon_0 A}{d/3}$$

$$= \frac{3 \times 3 \epsilon_0 A}{d} = \frac{9 \epsilon_0 A}{d}$$

and

$$C_2 = \frac{k_2 \epsilon_0 A}{2d/3} = \frac{3k_2 \epsilon_0 A}{2d}$$

$$= \frac{3 \times 6 \epsilon_0 A}{2d} = \frac{9 \epsilon_0 A}{d}$$

The equivalent capacitance C_{eq} is

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$= \frac{d}{9 \epsilon_0 A} + \frac{d}{9 \epsilon_0 A} = \frac{2d}{9 \epsilon_0 A}$$

$$\therefore C_{eq} = \frac{9}{2} \frac{\epsilon_0 A}{d} = \frac{9}{2} \times 9 \text{ pF} = 40.5 \text{ pF}$$

101. (a) The potential energy of a charged capacitor is given

$$\text{by } U = \frac{Q^2}{2C}.$$

When a dielectric slab is introduced between the plates

the energy is given by $\frac{Q^2}{2KC}$, where K is the dielectric constant.

Again, when the dielectric slab is removed slowly its energy increases to initial potential energy. Thus, work done is zero.

102. (b) As n plates are joined alternately positive plate of all $(n-1)$ capacitor are connected to one point and negative plate of all $(n-1)$ capacitors are connected to other point. It means $(n-1)$ capacitors joined in parallel.
 $\therefore$ Resultant capacitance $= (n-1)C$

103. (c) Applying conservation of energy,

Electric potential energy of capacitor = heat absorbed

$$\frac{1}{2} CV^2 = m.s \Delta t; \quad V = \sqrt{\frac{2m.s.\Delta t}{C}}$$

104. (b) The capacitance without aluminium foil is

$$C = \frac{\epsilon_0 A}{d}$$

Here, d is distance between the plates of a capacitor

A = Area of plates of capacitor

When an aluminium foil of thickness t is introduced between the plates.

$$\text{Capacitance, } C' = \frac{\epsilon_0 A}{d-t}$$

If thickness of foil is negligible $50 d - t \sim d$. Hence, $C = C'$.

105. (d) The work done is stored in the form of potential energy which is given by

$$U = \frac{1}{2} \frac{Q^2}{C}$$

$$\therefore U = \frac{1}{2} \times \frac{(8 \times 10^{-18})^2}{100 \times 10^{-6}} = 32 \times 10^{-32} \text{ J}$$

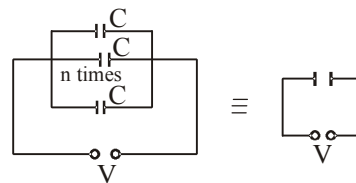
106. (b) In parallel, equivalent capacitance of n capacitor of capacitance C

$$C' = nC$$

Energy stored in this capacitor

$$E = \frac{1}{2} C' V^2$$

$$\Rightarrow E = \frac{1}{2} (nC) V^2 = \frac{1}{2} n C V^2$$



Alternatively

Each capacitor has a potential difference of V between the plates.

So, energy stored in each capacitor

$$= \frac{1}{2} C V^2.$$

$\therefore$ Energy stored in n capacitor

$$= \left[\frac{1}{2} C V^2 \right] \times n$$

107. (a) Capacitance of spherical conductor $= 4\pi \epsilon_0 R$

Here, R is radius of conductor

$$\therefore C = 4\pi \epsilon_0 R = \frac{1}{9 \times 10^9} \times 1.1 \times 10^{-10} \text{ F}$$

Current Electricity

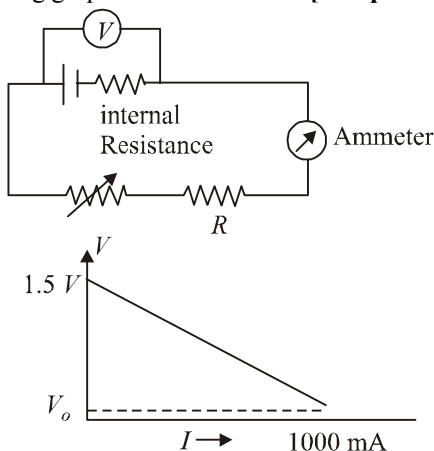


TOPIC 1

Electric Current, Drift of Electrons, Ohm's Law, Resistance and Resistivity

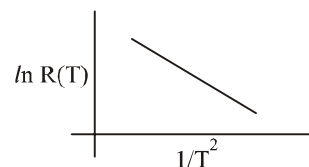


- A circuit to verify Ohm's law uses ammeter and voltmeter in series or parallel connected correctly to the resistor. In the circuit : **[Sep. 06, 2020 (II)]**
 - ammeter is always used in parallel and voltmeter is series
 - Both ammeter and voltmeter must be connected in parallel
 - ammeter is always connected in series and voltmeter in parallel
 - Both, ammeter and voltmeter must be connected in series
- Consider four conducting materials copper, tungsten, mercury and aluminium with resistivity ρ_C , ρ_T , ρ_M and ρ_A respectively. Then : **[Sep. 02, 2020 (I)]**
 - $\rho_C > \rho_A > \rho_T$
 - $\rho_M > \rho_A > \rho_C$
 - $\rho_A > \rho_T > \rho_C$
 - $\rho_A > \rho_M > \rho_C$
- To verify Ohm's law, a student connects the voltmeter across the battery as, shown in the figure. The measured voltage is plotted as a function of the current, and the following graph is obtained : **[12 Apr. 2019 I]**



If V_o is almost zero, identify the correct statement:

- The emf of the battery is 1.5 V and its internal resistance is 1.5Ω
 - The value of the resistance R is 1.5Ω
 - The potential difference across the battery is 1.5 V when it sends a current of 1000 mA
 - The emf of the battery is 1.5 V and the value of R is 1.5Ω
- A current of 5 A passes through a copper conductor (resistivity) = $1.7 \times 10^{-8} \Omega \text{m}$ of radius of cross-section 5 mm. Find the mobility of the charges if their drift velocity is $1.1 \times 10^{-3} \text{ m/s}$. **[10 Apr. 2019 I]**
 - $1.8 \text{ m}^2/\text{Vs}$
 - $1.5 \text{ m}^2/\text{Vs}$
 - $1.3 \text{ m}^2/\text{Vs}$
 - $1.0 \text{ m}^2/\text{Vs}$
 - In an experiment, the resistance of a material is plotted as a function of temperature (in some range). As shown in the figure, it is a straight line. **[10 Apr. 2019 I]**



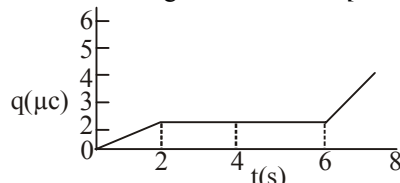
One may conclude that:

- $R(T) = \frac{R_0}{T^2}$
 - $R(T) = R_0 e^{-T_0^2/T^2}$
 - $R(T) = R_0 e^{-T^2/T_0^2}$
 - $R(T) = R_0 e^{T^2/T_0^2}$
- Space between two concentric conducting spheres of radii a and b ($b > a$) is filled with a medium of resistivity ρ . The resistance between the two spheres will be : **[10 Apr. 2019 II]**
 - $\frac{\rho}{4\pi} \left(\frac{1}{a} - \frac{1}{b} \right)$
 - $\frac{\rho}{2\pi} \left(\frac{1}{a} - \frac{1}{b} \right)$
 - $\frac{\rho}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right)$
 - $\frac{\rho}{4\pi} \left(\frac{1}{a} + \frac{1}{b} \right)$
 - In a conductor, if the number of conduction electrons per unit volume is $8.5 \times 10^{28} \text{ m}^{-3}$ and mean free time is 25 fs (femto second), its approximate resistivity is: ($m_e = 9.1 \times 10^{-31} \text{ kg}$) **[9 Apr. 2019 II]**
 - $10^{-6} \Omega \text{m}$
 - $10^{-7} \Omega \text{m}$
 - $10^{-8} \Omega \text{m}$
 - $10^{-5} \Omega \text{m}$

8. A $200\ \Omega$ resistor has a certain color code. If one replaces the red color by green in the code, the new resistance will be : [8 April 2019 I]

(a) $100\ \Omega$ (b) $400\ \Omega$ (c) $300\ \Omega$ (d) $500\ \Omega$

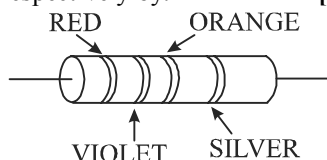
9. The charge on a capacitor plate in a circuit, as a function of time, is shown in the figure: [12 Jan. 2019 II]



What is the value of current at $t = 4\text{ s}$?

(a) Zero (b) $3\ \mu\text{A}$ (c) $2\ \mu\text{A}$ (d) $1.5\ \mu\text{A}$

10. A resistance is shown in the figure. Its value and tolerance are given respectively by: [9 Jan. 2019 I]



(a) $270\ \Omega, 10\%$ (b) $27\text{ k}\Omega, 10\%$
(c) $27\text{ k}\Omega, 20\%$ (d) $270\ \Omega, 5\%$

11. Drift speed of electrons, when 1.5 A of current flows in a copper wire of cross section 5 mm^2 , is v . If the electron density in copper is $9 \times 10^{28}/\text{m}^3$ the value of v in mm/s close to (Take charge of electron to be $= 1.6 \times 10^{-19}\text{ C}$) [9 Jan. 2019 I]

(a) 0.02 (b) 3 (c) 2 (d) 0.2

12. A copper wire is stretched to make it 0.5% longer. The percentage change in its electrical resistance if its volume remains unchanged is: [9 Jan. 2019 I]

(a) 2.0% (b) 2.5% (c) 1.0% (d) 0.5%

13. A carbon resistance has following colour code. What is the value of the resistance? [9 Jan. 2019 II]



(a) $530\text{ k}\Omega \pm 5\%$ (b) $5.3\text{ k}\Omega \pm 5\%$
(c) $6.4\text{ M}\Omega \pm 5\%$ (d) $64\text{ M}\Omega \pm 10\%$

14. A heating element has a resistance of $100\ \Omega$ at room temperature. When it is connected to a supply of 220 V , a steady current of 2 A passes in it and temperature is 500°C more than room temperature. What is the temperature coefficient of resistance of the heating element? [Online April 16, 2018]

(a) $1 \times 10^{-4}^\circ\text{C}^{-1}$ (b) $5 \times 10^{-4}^\circ\text{C}^{-1}$
(c) $2 \times 10^{-4}^\circ\text{C}^{-1}$ (d) $0.5 \times 10^{-4}^\circ\text{C}^{-1}$

15. A copper rod of cross-sectional area A carries a uniform current I through it. At temperature T , if the volume charge density of the rod is ρ , how long will the charges take to travel a distance d ? [Online April 15, 2018]

(a) $\frac{2\rho dA}{IT}$ (b) $\frac{2\rho dA}{I}$ (c) $\frac{\rho dA}{I}$ (d) $\frac{\rho dA}{IT}$

16. A uniform wire of length l and radius r has a resistance of $100\ \Omega$. It is recast into a wire of radius $\frac{r}{2}$. The resistance

of new wire will be :

[Online April 9, 2017]

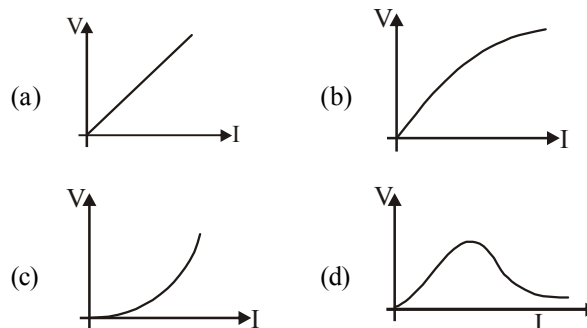
(a) $1600\ \Omega$ (b) $400\ \Omega$ (c) $200\ \Omega$ (d) $100\ \Omega$

17. When 5 V potential difference is applied across a wire of length 0.1 m , the drift speed of electrons is $2.5 \times 10^{-4}\text{ ms}^{-1}$. If the electron density in the wire is $8 \times 10^{28}\text{ m}^{-3}$, the resistivity of the material is close to : [2015]

(a) $1.6 \times 10^{-6}\ \Omega\text{m}$ (b) $1.6 \times 10^{-5}\ \Omega\text{m}$
(c) $1.6 \times 10^{-8}\ \Omega\text{m}$ (d) $1.6 \times 10^{-7}\ \Omega\text{m}$

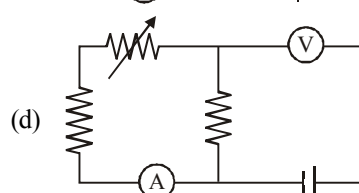
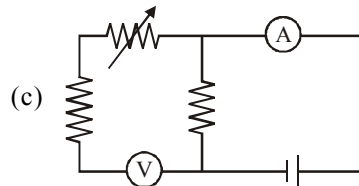
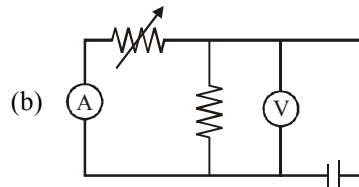
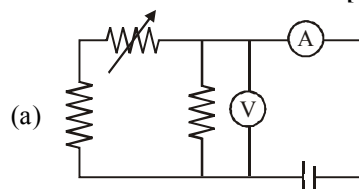
18. Suppose the drift velocity v_d in a material varied with the applied electric field E as $v_d \propto \sqrt{E}$. Then $V-I$ graph for a wire made of such a material is best given by :

[Online April 10, 2015]



19. Correct set up to verify Ohm's law is :

[Online April 23, 2013]

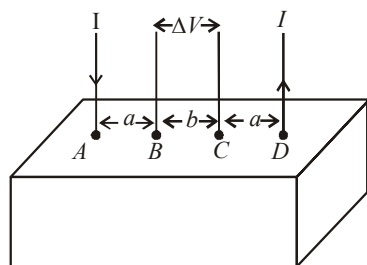


20. The resistance of a wire is R . It is bent at the middle by 180° and both the ends are twisted together to make a shorter wire. The resistance of the new wire is [Online May 26, 2012]
 (a) $2R$ (b) $R/2$ (c) $R/4$ (d) $R/8$
21. If a wire is stretched to make it 0.1% longer, its resistance will : [2011]
 (a) increase by 0.2% (b) decrease by 0.2%
 (c) decrease by 0.05% (d) increase by 0.05%

DIRECTIONS : Question No. 22 and 23 are based on the following paragraph.

Consider a block of conducting material of resistivity ' ρ ' shown in the figure. Current ' I ' enters at 'A' and leaves from 'D'. We apply superposition principle to find voltage ' ΔV ' developed between 'B' and 'C'. The calculation is done in the following steps:

- Take current ' I ' entering from 'A' and assume it to spread over a hemispherical surface in the block.
- Calculate field $E(r)$ at distance ' r ' from A by using Ohm's law $E = \rho j$, where j is the current per unit area at ' r '.
- From the ' r ' dependence of $E(r)$, obtain the potential $V(r)$ at r .
- Repeat (i), (ii) and (iii) for current ' I ' leaving 'D' and superpose results for 'A' and 'D'.



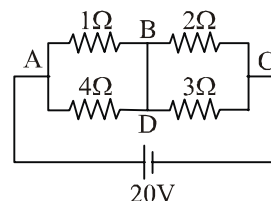
22. ΔV measured between B and C is [2008]
 (a) $\frac{\rho I}{\pi a} - \frac{\rho I}{\pi(a+b)}$ (b) $\frac{\rho I}{a} - \frac{\rho I}{(a+b)}$
 (c) $\frac{\rho I}{2\pi a} - \frac{\rho I}{2\pi(a+b)}$ (d) $\frac{\rho I}{2\pi(a-b)}$
23. For current entering at A, the electric field at a distance ' r ' from A is [2008]
 (a) $\frac{\rho I}{8\pi r^2}$ (b) $\frac{\rho I}{r^2}$ (c) $\frac{\rho I}{2\pi r^2}$ (d) $\frac{\rho I}{4\pi r^2}$
24. The resistance of a wire is 5 ohm at 50°C and 6 ohm at 100°C . The resistance of the wire at 0°C will be [2007]
 (a) 3 ohm (b) 2 ohm (c) 1 ohm (d) 4 ohm
25. A material 'B' has twice the specific resistance of 'A'. A circular wire made of 'B' has twice the diameter of a wire made of 'A'. then for the two wires to have the same resistance, the ratio l_B/l_A of their respective lengths must be [2006]
 (a) 1 (b) $\frac{1}{2}$ (c) $\frac{1}{4}$ (d) 2
26. An electric current is passed through a circuit containing two wires of the same material, connected in parallel. If the lengths and radii are in the ratio of $\frac{4}{3}$ and $\frac{2}{3}$, then the ratio of the current passing through the wires will be [2004]
 (a) $8/9$ (b) $1/3$ (c) 3 (d) 2

27. The length of a given cylindrical wire is increased by 100% . Due to the consequent decrease in diameter the change in the resistance of the wire will be [2003]
 (a) 200% (b) 100% (c) 50% (d) 300%

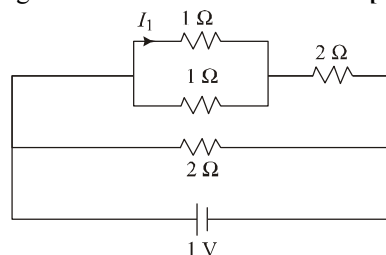
TOPIC 2 Combination of Resistances



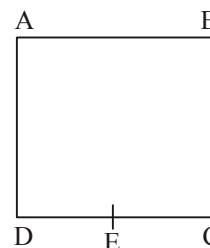
28. In the given circuit diagram, a wire is joining points B and D. The current in this wire is: [9 Jan. 2020 I]



- (a) 0.4A (b) 2A (c) 4A (d) zero
29. The series combination of two batteries, both of the same emf 10 V , but different internal resistance of 20Ω and 5Ω , is connected to the parallel combination of two resistors 30Ω and $R \Omega$. The voltage difference across the battery of internal resistance 20Ω is zero, the value of R (in Ω) is [NA. 8 Jan. 2020 II]
30. The current I_1 (in A) flowing through 1Ω resistor in the following circuit is: [7 Jan. 2020 I]



- (a) 0.4 (b) 0.5 (c) 0.2 (d) 0.25
31. A wire of resistance R is bent to form a square ABCD as shown in the figure. The effective resistance between E and C is: (E is mid-point of arm CD) [9 April 2019 I]

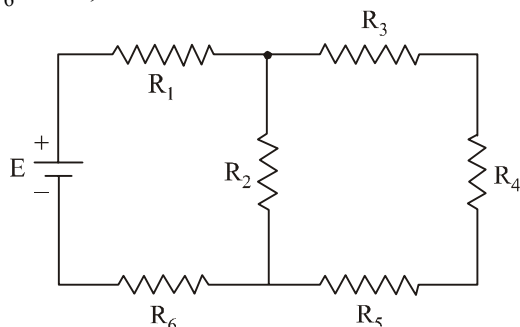


- (a) R (b) $\frac{7}{64}R$ (c) $\frac{3}{4}R$ (d) $\frac{1}{16}R$
32. A metal wire of resistance 3Ω is elongated to make a uniform wire of double its previous length. This new wire is now bent and the ends joined to make a circle. If two points on the circle make an angle 60° at the centre, the equivalent resistance between these two points will be: [9 Apr. 2019 II]

- (a) $\frac{12}{5}\Omega$ (b) $\frac{5}{2}\Omega$ (c) $\frac{5}{3}\Omega$ (d) $\frac{7}{2}\Omega$

33. In the figure shown, what is the current (in Ampere) drawn from the battery? You are given : [8 Apr. 2019 II]

$R_1 = 15 \Omega$, $R_2 = 10 \Omega$, $R_3 = 20 \Omega$, $R_4 = 5 \Omega$, $R_5 = 25 \Omega$, $R_6 = 30 \Omega$, $E = 15 \text{ V}$



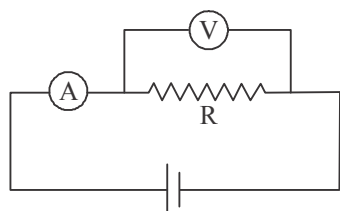
- (a) $13/24$ (b) $7/18$ (c) $9/32$ (d) $20/3$
34. A uniform metallic wire has a resistance of 18Ω and is bent into an equilateral triangle. Then, the resistance between any two vertices of the triangle is:

[10 Jan. 2019 I]

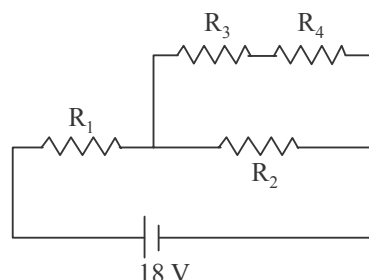
- (a) 4Ω (b) 8Ω (c) 12 W (d) 2 W
35. The actual value of resistance R , shown in the figure is 30Ω . This is measured in an experiment as shown using the

standard formula $R = \frac{V}{I}$, where V and I are the reading of the voltmeter and ammeter, respectively. If the measured value of R is 5% less, then the internal resistance of the voltmeter is:

[10 Jan. 2019 II]

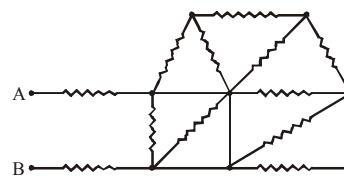


- (a) 600Ω (b) 570Ω (c) 35 W (d) 350 W
36. In the given circuit the internal resistance of the 18 V cell is negligible. If $R_1 = 400 \Omega$, $R_3 = 100 \Omega$ and $R_4 = 500 \Omega$ and the reading of an ideal voltmeter across R_4 is 5 V , then the value of R_2 will be: [9 Jan. 2019 II]



- (a) 300 W (b) 450 W
(c) 550 W (d) 230 W
37. In the given circuit all resistances are of value R ohm each. The equivalent resistance between A and B is :

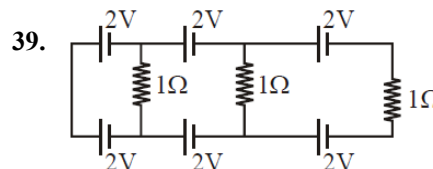
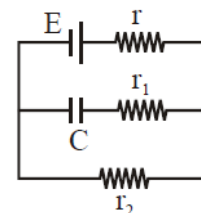
[Online April 15, 2018]



- (a) $2R$ (b) $\frac{5R}{2}$ (c) $\frac{5R}{3}$ (d) $3R$

38. In the given circuit diagram when the current reaches steady state in the circuit, the charge on the capacitor of capacitance C will be : [2017]

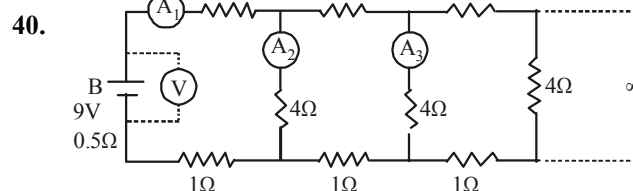
- (a) $CE \frac{r_2}{(r_1 + r_2)}$
(b) $CE \frac{r_1}{(r_1 + r)}$
(c) CE
(d) $CE \frac{r_1}{(r_2 + r)}$



In the above circuit the current in each resistance is

[2017]

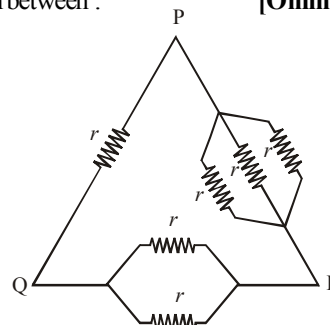
- (a) 0.5 A (b) 0 A (c) 1 A (d) 0.25 A



A 9 V battery with internal resistance of 0.5Ω is connected across an infinite network as shown in the figure. All ammeters A_1 , A_2 , A_3 and voltmeter V are ideal.

Choose correct statement. [Online April 8, 2017]

- (a) Reading of A_1 is 2 A (b) Reading of A_1 is 18 A
(c) Reading of V is 9 V (d) Reading of V is 7 V
41. Six equal resistances are connected between points P , Q and R as shown in figure. Then net resistance will be maximum between : [Online April 25, 2013]



- (a) P and R (b) P and Q
(c) Q and R (d) Any two points

42. A letter 'A' is constructed of a uniform wire with resistance 1.0Ω per cm. The sides of the letter are 20 cm and the cross piece in the middle is 10 cm long. The apex angle is 60° . The resistance between the ends of the legs is close to:

[Online April 9, 2013]

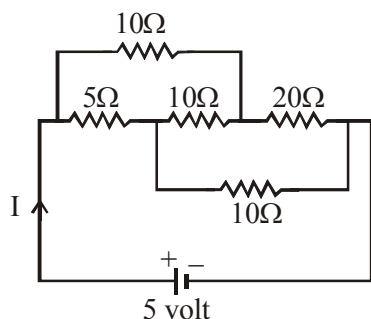
- (a) 50.0Ω (b) 10Ω (c) 36.7Ω (d) 26.7Ω
43. Two conductors have the same resistance at 0°C but their temperature coefficients of resistance are α_1 and α_2 . The respective temperature coefficients of their series and parallel combinations are nearly

[2010]

- (a) $\frac{\alpha_1 + \alpha_2}{2}, \alpha_1 + \alpha_2$ (b) $\alpha_1 + \alpha_2, \frac{\alpha_1 + \alpha_2}{2}$
 (c) $\alpha_1 + \alpha_2, \frac{\alpha_1 \alpha_2}{\alpha_1 + \alpha_2}$ (d) $\frac{\alpha_1 + \alpha_2}{2}, \frac{\alpha_1 + \alpha_2}{2}$

44. The current I drawn from the 5 volt source will be

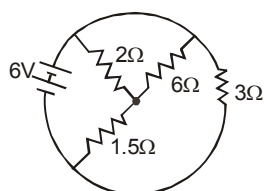
[2006]



- (a) 0.33 A (b) 0.5 A (c) 0.67 A (d) 0.17 A

45. The total current supplied to the circuit by the battery is

[2004]



- (a) 4 A (b) 2 A (c) 1 A (d) 6 A

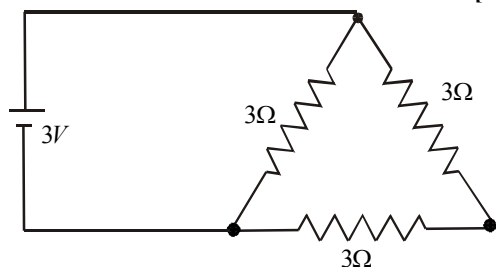
46. The resistance of the series combination of two resistances is S . When they are joined in parallel the total resistance is P . If $S = nP$ then the minimum possible value of n is

[2004]

- (a) 2 (b) 3 (c) 4 (d) 1

47. A 3 volt battery with negligible internal resistance is connected in a circuit as shown in the figure. The current I , in the circuit will be

[2003]

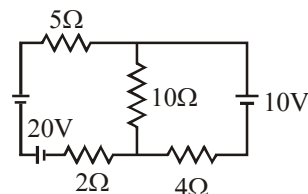


- (a) 1 A (b) 1.5 A (c) 2 A (d) $1/3$ A

TOPIC 3 Kirchhoff's Laws, Cells, Thermo e.m.f. & Electrolysis



48.



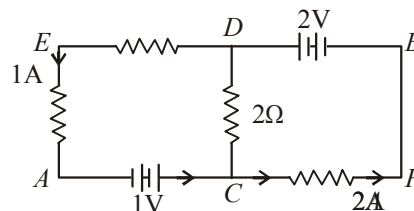
In the figure shown, the current in the 10 V battery is close to :

[Sep. 06, 2020 (II)]

- (a) 0.71 A from positive to negative terminal
 (b) 0.42 A from positive to negative terminal
 (c) 0.21 A from positive to negative terminal
 (d) 0.36 A from negative to positive terminal

49. In the circuit, given in the figure currents in different branches and value of one resistor are shown. Then potential at point B with respect to the point A is :

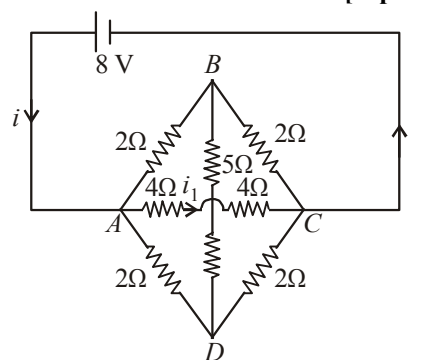
[Sep. 05, 2020 (II)]



- (a) +2 V (b) -2 V (c) -1 V (d) +1 V

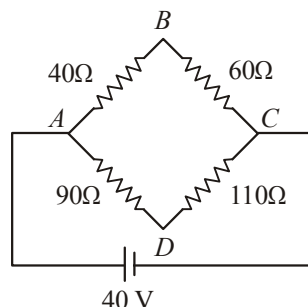
50. The value of current i_1 flowing from A to C in the circuit diagram is :

[Sep. 04, 2020 (II)]



- (a) 2 A (b) 4 A (c) 1 A (d) 5 A

51.

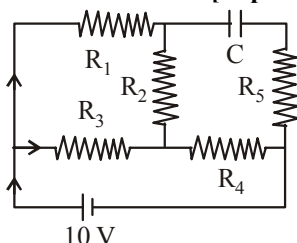


Four resistances 40Ω , 60Ω , 90Ω and 110Ω make the arms of a quadrilateral $ABCD$. Across AC is a battery of emf 40 V and internal resistance negligible. The potential difference across BD in V is _____.

[NA. Sep. 04, 2020 (II)]

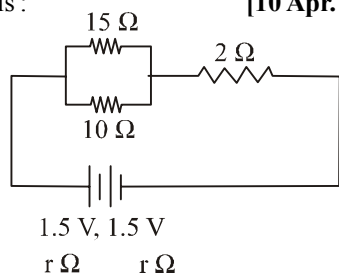
52. An ideal cell of emf 10 V is connected in circuit shown in figure. Each resistance is $2\ \Omega$. The potential difference (in V) across the capacitor when it is fully charged is _____.

[Sep. 02, 2020 (II)]



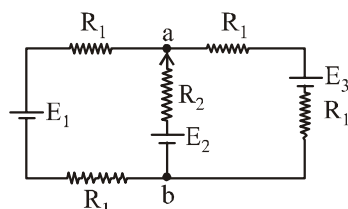
53. In the given circuit, an ideal voltmeter connected across the $10\ \Omega$ resistance reads 2V. The internal resistance r , of each cell is : _____

[10 Apr. 2019 I]



- (a) $1\ \Omega$ (b) $0.5\ \Omega$ (c) $1.5\ \Omega$ (d) $0\ \Omega$
54. For the circuit shown, with $R_1 = 1.0\ \Omega$, $R_2 = 2.0\ \Omega$, $E_1 = 2\text{ V}$ and $E_2 = E_3 = 4\text{ V}$, the potential difference between the points 'a' and 'b' is approximately (in V) : _____

[8 April 2019 I]

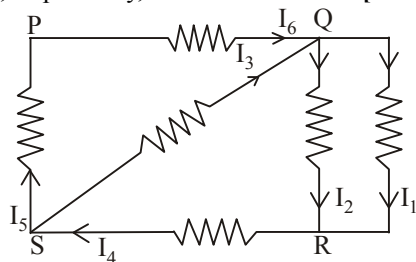


- (a) 2.7 (b) 2.3 (c) 3.7 (d) 3.3
55. A cell of internal resistance r drives current through an external resistance R . The power delivered by the cell to the external resistance will be maximum when : _____

[8 Apr. 2019 II]

- (a) $R = 0.001 r$ (b) $R = 1000 r$
(c) $R = 2r$ (d) $R = r$
56. In the given circuit diagram, the currents, $I_1 = -0.3\text{ A}$, $I_4 = 0.8\text{ A}$ and $I_5 = 0.4\text{ A}$, are flowing as shown. The currents I_2 , I_3 and I_6 , respectively, are : _____

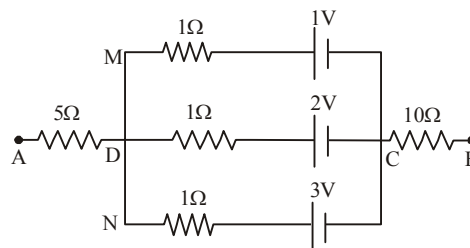
[12 Jan. 2019 II]



- (a) 1.1 A, -0.4 A , 0.4 A (b) 1.1 A, 0.4 A , 0.4 A
(c) 0.4 A , 1.1 A, 0.4 A (d) -0.4 A , 0.4 A , 1.1 A

57. In the circuit shown, the potential difference between A and B is : _____

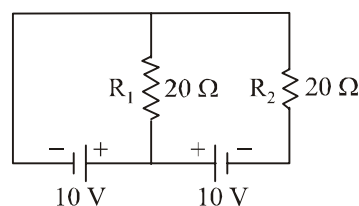
[11 Jan. 2019 II]



- (a) 1V (b) 2V (c) 3V (d) 6V

58. In the given circuit the cells have zero internal resistance. The currents (in Amperes) passing through resistance R_1 and R_2 respectively, are: _____

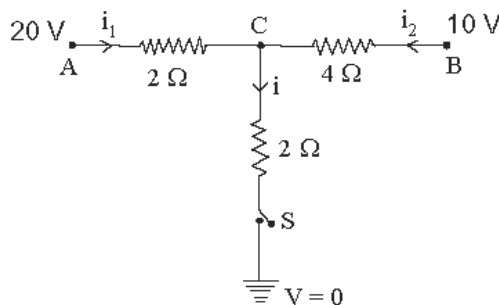
[10 Jan. 2019 I]



- (a) 1, 2 (b) 2, 2 (c) 0.5, 0 (d) 0, 1

59. When the switch S , in the circuit shown, is closed then the value of current i will be: _____

[9 Jan. 2019 I]



- (a) 3A (b) 5A (c) 4A (d) 2A

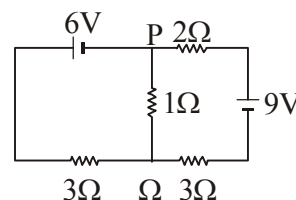
60. Two batteries with e.m.f. 12 V and 13 V are connected in parallel across a load resistor of $10\ \Omega$. The internal resistances of the two batteries are $1\ \Omega$ and $2\ \Omega$ respectively. The voltage across the load lies between: _____

[2018]

- (a) 11.6 V and 11.7 V (b) 11.5 V and 11.6 V
(c) 11.4 V and 11.5 V (d) 11.7 V and 11.8 V

61. In the circuit shown, the current in the $1\ \Omega$ resistor is: _____

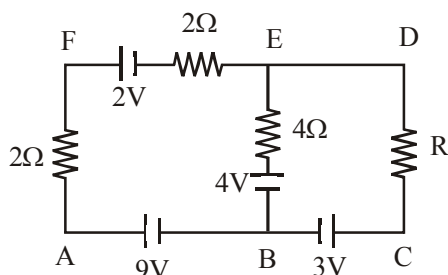
[2015]



- (a) 0.13 A, from Q to P (b) 0.13 A, from P to Q
(c) 1.3 A from P to Q (d) 0A

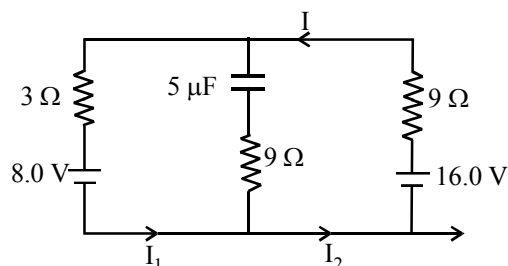
62. In the electric network shown, when no current flows through the 4Ω resistor in the arm EB, the potential difference between the points A and D will be :

[Online April 11, 2015]



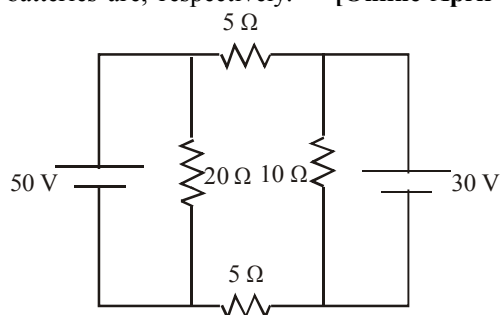
- (a) 6 V (b) 3 V (c) 5 V (d) 4 V
63. The circuit shown here has two batteries of 8.0 V and 16.0 V and three resistors 3Ω , 9Ω and 9Ω and a capacitor of $5.0\mu\text{F}$.

[Online April 12, 2014]



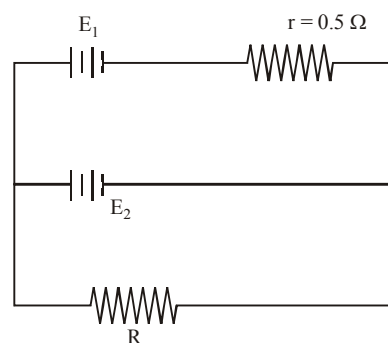
How much is the current I in the circuit in steady state?

- (a) 1.6 A (b) 0.67 A
(c) 2.5 A (d) 0.25 A
64. In the circuit shown, current (in A) through 50 V and 30 V batteries are, respectively. [Online April 11, 2014]

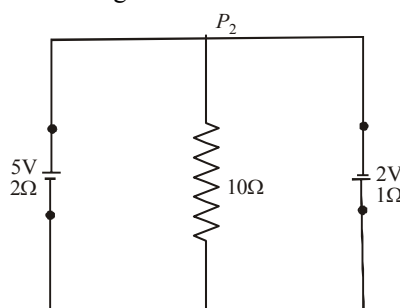


- (a) 2.5 and 3 (b) 3.5 and 2
(c) 4.5 and 1 (d) 3 and 2.5
65. A d.c. main supply of e.m.f. 220 V is connected across a storage battery of e.m.f. 200 V through a resistance of 1Ω . The battery terminals are connected to an external resistance 'R'. The minimum value of 'R', so that a current passes through the battery to charge it is: [Online April 9, 2014]
- (a) 7Ω (b) 9Ω (c) 11Ω (d) Zero
66. A dc source of emf $E_1 = 100\text{ V}$ and internal resistance $r = 0.5\Omega$, a storage battery of emf $E_2 = 90\text{ V}$ and an external resistance R are connected as shown in figure. For what value of R no current will pass through the battery ?

[Online April 22, 2013]



- (a) 5.5Ω (b) 3.5Ω
(c) 4.5Ω (d) 2.5Ω
67. A 5V battery with internal resistance 2Ω and a 2V battery with internal resistance 1Ω are connected to a 10Ω resistor as shown in the figure. [2008]



The current in the 10Ω resistor is

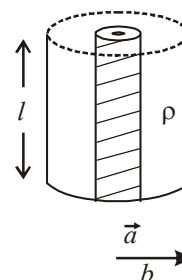
- (a) $0.27\text{ A } P_2 \text{ to } P_1$ (b) $0.03\text{ A } P_1 \text{ to } P_2$
(c) $0.03\text{ A } P_2 \text{ to } P_1$ (d) $0.27\text{ A } P_1 \text{ to } P_2$
68. A battery is used to charge a parallel plate capacitor till the potential difference between the plates becomes equal to the electromotive force of the battery. The ratio of the energy stored in the capacitor and the work done by the battery will be [2007]
- (a) $1/2$ (b) 1 (c) 2 (d) $1/4$
69. The Kirchhoff's first law ($\sum i = 0$) and second law ($\sum iR = \sum E$), where the symbols have their usual meanings, are respectively based on [2006]
- (a) conservation of charge, conservation of momentum
(b) conservation of energy, conservation of charge
(c) conservation of momentum, conservation of charge
(d) conservation of charge, conservation of energy
70. A thermocouple is made from two metals, Antimony and Bismuth. If one junction of the couple is kept hot and the other is kept cold, then, an electric current will [2006]
- (a) flow from Antimony to Bismuth at the hot junction
(b) flow from Bismuth to Antimony at the cold junction
(c) now flow through the thermocouple
(d) flow from Antimony to Bismuth at the cold junction
71. Two sources of equal emf are connected to an external resistance R . The internal resistance of the two sources are R_1 and R_2 ($R_1 > R_2$). If the potential difference across the source having internal resistance R_2 is zero, then [2005]

- (a) $R = R_2 - R_1$
 (b) $R = R_2 \times (R_1 + R_2) / (R_2 - R_1)$
 (c) $R = R_1 R_2 / (R_2 - R_1)$
 (d) $R = R_1 R_2 / (R_1 - R_2)$
72. Two voltmeters, one of copper and another of silver, are joined in parallel. When a total charge q flows through the voltmeters, equal amount of metals are deposited. If the electrochemical equivalents of copper and silver are Z_1 and Z_2 respectively the charge which flows through the silver voltmeter is [2005]
- (a) $\frac{q}{1 + \frac{Z_2}{Z_1}}$ (b) $\frac{q}{1 + \frac{Z_1}{Z_2}}$ (c) $q \frac{Z_2}{Z_1}$ (d) $q \frac{Z_1}{Z_2}$
73. An energy source will supply a constant current into the load if its internal resistance is [2005]
- (a) very large as compared to the load resistance
 (b) equal to the resistance of the load
 (c) non-zero but less than the resistance of the load
 (d) zero
74. The thermo emf of a thermocouple varies with the temperature θ of the hot junction as $E = a\theta + b\theta^2$ in volts where the ratio a/b is 700°C . If the cold junction is kept at 0°C , then the neutral temperature is [2004]
- (a) 1400°C
 (b) 350°C
 (c) 700°C
 (d) No neutral temperature is possible for this thermocouple.
75. The electrochemical equivalent of a metal is 3.35×10^{-7} kg per Coulomb. The mass of the metal liberated at the cathode when a 3A current is passed for 2 seconds will be [2004]
- (a) $6.6 \times 10^{-5} \text{ kg}$ (b) $9.9 \times 10^{-7} \text{ kg}$
 (c) $19.8 \times 10^{-7} \text{ kg}$ (d) $1.1 \times 10^{-7} \text{ kg}$
76. The thermo e.m.f. of a thermo-couple is $25 \mu\text{V}/^\circ\text{C}$ at room temperature. A galvanometer of 40 ohm resistance, capable of detecting current as low as 10^{-5} A, is connected with the thermo couple. The smallest temperature difference that can be detected by this system is [2003]
- (a) 16°C (b) 12°C (c) 8°C (d) 20°C
77. The negative Zn pole of a Daniell cell, sending a constant current through a circuit, decreases in mass by 0.13g in 30 minutes. If the electrochemical equivalent of Zn and Cu are 32.5 and 31.5 respectively, the increase in the mass of the positive Cu pole in this time is [2003]
- (a) 0.180g (b) 0.141g (c) 0.126g (d) 0.242g
78. The mass of product liberated on anode in an electrochemical cell depends on [2002]
- (a) $(It)^{1/2}$ (b) It (c) I/t (d) $I^2 t$
 (where t is the time period for which the current is passed).

TOPIC 4 Heating Effect of Current



79. An electrical power line, having a total resistance of 2Ω , delivers 1 kW at 220 V. The efficiency of the transmission line is approximately : [Sep. 05, 2020 (I)]
- (a) 72% (b) 91% (c) 85% (d) 96%
80. Model a torch battery of length l to be made up of a thin cylindrical bar of radius ' a ' and a concentric thin cylindrical shell of radius ' b ' filled in between with an electrolyte of resistivity ρ (see figure). If the battery is connected to a resistance of value R , the maximum Joule heating in R will take place for : [Sep. 03, 2020 (I)]



- (a) $R = \frac{\rho}{2\pi l} \left(\frac{b}{a} \right)$ (b) $R = \frac{\rho}{2\pi l} \ln \left(\frac{b}{a} \right)$
 (c) $R = \frac{\rho}{\pi l} \ln \left(\frac{b}{a} \right)$ (d) $R = \frac{2\rho}{\pi l} \ln \left(\frac{b}{a} \right)$
81. In a building there are 15 bulbs of 45 W, 15 bulbs of 100 W, 15 small fans of 10 W and 2 heaters of 1 kW. The voltage of electric main is 220 V. The minimum fuse capacity (rated value) of the building will be: [7 Jan. 2020 II]
- (a) 10 A (b) 25 A (c) 15 A (d) 20 A
82. The resistive network shown below is connected to a D.C. source of 16 V. The power consumed by the network is 4 Watt. The value of R is : [12 Apr. 2019 I]
-
- (a) 6Ω (b) 8Ω (c) 1Ω (d) 16Ω
83. One kg of water, at 20°C , is heated in an electric kettle whose heating element has a mean (temperature averaged) resistance of 20Ω . The rms voltage in the mains is 200 V. Ignoring heat loss from the kettle, time taken for water to evaporate fully, is close to : [Specific heat of water = $4200 \text{ J}/(\text{kg}^\circ\text{C})$, Latent heat of water = 2260 kJ/kg] [12 Apr. 2019 II]
- (a) 16 minutes (b) 22 minutes
 (c) 3 minutes (d) 3 minutes

84. Two electric bulbs, rated at (25 W, 220 V) and (100 W, 220 V), are connected in series across a 220 V voltage source. If the 25 W and 100 W bulbs draw powers P_1 and P_2 respectively, then: [12 Jan. 2019 I]

(a) $P_1=16$ W, $P_2=4$ W (b) $P_1=16$ W, $P_2=9$ W
(c) $P_1=9$ W, $P_2=16$ W (d) $P_1=4$ W, $P_2=16$ W

85. Two equal resistances when connected in series to a battery, consume electric power of 60 W. If these resistance are now connected in parallel combination to the same battery, the electric power consumed will be: [11 Jan. 2019 I]

(a) 60 W (b) 240 W (c) 120 W (d) 30 W

86. A 2 W carbon resistor is color coded with green, black, red and brown respectively. The maximum current which can be passed through this resistor is: [10 Jan. 2019 I]

(a) 20 mA (b) 100 mA (c) 0.4 mA (d) 63 mA

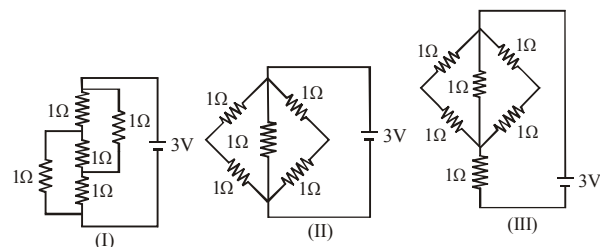
87. A current of 2 mA was passed through an unknown resistor which dissipated a power of 4.4 W. Dissipated power when an ideal power supply of 11 V is connected across it is: [10 Jan. 2019 II]

(a) 11×10^{-5} W (b) 11×10^{-3} W
(c) 11×10^{-4} W (d) 11×10^5 W

88. A constant voltage is applied between two ends of a metallic wire. If the length is halved and the radius of the wire is doubled, the rate of heat developed in the wire will be: [Online April 15, 2018]

(a) Increased 8 times (b) Doubled
(c) Halved (d) Unchanged

89. The figure shows three circuits I, II and III which are connected to a 3V battery. If the powers dissipated by the configurations I, II and III are P_1 , P_2 and P_3 respectively, then: [Online April 9, 2017]

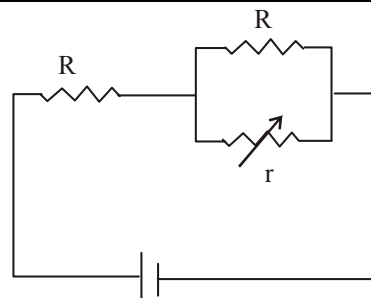


(a) $P_1 > P_2 > P_3$ (b) $P_1 > P_3 > P_2$
(c) $P_2 > P_1 > P_3$ (d) $P_3 > P_2 > P_1$

90. The resistance of an electrical toaster has a temperature dependence given by $R(T) = R_0 [1 + \alpha(T - T_0)]$ in its range of operation. At $T_0 = 300$ K, $R = 100 \Omega$ and at $T = 500$ K, $R = 120 \Omega$. The toaster is connected to a voltage source at 200 V and its temperature is raised at a constant rate from 300 to 500 K in 30 s. The total work done in raising the temperature is: [Online April 10, 2016]

(a) $400 \ln \frac{5}{6}$ J (b) $200 \ln \frac{2}{3}$ J
(c) 300 J (d) $400 \ln \frac{1.5}{1.3}$ J

91.



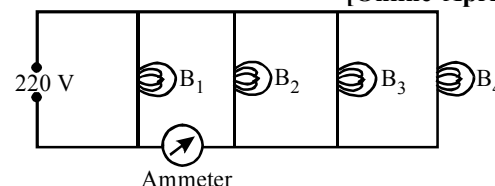
In the circuit shown, the resistance r is a variable resistance. If for $r = fR$, the heat generation in r is maximum then the value of f is: [Online April 9, 2016]

(a) $\frac{1}{2}$ (b) 1 (c) $\frac{1}{4}$ (d) $\frac{3}{4}$

92. In a large building, there are 15 bulbs of 40 W, 5 bulbs of 100 W, 5 fans of 80 W and 1 heater of 1 kW. The voltage of electric mains is 220 V. The minimum capacity of the main fuse of the building will be: [2014]

(a) 8 A (b) 10 A (c) 12 A (d) 14 A

93. Four bulbs B_1 , B_2 , B_3 and B_4 of 100 W each are connected to 220 V main as shown in the figure. [Online April 19, 2014]



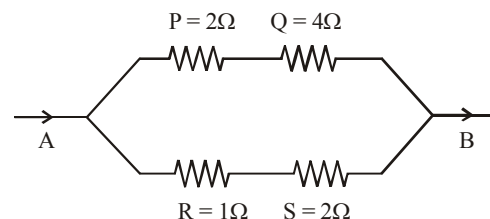
The reading in an ideal ammeter will be:

(a) 0.45 A (b) 0.90 A (c) 1.35 A (d) 1.80 A

94. The supply voltage to room is 120V. The resistance of the lead wires is 6Ω . A 60 W bulb is already switched on. What is the decrease of voltage across the bulb, when a 240 W heater is switched on in parallel to the bulb? [2013]

(a) zero (b) 2.9 Volt
(c) 13.3 Volt (d) 10.04 Volt

95. Which of the four resistances P, Q, R and S generate the greatest amount of heat when a current flows from A to B? [Online April 23, 2013]

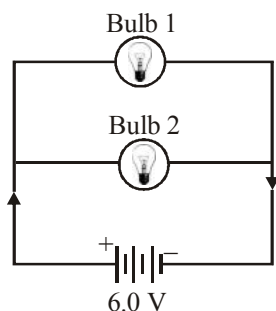


(a) Q (b) S (c) P (d) R

96. Two electric bulbs rated 25W – 220 V and 100W – 220V are connected in series to a 440 V supply. Which of the bulbs will fuse? [2012]

(a) Both (b) 100 W (c) 25 W (d) Neither

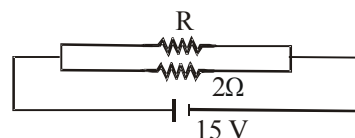
97. A 6.0 volt battery is connected to two light bulbs as shown in figure. Light bulb 1 has resistance 3 ohm while light bulb 2 has resistance 6 ohm. Battery has negligible internal resistance. Which bulb will glow brighter? [Online May 19, 2012]



- (a) Bulb 1 will glow more first and then its brightness will become less than bulb 2
 (b) Bulb 1
 (c) Bulb 2
 (d) Both glow equally
98. Three resistors of $4\ \Omega$, $6\ \Omega$ and $12\ \Omega$ are connected in parallel and the combination is connected in series with a $1.5\ \text{V}$ battery of $1\ \Omega$ internal resistance. The rate of Joule heating in the $4\ \Omega$ resistor is [Online May 12, 2012]
 (a) $0.55\ \text{W}$ (b) $0.33\ \text{W}$ (c) $0.25\ \text{W}$ (d) $0.86\ \text{W}$
99. This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement 1: The possibility of an electric bulb fusing is higher at the time of switching ON.
Statement 2: Resistance of an electric bulb when it is not lit up is much smaller than when it is lit up.
 [Online May 7, 2012]
 (a) Statement 1 is true, Statement 2 is false
 (b) Statement 1 is false, Statement 2 is true, Statement 2 is not a correct explanation of Statement 1.
 (c) Statement 1 is true, Statement 2 is true, Statement 2 is a correct explanation of Statement 1.
 (d) Statement 1 is false, Statement 2 is true.
100. The resistance of a bulb filament is $100\ \Omega$ at a temperature of 100°C . If its temperature coefficient of resistance be $0.005\ \text{per } ^\circ\text{C}$, its resistance will become $200\ \Omega$ at a temperature of [2006]
 (a) 300°C (b) 400°C (c) 500°C (d) 200°C
101. An electric bulb is rated $220\ \text{V} - 100\ \text{W}$. The power consumed by it when operated on $110\ \text{V}$ will be [2006]
 (a) $75\ \text{W}$ (b) $40\ \text{W}$ (c) $25\ \text{W}$ (d) $50\ \text{W}$
102. A heater coil is cut into two equal parts and only one part is now used in the heater. The heat generated will now be [2005]
 (a) four times (b) doubled
 (c) halved (d) one fourth
103. The resistance of hot tungsten filament is about 10 times the cold resistance. What will be the resistance of $100\ \text{W}$ and $200\ \text{V}$ lamp when not in use? [2005]
 (a) $20\ \Omega$ (b) $40\ \Omega$ (c) $200\ \Omega$ (d) $400\ \Omega$
104. The thermistors are usually made of [2004]
 (a) metal oxides with high temperature coefficient of resistivity
 (b) metals with high temperature coefficient of resistivity

- (c) metals with low temperature coefficient of resistivity
 (d) semiconducting materials having low temperature coefficient of resistivity

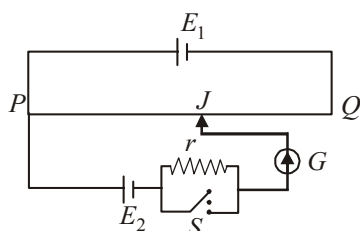
105. Time taken by a $836\ \text{W}$ heater to heat one litre of water from 10°C to 40°C is [2004]
 (a) $150\ \text{s}$ (b) $100\ \text{s}$ (c) $50\ \text{s}$ (d) $200\ \text{s}$
106. A $220\ \text{V}$, $1000\ \text{W}$ bulb is connected across a $110\ \text{V}$ mains supply. The power consumed will be [2003]
 (a) $750\ \text{W}$ (b) $500\ \text{W}$
 (c) $250\ \text{W}$ (d) $1000\ \text{W}$
107. A wire when connected to $220\ \text{V}$ mains supply has power dissipation P_1 . Now the wire is cut into two equal pieces which are connected in parallel to the same supply. Power dissipation in this case is P_2 . Then $P_2 : P_1$ is [2002]
 (a) 1 (b) 4 (c) 2 (d) 3
108. If in the circuit, power dissipation is $150\ \text{W}$, then R is [2002]



- (a) $2\ \Omega$ (b) $6\ \Omega$ (c) $5\ \Omega$ (d) $4\ \Omega$

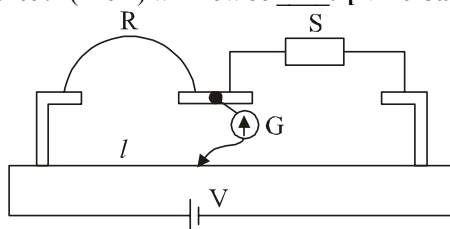
TOPIC 5 Wheatstone Bridge and Different Measuring Instruments

109. Two resistors $400\ \Omega$ and $800\ \Omega$ are connected in series across a $6\ \text{V}$ battery. The potential difference measured by a voltmeter of $10\ \text{k}\Omega$ across $400\ \Omega$ resistor is close to: [Sep. 03, 2020 (II)]
 (a) $2\ \text{V}$ (b) $1.8\ \text{V}$ (c) $2.05\ \text{V}$ (d) $1.95\ \text{V}$
110. Which of the following will NOT be observed when a multimeter (operating in resistance measuring mode) probes connected across a component, are just reversed? [Sep. 03, 2020 (II)]
 (a) Multimeter shows an equal deflection in both cases i.e. before and after reversing the probes if the chosen component is resistor.
 (b) Multimeter shows NO deflection in both cases i.e. before and after reversing the probes if the chosen component is capacitor.
 (c) Multimeter shows a deflection, accompanied by a splash of light out of connected and NO deflection on reversing the probes if the chosen component is LED.
 (d) Multimeter shows NO deflection in both cases i.e. before and after reversing the probes if the chosen component is metal wire.
111. A potentiometer wire PQ of $1\ \text{m}$ length is connected to a standard cell E_1 . Another cell E_2 of emf $1.02\ \text{V}$ is connected with a resistance ' r ' and switch S (as shown in figure). With switch S open, the null position is obtained at a distance of $49\ \text{cm}$ from Q . The potential gradient in the potentiometer wire is : [Sep. 02, 2020 (II)]



- (a) 0.02 V/cm (b) 0.01 V/cm
(c) 0.03 V/cm (d) 0.04 V/cm

112. In a meter bridge experiment S is a standard resistance. R is a resistance wire. It is found that balancing length is $l = 25$ cm. If R is replaced by a wire of half length and half diameter that of R of same material, then the balancing distance l' (in cm) will now be _____. [NA. 9 Jan. 2020 II]



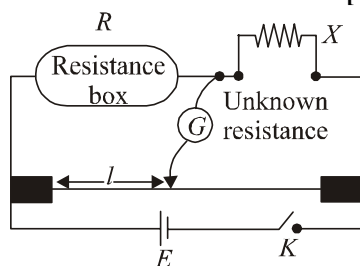
113. The length of a potentiometer wire is 1200 cm and it carries a current of 60 mA. For a cell of emf 5 V and internal resistance of 20Ω , the null point on it is found to be at 1000 cm. The resistance of whole wire is: [8 Jan. 2020 I]

- (a) 80Ω (b) 120Ω (c) 60Ω (d) 100Ω

114. Four resistances of 15Ω , 12Ω , 4Ω and 10Ω respectively in cyclic order to form Wheatstone's network. The resistance that is to be connected in parallel with the resistance of 10Ω to balance the network is _____ Ω . [NA. 8 Jan. 2020 I]

115. The balancing length for a cell is 560 cm in a potentiometer experiment. When an external resistance of 10Ω is connected in parallel to the cell, the balancing length changes by 60 cm. If the internal resistance of the cell is $\frac{N}{10} \Omega$, where N is an integer then value of N is _____. [NA. 7 Jan. 2020 II]

116. In a meter bridge experiment, the circuit diagram and the corresponding observation table are shown in figure. [10 Apr. 2019 I]

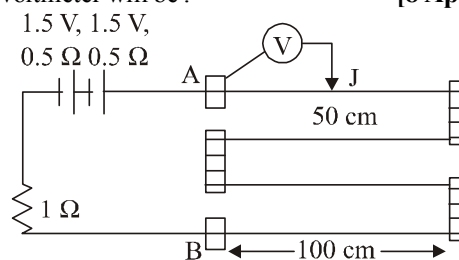


Sl.No.	$R \Omega$	l (cm)
1.	1000	60
2.	100	13
3.	10	1.5
4.	1	1.0

Which of the reading is consistent ?

- (a) 3 (b) 2 (c) 4 (d) 1

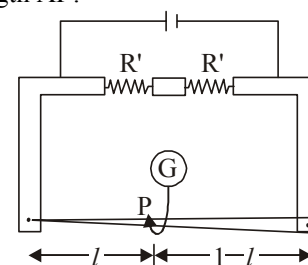
117. In the circuit shown, a four-wire potentiometer is made of a 400 cm long wire, which extends between A and B. The resistance per unit length of the potentiometer wire is $r = 0.01 \Omega/\text{cm}$. If an ideal voltmeter is connected as shown with jockey J at 50 cm from end A, the expected reading of the voltmeter will be : [8 Apr. 2019 II]



- (a) 0.50 V (b) 0.75 V (c) 0.25 V (d) 0.20 V

118. In a meter bridge, the wire of length 1 m has a non-uniform cross-section such that, the variation $\frac{dR}{dl}$ of its resistance

R with length l is $\frac{dR}{dl} \propto \frac{1}{\sqrt{l}}$. Two equal resistances are connected as shown in the figure. The galvanometer has zero deflection when the jockey is at point P. What is the length AP? [12 Jan. 2019 I]

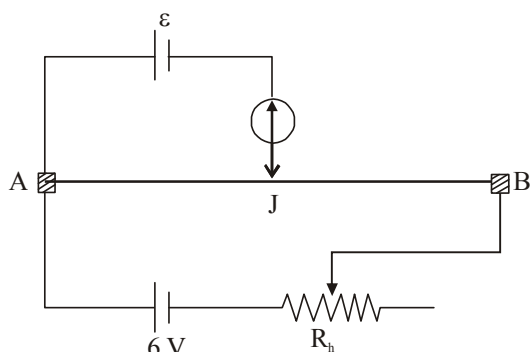


- (a) 0.2 m (b) 0.3 m (c) 0.25 m (d) 0.35 m

119. An ideal battery of 4 V and resistance R are connected in series in the primary circuit of a potentiometer of length 1 m and resistance 5Ω . The value of R, to give a potential difference of 5 mV across 10 cm of potentiometer wire is: [12 Jan. 2019 I]

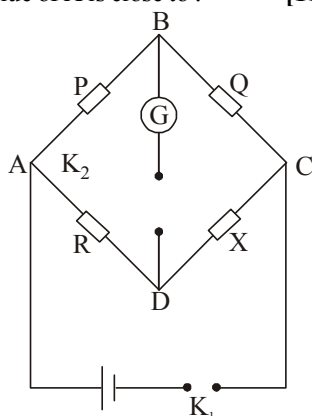
- (a) 490Ω (b) 480Ω (c) 395Ω (d) 495Ω

120. The resistance of the meter bridge AB in given figure is 4Ω . With a cell of emf $\epsilon = 0.5$ V and rheostat resistance $R_h = 2 \Omega$ the null point is obtained at some point J. When the cell is replaced by another one of emf $\epsilon = \epsilon_2$ the same null point J is found for $R_h = 6 \Omega$. The emf ϵ_2 is: [11 Jan. 2019 I]



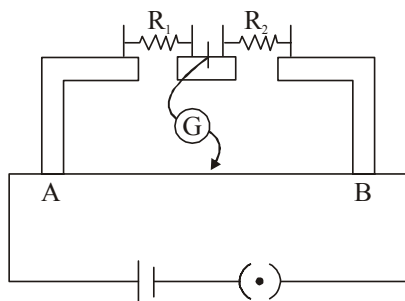
- (a) 0.4V (b) 0.3V (c) 0.6V (d) 0.5V

121. In a Wheatstone bridge (see fig.), Resistances P and Q are approximately equal. When $R = 400\ \Omega$, the bridge is balanced. On interchanging P and Q, the value of R, for balance, is $405\ \Omega$. The value of X is close to : [11 Jan. 2019 I]



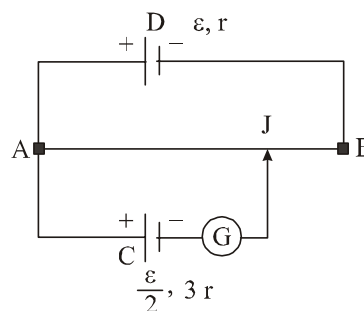
- (a) 401.5 ohm (b) 404.5 ohm
(c) 403.5 ohm (d) 402.5 ohm

122. In the experimental set up of metre bridge shown in the figure, the null point is obtained at a distance of 40 cm from A. If a $10\ \Omega$ resistor is connected in series with R_1 , the null point shifts by 10 cm. The resistance that should be connected in parallel with $(R_1 + 10)\ \Omega$ such that the null point shifts back to its initial position is : [11 Jan. 2019 II]



- (a) 20 Ω (b) 40 Ω (c) 60 Ω (d) 30 Ω

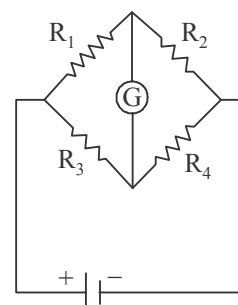
123. A potentiometer wire AB having length L and resistance $12r$ is joined to a cell D of emf ε and internal resistance r . A cell C having emf $\varepsilon/2$ and internal resistance $3r$ is connected. The length AJ at which the galvanometer as shown in fig. shows no deflection is: [10 Jan. 2019 I]



- (a) $\frac{11}{12}L$ (b) $\frac{11}{24}L$ (c) $\frac{13}{24}L$ (d) $\frac{5}{12}L$

124. The Wheatstone bridge shown in Fig. here, gets balanced when the carbon resistor used as R_1 has the colour code (Orange, Red, Brown). The resistors R_2 and R_4 are $80\ \Omega$ and $40\ \Omega$, respectively.

Assuming that the colour code for the carbon resistors gives their accurate values, the colour code for the carbon resistor, used as R_3 , would be: [10 Jan. 2019 II]



- (a) Brown, Blue, Brown (b) Brown, Blue, Black
(c) Red, Green, Brown (d) Grey, Black, Brown

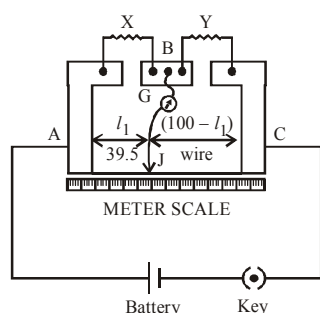
125. In a potentiometer experiment, it is found that no current passes through the galvanometer when the terminals of the cell are connected across 52 cm of the potentiometer wire. If the cell is shunted by a resistance of $5\ \Omega$, a balance is found when the cell is connected across 40 cm of the wire. Find the internal resistance of the cell. [2018]

- (a) $1\ \Omega$ (b) $1.5\ \Omega$ (c) $2\ \Omega$ (d) $2.5\ \Omega$

126. On interchanging the resistances, the balance point of a meter bridge shifts to the left by 10 cm. The resistance of their series combination is $1\text{ k}\Omega$. How much was the resistance on the left slot before interchanging the resistances? [2018]

- (a) $990\ \Omega$ (b) $505\ \Omega$ (c) $550\ \Omega$ (d) $910\ \Omega$

127. In a meter bridge, as shown in the figure, it is given that resistance $Y = 12.5\ \Omega$ and that the balance is obtained at a distance 39.5 cm from end A (by jockey J). After interchanging the resistances X and Y, a new balance point is found at a distance l_2 from end A. What are the values of X and l_2 ? [Online April 15, 2018]

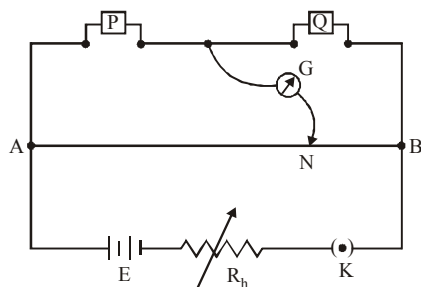


- (a) 19.15Ω and 39.5 cm (b) 8.16Ω and 60.5 cm
 (c) 19.15Ω and 60.5 cm (d) 8.16Ω and 39.5 cm

128. Which of the following statements is false ? [2017]

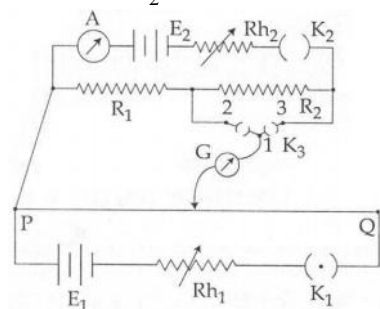
- (a) A rheostat can be used as a potential divider
 (b) Kirchhoff's second law represents energy conservation
 (c) Wheatstone bridge is the most sensitive when all the four resistances are of the same order of magnitude
 (d) In a balanced wheatstone bridge if the cell and the galvanometer are exchanged, the null point is disturbed.

129. In a meter bridge experiment resistances are connected as shown in the figure. Initially resistance $P = 4 \Omega$ and the neutral point N is at 60 cm from A. Now an unknown resistance R is connected in series to P and the new position of the neutral point is at 80 cm from A. The value of unknown resistance R is : [Online April 9, 2017]



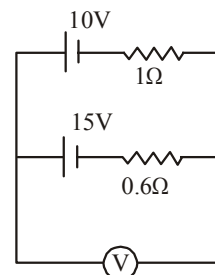
- (a) $\frac{33}{5} \Omega$ (b) 6Ω (c) 7Ω (d) $\frac{20}{3} \Omega$

130. A potentiometer PQ is set up to compare two resistances as shown in the figure. The ammeter A in the circuit reads 1.0 A when two way key K_3 is open. The balance point is at a length $l_1 \text{ cm}$ from P when two way key K_3 is plugged in between 2 and 1, while the balance point is at a length $l_2 \text{ cm}$ from P when key K_3 is plugged in between 3 and 1. The ratio of two resistances $\frac{R_1}{R_2}$ is found to be : [Online April 8, 2017]



- (a) $\frac{l_1}{l_1 + l_2}$ (b) $\frac{l_2}{l_2 - l_1}$ (c) $\frac{l_1}{l_1 - l_2}$ (d) $\frac{l_1}{l_2 - l_1}$

131. A 10 V battery with internal resistance 1Ω and a 15 V battery with internal resistance 0.6Ω are connected in parallel to a voltmeter (see figure). The reading in the voltmeter will be close to : [Online April 10, 2015]



- (a) 12.5 V (b) 24.5 V (c) 13.1 V (d) 11.9 V

132. In an experiment of potentiometer for measuring the internal resistance of primary cell a balancing length ℓ is obtained on the potentiometer wire when the cell is open circuit. Now the cell is short circuited by a resistance R. If R is to be equal to the internal resistance of the cell the balancing length on the potentiometer wire will be [Online May 26, 2012]

- (a) ℓ (b) 2ℓ (c) $\ell/2$ (d) $\ell/4$

133. It is preferable to measure the e.m.f. of a cell by potentiometer than by a voltmeter because of the following possible reasons. [Online May 12, 2012]

- (i) In case of potentiometer, no current flows through the cell.
 (ii) The length of the potentiometer allows greater precision.
 (iii) Measurement by the potentiometer is quicker.
 (iv) The sensitivity of the galvanometer, when using a potentiometer is not relevant.

Which of these reasons are correct?

- (a) (i), (iii), (iv) (b) (i), (iii), (iv)
 (c) (i), (ii) (d) (i), (ii), (iii), (iv)

134. In a sensitive meter bridge apparatus the bridge wire should possess [Online May 12, 2012]

- (a) high resistivity and low temperature coefficient.
 (b) low resistivity and high temperature coefficient.
 (c) low resistivity and low temperature coefficient.
 (d) high resistivity and high temperature coefficient.

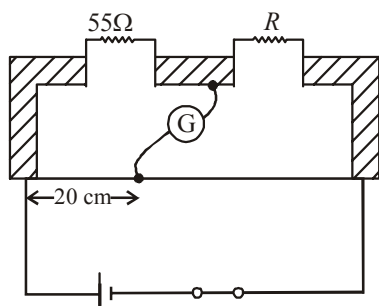
135. In a metre bridge experiment null point is obtained at 40 cm from one end of the wire when resistance X is balanced against another resistance Y. If $X < Y$, then the new position of the null point from the same end, if one decides to balance a resistance of $3X$ against Y, will be close to : [Online April 9, 2013]

- (a) 80 cm (b) 75 cm (c) 67 cm (d) 50 cm

136. The current in the primary circuit of a potentiometer is 0.2 A . The specific resistance and cross-section of the potentiometer wire are $4 \times 10^{-7} \text{ ohm metre}$ and $8 \times 10^{-7} \text{ m}^2$, respectively. The potential gradient will be equal to [2011 RS]

- (a) 1 V/m (b) 0.5 V/m (c) 0.1 V/m (d) 0.2 V/m

137. Shown in the figure below is a meter-bridge set up with null deflection in the galvanometer.



The value of the unknown resistor R is [2008]

- (a) 13.75Ω (b) 220Ω (c) 110Ω (d) 55Ω

138. In a Wheatstone's bridge, three resistances P , Q and R connected in the three arms and the fourth arm is formed by two resistances S_1 and S_2 connected in parallel. The condition for the bridge to be balanced will be [2006]

(a) $\frac{P}{Q} = \frac{2R}{S_1 + S_2}$ (b) $\frac{P}{Q} = \frac{R(S_1 + S_2)}{S_1 S_2}$

(c) $\frac{P}{Q} = \frac{R(S_1 + S_2)}{2S_1 S_2}$ (d) $\frac{P}{Q} = \frac{R}{S_1 + S_2}$

139. In a potentiometer experiment the balancing with a cell is at length 240 cm. On shunting the cell with a resistance of 2Ω , the balancing length becomes 120 cm. The internal resistance of the cell is [2005]

- (a) 0.5Ω (b) 1Ω (c) 2Ω (d) 4Ω

140. In a meter bridge experiment null point is obtained at 20 cm. from one end of the wire when resistance X is balanced against another resistance Y . If $X < Y$, then where will be the new position of the null point from the same end, if one decides to balance a resistance of $4X$ against Y [2004]

- (a) 40 cm (b) 80 cm (c) 50 cm (d) 70 cm

141. The length of a wire of a potentiometer is 100 cm, and the e. m. f. of its standard cell is E volt. It is employed to measure the e. m. f. of a battery whose internal resistance is 0.5Ω . If the balance point is obtained at $\ell = 30$ cm from the positive end, the e. m. f. of the battery is [2003]

(a) $\frac{30E}{100.5}$ (b) $\frac{30E}{(100 - 0.5)}$

(c) $\frac{30(E - 0.5i)}{100}$ (d) $\frac{30E}{100}$

where i is the current in the potentiometer wire.

142. An ammeter reads upto 1 ampere. Its internal resistance is 0.81ohm . To increase the range to 10 A the value of the required shunt is [2003]

- (a) 0.03Ω (b) 0.3Ω (c) 0.9Ω (d) 0.09Ω

143. If an ammeter is to be used in place of a voltmeter, then we must connect with the ammeter a [2002]

- (a) low resistance in parallel
(b) high resistance in parallel
(c) high resistance in series
(d) low resistance in series.



Hints & Solutions



1. (c) **Ammeter** : In series connection, the same current flows through all the components. It aims at measuring the current flowing through the circuit and hence, it is connected in series.

Voltmeter : A voltmeter measures voltage change between two points in a circuit. So we have to place the voltmeter in parallel with the circuit component.

2. (b) $\rho_M = 98 \times 10^{-8}$

$$\rho_A = 2.65 \times 10^{-8}$$

$$\rho_C = 1.724 \times 10^{-8}$$

$$\rho_T = 5.65 \times 10^{-8}$$

$$\therefore \rho_M > \rho_T > \rho_A > \rho_C$$

3. (a) When $i = 0$, $V = \varepsilon = 1.5$ volt

4. (d) Charge mobility

$$(\mu) = \frac{V_d}{E} \quad [\text{Where } V_d = \text{drift velocity}]$$

$$\text{and resistivity } (\rho) = \frac{E}{j} = \frac{EA}{I} \Rightarrow E = \frac{I(\rho)}{A}$$

$$\Rightarrow \mu = \frac{V_d}{E} = \frac{V_d A}{I \rho}$$

$$= \frac{1.1 \times 10^{-3} \times \pi \times (5 \times 10^{-3})^2}{5 \times 1.7 \times 10^{-8}}$$

$$\mu = 1.0 \frac{\text{m}^2}{\text{Vs}}$$

5. (b) Equation of straight line from graph

$$y = -mx + c$$

$$\Rightarrow \ln R = -m \left(\frac{1}{T^2} \right) + c$$

here, m & c are constants

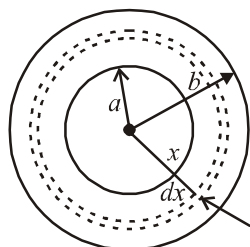
$$R = e^{\left[-m \left(\frac{1}{T^2} \right) + c \right]} = e^{-m \left(\frac{1}{T^2} \right)} \times e^c$$

$$R(T) = R_0 e^{\frac{-T_0^2}{T^2}}$$

6. (a) $dR = \frac{(\rho)(dx)}{4\pi x^2}$

$$R = \int dR$$

$$\int dR = \rho \int_a^b \frac{dx}{4\pi x^2}$$



$$\Rightarrow R = \frac{\rho}{4\pi} \left[\frac{-1}{x} \right]_a^b$$

$$R = \left(\frac{\rho}{4\pi} \right) \cdot \left(\frac{1}{a} - \frac{1}{b} \right)$$

7. (c) $\rho = \frac{m}{ne^2 \tau}$

$$= \frac{9.1 \times 10^{-31}}{8.5 \times 10^{28} \times (1.6 \times 10^{-19})^2 \times 25 \times 10^{-15}}$$

$$= 10^{-8} \Omega\text{-m}$$

8. (d) Number 2 is associated with the red colour. This colour is replaced by green.

$\therefore$ Colour code figure for green is 5

$\therefore$ New resistance = 500 Ω

9. (a) Clearly, from graph

$$\text{Current, } I = \frac{dq}{dt} = 0 \text{ at } t = 4\text{s} \quad [\text{Since } q \text{ is constant}]$$

10. (b) Color code : Bl, Br, R, O, Y, G, B, V, Gr, W
0, 1, 2, 3, 4, 5, 6, 7, 8, 9

$R = AB \times C \pm D\%$ where D = tolerance

$D_{\text{gold}} = \pm 5\%$, $D_{\text{silver}} = \pm 10\%$; $D_{\text{no colour}} = \pm 20\%$

Red violet orange silver

$$R = 27 \times 10^3 \Omega \pm 10\% = 27 \text{ k}\Omega \pm 10\%$$

11. (a) Using, $I = neAv_d$

$$\therefore \text{Drift speed } v_d = \frac{I}{neA}$$

$$= \frac{1.5}{9 \times 10^{28} \times 1.6 \times 10^{-19} \times 5 \times 10^{-6}} = 0.02 \text{ mms}^{-1}$$

12. (c) Resistance, $R = \frac{\rho \ell}{A}$

$$R = \rho \frac{\ell}{A} \times \frac{\ell}{\ell} = \frac{\rho \ell^2}{V} \quad [V \gg \text{Volume (V)} = A \times \ell]$$

Since resistivity and volume remains constant therefore % change in resistance

$$\frac{\Delta R}{R} = \frac{2\Delta \ell}{\ell} = 2 \times (0.5) = 1\%$$

13. (a) Colour code for carbon resistor

Bl, Br, R, O, Y, G, Blue, V, Gr, W

0 1 2 3 4 5 6 7 8 9

Resistance, $R = AB \times C \pm D$

Bands A and B are the first two significant figures of resistance

B and C indicates the decimal multiplier or the number of zeros that follow A and B

B and D is tolerance: Gold = $\pm 5\%$,

Silver = $\pm 10\%$ No colour = $\pm 20\%$

$$R = 53 \times 10^4 \pm 5\% = 530 \text{ k}\Omega \pm 5\%$$

14. (c) Resistance after temperature increases by 500°C i.e.,

$$R_t = \frac{V}{I} = \frac{220}{2} = 110\Omega$$

$R_0 = 100$ (given) temperature coefficient of resistance, $\alpha = ?$
using $R_t = R_0(1 + \alpha t)$
 $110 = 100(1 + \alpha 500)$

$$\alpha = \frac{10}{100 \times 500}$$

$$\text{or, } \alpha = 2 \times 10^{-4}^\circ\text{C}^{-1}$$

15. (c) Charge density $\rho = \frac{\text{charge}}{\text{volume}} = \frac{q}{Ad} \Rightarrow q = \rho Ad$

$$\text{Also, } q = IT \Rightarrow T = \frac{q}{I} = \frac{\rho Ad}{I}$$

16. (a) Given, $R_1 = 100\Omega$, $r' = r/2$, $R_2 = ?$

Resistivity of wire, $R = \frac{\rho l}{A} \quad \because \text{Area} \times \text{length} = \text{volume}$

$$\text{Hence, } R = \frac{\rho V}{A^2}$$

Since, $\rho \rightarrow \text{constant}$, $V \rightarrow \text{constant}$

$$R \propto \frac{1}{A^2}$$

$$\text{or } R \propto \frac{1}{r^4} \quad \because A = \pi r^2$$

$$\frac{R_2}{R_1} = 16 \Rightarrow R_2 = 16 \times 100 = 1600\Omega, \text{ Resistance of new wire.}$$

17. (b) $V = IR = (n e A v_d) \rho \frac{\ell}{A}$

$$\therefore \rho = \frac{V}{V_d l n e}$$

Here V = potential difference

l = length of wire

n = no. of electrons per unit volume of conductor.

e = no. of electrons

Placing the value of above parameters we get resistivity

$$\rho = \frac{5}{8 \times 10^{28} \times 1.6 \times 10^{-19} \times 2.5 \times 10^{-4} \times 0.1} \\ = 1.6 \times 10^{-5} \Omega\text{m}$$

18. (c) $i = n e A V_d$ and $V_d \propto \sqrt{E}$ (Given)

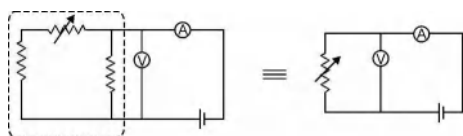
$$\text{or, } i \propto \sqrt{E}$$

$$i^2 \propto E$$

$$i^2 \propto V$$

Hence graph (c) correctly depicts the V - I graph for a wire made of such type of material.

19. (a) In ohm's law, we check $V = IR$ where I is the current flowing through a resistor and V is the potential difference across that resistor. Only option (a) fits the above criteria. Remember that ammeter is connected in series with resistance and voltmeter parallel with the resistance.



20. (c) Resistance of wire (R) = $\rho \frac{l}{A}$

If wire is bent in the middle then

$$l' = \frac{l}{2}, A' = 2A$$

$$\therefore \text{New resistance, } R' = \rho \frac{l'}{A'}$$

$$= \frac{\rho \frac{l}{2}}{2A} = \frac{\rho l}{4A} = \frac{R}{4}$$

21. (a) Resistance of wire

$$R = \frac{\rho l}{A} = \frac{\rho l^2}{V} \quad (\because V = Al)$$

$$\text{Hence, } R = \rho \frac{\ell^2}{V} = \text{constant} \times \ell^2$$

$\therefore$ Fractional change in resistance

$$\frac{\Delta R}{R} = 2 \frac{\Delta l}{l}$$

$$100 \times \frac{\Delta R}{R} = 200 \times \left(\frac{d\ell}{\ell} \right)$$

$$\therefore d\ell/\ell = 0.1\%$$

$$\therefore \% \text{ change in } R = \left[200 \times \left(\frac{0.1}{100} \right) \right] = 0.2\%$$

$\therefore$ Resistance will increase by 0.2%.

22. (a) Let j be the current density.

$$\text{Then } j \times 2\pi r^2 = I \Rightarrow j = \frac{I}{2\pi r^2}$$

$$\therefore E = \rho j = \frac{\rho I}{2\pi r^2}$$

Now, $V_B - V_C$

$$= - \int_{a+b}^a \vec{E} \cdot d\vec{r} = - \int_{a+b}^a \frac{\rho I}{2\pi r^2} dr$$

$$= - \frac{\rho I}{2\pi} \left[-\frac{1}{r} \right]_{a+b}^a = \frac{\rho I}{2\pi a} - \frac{\rho I}{2\pi(a+b)}$$

On applying superposition as mentioned we get

$$\Delta V_{BC} = 2 \times \Delta V'_{BC} = \frac{\rho I}{\pi a} - \frac{\rho I}{\pi(a+b)}$$

23. (c) As shown in Answer (a) $E = \frac{\rho I}{2\pi r^2}$

24. (d) Resistance of a metal conductor at temperature $t^\circ\text{C}$ is given by

$$R_t = R_0(1 + \alpha t),$$

R_0 is the resistance of the wire at 0°C

and α is the temperature coefficient of resistance.

$$\text{Resistance at } 50^\circ\text{C}, R_{50} = R_0(1 + 50\alpha) \quad \dots (i)$$

$$\text{Resistance at } 100^\circ\text{C}, R_{100} = R_0(1 + 100\alpha) \quad \dots (ii)$$

$$\text{From (i), } R_{50} - R_0 = 50\alpha R_0 \quad \dots (iii)$$

$$\text{From (ii), } R_{100} - R_0 = 100\alpha R_0 \quad \dots (iv)$$

Dividing (iii) by (iv), we get

$$\frac{R_{50} - R_0}{R_{100} - R_0} = \frac{1}{2}$$

Here, $R_{50} = 5\Omega$ and $R_{100} = 6\Omega$

$$\therefore \frac{5 - R_0}{6 - R_0} = \frac{1}{2}$$

or, $6 - R_0 = 10 - 2R_0$ or, $R_0 = 4\Omega$.

25. (d) Let d_A and d_B be the diameter of wire A and B respectively. Let ρ_B and ρ_A be the resistivity of wire A and B. We have given

$$\rho_B = 2\rho_A$$

$$d_B = 2d_A$$

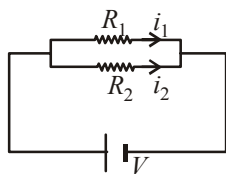
If both resistances are equal

$$R_B = R_A$$

$$\Rightarrow \frac{\rho_B \ell_B}{A_B} = \frac{\rho_A \ell_A}{A_A}$$

$$\therefore \frac{\ell_B}{\ell_A} = \frac{\rho_A}{\rho_B} \times \frac{d_B^2}{d_A^2} = \frac{\rho_A}{2\rho_A} \times \frac{4d_A^2}{d_A^2} = 2$$

26. (b)



Given,

$$\frac{\ell_1}{\ell_2} = \frac{4}{3} \text{ and } \frac{r_1}{r_2} = \frac{2}{3}$$

$$R_1 = \frac{\rho \ell_1}{\pi r_1^2}; R_2 = \frac{\rho \ell_2}{\pi r_2^2}$$

When wires are in parallel to the circuit potential difference across each wire is same

$$i_1 R_1 = i_2 R_2$$

$$\therefore \frac{i_1}{i_2} = \frac{R_2}{R_1} = \frac{\rho \ell_2}{\pi r_2^2} \times \frac{\pi r_1^2}{\rho \ell_1} = \frac{\ell_2}{\ell_1} \times \frac{r_1^2}{r_2^2}$$

$$= \frac{3}{4} \times \frac{4}{9} = \frac{1}{3}$$

27. (d) Since volume of wire remains unchanged on increasing length, hence

$$A \times \ell = A' \times \ell'$$

$$\Rightarrow \ell' = 2\ell$$

$$\therefore A' = \frac{A \times \ell}{\ell'} = \frac{A \times \ell}{2\ell} = \frac{A}{2}$$

Percentage change in resistance

$$= \frac{R_f - R_i}{R_i} \times 100 = \frac{\rho \frac{\ell'}{A'} - \rho \frac{\ell}{A}}{\rho \frac{\ell}{A}} \times 100$$

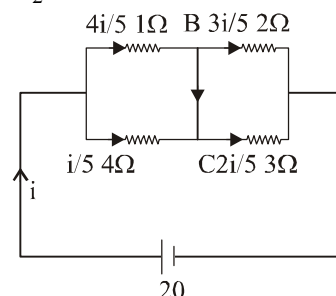
$$= \left[\left(\frac{\ell'}{A'} \times \frac{A}{\ell} \right) - 1 \right] \times 100$$

$$= \left[\left(\frac{2\ell}{A/2} \times \frac{A}{\ell} \right) - 1 \right] \times 100 = (4 - 1) \times 100 = 300\%$$

28. (b) From circuit diagram,

$$\frac{1}{R_1} = \frac{1}{1} + \frac{1}{4} \Rightarrow R_1 = \frac{4}{5}$$

$$\frac{1}{R_2} = \frac{1}{2} + \frac{1}{3} \Rightarrow R_2 = \frac{6}{5}$$

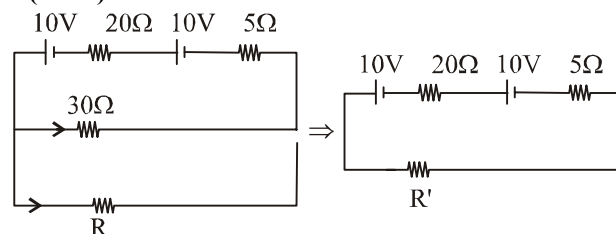


$$R_{\text{eff}} = R_1 + R_2 = \frac{4}{5} + \frac{6}{5} = 2\Omega$$

$$i = \frac{V}{R_{\text{eff}}} = \frac{20}{2} = 10A$$

$$\therefore I_{BC} = \frac{4i}{5} - \frac{3i}{5} = \frac{i}{5} = 2A$$

29. (30.00)



The resistance of 30Ω is in parallel with R . Their effective resistance

$$\frac{1}{R'} = \frac{1}{30} + \frac{1}{R}$$

$$R' = \frac{30R}{30 + R}$$

...(i)

$$\text{Also, } V = IR \Rightarrow 10 = \frac{20 \times 20}{R' + 25}$$

$$\Rightarrow R' + 25 = 40 \Rightarrow R' = 15$$

$$R' = 15 = \frac{30R}{30 + R}$$

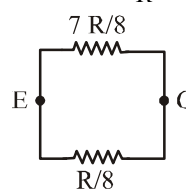
Using (i)

$$\Rightarrow 30 + R = 2R$$

$$\Rightarrow R = 30\Omega$$

30. (c)

$$31. (b) R_{\text{eq}} = \frac{\left(\frac{7R}{8}\right)\left(\frac{R}{8}\right)}{R} = \frac{7R}{64}$$

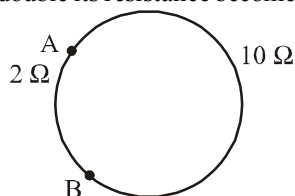


32. (c) When length becomes double its resistance becomes

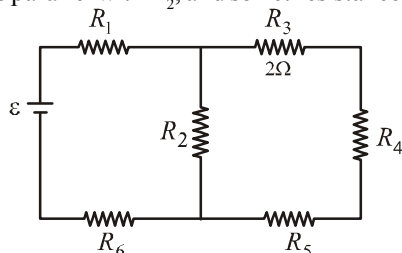
$$(R \propto l^2)$$

$$R = 4 \times 3 = 12 \Omega$$

$$R_{eq} = \frac{2 \times 10}{12} = \frac{5}{3} \Omega$$



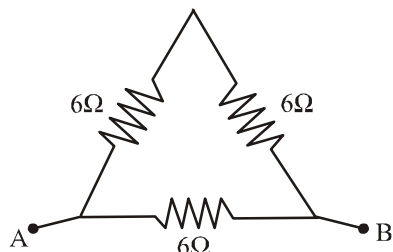
33. (c) R_3 , R_4 and R_5 are in series so their equivalent $R = 20 + 5 + 25 = 50 \Omega$
This is parallel with R_2 , and so net resistance of the circuit



$$R_{eq} = \left(\frac{10 \times 50}{10 + 50} \right) + 15 + 30 = \frac{160}{3} \Omega$$

$$\text{So, } i = \frac{\varepsilon}{R_{eq}} = \frac{15}{(100/3)} = \frac{9}{32} A$$

34. (a)



Resistance, $R \propto l$ so resistance of each side of the equilateral triangle = 6Ω

Resistance R_{eq} between any two vertices

$$\frac{1}{R_{eq}} = \frac{1}{12} + \frac{1}{6} \Rightarrow R_{eq} = 4 \Omega$$

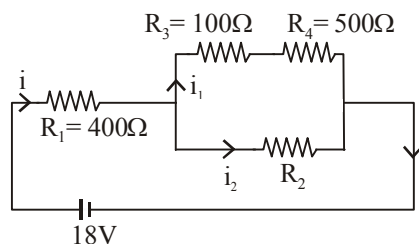
35. (b) Using, $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$

$$0.95 R = \frac{R R_0}{R + R_0} \text{ (measured value 5\% less than internal resistance of voltmeter)}$$

$$\text{or, } 0.95 \times 30 = 0.05 R_0$$

$$\therefore R_0 = 19 \times 30 = 570 \Omega$$

36. (a)



Across R_4 reading of voltmeter, $V_4 = 5V$

$$\text{Current, } i_4 = \frac{V_4}{R_4} = 0.01 A$$

$$V_3 = i_1 R_3 = 1V$$

$$V_3 + V_4 = 6V = V_2$$

$$V_1 + V_3 + V_4 = 18V$$

$$\Rightarrow V_1 = 12V$$

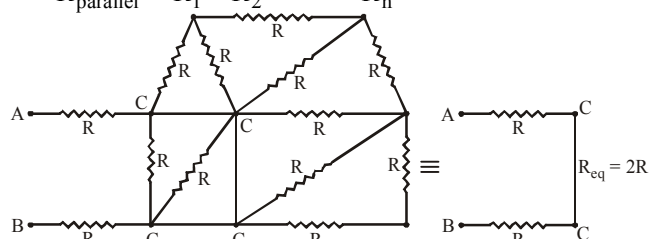
$$i = \frac{V_1}{R_1} = 0.03 A$$

$$i = i_1 + i_2 \Rightarrow i_2 = i - i_1 = 0.03 - 0.01 A = 0.02 A$$

$$\therefore R_2 = \frac{V_2}{i_2} = \frac{6}{0.02} = 300 \Omega$$

37. (a) $R_{series} = R_1 + R_2 + \dots + R_n$

$$\frac{1}{R_{parallel}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$$



38. (a) In steady state, flow of current through capacitor will be zero.

Current through the circuit,

$$i = \frac{E}{r + r_2}$$

Potential difference through capacitor

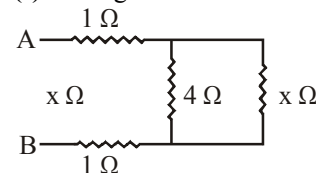
$$V_c = \frac{Q}{C} = E - ir = E - \left(\frac{E}{r + r_2} \right) r$$

$$\therefore Q = CE \frac{r_2}{r + r_2}$$

39. (b) The potential difference in each loop is zero.

$\therefore$ No current will flow or current in each resistance is Zero.

40. (a) The given circuit can be redrawn as,



$$\text{as } 4 \Omega \text{ and } x \Omega \text{ are parallel } x' = \frac{1}{4} + \frac{1}{x} = \frac{(4+x)}{4x}$$

$$x' = \frac{4x}{4+x}$$

& 1Ω and 1Ω are also parallel $x'' = 2 \Omega$

Now equivalent resistance of circuit

$$x = \frac{4x}{4+x} + 2 = \frac{8+6x}{4+x}$$

$$4x + x^2 = 8 + 6x$$

$$x^2 - 2x - 8 = 0$$

$$x = \frac{2 \pm \sqrt{4 - 4(1)(-8)}}{2} = \frac{2 \pm \sqrt{36}}{2} = \frac{2 \pm 6}{2} = 4\Omega$$

$$\text{Reading of Ammeter } A_1 = \frac{V}{(R+r)}$$

$$A_1 = \frac{9}{4+0.5} = 2 \text{ Ampere}$$

41. (b) Resistance between P and Q

$$r_{PQ} = r \parallel \left(\frac{r}{3} + \frac{r}{2} \right) = \frac{r \times \frac{5}{6}r}{r + \frac{5}{6}r} = \frac{5}{11}r$$

Resistance between Q and R

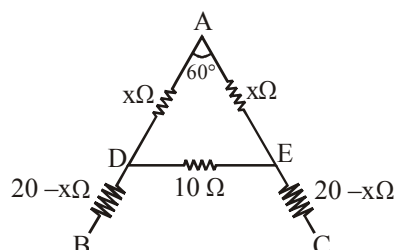
$$r_{QR} = \frac{r}{2} \parallel \left(r + \frac{r}{3} \right) = \frac{\frac{r}{2} \times \frac{4}{3}r}{\frac{r}{2} + \frac{4}{3}r} = \frac{4}{11}r$$

Resistance between P and R

$$r_{PR} = \frac{r}{3} \parallel \left(\frac{r}{2} + r \right) = \frac{\frac{r}{3} \times \frac{3}{2}r}{\frac{r}{3} + \frac{3}{2}r} = \frac{3}{11}r$$

Hence, it is clear that r_{PQ} is maximum

42. (d)



$$\text{For ADE } \frac{1}{R'} = \frac{1}{2x} + \frac{1}{10}$$

$$\text{or } R' = \frac{20x}{10+2x}$$

$$R_{BC} = \frac{20x}{10+2x} + 20 - x + 20 - x \dots (i)$$

$$\text{or } \frac{20x}{10+2x} + 40 = 2x$$

Solving we get

$$x = 10\Omega$$

Putting the value of $x = 10\Omega$ in equation (i)

We get

$$R_{BC} = \frac{20 \times 10}{10 + 2 \times 10} + 20 - 10 + 20 - 10$$

$$= \frac{80}{3} = 26.7\Omega$$

43. (d) Let R_1 and R_2 be the resistances of two conductors, then

$$R_1 = R_0 [1 + \alpha_1 \Delta t]$$

$$R_2 = R_0 [1 + \alpha_2 \Delta t]$$

Here, R_0 is the resistance of conductor at 0°C

In Series, $R = R_1 + R_2$

$$= R_0 [2 + (\alpha_1 + \alpha_2) \Delta t]$$

$$= 2R_0 \left[1 + \left(\frac{\alpha_1 + \alpha_2}{2} \right) \Delta t \right]$$

$$\therefore \alpha_{eq} = \frac{\alpha_1 + \alpha_2}{2}$$

$$\text{In Parallel, } \frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$= \frac{1}{R_0 [1 + \alpha_1 \Delta t]} + \frac{1}{R_0 [1 + \alpha_2 \Delta t]}$$

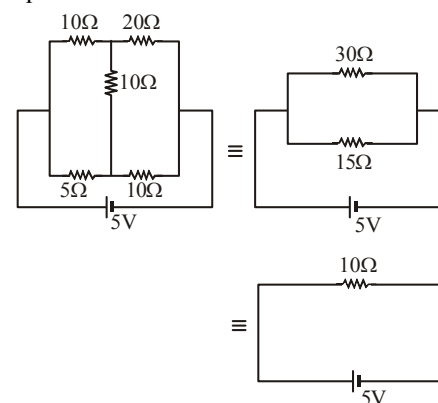
$$\Rightarrow \frac{1}{\frac{R_0}{2} (1 + \alpha_{eq} \Delta t)}$$

$$= \frac{1}{R_0 (1 + \alpha_1 \Delta t)} + \frac{1}{R_0 (1 + \alpha_2 \Delta t)}$$

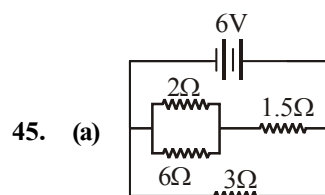
$$2(1 - \alpha_{eq} \Delta t) = (1 - \alpha_1 \Delta t)(1 - \alpha_2 \Delta t)$$

$$\therefore \alpha_{eq} = \frac{\alpha_1 + \alpha_2}{2}$$

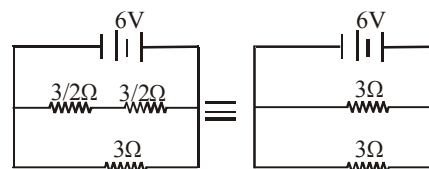
44. (b) The network of resistors is a balanced wheatstone bridge. Hence, no current will flow through centre resistor. The equivalent circuit is



$$R_{eq} = \frac{15 \times 30}{15 + 30} = 10\Omega \Rightarrow I = \frac{V}{R} = \frac{5}{10} = 0.5 \text{ A}$$



45. (a)



$$\text{hence } R_{eq} = 3/2; \therefore I = \frac{6}{3/2} = 4 \text{ A}$$

46. (c) Let R_1 and R_2 be the two given resistances
Resistance of the series combination,
 $S = R_1 + R_2$
Resistance of the parallel combination,

$$P = \frac{R_1 R_2}{R_1 + R_2}$$

As per question $S = nP$

$$\Rightarrow R_1 + R_2 = \frac{n(R_1 R_2)}{(R_1 + R_2)}$$

$$\Rightarrow (R_1 + R_2)^2 = nR_1 R_2$$

Minimum value of n is 4 for that

$$(R_1 + R_2)^2 = 4R_1 R_2$$

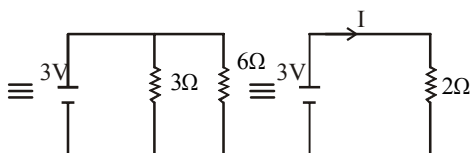
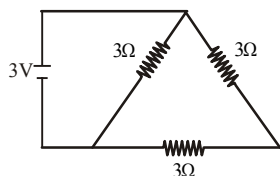
$$\Rightarrow (R_1 - R_2)^2 = 0$$

47. (b) In the given circuit, resistance of 3Ω is in parallel with series combination of two 3Ω resistance.

$$R_p = \frac{3 \times 6}{3 + 6} = \frac{18}{9} = 2\Omega$$

Using ohm's law $V = IR$

$$\Rightarrow I = \frac{V}{R} = \frac{3}{2} = 1.5A$$



48. (c)

Using Kirchhoff's loop law in loop ABCD

$$-5i_2 - 10(i_1 + i_2) - 2i_2 + 20 = 0$$

$$\Rightarrow -10i_1 - 17i_2 + 20 = 0 \quad \dots(i)$$

Using Kirchhoff's loop law in loop BEFC

$$\Rightarrow -10 + 4i_1 + 10(i_1 + i_2) = 0$$

$$\Rightarrow 14i_1 + 10i_2 + 10 = 0 \quad \dots(ii)$$

Multiplying equation (i) by 10, we have

$$(10i_1 + 17i_2 = 20) \times 10$$

$$\Rightarrow 100i_1 - 170i_2 = 200 \quad \dots(iii)$$

Multiplying equation (ii) by 17, we have

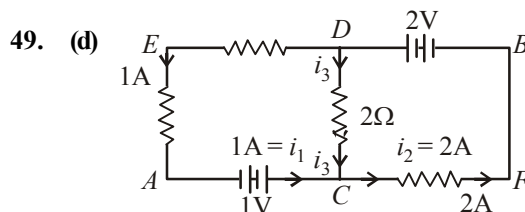
$$(14i_1 + 10i_2 = 10) \times 17$$

$$\Rightarrow 238i_1 - 170i_2 = 170 \quad \dots(iv)$$

On solving equations (iii) and (iv), we get

$$-138i_1 = 30 \Rightarrow i_1 = -\frac{30}{138} = -0.217$$

i_1 is negative it means current flows from positive to negative terminal.



Let us assume the potential at $A = V_A = 0$

Using Kirchhoff's junction rule at C, we get

$$i_1 + i_3 = i_2$$

$$1A + i_3 = 2A \Rightarrow i_3 = 1A$$

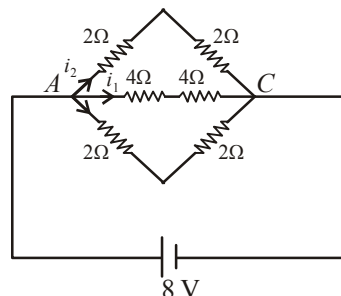
Now using Kirchhoff's loop law along ACDB

$$V_A + 1 + i_3(2) - 2 = V_B$$

$$\Rightarrow V_A + 1 + i_3(1) - 2 = V_B$$

$$\Rightarrow V_B - V_A = 3 - 2 = 1 \text{ volt}$$

50. (c) The equivalent circuit can be drawn as

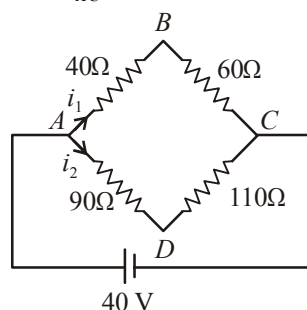


Voltage across AC = 8 V

Resistance $R_{AC} = 4 + 4 = 8\Omega$

$$i_1 = \frac{V}{R_{AC}} = \frac{8}{4 + 4} = 1 \text{ Amp}$$

51. (2)



$$\text{Current through AB, } i_1 = \frac{40}{40 + 60} = 0.4$$

$$\text{Current through AD, } i_2 = \frac{40}{90 + 110} = \frac{1}{5}$$

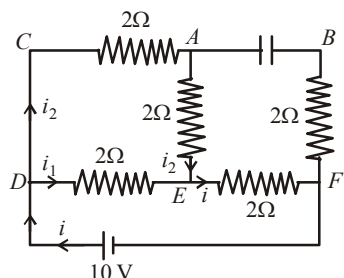
Using KVL in BAD loop

$$V_B + i_1(40) - i_2(90) = V_D$$

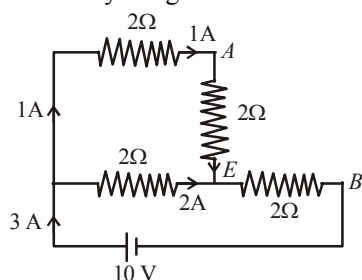
$$\Rightarrow V_B - V_D = \frac{1}{5}(90) - \frac{4}{10}(40)$$

$$\Rightarrow V_B - V_D = 18 - 16 = 2 \text{ V}$$

52. (08.00)



As capacitor is fully charged no current will flow through it.



We have the current distribution as shown in the figure.

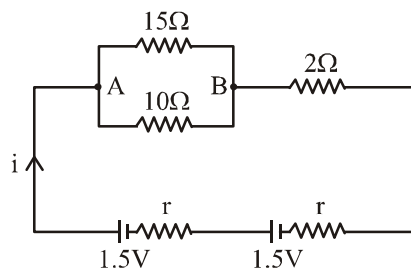
$$\text{Equivalent resistance, } R_{\text{eq}} = \left(\frac{4 \times 2}{4 + 2} \right) + 2$$

$$\text{Net current, } i = \frac{10}{\frac{4}{3} + 2} = \frac{10 \times 3}{10} = 3 \text{ Amp}$$

$$i_1 = 2 \text{ A and } i_2 = 1 \text{ A}$$

$$V_{AEB} = 1 \times 2 + 3 \times 2 = 8 \text{ V}$$

53. (b) For the given circuit



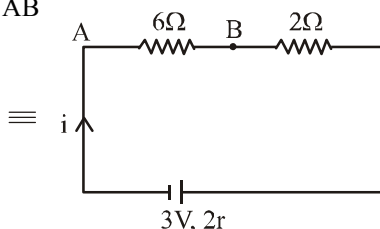
$$i = \frac{3}{8 + 2r}$$

Now voltage across AB

$$i \times 6 = \frac{3}{8 + 2r} \times 6 = 2$$

$$\Rightarrow 9 = 8 + 2r$$

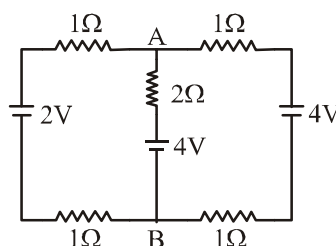
$$\Rightarrow r = \frac{1}{2} \Omega$$



54. (d) Applying parallel combination of batteries

$$\frac{E_1}{1+1} + \frac{E_2}{2} + \frac{E_3}{1+1}$$

$$\frac{1}{1+1} + \frac{1}{2} + \frac{1}{1+1}$$



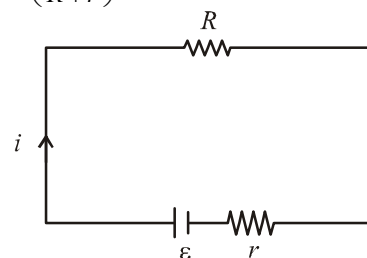
$$\frac{\frac{2}{2} + \frac{4}{2} + \frac{4}{2}}{\frac{1}{2} + \frac{1}{2} + \frac{1}{2}} = \frac{5 \times 2}{3}$$

$$= \frac{10}{3} = 3.3 \text{ Volt}$$

55. (d) $i = \left(\frac{\varepsilon}{R+r} \right)$

Power delivered to R.

$$P = i^2 R = \left(\frac{\varepsilon}{R+r} \right)^2 R$$

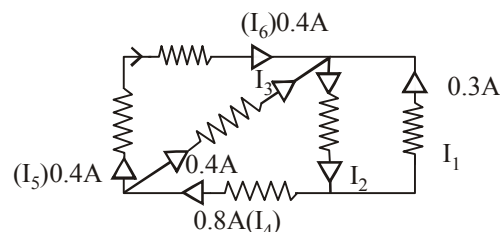


$$P \text{ to be maximum, } \frac{dP}{dR} = 0$$

$$\text{or } \frac{d}{dR} \left[\left(\frac{\varepsilon}{R+r} \right)^2 R \right] = 0$$

$$\text{or } R = r$$

56. (b)



$$\text{From KCL, } I_3 = 0.8 - 0.4 = 0.4 \text{ A}$$

$$I_2 = 0.4 + 0.4 + 0.3$$

$$= 1.1 \text{ A}$$

$$\text{and } I_6 = 0.4 \text{ A}$$

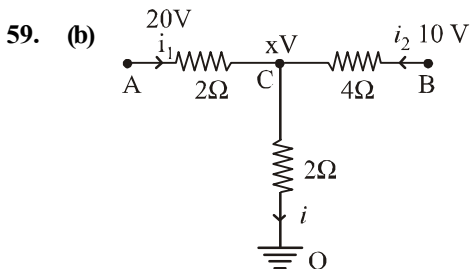
57. (b) Given, $E_1 = 1 \text{ V}, E_2 = 2 \text{ V}, E_3 = 3 \text{ V}, r_1 = 1 \Omega,$
 $r_2 = 1 \Omega$ and $r_3 = 1 \Omega$

$$V_{AB} = V_{CD} = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2} + \frac{E_3}{r_3}}{\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}} = \frac{\frac{1}{1} + \frac{2}{1} + \frac{3}{1}}{\frac{1}{1} + \frac{1}{1} + \frac{1}{1}} = \frac{6}{3} = 2V$$

58. (c) Current passing through resistance R_1 ,

$$i_1 = \frac{V}{R_1} = \frac{10}{20} = 0.5A$$

and, $i_2 = 0$



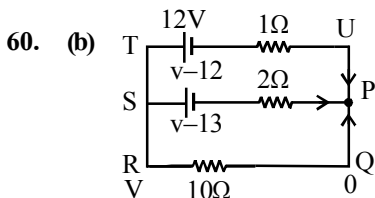
Let voltage at C = xV

From Kirchhoff's current law,

KCL: $i_1 + i_2 = i$

$$\frac{20-x}{2} + \frac{10-x}{4} = \frac{x-0}{2} \Rightarrow x = 10$$

$$\therefore i = \frac{V}{R} = \frac{X}{R} = \frac{10}{2} = 5A$$



Using Kirchhoff's law at P we get

$$\frac{V-12}{1} + \frac{V-13}{2} + \frac{V-0}{10} = 0$$

[Let potential at P, Q, U = 0 and at R = V]

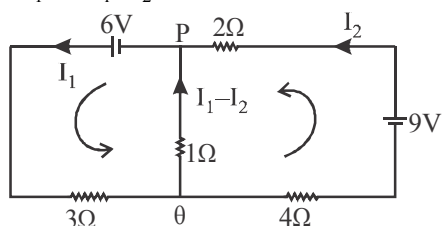
$$\Rightarrow \frac{V}{1} + \frac{V}{2} + \frac{V}{10} = \frac{12}{1} + \frac{13}{2} + \frac{0}{10}$$

$$\Rightarrow \frac{10+5+1}{10} V = \frac{24+13}{2} \Rightarrow V \left(\frac{16}{10} \right) = \frac{37}{2}$$

$$\Rightarrow V = \frac{37 \times 10}{16 \times 2} = \frac{370}{32} = 11.56 \text{ volt}$$

61. (a) From KVL

$$-6 + 3I_1 + 1(I_1 - I_2) = 0$$



$$6 = 3I_1 + I_1 - I_2; 4I_1 - I_2 = 6 \quad \dots(1)$$

$$-9 + 2I_2 - (I_1 - I_2) + 3I_2 = 0$$

$$-I_1 + 6I_2 = 9 \quad \dots(2)$$

On solving (1) and (2)

$$I_1 = 0.13A$$

Direction Q to P, since $I_1 > I_2$.

62. (c) As no current flows through arm EB then

$$V_D = 0V$$

$$V_E = 0V$$

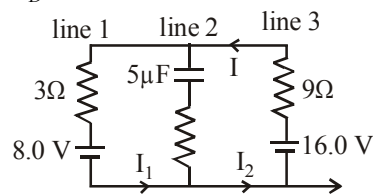
$$V_B = -4V$$

$$V_A = 5V$$

So, potential difference between the points A and D

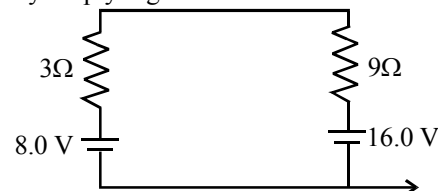
$$V_A - V_D = 5V$$

63. (b)



In steady state capacitor is fully charged hence no current will flow through line 2.

By simplifying the circuit



Hence resultant potential difference across resistances will be 8.0 V.

$$\text{Thus current } I = \frac{V}{R} = \frac{8.0}{3+9} = \frac{8}{12}$$

$$\text{or, } I = \frac{2}{3} = 0.67 A$$

64. (a) Current through 50 V and 30 V batteries are respectively 2.5 A and 3 A.

65. (c) Given, emf of cell E = 200 V

Internal resistance of cells = 1 Ω

D. C. main supply voltage V = 220 V

External resistance R = ?

$$r = \left(\frac{E - V}{V} \right) R$$

$$1 = \left(\frac{20}{220} \right) \times R \quad \therefore R = 11 \Omega.$$

66. (c) $\frac{100}{R+r} = \frac{90}{R} \Rightarrow \frac{R+r}{R} = \frac{10}{9} \Rightarrow 1 + \frac{0.5}{R} = \frac{10}{9}$

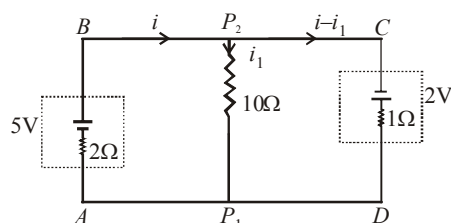
$$\Rightarrow \frac{0.5}{R} = \frac{1}{9} \therefore R = 4.5 \Omega$$

67. (c) Applying Kirchhoff's second law in ABP_2P_1A , we get

$$-2i + 5 - 10i_1 = 0$$

$$2i + 10i_1 = 5$$

.....(i)



Again applying Kirchhoff's second law in $P_2CDP_1P_2$ we get,

$$10i_1 + 2 - i + i_1 = 0$$

$$2i - 22i_1 = 4$$

From (i) and (ii)

$$32i_1 = 1$$

$$\Rightarrow i_1 = \frac{1}{32} \text{ A from } P_2 \text{ to } P_1$$

68. (a) Energy in capacitor = $\frac{1}{2}CV^2$

Work done by battery = $QV = CV^2$

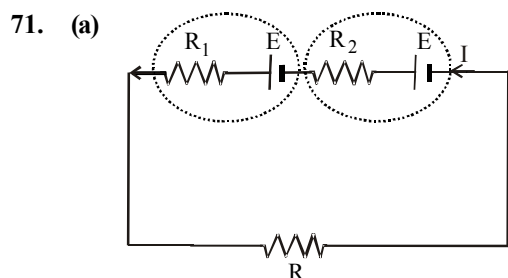
where C = Capacitance of capacitor

V = Potential difference,

e = emf of battery

$$\text{Required ratio} = \frac{\frac{1}{2}CV^2}{CV^2} = \frac{1}{2} (\because V=e)$$

69. (d) **Note :** Kirchhoff's first law is based on conservation of charge and Kirchhoff's second law is based on conservation of energy.
70. (d) At cold junction, current flows from Antimony to Bismuth because current flows from metal occurring later in the series to metal occurring earlier in the thermoelectric series. In thermoelectric series, Bismuth comes earlier than Antimony so at cold junction, current flows from Antimony to Bismuth.



Let E be the emf of each source of current

$$\text{Current in the circuit } I = \frac{2E}{R + R_1 + R_2}$$

Potential difference across cell having internal resistance R_2

$$V = E - iR_2 = 0$$

$$E - \frac{2E}{R + R_1 + R_2} \cdot R_2 = 0$$

$$\Rightarrow R + R_1 + R_2 - 2R_2 = 0$$

$$\Rightarrow R + R_1 - R_2 = 0$$

$$\Rightarrow R = R_2 - R_1$$

72. (a) From Faraday's first law of electrolysis, mass deposited

$$m = Zq$$

$$\Rightarrow Z \propto \frac{1}{q} \Rightarrow \frac{Z_1}{Z_2} = \frac{q_2}{q_1} \quad \dots (i)$$

$$\text{Also } q = q_1 + q_2 \quad \dots (ii)$$

$$\Rightarrow \frac{q}{q_2} = \frac{q_1}{q_2} + 1 \quad (\text{Dividing (ii) by } q_2)$$

$$\Rightarrow q_2 = \frac{q}{1 + \frac{q_1}{q_2}} \quad \dots (iii)$$

From equation (i) and (iii),

$$q_2 = \frac{q}{1 + \frac{Z_2}{Z_1}}$$

73. (d) Current is given by

$$I = \frac{E}{R + r},$$

If internal resistance (r) is zero,

$$I = \frac{E}{R} = \text{constant.}$$

Thus, energy source will supply a constant current if its internal resistance is zero.

74. (d) Given $E = a\theta + b\theta^2$

$$\Rightarrow \frac{dE}{d\theta} = a + 2b\theta$$

At neutral temperature

$$\theta = \theta_n : \frac{dE}{d\theta} = 0$$

$$\Rightarrow \theta_n = \frac{-a}{2b} = -350 \Rightarrow \frac{d^2E}{d\theta^2} = 2b$$

hence no θ is possible for E to be maximum no neutral temperature is possible.

75. (c) From the Faraday's first law of electrolysis,

$$m = Zit$$

$$\Rightarrow m = 3.3 \times 10^{-7} \times 3 \times 2$$

$$= 19.8 \times 10^{-7} \text{ kg}$$

76. (a) Let the smallest temperature difference be $\theta^\circ\text{C}$ that can be detected by the thermocouple, then

$$\text{Thermo emf} = (25 \times 10^{-6})\theta$$

Let I is the smallest current which can be detected by the galvanometer of resistance R .

Potential difference across galvanometer

$$IR = 10^{-5} \times 40$$

$$\therefore 10^{-5} \times 40 = 25 \times 10^{-6} \times \theta$$

$$\Rightarrow \theta = 16^\circ\text{C.}$$

77. (c) According to Faraday's first law of electrolysis

$$m = Z \times I \times t$$

When I and t is same, $m \propto Z$

$$\therefore \frac{m_{\text{Cu}}}{m_{\text{Zn}}} = \frac{Z_{\text{Cu}}}{Z_{\text{Zn}}} \Rightarrow m_{\text{Cu}} = \frac{Z_{\text{Cu}}}{Z_{\text{Zn}}} \times m_{\text{Zn}}$$

$$\Rightarrow m_{\text{Cu}} = \frac{31.5}{32.5} \times 0.13 = 0.126 \text{ g}$$

78. (b) From the Faraday's first law of electrolysis

$$m = ZIt \Rightarrow m \propto It$$

79. (b) Given : Power, $P = 1 \text{ kW} = 1000 \text{ W}$

$$R = 2\Omega, V = 220 \text{ V}$$

$$\text{Current, } I = \frac{P}{V} = \frac{1000}{220}$$

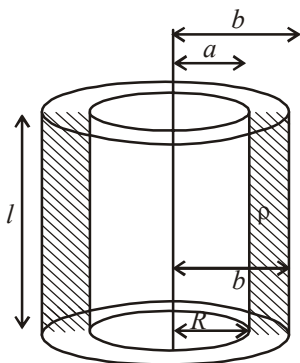
$$P_{\text{loss}} = I^2 R = \left(\frac{1000}{220}\right)^2 \times 2$$

$$\therefore \text{Efficiency} = \frac{1000}{1000 + P_{\text{loss}}} \times 100 = 96\%.$$

80. (b) Maximum power in external resistance is generated when it is equal to internal resistance of battery i.e., P_R maximum when $r = R$

The maximum Joule heating in R will take place for, the resistance of small element

$$\Delta R = \frac{\rho dr}{2\pi r l} \Rightarrow R = \frac{\rho}{2\pi l} \int_a^b \frac{dr}{r}$$



$$\text{or, } R = \frac{\rho}{2\pi l} \ln \frac{b}{a}$$

81. (d) Net Power, P
 $= 15 \times 45 + 15 \times 100 + 15 \times 10 + 2 \times 1000$
 $= 15 \times 155 + 2000 \text{ W}$

$$\text{Power, } P = VI$$

$$\Rightarrow I = \frac{P}{V}$$

$$\therefore I_{\text{main}} = \frac{15 \times 155 + 2000}{220} = 19.66 \text{ A} \approx 20 \text{ A}$$

82. (b) Equivalent resistance,

$$R_{\text{eq}} = \frac{4R \times 4R}{4R + 4R} + R + \frac{6R \times 12R}{6R + 12R} + R$$

$$= 2R + R + 4R + R$$

$$= 8R.$$

$$\text{Using, } P = \frac{V^2}{R_{\text{eq}}} \Rightarrow 4 = \frac{16^2}{8R}$$

$$\therefore R = \frac{16^2}{4 \times 8} = 8 \Omega$$

83. (b)

84. (a) As $R = \frac{V^2}{P}$, so $R_1 = \frac{220^2}{25}$ and $R_2 = \frac{220^2}{100}$

$$\text{Current flown } i = \frac{220}{R_1 + R_2}$$

$$P_1 = i^2 R_1 = \frac{220^2}{\left(\frac{220^2}{25} + \frac{220^2}{100}\right)} \times \frac{220^2}{25} = 16 \text{ W}$$

$$\text{Similarly, } P_2 = i^2 R_2 = 4 \text{ W}$$

85. (b) When two resistances are connected in series,
 $R_{\text{eq}} = 2R$

$$\text{Power consumed, } P = \frac{\varepsilon^2}{R_{\text{eq}}} = \frac{\varepsilon^2}{2R}$$

$$\text{In parallel condition, } R_{\text{eq}} = R/2.$$

$$\text{New power, } P' = \frac{\varepsilon^2}{(R/2)}$$

$$\text{or } P' = 4P = 240 \text{ W } (\because P = 60 \text{ W})$$

86. (a) Colour code for carbon resistor

Bl,	Br,	R,	O,	Y,	G	Blue,	V,	Gr,	W
0	1	2	3	4	5	6	7	8	9

$$\text{Resistance, } R = AB \times C \pm D$$

$$\therefore \text{Resistance, } R = 50 \times 10^2 \Omega$$

$$\text{Now using formula, Power, } P = i^2 R$$

$$\therefore i = \sqrt{\frac{P}{R}} = \sqrt{\frac{2}{50 \times 10^2}} = 20 \text{ mA}$$

87. (a) Power, $P = I^2 R$

$$4.4 = 4 \times 10^{-6} \times R$$

$$\Rightarrow R = 1.1 \times 10^6 \Omega$$

When supply of 11 v is connected

$$\text{Power, } P' = \frac{V^2}{R} = \frac{11^2}{1.1} \times \frac{11^2}{1.1} \times 10^{-6}$$

$$= 11 \times 10^{-5} \text{ W}$$

88. (a) Rate of heat i.e., Power developed in the wire =

$$P = \frac{V^2}{R}$$

$$\text{Resistance of the wire of length, } L \quad R_1 = \frac{\rho L}{A} = \frac{\rho L}{\pi r^2}$$

$$\therefore \text{Power, } P_1 = \frac{V^2}{R_1}$$

Resistance of the wire when length is halved i.e., $L/2$

$$R_2 = \frac{\rho \frac{L}{2}}{\pi (2r)^2} = \frac{\rho L}{\pi 8r^2} = \frac{R_1}{8}$$

$$\therefore \text{Power, } P_2 = \frac{V^2}{\frac{R_1}{8}} = \frac{8V^2}{R_1}$$

or, $P_2 = 8P_1$ i.e., power increased 8 times of previous or original wire.

89. (c) From the given circuit, net resistances

$$R_I = 1 \Omega, R_{II} = 1/2 \Omega, R_{III} = 3/2 \Omega$$

It is clear that $R_3 > R_1 > R_2$

$$\text{Hence, } P_3 < P_1 < P_2$$

$$\text{As Power (P)} = \frac{V^2}{R} \Rightarrow P \propto \frac{1}{R}$$

90. (None) Work done in 30s, $W = \int_0^{30} \frac{V^2}{R} dt$

$$\text{or, } W = \int_0^{30} \frac{(200)^2}{100(1 + \alpha \frac{20t}{3})} dt = \frac{(200)^2}{100} \int_0^{30} \frac{dt}{1 + \frac{20\alpha}{3}t}$$

$$= \frac{400 \times 3}{20\alpha} \ln \left(\frac{1 + \frac{20\alpha}{3} \times 30}{1} \right) = 60,000 \ln \left(\frac{6}{5} \right)$$

$$\therefore 120 = 100 \left[1 + \alpha(200) \right]$$

$$\therefore \alpha = \frac{1}{1000}$$

91. (c) Heat energy will be maximum when resistance will be minimum.

92. (c) Total power consumed by electrical appliances in the building, $P_{\text{total}} = 2500 \text{ W}$

$$\text{Watt} = \text{Volt} \times \text{ampere}$$

$$\Rightarrow 2500 = V \times I \Rightarrow 2500 = 220 I$$

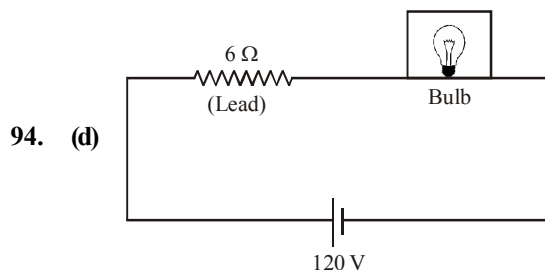
$$\Rightarrow I = \frac{2500}{220} = 11.36 \approx 12 \text{ A}$$

(Minimum capacity of main fuse)

93. (c) Current in each bulb = $\frac{\text{Power}}{\text{Voltage}}$

$$= \frac{100}{220} = 0.45 \text{ A}$$

$$\text{Current through ammeter} = 0.45 \times 3 = 1.35 \text{ A}$$



$$\text{Power of bulb} = 60 \text{ W (given)}$$

$$\text{Resistance of bulb} = \frac{120 \times 120}{60} = 240 \Omega$$

$$\left[\because P = \frac{V^2}{R} \right]$$

$$\text{Power of heater} = 240 \text{ W (given)}$$

$$\text{Resistance of heater} = \frac{120 \times 120}{240} = 60 \Omega$$

Voltage across bulb before heater is switched on,

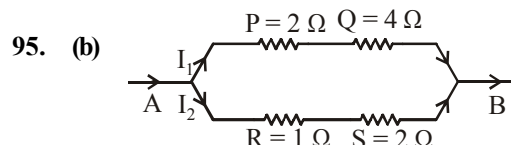
$$V_1 = \frac{240}{246} \times 120 = 117.73 \text{ volt}$$

Voltage across bulb after heater is switched on,

$$V_2 = \frac{48}{54} \times 120 = 106.66 \text{ volt}$$

Hence decrease in voltage

$$V_1 - V_2 = 117.073 - 106.66 = 10.04 \text{ Volt (approximately)}$$



$$R_1 = P + Q = 2 \Omega + 4 \Omega = 6 \Omega$$

$$R_2 = R + S = 1 \Omega + 2 \Omega = 3 \Omega$$

$$I_1 R_1 = I_2 R_2$$

$$I_1 = \frac{R_2}{R_1} I_2 = \frac{3}{6} I_2 = \frac{I_2}{2}$$

$$\text{or } I_2 = 2I_1$$

$$\text{Heat flow } H = I^2 R t$$

$$\text{For Q, } H_Q = I_1^2 Q t = \frac{I_2^2}{4} \times 4 t = I_2^2 t$$

$$\text{For S, } H_S = I_2^2 S t = I_2^2 \cdot 2 t = 2I_2^2 t$$

$\therefore$ Greatest amount of heat generated by S.

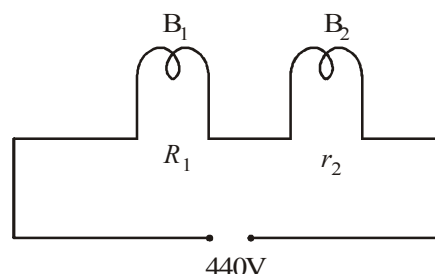
96. (c) Current capacity of 25 W bulb

$$I_1 = \frac{W_1}{V_1} = \frac{25}{220} \text{ Amp}$$

Current capacity of 100 W bulb

$$I_2 = \frac{W_2}{V_2} = \frac{100}{220} \text{ Amp}$$

The current flowing through the circuit



Resistance of 25 W bulb,

$$R_1 = \frac{V_1^2}{P_1} = \frac{(220)^2}{25};$$

Resistance of 100 W bulb

$$R_2 = \frac{V_2^2}{P} = \frac{(220)^2}{100}$$

$$R_{\text{eff}} = R_1 + R_2$$

Current flowing through circuit

$$I = \frac{440}{R_{eff}}$$

$$I = \frac{440}{\frac{(220)^2}{25} + \frac{(220)^2}{100}}$$

$$= \frac{440}{(220)^2 \left[\frac{1}{25} + \frac{1}{100} \right]}; \quad I = \frac{40}{220} \text{ Amp}$$

$$\therefore I_1 \left(= \frac{25}{220} A \right) < I \left(= \frac{40}{220} A \right) < I_2 \left(= \frac{100}{220} A \right)$$

Thus the bulb rated 25 W–220 will fuse.

97. (b) Total resistance = $\frac{6 \times 3}{6 + 3} = 2\Omega$

Current in circuit = $\frac{6}{2} = 3A$

Therefore current through bulb 1 is 2A and bulb 2 is 1A.
So bulb 1 will glow more

98. (c) Resistors 4 Ω , 6 Ω and 12 Ω are connected in parallel, its equivalent resistance (R) is given by

$$\frac{1}{R} = \frac{1}{4} + \frac{1}{6} + \frac{1}{12} \Rightarrow R = \frac{12}{6} = 2\Omega$$

Again R is connected to 1.5 V battery whose internal resistance $r = 1\Omega$.

Equivalent resistance now,

$$R' = 2\Omega + 1\Omega = 3\Omega$$

Current, $I_{total} = \frac{V}{R'} = \frac{1.5}{3} = \frac{1}{2} A$

$$I_{total} = \frac{1}{2} = 3x + 2x + x = 6x$$

$$\Rightarrow x = \frac{1}{12}$$

$\therefore$ Current through 4 Ω resistor = $3x$

$$= 3 \times \frac{1}{12} = \frac{1}{4} A$$

Therefore, rate of Joule heating in the 4 Ω resistor

$$= I^2 R = \left(\frac{1}{4} \right)^2 \times 4 = \frac{1}{4} = 0.25 W$$

99. (c)

100. (b) Let resistance of bulb filament be R_0 at $0^\circ C$ using $R = R_0(1 + \alpha \Delta T)$ we have

$$R_1 = R_0[1 + \alpha \times 100] = 100 \dots (1)$$

$$R_2 = R_0[1 + \alpha \times T] = 200 \dots (2)$$

On dividing we get

$$\frac{200}{100} = \frac{1 + \alpha T}{1 + 100\alpha} \Rightarrow 2 = \frac{1 + 0.005 T}{1 + 100 \times 0.005}$$

$$\Rightarrow T = 400^\circ C$$

Note : We may use this expression as an approximation because the difference in the answers is appreciable. For accurate results one should use $R = R_0 e^{\alpha \Delta T}$

101. (c) The resistance of the electric bulb is

$$R = \frac{V^2}{P} = \frac{(220)^2}{100}$$

The power consumed when operated at 110 V is

$$P' = \frac{V^2}{R}$$

$$\Rightarrow P = \frac{(110)^2}{(220)^2 / 100} = \frac{100}{4} = 25 W$$

102. (b) Heat generated,

$$H = \frac{V^2 t}{R}$$

After cutting equal length of heater coil will become half.

As $R \propto \ell$

Resistance of half the coil = $\frac{R}{2}$

$$H' = \frac{V^2 t}{\frac{R}{2}} = 2H$$

$\therefore$ As R reduces to half, ' H ' will be doubled.

103. (b) Power, $P = Vi = \frac{V^2}{R}$

$\therefore$ Resistance of tungsten filament when in use

$$R_{hot} = \frac{V^2}{P} = \frac{200 \times 200}{100} = 400\Omega$$

Resistance when not in use *i.e.*, cold resistance

$$R_{cold} = \frac{400}{10} = 40\Omega$$

104. (a) Thermistors are usually made of metaloxides with high temperature coefficient of resistivity.

105. (a) Heat supplied in time t for heating 1L water from $10^\circ C$ to $40^\circ C$

$$\Delta Q = m C_p \times \Delta T$$

$$= 1 \times 4180 \times (40 - 10) = 4180 \times 30$$

$$\text{But } \Delta Q = P \times t = 836 \times t$$

$$\Rightarrow t = \frac{4180 \times 30}{836} = 150s$$

106. (c) We know that resistance,

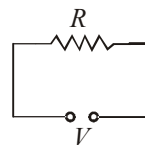
$$R = \frac{V_{rated}^2}{P_{rated}} = \frac{(220)^2}{1000} = 48.4\Omega$$

When this bulb is connected to 110 volt mains supply we get

$$P = \frac{V^2}{R} = \frac{(110)^2}{48.4} = 250W$$

107. (b) **Case 1** Initial power dissipation,

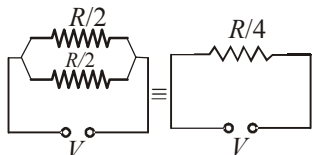
$$P_1 = \frac{V^2}{R}$$



Case 2

When wire is cut into two equal pieces, the resistance of each piece is $\frac{R}{2}$. When they are connected in parallel

Equivalent resistance, $R_{eq} = \frac{R/2}{2} = \frac{R}{4}$



Power dissipated,

$$P_2 = \frac{V^2}{R/4} = 4 \left(\frac{V^2}{R} \right) = 4P_1$$

- 108. (b)** The equivalent resistance of parallel combination of 2Ω and R is

$$R_{eq} = \frac{2 \times R}{2 + R}$$

$$\therefore \text{Power dissipation } P = \frac{V^2}{R_{eq}} \quad \therefore 150 = \frac{(15)^2}{R_{eq}}$$

$$\Rightarrow 150 = \frac{225 \times (R + 2)}{2R} \Rightarrow \frac{2R}{2 + R} = \frac{3}{2}$$

$$\Rightarrow 4R = 6 + 3R \Rightarrow R = 6\Omega$$

- 109. (d)** The voltmeter of resistance $10k\Omega$ is parallel to the resistance of 400Ω . So, their equivalent resistance is

$$\frac{1}{R'} = \frac{1}{10k\Omega} + \frac{1}{400\Omega} = \frac{1}{10000} + \frac{1}{400}$$

$$\Rightarrow \frac{1}{R'} = \frac{1 + 25}{10000} = \frac{26}{10000}$$

$$\Rightarrow R' = \frac{10000}{26}\Omega$$

Using Ohm's law, current in the circuit

$$I = \frac{\text{Voltage}}{\text{Net Resistance}} = \frac{6}{\frac{10000}{26} + 800}$$

Potential difference measured by voltmeter

$$V = IR' = \frac{6}{\frac{10000}{26} + 800} \times \frac{10000}{26}$$

$$\Rightarrow V = \frac{150}{77} = 1.95 \text{ volt}$$

- 110. (b)** Multimeter shows deflection in both cases i.e. before and after reversing the probes if the chosen component is capacitor.

- 111. (a)** Potential gradient, $x = \frac{\text{Potential drop}}{\text{length}}$

Here, Potential drop = 1.02

Balancing length from P = $100 - 49$

$$\therefore x = \frac{1.02}{100 - 49} = 0.02 \text{ volt/cm}$$

- 112. (40)** For the given meter bridge

$$\frac{R}{S} = \frac{\ell_1}{100 - \ell_1} \quad \text{Where, } \ell_1 = \text{balancing length}$$

$$\Rightarrow \frac{R}{S} = \frac{25}{75} = \frac{1}{3} \quad \dots(i)$$

New resistance,

$$R' = \frac{\rho \frac{\ell}{2}}{\frac{A}{4}} = \rho \frac{\ell \times 2}{A} \quad \left(\because R = \rho \frac{\ell}{A} \right)$$

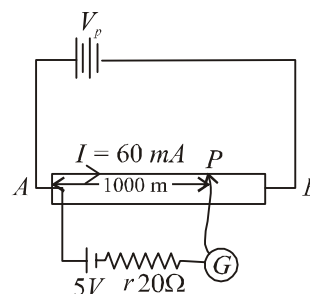
$$\Rightarrow R' = 2R$$

$$\frac{R'}{S} = \frac{\ell_2}{100 - \ell_2} \Rightarrow \frac{2R}{S} = \frac{\ell_2}{100 - \ell_2}$$

$$\Rightarrow 2 \times \frac{1}{3} = \frac{\ell_2}{100 - \ell_2} \quad \text{Using (i)}$$

$$\Rightarrow \ell_2 = 40 \text{ cm}$$

- 113. (d)**



Let R be the resistance of the whole wire
Potential gradient for the potentiometer wire

$$'AB' = -\frac{dV}{d\ell} = \frac{I \times R}{\ell} = \left[\frac{60 \times R}{\ell_{AB}} \right] \text{ mv / m}$$

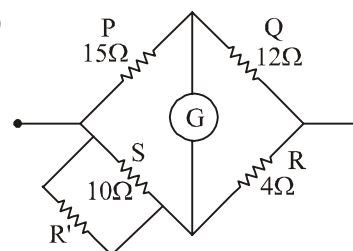
$$V_{AP} = \left(\frac{dV}{d\ell_{AB}} \right) \ell_{AP} = \frac{60 \times R}{1200} \times 1000 \text{ mV}$$

$$\Rightarrow V_{AP} = 50 \text{ mV}$$

Also, $V_{AP} = 5 \text{ V}$ (for balance point at P)

$$\therefore R = \frac{V_{AP}}{50 \times 10^{-3}} = \frac{5}{50 \times 10^{-3}} = 100\Omega$$

- 114. (10)**



As per Wheatstone bridge balance condition $\frac{P}{Q} = \frac{S}{R}$

Let resistance R' is connected in parallel with resistance S of 10Ω

$$\therefore \frac{15}{12} = \frac{10R'}{10+R'} \Rightarrow 5 = \frac{10R'}{10+R'}$$

$$\Rightarrow 50 + 5R' = 10R'$$

$$\therefore R' = \frac{50}{5} = 10\Omega$$

115. (12) We know that

$E \propto \ell$ where ℓ is the balancing length

$$\therefore E = k(560) \quad \dots(i)$$

When the balancing length changes by 60 cm

$$\frac{E}{r+10} \cdot 10 = k(500) \quad \dots(ii)$$

Dividing (i) by (ii) we get

$$\Rightarrow \frac{r+10}{10} = \frac{56}{50} \Rightarrow 50r + 500 = 560$$

$$\Rightarrow r = \frac{6}{5}\Omega = \frac{N}{10}\Omega \Rightarrow N = 12$$

116. (c) For a balanced bridge

$$\frac{R_1}{R_2} = \frac{l_2}{l_1}$$

$$\text{So } \frac{R}{X} = \frac{l}{100-l}$$

Using the above expression

$$X = \frac{R(100-l)}{l}$$

$$\text{for observation (1) } X = \frac{100 \times 40}{60} = \frac{2000}{3}\Omega$$

$$\text{for observation (2) } X = \frac{100 \times 87}{13} = \frac{8700}{13}\Omega$$

$$\text{for observation (3) } X = \frac{10 \times 98.5}{1.5} = \frac{1970}{3}\Omega$$

$$\text{for observation (4) } X = \frac{1 \times 99}{1} = 99\Omega$$

Clearly we can see that the value of x calculated in observation (4) is inconsistent than other.

117. (c) The resistance of potentiometer wire

$$R = 0.01 \times 400 = 4\Omega$$

Current in the wire

$$i = \frac{V}{R_T} = \frac{3}{4 + 0.5 + 0.57 + 1} = \frac{1}{2}A$$

$$\text{Now } V = iR_{AJ} = \frac{1}{2} \times (0.01 \times 50) = 0.25V.$$

118. (c) We have given

$$\frac{dR}{d\ell} \propto \frac{1}{\sqrt{\ell}} \Rightarrow \frac{dR}{d\ell} = k \times \frac{1}{\sqrt{\ell}} \quad (\text{where } k \text{ is constant})$$

$$dR = k \frac{d\ell}{\sqrt{\ell}}$$

Let R_1 and R_2 be the resistance of AP and PB respectively.

Using wheatstone bridge principle

$$\therefore \frac{R'}{R_1} = \frac{R_2}{R_1} \text{ or } R_1 = R_2$$

$$\text{Now, } \int dR = k \int \frac{d\ell}{\sqrt{\ell}}$$

$$\therefore R_1 = k \int_0^{\ell} \ell^{-1/2} d\ell = k \cdot 2\sqrt{\ell}$$

$$R_2 = k \int_{\ell}^1 \ell^{-1/2} d\ell = k \cdot (2 - 2\sqrt{\ell})$$

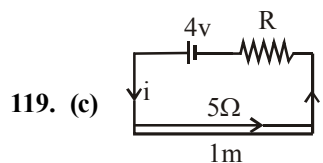
Putting $R_1 = R_2$

$$k2\sqrt{\ell} = k(2 - 2\sqrt{\ell})$$

$$\therefore 2\sqrt{\ell} = 1$$

$$\sqrt{\ell} = \frac{1}{2}$$

$$\text{i.e., } \ell = \frac{1}{4}m \Rightarrow 0.25m$$



119. (c)

Current flowing through the circuit (I) is given by

$$I = \left(\frac{4}{R+5} \right) A$$

Resistance of length 10 cm of wire

$$= 5 \times \frac{10}{100} = 0.5\Omega$$

According to question,

$$5 \times 10^{-3} = \left(\frac{4}{R+5} \right) \cdot (0.5)$$

$$\therefore \frac{4}{R+5} = 10^{-2} \text{ or } R+5 = 400\Omega$$

$$\therefore R = 395\Omega$$

120. (b) Given, Emf of cell, $\varepsilon = 0.5V$

Rheostat resistance, $R_h = 2\Omega$

Potential gradient is

$$\frac{dv}{dL} = \left(\frac{6}{2+4} \right) \times \frac{4}{L}$$

Let null point be at ℓ cm when cell of emf $\varepsilon = 0.5V$ is used.

$$\text{thus } \varepsilon_1 = 0.5V = \left(\frac{6}{2+4} \right) \times \frac{4}{L} \times \ell \quad \dots(i)$$

For resistance $R_h = 6\Omega$ new potential gradient is

$$\left(\frac{6}{4+6}\right) \times \frac{4}{L} \text{ and at null point}$$

$$\left(\frac{6}{4+6}\right) \left(\frac{4}{L}\right) \times \ell = \varepsilon_2 \quad \dots (ii)$$

Dividing equation (i) by (ii) we get

$$\frac{0.5}{\varepsilon_2} = \frac{10}{6} \text{ thus } \varepsilon_2 = 0.3\text{V}$$

121. (d)

122. (c) Initially at null deflection $\frac{R_1}{R_2} = \frac{2}{3} \quad \dots (i)$

Finally at null deflection, when null point is shifted

$$\frac{R_1+10}{R_2} = 1 \Rightarrow R_1+10=R_2 \quad \dots (ii)$$

Solving equations (i) and (ii) we get

$$\frac{2R_2}{3} + 10 = R_2$$

$$10 = \frac{R_2}{3} \Rightarrow R_2 = 30\Omega$$

& $R_1 = 20\Omega$

Now if required resistance is R then

$$\frac{30 \times R}{30+R} = \frac{2}{3}$$

$$R = 60\Omega$$

123. (c) Let x be the length AJ at which galvanometer shows null deflection current,

$$i = \frac{\varepsilon}{12r+r} = \frac{3}{13r} \text{ or, } i \left(\frac{x}{L} \cdot 12r \right) = \frac{\varepsilon}{2}$$

$$\Rightarrow \frac{\varepsilon}{13r} \left[\frac{x}{L} \cdot 12r \right] = \frac{\varepsilon}{2} \Rightarrow \frac{\varepsilon}{13r} \left[\frac{x}{L} \cdot 12r \right] = \frac{\varepsilon}{2}$$

$$\text{or, } x = \frac{13L}{24}$$

124. (a) Given, colour code of resistance,

R_1 = Orange, Red and Brown

$$\therefore R_1 = 32 \times 10 = 320$$

using balanced wheatstone bridge principle,

$$\frac{R_1}{R_3} = \frac{R_2}{R_4} \Rightarrow \frac{320}{R_3} = \frac{80}{40}$$

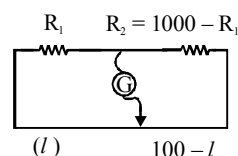
$\therefore R_3 = 160$ i.e. colour code for R_3 Brown, Blue and Brown

125. (b) Using formula, internal resistance,

$$r = \left(\frac{l_1 - l_2}{l_2} \right) s = \left(\frac{52 - 40}{40} \right) \times 5 = 1.5\Omega$$

126. (c) $R_1 + R_2 = 1000$

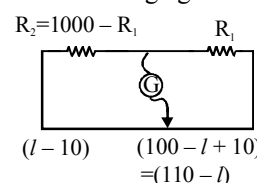
$$\Rightarrow R_2 = 1000 - R_1$$



On balancing condition

$$R_1(100-l) = (1000-R_1)l \quad \dots (i)$$

On Interchanging resistance balance point shifts left by 10 cm



On balancing condition

$$(1000-R_1)(110-l) = R_1(l-10)$$

$$\text{or, } R_1(l-10) = (1000-R_1)(110-l) \quad \dots (ii)$$

Dividing eqn (i) by (ii)

$$\frac{100-l}{l-10} = \frac{l}{110-l}$$

$$\Rightarrow (100-l)(110-l) = l(l-10)$$

$$\Rightarrow 11000 - 100l - 110l + l^2 = l^2 - 10l$$

$$\Rightarrow 11000 = 200l$$

$$\text{or, } l = 55$$

Putting the value of 'l' in eqn (i)

$$R_1(100-55) = (1000-R_1)55$$

$$\Rightarrow R_1(45) = (1000-R_1)55$$

$$\Rightarrow R_1(9) = (1000-R_1)11$$

$$\Rightarrow 20R_1 = 11000$$

$$\therefore R_1 = 550K\Omega$$

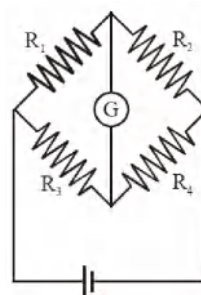
127. (b) For a balanced meter bridge,

$$\frac{X}{39.5} = \frac{Y}{(100-39.5)} \Rightarrow Y = 39.5 = X \times (100-39.5)$$

$$\text{or, } X = \frac{12.5 \times 39.5}{60.5} = 8.16\Omega$$

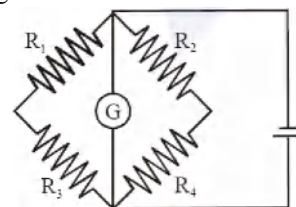
When X and Y are interchanged l_1 and $(100-l_1)$ will also interchange so, $l_2 = 60.5$ cm

128. (d) There is no change in null point, if the cell and the galvanometer are exchanged in a balanced wheatstone bridge.



$$\text{On balancing condition } \frac{R_1}{R_3} = \frac{R_2}{R_4}$$

After exchange



$$\text{On balancing condition } \frac{R_1}{R_2} = \frac{R_3}{R_4}$$

129. (d) In balance position of bridge, $\frac{P}{Q} = \frac{l}{(100-l)}$

Initially neutral position is 60 cm. from A, so

$$\frac{4}{60} = \frac{Q}{40} \Rightarrow Q = \frac{16}{6} = \frac{8}{3} \Omega$$

Now, when unknown resistance R is connected in series to P, neutral point is 80 cm from A then,

$$\frac{4+R}{80} = \frac{Q}{20}$$

$$\frac{4+R}{80} = \frac{8}{60}$$

$$R = \frac{64}{6} - 4 = \frac{64-24}{6} = \frac{40}{6} \Omega$$

Hence, the value of unknown resistance R is $= \frac{20}{3} \Omega$

130. (d) When key is at point (1)

$$V_1 = iR_1 = x l_1$$

When key is at (3)

$$V_2 = i(R_1 + R_2) = x l_2$$

$$\frac{R_1}{R_1 + R_2} = \frac{l_1}{l_2} \Rightarrow \frac{R_1}{R_2} = \frac{l_1}{l_2 - l_1}$$

131. (c) As the two cells oppose each other hence, the effective emf in closed circuit is $15 - 10 = 5$ V and net resistance is $1 + 0.6 = 1.6 \Omega$ (because in the closed circuit the internal resistance of two cells are in series. Current in the circuit,

$$I = \frac{\text{effective emf}}{\text{total resistance}} = \frac{5}{1.6} A$$

The potential difference across voltmeter will be same as the terminal voltage of either cell.

Since the current is drawn from the cell of 15 V

$$\begin{aligned} \therefore V_1 &= E_1 - I r_1 \\ &= 15 - \frac{5}{1.6} \times 0.6 = 13.1 \text{ V} \end{aligned}$$

132. (c) Balancing length l will give emf of cell

$$\therefore E = K l$$

Here K is potential gradient.

If the cell is short circuited by resistance 'R'

Let balancing length obtained be l' then

$$V = K l'$$

$$r = \left(\frac{E - V}{V} \right) R$$

$$\Rightarrow V = E - V \quad [\because r = R \text{ given}]$$

$$\Rightarrow 2V = E$$

$$\text{or, } 2K l' = K l$$

$$\therefore l' = \frac{l}{2}$$

133. (c) To measure the emf of a cell we prefer potentiometer rather than voltmeter because

- (i) the length of potentiometer which allows greater precision.
- (ii) in case of potentiometer, no current flows through the cell.
- (iii) of high sensitivity.

134. (a) Bridge wire in a sensitive meter bridge wire should be of high resistivity and low temperature coefficient.

135. (c) From question, $\frac{x}{y} = \frac{40}{100-40} = \frac{2}{3}$

$$\Rightarrow x = \frac{2}{3} y$$

$$\text{Again, } \frac{3x}{y} = \frac{Z}{100-Z}$$

$$\text{or } \frac{3 \times \frac{2y}{3}}{y} = \frac{Z}{100-Z}$$

Solving we get $Z = 67$ cm

Therefore new position of null point $\cong 67$ cm

136. (c) Potential gradient

$$\Rightarrow k = \frac{V}{\ell} = \frac{IR}{\ell} = \frac{I}{\ell} \left(\frac{\rho \ell}{A} \right) = \frac{I \rho}{A}$$

$$k = \frac{0.2 \times 4 \times 10^{-7}}{8 \times 10^{-7}} = \frac{0.8}{8} = 0.1 \text{ V/m}$$

137. (b) Given,

Balance point from one end, $\ell_1 = 20$ cm

From the condition for balance of metre bridge, we have

$$\frac{55}{R} = \frac{\ell_1}{100 - \ell_1}$$

$$\frac{55}{R} = \frac{20}{80}$$

$$\Rightarrow R = 220 \Omega$$

138. (b) From balanced wheat stone bridge $\frac{P}{Q} = \frac{R}{S}$ where

$$S = \frac{S_1 S_2}{S_1 + S_2}$$

139. (c) Initial balancing length, $\ell_1 = 240$ cm New balancing length, $\ell_2 = 120$ cm.

The internal resistance of the cell,

$$r = \left(\frac{\ell_1 - \ell_2}{\ell_2} \right) \times R = \frac{240 - 120}{120} \times 2 = 2 \Omega$$

140. (c) From the balanced wheat stone bridge

$$\frac{R_1}{R_2} = \frac{\ell_1}{\ell_2}$$

$$\text{where } \ell_2 = 100 - \ell_1$$

$$\text{In the first case } \frac{X}{Y} = \frac{20}{80}$$

$$Y = 4X$$

In the second case

$$\frac{4X}{Y} = \frac{\ell}{100 - \ell}$$

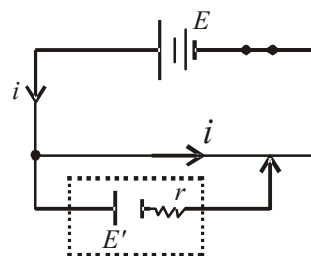
$$\Rightarrow \frac{4X}{4X} = \frac{\ell}{100 - \ell}$$

$$\Rightarrow \ell = 50$$

141. (d) From the principle of potentiometer, $V \propto l$
If a cell of emf E is employed in the circuit between the ends of potentiometer wire of length L , then

$$\frac{V}{E} = \frac{l}{L};$$

$$\Rightarrow V = \frac{El}{L} = \frac{30E}{100}$$



Note : In this arrangement, the internal resistance of the battery E does not play any role as current is not passing through the battery.

$$142. (d) \quad i_g \times G = (i - i_g) S$$

$$\therefore S = \frac{i_g \times G}{i - i_g} = \frac{1 \times 0.81}{10 - 1} = 0.09 \Omega$$

143. (c) To use an ammeter in place of voltmeter, we must connect a high resistance in series with the ammeter. Connecting high resistance in series makes its resistance much higher.

Moving Charges and Magnetism



TOPIC 1 Motion of Charged Particle in Magnetic Field



1. An electron is moving along $+x$ direction with a velocity of $6 \times 10^6 \text{ ms}^{-1}$. It enters a region of uniform electric field of 300 V/cm pointing along $+y$ direction. The magnitude and direction of the magnetic field set up in this region such that the electron keeps moving along the x direction will be : **[Sep. 06, 2020 (I)]**

- (a) $3 \times 10^{-4} \text{ T}$, along $+z$ direction
 (b) $5 \times 10^{-3} \text{ T}$, along $-z$ direction
 (c) $5 \times 10^{-3} \text{ T}$, along $+z$ direction
 (d) $3 \times 10^{-4} \text{ T}$, along $-z$ direction

2. A particle of charge q and mass m is moving with a velocity $-v\hat{i}$ ($v \neq 0$) towards a large screen placed in the $Y-Z$ plane at a distance d . If there is a magnetic field $\vec{B} = B_0\hat{k}$, the minimum value of v for which the particle will not hit the screen is: **[Sep. 06, 2020 (I)]**

- (a) $\frac{qdB_0}{3m}$ (b) $\frac{2qdB_0}{m}$
 (c) $\frac{qdB_0}{m}$ (d) $\frac{qdB_0}{2m}$

3. A charged particle carrying charge $1 \mu\text{C}$ is moving with velocity $(2\hat{i} + 3\hat{j} + 4\hat{k}) \text{ ms}^{-1}$. If an external magnetic field of $(5\hat{i} + 3\hat{j} - 6\hat{k}) \times 10^{-3} \text{ T}$ exists in the region where the particle is moving then the force on the particle is $\vec{F} \times 10^{-9} \text{ N}$. The vector $\vec{F}$ is : **[Sep. 03, 2020 (I)]**

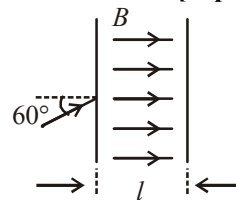
- (a) $-0.30\hat{i} + 0.32\hat{j} - 0.09\hat{k}$
 (b) $-30\hat{i} + 32\hat{j} - 9\hat{k}$
 (c) $-300\hat{i} + 320\hat{j} - 90\hat{k}$
 (d) $-3.0\hat{i} + 3.2\hat{j} - 0.9\hat{k}$

4. A beam of protons with speed $4 \times 10^5 \text{ ms}^{-1}$ enters a uniform magnetic field of 0.3 T at an angle of 60° to the magnetic field. The pitch of the resulting helical path of protons is close to : (Mass of the proton = $1.67 \times 10^{-27} \text{ kg}$, charge of the proton = $1.69 \times 10^{-19} \text{ C}$) **[Sep. 02, 2020 (I)]**

- (a) 2 cm (b) 5 cm (c) 12 cm (d) 4 cm

5. The figure shows a region of length ' l ' with a uniform magnetic field of 0.3 T in it and a proton entering the region with velocity $4 \times 10^5 \text{ ms}^{-1}$ making an angle 60° with the field. If the proton completes 10 revolution by the time it cross the region shown, ' l ' is close to (mass of proton = $1.67 \times 10^{-27} \text{ kg}$, charge of the proton = $1.6 \times 10^{-19} \text{ C}$) **[Sep. 02, 2020 (II)]**

- (a) 0.11 m
 (b) 0.88 m
 (c) 0.44 m
 (d) 0.22 m



6. Proton with kinetic energy of 1 MeV moves from south to north. It gets an acceleration of 10^{12} m/s^2 by an applied magnetic field (west to east). The value of magnetic field: (Rest mass of proton is $1.6 \times 10^{-27} \text{ kg}$) **[8 Jan 2020, I]**

- (a) 0.71 mT (b) 7.1 mT
 (c) 0.071 mT (d) 71 mT

7. A particle having the same charge as of electron moves in a circular path of radius 0.5 cm under the influence of a magnetic field of 0.5 T . If an electric field of 100 V/m makes it to move in a straight path then the mass of the particle is (Given charge of electron = $1.6 \times 10^{-19} \text{ C}$) **[12 April 2019, I]**

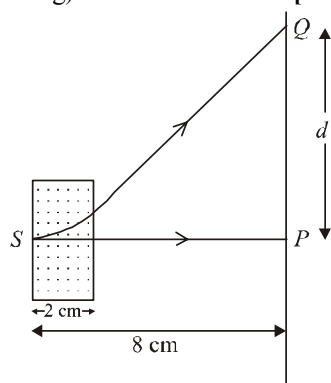
- (a) $9.1 \times 10^{-31} \text{ kg}$ (b) $1.6 \times 10^{-27} \text{ kg}$
 (c) $1.6 \times 10^{-19} \text{ kg}$ (d) $2.0 \times 10^{-24} \text{ kg}$

8. An electron, moving along the x -axis with an initial energy of 100 eV , enters a region of magnetic field $\vec{B} = (1.5 \times 10^{-3} \text{ T})\hat{k}$ at S (see figure). The field extends between $x = 0$ and $x = 2 \text{ cm}$. The electron is detected at the point Q on a screen

placed 8 cm away from the point S. The distance d between P and Q (on the screen) is :

(Electron's charge = 1.6×10^{-19} C, mass of electron = 9.1×10^{-31} kg)

[12 April 2019, II]



- (a) 11.65 cm (b) 12.87 cm
(c) 1.22 cm (d) 2.25 cm

9. A proton, an electron, and a Helium nucleus, have the same energy. They are in circular orbits in a plane due to magnetic field perpendicular to the plane. Let r_p , r_e and r_{He} be their respective radii, then,

[10 April 2019, I]

- (a) $r_e > r_p = r_{He}$ (b) $r_e < r_p = r_{He}$
(c) $r_e < r_p < r_{He}$ (d) $r_e > r_p > r_{He}$

10. A proton and an α -particle (with their masses in the ratio of 1 : 4 and charges in the ratio 1 : 2) are accelerated from rest through a potential difference V . If a uniform magnetic field (B) is set up perpendicular to their velocities, the ratio of the radii $r_p : r_\alpha$ of the circular paths described by them will be:

[12 Jan 2019, I]

- (a) $1:\sqrt{2}$ (b) 1:2 (c) 1:3 (d) $1:\sqrt{3}$

11. In an experiment, electrons are accelerated, from rest, by applying a voltage of 500 V. Calculate the radius of the path if a magnetic field 100 mT is then applied.

[Charge of the electron = 1.6×10^{-19} C

Mass of the electron = 9.1×10^{-31} kg] [11 Jan 2019, I]

- (a) 7.5×10^{-3} m (b) 7.5×10^{-2} m
(c) 7.5m (d) 7.5×10^{-4} m

12. The region between $y = 0$ and $y = d$ contains a magnetic field $\vec{B} = B\hat{z}$. A particle of mass m and charge q enters the region with a velocity $\vec{v} = v\hat{i}$. if $d = \frac{mv}{2qB}$, the

acceleration of the charged particle at the point of its emergence at the other side is :

[11 Jan 2019, II]

- (a) $\frac{qvB}{m} \left(\frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{j} \right)$ (b) $\frac{qvB}{m} \left(\frac{\sqrt{3}}{2}\hat{i} + \frac{1}{2}\hat{j} \right)$
(c) $\frac{qvB}{m} \left(-\frac{\hat{j} + \hat{i}}{\sqrt{2}} \right)$ (d) $\frac{qvB}{m} \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} \right)$

13. An electron, a proton and an alpha particle having the same kinetic energy are moving in circular orbits of radii r_e , r_p , r_α respectively in a uniform magnetic field B . The relation between r_e , r_p , r_α is :

[2018]

- (a) $r_e > r_p = r_\alpha$ (b) $r_e < r_p = r_\alpha$
(c) $r_e < r_p < r_\alpha$ (d) $r_e < r_\alpha < r_p$

14. A negative test charge is moving near a long straight wire carrying a current. The force acting on the test charge is parallel to the direction of the current. The motion of the charge is :

[Online April 9, 2017]

- (a) away from the wire
(b) towards the wire
(c) parallel to the wire along the current
(d) parallel to the wire opposite to the current

15. In a certain region static electric and magnetic fields exist.

The magnetic field is given by $\vec{B} = B_0(\hat{i} + 2\hat{j} - 4\hat{k})$. If a

test charge moving with a velocity $\vec{v} = v_0(3\hat{i} - \hat{j} + 2\hat{k})$ experiences no force in that region, then the electric field in the region, in SI units, is :

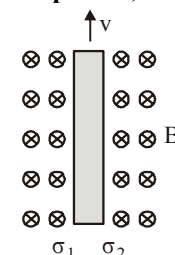
[Online April 8, 2017]

- (a) $\vec{E} = -v_0 B_0(3\hat{i} - 2\hat{j} - 4\hat{k})$ (b) $\vec{E} = -v_0 B_0(\hat{i} + \hat{j} + 7\hat{k})$
(c) $\vec{E} = v_0 B_0(14\hat{j} + 7\hat{k})$ (d) $\vec{E} = -v_0 B_0(14\hat{j} + 7\hat{k})$

16. Consider a thin metallic sheet perpendicular to the plane of the paper moving with speed ' v ' in a uniform magnetic field B going into the plane of the paper (See figure). If charge densities σ_1 and σ_2 are induced on the left and right surfaces, respectively, of the sheet then (ignore fringe effects):

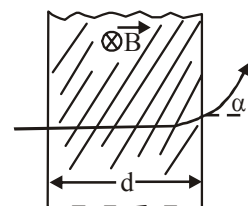
[Online April 10, 2016]

- (a) $\sigma_1 = \frac{-\epsilon_0 vB}{2}$, $\sigma_2 = \frac{\epsilon_0 vB}{2}$
(b) $\sigma_1 = \epsilon_0 vB$, $\sigma_2 = -\epsilon_0 vB$
(c) $\sigma_1 = \frac{\epsilon_0 vB}{2}$, $\sigma_2 = \frac{-\epsilon_0 vB}{2}$
(d) $\sigma_1 = \sigma_2 = \epsilon_0 vB$



17. A proton (mass m) accelerated by a potential difference V flies through a uniform transverse magnetic field B . The field occupies a region of space by width ' d '. If α be the angle of deviation of proton from initial direction of motion (see figure), the value of $\sin \alpha$ will be :

[Online April 10, 2015]



- (a) $qV\sqrt{\frac{Bd}{2m}}$ (b) $\frac{B}{2}\sqrt{\frac{qd}{mV}}$
(c) $\frac{B}{d}\sqrt{\frac{q}{2mV}}$ (d) $Bd\sqrt{\frac{q}{2mV}}$

18. A positive charge 'q' of mass 'm' is moving along the + x axis. We wish to apply a uniform magnetic field B for time Δt so that the charge reverses its direction crossing the y axis at a distance d. Then: [Online April 12, 2014]
- (a) $B = \frac{mv}{qd}$ and $\Delta t = \frac{\pi d}{v}$ (b) $B = \frac{mv}{2qd}$ and $\Delta t = \frac{\pi d}{2v}$
 (c) $B = \frac{2mv}{qd}$ and $\Delta t = \frac{\pi d}{2v}$ (d) $B = \frac{2mv}{qd}$ and $\Delta t = \frac{\pi d}{v}$
19. A particle of charge $16 \times 10^{-16} \text{ C}$ moving with velocity 10 ms^{-1} along x-axis enters a region where magnetic field of induction $\vec{B}$ is along the y-axis and an electric field of magnitude 10^4 Vm^{-1} is along the negative z-axis. If the charged particle continues moving along x-axis, the magnitude of $\vec{B}$ is : [Online April 23, 2013]
- (a) $16 \times 10^3 \text{ Wb m}^{-2}$ (b) $2 \times 10^3 \text{ Wb m}^{-2}$
 (c) $1 \times 10^3 \text{ Wb m}^{-2}$ (d) $4 \times 10^3 \text{ Wb m}^{-2}$
20. Proton, deuteron and alpha particle of same kinetic energy are moving in circular trajectories in a constant magnetic field. The radii of proton, deuteron and alpha particle are respectively r_p , r_d and r_α . Which one of the following relation is correct? [2012]
- (a) $r_\alpha = r_p = r_d$ (b) $r_\alpha = r_p < r_d$
 (c) $r_\alpha > r_d > r_p$ (d) $r_\alpha = r_d > r_p$
21. This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
- Statement 1:** A charged particle is moving at right angle to a static magnetic field. During the motion the kinetic energy of the charge remains unchanged.
- Statement 2:** Static magnetic field exert force on a moving charge in the direction perpendicular to the magnetic field.
- [Online May 26, 2012]
- (a) Statement 1 is false, Statement 2 is true.
 (b) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.
 (c) Statement 1 is true, Statement 2 is false.
 (d) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.
22. A proton and a deuteron are both accelerated through the same potential difference and enter in a magnetic field perpendicular to the direction of the field. If the deuteron follows a path of radius R, assuming the neutron and proton masses are nearly equal, the radius of the proton's path will be [Online May 19, 2012]
- (a) $\sqrt{2}R$ (b) $\frac{R}{\sqrt{2}}$ (c) $\frac{R}{2}$ (d) R
23. The magnetic force acting on charged particle of charge $2 \mu\text{C}$ in magnetic field of 2 T acting in y-direction, when the particle velocity is $(2\hat{i} + 3\hat{j}) \times 10^6 \text{ ms}^{-1}$ is [Online May 12, 2012]
- (a) 8 N in z-direction (b) 8 N in y-direction
 (c) 4 N in y-direction (d) 4 N in z-direction
24. The velocity of certain ions that pass undeflected through crossed electric field $E = 7.7 \text{ k V/m}$ and magnetic field $B = 0.14 \text{ T}$ is [Online May 7, 2012]
- (a) 18 km/s (b) 77 km/s
 (c) 55 km/s (d) 1078 km/s
25. An electric charge +q moves with velocity $\vec{v} = 3\hat{i} + 4\hat{j} + \hat{k}$ in an electromagnetic field given by $\vec{E} = 3\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{B} = \hat{i} + \hat{j} - 3\hat{k}$. The y - component of the force experienced by + q is : [2011 RS]
- (a) $11q$ (b) $5q$ (c) $3q$ (d) $2q$
26. A charged particle with charge q enters a region of constant, uniform and mutually orthogonal fields $\vec{E}$ and $\vec{B}$ with a velocity $\vec{v}$ perpendicular to both $\vec{E}$ and $\vec{B}$, and comes out without any change in magnitude or direction of $\vec{v}$. Then [2007]
- (a) $\vec{v} = \vec{B} \times \vec{E} / E^2$ (b) $\vec{v} = \vec{E} \times \vec{B} / B^2$
 (c) $\vec{v} = \vec{B} \times \vec{E} / B^2$ (d) $\vec{v} = \vec{E} \times \vec{B} / E^2$
27. A charged particle moves through a magnetic field perpendicular to its direction. Then [2007]
- (a) kinetic energy changes but the momentum is constant
 (b) the momentum changes but the kinetic energy is constant
 (c) both momentum and kinetic energy of the particle are not constant
 (d) both momentum and kinetic energy of the particle are constant
28. In a region, steady and uniform electric and magnetic fields are present. These two fields are parallel to each other. A charged particle is released from rest in this region. The path of the particle will be a [2006]
- (a) helix (b) straight line
 (c) ellipse (d) circle
29. A charged particle of mass m and charge q travels on a circular path of radius r that is perpendicular to a magnetic field B. The time taken by the particle to complete one revolution is [2005]
- (a) $\frac{2\pi q^2 B}{m}$ (b) $\frac{2\pi m q}{B}$ (c) $\frac{2\pi m}{qB}$ (d) $\frac{2\pi qB}{m}$
30. A uniform electric field and a uniform magnetic field are acting along the same direction in a certain region. If an electron is projected along the direction of the fields with a certain velocity then [2005]
- (a) its velocity will increase
 (b) Its velocity will decrease
 (c) it will turn towards left of direction of motion
 (d) it will turn towards right of direction of motion

31. A particle of mass M and charge Q moving with velocity $\vec{v}$ describe a circular path of radius R when subjected to a uniform transverse magnetic field of induction B . The work done by the field when the particle completes one full circle is [2003]
- (a) $\left(\frac{Mv^2}{R}\right) 2\pi R$ (b) zero
(c) $BQ2\pi R$ (d) $BQv2\pi R$
32. If an electron and a proton having same momenta enter perpendicular to a magnetic field, then [2002]
- (a) curved path of electron and proton will be same (ignoring the sense of revolution)
(b) they will move undeflected
(c) curved path of electron is more curved than that of the proton
(d) path of proton is more curved.
33. The time period of a charged particle undergoing a circular motion in a uniform magnetic field is independent of its
(a) speed (b) mass [2002]
(c) charge (d) magnetic induction
36. Magnitude of magnetic field (in SI units) at the centre of a hexagonal shape coil of side 10 cm, 50 turns and carrying current I (Ampere) in units of $\frac{\mu_0 I}{\pi}$ is : [Sep. 03, 2020 (I)]
- (a) $250\sqrt{3}$ (b) $50\sqrt{3}$ (c) $500\sqrt{3}$ (d) $5\sqrt{3}$
37. A long, straight wire of radius a carries a current distributed uniformly over its cross-section. The ratio of the magnetic fields due to the wire at distance $\frac{a}{3}$ and $2a$, respectively from the axis of the wire is: [9 Jan 2020, I]
- (a) $\frac{2}{3}$ (b) 2 (c) $\frac{1}{2}$ (d) $\frac{3}{2}$
38. An electron gun is placed inside a long solenoid of radius R on its axis. The solenoid has n turns/length and carries a current I . The electron gun shoots an electron along the radius of the solenoid with speed v . If the electron does not hit the surface of the solenoid, maximum possible value of v is (all symbols have their standard meaning): [9 Jan 2020, II]

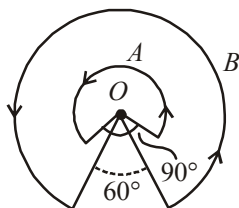
TOPIC 2 Magnetic Field Lines, Biot-Savart's Law and Ampere's Circuital



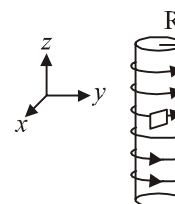
34. A charged particle going around in a circle can be considered to be a current loop. A particle of mass m carrying charge q is moving in a plane with speed v under the

influence of magnetic field $\vec{B}$. The magnetic moment of this moving particle : [Sep. 06, 2020 (II)]

- (a) $\frac{mv^2 \vec{B}}{2 B^2}$ (b) $-\frac{mv^2 \vec{B}}{2 \pi B^2}$
(c) $-\frac{mv^2 \vec{B}}{B^2}$ (d) $-\frac{mv^2 \vec{B}}{2 B^2}$
35. A wire A , bent in the shape of an arc of a circle, carrying a current of 2 A and having radius 2 cm and another wire B , also bent in the shape of arc of a circle, carrying a current of 3 A and having radius of 4 cm, are placed as shown in the figure. The ratio of the magnetic fields due to the wires A and B at the common centre O is : [Sep. 04, 2020 (I)]

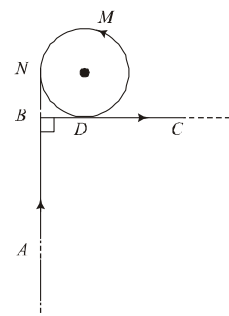


- (a) 4 : 6 (b) 6 : 4
(c) 2 : 5 (d) 6 : 5

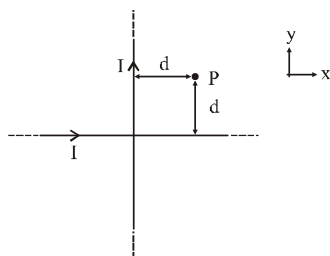


- (a) $\frac{e\mu_0 nIR}{m}$ (b) $\frac{e\mu_0 nIR}{2m}$
(c) $\frac{e\mu_0 nIR}{4m}$ (d) $\frac{2e\mu_0 nIR}{m}$
39. A very long wire $ABDMNDC$ is shown in figure carrying current I . AB and BC parts are straight, long and at right angle. At D wire forms a circular turn $DMND$ of radius R . AB , BC parts are tangential to circular turn at N and D . Magnetic field at the centre of circle is: [8 Jan 2020, II]

- (a) $\frac{\mu_0 I}{2\pi R} \left(\pi + \frac{1}{\sqrt{2}} \right)$
(b) $\frac{\mu_0 I}{2\pi R} \left(\pi - \frac{1}{\sqrt{2}} \right)$
(c) $\frac{\mu_0 I}{2\pi R} (\pi + 1)$
(d) $\frac{\mu_0 I}{2R}$

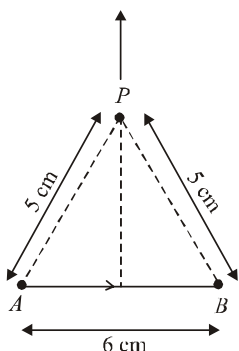


40. Two very long, straight, and insulated wires are kept at 90° angle from each other in xy -plane as shown in the figure.



These wires carry currents of equal magnitude I , whose directions are shown in the figure. The net magnetic field at point P will be : [12 April 2019, I]

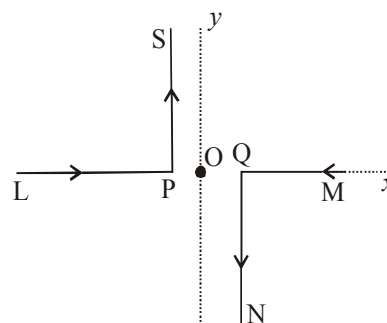
- (a) Zero (b) $-\frac{\mu_0 I}{2\pi d}(\hat{x} + \hat{y})$
 (c) $\frac{+\mu_0 I}{\pi d}(\hat{z})$ (d) $\frac{\mu_0 I}{2\pi d}(\hat{x} + \hat{y})$
41. A thin ring of 10 cm radius carries a uniformly distributed charge. The ring rotates at a constant angular speed of 40 rad s^{-1} about its axis, perpendicular to its plane. If the magnetic field at its centre is $3.8 \times 10^{-9} \text{ T}$, then the charge carried by the ring is close to ($\mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$). [12 April 2019, I]
- (a) $2 \times 10^{-6} \text{ C}$ (b) $3 \times 10^{-5} \text{ C}$ (c) $4 \times 10^{-5} \text{ C}$ (d) $7 \times 10^{-6} \text{ C}$
42. Find the magnetic field at point P due to a straight line segment AB of length 6 cm carrying a current of 5 A. (See figure) ($\mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$) [12 April 2019, II]



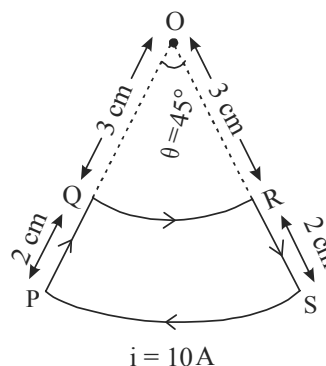
- (a) $2.0 \times 10^{-5} \text{ T}$ (b) $1.5 \times 10^{-5} \text{ T}$
 (c) $3.0 \times 10^{-5} \text{ T}$ (d) $2.5 \times 10^{-5} \text{ T}$
43. The magnitude of the magnetic field at the center of an equilateral triangular loop of side 1 m which is carrying a current of 10 A is : [10 April 2019, II]
- [Take $\mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$]
- (a) $18 \mu\text{T}$ (b) $9 \mu\text{T}$ (c) $3 \mu\text{T}$ (d) $1 \mu\text{T}$
44. A square loop is carrying a steady current I and the magnitude of its magnetic dipole moment is m . If this square loop is changed to a circular loop and it carries the same current, the magnitude of the magnetic dipole moment of circular loop will be : [10 April 2019, II]

- (a) $\frac{m}{\pi}$ (b) $\frac{3m}{\pi}$ (c) $\frac{2m}{\pi}$ (d) $\frac{4m}{\pi}$

45. As shown in the figure, two infinitely long, identical wires are bent by 90° and placed in such a way that the segments LP and QM are along the x -axis, while segments PS and QN are parallel to the y -axis. If $OP = OQ = 4 \text{ cm}$, and the magnitude of the magnetic field at O is 10^{-4} T , and the two wires carry equal currents (see figure), the magnitude of the current in each wire and the direction of the magnetic field at O will be ($\mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$): [12 Jan 2019, I]



- (a) 20 A, perpendicular out of the page
 (b) 40 A, perpendicular out of the page
 (c) 20 A, perpendicular into the page
 (d) 40 A, perpendicular into the page
46. A current loop, having two circular arcs joined by two radial lines is shown in the figure. It carries a current of 10 A. The magnetic field at point O will be close to: [9 Jan. 2019 I]



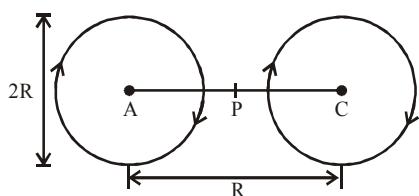
- (a) $1.0 \times 10^{-7} \text{ T}$ (b) $1.5 \times 10^{-7} \text{ T}$
 (c) $1.5 \times 10^{-5} \text{ T}$ (d) $1.0 \times 10^{-5} \text{ T}$
47. One of the two identical conducting wires of length L is bent in the form of a circular loop and the other one into a circular coil of N identical turns. If the same current is passed in both, the ratio of the magnetic field at the central of the loop (B_L) to that at the centre of the coil (B_C), i.e., $\frac{B_L}{B_C}$ will be: [9 Jan 2019, II]
- (a) N (b) $\frac{1}{N}$
 (c) N^2 (d) $\frac{1}{N^2}$
48. The dipole moment of a circular loop carrying a current I , is m and the magnetic field at the centre of the loop is B_1 . When the dipole moment is doubled by keeping the current

current constant, the magnetic field at the centre of the loop is B_2 . The ratio $\frac{B_1}{B_2}$ is: [2018]

- (a) 2 (b) $\sqrt{3}$ (c) $\sqrt{2}$ (d) $\frac{1}{\sqrt{2}}$

49. A Helmholtz coil has pair of loops, each with N turns and radius R . They are placed coaxially at distance R and the same current I flows through the loops in the same direction. The magnitude of magnetic field at P , midway between the centres A and C , is given by (Refer to figure):

[Online April 15, 2018]



- (a) $\frac{4N\mu_0 I}{5^{3/2}R}$ (b) $\frac{8N\mu_0 I}{5^{3/2}R}$ (c) $\frac{4N\mu_0 I}{5^{1/2}R}$ (d) $\frac{8N\mu_0 I}{5^{1/2}R}$

50. A current of 1A is flowing on the sides of an equilateral triangle of side $4.5 \times 10^{-2}\text{m}$. The magnetic field at the centre of the triangle will be: [Online April 15, 2018]

- (a) $4 \times 10^{-5}\text{Wb/m}^2$ (b) Zero
(c) $2 \times 10^{-5}\text{Wb/m}^2$ (d) $8 \times 10^{-5}\text{Wb/m}^2$

51. Two identical wires A and B, each of length ' l ', carry the same current I . Wire A is bent into a circle of radius R and wire B is bent to form a square of side ' a '. If B_A and B_B are the values of magnetic field at the centres of the circle and square respectively, then the ratio $\frac{B_A}{B_B}$ is:

[2016]

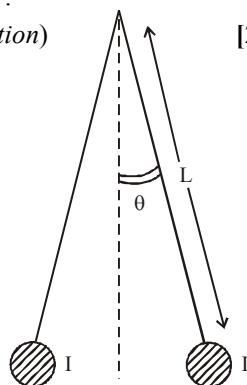
- (a) $\frac{\pi^2}{16}$ (b) $\frac{\pi^2}{8\sqrt{2}}$ (c) $\frac{\pi^2}{8}$ (d) $\frac{\pi^2}{16\sqrt{2}}$

52. Two long current carrying thin wires, both with current I , are held by insulating threads of length L and are in equilibrium as shown in the figure, with threads making an angle ' θ ' with the vertical. If wires have mass λ per unit length then the value of I is:

(g = gravitational acceleration)

[2015]

- (a) $2\sqrt{\frac{\pi g L}{\mu_0}} \tan \theta$
(b) $\sqrt{\frac{\pi g L}{\mu_0}} \tan \theta$
(c) $\sin \theta \sqrt{\frac{\pi g L}{\mu_0 \cos \theta}}$
(d) $2 \sin \theta \sqrt{\frac{\pi g L}{\mu_0 \cos \theta}}$



53. Consider two thin identical conducting wires covered with very thin insulating material. One of the wires is bent into a loop and produces magnetic field B_1 at its centre when a current I passes through it. The ratio $B_1 : B_2$ is: [Online April 12, 2014]

- (a) 1 : 1 (b) 1 : 3 (c) 1 : 9 (d) 9 : 1

54. A parallel plate capacitor of area 60 cm^2 and separation 3 mm is charged initially to $90\text{ }\mu\text{C}$. If the medium between the plate gets slightly conducting and the plate loses the charge initially at the rate of $2.5 \times 10^{-8}\text{ C/s}$, then what is the magnetic field between the plates?

[Online April 23, 2013]

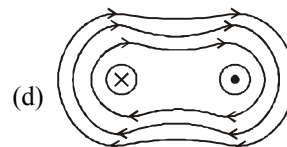
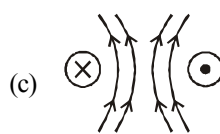
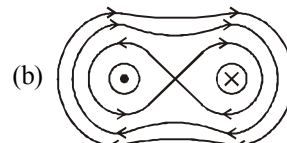
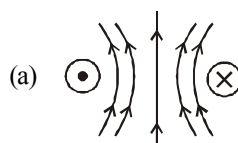
- (a) $2.5 \times 10^{-8}\text{ T}$ (b) $2.0 \times 10^{-7}\text{ T}$
(c) $1.63 \times 10^{-11}\text{ T}$ (d) Zero

55. A current i is flowing in a straight conductor of length L . The magnetic induction at a point on its axis at a distance

$\frac{L}{4}$ from its centre will be: [Online April 22, 2013]

- (a) Zero (b) $\frac{\mu_0 i}{2\pi L}$
(c) $\frac{\mu_0 i}{\sqrt{2}L}$ (d) $\frac{4\mu_0 i}{\sqrt{5}\pi L}$

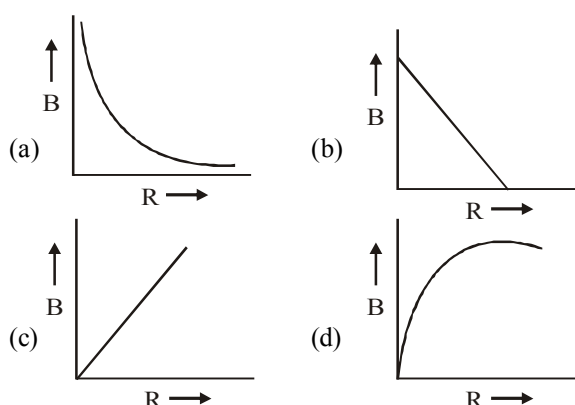
56. Choose the correct sketch of the magnetic field lines of a circular current loop shown by the dot $\odot$ and the cross $\otimes$. [Online April 22, 2013]



57. An electric current is flowing through a circular coil of radius R . The ratio of the magnetic field at the centre of the coil and that at a distance $2\sqrt{2}R$ from the centre of the coil and on its axis is: [Online April 9, 2013]

- (a) $2\sqrt{2}$ (b) 27 (c) 36 (d) 8

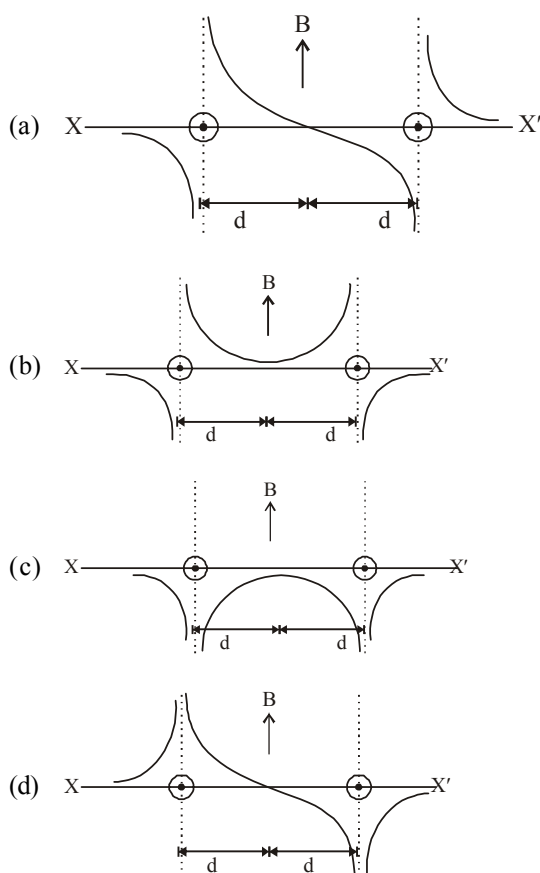
58. A charge Q is uniformly distributed over the surface of non-conducting disc of radius R . The disc rotates about an axis perpendicular to its plane and passing through its centre with an angular velocity ω . As a result of this rotation a magnetic field of induction B is obtained at the centre of the disc. If we keep both the amount of charge placed on the disc and its angular velocity to be constant and vary the radius of the disc then the variation of the magnetic induction at the centre of the disc will be represented by the figure: [2012]



59. A current I flows in an infinitely long wire with cross section in the form of a semi-circular ring of radius R . The magnitude of the magnetic induction along its axis is: [2011]

(a) $\frac{\mu_0 I}{2\pi^2 R}$ (b) $\frac{\mu_0 I}{2\pi R}$ (c) $\frac{\mu_0 I}{4\pi R}$ (d) $\frac{\mu_0 I}{\pi^2 R}$

60. Two long parallel wires are at a distance $2d$ apart. They carry steady equal currents flowing out of the plane of the paper as shown. The variation of the magnetic field B along the line XX' is given by [2010]



61. A horizontal overhead powerline is at height of 4m from the ground and carries a current of 100A from east to west. The magnetic field directly below it on the ground is ($\mu_0 = 4\pi \times 10^{-7} \text{ Tm A}^{-1}$) [2008]

- (a) $2.5 \times 10^{-7} \text{ T}$ southward
(b) $5 \times 10^{-6} \text{ T}$ northward
(c) $5 \times 10^{-6} \text{ T}$ southward
(d) $2.5 \times 10^{-7} \text{ T}$ northward

62. A long straight wire of radius a carries a steady current i . The current is uniformly distributed across its cross section. The ratio of the magnetic field at $a/2$ and $2a$ is [2007]

- (a) $1/2$ (b) $1/4$ (c) 4 (d) 1

63. A current I flows along the length of an infinitely long, straight, thin walled pipe. Then [2007]

- (a) the magnetic field at all points inside the pipe is the same, but not zero
(b) the magnetic field is zero only on the axis of the pipe
(c) the magnetic field is different at different points inside the pipe
(d) the magnetic field at any point inside the pipe is zero

64. Two identical conducting wires AOB and COD are placed at right angles to each other. The wire AOB carries an electric current I_1 and COD carries a current I_2 . The magnetic field on a point lying at a distance d from O , in a direction perpendicular to the plane of the wires AOB and COD , will be given by [2007]

(a) $\frac{\mu_0}{2\pi d} (I_1^2 + I_2^2)$ (b) $\frac{\mu_0}{2\pi} \left(\frac{I_1 + I_2}{d} \right)^2$
(c) $\frac{\mu_0}{2\pi d} \left(I_1^2 + I_2^2 \right)^{1/2}$ (d) $\frac{\mu_0}{2\pi d} (I_1 + I_2)$

65. A long solenoid has 200 turns per cm and carries a current i . The magnetic field at its centre is $6.28 \times 10^{-2} \text{ Weber/m}^2$. Another long solenoid has 100 turns per cm and it carries a current $\frac{i}{3}$. The value of the magnetic field at its centre is

[2006]

- (a) $1.05 \times 10^{-2} \text{ Weber/m}^2$ (b) $1.05 \times 10^{-5} \text{ Weber/m}^2$
(c) $1.05 \times 10^{-3} \text{ Weber/m}^2$ (d) $1.05 \times 10^{-4} \text{ Weber/m}^2$

66. Two concentric coils each of radius equal to $2\pi \text{ cm}$ are placed at right angles to each other. 3 ampere and 4 ampere are the currents flowing in each coil respectively. The magnetic induction in Weber/m^2 at the centre of the coils will be ($\mu_0 = 4\pi \times 10^{-7} \text{ Wb/A.m}$) [2005]

- (a) 10^{-5} (b) 12×10^{-5}
(c) 7×10^{-5} (d) 5×10^{-5}

67. A current i ampere flows along an infinitely long straight thin walled tube, then the magnetic induction at any point inside the tube is [2004]

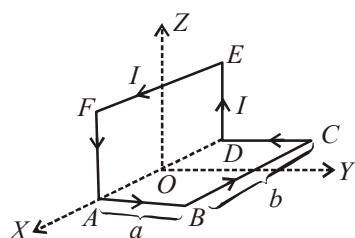
- (a) $\frac{\mu_0}{4\pi} \frac{2i}{r}$ tesla (b) zero
(c) infinite (d) $\frac{2i}{r}$ tesla

68. A long wire carries a steady current. It is bent into a circle of one turn and the magnetic field at the centre of the coil is B . It is then bent into a circular loop of n turns. The magnetic field at the centre of the coil will be [2004]
- (a) $2nB$ (b) n^2B (c) nB (d) $2n^2B$
69. The magnetic field due to a current carrying circular loop of radius 3 cm at a point on the axis at a distance of 4 cm from the centre is $54 \mu\text{T}$. What will be its value at the centre of loop? [2004]
- (a) $125 \mu\text{T}$ (b) $150 \mu\text{T}$ (c) $250 \mu\text{T}$ (d) $75 \mu\text{T}$
70. If in a circular coil A of radius R , current I is flowing and in another coil B of radius $2R$ a current $2I$ is flowing, then the ratio of the magnetic fields B_A and B_B , produced by them will be [2002]
- (a) 1 (b) 2 (c) $1/2$ (d) 4

TOPIC 3 Force and Torque on Current Carrying Conductor



71. A square loop of side $2a$ and carrying current I is kept in xz plane with its centre at origin. A long wire carrying the same current I is placed parallel to z -axis and passing through point $(0, b, 0)$, ($b \gg a$). The magnitude of torque on the loop about z -axis will be : [Sep. 06, 2020 (II)]
- (a) $\frac{2\mu_0 I^2 a^2}{\pi b}$ (b) $\frac{2\mu_0 I^2 a^2 b}{\pi(a^2 + b^2)}$
- (c) $\frac{\mu_0 I^2 a^2 b}{2\pi(a^2 + b^2)}$ (d) $\frac{\mu_0 I^2 a^2}{2\pi b}$
72. A square loop of side $2a$, and carrying current I , is kept in XZ plane with its centre at origin. A long wire carrying the same current I is placed parallel to the z -axis and passing through the point $(0, b, 0)$, ($b \gg a$). The magnitude of the torque on the loop about z -axis is given by : [Sep. 05, 2020 (I)]
- (a) $\frac{\mu_0 I^2 a^2}{2\pi b}$ (b) $\frac{\mu_0 I^2 a^3}{2\pi b^2}$
- (c) $\frac{2\mu_0 I^2 a^2}{\pi b}$ (d) $\frac{2\mu_0 I^2 a^3}{\pi b^2}$
73. A wire carrying current I is bent in the shape $ABCDEF$ as shown, where rectangle $ABCD$ and $ADEFA$ are perpendicular to each other. If the sides of the rectangles are of lengths a and b , then the magnitude and direction of magnetic moment of the loop $ABCDEF$ is : [Sep. 02, 2020 (II)]

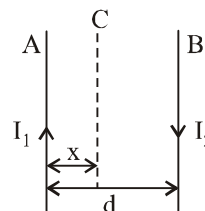


- (a) abI , along $\left(\frac{\hat{j}}{\sqrt{2}} + \frac{\hat{k}}{\sqrt{2}}\right)$
- (b) $\sqrt{2}abI$, along $\left(\frac{\hat{j}}{\sqrt{2}} + \frac{\hat{k}}{\sqrt{2}}\right)$
- (c) $\sqrt{2}abI$, along $\left(\frac{\hat{j}}{\sqrt{5}} + \frac{2\hat{k}}{\sqrt{5}}\right)$
- (d) abI , along $\left(\frac{\hat{j}}{\sqrt{5}} + \frac{2\hat{k}}{\sqrt{5}}\right)$

74. A small circular loop of conducting wire has radius a and carries current I . It is placed in a uniform magnetic field B perpendicular to its plane such that when rotated slightly about its diameter and released, it starts performing simple harmonic motion of time period T . If the mass of the loop is m then : [9 Jan 2020, II]

- (a) $T = \sqrt{\frac{2m}{IB}}$ (b) $T = \sqrt{\frac{\pi m}{2IB}}$
- (c) $T = \sqrt{\frac{2\pi m}{IB}}$ (d) $T = \sqrt{\frac{\pi m}{IB}}$

75. Two wires A & B are carrying currents I_1 and I_2 as shown in the figure. The separation between them is d . A third wire C carrying a current I is to be kept parallel to them at a distance x from A such that the net force acting on it is zero. The possible values of x are : [10 April 2019, I]

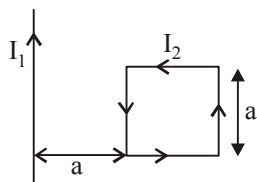


- (a) $x = \left(\frac{I_1}{I_1 - I_2}\right)d$ and $x = \left(\frac{I_2}{I_1 + I_2}\right)d$
- (b) $x = \left(\frac{I_2}{I_1 + I_2}\right)d$ and $x = \left(\frac{I_2}{I_1 - I_2}\right)d$
- (c) $x = \left(\frac{I_1}{I_1 + I_2}\right)d$ and $x = \left(\frac{I_2}{I_1 - I_2}\right)d$
- (d) $x = \pm \frac{I_1 d}{(I_1 - I_2)}$

76. A rectangular coil (Dimension $5 \text{ cm} \times 2.5 \text{ cm}$) with 100 turns, carrying a current of 3 A in the clock-wise direction, is kept centered at the origin and in the $X-Z$ plane. A magnetic field of 1 T is applied along X -axis. If the coil is tilted through 45° about Z -axis, then the torque on the coil is : [9 April 2019 I]
- (a) 0.38 Nm (b) 0.55 Nm
- (c) 0.42 Nm (d) 0.27 Nm

77. A rigid square of loop of side 'a' and carrying current I_2 is lying on a horizontal surface near a long current I_1 carrying wire in the same plane as shown in figure. The net force on the loop due to the wire will be:

[9 April 2019 I]



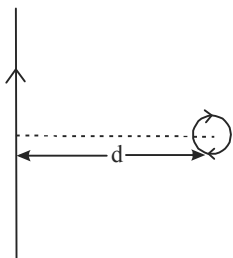
- (a) Repulsive and equal to $\frac{\mu_0 I_1 I_2}{2\pi}$
 (b) Attractive and equal to $\frac{\mu_0 I_1 I_2}{3\pi}$
 (c) Repulsive and equal to $\frac{\mu_0 I_1 I_2}{4\pi}$
 (d) Zero
78. A circular coil having N turns and radius r carries a current I . It is held in the XZ plane in a magnetic field B . The torque on the coil due to the magnetic field is :

[8 April 2019 I]

- (a) $\frac{Br^2 I}{\pi N}$ (b) $B\pi r^2 IN$
 (c) $\frac{B\pi r^2 I}{N}$ (d) Zero

79. An infinitely long current carrying wire and a small current carrying loop are in the plane of the paper as shown. The radius of the loop is a and distance of its centre from the wire is d ($d \gg a$). If the loop applies a force F on the wire then:

[9 Jan. 2019 I]



- (a) $F = 0$ (b) $F \propto \left(\frac{a}{d}\right)$
 (c) $F \propto \left(\frac{a^2}{d^3}\right)$ (d) $F \propto \left(\frac{a}{d}\right)^2$
80. A charge q is spread uniformly over an insulated loop of radius r . If it is rotated with an angular velocity ω with respect to normal axis then the magnetic moment of the loop is

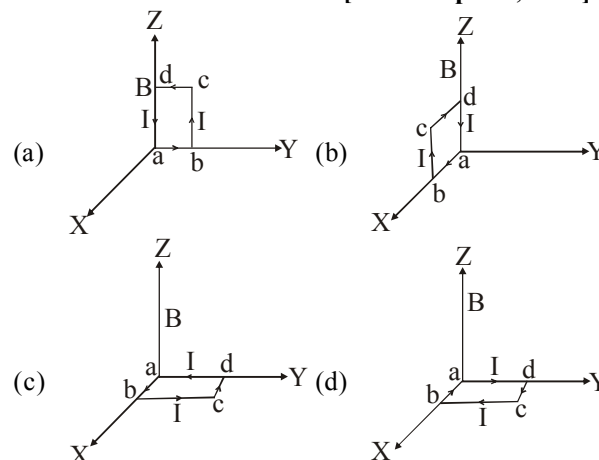
[Online April 16, 2018]

- (a) $\frac{1}{2}q\omega r^2$ (b) $\frac{4}{3}q\omega r^2$ (c) $\frac{3}{2}q\omega r^2$ (d) $q\omega r^2$

81. A uniform magnetic field B of 0.3 T is along the positive Z -direction. A rectangular loop (abcd) of sides $10 \text{ cm} \times 5$

cm carries a current I of 12 A. Out of the following different orientations which one corresponds to stable equilibrium?

[Online April 9, 2017]

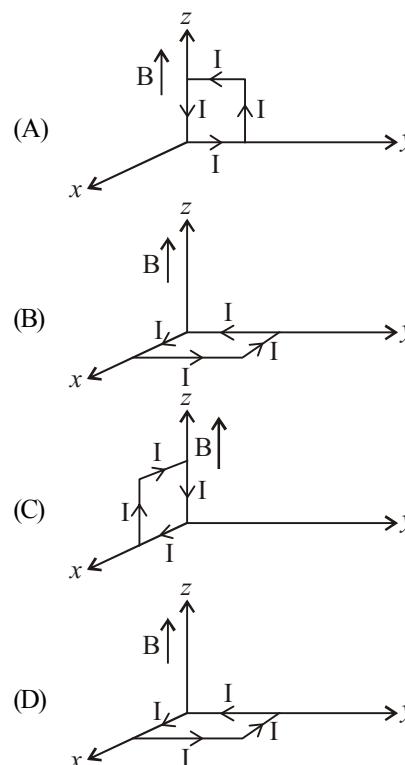


82. Two coaxial solenoids of different radius carry current I in the same direction. $\vec{F}_1$ be the magnetic force on the inner solenoid due to the outer one and $\vec{F}_2$ be the magnetic force on the outer solenoid due to the inner one. Then :

[2015]

- (a) $\vec{F}_1$ is radially inwards and $\vec{F}_2 = 0$
 (b) $\vec{F}_1$ is radially outwards and $\vec{F}_2 = 0$
 (c) $\vec{F}_1 = \vec{F}_2 = 0$
 (d) $\vec{F}_1$ is radially inwards and $\vec{F}_2$ is radially outwards

83. A rectangular loop of sides 10 cm and 5 cm carrying a current I of 12 A is placed in different orientations as shown in the figures below :



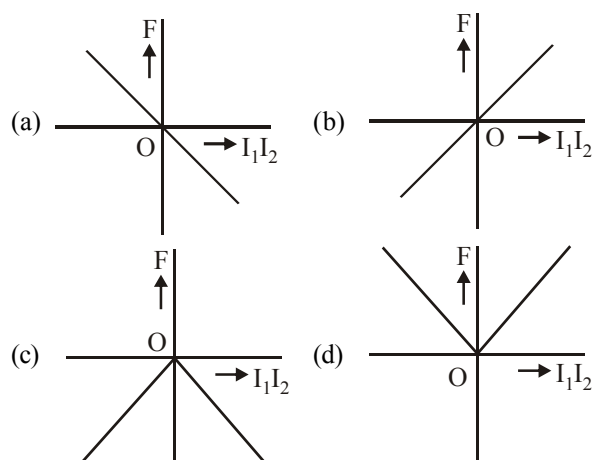
If there is a uniform magnetic field of 0.3 T in the positive z direction, in which orientations the loop would be in (i) stable equilibrium and (ii) unstable equilibrium ?

[2015]

- (a) (B) and (D), respectively
- (b) (B) and (C), respectively
- (c) (A) and (B), respectively
- (d) (A) and (C), respectively

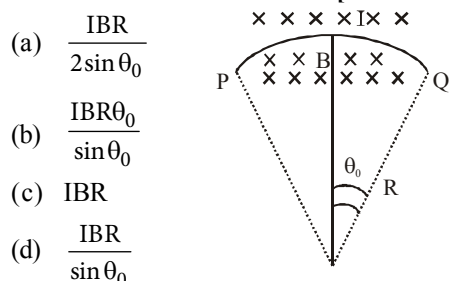
84. Two long straight parallel wires, carrying (adjustable) current I_1 and I_2 , are kept at a distance d apart. If the force 'F' between the two wires is taken as 'positive' when the wires repel each other and 'negative' when the wires attract each other, the graph showing the dependence of 'F', on the product $I_1 I_2$, would be :

[Online April 11, 2015]

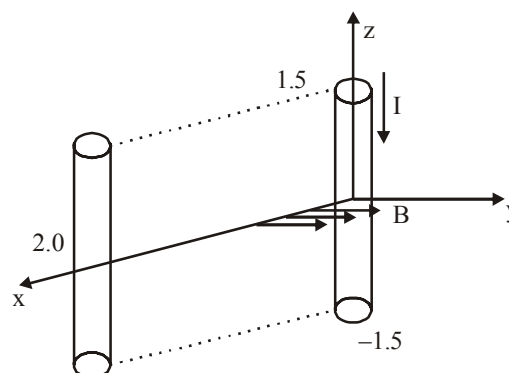


85. A wire carrying current I is tied between points P and Q and is in the shape of a circular arc of radius R due to a uniform magnetic field B (perpendicular to the plane of the paper, shown by xxx) in the vicinity of the wire. If the wire subtends an angle $2\theta_0$ at the centre of the circle (of which it forms an arc) then the tension in the wire is :

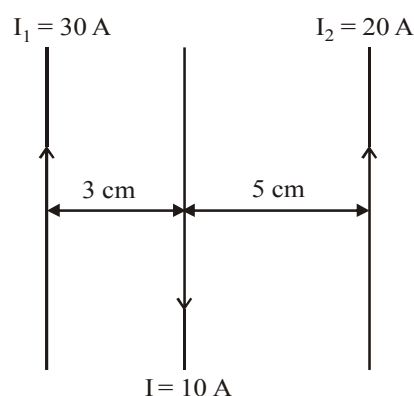
[Online April 11, 2015]



86. A conductor lies along the z-axis at $-1.5 \leq z < 1.5$ m and carries a fixed current of 10.0 A in $-\hat{a}_z$ direction (see figure). For a field $\vec{B} = 3.0 \times 10^{-4} e^{-0.2x} \hat{a}_y$ T, find the power required to move the conductor at constant speed to $x = 2.0$ m, $y = 0$ m in 5×10^{-3} s. Assume parallel motion along the x-axis. [2014]

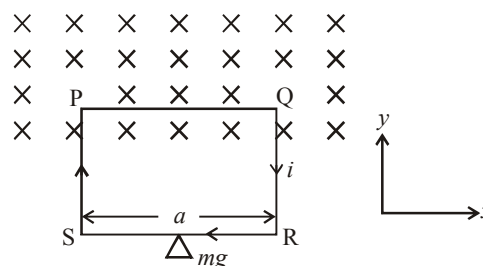


- (a) 1.57 W (b) 2.97 W (c) 14.85 W (d) 29.7 W
87. Three straight parallel current carrying conductors are shown in the figure. The force experienced by the middle conductor of length 25 cm is: [Online April 11, 2014]



- (a) 3×10^{-4} N toward right
 (b) 6×10^{-4} N toward right
 (c) 9×10^{-4} N toward right
 (d) Zero
88. A rectangular loop of wire, supporting a mass m , hangs with one end in a uniform magnetic field $\vec{B}$ pointing out of the plane of the paper. A clockwise current is set up such that $i > mg/Ba$, where a is the width of the loop. Then :

[Online April 23, 2013]

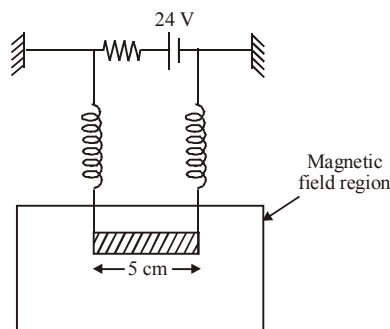


- (a) The weight rises due to a vertical force caused by the magnetic field and work is done on the system.
 (b) The weight do not rise due to vertical for caused by the magnetic field and work is done on the system.

- (c) The weight rises due to a vertical force caused by the magnetic field but no work is done on the system.
 (d) The weight rises due to a vertical force caused by the magnetic field and work is extracted from the magnetic field.
89. Currents of a 10 ampere and 2 ampere are passed through two parallel thin wires A and B respectively in opposite directions. Wire A is infinitely long and the length of the wire B is 2 m. The force acting on the conductor B , which is situated at 10 cm distance from A will be

[Online May 26, 2012]

- (a) 8×10^{-5} N (b) 5×10^{-5} N
 (c) $8\pi \times 10^{-7}$ N (d) $4\pi \times 10^{-7}$ N
90. The circuit in figure consists of wires at the top and bottom and identical springs as the left and right sides. The wire at the bottom has a mass of 10 g and is 5 cm long. The wire is hanging as shown in the figure. The springs stretch 0.5 cm under the weight of the wire and the circuit has a total resistance of 12Ω . When the lower wire is subjected to a static magnetic field, the springs stretch an additional 0.3 cm. The magnetic field is [Online May 12, 2012]

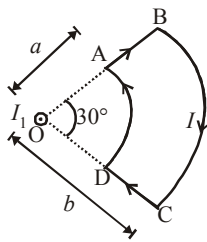


- (a) 0.6 T and directed out of page
 (b) 1.2 T and directed into the plane of page
 (c) 0.6 T and directed into the plane of page
 (d) 1.2 T and directed out of page

Directions : Question numbers 91 and 92 are based on the following paragraph.

A current loop $ABCD$ is held fixed on the plane of the paper as shown in the figure. The arcs BC (radius = b) and DA (radius = a) of the loop are joined by two straight wires AB and CD . A steady current I is flowing in the loop. Angle made by AB and CD at the origin O is 30° . Another straight thin wire with steady current I_1 flowing out of the plane of the paper is kept at the origin.

[2009]



91. The magnitude of the magnetic field (B) due to the loop $ABCD$ at the origin (O) is :

(a) $\frac{\mu_0 I (b - a)}{24ab}$

(b) $\frac{\mu_0 I}{4\pi} \left[\frac{b - a}{ab} \right]$

(c) $\frac{\mu_0 I}{4\pi} [2(b - a) + p / 3(a + b)]$

(d) zero

92. Due to the presence of the current I_1 at the origin:

- (a) The forces on AD and BC are zero.
 (b) The magnitude of the net force on the loop is given

by $\frac{I_1 I}{4\pi} m_0 [2(b - a) + p / 3(a + b)]$.

- (c) The magnitude of the net force on the loop is given

by $\frac{\mu_0 I I_1}{24ab} (b - a)$.

- (d) The forces on AB and DC are zero.

93. Two long conductors, separated by a distance d carry current I_1 and I_2 in the same direction. They exert a force F on each other. Now the current in one of them is increased to two times and its direction is reversed. The distance is also increased to $3d$. The new value of the force between them is [2004]

(a) $-\frac{2F}{3}$ (b) $\frac{F}{3}$ (c) $-2F$ (d) $-\frac{F}{3}$

94. If a current is passed through a spring then the spring will [2002]

- (a) expand (b) compress
 (c) remains same (d) none of these

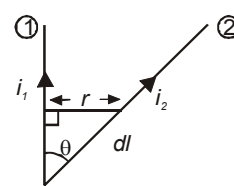
95. Wires 1 and 2 carrying currents i_1 and i_2 respectively are inclined at an angle θ to each other. What is the force on a small element dl of wire 2 at a distance of r from wire 1 (as shown in figure) due to the magnetic field of wire 1? [2002]

(a) $\frac{\mu_0}{2\pi r} i_1 i_2 dl \tan \theta$

(b) $\frac{\mu_0}{2\pi r} i_1 i_2 dl \sin \theta$

(c) $\frac{\mu_0}{2\pi r} i_1 i_2 dl \cos \theta$

(d) $\frac{\mu_0}{4\pi r} i_1 i_2 dl \sin \theta$

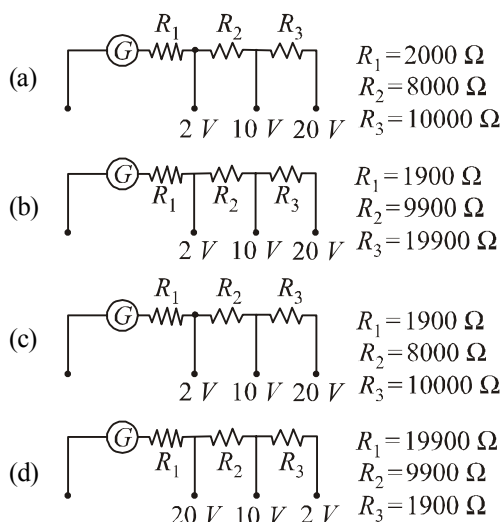


TOPIC 4 Galvanometer and its Conversion into Ammeter and Voltmeter

96. A galvanometer of resistance G is converted into a voltmeter of range $0 - 1V$ by connecting a resistance R_1 in series with it. The additional resistance R_1 in series with it. The additional resistance that should be connected in series with R_1 to increase the range of the voltmeter to $0 - 2V$ will be: [Sep. 05, 2020 (I)]

- (a) G (b) R_1
 (c) $R_1 - G$ (d) $R_1 + G$

97. A galvanometer is used in laboratory for detecting the null point in electrical experiments. If, on passing a current of 6 mA it produces a deflection of 2° , its figure of merit is close to : **[Sep. 05, 2020 (II)]**
 (a) 333° A/div. (b) $6 \times 10^{-3} \text{ A/div.}$
 (c) 666° A/div. (d) $3 \times 10^{-3} \text{ A/div.}$
98. A galvanometer coil has 500 turns and each turn has an average area of $3 \times 10^{-4} \text{ m}^2$. If a torque of 1.5 Nm is required to keep this coil parallel to a magnetic field when a current of 0.5 A is flowing through it, the strength of the field (in T) is _____. **[NA Sep. 03, 2020 (II)]**
99. A galvanometer of resistance 100Ω has 50 divisions on its scale and has sensitivity of $20 \mu\text{A/division}$. It is to be converted to a voltmeter with three ranges, of 0–2 V, 0–10 V and 0–20 V. The appropriate circuit to do so is : **[12 April 2019, II]**



100. A moving coil galvanometer, having a resistance G , produces full scale deflection when a current I_g flows through it. This galvanometer can be converted into (i) an ammeter of range 0 to I_0 ($I_0 > I_g$) by connecting a shunt resistance R_A to it and (ii) into a voltmeter of range 0 to V ($V = GI_0$) by connecting a series resistance R_V to it. Then, **[12 April 2019, II]**

- (a) $R_A R_V = G^2 \left(\frac{I_0 - I_g}{I_g} \right)$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g} \right)^2$
- (b) $R_A R_V = G^2$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g} \right)^2$
- (c) $R_A R_V = G^2 \left(\frac{I_g}{I_0 - I_g} \right)$ and $\frac{R_A}{R_V} = \left(\frac{I_0 - I_g}{I_g} \right)^2$
- (d) $R_A R_V = G^2$ and $\frac{R_A}{R_V} = \frac{I_g}{(I_0 - I_g)}$

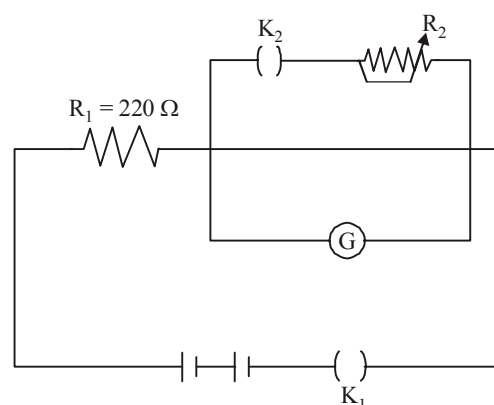
101. A moving coil galvanometer allows a full scale current of 10^{-4} A . A series resistance of $2 \text{ M}\Omega$ is required to convert the above galvanometer into a voltmeter of range

0 – 5 V. Therefore the value of shunt resistance required to convert the above galvanometer into an ammeter of range 0 – 10 mA is : **[10 April 2019, I]**

- (a) 500Ω (b) 100Ω (c) 200Ω (d) 10Ω
102. A moving coil galvanometer has resistance 50Ω and it indicates full deflection at 4 mA current. A voltmeter is made using this galvanometer and a $5 \text{ k}\Omega$ resistance. The maximum voltage, that can be measured using this voltmeter, will be close to: **[9 April 2019 I]**
 (a) 40 V (b) 15 V (c) 20 V (d) 10 V
103. A moving coil galvanometer has a coil with 175 turns and area 1 cm^2 . It uses a torsion band of torsion constant 10^{-6} N-m/rad . The coil is placed in a magnetic field B parallel to its plane. The coil deflects by 1° for a current of 1 mA. The value of B (in Tesla) is approximately: **[9 April 2019, II]**

- (a) 10^{-4} (b) 10^{-2} (c) 10^{-1} (d) 10^{-3}
104. The resistance of a galvanometer is 50 ohm and the maximum current which can be passed through it is 0.002 A . What resistance must be connected to it order to convert it into an ammeter of range 0 – 0.5 A? **[9 April 2019, II]**

- (a) 0.5 ohm (b) 0.002 ohm
 (c) 0.02 ohm (d) 0.2 ohm
105. The galvanometer deflection, when key K_1 is closed but K_2 is open, equals θ_0 (see figure). On closing K_2 also and adjusting R_2 to 5Ω , the deflection in galvanometer becomes $\frac{\theta_0}{5}$. The resistance of the galvanometer is, then, given by [Neglect the internal resistance of battery]: **[12 Jan 2019, I]**



- (a) 5Ω (b) 22Ω
 (c) 25Ω (d) 12Ω
106. A galvanometer, whose resistance is 50 ohm , has 25 divisions in it. When a current of $4 \times 10^{-4} \text{ A}$ passes through it, its needle (pointer) deflects by one division. To use this galvanometer as a voltmeter of range 2.5 V, it should be connected to a resistance of : **[12 Jan 2019, II]**

- (a) 250 ohm (b) 200 ohm
 (c) 6200 ohm (d) 6250 ohm

107. A galvanometer having a resistance of $20\ \Omega$ and 30 division on both sides has figure of merit 0.005 ampere/division. The resistance that should be connected in series such that it can be used as a voltmeter upto 15 volt, is:

[11 Jan 2019, II]

- (a) $100\ \Omega$ (b) $120\ \Omega$ (c) $80\ \Omega$ (d) $125\ \Omega$

108. A galvanometer having a coil resistance $100\ \Omega$ gives a full scale deflection when a current of 1 mA is passed through it. What is the value of the resistance which can convert this galvanometer into a voltmeter giving full scale deflection for a potential difference of 10 V ?

[8 Jan 2019, II]

- (a) $10\text{ k}\Omega$ (b) $8.9\text{ k}\Omega$
(c) $7.9\text{ k}\Omega$ (d) $9.9\text{ k}\Omega$

109. In a circuit for finding the resistance of a galvanometer by half deflection method, a 6 V battery and a high resistance of $1\text{ k}\Omega$ are used. The figure of merit of the galvanometer $60\ \mu\text{A}/\text{division}$. In the absence of shunt resistance, the galvanometer produces a deflection of $\theta = 9$ divisions when current flows in the circuit. The value of the shunt resistance that can cause the deflection of $\theta/2$, is closest to

[Online April 16, 2018]

- (a) $55\ \Omega$ (b) $110\ \Omega$ (c) $220\ \Omega$ (d) $550\ \Omega$

110. A galvanometer with its coil resistance $25\ \Omega$ requires a current of 1 mA for its full deflection. In order to construct an ammeter to read up to a current of 2 A , the approximate value of the shunt resistance should be

[Online April 16, 2018]

- (a) $2.5 \times 10^{-2}\ \Omega$ (b) $1.25 \times 10^{-3}\ \Omega$
(c) $2.5 \times 10^{-3}\ \Omega$ (d) $1.25 \times 10^{-2}\ \Omega$

111. When a current of 5 mA is passed through a galvanometer having a coil of resistance $15\ \Omega$, it shows full scale deflection. The value of the resistance to be put in series with the galvanometer to convert it into a voltmeter of range $0 - 10\text{ V}$ is

[2017]

- (a) $2.535 \times 10^3\ \Omega$ (b) $4.005 \times 10^3\ \Omega$
(c) $1.985 \times 10^3\ \Omega$ (d) $2.045 \times 10^3\ \Omega$

112. A galvanometer having a coil resistance of $100\ \Omega$ gives a full scale deflection, when a current of 1 mA is passed through it. The value of the resistance, which can convert this galvanometer into ammeter giving a full scale deflection for a current of 10 A , is:

[2016]

- (a) $0.1\ \Omega$ (b) $3\ \Omega$ (c) $0.01\ \Omega$ (d) $2\ \Omega$

113. A $50\ \Omega$ resistance is connected to a battery of 5 V . A galvanometer of resistance $100\ \Omega$ is to be used as an ammeter to measure current through the resistance, for this a resistance r_s is connected to the galvanometer. Which of the following connections should be employed if the measured current is within 1% of the current without the ammeter in the circuit?

[Online April 9, 2016]

- (a) $r_s = 0.5\ \Omega$ in series with the galvanometer
(b) $r_s = 1\ \Omega$ in series with galvanometer
(c) $r_s = 1\ \Omega$ in parallel with galvanometer
(d) $r_s = 0.5\ \Omega$ in parallel with the galvanometer.

114. To know the resistance G of a galvanometer by half deflection method, a battery of emf V_E and resistance R is used to deflect the galvanometer by angle θ . If a shunt of resistance S is needed to get half deflection then G , R and S related by the equation:

[Online April 9, 2016]

- (a) $S(R+G)=RG$ (b) $2S(R+G)=RG$
(c) $2G=S$ (d) $2S=G$

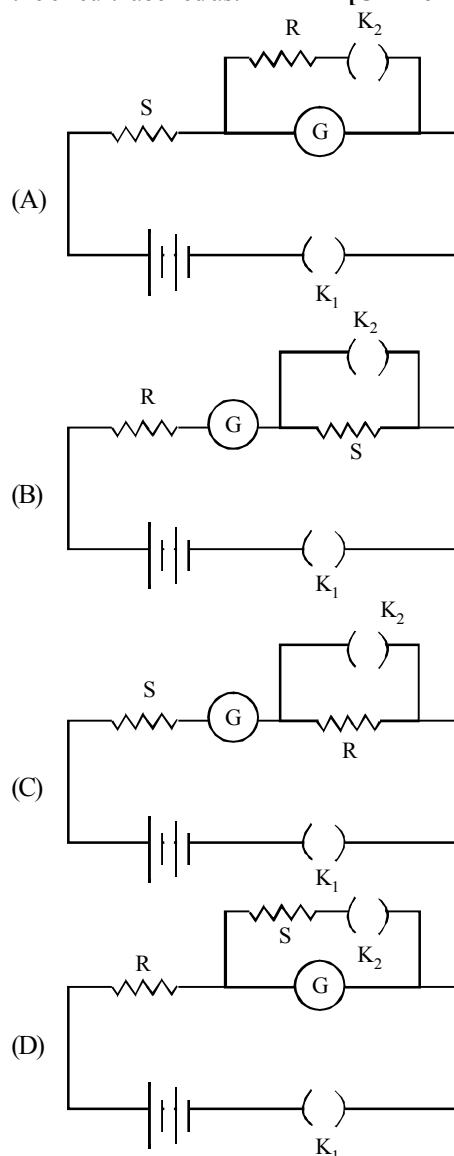
115. The AC voltage across a resistance can be measured using a :

[Online April 11, 2015]

- (a) hot wire voltmeter
(b) moving coil galvanometer
(c) potential coil galvanometer
(d) moving magnet galvanometer

116. In the circuit diagrams (A, B, C and D) shown below, R is a high resistance and S is a resistance of the order of galvanometer resistance G . The correct circuit, corresponding to the half deflection method for finding the resistance and figure of merit of the galvanometer, is the circuit labelled as:

[Online April 11, 2014]



- (a) Circuit A with $G = \frac{RS}{(R-S)}$
 (b) Circuit B with $G = S$
 (c) Circuit C with $G = S$
 (d) Circuit D with $G = \frac{RS}{(R-S)}$

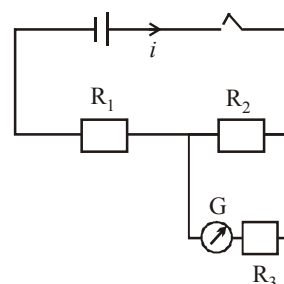
117. This questions has Statement I and Statement II. Of the four choices given after the Statements, choose the one that best describes into two Statements.

Statement-I : Higher the range, greater is the resistance of ammeter.

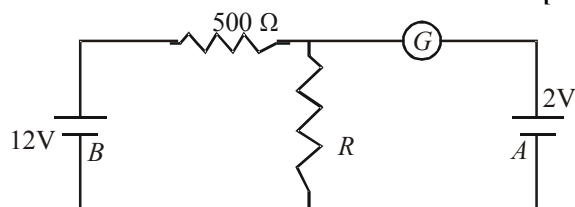
Statement-II : To increase the range of ammeter, additional shunt needs to be used across it. [2013]

- (a) Statement-I is true, Statement-II is true, Statement-II is the correct explanation of Statement-I.
 (b) Statement-I is true, Statement-II is true, Statement-II is not the correct explanation of Statement-I.
 (c) Statement-I is true, Statement-II is false.
 (d) Statement-I is false, Statement-II is true.
118. To find the resistance of a galvanometer by the half deflection method the following circuit is used with resistances $R_1 = 9970 \text{ W}$, $R_2 = 30 \text{ W}$ and $R_3 = 0$. The deflection in the galvanometer is d . With $R_3 = 107 \text{ W}$ the deflection changed to $\frac{d}{2}$. The galvanometer resistance is approximately :

[Online April 22, 2013]



- (a) 107Ω (b) 137Ω (c) $107/2 \Omega$ (d) 77Ω
119. A shunt of resistance 1Ω is connected across a galvanometer of 120Ω resistance. A current of 5.5 ampere gives full scale deflection in the galvanometer. The current that will give full scale deflection in the absence of the shunt is nearly : [Online April 9, 2013]
 (a) 5.5 ampere (b) 0.5 ampere
 (c) 0.004 ampere (d) 0.045 ampere
120. In the circuit , the galvanometer G shows zero deflection. If the batteries A and B have negligible internal resistance, the value of the resistor R will be - [2005]



- (a) 100Ω (b) 200Ω (c) 1000Ω (d) 500Ω
121. A moving coil galvanometer has 150 equal divisions. Its current sensitivity is 10-divisions per milliampere and voltage sensitivity is 2 divisions per millivolt. In order that each division reads 1 volt, the resistance in ohms needed to be connected in series with the coil will be - [2005]
 (a) 10^5 (b) 10^3 (c) 9995 (d) 99995

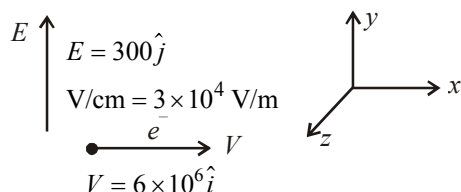


Hints & Solutions



1. (c) $\vec{E} = 300\hat{j}$ V/cm = 3×10^4 V/m

$$\vec{V} = 6 \times 10^6 \hat{i}$$



$\vec{B}$ must be in $+z$ axis.

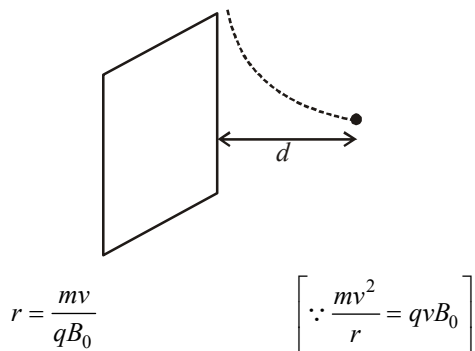
$$q\vec{E} + q\vec{V} \times \vec{B} = 0$$

$$E = VB$$

$$\therefore B = \frac{E}{V} = \frac{3 \times 10^4}{6 \times 10^6} = 5 \times 10^{-3} \text{ T}$$

Hence, magnetic field $B = 5 \times 10^{-3}$ T along $+z$ direction.

2. (c) In uniform magnetic field particle moves in a circular path, if the radius of the circular path is ' r ', particle will not hit the screen.



Hence, minimum value of v for which the particle will not hit the screen.

$$v = \frac{qB_0 d}{m}$$

3. (a) [Given: $q = 1 \mu\text{C} = 1 \times 10^{-6} \text{ C}$;

$$\vec{V} = (2\hat{i} + 3\hat{j} + 4\hat{k}) \text{ m/s and}$$

$$\vec{B} = (5\hat{i} + 3\hat{j} - 6\hat{k}) \times 10^{-3} \text{ T}]$$

$$\vec{F} = q(\vec{V} \times \vec{B}) = 10^{-6} \times 10^{-3} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 5 & 3 & -6 \end{vmatrix}$$

$$= (-30\hat{i} + 32\hat{j} - 9\hat{k}) \times 10^{-9} \text{ N}$$

$$\therefore \vec{F} = (-30\hat{i} + 32\hat{j} - 9\hat{k})$$

4. (d) Pitch = $(v \cos \theta)T$ and $T = \frac{2\pi m}{qB}$

$$\therefore \text{Pitch} = (V \cos \theta) \frac{2\pi m}{qB}$$

$$= (4 \times 10^5 \cos 60^\circ) \frac{2\pi \left(\frac{1.67 \times 10^{-27}}{1.69 \times 10^{-19}} \right)}{0.3} = 4 \text{ cm}$$

5. (c) Time period of one revolution of proton, $T = \frac{2\pi m}{qB}$

Here, m = mass of proton

q = charge of proton

B = magnetic field.

Linear distance travelled in one revolution,

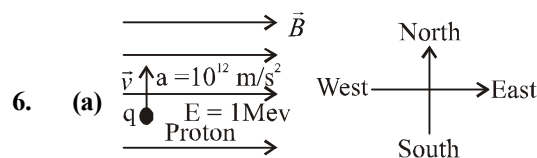
$p = T(v \cos \theta)$ (Here, v = velocity of proton)

$\therefore$ Length of region, $l = 10 \times (v \cos \theta)T$

$$\Rightarrow l = 10 \times v \cos 60^\circ \times \frac{2\pi m}{qB}$$

$$\Rightarrow l = \frac{20\pi mv}{qB} = \frac{20 \times 3.14 \times 1.67 \times 10^{-27} \times 4 \times 10^5}{1.6 \times 10^{-19} \times 0.3}$$

$$\Rightarrow l = 0.44 \text{ m}$$



6.

- (a) $\vec{v} \uparrow a = 10^{12} \text{ m/s}^2$
Proton $E = 1 \text{ MeV}$

As we know, magnetic force $F = qvB = ma$

$$\therefore \vec{a} = \left(\frac{qvB}{m} \right) \text{ perpendicular to velocity.}$$

$$\therefore \text{Also } v = \sqrt{\frac{2KE}{m}} = \sqrt{\frac{2 \times e \times 10^6}{m}}$$

$$\therefore a = \frac{qvB}{m} = \frac{eB}{m} \sqrt{\frac{2 \times e \times 10^6}{m}}$$

$$\therefore 10^{12} = \left(\frac{1.6 \times 10^{-19}}{1.67 \times 10^{-27}} \right)^{\frac{3}{2}} \cdot \sqrt{2} \times 10^3 B$$

$$\therefore B \approx \frac{1}{\sqrt{2}} \times 10^{-3} \text{ T} = 0.71 \text{ mT (approx)}$$

7. (d) As particle is moving along a circular path

$$\therefore R = \frac{mv}{qB} \quad \dots(i)$$

Path is straight line, then

$$qE = qvB$$

$$E = vB \Rightarrow v = \frac{E}{B} \quad \dots(ii)$$

From equation (i) and (ii)

$$m = \frac{qB^2 R}{E} = \frac{1.6 \times 10^{-19} \times (0.5)^2 \times 0.5 \times 10^{-2}}{100}$$

$$\therefore m = 2.0 \times 10^{-24} \text{ kg}$$

8. (b)

9. (b) As $mvr = qvB \Rightarrow r = \frac{mv}{qB} = \frac{\sqrt{2mK.E.}}{qB}$

$$[\text{As: } \frac{1}{2}mv^2 = K.E.]$$

$$\Rightarrow m^2 v^2 = 2m K.E.$$

$$\Rightarrow mv = \sqrt{2m K.E.}$$

For proton, electron and α -particle,

$$m_{He} = 4m_p \text{ and } m_p \gg m_e$$

$$\text{Also } a_{He} = 2q_p \text{ and } q_p = q_e$$

$\therefore$ As KE of all the particles is same then,

$$r \propto \frac{\sqrt{m}}{q}$$

$$\therefore r_{He} = r_p > r_e$$

10. (a) Radius of the circular path will be $r = \frac{mv}{qB}$

$$\Rightarrow r = \frac{\sqrt{2mKE}}{qB} \quad (\because p = mv = \sqrt{2mKE})$$

$$\therefore KE = q\Delta V$$

$$\therefore r = \frac{\sqrt{2mq\Delta V}}{qB} \Rightarrow r \propto \sqrt{\frac{m}{q}}$$

$$\therefore \frac{r_p}{r_\alpha} = \frac{1}{\sqrt{2}}$$

11. (d) Radius of the path (r) is given by $r = \frac{mv}{qB}$

$$r = \frac{\sqrt{2mk}}{eB} \quad (\because p = mv = \sqrt{2mk})$$

$$= \frac{\sqrt{2meV}}{eB} \quad (\because k = eV)$$

$$r = \frac{\sqrt{\frac{2m}{e}} V}{B} = \frac{\sqrt{\frac{2 \times 9.1 \times 10^{-31}}{1.6 \times 10^{-19}}} (500)}{100 \times 10^{-3}}$$

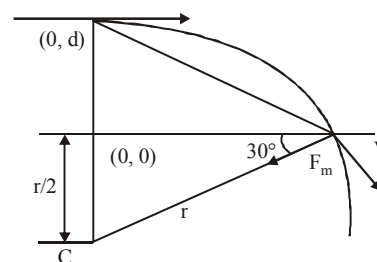
$$r = \frac{\sqrt{\frac{9.1}{0.16} \times 10^{-10}}}{10^{-1}} = \frac{3}{.4} \times 10^{-4}$$

$$= 7.5 \times 10^{-4}$$

12. (BONUS)

Assuming particle enters from (0, d)

$$r = \frac{mv}{qB}, \quad d = \frac{r}{2}$$



$$a = \frac{qVB}{m} \left[\frac{-\sqrt{3}i - j}{2} \right]$$

this option is not given in the all above four choices.

13. (b) As we know, radius of circular path in magnetic field

$$r = \frac{\sqrt{2Km}}{qB}$$

$$\text{For electron, } r_e = \frac{\sqrt{2Km_e}}{eB} \quad \dots(i)$$

$$\text{For proton, } r_p = \frac{\sqrt{2Km_p}}{eB} \quad \dots(ii)$$

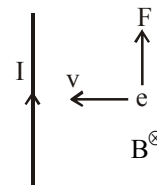
$$\text{For } \alpha \text{ particle, } r_\alpha = \frac{\sqrt{2Km_a}}{q_\alpha B} = \frac{\sqrt{2K4m_p}}{2eB} = \frac{\sqrt{2Km_p}}{eB} \quad \dots(iii)$$

$$\therefore r_e < r_p = r_\alpha \quad (\because m_e < m_p)$$

14. (b) The force is parallel to the direction of current in magnetic field,

$$\text{hence } F = q(v \times \vec{B})$$

According to Fleming's left hand rule,



we have, the direction of motion of charge is towards the wire.

15. (d) According to question, as the test charge experiences no net force in that region i.e., sum of electric force ($F_e = q\vec{E}$) and magnetic forces [$F_m = q(\vec{v} \times \vec{B})$] will be zero. Hence, $F_e + F_m = 0$

$$F_e = -q(\vec{v} \times \vec{B})$$

$$= -B_0 v_0 [(3\hat{i} - \hat{j} + 2\hat{k}) \times (\hat{i} + 2\hat{j} - 4\hat{k})]$$

$$= -B_0 v_0 (14\hat{j} + 7\hat{k})$$

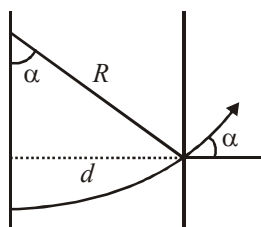
16. (b) $\therefore F = qE$ and $F = qvB$
 $\therefore E = vB$

And Gauss's law in Electrostatics $E = \frac{\sigma}{\epsilon_0}$

$$E = \frac{\sigma}{\epsilon_0} = vB \Rightarrow \sigma = \epsilon_0 vB$$

$$\sigma_1 = -\sigma_2$$

17. (d) From figure, $\sin \alpha = d/R$



And we know, $\frac{mv^2}{R} = qvB$

$$\Rightarrow R = \frac{mv}{qB}$$

$$\therefore \sin \alpha = \frac{dqB}{mv}$$

$$\sin \alpha = Bd \sqrt{\frac{q}{2mV}} \left[\because qV = \frac{1}{2}mv^2 \right]$$

18. (c) The applied magnetic field provides the required centripetal force to the charge particle, so it can move in circular path of radius $\frac{d}{2}$

$$\therefore Bqv = \frac{mv^2}{d/2}$$

$$\text{or, } B = \frac{2mv}{qd}$$

Time interval for which a uniform magnetic field is applied

$$\Delta t = \frac{\pi \cdot \frac{d}{2}}{v}$$

(particle reverses its direction after time Δt by covering semi circle).

$$\Delta t = \frac{\pi d}{2v}$$

19. (c) Since particle is moving undeflected

$$\text{So, } qE = qvB$$

$$\Rightarrow B = \frac{E}{v} = \frac{10^4}{10} = 10^3 \text{ wb/m}^2$$

20. (b) The centripetal force is provided by the magnetic force

$$\therefore \frac{mv^2}{R} = qvB \Rightarrow r = \frac{mv}{Bq} \therefore r \propto \frac{\sqrt{m}}{q}$$

$$\therefore r_p : r_d : r_\alpha = \frac{\sqrt{m_p}}{q_p} : \frac{\sqrt{m_d}}{q_d} : \frac{\sqrt{m_\alpha}}{q_\alpha}$$

$$= 1 : \sqrt{2} : 1$$

Thus we have, $r_\alpha = r_p < r_d$

21. (d) When a charged particle enters the magnetic field in perpendicular direction then it experience a force in perpendicular direction.

$$\text{i.e. } F = Bqv \sin \theta$$

Due to which it moves in a circular path.

22. (b) As charge on both proton and deuteron is same i.e. 'e'
 Energy acquired by both, $E = eV$
 For Deuteron.

$$\text{Kinetic energy, } \frac{1}{2} mV^2 = eV$$

[V is the potential difference]

$$v = \sqrt{\frac{2eV}{m_d}}$$

$$\text{But } m_d = 2m$$

$$\text{Therefore, } v = \sqrt{\frac{2eV}{2m}} = \sqrt{\frac{eV}{m}}$$

$$\text{Radius of path, } R = \frac{mv}{eB}$$

Substituting value of 'v' we get

$$R = \frac{2m \sqrt{\frac{ev}{m}}}{eB}$$

$$\frac{R}{2} = \frac{m \sqrt{\frac{ev}{m}}}{eB}$$

...(i)

For proton :

$$\frac{1}{2} mV^2 = eV$$

$$V = \sqrt{\frac{2eV}{m}}$$

$$\text{Radius of path, } R' = \frac{mV}{eB} = \frac{m \sqrt{\frac{2eV}{m}}}{eB}$$

$$R' = \sqrt{2} \times \frac{R}{2} \text{ [From eq. (i)]}$$

$$R' = \frac{R}{\sqrt{2}}$$

23. (a) $\vec{F} = q(\vec{V} \times \vec{B})$

$$= 2 \times 10^{-6} [(2\hat{i} + 3\hat{j}) \times 10^6 \times 2\hat{j}]$$

$$= 2 \times 4\hat{k} = 8N \text{ in Z-direction.}$$

24. (c) As velocity $v = \frac{E}{B} = \frac{7.7 \times 10^3}{0.14} = 55 \text{ km/s}$
25. (a) The charge experiences both electric and magnetic force.
Electric force, $F_e = qE$
Magnetic force, $F_m = q(\vec{v} \times \vec{B})$
 $\therefore$ Net force, $\vec{F} = q[\vec{E} + \vec{v} \times \vec{B}]$
$$= q \left[3\hat{i} + \hat{j} + 2\hat{k} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 4 & 1 \\ 1 & 1 & -3 \end{vmatrix} \right]$$
$$= q[3\hat{i} + \hat{j} + 2\hat{k} + \hat{i}(-12-1) - \hat{j}(-9-1) + \hat{k}(3-4)]$$
$$= q[3\hat{i} + \hat{j} + 2\hat{k} - 13\hat{i} + 10\hat{j} - \hat{k}]$$
$$= q[-10\hat{i} + 11\hat{j} + \hat{k}]$$
$$F_y = 11q\hat{j}$$

Thus, the y component of the force.

26. (b) As velocity is not changing, charge particle must go undeflected, then
 $qE = qvB$
 $\Rightarrow v = \frac{E}{B}$
Also,
$$\left| \frac{\vec{E} \times \vec{B}}{B^2} \right| = \frac{E B \sin \theta}{B^2}$$
$$= \frac{E B \sin 90^\circ}{B^2} = \frac{E}{B} = |\vec{v}| = v$$

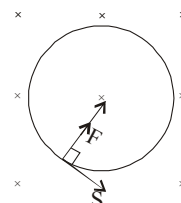
27. (b) When a charged particle enters a magnetic field at a direction perpendicular to the direction of motion, the path of the motion is circular. In circular motion the direction of velocity changes at every point (the magnitude remains constant).
Therefore, the tangential momentum will change at every point. But kinetic energy will remain constant as it is given by $\frac{1}{2}mv^2$ and v^2 is the square of the magnitude of velocity which does not change.

28. (b) The charged particle will move along the lines of electric field (and magnetic field). Magnetic field will exert no force. The force by electric field will be along the lines of uniform electric field. Hence the particle will move in a straight line.

29. (c) Equating magnetic force to centripetal force,
$$\frac{mv^2}{r} = qvB \sin 90^\circ$$
$$\Rightarrow \frac{mv}{r} = Bq \Rightarrow v = \frac{Bqr}{m}$$

Time to complete one revolution,
$$T = \frac{2\pi r}{v} = \frac{2\pi m}{qB}$$

30. (b) Due to electric field, it experiences force and accelerates i.e. its velocity decreases.
31. (b) The workdone, $dW = Fds \cos \theta$
The angle between force and displacement is 90° .
Therefore work done is zero.

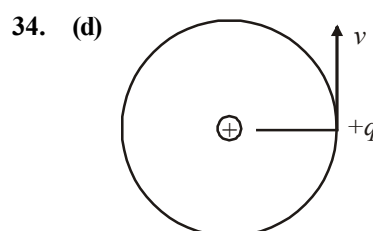


32. (a) When a moving charged particle is subjected to a perpendicular magnetic field, then it describes a circular path of radius.
$$r = \frac{p}{qB}$$

where q = Charge of the particle
 p = Momentum of the particle
 B = Magnetic field

Here p , q and B are constant for electron and proton, therefore the radius will be same.

33. (a) The time period of a charged particle of charge q and mass m moving in a magnetic field (B) is $T = \frac{2\pi m}{qB}$
Clearly time period is independent of speed of the particle.



Length of the circular path, $l = 2\pi r$

$$\text{Current, } i = \frac{q}{T} = \frac{qv}{2\pi r}$$

Magnetic moment $M = \text{Current} \times \text{Area}$

$$= i \times \pi r^2 = \frac{qv}{2\pi r} \times \pi r^2$$

$$M = \frac{1}{2} q \cdot v \cdot r$$

Radius of circular path in magnetic field, $r = \frac{mv}{qB}$

$$\therefore M = \frac{1}{2} qv \times \frac{mv}{qB} \Rightarrow M = \frac{mv^2}{2B}$$

Direction of $\vec{M}$ is opposite of $\vec{B}$ therefore

$$\vec{M} = \frac{-mv^2 \vec{B}}{2B^2}$$

(By multiplying both numerator and denominator by B).

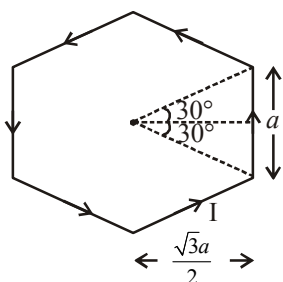
35. (d) Given : $I_A = 2 \text{ A}$, $R_A = 2 \text{ cm}$, $\theta_A = 2\pi - \frac{\pi}{2} = \frac{3\pi}{2}$

$$I_B = 3 \text{ A}, R_B = 4 \text{ cm}, \theta_B = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$$

Using, magnetic field, $B = \frac{\mu_0 I \theta}{4\pi R}$

$$\frac{B_A}{B_B} = \frac{I_A}{I_B} \times \frac{\theta_A R_B}{\theta_B R_A} = \frac{2 \times \frac{3\pi}{2} \times 4}{3 \times \frac{5\pi}{3} \times 2} = \frac{6}{5}$$

36. (c)



Magnetic field due to one side of hexagon

$$B = \frac{\mu_0 I}{4\pi \frac{\sqrt{3}a}{2}} (\sin 30^\circ + \sin 30^\circ)$$

$$\Rightarrow B = \frac{\mu_0 I}{2\sqrt{3}a} \left(\frac{1}{2} + \frac{1}{2} \right) = \frac{\mu_0 I}{2\sqrt{3}a\pi}$$

Now, magnetic field due to one hexagon coil

$$B = 6 \times \frac{\mu_0 I}{2\sqrt{3}a\pi}$$

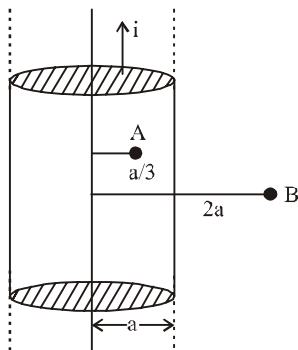
Again magnetic field at the centre of hexagonal shape coil of 50 turns,

$$B = 50 \times 6 \times \frac{\mu_0 I}{2\sqrt{3}a\pi} \quad \left[\because a = \frac{10}{100} = 0.1 \text{ m} \right]$$

$$\text{or, } B = \frac{150\mu_0 I}{\sqrt{3} \times 0.1 \times \pi} = 500\sqrt{3} \frac{\mu_0 I}{\pi}$$

37. (a) Let a be the radius of the wire
Magnetic field at point A (inside)

$$B_A = \frac{\mu_0 i r}{2\pi a^2} = \frac{\mu_0 i \frac{a}{3}}{2\pi a^2} = \frac{\mu_0 i}{\pi a^2} \frac{a}{6} = \frac{\mu_0 i}{6\pi a}$$



Magnetic field at point B (outside)

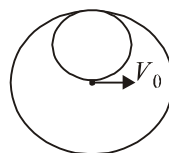
$$B_B = \frac{\mu_0 i}{2\pi(2a)}$$

$$\therefore \frac{B_A}{B_B} = \frac{\frac{\mu_0 i}{6\pi a}}{\frac{\mu_0 i}{2\pi(2a)}} = \frac{4}{6} = \frac{2}{3}$$

38. (b) Magnetic field inside the solenoid is given by

$$B = \mu_0 n I \quad \dots (i)$$

Here, n = number of turns per unit length

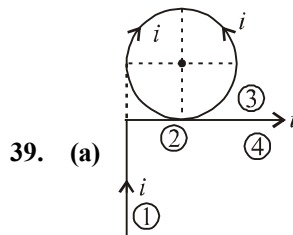


The path of charge particle is circular. The maximum

possible radius of electron = $\frac{R}{2}$

$$\therefore \frac{mV_{\max}}{qB} = \frac{R}{2}$$

$$\Rightarrow V_{\max} = \frac{qBR}{2m} = \frac{eR\mu_0 n I}{2m} \quad (\text{using (i)})$$



39. (a)

$$B_0 = B_1 + B_2 + B_3 + B_4$$

$$= \frac{\mu_0 I}{4\pi R} [\sin 90^\circ - \sin 45^\circ] + \frac{\mu_0 I}{2R} + \frac{\mu_0 I}{4\pi R} [\sin 45^\circ + \sin 90^\circ]$$

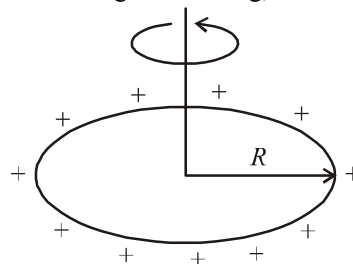
$$= -\frac{\mu_0 I}{4\pi R} \left(1 - \frac{1}{\sqrt{2}} \right) + \frac{\mu_0 I}{2R} + \frac{\mu_0 I}{4\pi R} \left(1 + \frac{1}{\sqrt{2}} \right)$$

$$\vec{B}_0 = \frac{\mu_0 I}{2\pi R} \left(\pi + \frac{1}{\sqrt{2}} \right) \hat{\phi}$$

40. (a) $\vec{B} = \vec{B}_1 + \vec{B}_2$

$$= \frac{\mu_0}{2\pi} \cdot \left(\frac{i^\circ}{d} \hat{k} + \frac{i^\circ}{d} (-\hat{k}) \right) = 0$$

41. (b) If q is the charge on the ring, then



$$i = \frac{q}{T} = \frac{q\omega}{2\pi}$$

Magnetic field,

$$B = \frac{\mu_0 i}{2R}$$

$$= \frac{\mu_0 \left(\frac{q\omega}{2\pi} \right)}{2R}$$

$$\text{or } 3.8 \times 10^{-9} = \left(\frac{\mu_0}{4\pi} \right) \frac{q\omega}{R} = (10^{-7}) \frac{q \times 40\pi}{0.10}$$

$$\therefore q = 3 \times 10^{-5} \text{ C.}$$

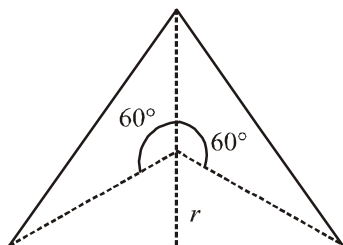
$$42. (b) B = \frac{\mu_0}{4\pi} \cdot \frac{i}{r} (\sin \alpha + \sin \beta)$$

$$\text{Here } r = \sqrt{5^2 - 3^2} = 4 \text{ cm}$$

$$\alpha = \beta = 37^\circ$$

$$\therefore B = 10^{-7} \times \frac{5}{4} 2 \sin 37^\circ = 1.5 \times 10^{-5} \text{ T}$$

43. (a)



$$r = \left(\frac{1}{3} \right) (a \sin 60^\circ)$$

$$r = \frac{a}{3} \times \frac{\sqrt{3}}{2} = \left(\frac{a}{2\sqrt{3}} \right)$$

$$B_0 = 3 \left[\frac{\mu_0 I}{4\pi r} (\sin 60^\circ + \sin 60^\circ) \right]$$

$$= \frac{3\mu_0 I}{4\pi \left(\frac{a}{2\sqrt{3}} \right)} \times (2) \left(\frac{\sqrt{3}}{2} \right) = \frac{9}{2} \left(\frac{\mu_0 I}{\pi a} \right)$$

$$= \frac{9 \times 2 \times 10^{-7} \times 10}{1} = 18 \mu\text{T}$$

44. (d) Let a be the area of the square and r be the radius of circular loop.

$$2\pi r = 4a \Rightarrow r = \left(\frac{2a}{\pi} \right)$$

For square

$$M = (I) a^2$$

For circular loop

$$M_1 = (I) \pi r^2$$

$$M_1 = (I)(\pi) \left(\frac{4a^2}{\pi^2} \right)$$

$$M_1 = \frac{4Ia^2}{\pi}$$

$$M_1 = \frac{4M}{\pi} \quad (\because M = Ia^2)$$

45. (c) Let I be the current in each wire. (directed inwards)
Magnetic field at 'O' due to LP and QM will be zero.

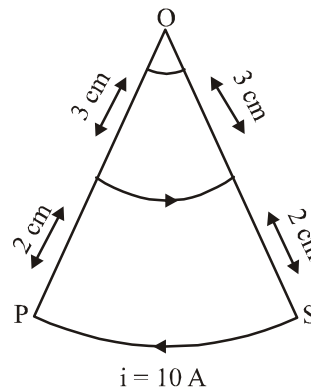
$$\text{i.e., } B_0 = B_{PS} + B_{QN}$$

$$\therefore \text{Net magnetic field } B_0 = \frac{\mu_0 i}{4\pi d} + \frac{\mu_0 i}{4\pi d}$$

$$\text{or } 10^{-4} = \frac{\mu_0 i}{2\pi d} + \frac{2 \times 10^{-7} \times i}{4 \times 10^{-2}}$$

$\therefore i = 20 \text{ A}$ and the direction of magnetic field is perpendicular into the plane

46. (d)



There will be no magnetic field at O due to wire PQ and RS

Magnetic field at 'O' due to arc QR

$$= \frac{\mu_0}{4\pi} \left(\frac{\pi}{4} \right) \cdot \frac{I}{r_1}$$

Magnetic field at 'O' due to arc PS

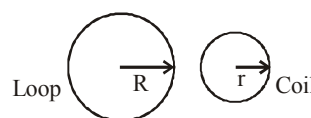
$$= \frac{\mu_0}{4\pi} \left(\frac{\pi}{4} \right) \cdot \frac{I}{r_2}$$

$\therefore$ Net magnetic field at 'O'

$$B = \frac{\mu_0}{4\pi} (\pi/4) \times 10 \left[\frac{1}{(3 \times 10^{-2})} - \frac{1}{(5 \times 10^{-2})} \right]$$

$$\Rightarrow |\vec{B}| = \frac{\pi}{3} \times 10^{-5} \text{ T} \approx 1 \times 10^{-5} \text{ T}$$

47. (d)



$$L = 2\pi R$$

$$L = N \times 2\pi r$$

$$R = Nr \Rightarrow r = \frac{R}{N}$$

$$B_{\text{Loop}} = \frac{\mu_0 i}{2R} \quad B_{\text{coil}} = \frac{\mu_0 Ni}{2r} = \frac{\mu_0 Ni}{2\left(\frac{R}{N}\right)} = \frac{\mu_0 N^2 i}{2R}$$

$$\therefore \frac{B_L}{B_C} = \frac{1}{N^2}$$

48. (c) Magnetic field at the centre of loop, $B_1 = \frac{\mu_0 I}{2R}$

Dipole moment of circular loop is $m = IA$

$m_1 = I \cdot A = I \cdot \pi R^2$ { R = Radius of the loop}

If moment is doubled (keeping current constant) R becomes $\sqrt{2}R$

$$m_2 = I \cdot \pi (\sqrt{2}R)^2 = 2 \cdot I \pi R^2 = 2m_1$$

$$B_2 = \frac{\mu_0 I}{2(\sqrt{2}R)}$$

$$\therefore \frac{B_1}{B_2} = \frac{\frac{\mu_0 I}{2R}}{\frac{\mu_0 I}{2(\sqrt{2}R)}} = \sqrt{2}$$

49. (b) Point P is situated at the mid-point of the line joining the centres of the circular wires which have same radii (R).

The magnetic fields ($\vec{B}$) at P due to the currents in the wires are in same direction.

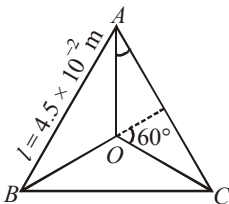
Magnitude of magnetic field at point, P

$$B = 2 \left\{ \frac{\mu_0 N I R^2}{2 \left(R^2 + \frac{R^2}{4} \right)^{3/2}} \right\} = \frac{\mu_0 N I R^2}{\frac{5^{3/2}}{8}} = \frac{8\mu_0 N I}{5^{3/2} R}$$

50. (a) Here, side of the triangle, $l = 4.5 \times 10^{-2} \text{ m}$, current, $I = 1 \text{ A}$
magnetic field at the centre of the triangle ' O ' $B = ?$

$$\text{From figure, } \tan 60^\circ = \sqrt{3} = \frac{1}{2d}$$

$$\Rightarrow d = \frac{l}{2\sqrt{3}} = \left(\frac{4.5 \times 10^{-2}}{2\sqrt{3}} \right) \text{ m}$$



$$\text{Magnetic field, } B = \frac{\mu_0 i}{4\pi d} (\cos \theta_1 + \cos \theta_2)$$

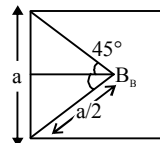
Putting value of $\mu = 4\pi \times 10^{-7}$ and θ_1 and θ_2 we will get $B = 4 \times 10^{-5} \text{ Wb/m}^2$

51. (b) Case (a):

$$B_A = \frac{\mu_0}{4\pi} \frac{I}{R} \times 2\pi = \frac{\mu_0}{4\pi} \frac{I}{\ell/2\pi} \times 2\pi (\because 2\pi R = \ell)$$

$$= \frac{\mu_0}{4\pi} \frac{I}{\ell} \times (2\pi)^2$$

Case (b):



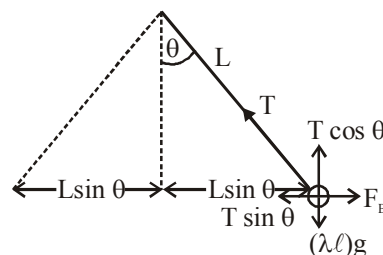
$$B_B = 4 \times \frac{\mu_0}{4\pi} \frac{I}{a/2} [\sin 45^\circ + \sin 45^\circ]$$

$$= 4 \times \frac{\mu_0}{4\pi} \times \frac{I}{\ell/8} \times \frac{2}{\sqrt{2}} = \frac{\mu_0}{4\pi} \frac{I}{\ell} \times \frac{64}{\sqrt{2}} = \frac{\mu_0 I}{4\pi \ell} 32\sqrt{2} \quad [4a = \ell]$$

$$\Rightarrow \frac{B_A}{B_B} = \frac{\pi^2}{8\sqrt{2}}$$

52. (d) Let us consider ' ℓ ' length of current carrying wire.
At equilibrium

$$T \cos \theta = \lambda g \ell$$



$$\text{and } T \sin \theta = \frac{m_0}{2p} \frac{I' I \ell}{2L \sin \theta} \left[\because \frac{F_B}{\ell} = \frac{\mu_0}{4\pi} \frac{2I \times I}{2\ell \sin \theta} \right]$$

$$\text{Therefore, } I = 2 \sin \theta \sqrt{\frac{p \ell g L}{\mu_0 \cos \theta}}$$

53. (b) For loop $B = \frac{\mu_0 n I}{2a}$
where, a is the radius of loop.

$$\text{Then, } B_1 = \frac{\mu_0 I}{2a}$$

$$\text{Now, for coil } B = \frac{\mu_0 I}{4\pi} \cdot \frac{2nA}{x^3}$$

at the centre x = radius of loop

$$B_2 = \frac{\mu_0}{4\pi} \cdot \frac{2 \times 3 \times (I/3) \times \pi (a/3)^2}{(a/3)^3} = \frac{\mu_0 \cdot 3I}{2a}$$

$$\therefore \frac{B_1}{B_2} = \frac{\mu_0 I / 2a}{\mu_0 \cdot 3I / 2a}$$

$$B_1 : B_2 = 1 : 3$$

54. (d) Magnetic field between the plates in this case is zero.

55. (a) Magnetic field at any point lies on axial position of current carrying conductor $B = 0$

56. (a) If magnetic field is perpendicular and into the plane of the paper, it is represented by cross $\otimes$ and if the direction of the magnetic field is perpendicular out of the plane of the paper it is represented by dot $\odot$.

57. (b) Given : Radius = R

Distance $x = 2\sqrt{2}R$

$$\frac{B_{\text{centre}}}{B_{\text{axis}}} = \left(1 + \frac{x^2}{R^2}\right)^{3/2} = \left(1 + \frac{(2\sqrt{2}R)^2}{R^2}\right)^{3/2} \\ = (9)^{3/2} = 27$$

58. (a) The magnetic field due to a disc is given as

$$B = \frac{\mu_0 \omega Q}{2\pi R} \quad \text{i.e., } B \propto \frac{1}{R}$$

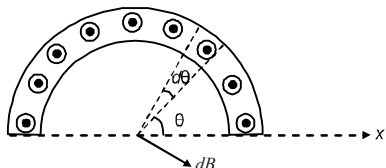
59. (d) Let R be the radius of semicircular ring. Let an elementary length dl is cut for finding magnetic field. So,

$$dl = R d\theta. \text{ Current in a small element, } dI = \frac{d\theta}{\pi} I$$

Magnetic field due to the element

$$dB = \frac{\mu_0}{4\pi} \frac{2dI}{R} = \frac{\mu_0 I}{2\pi^2 R}$$

The component $dB \cos \theta$, of the field is cancelled by another opposite component. Therefore,



$$B_{\text{net}} = \int dB \sin \theta = \frac{\mu_0 I}{2\pi^2 R} \int_0^\pi \sin \theta d\theta = \frac{\mu_0 I}{\pi^2 R}$$

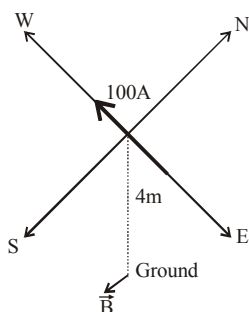
60. (a) The magnetic field varies inversely with the distance

for a long conductor. That is, $B \propto \frac{1}{d}$

so, graph in option (a) is the correct one.

61. (c) The magnetic field is

$$B = \frac{\mu_0}{4\pi} \frac{2I}{r} = 10^{-7} \times \frac{2 \times 100}{4} = 5 \times 10^{-6} \text{ T}$$



Current flows from east to west. Point is below the power line, using right hand thumb rule, the magnetic field is directed towards south.

62. (d) Since uniform current is flowing through a straight wire, current enclosed in the amperian path formed at a

distance $r_1 \left(= \frac{a}{2} \right)$ is

$$i = \left(\frac{\pi r_1^2}{\pi a^2} \right) \times I,$$

where I is total current

Using Ampere circuital law,

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i$$

$$\Rightarrow B_1 = \frac{\mu_0 \times \text{current enclosed}}{\text{Path}}$$

$$\Rightarrow B_1 = \frac{\mu_0 \times \left(\frac{\pi r_1^2}{\pi a^2} \right) \times I}{2\pi r_1} = \frac{\mu_0 \times I r_1}{2\pi a^2}$$

Now, magnetic field induction at point P_2 ,

$$B_2 = \frac{\mu_0}{2\pi} \cdot \frac{I}{(2a)} = \frac{\mu_0 I}{4\pi a}$$

$$\therefore \frac{B_1}{B_2} = \frac{\mu_0 I r_1}{2\pi a^2} \times \frac{4\pi a}{\mu_0 I}$$

$$\Rightarrow \frac{B_1}{B_2} = \frac{2r_1}{a} = \frac{2 \times \frac{a}{2}}{a} = 1.$$

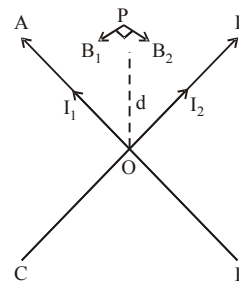
63. (d) There is no current inside the pipe. From Ampere's

$$\text{circuital law } \oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

$$\therefore I = 0$$

$$\therefore B = 0$$

64. (c) The direction of magnetic field induction due to current through AB and CD at P are indicated as B_1 and B_2 . The magnetic fields at a point P , equidistant from AOB and COD will have directions perpendicular to each other, as they are placed normal to each other.



Magnetic field at P due to current through AB ,

$$B_1 = \frac{\mu_0 I_1}{2\pi d}$$

Magnetic field at P due to current through CD ,

$$B_2 = \frac{\mu_0 I_2}{2\pi d}$$

$$\therefore \text{Resultant field, } B = \sqrt{B_1^2 + B_2^2}$$

$$\therefore B = \sqrt{\left(\frac{\mu_0}{2\pi d} \right)^2 (I_1^2 + I_2^2)}$$

$$\text{or, } B = \frac{\mu_0}{2\pi d} (I_1^2 + I_2^2)^{1/2}$$

65. (a) Magnetic field due to long solenoid is given by $B = \mu_0 n I$

In first case $B_1 = \mu_0 n_1 I_1$

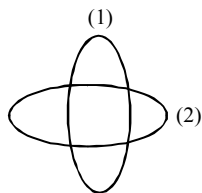
In second case, $B_2 = \mu_0 n_2 I_2$

$$\therefore \frac{B_2}{B_1} = \frac{\mu_0 n_2 I_2}{\mu_0 n_1 I_1}$$

$$\Rightarrow \frac{B_2}{6.28 \times 10^{-2}} = \frac{100 \times \frac{i}{3}}{200 \times i}$$

$$\Rightarrow B_2 = \frac{6.28 \times 10^{-2}}{6} = 1.05 \times 10^{-2} \text{ Wb/m}^2$$

66. (d)



The magnetic field due to circular coil (1) is

$$B_1 = \frac{\mu_0 i_1}{2r} = \frac{\mu_0 i_1}{2(2\pi \times 10^{-2})} = \frac{\mu_0 \times 3 \times 10^2}{4\pi}$$

Magnetic field due to coil (2)

Total magnetic field

$$B_2 = \frac{\mu_0 i_2}{2(2\pi \times 10^{-2})} = \frac{\mu_0 \times 4 \times 10^2}{4\pi}$$

$$\text{Total magnetic field, } B = \sqrt{B_1^2 + B_2^2}$$

$$= \frac{\mu_0}{4\pi} \cdot 5 \times 10^2$$

$$\Rightarrow B = 10^{-7} \times 5 \times 10^2$$

$$\Rightarrow B = 5 \times 10^{-5} \text{ Wb/m}^2$$

67. (b) From Ampere's circuital law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i$$

$$\Rightarrow B \times 2\pi r = \mu_0 i$$

Here i is zero, for $r < R$, whereas R is the radius

$$\therefore B = 0$$

68. (b) Magnetic field at the centre of a circular coil of radius

$$R \text{ carrying current } i \text{ is } B = \frac{\mu_0 i}{2R}$$

The circumference of the first loop $= 2\pi R$. If it is bent into n circular coil of radius r' .

$$n \times (2\pi r') = 2\pi R$$

$$\Rightarrow nr' = R \quad \dots(1)$$

$$\text{New magnetic field, } B' = \frac{n \cdot \mu_0 i}{2r'} \quad \dots(2)$$

From (1) and (2),

$$B' = \frac{n\mu_0 i \cdot n}{2\pi R} = n^2 B$$

69. (c) The magnetic field at a point on the axis of a circular loop at a distance x from centre is,

$$B = \frac{\mu_0 i a^2}{2(x^2 + a^2)^{3/2}}$$

Magnetic field at the centre of loop is

$$B' = \frac{\mu_0 i}{2a}$$

$$\therefore B' = \frac{B \cdot (x^2 + a^2)^{3/2}}{a^3}$$

Put $x = 4$ & $a = 3$

$$\Rightarrow B' = \frac{54(5^3)}{3 \times 3 \times 3} = 250 \mu T$$

70. (a) Magnetic field induction at the centre of current carrying circular coil of radius r is

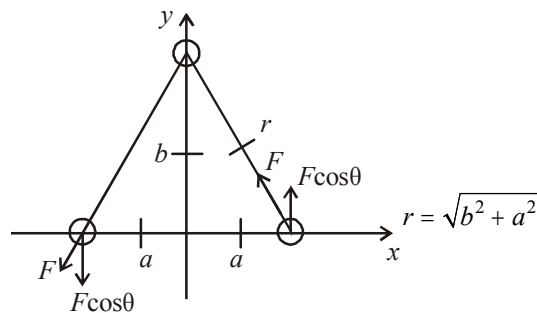
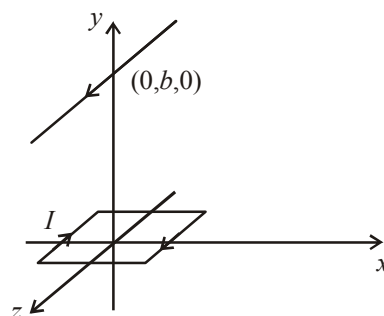
$$B = \frac{\mu_0}{4\pi} \frac{I}{R} \times 2\pi$$

$$\text{Here } B_A = \frac{\mu_0}{4\pi} \frac{I}{R} \times 2\pi$$

$$\text{and } B_B = \frac{\mu_0}{4\pi} \frac{2I}{2R} \times 2\pi$$

$$\Rightarrow \frac{B_A}{B_B} = \frac{I/R}{2I/2R} = 1$$

71. (b)



$$\text{Force, } F = B I 2a = \frac{\mu_0 I}{2\pi r} I \times 2a$$

$$\text{Force, } F = \frac{\mu_0 I^2 a}{\pi \sqrt{b^2 + a^2}}$$

$$\text{Torque, } \tau = F_1 \times \text{Perpendicular distance} = F \cos \theta \times 2a$$

$$= \frac{\mu_0 I^2 a}{\pi \sqrt{b^2 + a^2}} \times \frac{b}{\sqrt{b^2 + a^2}} \times 2a$$

$$\Rightarrow \tau = \frac{2\mu_0 I^2 a^2 b}{\pi(a^2 + b^2)}$$

$$\text{If } b \gg a \text{ then } \tau = \frac{2\mu_0 I^2 a^2}{\pi b}$$

72. (c) Torque on the loop,

$$\vec{\tau} = \vec{M} \times \vec{B} = MB \sin \theta = MB \sin 90^\circ$$

$$\text{Magnetic field, } B = \frac{\mu_0 I}{2\pi d}$$

$$\therefore \tau = I_1 (2a)^2 \left(\frac{\mu_0 I_2}{2\pi d} \right) \sin 90^\circ$$

$$= \frac{2\mu_0 I_1 I_2}{\pi d} \times a^2 = \frac{2\mu_0 I^2 a^2}{\pi d}$$

73. (b) Magnetic moment of loop ABCD,

$$M_1 = \text{area of loop} \times \text{current}$$

$$\vec{M}_1 = (abI)(\hat{j}) \quad (\text{Here, } ab = \text{area of rectangle})$$

$$\text{Magnetic moment of loop DEFA,}$$

$$\vec{M}_2 = (abI)(\hat{i})$$

$$\text{Net magnetic moment,}$$

$$\vec{M} = \vec{M}_1 + \vec{M}_2 \Rightarrow \vec{M} = abI(\hat{i} + \hat{j})$$

$$\Rightarrow |\vec{M}| = \sqrt{2}abI \left(\frac{\hat{j}}{\sqrt{2}} + \frac{\hat{k}}{\sqrt{2}} \right)$$

74. (c) Torque on circular loop, $\tau = MB \sin \theta$

$$\text{where, } M = \text{magnetic moment}$$

$$B = \text{magnetic field}$$

$$\text{Now, using } \tau = I\alpha$$

$$\therefore \tau = MB \sin \theta = I\alpha$$

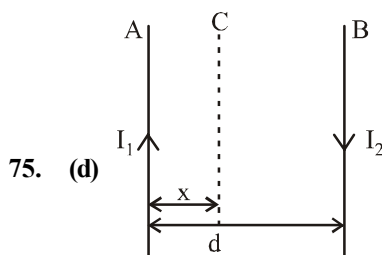
$$\Rightarrow \pi R^2 IB \theta = \frac{mR^2 \alpha}{2}$$

$$(\because m = IA \text{ and moment of inertia of circular loop, } I = \frac{mR^2}{2})$$

$$\Rightarrow \pi R^2 IB \theta = \frac{mR^2}{2} \omega \theta$$

$$\Rightarrow \omega = \sqrt{\frac{2\pi IB}{m}} \Rightarrow \frac{2\pi}{T} = \sqrt{\frac{2\pi IB}{m}}$$

$$\Rightarrow T = \sqrt{\frac{2\pi m}{IB}}$$



As net force on the third wire C is zero.

$$\Rightarrow \vec{F} = \frac{\mu_0 I_1}{2\pi x} + \frac{\mu_0 I_2}{2\pi(d-x)} = 0$$

$$\frac{\mu_0 I_1}{2\pi x} = \frac{\mu_0 I_2}{2\pi(x-d)}$$

$$I_1 x - I_1 d = I_2 x$$

$$x = \frac{I_1 d}{I_1 - I_2}$$

Two cases may be possible if $I_1 > I_2$ or $I_2 > I_1$

76. (d) $\tau = MB \sin 45^\circ = N(iA) B \sin 45^\circ$

$$= 100 \times 3(5 \times 2.5) \times 10^{-4} \times 1 \times \frac{1}{\sqrt{2}}$$

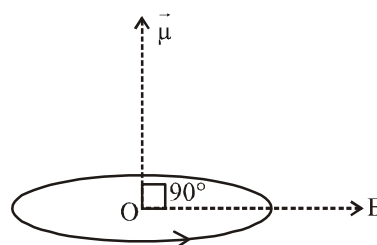
$$= 0.27 \text{ N-m}$$

77. (c) $F = \frac{\mu_0}{2\pi} \left(\frac{i_1 i_2}{a} - \frac{i_1 i_2}{2a} \right) \times a = \frac{\mu_0 i_1 i_2}{4\pi}$

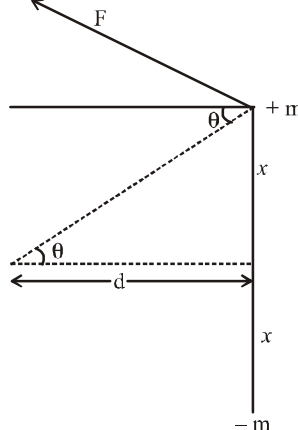
78. (b) $|\vec{\tau}| = |\vec{\mu} \times \vec{B}| \quad [\mu = NIA]$

$$= NIA \times B \sin 90^\circ \quad [A = \pi r^2]$$

$$\Rightarrow \tau = NI\pi^2 B$$



79. (d)



Force on one pole,

$$F = m \times \frac{\mu_0 I}{2\pi \sqrt{d^2 + x^2}}$$

$$\text{Total force, } F_{\text{total}} = 2F \sin \theta$$

$$= 2 \times \frac{\mu_0 Im}{2\pi \sqrt{d^2 + a^2}} \times \frac{x}{\sqrt{d^2 + a^2}}$$

$$= \frac{\mu_0 Im x}{\pi(d^2 + a^2)}$$

$$\text{Magnetic moment, } M = I\pi a^2 = m \times 2$$

$$\text{or, Total force, } F_{\text{total}} = \frac{\mu_0 I a^2}{2(d^2 + a^2)}$$

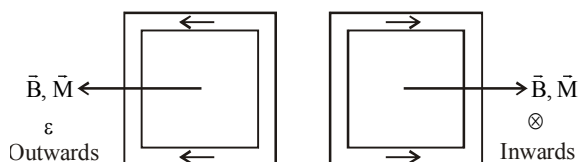
$$= \frac{\mu_0 I a^2}{2d^2} \quad [\because d \gg a]$$

$$\text{Clearly } F_{\text{total}} \propto \frac{a^2}{d^2}$$

80. (a) Magnetic moment, $\mu = IA = \frac{qv}{2\pi r}(\pi r^2)$

$$\text{or, } \mu = \frac{qr\omega}{2\pi r}(\pi r^2) = \frac{1}{2}qr^2\omega$$

81. (c) Magnetic moment of current carrying rectangular loop of area A is given by $M = NIA$
magnetic moment of current carrying coil is a vector and its direction is given by right hand thumb rule, for rectangular loop, $\vec{B}$ at centre due to current in loop and $\vec{M}$ are always parallel.



Hence, (c) corresponds to stable equilibrium.

82. (c) $\vec{F}_1 = \vec{F}_2 = 0$
because of action and reaction pair

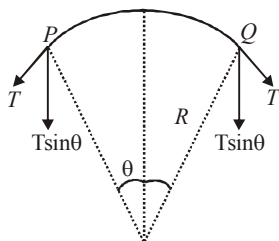
83. (a) For stable equilibrium $\vec{M} \parallel \vec{B}$

For unstable equilibrium $\vec{M} \parallel (-\vec{B})$

84. (a) $I_1 I_2 = \text{Positive}$
(attract) $F = \text{Negative}$
 $I_1 I_2 = \text{Negative}$
(repell) $F = \text{Positive}$

Hence, option (a) is the correct answer.

85. (c) For small arc length
 $2T \sin \theta = BIR \cdot 2\theta$ (As $F = BIL$ and $L = RZ\theta$)
 $T = BIR$

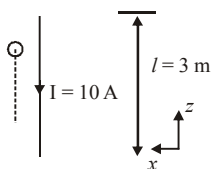


86. (b) Work done in moving the conductor is,

$$W = \int_0^2 F dx = \int_0^2 3.0 \times 10^{-4} e^{-0.2x} \times 10 \times 3 dx$$

$$= 9 \times 10^{-3} \int_0^2 e^{-0.2x} dx$$

$$= \frac{9 \times 10^{-3}}{0.2} [-e^{-0.2 \times 2} + 1]$$



$$= \frac{9 \times 10^{-3}}{0.2} \times [1 - e^{-0.4}]$$

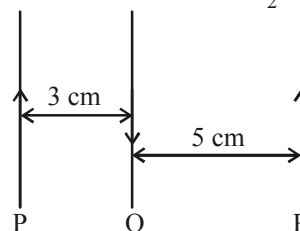
$$= \frac{9 \times 10^{-3} \times (0.33)}{2} = \frac{2.97 \times 10^{-3}}{2}$$

Power required to move the conductor is,

$$P = \frac{W}{t}$$

$$P = \frac{2.97 \times 10^{-3}}{(0.2) \times 5 \times 10^{-3}} = 2.97 \text{ W}$$

87. (a) $I_1 = 30 \text{ A}$ $I = 10 \text{ A}$ $I_2 = 20 \text{ A}$



Also given; length of wire Q

$$= 25 \text{ cm} = 0.25 \text{ m}$$

Force on wire Q due to wire R

$$F_{QR} = 10^{-7} \times \frac{2 \times 20 \times 10}{0.05} \times 0.25$$

$$= 20 \times 10^{-5} \text{ N (Towards left)}$$

Force on wire Q due to wire P

$$F_{QP} = 10^{-7} \times \frac{2 \times 30 \times 10}{0.03} \times 0.25$$

$$= 50 \times 10^{-5} \text{ N (Towards right)}$$

Hence, $F_{\text{net}} = F_{QP} - F_{QR}$

$$= 50 \times 10^{-5} \text{ N} - 20 \times 10^{-5} \text{ N}$$

$$= 3 \times 10^{-4} \text{ N towards right}$$

88. (c)

89. (a) Force acting on conductor B due to conductor A is given by relation

$$F = \frac{\mu_0 I_1 I_2 l}{2\pi r}$$

l -length of conductor B

r -distance between two conductors

$$\therefore F = \frac{4\pi \times 10^{-7} \times 10 \times 2 \times 2}{2 \times \pi \times 0.1} = 8 \times 10^{-5} \text{ N}$$

90. (a)

91. (a) The magnetic field at O due to current in DA is

$$B_1 = \frac{\mu_0 I}{4\pi a} \times \frac{\pi}{6} \quad (\text{directed vertically upwards})$$

The magnetic field at O due to current in BC is

$$B_2 = \frac{\mu_0 I}{4\pi b} \times \frac{\pi}{6} \quad (\text{directed vertically downwards})$$

The magnetic field due to current AB and CD at O is zero.
Therefore the net magnetic field is

$$B = B_1 - B_2 \text{ (directed vertically upwards)}$$

$$= \frac{\mu_0 I \pi}{4\pi a 6} - \frac{\mu_0 I \pi}{4\pi b 6} \times \frac{\pi}{6}$$

$$= \frac{\mu_0 I}{24} \left(\frac{1}{a} - \frac{1}{b} \right) = \frac{\mu_0 I}{24ab} (b - a)$$

92. (d) $\vec{F} = I(\vec{\ell} \times \vec{B})$

The force on AD and BC due to current I_1 is zero. This is because the directions of current element $I d\vec{\ell}$ and magnetic field $\vec{B}$ are parallel.

93. (a) Force acting between two long conductor carrying current,

$$F = \frac{\mu_0}{4\pi} \frac{2I_1 I_2}{d} \times \ell \quad \dots(i)$$

Where d = distance between the conductors
 ℓ = length of conductor

$$\text{In second case, } F' = -\frac{\mu_0}{4\pi} \frac{2(2I_1)I_2}{3d} \ell \quad \dots(ii)$$

From equation (i) and (ii), we have

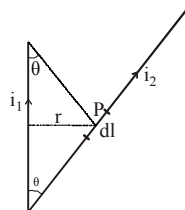
$$\therefore \frac{F'}{F} = \frac{-2}{3}$$

94. (b) When current is passed through a spring then current flows parallel in the adjacent turns in the same direction. As a result the various turn attract each other and spring get compress.

95. (c) Magnetic field due to current in wire 1 at point P distant r from the wire is

$$B = \frac{\mu_0}{4\pi} \frac{i_1}{r} [\cos \theta + \cos \theta]$$

$$B = \frac{\mu_0}{2\pi} \frac{i_1 \cos \theta}{r}$$



This magnetic field is directed perpendicular to the plane of paper, inwards.

The force exerted due to this magnetic field on current element $i_2 dl$ is

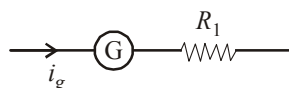
$$dF = i_2 dl B \sin 90^\circ$$

$$\therefore dF = i_2 dl B$$

$$\Rightarrow dF = i_2 dl \left(\frac{\mu_0}{4\pi} \frac{i_1 \cos \theta}{r} \right)$$

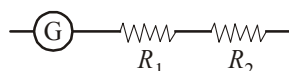
$$= \frac{\mu_0}{2\pi r} i_1 i_2 dl \cos \theta$$

96. (d) Galvanometer of resistance (G) converted into a voltmeter of range 0-1 V.



$$V = 1 = i_g (G + R_1) \quad \dots(i)$$

To increase the range of voltmeter 0-2 V



$$2 = i_g (R_1 + R_2 + G) \quad \dots(ii)$$

Dividing eq. (i) by (ii),

$$\Rightarrow \frac{1}{2} = \frac{G + R_1}{G + R_1 + R_2}$$

$$\Rightarrow G + R_1 + R_2 = 2G + 2R_1$$

$$\therefore R_2 = G + R_1$$

97. (d) Given,

Current passing through galvanometer, $I = 6 \text{ mA}$

Deflection, $\theta = 2^\circ$

Figure of merit of galvanometer

$$= \frac{I}{\theta} = \frac{6 \times 10^{-3}}{2} = 3 \times 10^{-3} \text{ A/div}$$

98. (20)

Given,

Area of galvanometer coil, $A = 3 \times 10^{-4} \text{ m}^2$

Number of turns in the coil, $N = 500$

Current in the coil, $I = 0.5 \text{ A}$

Torque $\tau = |\vec{M} \times \vec{B}| = NiAB \sin(90^\circ) = NiAB$

$$\Rightarrow B = \frac{\tau}{NiA} = \frac{1.5}{500 \times 0.5 \times 3 \times 10^{-4}} = 20 \text{ T}$$

99. (c) $i_g = 20 \times 50 = 1000 \mu\text{A} = 1 \text{ mA}$

Using, $V = i_g (G + R)$, we have

$$2 = 10^{-3} (100 + R_1)$$

$$R_1 = 1900 \Omega$$

when, $V = 10 \text{ volt}$

$$10 = 10^{-3} (100 + R_2 + R_1)$$

$$10000 = (100 + R_2 + 1900)$$

$$\therefore R_2 = 8000 \Omega$$

100. (b) In an ammeter,

$$i_g = i_0 \frac{R_A}{R_A + G}$$

and for voltmeter,

$$V = i_g (G + R_V) = G i_0$$

On solving above equations, we get

$$R_A R_V = G^2$$

$$\text{and } \frac{R_A}{R_V} = \left(\frac{i_g}{i_0 - i_g} \right)^2$$

101. (Bonus) $v = i_g (R + G)$

$$\Rightarrow 5 = 10^{-4} (2 \times 10^6 + x)$$

$$x = -195 \times 10^4 \Omega$$

102. (c) $V = i_g(G + R) = 4 \times 10^{-3}(50 + 5000) = 20V$

103. (d) $C\theta = NBI A \sin 90^\circ$

$$\text{or } 10^{-6} \left(\frac{\pi}{180} \right) = 175B(10^{-3}) \times 10^{-4}$$

$$\therefore B = 10^{-3} T$$

104. (d) Using, $i_g = i \frac{S}{S + G}$

$$0.002 = 0.5 \frac{S}{S + 50}$$

On solving, we get

$$S = \frac{100}{498} \approx 0.2 \Omega$$

105. (b) When key K_1 is closed and key K_2 is open

$$i_g = \frac{E}{220 + R_g} = C\theta_0 \quad \dots (i)$$

When both the keys are closed

$$i_g = \left(\frac{E}{220 + \frac{5R_g}{5 + R_g}} \right) \times \frac{5}{(R_g + 5)} = \frac{C\theta_0}{5}$$

$$\Rightarrow \frac{5E}{225R_g + 1100} = \frac{C\theta_0}{5} \quad \dots (ii)$$

$$\frac{E}{220 + R_g} = C\theta_0 \quad \dots (i)$$

Dividing (i) by (ii), we get

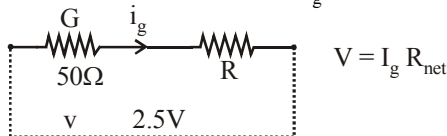
$$\Rightarrow \frac{225R_g + 1100}{1100 + 5R_g} = 5$$

$$\Rightarrow 5500 + 25R_g = 225R_g + 1100$$

$$200R_g = 4400$$

$$R_g = 22 \Omega$$

106. (b) Galvanometer has 25 divisions $I_g = 4 \times 10^{-4} \times 25 = 10^{-2} A$



$$v = I_g (G + R)$$

$$2.5 = (50 + R) 10^{-2} \therefore R = 200 \Omega$$

107. (c) Deflection current

$$= I_{g_{\max}} = nxk = 0.005 \times 30$$

Where, n = Number of divisions = 30 and $k = 0.005$ amp/division

$$= 15 \times 10^{-2} = 0.15$$

$$v = I_g [20 + R]$$

$$15 = 0.15 [20 + R]$$

$$100 = 20 + R$$

$$R = 80 \Omega$$

108. (d) Given,

Resistance of galvanometer, $G = 100 \Omega$

Current, $i_g = 1 mA$

A galvanometer can be converted into voltmeter by connecting a large resistance R in series with it.

Total resistance of the combination = $G + R$

According to Ohm's law, $V = i_g (G + R)$

$$\therefore 10 = 1 \times 10^{-3} (100 + R_0)$$

$$\Rightarrow 10000 - 100 = 9900 \Omega = R_0$$

$$\Rightarrow R_0 = 9.9 k\Omega$$

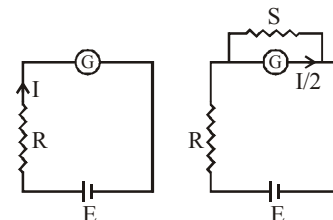
109. (b) Figure of merit of a galvanometer is the current required to produce a deflection of one division in the galvanometer

$$\text{i.e., figure of merit} = \frac{I}{\theta}$$

$$I = \frac{\varepsilon}{R + G} \quad G = \frac{1}{9} K\Omega$$

$$\frac{1}{2} = \frac{\varepsilon}{R + \frac{GS}{G + S}} \times \frac{S}{S + G} \Rightarrow \frac{1}{2} = \frac{\varepsilon S}{R(S + G) + GS}$$

$$S = \frac{RG \times \frac{1}{2}}{\varepsilon - \frac{(R + G)I}{2}}$$



$$S = \frac{11 \times 10^3 \times \frac{1}{2} \times 10^2 \times 270 \times 10^{-6}}{6 - \left(\frac{6}{2} \right)} = 110 \Omega$$

110. (d) According to question, current through galvanometer,

$$I_g = 1 mA$$

$$\text{Current through shunt } (I - I_g) = 2 A$$

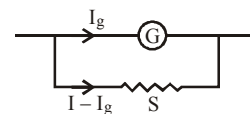
$$\text{Galvanometer resistance } R_g = 25 \Omega$$

$$\text{Resistance of shunt, } S = ?$$

$$I_0 R_0 = (I - I_g) S$$

$$\Rightarrow S = \frac{10^{-3} \times 25}{2}$$

$$S \approx 1.25 \times 10^{-2} \Omega$$



111. (c) Given : Current through the galvanometer,

$$i_g = 5 \times 10^{-3} A$$

$$\text{Galvanometer resistance, } G = 15 \Omega$$

Let resistance R to be put in series with the galvanometer to convert it into a voltmeter.

$$V = i_g (R + G)$$

$$10 = 5 \times 10^{-3} (R + 15)$$

$$\therefore R = 2000 - 15 = 1985 = 1.985 \times 10^3 \Omega$$

112. (c) $I_g G = (I - I_g) S$

$$\therefore 10^{-3} \times 100 = (10 - 10^{-3}) \times S$$

$$\therefore S \approx 0.01 \Omega$$

113. (d) As we know, $I = \frac{V}{R} = \frac{5}{50} = 0.1$

$I' = 0.099$

When Galvanometer is connected

$$R_{eq} = 50 + \frac{100S}{100+S} = \frac{V}{I}$$

$$\Rightarrow \frac{100S}{100+S} = \frac{5}{0.099} - 50$$

$$\Rightarrow \frac{100S}{100+S} = 50.50 - 50 \Rightarrow \frac{100S}{100+S} = 0.5$$

$$\Rightarrow 100S = 50 + 0.5S \Rightarrow 99.5S = 50$$

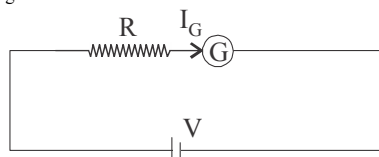
$$S = \frac{50}{99.05} = 0.5 \Omega$$

So, shunt of resistance $= 0.5 \Omega$ is connected in parallel with the galvanometer.

114. (a) According to Ohm's Law, $I = \frac{V}{R}$

$$I_g = \frac{V}{R+G}$$

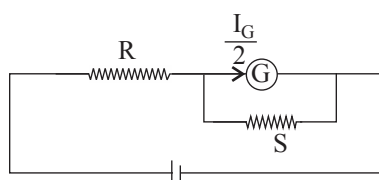
where, I_g - Galvanometer current, G - Galvanometer resistance



When shunt of resistance S is connected parallel to the

Galvanometer then $G = \frac{GS}{G+S}$

$$\therefore I = \frac{V}{R + \frac{GS}{G+S}}$$



Equal potential difference is given by

$$I'_g G = (I - I'_g)S$$

$$I'_g (G + S) = IS$$

$$\Rightarrow \frac{I_g}{2} = \frac{IS}{G+S}$$

$$\Rightarrow \frac{V}{2(R+G)} = \frac{V}{R + \frac{GS}{G+S}} \times \frac{S}{G+S}$$

$$\Rightarrow \frac{1}{2(R+G)} = \frac{S}{R(G+S)+GS}$$

$$\Rightarrow R(G+S) + GS = 2S(R+G)$$

$$\Rightarrow RG + RS + GS = 2S(R+G)$$

$$\Rightarrow RG = 2S(R+G) - S(R+G)$$

$$\therefore RG = S(R+G)$$

115. (b) To measure AC voltage across a resistance a moving coil galvanometer is used.

116. (d) The correct circuit diagram is D with galvanometer resistance

$$G = \frac{RS}{R-S}$$

117. (d) Statement I is false and Statement II is true

For ammeter, shunt resistance, $S = \frac{I_g G}{I - I_g}$

Therefore for I to increase, S should decrease, So additional S can be connected across it.

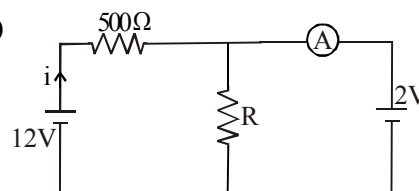
118. (d)

119. (d) The current that will give full scale deflection in the absence of the shunt is nearly equal to the current through the galvanometer when shunt is connected i.e. I_g

$$\text{As } I_g = \frac{IS}{G+S}$$

$$= \frac{5.5 \times 1}{120 + 1} = 0.045 \text{ ampere.}$$

120. (a)



$$12 - 2 = (500 \Omega)i \Rightarrow i = \frac{10}{500} = \frac{1}{50}$$

$$\text{Again, } i = \frac{12}{500 + R} = \frac{1}{50}$$

$$\Rightarrow 500 + R = 600$$

$$\Rightarrow R = 100 \Omega$$

121. (c) Resistance of Galvanometer,

$$G = \frac{\text{Current sensitivity}}{\text{Voltage sensitivity}} \Rightarrow G = \frac{10}{2} = 5 \Omega$$

Here i_g = Full scale deflection current $= \frac{150}{10} = 15 \text{ mA}$

V = voltage to be measured = 150 volts
(such that each division reads 1 volt)

$$\Rightarrow R = \frac{150}{15 \times 10^{-3}} - 5 = 9995 \Omega$$

Magnetism and Matter



TOPIC 1

Magnetism, Gauss's Law, Magnetic Moment, Properties of Magnet



1. A small bar magnet placed with its axis at 30° with an external field of 0.06 T experiences a torque of 0.018 Nm. The minimum work required to rotate it from its stable to unstable equilibrium position is :

[Sep. 04, 2020 (I)]

- (a) 6.4×10^{-2} J (b) 9.2×10^{-3} J
(c) 7.2×10^{-2} J (d) 11.7×10^{-3} J

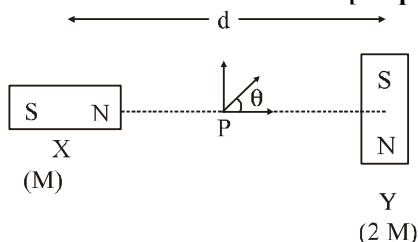
2. A circular coil has moment of inertia 0.8 kg m^2 around any diameter and is carrying current to produce a magnetic moment of 20 Am^2 . The coil is kept initially in a vertical position and it can rotate freely around a horizontal diameter. When a uniform magnetic field of 4 T is applied along the vertical, it starts rotating around its horizontal diameter. The angular speed the coil acquires after rotating by 60° will be :

[Sep. 04, 2020 (II)]

- (a) 10 rad s^{-1} (b) $10\pi \text{ rad s}^{-1}$
(c) $20\pi \text{ rad s}^{-1}$ (d) 20 rad s^{-1}

3. Two magnetic dipoles X and Y are placed at a separation d , with their axes perpendicular to each other. The dipole moment of Y is twice that of X. A particle of charge q is passing through their midpoint P, at angle $\theta = 45^\circ$ with the horizontal line, as shown in figure. What would be the magnitude of force on the particle at that instant? (d is much larger than the dimensions of the dipole)

[8 April 2019 II]



- (a) $\left(\frac{\mu_0}{4\pi}\right) \frac{M}{(d/2)^3} \times qv$ (b) 0

(c) $\sqrt{2} \left(\frac{\mu_0}{4\pi}\right) \frac{M}{(d/2)^3} \times qv$ (d) $\left(\frac{\mu_0}{4\pi}\right) \frac{2M}{(d/2)^3} \times qv$

4. A magnet of total magnetic moment $10^{-2} \hat{i} \text{ A-m}^2$ is placed in a time varying magnetic field, $B \hat{i} (\cos \omega t)$ where $B = 1 \text{ Tesla}$ and $\omega = 0.125 \text{ rad/s}$. The work done for reversing the direction of the magnetic moment at $t = 1 \text{ second}$, is: [10 Jan. 2019 I]

- (a) 0.01 J (b) 0.007 J
(c) 0.028 J (d) 0.014 J

5. A magnetic dipole in a constant magnetic field has :

[Online April 8, 2017]

- (a) maximum potential energy when the torque is maximum
(b) zero potential energy when the torque is minimum.
(c) zero potential energy when the torque is maximum.
(d) minimum potential energy when the torque is maximum.

6. A magnetic dipole is acted upon by two magnetic fields which are inclined to each other at an angle of 75° . One of the fields has a magnitude of 15 mT. The dipole attains stable equilibrium at an angle of 30° with this field. The magnitude of the other field (in mT) is close to :

[Online April 9, 2016]

- (a) 1 (b) 11 (c) 36 (d) 1060

7. A 25 cm long solenoid has radius 2 cm and 500 total number of turns. It carries a current of 15 A. If it is equivalent to a magnet of the same size and magnetization $\vec{M}$ (magnetic moment/volume), then $|\vec{M}|$ is : [Online April 10, 2015]

- (a) $30000\pi \text{ Am}^{-1}$ (b) $3\pi \text{ Am}^{-1}$
(c) 30000 Am^{-1} (d) 300 Am^{-1}

8. A bar magnet of length 6 cm has a magnetic moment of 4 J T^{-1} . Find the strength of magnetic field at a distance of 200 cm from the centre of the magnet along its equatorial line. [Online May 7, 2012]

- (a) $4 \times 10^{-8} \text{ tesla}$ (b) $3.5 \times 10^{-8} \text{ tesla}$
(c) $5 \times 10^{-8} \text{ tesla}$ (d) $3 \times 10^{-8} \text{ tesla}$

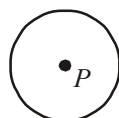
9. A thin circular disc of radius R is uniformly charged with density $\sigma > 0$ per unit area. The disc rotates about its axis with a uniform angular speed ω . The magnetic moment of the disc is [2011 RS]

- (a) $\pi R^4 \sigma \omega$ (b) $\frac{\pi R^4}{2} \sigma \omega$
 (c) $\frac{\pi R^4}{4} \sigma \omega$ (d) $2\pi R^4 \sigma \omega$
10. A magnetic needle is kept in a non-uniform magnetic field. It experiences [2005]
 (a) neither a force nor a torque
 (b) a torque but not a force
 (c) a force but not a torque
 (d) a force and a torque
11. The length of a magnet is large compared to its width and breadth. The time period of its oscillation in a vibration magnetometer is 2s. The magnet is cut along its length into three equal parts and these parts are then placed on each other with their like poles together. The time period of this combination will be [2004]
 (a) $2\sqrt{3}$ s (b) $\frac{2}{3}$ s (c) 2 s (d) $\frac{2}{\sqrt{3}}$ s
12. A magnetic needle lying parallel to a magnetic field requires W units of work to turn it through 60° . The torque needed to maintain the needle in this position will be [2003]
 (a) $\sqrt{3}W$ (b) W (c) $\frac{\sqrt{3}}{2}W$ (d) $2W$
13. The magnetic lines of force inside a bar magnet [2003]
 (a) are from north-pole to south-pole of the magnet
 (b) do not exist
 (c) depend upon the area of cross-section of the bar magnet
 (d) are from south-pole to north-pole of the Magnet

TOPIC 2 The Earth Magnetism, Magnetic Materials and their properties

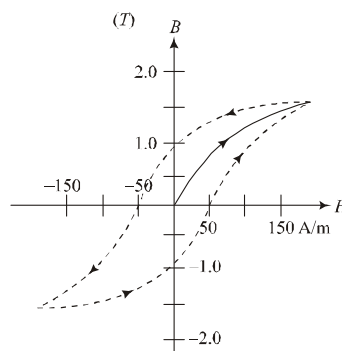


14. An iron rod of volume 10^{-3} m^3 and relative permeability 1000 is placed as core in a solenoid with 10 turns/cm. If a current of 0.5 A is passed through the solenoid, then the magnetic moment of the rod will be : [Sep. 05, 2020 (II)]
 (a) $50 \times 10^2 \text{ Am}^2$ (b) $5 \times 10^2 \text{ Am}^2$
 (c) $500 \times 10^2 \text{ Am}^2$ (d) $0.5 \times 10^2 \text{ Am}^2$
15. A paramagnetic sample shows a net magnetisation of 6 A/m when it is placed in an external magnetic field of 0.4 T at a temperature of 4 K. When the sample is placed in an external magnetic field of 0.3 T at a temperature of 24 K, then the magnetisation will be : [Sep. 04, 2020 (II)]
 (a) 1 A/m (b) 4 A/m
 (c) 2.25 A/m (d) 0.75 A/m
16. A perfectly diamagnetic sphere has a small spherical cavity at its centre, which is filled with a paramagnetic substance. The whole system is placed in a uniform magnetic field $\vec{B}$. Then the field inside the paramagnetic substance is : [Sep. 03, 2020 (II)]



- (a) $\vec{B}$
 (b) zero
 (c) much large than $|\vec{B}|$ and parallel to $\vec{B}$
 (d) much large than $|\vec{B}|$ but opposite to $\vec{B}$
17. Magnetic materials used for making permanent magnets (P) and magnets in a transformer (T) have different properties of the following, which property best matches for the type of magnet required? [Sep. 02, 2020 (I)]
 (a) T : Large retentivity, small coercivity
 (b) P : Small retentivity, large coercivity
 (c) T : Large retentivity, large coercivity
 (d) P : Large retentivity, large coercivity

18.



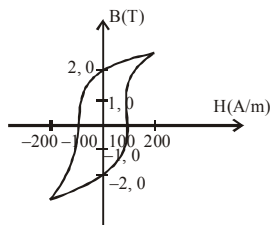
The figure gives experimentally measured B vs. H variation in a ferromagnetic material. The retentivity, co-ercivity and saturation, respectively, of the material are:

[7 Jan. 2020 II]

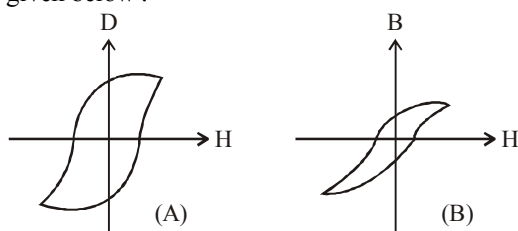
- (a) 1.5 T, 50 A/m and 1.0 T
 (b) 1.5 T, 50 A/m and 1.0 T
 (c) 150 A/m, 1.0 T and 1.5 T
 (d) 1.0 T, 50 A/m and 1.5 T
19. A paramagnetic material has 10^{28} atoms/ m^3 . Its magnetic susceptibility at temperature 350 K is 2.8×10^{-4} . Its susceptibility at 300 K is: [12 Jan. 2019 II]
 (a) 3.267×10^{-4} (b) 3.672×10^{-4}
 (c) 3.726×10^{-4} (d) 2.672×10^{-4}
20. A paramagnetic substance in the form of a cube with sides 1 cm has a magnetic dipole moment of $20 \times 10^{-6} \text{ J/T}$ when a magnetic intensity of $60 \times 10^3 \text{ A/m}$ is applied. Its magnetic susceptibility is: [11 Jan. 2019 II]
 (a) 3.3×10^{-2} (b) 4.3×10^{-2}
 (c) 2.3×10^{-2} (d) 3.3×10^{-4}
21. At some location on earth the horizontal component of earth's magnetic field is $18 \times 10^{-6} \text{ T}$. At this location, magnetic needle of length 0.12 m and pole strength 1.8 Am is suspended from its mid-point using a thread, it makes 45° angle with horizontal in equilibrium. To keep this needle horizontal, the vertical force that should be applied at one of its ends is: [10 Jan. 2019 II]
 (a) $3.6 \times 10^{-5} \text{ N}$ (b) $1.8 \times 10^{-5} \text{ N}$
 (c) $1.3 \times 10^{-5} \text{ N}$ (d) $6.5 \times 10^{-5} \text{ N}$
22. A bar magnet is demagnetized by inserting it inside a solenoid of length 0.2 m, 100 turns, and carrying a current of 5.2 A. The coercivity of the bar magnet is: [9 Jan. 2019 I]

- (a) 285 A/m (b) 2600 A/m
 (c) 520 A/m (d) 1200 A/m

23. The B-H curve for a ferromagnet is shown in the figure. The ferromagnet is placed inside a long solenoid with 1000 turns/cm. The current that should be passed in the solenoid to demagnetise the ferromagnet completely is: [Online April 15, 2018]



- (a) 2 mA (b) 1 mA (c) 40 μ A (d) 20 μ A
24. Hysteresis loops for two magnetic materials A and B are given below :



These materials are used to make magnets for electric generators, transformer core and electromagnet core. Then it is proper to use : [2016]

- (a) A for transformers and B for electric generators.
 (b) B for electromagnets and transformers.
 (c) A for electric generators and transformers.
 (d) A for electromagnets and B for electric generators.
25. A fighter plane of length 20 m, wing span (distance from tip of one wing to the tip of the other wing) of 15 m and height 5 m is lying towards east over Delhi. Its speed is 240 ms^{-1} . The earth's magnetic field over Delhi is $5 \times 10^{-5} \text{ T}$ with the declination angle $\sim 0^\circ$ and dip of θ such

that $\sin \theta = \frac{2}{3}$. If the voltage developed is V_B between the lower and upper side of the plane and V_W between the tips of the wings then V_B and V_W are close to :

[Online April 10, 2016]

- (a) $V_B = 40 \text{ mV}$; $V_W = 135 \text{ mV}$ with left side of pilot at higher voltage
 (b) $V_B = 45 \text{ mV}$; $V_W = 120 \text{ mV}$ with right side of pilot at higher voltage
 (c) $V_B = 40 \text{ mV}$; $V_W = 135 \text{ mV}$ with right side of pilot at higher voltage
 (d) $V_B = 45 \text{ mV}$; $V_W = 120 \text{ mV}$ with left side of pilot at higher voltage
26. A short bar magnet is placed in the magnetic meridian of the earth with north pole pointing north. Neutral points are found at a distance of 30 cm from the magnet on the East – West line, drawn through the middle point of the magnet. The magnetic moment of the magnet in

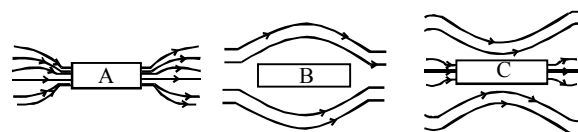
Am^2 is close to: (Given $\frac{\mu_0}{4\pi} = 10^{-7}$ in SI units and B_H

= Horizontal component of earth's magnetic field = $3.6 \times 10^{-5} \text{ tesla}$) [Online April 11, 2015]

- (a) 14.6 (b) 19.4 (c) 9.7 (d) 4.9
27. The coercivity of a small magnet where the ferromagnet gets demagnetized is $3 \times 10^3 \text{ Am}^{-1}$. The current required to be passed in a solenoid of length 10 cm and number of turns 100, so that the magnet gets demagnetized when inside the solenoid, is: [2014]
- (a) 30 mA (b) 60 mA (c) 3 A (d) 6 A
28. An example of a perfect diamagnet is a superconductor. This implies that when a superconductor is put in a magnetic field of intensity B, the magnetic field B_s inside the superconductor will be such that:

[Online April 19, 2014]

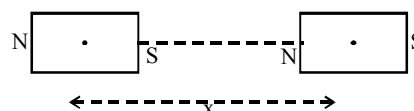
- (a) $B_s = -B$ (b) $B_s = 0$
 (c) $B_s = B$ (d) $B_s < B$ but $B_s \neq 0$
29. Three identical bars A, B and C are made of different magnetic materials. When kept in a uniform magnetic field, the field lines around them look as follows:



Make the correspondence of these bars with their material being diamagnetic (D), ferromagnetic (F) and paramagnetic (P): [Online April 11, 2014]

- (a) $A \leftrightarrow D, B \leftrightarrow P, C \leftrightarrow F$
 (b) $A \leftrightarrow F, B \leftrightarrow D, C \leftrightarrow P$
 (c) $A \leftrightarrow P, B \leftrightarrow F, C \leftrightarrow D$
 (d) $A \leftrightarrow F, B \leftrightarrow P, C \leftrightarrow D$
30. The magnetic field of earth at the equator is approximately $4 \times 10^{-5} \text{ T}$. The radius of earth is $6.4 \times 10^6 \text{ m}$. Then the dipole moment of the earth will be nearly of the order of: [Online April 9, 2014]
- (a) 10^{23} Am^2 (b) 10^{20} Am^2
 (c) 10^{16} Am^2 (d) 10^{10} Am^2

31. The mid points of two small magnetic dipoles of length d in end-on positions, are separated by a distance x , ($x \gg d$). The force between them is proportional to x^{-n} where n is: [Online April 9, 2014]



- (a) 1 (b) 2 (c) 3 (d) 4
32. Two short bar magnets of length 1 cm each have magnetic moments 1.20 Am^2 and 1.00 Am^2 respectively. They are placed on a horizontal table parallel to each other with their N poles pointing towards the South. They have a common magnetic equator and are separated by a distance of 20.0 cm. The value of the resultant horizontal magnetic induction at the mid-point O of the line joining their centres is close to (Horizontal component of earth's magnetic induction is $3.6 \times 10^{-5} \text{ Wb/m}^2$) [2013]
- (a) $3.6 \times 10.5 \text{ Wb/m}^2$ (b) $2.56 \times 10.4 \text{ Wb/m}^2$
 (c) $3.50 \times 10.4 \text{ Wb/m}^2$ (d) $5.80 \times 10.4 \text{ Wb/m}^2$

33. The earth's magnetic field lines resemble that of a dipole at the centre of the earth. If the magnetic moment of this dipole is close to $8 \times 10^{22} \text{ Am}^2$, the value of earth's magnetic field near the equator is close to (radius of the earth = $6.4 \times 10^6 \text{ m}$)

[Online April 25, 2013]

- (a) 0.6 Gauss (b) 1.2 Gauss
(c) 1.8 Gauss (d) 0.32 Gauss
34. Relative permittivity and permeability of a material ϵ_r and μ_r , respectively. Which of the following values of these quantities are allowed for a diamagnetic material? [2008]

- (a) $\epsilon_r = 0.5, \mu_r = 1.5$ (b) $\epsilon_r = 1.5, \mu_r = 0.5$
(c) $\epsilon_r = 0.5, \mu_r = 0.5$ (d) $\epsilon_r = 1.5, \mu_r = 1.5$

35. Needles N_1 , N_2 and N_3 are made of a ferromagnetic, a paramagnetic and a diamagnetic substance respectively. A magnet when brought close to them will [2006]

- (a) attract N_1 and N_2 strongly but repel N_3
(b) attract N_1 strongly, N_2 weakly and repel N_3 weakly
(c) attract N_1 strongly, but repel N_2 and N_3 weakly
(d) attract all three of them

36. The materials suitable for making electromagnets should have [2004]

- (a) high retentivity and low coercivity
(b) low retentivity and low coercivity
(c) high retentivity and high coercivity
(d) low retentivity and high coercivity

37. A thin rectangular magnet suspended freely has a period of oscillation equal to T . Now it is broken into two equal halves (each having half of the original length) and one piece is made to oscillate freely in the same field. If its

period of oscillation is T' , the ratio $\frac{T'}{T}$ is [2003]

- (a) $\frac{1}{2\sqrt{2}}$ (b) $\frac{1}{2}$
(c) 2 (d) $\frac{1}{4}$

38. Curie temperature is the temperature above which

[2003]

- (a) a ferromagnetic material becomes paramagnetic
(b) a paramagnetic material becomes diamagnetic
(c) a ferromagnetic material becomes diamagnetic
(d) a paramagnetic material becomes ferromagnetic

TOPIC 3 Magnetic Equipment



39. A ring is hung on a nail. It can oscillate, without slipping or sliding (i) in its plane with a time period T_1 and, (ii) back and forth in a direction perpendicular to its plane, with a

period T_2 . The ratio $\frac{T_1}{T_2}$ will be : [Sep. 05, 2020 (II)]

- (a) $\frac{2}{\sqrt{3}}$ (b) $\frac{2}{3}$
(c) $\frac{3}{\sqrt{2}}$ (d) $\frac{\sqrt{2}}{3}$

40. A magnetic compass needle oscillates 30 times per minute at a place where the dip is 45° , and 40 times per minute where the dip is 30° . If B_1 and B_2 are respectively the total magnetic field due to the earth and the two places, then the ratio B_1/B_2 is best given by :

[12 April 2019 I]

- (a) 1.8 (b) 0.7 (c) 3.6 (d) 2.2

41. A hoop and a solid cylinder of same mass and radius are made of a permanent magnetic material with their magnetic moment parallel to their respective axes. But the magnetic moment of hoop is twice of solid cylinder. They are placed in a uniform magnetic field in such a manner that their magnetic moments make a small angle with the field. If the oscillation periods of hoop and cylinder are T_h and T_c respectively, then:

[10 Jan. 2019 II]

- (a) $T_h = T_c$ (b) $T_h = 2 T_c$
(c) $T_h = 1.5 T_c$ (d) $T_h = 0.5 T_c$

42. A magnetic needle of magnetic moment $6.7 \times 10^{-2} \text{ Am}^2$ and moment of inertia $7.5 \times 10^{-6} \text{ kg m}^2$ is performing simple harmonic oscillations in a magnetic field of 0.01 T. Time taken for 10 complete oscillations is : [2017]

- (a) 6.98 s (b) 8.76 s
(c) 6.65 s (d) 8.89 s



Hints & Solutions



1. (c) Here, $\theta = 30^\circ$, $\tau = 0.018 \text{ N-m}$, $B = 0.06 \text{ T}$

Torque on a bar magnet :

$$\tau = MB \sin \theta$$

$$0.018 = M \times 0.06 \times \sin 30^\circ$$

$$\Rightarrow 0.018 = M \times 0.06 \times \frac{1}{2} \Rightarrow M = 0.6 \text{ A-m}^2$$

Position of stable equilibrium ($\theta = 0^\circ$)

Position of unstable equilibrium ($\theta = 180^\circ$)

Minimum work required to rotate bar magnet from stable to unstable equilibrium

$$\Delta U = U_f - U_i = -MB \cos 180^\circ - (-MB \cos 0^\circ)$$

$$W = 2MB = 2 \times 0.6 \times 0.06$$

$$\therefore W = 7.2 \times 10^{-2} \text{ J}$$

2. (a) Given,

Moment of inertia of circular coil, $I = 0.8 \text{ kg m}^2$

Magnetic moment of circular coil, $M = 20 \text{ Am}^2$

Rotational kinetic energy of circular coil,

$$\text{KE} = \frac{1}{2} I \omega^2$$

Here, ω = angular speed of coil

Potential energy of bar magnet = $-MB \cos \phi$

From energy conservation

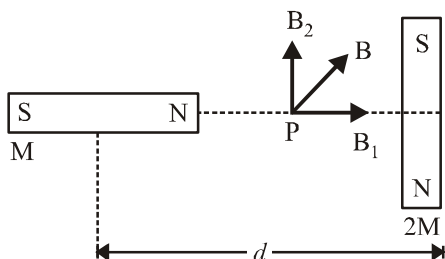
$$\frac{1}{2} I \omega^2 = U_{\text{in}} - U_f = -MB \cos 60^\circ - (-MB)$$

$$\Rightarrow \frac{MB}{2} = \frac{1}{2} I \omega^2$$

$$\Rightarrow \frac{20 \times 4}{2} = \frac{1}{2} (0.8) \omega^2$$

$$\Rightarrow 100 = \omega^2 \Rightarrow \omega = 10 \text{ rad/s}$$

3. (b) $B_1 = \frac{\mu_0}{4\pi} \frac{2K}{(d/2)^3}$



$$\text{and } B_2 = \frac{\mu_0}{4\pi} \frac{2M}{(d/2)^3}$$

$$\therefore \tan \theta = \frac{B_2}{B_1} = \frac{\frac{\mu_0}{4\pi} \frac{2M}{(d/2)^3}}{\frac{\mu_0}{4\pi} \frac{2M}{(d/2)^3}} = 1$$

$$\text{or } \theta = 45^\circ$$

The resultant field is 45° from B_1 . The angle between $\vec{B}$ and $\vec{v}$ is zero, so force on the particle is zero.

4. (c)

Work done, $W = 2 \text{ m} \cdot B$

$$= 2 \times 10^{-2} \times 1 \cos(0.125)$$

$$= 0.02 \text{ J}$$

5. (c) Potential energy of dipole,

$$U = -pE \cos \theta$$

Torque experienced by dipole

$$\tau = pE \sin \theta$$

Torque will be maximum (τ_{max}) when $\theta = 90^\circ$ then potential energy $U = 0$

6. (b) We know that, magnetic dipole moment

$$M = NiA \cos \theta \text{ i.e., } M \propto \cos \theta$$

When two magnetic fields are inclined at an angle of 75° the equilibrium will be at 30° , so

$$\cos \theta = \cos(75^\circ - 30^\circ) = \cos 45^\circ = \frac{1}{\sqrt{2}}$$

$$\frac{x}{\sqrt{2}} = \frac{15}{2} \therefore x \approx 11$$

7. (c) $\vec{M}$ (mag. moment/volume) = $\frac{NiA}{\ell}$

$$= \frac{Ni}{\ell} = \frac{(500)15}{25 \times 10^{-2}} = 30000 \text{ Am}^{-1}$$

8. (c) Along the equatorial line, magnetic field strength

$$B = \frac{\mu_0}{4\pi} \frac{M}{(r^2 + \ell^2)^{3/2}}$$

Given: $M = 4 \text{ JT}^{-1}$

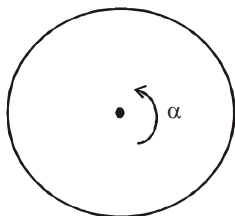
$$r = 200 \text{ cm} = 2 \text{ m}$$

$$\ell = \frac{6 \text{ cm}}{2} = 3 \text{ cm} = 3 \times 10^{-2} \text{ m}$$

$$\therefore B = \frac{4\pi \times 10^{-7}}{4\pi} \times \frac{4}{\left[2^2 + (3 \times 10^{-2})^2\right]^{3/2}}$$

Solving we get, $B = 5 \times 10^{-8} \text{ tesla}$

9. (c) $\frac{q}{2m} = \frac{\text{Magnetic dipole moment}}{\text{Angular momentum}}$



∴ Magnetic dipole moment (M)

$$M = \frac{q}{2m} \cdot \left(\frac{mR^2}{2} \right) \cdot \omega = \frac{1}{4} \sigma \pi R^4 \omega.$$

10. (d) A magnetic needle kept in non uniform magnetic field experience a force and torque due to unequal forces acting on poles.

11. (b) Initially, time period of magnet

$$T = 2\pi \sqrt{\frac{I}{MB}} = 25 \text{ where } I = \frac{1}{12} m \ell^2$$

When the magnet is cut into three pieces the pole strength will remain the same and

Moment of inertia of each part,

$$(I') = \frac{1}{12} \left(\frac{m}{3} \right) \left(\frac{\ell}{3} \right)^2 \times 3 = \frac{I}{9}$$

We have, Magnetic moment (M)

= Pole strength (m) × ℓ

∴ New magnetic moment,

$$M' = m \times \left(\frac{\ell}{3} \right) \times 3 = m\ell = M$$

$$\text{New time period, } T' = 2\pi \sqrt{\frac{I'}{MB}}$$

$$= 2\pi \sqrt{\frac{I}{9MB}} \Rightarrow T' = \frac{T}{3} = \frac{25}{3} \text{ s.}$$

12. (a) Workdone to turn a magnetic needle from angle θ_1 to θ_2 is given by

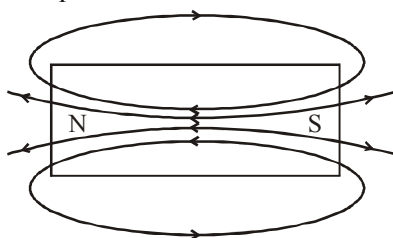
$$W = MB(\cos \theta_1 - \cos \theta_2)$$

$$\therefore W = MB(\cos 0^\circ - \cos 60^\circ)$$

$$= MB \left(1 - \frac{1}{2} \right) = \frac{MB}{2}$$

$$\therefore \text{Torque, } \tau = MB \sin \theta = MB \sin 60^\circ = \sqrt{3} \frac{MB}{2} = \sqrt{3} W$$

13. (d) The magnetic field lines of bar magnet form closed lines. As shown in the figure, the magnetic lines of force are directed from south to north inside a bar magnet. Outside the bar magnet magnetic field lines directed from north to south pole.



14. (b) Given,

$$\text{Volume of iron rod, } V = 10^{-3} \text{ m}^3$$

$$\text{Relative permeability, } \mu_r = 1000$$

$$\text{Number of turns per unit length, } n = 10$$

Magnetic moment of an iron core solenoid,

$$M = (\mu_r - 1) \times NiA$$

$$\Rightarrow M = (\mu_r - 1) \times Ni \frac{V}{l} \Rightarrow M = (\mu_r - 1) \times \frac{N}{l} iV$$

$$\Rightarrow M = 999 \times \frac{10}{10^{-2}} \times 0.5 \times 10^{-3} = 499.5 \approx 500.$$

15. (d) For paramagnetic material. According to Curie's law

$$\chi \propto \frac{1}{T}$$

For two temperatures T_1 and T_2

$$\chi_1 T_1 = \chi_2 T_2$$

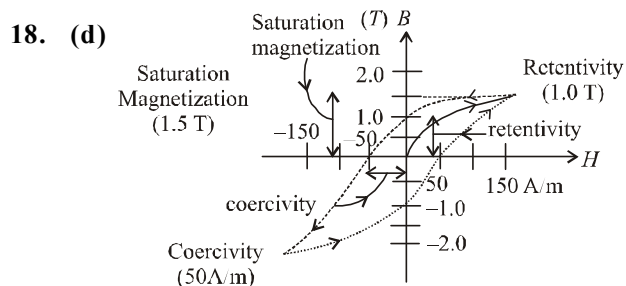
$$\text{But } \chi = \frac{I}{B}$$

$$\therefore \frac{I_1}{B_1} T_1 = \frac{I_2}{B_2} T_2$$

$$\Rightarrow \frac{6}{0.4} \times 4 = \frac{I_2}{0.3} \times 24 \Rightarrow I_2 = \frac{0.3}{0.4} = 0.75 \text{ A/m}$$

16. (b) When magnetic field is applied to a diamagnetic substance, it produces magnetic field in opposite direction so net magnetic field inside the cavity of sphere will be zero. So, field inside the paramagnetic substance kept inside the cavity is zero.

17. (d) Permanent magnets (P) are made of materials with large retentivity and large coercivity. Transformer cores (T) are made of materials with low retentivity and low coercivity.



19. (a) According to Curie law for paramagnetic substance,

$$\chi \propto \frac{1}{T_C} \Rightarrow \frac{\chi_1}{\chi_2} = \frac{T_{C2}}{T_{C1}}$$

$$\frac{2.8 \times 10^{-4}}{\chi_2} = \frac{300}{350}$$

$$\chi_2 = \frac{2.8 \times 350 \times 10^{-4}}{300} = 3.266 \times 10^{-4}$$

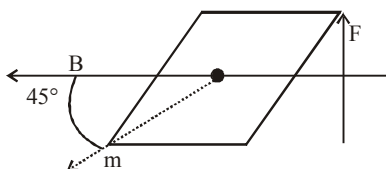
20. (d) Magnetic susceptibility,

$$\chi = \frac{I}{H}$$

$$\text{where, } I = \frac{\text{Magnetic moment}}{\text{Volume}} = \frac{20 \times 10^{-6}}{10^{-6}} = 20 \text{ N/m}^2$$

$$\text{Now, } \chi = \frac{20}{60 \times 10^3} = \frac{1}{3} \times 10^{-3} = 3.3 \times 10^{-4}$$

21. (d) using, $MB \sin \theta = F \ell \sin \theta$ (τ)



$$MB \sin 45^\circ = F \frac{\ell}{2} \sin 45^\circ$$

$$F = 2MB = 2 \times 1.8 \times 18 \times 10^{-6} = 6.5 \times 10^{-5} \text{ N}$$

22. (b) Corecivity, $H = \frac{B}{\mu_0}$ and $B = \mu_0 n i$ ($n = \frac{N}{\ell}$)

$$\text{or, } H = \frac{N}{\ell} i = \frac{100}{0.2} \times 5.2 = 2600 \text{ A/m}$$

23. (b) Given Number of turns,
 $n = 1000 \text{ turns/cm} = 1000 \times 100 \text{ turns/m}$
 Coercivity of ferromagnet, $H = 100 \text{ A/m}$
 Current to demagnetise the ferromagnet, $I = ?$

$$\text{Using, } H = nI$$

$$\text{or, } 100 = 10^5 \times I$$

$$\therefore I = \frac{100}{10^5} = 1 \text{ mA}$$

24. (b) Graph [A] is for material used for making permanent magnets (high coercivity)

Graph [B] is for making electromagnets and transformers.

25. (d) $V_B = V_{B_H} l = 240 \times 5 \times 10^{-5} \cos(\theta) \times 5 = 44.7 \text{ mV}$
 By right hand rule, the charge moves to the left of pilot.

26. (c) Here, $r = 30 \text{ cm} = 0.3 \text{ m}$

$$\text{we know } \frac{\mu_0 M}{4\pi r^3} = B_H = 3.6 \times 10^{-5}$$

$$\Rightarrow M = \frac{3.6 \times 10^{-5}}{10^{-7}} (0.3)^3$$

$$\text{Hence, } M = 9.7 \text{ Am}^2$$

27. (c) Magnetic field in solenoid $B = \mu_0 n i$

$$\Rightarrow \frac{B}{\mu_0} = n i$$

(Where n = number of turns per unit length)

$$\Rightarrow \frac{B}{\mu_0} = \frac{N i}{L} \Rightarrow 3 \times 10^3 = \frac{100 i}{10 \times 10^{-2}}$$

$$\Rightarrow i = 3 \text{ A}$$

28. (b) Magnetic field inside the superconductor is zero. Diamagnetic substances are repelled in external magnetic field.

29. (b) Diamagnetic materials are repelled in an external magnetic field.

$\bar{B}$ represents diamagnetic materials.

30. (a) Given, $B = 4 \times 10^{-5} \text{ T}$
 $R_E = 6.4 \times 10^6 \text{ m}$
 Dipole moment of the earth $M = ?$

$$B = \frac{\mu_0 M}{4\pi d^3}$$

$$4 \times 10^{-5} = \frac{4\pi \times 10^{-7} \times M}{4\pi \times (6.4 \times 10^6)^3}$$

$$\therefore M \cong 10^{23} \text{ Am}^2$$

31. (d) In magnetic dipole

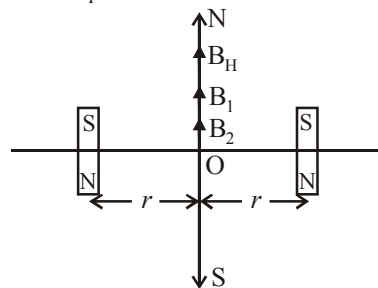
$$\text{Force} \propto \frac{1}{r^4}$$

In the given question,

$$\text{Force} \propto x^{-n}$$

$$\text{Hence, } n = 4$$

32. (b) Given : $M_1 = 1.20 \text{ Am}^2$



$$M_2 = 1.00 \text{ Am}^2; r = \frac{20}{2} \text{ cm} = 0.1 \text{ m}$$

$$B_{\text{net}} = B_1 + B_2 + B_H$$

$$B_{\text{net}} = \frac{\mu_0 (M_1 + M_2)}{4\pi r^3} + B_H$$

$$= \frac{10^{-7} (1.2 + 1)}{(0.1)^3} + 3.6 \times 10^{-5} = 2.56 \times 10^{-4} \text{ wb/m}^2$$

33. (a) Given $M = 8 \times 10^{22} \text{ Am}^2$

$$d = R_E = 6.4 \times 10^6 \text{ m}$$

$$\text{Earth's magnetic field, } B = \frac{\mu_0}{4\pi} \cdot \frac{2M}{d^3}$$

$$= \frac{4\pi \times 10^{-7}}{4\pi} \times \frac{2 \times 8 \times 10^{22}}{(6.4 \times 10^6)^3} \cong 0.6 \text{ Gauss}$$

34. (b) For a diamagnetic material, the value of μ_r is slightly less than one. For any material, the value of ϵ_r is always greater than 1.

35. (b) Ferromagnetic substance has magnetic domains whereas paramagnetic substances have magnetic dipoles which get attracted to a magnetic field. Ferromagnetic material magnetised strongly in the direction of magnetism field. Hence, N_1 will be attracted paramagnetic substance attract weakly in the direction of field. Hence, N_2 will weakly attracted. Diamagnetic substances do not have magnetic dipole but in the presence of external magnetic field due to their orbital motion of electrons these substances are repelled. Hence, N_3 will be repelled.

36. (b) Electromagnet should be amenable to magnetisation & demagnetization.
 $\therefore$ Materials suitable for making electromagnets should have low retentivity and low coercivity should be low.
37. (b) The time period of a rectangular magnet oscillating in

earth's magnetic field is given by $T = 2\pi \sqrt{\frac{I}{MB_H}}$

where I = Moment of inertia of the rectangular magnet

M = Magnetic moment

B_H = Horizontal component of the earth's magnetic field

Initially, the time period of the magnet

$$T = 2\pi \sqrt{\frac{I}{MB_H}} \text{ where } I = \frac{1}{12} M \ell^2$$

Case 2

Magnet is cut into two identical pieces such that each piece has half the original length.

$$\text{Then } T' = 2\pi \sqrt{\frac{I'}{M'B_H}}$$

Moment of inertia of each part

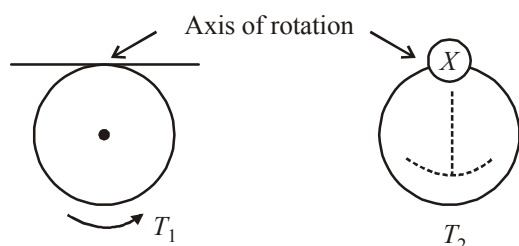
$$I' = \frac{1}{12} \left(\frac{M}{2} \right) \left(\frac{\ell}{2} \right)^2 = \frac{I}{8} \text{ and } M' = \frac{M}{2}$$

$$\therefore \frac{T'}{T} = \sqrt{\frac{I'}{M} \times \frac{M}{I}} = \sqrt{\frac{I/8}{M/2} \times \frac{M}{I}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

38. (a) The temperature above which a ferromagnetic substance becomes paramagnetic is called Curie's temperature.
39. (a) Let I_1 and I_2 be the moment of inertia in first and second case respectively.

$$I_1 = 2MR^2$$

$$I_2 = MR^2 + \frac{MR^2}{2} = \frac{3}{2} MR^2$$



$$\text{Time period, } T = 2\pi \sqrt{\frac{I}{mgd}}$$

$$T \propto I$$

$$\therefore \frac{T_1}{T_2} = \sqrt{\frac{I_1}{I_2}} = \sqrt{\frac{2MR^2}{\frac{3}{2}MR^2}} = \frac{2}{\sqrt{3}}$$

40. (Bonus) We have, $T = 2\pi \sqrt{\frac{I}{MB_x}}$

$$\therefore \frac{T_1^2}{T_2^2} = \frac{B_{x2}}{B_{x1}}$$

$$\text{or } \left(\frac{2}{1.5} \right)^2 = \frac{B_2 \cos 45^\circ}{B_1 \cos 30^\circ} = \frac{B_2 \times 2}{\sqrt{2} \times B_1 \times \sqrt{3}}$$

$$\left(\frac{4}{3} \right)^2 = \frac{B_2}{B_1} \times \frac{2}{\sqrt{6}}$$

$$\therefore \frac{B_1}{B_2} = \frac{9}{8\sqrt{6}} = 0.46$$

41. (a) Using, time / oscillation period,

$$T = 2\pi \sqrt{\frac{I}{MB}}$$

Where, M = magnetic moment, I moment of inertia and B = magnetic field

$$T_h = 2\pi \sqrt{\frac{mR^2}{(2MB)}}$$

$$T_c = 2\pi \sqrt{\frac{1/2 mR^2}{MB}}$$

Clearly, $T_h = T_c$

42. (c) Given : Magnetic moment, $M = 6.7 \times 10^{-2} \text{ Am}^2$
 Magnetic field, $B = 0.01 \text{ T}$
 Moment of inertia, $I = 7.5 \times 10^{-6} \text{ Kg m}^2$

$$\text{Using, } T = 2\pi \sqrt{\frac{I}{MB}}$$

$$= 2\pi \sqrt{\frac{7.5 \times 10^{-6}}{6.7 \times 10^{-2} \times 0.01}} = \frac{2\pi}{10} \times 1.06 \text{ s}$$

Time taken for 10 complete oscillations

$$t = 10T = 2\pi \times 1.06$$

$$= 6.6568 \approx 6.65 \text{ s}$$

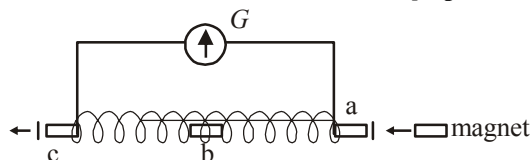
Electromagnetic Induction



TOPIC 1 Magnetic Flux, Faraday's and Lenz's Law



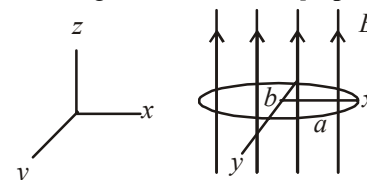
- Two concentric circular coils, C_1 and C_2 , are placed in the XY plane. C_1 has 500 turns, and a radius of 1 cm. C_2 has 200 turns and radius 20 cm. C_2 carries a time dependent current $I(t) = (5t^2 - 2t + 3)A$ Where t is in s. The emf induced in C_1 (in mV), at the instant $t = 1$ s is $\frac{4}{x}$. The value of x is _____. [NA Sep. 05, 2020 (I)]
- A small bar magnet is moved through a coil at constant speed from one end to the other. Which of the following series of observations will be seen on the galvanometer G attached across the coil ? [Sep. 04, 2020 (I)]



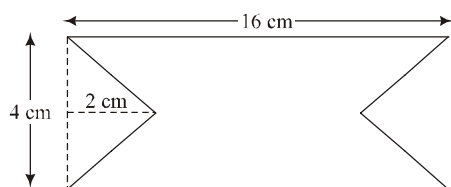
Three positions shown describe : (1) the magnet's entry (2) magnet is completely inside and (3) magnet's exit.

- (1) (2) (3)
- (a) → →
- (b) → →
- (c) → →
- (d) → →

- An elliptical loop having resistance R , of semi major axis a , and semi minor axis b is placed in a magnetic field as shown in the figure. If the loop is rotated about the x -axis with angular frequency ω , the average power loss in the loop due to Joule heating is : [Sep. 03, 2020 (I)]



- (a) $\frac{\pi^2 a^2 b^2 B^2 \omega^2}{2R}$ (b) zero
- (c) $\frac{\pi ab B \omega}{R}$ (d) $\frac{\pi^2 a^2 b^2 B^2 \omega^2}{R}$
- A uniform magnetic field B exists in a direction perpendicular to the plane of a square loop made of a metal wire. The wire has a diameter of 4 mm and a total length of 30 cm. The magnetic field changes with time at a steady rate $dB/dt = 0.032 \text{ Ts}^{-1}$. The induced current in the loop is close to (Resistivity of the metal wire is $1.23 \times 10^{-8} \Omega \text{ m}$) [Sep. 03, 2020 (II)]
- (a) 0.43 A (b) 0.61 A (c) 0.34 A (d) 0.53 A
- A circular coil of radius 10 cm is placed in a uniform magnetic field of $3.0 \times 10^{-5} \text{ T}$ with its plane perpendicular to the field initially. It is rotated at constant angular speed about an axis along the diameter of coil and perpendicular to magnetic field so that it undergoes half of rotation in 0.2 s. The maximum value of EMF induced (in μV) in the coil will be close to the integer _____. [NA Sep. 02, 2020 (I)]
 - In a fluorescent lamp choke (a small transformer) 100 V of reverse voltage is produced when the choke current changes uniformly from 0.25 A to 0 in a duration of 0.025 ms. The self-inductance of the choke (in mH) is estimated to be _____. [NA 9 Jan. 2020 I]
 - At time $t = 0$ magnetic field of 1000 Gauss is passing perpendicularly through the area defined by the closed loop shown in the figure. If the magnetic field reduces linearly to 500 Gauss, in the next 5 s, then induced EMF in the loop is : [NA 8 Jan. 2020 I]



- (a) $56\mu\text{V}$ (b) $28\mu\text{V}$ (c) $48\mu\text{V}$ (d) $36\mu\text{V}$

8. Consider a circular coil of wire carrying constant current I , forming a magnetic dipole. The magnetic flux through an infinite plane that contains the circular coil and excluding the circular coil area is given by ϕ_i . The magnetic flux through the area of the circular coil area is given by ϕ_0 . Which of the following option is correct?

[7 Jan. 2020 I]

- (a) $\phi_i = \phi_0$ (b) $\phi_i > \phi_0$ (c) $\phi_i < \phi_0$ (d) $\phi_i = -\phi_0$

9. A long solenoid of radius R carries a time (t)-dependent current $I(t) = I_0 t(1-t)$. A ring of radius $2R$ is placed coaxially near its middle. During the time interval $0 \leq t \leq 1$, the induced current (I_R) and the induced EMF (V_R) in the ring change as:

[7 Jan. 2020 I]

- (a) Direction of I_R remains unchanged and V_R is maximum at $t = 0.5$
 (b) At $t = 0.25$ direction of I_R reverses and V_R is maximum
 (c) Direction of I_R remains unchanged and V_R is zero at $t = 0.25$
 (d) At $t = 0.5$ direction of I_R reverses and V_R is zero

10. A loop ABCDEFA of straight edges has six corner points $A(0, 0, 0)$, $B\{5, 0, 0\}$, $C(5, 5, 0)$, $D(0, 5, 0)$, $E(0, 5, 5)$ and $F(0, 0, 5)$. The magnetic field in this region is

$\vec{B} = (3\hat{i} + 4\hat{k})\text{T}$. The quantity of flux through the loop ABCDEFA (in Wb) is _____. [NA 7 Jan. 2020 I]

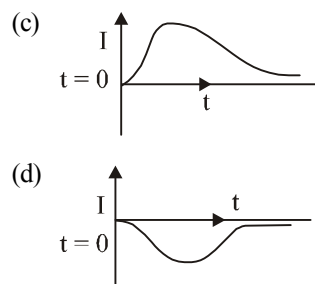
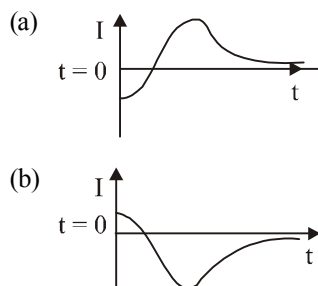
11. A planar loop of wire rotates in a uniform magnetic field. Initially, at $t = 0$, the plane of the loop is perpendicular to the magnetic field. If it rotates with a period of 10 s about an axis in its plane then the magnitude of induced emf will be maximum and minimum, respectively at:

[7 Jan. 2020 II]

- (a) 2.5 s and 7.5 s (b) 2.5 s and 5.0 s
 (c) 5.0 s and 7.5 s (d) 5.0 s and 10.0 s

12. A very long solenoid of radius R is carrying current $I(t) = kte^{-at}$ ($k > 0$), as a function of time ($t \geq 0$). Counter clockwise current is taken to be positive. A circular conducting coil of radius $2R$ is placed in the equatorial plane of the solenoid and concentric with the solenoid. The current induced in the outer coil is correctly depicted, as a function of time, by:

[9 Apr. 2019 II]



13. Two coils 'P' and 'Q' are separated by some distance. When a current of 3A flows through coil 'P', a magnetic flux of 10^{-3} Wb passes through 'Q'. No current is passed through 'Q'. When no current passes through 'P' and a current of 2A passes through 'Q', the flux through 'P' is:

[9 Apr. 2019 II]

- (a) 6.67×10^{-4} Wb (b) 3.67×10^{-3} Wb
 (c) 6.67×10^{-3} Wb (d) 3.67×10^{-4} Wb

14. The self induced emf of a coil is 25 volts. When the current in it is changed at uniform rate from 10 A to 25 A in 1s, the change in the energy of the inductance is:

[9 Jan. 2019 II]

- (a) 740 J (b) 437.5 J
 (c) 540 J (d) 637.5 J

15. A conducting circular loop made of a thin wire, has area $3.5 \times 10^{-3}\text{m}^2$ and resistance 10Ω . It is placed perpendicular to a time dependent magnetic field $B(t) = (0.4\text{T}) \sin(50\pi t)$. The net charge flowing through the loop during $t = 0$ s and $t = 10$ ms is close to:

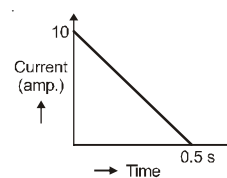
[9 Jan. 2019 I]

- (a) 14mC (b) 7mC (c) 21mC (d) 6mC

16. In a coil of resistance 100Ω , a current is induced by changing the magnetic flux through it as shown in the figure. The magnitude of change in flux through the coil is

[2017]

- (a) 250 Wb
 (b) 275 Wb
 (c) 200 Wb
 (d) 225 Wb



17. A conducting metal circular-wire-loop of radius r is placed perpendicular to a magnetic field which varies with time as

$B = B_0 e^{-t/\tau}$, where B_0 and τ are constants, at time $t = 0$. If the resistance of the loop is R then the heat generated in the loop after a long time ($t \rightarrow \infty$) is;

[Online April 10, 2016]

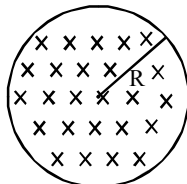
- (a) $\frac{\pi^2 r^4 B_0^4}{2\tau R}$ (b) $\frac{\pi^2 r^4 B_0^2}{2\tau R}$
 (c) $\frac{\pi^2 r^4 B_0^2 R}{\tau}$ (d) $\frac{\pi^2 r^4 B_0^2}{\tau R}$

18. When current in a coil changes from 5 A to 2 A in 0.1 s, average voltage of 50 V is produced. The self-inductance of the coil is : **[Online April 10, 2015]**

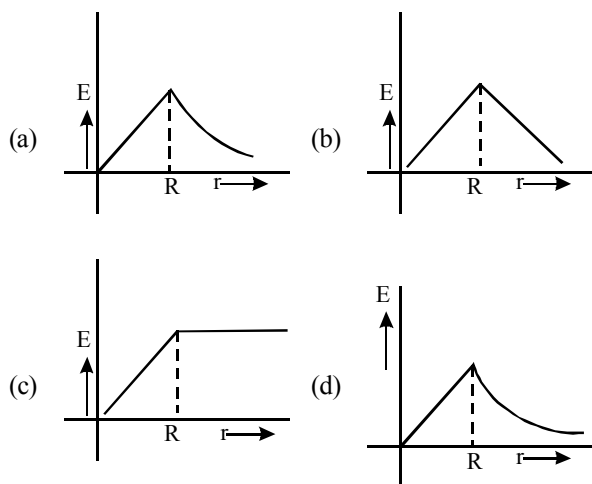
(a) 6 H (b) 0.67 H
(c) 3 H (d) 1.67 H

19. Figure shows a circular area of radius

R where a uniform magnetic field $\vec{B}$ is going into the plane of paper and increasing in magnitude at a constant rate.



In that case, which of the following graphs, drawn schematically, correctly shows the variation of the induced electric field $E(r)$? **[Online April 19, 2014]**



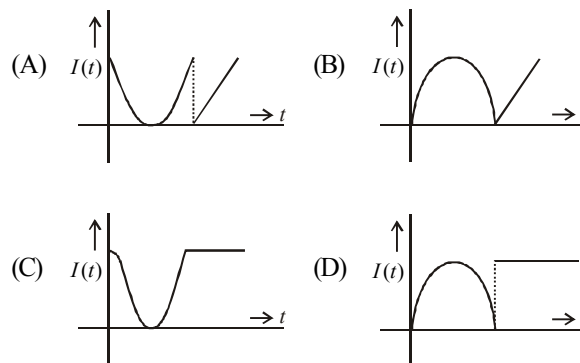
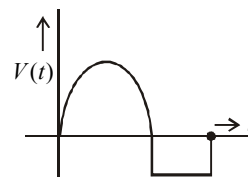
20. A coil of circular cross-section having 1000 turns and 4 cm² face area is placed with its axis parallel to a magnetic field which decreases by 10^{-2} Wb m⁻² in 0.01 s. The e.m.f. induced in the coil is: **[Online April 11, 2014]**

(a) 400 mV (b) 200 mV
(c) 4 mV (d) 0.4 mV

21. A circular loop of radius 0.3 cm lies parallel to a much bigger circular loop of radius 20 cm. The centre of the small loop is on the axis of the bigger loop. The distance between their centres is 15 cm. If a current of 2.0 A flows through the smaller loop, then the flux linked with bigger loop is **[2013]**

(a) 9.1×10^{-11} weber (b) 6×10^{-11} weber
(c) 3.3×10^{-11} weber (d) 6.6×10^{-9} weber

22. Two coils, X and Y, are kept in close vicinity of each other. When a varying current, $I(t)$, flows through coil X, the induced emf ($V(t)$) in coil Y, varies in the manner shown here. The variation of $I(t)$, with time, can then be represented by the graph labelled as graph : **[Online April 9, 2013]**

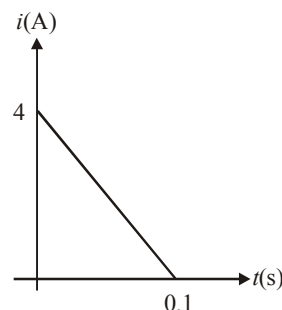


(a) A (b) C
(c) B (d) D

23. A coil is suspended in a uniform magnetic field, with the plane of the coil parallel to the magnetic lines of force. When a current is passed through the coil it starts oscillating; It is very difficult to stop. But if an aluminium plate is placed near to the coil, it stops. This is due to : **[2012]**

(a) development of air current when the plate is placed
(b) induction of electrical charge on the plate
(c) shielding of magnetic lines of force as aluminium is a paramagnetic material.
(d) electromagnetic induction in the aluminium plate giving rise to electromagnetic damping.

24. Magnetic flux through a coil of resistance 10Ω is changed by $\Delta\phi$ in 0.1 s. The resulting current in the coil varies with time as shown in the figure. Then $|\Delta\phi|$ is equal to (in weber) **[Online May 12, 2012]**



(a) 6 (b) 4
(c) 2 (d) 8

25. The flux linked with a coil at any instant 't' is given by $\phi = 10t^2 - 50t + 250$. The induced emf at $t = 3$ s is **[2006]**

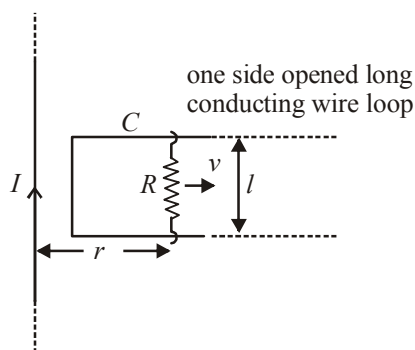
(a) -190 V (b) -10 V
(c) 10 V (d) 190 V

TOPIC

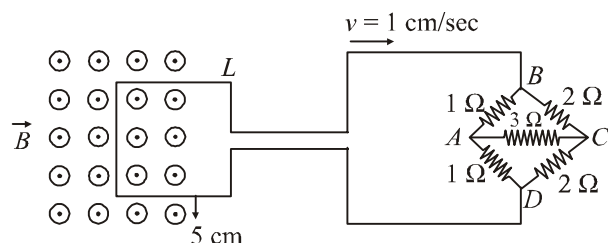
2 Motional and Static EMI and Application of EMI



26. An infinitely long straight wire carrying current I , one side opened rectangular loop and a conductor C with a sliding connector are located in the same plane, as shown in the figure. The connector has length l and resistance R . It slides to the right with a velocity v . The resistance of the conductor and the self inductance of the loop are negligible. The induced current in the loop, as a function of separation r , between the connector and the straight wire is : [Sep. 05, 2020 (II)]

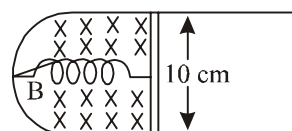


- (a) $\frac{\mu_0 I v l}{4\pi R r}$ (b) $\frac{\mu_0 I v l}{\pi R r}$
 (c) $\frac{2\mu_0 I v l}{\pi R r}$ (d) $\frac{\mu_0 I v l}{2\pi R r}$
27. The figure shows a square loop L of side 5 cm which is connected to a network of resistances. The whole setup is moving towards right with a constant speed of 1 cm s^{-1} . At some instant, a part of L is in a uniform magnetic field of 1 T, perpendicular to the plane of the loop. If the resistance of L is 1.7Ω , the current in the loop at that instant will be close to : [12 Apr. 2019 I]



- (a) $60 \mu\text{A}$ (b) $170 \mu\text{A}$
 (c) $150 \mu\text{A}$ (d) $115 \mu\text{A}$
28. The total number of turns and cross-section area in a solenoid is fixed. However, its length L is varied by adjusting the separation between windings. The inductance of solenoid will be proportional to: [9 April 2019 I]
- (a) L (b) L^2 (c) $1/L^2$ (d) $1/L$

29. A thin strip 10 cm long is on a U shaped wire of negligible resistance and it is connected to a spring of spring constant 0.5 Nm^{-1} (see figure). The assembly is kept in a uniform magnetic field of 0.1 T. If the strip is pulled from its equilibrium position and released, the number of oscillations it performs before its amplitude decreases by a factor of e is N . If the mass of strip is 50 grams, its resistance 10Ω and air drag negligible, N will be close to : [8 April 2019 I]



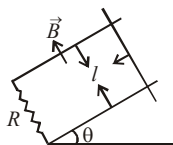
- (a) 1000 (b) 50000 (c) 5000 (d) 10000
30. A 10 m long horizontal wire extends from North East to South West. It is falling with a speed of 5.0 ms^{-1} , at right angles to the horizontal component of the earth's magnetic field, of $0.3 \times 10^{-4} \text{ Wb/m}^2$. The value of the induced emf in wire is : [12 Jan. 2019 II]
- (a) $1.5 \times 10^{-3} \text{ V}$ (b) $1.1 \times 10^{-3} \text{ V}$
 (c) $2.5 \times 10^{-3} \text{ V}$ (d) $0.3 \times 10^{-3} \text{ V}$
31. There are two long co-axial solenoids of same length l . The inner and outer coils have radii r_1 and r_2 and number of turns per unit length n_1 and n_2 , respectively. The ratio of mutual inductance to the self-inductance of the inner-coil is : [11 Jan. 2019 I]
- (a) $\frac{n_1}{n_2}$ (b) $\frac{n_2}{n_1} \cdot \frac{r_1}{r_2}$
 (c) $\frac{n_2}{n_1} \cdot \frac{r_2^2}{r_1^2}$ (d) $\frac{n_2}{n_1}$
32. A copper wire is wound on a wooden frame, whose shape is that of an equilateral triangle. If the linear dimension of each side of the frame is increased by a factor of 3, keeping the number of turns of the coil per unit length of the frame the same, then the self inductance of the coil: [11 Jan. 2019 II]
- (a) decreases by a factor of 9
 (b) increases by a factor of 27
 (c) increases by a factor of 3
 (d) decreases by a factor of $9\sqrt{3}$
33. A solid metal cube of edge length 2 cm is moving in a positive y -direction at a constant speed of 6 m/s. There is a uniform magnetic field of 0.1 T in the positive z -direction. The potential difference between the two faces of the cube perpendicular to the x -axis, is: [10 Jan. 2019 I]

- (a) 12 mV (b) 6 mV
 (c) 1 mV (d) 2 mV

34. An insulating thin rod of length l has a linear charge density $\rho(x) = \rho_0 \frac{x}{l}$ on it. The rod is rotated about an axis passing through the origin ($x = 0$) and perpendicular to the rod. If the rod makes n rotations per second, then the time averaged magnetic moment of the rod is:

[10 Jan. 2019 I]

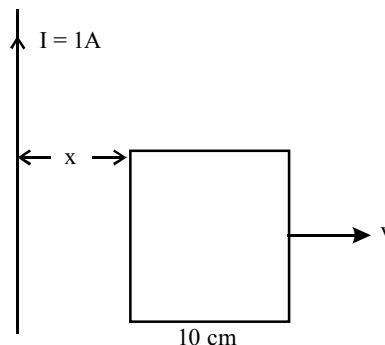
- (a) $\pi n \rho l^3$ (b) $\frac{\pi}{3} n \rho l^3$
 (c) $\frac{\pi}{4} n \rho l^3$ (d) $n \rho l^3$
35. A coil of cross-sectional area A having n turns is placed in a uniform magnetic field B . When it is rotated with an angular velocity ω , the maximum e.m.f. induced in the coil will be
 [Online April 16, 2018]
- (a) $nBA\omega$ (b) $\frac{3}{2} nBA\omega$
 (c) $3nBA\omega$ (d) $\frac{1}{2} nBA\omega$
36. An ideal capacitor of capacitance $0.2 \mu\text{F}$ is charged to a potential difference of 10V . The charging battery is then disconnected. The capacitor is then connected to an ideal inductor of self inductance 0.5mH . The current at a time when the potential difference across the capacitor is 5V , is:
 [Online April 15, 2018]
- (a) 0.17A (b) 0.15A (c) 0.34A (d) 0.25A
37. A copper rod of mass m slides under gravity on two smooth parallel rails, with separation l and set at an angle of θ with the horizontal. At the bottom, rails are joined by a resistance R . There is a uniform magnetic field B normal to the plane of the rails, as shown in the figure. The terminal speed of the copper rod is:
 [Online April 15, 2018]



- (a) $\frac{mgR \cos \theta}{B^2 l^2}$ (b) $\frac{mgR \sin \theta}{B^2 l^2}$
 (c) $\frac{mgR \tan \theta}{B^2 l^2}$ (d) $\frac{mgR \cot \theta}{B^2 l^2}$
38. At the centre of a fixed large circular coil of radius R , a much smaller circular coil of radius r is placed. The two coils are concentric and are in the same plane. The larger coil carries a current I . The smaller coil is set to rotate with a constant angular velocity ω about an axis along their common diameter. Calculate the emf induced in the smaller coil after a time t of its start of rotation.
 [Online April 15, 2018]

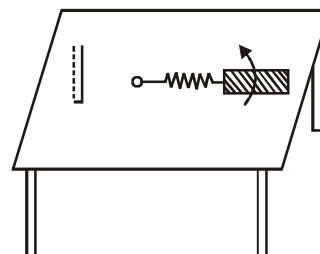
- (a) $\frac{\mu_0 I}{2R} \omega r^2 \sin \omega t$ (b) $\frac{\mu_0 I}{4R} \omega \pi r^2 \sin \omega t$
 (c) $\frac{\mu_0 I}{2R} \omega \pi r^2 \sin \omega t$ (d) $\frac{\mu_0 I}{4R} \omega r^2 \sin \omega t$

39. A square frame of side 10 cm and a long straight wire carrying current 1 A are in the plane of the paper. Starting from close to the wire, the frame moves towards the right with a constant speed of 10 ms^{-1} (see figure).

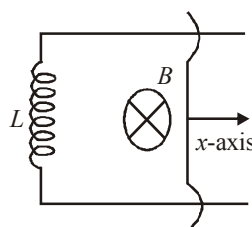


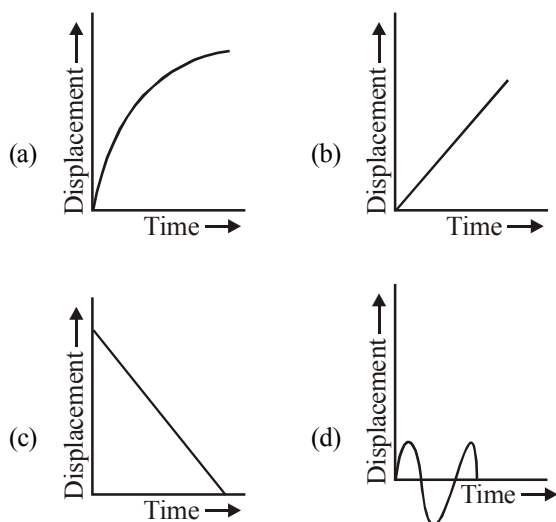
The e.m.f. induced at the time the left arm of the frame is at $x = 10\text{ cm}$ from the wire is:
 [Online April 19, 2014]

- (a) $2\mu\text{V}$ (b) $1\mu\text{V}$
 (c) $0.75\mu\text{V}$ (d) $0.5\mu\text{V}$
40. A metallic rod of length ' ℓ ' is tied to a string of length 2ℓ and made to rotate with angular speed ω on a horizontal table with one end of the string fixed. If there is a vertical magnetic field ' B ' in the region, the e.m.f. induced across the ends of the rod is
 [2013]



- (a) $\frac{2B\omega\ell^2}{2}$ (b) $\frac{3B\omega\ell^2}{2}$
 (c) $\frac{4B\omega\ell^2}{2}$ (d) $\frac{5B\omega\ell^2}{2}$
41. A coil of self inductance L is connected at one end of two rails as shown in figure. A connector of length l , mass m can slide freely over the two parallel rails. The entire set up is placed in a magnetic field of induction B going into the page. At an instant $t = 0$ an initial velocity v_0 is imparted to it and as a result of that it starts moving along x -axis. The displacement of the connector is represented by the figure.
 [Online May 19, 2012]





42. This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.

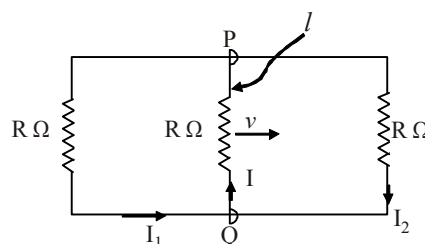
Statement 1: Self inductance of a long solenoid of length L , total number of turns N and radius r is less than

$$\frac{\pi\mu_0 N^2 r^2}{L}.$$

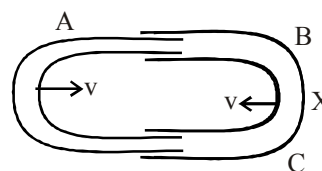
Statement 2: The magnetic induction in the solenoid in Statement 1 carrying current I is $\frac{\mu_0 NI}{L}$ in the middle of the solenoid but becomes less as we move towards its ends.

[Online May 19, 2012]

- (a) Statement 1 is true, Statement 2 is false.
 (b) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.
 (c) Statement 1 is false, Statement 2 is true.
 (d) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.
43. A boat is moving due east in a region where the earth's magnetic field is $5.0 \times 10^{-5} \text{ NA}^{-1} \text{ m}^{-1}$ due north and horizontal. The boat carries a vertical aerial 2 m long. If the speed of the boat is 1.50 ms^{-1} , the magnitude of the induced emf in the wire of aerial is: [2011]
 (a) 0.75 mV (b) 0.50 mV
 (c) 0.15 mV (d) 1 mV
44. A horizontal straight wire 20 m long extending from east to west falling with a speed of 5.0 m/s, at right angles to the horizontal component of the earth's magnetic field $0.30 \times 10^{-4} \text{ Wb/m}^2$. The instantaneous value of the e.m.f. induced in the wire will be [2011 RS]
 (a) 3 mV (b) 4.5 mV (c) 1.5 mV (d) 6.0 mV
45. A rectangular loop has a sliding connector PQ of length l and resistance $R \Omega$ and it is moving with a speed v as shown. The set-up is placed in a uniform magnetic field going into the plane of the paper. The three currents I_1 , I_2 and I are [2010]



- (a) $I_1 = -I_2 = \frac{Blv}{6R}$, $I = \frac{2Blv}{6R}$
 (b) $I_1 = I_2 = \frac{Blv}{3R}$, $I = \frac{2Blv}{3R}$
 (c) $I_1 = I_2 = I = \frac{Blv}{R}$
 (d) $I_1 = I_2 = \frac{Blv}{6R}$, $I = \frac{Blv}{3R}$
46. Two coaxial solenoids are made by winding thin insulated wire over a pipe of cross-sectional area $A = 10 \text{ cm}^2$ and length = 20 cm. If one of the solenoid has 300 turns and the other 400 turns, their mutual inductance is [2008]
 ($\mu_0 = 4\pi \times 10^{-7} \text{ Tm A}^{-1}$)
 (a) $2.4\pi \times 10^{-5} \text{ H}$ (b) $4.8\pi \times 10^{-4} \text{ H}$
 (c) $4.8\pi \times 10^{-5} \text{ H}$ (d) $2.4\pi \times 10^{-4} \text{ H}$
47. One conducting U tube can slide inside another as shown in figure, maintaining electrical contacts between the tubes. The magnetic field B is perpendicular to the plane of the figure. If each tube moves towards the other at a constant speed v , then the emf induced in the circuit in terms of B , l and v where l is the width of each tube, will be [2005]

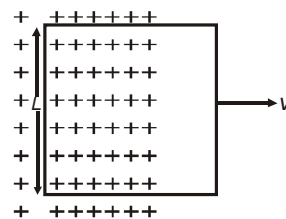


- (a) $-Blv$ (b) Blv
 (c) $2Blv$ (d) zero
48. A metal conductor of length 1 m rotates vertically about one of its ends at angular velocity 5 radians per second. If the horizontal component of earth's magnetic field is $0.2 \times 10^{-4} \text{ T}$, then the e.m.f. developed between the two ends of the conductor is [2004]
 (a) 5 mV (b) 50 μV
 (c) 5 μV (d) 50 mV
49. A coil having n turns and resistance $R \Omega$ is connected with a galvanometer of resistance $4R \Omega$. This combination is moved in time t seconds from a magnetic field W_1 weber to W_2 weber. The induced current in the circuit is [2004]

- (a) $-\frac{(W_2 - W_1)}{Rnt}$ (b) $-\frac{n(W_2 - W_1)}{5 R t}$
 (c) $-\frac{(W_2 - W_1)}{5 Rnt}$ (d) $-\frac{n(W_2 - W_1)}{Rt}$

50. Two coils are placed close to each other. The mutual inductance of the pair of coils depends upon [2003]
 (a) the rates at which currents are changing in the two coils
 (b) relative position and orientation of the two coils
 (c) the materials of the wires of the coils
 (d) the currents in the two coils
51. When the current changes from +2 A to -2 A in 0.05 second, an e.m.f. of 8 V is induced in a coil. The coefficient of self-induction of the coil is [2003]
 (a) 0.2 H (b) 0.4 H (c) 0.8 H (d) 0.1 H

52. A conducting square loop of side L and resistance R moves in its plane with a uniform velocity v perpendicular to one of its sides. A magnetic induction B constant in time and space, pointing perpendicular and into the plane at the loop exists everywhere with half the loop outside the field, as shown in figure. The induced emf is [2002]



- (a) zero (b) RvB (c) vBL/R (d) vBL



Hints & Solutions



1. (5)

For coil C_1 , No. of turns $N_1 = 500$ and radius, $r = 1$ cm.

For coil C_2 , No. of turns $N_2 = 200$ and radius, $R = 20$ cm

$$I = (5t^2 - 2t + 3) \Rightarrow \frac{dI}{dt} = (10t - 2)$$

$$\phi_{\text{small}} = BA = \left(\frac{\mu_0 IN_2}{2R} \right) (\pi r^2)$$

Induced emf in small coil,

$$e = \frac{d\phi}{dt} = \left(\frac{\mu_0 N_2}{2r} \right) \pi r^2 N_1 \frac{dI}{dt} = \left(\frac{\mu_0 N_1 N_2 \pi r^2}{2R} \right) (10t - 2)$$

At $t = 1$ s

$$\begin{aligned} e &= \left(\frac{\mu_0 N_1 N_2 \pi r^2}{2R} \right) 8 = 4 \frac{\mu_0 N_1 N_2 \pi r^2}{R} \\ &= \frac{4(4\pi)10^{-7} \times 200}{20} \times 500 \times \frac{10^{-4}}{10^{-2}} \pi \\ &= 80 \times \pi^2 \times 10^{-7} \times 10 \times 10^2 \times 10^{-2} \\ &= 8 \times 10^{-4} \text{ volt} = 0.8 \text{ mV} = \frac{4}{x} \Rightarrow x = 5. \end{aligned}$$

2. (b) **Case (a)** : When bar magnet is entering with constant speed, flux (ϕ) will change and an e.m.f. is induced, so galvanometer will deflect in positive direction.

Case (b) : When magnet is completely inside, flux (ϕ) will not change, so galvanometer will show null deflection.

Case (c) : When bar magnet is making on exit, again flux (ϕ) will change and an e.m.f. is induced in opposite direction so galvanometer will deflect in negative direction i.e. reverse direction.

3. (a) As we know, emf $\varepsilon = NAB\omega \cos \omega t$, Here $N = 1$

Average power,

$$\langle P \rangle = \langle \frac{\varepsilon^2}{R} \rangle = \langle \frac{A^2 B^2 \omega^2 \cos^2 \omega t}{R} \rangle = \frac{A^2 B^2 \omega^2}{R} \left(\frac{1}{2} \right)$$

Therefore average power loss in the loop due to Joule heating

$$\langle P \rangle = \frac{\pi^2 a^2 b^2 B^2}{2R} (\omega^2)$$

4. (b) Given,

Length of wire, $l = 30$ cm

Radius of wire, $r = 2$ mm $= 2 \times 10^{-3}$ m

Resistivity of metal wire, $\rho = 1.23 \times 10^{-8} \Omega \text{m}$

$$\text{Emf generated, } |e| = \frac{d\phi}{dt} = \frac{dB}{dt} (A) \quad (\because \phi = B.A.)$$

$$\text{Current, } i = \frac{e}{R}$$

$$\text{But, resistance of wire, } R = \rho \frac{l}{A}$$

$$\therefore i = \left| \frac{dB}{dt} \right| \frac{(A)^2}{\rho l} = \frac{0.032 \times \{\pi \times 2 \times 10^{-3}\}^2}{1.23 \times 10^{-8} \times 0.3} = 0.61 \text{ A.}$$

5. (15)

Here, $B = 3.0 \times 10^{-5} \text{ T}$, $R = 10 \text{ cm} = 0.1 \text{ m}$

$$\omega = \frac{2\pi}{2T} = \frac{\pi}{0.2}$$

Flux as a function of time $\phi = \vec{B} \cdot \vec{A} = AB \cos(\omega t)$

$$\text{Emf induced, } e = \frac{-d\phi}{dt} = AB\omega \sin(\omega t)$$

Max. value of Emf $= AB\omega = \pi R^2 B\omega$

$$= 3.14 \times 0.1 \times 0.1 \times 3 \times 10^{-5} \times \frac{\pi}{0.2}$$

$$= 15 \times 10^{-6} \text{ V} = 15 \mu\text{V}$$

6. (10) Given $dI = 0.25 - 0 = 0.25 \text{ A}$

$dt = 0.025 \text{ ms}$

Induced voltage

$$E_{\text{ind}} = 100 \text{ v}$$

Self-inductance, $L = ?$

$$\text{Using, } E_{\text{ind}} = \frac{\Delta\phi}{\Delta t} \Rightarrow 100 = \frac{L(0.25 - 0)}{0.025 \times 10^{-3}}$$

$$\Rightarrow L = 10^{-3} \text{ H} = 10 \text{ mH}$$

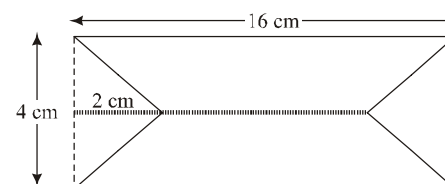
7. (a) According to question, $dB = 1000 - 500 = 500$ gauss $= 500 \times 10^{-4} \text{ T}$

Time $dt = 5 \text{ s}$

Using faraday law

$$\text{Induced EMF, } e = \left| -\frac{d\phi}{dt} \right| = \left| A \frac{dB}{dt} \right|$$

$$\frac{dB}{dt} = \frac{1000 - 500}{5} \times 10^{-4} = 10^{-2} \text{ T/sec}$$



Area, $A = \text{ar of } \square - 2 \text{ ar of } \Delta = (16 \times 4 - 2 \times \text{Area of triangle}) \text{ cm}^2$

$$= \left(64 - 2 \times \frac{1}{2} \times 2 \times 4 \right) \text{ cm}^2$$

$$= 56 \times 10^{-4} \text{ m}^2$$

$$\therefore \varepsilon_{\text{induced}} = \left| A \frac{dB}{dt} \right| = 56 \times 10^{-4} \times 10^{-2} = 56 \times 10^{-6} \text{ V} = 56 \mu\text{V}$$

8. (d) As magnetic field lines form close loop, hence every magnetic field line creating magnetic flux through the inner region (ϕ_i) must be passing through the outer region. Since flux in two regions are in opposite region.

$$\therefore \phi_i = -\phi_o$$



9. (d) According to question,

$$I(t) = I_0 t(1-t)$$

$$\therefore I = I_0 t - I_0 t^2$$

$$\phi = B \cdot A$$

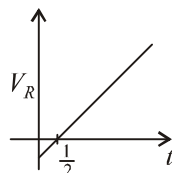
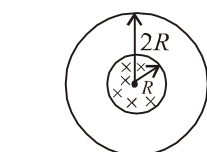
$$\phi = (\mu_0 n I) \times (\pi R^2)$$

$$(\because B = \mu_0 n I \text{ and } A = \pi R^2)$$

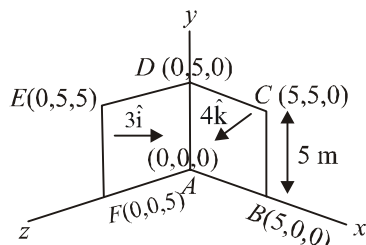
$$V_R = \frac{-d\phi}{dt}$$

$$V_R = \mu_0 n \pi R^2 (I_0 - 2I_0 t)$$

$$\Rightarrow V_R = 0 \text{ at } t = \frac{1}{2} s$$



10. (175.00)



Flux through the loop ABCDEFA,

$$\phi = \vec{B} \cdot \vec{A} = (3\hat{i} + 4\hat{k}) \cdot (25\hat{i} + 25\hat{k})$$

$$\Rightarrow \phi = (3 \times 25) + (4 \times 25) = 175 \text{ weber}$$

11. (b) We have given, time period, $T = 10s$

$$\therefore \text{Angular velocity, } \omega = \frac{2\pi}{10} = \frac{\pi}{5}$$

Magnetic flux, $\phi(t) = BA \cos \omega t$

$$\text{Emf induced, } E = \frac{-d\phi}{dt} = BA\omega \sin \omega t = BA\omega \sin(\omega t)$$

Induced emf, $|\varepsilon|$ is maximum when $\omega t = \frac{\pi}{2}$

$$\Rightarrow t = \frac{\pi}{\frac{\pi}{5}} = 5 s$$

For induced emf to be minimum i.e zero

$$\omega t = \pi \Rightarrow t = \frac{\pi}{\frac{\pi}{5}} = 5 s$$

$\therefore$ Induced emf is zero at $t = 5 s$

12. (a) $Q = BA$

$$= (\mu_0 n_i) A$$

$$= \mu_0 n (kt e^{-\alpha t}) A$$

$$e = -\frac{dQ}{dt} = -\mu_0 n A k \frac{d}{dt}(te^{-\alpha t})$$

$$= -\mu_0 n A k [t(-1)e^{-\alpha t} + e^{-\alpha t} \times 1]$$

$$= -\mu_0 n A k [e^{-\alpha t} (1-t)]$$

$$i = \frac{e}{R} = \frac{-\mu_0 n A k}{R} [e^{-\alpha t} (1-t)]$$

At $t = 0$, $i \Rightarrow -ve$

13. (a) $Q_{\text{coil}} = (NQ) \propto i$

$$\text{So, } \frac{Q_1}{Q_2} = \frac{i_1}{i_2} = \frac{3}{2}$$

$$\text{or } Q_2 = \frac{2}{3} Q_1 = \frac{2}{3} \times 10^{-3} = 6.67 \times 10^{-4} \text{ Wb}$$

14. (b) According to faraday's law of electromagnetic induction,

$$e = \frac{-d\phi}{dt}$$

$$L \times \frac{di}{dt} = 25 \Rightarrow L \times \frac{15}{1} = 25 \text{ or } L = \frac{5}{3} \text{ H}$$

Change in the energy of the inductance,

$$\Delta E = \frac{1}{2} L (i_1^2 - i_2^2) = \frac{1}{2} \times \frac{5}{3} \times (25^2 - 10^2)$$

$$= \frac{5}{6} \times 525 = 437.5 \text{ J}$$

15. [Bonus]

$$\text{Net charge } Q = \frac{\Delta\phi}{R} = \frac{1}{10} A (B_f - B_i) = \frac{1}{10} \times 3.5 \times 10^{-3}$$

$$\left(0.4 \sin \frac{\pi}{2} - 0 \right)$$

$$= \frac{1}{10} (3.5 \times 10^{-3}) (0.4 - 0)$$

$$= 1.4 \times 10^{-4}$$

No option matches, So it should be a bonus.

16. (a) According to Faraday's law of electromagnetic

$$\text{induction, } \varepsilon = \frac{d\phi}{dt}$$

$$\text{Also, } \varepsilon = iR$$

$$\therefore iR = \frac{d\phi}{dt} \Rightarrow \int d\phi = R \int i dt$$

Magnitude of change in flux ($d\phi$) = $R \times$ area under current vs time graph

$$\text{or, } d\phi = 100 \times \frac{1}{2} \times \frac{1}{2} \times 10 = 250 \text{ Wb}$$

17. (b) Electric flux is given by

$$\phi = B \cdot A$$

$$\phi = B_0 \pi r^2 e^{-t/\tau} \quad (\because B = B_0 e^{-t/\tau})$$

$$\text{Induced E.m.f. } \varepsilon = \frac{d\phi}{dt} = \frac{B_0 \pi r^2}{\tau^2} e^{-t/\tau}$$

$$\text{Heat} = \int_0^\infty \frac{\varepsilon^2}{R} = \frac{\pi^2 r^4 B_0^2}{2\tau R}$$

18. (d) According to Faraday's law of electromagnetic induction,

$$\text{Induced emf, } e = \frac{L di}{dt}$$

$$50 = L \left(\frac{5-2}{0.1 \text{ sec}} \right)$$

$$\Rightarrow L = \frac{50 \times 0.1}{3} = \frac{5}{3} = 1.67 \text{ H}$$

19. (a) Inside the sphere field varies linearly i.e., $E \propto r$ with distance and outside varies according to $E \propto \frac{1}{r^2}$

Hence the variation is shown by curve (a)

20. (a) **Given:** No. of turns $N = 1000$

$$\text{Face area, } A = 4 \text{ cm}^2 = 4 \times 10^{-4} \text{ m}^2$$

Change in magnetic field,

$$\Delta B = 10^{-2} \text{ wb m}^{-2}$$

$$\text{Time taken, } t = 0.01 \text{ s} = 10^{-2} \text{ sec}$$

Emf induced in the coil $e = ?$

Applying formula,

$$\text{Induced emf, } e = \frac{-d\phi}{dt} = N \left(\frac{\Delta B}{\Delta t} \right) A \cos \theta$$

$$= \frac{1000 \times 10^{-2} \times 4 \times 10^{-4}}{10^{-2}} = 400 \text{ mV}$$

21. (a) As we know, Magnetic flux, $\phi = B \cdot A$

$$\frac{\mu_0 (2)(20 \times 10^{-2})^2}{2[(0.2)^2 + (0.15)^2]} \times \pi (0.3 \times 10^{-2})^2$$

On solving

$$= 9.216 \times 10^{-11} = 9.2 \times 10^{-11} \text{ weber}$$

22. (a) Induced emf

$$\varepsilon \propto \frac{-di}{dt}$$

23. (d) Because of the Lenz's law of conservation of energy. Length of straight wire, $\ell = 20 \text{ m}$ Earth's Magnetic field, $B = 0.30 \times 10^{-4} \text{ Wb/m}^2$.

24. (c) As $e = \frac{\Delta\phi}{\Delta t}$ or $Ri = \frac{\Delta\phi}{\Delta t}$ ($\because e = Ri$)

$$\Rightarrow \Delta\phi = R(i \Delta t) = R \times \text{area under } i-t \text{ graph} = 10 \times \frac{1}{2} \times 4 \times 0.1 = 2 \text{ weber}$$

25. (b) Electric flux, $\phi = 10t^2 - 50t + 250$

$$\text{Induced emf, } e = -\frac{d\phi}{dt} = -(20t - 50)$$

$$e_{t=3} = -10 \text{ V}$$

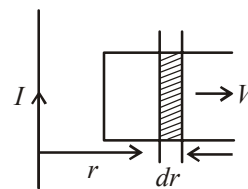
26. (d) Magnetic field at a distance r from the wire

$$B = \frac{\mu_0 I}{2\pi r}$$

Magnetic flux for small displacement dr ;

$$\phi = B \cdot A = B l dr \quad [\because A = l dr \text{ and } B \cdot A = BA \cos 0^\circ]$$

$$\Rightarrow \phi = \frac{\mu_0 I}{2\pi r} l dr$$



$$\text{Emf, } e = \frac{d\phi}{dt} = \frac{\mu_0 I l}{2\pi r} \cdot \frac{dr}{dt} \Rightarrow e = \frac{\mu_0}{2\pi} \cdot \frac{I v l}{r}$$

$$\text{Induced current in the loop, } i = \frac{e}{R} = \frac{\mu_0}{2\pi} \cdot \frac{I v l}{R r}$$

27. (b) Induced emf,

$$e = B v \ell = 1 \times 10^{-2} \times 0.05 = 5 \times 10^{-4} \text{ V}$$

Equivalent resistance,

$$R = \frac{4 \times 2}{4 + 2} + 1.7 = \frac{4}{3} + 1.7 \approx 3 \Omega$$

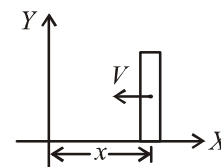
$$\text{Current, } i = \frac{e}{R} = \frac{5 \times 10^{-4}}{3} \approx 170 \mu\text{A}$$

28. (d) Inductance = $\frac{\mu_0 N^2 A}{L}$

29. (c) Force on the strip when it is at stretched position x from mean position is

$$F = -kx - i l B = -kx - \frac{B I v}{R} \times l B$$

$$F = -kx - \frac{B^2 I^2}{R} \times v$$



Above expression shows that it is case of damped oscillation, so its amplitude can be given by

$$\Rightarrow A = A_0 e^{-\frac{bt}{2m}}$$

$$\Rightarrow \frac{A_0}{e} = A_0 e^{-\frac{bt}{2m}} \quad [\text{as per question } A = \frac{A_0}{e}]$$

$$\Rightarrow t = \frac{2m}{\left(\frac{B^2 I^2}{R}\right)} = \frac{2 \times 50 \times 10^{-3} \times 10}{0.01 \times 0.01}$$

Given, $m = 50 \times 10^{-3} \text{ kg}$

$$B = 0.1 \text{ T}$$

$$l = 0.1 \text{ m}$$

$$R = 10 \Omega$$

$$k = 0.5 \text{ N}$$

$$\text{Time period, } T = 2\pi \sqrt{\frac{m}{k}} \approx 2 \text{ s}$$

so, required number of oscillations,

$$N = \frac{10000}{2} = 5000$$

30. (a) Induced emf, $\varepsilon = Bv\ell$
 $= 0.3 \times 10^{-4} \times 5 \times 10$
 $= 1.5 \times 10^{-3} \text{ V}$

31. (d) The rate of mutual inductance is given by

$$M = \mu_0 n_1 n_2 \pi r_1^2 \quad \dots (i)$$

The rate of self inductance is given by

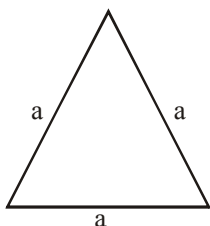
$$L = \mu_0 n_1^2 \pi r_1^2 \quad \dots (ii)$$

$$\text{Dividing (i) by (ii)} \Rightarrow \frac{M}{L} = \frac{n_2}{n_1}$$

32. (c) As total length L of the wire will remain constant

$$L = (3a)N \quad (N = \text{total turns})$$

and length of winding = $(d)N$



(d = diameter of wire)

$$\text{self inductance} = \mu_0 n^2 A \ell$$

$$= \mu_0 n^2 \left(\frac{\sqrt{3}a^2}{4} \right) dN$$

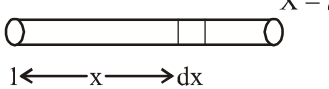
$$\propto a^2 N \propto a \quad [\text{as } N = L/3a \Rightarrow N \propto \frac{1}{a}]$$

Now 'a' increased to '3a'

So self inductance will become 3 times

33. (a) Potential difference between two faces perpendicular to x-axis

$$= lVB = 2 \times 10^{-2} (6 \times 0.1) = 12 \text{ mV}$$

34. (c) 

$$l \leftarrow x \rightarrow dx$$

Magnetic moment, $M = NIA$

$$dQ = \rho dx$$

$$dI = \frac{dQ}{2\pi} \cdot \omega$$

$$dM = dI \times A$$

$$= \frac{\omega}{2\pi} \cdot \frac{\rho_0}{\ell} \cdot x \pi x^2 dx \Rightarrow M = \frac{\rho_0}{\ell} n \pi \int_0^\ell x^3 dx$$

$$= \frac{\pi}{4} \cdot n \rho \ell^3$$

35. (a) Induced emf in a coil, $e = -\frac{d\phi}{dt} = NBA \sin \omega t$

Also, $e = e_0 \sin \omega t$

$\therefore$ Maximum emf induced, $e_0 = nBA\omega$

36. (a) Given: Capacitance, $C = 0.2 \mu\text{F} = 0.2 \times 10^{-6} \text{ F}$

$$\text{Inductance } L = 0.5 \text{ mH} = 0.5 \times 10^{-3} \text{ H}$$

Current $I = ?$

Using energy conservation

$$\frac{1}{2} CV^2 = \frac{1}{2} CV_1^2 + \frac{1}{2} LI^2$$

$$\frac{1}{2} \times 0.2 \times 10^{-6} \times 10^2 + 0$$

$$= \frac{1}{2} \times 0.2 \times 10^{-6} \times 5^2 + \frac{1}{2} \times 0.5 \times 10^{-3} I^2$$

$$\therefore I = \sqrt{3} \times 10^{-1} \text{ A} = 0.17 \text{ A}$$

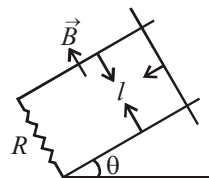
37. (b) From Faraday's law of electromagnetic induction,

$$e = \frac{d\phi}{dt} = \frac{d(BA)}{dt} = \frac{d(Bl)}{dt}$$

$$= \frac{Bdl \times l}{dt} = BVI$$

$$\text{Also, } F = ilB = \left(\frac{BV}{R} \right) (l^2 B) = \frac{B^2 l^2 V}{R}$$

At equilibrium



$$mg \sin \theta = \frac{B^2 l V}{R} \Rightarrow V = \frac{mgR \sin \theta}{B^2 l^2}$$

38. (c) According to Faraday's law of electromagnetic induction,

$$e = -\frac{d\phi}{dt} \text{ and } \phi = BA \cos \omega t = B\pi r^2 \cos \omega t$$

$$\Rightarrow e = -\frac{d}{dt} (\pi r^2 B \cos \omega t) = \pi r^2 B \sin \omega t (\omega)$$

$$\therefore e = \frac{\mu_0 I}{2R} \pi \omega r^2 \sin \omega t \left(\because B = \frac{\mu_0 I}{2R} \right)$$

39. (b) In the given question,

Current flowing through the wire, $I = 1 \text{ A}$

Speed of the frame, $v = 10 \text{ ms}^{-1}$

Side of square loop, $l = 10 \text{ cm}$

Distance of square frame from current carrying wires $x = 10 \text{ cm}$.

We have to find, e.m.f induced $e = ?$

According to Biot-Savart's law

$$B = \frac{\mu_0}{4\pi} \frac{Idl \sin \theta}{x^2}$$

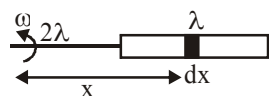
$$= \frac{4\pi \times 10^{-7}}{4\pi} \times \frac{1 \times 10^{-1}}{(10^{-1})^2}$$

$$= 10^{-6}$$

Induced e.m.f. $e = Blv$

$$= 10^{-6} \times 10^{-1} \times 10 = 1 \mu\text{V}$$

40. (d) Here, induced e.m.f.



$$e = \int_{2\ell}^{3\ell} (\omega x) B dx = B\omega \frac{[(3\ell)^2 - (2\ell)^2]}{2}$$

$$= \frac{5B\ell^2 \omega}{2}$$

41. (d)

42. (b) Self inductance of a long solenoid is given by

$$L = \frac{\mu_0 N^2 A}{l}$$

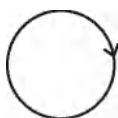
Magnetic field at the centre of solenoid

$$B = \frac{\mu_0 NI}{l}$$

So both the statements are correct and statement 2 is correct explanation of statement 1

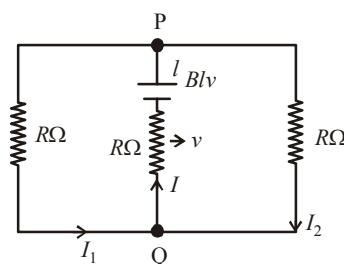
43. (d) As magnetic field lines form close loop, hence every magnetic field line creating magnetic flux through the inner region (ϕ_i) must be passing through the outer region. Since flux in two regions are in opposite region.

$$\therefore \phi_i = -\phi_o$$



44. (a) Induced, emf, $\varepsilon = Bv\ell$
 $= 0.3 \times 10^{-4} \times 5 \times 20$
 $= 3 \times 10^{-3} \text{ V} = 3 \text{ mV}$

45. (b) Due to the movement of resistor R , an emf equal to Blv will be induced in it as shown in figure clearly,



$$I = I_1 + I_2 \quad \text{Also, } I_1 = I_2$$

Solving the circuit,

$$\text{we get } I_1 = I_2 = \frac{Blv}{3R}$$

$$\text{and } I = 2I_1 = \frac{2Blv}{3R}$$

46. (d) Given, Area of cross-section of pipe,
 $A = 10 \text{ cm}^2$

Length of pipe, $\ell = 20 \text{ cm}$

$$M = \frac{\mu_0 N_1 N_2 A}{\ell}$$

$$= \frac{4\pi \times 10^{-7} \times 300 \times 400 \times 100 \times 10^{-4}}{0.2}$$

$$M = \frac{\mu_0 N_1 N_2 A}{\ell}$$

$$= 2.4\pi \times 10^{-4} \text{ H}$$

47. (c) Relative velocity of the tube of width l ,

$$= v - (-v) \quad v = 2v$$

$$\therefore \text{Induced emf.} = B.l(2v)$$

48. (b) Given, length of conductor $\ell = 1 \text{ m}$,

Angular speed, $\omega = 5 \text{ rad/s}$,

Magnetic field, $B = 0.2 \times 10^{-4} \text{ T}$

Emf generated between two ends of conductor

$$\varepsilon = \frac{B\omega\ell^2}{2} = \frac{0.2 \times 10^{-4} \times 5 \times 1}{2} = 50 \mu\text{V}$$

49. (b) $\frac{\Delta\phi}{\Delta t} = \frac{(W_2 - W_1)}{t}$

$$R_{\text{tot}} = (R + 4R)\Omega = 5R\Omega$$

$$i = \frac{nd\phi}{R_{\text{tot}} dt} = \frac{-n(W_2 - W_1)}{5Rt}$$

($\because W_2$ & W_1 are magnetic flux)

50. (b) Mutual inductance depends on the relative position and orientation of the two coils.

51. (d) Induced emf,

$$e = -\frac{\Delta\phi}{\Delta t} = -\frac{\Delta(LI)}{\Delta t} = -L \frac{\Delta I}{\Delta t}$$

$$\therefore |e| = L \frac{\Delta I}{\Delta t}$$

$$\Rightarrow 8 = L \times \frac{[2 - (-2)]}{0.05}$$

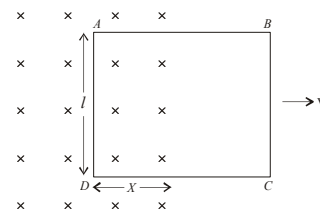
$$\Rightarrow L = \frac{8 \times 0.05}{4} = 0.1 \text{H}$$

52. (d) As the side BC is outside the field, no emf is induced across BC. Further, sides AB and CD are not cutting any flux. So, they will not contribute in flux.

Only side AD is cutting the flux so emf will be induced due to AD only.

The induced emf is

$$e = \frac{-d\phi}{dt} = -\frac{d(\vec{B} \cdot \vec{A})}{dt} = \frac{-d(BA \cos 0^\circ)}{dt}$$



$$\therefore e = -B \frac{dA}{dt} = -B \frac{d(\ell \times x)}{dt}$$

$$\therefore e = -B\ell \frac{dx}{dt} = -B\ell v$$

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Alternating Current

TOPIC 1 Alternating Current, Voltage and Power



- An alternating voltage $v(t) = 220 \sin 100\pi t$ volt is applied to a purely resistive load of 50Ω . The time taken for the current to rise from half of the peak value to the peak value is : **[8 April 2019 I]**
(a) 5 ms (b) 2.2 ms (c) 7.2 ms (d) 3.3 ms
- A small circular loop of wire of radius a is located at the centre of a much larger circular wire loop of radius b . The two loops are in the same plane. The outer loop of radius b carries an alternating current $I = I_0 \cos(\omega t)$. The emf induced in the smaller inner loop is nearly : **[Online April 8, 2017]**

- (a) $\frac{\pi\mu_0 I_0}{2} \cdot \frac{a^2}{b} \omega \sin(\omega t)$ (b) $\frac{\pi\mu_0 I_0}{2} \cdot \frac{a^2}{b} \omega \cos(\omega t)$
(c) $\pi\mu_0 I_0 \frac{a^2}{b} \omega \sin(\omega t)$ (d) $\frac{\pi\mu_0 I_0 b^2}{a} \omega \cos(\omega t)$

- A sinusoidal voltage $V(t) = 100 \sin(500t)$ is applied across a pure inductance of $L = 0.02$ H. The current through the coil is : **[Online April 12, 2014]**
(a) $10 \cos(500t)$ (b) $-10 \cos(500t)$
(c) $10 \sin(500t)$ (d) $-10 \sin(500t)$
- In an a.c. circuit the voltage applied is $E = E_0 \sin \omega t$. The resulting current in the circuit is $I = I_0 \sin\left(\omega t - \frac{\pi}{2}\right)$. The power consumption in the circuit is given by **[2007]**
(a) $P = \sqrt{2}E_0 I_0$ (b) $P = \frac{E_0 I_0}{\sqrt{2}}$
(c) $P = \text{zero}$ (d) $P = \frac{E_0 I_0}{2}$
- In a uniform magnetic field of induction B a wire in the form of a semicircle of radius r rotates about the diameter of the circle with an angular frequency ω . The axis of rotation is perpendicular to the field. If the total resistance of the circuit is R , the mean power generated per period of rotation is **[2004]**

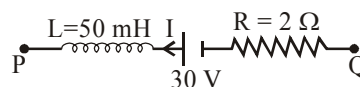
- (a) $\frac{(B\pi r\omega)^2}{2R}$ (b) $\frac{(B\pi r^2\omega)^2}{8R}$
(c) $\frac{B\pi r^2\omega}{2R}$ (d) $\frac{(B\pi r\omega^2)^2}{8R}$

- Alternating current can not be measured by D.C. ammeter because **[2004]**
(a) Average value of current for complete cycle is zero
(b) A.C. Changes direction
(c) A.C. can not pass through D.C. Ammeter
(d) D.C. Ammeter will get damaged.

TOPIC 2 AC Circuit, LCR Circuit, Quality and Power Factor



- A part of a complete circuit is shown in the figure. At some instant, the value of current I is 1 A and it is decreasing at a rate of 10^2 A s^{-1} . The value of the potential difference $V_P - V_Q$ (in volts) at that instant, is _____. **[NA Sep. 06, 2020 (I)]**



- An AC circuit has $R = 100\Omega$, $C = 2\mu\text{F}$ and $L = 80\text{ mH}$, connected in series. The quality factor of the circuit is : **[Sep. 06, 2020 (I)]**
(a) 2 (b) 0.5
(c) 20 (d) 400
- In a series LR circuit, power of 400 W is dissipated from a source of 250 V, 50 Hz. The power factor of the circuit is 0.8. In order to bring the power factor to unity, a capacitor of value C is added in series to the L and R . Taking the value C as $\left(\frac{n}{3\pi}\right)\mu\text{F}$, then value of n is _____. **[NA Sep. 06, 2020 (II)]**
- A series L - R circuit is connected to a battery of emf V . If the circuit is switched on at $t = 0$, then the time at which the energy stored in the inductor reaches $\left(\frac{1}{n}\right)$ times of its maximum value, is : **[Sep. 04, 2020 (II)]**

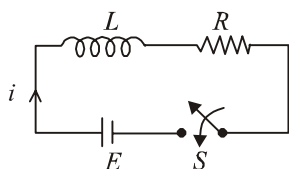
- (a) $\frac{L}{R} \ln \left(\frac{\sqrt{n}}{\sqrt{n}-1} \right)$ (b) $\frac{L}{R} \ln \left(\frac{\sqrt{n}+1}{\sqrt{n}-1} \right)$
 (c) $\frac{L}{R} \ln \left(\frac{\sqrt{n}}{\sqrt{n}+1} \right)$ (d) $\frac{L}{R} \ln \left(\frac{\sqrt{n}-1}{\sqrt{n}} \right)$

11. A 750 Hz, 20 V (rms) source is connected to a resistance of 100Ω , an inductance of 0.1803 H and a capacitance of $10 \mu\text{F}$ all in series. The time in which the resistance (heat capacity $2 \text{ J}^\circ\text{C}$) will get heated by 10°C . (assume no loss of heat to the surroundings) is close to :

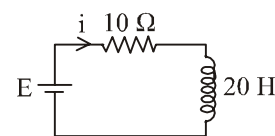
[Sep. 03, 2020 (I)]

- (a) 418 s (b) 245 s
 (c) 365 s (d) 348 s
12. An inductance coil has a reactance of 100Ω . When an AC signal of frequency 1000 Hz is applied to the coil, the applied voltage leads the current by 45° . The self-inductance of the coil is : [Sep. 02, 2020 (II)]
- (a) $1.1 \times 10^{-2} \text{ H}$ (b) $1.1 \times 10^{-1} \text{ H}$
 (c) $5.5 \times 10^{-5} \text{ H}$ (d) $6.7 \times 10^{-7} \text{ H}$
13. Consider the LR circuit shown in the figure. If the switch S is closed at $t = 0$ then the amount of charge that passes through the battery between $t = 0$ and $t = \frac{L}{R}$ is :

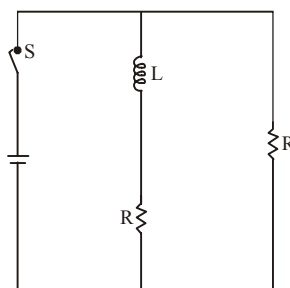
[12 April 2019 II]



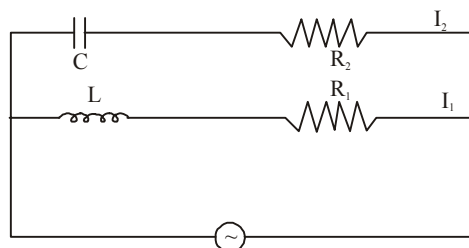
- (a) $\frac{2.7EL}{R^2}$ (b) $\frac{EL}{2.7R^2}$
 (c) $\frac{7.3EL}{R^2}$ (d) $\frac{EL}{7.3R^2}$
14. A coil of self inductance 10 mH and resistance 0.1Ω is connected through a switch to a battery of internal resistance 0.9Ω . After the switch is closed, the time taken for the current to attain 80% of the saturation value is [take $\ln 5 = 1.6$] [10 April 2019 II]
- (a) 0.324 s (b) 0.103 s
 (c) 0.002 s (d) 0.016 s
15. A 20 Henry inductor coil is connected to a 10 ohm resistance in series as shown in figure. The time at which rate of dissipation of energy (Joule's heat) across resistance is equal to the rate at which magnetic energy is stored in the inductor, is : [8 April 2019 I]



- (a) $\frac{2}{\ln 2}$ (b) $\frac{1}{2} \ln 2$
 (c) $2 \ln 2$ (d) $\ln 2$
16. A circuit connected to an ac source of $\text{emf } e = e_0 \sin(100t)$ with t in seconds, gives a phase difference of $\frac{\pi}{4}$ between the $\text{emf } e$ and current i . Which of the following circuits will exhibit this? [8 April 2019 II]
- (a) RL circuit with $R = 1 \text{ k}\Omega$ and $L = 10 \text{ mH}$
 (b) RL circuit with $R = 1 \text{ k}\Omega$ and $L = 1 \text{ mH}$
 (c) RC circuit with $R = 1 \text{ k}\Omega$ and $C = 1 \mu\text{F}$
 (d) RC circuit with $R = 1 \text{ k}\Omega$ and $C = 10 \mu\text{F}$
17. In the figure shown, a circuit contains two identical resistors with resistance $R = 5 \Omega$ and an inductance with $L = 2 \text{ mH}$. An ideal battery of 15 V is connected in the circuit. What will be the current through the battery long after the switch is closed? [12 Jan. 2019 I]

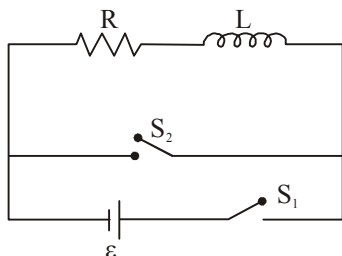


- (a) 5.5 A (b) 7.5 A
 (c) 3 A (d) 6 A
18. In the above circuit, $C = \frac{\sqrt{3}}{2} \mu\text{F}$, $R_2 = 20 \Omega$, $L = \frac{\sqrt{3}}{10} \text{ H}$ and $R_1 = 10 \Omega$. Current in L - R_1 path is I_1 and in C - R_2 path it is I_2 . The voltage of A.C source is given by, $V = 200\sqrt{2} \sin(100t)$ volts. The phase difference between I_1 and I_2 is : [12 Jan. 2019 II]

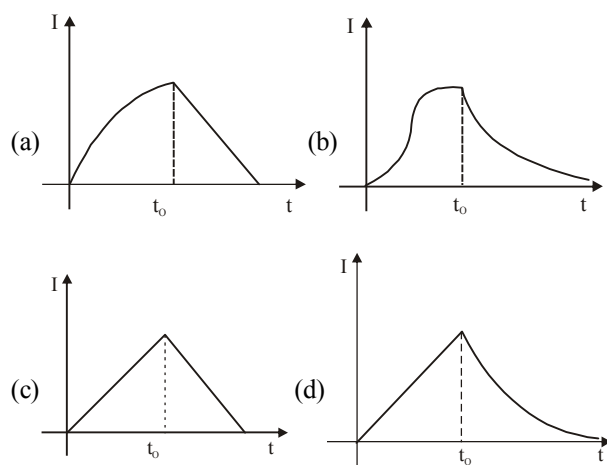


- (a) 60° (b) 30°
 (c) 90° (d) 0

19. In the circuit shown,



the switch S_1 is closed at time $t = 0$ and the switch S_2 is kept open. At some later time (t_0), the switch S_1 is opened and S_2 is closed. the behaviour of the current I as a function of time ' t ' is given by: [11 Jan. 2019 II]

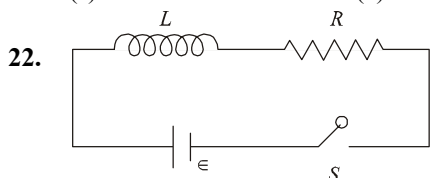


20. A series AC circuit containing an inductor (20 mH), a capacitor (120 μ F) and a resistor (60 Ω) is driven by an AC source of 24 V/50 Hz. The energy dissipated in the circuit in 60 s is: [9 Jan. 2019 I]

- (a) 5.65×10^2 J (b) 2.26×10^3 J
(c) 5.17×10^2 J (d) 3.39×10^3 J

21. In LC circuit the inductance $L = 40$ mH and capacitance $C = 100$ μ F. If a voltage $V(t) = 10 \sin(314 t)$ is applied to the circuit, the current in the circuit is given as: [9 Jan. 2019 II]

- (a) $0.52 \cos 314 t$ (b) $10 \cos 314 t$
(c) $5.2 \cos 314 t$ (d) $0.52 \sin 314 t$



As shown in the figure, a battery of emf ϵ is connected to an inductor L and resistance R in series. The switch is closed at $t = 0$. The total charge that flows from the battery, between $t = 0$ and $t = t_c$ (t_c is the time constant of the circuit) is:

[8 Jan. 2020 II]

- (a) $\frac{\epsilon R}{eL^2}$ (b) $\frac{\epsilon L}{R^2} \left(1 - \frac{1}{e}\right)$
(c) $\frac{\epsilon L}{R^2}$ (d) $\frac{\epsilon R}{eL^2}$

23. A LCR circuit behaves like a damped harmonic oscillator. Comparing it with a physical spring-mass damped oscillator having damping constant ' b ', the correct equivalence would be: [7 Jan. 2020 I]

- (a) $L \leftrightarrow m, C \leftrightarrow k, R \leftrightarrow b$
(b) $L \leftrightarrow \frac{1}{b}, C \leftrightarrow \frac{1}{m}, R \leftrightarrow \frac{1}{k}$
(c) $L \leftrightarrow k, C \leftrightarrow b, R \leftrightarrow m$
(d) $L \leftrightarrow m, C \leftrightarrow \frac{1}{k}, R \leftrightarrow b$

24. An emf of 20 V is applied at time $t = 0$ to a circuit containing in series 10 mH inductor and 5 Ω resistor. The ratio of the currents at time $t = \infty$ and at $t = 40$ s is close to:

(Take $e^2 = 7.389$)

[7 Jan. 2020 II]

- (a) 1.06 (b) 1.15
(c) 1.46 (d) 0.84

25. In an a.c. circuit, the instantaneous e.m.f. and current are given by
 $e = 100 \sin 30 t$

$$i = 20 \sin \left(30 t - \frac{\pi}{4} \right)$$

In one cycle of a.c., the average power consumed by the circuit and the wattless current are, respectively: [2018]

- (a) 50W, 10A (b) $\frac{1000}{\sqrt{2}}$ W, 10A
(c) $\frac{50}{\sqrt{2}}$ W, 0 (d) 50W, 0

26. For an RLC circuit driven with voltage of amplitude v_m and frequency $\omega_0 = \frac{1}{\sqrt{LC}}$ the current exhibits resonance. The quality factor, Q is given by: [2018]

- (a) $\frac{\omega_0 L}{R}$ (b) $\frac{\omega_0 R}{L}$ (c) $\frac{R}{(\omega_0 C)}$ (d) $\frac{CR}{\omega_0}$

27. A sinusoidal voltage of peak value 283 V and angular frequency 320/s is applied to a series LCR circuit. Given that $R = 5 \Omega$, $L = 25$ mH and $C = 1000 \mu$ F. The total impedance, and phase difference between the voltage across the source and the current will respectively be:

[Online April 9, 2017]

- (a) 10 Ω and $\tan^{-1} \left(\frac{5}{3} \right)$ (b) 7 Ω and 45°
(c) 10 Ω and $\tan^{-1} \left(\frac{8}{3} \right)$ (d) 7 Ω and $\tan^{-1} \left(\frac{5}{3} \right)$

28. An arc lamp requires a direct current of 10 A at 80 V to function. If it is connected to a 220 V (rms), 50 Hz AC supply, the series inductor needed for it to work is close to :

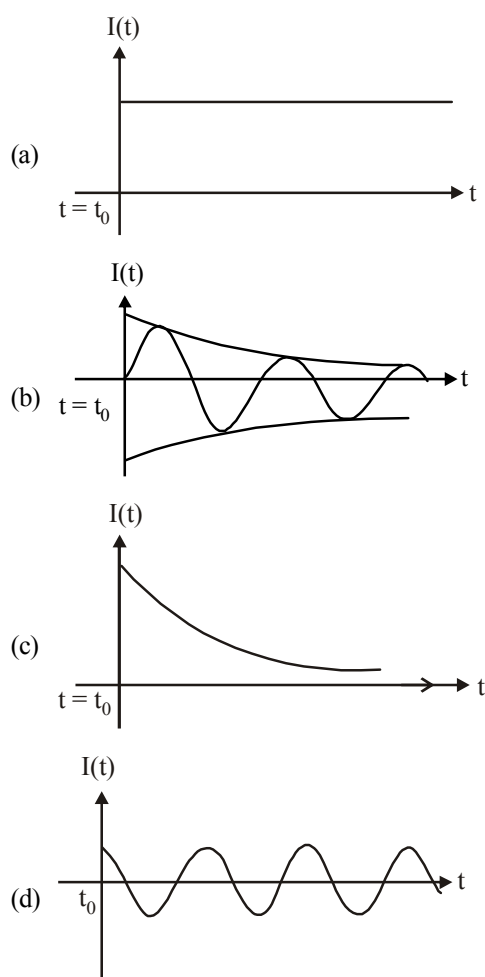
[2016]

- (a) 0.044 H (b) 0.065 H
(c) 80 H (d) 0.08 H

29. A series LR circuit is connected to a voltage source with $V(t) = V_0 \sin \omega t$. After very large time, current $I(t)$ behaves

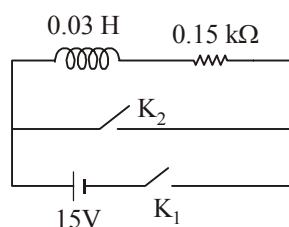
as $\left(t_0 \gg \frac{L}{R}\right)$:

[Online April 9, 2016]



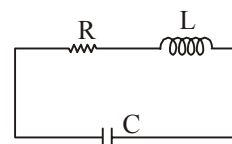
30. An inductor ($L = 0.03$ H) and a resistor ($R = 0.15$ k Ω) are connected in series to a battery of 15V emf in a circuit shown below. The key K_1 has been kept closed for a long time. Then at $t = 0$, K_1 is opened and key K_2 is closed simultaneously. At $t = 1$ ms, the current in the circuit will be : ($e^5 \approx 150$)

[2015]



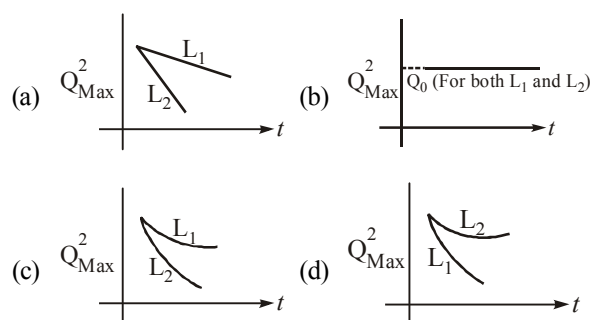
- (a) 6.7 mA (b) 0.67 mA
(c) 100 mA (d) 67 mA

31. An LCR circuit is equivalent to a damped pendulum. In an LCR circuit the capacitor is charged to Q_0 and then connected to the L and R as shown below :



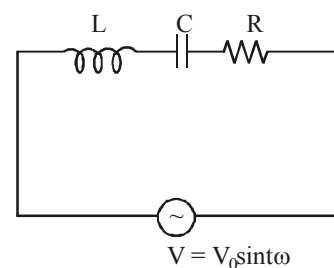
If a student plots graphs of the square of maximum charge (Q_{Max}^2) on the capacitor with time (t) for two different values L_1 and L_2 ($L_1 > L_2$) of L then which of the following represents this graph correctly ? (plots are schematic and not drawn to scale)

[2015]



32. For the LCR circuit, shown here, the current is observed to lead the applied voltage. An additional capacitor C' , when joined with the capacitor C present in the circuit, makes the power factor of the circuit unity. The capacitor C' , must have been connected in :

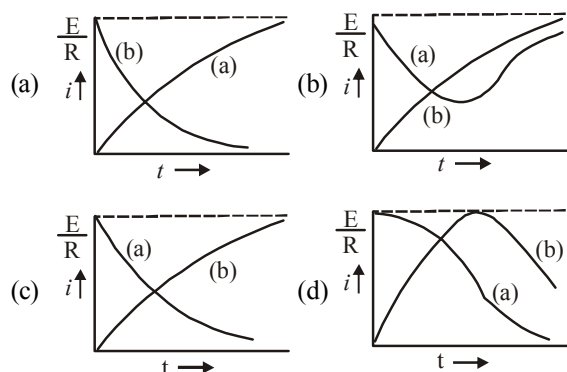
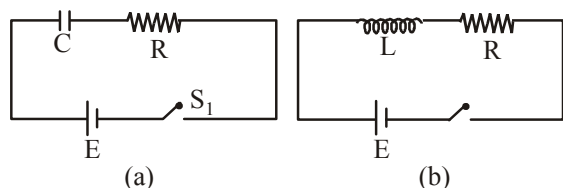
[Online April 11, 2015]



- (a) series with C and has a magnitude $\frac{C}{(\omega^2 LC - 1)}$
(b) series with C and has a magnitude $\frac{1 - \omega^2 LC}{\omega^2 L}$
(c) parallel with C and has a magnitude $\frac{1 - \omega^2 LC}{\omega^2 L}$
(d) parallel with C and has a magnitude $\frac{C}{(\omega^2 LC - 1)}$

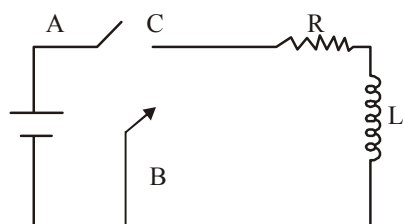
33. In the circuits (a) and (b) switches S_1 and S_2 are closed at $t=0$ and are kept closed for a long time. The variation of current in the two circuits for $t \geq 0$ are roughly shown by figure (figures are schematic and not drawn to scale) :

[Online April 10, 2015]



34. In the circuit shown here, the point 'C' is kept connected to point 'A' till the current flowing through the circuit becomes constant. Afterward, suddenly, point 'C' is disconnected from point 'A' and connected to point 'B' at time $t = 0$. Ratio of the voltage across resistance and the inductor at $t = L/R$ will be equal to:

[2014]



- (a) $\frac{e}{1-e}$ (b) 1 (c) -1 (d) $\frac{1-e}{e}$

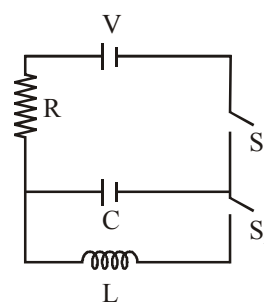
35. When the rms voltages V_L , V_C and V_R are measured respectively across the inductor L, the capacitor C and the resistor R in a series LCR circuit connected to an AC source, it is found that the ratio $V_L : V_C : V_R = 1 : 2 : 3$. If the rms voltage of the AC sources is 100 V, the V_R is close to:

[Online April 9, 2014]

- (a) 50V (b) 70V (c) 90V (d) 100V

36. In an LCR circuit as shown below both switches are open initially. Now switch S_1 is closed, S_2 kept open. (q is charge on the capacitor and $\tau = RC$ is Capacitive time constant). Which of the following statement is correct ?

[2013]



- (a) Work done by the battery is half of the energy dissipated in the resistor
(b) At, $t = \tau$, $q = CV/2$
(c) At, $t = 2\tau$, $q = CV(1 - e^{-2})$
(d) At, $t = 2\tau$, $q = CV(1 - e^{-1})$
37. A series LR circuit is connected to an ac source of frequency ω and the inductive reactance is equal to $2R$. A capacitance of capacitive reactance equal to R is added in series with L and R. The ratio of the new power factor to the old one is :

[Online April 25, 2013]

- (a) $\sqrt{\frac{2}{3}}$ (b) $\sqrt{\frac{2}{5}}$ (c) $\sqrt{\frac{3}{2}}$ (d) $\sqrt{\frac{5}{2}}$

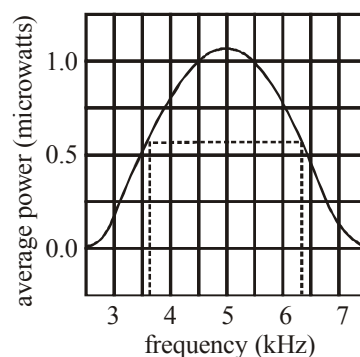
38. When resonance is produced in a series LCR circuit, then which of the following is not correct ?

[Online April 25, 2013]

- (a) Current in the circuit is in phase with the applied voltage.
(b) Inductive and capacitive reactances are equal.
(c) If R is reduced, the voltage across capacitor will increase.
(d) Impedance of the circuit is maximum.

39. The plot given below is of the average power delivered to an LRC circuit versus frequency. The quality factor of the circuit is :

[Online April 23, 2013]

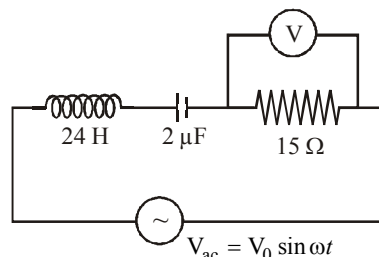


- (a) 5.0 (b) 2.0 (c) 2.5 (d) 0.4
40. In a series L-C-R circuit, $C = 10^{-11}$ Farad, $L = 10^{-5}$ Henry and $R = 100$ Ohm, when a constant D.C. voltage E is applied to the circuit, the capacitor acquires a charge 10^{-9} C. The D.C. source is replaced by a sinusoidal voltage source in

which the peak voltage E_0 is equal to the constant D.C. voltage E . At resonance the peak value of the charge acquired by the capacitor will be : [Online April 22, 2013]

- (a) 10^{-15} C (b) 10^{-6} C (c) 10^{-10} C (d) 10^{-8} C

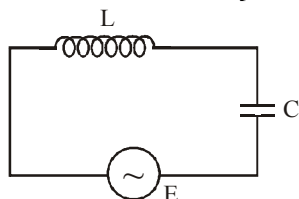
41. An LCR circuit as shown in the figure is connected to a voltage source V_{ac} whose frequency can be varied.



The frequency, at which the voltage across the resistor is maximum, is : [Online April 22, 2013]

- (a) 902 Hz (b) 143 Hz (c) 23 Hz (d) 345 Hz

42. In the circuit shown here, the voltage across E and C are respectively 300 V and 400 V. The voltage E of the ac source is : [Online April 9, 2013]

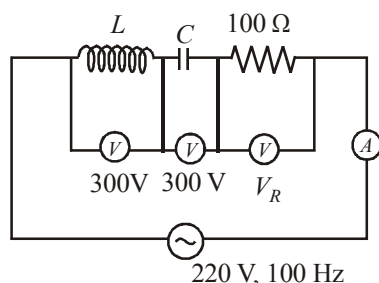


- (a) 400 Volt (b) 500 Volt (c) 100 Volt (d) 700 Volt

43. A resistance R and a capacitance C are connected in series to a battery of negligible internal resistance through a key. The key is closed at $t = 0$. If after t sec the voltage across the capacitance was seven times the voltage across R , the value of t is [Online May 12, 2012]

- (a) $3RC \ln 2$ (b) $2RC \ln 2$
(c) $2RC \ln 7$ (d) $3RC \ln 7$

44. In an LCR circuit shown in the following figure, what will be the readings of the voltmeter across the resistor and ammeter if an a.c. source of 220V and 100 Hz is connected to it as shown? [Online May 7, 2012]



- (a) 800 V, 8 A (b) 110 V, 1.1 A
(c) 300 V, 3 A (d) 220 V, 2.2 A

45. A fully charged capacitor C with initial charge q_0 is connected to a coil of self inductance L at $t = 0$. The time at which the energy is stored equally between the electric and the magnetic fields is: [2011]

- (a) $\frac{\pi}{4} \sqrt{LC}$ (b) $2\pi \sqrt{LC}$
(c) $\sqrt{LC}$ (d) $\pi \sqrt{LC}$

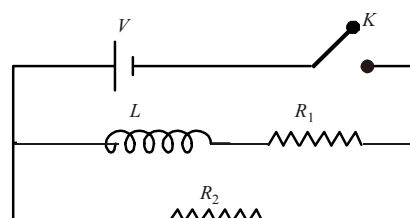
46. A resistor ' R ' and $2\mu\text{F}$ capacitor in series is connected through a switch to 200 V direct supply. Across the capacitor is a neon bulb that lights up at 120 V. Calculate the value of R to make the bulb light up 5 s after the switch has been closed. ($\log_{10} 2.5 = 0.4$) [2011]

- (a) $1.7 \times 10^5 \Omega$ (b) $2.7 \times 10^6 \Omega$
(c) $3.3 \times 10^7 \Omega$ (d) $1.3 \times 10^4 \Omega$

47. Combination of two identical capacitors, a resistor R and a dc voltage source of voltage 6V is used in an experiment on a (C-R) circuit. It is found that for a parallel combination of the capacitor the time in which the voltage of the fully charged combination reduces to half its original voltage is 10 second. For series combination the time for needed for reducing the voltage of the fully charged series combination by half is [2011 RS]

- (a) 10 second (b) 5 second
(c) 2.5 second (d) 20 second

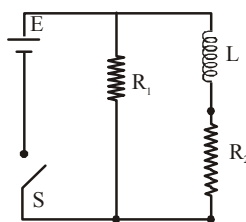
48. In the circuit shown below, the key K is closed at $t = 0$. The current through the battery is [2010]



- (a) $\frac{VR_1R_2}{\sqrt{R_1^2 + R_2^2}}$ at $t = 0$ and $\frac{V}{R_2}$ at $t = \infty$
(b) $\frac{V}{R_2}$ at $t = 0$ and $\frac{V(R_1 + R_2)}{R_1R_2}$ at $t = \infty$
(c) $\frac{V}{R_2}$ at $t = 0$ and $\frac{VR_1R_2}{\sqrt{R_1^2 + R_2^2}}$ at $t = \infty$
(d) $\frac{V(R_1 + R_2)}{R_1R_2}$ at $t = 0$ and $\frac{V}{R_2}$ at $t = \infty$

49. In a series LCR circuit $R = 200\Omega$ and the voltage and the frequency of the main supply is 220V and 50 Hz respectively. On taking out the capacitance from the circuit the current lags behind the voltage by 30° . On taking out the inductor from the circuit the current leads the voltage by 30° . The power dissipated in the LCR circuit is [2010]
- (a) 305 W (b) 210 W (c) Zero W (d) 242 W

50.



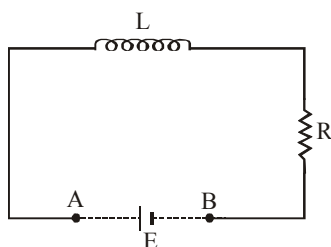
An inductor of inductance $L = 400 \text{ mH}$ and resistors of resistance $R_1 = 2\Omega$ and $R_2 = 2\Omega$ are connected to a battery of emf 12 V as shown in the figure. The internal resistance of the battery is negligible. The switch S is closed at $t = 0$. The potential drop across L as a function of time is [2009]

- (a) $\frac{12}{t}e^{-3t} \text{ V}$ (b) $6(1 - e^{-t/0.2}) \text{ V}$
 (c) $12e^{-5t} \text{ V}$ (d) $6e^{-5t} \text{ V}$

51. In a series resonant LCR circuit, the voltage across R is 100 volts and $R = 1 \text{ k}\Omega$ with $C = 2\mu\text{F}$. The resonant frequency ω is 200 rad/s . At resonance the voltage across L is [2006]

- (a) $2.5 \times 10^{-2} \text{ V}$ (b) 40 V
 (c) 250 V (d) $4 \times 10^{-3} \text{ V}$

52. An inductor ($L = 100 \text{ mH}$), a resistor ($R = 100 \Omega$) and a battery ($E = 100 \text{ V}$) are initially connected in series as shown in the figure. After a long time the battery is disconnected after short circuiting the points A and B . The current in the circuit 1 ms after the short circuit is [2006]



- (a) $1/e \text{ A}$ (b) $e \text{ A}$ (c) 0.1 A (d) 1 A

53. In an AC generator, a coil with N turns, all of the same area A and total resistance R , rotates with frequency ω in a magnetic field B . The maximum value of emf generated in the coil is [2006]

- (a) $N.A.B.R.\omega$ (b) $N.A.B$
 (c) $N.A.B.R$ (d) $N.A.B.\omega$

54. The phase difference between the alternating current and emf is $\frac{\pi}{2}$. Which of the following cannot be the constituent of the circuit? [2005]

- (a) R, L (b) C alone (c) L alone (d) L, C

55. A circuit has a resistance of 12 ohm and an impedance of 15 ohm . The power factor of the circuit will be [2005]

- (a) 0.4 (b) 0.8 (c) 0.125 (d) 1.25

56. A coil of inductance 300 mH and resistance 2Ω is connected to a source of voltage 2 V . The current reaches half of its steady state value in [2005]

- (a) 0.1 s (b) 0.05 s (c) 0.3 s (d) 0.15 s

57. The self inductance of the motor of an electric fan is 10 H . In order to impart maximum power at 50 Hz , it should be connected to a capacitance of [2005]

- (a) $8\mu\text{F}$ (b) $4\mu\text{F}$ (c) $2\mu\text{F}$ (d) $1\mu\text{F}$

58. In an LCR series a.c. circuit, the voltage across each of the components, L, C and R is 50 V . The voltage across the LC combination will be [2004]

- (a) 100 V (b) $50\sqrt{2} \text{ V}$
 (c) 50 V (d) 0 V (zero)

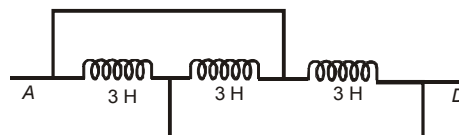
59. In a LCR circuit capacitance is changed from C to $2C$. For the resonant frequency to remain unchanged, the inductance should be changed from L to [2004]

- (a) $L/2$ (b) $2L$ (c) $4L$ (d) $L/4$

60. The power factor of an AC circuit having resistance (R) and inductance (L) connected in series and an angular velocity ω is [2002]

- (a) $R/\omega L$ (b) $R/(R^2 + \omega^2 L^2)^{1/2}$
 (c) $\omega L/R$ (d) $R/(R^2 - \omega^2 L^2)^{1/2}$

61. The inductance between A and D is [2002]

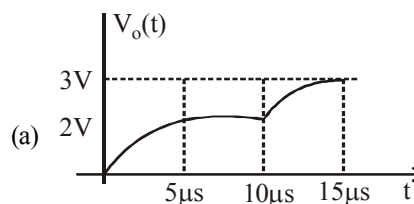
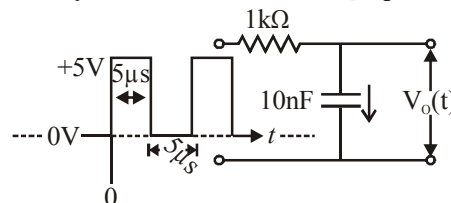


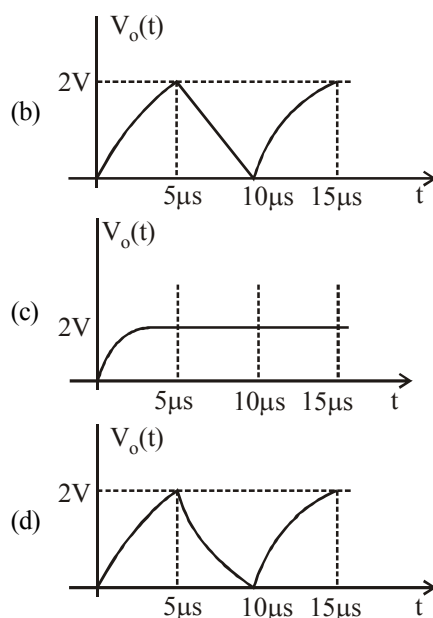
- (a) 3.66 H (b) 9 H
 (c) 0.66 H (d) 1 H

TOPIC 3 Transformers and LC Oscillations



62. For the given input voltage waveform $V_{in}(t)$, the output voltage waveform $V_o(t)$, across the capacitor is correctly depicted by : [Sep. 06, 2020 (I)]





63. A transformer consisting of 300 turns in the primary and 150 turns in the secondary gives output power of $2.2kW$. If the current in the secondary coil is $10A$, then the input voltage and current in the primary coil are :

[10 April 2019 I]

- (a) $220V$ and $20A$ (b) $440V$ and $20A$
(c) $440V$ and $5A$ (d) $220V$ and $10A$

64. A power transmission line feeds input power at $2300V$ to a step down transformer with its primary windings

having 4000 turns. The output power is delivered at $230V$ by the transformer. If the current in the primary of the transformer is $5A$ and its efficiency is 90% , the output current would be:

[9 Jan. 2019 II]

- (a) $50A$ (b) $45A$ (c) $35A$ (d) $25A$

65. A power transmission line feeds input power at $2300V$ to a step down transformer with its primary windings having 4000 turns, giving the output power at $230V$. If the current in the primary of the transformer is $5A$, and its efficiency is 90% , the output current would be: [Online April 16, 2018]

- (a) $20A$ (b) $40A$ (c) $45A$ (d) $25A$

66. In an oscillating LC circuit the maximum charge on the capacitor is Q . The charge on the capacitor when the energy is stored equally between the electric and magnetic field is [2003]

- (a) $\frac{Q}{2}$ (b) $\frac{Q}{\sqrt{3}}$ (c) $\frac{Q}{\sqrt{2}}$ (d) Q

67. The core of any transformer is laminated so as to [2003]

- (a) reduce the energy loss due to eddy currents
(b) make it light weight
(c) make it robust and strong
(d) increase the secondary voltage

68. In a transformer, number of turns in the primary coil are 140 and that in the secondary coil are 280. If current in primary coil is $4A$, then that in the secondary coil is [2002]

- (a) $4A$ (b) $2A$ (c) $6A$ (d) $10A$



Hints & Solutions



1. (d) As $V(t) = 220 \sin 100 \pi t$

$$\text{so, } I(t) = \frac{220}{50} \sin 100 \pi t$$

$$\text{i.e., } I = I_m \sin(100 \pi t)$$

$$\text{For } I = I_m$$

$$t_1 = \frac{\pi}{2} \times \frac{1}{100\pi} = \frac{1}{200} \text{ sec.}$$

$$\text{and for } I = \frac{I_m}{2}$$

$$\Rightarrow \frac{I_m}{2} = I_m \sin(100 \pi t_2) \Rightarrow \frac{\pi}{6} = 100 \pi t_2$$

$$\Rightarrow t_2 = \frac{1}{600} \text{ s}$$

$$\therefore t_{\text{req}} = \frac{1}{200} - \frac{1}{600} = \frac{2}{600} = \frac{1}{300} \text{ s} = 3.3 \text{ ms}$$

2. (a) For two concentric circular coil,

$$\text{Mutual Inductance } M = \frac{\mu_0 \pi N_1 N_2 a^2}{2b}$$

$$\text{here, } N_1 = N_2 = 1$$

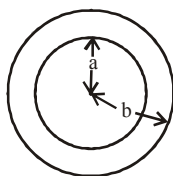
$$\text{Hence, } M = \frac{\mu_0 \pi a^2}{2b} \quad \dots (i)$$

$$\text{and given } I = I_0 \cos \omega t \quad \dots (ii)$$

Now according to Faraday's second law induced emf

$$e = -M \frac{dI}{dt}$$

From eq. (ii),



$$e = \frac{-\mu_0 \pi a^2}{2b} \frac{d}{dt} (I_0 \cos \omega t)$$

$$e = \frac{\mu_0 \pi a^2}{2b} I_0 \sin \omega t (\omega)$$

$$e = \frac{\pi \mu_0 I_0}{2} \cdot \frac{a^2}{b} \omega \sin \omega t$$

3. (b) In a pure inductive circuit current always lags behind

the emf by $\frac{\pi}{2}$.

$$\text{If } v(t) = v_0 \sin \omega t$$

$$\text{then } I = I_0 \sin \left(\omega t - \frac{\pi}{2} \right)$$

$$\text{Now, given } v(t) = 100 \sin(500 t)$$

$$\text{and } I_0 = \frac{E_0}{\omega L} = \frac{100}{500 \times 0.02} [\because L = 0.02 \text{ H}]$$

$$I_0 = 10 \sin \left(500 t - \frac{\pi}{2} \right)$$

$$I_0 = -10 \cos(500 t)$$

4. (c) We know that power consumed in a.c. circuit is given by,

$$P = E_{\text{rms}} I_{\text{rms}} \cos \phi$$

$$\text{Here, } E = E_0 \sin \omega t$$

$$I = I_0 \sin \left(\omega t - \frac{\pi}{2} \right)$$

This means the phase difference, is $\phi = \frac{\pi}{2}$

$$\therefore \cos \phi = \cos \frac{\pi}{2} = 0$$

$$\therefore P = E_{\text{rms}} I_{\text{rms}} \cos \frac{\pi}{2} = 0$$

5. (b) $\phi = \vec{B} \cdot \vec{A}$; $\phi = BA \cos \omega t$

$$\varepsilon = -\frac{d\phi}{dt} = \omega BA \sin \omega t; \quad i = \frac{\omega BA}{R} \sin \omega t$$

$$P_{\text{inst}} = i^2 R = \left(\frac{\omega BA}{R} \right)^2 \times R \sin^2 \omega t$$

$$P_{\text{avg}} = \frac{\int_0^T P_{\text{inst}} \times dt}{\int_0^T dt} = \frac{(\omega BA)^2}{R} \frac{\int_0^T \sin^2 \omega t dt}{\int_0^T dt} = \frac{1}{2} \frac{(\omega BA)^2}{R}$$

$$\therefore P_{\text{avg}} = \frac{(\omega B \pi r^2)^2}{8R} \quad \left[A = \frac{\pi r^2}{2} \right]$$

6. (a) D.C. ammeter measure average value of current. In AC current, average value of current in complete cycle is zero. Hence reading will be zero.

7. (33)

$$\text{Here, } L = 50 \text{ mH} = 50 \times 10^{-3} \text{ H}; I = 1 \text{ A}, R = 2 \Omega$$

$$V_P - L \frac{dI}{dt} - 30 + RI = V_Q$$

$$\Rightarrow V_P - V_Q = 50 \times 10^{-3} \times 10^2 + 30 - 1 \times 2$$

$$= 5 + 30 - 2 = 33 \text{ V.}$$

8. (a) Quality factor,

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{1}{100} \sqrt{\frac{80 \times 10^{-3}}{2 \times 10^{-6}}} \\ = \frac{1}{100} \sqrt{40 \times 10^3} = \frac{200}{100} = 2$$

9. (400)

Given: Power $P = 400$ W, Voltage $V = 250$ V

$$P = V_m \cdot I_{\text{rms}} \cdot \cos \phi$$

$$\Rightarrow 400 = 250 \times I_{\text{rms}} \times 0.8 \Rightarrow I_{\text{rms}} = 2 \text{ A}$$

Using $P = I_{\text{rms}}^2 R$

$$(I_{\text{rms}})^2 \cdot R = P \Rightarrow 4 \times R = 400$$

$$\Rightarrow R = 100 \Omega$$

Power factor is,

$$\cos \phi = \frac{R}{\sqrt{R^2 + X_L^2}}$$

$$\Rightarrow 0.8 = \frac{100}{\sqrt{100^2 + X_L^2}} \Rightarrow 100^2 + X_L^2 = \left(\frac{100}{0.8}\right)^2$$

$$\Rightarrow X_L = \sqrt{-100^2 + \left(\frac{100}{0.8}\right)^2} \Rightarrow X_L = 75 \Omega$$

When power factor is unity,

$$X_C = X_L = 75 \Rightarrow \frac{1}{\omega C} = 75$$

$$\Rightarrow C = \frac{1}{75 \times 2\pi \times 50} = \frac{1}{7500\pi} \text{ F}$$

$$= \left(\frac{10^6}{2500} \times \frac{1}{3\pi}\right) \mu\text{F} = \frac{400}{3\pi} \mu\text{F}$$

$$N = 400$$

10. (a) Potential energy stored in the inductor

$$U = \frac{1}{2} LI^2$$

During growth of current,

$$i = I_{\text{max}} (1 - e^{-Rt/L})$$

For U to be $\frac{U_{\text{max}}}{n}$; i has to be $\frac{I_{\text{max}}}{\sqrt{n}}$

$$\therefore \frac{I_{\text{max}}}{\sqrt{n}} = I_{\text{max}} (1 - e^{-Rt/L})$$

$$\Rightarrow e^{-Rt/L} = 1 - \frac{1}{\sqrt{n}} = \frac{\sqrt{n} - 1}{\sqrt{n}}$$

$$\Rightarrow -\frac{Rt}{L} = \ln\left(\frac{\sqrt{n} - 1}{\sqrt{n}}\right)$$

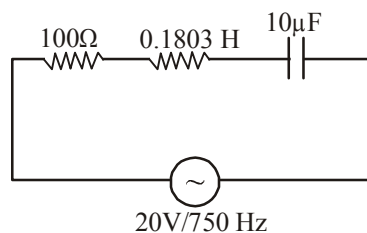
$$\Rightarrow t = \frac{L}{R} \ln\left(\frac{\sqrt{n}}{\sqrt{n} - 1}\right)$$

11. (d) Here, $R = 100$, $X_L = L\omega = 0.1803 \times 750 \times 2\pi = 850 \Omega$,

$$X_C = \frac{1}{C\omega} = \frac{1}{10^{-5} \times 2\pi \times 750} = 21.23 \Omega$$

$$\text{Impedance } Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$= \sqrt{100^2 + (850 - 21.23)^2} = 834.77 \approx 835$$



$$H = i_{\text{rms}}^2 R t = \left(\frac{V_{\text{rms}}}{|Z|}\right)^2 R t = (ms) \Delta t$$

$$\Rightarrow \frac{20}{835} \times \frac{20}{835} \times 100t = (2) \times 10$$

$$\therefore V_{\text{rms}} = 20 \text{ V and } \Delta t = 10^\circ \text{C}$$

$$\therefore \text{Time, } t = 348.61 \text{ s.}$$

12. (a) Given,

Reactance of inductance coil, $Z = 100 \Omega$

Frequency of AC signal, $\nu = 1000$ Hz

Phase angle, $\phi = 45^\circ$

$$\tan \phi = \frac{X_L}{R} = \tan 45^\circ = 1$$

$$\Rightarrow X_L = R$$

$$\text{Reactance, } Z = 100 = \sqrt{X_L^2 + R^2}$$

$$\Rightarrow 100 = \sqrt{R^2 + R^2}$$

$$\Rightarrow \sqrt{2}R = 100 \Rightarrow R = 50\sqrt{2}$$

$$\therefore X_L = 50\sqrt{2}$$

$$\Rightarrow L\omega = 50\sqrt{2} \quad (\because X_L = \omega L)$$

$$\Rightarrow L = \frac{50\sqrt{2}}{2\pi \times 1000} \quad (\because \omega = 2\pi\nu)$$

$$= \frac{25\sqrt{2}}{\pi} \text{ mH} \\ = 1.1 \times 10^{-2} \text{ H}$$

13. (b) We have, $i = i_0(1 - e^{-t/\tau}) = \frac{\varepsilon}{R}(1 - e^{-t/\tau})$

$$\text{Charge, } q = \int_0^{\tau} i dt$$

$$= \frac{\varepsilon}{R} \int_0^{\tau} (1 - e^{-t/\tau}) dt = \frac{E}{R} \frac{\tau}{e} = \frac{E}{R} \times \frac{(L/R)}{e} = \frac{EL}{2.7R^2}$$

14. (d) $I = I_0 \left(1 - e^{-\frac{Rt}{L}} \right)$ Here $R = R_L + r = 1\Omega$

$$0.8I_0 = I_0 \left(1 - e^{-\frac{t}{.01}} \right)$$

$$\Rightarrow 0.8 = 1 - e^{-100t}$$

$$\Rightarrow e^{-100t} = 0.2 = \left(\frac{1}{5} \right)$$

$$\Rightarrow 100t = \ln 5 \Rightarrow t = \frac{1}{100} \ln 5 = 0.016 \text{ s}$$

15. (c) $i^2 R = \left(\tau \frac{di}{dt} \right) i$

$$\Rightarrow \frac{di}{dt} = \frac{i}{\tau}$$

$$\Rightarrow t = \tau \ln 2 = 2 \ln 2 \left[\text{as } \tau = \frac{L}{R} = \frac{20}{10} = 2 \right]$$

16. (d) $\omega = 100 \text{ rad/s}$
We know that

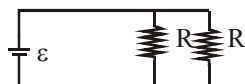
$$\tan \phi = \frac{X_C}{R} = \frac{1}{\omega CR}$$

$$\text{or } \tan 45^\circ = \frac{1}{\omega CR}$$

$$\text{or } \omega CR = 1$$

$$\text{LHS: } \omega CR = 10 \times 10 \times 10^{-6} \times 10^3 = 1$$

17. (d) Long time after switch is closed, the inductor will be idle so, the equivalent diagram will be as below



$$I = \frac{\varepsilon}{\left(\frac{R \times R}{R + R} \right)} = \frac{2\varepsilon}{R} = \frac{2 \times 15}{5} = 6 \text{ A}$$

18. (Bonus)

Capacitive reactance,

$$X_C = \frac{1}{\omega C} = \frac{4}{10^{-6} \times \sqrt{3} \times 100} = \frac{2 \times 10^4}{\sqrt{3}}$$

$$\tan \theta_1 = \frac{X_C}{R_2} = \frac{10^3}{\sqrt{3}}$$

θ_1 is close to 90°

For L-R circuit

$$X_L = \omega L = 100 \times \frac{\sqrt{3}}{10} = 10\sqrt{3}$$

$$R_1 = 10$$

$$\tan \theta_2 = \frac{X_L}{R_1}$$

$$\tan \theta_2 = \sqrt{3} \Rightarrow \theta_2 = \tan^{-1}(\sqrt{3})$$

$$\theta_2 = 60^\circ$$

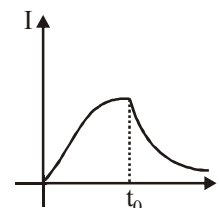
So, phase difference comes out $90^\circ + 60^\circ = 150^\circ$

If R_2 is $20 \text{ K}\Omega$

then phase difference comes out to be $60 + 30 = 90^\circ$.

Therefore Ans. is **Bonus**

19. (b)



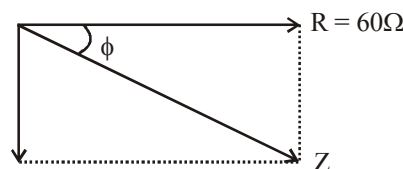
The current will grow for the time $t = 0$ to $t = t_0$ and after that decay of current takes place.

20. (c) Given: $R = 60\Omega$, $f = 50 \text{ Hz}$, $\omega = 2\pi f = 100\pi$ and $v = 24 \text{ V}$
 $C = 120 \mu\text{F} = 120 \times 10^{-6} \text{ F}$

$$X_C = \frac{1}{\omega C} = \frac{1}{100\pi \times 120 \times 10^{-6}} = 26.52\Omega$$

$$X_L = \omega L = 100\pi \times 20 \times 10^{-3} = 2\pi\Omega$$

$$X_C - X_L = 20.24 \approx 20$$



$$Z = \sqrt{R^2 + (X_C - X_L)^2}$$

$$Z = 20\sqrt{10}\Omega$$

$$\cos \phi = \frac{R}{Z} = \frac{60}{20\sqrt{10}} = \frac{3}{\sqrt{10}}$$

$$P_{\text{avg}} = VI \cos \phi, I = \frac{V}{Z} = \frac{V^2}{Z} \cos \phi = 8.64 \text{ watt}$$

Energy dissipated (Q) in time $t = 60 \text{ s}$ is

$$Q = Pt = 8.64 \times 60 = 5.17 \times 10^2 \text{ J}$$

21. (a) Given, Inductance, $L = 40 \text{ mH}$

Capacitance, $C = 100 \mu\text{F}$

Impedance, $Z = X_C - X_L$

$$\Rightarrow Z = \frac{1}{\omega C} - \omega L \quad \left(\because X_C = \frac{1}{\omega C} \text{ and } X_L = \omega L \right)$$

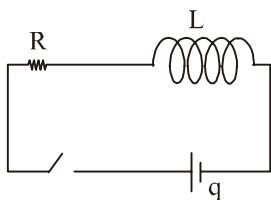
$$= \frac{1}{314 \times 100 \times 10^{-6}} - 314 \times 40 \times 10^{-3}$$

$$= 19.28 \Omega$$

Current, $i = \frac{V_0}{Z} \sin(\omega t + \pi/2)$

$$\Rightarrow i = \frac{10}{19.28} \cos \omega t = 0.52 \cos(314 t)$$

22. (a) For series connection of a resistor and inductor, time variation of current is $I = I_0(1 - e^{-t/T_C})$



Here, $T_C = \frac{L}{R}$

$$q = \int_0^{T_C} i dt$$

$$\Rightarrow \int dq = \int \frac{E}{R} (1 - e^{-t/t_C}) dt$$

$$\Rightarrow q = \frac{E}{R} \left[t + t_C e^{-t/t_C} \right]_0^{t_C}$$

$$\Rightarrow q = \frac{E}{R} \left[t_C + \frac{t_C}{e} - t_C \right]$$

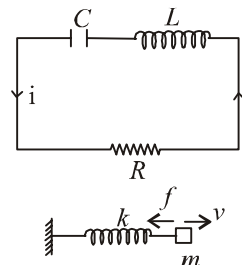
$$\Rightarrow q = \frac{E}{R} \frac{L}{R e}$$

$$\therefore q = \frac{E L}{R^2 e}$$

23. (d) In damped harmonic oscillation,

$$\frac{m d^2 x}{dt^2} = -kx - bv$$

$$\Rightarrow \frac{m d^2 x}{dt^2} + b \frac{dx}{dt} + kx = 0 \quad \dots(i)$$



In LCR circuit, $\frac{-q}{C} - iR - \frac{L di}{dt} = 0$

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = 0 \quad \dots(ii)$$

Comparing equations (i) & (ii)

$$L \leftrightarrow m, C \leftrightarrow \frac{1}{k}, R \leftrightarrow b$$

24. (a) The current (I) in LR series circuit is given by

$$I = \frac{V}{R} \left(1 - e^{-\frac{tR}{L}} \right)$$

At $t = \infty$,

$$I_\infty = \frac{20}{5} \left(1 - e^{-\frac{\infty}{L/R}} \right) = 4 \quad \dots(i)$$

At $t = 40s$,

$$\left(1 - e^{-\frac{40 \times 5}{10 \times 10^{-3}}} \right) = 4(1 - e^{-20,000}) \quad \dots(ii)$$

Dividing (i) by (ii) we get

$$\Rightarrow \frac{I_\infty}{I_{40}} = \frac{1}{1 - e^{-20,000}}$$

25. (b) As we know, average power $P_{avg} = V_{rms} I_{rms} \cos \theta$
- $$= \left(\frac{V_0}{\sqrt{2}} \right) \left(\frac{I_0}{\sqrt{2}} \right) \cos \theta = \left(\frac{100}{\sqrt{2}} \right) \left(\frac{20}{\sqrt{2}} \right) \cos 45^\circ \quad (\because \theta = 45^\circ)$$

$$P_{avg} = \frac{1000}{\sqrt{2}} \text{ watt}$$

Wattless current $I = I_{rms} \sin \theta$

$$= \frac{I_0}{\sqrt{2}} \sin \theta = \frac{20}{\sqrt{2}} \sin 45^\circ = 10 \text{ A}$$

26. (a) Quality factor $Q = \frac{\omega_0}{2\Delta\omega} = \frac{\omega_0 L}{R}$

27. (b) Given,

$$V_0 = 283 \text{ volt}, \omega = 320, R = 5 \Omega, L = 25 \text{ mH}, C = 1000 \mu\text{F}$$

$$X_L = \omega L = 320 \times 25 \times 10^{-3} = 8 \Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{320 \times 1000 \times 10^{-6}} = 3.1 \Omega$$

Total impedance of the circuit :

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{25 + (4.9)^2} = 7 \Omega$$

Phase difference between the voltage and current

$$\tan \phi = \frac{X_L - X_C}{R}$$

$$\tan \phi = \frac{4.9}{5} \approx 1 \Rightarrow \phi = 45^\circ$$

28. (b) Here

$$i = \frac{e}{\sqrt{R^2 + X_L^2}} = \frac{e}{\sqrt{R^2 + \omega^2 L^2}} = \frac{e}{\sqrt{R^2 + 4\pi^2 v^2 L^2}}$$

$$10 = \frac{220}{\sqrt{64 + 4\pi^2 (50)^2 L}}$$

$$[\because R = \frac{V}{I} = \frac{80}{10} = 8]$$

On solving we get

$$L = 0.065 \text{ H}$$

29. (d)

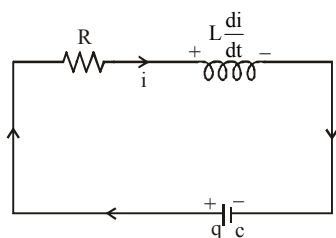
30. (b) $I(0) = \frac{15 \times 100}{0.15 \times 10^3} = 0.1A$

$$I(\infty) = 0$$

$$I(t) = [I(0) - I(\infty)] e^{-t/L/R} + I(\infty)$$

$$I(t) = 0.1 e^{-t/L/R} = 0.1 e^{-t/R}$$

$$I(t) = 0.1 e^{-\frac{0.15 \times 1000}{0.03}} = 0.67mA$$

 31. (c) From KVL at any time t


$$\frac{q}{c} - iR - L \frac{di}{dt} = 0$$

$$i = -\frac{dq}{dt} \Rightarrow \frac{q}{c} + \frac{dq}{dt}R + \frac{Ld^2q}{dt^2} = 0$$

$$\frac{d^2q}{dt^2} + \frac{R}{L} \frac{dq}{dt} + \frac{q}{Lc} = 0$$

From damped harmonic oscillator, the amplitude is given

$$\text{by } A = A_0 e^{-\frac{dt}{2m}}$$

Double differential equation

$$\frac{d^2x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \frac{k}{m} x = 0$$

$$Q_{\max} = Q_0 e^{-\frac{Rt}{2L}} \Rightarrow Q_{\max}^2 = Q_0^2 e^{-\frac{Rt}{L}}$$

Hence damping will be faster for lesser self inductance.

32. (c) Power factor

$$\cos \phi = \frac{R}{\sqrt{R^2 + \left[\omega L - \frac{1}{\omega(C+C')} \right]^2}} = 1$$

On solving we get,

$$\omega L = \frac{1}{\omega(C+C')}$$

$$C' = \frac{1 - \omega^2 LC}{\omega^2 L}$$

Hence option (c) is the correct answer.

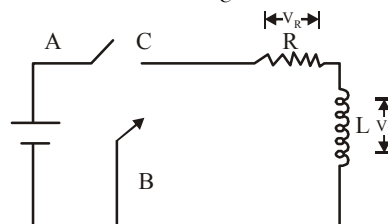
 33. (c) For capacitor circuit, $i = i_0 e^{-t/RC}$

$$\text{For inductor circuit, } i = i_0 \left(1 - e^{-\frac{Rt}{L}} \right)$$

 Hence graph (c) correctly depicts i versus t graph.

34. (c) Applying Kirchhoff's law of voltage in closed loop

$$-V_R - V_C = 0 \Rightarrow \frac{V_R}{V_C} = -1$$


 35. (c) Given, $V_L : V_C : V_R = 1 : 2 : 3$

$$V = 100V$$

$$V_R = ?$$

As we know,

$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$\text{Solving we get, } V_R \approx 90V$$

 36. (c) Charge on the capacitor at any time t is given by $q = CV(1 - e^{-t/\tau})$

$$\text{at } t = 2\tau$$

$$q = CV(1 - e^{-2})$$

37. (d) Power factor (old)

$$= \frac{R}{\sqrt{R^2 + X_L^2}} = \frac{R}{\sqrt{R^2 + (2R)^2}} = \frac{R}{\sqrt{5}R}$$

Power factor (new)

$$= \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}} = \frac{R}{\sqrt{R^2 + (2R - R)^2}} = \frac{R}{\sqrt{2}R}$$

$$\therefore \frac{\text{New power factor}}{\text{Old power factor}} = \frac{\frac{R}{\sqrt{2}R}}{\frac{R}{\sqrt{5}R}} = \frac{\sqrt{5}}{\sqrt{2}}$$

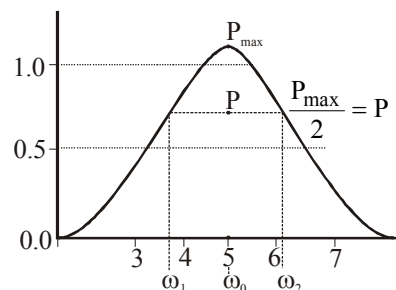
 38. (d) Impedance (Z) of the series LCR circuit is

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

 At resonance, $X_L = X_C$

 Therefore, $Z_{\text{minimum}} = R$

39. (b)



Quality factor of the circuit

$$= \frac{\omega_0}{\omega_2 - \omega_1} = \frac{5}{2.5} = 2.0$$

40. (d)

$$41. \text{ (c) Frequency } f = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2 \times 3.4\sqrt{24 \times 2 \times 10^{-6}}}$$

$$\approx 23\text{Hz}$$

42. (c) Voltage E of the ac source

$$E = V_c - V_L = 400\text{ V} - 300\text{ V} = 100\text{ V}$$

43. (a) $t = 3RC \ln 2$

44. (d) In case of series RLC circuit, Equation of voltage is given by

$$V^2 = V_R^2 + (V_L - V_C)^2$$

$$\text{Here, } V = 220\text{ V}; V_L = V_C = 300\text{ V}$$

$$\therefore V_R = \sqrt{V^2} = 220\text{ V}$$

$$\text{Current } i = \frac{V}{R} = \frac{220}{100} = 2.2\text{ A}$$

45. (a) Energy stored in magnetic field = $\frac{1}{2} Li^2$

$$\text{Energy stored in electric field} = \frac{1}{2} \frac{q^2}{C}$$

Energy will be equal when

$$\therefore \frac{1}{2} Li^2 = \frac{1}{2} \frac{q^2}{C}$$

$$\tan \omega t = 1$$

$$q = q_0 \cos \omega t$$

$$\Rightarrow \frac{1}{2} L(\omega q_0 \sin \omega t)^2 = \frac{(q_0 \cos \omega t)^2}{2C}$$

$$\Rightarrow \omega = \frac{1}{\sqrt{LC}} \Rightarrow \omega t = \frac{\pi}{4}$$

$$\Rightarrow t = \frac{\pi}{4} \sqrt{LC}$$

46. (b) We have, $V = V_0(1 - e^{-t/RC})$

$$\Rightarrow 120 = 200(1 - e^{-t/RC})$$

$$e^{-t/r} = \frac{200 - 120}{200} = \frac{80}{200}$$

$$t = \log_e(2.5)$$

$$\Rightarrow t = RC \ln(2.5) \quad [\because r = RC]$$

$$\Rightarrow R = 2.71 \times 10^6 \Omega$$

47. (c) Time constant for parallel combination = $2RC$

$$\text{Time constant for series combination} = \frac{RC}{2}$$

In first case :

$$V = V_0 \left(1 - e^{-\frac{t}{CR}}\right) \Rightarrow \frac{V_0}{2} = V_0 - V_0 e^{-\frac{t}{CR}}$$

$$V = V_0 e^{-\frac{t_1}{2RC}} = \frac{V_0}{2} \quad \dots(1)$$

In second case :

$$\text{In series grouping, equivalent capacitance} = \frac{C}{2}$$

$$V = V_0 e^{-\frac{t_2}{(RC/2)}} = \frac{V_0}{2} \quad \dots(2)$$

From (1) and (2)

$$\frac{t_1}{2RC} = \frac{t_2}{(RC/2)}$$

$$\Rightarrow t_2 = \frac{t_1}{4} = \frac{10}{4} = 2.5 \text{ sec.}$$

48. (c) At $t = 0$, no current will flow through L and R_1 as inductor will offer infinite resistance.

$$\therefore \text{Current through battery, } i = \frac{V}{R_2}$$

At $t = \infty$, inductor behave as conducting wire

$$\text{Effective resistance, } R_{\text{eff}} = \frac{R_1 R_2}{R_1 + R_2}$$

$$\therefore \text{Current through battery} = \frac{V}{R_{\text{eff}}} = \frac{V(R_1 + R_2)}{R_1 R_2}$$

49. (d) When only the capacitance is removed phase difference between current and voltage is

$$\tan \phi = \frac{X_L}{R}$$

$$\Rightarrow \tan \phi = \frac{\omega L}{R}$$

$$\Rightarrow \omega L = R \tan \phi = 200 \times \frac{1}{\sqrt{3}} = \frac{200}{\sqrt{3}}$$

When only inductor is removed, phase difference between current and voltage is

$$\therefore \tan \phi = \frac{1}{\omega CR}$$

$$\Rightarrow \frac{1}{\omega C} = R \tan \phi = 200 \times \frac{1}{\sqrt{3}} = \frac{200}{\sqrt{3}}$$

Impedance of the circuit,

$$Z = \sqrt{R^2 + \left(\frac{1}{\omega C} - \omega L\right)^2}$$

$$= \sqrt{(200)^2 + \left(\frac{200}{\sqrt{3}} - \frac{200}{\sqrt{3}}\right)^2} = 200 \Omega$$

$$\text{Power dissipated in the circuit} = V_{\text{rms}} I_{\text{rms}} \cos \phi$$

$$= V_{\text{rms}} \cdot \frac{V_{\text{rms}}}{Z} \cdot \frac{R}{Z} \left(\because \cos \phi = \frac{R}{Z} \right) = \frac{V_{\text{rms}}^2 R}{Z^2}$$

$$= \frac{(220)^2 \times 200}{(200)^2} = \frac{220 \times 220}{200} = 242\text{ W}$$

50. (c) Growth in current in branch containing L and R_2 when switch is closed is given by

$$i = \frac{E}{R_2} [1 - e^{-R_2 t / L}]$$

$$\Rightarrow \frac{di}{dt} = \frac{E}{R_2} \cdot \frac{R_2}{L} \cdot e^{-R_2 t / L} = \frac{E}{L} e^{-\frac{R_2 t}{L}}$$

Hence, potential drop across L

$$V_L = \frac{L di}{dt} = \left(\frac{E}{L} e^{-R_2 t/L} \right) L$$

$$= E e^{-R_2 t/L} = 12 e^{-\frac{2t}{400 \times 10^{-3}}} = 12 e^{-5t} \text{V}$$

51. (c) Across resistor, $I = \frac{V}{R} = \frac{100}{1000} = 0.1 \text{ A}$

At resonance,

$$X_L = X_C = \frac{1}{\omega C} = \frac{1}{200 \times 2 \times 10^{-6}} = 2500$$

Voltage across L is

$$I X_L = 0.1 \times 2500 = 250 \text{ V}$$

52. (a) Initially, when steady state is achieved,

$$i = \frac{E}{R}$$

Let E is short circuited at $t = 0$. Then

At $t = 0$

$$\text{Maximum current, } i_0 = \frac{E}{R} = \frac{100}{100} = 1 \text{ A}$$

Let during decay of current at any time the current flowing

$$\text{is } -L \frac{di}{dt} - iR = 0$$

$$\Rightarrow \frac{di}{i} = -\frac{R}{L} dt$$

$$\Rightarrow \int_{i_0}^i \frac{di}{i} = \int_0^t -\frac{R}{L} dt$$

$$\Rightarrow \log_e \frac{i}{i_0} = -\frac{R}{L} t$$

$$\Rightarrow i = i_0 e^{-\frac{R}{L} t}$$

$$\Rightarrow i = \frac{E}{R} e^{-\frac{R}{L} t} = 1 \times e^{-\frac{100 \times 10^{-3}}{100 \times 10^{-3}}} = \frac{1}{e}$$

53. (d) $e = -\frac{d\phi}{dt} = -\frac{d(N\vec{B} \cdot \vec{A})}{dt}$
- $$= -N \frac{d}{dt} (BA \cos \omega t) = NBA \omega \sin \omega t$$
- $$\Rightarrow e_{\max} = NBA \omega$$

54. (a) Phase difference for R - L circuit lies between

$$\left(0, \frac{\pi}{2} \right) \text{ but } 0 \text{ or } \pi/2$$

55. (b) Given, Resistance of circuit, $R = 12 \Omega$
Imedance of circuit, $Z = 15 \Omega$

$$\text{Power factor} = \cos \phi = \frac{R}{Z} = \frac{12}{15} = \frac{4}{5} = 0.8$$

56. (a) Current in inductor circuit is given by,

$$i = i_0 \left(1 - e^{-\frac{Rt}{L}} \right)$$

$$\frac{i_0}{2} = i_0 \left(1 - e^{-\frac{Rt}{L}} \right) \Rightarrow e^{-\frac{Rt}{L}} = \frac{1}{2}$$

Taking log on both the sides,

$$-\frac{Rt}{L} = \log 1 - \log 2$$

$$\Rightarrow t = \frac{L}{R} \log 2 = \frac{300 \times 10^{-3}}{2} \times 0.69$$

$$\Rightarrow t = 0.1 \text{ sec.}$$

57. (d) For maximum power, $X_L = X_C$, which yields

$$C = \frac{1}{(2\pi n)^2 L} = \frac{1}{4\pi^2 \times 50 \times 50 \times 10}$$

$$\therefore C = 0.1 \times 10^{-5} \text{ F} = 1 \mu\text{F}$$

58. (d) In a series LCR circuit voltage across the inductor and capacitor are in opposite phase

$\therefore$ Net voltage difference across

$$LC = 50 - 50 = 0$$

59. (a) Resonant frequency, $F_r = \frac{1}{2\pi\sqrt{LC}}$

For resonant frequency to remain same

$$LC = \text{constant}$$

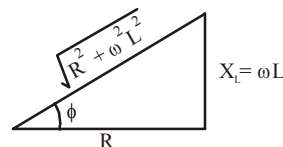
$$\therefore LC = L' C'$$

$$\Rightarrow LC = L' \times 2C$$

$$\Rightarrow L' = \frac{L}{2}$$

60. (b) Resistance of the inductor, $X_L = \omega L$

The impedance triangle for resistance (R) and inductor (L) connected in series is shown in the figure.



$$\text{Net impedance of circuit } Z = \sqrt{X_L^2 + R^2}$$

$$\text{Power factor, } \cos \phi = \frac{R}{Z}$$

$$\Rightarrow \cos \phi = \frac{R}{\sqrt{R^2 + \omega^2 L^2}}$$

61. (d) All three inductors are connected in parallel. The equivalent inductance L_p is given by

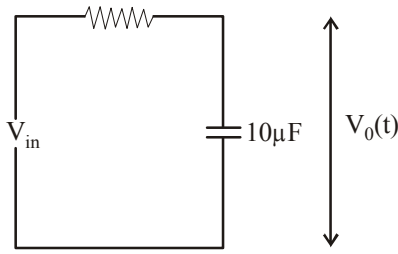
$$\frac{1}{L_p} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = \frac{3}{3} = 1$$

$$\therefore L_p = 1$$

62. (a) When first pulse is applied, the potential across capacitor

$$V_0(t) = V_{\text{in}} \left(1 - e^{-\frac{t}{RC}} \right)$$

$$\text{At } t = 5 \mu\text{s} = 5 \times 10^{-6} \text{ s}$$



$$V_0(t) = 5 \left(1 - e^{\frac{5 \times 10^{-6}}{10^3 \times 10 \times 10^{-9}}} \right) = 5(1 - e^{-0.5}) = 2V$$

When no pulse is applied, capacitor will discharge.

Now, $V_{in} = 0$ means discharging.

$$V_0(t) = 2e^{\frac{1}{RC}} = 2e^{-0.5} = 1.21 V$$

Now for next 5 μs

$$V_0(t) = 5 - 3.79e^{\frac{1}{RC}}$$

After 5 μs again, $V_0(t) = 2.79 \text{ Volt} \approx 3 V$

Hence, graph (a) correctly depicts.

63. (c) Power output ($V_2 I_2$) = 2.2 kW

$$\therefore V_2 = \frac{2.2 \text{ kW}}{(10 \text{ A})} = 220 \text{ volts}$$

$\therefore$ Input voltage for step-down transformer

$$\frac{V_1}{V_2} = \frac{N_1}{N_2} = 2$$

$$V_{\text{input}} = 2 \times V_{\text{output}} = 2 \times 220 = 440 \text{ V}$$

$$\text{Also } \frac{I_1}{I_2} = \frac{N_2}{N_1}$$

$$\therefore I_1 = \frac{1}{2} \times 10 = 5 \text{ A}$$

64. (b) Efficiency, $\eta = \frac{P_{\text{out}}}{P_{\text{in}}} = \frac{V_s I_s}{V_p I_p}$

$$\Rightarrow 0.9 = \frac{230 \times I_s}{2300 \times 5}$$

$$\Rightarrow I_s = 0.9 \times 50 = 45 \text{ A}$$

Output current = 45 A

65. (c) Given: $V_p = 2300 \text{ V}$, $V_s = 230 \text{ V}$, $I_p = 5 \text{ A}$, $\eta = 90\% = 0.9$

$$\text{Efficiency } \eta = 0.9 = \frac{P_s}{P_p} \Rightarrow P_s = 0.9 P_p$$

$$V_s I_s = 0.9 \times V_p I_p \quad (\because P = VI)$$

$$I_s = \frac{0.9 \times 2300 \times 5}{230} = 45 \text{ A}$$

66. (c) When the capacitor is completely charged, the total energy in the LC circuit is with the capacitor and that energy is given by

$$U_{\text{max}} = \frac{1}{2} \frac{Q^2}{C}$$

When half energy is with the capacitor in the form of electric field between the plates of the capacitor we get

$$\frac{U_{\text{max}}}{2} = \frac{1}{2} \frac{q'^2}{C}$$

Here q' is the charge on the plate of capacitor when energy is shared equally.

$$\therefore \frac{1}{2} \times \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} \frac{q'^2}{C} \Rightarrow q' = \frac{Q}{\sqrt{2}}$$

67. (a) Laminated core provide less area of cross-section for the current to flow. Because of this, resistance of the core increases and current decreases there by decreasing the energy loss due to eddy current.

68. (b) Number of turns in primary

$$N_p = 140$$

Number of turns in secondary $N_s = 280$, $I_p = 4 \text{ A}$, $I_s = ?$

$$\text{Using transformation ratio for a transformer } \frac{I_s}{I_p} = \frac{N_p}{N_s}$$

$$\Rightarrow \frac{I_s}{4} = \frac{140}{280}$$

$$\Rightarrow I_s = 2 \text{ A}$$

Electromagnetic Waves



TOPIC 1 Electromagnetic Waves, Conduction and Displacement Current



1. For a plane electromagnetic wave, the magnetic field at a point x and time t is

$$\vec{B}(x, t) = [1.2 \times 10^{-7} \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t) \hat{k}] \text{ T}$$

The instantaneous electric field $\vec{E}$ corresponding to $\vec{B}$ is: (speed of light $c = 3 \times 10^8 \text{ ms}^{-1}$) [Sep. 06, 2020 (II)]

- (a) $\vec{E}(x, t) = [-36 \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t) \hat{j}] \frac{\text{V}}{\text{m}}$
 (b) $\vec{E}(x, t) = [36 \sin(1 \times 10^3 x + 0.5 \times 10^{11} t) \hat{j}] \frac{\text{V}}{\text{m}}$
 (c) $\vec{E}(x, t) = [36 \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t) \hat{k}] \frac{\text{V}}{\text{m}}$
 (d) $\vec{E}(x, t) = [36 \sin(1 \times 10^3 x + 1.5 \times 10^{11} t) \hat{i}] \frac{\text{V}}{\text{m}}$

2. An electron is constrained to move along the y -axis with a speed of $0.1 c$ (c is the speed of light) in the presence of electromagnetic wave, whose electric field is $E = 30 \hat{j} \sin(1.5 \times 10^7 t - 5 \times 10^{-2} x) \text{ V/m}$. The maximum magnetic force experienced by the electron will be:

(given $c = 3 \times 10^8 \text{ ms}^{-1}$ & electron charge $= 1.6 \times 10^{-19} \text{ C}$)

[Sep. 05, 2020 (I)]

- (a) $3.2 \times 10^{-18} \text{ N}$ (b) $2.4 \times 10^{-18} \text{ N}$
 (c) $4.8 \times 10^{-19} \text{ N}$ (d) $1.6 \times 10^{-19} \text{ N}$
3. The electric field of a plane electromagnetic wave is given by $\vec{E} = E_0(\hat{x} + \hat{y}) \sin(kz - \omega t)$

Its magnetic field will be given by: [Sep. 04, 2020 (II)]

- (a) $\frac{E_0}{c}(-\hat{x} + \hat{y}) \sin(kz - \omega t)$
 (b) $\frac{E_0}{c}(\hat{x} + \hat{y}) \sin(kz - \omega t)$

(c) $\frac{E_0}{c}(\hat{x} - \hat{y}) \sin(kz - \omega t)$

(d) $\frac{E_0}{c}(\hat{x} - \hat{y}) \cos(kz - \omega t)$

4. The magnetic field of a plane electromagnetic wave is

$$\vec{B} = 3 \times 10^{-8} \sin[200\pi(y + ct)] \hat{i} \text{ T}$$

where $c = 3 \times 10^8 \text{ ms}^{-1}$ is the speed of light.

The corresponding electric field is: [Sep. 03, 2020 (I)]

- (a) $\vec{E} = 9 \sin[200\pi(y + ct)] \hat{k} \text{ V/m}$
 (b) $\vec{E} = -10^{-6} \sin[200\pi(y + ct)] \hat{k} \text{ V/m}$
 (c) $\vec{E} = 3 \times 10^{-8} \sin[200\pi(y + ct)] \hat{k} \text{ V/m}$
 (d) $\vec{E} = -9 \sin[200\pi(y + ct)] \hat{k} \text{ V/m}$

5. The electric field of a plane electromagnetic wave propagating along the x direction in vacuum is

$\vec{E} = E_0 \hat{j} \cos(\omega t - kx)$. The magnetic field $\vec{B}$, at the moment $t = 0$ is: [Sep. 03, 2020 (II)]

- (a) $\vec{B} = \frac{E_0}{\sqrt{\mu_0 \epsilon_0}} \cos(kx) \hat{k}$
 (b) $\vec{B} = E_0 \sqrt{\mu_0 \epsilon_0} \cos(kx) \hat{j}$
 (c) $\vec{B} = E_0 \sqrt{\mu_0 \epsilon_0} \cos(kx) \hat{k}$
 (d) $\vec{B} = \frac{E_0}{\sqrt{\mu_0 \epsilon_0}} \cos(kx) \hat{j}$

6. A plane electromagnetic wave, has frequency of $2.0 \times 10^{10} \text{ Hz}$ and its energy density is $1.02 \times 10^{-8} \text{ J/m}^3$ in vacuum. The amplitude of the magnetic field of the wave is close to

$(\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \frac{\text{Nm}^2}{\text{C}^2} \text{ and speed of light } = 3 \times 10^8 \text{ ms}^{-1})$:

[Sep. 02, 2020 (I)]

- (a) 150 nT (b) 160 nT
 (c) 180 nT (d) 190 nT

7. In a plane electromagnetic wave, the directions of electric field and magnetic field are represented by $\hat{k}$ and $2\hat{i} - 2\hat{j}$, respectively. What is the unit vector along direction of propagation of the wave. [Sep. 02, 2020 (II)]

(a) $\frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$ (b) $\frac{1}{\sqrt{2}}(\hat{j} + \hat{k})$

(c) $\frac{1}{\sqrt{5}}(\hat{i} + 2\hat{j})$ (d) $\frac{1}{\sqrt{5}}(2\hat{i} + \hat{j})$

8. The electric fields of two plane electromagnetic plane waves in vacuum are given by

$$\vec{E}_1 = E_0 \hat{j} \cos(\omega t - kx) \text{ and } \vec{E}_2 = E_0 \hat{k} \cos(\omega t - ky).$$

At $t = 0$, a particle of charge q is at origin with a velocity $\vec{v} = 0.8c\hat{j}$ (c is the speed of light in vacuum). The instantaneous force experienced by the particle is:

[9 Jan 2020, I]

(a) $E_0 q(0.8\hat{i} - \hat{j} + 0.4\hat{k})$ (b) $E_0 q(0.4\hat{i} - 3\hat{j} + 0.8\hat{k})$

(c) $E_0 q(-0.8\hat{i} + \hat{j} + \hat{k})$ (d) $E_0 q(0.8\hat{i} + \hat{j} + 0.2\hat{k})$

9. A plane electromagnetic wave is propagating along the direction $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$, with its polarization along the direction

$\hat{k}$. The correct form of the magnetic field of the wave would be (here B_0 is an appropriate constant):

[9 Jan 2020, II]

(a) $B_0 \frac{\hat{i} - \hat{j}}{\sqrt{2}} \cos\left(\omega t - k \frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$

(b) $B_0 \frac{\hat{j} - \hat{i}}{\sqrt{2}} \cos\left(\omega t + k \frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$

(c) $B_0 \hat{k} \cos\left(\omega t - k \frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$

(d) $B_0 \frac{\hat{i} + \hat{j}}{\sqrt{2}} \cos\left(\omega t - k \frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$

10. A plane electromagnetic wave of frequency 25 GHz is propagating in vacuum along the z -direction. At a particular point in space and time, the magnetic field is given by $\vec{B} = 5 \times 10^{-8} \hat{j} \text{ T}$. The corresponding electric field $\vec{E}$ is (speed of light $c = 3 \times 10^8 \text{ ms}^{-1}$)

[8 Jan 2020, II]

(a) $1.66 \times 10^{-16} \hat{i} \text{ V/m}$ (b) $-1.66 \times 10^{-16} \hat{i} \text{ V/m}$

(c) $-15 \hat{i} \text{ V/m}$ (d) $15 \hat{i} \text{ V/m}$

11. If the magnetic field in a plane electromagnetic wave is given by $\vec{B} = 3 \times 10^{-8} \sin(1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{j} \text{ T}$, then what will be expression for electric field?

[7 Jan 2020, I]

(a) $\vec{E} = (60 \sin(1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{k} \text{ v/m})$

(b) $\vec{E} = (9 \sin(1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{k} \text{ v/m})$

(c) $\vec{E} = (3 \times 10^{-8} \sin(1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{k} \text{ v/m})$

(d) $\vec{E} = (3 \times 10^{-8} \sin(1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{k} \text{ v/m})$

12. The electric field of a plane electromagnetic wave is given by

$$\vec{E} = E_0 \frac{\hat{i} + \hat{j}}{\sqrt{2}} \cos(kz + \omega t)$$

At $t = 0$, a positively charged particle is at the point

$$(x, y, z) = \left(0, 0, \frac{\pi}{k}\right).$$

If its instantaneous velocity at $(t = 0)$ is $v_0 \hat{k}$, the force acting on it due to the wave is:

[7 Jan 2020, II]

(a) parallel to $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$ (b) zero

(c) antiparallel to $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$ (d) parallel to $\hat{k}$

13. An electromagnetic wave is represented by the electric field $\vec{E} = E_0 \hat{n} \sin[\omega t + (6y - 8z)]$. Taking unit vectors in x , y and z directions to be $\hat{i}, \hat{j}, \hat{k}$, the direction of propagation $\hat{s}$ is :

[12 April 2019, I]

(a) $\hat{s} = \frac{3\hat{i} - 4\hat{j}}{5}$ (b) $\hat{s} = \frac{-4\hat{k} + 3\hat{j}}{5}$

(c) $\hat{s} = \left(\frac{-3\hat{j} + 4\hat{k}}{5}\right)$ (d) $\hat{s} = \frac{3\hat{j} - 3\hat{k}}{5}$

14. A plane electromagnetic wave having a frequency $\nu = 23.9 \text{ GHz}$ propagates along the positive z -direction in free space. The peak value of the Electric Field is 60 V/m . Which among the following is the acceptable magnetic field component in the electromagnetic wave?

[12 April 2019, II]

(a) $\vec{B} = 2 \times 10^{-7} \sin(0.5 \times 10^3 z + 1.5 \times 10^{11} t) \hat{i}$

(b) $\vec{B} = 2 \times 10^{-7} \sin(0.5 \times 10^3 z - 1.5 \times 10^{11} t) \hat{i}$

(c) $\vec{B} = 60 \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t) \hat{k}$

(d) $\vec{B} = 2 \times 10^{-7} \sin(1.5 \times 10^2 x + 0.5 \times 10^{11} t) \hat{j}$

15. The electric field of a plane electromagnetic wave is given by $\vec{E} = E_0 \hat{i} \cos(kz) \cos(\omega t)$

The corresponding magnetic field is then given by :

[10 April 2019, I]

(a) $\vec{B} = \frac{E_0}{C} \hat{j} \sin(kz) \sin(\omega t)$

(b) $\vec{B} = \frac{E_0}{C} \hat{j} \sin(kz) \cos(\omega t)$

(c) $\vec{B} = \frac{E_0}{C} \hat{j} \cos(kz) \sin(\omega t)$

(d) $\vec{B} = \frac{E_0}{C} \hat{k} \sin(kz) \cos(\omega t)$

16. Light is incident normally on a completely absorbing surface with an energy flux of 25 W cm^{-2} . If the surface has an area of 25 cm^2 , the momentum transferred to the surface in 40 min time duration will be: [10 April 2019, II]
 (a) $6.3 \times 10^{-4} \text{ N s}$ (b) $1.4 \times 10^{-6} \text{ N s}$
 (c) $5.0 \times 10^{-3} \text{ N s}$ (d) $3.5 \times 10^{-6} \text{ N s}$
17. The magnetic field of a plane electromagnetic wave is given by:
 $\vec{B} = B_0 \hat{i} [\cos(kz - \omega t)] + B_1 \hat{j} \cos(kz + \omega t)$
 Where $B_0 = 3 \times 10^{-5} \text{ T}$ and $B_1 = 2 \times 10^{-6} \text{ T}$.
 The rms value of the force experienced by a stationary charge $Q = 10^{-4} \text{ C}$ at $z = 0$ is closest to: [9 April 2019 I]
 (a) 0.6 N (b) 0.1 N
 (c) 0.9 N (d) $3 \times 10^{-2} \text{ N}$
18. A plane electromagnetic wave of frequency 50 MHz travels in free space along the positive x -direction. At a particular point in space and time, $\vec{E} = 6.3 \hat{j} \text{ V/m}$. The corresponding magnetic field $\vec{B}$, at that point will be: [9 April 2019 I]
 (a) $18.9 \times 10^{-8} \hat{k} \text{ T}$ (b) $2.1 \times 10^{-8} \hat{k} \text{ T}$
 (c) $6.3 \times 10^{-8} \hat{k} \text{ T}$ (d) $18.9 \times 10^8 \hat{k} \text{ T}$
19. 50 W/m^2 energy density of sunlight is normally incident on the surface of a solar panel. Some part of incident energy (25%) is reflected from the surface and the rest is absorbed. The force exerted on 1 m^2 surface area will be close to ($c = 3 \times 10^8 \text{ m/s}$): [9 April 2019, II]
 (a) $15 \times 10^{-8} \text{ N}$ (b) $20 \times 10^{-8} \text{ N}$
 (c) $10 \times 10^{-8} \text{ N}$ (d) $35 \times 10^{-8} \text{ N}$
20. A plane electromagnetic wave travels in free space along the x -direction. The electric field component of the wave at a particular point of space and time is $E = 6 \text{ Vm}^{-1}$ along y -direction. Its corresponding magnetic field component, B would be: [8 April 2019 I]
 (a) $2 \times 10^{-8} \text{ T}$ along z -direction
 (b) $6 \times 10^{-8} \text{ T}$ along x -direction
 (c) $6 \times 10^{-8} \text{ T}$ along z -direction
 (d) $2 \times 10^{-8} \text{ T}$ along y -direction
21. The magnetic field of an electromagnetic wave is given by:
 $\vec{B} = 1.6 \times 10^{-6} \cos(2 \times 10^7 z + 6 \times 10^{15} t) (2\hat{i} + \hat{j}) \frac{\text{Wb}}{\text{m}^2}$
 The associated electric field will be: [8 April 2019, II]
 (a) $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z - 6 \times 10^{15} t) (2\hat{i} + \hat{j}) \frac{\text{V}}{\text{m}}$
 (b) $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z - 6 \times 10^{15} t) (-2\hat{j} + \hat{i}) \frac{\text{V}}{\text{m}}$
 (c) $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z + 6 \times 10^{15} t) (-\hat{i} + 2\hat{j}) \frac{\text{V}}{\text{m}}$
 (d) $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z + 6 \times 10^{15} t) (\hat{i} - 2\hat{j}) \frac{\text{V}}{\text{m}}$
22. The mean intensity of radiation on the surface of the Sun is about 10^8 W/m^2 . The rms value of the corresponding magnetic field is closest to: [12 Jan 2019, II]
 (a) 1 T (b) 10^2 T (c) 10^{-2} T (d) 10^{-4} T
23. An electromagnetic wave of intensity 50 Wm^{-2} enters in a medium of refractive index ' n ' without any loss. The ratio of the magnitudes of electric fields, and the ratio of the magnitudes of magnetic fields of the wave before and after entering into the medium are respectively, given by: [11 Jan 2019, I]
 (a) $\left(\frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}}\right)$ (b) $(\sqrt{n}, \sqrt{n})$
 (c) $\left(\sqrt{n}, \frac{1}{\sqrt{n}}\right)$ (d) $\left(\frac{1}{\sqrt{n}}, \sqrt{n}\right)$
24. A 27 mW laser beam has a cross-sectional area of 10 mm^2 . The magnitude of the maximum electric field in this electromagnetic wave is given by:
 [Given permittivity of space $\epsilon_0 = 9 \times 10^{-12} \text{ SI units}$, Speed of light $c = 3 \times 10^8 \text{ m/s}$] [11 Jan 2019, II]
 (a) 2 kV/m (c) 0.7 kV/m
 (b) 1 kV/m (d) 1.4 kV/m
25. If the magnetic field of a plane electromagnetic wave is given by (The speed of light = $3 \times 10^8 \text{ m/s}$)
 $B = 100 \times 10^{-6} \sin \left[2\pi \times 2 \times 10^{15} \left(t - \frac{x}{c} \right) \right]$
 then the maximum electric field associated with it is: [10 Jan. 2019 I]
 (a) $6 \times 10^4 \text{ N/C}$ (b) $3 \times 10^4 \text{ N/C}$
 (c) $4 \times 10^4 \text{ N/C}$ (d) $4.5 \times 10^4 \text{ N/C}$
26. The electric field of a plane polarized electromagnetic wave in free space at time $t = 0$ is given by an expression $\vec{E}(x, y) = 10 \hat{j} \cos[(6x + 8z)]$
 The magnetic field $\vec{B}(x, z, t)$ is given by: (c is the velocity of light) [10 Jan 2019, II]
 (a) $\frac{1}{c} (6\hat{k} + 8\hat{i}) \cos[(6x - 8z + 10ct)]$
 (b) $\frac{1}{c} (6\hat{k} - 8\hat{i}) \cos[(6x + 8z - 10ct)]$
 (c) $\frac{1}{c} (6\hat{k} + 8\hat{i}) \cos[(6x + 8z - 10ct)]$
 (d) $\frac{1}{c} (6\hat{k} - 8\hat{i}) \cos[(6x + 8z + 10ct)]$
27. An EM wave from air enters a medium. The electric fields are $\vec{E}_1 = E_{01} \hat{x} \cos \left[2\pi \nu \left(\frac{z}{c} - t \right) \right]$ in air and $\vec{E}_2 = E_{02} \hat{x} \cos[k(2z - ct)]$ in medium, where the wave number k and frequency ν refer to their values in air. The medium is nonmagnetic. If ϵ_1 and ϵ_2 refer to relative permittivities of air and medium respectively, which of the following options is correct? [9 Jan 2019, I]
 (a) $\frac{\epsilon_1}{\epsilon_2} = 4$ (b) $\frac{\epsilon_1}{\epsilon_2} = 2$
 (c) $\frac{\epsilon_1}{\epsilon_2} = \frac{1}{4}$ (d) $\frac{\epsilon_1}{\epsilon_2} = \frac{1}{2}$

28. The energy associated with electric field is (U_E) and with magnetic fields is (U_B) for an electromagnetic wave in free space. Then : **[9 Jan 2019, II]**

(a) $U_E = \frac{U_B}{2}$ (b) $U_E > U_B$
 (c) $U_E < U_B$ (d) $U_E = U_B$

29. A plane electromagnetic wave of wavelength λ has an intensity I . It is propagating along the positive Y-direction. The allowed expressions for the electric and magnetic fields are given by **[Online April 16, 2018]**

(a) $\vec{E} = \sqrt{\frac{I}{\epsilon_0 C}} \cos\left[\frac{2\pi}{\lambda}(y - ct)\right] \hat{i}; \vec{B} = \frac{1}{c} E \hat{k}$
 (b) $\vec{E} = \sqrt{\frac{I}{\epsilon_0 C}} \cos\left[\frac{2\pi}{\lambda}(y - ct)\right] \hat{k}; \vec{B} = -\frac{1}{c} E \hat{i}$
 (c) $\vec{E} = \sqrt{\frac{2I}{\epsilon_0 C}} \cos\left[\frac{2\pi}{\lambda}(y - ct)\right] \hat{k}; \vec{B} = +\frac{1}{c} E \hat{i}$
 (d) $\vec{E} = \sqrt{\frac{2I}{\epsilon_0 C}} \cos\left[\frac{2\pi}{\lambda}(y + ct)\right] \hat{k}; \vec{B} = \frac{1}{c} E \hat{i}$

30. A monochromatic beam of light has a frequency $\nu = \frac{3}{2\pi} \times 10^{12}$ Hz and is propagating along the direction $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$. It is polarized along the $\hat{k}$ direction. The acceptable form for the magnetic field is: **[Online April 15, 2018]**

(a) $k \frac{E_0}{C} \left(\frac{\hat{i} - \hat{j}}{\sqrt{2}}\right) \cos\left[10^4 \left(\frac{\hat{i} - \hat{j}}{\sqrt{2}}\right) \cdot \vec{r} - (3 \times 10^{12})t\right]$
 (b) $\frac{E_0}{C} \left(\frac{\hat{i} - \hat{j}}{\sqrt{2}}\right) \cos\left[10^4 \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}}\right) \cdot \vec{r} - (3 \times 10^{12})t\right]$
 (c) $\frac{E_0}{C} \hat{k} \cos\left[10^4 \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}}\right) \cdot \vec{r} + (3 \times 10^{12})t\right]$
 (d) $\frac{E_0}{C} \frac{(\hat{i} + \hat{j} + \hat{k})}{\sqrt{3}} \cos\left[10^4 \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}}\right) \cdot \vec{r} + (3 \times 10^{12})t\right]$

31. The electric field component of a monochromatic radiation is given by

$$\vec{E} = 2 E_0 \hat{i} \cos kz \cos \omega t$$

Its magnetic field $\vec{B}$ is then given by :

[Online April 9, 2017]

(a) $\frac{2E_0}{c} \hat{j} \sin kz \cos \omega t$ (b) $-\frac{2E_0}{c} \hat{j} \sin kz \sin \omega t$
 (c) $\frac{2E_0}{c} \hat{j} \sin kz \sin \omega t$ (d) $\frac{2E_0}{c} \hat{j} \cos kz \cos \omega t$

32. Magnetic field in a plane electromagnetic wave is given by $\vec{B} = B_0 \sin(kx + \omega t) \hat{j}$ T

Expression for corresponding electric field will be :

Where c is speed of light. **[Online April 8, 2017]**

(a) $\vec{E} = B_0 c \sin(kx + \omega t) \hat{k}$ V/m
 (b) $\vec{E} = \frac{B_0}{c} \sin(kx + \omega t) \hat{k}$ V/m
 (c) $\vec{E} = -B_0 c \sin(kx + \omega t) \hat{k}$ V/m
 (d) $\vec{E} = B_0 c \sin(kx - \omega t) \hat{k}$ V/m

33. Consider an electromagnetic wave propagating in vacuum. Choose the correct statement : **[Online April 10, 2016]**

(a) For an electromagnetic wave propagating in +y direction the electric field is $\vec{E} = \frac{1}{\sqrt{2}} E_{yz}(x, t) \hat{z}$ and

the magnetic field is $\vec{B} = \frac{1}{\sqrt{2}} B_z(x, t) \hat{y}$

(b) For an electromagnetic wave propagating in +y direction the electric field is $\vec{E} = \frac{1}{\sqrt{2}} E_{yz}(x, t) \hat{y}$ and

the magnetic field is $\vec{B} = \frac{1}{\sqrt{2}} B_{yz}(x, t) \hat{z}$

(c) For an electromagnetic wave propagating in +x direction the electric field is $\vec{E} = \frac{1}{\sqrt{2}} E_{yz}(y, z, t)$

$(\hat{y} + \hat{z})$ and the magnetic field is

$$\vec{B} = \frac{1}{\sqrt{2}} B_{yz}(y, z, t) (\hat{y} + \hat{z})$$

(d) For an electromagnetic wave propagating in +x direction the electric field is $\vec{E} = \frac{1}{\sqrt{2}} E_{yz}(x, t) (\hat{y} - \hat{z})$

and the magnetic field is $\vec{B} = \frac{1}{\sqrt{2}} B_{yz}(x, t) (\hat{y} + \hat{z})$

34. For plane electromagnetic waves propagating in the z-direction, which one of the following combination gives the correct possible direction for $\vec{E}$ and $\vec{B}$ field respectively? **[Online April 11, 2015]**

(a) $(2\hat{i} + 3\hat{j})$ and $(\hat{i} + 2\hat{j})$ (b) $(-2\hat{i} - 3\hat{j})$ and $(3\hat{i} - 2\hat{j})$
 (c) $(3\hat{i} + 4\hat{j})$ and $(4\hat{i} - 3\hat{j})$ (d) $(\hat{i} + 2\hat{j})$ and $(2\hat{i} - \hat{j})$

35. An electromagnetic wave travelling in the x-direction has frequency of 2×10^{14} Hz and electric field amplitude of 27 Vm^{-1} . From the options given below, which one describes the magnetic field for this wave ? **[Online April 10, 2015]**

(a) $\vec{B}(x, t) = (3 \times 10^{-8} \text{ T}) \hat{j} \sin[2\pi(1.5 \times 10^{-8} x - 2 \times 10^{14} t)]$
 (b) $\vec{B}(x, t) = (9 \times 10^{-8} \text{ T}) \hat{i} \sin[2\pi(1.5 \times 10^{-8} x - 2 \times 10^{14} t)]$
 (c) $\vec{B}(x, t) = (9 \times 10^{-8} \text{ T}) \hat{j} \sin[1.5 \times 10^{-6} x - 2 \times 10^{14} t]$
 (d) $\vec{B}(x, t) = (9 \times 10^{-8} \text{ T}) \hat{k} \sin[2\pi(1.5 \times 10^{-6} x - 2 \times 10^{14} t)]$

36. During the propagation of electromagnetic waves in a medium: [2014]
 (a) Electric energy density is double of the magnetic energy density.
 (b) Electric energy density is half of the magnetic energy density.
 (c) Electric energy density is equal to the magnetic energy density.
 (d) Both electric and magnetic energy densities are zero.
37. A lamp emits monochromatic green light uniformly in all directions. The lamp is 3% efficient in converting electrical power to electromagnetic waves and consumes 100 W of power. The amplitude of the electric field associated with the electromagnetic radiation at a distance of 5 m from the lamp will be nearly: [Online April 12, 2014]
 (a) 1.34 V/m (b) 2.68 V/m
 (c) 4.02 V/m (d) 5.36 V/m
38. An electromagnetic wave of frequency 1×10^{14} hertz is propagating along z-axis. The amplitude of electric field is 4 V/m. If $\epsilon_0 = 8.8 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2$, then average energy density of electric field will be: [Online April 11, 2014]
 (a) $35.2 \times 10^{-10} \text{ J/m}^3$ (b) $35.2 \times 10^{-11} \text{ J/m}^3$
 (c) $35.2 \times 10^{-12} \text{ J/m}^3$ (d) $35.2 \times 10^{-13} \text{ J/m}^3$
39. The magnetic field in a travelling electromagnetic wave has a peak value of 20 nT. The peak value of electric field strength is : [2013]
 (a) 3 V/m (b) 6 V/m (c) 9 V/m (d) 12 V/m
40. A plane electromagnetic wave in a non-magnetic dielectric medium is given by $\vec{E} = \vec{E}_0 (4 \times 10^{-7} x - 50t)$ with distance being in meter and time in seconds. The dielectric constant of the medium is : [Online April 22, 2013]
 (a) 2.4 (b) 5.8 (c) 8.2 (d) 4.8
41. Select the correct statement from the following : [Online April 9, 2013]
 (a) Electromagnetic waves cannot travel in vacuum.
 (b) Electromagnetic waves are longitudinal waves.
 (c) Electromagnetic waves are produced by charges moving with uniform velocity.
 (d) Electromagnetic waves carry both energy and momentum as they propagate through space.
42. An electromagnetic wave in vacuum has the electric and magnetic field $\vec{E}$ and $\vec{B}$, which are always perpendicular to each other. The direction of polarization is given by $\vec{X}$ and that of wave propagation by $\vec{k}$. Then [2012]
 (a) $\vec{X} \parallel \vec{B}$ and $\vec{k} \parallel \vec{B} \times \vec{E}$
 (b) $\vec{X} \parallel \vec{E}$ and $\vec{k} \parallel \vec{E} \times \vec{B}$
 (c) $\vec{X} \parallel \vec{B}$ and $\vec{k} \parallel \vec{E} \times \vec{B}$
 (d) $\vec{X} \parallel \vec{E}$ and $\vec{k} \parallel \vec{B} \times \vec{E}$
43. An electromagnetic wave with frequency ω and wavelength λ travels in the +y direction. Its magnetic field is along +x-axis. The vector equation for the associated electric field (of amplitude E_0) is [Online May 19, 2012]
 (a) $\vec{E} = -E_0 \cos\left(\omega t + \frac{2\pi}{\lambda} y\right) \hat{x}$

$$(b) \vec{E} = E_0 \cos\left(\omega t - \frac{2\pi}{\lambda} y\right) \hat{x}$$

$$(c) \vec{E} = E_0 \cos\left(\omega t - \frac{2\pi}{\lambda} y\right) \hat{z}$$

$$(d) \vec{E} = -E_0 \cos\left(\omega t + \frac{2\pi}{\lambda} y\right) \hat{z}$$

44. An electromagnetic wave of frequency $\nu = 3.0 \text{ MHz}$ passes from vacuum into a dielectric medium with permittivity $\epsilon = 4.0$. Then [2004]
 (a) wave length is halved and frequency remains unchanged
 (b) wave length is doubled and frequency becomes half
 (c) wave length is doubled and the frequency remains unchanged
 (d) wave length and frequency both remain unchanged.
45. Electromagnetic waves are transverse in nature is evident by [2002]
 (a) polarization (b) interference
 (c) reflection (d) diffraction

TOPIC 2 Electromagnetic Spectrum



46. The correct match between the entries in column I and column II are : [Sep. 05, 2020 (II)]

I	II
Radiation	Wavelength
(A) Microwave	(i) 100m
(B) Gamma rays	(ii) 10^{-15} m
(C) A.M. radio waves	(iii) 10^{-10} m
(D) X-rays	(iv) 10^{-3} m
(a) (A)-(ii), (B)-(i), (C)-(iv), (D)-(iii)	
(b) (A)-(i), (B)-(iii), (C)-(iv), (D)-(ii)	
(c) (A)-(iii), (B)-(ii), (C)-(i), (D)-(iv)	
(d) (A)-(iv), (B)-(ii), (C)-(i), (D)-(iii)	

47. Chosse the **correct** option relating wavelengths of different parts of electromagnetic wave spectrum : [Sep. 04, 2020 (I)]

- (a) $\lambda_{\text{visible}} < \lambda_{\text{micro waves}} < \lambda_{\text{radio waves}} < \lambda_{\text{x-rays}}$
 (b) $\lambda_{\text{radio waves}} > \lambda_{\text{micro waves}} > \lambda_{\text{visible}} > \lambda_{\text{x-rays}}$
 (c) $\lambda_{\text{x-rays}} < \lambda_{\text{micro waves}} < \lambda_{\text{radio waves}} < \lambda_{\text{visible}}$
 (d) $\lambda_{\text{visible}} > \lambda_{\text{x-rays}} > \lambda_{\text{radio waves}} > \lambda_{\text{micro waves}}$

48. Given below in the left column are different modes of communication using the kinds of waves given in the right column. [10 April 2019, I]

A. Optical Fibre Communication	P. Ultrasound
B. Radar	Q. Infrared Light
C. Sonar	R. Microwaves
D. Mobile Phones	S. Radio Waves

From the options given below, find the most appropriate match between entries in the left and the right column.

- (a) A – Q, B – S, C – R, D – P
 (b) A – S, B – Q, C – R, D – P
 (c) A – Q, B – S, C – P, D – R
 (d) A – R, B – P, C – S, D – Q

49. Arrange the following electromagnetic radiations per quantum in the order of increasing energy : [2016]

A : Blue light B : Yellow light

C : X-ray D : Radiowave.

- (a) C, A, B, D (b) B, A, D, C
 (c) D, B, A, C (d) A, B, D, C

50. Microwave oven acts on the principle of :

[Online April 9, 2016]

- (a) giving rotational energy to water molecules
 (b) giving translational energy to water molecules
 (c) giving vibrational energy to water molecules
 (d) transferring electrons from lower to higher energy levels in water molecule

51. Match List - I (Electromagnetic wave type) with List - II (Its association/application) and select the correct option from the choices given below the lists: [2014]

List 1	List 2
1. Infrared waves	(i) To treat muscular strain
2. Radio waves	(ii) For broadcasting
3. X-rays	(iii) To detect fracture of bones
4. Ultraviolet rays	(iv) Absorbed by the ozone layer of the atmosphere

1	2	3	4
(a) (iv)	(iii)	(ii)	(i)
(b) (i)	(ii)	(iv)	(iii)
(c) (iii)	(ii)	(i)	(iv)
(d) (i)	(ii)	(iii)	(iv)

52. If microwaves, X rays, infrared, gamma rays, ultra-violet, radio waves and visible parts of the electromagnetic spectrum are denoted by M, X, I, G, U, R and V then which of the following is the arrangement in ascending order of wavelength ? [Online April 19, 2014]

- (a) R, M, I, V, U, X and G
 (b) M, R, V, X, U, G and I
 (c) G, X, U, V, I, M and R
 (d) I, M, R, U, V, X and G

53. Match the List-I (Phenomenon associated with electromagnetic radiation) with List-II (Part of electromagnetic spectrum) and select the correct code from the choices given below this lists: [Online April 11, 2014]

List I		List II	
I	Doublet of sodium	(A)	Visible radiation
II	Wavelength corresponding to temperature associated with the isotropic radiation filling all space	(B)	Microwave
III	Wavelength emitted by atomic hydrogen in interstellar space	(C)	Short radio wave
IV	Wavelength of radiation arising from two close energy levels in hydrogen	(D)	X-rays

- (a) (I)-(A), (II)-(B), (III)-(B), (IV)-(C)
 (b) (I)-(A), (II)-(B), (III)-(C), (IV)-(C)
 (c) (I)-(D), (II)-(C), (III)-(A), (IV)-(B)
 (d) (I)-(B), (II)-(A), (III)-(D), (IV)-(A)

54. Match List I (Wavelength range of electromagnetic spectrum) with List II (Method of production of these waves) and select the correct option from the options given below the lists. [Online April 9, 2014]

List I		List II	
(1)	700 nm to 1 mm	(i)	Vibration of atoms and molecules.
(2)	1 nm to 400 nm	(ii)	Inner shell electrons in atoms moving from one energy level to a lower level.
(3)	$< 10^{-3}$ nm	(iii)	Radioactive decay of the nucleus.
(4)	1 mm to 0.1 m	(iv)	Magnetron valve.

- (a) (1)-(iv), (2)-(iii), (3)-(ii), (4)-(i)
 (b) (1)-(iii), (2)-(iv), (3)-(i), (4)-(ii)
 (c) (1)-(ii), (2)-(iii), (3)-(iv), (4)-(i)
 (d) (1)-(i), (2)-(ii), (3)-(iii), (4)-(iv)

55. Photons of an electromagnetic radiation has an energy 11 keV each. To which region of electromagnetic spectrum does it belong ? [Online April 9, 2013]

- (a) X-ray region (b) Ultra violet region
 (c) Infrared region (d) Visible region

56. The frequency of X-rays; γ -rays and ultraviolet rays are respectively a , b and c then [Online May 26, 2012]

- (a) $a < b$; $b > c$ (b) $a > b$; $b > c$
 (c) $a < b < c$ (d) $a = b = c$



Hints & Solutions



1. (a) Relation between electric field E_0 and magnetic field B_0 of an electromagnetic wave is given by

$$c = \frac{E_0}{B_0} \quad (\text{Here, } c = \text{Speed of light})$$

$$\Rightarrow E_0 = B_0 \times c = 1.2 \times 10^{-7} \times 3 \times 10^8 = 36$$

As the wave is propagating along x -direction, magnetic field is along z -direction

$$\text{and } (\hat{E} \times \hat{B}) \parallel \hat{C}$$

$\therefore \vec{E}$ should be along y -direction.

So, electric field $\vec{E} = E_0 \sin \vec{E} \cdot (x, t)$

$$= [-36 \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t) \hat{j}] \frac{V}{m}$$

2. (c) In electromagnetic wave, $\frac{E_0}{B_0} = C$

$$\therefore \text{Maximum value of magnetic field, } B_0 = \frac{E_0}{C}$$

$$F_{\max} = qVB_{\max} \sin 90^\circ = \frac{qV_0 E_0}{C}$$

(Given $V_0 = 0.1$ C and $E_0 = 30$)

$$= \frac{1.6 \times 10^{-19} \times 0.1 \times 3 \times 10^8 \times 30}{3 \times 10^8} = 4.8 \times 10^{-19} \text{ N}$$

3. (a) $\vec{E} = E_0(\hat{x} + \hat{y}) \sin(kz - \omega t)$

Direction of propagation of em wave $= +\hat{k}$

$$\text{Unit vector in the direction of electric field, } \hat{E} = \frac{\hat{i} + \hat{j}}{\sqrt{2}}$$

The direction of electromagnetic wave is perpendicular to both electric and magnetic field.

$$\therefore \hat{k} = \hat{E} \times \hat{B}$$

$$\Rightarrow \hat{k} = \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} \right) \times \hat{B} \Rightarrow \hat{B} = \frac{-\hat{i} + \hat{j}}{\sqrt{2}}$$

$$\therefore \vec{B} = \frac{E_0}{c} (-\hat{x} + \hat{y}) \sin(kz - \omega t)$$

4. (d) Given: $\vec{B} = 3 \times 10^{-8} \sin[200\pi(y + ct)] \hat{i} T$

$$\therefore B_0 = 3 \times 10^{-8}$$

$$E_0 = CB_0 \Rightarrow E_0 = 3 \times 10^8 \times 3 \times 10^{-8} = 9 \text{ V/m}$$

Direction of wave propagation

$$(\vec{E} \times \vec{B}) \parallel \vec{C}$$

$$\hat{B} = \hat{i} \text{ and } \hat{C} = -\hat{j} \therefore \hat{E} = -\hat{k}$$

$$\therefore \vec{E} = E_0 \sin[200\pi(y + ct)](-\hat{k}) \text{ V/m}$$

$$\text{or, } \vec{E} = -9 \sin[200\pi(y + ct)] \hat{k} \text{ V/m}$$

5. (c) Relation between electric field and magnetic field for an electromagnetic wave in vacuum is $B_0 = \frac{E_0}{c}$.

$$\text{In free space, its speed } c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

Here, μ_0 = absolute permeability, ϵ_0 = absolute permittivity

$$\therefore B_0 = \frac{E_0}{c} = \frac{E_0}{1/\sqrt{\mu_0 \epsilon_0}} = E_0 \sqrt{\mu_0 \epsilon_0}$$

As the electromagnetic wave is propagating along x direction and electric field is along y direction.

$$\therefore \hat{E} \times \hat{B} \parallel \hat{C} \text{ (Here, } \hat{C} = \text{direction of propagation of wave)}$$

$$\therefore \vec{B} \text{ should be in } \hat{k} \text{ direction.}$$

$$\therefore B = E_0 \sqrt{\mu_0 \epsilon_0} \cos(\omega t - kx) \hat{k}$$

At $t = 0$

$$B = E_0 \sqrt{\mu_0 \epsilon_0} \cos(kx) \hat{k}$$

6. (b) Energy density $= \frac{1}{2} \frac{B^2}{\mu_0}$

$$\Rightarrow B = \sqrt{2 \times \mu_0 \times \text{Energy density}}$$

$$\mu_0 = \frac{1}{C^2 \epsilon_0} = 4\pi \times 10^{-7}$$

$$\therefore B = \sqrt{2 \times 4\pi \times 10^{-7} \times 1.02 \times 10^{-8}} = 160 \times 10^{-9} = 160 \text{ nT}$$

7. (a) Electromagnetic wave will propagate perpendicular to the direction of Electric and Magnetic fields

$$\hat{C} = \hat{E} \times \hat{B}$$

Here unit vector $\hat{C}$ is perpendicular to both $\hat{E}$ and $\hat{B}$

$$\text{Given, } \vec{E} = \hat{k}, \vec{B} = 2\hat{i} - 2\hat{j}$$

$$\therefore \hat{C} = \hat{E} \times \hat{B} = \frac{1}{\sqrt{2}} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 1 \\ 1 & -1 & 0 \end{vmatrix} = \frac{\hat{i} + \hat{j}}{\sqrt{2}}$$

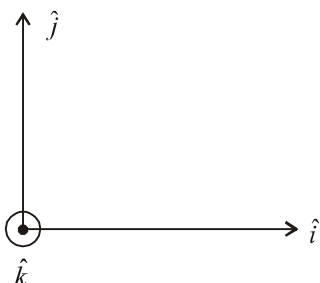
$$\Rightarrow \hat{C} = \frac{\hat{i} + \hat{j}}{\sqrt{2}}$$

8. (d) Given: $\vec{E}_1 = E_0 \hat{j} \cos(\omega t - kx)$

i.e., Travelling in +ve x-direction $\vec{E} \times \vec{B}$ should be in x-direction

$\therefore \vec{B}$ is in $\hat{k}$

$$\therefore \vec{B}_1 = \frac{E_0}{C} \cos(\omega t - kx) \hat{k} \quad \left(\because B_0 = \frac{E_0}{C} \right)$$



$$\vec{E}_2 = E_0 \hat{k} \cos(\omega t - ky)$$

$$\vec{B}_2 = \frac{E_0}{C} \hat{i} \cos(\omega t - ky)$$

$\therefore$ Travelling in +ve y-axis

$\vec{E} \times \vec{B}$ should be in y-axis

$\therefore$ Net force $\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})$

$$q(\vec{E}_1 + \vec{E}_2) + q(0.8c\hat{j} \times (\vec{B}_1 + \vec{B}_2))$$

If $t = 0$ and $x = y = 0$

$$\vec{E}_1 = E_0 \hat{j} \quad \vec{E}_2 = E_0 \hat{k}$$

$$\vec{B}_1 = \frac{E_0}{c} \hat{k} \quad \vec{B}_2 = \frac{E_0}{c} \hat{i}$$

$$\therefore \vec{F}_{\text{net}} = qE_0(\hat{j} + \hat{k}) + q \times 0.8c \times \frac{E_0}{C} \hat{j} \times (\hat{k} + \hat{i})$$

$$= qE_0(\hat{j} + \hat{k}) + 0.8qE_0(\hat{i} - \hat{k})$$

$$= qE_0(0.8\hat{i} + \hat{j} + 0.2\hat{k})$$

9. (a) Direction of polarisation = $\hat{E} = \hat{k}$

$$\text{Direction of propagation} = \hat{E} \times \hat{B} = \frac{\hat{i} + \hat{j}}{\sqrt{2}}$$

$$\text{But } \vec{E} \cdot \vec{B} = 0 \therefore \hat{B} = \frac{\hat{i} - \hat{j}}{\sqrt{2}}$$

10. (d) Amplitude of electric field (E) and Magnetic field (B) of an electromagnetic wave are related by the relation

$$\frac{E}{B} = c$$

$$\Rightarrow E = Bc$$

$$\Rightarrow E = 5 \times 10^{-8} \times 3 \times 10^8 = 15 \text{ N/C}$$

$$\Rightarrow \vec{E} = 15\hat{i} \text{ V/m}$$

11. (b) Given, $\vec{B} = 3 \times 10^{-8} \sin(1.6 \times 10^3 x + 48 \times 10^{10} t)$

$$\text{Using, } E_0 = B_0 \times C = 3 \times 10^{-8} \times 3 \times 10^8 = 9 \text{ V/m}$$

$\therefore$ Electric field,

$$\vec{E} = 9 \sin(1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{k} \text{ V/m}$$

12. (c) At $t = 0, z = \frac{\pi}{k}$

$$\therefore \vec{E} = \frac{E_0}{\sqrt{2}} (\hat{i} + \hat{j}) \cos[\pi] = -\frac{E_0}{\sqrt{2}} (\hat{i} + \hat{j})$$

$$\vec{F}_E = q\vec{E}$$

Force due to electric field will be in the direction $\frac{-(\hat{i} + \hat{j})}{\sqrt{2}}$

Force due to magnetic field is in direction

$q(\vec{v} \times \vec{B})$ and $\vec{v} \parallel \vec{k}$. Therefore, it is parallel to $\vec{E}$.

$$\Rightarrow \vec{F}_{\text{net}} = \vec{F}_E + \vec{F}_B \text{ is antiparallel to } \frac{\hat{i} + \hat{j}}{\sqrt{2}}$$

13. (c) $\hat{S} = \frac{6\hat{j} + 8\hat{k}}{\sqrt{6^2 + 8^2}} = \frac{-3\hat{j} + 4\hat{k}}{5}$

14. (b) $B_0 = \frac{E_0}{C} = \frac{60}{3 \times 10^8}$
 $= 20 \times 10^{-8} \text{ T} = 2 \times 10^{-7} \text{ T}$

$$K = \frac{\omega}{v} = \frac{2\pi f}{v} = \frac{2\pi \times 23.9 \times 10^9}{3 \times 10^8} = 500$$

$$\text{Therefore, } \vec{B} = B_0 \sin(kz - \omega t)$$

$$= 2 \times 10^{-7} \sin(0.5 \times 10^3 z - 1.5 \times 10^{11} t) \hat{i}$$

15. (a)

$$\frac{E_0}{B_0} = C$$

$$\Rightarrow B_0 = \frac{E_0}{C}$$

$$\text{Given that } \vec{E} = E_0 \cos(kz) \cos(\omega t) \hat{i}$$

$$\vec{E} = \frac{E_0}{2} [\cos(kz - \omega t) \hat{i} - \cos(kz + \omega t) \hat{i}]$$

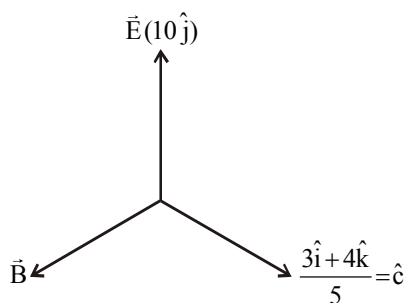
Correspondingly

$$\vec{B} = \frac{B_0}{2} [\cos(kz - \omega t) \hat{j} - \cos(kz + \omega t) \hat{j}]$$

$$\vec{B} = \frac{B_0}{2} \times 2 \sin kz \sin \omega t$$

$$\vec{B} = \left(\frac{E_0}{C} \sin kz \sin \omega t \right) \hat{j}$$

16. (c) Pressure, $P = \frac{I}{C}$
- $$\Rightarrow \frac{F}{A} = \frac{I}{C}$$
- $$\Rightarrow F = \frac{IA}{C} = \frac{\Delta p}{\Delta t}$$
- $$\Rightarrow \Delta p = \frac{I}{C} A \Delta t$$
- $$= \frac{(25 \times 25) \times 10^4 \times 10^{-4} \times 40 \times 60}{3 \times 10^8} \text{ N-s}$$
- $$= 5 \times 10^{-3} \text{ N-s}$$
17. (a) $B_0 = \sqrt{B_0^2 + B_1^2} = \sqrt{30^2 + 2^2} \times 10^{-6}$
- $$= 30 \times 10^{-6} \text{ T}$$
- $$\therefore E_0 = CB = 3 \times 10^8 \times 30 \times 10^{-6}$$
- $$= 9 \times 10^3 \text{ V/m}$$
- $$\frac{E_0}{V_2} = \frac{9}{\sqrt{2}} \times 10^3 \text{ V/m}$$
- Force on the charge,
- $$F = EQ = \frac{9}{\sqrt{2}} \times 10^3 \times 10^{-4} \simeq 0.64 \text{ N}$$
18. (b) As we know,
- $$|\vec{B}| = \frac{|\vec{E}|}{C} = \frac{6.3}{3 \times 10^8} = 2.1 \times 10^{-8} \text{ T}$$
- and $\hat{E} \times \hat{B} = \hat{C}$
- $$\hat{j} \times \hat{B} = \hat{i} \quad [\because \text{EM wave travels along } +(\text{ve}) x\text{-direction.}]$$
- $$\therefore \hat{B} = \hat{k} \text{ or } \vec{B} = 2.1 \times 10^{-8} \hat{k} \text{ T}$$
19. (b) $F = (1+r) \frac{IA}{C}$
- $$= \frac{(1+0.25) \times 50 \times 1}{3 \times 10^8}$$
- $$\simeq 20 \times 10^{-8} \text{ N}$$
20. (a) The relation between amplitudes of electric and magnetic field in free space is given by
- $$B_0 = \frac{E_0}{c} = \frac{6}{3 \times 10^8} = 2 \times 10^{-8} \text{ T}$$
- Propagation direction $= \hat{E} \times \hat{B}$
- $$\hat{i} = \hat{j} \times \hat{B}$$
- $$\Rightarrow \hat{B} = \hat{k}$$
- $$\therefore \text{The magnetic field component will be along } z \text{ direction.}$$
21. (c) $E_0 = cB_0 = 3 \times 10^8 \times 1.6 \times 10^{-6} = 4.8 \times 10^2 \text{ V/m}$
- Also $\vec{S} \Rightarrow \vec{E} \times \vec{B}$
- or $-\vec{K} \Rightarrow \vec{E} \times (2\hat{i} + \hat{j})$
- Therefore direction of $\vec{E} \rightarrow (-\hat{i} + 2\hat{j})$
22. (d) $I = \frac{B_0^2}{2\mu_0} \cdot C$
- $$\Rightarrow \frac{B_0^2}{2} = \frac{I\mu_0}{C}$$
- $$\Rightarrow B_{\text{rms}} = \sqrt{\frac{I\mu_0}{C}}$$
- $$= \sqrt{\frac{10^8 \times 4\pi \times 10^{-7}}{3 \times 10^8}}$$
- $$\simeq 6 \times 10^{-4} \text{ T}$$
- Which is closest to 10^{-4} .
23. (c) The speed of electromagnetic wave in free space is given by
- $$C = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \quad \dots(i)$$
- In medium, $v = \frac{1}{\sqrt{k \epsilon_0 \mu_0}} \quad \dots(ii)$
- Dividing equation (i) by (ii), we get
- $$\therefore \frac{C}{v} = \sqrt{k} = n$$
- $$\frac{1}{2} \epsilon_0 E_0^2 C = \text{intensity} = \frac{1}{2} \epsilon_0 k E^2 v$$
- $$\therefore E_0^2 C = k E^2 v$$
- $$\Rightarrow \frac{E_0^2}{E^2} = \frac{kV}{C} = \frac{n^2}{n} \Rightarrow \frac{E_0}{E} = \sqrt{n}$$
- similarly
- $$\frac{B_0^2 C}{2\mu_0} = \frac{B^2 v}{2\mu_0} \Rightarrow \frac{B_0}{B} = \frac{1}{\sqrt{n}}$$
24. (d) EM wave intensity
- $$\Rightarrow I = \frac{\text{Power}}{\text{Area}} = \frac{1}{2} \epsilon_0 E_0^2 c$$
- [where E_0 = maximum electric field]
- $$\Rightarrow \frac{27 \times 10^{-3}}{10 \times 10^{-6}} = \frac{1}{2} \times 9 \times 10^{-12} \times E_0^2 \times 3 \times 10^8$$
- $$\Rightarrow E_0 = \sqrt{2} \times 10^3 \text{ kV/m} = 1.4 \text{ kV/m}$$
25. (b) Using, formula $E_0 = B_0 \times C$
- $$= 100 \times 10^{-6} \times 3 \times 10^8$$
- $$= 3 \times 10^4 \text{ N/C}$$
- Here we assumed that $B_0 = 100 \times 10^{-6}$ is in tesla (T) units
26. (b) $\vec{E} = 10\hat{j} \cos[(6\hat{i} + 8\hat{k}) \cdot (x\hat{i} + z\hat{k})]$
- $$= 10\hat{j} \cos[\vec{K} \cdot \vec{r}]$$
- $$\therefore \vec{K} = 6\hat{i} + 8\hat{k}; \text{ direction of waves travel}$$
- i. e. direction of 'c'.



$$\hat{c} \times \hat{E} = \frac{-4\hat{i} + 3\hat{k}}{5}$$

$$\vec{B} = \frac{E}{C} = \frac{10}{C}$$

$$\therefore \vec{B} \frac{10}{C} \left(\frac{-4\hat{i} + 3\hat{k}}{5} \right) = \left(\frac{-8\hat{i} + 6\hat{k}}{C} \right)$$

$$\text{or, magnetic field } \vec{B}(x, z, t) = \frac{1}{C}$$

$$(6\hat{k} - 8\hat{i}) \cos(6x + 8z - 10ct)$$

27. (c) Velocity of EM wave is given by $v = \frac{1}{\sqrt{\mu \epsilon}}$

$$\text{Velocity in air} = \frac{\omega}{k} = C$$

$$\text{Velocity in medium} = \frac{C}{2}$$

Here, $\mu_1 = \mu_2 = 1$ as medium is non-magnetic

$$\therefore \frac{\frac{1}{\sqrt{\epsilon_1}}}{\frac{1}{\sqrt{\epsilon_2}}} = \frac{C}{\left(\frac{C}{2}\right)} = 2 \Rightarrow \frac{\epsilon_1}{\epsilon_2} = \frac{1}{4}$$

28. (d) Average energy density of magnetic field,

$$u_B = \frac{B_0^2}{4\mu_0}$$

Average energy density of electric field,

$$u_E = \frac{\epsilon_0 E_0^2}{4}$$

$$\text{Now, } E_0 = CB_0 \text{ and } C^2 = \frac{1}{\mu_0 \epsilon_0}$$

$$u_E = \frac{\epsilon_0}{4} \times C^2 B_0^2 = \frac{\epsilon_0}{4} \times \frac{1}{\mu_0 \epsilon_0} \times B_0^2 = \frac{B_0^2}{4\mu_0} = u_B$$

$$\therefore u_E = u_B$$

Since energy density of electric and magnetic field is same, so energy associated with equal volume will be equal i.e., $u_E = u_B$

29. (c) If E_0 is magnitude of electric field then

$$\frac{1}{2} \epsilon_0 E^2 \times C = 1 \Rightarrow E_0 = \sqrt{\frac{2I}{C \epsilon_0}}$$

$$E_0 = \frac{E_0}{C}$$

Direction of $\vec{E} \times \vec{B}$ will be along $+\hat{j}$.

30. (c) $\hat{E} \times \hat{B}$ should give the direction of wave propagation

$$\Rightarrow \vec{K} \times \vec{B} \parallel \frac{\hat{i} \times \hat{j}}{\sqrt{2}} \Rightarrow \vec{K} \times \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} \right) = \frac{\hat{j} - (-\hat{i})}{\sqrt{2}} = \frac{\hat{i} + \hat{j}}{\sqrt{2}} \parallel \frac{\hat{i} + \hat{j}}{\sqrt{2}}$$

Option (a), option (b) and option (d) does not satisfy.

Wave propagation vector $\vec{K}$ should along $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$.

31. (c) Given, Electric field component of monochromatic radiation, $(\vec{E}) = 2E_0 \hat{i} \cos kz \cos \omega t$

$$\text{We know that, } \frac{dE}{dz} = -\frac{dB}{dt}$$

$$\frac{dE}{dz} = -2E_0 k \sin kz \cos \omega t = -\frac{dB}{dt}$$

$$dB = +2E_0 k \sin kz \cos \omega t dt \quad \dots (i)$$

Integrating eqⁿ (i), we have

$$B = +2E_0 k \sin kz \int \cos \omega t dt$$

Magnetic field is given by,

$$= +2E_0 \frac{k}{\omega} \sin kz \sin \omega t$$

We also know that,

$$\frac{E_0}{B_0} = \frac{\omega}{k} = c$$

Magnetic field vector,

$$\vec{B} = \frac{2E_0}{c} \hat{j} \sin kz \sin \omega t$$

32. (a) Speed of EM wave in force space (c) = $\frac{E_0}{B_0}$

$$\text{or } \vec{E} = cB_0 \sin(kx + \omega t) \hat{k}$$

33. (d) Wave in X-direction means E and B should be function of x and t.

$$\hat{y} - \hat{z} \perp \hat{y} + \hat{z}$$

34. (b) As we know, $\vec{E} \cdot \vec{B} = 0 \therefore [\vec{E} \perp \vec{B}]$

and $\vec{E} \times \vec{B}$ should be along Z direction

$$\text{As } (-2\hat{i} - 3\hat{j}) \times (3\hat{i} - 2\hat{j}) = 5\hat{k}$$

Hence option (b) is the correct answer.

35. (d) As we know,

$$B_0 = \frac{E_0}{C} = \frac{27}{3 \times 10^8} = 9 \times 10^{-8} \text{ tesla}$$

Oscillation of B can be only along $\hat{j}$ or $\hat{k}$ direction.

$$\omega = 2\pi f = 2\pi \times 2 \times 10^{14} \text{ Hz}$$

$$\therefore \vec{B}(x, t) = (9 \times 10^{-8} T) \hat{k} \sin[2\pi(1.5 \times 10^{-6} \times -2 \times 10^4 t)]$$

36. (c) $E_0 = CB_0$ and $C = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$

$$\text{Electric energy density} = \frac{1}{2} \epsilon_0 E_0^2 = \mu_E$$

$$\text{Magnetic energy density} = \frac{1}{2} \frac{B_0^2}{\mu_0} = \mu_B$$

Thus, $\mu_E = \mu_B$

Energy is equally divided between electric and magnetic field.

37. (b) Wavelength of monochromatic green light
 $= 5.5 \times 10^{-5} \text{ cm}$

$$\text{Intensity } I = \frac{\text{Power}}{\text{Area}}$$

$$= \frac{100 \times (3/100)}{4\pi(5)^2} = \frac{3}{100\pi} \text{ Wm}^{-2}$$

Now, half of this intensity (I) belongs to electric field and half of that to magnetic field, therefore,

$$\frac{I}{2} = \frac{1}{4} \epsilon_0 E_0^2 C$$

$$\text{or } E_0 = \sqrt{\frac{2I}{\epsilon_0 C}}$$

$$= \sqrt{\frac{2 \times \left(\frac{3}{100} \pi\right)}{\left(\frac{1}{4\pi \times 9 \times 10^9}\right) \times (3 \times 10^8)}} \\ = \sqrt{\frac{6}{25}} \times 30 = \sqrt{7.2}$$

$$\therefore E_0 = 2.68 \text{ V/m}$$

38. (c) **Given:** Amplitude of electric field,

$$E_0 = 4 \text{ v/m}$$

Absolute permittivity,

$$\epsilon_0 = 8.8 \times 10^{-12} \text{ C}^2/\text{N-m}^2$$

Average energy density $u_E = ?$

Applying formula,

$$\text{Average energy density } u_E = \frac{1}{4} \epsilon_0 E^2$$

$$\Rightarrow u_E = \frac{1}{4} \times 8.8 \times 10^{-12} \times (4)^2 \\ = 35.2 \times 10^{-12} \text{ J/m}^3$$

39. (b) From question,

$$B_0 = 20 \text{ nT} = 20 \times 10^{-9} \text{ T}$$

($\because$ velocity of light in vacuum $C = 3 \times 10^8 \text{ ms}^{-1}$)

$$\vec{E}_0 = \vec{B}_0 \times \vec{C}$$

$$|\vec{E}_0| = |\vec{B}| \cdot |\vec{C}| = 20 \times 10^{-9} \times 3 \times 10^8 \\ = 6 \text{ V/m.}$$

40. (b)

41. (d) Electromagnetic waves do not required any medium to propagate. They can travel in vacuum. They are transverse in nature like light. They carry both energy and momentum.

A changing electric field produces a changing magnetic field and vice-versa. Which gives rise to a transverse wave known as electromagnetic wave.

42. (b) $\because$ The E.M. wave are transverse in nature *i.e.*,

$$= \frac{\vec{k} \times \vec{E}}{\mu} = \vec{H} \quad \dots(i)$$

$$\text{where } \vec{H} = \frac{\vec{B}}{\mu}$$

$$\text{and } \frac{\vec{k} \times \vec{H}}{\omega \epsilon} = -\vec{E} \quad \dots(ii)$$

$\vec{k}$ is $\perp$ $\vec{H}$ and $\vec{k}$ is also $\perp$ to $\vec{E}$

The direction of wave propagation is parallel to $\vec{E} \times \vec{B}$.

The direction of polarization is parallel to electric field.

43. (c) In an electromagnetic wave electric field and magnetic field are perpendicular to the direction of propagation of wave. The vector equation for the electric field is

$$\vec{E} = E_0 \cos \left(\omega t - \frac{2\pi}{\lambda} y \right) \hat{z}$$

44. (a) Frequency remains unchanged during refraction
 Velocity of EM wave in vacuum

$$V_{\text{vacuum}} = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = C$$

$$v_{\text{med}} = \frac{1}{\sqrt{\mu_0 \epsilon_0 \times 4}} = \frac{c}{2}$$

$$\frac{\lambda_{\text{med}}}{\lambda_{\text{vacuum}}} = \frac{v_{\text{med}}}{v_{\text{vacuum}}} = \frac{c/2}{c} = \frac{1}{2}$$

$\therefore$ Wavelength is halved and frequency remains unchanged

45. (a) The phenomenon of polarisation is shown only by transverse waves. The vibration of electromagnetic wave are restricted through polarization in a direction perpendicular to wave propagation.

46. (d) Energy sequence of radiations is

$$E_{\gamma\text{-Rays}} > E_{\text{X-Rays}} > E_{\text{microwave}} > E_{\text{AM Radiowaves}}$$

$$\therefore \lambda_{\gamma\text{-Rays}} < \lambda_{\text{X-Rays}} < \lambda_{\text{microwave}} < \lambda_{\text{AM Radiowaves}}$$

From the above sequence, we have

(a) Microwave $\rightarrow 10^{-3} \text{ m}$ (iv)

(b) Gamma Rays $\rightarrow 10^{-15} \text{ m}$ (ii)

(c) AM Radio wave $\rightarrow 100 \text{ m}$ (i)

(d) X-Rays $\rightarrow 10^{-10} \text{ m}$ (iii)

47. (b) The orderly arrangement of different parts of EM wave in decreasing order of wavelength is as follows:

$$\lambda_{\text{radiowaves}} > \lambda_{\text{microwaves}} > \lambda_{\text{visible}} > \lambda_{\text{X-rays}}$$

48. (c) Optical Fibre Communication – Infrared Light
Radar – Radio Waves
Sonar – Ultrasound
Mobile Phones – Microwaves

49. (c) $\xrightarrow{\text{E, Decreases}}$
 $\gamma\text{-rays}$ X-rays uv-rays Visible rays IR rays Radio waves
 VIBGYOR Microwaves

Radio wave < yellow light < blue light < X-rays

(Increasing order of energy)

50. (c) Microwave oven acts on the principle of giving vibrational energy to water molecules.
51. (d)
 (1) Infrared rays are used to treat muscular strain because these are heat rays.
 (2) Radio waves are used for broadcasting because these waves have very long wavelength ranging from few centimeters to few hundred kilometers.
 (3) X-rays are used to detect fracture of bones because they have high penetrating power but they can't penetrate through denser medium like bones.
 (4) Ultraviolet rays are absorbed by ozone of the atmosphere.
52. (c) Gamma rays < X-rays < Ultra violet < Visible rays < Infrared rays < Microwaves < Radio waves.

53. (d) Wavelength emitted by atomic hydrogen in interstellar space - Part of short radio wave of electromagnetic spectrum.

Doublet of sodium - visible radiation.

54. (d) Vibration of atoms and molecules 700 nm to 1 mm
 Radioactive decay of the nucleus < 10^{-3} nm
 Magnetron valve 1 mm to 0.1 m

55. (a) $E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{E}$

$$\Rightarrow \lambda = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{11 \times 1000 \times 1.6 \times 10^{-19}}$$

$$= 12.4 \text{ \AA}$$

$\xrightarrow{\text{Increasing order of frequency}}$
 x-rays u-v rays visible Infrared

wavelength range of visible region is 4000 Å to 7800 Å.

56. (a) Frequency range of γ -ray,

$$b = 10^{18} - 10^{23} \text{ Hz}$$

Frequency range of X-ray,

$$a = 10^{16} - 10^{20} \text{ Hz}$$

Frequency range of ultraviolet ray,

$$c = 10^{15} - 10^{17} \text{ Hz}$$

$$\therefore a < b; b > c$$

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Ray Optics and Optical Instruments

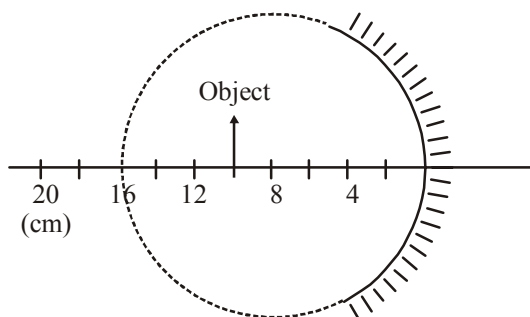


TOPIC 1 Plane Mirror, Spherical Mirror and Reflection of Light



1. When an object is kept at a distance of 30 cm from a concave mirror, the image is formed at a distance of 10 cm from the mirror. If the object is moved with a speed of 9 cm s^{-1} , the speed (in cm s^{-1}) with which image moves at that instant is _____. [NA Sep. 03, 2020 (II)]

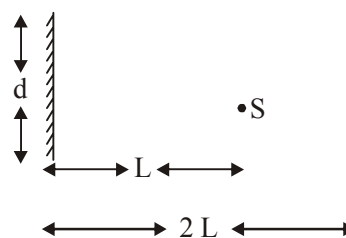
2.



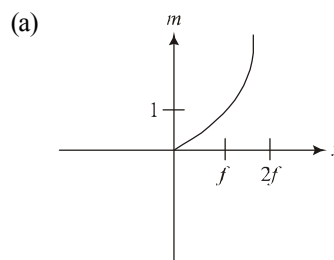
A spherical mirror is obtained as shown in the figure from a hollow glass sphere. If an object is positioned in front of the mirror, what will be the nature and magnification of the image of the object? (Figure drawn as schematic and not to scale) [Sep. 02, 2020 (I)]

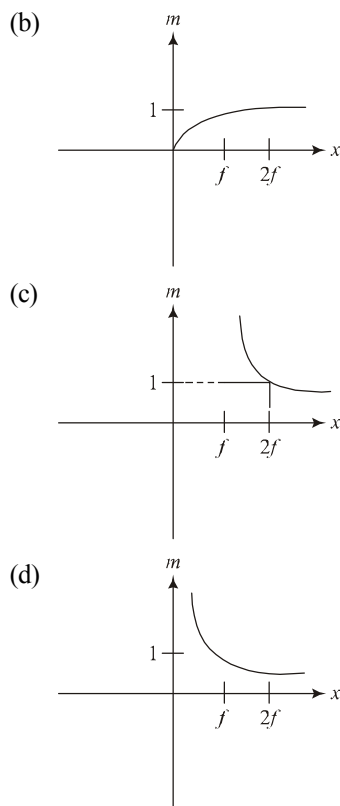
- (a) Inverted, real and magnified
(b) Erect, virtual and magnified
(c) Erect, virtual and unmagnified
(d) Inverted, real and unmagnified
3. A concave mirror for face viewing has focal length of 0.4 m. The distance at which you hold the mirror from your face in order to see your image upright with a magnification of 5 is: [9 April 2019 I]
(a) 0.24m (b) 1.60m (c) 0.32m (d) 0.16m
4. A point source of light, S is placed at a distance L in front of the centre of plane mirror of width d which is hanging vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror, at a distance 2L as shown

below. The distance over which the man can see the image of the light source in the mirror is: [12 Jan. 2019 I]



- (a) d (b) 2d
(c) 3d (d) $\frac{d}{2}$
5. Two plane mirrors are inclined to each other such that a ray of light incident on the first mirror (M_1) and parallel to the second mirror (M_2) is finally reflected from the second mirror (M_2) parallel to the first mirror (M_1). The angle between the two mirrors will be: [9 Jan. 2019 II]
(a) 45° (b) 60°
(c) 75° (d) 90°
6. An object is gradually moving away from the focal point of a concave mirror along the axis of the mirror. The graphical representation of the magnitude of linear magnification (m) versus distance of the object from the mirror (x) is correctly given by (Graphs are drawn schematically and are not to scale) [8 Jan. 2020 II]

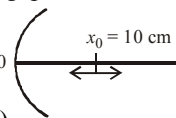




7. A particle is oscillating on the X-axis with an amplitude 2 cm about the point $x_0 = 10$ cm with a frequency ω . A concave mirror of focal length 5 cm is placed at the origin (see figure) Identify the correct statements: [Online April 15, 2018]

- (A) The image executes periodic motion
 (B) The image executes non-periodic motion
 (C) The turning points of the image are asymmetric w.r.t the image of the point at $x = 10$ cm
 (D) The distance between the turning points of the

oscillation of the image is $\frac{100}{21}$ $x = 0$



- (a) (B), (D) (b) (B), (C)
 (c) (A), (C), (D) (d) (A), (D)

8. You are asked to design a shaving mirror assuming that a person keeps it 10 cm from his face and views the magnified image of the face at the closest comfortable distance of 25 cm. The radius of curvature of the mirror would then be : [Online April 10, 2015]

- (a) 60 cm (b) -24 cm
 (c) -60 cm (d) 24 cm

9. A car is fitted with a convex side-view mirror of focal length 20 cm. A second car 2.8 m behind the first car is overtaking the first car at a relative speed of 15 m/s. The speed of the image of the second car as seen in the mirror of the first one is : [2011]

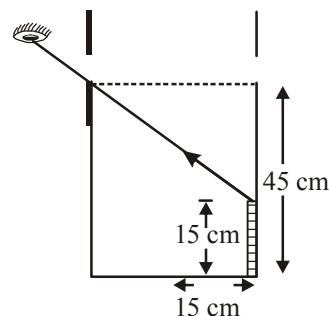
- (a) $\frac{1}{15}$ m/s (b) 10 m/s (c) 15 m/s (d) $\frac{1}{10}$ m/s

10. To get three images of a single object, one should have two plane mirrors at an angle of [2003]
 (a) 60° (b) 90° (c) 120° (d) 30°
 11. If two plane mirrors are kept at 60° to each other, then the number of images formed by them is [2002]
 (a) 5 (b) 6 (c) 7 (d) 8

TOPIC 2 Refraction of Light at Plane Surface and Total Internal Reflection



12. An observer can see through a small hole on the side of a jar (radius 15 cm) at a point at height of 15 cm from the bottom (see figure). The hole is at a height of 45 cm. When the jar is filled with a liquid up to a height of 30 cm the same observer can see the edge at the bottom of the jar. If the refractive index of the liquid is $N/100$, where N is an integer, the value of N is [NA Sep. 03, 2020 (I)]



13. A light ray enters a solid glass sphere of refractive index $\mu = \sqrt{3}$ at an angle of incidence 60° . The ray is both reflected and refracted at the farther surface of the sphere. The angle (in degrees) between the reflected and refracted rays at this surface is [NA Sep. 02, 2020 (II)]

14. A vessel of depth $2h$ is half filled with a liquid of refractive index $2\sqrt{2}$ and the upper half with another liquid of refractive index $\sqrt{2}$. The liquids are immiscible. The apparent depth of the inner surface of the bottom of vessel will be: [9 Jan. 2020 I]

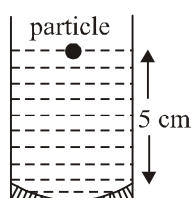
- (a) $\frac{h}{\sqrt{2}}$ (b) $\frac{h}{2(\sqrt{2} + 1)}$
 (c) $\frac{h}{3\sqrt{2}}$ (d) $\frac{3}{4}h\sqrt{2}$

15. There is a small source of light at some depth below the surface of water (refractive index $= \frac{4}{3}$) in a tank of large cross sectional surface area. Neglecting any reflection from the bottom and absorption by water, percentage of light that emerges out of surface is (nearly): [Use the fact that surface area of a spherical cap of height h and radius of curvature r is $2\pi rh$] [9 Jan. 2020 II]
 (a) 21% (b) 34% (c) 17% (d) 50%

16. The critical angle of a medium for a specific wavelength, if the medium has relative permittivity 3 and relative permeability $\frac{4}{3}$ for this wavelength, will be: [8 Jan. 2020 I]

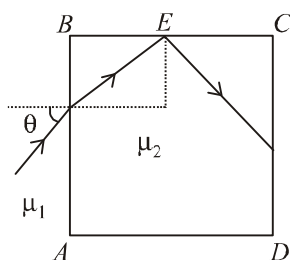
(a) 15° (b) 30°
(c) 45° (d) 60°

17. A concave mirror has radius of curvature of 40 cm. It is at the bottom of a glass that has water filled up to 5 cm (see figure). If a small particle is floating on the surface of water, its image as seen, from directly above the glass, is at a distance d from the surface of water. The value of d is close to : [12 Apr. 2019 I]
(Refractive index of water = 1.33)



(a) 6.7 cm (b) 13.4 cm (c) 8.8 cm (d) 11.7 cm

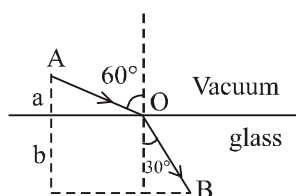
18. A transparent cube of side d , made of a material of refractive index μ_2 , is immersed in a liquid of refractive index μ_1 ($\mu_1 < \mu_2$). A ray is incident on the face AB at an angle θ (shown in the figure). Total internal reflection takes place at point E on the face BC.



Then θ must satisfy : [12 Apr. 2019 II]

(a) $\theta < \sin^{-1} \frac{\mu_1}{\mu_2}$ (b) $\theta > \sin^{-1} \sqrt{\frac{\mu_2^2}{\mu_1^2} - 1}$
(c) $\theta < \sin^{-1} \sqrt{\frac{\mu_2^2}{\mu_1^2} - 1}$ (d) $\theta > \sin^{-1} \frac{\mu_1}{\mu_2}$

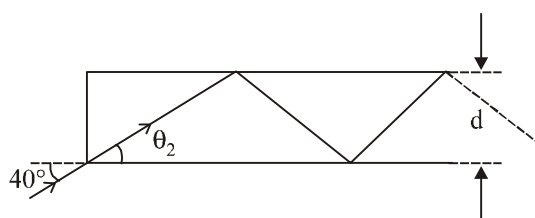
19. A ray of light AO in vacuum is incident on a glass slab at angle 60° and refracted at angle 30° along OB as shown in the figure. The optical path length of light ray from A to B is : [10 Apr. 2019 I]



(a) $\frac{2\sqrt{3}}{a} + 2b$ (b) $2a + \frac{2b}{3}$
(c) $2a + \frac{2b}{\sqrt{3}}$ (d) $2a + 2b$

20. In figure, the optical fiber is $l = 2$ m long and has a diameter of $d = 20 \mu\text{m}$. If a ray of light is incident on one end of the fiber at angle $\theta_1 = 40^\circ$, the number of reflections it makes before emerging from the other end is close to : (refractive index of fiber is 1.31 and $\sin 40^\circ = 0.64$)

[8 April 2019 I]



(a) 55000 (b) 66000 (c) 45000 (d) 57000

21. A light wave is incident normally on a glass slab of refractive index 1.5. If 4% of light gets reflected and the amplitude of the electric field of the incident light is 30 V/m, then the amplitude of the electric field for the wave propagating in the glass medium will be: [12 Jan. 2019 I]

(a) 30 V/m (b) 10 V/m
(c) 24 V/m (d) 6 V/m

22. Let the refractive index of a denser medium with respect to a rarer medium be n_{12} and its critical angle be θ_C . At an angle of incidence A when light is travelling from denser medium to rarer medium, a part of the light is reflected and the rest is refracted and the angle between reflected and refracted rays is 90° . Angle A is given by :

[Online April 8, 2017]

(a) $\frac{1}{\cos^{-1}(\sin \theta_C)}$ (b) $\frac{1}{\tan^{-1}(\sin \theta_C)}$
(c) $\cos^{-1}(\sin \theta_C)$ (d) $\tan^{-1}(\sin \theta_C)$

23. A diver looking up through the water sees the outside world contained in a circular horizon. The refractive index of water is $\frac{4}{3}$, and the diver's eyes are 15 cm below the surface of water. Then the radius of the circle is:

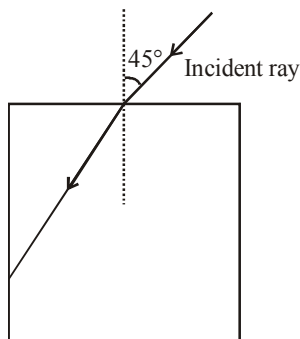
[Online April 9, 2014]

(a) $15 \times 3 \times \sqrt{5}$ cm (b) $15 \times 3 \sqrt{7}$ cm
(c) $\frac{15 \times \sqrt{7}}{3}$ cm (d) $\frac{15 \times 3}{\sqrt{7}}$ cm

24. A printed page is pressed by a glass of water. The refractive index of the glass and water is 1.5 and 1.33, respectively. If the thickness of the bottom of glass is 1 cm and depth of water is 5 cm, how much the page will appear to be shifted if viewed from the top ? **[Online April 25, 2013]**

(a) 1.033 cm (b) 3.581 cm
(c) 1.3533 cm (d) 1.90 cm

25. A light ray falls on a square glass slab as shown in the diagram. The index of refraction of the glass, if total internal reflection is to occur at the vertical face, is equal to : **[Online April 23, 2013]**



(a) $\frac{(\sqrt{2}+1)}{2}$ (b) $\sqrt{\frac{5}{2}}$ (c) $\frac{3}{2}$ (d) $\sqrt{\frac{3}{2}}$

26. Light is incident from a medium into air at two possible angles of incidence (A) 20° and (B) 40° . In the medium light travels 3.0 cm in 0.2 ns. The ray will : **[Online April 9, 2013]**

(a) suffer total internal reflection in both cases (A) and (B)
(b) suffer total internal reflection in case (B) only
(c) have partial reflection and partial transmission in case (B)
(d) have 100% transmission in case (A)

27. Let the x - z plane be the boundary between two transparent media. Medium 1 in $z \geq 0$ has a refractive index of $\sqrt{2}$ and medium 2 with $z < 0$ has a refractive index of $\sqrt{3}$. A ray of light in medium 1 given by the vector $\vec{A} = 6\sqrt{3}\hat{i} + 8\sqrt{3}\hat{j} - 10\hat{k}$ is incident on the plane of separation. The angle of refraction in medium 2 is: **[2011]**

(a) 45° (b) 60° (c) 75° (d) 30°

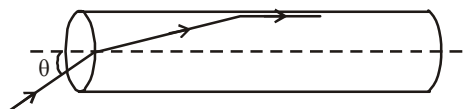
28. A beaker contains water up to a height h_1 and kerosene of height h_2 above water so that the total height of (water + kerosene) is $(h_1 + h_2)$. Refractive index of water is μ_1 and that of kerosene is μ_2 . The apparent shift in the position of the bottom of the beaker when viewed from above is **[2011 RS]**

(a) $\left(1 + \frac{1}{\mu_1}\right)h_1 - \left(1 + \frac{1}{\mu_2}\right)h_2$
(b) $\left(1 - \frac{1}{\mu_1}\right)h_1 + \left(1 - \frac{1}{\mu_2}\right)h_2$

(c) $\left(1 + \frac{1}{\mu_1}\right)h_2 - \left(1 + \frac{1}{\mu_2}\right)h_1$

(d) $\left(1 - \frac{1}{\mu_1}\right)h_2 + \left(1 - \frac{1}{\mu_2}\right)h_1$

29. A transparent solid cylindrical rod has a refractive index of $\frac{2}{\sqrt{3}}$. It is surrounded by air. A light ray is incident at the mid-point of one end of the rod as shown in the figure.



The incident angle θ for which the light ray grazes along the wall of the rod is : **[2009]**

(a) $\sin^{-1}(\sqrt{3}/2)$ (b) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$

(c) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (d) $\sin^{-1}(1/2)$

30. A fish looking up through the water sees the outside world contained in a circular horizon. If the refractive index of water is $\frac{4}{3}$ and the fish is 12 cm below the surface, the radius of this circle in cm is **[2005]**

(a) $\frac{36}{\sqrt{7}}$ (b) $36\sqrt{7}$ (c) $4\sqrt{5}$ (d) $36\sqrt{5}$

31. Consider telecommunication through optical fibres. Which of the following statements is **not** true? **[2003]**

(a) Optical fibres can be of graded refractive index
(b) Optical fibres are subject to electromagnetic interference from outside
(c) Optical fibres have extremely low transmission loss
(d) Optical fibres may have homogeneous core with a suitable cladding.

32. Which of the following is used in optical fibres? **[2002]**

(a) total internal reflection (b) scattering
(c) diffraction (d) refraction.

TOPIC 3

Refraction at Curved Surface Lenses and Power of Lens



33. A point like object is placed at a distance of 1 m in front of a convex lens of focal length 0.5 m. A plane mirror is placed at a distance of 2 m behind the lens. The position and nature of the final image formed by the system is :

[Sep. 06, 2020 (I)]

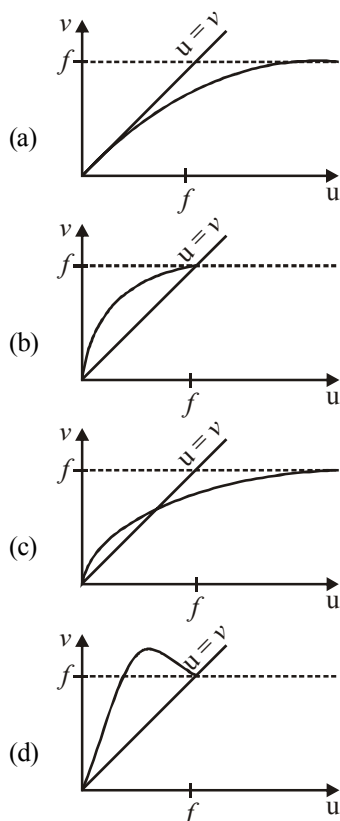
(a) 2.6 m from the mirror, real
(b) 1 m from the mirror, virtual
(c) 1 m from the mirror, real
(d) 2.6 m from the mirror, virtual

34. A double convex lens has power P and same radii of curvature R of both the surfaces. The radius of curvature of a surface of a plano-convex lens made of the same material with power $1.5P$ is : [Sep. 06, 2020 (II)]

(a) $2R$ (b) $\frac{R}{2}$ (c) $\frac{3R}{2}$ (d) $\frac{R}{3}$

35. For a concave lens of focal length f , the relation between object and image distances u and v , respectively, from its pole can best be represented by ($u = v$ is the reference line) :

[Sep. 05, 2020 (I)]



36. The distance between an object and a screen is 100 cm. A lens can produce real image of the object on the screen for two different positions between the screen and the object. The distance between these two positions is 40 cm. If the power of the lens is close to $\left(\frac{N}{100}\right)D$ where N is an integer, the value of N is _____.

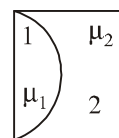
[NA Sep. 04, 2020 (II)]

37. A point object in air is in front of the curved surface of a *plano-convex* lens. The radius of curvature of the curved surface is 30 cm and the refractive index of the lens material is 1.5, then the focal length of the lens (in cm) is _____. [NA 8 Jan. 2020 I]

38. A thin lens made of glass (refractive index = 1.5) of focal length $f = 16$ cm is immersed in a liquid of refractive index 1.42. If its focal length in liquid is f_l , then the ratio f_l/f is closest to the integer: [7 Jan. 2020 II]

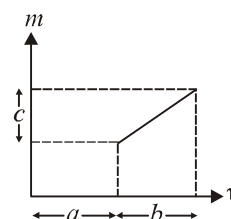
(a) 1 (b) 9 (c) 5 (d) 17

39. One plano-convex and one plano-concave lens of same radius of curvature ' R ' but of different materials are joined side by side as shown in the figure. If the refractive index of the material of 1 is μ_1 and that of 2 is μ_2 , then the focal length of the combination is : [10 Apr. 2019 I]



(a) $\frac{R}{\mu_1 - \mu_2}$ (b) $\frac{2R}{\mu_1 - \mu_2}$
(c) $\frac{2R}{2(\mu_1 - \mu_2)}$ (d) $\frac{R}{2 - (\mu_1 - \mu_2)}$

40. The graph shows how the magnification m produced by a thin lens varies with image distance v . What is the focal length of the lens used ? [10 Apr. 2019 II]

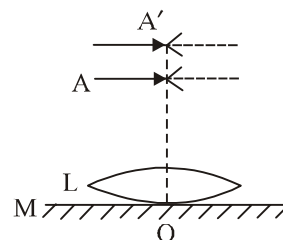


(a) $\frac{b^2}{ac}$ (b) $\frac{b^2c}{a}$ (c) $\frac{a}{c}$ (d) $\frac{b}{c}$

41. A convex lens of focal length 20 cm produces images of the same magnification 2 when an object is kept at two distances x_1 and x_2 ($x_1 > x_2$) from the lens. The ratio of x_1 and x_2 is: [9 Apr. 2019 II]

(a) 2 : 1 (b) 3 : 1 (c) 5 : 3 (d) 4 : 3

42. A thin convex lens L (refractive index = 1.5) is placed on a plane mirror M . When a pin is placed at A , such that $OA = 18$ cm, its real inverted image is formed at A itself, as shown in figure. When a liquid of refractive index μ_l is put between the lens and the mirror, the pin has to be moved to A' , such that $OA' = 27$ cm, to get its inverted real image at A' itself. The value of μ_l will be: [9 Apr. 2019 II]



(a) $\frac{4}{3}$ (b) $\frac{3}{2}$ (c) $\sqrt{3}$ (d) $\sqrt{2}$

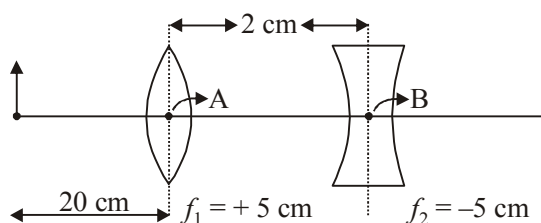
43. An upright object is placed at a distance of 40 cm in front of a convergent lens of focal length 20 cm. A convergent mirror of focal length 10 cm is placed at a distance of 60 cm on the other side of the lens. The position and size of the final image will be : **[8 April 2019 I]**

(a) 20 cm from the convergent mirror, same size as the object
 (b) 40 cm from the convergent mirror, same size as the object
 (c) 40 cm from the convergent lens, twice the size of the object
 (d) 20 cm from the convergent mirror, twice the size of the object

44. A convex lens (of focal length 20 cm) and a concave mirror, having their principal axes along the same lines, are kept 80 cm apart from each other. The concave mirror is to the right of the convex lens. When an object is kept at a distance of 30 cm to the left of the convex lens, its image remains at the same position even if the concave mirror is removed. The maximum distance of the object for which this concave mirror, by itself would produce a virtual image would be : **[8 Apr. 2019 II]**

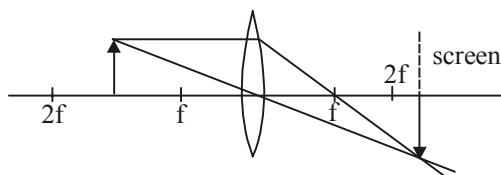
(a) 30 cm (b) 25 cm (c) 10 cm (d) 20 cm

45. What is the position and nature of image formed by lens combination shown in figure? (f_1, f_2 are focal lengths) **[12 Jan. 2019 I]**



(a) 70 cm from point B at left; virtual
 (b) 40 cm from point B at right; real
 (c) $\frac{20}{3}$ cm from point B at right; real
 (d) 70 cm from point B at right; real

46. Formation of real image using a biconvex lens is shown below : **[12 Jan. 2019 II]**



If the whole set up is immersed in water without disturbing the object and the screen positions, what will one observe on the screen ?

(a) Image disappears (b) Magnified image
 (c) Erect real image (d) No change

47. A plano-convex lens (focal length f_2 , refractive index μ_2 , radius of curvature R) fits exactly into a plano-concave lens (focal length f_1 , refractive index μ_1 , radius of curvature R). Their plane surfaces are parallel to each other. Then, the focal length of the combination will be : **[12 Jan. 2019 II]**

(a) $f_1 - f_2$ (b) $\frac{R}{\mu_2 - \mu_1}$
 (c) $\frac{2f_1f_2}{f_1 + f_2}$ (d) $f_1 + f_2$

48. An object is at a distance of 20 m from a convex lens of focal length 0.3 m. The lens forms an image of the object. If the object moves away from the lens at a speed of 5 m/s, the speed and direction of the image will be :

[11 Jan. 2019 I]

(a) 2.26×10^{-3} m/s away from the lens
 (b) 0.92×10^{-3} m/s away from the lens
 (c) 3.22×10^{-3} m/s towards the lens
 (d) 1.16×10^{-3} m/s towards the lens

49. A plano convex lens of refractive index μ_1 and focal length f_1 is kept in contact with another plano concave lens of refractive index μ_2 and focal length f_2 . If the radius of curvature of their spherical faces is R each and $f_1 = 2f_2$, then μ_1 and μ_2 are related as:

[10 Jan. 2019 I]

(a) $\mu_1 + \mu_2 = 3$ (b) $2\mu_1 - \mu_2 = 1$
 (c) $3\mu_2 - 2\mu_1 = 1$ (d) $2\mu_2 - \mu_1 = 1$

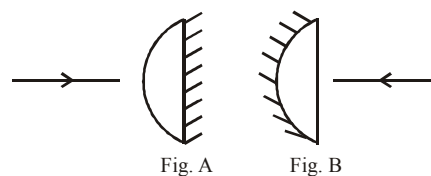
50. The eye can be regarded as a single refracting surface. The radius of curvature of this surface is equal to that of cornea (7.8 mm). This surface separates two media of refractive indices 1 and 1.34. Calculate the distance from the refracting surface at which a parallel beam of light will come to focus. **[10 Jan. 2019 II]**

(a) 1 cm (b) 2 cm
 (c) 4.0 cm (d) 3.1 cm

51. A convex lens is put 10 cm from a light source and it makes a sharp image on a screen, kept 10 cm from the lens. Now a glass block (refractive index 1.5) of 1.5 cm thickness is placed in contact with the light source. To get the sharp image again, the screen is shifted by a distance d. Then d is: **[9 Jan. 2019 I]**

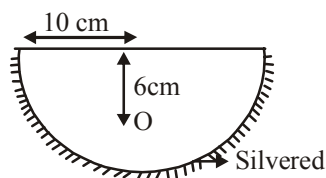
(a) 1.1 cm away from the lens
 (b) 0
 (c) 0.55 cm towards the lens
 (d) 0.55 cm away from the lens

52. A planoconvex lens becomes an optical system of 28 cm focal length when its plane surface is silvered and illuminated from left to right as shown in Fig-A. If the same lens is instead silvered on the curved surface and illuminated from other side as in Fig. B, it acts like an optical system of focal length 10 cm. The refractive index of the material of lens is: **[Online April 15, 2018]**

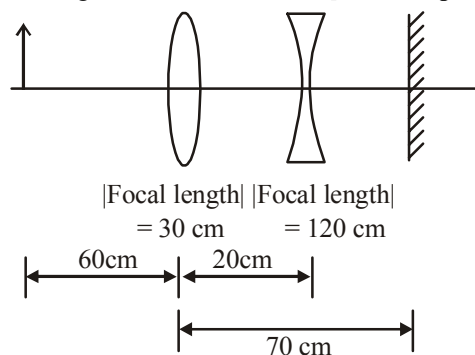


(a) 1.50 (b) 1.55 (c) 1.75 (d) 1.51

53. A convergent doublet of separated lenses, corrected for spherical aberration, has resultant focal length of 10cm. The separation between the two lenses is 2cm. The focal lengths of the component lenses [Online April 15, 2018]
 (a) 18cm, 20cm (b) 10cm, 12cm
 (c) 12cm, 14cm (d) 16cm, 18cm
54. In an experiment a convex lens of focal length 15 cm is placed coaxially on an optical bench in front of a convex mirror at a distance of 5 cm from it. It is found that an object and its image coincide, if the object is placed at a distance of 20 cm from the lens. The focal length of the convex mirror is : [Online April 9, 2017]
 (a) 27.5 cm (b) 20.0 cm (c) 25.0 cm (d) 30.5 cm
55. A hemispherical glass body of radius 10 cm and refractive index 1.5 is silvered on its curved surface. A small air bubble is 6 cm below the flat surface inside it along the axis. The position of the image of the air bubble made by the mirror is seen : [Online April 10, 2016]



- (a) 14 cm below flat surface
 (b) 20 cm below flat surface
 (c) 16 cm below flat surface
 (d) 30 cm below flat surface
56. A convex lens, of focal length 30 cm, a concave lens of focal length 120 cm, and a plane mirror are arranged as shown. For an object kept at a distance of 60 cm from the convex lens, the final image, formed by the combination, is a real image, at a distance of : [Online April 9, 2016]



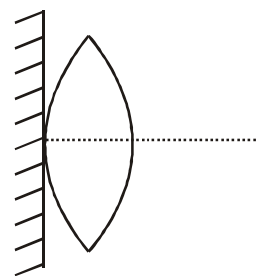
- (a) 60 cm from the convex lens
 (b) 60 cm from the concave lens
 (c) 70 cm from the convex lens
 (d) 70 cm from the concave lens
57. To find the focal length of a convex mirror, a student records the following data : [Online April 9, 2016]

Object Pin	Convex Lens	Convex Mirror	Image Pin
22.2cm	32.2cm	45.8cm	71.2cm

The focal length of the convex lens is f_1 and that of mirror is f_2 . Then taking index correction to be negligibly small,

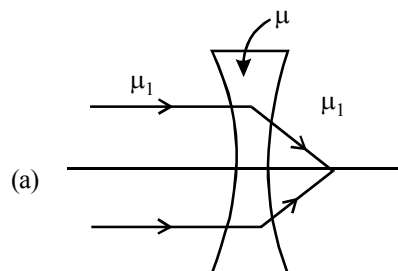
f_1 and f_2 are close to :

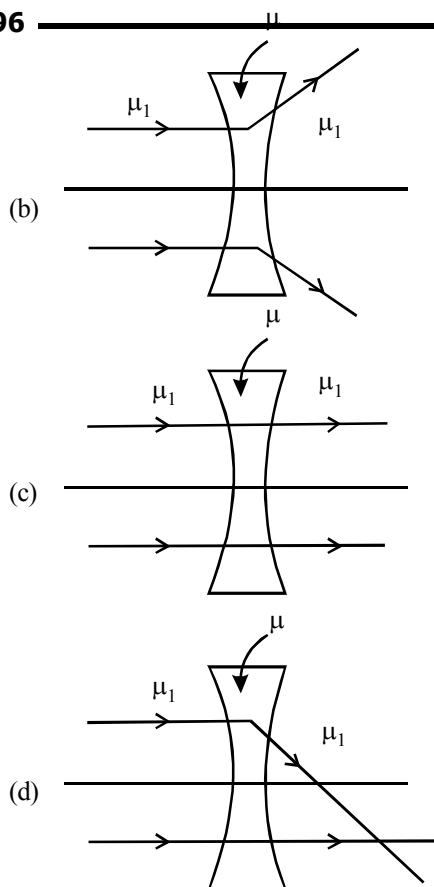
- (a) $f_1 = 7.8$ cm $f_2 = 12.7$ cm
 (b) $f_1 = 12.7$ cm $f_2 = 7.8$ cm
 (c) $f_1 = 15.6$ cm $f_2 = 25.4$ cm
 (d) $f_1 = 7.8$ cm $f_2 = 25.4$ cm
58. A thin convex lens of focal length ' f ' is put on a plane mirror as shown in the figure. When an object is kept at a distance ' a ' from the lens - mirror combination, its image is formed at a distance $\frac{a}{3}$ in front of the combination. The value of ' a ' is : [Online April 11, 2015]



- (a) $3f$ (b) $\frac{3}{2}f$ (c) f (d) $2f$
59. A thin convex lens made from crown glass ($\mu = \frac{3}{2}$) has focal length f . When it is measured in two different liquids having refractive indices $\frac{4}{3}$ and $\frac{5}{3}$, it has the focal lengths f_1 and f_2 respectively. The correct relation between the focal lengths is: [2014]
 (a) $f_1 = f_2 < f$
 (b) $f_1 > f$ and f_2 becomes negative
 (c) $f_2 > f$ and f_1 becomes negative
 (d) f_1 and f_2 both become negative
60. The refractive index of the material of a concave lens is μ . It is immersed in a medium of refractive index μ_1 . A parallel beam of light is incident on the lens. The path of the emergent rays when $\mu_1 > \mu$ is:

[Online April 12, 2014]





61. An object is located in a fixed position in front of a screen. Sharp image is obtained on the screen for two positions of a thin lens separated by 10 cm. The size of the images in two situations are in the ratio 3 : 3. What is the distance between the screen and the object?

[Online April 11, 2014]

- (a) 124.5 cm (b) 144.5 cm
(c) 65.0 cm (d) 99.0 cm
62. Diameter of a plano-convex lens is 6 cm and thickness at the centre is 3 mm. If speed of light in material of lens is 2×10^8 m/s, the focal length of the lens is [2013]
- (a) 15 cm (b) 20 cm (c) 30 cm (d) 10 cm
63. The image of an illuminated square is obtained on a screen with the help of a converging lens. The distance of the square from the lens is 40 cm. The area of the image is 9 times that of the square. The focal length of the lens is :

[Online April 22, 2013]

- (a) 36 cm (b) 27 cm (c) 60 cm (d) 30 cm
64. An object at 2.4 m in front of a lens forms a sharp image on a film 12 cm behind the lens. A glass plate 1 cm thick, of refractive index 1.50 is interposed between lens and film with its plane faces parallel to film. At what distance (from lens) should object shifted to be in sharp focus of film? [2012]
- (a) 7.2 m (b) 2.4 m (c) 3.2 m (d) 5.6 m
65. When monochromatic red light is used instead of blue light in a convex lens, its focal length will [2011 RS]
- (a) increase
(b) decrease

(c) remain same

(d) does not depend on colour of light

66. In an optics experiment, with the position of the object fixed, a student varies the position of a convex lens and for each position, the screen is adjusted to get a clear image of the object. A graph between the object distance u and the image distance v , from the lens, is plotted using the same scale for the two axes. A straight line passing through the origin and making an angle of 45° with the x -axis meets the experimental curve at P . The coordinates of P will be [2009]

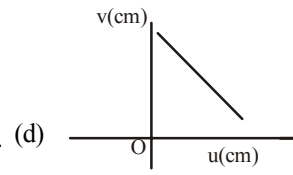
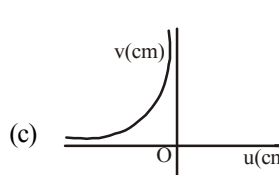
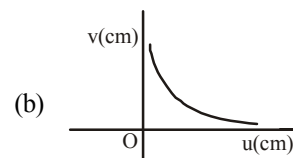
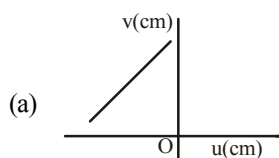
(a) $\left(\frac{f}{2}, \frac{f}{2}\right)$

(b) (f, f)

(c) $(4f, 4f)$

(d) $(2f, 2f)$

67. A student measures the focal length of a convex lens by putting an object pin at a distance ' u ' from the lens and measuring the distance ' v ' of the image pin. The graph between ' u ' and ' v ' plotted by the student should look like [2008]



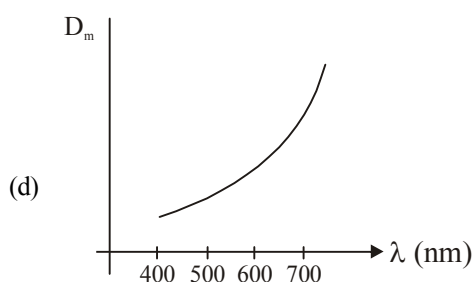
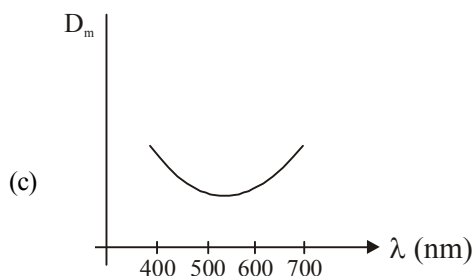
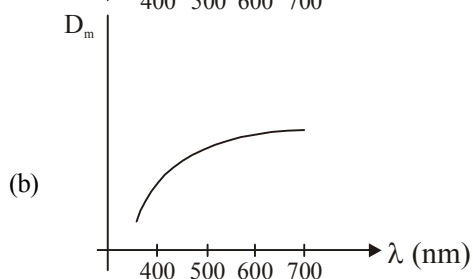
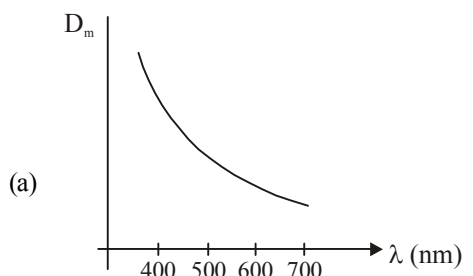
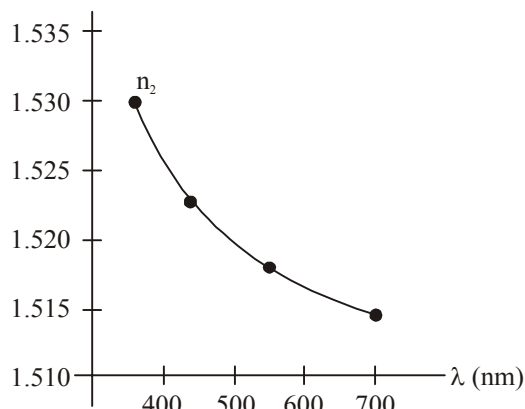
68. Two lenses of power $-15 D$ and $+5 D$ are in contact with each other. The focal length of the combination is [2007]
- (a) $+10$ cm (b) -20 cm
(c) -10 cm (d) $+20$ cm
69. A thin glass (refractive index 1.5) lens has optical power of $-5 D$ in air. Its optical power in a liquid medium with refractive index 1.6 will be [2005]
- (a) $-1D$ (b) $1D$ (c) $-25D$ (d) $25D$
70. A plano convex lens of refractive index 1.5 and radius of curvature 30 cm. Is silvered at the curved surface. Now this lens has been used to form the image of an object. At what distance from this lens an object be placed in order to have a real image of size of the object [2004]
- (a) 60 cm (b) 30 cm (c) 20 cm (d) 80 cm

TOPIC 4 Prism and Dispersion of Light



71. The surface of a metal is illuminated alternately with photons of energies $E_1 = 4$ eV and $E_2 = 2.5$ eV respectively. The ratio of maximum speeds of the photoelectrons emitted in the two cases is 2. The work function of the metal in (eV) is [NA Sep. 05, 2020 (II)]

72. The variation of refractive index of a crown glass thin prism with wavelength of the incident light is shown. Which of the following graphs is the correct one, if D_m is the angle of minimum deviation ? [11 Jan. 2019, I]



73. A monochromatic light is incident at a certain angle on an equilateral triangular prism and suffers minimum deviation. If the refractive index of the material of the prism is $\sqrt{3}$, then the angle of incidence is : [11 Jan. 2019 II]

(a) 90° (b) 30°
(c) 60° (d) 45°

74. A ray of light is incident at an angle of 60° on one face of a prism of angle 30° . The emergent ray of light makes an angle of 30° with incident ray. The angle made by the emergent ray with second face of prism will be:

[Online April 16, 2018]

(a) 30° (b) 90° (c) 0° (d) 45°

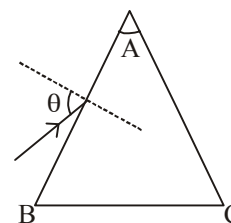
75. In an experiment for determination of refractive index of glass of a prism by $i - \delta$, plot it was found that a ray incident at angle 35° , suffers a deviation of 40° and that it emerges at angle 79° . In that case which of the following is closest to the maximum possible value of the refractive index?

[2016]

(a) 1.7 (b) 1.8 (c) 1.5 (d) 1.6

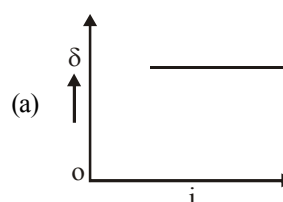
76. Monochromatic light is incident on a glass prism of angle A. If the refractive index of the material of the prism is μ , a ray, incident at an angle θ , on the face AB would get transmitted through the face AC of the prism provided :

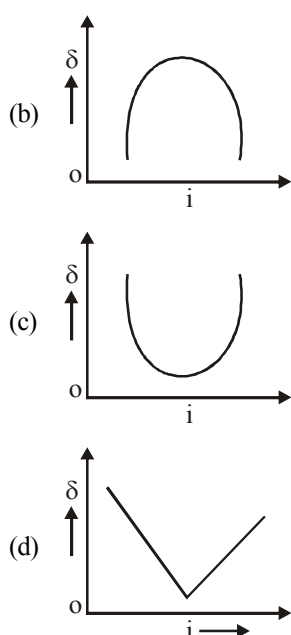
[2015]



- (a) $\theta > \cos^{-1} \left[\mu \sin \left(A + \sin^{-1} \left(\frac{1}{\mu} \right) \right) \right]$
(b) $\theta < \cos^{-1} \left[\mu \sin \left(A + \sin^{-1} \left(\frac{1}{\mu} \right) \right) \right]$
(c) $\theta > \sin^{-1} \left[\mu \sin \left(A - \sin^{-1} \left(\frac{1}{\mu} \right) \right) \right]$
(d) $\theta < \sin^{-1} \left[\mu \sin \left(A - \sin^{-1} \left(\frac{1}{\mu} \right) \right) \right]$

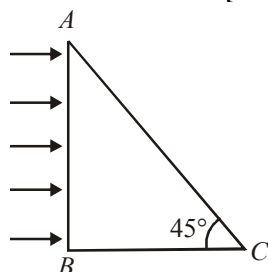
77. The graph between angle of deviation (δ) and angle of incidence (i) for a triangular prism is represented by [2013]





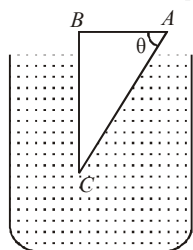
78. A beam of light consisting of red, green and blue colours is incident on a right-angled prism on face AB . The refractive indices of the material for the above red, green and blue colours are 1.39, 1.44 and 1.47 respectively. A person looking on surface AC of the prism will see

[Online May 26, 2012]



- (a) no light
(b) green and blue colours
(c) red and green colours
(d) red colour only
79. A glass prism of refractive index 1.5 is immersed in water (refractive index $\frac{4}{3}$) as shown in figure. A light beam incident normally on the face AB is totally reflected to reach the face BC , if

[Online May 19, 2012]



- (a) $\sin \theta > \frac{5}{9}$ (b) $\sin \theta > \frac{2}{3}$
(c) $\sin \theta > \frac{8}{9}$ (d) $\sin \theta > \frac{1}{3}$

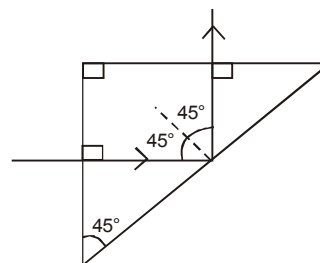
80. Which of the following processes play a part in the formation of a rainbow? [Online May 7, 2012]

- (i) Refraction (ii) Total internal reflection
(iii) Dispersion (iv) Interference
(a) (i), (ii) and (iii) (b) (i) and (ii)
(c) (i), (ii) and (iv) (d) (iii) and (iv)

81. The refractive index of a glass is 1.520 for red light and 1.525 for blue light. Let D_1 and D_2 be angles of minimum deviation for red and blue light respectively in a prism of this glass. Then, [2006]

- (a) $D_1 < D_2$
(b) $D_1 = D_2$
(c) D_1 can be less than or greater than D_2 depending upon the angle of prism
(d) $D_1 > D_2$

82. A light ray is incident perpendicularly to one face of a 90° prism and is totally internally reflected at the glass-air interface. If the angle of reflection is 45° , we conclude that the refractive index n [2004]



- (a) $n > \frac{1}{\sqrt{2}}$ (b) $n > \sqrt{2}$
(c) $n < \frac{1}{\sqrt{2}}$ (d) $n < \sqrt{2}$

TOPIC 5 Optical Instruments



83. A compound microscope consists of an objective lens of focal length 1 cm and an eye piece of focal length 5 cm with a separation of 10 cm.

The distance between an object and the objective lens, at which the strain on the eye is minimum is $\frac{n}{40}$ cm.

The value of n is _____. [NA Sep. 05, 2020 (I)]

84. In a compound microscope, the magnified virtual image is formed at a distance of 25 cm from the eye-piece. The focal length of its objective lens is 1 cm. If the magnification is 100 and the tube length of the microscope is 20 cm, then the focal length of the eye-piece lens (in cm) is _____. [NA Sep. 04, 2020 (I)]

85. The magnifying power of a telescope with tube length 60 cm is 5. What is the focal length of its eye piece?

[8 Jan. 2020 I]

- (a) 20 cm (b) 40 cm
(c) 30 cm (d) 10 cm

86. If we need a magnification of 375 from a compound microscope of tube length 150 mm and an objective of focal length 5 mm, the focal length of the eye-piece, should be close to: [7 Jan. 2020 I]
 (a) 22mm (b) 12mm
 (c) 2mm (d) 33mm
87. An observer looks at a distant tree of height 10 m with a telescope of magnifying power of 20. To the observer the tree appears : [2016]
 (a) 20 times taller (b) 20 times nearer
 (c) 10 times taller (d) 10 times nearer
88. To determine refractive index of glass slab using a travelling microscope, minimum number of readings required are : [Online April 10, 2016]
 (a) Two (b) Four (c) Three (d) Five
89. A telescope has an objective lens of focal length 150 cm and an eyepiece of focal length 5 cm. If a 50 m tall tower at a distance of 1 km is observed through this telescope in normal setting, the angle formed by the image of the tower is θ , then θ is close to : [Online April 10, 2015]
 (a) 30° (b) 15° (c) 60° (d) 1°
90. In a compound microscope, the focal length of objective lens is 1.2 cm and focal length of eye piece is 3.0 cm. When object is kept at 1.25 cm in front of objective, final image is formed at infinity. Magnifying power of the compound microscope should be: [Online April 11, 2014]
 (a) 200 (b) 100
 (c) 400 (d) 150
91. The focal lengths of objective lens and eye lens of a Galilean telescope are respectively 30 cm and 3.0 cm. telescope produces virtual, erect image of an object situated far away from it at least distance of distinct vision from the eye lens. In this condition, the magnifying power of the Galilean telescope should be:
 [Online April 9, 2014]
 (a) +11.2 (b) -11.2 (c) -8.8 (d) +8.8
92. This question has Statement-1 and Statement-2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement 1: Very large size telescopes are reflecting telescopes instead of refracting telescopes.
Statement 2: It is easier to provide mechanical support to large size mirrors than large size lenses.
 [Online April 23, 2013]
 (a) Statement-1 is true and Statement-2 is false.
 (b) Statement-1 is false and Statement-2 is true.
 (c) Statement-1 and statement-2 are true and Statement-2 is correct explanation for statement-1.
 (d) Statements-1 and statement-2 are true and Statement-2 is not the correct explanation for statement-1.
93. The focal length of the objective and the eyepiece of a telescope are 50 cm and 5 cm respectively. If the telescope is focussed for distinct vision on a scale distant 2 m from its objective, then its magnifying power will be:
 [Online April 22, 2013]
 (a) -4 (b) -8 (c) +8 (d) -2
94. A telescope of aperture 3×10^{-2} m diameter is focused on a window at 80 m distance fitted with a wire mesh of spacing 2×10^{-3} m. Given: $\lambda = 5.5 \times 10^{-7}$ m, which of the following is true for observing the mesh through the telescope?
 [Online May 26, 2012]
 (a) Yes, it is possible with the same aperture size.
 (b) Possible also with an aperture half the present diameter.
 (c) No, it is not possible.
 (d) Given data is not sufficient.
95. We wish to make a microscope with the help of two positive lenses both with a focal length of 20 mm each and the object is positioned 25 mm from the objective lens. How far apart the lenses should be so that the final image is formed at infinity? [Online May 12, 2012]
 (a) 20mm (b) 100mm
 (c) 120mm (d) 80mm
96. An experiment is performed to find the refractive index of glass using a travelling microscope. In this experiment distances are measured by [2008]
 (a) a vernier scale provided on the microscope
 (b) a standard laboratory scale
 (c) a meter scale provided on the microscope
 (d) a screw gauge provided on the microscope
97. The image formed by an objective of a compound microscope is [2003]
 (a) virtual and diminished
 (b) real and diminished
 (c) real and enlarged
 (d) virtual and enlarged
98. An astronomical telescope has a large aperture to [2002]
 (a) reduce spherical aberration
 (b) have high resolution
 (c) increase span of observation
 (d) have low dispersion



Hints & Solutions



1. (1)

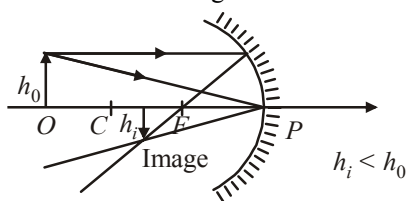
Distance of object, $u = -30$ cm

Distance of image, $v = 10$ cm

$$\text{Magnification, } m = \frac{-v}{u} = \frac{(-10)}{-30} = \frac{1}{3}$$

$$\text{Speed of image} = m^2 \times \text{speed of object} = \frac{1}{9} \times 9 = 1 \text{ cm s}^{-1}$$

2. (d) Object is placed beyond radius of curvature (R) of concave mirror hence image formed is real, inverted and diminished or unmagnified.



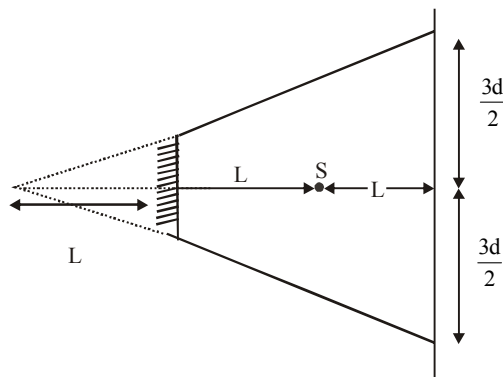
3. (c) $+5 = -\frac{v}{u} \Rightarrow v = -5u$

$$\text{Using } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\text{or } \frac{1}{-5u} + \frac{1}{u} = \frac{1}{0.4}$$

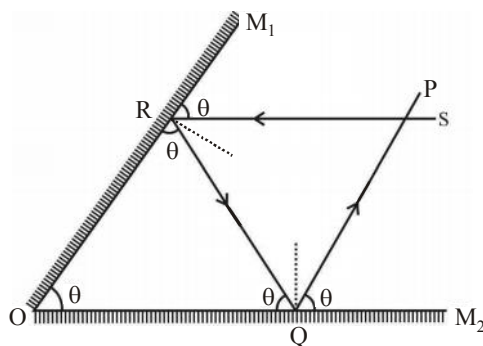
$$\therefore u = 0.32 \text{ m}$$

4. (c)



$$\text{Total distance} = \frac{3d}{2} + \frac{3d}{2} = 3d$$

5. (b)



Let angle between the two mirrors be θ .

Ray PQ $\parallel$ mirror M_1 and Rs $\parallel$ mirror M_2

$$\therefore \angle M_1 R s = \angle O R Q = \angle M_1 O M_2 = \theta$$

$$\text{Similarly, } \angle M_2 Q P = \angle O Q R = \angle M_2 O M_1 = \theta$$

$$\therefore \text{In } \triangle O R Q, 3\theta = 180^\circ \Rightarrow \theta = \frac{180^\circ}{3} = 60^\circ$$

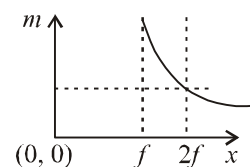
6. (c) Using mirror formula, magnification is given by

$$m = \frac{f}{u-f} = \frac{-1}{1-\frac{u}{f}}$$

At focus magnification is ∞

And at $u = 2f$, magnification is 1.

Hence graph (d) correctly depicts ' m ' versus distance of object ' x ' graph.



7. (c) When object is at 8 cm

$$\text{Image } V_1 = \frac{f \times u}{u-f} = \frac{5 \times 8}{8-5} = -\frac{40}{3} \text{ cm}$$

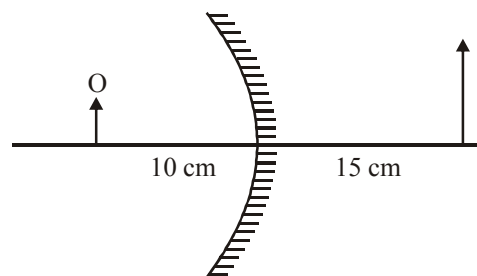
When object is at 12 cm

$$\text{Image } V_2 = \frac{f \times u}{u-f} = \frac{5 \times 12}{12-5} = -\frac{60}{7} \text{ cm}$$

$$\text{Separation} = |V_1 - V_2| = \frac{40}{3} - \frac{60}{7} = \frac{100}{21} \text{ cm}$$

So A, C and D are correct statements.

8. (c) Convex mirror is used as a shaving mirror.



From question : $v = 15$ cm, $u = -10$ cm

Radius of curvature, $R = 2f = ?$

$$\text{Using mirror formula, } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{15} + \frac{1}{(-10)} = \frac{1}{f} \Rightarrow f = -30 \text{ cm}$$

Therefore radius of curvature,
 $R = 2f = -60$ cm

9. (a) From mirror formula

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

Differentiating the above equation, we get

$$\frac{dv}{dt} = -\frac{v^2}{u^2} \left(\frac{du}{dt} \right)$$

Also,

$$\frac{v}{u} = \frac{f}{u-f}$$

$$\Rightarrow \frac{dv}{dt} = -\left(\frac{f}{u-f} \right)^2 \frac{du}{dt}$$

$$\Rightarrow \frac{dv}{dt} = \left(\frac{0.2}{2.8-0.2} \right)^2 \times 15$$

$$\Rightarrow \frac{dv}{dt} = \frac{1}{15} \text{ m/s}$$

10. (b) The number of images formed is given by

$$n = \frac{360}{\theta} - 1$$

$$\Rightarrow \frac{360}{\theta} - 1 = 3$$

$$\Rightarrow \theta = \frac{360^\circ}{4} = 90^\circ$$

11. (a) When two plane mirrors are inclined at each other at an angle θ then the number of the images (n) of a point object kept between the plane mirrors is

$$n = \frac{360^\circ}{\theta} - 1,$$

(if $\frac{360^\circ}{\theta}$ is even integer)

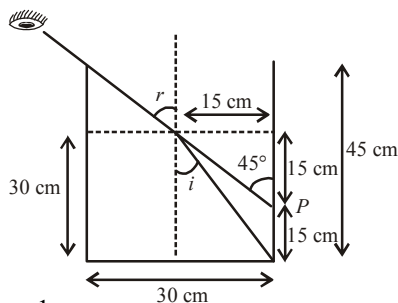
$$\therefore \text{Number of images formed} = \frac{360^\circ}{60^\circ} - 1 = 5$$

12. (158)

From figure, $\sin i = \frac{15}{\sqrt{15^2 + 30^2}}$ and $\sin r = \sin 45^\circ$

From Snell's law, $\mu \times \sin i = 1 \times \sin r$

$$\Rightarrow \mu \times \frac{15}{\sqrt{15^2 + 30^2}} = 1 \times \sin 45^\circ = \frac{1}{\sqrt{2}}$$

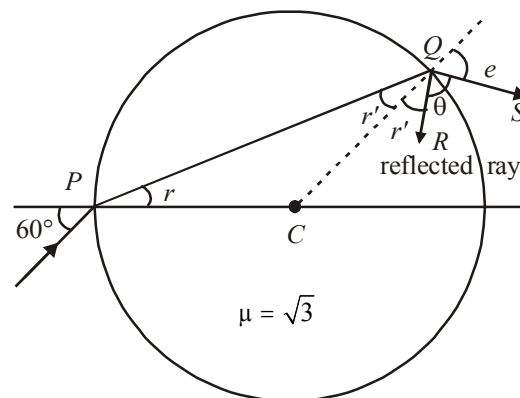


$$\therefore \mu = \frac{\frac{1}{\sqrt{2}}}{\frac{15}{\sqrt{1125}}} = 158 \times 10^{-2} = \frac{N}{100}$$

Hence, value of $N \approx 158$.

13. (90.00)

In the figure, QR is the reflected ray and QS is refracted ray. CQ is normal.



Apply Snell's law at P

$$1 \sin 60^\circ = \sqrt{3} \sin r$$

$$\Rightarrow \sin r = \frac{1}{2} \Rightarrow r = 30^\circ$$

From geometry, $CP = CQ$

$$\therefore r' = 30^\circ$$

Again apply Snell's law at Q ,

$$\sqrt{3} \sin r' = 1 \sin e$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \sin e \Rightarrow e = 60^\circ$$

From geometry

$$r' + \theta + e = 180^\circ \quad (\text{As angles lie on a straight line})$$

$$\Rightarrow 30^\circ + \theta + 60^\circ = 180^\circ \Rightarrow \theta = 90^\circ.$$

14. (d) Apparent depth,

$$h \updownarrow \begin{matrix} \mu_1 = \sqrt{2} \\ \mu_2 = 2\sqrt{2} \end{matrix} \updownarrow h$$

$$D = \frac{t_1}{\mu_1} + \frac{t_2}{\mu_2} = \frac{h}{\sqrt{2}} + \frac{h}{2\sqrt{2}} = \frac{3h}{2\sqrt{2}} = \frac{3h\sqrt{2}}{4}$$

15. (c) Given,

Refractive index, $\mu = \frac{4}{3}$

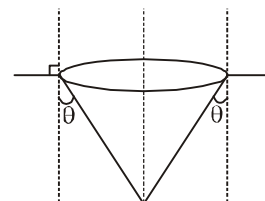
$$\frac{4}{3} \sin \theta = 1 \sin 90^\circ$$

$$\Rightarrow \sin \theta = \frac{3}{4}$$

$$\cos \theta = \frac{\sqrt{7}}{4}$$

$$\text{Solid angle, } \Omega = 2\pi(1 - \cos \theta) = 2\pi(1 - \sqrt{7}/4)$$

Fraction of energy transmitted



$$= \frac{2\pi(1 - \cos \theta)}{4\pi} = \frac{1 - \sqrt{7}/4}{2} = 0.17$$

Percentage of light emerges out of surface
 $= 0.17 \times 100 = 17\%$

16. (b) Here, from question, relative permittivity

$$\epsilon_r = \frac{\epsilon}{\epsilon_0} = 3 \Rightarrow \epsilon = 3\epsilon_0$$

$$\text{Relative permeability } \mu_r = \frac{\mu}{\mu_0} = \frac{4}{3} \Rightarrow \mu = \frac{4}{3}\mu_0$$

$$\therefore \mu\epsilon = 4\mu_0\epsilon_0$$

$$\sqrt{\frac{\mu_0\epsilon_0}{\mu\epsilon}} = \frac{v}{c} = \frac{1}{2} \left(\because c = \frac{1}{\sqrt{\mu_0\epsilon_0}} \right)$$

$$n = \sqrt{\mu_r\epsilon_r} = \sqrt{\frac{4}{3} \times 3} = 2$$

$$\text{And } n = \frac{1}{\sin \theta_c}$$

$$\Rightarrow \sin \theta_c = \frac{1}{n} = \frac{1}{2}$$

$$\therefore \text{Critical angle, } \theta_c = 30^\circ$$

17. (c) If v is the distance of image formed by mirror, then

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\text{or } \frac{1}{v} + \frac{1}{-5} = \frac{1}{-20}$$

$$\therefore v = \frac{20}{3} \text{ cm}$$

Distance of this image from water surface

$$= \frac{20}{3} + 5 = \frac{35}{3} \text{ cm}$$

$$\text{Using, } \frac{RD}{AD} = \mu$$

$$\therefore AD = d = \frac{RD}{\mu} = \frac{(35/3)}{1.33} = 8.8 \text{ cm}$$

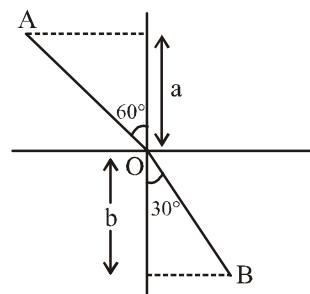
18. (c) Using, $\sin \theta_{\max} = \mu_1 \sqrt{\mu_2^2 - \mu_1^2} = \sqrt{\frac{\mu_2^2}{\mu_1^2} - 1}$

$$\text{or } \theta_{\max} = \sin^{-1} \left(\sqrt{\frac{\mu_2^2}{\mu_1^2} - 1} \right)$$

$$\text{For } T_1 R, \theta < \sin^{-1} \left(\sqrt{\frac{\mu_2^2}{\mu_1^2} - 1} \right)$$

19. (d) From the given figure

$$\text{As } \sin 60^\circ = \mu \sin 30^\circ$$



$$\Rightarrow \mu = \frac{\sin 60^\circ}{\sin 30^\circ} = \sqrt{3}$$

$$\frac{a}{AO} = \cos 60^\circ \Rightarrow AO = 2a$$

$$\frac{b}{BO} = \cos 30^\circ \Rightarrow BO = \frac{2b}{\sqrt{3}}$$

Optical path length = $AO + \mu BO$

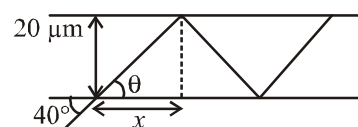
$$= 2a + (\sqrt{3}) \frac{2b}{\sqrt{3}} = 2a + 2b$$

20. (d) Using Snell's law of refraction,

$$1 \times \sin 40^\circ = 1.31 \sin \theta$$

$$\Rightarrow \sin \theta = \frac{0.64}{1.31} = 0.49 \approx 0.5$$

$$\Rightarrow \theta = 30^\circ$$



$$x = 20 \mu\text{m} \times \cot \theta$$

$$\therefore \text{Number of reflections} = \frac{2}{20 \times 10^{-6} \times \cot \theta}$$

$$= \frac{2 \times 10^6}{20 \times \sqrt{3}} = 57735 \approx 57000$$

21. (c) As 4% of light gets reflected, so only $(100 - 4 = 96\%)$ of light comes after refraction so,

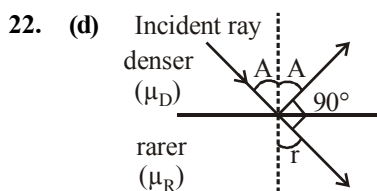
$$P_{\text{refracted}} = \frac{96}{100} P_i$$

$$\Rightarrow K_2 A_t^2 = \frac{96}{100} K_1 A_i^2$$

$$\Rightarrow r_2 A_t^2 = \frac{96}{100} r_1 A_i^2$$

$$\Rightarrow A_t^2 = \frac{96}{100} \times \frac{1}{3} \times (30)^2$$

$$A_t \sqrt{\frac{64}{100}} \times (30)^2 = 24$$



From Snell's law, $\frac{\mu_R}{\mu_D} = \frac{\sin i}{\sin r}$ (i)

$\therefore \angle i = A$ and $\angle r = (90^\circ - A)$

We also know that, $\sin \theta_C = \frac{\mu_R}{\mu_D}$

From eqⁿ (i), $\sin \theta_C = \frac{\sin A}{\sin(90^\circ - A)}$

$\sin \theta_C = \frac{\sin A}{\cos A}$

$\sin \theta_C = \tan A$

or $A = \tan^{-1}(\sin \theta_C)$

23. (d) Given, $\mu = \frac{4}{3}$

$h = 15 \text{ cm}$

$R = ?$

$\frac{\sin 90^\circ}{\sin C} = \mu$

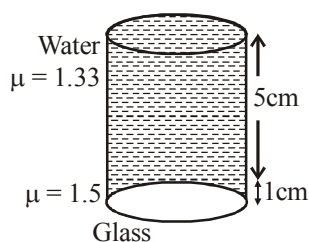
$\Rightarrow \sin C = \frac{1}{\mu} = \frac{R}{\sqrt{R^2 + h^2}} = \frac{3}{4}$

$\Rightarrow 16R^2 = 9R^2 + 9h^2$

or, $7R^2 = 9h^2$

or, $R = \frac{3}{\sqrt{7}}h = \frac{3}{\sqrt{7}} \times 15 \text{ cm}$

24. (c) Real depth = $5 \text{ cm} + 1 \text{ cm} = 6 \text{ cm}$



Apparent depth = $\frac{d_1}{\mu_1} + \frac{d_2}{\mu_2} + \dots$

$= \frac{5}{1.33} + \frac{1}{1.5}$

$\approx 3.8 + 0.7 \approx 4.5 \text{ cm}$

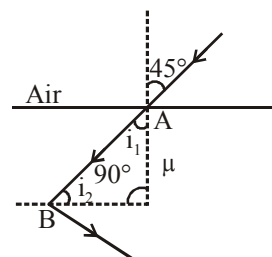
$\therefore \text{Shift} = 6 \text{ cm} - 4.5 \text{ cm} \approx 1.5 \text{ cm}$

25. (d) At point A by Snell's law

$\mu = \frac{\sin 45^\circ}{\sin r} \Rightarrow \sin r = \frac{1}{\mu\sqrt{2}}$... (i)

At point B, for total internal reflection,

$\sin i_1 = \frac{1}{\mu}$



From figure, $i_1 = 90^\circ - r$

$\therefore (\sin 90^\circ - r) = \frac{1}{\mu}$

$\Rightarrow \cos r = \frac{1}{\mu}$... (ii)

Now $\cos r = \sqrt{1 - \sin^2 r} = \sqrt{1 - \frac{1}{2\mu^2}}$

$= \sqrt{\frac{2\mu^2 - 1}{2\mu^2}}$... (iii)

From eqs (ii) and (iii)

$\frac{1}{\mu} = \sqrt{\frac{2\mu^2 - 1}{2\mu^2}}$

Squaring both sides and then solving, we get

$\mu = \sqrt{\frac{3}{2}}$

26. (b) Velocity of light in medium

$V_{\text{med}} = \frac{3 \text{ cm}}{0.2 \text{ ns}} = \frac{3 \times 10^{-2} \text{ m}}{0.2 \times 10^{-9} \text{ s}} = 1.5 \text{ m/s}$

Refractive index of the medium

$\mu = \frac{V_{\text{air}}}{V_{\text{med}}} = \frac{3 \times 10^8}{1.5} = 2 \text{ m/s}$

As $\mu = \frac{1}{\sin C}$

$\therefore \sin C = \frac{1}{\mu} = \frac{1}{2} = 30^\circ$

Condition of TIR is angle of incidence i must be greater than critical angle. Hence ray will suffer TIR in case of (B) ($i = 40^\circ > 30^\circ$) only.

27. (a) As refractive index for $z > 0$ and $z \leq 0$ is different xy plane should be the boundary between two media.

Angle of incidence is given by

$\cos(\pi - i) = \frac{(6\sqrt{3}\hat{i} + 8\sqrt{3}\hat{j} - 10\hat{k}) \cdot \hat{k}}{20}$

$$-\cos i = -\frac{1}{2}$$

$$\Rightarrow \angle i = 60^\circ$$

From Snell's law,

$$\frac{\sin i}{\sin r} = \frac{u_2}{u_1}$$

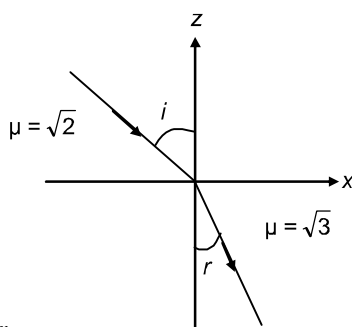
$$\Rightarrow \frac{\sin i}{\sin r} = \frac{\sqrt{3}}{\sqrt{2}}$$

$$\Rightarrow \sqrt{2} \sin i = \sqrt{3} \sin r$$

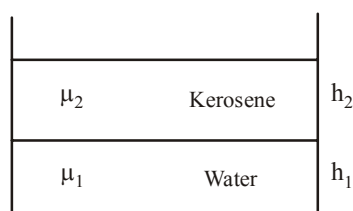
$$\Rightarrow \sqrt{2} \sin 60^\circ = \sqrt{3}$$

$$\Rightarrow \sqrt{2} \times \frac{\sqrt{3}}{2} = \sqrt{3} \sin r$$

$$\Rightarrow \angle r = 45^\circ$$



28. (b)



Apparent shift of the bottom due to water,

$$\Delta h_1 = h_1 \left[1 - \frac{1}{\mu_1} \right]$$

Apparent shift of the bottom due to kerosene, Δh_2

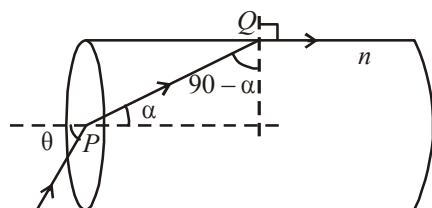
$$= h_2 \left[1 - \frac{1}{\mu_2} \right]$$

Thus, total apparent shift :

$$= \Delta h_1 + \Delta h_2$$

$$= h_1 \left(1 - \frac{1}{\mu_1} \right) + h_2 \left(1 - \frac{1}{\mu_2} \right)$$

29. (c)



Applying Snell's law

for medium inside the cylinder and air at Q we get

$$n = \frac{\sin 90^\circ}{\sin(90^\circ - \alpha)} = \frac{1}{\cos \alpha}$$

$$\therefore \cos \alpha = \frac{1}{n}$$

$$\therefore \sin \alpha = \sqrt{1 - \cos^2 \alpha} = \sqrt{1 - \frac{1}{n^2}} = \frac{\sqrt{n^2 - 1}}{n} \dots (i)$$

Applying Snell's Law for air and medium inside the cylinder at P we get

$$n = \frac{\sin \theta}{\sin \alpha}$$

$$\Rightarrow \sin \theta = n \times \sin \alpha = \sqrt{n^2 - 1}; \text{ [from (i)]}$$

$$\therefore \sin \theta = \sqrt{\left(\frac{2}{\sqrt{3}}\right)^2 - 1} = \sqrt{\frac{4}{3} - 1} = \frac{1}{\sqrt{3}}$$

$$\text{or } \theta = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

30. (a) From the figure it is clear that

$$\tan \theta_c = \frac{AB}{OA}$$

$$\Rightarrow R = OA \tan \theta_c$$

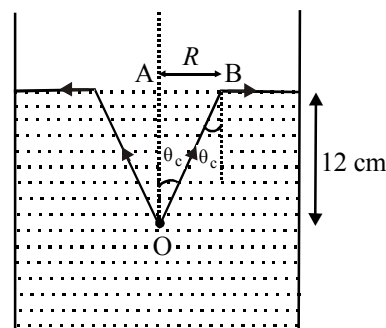
$$\Rightarrow R = \frac{OA \sin \theta_c}{\cos \theta_c}$$

$$\Rightarrow R = \frac{OA \sin \theta_c}{\sqrt{1 - \sin^2 \theta_c}}$$

$$\Rightarrow \tan \theta_c = \frac{R}{12} = \frac{\sin \theta_c}{\sqrt{1 - \sin^2 \theta_c}}$$

$$\therefore \sin \theta_c = \frac{1}{\mu} = \frac{3}{4}$$

$$\Rightarrow \tan \theta_c = \frac{3}{\sqrt{16 - 9}} = \frac{3}{\sqrt{7}} = \frac{R}{12}$$

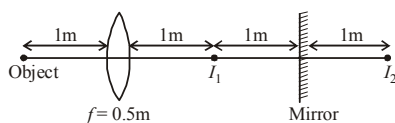


$$\Rightarrow R = \frac{36}{\sqrt{7}} \text{ cm}$$

31. (b) Optical fibres form a dielectric wave guide and are free from electromagnetic interference or radio frequency interference. There is extremely low transmission loss in optical fibre.

32. (a) In an optical fibre, light is sent through the fibre without any loss by the phenomenon of total internal reflection. Total internal reflection of light waves confine the light rays inside the optical fiber.

33. (d) Focal length of the convex lens, $f = 0.5 \text{ m}$
Object is at $2f$ so, image (I_1) will also be at $2f$.
Image of I_1 i.e., I_2 will be 1 m behind mirror.
Now I_2 will be object for lens.



$$\therefore u = (-1) + (-1) + (-1) = -3 \text{ m}$$

Using lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{+0.5} + \frac{1}{-3} \text{ or } v = \frac{3}{5} = 0.6 \text{ m}$$

Hence, distance of image from mirror
 $= 2 + 0.6 = 2.6 \text{ m}$ and real.

34. (d) Given, using lens maker's formula

$$\frac{1}{f} = (k-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

Here, $R_1 = R_2 = R$ (For double convex lens)

$$\therefore \frac{1}{f} = (\mu - 1) \left(\frac{1}{R} - \frac{1}{-R} \right)$$

$$\Rightarrow P = \frac{1}{f} = (\mu - 1) \frac{2}{R} \quad \dots(i)$$

For plano convex lens,

$$R_1 = R', R_2 = \infty$$

Using lens maker's formula again, we have

$$1.5P = (\mu - 1) \left(\frac{1}{R'} - \frac{1}{\infty} \right) \quad \dots(ii)$$

$$\Rightarrow \frac{3}{2}P = \frac{\mu - 1}{R'}$$

From (i) and (ii),

$$\frac{3}{2} = \frac{R'}{2R} \Rightarrow R' = \frac{R}{3}$$

35. (d) From lens formula,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow v = \frac{uf}{u + f}$$

Case-I : If $v = u \Rightarrow f + u = f \Rightarrow u = 0$

Case-II : If $u = \infty$ then $v = f$.

Hence, correct u versus v graph, that satisfies this condition is (a).

36. (476.19)

Given,

Distance between an object and screen, $D = 100 \text{ cm}$

Distance between the two position of lens, $d = 40 \text{ cm}$

Focal length of lens,

$$f = \frac{D^2 - d^2}{4D} = \frac{100^2 - 40^2}{4(100)} = \frac{(100 + 40)(100 - 40)}{4(100)} = 21 \text{ cm}$$

$$\text{Power, } P = \frac{1}{f} = \frac{100}{21} = \frac{N}{100}$$

$$\therefore N = 476.19.$$

37. (60) Given : $\mu = 1.5$; $R_{\text{curved}} = 30 \text{ cm}$
 Using, Lens-maker formula

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

For plano-convex lens

$$R_1 \rightarrow \infty \text{ then } R_2 = -R$$

$$\therefore f = \frac{R}{\mu - 1} = \frac{30}{1.5 - 1} = 60 \text{ cm}$$

38. (b) Using lens maker's formula

$$\frac{1}{f} = \left(\frac{\mu_g}{\mu_a} - 1 \right) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

Here, μ_g and μ_a are the refractive index of glass and air respectively

$$\Rightarrow \frac{1}{f} = (1.5 - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \dots(i)$$

When immersed in liquid

$$\frac{1}{f_l} = \left(\frac{\mu_g}{\mu_l} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

[Here, μ_l = refractive index of liquid]

$$\Rightarrow \frac{1}{f_l} = \left(\frac{1.5}{1.42} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \dots(ii)$$

Dividing (i) by (ii)

$$\Rightarrow \frac{f_l}{f} = \frac{(1.5 - 1)1.42}{0.08} = \frac{1.42}{0.16} = \frac{142}{16} \approx 9$$

39. (a) Focal length of plano-convex lens-

$$\frac{1}{f_1} = (\mu_1 - 1) \left(\frac{1}{\infty} - \frac{1}{-R} \right) = \frac{\mu_1 - 1}{R}$$

$$\Rightarrow f_1 = \frac{R}{(\mu_1 - 1)}$$

Focal length of plano-concave lens -

$$\frac{1}{f_2} = (\mu_2 - 1) \left(\frac{1}{-R} - \frac{1}{\infty} \right) = \frac{\mu_2 - 1}{-R}$$

$$\Rightarrow f_2 = \frac{-R}{(\mu_2 - 1)}$$

For the combination of two lens-

$$\frac{1}{f_{\text{eq}}} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{\mu_1 - 1}{R} - \frac{\mu_2 - 1}{R}$$

$$= \frac{\mu_1 - \mu_2}{R}$$

$$\Rightarrow f_{eq} = \frac{R}{\mu_1 - \mu_2}$$

40. (d) From the equation of line
 $m = k_1 v + k_2 \quad (\because y = mx + c)$

$$\Rightarrow \frac{v}{u} = k_1 v + k_2 \quad \left(\because m = \frac{v}{u} \right)$$

$$\Rightarrow \frac{1}{u} = k_1 + \frac{k_2}{v} \quad (\text{Dividing both sides by } v)$$

$$\Rightarrow \frac{k_2}{v} - \frac{1}{u} = -k_1$$

Comparing with lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get

$$k_1 = \frac{1}{-f} \text{ and } k_2 = 1$$

$$\therefore f = \frac{1}{\text{slope of } m-v \text{ graph}} = -\frac{b}{c}$$

41. (b) Using, $M = \frac{v}{u}$

$$\text{or } -2 = \frac{v_1}{x_1} \Rightarrow v_1 = -2x_1$$

$$\text{We have } \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\text{or } \frac{1}{-2x_1} - \frac{1}{x_1} = \frac{1}{20}$$

$$x_1 = 30 \text{ cm}$$

$$\text{And } \frac{1}{2x_2} - \frac{1}{x_2} = \frac{1}{20}$$

$$\text{or } x_2 = -10 \text{ cm}$$

$$\text{So, } \frac{x_1}{x_2} = \frac{30}{-10} = -3$$

42. (a) $\frac{1}{f_1} = \frac{2}{f_\ell}$

$$\text{Here } 2f_1 = 18 \text{ cm or } f_1 = 9 \text{ cm}$$

$$\text{So, } \frac{1}{9} = \frac{2}{f_\ell} \text{ or } f_\ell = 18 \text{ cm}$$

$$\text{Using, } \frac{1}{f_\ell} = (\mu - 1) \left(\frac{2}{R} \right)$$

$$\text{or } \frac{1}{18} = (1.5 - 1) \left(\frac{2}{R} \right)$$

$$\therefore R = 18 \text{ cm}$$

when liquid is put between, then

$$\frac{1}{f_2} = \frac{2}{f_\ell} + \frac{2}{f}$$

$$\text{or } \frac{1}{(27/2)} = \frac{2}{18} + \frac{2}{f}$$

$$\text{or } f = -54 \text{ cm}$$

$$\text{Now } -\frac{1}{54} = (\mu_1 - 1) \times \frac{1}{R}$$

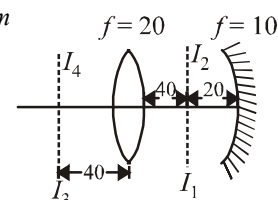
$$= (\mu_1 - 1) \times \left(\frac{1}{-18} \right)$$

$$\therefore \mu_1 = \frac{1}{3} + 1 = \frac{4}{3}$$

43. (Bouns) $v_1 = \frac{40 \times 20}{(40 - 20)} = 40 \text{ cm}$

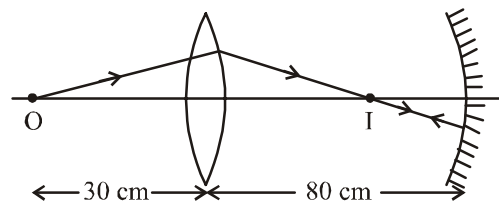
$$u_2 = 60 - 40 = 20 \text{ cm}$$

$$\therefore v_2 = \frac{20 \times 10}{(20 - 10)} = 20 \text{ cm}$$



$\therefore$ Image traces back to object itself as image formed by lens is a centre of curvature of mirror.

44. (c) For lens



$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\text{or } \frac{1}{v} - \frac{1}{-30} = \frac{1}{20}$$

$$\therefore v = +60 \text{ cm}$$

According to the condition, image formed by lens should be the centre of curvature of the mirror, and so $2f = 20$ or $f = 10 \text{ cm}$

45. (d) By lens's formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$
 For first lens, $[u_1 = -20]$

$$\frac{1}{V_1} - \frac{1}{-20} = \frac{1}{5} \Rightarrow V_1 = \frac{20}{3}$$

Image formed by first lense will behave as an object for second lens

$$\text{so, } u_2 = \frac{20}{3} - 2 = \frac{14}{3}$$

$$\frac{1}{V_2} - \frac{1}{\frac{14}{3}} = \frac{1}{-5} \Rightarrow V_2 = 70 \text{ cm}$$

46. (a) According to lens maker's formula,

$$\frac{1}{f} = (\mu_{\text{rel}} - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

Focal length of lens will change due to change in refractive index μ_{rel} . So, image will be formed at new position. Hence image disappears

47. (b) $\frac{1}{f_2} = (\mu_2 - 1) \left(\frac{+1}{R} \right)$

$$\frac{1}{f_1} = (\mu_1 - 1) \left(\frac{-1}{R} \right)$$

Now when combined the focal length is given by

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$= (\mu_1 - 1) \frac{(-1)}{R} + (\mu_2 - 1) \frac{+1}{R}$$

$$= \frac{1}{R} [\mu_2 - 1 - \mu_1 + 1]$$

$$= \frac{\mu_2 - \mu_1}{R}$$

$$\Rightarrow f = \frac{R}{\mu_2 - \mu_1}$$

48. (d) By lens formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} - \frac{1}{(-20)} = \frac{10}{3}$$

$$\frac{1}{v} = \frac{10}{3} - \frac{1}{20}$$

$$\frac{1}{v} = \frac{197}{60}; v = \frac{60}{197}$$

Magnification of lens (m) is given by

$$m = \left(\frac{v}{u} \right) = \frac{\left(\frac{60}{197} \right)}{20}$$

velocity of image wrt. to lens is given by

$$v_{I/L} = m^2 v_{O/L}$$

direction of velocity of image is same as that of object

$$v_{O/L} = 5 \text{ m/s}$$

$$v_{I/L} = \left(\frac{60 \times 1}{197 \times 20} \right)^2 (5)$$

$$= 1.16 \times 10^{-3} \text{ m/s towards the lens}$$

49. (b) From lens maker's formula,

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\frac{1}{f_1} = (\mu_1 - 1) \left(\frac{1}{\infty} - \frac{1}{-R} \right) = \frac{1}{2f_2}$$

Similarly, for plano-concave lens

$$\frac{1}{f_2} = (\mu_2 - 1) \left(\frac{1}{-R} - \frac{1}{\infty} \right)$$

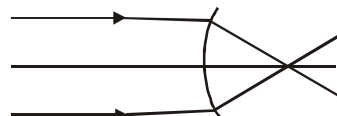
Dividing $\frac{1}{f_1}$ by $\frac{1}{f_2}$ we get,

$$\frac{(\mu_1 - 1)}{R} = \frac{(\mu_2 - 1)}{2R}$$

$$\text{or, } 2\mu_1 - \mu_2 = 1$$

50. (d) using, $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

$$R = 7.8 \text{ mm}$$

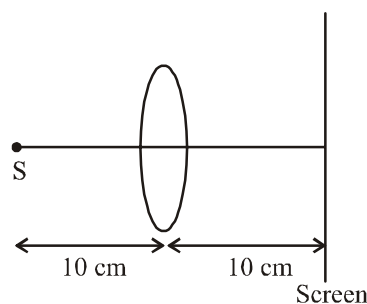


$$\mu_1 = 1, \mu_2 = 1.34$$

$$\Rightarrow \frac{1.34}{V} - \frac{1}{\infty} = \frac{1.34 - 1}{7.8} [\because u = \infty]$$

$$\therefore V = 30.7 \text{ mm} = 3.07 \text{ cm} \approx 3.1 \text{ cm}$$

51. (d)



Using lens formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{10} - \frac{1}{-10} = \frac{1}{f} \Rightarrow f = 5 \text{ cm}$$

Shift due to slab, $= t \left(1 - \frac{1}{\mu} \right)$ in the direction of incident ray

$$\text{or, } d = 1.5 \left(1 - \frac{2}{3} \right) = 0.5$$

$$\text{Now, } u = -9.5$$

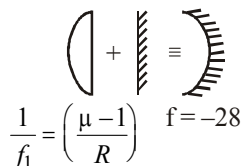
$$\text{Again using lens formula } \frac{1}{v} - \frac{1}{-9.5} = \frac{1}{5}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{5} - \frac{2}{19} = \frac{9}{95}$$

$$\text{or, } v = \frac{95}{9} = 10.55 \text{ cm}$$

Thus, screen is shifted by a distance
 $d = 10.55 - 10 = 0.55 \text{ cm}$ away from the lens.

52. (b) Case-1

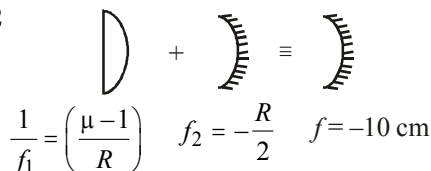


$$\frac{1}{f_1} = \left(\frac{\mu - 1}{R} \right) \quad f = -28$$

$$P = 2P_1 + P_2 \Rightarrow \frac{1}{28} = 2 \left(\frac{\mu - 1}{R} \right)$$

$$(\because \text{Power, } P = \frac{1}{f} \text{ \& } f_{\text{plane mirror}} = \infty)$$

Case-2



$$\frac{1}{f_1} = \left(\frac{\mu - 1}{R} \right) \quad f_2 = -\frac{R}{2} \quad f = -10 \text{ cm}$$

$$P = 2P_1 + P_2 \Rightarrow \frac{1}{10} = 2 \left(\frac{\mu - 1}{2} \right) + \frac{2}{R}$$

$$\text{or, } \frac{1}{10} = \frac{1}{28} + \frac{2}{R} \Rightarrow \frac{2}{R} = \frac{1}{10} - \frac{2}{28} = \frac{18}{280}$$

$$\text{or, } R = \frac{280}{9} \text{ cm}$$

$$\text{or, } \frac{1}{28} = 2 \left(\frac{\mu - 1}{280} \right) \Rightarrow \mu - 1 = \frac{5}{9}$$

$$\therefore \mu = 1 + \frac{5}{9} = \frac{14}{9} = 1.55$$

53. (a) For minimum spherical aberration separation,

$$d = f_1 - f_2 = 2 \text{ cm}$$

$$\text{Resultant focal length} = F = 10 \text{ cm}$$

$$\text{Using } \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2} \text{ and solving, we get } f_1, f_2 \text{ 18}$$

cm and 20 cm respectively.

54. (a) Given, focal length of lens (f) = 15 cm

object is placed at a distance (u) = -20 cm

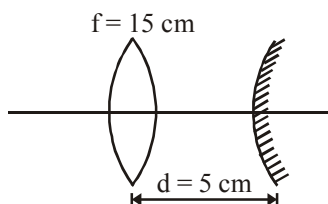
By lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{15} - \frac{1}{20}$$

$$\frac{1}{v} = \frac{4 - 3}{60}$$

$$v = 60 \text{ cm}$$



The image I gets formed at 60 cm to the right of the lens and it will be inverted.

The rays from the image (I) formed further falls on the convex mirror forms another image. This image should be formed in such a way that it coincide with object at the same point due to reflection takes place by convex mirror.

Distance between lens and mirror will be

$d = \text{image distance (v)} - \text{radius of curvature of convex mirror.}$

$$5 = 60 - 2f$$

$$2f = 60 - 5$$

$$f = \frac{55}{2} = 27.5 \text{ cm (convex mirror)}$$

55. (b) Given, radius of hemispherical glass $R = 10 \text{ cm}$

$$\therefore \text{Focal length } f = \frac{10}{2} = -5 \text{ cm}$$

$$u = (10 - 6) = -4 \text{ cm.}$$

By using mirror formula,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} + \frac{1}{-4} = \frac{1}{-5} \Rightarrow v = 20 \text{ cm.}$$

$$\text{Apparent height, } h_a = h_r \frac{\mu_1}{\mu_2} = 30 \times \frac{1}{1.5} = 20 \text{ cm below}$$

flat surface.

56. (a) Len's formula is given by

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

For convex lens,

$$\frac{1}{30} = \frac{1}{v} + \frac{1}{60} \Rightarrow \frac{1}{60} = \frac{1}{v}$$

Similarly for concave lens

$$\frac{1}{-120} = \frac{1}{v} - \frac{1}{40} \Rightarrow \frac{1}{v} = \frac{1}{60}$$

Virtual object 10 cm behind plane mirror.

Hence real image 10 cm in front of mirror or, 60 cm from convex lens.

57. (a) Taking $f_2 = 12.07$

Using Mirror's formula

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\Rightarrow \frac{1}{12.7} = \frac{1}{25.4} + \frac{1}{u} \Rightarrow \frac{1}{12.7} - \frac{1}{25.4} = \frac{1}{u}$$

$$u = 25.4 = v'$$

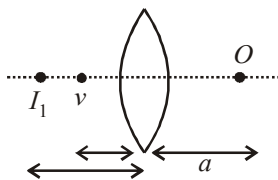
Now using Len's formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{1}{f_1} = \frac{1}{25.4 + 13.6} + \frac{1}{10}$$

$$\Rightarrow \frac{1}{f_1} = \frac{1}{39} + \frac{1}{10} \Rightarrow f_1 = \frac{390}{49} = 7.96$$

The closest answers is (a) as option (c) and (d) are not possible.

58. (d) When object is kept at a distance 'a' from thin convex lens

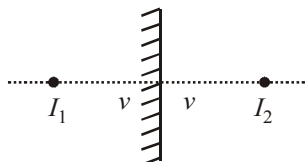


By lens formula : $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

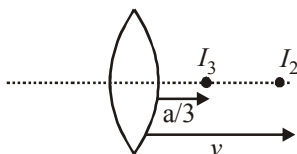
$$\frac{1}{V} - \frac{1}{(-a)} = \frac{1}{f}$$

$$\text{or, } \frac{1}{v} = \frac{1}{f} - \frac{1}{a} \quad \dots(i)$$

Mirror forms image at equal distance from mirror



Now, again from lens formula



$$\frac{3}{a} - \frac{1}{V} = \frac{1}{f}$$

$$\frac{3}{a} - \frac{1}{f} + \frac{1}{a} = \frac{1}{f}$$

[From eqn. (i)]

Hence, $a = 2f$

59. (b) By Lens maker's formula for convex lens

$$\frac{1}{f} = \left(\frac{\mu}{\mu_L} - 1 \right) \left(\frac{2}{R} \right)$$

$$\text{for, } \mu_{L1} = \frac{4}{3}, f_1 = 4R$$

$$\text{for } \mu_{L2} = \frac{5}{3}, f_2 = -5R$$

$$\Rightarrow f_2 = (-) \text{ ve}$$

60. (a) If a lens of refractive index μ is immersed in a medium of refractive index μ_1 , then its focal length in medium is given by

$$\frac{1}{f_m} = (\mu_{\mu_1} - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

If f_a is the focal length of lens in air, then

$$\frac{1}{f_a} = (\mu_{\mu_1} - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{f_m}{f_a} = \frac{(\mu_{\mu_1} - 1)}{(\mu_{\mu_1} - 1)}$$

If $\mu_1 > \mu$, then f_m and f_a have opposite signs and the nature of lens changes i.e. a convex lens diverges the light rays and concave lens converges the light rays. Thus given option (a) is correct.

61. (d) **Given:** Separation of lens for two of its position, $d = 10 \text{ cm}$

Ratio of size of the images in two positions

$$\frac{I_1}{I_2} = \frac{3}{2}$$

Distance of object from the screen, $D = ?$

Applying formula,

$$\frac{I_1}{I_2} = \frac{(D+d)^2}{(D-d)^2}$$

$$\Rightarrow \frac{3}{2} = \frac{(D+10)^2}{(D-10)^2}$$

$$\Rightarrow \frac{3}{2} = \frac{D^2 + 100 + 20D}{D^2 + 100 - 20D}$$

$$\Rightarrow 3D^2 + 300 - 60D = 2D^2 + 200 + 40D$$

$$\Rightarrow D^2 - 100D + 100 = 0$$

On solving, we get $D = 99 \text{ cm}$

Hence the distance between the screen and the object is 99 cm.

62. (c) $\therefore n = \frac{\text{Velocity of light in vacuum}}{\text{Velocity of light in medium}}$

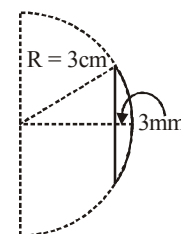
$$\therefore n = \frac{3}{2}$$

$$3^2 + (R - 3\text{mm})^2 = R^2$$

$$\Rightarrow 3^2 + R^2 - 2R(3\text{mm}) + (3\text{mm})^2 = R^2$$

$$\Rightarrow R \approx 15 \text{ cm}$$

$$\frac{1}{f} = \left(\frac{3}{2} - 1 \right) \left(\frac{1}{15} \right) \Rightarrow f = 30 \text{ cm}$$



63. (d) If side of object square = ℓ
and side of image square = ℓ'

$$\text{From question, } \frac{\ell'^2}{\ell} = 9$$

$$\text{or } \frac{\ell'}{\ell} = 3$$

i.e., magnification $m = 3$

$$u = -40 \text{ cm}$$

$$v = 3 \times 40 = 120 \text{ cm}$$

$$f = ?$$

From formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$$\frac{1}{120} - \frac{1}{-40} = \frac{1}{f}$$

$$\text{or, } \frac{1}{f} = \frac{1}{120} + \frac{1}{40} = \frac{1+3}{120} \therefore f = 30 \text{ cm}$$

64. (d) The focal length of the lens

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$= \frac{1}{12} + \frac{1}{240}$$

$$= \frac{20+1}{240} = \frac{21}{240}$$

$$f = \frac{240}{21} \text{ cm}$$

When glass plate is interposed between lens and film, so shift produced will be

$$\text{Shift} = t \left(1 - \frac{1}{\mu} \right)$$

$$1 \left(1 - \frac{1}{3/2} \right) = 1 \times \frac{1}{3}$$

Now image should be form at

$$v' = 12 - \frac{1}{3} = \frac{35}{3} \text{ cm}$$

Now the object distance u .

Using lens formula again

$$\frac{1}{f} = \frac{1}{v'} - \frac{1}{u}$$

$$\Rightarrow \frac{1}{u} = \frac{1}{v'} - \frac{1}{f}$$

$$\Rightarrow \frac{1}{u} = \frac{3}{35} - \frac{21}{240} = \frac{1}{5} \left[\frac{3}{7} - \frac{21}{48} \right]$$

$$\Rightarrow \frac{1}{u} = \frac{1}{5} \left[\frac{48-49}{7 \times 16} \right]$$

$$\Rightarrow u = -7 \times 16 \times 5 = -560 \text{ cm} = -5.6 \text{ m}$$

65. (a) From the Cauchy

$$\text{Formula, } \mu = A + \frac{B}{\lambda^2} + \frac{C}{\lambda^4}$$

$$\therefore \mu \propto \frac{1}{\lambda}$$

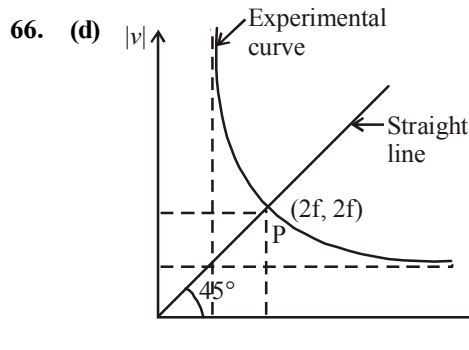
$$\text{As, } \lambda_{\text{blue}} < \lambda_{\text{red}}$$

$$\therefore \lambda_{\text{blue}} > \mu_{\text{red}}$$

From lens maker's formula

$$\text{and } \frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{1}{f_B} > \frac{1}{f_R} \Rightarrow f_R > f_B$$



For the graph to intersect $y = x$ line. The value of $|v|$ and $|u|$ must be equal.

From lens formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\text{When } u = -2f, v = 2f$$

$$\text{Also } v = \frac{f}{1 + \frac{f}{u}}$$

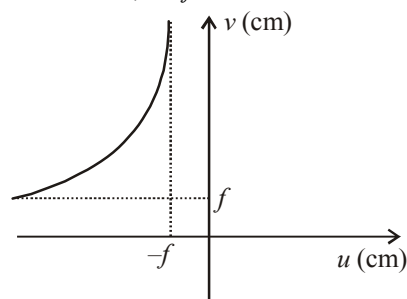
As $|u|$ increases, v decreases for $|u| > f$. The graph between $|v|$ and $|u|$ is shown in the figure. A straight line passing through the origin and making an angle of 45° with the x-axis meets the experimental curve at $P(2f, 2f)$.

67. (c) From the lens formula $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

This graph suggest that when

$$u = -f, v = +\infty$$

$$\text{When } u \text{ is at } -\infty, v = f.$$



When the object is moved further away from the lens, v decreases but remains positive.

68. (c) When two thin lenses are in contact coaxially, power of combination is given by

$$P = P_1 + P_2$$

$$= (-15 + 5)D$$

$$= -10D.$$

$$\text{Also, } P = \frac{1}{f}$$

$$\Rightarrow f = \frac{1}{P} = \frac{1}{-10} \text{ metre}$$

$$\therefore f = -\left(\frac{1}{10} \times 100\right) \text{ cm} = -10 \text{ cm}.$$

69. (b) According to lens maker's formula in air

$$\frac{1}{f_a} = (\mu_g - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{1}{f_a} = \left(\frac{1.5}{1} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \dots (i)$$

Using lens maker's formula in liquid medium,

$$\frac{1}{f_m} = \left(\frac{\mu_g}{\mu_m} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{1}{f_m} = \left(\frac{1.5}{1.6} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \dots (ii)$$

Dividing (i) by (ii),

$$\frac{f_m}{f_a} = \left(\frac{1.5 - 1}{\frac{1.5}{1.6} - 1} \right) = -8$$

$$P_a = -5 = \frac{1}{f_a}$$

$$\Rightarrow f_a = -\frac{1}{5}$$

$$\Rightarrow f_m = -8 \times f_a = -8 \times -\frac{1}{5} = \frac{8}{5}$$

$$P_m = \frac{\mu}{f_m} = \frac{1.6}{8} \times 5 = 1D$$

70. (c) The focal length (F) of the final mirror is

$$\frac{1}{F} = \frac{2}{f_\ell} + \frac{1}{f_m}$$

Using lens maker's formula

$$\text{Here } \frac{1}{f_\ell} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

Here, $R_1 = \infty$

$R_2 = 30 \text{ cm}$

$$= (1.5 - 1) \left[\frac{1}{\infty} - \frac{1}{-30} \right] = \frac{1}{60}$$

$$\therefore \frac{1}{F} = 2 \times \frac{1}{60} + \frac{1}{30/2} = \frac{1}{10}$$

$$\therefore F = 10 \text{ cm}$$

Real image will be equal to the size of the object if the object distance

$$u = 2F = 20 \text{ cm}$$

71. 2

From the Einstein's photoelectric equation

Energy of photon

= Kinetic energy of photoelectrons + Work function

$\Rightarrow$ Kinetic energy = Energy of Photon - Work Function

Let ϕ_0 be the work function of metal and v_1 and v_2 be the velocity of photoelectrons. Using Einstein's photoelectric equation we have

$$\frac{1}{2}mv_1^2 = 4 - \phi_0 \dots (i)$$

$$\frac{1}{2}mv_2^2 = 2.5 - \phi_0 \dots (ii)$$

$$\Rightarrow \frac{\frac{1}{2}mv_1^2}{\frac{1}{2}mv_2^2} = \frac{4 - \phi_0}{2.5 - \phi_0}$$

$$\Rightarrow (2)^2 = \frac{4 - \phi_0}{2.5 - \phi_0} \Rightarrow 10 - 4\phi_0 = 4 - \phi_0$$

$$\phi_0 = 2eV$$

72. (a) When angle of prism is small, then angle of deviation is given by $D_m = (\mu - 1)A$

So, if wavelength of incident light is increased, μ decreases and hence D_m decreases.

73. (c) For minimum deviation:

$$r_1 = r_2 = \frac{A}{2} = 30^\circ$$

by Snell's law $\mu_1 \sin i = \mu_2 \sin r$

$$1 \times \sin i = \sqrt{3} \times \frac{1}{2} = \frac{\sqrt{3}}{2} \Rightarrow i = 60^\circ$$

74. (c) Angle of prism, $A = 30^\circ$, $i = 60^\circ$, angle of deviation, $\delta = 30^\circ$

Using formula, $\delta = i + e - A$

$$\Rightarrow e = \delta + A - i$$

$$= 30^\circ + 30^\circ - 60^\circ = 0^\circ$$

$\therefore$ Emergent ray will be perpendicular to the face

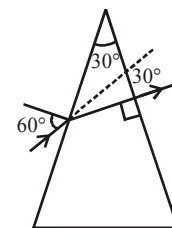
So it will make angle 90° with the face through which it emerges.

75. (c) We know that $i + e - A = \delta$

$$35^\circ + 79^\circ - A = 40^\circ \therefore A = 74^\circ$$

$$\text{But } \mu = \frac{\sin \left(\frac{A + \delta_m}{2} \right)}{\sin \frac{A}{2}} = \frac{\sin \left(\frac{74 + \delta_m}{2} \right)}{\sin \frac{74}{2}}$$

$$= \frac{5}{3} \sin \left(37^\circ + \frac{\delta_m}{2} \right)$$



$\mu_{\max}$ can be $\frac{5}{3}$. That is $\mu_{\max}$ is less than $\frac{5}{3} = 1.67$

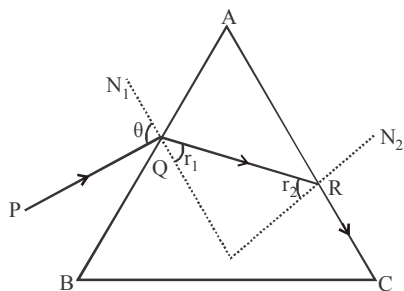
But δ_m will be less than 40° so

$$\mu < \frac{5}{3} \sin 57^\circ < \frac{5}{3} \sin 60^\circ \Rightarrow \mu = 1.5$$

76. (c) When $r_2 = C$, $\angle N_2RC = 90^\circ$

Where C = critical angle

$$\text{As } \sin C = \frac{1}{\mu} = \sin r_2$$



Applying Snell's law at 'R'

$$\mu \sin r_2 = 1 \sin 90^\circ \quad \dots(i)$$

Applying Snell's law at 'Q'

$$1 \times \sin \theta = \mu \sin r_1 \quad \dots(ii)$$

But $r_1 = A - r_2$

$$\text{So, } \sin \theta = \mu \sin (A - r_2)$$

$$\sin \theta = \mu \sin A \cos r_2 - \cos A \quad \dots(iii) \quad [\text{using (i)}]$$

From (i)

$$\cos r_2 = \sqrt{1 - \sin^2 r_2} = \sqrt{1 - \frac{1}{\mu^2}} \quad \dots(iv)$$

By eq. (iii) and (iv)

$$\sin \theta = \mu \sin A \sqrt{1 - \frac{1}{\mu^2}} - \cos A$$

on further solving we can show for ray not to be transmitted through face AC

$$\theta = \sin^{-1} \left[\mu \sin(A - \sin^{-1}(\frac{1}{\mu})) \right]$$

So, for transmission through face AC

$$\theta > \sin^{-1} \left[\mu \sin(A - \sin^{-1}(\frac{1}{\mu})) \right]$$

77. (c) For the prism as the angle of incidence (i) increases, the angle of deviation (δ) first decreases goes to minimum value and then increases.

78. (d) For light to come out through face 'AC', total internal reflection must not take place.

$$\text{i.e., } \theta < c \Rightarrow \sin \theta < \sin c$$

$$\Rightarrow \sin \theta < \frac{1}{\mu}$$

$$\text{or } \mu < \frac{1}{\sin \theta} \Rightarrow \mu < \frac{1}{\sin 45^\circ}$$

$$\Rightarrow \mu < \sqrt{2} \Rightarrow \mu < 1.414$$

79. (c) For total internal reflection on face AC
 $\theta > \text{critical angle } (C)$
 and $\sin \theta \geq \sin C$

$$\sin \theta \geq \frac{1}{\mu_g}$$

$$\sin \theta \geq \frac{\mu_w}{\mu_g} \Rightarrow \sin \theta \geq \frac{\frac{4}{3}}{\frac{3}{2}}$$

$$\therefore \sin \theta \geq \frac{8}{9}$$

80. (a) Rainbow is formed due to the dispersion of light suffering refraction and total internal reflection (TIR) in the droplets present in the atmosphere.

81. (a) When angle of prism is small,

$$\text{Angle of deviation, } D = (\mu - 1) A$$

$$\text{Since } \lambda_b < \lambda_r$$

$$\Rightarrow \mu_r < \mu_b \Rightarrow D_1 < D_2$$

82. (b) For total internal reflection
 Incident angle (i) > critical angle (i_c),

$$\therefore \sin i > \sin i_c$$

$$\Rightarrow \sin 45^\circ > \sin i_c \Rightarrow \sin i_c = \frac{1}{n}$$

$$\therefore \sin 45^\circ > \frac{1}{n}$$

$$\Rightarrow \frac{1}{\sqrt{2}} > \frac{1}{n} \Rightarrow n > \sqrt{2}$$

83. (50)

Given : Length of compound microscope, $L = 10$ cm

Focal length of objective $f_o = 1$ cm and of eye-piece,

$$f_e = 5$$

$$u_o = f_e = 5$$

Final image formed at infinity (∞), $v_e = \infty$

$$v_o = 10 - 5 = 5$$

$$\text{Using lens formula, } \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

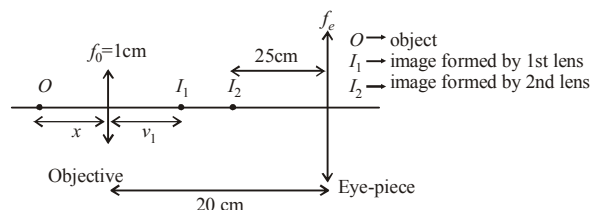
$$\frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o} \Rightarrow \frac{1}{5} - \frac{1}{u_o} = \frac{1}{1} \Rightarrow u_o = -\frac{5}{4}$$

$$\text{or, } \frac{5}{4} = \frac{N}{40}$$

$$\therefore N = \frac{200}{4} = 50 \text{ cm.}$$

84. (4.48)

According to question, final image i.e., $v_2 = 25$ cm,
 $f_0 = 1$ cm, magnification, $m = m_1 m_2 = 100$



Using lens formula,

$$\text{For first lens or objective} = \frac{1}{v_1} - \frac{1}{-x} = \frac{1}{1} \Rightarrow v_1 = \frac{x}{x-1}$$

$$\text{Also magnification } |m_1| = \left| \frac{v_1}{u_1} \right| = \frac{1}{x-1}$$

For 2nd lens or eye-piece, this is acting as object

$$\therefore u_2 = -(20 - v_1) = -\left(20 - \frac{x}{x-1}\right) \text{ and } v_2 = -25 \text{ cm}$$

$$\text{Angular magnification } |m_A| = \left| \frac{D}{u_2} \right| = \frac{25}{|u_2|}$$

$$\text{Total magnification } m = m_1 m_A = 100$$

$$\left(\frac{1}{x-1} \right) \left(\frac{25}{20 - \frac{x}{x-1}} \right) = 100$$

$$\Rightarrow \frac{25}{20(x-1) - x} = 100 \Rightarrow 1 = 80(x-1) - 4x$$

$$\Rightarrow 76x = 81 \Rightarrow x = \frac{81}{76}$$

$$\Rightarrow u_2 = -\left(20 - \frac{\frac{81}{76}}{\frac{81}{76} - 1}\right) = \frac{-19}{5}$$

Again using lens formula for eye-piece

$$\frac{1}{-25} - \frac{1}{-\frac{19}{5}} = \frac{1}{f_e} \Rightarrow f_e = \frac{25 \times 19}{106} \approx 4.48 \text{ cm}$$

85. (d) For telescope

$$\text{Tube length (L)} = f_o + f_e = 60$$

$$\text{and magnification (m)} = \frac{f_o}{f_e} = 5 \Rightarrow f_o = 5f_e$$

$$\therefore f_o = 50 \text{ cm and } f_e = 10 \text{ cm}$$

Hence focal length of eye-piece, $f_e = 10$ cm

86. (a) According question, $M = 375$

$$L = 150 \text{ mm}, f_0 = 5 \text{ mm and } f_e = ?$$

$$\text{Using, magnification, } M \approx \frac{L}{f_0} \left(1 + \frac{D}{f_e} \right)$$

$$\Rightarrow 375 = \frac{150}{5} \left(1 + \frac{250}{f_e} \right) \quad (\because D = 25 \text{ cm} = 250 \text{ mm})$$

$$\Rightarrow 12.5 = 1 + \frac{250}{f_e}$$

$$\Rightarrow f_e = \frac{250}{11.5} = 21.7 \approx 22 \text{ mm}$$

87. (b) A telescope magnifies by making the object appearing closer.

88. (c) Reading one $\Rightarrow$ without slab

Reading two $\Rightarrow$ with slab

Reading three $\Rightarrow$ with saw dust

Minimum three readings are required to determine refractive index of glass slab using a travelling microscope.

89. (c) Magnifying power of telescope,

$$MP = \frac{\beta \text{ (angle subtended by image at eye piece)}}{\alpha \text{ (angle subtended by object on objective)}}$$

$$\text{Also, } MP = \frac{f_o}{f_e} = \frac{150}{5} = 30$$

$$\alpha = \frac{50}{1000} = \frac{1}{20} \text{ rad}$$

$$\therefore \beta = \theta = MP \times \alpha = 30 \times \frac{1}{20} = \frac{3}{2} = 1.5 \text{ rad}$$

$$\text{or, } \beta = 1.5 \times \frac{180^\circ}{\pi} \approx 84^\circ$$

90. (a) Given : $f_0 = 1.2$ cm; $f_e = 3.0$ cm

$$u_0 = 1.25 \text{ cm}; M_\infty = ?$$

$$\text{From } \frac{1}{f_0} = \frac{1}{v_0} - \frac{1}{u_0}$$

$$\Rightarrow \frac{1}{1.2} = \frac{1}{v_0} - \frac{1}{(-1.25)}$$

$$\Rightarrow \frac{1}{v_0} = \frac{1}{1.2} - \frac{1}{1.25}$$

$$\Rightarrow v_0 = 30 \text{ cm}$$

Magnification at infinity,

$$M_\infty = -\frac{v_0}{u_0} \times \frac{D}{f_e}$$

$$= \frac{30}{1.25} \times \frac{25}{3}$$

($\because D = 25$ cm least distance of distinct vision) = 200
Hence the magnifying power of the compound microscope is 200

91. (d) Given, Focal length of objective, $f_0 = 30$ cm
focal length of eye lens, $f_e = 3.0$ cm
Magnifying power, $M = ?$
Magnifying power of the Galilean telescope,

$$M_D = \frac{f_0}{f_e} \left(1 - \frac{f_e}{D} \right)$$

$$= \frac{30}{3} \left(1 - \frac{3}{25} \right) [\because D = 25 \text{ cm}]$$

$$= 10 \times \frac{22}{25} = 8.8 \text{ cm}$$

92. (c) One side of mirror is opaque and another side is reflecting this is not in case of lens hence, it is easier to provide mechanical support to large size mirrors than large size lenses. Reflecting telescopes are based on the same principle except that the formation of images takes place by reflection instead of refraction.

93. (d) Given: $f_0 = 50$ cm, $f_e = 5$ cm
 $d = 25$ cm, $u_0 = -200$ cm
Magnification $M = ?$

$$\text{As } \frac{1}{v_0} - \frac{1}{u_0} = \frac{1}{f_0}$$

$$\Rightarrow \frac{1}{v_0} = \frac{1}{f_0} + \frac{1}{u_0} = \frac{1}{50} - \frac{1}{200} = \frac{4-1}{200} = \frac{3}{200}$$

$$\text{or } v_0 = \frac{200}{3} \text{ cm}$$

Now $v_e = d = -25$ cm

$$\text{From, } \frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e}$$

$$-\frac{1}{u_e} = \frac{1}{f_e} - \frac{1}{v_e}$$

$$= \frac{1}{5} + \frac{1}{25} = \frac{6}{25}$$

$$\text{or, } v_e = \frac{-25}{6} \text{ cm}$$

$$\text{Magnification } M = M_0 \times M_e$$

$$= \frac{v_0}{u_0} \times \frac{v_e}{u_e} = \frac{-200/3}{200} \times \frac{-25}{-25/6}$$

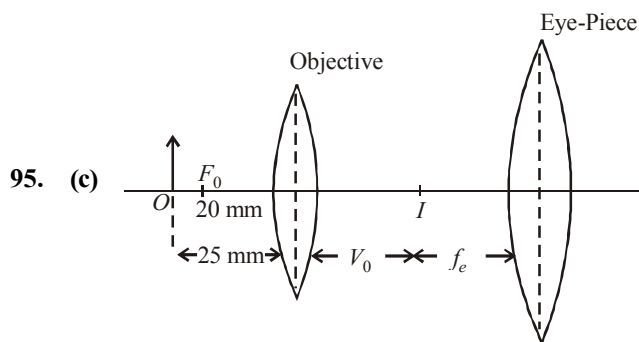
$$= -\frac{1}{3} \times 6 = -2$$

94. (a) Given: $d = 3 \times 10^{-2}$ m
 $\lambda = 5.5 \times 10^{-7}$ m

$$\text{Limit of resolution, } \Delta\theta = \frac{1.22\lambda}{d}$$

$$= \frac{1.22 \times 5.5 \times 10^{-7}}{3 \times 10^{-2}} = 2.23 \times 10^{-5} \text{ rad.}$$

At a distance of 80 m, the telescope is able to resolve between two points which are separated by $2.23 \times 10^{-5} \times 80$ m
 $= 1.78 \times 10^{-3}$ m



To obtain final image at infinity, object which is the image formed by objective should be at focal distance of eye-piece.

By lens formula (for objective)

$$\frac{1}{v_0} - \frac{1}{u_0} = \frac{1}{f_0}$$

$$\text{or, } \frac{1}{v_0} - \frac{1}{-25} = \frac{1}{20}$$

$$\Rightarrow \frac{1}{v_0} = \frac{1}{20} - \frac{1}{25} = \frac{5-4}{100} = \frac{1}{100} \text{ mm}$$

$$\therefore v_0 = 100 \text{ mm}$$

Therefore the distance between the lenses

$$= v_0 + f_e = 100 \text{ mm} + 20 \text{ mm} = 120 \text{ mm}$$

96. (a) To find the refractive index of glass using a travelling microscope, a vernier scale is provided on the microscope
97. (c) A real, inverted and enlarged image of the object is formed by the objective lens of a compound microscope.
98. (b) The resolving power of a telescope is

$$R.P. = \frac{D}{1.22\lambda}$$

where D = diameter of the objective lens

λ = wavelength of light.

$$\text{Clearly, } R.P. \propto \frac{D}{\lambda}$$

Resolving power of telescope resolution will be high if its objective is of large aperture.

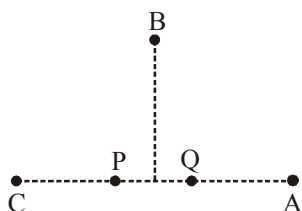
Wave Optics



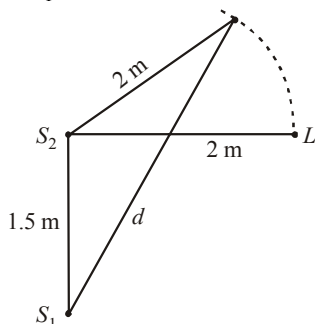
TOPIC 1 Wavefront, Interference of Light, Coherent and Incoherent Sources



1. In the figure below, P and Q are two equally intense coherent sources emitting radiation of wavelength 20 m. The separation between P and Q is 5 m and the phase of P is ahead of that of Q by 90° . A, B and C are three distinct points of observation, each equidistant from the midpoint of PQ. The intensities of radiation at A, B, C will be in the ratio : **[Sep. 06, 2020 (I)]**



- (a) 0 : 1 : 4 (b) 2 : 1 : 0
(c) 0 : 1 : 2 (d) 4 : 1 : 0
2. Two coherent sources of sound, S_1 and S_2 , produce sound waves of the same wavelength, $\lambda = 1$ m, in phase. S_1 and S_2 are placed 1.5 m apart (see fig.). A listener, located at L, directly in front of S_2 finds that the intensity is at a minimum when he is 2 m away from S_2 . The listener moves away from S_1 , keeping his distance from S_2 fixed. The adjacent maximum of intensity is observed when the listener is at a distance d from S_1 . Then, d is : **[Sep. 05, 2020 (II)]**



- (a) 12 m (b) 5 m (c) 2 m (d) 3 m

3. Two light waves having the same wavelength λ in vacuum are in phase initially. Then the first wave travels a path L_1 through a medium of refractive index n_1 while the second wave travels a path of length L_2 through a medium of refractive index n_2 . After this the phase difference between the two waves is : **[Sep. 03, 2020 (II)]**

- (a) $\frac{2\pi}{\lambda} \left(\frac{L_2}{n_1} - \frac{L_1}{n_2} \right)$ (b) $\frac{2\pi}{\lambda} \left(\frac{L_1}{n_1} - \frac{L_2}{n_2} \right)$
(c) $\frac{2\pi}{\lambda} (n_1 L_1 - n_2 L_2)$ (d) $\frac{2\pi}{\lambda} (n_2 L_1 - n_1 L_2)$

4. In an interference experiment the ratio of amplitudes of coherent waves is $\frac{a_1}{a_2} = \frac{1}{3}$. The ratio of maximum and minimum intensities of fringes will be : **[8 April 2019 I]**

- (a) 2 (b) 18 (c) 4 (d) 9

5. Two coherent sources produce waves of different intensities which interfere. After interference, the ratio of the maximum intensity to the minimum intensity is 16. The intensity of the waves are in the ratio : **[9 Jan. 2019 I]**

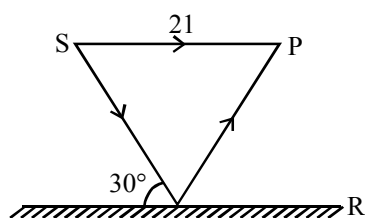
- (a) 16 : 9 (b) 25 : 9 (c) 4 : 1 (d) 5 : 3

6. On a hot summer night, the refractive index of air is smallest near the ground and increases with height from the ground. When a light beam is directed horizontally, the Huygens' principle leads us to conclude that as it travels, the light beam : **[2015]**

- (a) bends downwards
(b) bends upwards
(c) becomes narrower
(d) goes horizontally without any deflection

7. Interference pattern is observed at 'P' due to superimposition of two rays coming out from a source 'S' as shown in the figure. The value of 'l' for which maxima is obtained at 'P' is:

(R is perfect reflecting surface) **[Online April 12, 2014]**



- (a) $l = \frac{2n\lambda}{\sqrt{3}-1}$ (b) $l = \frac{(2n-1)\lambda}{2(\sqrt{3}-1)}$
 (c) $l = \frac{(2n-1)\lambda\sqrt{3}}{4(2-\sqrt{3})}$ (d) $l = \frac{(2n-1)\lambda}{\sqrt{3}-1}$

8. Two monochromatic light beams of intensity 16 and 9 units are interfering. The ratio of intensities of bright and dark parts of the resultant pattern is: [Online April 11, 2014]

- (a) $\frac{16}{9}$ (b) $\frac{4}{3}$ (c) $\frac{7}{1}$ (d) $\frac{49}{1}$

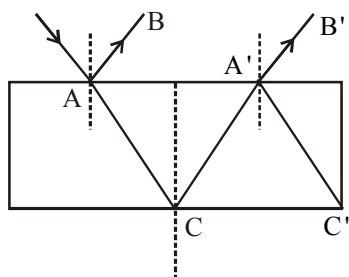
9. n identical waves each of intensity I_0 interfere with each other. The ratio of maximum intensities if the interference is (i) coherent and (ii) incoherent is:

[Online April 23, 2013]

- (a) n^2 (b) $\frac{1}{n}$ (c) $\frac{1}{n^2}$ (d) n

10. A ray of light of intensity I is incident on a parallel glass slab at point A as shown in diagram. It undergoes partial reflection and refraction. At each reflection, 25% of incident energy is reflected. The rays AB and A'B' undergo interference. The ratio of $I_{\max}$ and $I_{\min}$ is:

[Online April 9, 2013]



- (a) 49:1 (b) 7:1 (c) 4:1 (d) 8:1

11. Two coherent plane light waves of equal amplitude makes a small angle α ($\alpha \ll 1$) with each other. They fall almost normally on a screen. If λ is the wavelength of light waves, the fringe width Δx of interference patterns of the two sets of waves on the screen is [Online May 19, 2012]

- (a) $\frac{2\lambda}{\alpha}$ (b) $\frac{\lambda}{\alpha}$ (c) $\frac{\lambda}{(2\alpha)}$ (d) $\frac{\lambda}{\sqrt{\alpha}}$

12. This question has a paragraph followed by two statements, Statement – 1 and Statement – 2. Of the given four alternatives after the statements, choose the one that describes the statements.

A thin air film is formed by putting the convex surface of a plane-convex lens over a plane glass plate. With

monochromatic light, this film gives an interference pattern due to light reflected from the top (convex) surface and the bottom (glass plate) surface of the film.

Statement – 1 : When light reflects from the air-glass plate interface, the reflected wave suffers a phase change of π .

Statement – 2 : The centre of the interference pattern is dark. [2011]

- (a) Statement – 1 is true, Statement – 2 is true, Statement – 2 is the correct explanation of Statement – 1.
 (b) Statement – 1 is true, Statement – 2 is true, Statement – 2 is not the correct explanation of Statement – 1.
 (c) Statement – 1 is false, Statement – 2 is true.
 (d) Statement – 1 is true, Statement – 2 is false.

Directions : Questions number 13-15 are based on the following paragraph.

An initially parallel cylindrical beam travels in a medium of refractive index $\mu(I) = \mu_0 + \mu_2 I$, where μ_0 and μ_2 are positive constants and I is the intensity of the light beam. The intensity of the beam is decreasing with increasing radius

13. As the beam enters the medium, it will [2010]

- (a) diverge
 (b) converge
 (c) diverge near the axis and converge near the periphery
 (d) travel as a cylindrical beam

14. The initial shape of the wavefront of the beam is [2010]

- (a) convex
 (b) concave
 (c) convex near the axis and concave near the periphery
 (d) planar

15. The speed of light in the medium is [2010]

- (a) minimum on the axis of the beam
 (b) the same everywhere in the beam
 (c) directly proportional to the intensity I
 (d) maximum on the axis of the beam

16. To demonstrate the phenomenon of interference, we require two sources which emit radiation [2003]

- (a) of nearly the same frequency
 (b) of the same frequency
 (c) of different wavelengths
 (d) of the same frequency and having a definite phase relationship

TOPIC 2 Young's Double Slit Experiment



17. A young's double-slit experiment is performed using monochromatic light of wavelength λ . The intensity of light at a point on the screen, where the path difference is λ , is K units. The intensity of light at a point where the path difference is $\frac{\lambda}{6}$ is given by $\frac{nK}{12}$, where n is an integer. The value of n is _____. [NA Sep. 06, 2020 (II)]

18. In a Young's double slit experiment, light of 500 nm is used to produce an interference pattern. When the distance between the slits is 0.05 mm, the angular width (in degree) of the fringes formed on the distance screen is close to :

[Sep. 03, 2020 (I)]

- (a) 0.17° (b) 0.57°
(c) 1.7° (d) 0.07°
19. Interference fringes are observed on a screen by illuminating two thin slits 1 mm apart with a light source ($\lambda = 632.8$ nm). The distance between the screen and the slits is 100 cm. If a bright fringe is observed on a screen at a distance of 1.27 mm from the central bright fringe, then the path difference between the waves, which are reaching this point from the slits is close to : [Sep. 02, 2020 (I)]
- (a) $1.27 \mu\text{m}$ (b) 2.87 nm
(c) 2 nm (d) $2.05 \mu\text{m}$

20. In a Young's double slit experiment, 16 fringes are observed in a certain segment of the screen when light of wavelength 700 nm is used. If the wavelength of light is changed to 400 nm, the number of fringes observed in the same segment of the screen would be : [Sep. 02, 2020 (II)]
- (a) 24 (b) 30
(c) 18 (d) 28

21. In a Young's double slit experiment 15 fringes are observed on a small portion of the screen when light of wavelength 500 nm is used. Ten fringes are observed on the same section of the screen when another light source of wavelength λ is used. Then the value of λ is (in nm)

[NA 9 Jan 2020 II]

22. In a double-slit experiment, at a certain point on the screen the path difference between the two interfering waves is $\frac{1}{8}$ th of a wavelength. The ratio of the intensity of light at that point to that at the centre of a bright fringe is:

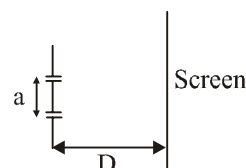
[8 Jan 2020 II]

- (a) 0.853 (b) 0.672 (c) 0.568 (d) 0.760
23. In a Young's double slit experiment, the separation between the slits is 0.15 mm. In the experiment, a source of light of wavelength 589 nm is used and the interference pattern is observed on a screen kept 1.5 m away. The separation between the successive bright fringes on the screen is: [7 Jan 2020 II]

- (a) 6.9 mm (b) 3.9 mm (c) 5.9 mm (d) 4.9 mm
24. In a double slit experiment, when a thin film of thickness t having refractive index μ is introduced in front of one of the slits, the maximum at the centre of the fringe pattern shifts by one fringe width. The value of t is ($\approx$ is the wavelength of the light used) : [12 April 2020 I]

- (a) $\frac{2\lambda}{(\mu-1)}$ (b) $\frac{\lambda}{2(\mu-1)}$
(c) $\frac{\lambda}{(\mu-1)}$ (d) $\frac{\lambda}{(2\mu-1)}$

25. The figure shows a Young's double slit experimental setup. It is observed that when a thin transparent sheet of thickness t and refractive index $\frac{3}{4}$ is put in front of one of the slits, the central maximum gets shifted by a distance equal to n fringe widths. If the wavelength of light used is λ , t will be: [9 April 2019 I]



- (a) $\frac{2nD\lambda}{a(\mu-1)}$ (b) $\frac{nD\lambda}{a(\mu-1)}$
(c) $\frac{D\lambda}{a(\mu-1)}$ (d) $\frac{2D\lambda}{a(\mu-1)}$

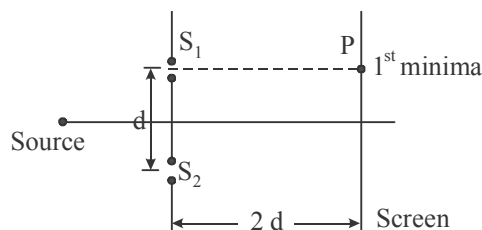
26. In a Young's double slit experiment, the path difference, at a certain point on the screen, between two interfering waves is $\frac{1}{8}$ th of wavelength. The ratio of the intensity at this point to that at the centre of a bright fringe is close to:

[11 Jan 2019 I]

- (a) 0.74 (b) 0.85 (c) 0.94 (d) 0.80
27. In a Young's double slit experiment with slit separation 0.1 mm, one observes a bright fringe at angle $\frac{1}{40}$ rad by using light of wavelength λ_1 . When the light of wavelength λ_2 is used a bright fringe is seen at the same angle in the same set up. Given that λ_1 and λ_2 are in visible range (380 nm to 740 nm), their values are:

[10 Jan. 2019 I]

- (a) 625 nm, 500 nm (b) 380 nm, 525 nm
(c) 380 nm, 500 nm (d) 400 nm, 500 nm
28. Consider a Young's double slit experiment as shown in figure. What should be the slit separation d in terms of wavelength λ such that the first minima occurs directly in front of the slit (S_1)? [10 Jan 2019 II]



- (a) $\frac{\lambda}{2(\sqrt{5}-2)}$ (b) $\frac{\lambda}{(\sqrt{5}-2)}$
(c) $\frac{\lambda}{2(5-\sqrt{2})}$ (d) $\frac{\lambda}{(5-\sqrt{2})}$

29. In a Young's double slit experiment, the slits are placed 0.320 mm apart. Light of wavelength $\lambda = 500$ nm is incident on the slits. The total number of bright fringes that are observed in the angular range $-30^\circ \leq \theta \leq 30^\circ$ is

[9 Jan 2019 II]

- (a) 640 (b) 320 (c) 321 (d) 641
30. In a Young's double slit experiment, slits are separated by 0.5 mm, and the screen is placed 150 cm away. A beam of light consisting of two wavelengths, 650 nm and 520 nm, is used to obtain interference fringes on the screen. The least distance from the common central maximum to the point where the bright fringes due to both the wavelengths coincide is :

[2017]

- (a) 9.75 mm (b) 15.6 mm (c) 1.56 mm (d) 7.8 mm
31. In a Young's double slit experiment with light of wavelength λ the separation of slits is d and distance of screen is D such that $D \gg d \gg \lambda$. If the fringe width is β , the distance from point of maximum intensity to the point where intensity falls to half of maximum intensity on either side is:

[Online April 11, 2015]

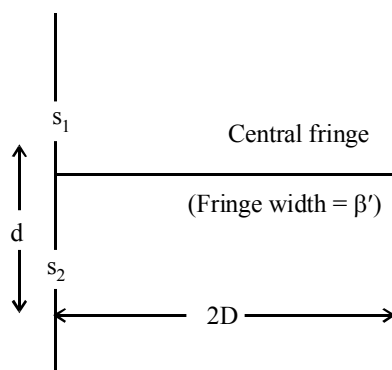
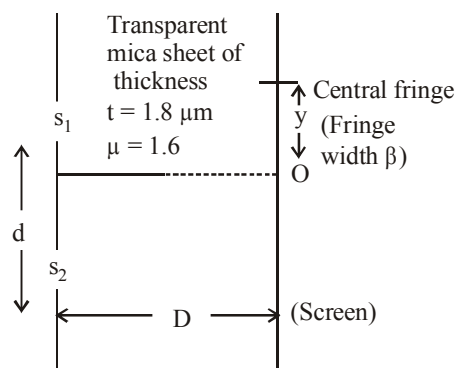
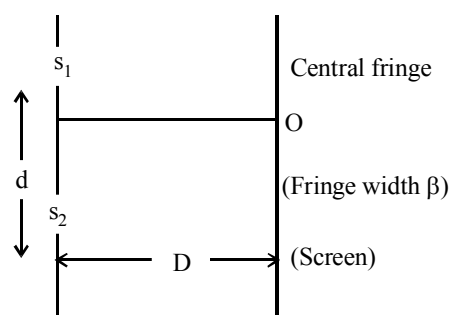
- (a) $\frac{\beta}{6}$ (b) $\frac{\beta}{3}$ (c) $\frac{\beta}{4}$ (d) $\frac{\beta}{2}$
32. In a Young's double slit experiment, the distance between the two identical slits is 6.1 times larger than the slit width. Then the number of intensity maxima observed within the central maximum of the single slit diffraction pattern is:

[Online April 19, 2014]

- (a) 3 (b) 6 (c) 12 (d) 24
33. Using monochromatic light of wavelength λ , an experimentalist sets up the Young's double slit experiment in three ways as shown.

If she observes that $y = \beta'$, the wavelength of light used is:

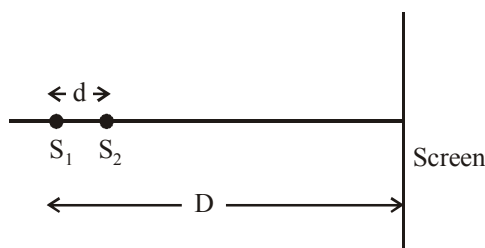
[Online April 9, 2014]



- (a) 520 nm (b) 540 nm (c) 560 nm (d) 580 nm

34. Two coherent point sources S_1 and S_2 are separated by a small distance ' d ' as shown. The fringes obtained on the screen will be

[2013]



- (a) points (b) straight lines
(c) semi-circles (d) concentric circles
35. The source that illuminates the double-slit in 'double-slit interference experiment' emits two distinct monochromatic waves of wavelength 500 nm and 600 nm, each of them producing its own pattern on the screen. At the central point of the pattern when path difference is zero, maxima of both the patterns coincide and the resulting interference pattern is most distinct at the region of zero path difference. But as one moves out of this central region, the two fringe systems are gradually out of step such that maximum due to one wavelength coincides with the minimum due to the other and the combined fringe system becomes completely indistinct. This may happen when path difference in nm is:

[Online April 25, 2013]

- (a) 2000 (b) 3000 (c) 1000 (d) 1500
36. A thin glass plate of thickness is $\frac{2500}{3} \lambda$ (λ is wavelength

of light used) and refractive index $\mu = 1.5$ is inserted between one of the slits and the screen in Young's double slit experiment. At a point on the screen equidistant from the slits, the ratio of the intensities before and after the introduction of the glass plate is :

[Online April 25, 2013]

- (a) 2 : 1 (b) 1 : 4 (c) 4 : 1 (d) 4 : 3
37. This question has Statement-1 and Statement-2. Of the four choices given after the Statements, choose the one that best describes the two Statements.

Statement-1: In Young's double slit experiment, the number of fringes observed in the field of view is small with longer wavelength of light and is large with shorter wavelength of light.

Statement-2: In the double slit experiment the fringe width depends directly on the wavelength of light.

[Online April 22, 2013]

- (a) Statement-1 is true, Statement-2 is true and the Statement-2 is correct explanation of the Statement-1.
 (b) Statement-1 is false and the Statement-2 is true.
 (c) Statement-1 is true Statement-2 is true and the Statement-2 is not correct explanation of the Statement-1.
 (d) Statement-1 is true and the Statement-2 is false.
38. In Young's double slit experiment, one of the slit is wider than other, so that amplitude of the light from one slit is double of that other slit. If I_m be the maximum intensity, the resultant intensity I when they interfere at phase difference ϕ is given by : [2012]

- (a) $\frac{I_m}{9}(4 + 5 \cos \phi)$ (b) $\frac{I_m}{3}\left(1 + 2 \cos^2 \frac{\phi}{2}\right)$
 (c) $\frac{I_m}{5}\left(1 + 4 \cos^2 \frac{\phi}{2}\right)$ (d) $\frac{I_m}{9}\left(1 + 8 \cos^2 \frac{\phi}{2}\right)$

39. In Young's double slit interference experiment, the slit widths are in the ratio 1 : 25. Then the ratio of intensity at the maxima and minima in the interference pattern is

[Online May 26, 2012]

- (a) 3 : 2 (b) 1 : 25 (c) 9 : 4 (d) 1 : 5
40. The maximum number of possible interference maxima for slit separation equal to 1.8λ , where λ is the wavelength of light used, in a Young's double slit experiment is

[Online May 12, 2012]

- (a) zero (b) 3 (c) infinite (d) 5
41. In a Young's double slit experiment with light of wavelength λ , fringe pattern on the screen has fringe width β . When two thin transparent glass (refractive index μ) plates of thickness t_1 and t_2 ($t_1 > t_2$) are placed in the path of the two beams respectively, the fringe pattern will shift by a distance [Online May 7, 2012]

- (a) $\frac{\beta(\mu-1)}{\lambda}\left(\frac{t_1}{t_2}\right)$ (b) $\frac{\mu\beta}{\lambda}\frac{t_1}{t_2}$
 (c) $\frac{\beta(\mu-1)}{\lambda}(t_1 - t_2)$ (d) $(\mu-1)\frac{\lambda}{\beta}(t_1 + t_2)$

42. At two points P and Q on screen in Young's double slit experiment, waves from slits S_1 and S_2 have a path difference of 0 and $\frac{\lambda}{4}$, respectively. The ratio of intensities at P and Q will be : [2011 RS]

- (a) 2 : 1 (b) $\sqrt{2} : 1$ (c) 4 : 1 (d) 3 : 2

43. In a Young's double slit experiment, the two slits act as coherent sources of wave of equal amplitude A and wavelength λ . In another experiment with the same arrangement the two slits are made to act as incoherent sources of waves of same amplitude and wavelength. If the intensity at the middle point of the screen in the first

case is I_1 and in the second case is I_2 , then the ratio $\frac{I_1}{I_2}$ is

[2011 RS]

- (a) 2 (b) 1 (c) 0.5 (d) 4
44. A mixture of light, consisting of wavelength 590 nm and an unknown wavelength, illuminates Young's double slit and gives rise to two overlapping interference patterns on the screen. The central maximum of both lights coincide. Further, it is observed that the third bright fringe of known light coincides with the 4th bright fringe of the unknown light. From this data, the wavelength of the unknown light is: [2009]

- (a) 885.0 nm (b) 442.5 nm
 (c) 776.8 nm (d) 393.4 nm

45. In a Young's double slit experiment the intensity at a point where the path difference is $\frac{\lambda}{6}$ (λ being the wavelength of

light used) is I . If I_0 denotes the maximum intensity, $\frac{I}{I_0}$ is equal to [2007]

- (a) $\frac{3}{4}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{1}{2}$

46. A Young's double slit experiment uses a monochromatic source. The shape of the interference fringes formed on a screen is [2005]

- (a) circle (b) hyperbola
 (c) parabola (d) straight line

47. The maximum number of possible interference maxima for slit-separation equal to twice the wavelength in Young's double-slit experiment is [2004]

- (a) three (b) five (c) infinite (d) zero

TOPIC 3 Diffraction, Polarisation of Light and Resolving Power



48. A beam of plane polarised light of large cross-sectional area and uniform intensity of 3.3 W m^{-2} falls normally on a polariser (cross sectional area $3 \times 10^{-4} \text{ m}^2$) which rotates about its axis with an angular speed of 31.4 rad/s . The energy of light passing through the polariser per revolution, is close to : [Sep. 04, 2020 (I)]

- (a) $1.0 \times 10^{-5} \text{ J}$ (b) $1.0 \times 10^{-4} \text{ J}$
 (c) $1.5 \times 10^{-4} \text{ J}$ (d) $5.0 \times 10^{-4} \text{ J}$

49. Orange light of wavelength 6000×10^{-10} m illuminates a single slit of width 0.6×10^{-4} m. The maximum possible number of diffraction minima produced on both sides of the central maximum is _____.

[NA Sep. 04, 2020 (II)]

50. The aperture diameter of telescope is 5m. The separation between the moon and the earth is 4×10^5 km. With light of wavelength of $5500 \text{ \AA}$, the minimum separation between objects on the surface of moon, so that they are just resolved, is close to:

[9 Jan. 2020 I]

- (a) 60m (b) 20m (c) 200m (d) 600m
51. A polarizer - analyser set is adjusted such that the intensity of light coming out of the analyser is just 10% of the original intensity. Assuming that the polarizer - analyser set does not absorb any light, the angle by which the analyser need to be rotated further to reduce the output intensity to be zero, is:

[7 Jan. 2020 I]

- (a) 71.6° (b) 18.4° (c) 90° (d) 45°
52. The value of numerical aperture of the objective lens of a microscope is 1.25. If light of wavelength $5000 \text{ \AA}$ is used, the minimum separation between two points, to be seen as distinct, will be:

[12 April 2019 I]

- (a) $0.24 \mu\text{m}$ (b) $0.38 \mu\text{m}$ (c) $0.12 \mu\text{m}$ (d) $0.48 \mu\text{m}$
53. A system of three polarizers P_1, P_2, P_3 is set up such that the pass axis of P_3 is crossed with respect to that of P_1 . The pass axis of P_2 is inclined at 60° to the pass axis of P_3 . When a beam of unpolarized light of intensity I_0 is incident on P_1 , the intensity of light transmitted by the three polarizers is I . The ratio (I_0/I) equals (nearly):

[12 April 2019 II]

- (a) 5.33 (b) 16.00 (c) 10.67 (d) 1.80
54. Diameter of the objective lens of a telescope is 250 cm. For light of wavelength 600 nm. Coming from a distant object, the limit of resolution of the telescope is close to:

[9 April 2019 II]

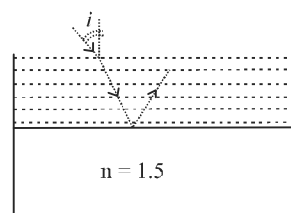
- (a) $1.5 \times 10^{-7} \text{ rad}$ (b) $2.0 \times 10^{-7} \text{ rad}$
- (c) $3.0 \times 10^{-7} \text{ rad}$ (d) $4.5 \times 10^{-7} \text{ rad}$
55. Calculate the limit of resolution of a telescope objective having a diameter of 200 cm, if it has to detect light of wavelength 500 nm coming from a star. [8 April 2019 II]

- (a) $305 \times 10^{-9} \text{ radian}$ (b) $610 \times 10^{-9} \text{ radian}$
- (c) $152.5 \times 10^{-9} \text{ radian}$ (d) $457.5 \times 10^{-9} \text{ radian}$
56. In a double-slit experiment, green light ($5303 \text{ \AA}$) falls on a double slit having a separation of $19.44 \mu\text{m}$ and a width of $4.05 \mu\text{m}$. The number of bright fringes between the first and the second diffraction minima is:

[11 Jan 2019 II]

- (a) 10 (b) 05 (c) 04 (d) 09
57. Consider a tank made of glass (refractive index 1.5) with a thick bottom. It is filled with a liquid of refractive index μ . A student finds that, irrespective of what the incident angle i (see figure) is for a beam of light entering the liquid, the light reflected from the liquid glass interface is never completely polarized. For this to happen, the minimum value of μ is:

[9 Jan. 2019 I]



- (a) $\sqrt{\frac{5}{3}}$ (b) $\frac{3}{\sqrt{5}}$ (c) $\frac{5}{\sqrt{3}}$ (d) $\frac{4}{3}$
58. The angular width of the central maximum in a single slit diffraction pattern is 60° . The width of the slit is $1 \mu\text{m}$. The slit is illuminated by monochromatic plane waves. If another slit of same width is made near it, Young's fringes can be observed on a screen placed at a distance 50 cm from the slits. If the observed fringe width is 1 cm, what is slit separation distance?

(i.e. distance between the centres of each slit.) [2018]

- (a) $25 \mu\text{m}$ (b) $50 \mu\text{m}$ (c) $75 \mu\text{m}$ (d) $100 \mu\text{m}$
59. Unpolarized light of intensity I passes through an ideal polarizer A. Another identical polarizer B is placed behind

A. The intensity of light beyond B is found to be $\frac{I}{2}$. Now another identical polarizer C is placed between A and B.

The intensity beyond B is now found to be $\frac{I}{8}$. The angle between polarizer A and C is:

[2018]

- (a) 0° (b) 30° (c) 45° (d) 60°
60. Light of wavelength 550 nm falls normally on a slit of width $22.0 \times 10^{-5} \text{ cm}$. The angular position of the second minima from the central maximum will be (in radians)

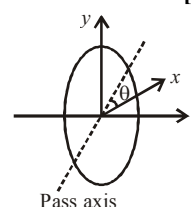
[Online April 15, 2018]

- (a) $\frac{\pi}{8}$ (b) $\frac{\pi}{12}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$
61. Unpolarized light of intensity I is incident on a system of two polarizers, A followed by B. The intensity of emergent light is $I/2$. If a third polarizer C is placed between A and B, the intensity of emergent light is reduced to $I/3$. The angle between the polarizers A and C is θ . Then

[Online April 16, 2018]

- (a) $\cos \theta = \left(\frac{2}{3}\right)^{1/4}$ (b) $\cos \theta = \left(\frac{1}{3}\right)^{1/4}$
- (c) $\cos \theta = \left(\frac{1}{3}\right)^{1/2}$ (d) $\cos \theta = \left(\frac{2}{3}\right)^{1/2}$
62. A plane polarized light is incident on a polariser with its pass axis making angle θ with x-axis, as shown in the figure. At four different values of θ , $\theta = 8^\circ, 38^\circ, 188^\circ$ and 218° , the observed intensities are same. What is the angle between the direction of polarization and x-axis

[Online April 15, 2018]



- (a) 203° (b) 45° (c) 98° (d) 128°

63. An observer is moving with half the speed of light towards a stationary microwave source emitting waves at frequency 10 GHz. What is the frequency of the microwave measured by the observer? (speed of light = $3 \times 10^8 \text{ ms}^{-1}$) [2017]
 (a) 17.3 GHz (b) 15.3 GHz
 (c) 10.1 GHz (d) 12.1 GHz
64. A single slit of width 0.1 mm is illuminated by a parallel beam of light of wavelength 6000 Å and diffraction bands are observed on a screen 0.5 m from the slit. The distance of the third dark band from the central bright band is : [Online April 9, 2017]
 (a) 3mm (b) 9mm (c) 4.5mm (d) 1.5mm
65. A single slit of width b is illuminated by a coherent monochromatic light of wavelength λ . If the second and fourth minima in the diffraction pattern at a distance 1 m from the slit are at 3 cm and 6 cm respectively from the central maximum, what is the width of the central maximum? (i.e. distance between first minimum on either side of the central maximum) [Online April 8, 2017]
 (a) 1.5 cm (b) 3.0 cm (c) 4.5 cm (d) 6.0 cm
66. The box of a pin hole camera, of length L , has a hole of radius a . It is assumed that when the hole is illuminated by a parallel beam of light of wavelength λ the spread of the spot (obtained on the opposite wall of the camera) is the sum of its geometrical spread and the spread due to diffraction. The spot would then have its minimum size (say $b_{\min}$) when : [2016]
 (a) $a = \sqrt{\lambda L}$ and $b_{\min} = \sqrt{4\lambda L}$
 (b) $a = \frac{\lambda^2}{L}$ and $b_{\min} = \sqrt{4\lambda L}$
 (c) $a = \frac{\lambda^2}{L}$ and $b_{\min} = \left(\frac{2\lambda^2}{L}\right)$
 (d) $a = \sqrt{\lambda L}$ and $b_{\min} = \left(\frac{2\lambda^2}{L}\right)$
67. Two stars are 10 light years away from the earth. They are seen through a telescope of objective diameter 30 cm. The wavelength of light is 600 nm. To see the stars just resolved by the telescope, the minimum distance between them should be (1 light year = $9.46 \times 10^{15} \text{ m}$) of the order of : [Online April 10, 2016]
 (a) 10^8 km (b) 10^{10} km (c) 10^{11} km (d) 10^6 km
68. In Young's double slit experiment, the distance between slits and the screen is 1.0 m and monochromatic light of 600 nm is being used. A person standing near the slits is looking at the fringe pattern. When the separation between the slits is varied, the interference pattern disappears for a particular distance d_0 between the slits. If the angular resolution of the eye is $\frac{1^\circ}{60}$, the value of d_0 is close to : [Online April 9, 2016]
 (a) 1mm (b) 3mm (c) 2mm (d) 4mm
69. Assuming human pupil to have a radius of 0.25 cm and a comfortable viewing distance of 25 cm, the minimum separation between two objects that human eye can resolve at 500 nm wavelength is : [2015]
 (a) 100 μm (b) 300 μm (c) 1 μm (d) 30 μm
70. Unpolarized light of intensity I_0 is incident on surface of a block of glass at Brewster's angle. In that case, which one of the following statements is true ? [Online April 11, 2015]
 (a) reflected light is completely polarized with intensity less than $\frac{I_0}{2}$
 (b) transmitted light is completely polarized with intensity less than $\frac{I_0}{2}$
 (c) transmitted light is partially polarized with intensity $\frac{I_0}{2}$
 (d) reflected light is partially polarized with intensity $\frac{I_0}{2}$
71. Two beams, A and B, of plane polarized light with mutually perpendicular planes of polarization are seen through a polaroid. From the position when the beam A has maximum intensity (and beam B has zero intensity), a rotation of polaroid through 30° makes the two beams appear equally bright. If the initial intensities of the two beams are I_A and I_B respectively, then $\frac{I_A}{I_B}$ equals : [2014]
 (a) 3 (b) $\frac{3}{2}$ (c) 1 (d) $\frac{1}{3}$
72. The diameter of the objective lens of microscope makes an angle β at the focus of the microscope. Further, the medium between the object and the lens is an oil of refractive index n . Then the resolving power of the microscope [Online April 19, 2014]
 (a) increases with decreasing value of n
 (b) increases with decreasing value of β
 (c) increases with increasing value of $n \sin 2\beta$
 (d) increases with increasing value of $\frac{1}{n \sin 2\beta}$
73. A ray of light is incident from a denser to a rarer medium. The critical angle for total internal reflection is θ_{iC} and Brewster's angle of incidence is θ_{iB} , such that $\sin \theta_{iC} / \sin \theta_{iB} = \eta = 1.28$. The relative refractive index of the two media is : [Online April 19, 2014]
 (a) 0.2 (b) 0.4 (c) 0.8 (d) 0.9

74. In an experiment of single slit diffraction pattern, first minimum for red light coincides with first maximum of some other wavelength. If wavelength of red light is $6600 \text{ \AA}$, then wavelength of first maximum will be:

[Online April 12, 2014]

- (a) $3300 \text{ \AA}$ (b) $4400 \text{ \AA}$ (c) $5500 \text{ \AA}$ (d) $6600 \text{ \AA}$
75. A beam of unpolarised light of intensity I_0 is passed through a polaroid A and then through another polaroid B which is oriented so that its principal plane makes an angle of 45° relative to that of A. The intensity of the emergent light is [2013]
- (a) I_0 (b) $I_0/2$ (c) $I_0/4$ (d) $I_0/8$
76. This question has Statement-1 and Statements-2. Of the four choices given after the Statements, choose the one that best describes the two Statements.

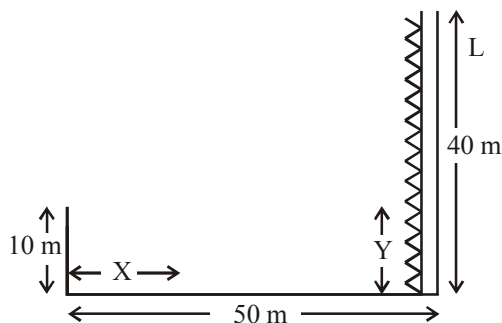
Statement-1 : Out of radio waves and microwaves, the radio waves undergo more diffraction.

Statement-2 : Radio waves have greater frequency compared to microwaves. [Online April 25, 2013]

- (a) Statement-1 is true, Statement-2 is true and Statement-2 is the correct explanation of Statement-1
- (b) Statement-1 is false, Statement-2 is true.
- (c) Statement-1 is true, Statement-2 is false.
- (d) Statement-1 is true, Statement-2 is true but Statement-2 is not the correct explanation of Statement-1
77. A person lives in a high-rise building on the bank of a river 50 m wide. Across the river is a well lit tower of height 40 m. When the person, who is at a height of 10 m, looks through a polarizer at an appropriate angle at light of the tower reflecting from the river surface, he notes that intensity of light coming from distance X from his building is the least and this corresponds to the light coming from light bulbs at height 'Y' on the tower. The values of X and Y are

respectively close to (refractive index of water $\approx \frac{4}{3}$)

[Online April 9, 2013]



- (a) 25 m, 10 m (b) 13 m, 27 m
- (c) 22 m, 13 m (d) 17 m, 20 m

78. The first diffraction minimum due to the single slit diffraction is seen at $\theta = 30^\circ$ for a light of wavelength $5000 \text{ \AA}$ falling perpendicularly on the slit. The width of the slit is [Online May 12, 2012]

- (a) $2.5 \times 10^{-5} \text{ cm}$ (b) $1.25 \times 10^{-5} \text{ cm}$
- (c) $10 \times 10^{-5} \text{ cm}$ (d) $5 \times 10^{-5} \text{ cm}$

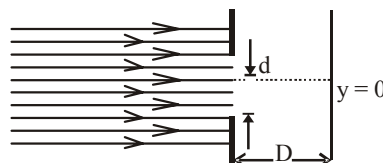
79. Two polaroids have their polarizing directions parallel so that the intensity of a transmitted light is maximum. The angle through which either polaroid must be turned if the intensity is to drop by one-half is [Online May 7, 2012]

- (a) 135° (b) 90° (c) 120° (d) 180°

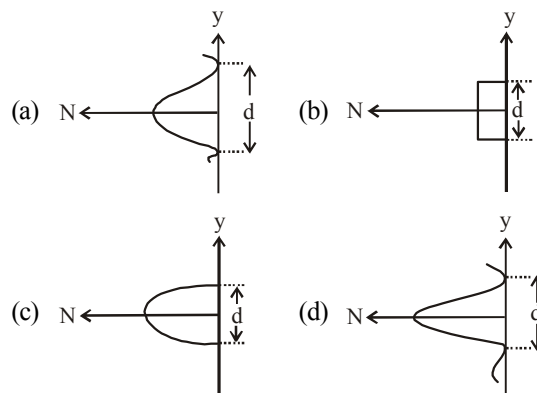
80. **Statement - 1 :** On viewing the clear blue portion of the sky through a Calcite Crystal, the intensity of transmitted light varies as the crystal is rotated.

Statement - 2 : The light coming from the sky is polarized due to scattering of sun light by particles in the atmosphere. The scattering is largest for blue light. [2011 RS]

- (a) Statement-1 is true, statement-2 is false.
- (b) Statement-1 is true, statement-2 is true, statement-2 is the correct explanation of statement-1
- (c) Statement-1 is true, statement-2 is true, statement-2 is not the correct explanation of statement-1
- (d) Statement-1 is false, statement-2 is true.
81. In an experiment, electrons are made to pass through a narrow slit of width ' d ' comparable to their de Broglie wavelength. They are detected on a screen at a distance ' D ' from the slit (see figure).



Which of the following graphs can be expected to represent the number of electrons ' N ' detected as a function of the detector position ' y ' ($y = 0$ corresponds to the middle of the slit) [2008]



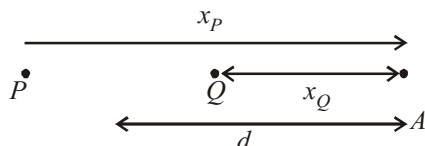
82. If I_0 is the intensity of the principal maximum in the single slit diffraction pattern, then what will be its intensity when the slit width is doubled? [2005]
(a) $4I_0$ (b) $2I_0$ (c) $\frac{I_0}{2}$ (d) I_0
83. When an unpolarized light of intensity I_0 is incident on a polarizing sheet, the intensity of the light which does not get transmitted is [2005]
(a) $\frac{1}{4}I_0$ (b) $\frac{1}{2}I_0$ (c) I_0 (d) zero
84. Two point white dots are 1 mm apart on a black paper. They are viewed by eye of pupil diameter 3 mm. Approximately, what is the maximum distance at which these dots can be resolved by the eye?
[Take wavelength of light = 500 nm] [2005]
(a) 1 m (b) 5 m (c) 3 m (d) 6 m
85. The angle of incidence at which reflected light is totally polarized for reflection from air to glass (refractive index n), is [2004]
(a) $\tan^{-1}(1/n)$ (b) $\sin^{-1}(1/n)$
(c) $\sin^{-1}(n)$ (d) $\tan^{-1}(n)$
86. Wavelength of light used in an optical instrument are $\lambda_1 = 4000 \text{ \AA}$ and $\lambda_2 = 5000 \text{ \AA}$, then ratio of their respective resolving powers (corresponding to λ_1 and λ_2) is [2002]
(a) 16:25 (b) 9:1 (c) 4:5 (d) 5:4



Hints & Solutions



1. (b) For (A)



$$x_P - x_Q = (d + 2.5) - (d - 2.5) = 5 \text{ m}$$

Phase difference $\Delta\phi$ due to path difference

$$= \frac{2\pi}{\lambda}(\Delta x) = \frac{2\pi}{20}(5) = \frac{\pi}{2}.$$

At A, Q is ahead of P by path, as wave emitted by Q reaches before wave emitted by P.

$$\therefore \text{Total phase difference at A } \frac{\pi}{2} - \frac{\pi}{2} = 0$$

(due to P being ahead of Q by 90°)

$$\begin{aligned} I_A &= I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2} \cos \Delta\phi \\ &= I + I + 2\sqrt{I}\sqrt{I} \cos(0) = 4I \end{aligned}$$

For C,

$$\text{Path difference, } x_Q - x_P = 5 \text{ m}$$

Phase difference $\Delta\phi$ due to path difference

$$= \frac{2\pi}{\lambda}(\Delta x) = \frac{2\pi}{20}(5) = \frac{\pi}{2}$$

$$\text{Total phase difference at C } = \frac{\pi}{2} + \frac{\pi}{2} = \pi$$

$$I_{\text{net}} = I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2} \cos(\Delta\phi)$$

$$= I + I + 2\sqrt{I}\sqrt{I} \cos(\pi) = 0$$

For B,

$$\text{Path difference, } x_P - x_Q = 0$$

$$\text{Phase difference, } \Delta\phi = \frac{\pi}{2}$$

(due to P being ahead of Q by 90°)

$$I_B = I + I + 2\sqrt{I}\sqrt{I} \cos \frac{\pi}{2} = 2I$$

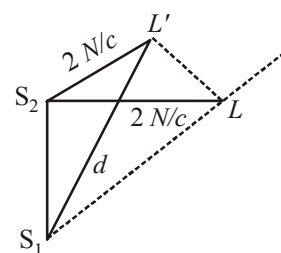
Therefore intensities of radiation at A, B and C will be in the ratio

$$I_A : I_B : I_C = 4I : 2I : 0 = 2 : 1 : 0.$$

2. (d) Initially, $S_2L = 2 \text{ m}$

$$S_1L = \sqrt{2^2 + \left(\frac{3}{2}\right)^2} = \frac{5}{2} = 2.5 \text{ m}$$

$$\text{Path difference, } \Delta x = S_1L - S_2L = 0.5 \text{ m} = \frac{\lambda}{2}$$



When the listener move from L, first maxima will appear if path difference is integral multiple of wavelength.

For example

$$\Delta x = n\lambda = 1\lambda \quad (n = 1 \text{ for first maxima})$$

$$\therefore \Delta x = \lambda = S_1L' - S_2L$$

$$\Rightarrow 1 = d - 2 \Rightarrow d = 3 \text{ m}$$

3. (c) The distance traversed by light in a medium of refractive index m in time t is given by

$$d = vt \quad \dots(i)$$

where v is velocity of light in the medium. The distance traversed by light in a vacuum in this time,

$$\Delta = ct = c \times \frac{d}{v} \quad [\text{from equation (i)}]$$

$$= d \frac{c}{v} = \mu d \quad \dots(ii) \quad (\because \mu = \frac{c}{v})$$

This distance is the equivalent distance in vacuum and is called optical path.

Optical path for first ray which travels a path L_1 through a medium of refractive index $n_1 = n_1L_1$

Optical path for second ray which travels a path L_2 through a medium of refractive index $n_2 = n_2L_2$

$$\text{Path difference} = n_1L_1 - n_2L_2$$

Now, phase difference

$$= \frac{2\pi}{\lambda} \times \text{path difference} = \frac{2\pi}{\lambda} \times (n_1L_1 - n_2L_2)$$

4. (c) Given amplitude ratio of waves is $\frac{a_1}{a_2} = \frac{3}{1}$

$$\text{so, } \frac{I_{\max}}{I_{\min}} = \left(\frac{a_2 + a_1}{a_2 - a_1} \right)^2 = \left(\frac{\frac{a_2}{a_1} + 1}{\frac{a_2}{a_1} - 1} \right)^2 = \left(\frac{3+1}{3-1} \right)^2 = \left(\frac{4}{2} \right)^2 = \frac{4}{1} = 4$$

5. (b) As we know, $\frac{I_{\max}}{I_{\min}} = \left(\frac{A_1 + A_2}{A_1 - A_2} \right)^2$

$$\text{and } \sqrt{\frac{I_1}{I_2}} = \frac{A_1}{A_2}$$

$$\frac{I_{\max}}{I_{\min}} = 16 \Rightarrow \frac{A_{\max}}{A_{\min}} = 4 \Rightarrow \frac{A_1 + A_2}{A_1 - A_2} = \frac{4}{1}$$

Using componendo and dividendo.

$$\frac{A_1}{A_2} = \frac{5}{3} \Rightarrow \frac{I_1}{I_2} = \left(\frac{5}{3} \right)^2 = \frac{25}{9}$$

6. (b) (Light bends upwards)
Refracted WF ←
-

7. (c)

8. (d) Intensity $\propto$ (amplitude)²

$$\frac{I_1}{I_2} = \frac{16}{9} = \frac{a_1^2}{a_2^2}$$

$$\Rightarrow a_1 = 4; a_2 = 3$$

Therefore the ratio of intensities of bright and dark parts

$$\frac{I_{\text{Bright}}}{I_{\text{Dark}}} = \frac{(a_1 + a_2)^2}{(a_1 - a_2)^2} = \frac{(4 + 3)^2}{(4 - 3)^2} = \frac{49}{1}$$

9. (d) $\frac{(\text{Maximum intensity}) \text{ coherent interference}}{(\text{Maximum intensity}) \text{ in coherent interference}}$

$$= \frac{n^2 I_0}{n I_0} = n$$

10. (a)
-

$$\text{From figure } I_1 = \frac{I}{4} \text{ and } I_2 = \frac{9I}{64}$$

$$\Rightarrow \frac{I_2}{I_1} = \frac{9}{16}$$

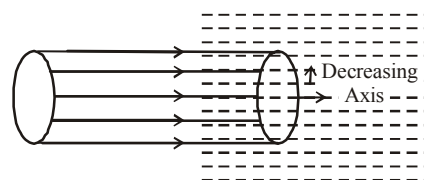
$$\text{By using } \frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{\frac{I_2}{I_1}} + 1}{\sqrt{\frac{I_2}{I_1}} - 1} \right)^2 = \left(\frac{\sqrt{\frac{9}{16}} + 1}{\sqrt{\frac{9}{16}} - 1} \right)^2 = \frac{49}{1}$$

11. (c) $\Delta x = \frac{\lambda}{(2\alpha)}$

12. (b) A phase change of π rad appears when the ray reflects at the glass-air interface. As a result, there will be a destructive interference at the centre. So, the centre of the interference pattern is dark.

13. (b) When light beam is moving and as it enters the medium, the refractive index will decrease from the axis towards the periphery of the beam.

Therefore, the beam will converge less distance as one move from the axis to the periphery and hence the beam will converge.



14. (d) Initially the parallel beam is cylindrical. Therefore, the wavefront will be planar.

15. (a) The speed of light (v) in a medium of refractive index (μ) is given by

$$\mu = \frac{c}{v}, \text{ where } c \text{ is the speed of light in vacuum}$$

$$\therefore v = \frac{c}{\mu} = \frac{c}{\mu_0 + \mu_2(I)}$$

As I is decreasing with increasing radius, it is maximum on the axis of the beam. Therefore, v is minimum on the axis of the beam.

16. (d) To demonstrate the phenomenon of interference we require two sources of light which emit radiation of same frequency and having a definite phase relationship (a phase relationship that does not change with time)

17. (9)

In young's double slit experiment, intensity at a point is given by

$$I = I_0 \cos^2 \frac{\phi}{2} \quad \dots(i)$$

where, ϕ = phase difference,

Using phase difference, $\phi = \frac{2\pi}{\lambda} \times \text{path difference}$

For path difference λ , phase difference $\phi_1 = 2\pi$

For path difference, $\frac{\lambda}{6}$, phase difference $\phi_2 = \frac{\pi}{3}$

Using equation (i),

$$\frac{I_1}{I_2} = \frac{\cos^2\left(\frac{\phi_1}{2}\right)}{\cos^2\left(\frac{\phi_2}{2}\right)} = \frac{\cos^2\left(\frac{2\pi}{2}\right)}{\cos^2\left(\frac{\pi}{3}\right)}$$

$$\Rightarrow \frac{K}{I_2} = \frac{1}{\frac{3}{4}} = \frac{4}{3} \Rightarrow I_2 = \frac{3K}{4} = \frac{9K}{12}$$

$\therefore n = 9$.

18. (b) Given : Wavelength of light, $\lambda = 500 \text{ nm}$
Distance between the slits, $d = 0.05 \text{ mm}$
Angular width of the fringe formed,

$$\theta = \frac{\lambda}{d} = \frac{500 \times 10^{-9}}{0.05 \times 10^{-3}} = 0.01 \text{ rad} = 0.57^\circ$$

19. (a) Path difference, $\Delta P = d \sin \theta = d\theta$
 $d = \text{distance between slits} = 1 \text{ mm} = 10^{-3} \text{ m}$
 $D = \text{distance between the slits and screen} = 100 \text{ cm} = 1 \text{ m}$
 $y = \text{distance between central bright fringe and observed fringe} = 1.27 \text{ mm}$

$$\therefore \Delta P = \frac{dy}{D} = \frac{10^{-3} \times 1.270 \text{ mm}}{1 \text{ m}} = 1.27 \mu\text{m}$$

20. (d) Let n_1 fringes are visible with light of wavelength λ_1 and n_2 with light of wavelength λ_2 . Then

$$\beta = \frac{n_1 D \lambda_1}{d} = \frac{n_2 D \lambda_2}{d} \quad \left(\because \beta = \frac{n \lambda D}{d} \right)$$

$$\Rightarrow \frac{n_2}{n_1} = \frac{\lambda_1}{\lambda_2}$$

$$\Rightarrow n_2 = \frac{700}{400} \times 16 = 28$$

21. (750) Fringe width, $\beta = \frac{\lambda D}{d}$ where, $\lambda = \text{wavelength}$, $D =$ distance of screen from slits, $d = \text{distance between slits}$

ATQ

$$15 \times \frac{\lambda_1 D}{d} = 10 \times \frac{\lambda_2 D}{d}$$

$$\Rightarrow 15\lambda_1 = 10\lambda_2$$

$$\Rightarrow \lambda_2 = 1.5\lambda_1 = 1.5 \times 500 \text{ nm}$$

$$\Rightarrow \lambda_2 = 750 \text{ nm}$$

22. (a) Given, Path difference, $\Delta x = \frac{\lambda}{8}$

Phase differences, $\Delta\phi = \frac{2\pi}{\lambda} \Delta x$

$$= \frac{2\pi}{\lambda} \times \frac{\lambda}{8} = \frac{\pi}{4}$$

$$I = I_0 \cos^2\left(\frac{\Delta\phi}{2}\right)$$

$$\Rightarrow \frac{I}{I_0} = \cos^2\left(\frac{\frac{\pi}{4}}{2}\right) = \cos^2\left(\frac{\pi}{8}\right)$$

$$\Rightarrow \frac{I}{I_0} = 0.853$$

23. (c) Given, distance between screen and slits, $D = 1.5 \text{ m}$
Separation between slits, $d = 0.15 \text{ mm}$
Wavelength of source of light, $\lambda = 589 \text{ nm}$
Fringe-width,

$$w = \frac{D}{d} \lambda = \frac{1.5}{0.15 \times 10^{-3}} \times 589 \times 10^{-9} \text{ m}$$

$$= 589 \times 10^{-2} \text{ mm} = 5.89 \text{ mm} \approx 5.9 \text{ mm}$$

24. (c) Given, $\Delta = \beta$

$$\text{or } \frac{D(\mu-1)t}{d} = \frac{D\lambda}{d}$$

$$\therefore t = \frac{\lambda}{(\mu-1)}$$

25. (Bonus) Shift $= n\beta$ (given)

$$\therefore D \frac{(\mu-1)t}{a} = \frac{n\lambda D}{a} \quad \left[\because \text{Shift} = \frac{D(\mu-1)t}{a} \right]$$

$$\text{or } t = \frac{n\lambda}{(\mu-1)}$$

26. (b) Given, path difference, $\Delta x = \frac{\lambda}{8}$

Phase difference ($\Delta\phi$) is given by

$$\Delta\phi = \frac{2\pi}{\lambda} (\Delta x)$$

$$\Delta\phi = \frac{(2\pi)\lambda}{\lambda} \frac{1}{8} = \frac{\pi}{4}$$

For two sources in different phases,

$$I = I_0 \cos^2\left(\frac{\pi}{8}\right)$$

$$\frac{I}{I_0} = \cos^2\left(\frac{\pi}{8}\right)$$

$$= \frac{1 + \cos \frac{\pi}{4}}{2} = \frac{1 + \frac{1}{\sqrt{2}}}{2} = 0.85$$

27. (a) Path difference = $d \sin \theta \approx d \theta$

$$= 0.1 \times \frac{1}{40} \text{ mm} = 2500 \text{ nm}$$

For bright fringe, path difference must be integral multiple of λ .

$$\therefore 2500 = n\lambda_1 = m\lambda_2$$

$$\therefore \lambda_1 = 625 \text{ (for } n = 4), \lambda_2 = 500 \text{ (for } m = 5)$$

28. (a) Here, $x_1 = 2d$ and $x_2 = \sqrt{5}d$

$$\text{For, first minima, } \Delta x = \frac{\lambda}{2}$$

$$\therefore \Delta x = x_2 - x_1 = \sqrt{5}d - 2d = \frac{\lambda}{2}$$

$$\Rightarrow d = \frac{\lambda}{2(\sqrt{5} - 2)}$$

29. (d) For 'n' number of maximas

$$d \sin \theta = n\lambda$$

$$0.32 \times 10^{-3} \sin 30^\circ = n \times 500 \times 10^{-9}$$

$$\therefore n = \frac{0.32 \times 10^{-3}}{500 \times 10^{-9}} \times \frac{1}{2} = 320$$

Hence total no. of maximas observed in angular range $-30^\circ \leq \theta \leq 30^\circ$

$$= 320 + 1 + 320 = 641$$

30. (d) For common maxima, $n_1\lambda_1 = n_2\lambda_2$

$$\Rightarrow \frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{520 \times 10^{-9}}{650 \times 10^{-9}} = \frac{4}{5}$$

For λ_1

$$y = \frac{n_1\lambda_1 D}{d}, \lambda_1 = 650 \text{ nm}$$

$$y = \frac{4 \times 650 \times 10^{-9} \times 1.5}{0.5 \times 10^{-3}} \quad \text{or, } y = 7.8 \text{ mm}$$

31. (c) $2I_0 = 4I_0 \cos^2\left(\frac{\Delta\phi}{2}\right)$ here, $\Delta\phi = \frac{\pi}{2}$

$$\text{But, } \Delta\phi = \frac{2\pi}{\lambda} \Delta x \quad \text{so, } \Delta x = \frac{\lambda}{4}$$

$$\frac{dy}{D} = \frac{\lambda}{4} \quad \dots(i)$$

$$\frac{\lambda D}{d} = \beta \quad \dots(ii)$$

Multiplying equation (i) and (ii) we get,

$$y = \frac{\beta}{4}$$

32. (c)

33. (b) Given $t = 1.8 \times 10^{-6} \text{ m}$

$$\mu = 1.6$$

$$n = 2 \text{ (from figure)}$$

$$\text{Applying formula } (\mu - 1)t = n\lambda$$

$$(1.6 - 1) \times 1.8 \times 10^{-6} = 2\lambda$$

$$\text{or, } \lambda = \frac{1.8 \times 10^{-6} \times 0.6}{2}$$

$$= 540 \text{ nm}$$

34. (d) It will be concentric circles.

35. (d)

36. (c)

37. (c) Fringe width $B = \frac{D}{d}\lambda$

And number of fringes observed in the field of view is

$$\text{obtained by } \frac{d}{\lambda}$$

38. (d) Let a_1 be the amplitude of light from first slit and a_2 be the amplitude of light from second slit.

$$a_1 = a, \text{ Then } a_2 = 2a$$

$$\text{Intensity } I \propto (\text{amplitude})^2$$

$$\therefore I_1 = a_1^2 = a^2$$

$$I_2 = a_2^2 = 4a^2 = 4I_1$$

$$I_r = a_1^2 + a_2^2 + 2a_1a_2 \cos \phi$$

$$= I_1 + I_2 + 2\sqrt{I_1I_2} \cos \phi$$

$$I_r = I_1 + 4I_1 + 2\sqrt{4I_1^2} \cos \phi$$

$$\Rightarrow I_r = 5I_1 + 4I_1 \cos \phi \quad \dots(1)$$

$$\text{Now, } I_{\max} = (a_1 + a_2)^2 = (a + 2a)^2 = 9a^2$$

$$I_{\max} = 9I_1 \Rightarrow I_1 = \frac{I_{\max}}{9}$$

Substituting in equation (1)

$$I_r = \frac{5I_{\max}}{9} + \frac{4I_{\max}}{9} \cos \phi$$

$$I_r = \frac{I_{\max}}{9} [5 + 4 \cos \phi]$$

$$I_r = \frac{I_{\max}}{9} \left[5 + 8 \cos^2 \frac{\phi}{2} - 4 \right]$$

$$I_r = \frac{I_{\max}}{9} \left[1 + 8 \cos^2 \frac{\phi}{2} \right]$$

39. (c) We know that,

$$\frac{I_{\max}}{I_{\min}} = \frac{\left(\sqrt{\frac{\omega_1}{\omega_2}} + 1\right)^2}{\left(\sqrt{\frac{\omega_1}{\omega_2}} - 1\right)^2}$$

$I_{\max}$ and $I_{\min}$ are maximum and minimum intensity

ω_1 and ω_2 are widths of two slits

$$\therefore \frac{I_{\max}}{I_{\min}} = \frac{\left(\sqrt{\frac{1}{25}} + 1\right)^2}{\left(\sqrt{\frac{1}{25}} - 1\right)^2} \left(\frac{\omega_1}{\omega_2} = \frac{1}{25} \text{ given}\right)$$

On solving we get,

$$\frac{I_{\max}}{I_{\min}} = \frac{36}{16} = \frac{9}{4} = 9 : 4$$

40. (b) As $\sin \theta = \frac{n\lambda}{d}$ and $\sin \theta$ cannot be $\neq 1$

$$\therefore 1 = \frac{n\lambda}{1.8\lambda}$$

or $n = 1.8$

Hence maximum number of possible interference maxima, 0, ± 1 i.e. 3

41. (c) Shift = $\frac{\beta(\mu-1)}{\lambda} t_1 - \frac{\beta(\mu-1)}{\lambda} t_2$
 $= \frac{\beta(\mu-1)}{\lambda} (t_1 - t_2)$

42. (a) Path difference at P $\Delta x_1 = 0$
 $\therefore$ Phase difference at P will be

$$\begin{aligned} \Delta \phi_1 &= \frac{2\pi}{\lambda} \Delta x_1 \\ &= \frac{2\pi}{\lambda} \times 0 \\ &= 0^\circ \end{aligned}$$

Resultant Intensity at P

$$I_1 = I_0 + I_0 + 2I_0 \cos 0^\circ = 4I_0$$

Path difference at Q

$$\Delta x_2 = \frac{\lambda}{4}$$

$\therefore$ Phase difference at Q

$$\Delta = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} = \left(\frac{\pi}{2}\right)$$

Resultant intensity at Q .

$$I_2 = I_0 + I_0 + 2I_0 \cos \frac{\pi}{2} = 2I_0$$

$$\text{Thus, } \frac{I_1}{I_2} = \frac{4I_0}{2I_0} = \frac{2}{1}$$

43. (a) For coherent sources, intensity at mid point

$$I_1 \propto (a+a)^2$$

$$\Rightarrow I_1 \propto (2a)^2$$

For incoherent sources, intensity of mid point is

$$I_2 \propto 2a^2$$

$$\therefore \frac{I_1}{I_2} = \frac{2}{1}$$

44. (b) Let λ be the wavelength of unknown light. Third bright fringe of known light coincides with the 4th bright fringe of the unknown light.

$$\therefore \frac{3\lambda_1 D}{d} = \frac{4\lambda D}{d}$$

$$\therefore \frac{3(590)D}{d} = \frac{4\lambda D}{d}$$

$$\Rightarrow \lambda = \frac{3}{4} \times 590 = 442.5 \text{ nm}$$

45. (a) For path difference of λ , the phase difference is 2π

For path difference of $\frac{\lambda}{6}$, the phase difference is

$$\frac{2\pi \times \lambda / 6}{\lambda} = \frac{\pi}{3}$$

Resultant intensity

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \frac{\pi}{3}$$

$$\therefore I = I_1 + I_2 + \sqrt{I_1 I_2}$$

For two identical source, $I_1 = I_2 = I'$ (say)

then $I = 3I'$

$$I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$$

Maximum resultant intensity,

$$= (\sqrt{I'} + \sqrt{I'})^2 = (2\sqrt{I'})^2 = 4I'$$

$$\therefore \frac{I}{I_{\max}} = \frac{3}{4}$$

✚ ALTERNATE SOLUTION

The intensity of light at any point of the screen where the phase difference due to light coming from the two slits is ϕ is given by

$$I = I_0 \cos^2 \left(\frac{\phi}{2}\right)$$

Where I_0 is the maximum intensity.

NOTE This formula is applicable when $I_1 = I_2$.

$$\text{Phase difference } \phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{6} = \pi/3$$

$$\therefore \frac{I}{I_0} = \cos^2 \frac{\pi}{6} = \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3}{4}$$

46. (d) The light passing through the slits interfere and produce dark and bright band on screen. The shape of interference fringes formed on a screen in case of a monochromatic source is a straight line.

47. (b) For constructive interference path difference (As $\sin \theta \leq 1$)

$$d \sin \theta = n\lambda$$

Given $d = 2\lambda$

$$\therefore 2\lambda \sin \theta = n\lambda \Rightarrow \sin \theta = \frac{n}{2}$$

$n = 0, 1, -1, 2, -2$ hence five maxima are possible.

48. (d) Given :

Intensity, $I_0 = 3.3 \text{ Wm}^{-2}$

Area, $A = 3 \times 10^{-4} \text{ m}^2$

Angular speed, $\omega = 31.4 \text{ rad/s}$

Average energy = $I_0 A < \cos^2 \theta >$

$$\therefore < \cos^2 \theta > = \frac{1}{2} \text{ per revolution}$$

$$\therefore \text{Average energy} = \frac{(3.3)(3 \times 10^{-4})}{2} \approx 5 \times 10^{-4} \text{ J}$$

49. (198)

For obtaining secondary minima at a point path difference should be integral multiple of wavelength

$$\therefore d \sin \theta = n\lambda$$

$$\therefore \sin \theta = \frac{n\lambda}{d}$$

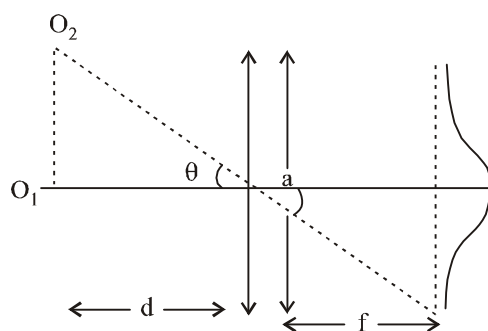
For n to be maximum $\sin \theta = 1$

$$n = \frac{d}{\lambda} = \frac{6 \times 10^{-5}}{6 \times 10^{-7}} = 100$$

Total number of minima on one side = 99

Total number of minima = 198.

50. (a)



Smallest angular separation between two distant objects here moon and earth,

$$\theta = 1.22 \frac{\lambda}{a}$$

a = aperture diameter of telescope

$$\text{Distance } O_1 O_2 = (\theta)d$$

Minimum separation between objects on the surface of moon,

$$= \left(1.22 \frac{\lambda}{a} \right) d$$

$$= \frac{(1.22)(5500 \times 10^{-10}) \times 4 \times 10^5 \times 10^3}{5}$$

$$= 5368 \times 10^{-2} \text{ m} = 53.68 \text{ m} \approx 60 \text{ m}$$

51. (b) According to question, the intensity of light coming out of the analyser is just 10% of the original intensity (I_0) Using, $I = I_0 \cos^2 \theta$

$$\Rightarrow \frac{I_0}{10} = I_0 \cos^2 \theta \Rightarrow \frac{1}{10} = \cos^2 \theta$$

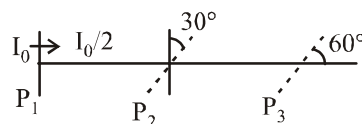
$$\Rightarrow \cos \theta = \frac{1}{\sqrt{10}} = 0.316 \Rightarrow \theta \approx 71.6^\circ$$

Therefore, the angle by which the analyser need to be rotated further to reduced the output intensity to be zero $\phi = 90^\circ - \theta = 90^\circ - 71.6^\circ = 18.4^\circ$

52. (a) $x = \frac{1.22\lambda}{2\mu \sin \theta}$

$$= \frac{1.22 \times 5000}{2 \times 1.25} = 0.24 \mu\text{m}$$

53. (c) $I = \left(\frac{I_0}{2} \right) \cos^2 30^\circ \cos^2 60^\circ$



$$= \frac{I_0}{2} \times \frac{3}{4} \times \frac{1}{4}$$

$$\therefore \frac{I_0}{I} = \frac{32}{3} = 10.67$$

54. (c) $\theta = \frac{1.22\lambda}{d} = \frac{1.22 \times 600 \times 10^{-9}}{250 \times 10^{-2}}$
 $= 3.0 \times 10^{-7} \text{ rad}$

55. (a) $\theta = \frac{1.22\lambda}{d} = \frac{1.22 \times 500 \times 10^{-9}}{2} = 305 \times 10^{-9} \text{ rad.}$

56. (b)

57. (b) According to Brewster's law, refractive index of material (μ) is equal to tangent of polarising angle

$$\therefore \tan i_b = \mu = \frac{1.5}{\mu}$$

$$\frac{1}{\mu} < \frac{1.5}{\sqrt{\mu^2 + (1.5)^2}} \left(\because \sin i_c < \sin i_b \right)$$

$$\therefore \sin i_b = \frac{1.5}{\sqrt{\mu^2 + (1.5)^2}}$$

$$\text{or, } \sqrt{\mu^2 + (1.5)^2} < 1.5 \times \mu$$

$$\Rightarrow \mu^2 + (1.5)^2 < (\mu \times 1.5)^2$$

$$\Rightarrow \mu < \frac{3}{\sqrt{5}} \text{ i.e. minimum value of } \mu \text{ should}$$

$$\text{be } \frac{3}{\sqrt{5}}$$

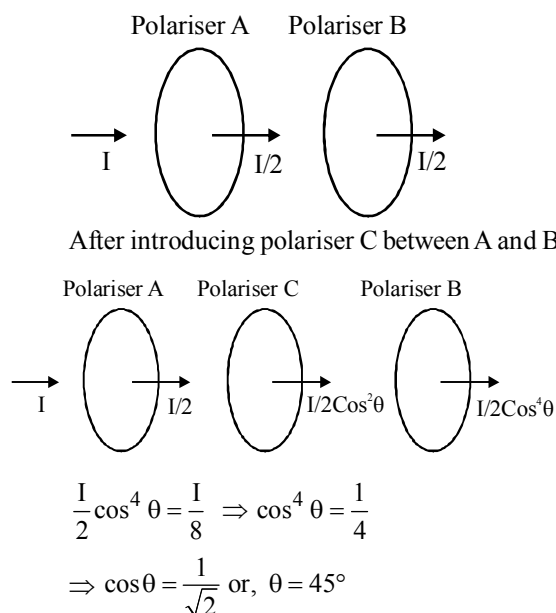
58. (a) Angular width of central maxima = $\frac{2\lambda}{d}$

or, $\lambda = \frac{d}{2}$; Fringe width, $\beta = \frac{\lambda \times D}{d'}$

$$10^{-2} = \frac{d}{2} \times \frac{50 \times 10^{-2}}{d'} = \frac{10^{-6} \times 50 \times 10^{-2}}{2 \times d'}$$

Therefore, slit separation distance, $d' = 25 \mu\text{m}$

59. (c) Axis of transmission of A & B are parallel.



60. (a) If angular position of 2nd maxima from central maxima is θ then

$$\sin \theta = \frac{(2n-1)\lambda}{2a} = \frac{3\lambda}{20} = \frac{3 \times 550 \times 10^{-9}}{2 \times 22 \times 10^{-7}}$$

$$\therefore \theta \approx \frac{\pi}{8} \text{ rad}$$

61. (a) Polariser A and B have same alignment of transmission axis.

Lets assume polariser c is introduced at θ angle

$$\frac{1}{2} \cos^2 \theta \times \cos^2 \theta = \frac{1}{3}$$

or, $\cos^4 \theta = \frac{2}{3} \Rightarrow \cos \theta = \left(\frac{2}{3}\right)^{1/4}$

62. (a)

63. (a) Use relativistic doppler's effect as velocity of observer is not small as compared to light

$$f = f_0 \sqrt{\frac{c+v}{c-v}}; \quad V = \text{relative speed of approach}$$

$$f_0 = 10 \text{ GHz}$$

$$f = 10 \sqrt{\frac{c + \frac{c}{2}}{c - \frac{c}{2}}} = 10\sqrt{3} = 17.3 \text{ GHz}$$

64. (b) $a = 0.1 \text{ mm} = 10^{-4} \text{ cm}$,
 $\lambda = 6000 \times 10^{-10} \text{ cm} = 6 \times 10^{-7} \text{ cm}$, $D = 0.5 \text{ m}$
 for 3rd dark band, $a \sin \theta = 3\lambda$

$$\text{or } \sin \theta = \frac{3\lambda}{a} = \frac{x}{D}$$

The distance of the third dark band from the central bright band

$$x = \frac{3\lambda D}{a} = \frac{3 \times 6 \times 10^{-7} \times 0.5}{10^{-4}} = 9 \text{ mm}$$

65. (b) For secondary minima,

$$b \sin \theta = n\lambda \Rightarrow \sin \theta = \frac{n\lambda}{b}$$

Distance of n^{th} secondary minima $x = D \sin \theta$

$$\text{or } \sin \theta_1 = \frac{x_1}{D}$$

$$\sin \theta_1 = \frac{2\lambda}{b}$$

$$n = 4$$

$$\sin \theta_2 = \frac{4\lambda}{b} = \frac{x_2}{D}$$

$$x_2 - x_1 = \frac{4\lambda}{b} - \frac{2\lambda}{b} = \frac{2\lambda}{b}$$

$$3 = \frac{2\lambda}{b} \Rightarrow b = \frac{2\lambda}{3} \quad \dots (i)$$

$$\begin{aligned} \text{Width of central maxima} &= \frac{2\lambda}{b} \\ &= \frac{2\lambda}{\frac{2\lambda}{3}} = 3 \text{ cm.} \end{aligned}$$

.. from eq. (i)

66. (a) Given geometrical spread = a

$$\text{Diffraction spread} = \frac{\lambda}{a} \times L = \frac{\lambda L}{a}$$

$$\text{The sum } b = a + \frac{\lambda L}{a}$$

For b to be minimum

$$\frac{db}{da} = 0 \quad \frac{d}{da} \left(a + \frac{\lambda L}{a} \right) = 0$$

$$a = \sqrt{\lambda L}$$

$$b_{\text{min}} = \sqrt{\lambda L} + \sqrt{\lambda L} = 2\sqrt{\lambda L} = \sqrt{4\lambda L}$$

67. (a) We know that $\Delta\theta = \frac{0.61\lambda}{4} = \frac{l}{R}$

The minimum distance between them

$$l = \frac{R}{9} 0.61 \times \lambda = \frac{9.46 \times 10^{15} \times 10 \times 0.61 \times 600 \times 10^{-9}}{0.3}$$

$$= 1.15 \times 10^{11} \text{ m}$$

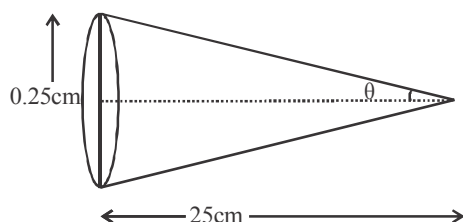
$$\Rightarrow 1.115 \times 10^8 \text{ km.}$$

68. (c) Given $D = 1.0 \text{ m}$, wavelength of monochromatic light $\lambda = 600 \text{ nm}$.

$$d : D\theta = 1 \times \frac{\pi}{180} \times \frac{1}{60}$$

$$d_0 = 2 \times 10^{-3} = 2 \text{ mm}$$

69. (d) $\sin \theta = \frac{0.25}{25} = \frac{1}{100}$



$$\text{Resolving power} = \frac{1.22\lambda}{2\mu \sin \theta} = 30 \mu\text{m.}$$

70. (a) When unpolarised light is incident at Brewster's angle then reflected light is completely polarized and the intensity of the reflected light is less than half of the incident light.

71. (d) According to Malus law, intensity of emerging beam is given by,

$$I = I_0 \cos^2 \theta$$

$$\text{Now, } I_{A'} = I_A \cos^2 30^\circ$$

$$I_{B'} = I_B \cos^2 60^\circ$$

$$\text{As } I_{A'} = I_{B'}$$

$$\Rightarrow I_A \times \frac{3}{4} = I_B \times \frac{1}{4}; \frac{I_A}{I_B} = \frac{1}{3}$$

72. (c) Resolving power of microscope,

$$\text{R.P.} = \frac{2n \sin \theta}{\lambda}$$

λ = Wavelength of light used to illuminate the object

n = Refractive index of the medium between object and objective

θ = Angle

73. (c) Here, $\sin \theta_{ic} / \sin \theta_{iB} = 1.28$

As we know,

$$\mu = \frac{\sin \theta_{iB}}{\sin \left(\frac{\pi}{2} - \theta_{iB} \right)}$$

where, θ_{iB} is Brewster's angle of incidence,

$$\text{And, } \mu = \frac{1}{\sin \theta_{ic}}$$

On solving we get, relative refractive index of the two media.

74. (b) In a single slit experiment, For diffraction maxima,

$$a \sin \theta = (2n + 1) \frac{\lambda}{2}$$

and for diffraction minima,

$$a \sin \theta = n\lambda$$

According to question,

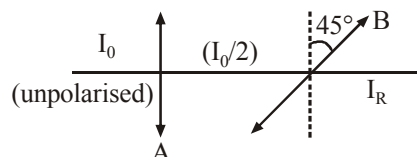
$$(2 \times 1 + 1) \frac{\lambda}{2} = 1 \times 6600$$

$$(\because \lambda_R = 6600 \text{ \AA})$$

$$\lambda = \frac{6600 \times 2}{3}$$

$$\lambda = 4400 \text{ \AA}$$

75. (c) Relation between intensities



$$I_r = \left(\frac{I_0}{2} \right) \cos^2(45^\circ) = \frac{I_0}{2} \times \frac{1}{2} = \frac{I_0}{4}$$

76. (c) Wavelength of radio waves is greater than microwaves hence frequency of radio waves is less than microwaves.

The degree of diffraction is greater whose wavelength is greater.

77. (b)

78. (c) For first minimum,

$$d \sin \theta = \lambda$$

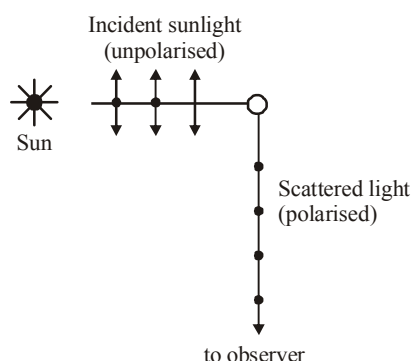
$$\Rightarrow d = \frac{\lambda}{\sin \theta} = \frac{5000 \times 10^{-8} \text{ cm}}{\sin 30^\circ} = \frac{5000 \times 10^{-8} \text{ cm}}{1/2} = 10 \times 10^{-5} \text{ cm}$$

79. (a) For $I = \frac{I_0}{2}$ and $I = I_0 \cos^2 \theta = \frac{I_0}{2}$

$$\therefore \theta = 45^\circ$$

Therefore the angle through which either polaroids turned is $135^\circ (= 180^\circ - 45^\circ)$

80. (b) When viewed through a polaroid which is rotated then the light from a clear blue portion of the sky shows a rise and fall of intensity. The light coming from the sky is polarised due to scattering of sunlight by particles in the atmosphere.



81. (d) The electron beam will be diffracted and the maxima is obtained at $y = 0$.
Also, the diffraction pattern, should be wider than the slit width.

82. (a) $I = I_0 \left(\frac{\sin \phi}{\phi} \right)^2$ and $\phi = \frac{\pi}{\lambda} (b \sin \theta)$

When the slit width is doubled, the amplitude of the wave at the centre of the screen is doubled, so the intensity at the centre is increased by a factor 4.

83. (b) From the law of Malus, $I = I_0 \cos^2 \theta$
When an unpolarised light is converted into plane polarised light by passing through polaroid, its intensity become half.

$$\therefore \text{Intensity of polarized light} = \frac{I_0}{2}$$

$$\Rightarrow \text{Intensity of untransmitted light}$$

$$= I_0 - \frac{I_0}{2} = \frac{I_0}{2}$$

84. (b) $\frac{y}{D} \geq 1.22 \frac{\lambda}{d}$

$$\Rightarrow D \leq \frac{yd}{(1.22)\lambda} = \frac{10^{-3} \times 3 \times 10^{-3}}{(1.22) \times 5 \times 10^{-7}} = \frac{30}{6.1} \approx 5\text{m}$$

$$\therefore D_{\max} = 5\text{m}$$

85. (d) From the Brewster's law, angle of incidence for total polarization is given by $\tan \theta = n$

$$\Rightarrow \theta = \tan^{-1} n$$

Where n is the refractive index of the glass.

86. (d) The resolving power of an optical instrument is inversely proportional to the wavelength of light used.

$$\frac{(R.P)_1}{(R.P)_2} = \frac{\lambda_2}{\lambda_1} = \frac{5}{4}$$

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Dual Nature of Radiation and Matter



TOPIC 1 Matter Waves, Cathode and Positive Rays



- An electron, a doubly ionized helium ion (He^{++}) and a proton are having the same kinetic energy. The relation between their respective de-Broglie wavelengths λ_e , $\lambda_{\text{He}^{++}}$ and λ_p is : **[Sep. 06, 2020 (I)]**
 - $\lambda_e > \lambda_{\text{He}^{++}} > \lambda_p$
 - $\lambda_e < \lambda_{\text{He}^{++}} = \lambda_p$
 - $\lambda_e > \lambda_p > \lambda_{\text{He}^{++}}$
 - $\lambda_e < \lambda_p < \lambda_{\text{He}^{++}}$
- Assuming the nitrogen molecule is moving with r.m.s. velocity at 400 K, the de-Broglie wavelength of nitrogen molecule is close to :
(Given : nitrogen molecule weight : 4.64×10^{-26} kg, Boltzman constant : 1.38×10^{-23} J/K, Planck constant : 6.63×10^{-34} J.s) **[Sep. 06, 2020 (II)]**
 - 0.24 Å
 - 0.20 Å
 - 0.34 Å
 - 0.44 Å
- Particle A of mass $m_A = \frac{m}{2}$ moving along the x-axis with velocity v_0 collides elastically with another particle B at rest having mass $m_B = \frac{m}{3}$. If both particles move along the x-axis after the collision, the change $\Delta\lambda$ in de-Broglie wavelength of particle A, in terms of its de-Broglie wavelength (λ_0) before collision is : **[Sep. 04, 2020 (I)]**
 - $\Delta\lambda = \frac{3}{2}\lambda_0$
 - $\Delta\lambda = \frac{5}{2}\lambda_0$
 - $\Delta\lambda = 2\lambda_0$
 - $\Delta\lambda = 4\lambda_0$
- A particle is moving 5 times as fast as an electron. The ratio of the de-Broglie wavelength of the particle to that of the electron is 1.878×10^{-4} . The mass of the particle is close to : **[Sep. 02, 2020 (II)]**
 - 4.8×10^{-27} kg
 - 9.1×10^{-31} kg
 - 1.2×10^{-28} kg
 - 9.7×10^{-28} kg

- A particle moving with kinetic energy E has de Broglie wavelength λ . If energy ΔE is added to its energy, the wavelength become $\frac{\lambda}{2}$. Value of ΔE , is: **[9 Jan. 2020 I]**
 - E
 - 4E
 - 3E
 - 2E
- An electron of mass m and magnitude of charge |e| initially at rest gets accelerated by a constant electric field E. The rate of change of de-Broglie wavelength of this electron at time t ignoring relativistic effects is: **[9 Jan. 2020 II]**
 - $-\frac{h}{|e|E\sqrt{t}}$
 - $\frac{|e|Et}{h}$
 - $-\frac{h}{|e|Et}$
 - $\frac{-h}{|e|Et^2}$
- An electron (mass m) with initial velocity $\vec{v} = v_0\hat{i} + v_0\hat{j}$ is in an electric field $\vec{E} = -E_0\hat{k}$. If λ_0 is initial de-Broglie wavelength of electron, its de-Broglie wave length at time t is given by: **[8 Jan. 2020 II]**
 - $\frac{\lambda_0\sqrt{2}}{\sqrt{1 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}}}$
 - $\frac{\lambda_0}{\sqrt{1 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}}}$
 - $\frac{\lambda_0}{\sqrt{1 + \frac{e^2 E^2 t^2}{2m^2 v_0^2}}}$
 - $\frac{\lambda_0}{\sqrt{2 + \frac{e^2 E^2 t^2}{m^2 v_0^2}}}$
- A particle 'P' is formed due to a completely inelastic collision of particles 'x' and 'y' having de-Broglie wavelengths ' γ_x ' and ' γ_y ' respectively. If x and y were moving in opposite directions, then the de-Broglie wavelength of 'P' is: **[9 Apr. 2019 II]**
 - $\frac{\gamma_x \gamma_y}{\gamma_x + \gamma_y}$
 - $\frac{\gamma_x \gamma_y}{|\gamma_x - \gamma_y|}$
 - $\gamma_x - \gamma_y$
 - $\gamma_x + \gamma_y$

9. Two particles move at right angle to each other. Their de Broglie wavelengths are λ_1 and λ_2 respectively. The particles suffer perfectly inelastic collision. The de Broglie wavelength λ , of the final particle, is given by:

[8 April 2019 I]

(a) $\frac{1}{\lambda^2} = \frac{1}{\lambda_1^2} + \frac{1}{\lambda_2^2}$ (b) $\lambda = \sqrt{\lambda_1 \lambda_2}$
 (c) $\lambda = \frac{\lambda_1 + \lambda_2}{2}$ (d) $\frac{2}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$

10. A particle A of mass 'm' and charge 'q' is accelerated by a potential difference of 50V. Another particle B of mass '4m' and charge 'q' is accelerated by a potential difference of 2500V. The ratio of de-Broglie wavelength $\frac{\lambda_A}{\lambda_B}$ is

[12 Jan. 2019 I]

(a) 10.00 (b) 0.07
 (c) 14.14 (d) 4.47

11. If the deBroglie wavelength of an electron is equal to 10^{-3} times the wavelength of a photon of frequency 6×10^{14} Hz, then the speed of electron is equal to :

(Speed of light = 3×10^8 m/s)Planck's constant = 6.63×10^{-34} J.sMass of electron = 9.1×10^{-31} kg) [11 Jan. 2019 I]

(a) 1.1×10^6 m/s (b) 1.7×10^6 m/s
 (c) 1.8×10^6 m/s (d) 1.45×10^6 m/s

12. In an electron microscope, the resolution that can be achieved is of the order of the wavelength of electrons used. To resolve a width of 7.5×10^{-12} m, the minimum electron energy required is close to: [10 Jan. 2019 I]

(a) 500 keV (b) 100 keV (c) 1 keV (d) 25 keV

13. Two electrons are moving with non-relativistic speeds perpendicular to each other. If corresponding de Broglie wavelengths are λ_1 and λ_2 , their de Broglie wavelength in the frame of reference attached to their centre of mass is:

[Online April 15, 2018]

(a) $\lambda_{CM} = \lambda_1 = \lambda_2$ (b) $\frac{1}{\lambda_1} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$
 (c) $\lambda_{CM} = \frac{2\lambda_1\lambda_2}{\sqrt{\lambda_1^2 + \lambda_2^2}}$ (d) $\lambda_{CM} = \left(\frac{\lambda_1 + \lambda_2}{2} \right)$

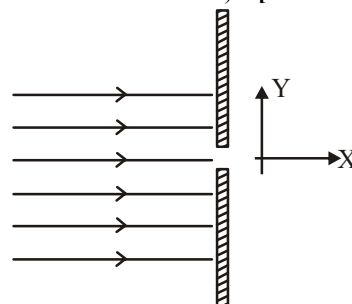
14. If the de Broglie wavelengths associated with a proton and an α -particle are equal, then the ratio of velocities of the proton and the α -particle will be: [Online April 15, 2018]

(a) 1 : 4 (b) 1 : 2 (c) 4 : 1 (d) 2 : 1

15. A particle A of mass m and initial velocity v collides with a particle B of mass $\frac{m}{2}$ which is at rest. The collision is head on, and elastic. The ratio of the de-Broglie wavelengths λ_A to λ_B after the collision is [2017]

(a) $\frac{\lambda_A}{\lambda_B} = \frac{2}{3}$ (b) $\frac{\lambda_A}{\lambda_B} = \frac{1}{2}$
 (c) $\frac{\lambda_A}{\lambda_B} = \frac{1}{3}$ (d) $\frac{\lambda_A}{\lambda_B} = 2$

16. A parallel beam of electrons travelling in x-direction falls on a slit of width d (see figure). If after passing the slit, an electron acquires momentum p_y in the y-direction then for a majority of electrons passing through the slit (h is Planck's constant) : [Online April 10, 2015]



(a) $|P_y|d > h$ (b) $|P_y|d < h$
 (c) $|P_y|d \approx h$ (d) $|P_y|d >> h$

17. de-Broglie wavelength of an electron accelerated by a voltage of 50 V is close to ($|e| = 1.6 \times 10^{-19}$ C, $m_e = 9.1 \times 10^{-31}$ kg, $h = 6.6 \times 10^{-34}$ Js) : [Online April 10, 2015]

(a) 2.4 Å (b) 0.5 Å (c) 1.7 Å (d) 1.2 Å

18. For which of the following particles will it be most difficult to experimentally verify the de-Broglie relationship?

[Online April 9, 2014]

(a) an electron (b) a proton
 (c) an α -particle (d) a dust particle

19. Electrons are accelerated through a potential difference V and protons are accelerated through a potential difference 4 V. The de-Broglie wavelengths are λ_e and λ_p for electrons

and protons respectively. The ratio of $\frac{\lambda_e}{\lambda_p}$ is given by : (given m_e is mass of electron and m_p is mass of proton).

[Online April 23, 2013]

(a) $\frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_p}{m_e}}$ (b) $\frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_e}{m_p}}$
 (c) $\frac{\lambda_e}{\lambda_p} = \frac{1}{2} \sqrt{\frac{m_e}{m_p}}$ (d) $\frac{\lambda_e}{\lambda_p} = 2 \sqrt{\frac{m_p}{m_e}}$

20. If the kinetic energy of a free electron doubles, its de-Broglie wavelength changes by the factor [2005]

(a) 2 (b) $\frac{1}{2}$ (c) $\sqrt{2}$ (d) $\frac{1}{\sqrt{2}}$

21. Formation of covalent bonds in compounds exhibits [2002]

(a) wave nature of electron
 (b) particle nature of electron
 (c) both wave and particle nature of electron
 (d) none of these

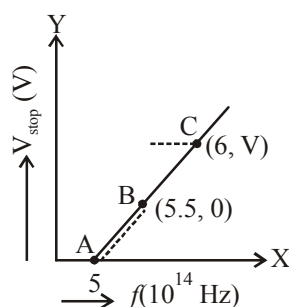
TOPIC 2 Photon, Photoelectric Effect X-rays and Davisson-Germer Experiment



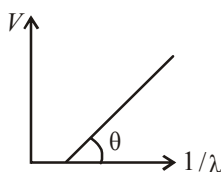
22. A beam of electrons of energy E scatters from a target having atomic spacing of $1 \text{ \AA}$. The first maximum intensity occurs at $\theta = 60^\circ$. Then E (in eV) is _____.

(Plank constant $h = 6.64 \times 10^{-34} \text{ Js}$, $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$, electron mass $m = 9.1 \times 10^{-31} \text{ kg}$) [NA Sep. 05, 2020 (I)]

23. The surface of a metal is illuminated alternately with photons of energies $E_1 = 4 \text{ eV}$ and $E_2 = 2.5 \text{ eV}$ respectively. The ratio of maximum speeds of the photoelectrons emitted in the two cases is 2. The work function of the metal (in eV) is _____. [NA Sep. 05, 2020 (II)]
24. Given figure shows few data points in a photo electric effect experiment for a certain metal. The minimum energy for ejection of electron from its surface is : (Plancks constant $h = 6.62 \times 10^{-34} \text{ J.s}$) [Sep. 04, 2020 (I)]



- (a) 2.27 eV (b) 2.59 eV
(c) 1.93 eV (d) 2.10 eV
25. In a photoelectric effect experiment, the graph of stopping potential V versus reciprocal of wavelength obtained is shown in the figure. As the intensity of incident radiation is increased : [Sep. 04, 2020 (II)]



- (a) Straight line shifts to right
(b) Slope of the straight line get more steep
(c) Straight line shifts to left
(d) Graph does not change
26. When the wavelength of radiation falling on a metal is changed from 500 nm to 200 nm , the maximum kinetic energy of the photoelectrons becomes three times larger. The work function of the metal is close to : [Sep. 03, 2020 (I)]
- (a) 0.81 eV (b) 1.02 eV
(c) 0.52 eV (d) 0.61 eV

27. Two sources of light emit X-rays of wavelength 1 nm and visible light of wavelength 500 nm , respectively. Both the sources emit light of the same power 200 W . The ratio of the number density of photons of X-rays to the number density of photons of the visible light of the given wavelengths is : [Sep. 03, 2020 (II)]

(a) $\frac{1}{500}$ (b) 250 (c) $\frac{1}{250}$ (d) 500

28. When radiation of wavelength λ is used to illuminate a metallic surface, the stopping potential is V . When the same surface is illuminated with radiation of wavelength

3λ , the stopping potential is $\frac{V}{4}$. If the threshold wavelength for the metallic surface is $n\lambda$ then value of n will be _____. [NA Sep. 02, 2020 (I)]

29. Radiation, with wavelength $6561 \text{ \AA}$ falls on a metal surface to produce photoelectrons. The electrons are made to enter a uniform magnetic field of $3 \times 10^{-4} \text{ T}$. If the radius of the largest circular path followed by the electrons is 10 mm , the work function of the metal is close to: [9 Jan. 2020 I]

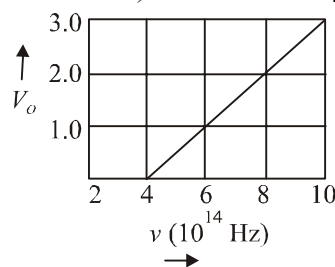
(a) 1.1 eV (b) 0.8 eV
(c) 1.6 eV (d) 1.8 eV

30. When photon of energy 4.0 eV strikes the surface of a metal A , the ejected photoelectrons have maximum kinetic energy $T_A \text{ eV}$ and de-Broglie wavelength λ_A . The maximum kinetic energy of photoelectrons liberated from another metal B by photon of energy 4.50 eV is $T_B = (T_A - 1.5) \text{ eV}$. If the de-Broglie wavelength of these photoelectrons $\lambda_B = 2\lambda_A$, then the work function of metal B is: [8 Jan. 2020 I]

(a) 4 eV (b) 2 eV
(c) 1.5 eV (d) 3 eV

31. A beam of electromagnetic radiation of intensity $6.4 \times 10^{-5} \text{ W/cm}^2$ is comprised of wavelength, $\lambda = 310 \text{ nm}$. It falls normally on a metal (work function $\phi = 2 \text{ eV}$) of surface area of 1 cm^2 . If one in 10^3 photons ejects an electron, total number of electrons ejected in 1 s is 10^x . ($hc = 1240 \text{ eVnm}$, $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$), then x is _____. [NA 7 Jan. 2020 I]

32. The stopping potential V_0 (in volt) as a function of frequency (ν) for a sodium emitter, is shown in the figure. The work function of sodium, from the data plotted in the figure, will be : (Given : Planck's constant (h) = $6.63 \times 10^{-34} \text{ Js}$, electron charge $e = 1.6 \times 10^{-19} \text{ C}$) [12 Apr. 2019 I]



(a) 1.82 eV (b) 1.66 eV (c) 1.95 eV (d) 2.12 eV

33. In a photoelectric effect experiment the threshold wavelength of light is 380 nm. If the wavelength of incident light is 260 nm, the maximum kinetic energy of emitted electrons will be:

Given E (in eV) = [10 Apr. 2019 I]

- (a) 1.5 eV (b) 3.0 eV (c) 4.5 eV (d) 15.1 eV

34. A 2 mW laser operates at a wavelength of 500 nm. The number of photons that will be emitted per second is :

[Given Planck's constant $h = 6.6 \times 10^{-34}$ Js, speed of light $c = 3.0 \times 10^8$ m/s] [10 Apr. 2019 II]

- (a) 5×10^{15} (b) 1.5×10^{16}
(c) 2×10^{16} (d) 1×10^{16}

35. The electric field of light wave is given as

$$\vec{E} = 10^3 \cos\left(\frac{2\pi x}{5 \times 10^{-7}} - 2\pi \times 6 \times 10^{14} t\right) \hat{x} \frac{N}{C}$$

This light falls on a metal plate of work function 2 eV. The stopping potential of the photo-electrons is:

Given, E (in eV) = $\frac{12375}{\lambda \text{ (in Å)}}$ [9 April 2019 I]

- (a) 2.0 V (b) 0.72 V (c) 0.48 V (d) 2.48 V

36. When a certain photosensitive surface is illuminated with monochromatic light of frequency ν , the stopping potential for the photo current is $-V_0/2$. When the surface is illuminated by monochromatic light of frequency $\nu/2$, the stopping potential is $-V_0$. The threshold frequency for photoelectric emission is : [12 Jan. 2019 II]

- (a) $\frac{5\nu}{3}$ (b) $\frac{4}{3}\nu$ (c) 2ν (d) $\frac{3\nu}{2}$

37. In a Frank-Hertz experiment, an electron of energy 5.6 eV passes through mercury vapour and emerges with an energy 0.7 eV. The minimum wavelength of photons emitted by mercury atoms is close to : [12 Jan. 2019 II]

- (a) 1700 nm (b) 2020 nm
(c) 220 nm (d) 250 nm

38. In a photoelectric experiment, the wavelength of the light incident on a metal is changed from 300 nm to 400 nm. The decrease in the stopping potential is close to : [11 Jan. 2019 II]

$$\left(\frac{hc}{e} = 1240 \text{ nm-eV}\right)$$

- (a) 0.5 V (b) 1.5 V
(c) 1.0 V (d) 2.0 V

39. A metal plate of area $1 \times 10^{-4} \text{ m}^2$ is illuminated by a radiation of intensity 16 mW/m^2 . The work function of the metal is 5 eV. The energy of the incident photons is 10 eV and only 10% of it produces photo electrons. The number of emitted photo electrons per second and their maximum energy, respectively, will be:

[1 eV = 1.6×10^{-19} J]

[10 Jan. 2019 II]

- (a) 10^{14} and 10 eV (b) 10^{12} and 5 eV
(c) 10^{11} and 5 eV (d) 10^{10} and 5 eV

40. Surface of certain metal is first illuminated with light of wavelength $\lambda_1 = 350 \text{ nm}$ and then, by light of wavelength $\lambda_2 = 540 \text{ nm}$. It is found that the maximum speed of the photo electrons in the two cases differ by a factor of (2). The work function of the metal (in eV) is close to:

$$\left(\text{Energy of photon} = \frac{1240}{\lambda \text{ (in nm)}} \text{ eV}\right) \quad [9 \text{ Jan. 2019 I}]$$

- (a) 1.8 (b) 2.5 (c) 5.6 (d) 1.4

41. The magnetic field associated with a light wave is given at the origin by

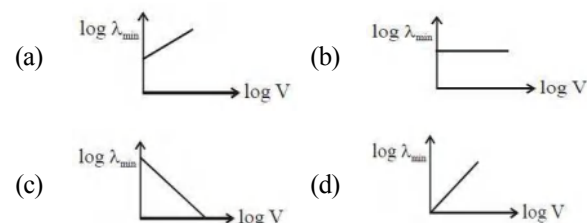
$$B = B_0 [\sin(3.14 \times 10^7)ct + \sin(6.28 \times 10^7)ct].$$

If this light falls on a silver plate having a work function of 4.7 eV, what will be the maximum kinetic energy of the photoelectrons? [9 Jan. 2019 II]

($c = 3 \times 10^8 \text{ ms}^{-1}$, $h = 6.6 \times 10^{-34} \text{ J-s}$)

- (a) 6.82 eV (b) 12.5 eV
(c) 8.52 eV (d) 7.72 eV

42. An electron beam is accelerated by a potential difference V to hit a metallic target to produce X-rays. It produces continuous as well as characteristic X-rays. If $\lambda_{\min}$ is the smallest possible wavelength of X-ray in the spectrum, the variation of $\log \lambda_{\min}$ with $\log V$ is correctly represented in : [2017]



43. A Laser light of wavelength 660 nm is used to weld Retina detachment. If a Laser pulse of width 60 ms and power 0.5 kW is used the approximate number of photons in the pulse are : [Take Planck's constant $h = 6.62 \times 10^{-34}$ Js]

[Online April 9, 2017]

- (a) 10^{20} (b) 10^{18} (c) 10^{22} (d) 10^{19}

44. The maximum velocity of the photoelectrons emitted from the surface is v when light of frequency n falls on a metal surface. If the incident frequency is increased to $3n$, the maximum velocity of the ejected photoelectrons will be :

[Online April 8, 2017]

- (a) less than $\sqrt{3}v$ (b) v
(c) more than $\sqrt{3}v$ (d) equal to $\sqrt{3}v$

45. Radiation of wavelength λ , is incident on a photocell. The fastest emitted electron has speed v . If the wavelength is

changed to $\frac{3\lambda}{4}$, the speed of the fastest emitted electron will be: [2016]

- (a) $= v \left(\frac{4}{3}\right)^{\frac{1}{2}}$ (b) $= v \left(\frac{3}{4}\right)^{\frac{1}{2}}$
(c) $> v \left(\frac{4}{3}\right)^{\frac{1}{2}}$ (d) $< v \left(\frac{4}{3}\right)^{\frac{1}{2}}$

46. A photoelectric surface is illuminated successively by monochromatic light of wavelengths λ and $\frac{\lambda}{2}$. If the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times that in the first case, the work function of the surface is : [Online April 10, 2016]

(a) $\frac{hc}{2\lambda}$ (b) $\frac{hc}{\lambda}$
(c) $\frac{hc}{3\lambda}$ (d) $\frac{3hc}{\lambda}$

47. When photons of wavelength λ_1 are incident on an isolated sphere, the corresponding stopping potential is found to be V . When photons of wavelength λ_2 are used, the corresponding stopping potential was thrice that of the above value. If light of wavelength λ_3 is used then find the stopping potential for this case : [Online April 9, 2016]

(a) $\frac{hc}{e} \left[\frac{1}{\lambda_3} + \frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right]$
(b) $\frac{hc}{e} \left[\frac{1}{\lambda_3} + \frac{1}{2\lambda_2} - \frac{1}{\lambda_1} \right]$
(c) $\frac{hc}{e} \left[\frac{1}{\lambda_3} - \frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right]$
(d) $\frac{hc}{e} \left[\frac{1}{\lambda_3} + \frac{1}{2\lambda_2} - \frac{3}{2\lambda_1} \right]$

48. Match **List - I** (Fundamental Experiment) with **List - II** (its conclusion) and select the correct option from the choices given below the list: [2015]

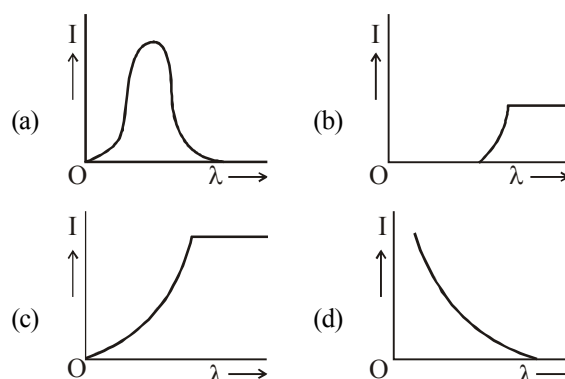
List-I	List-II
A. Franck-Hertz Experiment	(i) Particle nature of light
B. Photo-electric experiment	(ii) Discrete energy levels of atom
C. Davis on-Germer experiment	(iii) Wave nature of electron
	(iv) Structure of atom

- (a) (A)-(ii); (B)-(i); (C)-(iii)
(b) (A)-(iv); (B)-(iii); (C)-(ii)
(c) (A)-(i); (B)-(iv); (C)-(iii)
(d) (A)-(ii); (B)-(iv); (C)-(iii)
49. A beam of light has two wavelengths of $4972\text{\AA}$ and $6216\text{\AA}$ with a total intensity of $3.6 \times 10^{-3} \text{ Wm}^{-2}$ equally distributed among the two wavelengths. The beam falls normally on an area of 1 cm^2 of a clean metallic surface of work function 2.3 eV . Assume that there is no loss of light by reflection and that each capable photon ejects one electron. The number of photoelectrons liberated in 2s is approximately: [Online April 12, 2014]
- (a) 6×10^{11} (b) 9×10^{11}
(c) 11×10^{11} (d) 15×10^{11}

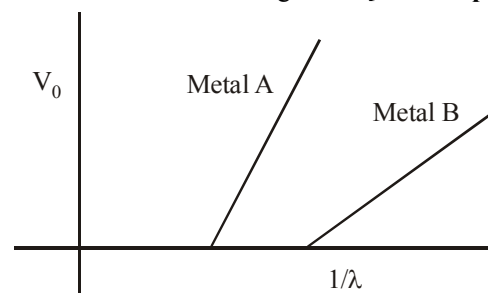
50. A photon of wavelength λ is scattered from an electron, which was at rest. The wavelength shift $\Delta\lambda$ is three times of λ and the angle of scattering θ is 60° . The angle at which the electron recoiled is ϕ . The value of $\tan \phi$ is : (electron speed is much smaller than the speed of light)

[Online April 11, 2014]

- (a) 0.16 (b) 0.22 (c) 0.25 (d) 0.28
51. The anode voltage of a photocell is kept fixed. The wavelength λ of the light falling on the cathode is gradually changed. The plate current I of the photocell varies as follows : [2013]



52. In an experiment on photoelectric effect, a student plots stopping potential V_0 against reciprocal of the wavelength λ of the incident light for two different metals A and B. These are shown in the figure. [Online April 25, 2013]



Looking at the graphs, you can most appropriately say that:

- (a) Work function of metal B is greater than that of metal A
(b) For light of certain wavelength falling on both metal, maximum kinetic energy of electrons emitted from A will be greater than those emitted from B.
(c) Work function of metal A is greater than that of metal B
(d) Students data is not correct
53. A copper ball of radius 1 cm and work function 4.47 eV is irradiated with ultraviolet radiation of wavelength $2500 \text{\AA}$. The effect of irradiation results in the emission of electrons from the ball. Further the ball will acquire charge and due to this there will be a finite value of the potential on the ball. The charge acquired by the ball is :

[Online April 25, 2013]

(a) $5.5 \times 10^{-13} \text{ C}$ (b) $7.5 \times 10^{-13} \text{ C}$
(c) $4.5 \times 10^{-12} \text{ C}$ (d) $2.5 \times 10^{-11} \text{ C}$

54. This equation has statement 1 and statement 2. Of the four choices given after the statements, choose the one that describes the two statements.

Statement 1: Davisson-Germer experiment established the wave nature of electrons.

Statement 2 : If electrons have wave nature, they can interfere and show diffraction. [2012]

- (a) Statement 1 is false, Statement 2 is true.
 (b) Statement 1 is true, Statement 2 is false
 (c) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of statement 1
 (d) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1
55. Photoelectrons are ejected from a metal when light of frequency ν falls on it. Pick out the wrong statement from the following. [Online May 26, 2012]
- (a) No electrons are emitted if ν is less than W/h , where W is the work function of the metal
 (b) The ejection of the photoelectrons is instantaneous.
 (c) The maximum energy of the photoelectrons is $h\nu$.
 (d) The maximum energy of the photoelectrons is independent of the intensity of the light.

56. This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.

Statement 1: A metallic surface is irradiated by a monochromatic light of frequency $\nu > \nu_0$ (the threshold frequency). If the incident frequency is now doubled, the photocurrent and the maximum kinetic energy are also doubled.

Statement 2: The maximum kinetic energy of photoelectrons emitted from a surface is linearly dependent on the frequency of the incident light. The photocurrent depends only on the intensity of the incident light. [Online May 19, 2012]

- (a) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.
 (b) Statement 1 is false, Statement 2 is true.
 (c) Statement 1 is true, Statement 2 is false.
 (d) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.
57. This question has Statement – 1 and Statement – 2. Of the four choices given after the statements, choose the one that best describes the two statements. [2011]

Statement – 1: A metallic surface is irradiated by a monochromatic light of frequency $\nu > \nu_0$ (the threshold frequency). The maximum kinetic energy and the stopping potential are $K_{\max}$ and V_0 respectively. If the frequency incident on the surface is doubled, both the $K_{\max}$ and V_0 are also doubled.

Statement – 2 : The maximum kinetic energy and the stopping potential of photoelectrons emitted from a surface are linearly dependent on the frequency of incident light.

- (a) Statement–1 is true, Statement–2 is true, Statement – 2 is the correct explanation of Statement – 1.
 (b) Statement–1 is true, Statement–2 is true, Statement – 2 is not the correct explanation of Statement – 1.
 (c) Statement – 1 is false, Statement – 2 is true.
 (d) Statement – 1 is true, Statement – 2 is false.

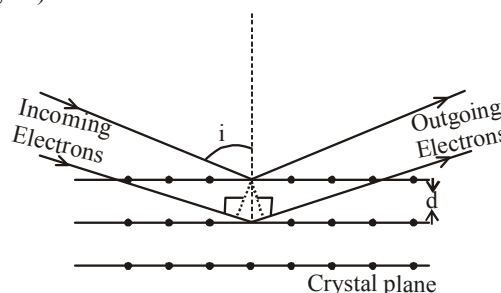
58. **Statement -1 :** When ultraviolet light is incident on a photocell, its stopping potential is V_0 and the maximum kinetic energy of the photoelectrons is $K_{\max}$. When the ultraviolet light is replaced by X-rays, both V_0 and $K_{\max}$ increase.

Statement -2 : Photoelectrons are emitted with speeds ranging from zero to a maximum value because of the range of frequencies present in the incident light. [2010]

- (a) Statement -1 is true, Statement -2 is true ; Statement -2 is the correct explanation of Statement -1.
 (b) Statement -1 is true, Statement -2 is true; Statement -2 is **not** the correct explanation of Statement -1
 (c) Statement -1 is false, Statement -2 is true.
 (d) Statement -1 is true, Statement -2 is false.
59. The surface of a metal is illuminated with the light of 400 nm. The kinetic energy of the ejected photoelectrons was found to be 1.68 eV. The work function of the metal is : ($hc = 1240 \text{ eV}\cdot\text{nm}$) [2009]
- (a) 1.41 eV (b) 1.51 eV
 (c) 1.68 eV (d) 3.09 eV

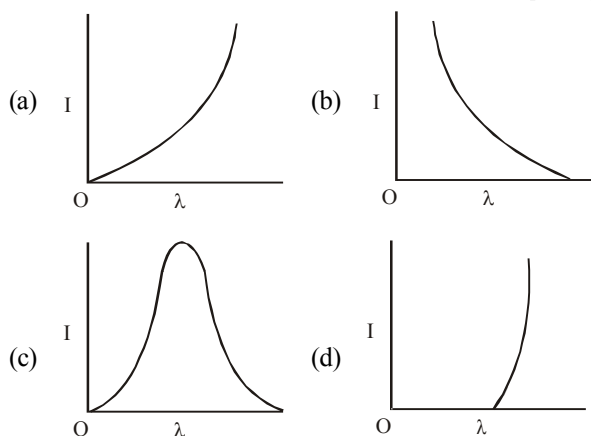
Directions: Question No. 60 and 61 are based on the following paragraph.

Wave property of electrons implies that they will show diffraction effects. Davisson and Germer demonstrated this by diffracting electrons from crystals. The law governing the diffraction from a crystal is obtained by requiring that electron waves reflected from the planes of atoms in a crystal interfere constructively (see figure).



60. Electrons accelerated by potential V are diffracted from a crystal. If $d = 1 \text{ \AA}$ and $i = 30^\circ$, V should be about [2008]
 ($h = 6.6 \times 10^{-34} \text{ Js}$, $m_e = 9.1 \times 10^{-31} \text{ kg}$, $e = 1.6 \times 10^{-19} \text{ C}$)
 (a) 2000 V (b) 50 V (c) 500 V (d) 1000 V
61. If a strong diffraction peak is observed when electrons are incident at an angle ' i ' from the normal to the crystal planes with distance ' d ' between them (see figure), de Broglie wavelength λ_{dB} of electrons can be calculated by the relationship (n is an integer) [2008]
- (a) $d \sin i = n\lambda_{dB}$ (b) $2d \cos i = n\lambda_{dB}$
 (c) $2d \sin i = n\lambda_{dB}$ (d) $d \cos i = n\lambda_{dB}$

62. Photon of frequency ν has a momentum associated with it. If c is the velocity of light, the momentum is [2007]
 (a) $h\nu/c$ (b) ν/c (c) $h\nu c$ (d) $h\nu/c^2$
63. The threshold frequency for a metallic surface corresponds to an energy of 6.2 eV and the stopping potential for a radiation incident on this surface is 5 V. The incident radiation lies in [2006]
 (a) ultra-violet region (b) infra-red region
 (c) visible region (d) X-ray region
64. The time taken by a photoelectron to come out after the photon strikes is approximately [2006]
 (a) 10^{-4} s (b) 10^{-10} s (c) 10^{-16} s (d) 10^{-1} s
65. The anode voltage of a photocell is kept fixed. The wavelength λ of the light falling on the cathode is gradually changed. The plate current I of the photocell varies as follows [2006]



66. A photocell is illuminated by a small bright source placed 1 m away. When the same source of light is placed $\frac{1}{2}$ m away, the number of electrons emitted by photocathode would [2005]
 (a) increase by a factor of 4
 (b) decrease by a factor of 4

- (c) increase by a factor of 2
 (d) decrease by a factor of 2
67. A radiation of energy E falls normally on a perfectly reflecting surface. The momentum transferred to the surface is [2004]
 (a) Ec (b) $2E/c$ (c) E/c (d) E/c^2
68. According to Einstein's photoelectric equation, the plot of the kinetic energy of the emitted photo electrons from a metal Versus the frequency, of the incident radiation gives a straight line whose slope [2004]
 (a) depends both on the intensity of the radiation and the metal used
 (b) depends on the intensity of the radiation
 (c) depends on the nature of the metal used
 (d) is the same for the all metals and independent of the intensity of the radiation
69. The work function of a substance is 4.0 eV. The longest wavelength of light that can cause photoelectron emission from this substance is approximately [2004]
 (a) 310 nm (b) 400 nm (c) 540 nm (d) 220 nm
70. Two identical photocathodes receive light of frequencies f_1 and f_2 . If the velocities of the photo electrons (of mass m) coming out are respectively v_1 and v_2 , then [2003]
 (a) $v_1^2 - v_2^2 = \frac{2h}{m}(f_1 - f_2)$
 (b) $v_1 + v_2 = \left[\frac{2h}{m}(f_1 + f_2) \right]^{1/2}$
 (c) $v_1^2 + v_2^2 = \frac{2h}{m}(f_1 + f_2)$
 (d) $v_1 - v_2 = \left[\frac{2h}{m}(f_1 - f_2) \right]^{1/2}$
71. Sodium and copper have work functions 2.3 eV and 4.5 eV respectively. Then the ratio of the wavelengths is nearest to [2002]
 (a) 1 : 2 (b) 4 : 1 (c) 2 : 1 (d) 1 : 4



Hints & Solutions



1. (c) de-Broglie wavelength, $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m(\text{KE})}}$

$$\therefore \lambda \propto \frac{1}{\sqrt{m}}$$

As $m_{\text{He}^{++}} > m_p > m_e$

$$\lambda_{\text{He}^{++}} > \lambda_p > \lambda_e \text{ or } \lambda_e > \lambda_p > \lambda_{\text{He}^{++}}$$

2. (a) Rms speed of gas molecule, $V_{\text{rms}} = \sqrt{\frac{3kT}{m}}$

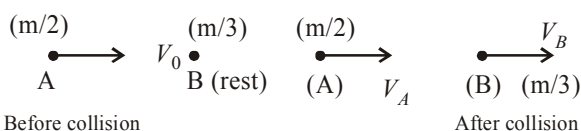
de Broglie wavelength, $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mk}}$

$$\therefore \lambda = \frac{h}{\sqrt{2m \times \frac{1}{2} m V_{\text{rms}}^2}} = \frac{h}{\sqrt{m \times \frac{3}{2} kT}} = \frac{h}{\sqrt{3mkT}}$$

Substituting the respective values we get

$$\lambda = \frac{6.63 \times 10^{-34}}{\sqrt{3 \times 4.64 \times 10^{-26} \times 1.38 \times 10^{-13} \times 400}} = 0.24 \text{ \AA}$$

3. (d)



Before collision

Applying momentum conservation

$$\frac{m}{2} \times V_0 + \frac{m}{3} \times (0) = \frac{m}{2} V_A + \frac{m}{3} V_B$$

$$= \frac{V_0}{2} = \frac{V_A}{2} + \frac{V_B}{3} \quad \dots(i)$$

Since, collision is elastic

$$e = 1 = \frac{V_B - V_A}{V_0} \Rightarrow V_0 = V_B - V_A \quad \dots(ii)$$

On solving equations (i) and (ii): $V_A = \frac{V_0}{5}$

Now, de-Broglie wavelength of A before collision :

$$\lambda_0 = \frac{h}{m_A V_0} = \frac{h}{\left(\frac{m}{2}\right) V_0} \Rightarrow \lambda_0 = \frac{2h}{m V_0}$$

Final de-Broglie wavelength :

$$\lambda_f = \frac{h}{m_A V_0} = \frac{h}{\frac{m}{2} \times \frac{V_0}{5}} \Rightarrow \lambda_f = \frac{10h}{m V_0}$$

$$\therefore \Delta\lambda = \lambda_f - \lambda_0 = \frac{10h}{m V_0} - \frac{2h}{m V_0}$$

$$\Rightarrow \Delta\lambda = \frac{8h}{m V_0} \Rightarrow \Delta\lambda = 4 \times \frac{2h}{m V_0}$$

$$\therefore \Delta\lambda = 4\lambda_0$$

4. (d) de Broglie wavelength

$$\lambda = \frac{h}{mv} \Rightarrow m = \frac{h}{\lambda v}$$

Clearly, $m \propto \frac{1}{\lambda v}$

If λ and v be the wavelength and velocity of electron and λ' and v' be the wavelength and velocity of the particle then

$$\Rightarrow \frac{m'}{m} = \frac{v\lambda}{v'\lambda'} = \frac{1}{5} \times \frac{1}{1.878} \times 10^{-4}$$

$$\Rightarrow m = 9.7 \times 10^{-28} \text{ kg}$$

5. (c) As per question, when KE of particle E , wavelength λ and when KE becomes $E + \Delta E$ wavelength becomes $\lambda/2$

Using, $\lambda = \frac{h}{\sqrt{2mKE}}$

$$\frac{\lambda}{2} = \frac{h}{\sqrt{2m(KE + \Delta E)}}$$

$$\Rightarrow \frac{\lambda}{\lambda/2} = \sqrt{\frac{KE + \Delta E}{KE}}$$

$$\Rightarrow 4 = \frac{KE + \Delta E}{KE}$$

$$\Rightarrow 4KE - KE = \Delta E$$

$$\therefore \Delta E = 3KE = 3E$$

6. (d) Acceleration of electron in electric field, $a = \frac{eE}{m}$

Using equation

$$v = u + at$$

$$\Rightarrow v = 0 + \frac{eE}{m} t$$

$$\Rightarrow v = \frac{eEt}{m} \quad \dots (i)$$

De-broglie wavelength λ is given by

$$\lambda = \frac{h}{mv} = \frac{h}{m\left(\frac{eEt}{m}\right)} \quad [\text{using (i)}]$$

$$\Rightarrow \lambda = \frac{h}{eEt}$$

Differentiating w.r.t. t

$$\frac{d\lambda}{dt} = \frac{d\left(\frac{h}{eEt}\right)}{dt} \Rightarrow \frac{d\lambda}{dt} = \frac{-h}{eEt^2}$$

7. (c) Given, Initial velocity, $u = v_0\hat{i} + v_0\hat{j}$

$$\text{Acceleration, } a = \frac{qE_0}{m} = \frac{eE_0}{m}$$

Using $v = u + at$

$$v = v_0\hat{i} + v_0\hat{j} + \frac{eE_0}{m}t\hat{k}$$

$$\therefore |\vec{v}| = \sqrt{2v_0^2 + \left(\frac{eE_0t}{m}\right)^2}$$

$$\text{de-Broglie wavelength, } \lambda = \frac{h}{p}$$

$$\Rightarrow \lambda = \frac{h}{mv} \quad (\because p = mv)$$

$$\text{Initial wavelength, } \lambda_0 = \frac{h}{mv_0\sqrt{2}}$$

Final wavelength,

$$\lambda = \frac{h}{m\sqrt{2v_0^2 + \left(\frac{eE_0t}{m}\right)^2}}$$

$$\frac{\lambda}{\lambda_0} = \frac{1}{\sqrt{1 + \left(\frac{eE_0t}{\sqrt{2}mv_0}\right)^2}}$$

$$\Rightarrow \lambda = \frac{\lambda_0}{\sqrt{1 + \frac{e^2 E_0^2 t^2}{2m^2 v_0^2}}}$$

8. (b) $P_1 - P_2 = (P_1 + P_2) = P$

$$\text{As } P \propto \frac{1}{\lambda}$$

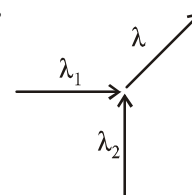
$$\text{or } \frac{1}{\lambda_x} - \frac{1}{\lambda_y} = \frac{1}{\lambda}$$

$$\text{or } \frac{\lambda_y - \lambda_x}{\lambda_x \lambda_y} = \frac{1}{\lambda}$$

9. (a) From the de-Broglie relation,

$$p_1 = \frac{h}{\lambda_1}$$

$$p_2 = \frac{h}{\lambda_2}$$



Momentum of the final particle (p_f) is given by

$$\therefore p_f = \sqrt{p_1^2 + p_2^2}$$

$$\Rightarrow \frac{h}{\lambda} = \sqrt{\frac{h^2}{\lambda_1^2} + \frac{h^2}{\lambda_2^2}}$$

$$\Rightarrow \frac{1}{\lambda^2} = \frac{1}{\lambda_1^2} + \frac{1}{\lambda_2^2}$$

10. (c) de Broglie wavelength (λ) is given by $K = qV$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2mqV}} \quad (\because p = \sqrt{2mK})$$

Substituting the values we get

$$\therefore \frac{\lambda_A}{\lambda_B} = \frac{\sqrt{2m_B q_B V_B}}{\sqrt{2m_A q_A V_A}} = \sqrt{\frac{4m.q.2500}{m.q.50}} = 2\sqrt{50} = 2 \times 7.07 = 14.14$$

11. (d) de-Broglie wavelength,

$$\lambda = \frac{h}{mv} = 10^{-3} \left(\frac{3 \times 10^8}{6 \times 10^{14}} \right) \quad \left[\because \lambda = \frac{c}{v} \right]$$

$$v = \frac{6.63 \times 10^{-34} \times 6 \times 10^{14}}{9.1 \times 10^{-31} \times 3 \times 10^5}$$

$$v = 1.45 \times 10^6 \text{ m/s}$$

12. (d) Using, $\lambda = \frac{h}{p}$ {given: $\lambda = 7.5 \times 10^{-12}$ }

$$\Rightarrow P = \frac{h}{\lambda}$$

Minimum energy required,

$$KE = \frac{p^2}{2m} = \frac{(h/\lambda)^2}{2m} = \frac{\left\{ \frac{6.6 \times 10^{-34}}{7.5 \times 10^{-12}} \right\}^2}{2 \times 9.1 \times 10^{-31}} \text{ J} = 25 \text{ keV}$$

13. (c) Momentum (p) of each electron $\frac{h}{\lambda_1} \hat{i}$ and $\frac{h}{\lambda_2} \hat{j}$

Velocity of centre of mass

$$V_{cm} = \frac{h}{2m\lambda_1} \hat{i} + \frac{h}{2m\lambda_2} \hat{j} \quad (\because p = mv)$$

Velocity of 1st particle about centre of mass

$$V_{1cm} = \frac{h}{2m\lambda_1} \hat{i} - \frac{h}{2m\lambda_2} \hat{j}$$

$$\lambda_{cm} = \frac{h}{\sqrt{\frac{h^2}{4\lambda_1^2} + \frac{h^2}{4\lambda_2^2}}} = \frac{2\lambda_1\lambda_2}{\sqrt{\lambda_1^2 + \lambda_2^2}} \quad \left(\because \lambda = \frac{h}{p} \right)$$

14. (c) According to question, $\lambda_p = \lambda_\alpha$

$$\text{Using, } \lambda = \frac{h}{p} = \frac{h}{mv}$$

$$\text{So, } \frac{h}{m_p \times v_p} = \frac{h}{m_\alpha \times v_\alpha}$$

$$\Rightarrow \frac{v_p}{v_\alpha} = \frac{m_\alpha}{m_p} = \frac{4m_p}{m_p}$$

($\because$ mass of α -particle is 4 times of mass of proton)

$$\text{So, } \frac{v_p}{v_\alpha} = \frac{4}{1}; \text{ i.e., } 4:1$$

15. (d) From question, $m_A = M; m_B = \frac{m}{2}$

$$u_A = V \quad u_B = 0$$

Let after collision velocity of A = V_1 and

velocity of B = V_2

Applying law of conservation of momentum,

$$mu = mv_1 + \left(\frac{m}{2}\right)v_2$$

$$\text{or, } 24 = 2v_1 + v_2 \dots (i)$$

By law of collision

$$e = \frac{v_2 - v_1}{u - 0}$$

$$\text{or, } u = v_2 - v_1 \dots (ii)$$

[$\because$ collision is elastic, $e = 1$]

using eqns (i) and (ii)

$$v_1 = \frac{4}{3} \text{ and } v_2 = \frac{4}{3}u$$

$$\text{de-Broglie wavelength } \lambda = \frac{h}{p}$$

$$\therefore \frac{\lambda_A}{\lambda_B} = \frac{P_B}{P_A} = \frac{\frac{m}{2} \times \frac{4}{3}u}{m \times \frac{4}{3}} = 2$$

16. (a) From Bragg's equation

$$d \sin \theta = \lambda$$

$$\sin \theta = \frac{\lambda}{d} < 1 \quad \therefore \lambda < d$$

$$\frac{h}{|p_y|} < d \quad \left[\because \lambda = \frac{h}{|p_y|} \right]$$

$$\therefore h < |p_y| d$$

17. (c) de-Broglie wavelength,

$$\lambda = \frac{h}{P} = \frac{h}{mv} = \frac{h}{\sqrt{2mqV}}$$

$$\text{or, } \lambda = \frac{6.6 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times 50}} = 1.7 \text{ \AA}$$

18. (d) Among the given particles most difficult to experimentally verify the de-broglie relationship is for a dust particle.

19. (d) Energy in joule (E)

= charge $\times$ potential diff. in volt

$$E_{\text{electron}} = q_e V \text{ and } E_{\text{proton}} = q_p 4V$$

$$\text{de-Broglie wavelength } \lambda = \frac{h}{P} = \frac{h}{\sqrt{2mE}}$$

$$\lambda_e = \frac{h}{\sqrt{2m_e eV}} \text{ and } \lambda_p = \frac{h}{\sqrt{2m_p e4V}} \quad (\because q_e = q_p)$$

$$\therefore \frac{\lambda_e}{\lambda_p} = \frac{\frac{h}{\sqrt{2m_e eV}}}{\frac{h}{\sqrt{2m_p e4V}}} = \sqrt{\frac{2m_p e4V}{2m_e eV}} = 2 \sqrt{\frac{m_p}{m_e}}$$

20. (d) de-Broglie wavelength,

$$\lambda = \frac{h}{p} = \frac{h}{mv} \quad \dots (i)$$

$$\text{but } K.E = \frac{1}{2}mv^2$$

$$\Rightarrow K.E = \frac{(mv)^2}{2m}$$

$$\Rightarrow mv = \sqrt{2m K.E}$$

$$\lambda = \frac{h}{\sqrt{2m K.E}}$$

$$\therefore \lambda \propto \frac{1}{\sqrt{K.E}}$$

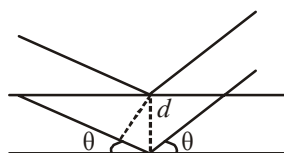
So, if K.E. is doubled, wavelength becomes $\frac{\lambda}{\sqrt{2}}$

21. (a) Covalent bonds are formed by sharing of electrons with different compounds. Formation of covalent bond is best explained by molecular orbital theory.

22. (50)

From Bragg's equation $2d \sin \theta = \lambda$ and de-Broglie

$$\text{wavelength, } \lambda = \frac{h}{P} = \frac{h}{\sqrt{2mE}}$$



$$2d \sin \theta = \lambda = \frac{h}{\sqrt{2mE}}$$

$$\Rightarrow 2 \times 10^{-10} \times \frac{\sqrt{3}}{2} = \frac{6.6 \times 10^{-34}}{\sqrt{2mE}}$$

$$[\because \theta = 60^\circ \text{ and } d = 1 \text{ \AA} = 1 \times 10^{-10} \text{ m}]$$

$$\therefore E = \frac{1}{2} \times \frac{6.64^2 \times 10^{-48}}{9.1 \times 10^{-31} \times 3 \times 1.6 \times 10^{-19}} \approx 50 \text{ eV}$$

23. 2

From the Einstein's photoelectric equation

Energy of photon

= Kinetic energy of photoelectrons + Work function

$\Rightarrow$ Kinetic energy = Energy of Photon – Work Function

Let ϕ_0 be the work function of metal and v_1 and v_2 be the velocity of photoelectrons. Using Einstein's photoelectric equation we have

$$\frac{1}{2}mv_1^2 = 4 - \phi_0 \quad \dots(i)$$

$$\frac{1}{2}mv_2^2 = 2.5 - \phi_0 \quad \dots(ii)$$

$$\Rightarrow \frac{\frac{1}{2}mv_1^2}{\frac{1}{2}mv_2^2} = \frac{4 - \phi_0}{2.5 - \phi_0}$$

$$\Rightarrow (2)^2 = \frac{4 - \phi_0}{2.5 - \phi_0} \Rightarrow 10 - 4\phi_0 = 4 - \phi_0$$

$$\phi_0 = 2 \text{ eV}$$

24. (a) Graph of V_s and f given at B (5.5, 0)
Minimum energy for ejection of electron
= Work function (ϕ).

$$\phi = hV \text{ joule or } \phi = \frac{hV}{e} \text{ eV (for } V=0)$$

$$\therefore \phi = \frac{6.62 \times 10^{-34} \times 5.5 \times 10^{14}}{1.6 \times 10^{-19}} \text{ eV} = 2.27 \text{ eV}$$

25. (d) According to Einstein's photoelectric equation

$$K_{\max} = h\nu - \phi_0$$

$$\Rightarrow eV_s = \frac{hc}{\lambda} - \phi_0$$

$$\Rightarrow V_s = \frac{hc}{\lambda e} - \frac{\phi_0}{e}$$

where λ = wavelength of incident light

ϕ_0 = work function

V_s = stopping potential

Comparing the above equation with $y = mx + c$, we get

$$\text{slope} = \frac{hc}{e}$$

Increasing the frequency of incident radiation has no effect on work function and frequency. So, graph will not change.

26. (d) Using equation, $= \frac{hc}{\lambda} - \phi$

$$KE_{\max} = \frac{hc}{\lambda} - \phi = \frac{hc}{500} - \phi \quad \dots(1)$$

$$\text{Again, } 3KE_{\max} = \frac{hc}{200} - \phi \quad \dots(2)$$

Dividing equation (2) by (1),

$$\frac{3KE_{\max}}{KE_{\max}} = \frac{3}{1} = \frac{\frac{hc}{200} - \phi}{\frac{hc}{500} - \phi}$$

Putting the value of $hc = 1237.5$ and solving we get, work function, $\phi = 0.61 \text{ eV}$.

27. (a) Given,

Wavelength of X-rays, $\lambda_1 = 1 \text{ nm} = 1 \times 10^{-9} \text{ m}$

Wavelength of visible light, $\lambda_2 = 500 \times 10^{-9} \text{ m}$

The number of photons emitted per second from a source of monochromatic radiation of wavelength λ and power P is given as

$$n = \frac{P}{E} = \frac{P}{h\nu} = \frac{P\lambda}{hc} \quad (\because E = h\nu \text{ and } \nu = \frac{c}{\lambda})$$

$\Rightarrow$ Clearly $n \propto \lambda$

$$\Rightarrow \frac{n_1}{n_2} = \frac{\lambda_1}{\lambda_2} = \frac{1}{500}$$

28. (9)

When radiation of wavelength λ_A is used to illuminate, stopping potential $V_A = V$

$$\frac{hc}{\lambda} = \phi + eV \quad \dots(i)$$

When radiation of wavelength B , λ_B is used to illuminate,

stopping potential, $V_B = \frac{V}{4}$

$$\frac{hc}{3\lambda} = \phi + \frac{eV}{4} \quad \dots(ii)$$

From eq. (i) – (ii),

$$\frac{hc}{\lambda} \left(1 - \frac{1}{3}\right) = \frac{3}{4} eV$$

$$\Rightarrow \frac{hc}{\lambda} \frac{2}{3} = \frac{3}{4} eV \Rightarrow eV = \frac{8}{9} \frac{hc}{\lambda}$$

$$\frac{hc}{\lambda} = \phi + \frac{8}{9} \frac{hc}{\lambda}$$

$$\therefore \phi = \frac{hc}{9\lambda} = \frac{hc}{n\lambda}, \text{ so, } n = 9.$$

29. (a) Using Einstein's photoelectric equation,

$$E = \omega_0 + KE_{\max}$$

$$\Rightarrow \omega_0 = KE_{\max} - E$$

$$p = \sqrt{2mKE} \Rightarrow KE = \frac{p^2}{2m}$$

$$r = \frac{p}{eB} \Rightarrow p = reB$$

$$K_{\max} = \frac{r^2 e^2 B^2}{2m} \quad KE_{\max} = \frac{12420}{\lambda} - \omega_0$$

$$\Rightarrow \omega_0 = \frac{12420}{6561} - \frac{r^2 e^2 B^2}{2m} \text{ (In eV)}$$

$$= 1.89 \text{ (eV)} - \frac{(10^{-4})(1.6 \times 10^{-19})^2 \times 9 \times 10^5}{2 \times 9.07 \times 10^{-31}}$$

$$= 1.89 \text{ (eV)} - \frac{(10^{-4})(1.6 \times 10^{-19})^2 \times 9 \times 10^5}{2 \times 9.07 \times 10^{-31}}$$

$$= (1.89 - 0.79) \text{ eV} = 1.1 \text{ eV}$$

30. (a) de-Broglie wavelength (λ),

$$\text{Momentum, } mv = \frac{h}{\lambda} = p = \sqrt{2m(KE)}$$

$$\therefore \lambda = \frac{h}{\sqrt{2mKE}} \Rightarrow \lambda \propto \frac{1}{\sqrt{KE}}$$

$$\therefore \frac{\lambda_A}{\lambda_B} = \sqrt{\frac{K_B}{K_A}} = \sqrt{\frac{T_A - 1.5}{T_A}} \quad (\text{as given})$$

$$\text{Also, } \frac{\lambda_A}{\lambda_B} = \frac{1}{2}$$

On solving we get, $T_A = 2 \text{ eV}$

$$\therefore KE_B = T_A - 1.5 = 2 - 1.5 = 0.5 \text{ eV}$$

$\therefore$ Work function of metal B is

$$\phi_B = E_B - KE_B = 4.5 - 0.5 = 4 \text{ eV}$$

31. (11.00) Energy of proton

$$E = \frac{hc}{\lambda} = \frac{1240}{310} = 4 \text{ eV} > 2 \text{ eV} [= \phi]$$

(so emission of photoelectron will take place)

$$= 4 \times 1.6 \times 10^{-19} = 6.4 \times 10^{-19} \text{ joule}$$

$$N = \frac{6.4 \times 10^{-5} \times 1}{4 \times 6.4 \times 10^{-19}} = 10^{14}$$

No. of photoelectrons emitted per second

$$= \frac{10^{14}}{10^3} = 10^{11} \quad (\because 1 \text{ in } 10^3 \text{ photons ejects an electron})$$

$\therefore$ Value of $X = 11.00$

32. (b) $f_0 = 4 \times 10^{14} \text{ Hz}$

$$W_0 = hf_0 = 6.63 \times 10^{-34} \times (4 \times 10^{14}) \text{ J}$$

$$= \frac{(6.63 \times 10^{-34}) \times (4 \times 10^{14})}{1.6 \times 10^{-19}}$$

$$= 1.66 \text{ eV}$$

33. (a) $KE_{\max} = E - \phi_0$

(where E = energy of incident light ϕ_0 = work function)

$$= \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$$

$$= 1237 \left[\frac{1}{260} - \frac{1}{380} \right]$$

$$= \frac{1237 \times 120}{380 \times 260} = 1.5 \text{ eV}$$

34. (a) Energy of photon (E) is given by

$$E = \frac{hc}{\lambda}$$

Number of photons of wavelength λ emitted in t second from laser of power P is given by

$$n = \frac{Pt\lambda}{hc}$$

$$\Rightarrow n = \frac{2 \times \lambda}{hc} = \frac{2 \times 10^{-3} \times 5 \times 10^{-7}}{2 \times 10^{-25}} \quad (\because t = 1 \text{ S})$$

$$\Rightarrow n = 5 \times 10^{15}$$

35. (c) Here $w = 2\pi \times 6 \times 10^{14}$ or $f = 6 \times 10^{14} \text{ Hz}$

$$\text{Wavelength } \lambda = \frac{c}{f} = \frac{3 \times 10^8}{6 \times 10^{14}} = 0.5 \times 10^{-6} \text{ m} = 5000 \text{ \AA}$$

$$\text{Now } E = \frac{12374}{5000} = 2.48 \text{ eV}$$

Using $E = w + eV_s$

$$2.48 = 2 + eV_s \text{ or } V_s = 0.48 \text{ V}$$

36. (BONUS)

37. (d) Using, wavelength, $\lambda = \frac{12375}{\Delta E}$

$$\text{or, } \lambda = \frac{12375}{4.9} \approx 250 \text{ nm}$$

38. (c) Let ϕ = work function of the metal,

$$\frac{hc}{\lambda_1} = \phi + eV_1 \quad \dots (i)$$

$$\frac{hc}{\lambda_2} = \phi + eV_2 \quad \dots (ii)$$

Subtracting (ii) from (i) we get

$$hc \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right) = e(V_1 - V_2)$$

$$\Rightarrow V_1 - V_2 = \frac{hc}{e} \left(\frac{\lambda_2 - \lambda_1}{\lambda_1 \lambda_2} \right) \left[\begin{array}{l} \lambda_1 = 300 \text{ nm} \\ \lambda_2 = 400 \text{ nm} \\ \frac{hc}{e} = 1240 \text{ nm} \cdot \text{V} \end{array} \right]$$

$$= (1240 \text{ nm} \cdot \text{V}) \left(\frac{100 \text{ nm}}{300 \text{ nm} \times 400 \text{ nm}} \right)$$

$$= 1.03 \text{ V} \approx 1 \text{ V}$$

39. (c) using, intensity $I = \frac{nE}{At}$

n = no. of photoelectrons

$$\Rightarrow 16 \times 10^{-3} = \left(\frac{n}{t} \right) \times \frac{10 \times 1.6 \times 10^{-19}}{10^{-4}} \quad \text{or, } \frac{n}{t} = 10^{12}$$

So, effective number of photoelectrons ejected per unit time = $10^{12} \times 10/100 = 10^{11}$

40. (a) From Einstein's photoelectric equation,

$$\frac{hc}{\lambda_1} = \phi + \frac{1}{2} m (2v)^2 \quad \dots (i)$$

$$\text{and } \frac{hc}{\lambda_2} = \phi + \frac{1}{2} mv^2 \quad \dots (ii)$$

As per question, maximum speed of photoelectrons in two cases differ by a factor 2

From eqn. (i) & (ii)

$$\Rightarrow \frac{\frac{hc}{\lambda_1} - \phi}{\frac{hc}{\lambda_2} - \phi} = 4 \Rightarrow \frac{hc}{\lambda_1} - \phi = \frac{4hc}{\lambda_2} - 4\phi$$

$$\Rightarrow \frac{4hc}{\lambda_2} - \frac{hc}{\lambda_1} = 3\phi \Rightarrow \phi = \frac{1}{3} hc \left(\frac{4}{\lambda_2} - \frac{1}{\lambda_1} \right)$$

$$= \frac{1}{3} \times 1240 \left(\frac{4 \times 350 - 540}{350 \times 540} \right) = 1.8 \text{ eV}$$

41. (d) According to question, there are two EM waves with different frequency,

$$B_1 = B_0 \sin(\pi \times 10^7 c)t$$

$$\text{and } B_2 = B_0 \sin(2\pi \times 10^7 c)t$$

To get maximum kinetic energy we take the photon with higher frequency

$$\text{using, } B = B_0 \sin \omega t \text{ and } \omega = 2\pi\nu \Rightarrow \nu = \frac{\omega}{2\pi}$$

$$B_1 = B_0 \sin(\pi \times 10^7 c)t \Rightarrow \nu_1 = \frac{10^7}{2} \times c$$

$$B_2 = B_0 \sin(2\pi \times 10^7 c)t \Rightarrow \nu_2 = 10^7 c$$

where c is speed of light $c = 3 \times 10^8 \text{ m/s}$

Clearly, $\nu_2 > \nu_1$

so KE of photoelectron will be maximum for photon of higher energy.

$$\nu_2 = 10^7 c \text{ Hz}$$

$$h\nu = \phi + \text{KE}_{\text{max}}$$

energy of photon

$$E_{\text{ph}} = h\nu = 6.6 \times 10^{-34} \times 10^7 \times 3 \times 10^8$$

$$E_{\text{ph}} = 6.6 \times 3 \times 10^{-19} \text{ J}$$

$$= \frac{6.6 \times 3 \times 10^{-19}}{1.6 \times 10^{-19}} \text{ eV} = 12.375 \text{ eV}$$

$$\text{KE}_{\text{max}} = E_{\text{ph}} - \phi$$

$$= 12.375 - 4.7 = 7.675 \text{ eV} \approx 7.7 \text{ eV}$$

42. (c) In X-ray tube, $\lambda_{\text{min}} = \frac{hc}{eV}$

$$\ln \lambda_{\text{min}} = \ln \left(\frac{hc}{e} \right) - \ln V$$

Clearly, $\log \lambda_{\text{min}}$ versus $\log V$ graph

slope is negative hence option (c) correctly depicts.

43. (a) Given, $\lambda = 660 \text{ nm}$, Power = 0.5 kW , $t = 60 \text{ ms}$

$$\text{Power } P = \frac{nhc}{\lambda t} \Rightarrow n = \frac{P\lambda t}{hc}$$

$$= 0.5 \times 10^3 \times \frac{660 \times 10^{-9} \times 60 \times 10^{-3}}{6.6 \times 10^{-34} \times 3 \times 10^8}$$

$$= 100 \times 10^{18} = 10^{20}$$

44. (c) As the metal surface is same, work function (ϕ) is same for both the case.

$$\text{Initially } \text{KE}_{\text{max}} = nh - \phi \quad \dots (i)$$

After increase

$$\text{KE}'_{\text{max}} = 3nh - \phi \quad \dots (ii)$$

For work function ϕ – not to be –ve or zero, $\nu' > \sqrt{3}\nu$

45. (c) $h \frac{c}{\lambda} - h\nu_0 = \frac{1}{2}mv^2$

$$\therefore \frac{4}{3} \frac{hc}{\lambda} - h\nu_0 = \frac{1}{2}mv'^2$$

$$\therefore \frac{v'^2}{v^2} = \frac{\frac{4}{3}v - v_0}{v - v_0} \therefore v' = v \sqrt{\frac{\frac{4}{3}v - v_0}{v - v_0}}$$

$$\therefore v' > v \sqrt{\frac{4}{3}}$$

46. (a) From Einstein's photoelectric equation

$$K.E._{\lambda} = \frac{hc}{\lambda} - \phi \quad \dots(i)$$

(for monochromatic light of wavelength λ)

where ϕ is work function

$$K.E._{\lambda/2} = \frac{hc}{\lambda/2} - \phi \quad \dots(ii)$$

(for monochromatic light of wavelength $\lambda/2$)

From question,

$$K.E._{\lambda/2} = 3(K.E._{\lambda}) \Rightarrow \frac{hc}{\lambda/2} - \phi = 3\left(\frac{hc}{\lambda} - \phi\right)$$

$$\frac{2hc}{\lambda} - \phi = 3\frac{hc}{\lambda} - 3\phi$$

$$\Rightarrow 2\phi = \frac{hc}{\lambda} \therefore \phi = \frac{hc}{2\lambda}$$

47. (None)

From Einstein's photoelectric equation, we have

$$\frac{hc}{\lambda_1} = \frac{hc}{\lambda_0} + eV \quad \dots(1)$$

$$\frac{hc}{\lambda_2} = \frac{hc}{\lambda_0} + eV \quad \dots(2)$$

$$\frac{hc}{\lambda_3} = \frac{hc}{\lambda_0} + 3eV' \quad \dots(3)$$

From equation (1) & (2)

$$\frac{3}{2\lambda_1} - \frac{2}{2\lambda_2} = \frac{1}{\lambda_0}$$

$$\frac{hc}{\lambda_1} - hc \left[\frac{3}{2\lambda_1} - \frac{1}{2\lambda_2} \right] = eV'$$

$$\frac{hc}{e} \left[\frac{1}{\lambda_3} - \frac{3}{2\lambda_1} + \frac{1}{2\lambda_2} \right] = V'$$

48. (a) Frank-Hertz experiment - Discrete energy levels of atom, Photoelectric effect - Particle nature of light.
Davison - Germer experiment - wave nature of electron.

49. (b) Given, $\lambda_1 = 4972 \text{ \AA}$

and $\lambda_2 = 6216 \text{ \AA}$

and $I = 3.6 \times 10^{-3} \text{ Wm}^{-2}$

Intensity associated with each wavelength

$$= \frac{3.6 \times 10^{-3}}{2}$$

$$= 1.8 \times 10^{-3} \text{ Wm}^{-2}$$

work function $\phi = h\nu$

$$= \frac{hc}{\lambda}$$

$$= \frac{(6.62 \times 10^{-34})(3 \times 10^8)}{\lambda}$$

$$= \frac{12.4 \times 10^3}{\lambda} \text{ eV}$$

for different wavelengths

$$\phi_1 = \frac{12.4 \times 10^3}{\lambda_1} = \frac{12.4 \times 10^3}{4972} = 2.493 \text{ eV} = 3.984 \times 10^{-19} \text{ J}$$

$$\phi_2 = \frac{12.4 \times 10^3}{\lambda_2} = \frac{12.4 \times 10^3}{6216} = 1.994 \text{ eV} = 3.184 \times 10^{-19} \text{ J}$$

J

Work function for metallic surface $\phi = 2.3 \text{ eV}$ (given)

$\phi_2 < \phi$

Therefore, ϕ_2 will not contribute in this process.

Now, no. of electrons per $\text{m}^2\text{-s}$ = no. of photons per $\text{m}^2\text{-s}$

$$\text{no. of electrons per } \text{m}^2\text{-s} = \frac{1.8 \times 10^{-3}}{3.984 \times 10^{-19}} \times 10^{-4}$$

$$\left(\because 1 \text{ cm}^2 = 10^{-4} \text{ m}^2 \right) = 0.45 \times 10^{12}$$

So, the number of photo electrons liberated in 2 sec.

$$= 0.45 \times 10^{12} \times 2$$

$$= 9 \times 10^{11}$$

50. (b)

51. (d) As λ is increased, there will be a value of λ above which photoelectrons will cease to come out so photocurrent will become zero. Hence (d) is correct answer.

52. (d) $\frac{hc}{\lambda} - \phi = eV_0$

$$v_0 = \frac{hc}{e\lambda} - \frac{\phi}{e}$$

For metal A

For metal B

$$\frac{\phi_A}{hc} = \frac{1}{\lambda}$$

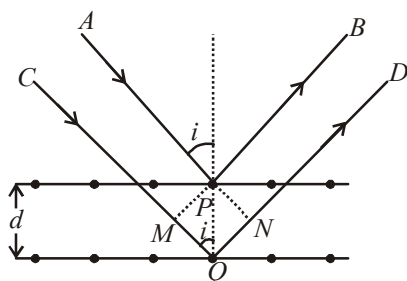
$$\frac{\phi_B}{hc} = \frac{1}{\lambda}$$

As the value of $\frac{1}{\lambda}$ (increasing and decreasing) is not specified hence we cannot say that which metal has comparatively greater or lesser work function (ϕ).

53. (a)

54. (a) Davisson Germer experiment showed that electron beams can undergo diffraction when passed through atomic crystal. This established wave nature of electron as waves can exhibit interference and diffraction.

55. (c) According to photo-electric equation :
 $K.E_{\max} = h\nu - h\nu_0$ (Work function)
 Some sort of energy is used in ejecting the photoelectrons.
56. (b) The maximum kinetic energy of photoelectrons depends upon frequency on incident light and photo current depends upon intensity of incident light.
57. (c) By Einstein photoelectric equation,
 $K_{\max} = eV_0 = h\nu - h\nu_0$
 When ν is doubled, $K_{\max}$ and V_0 become more than double.
58. (d) We know that
 $eV_0 = K_{\max} = h\nu - \phi$
 where, ϕ is the work function.
 X-rays have higher frequency (ν) than ultraviolet rays. Therefore as ν increases $K.E$ and V_0 both increases.
 The kinetic energy ranges from zero to maximum because of loss of energy due to subsequent collisions before getting ejected.
59. (a) Wavelength of incident light, $\lambda = 400 \text{ nm}$ $hc = 1240 \text{ eV}\cdot\text{nm}$
 $K.E = 1.68 \text{ eV}$
 Using Einstein's photoelectric equation
 $\frac{hc}{\lambda} - W = K.E$
 $\Rightarrow W = \frac{hc}{\lambda} - K.E$
 $\Rightarrow W = \frac{1240}{400} - 1.68$
 $= 3.1 - 1.68$
 $= 1.41 \text{ eV}$
60. (b) The path difference between the rays APB and CQD is
 $\Delta x = MQ + QN = d \cos i + d \cos i$
 $\Delta x = 2d \cos i$



For constructive interference the path difference is integral multiple of wavelength

$$\therefore n\lambda = 2d \cos i$$

From de-broglie concept

Wavelength,

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK.E}} = \frac{h}{\sqrt{2meV}}$$

$$\therefore \frac{nh}{\sqrt{2meV}} = 2d \cos i$$

Squaring both side

$$\frac{n^2 h^2}{2meV} = 4d^2 \cos^2 i$$

For first order interference $n = 1$

$$\therefore V = \frac{h^2}{8med^2 \cos^2 i}$$

$$= \frac{(6.6 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times (10^{-10})^2 \times \cos^2 30}$$

$$= 50 \text{ V}$$

61. (b) For constructive interference,
 $2d \cos i = n\lambda_{dB}$
62. (a) Energy of a photon of frequency ν is given by
 $E = h\nu$.
 Also, $E = mc^2$, $mc^2 = h\nu$
 $\Rightarrow mc = \frac{h\nu}{c} \Rightarrow p = \frac{h\nu}{c}$
63. (a) Work function, $\phi = 6.2 \text{ eV} = 6.2 \times 1.6 \times 10^{-19} \text{ J}$
 Stopping potential, $V = 5 \text{ volt}$
 From the Einstein's photoelectric equation
 $\frac{hc}{\lambda} - \phi = eV_0$
 $\Rightarrow \lambda = \frac{hc}{\phi + eV_0}$
 $= \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19} (6.2 + 5)} \approx 10^{-7} \text{ m}$

This range lies in ultra violet range.

64. (b) The photoelectric emission is an instantaneous process without any apparent time lag. It is known that emission starts in the time of the order of 10^{-9} second. So, the approximate time taken by a photoelectron to come out after the photon strikes is 10^{-10} second.
65. (b) As λ decreases, y increases and hence the speed of photoelectron increases. The chances of photo electron to meet the anode increases and hence photo electric current increases.
66. (a) $I \propto \frac{I}{r^2}$; $\frac{I_1}{I_2} = \left(\frac{r_2}{r_1}\right)^2 = \frac{1}{4}$
 $I_2 \rightarrow 4 \text{ times } I_1$
 When intensity becomes 4 times, no. of photoelectrons emitted would increase by 4 times, since number of electrons emitted per second is directly proportional to intensity.

67. (b) Momentum of photon of energy E is $= \frac{E}{c}$

When a photon hits a perfectly reflecting surface, it reflects back in opposite direction with same energy and momentum.

$$\therefore \text{Change in momentum} = \frac{E}{C} - \left(\frac{-E}{C} \right) = \frac{2E}{C}$$

This is equal to momentum transferred to the surface.

68. (d) From the Einstein photoelectric equation $K.E. = h\nu - \phi$

Here, ϕ = work function of metal

h = Planck's constant

slope of graph of $K.E.$ & ν is h (Planck's constant) which is same for all metals.

69. (a) Work function of metal (ϕ) is given by

$$\phi = \frac{hc}{\lambda}$$

$$\Rightarrow \lambda = \frac{hc}{\phi}$$

$$\Rightarrow \lambda = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{4 \times 1.6 \times 10^{-19}} = 310 \text{ nm}$$

70. (a) Let work function be W and ν_1 and ν_2 be the velocity of electrons for frequencies f_1 and f_2 .

Using Einstein's photo electric equation for one photodiode, we get

$$hf_1 - W = \frac{1}{2} m \nu_1^2 \quad \dots(i)$$

Using Einstein's photo electric equation for another photodiode we get,

$$hf_2 - W = \frac{1}{2} m \nu_2^2 \quad \dots(ii)$$

Subtracting (ii) from (i) we get

$$(hf_1 - W) - (hf_2 - W) = \frac{1}{2} m \nu_1^2 - \frac{1}{2} m \nu_2^2$$

$$\therefore h(f_1 - f_2) = \frac{m}{2} (\nu_1^2 - \nu_2^2)$$

$$\therefore \nu_1^2 - \nu_2^2 = \frac{2h}{m} (f_1 - f_2)$$

71. (c) We know that work function,

$$E = h\nu = \frac{hC}{\lambda}$$

where

h = Planck's constant

C = velocity of light

λ = wavelength of light

$$\therefore \frac{E_{\text{Na}}}{E_{\text{Cu}}} = \frac{\lambda_{\text{Cu}}}{\lambda_{\text{Na}}}$$

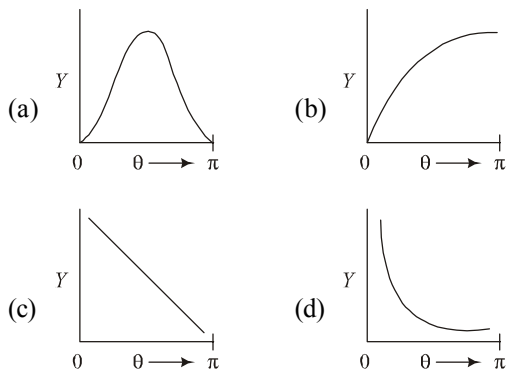
$$\Rightarrow \frac{\lambda_{\text{Na}}}{\lambda_{\text{Cu}}} = \frac{E_{\text{Cu}}}{E_{\text{Na}}} = \frac{4.5}{2.3} \approx \frac{2}{1}$$



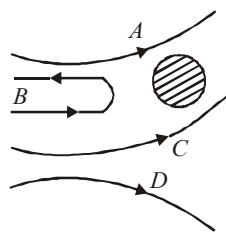
TOPIC 1 Atomic Structure and Rutherford's Nuclear Model



1. The graph which depicts the results of Rutherford gold foil experiment with α -particles is: [8 Jan. 2020 I]
 θ : Scattering angle
 Y : Number of scattered α -particles detected
 (Plots are schematic and not to scale)



2. In the Rutherford experiment, α -particles are scattered from a nucleus as shown. Out of the four paths, which path is not possible? [Online May 7, 2012]



- (a) D (b) B (c) C (d) A
3. An alpha nucleus of energy $\frac{1}{2}mv^2$ bombards a heavy nuclear target of charge Ze . Then the distance of closest approach for the alpha nucleus will be proportional to [2006]
- (a) v^2 (b) $\frac{1}{m}$ (c) $\frac{1}{v^2}$ (d) $\frac{1}{Ze}$

4. An α -particle of energy 5 MeV is scattered through 180° by a fixed uranium nucleus. The distance of closest approach is of the order of [2004]
- (a) 10^{-12} cm (b) 10^{-10} cm
 (c) 10^{-14} cm (d) 10^{-15} cm

TOPIC 2 Bohr's Model and the Spectra of the Hydrogen Atom

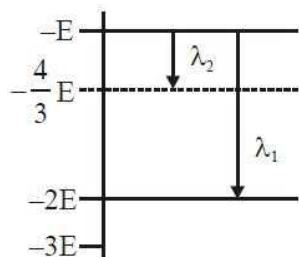


5. A particle of mass 200 MeV/ c^2 collides with a hydrogen atom at rest. Soon after the collision the particle comes to rest, and the atom recoils and goes to its first excited state. The initial kinetic energy of the particle (in eV) is $\frac{N}{4}$. The value of N is : [NA Sep. 05, 2020 (I)]
 (Given the mass of the hydrogen atom to be 1 GeV/ c^2)
6. In the line spectra of hydrogen atom, difference between the largest and the shortest wavelengths of the Lyman series is 304 Å. The corresponding difference for the Paschen series in Å is : [NA Sep. 04, 2020 (I)]
7. In a hydrogen atom the electron makes a transition from $(n+1)^{\text{th}}$ level to the n^{th} level. If $n \gg 1$, the frequency of radiation emitted is proportional to : [Sep. 02, 2020 (II)]
- (a) $\frac{1}{n}$ (b) $\frac{1}{n^3}$
 (c) $\frac{1}{n^2}$ (d) $\frac{1}{n^4}$
8. The energy required to ionise a hydrogen like ion in its ground state is 9 Rydbergs. What is the wavelength of the radiation emitted when the electron in this ion jumps from the second excited state to the ground state? [9 Jan. 2020 II]
- (a) 24.2 nm (b) 11.4 nm
 (c) 35.8 nm (d) 8.6 nm
9. The first member of the Balmer series of hydrogen atom has a wavelength of 6561 Å. The wavelength of the second member of the Balmer series (in nm) is : [NA 8 Jan. 2020 II]

10. The time period of revolution of electron in its ground state orbit in a hydrogen atom is 1.6×10^{-16} s. The frequency of revolution of the electron in its first excited state (in s^{-1}) is: [7 Jan. 2020 I]
 (a) 1.6×10^{14} (b) 7.8×10^{14}
 (c) 6.2×10^{15} (d) 5.6×10^{12}
11. An excited He^+ ion emits two photons in succession, with wavelengths 108.5 nm and 30.4 nm, in making a transition to ground state. The quantum number n , corresponding to its initial excited state is (for photon of wavelength λ),

$$\text{energy } E = \frac{1240 \text{ eV}}{\lambda (\text{in nm})}$$
 [12 April 2019 II]
 (a) $n = 4$ (b) $n = 5$ (c) $n = 7$ (d) $n = 6$
12. The electron in a hydrogen atom first jumps from the third excited state to the second excited state and subsequently to the first excited state. The ratio of the respective wavelengths, λ_1 / λ_2 , of the photons emitted in this process is: [12 April 2019 II]
 (a) 20/7 (b) 27/5 (c) 7/5 (d) 9/7
13. Consider an electron in a hydrogen atom, revolving in its second excited state (having radius 4.65 Å). The de-Broglie wavelength of this electron is: [12 April 2019 II]
 (a) 3.5 Å (b) 6.6 Å (c) 12.9 Å (d) 9.7 Å
14. In Li^{++} , electron in first Bohr orbit is excited to a level by a radiation of wavelength λ . When the ion gets deexcited to the ground state in all possible ways (including intermediate emissions), a total of six spectral lines are observed. What is the value of λ ? [10 April 2019 II]
 (Given: $h = 6.63 \times 10^{-34}$ Js; $c = 3 \times 10^8 \text{ ms}^{-1}$)
 (a) 11.4 nm (b) 9.4 nm (c) 12.3 nm (d) 10.8 nm
15. Taking the wavelength of first Balmer line in hydrogen spectrum ($n = 3$ to $n = 2$) as 660 nm, the wavelength of the 2nd Balmer line ($n = 4$ to $n = 2$) will be; [9 April 2019 I]
 (a) 889.2 nm (b) 488.9 nm
 (c) 642.7 nm (d) 388.9 nm
16. A He^+ ion is in its first excited state. Its ionization energy is: [9 April 2019 II]
 (a) 48.36 eV (b) 54.40 eV
 (c) 13.60 eV (d) 6.04 eV
17. Radiation coming from transitions $n = 2$ to $n = 1$ of hydrogen atoms fall on He^+ ions in $n = 1$ and $n = 2$ states. The possible transition of helium ions as they absorb energy from the radiation is: [8 April 2019 I]
 (a) $n = 2 \rightarrow n = 3$ (b) $n = 1 \rightarrow n = 4$
 (c) $n = 2 \rightarrow n = 5$ (d) $n = 2 \rightarrow n = 4$
18. A hydrogen atom, initially in the ground state is excited by absorbing a photon of wavelength 980 Å. The radius of the atom in the excited state, in terms of Bohr radius a_0 , will be: [11 Jan 2019 I]
 (a) $25a_0$ (b) $9a_0$ (c) $16a_0$ (d) $4a_0$
19. In a hydrogen like atom, when an electron jumps from the M-shell to the L-shell, the wavelength of emitted radiation is λ . If an electron jumps from N-shell to the L-shell, the wavelength of emitted radiation will be: [11 Jan 2019 II]
 (a) $\frac{27}{20}\lambda$ (b) $\frac{16}{25}\lambda$ (c) $\frac{25}{16}\lambda$ (d) $\frac{20}{27}\lambda$
20. An electron from various excited states of hydrogen atom emit radiation to come to the ground state. Let λ_n, λ_g be the de Broglie wavelength of the electron in the n^{th} state and the ground state respectively. Let Λ_n be the wavelength of the emitted photon in the transition from the n^{th} state to the ground state. For large n , A, B are constants) [2018]
 (a) $\Lambda_n \approx A + \frac{B}{\lambda_n^2}$ (b) $\Lambda_n \approx A + B\lambda_n$
 (c) $\Lambda_n^2 \approx A + B\lambda_n^2$ (d) $\Lambda_n^2 \approx \lambda$
21. If the series limit frequency of the Lyman series is ν_1 , then the series limit frequency of the P-fund series is: [2018]
 (a) $25 \nu_1$ (b) $16 \nu_1$ (c) $\nu_1/16$ (d) $\nu_1/25$
22. The de-Broglie wavelength (λ_B) associated with the electron orbiting in the second excited state of hydrogen atom is related to that in the ground state (λ_G) by [Online April 16, 2018]
 (a) $\lambda_B = \lambda_G/3$ (b) $\lambda_B = \lambda_G/2$
 (c) $\lambda_B = 2\lambda_G$ (d) $\lambda_B = 3\lambda_G$
23. The energy required to remove the electron from a singly ionized Helium atom is 2.2 times the energy required to remove an electron from Helium atom. The total energy required to ionize the Helium atom completely is: [Online April 15, 2018]
 (a) 20 eV (b) 79 eV (c) 109 eV (d) 34 eV
24. Muon (μ^{-1}) is negatively charged ($|q| = |e|$) with a mass $m_\mu = 200m_e$, where m_e is the mass of the electron and e is the electronic charge. If μ^{-1} is bound to a proton to form a hydrogen like atom, identify the correct statements [Online April 15, 2018]
 (A) Radius of the muonic orbit is 200 times smaller than that of the electron
 (B) the speed of the μ^{-1} in the n^{th} orbit is $\frac{1}{200}$ times that of the electron in the n^{th} orbit
 (C) The ionization energy of muonic atom is 200 times more than that of an hydrogen atom
 (D) The momentum of the muon in the n^{th} orbit is 200 times more than that of the electron
 (a) (A), (B), (D) (b) (B), (D)
 (c) (C), (D) (d) (A), (C), (D)

25. Some energy levels of a molecule are shown in the figure. The ratio of the wavelengths $r = \lambda_1/\lambda_2$, is given by



[2017]

- (a) $r = \frac{3}{4}$ (b) $r = \frac{1}{3}$ (c) $r = \frac{4}{3}$ (d) $r = \frac{2}{3}$
26. The acceleration of an electron in the first orbit of the hydrogen atom ($z = 1$) is : [Online April 9, 2017]

- (a) $\frac{h^2}{\pi^2 m^2 r^3}$ (b) $\frac{h^2}{8\pi^2 m^2 r^3}$
(c) $\frac{h^2}{4\pi^2 m^2 r^3}$ (d) $\frac{h^2}{4\pi m^2 r^3}$

27. According to Bohr's theory, the time averaged magnetic field at the centre (i.e. nucleus) of a hydrogen atom due to the motion of electrons in the n^{th} orbit is proportional to : (n = principal quantum number) [Online April 8, 2017]

- (a) n^{-4} (b) n^{-5} (c) n^{-3} (d) n^{-2}

28. A hydrogen atom makes a transition from $n = 2$ to $n = 1$ and emits a photon. This photon strikes a doubly ionized lithium atom ($z = 3$) in excited state and completely removes the orbiting electron. The least quantum number for the excited state of the ion for the process is : [Online April 9, 2016]

- (a) 2 (b) 4 (c) 5 (d) 3

29. As an electron makes a transition from an excited state to the ground state of a hydrogen-like atom/ion :

[2015]

- (a) kinetic energy decreases, potential energy increases but total energy remains same
(b) kinetic energy and total energy decrease but potential energy increases
(c) its kinetic energy increases but potential energy and total energy decrease
(d) kinetic energy, potential energy and total energy decrease

30. The de-Broglie wavelength associated with the electron in the $n = 4$ level is : [Online April 11, 2015]

- (a) $\frac{1}{4}$ th of the de-Broglie wavelength of the electron in the ground state.
(b) four times the de-Broglie wavelength of the electron in the ground state
(c) two times the de-Broglie wavelength of the electron in the ground state
(d) half of the de-Broglie wavelength of the electron in the ground state

31. If one were to apply Bohr model to a particle of mass ' m ' and charge ' q ' moving in a plane under the influence of a magnetic field ' B ', the energy of the charged particle in the n^{th} level will be : [Online April 10, 2015]

- (a) $n \left(\frac{hqB}{2\pi m} \right)$ (b) $n \left(\frac{hqB}{8\pi m} \right)$
(c) $n \left(\frac{hqB}{4\pi m} \right)$ (d) $n \left(\frac{hqB}{\pi m} \right)$

32. The radiation corresponding to $3 \rightarrow 2$ transition of hydrogen atom falls on a metal surface to produce photoelectrons. These electrons are made to enter a magnetic field of 3×10^{-4} T. If the radius of the largest circular path followed by these electrons is 10.0 mm, the work function of the metal is close to: [2014]

- (a) 1.8 eV (b) 1.1 eV
(c) 0.8 eV (d) 1.6 eV

33. Hydrogen (${}_1\text{H}^1$), Deuterium (${}_1\text{H}^2$), singly ionised Helium (${}_2\text{He}^4$)⁺ and doubly ionised lithium (${}_3\text{Li}^6$)⁺⁺ all have one electron around the nucleus. Consider an electron transition from $n = 2$ to $n = 1$. If the wavelengths of emitted radiation are $\lambda_1, \lambda_2, \lambda_3$ and λ_4 respectively then approximately which one of the following is correct? [2014]

- (a) $4\lambda_1 = 2\lambda_2 = 2\lambda_3 = \lambda_4$
(b) $\lambda_1 = 2\lambda_2 = 2\lambda_3 = \lambda_4$
(c) $\lambda_1 = \lambda_2 = 4\lambda_3 = 9\lambda_4$
(d) $\lambda_1 = 2\lambda_2 = 3\lambda_3 = 4\lambda_4$

34. Match List - I (Experiment performed) with List-II (Phenomena discovered/associated) and select the correct option from the options given below the lists:

[Online April 19, 2014]

List - I		List - II	
(1)	Davisson and Germer experiment	(i)	Wave nature of electrons
(2)	Millikan's oil drop experiment	(ii)	Charge of an electron
(3)	Rutherford experiment	(iii)	Quantisation of energy levels
(4)	Franck-Hertz experiment	(iv)	Existence of nucleus

- (a) (1)-(i), (2)-(ii), (3)-(iii), (4)-(iv)
(b) (1)-(i), (2)-(ii), (3)-(iv), (4)-(iii)
(c) (1)-(iii), (2)-(iv), (3)-(i), (4)-(ii)
(d) (1)-(iv), (2)-(iii), (3)-(ii), (4)-(i)

35. The binding energy of the electron in a hydrogen atom is 13.6 eV, the energy required to remove the electron from the first excited state of Li^{++} is: [Online April 9, 2014]

(a) 122.4 eV (b) 30.6 eV
(c) 13.6 eV (d) 3.4 eV

36. In a hydrogen like atom electron make transition from an energy level with quantum number n to another with quantum number $(n - 1)$. If $n \gg 1$, the frequency of radiation emitted is proportional to: [2013]

(a) $\frac{1}{n}$ (b) $\frac{1}{n^2}$ (c) $\frac{1}{n^3/2}$ (d) $\frac{1}{n^3}$

37. A 12.5 eV electron beam is used to bombard gaseous hydrogen at room temperature. It will emit:

[Online April 25, 2013]

(a) 2 lines in the Lyman series and 1 line in the Balmer series
(b) 3 lines in the Lyman series
(c) 1 line in the Lyman series and 2 lines in the Balmer series
(d) 3 lines in the Balmer series

38. In the Bohr's model of hydrogen-like atom the force between the nucleus and the electron is modified as

$$F = \frac{e^2}{4\pi\epsilon_0} \left(\frac{1}{r^2} + \frac{\beta}{r^3} \right), \text{ where } \beta \text{ is a constant. For this atom,}$$

the radius of the n^{th} orbit in terms of the Bohr radius

$$\left(a_0 = \frac{\epsilon_0 h^2}{m\pi e^2} \right) \text{ is: [Online April 23, 2013]}$$

(a) $r_n = a_0 n - \beta$ (b) $r_n = a_0 n^2 + \beta$
(c) $r_n = a_0 n^2 - \beta$ (d) $r_n = a_0 n + \beta$

39. Orbits of a particle moving in a circle are such that the perimeter of the orbit equals an integer number of de-Broglie wavelengths of the particle. For a charged particle moving in a plane perpendicular to a magnetic field, the radius of the n^{th} orbital will therefore be proportional to: [Online April 22, 2013]

(a) n^2 (b) n (c) $n^{1/2}$ (d) $n^{1/4}$

40. In the Bohr model an electron moves in a circular orbit around the proton. Considering the orbiting electron to be a circular current loop, the magnetic moment of the hydrogen atom, when the electron is in n^{th} excited state, is:

[Online April 9, 2013]

(a) $\left(\frac{e}{2m} \right) \frac{n^2 h}{2\pi}$ (b) $\left(\frac{e}{m} \right) \frac{nh}{2\pi}$
(c) $\left(\frac{e}{2m} \right) \frac{nh}{2\pi}$ (d) $\left(\frac{e}{m} \right) \frac{n^2 h}{2\pi}$

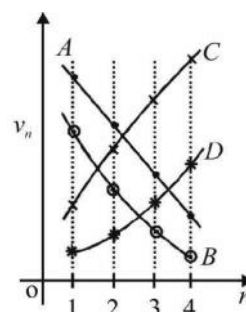
41. Hydrogen atom is excited from ground state to another state with principal quantum number equal to 4. Then the number of spectral lines in the emission spectra will be: [2012]

(a) 2 (b) 3 (c) 5 (d) 6

42. A diatomic molecule is made of two masses m_1 and m_2 which are separated by a distance r . If we calculate its rotational energy by applying Bohr's rule of angular momentum quantization, its energy will be given by: (n is an integer) [2012]

(a) $\frac{(m_1 + m_2)^2 n^2 h^2}{2m_1^2 m_2^2 r^2}$ (b) $\frac{n^2 h^2}{2(m_1 + m_2)r^2}$
(c) $\frac{2n^2 h^2}{(m_1 + m_2)r^2}$ (d) $\frac{(m_1 + m_2)n^2 h^2}{2m_1 m_2 r^2}$

43. Which of the plots shown in the figure represents speed (v_n) of the electron in a hydrogen atom as a function of the principal quantum number (n)? [Online May 26, 2012]



(a) B (b) D (c) C (d) A

44. A doubly ionised Li atom is excited from its ground state ($n = 1$) to $n = 3$ state. The wavelengths of the spectral lines are given by λ_{32} , λ_{31} and λ_{21} . The ratio $\lambda_{32}/\lambda_{31}$ and $\lambda_{21}/\lambda_{31}$ are, respectively [Online May 12, 2012]

(a) 8.1, 0.67 (b) 8.1, 1.2
(c) 6.4, 1.2 (d) 6.4, 0.67

45. A hypothetical atom has only three energy levels. The ground level has energy, $E_1 = -8$ eV. The two excited states have energies, $E_2 = -6$ eV and $E_3 = -2$ eV. Then which of the following wavelengths will not be present in the emission spectrum of this atom?

[Online May 12, 2012]

(a) 207 nm (b) 465 nm
(c) 310 nm (d) 620 nm

46. The electron of a hydrogen atom makes a transition from the $(n + 1)^{\text{th}}$ orbit to the n^{th} orbit. For large n the wavelength of the emitted radiation is proportional to

[Online May 7, 2012]

(a) n (b) n^3 (c) n^4 (d) n^2

47. Energy required for the electron excitation in Li^{++} from the first to the third Bohr orbit is: [2011]

(a) 36.3 eV (b) 108.8 eV
(c) 122.4 eV (d) 12.1 eV

48. The transition from the state $n = 4$ to $n = 3$ in a hydrogen like atom results in ultraviolet radiation. Infrared radiation will be obtained in the transition from: [2009]

(a) $3 \rightarrow 2$ (b) $4 \rightarrow 2$ (c) $5 \rightarrow 4$ (d) $2 \rightarrow 1$

49. Suppose an electron is attracted towards the origin by a

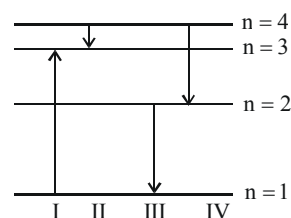
force $\frac{k}{r}$ where ' k ' is a constant and ' r ' is the distance of the electron from the origin. By applying Bohr model to this system, the radius of the n^{th} orbital of the electron is found to be ' r_n ' and the kinetic energy of the electron to be ' T_n '. Then which of the following is true? [2008]

- (a) $T_n \propto \frac{1}{n^2}, r_n \propto n^2$
 (b) T_n independent of $n, r_n \propto n$
 (c) $T_n \propto \frac{1}{n}, r_n \propto n$
 (d) $T_n \propto \frac{1}{n^3}, r_n \propto n^2$

50. Which of the following transitions in hydrogen atoms emit photons of highest frequency? [2007]

- (a) $n = 1$ to $n = 2$ (b) $n = 2$ to $n = 6$
 (c) $n = 6$ to $n = 2$ (d) $n = 2$ to $n = 1$

51. The diagram shows the energy levels for an electron in a certain atom. Which transition shown represents the emission of a photon with the most energy? [2005]



- (a) IV (b) III (c) II (d) I
52. The wavelengths involved in the spectrum of deuterium (2_1D) are slightly different from that of hydrogen spectrum, because [2003]
- (a) the size of the two nuclei are different
 (b) the nuclear forces are different in the two cases
 (c) the masses of the two nuclei are different
 (d) the attraction between the electron and the nucleus is different in the two cases
53. If 13.6 eV energy is required to ionize the hydrogen atom, then the energy required to remove an electron from $n = 2$ is [2002]
- (a) 10.2 eV (b) 0 eV
 (c) 3.4 eV (d) 6.8 eV



Hints & Solutions



- (c)
- (c) As α -particles are doubly ionised helium He^{++} i.e. Positively charged and nucleus is also positively charged and we know that like charges repel each other.
- (b) Work done to stop the α particle is equal to K.E.

$$\therefore qV = \frac{1}{2}mv^2 \Rightarrow q \times \frac{K(Ze)}{r} = \frac{1}{2}mv^2$$

$$\Rightarrow r = \frac{2(2e)K(Ze)}{mv^2} = \frac{4KZe^2}{mv^2}$$

$$\Rightarrow r \propto \frac{1}{v^2} \text{ and } r \propto \frac{1}{m}$$

- (a) Distance of closest approach

$$r_0 = \frac{Ze(2e)}{4\pi\epsilon_0 \left(\frac{1}{2}mv^2 \right)}$$

$$\text{Energy, } E = 5 \times 10^6 \times 1.6 \times 10^{-19} \text{ J}$$

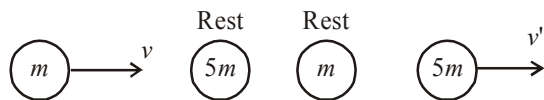
$$\therefore r_0 = \frac{9 \times 10^9 \times (92 \times 1.6 \times 10^{-19}) (2 \times 1.6 \times 10^{-19})}{5 \times 10^6 \times 1.6 \times 10^{-19}}$$

$$\Rightarrow r_h = 5.2 \times 10^{-14} \text{ m} = 5.3 \times 10^{-12} \text{ cm}$$

- (51)

Before collision

After collision



Particle Hydrogen Particle Hydrogen

From linear momentum conservation, $L_i = L_f$

$$mV + 0 = 0 + 5mV' \Rightarrow V' = \frac{v}{5}$$

$$\text{Loss of KE} = KE_i - KE_f = \frac{1}{2}mv^2 - \frac{1}{2}(5m)\left(\frac{v}{5}\right)^2$$

$$= \frac{1}{2}mv^2 \left(1 - \frac{1}{5} \right) = \frac{4}{5} \left(\frac{mv^2}{2} \right)$$

$$= \frac{4}{5} KE_i = 10.2 \text{ eV}$$

[$\therefore$ Energy in first excited state of atom = 10.2 eV]

$$KE_i = 12.75 \text{ eV} = \frac{N}{4} \Rightarrow N = 51$$

The value of $N = 51$.

- (10553.14)

From Bohr's formula for hydrogen atom,

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$R = 1.097 \times 10^7 \text{ m}^{-1}$$

For Lyman series :

$$\frac{1}{\lambda_{\min.}} = R(1) = R \quad \therefore n_2 = \infty \text{ and } n_1 = 1$$

$$\frac{1}{\lambda_{\max.}} = R \left\{ 1 - \frac{1}{4} \right\} = \frac{3R}{4} \quad \therefore n_1 = 2, n_2 = 1$$

$$\therefore \lambda_{\max.} - \lambda_{\min.} = \frac{4}{3R} - \frac{1}{R} = \frac{1}{3R} = 304 \text{ (Given)}$$

For Paschen series :

$$\lambda'_{\min.} = R \left(\frac{1}{9} \right) \text{ and } \lambda'_{\max.} = R \left(\frac{1}{9} - \frac{1}{16} \right) = \frac{7R}{16 \times 9}$$

$$\lambda'_{\max.} - \lambda'_{\min.} = \frac{16 \times 9}{7R} - \frac{9}{R} = \frac{81}{7R}$$

$$\text{or, } \lambda'_{\max.} - \lambda'_{\min.} = \frac{81}{7R} = \frac{81 \times 3}{7 \times 3R} = \frac{81 \times 3}{7} \times 304$$

$$\left(\therefore \frac{1}{3R} = 304 \text{ \AA} \right)$$

$$\therefore \text{For Paschen series, } \lambda'_{\max.} - \lambda'_{\min.} = 10553.14$$

- (b) Total energy of electron in n^{th} orbit of hydrogen atom

$$E_n = -\frac{Rhc}{n^2}$$

Total energy of electron in $(n+1)^{\text{th}}$ level of hydrogen atom

$$E_{n+1} = -\frac{Rhc}{(n+1)^2}$$

When electron makes a transition from $(n+1)^{\text{th}}$ level to n^{th} level

Change in energy,

$$\Delta E = E_{n+1} - E_n$$

$$h\nu = Rhc \cdot \left[\frac{1}{n^2} - \frac{1}{(n+1)^2} \right] \quad (\therefore E = h\nu)$$

$$\nu = R \cdot c \left[\frac{(n+1)^2 - n^2}{n^2(n+1)^2} \right]$$

$$v = R \cdot c \left[\frac{1+2n}{n^2(n+1)^2} \right]$$

For $n \gg 1$

$$\Rightarrow v = R \cdot c \left[\frac{2n}{n^2 \times n^2} \right] = \frac{2RC}{n^3}$$

$$\Rightarrow v \propto \frac{1}{n^3}$$

8. (b) According to Bohr's Theory the wavelength of the radiation emitted from hydrogen atom is given by

$$\frac{1}{\lambda} = RZ^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\therefore Z = 3$$

$$\therefore \frac{1}{\lambda} = 9R \left(1 - \frac{1}{9} \right)$$

$$\Rightarrow \lambda = \frac{1}{8R} = \frac{1}{8 \times 10973731.6 \text{ m}^{-1}} \quad (R = 10973731.6 \text{ m}^{-1})$$

$$\Rightarrow \lambda = 11.39 \text{ nm}$$

9. (486.00)

The wavelength of the spectral line of hydrogen spectrum is given by formula

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

Where, R = Rydberg constant

For the first member of Balmer series $n_f = 2$, $n_i = 3$

$$\therefore \frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) \quad \dots(i)$$

For last member of Balmer series, $n_f = 2$, $n_i = 4$

$$\text{So, } \frac{1}{\lambda'} = R \left[\frac{1}{4} - \frac{1}{16} \right] \quad \dots(ii)$$

Dividing (i) by (ii), we get

$$\Rightarrow \frac{\lambda'}{\lambda} = \frac{5 \times 16}{9 \times 4 \times 3}$$

$$\Rightarrow \lambda' = \frac{5 \times 4 \times 656.1}{9 \times 3} (\text{nm}) = 486 \text{ nm}$$

10. (b) For first excited state $n' = 3$

$$\text{Time period } T \propto \frac{n^3}{z^2}$$

$$\Rightarrow \frac{T_2}{T_1} = \frac{n^3}{n^3}$$

$$\therefore T_2 = 8T_1 = 8 \times 1.6 \times 10^{-16} \text{ s}$$

$$\therefore \text{Frequency, } \nu = \frac{1}{T_2} = \frac{1}{8 \times 1.6 \times 10^{-16}}$$

$$\approx 7.8 \times 10^{14} \text{ Hz}$$

11. (b) $E = E_1 + E_2$

$$13.6 \frac{z^2}{n^2} = \frac{1240}{\lambda_1} + \frac{1240}{\lambda_2}$$

$$\text{or } \frac{13.6(2)^2}{n^2} = 1240 \left(\frac{1}{108.5} + \frac{1}{30.4} \right) \times \frac{1}{10^{-9}}$$

On solving, $n = 5$

12. (a) $\frac{1}{\lambda_1} = R \left(\frac{1}{3^2} - \frac{1}{4^2} \right) = \frac{7R}{16 \times 9}$

$$\text{And } \frac{1}{\lambda_2} = R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = \frac{5R}{36}$$

$$\text{Now } \frac{\lambda_1}{\lambda_2} = \frac{(5R/36)}{7R/(16 \times 9)} = \frac{20}{7}$$

13. (d) $\nu = \frac{c}{137n} = \frac{c}{137 \times 3}$

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\left(\frac{m \times c}{3 \times 137} \right)} = \frac{h}{mc} \times (3 \times 137) = 9.7 \text{ \AA}$$

14. (d) Spectral lines obtained on account of transition from

n th orbit to various lower orbits is $\frac{n(n-1)}{2}$

$$\Rightarrow 6 = \frac{n(n-1)}{2}$$

$$\Rightarrow n = 4$$

$$\Delta E = \frac{hc}{\lambda} = \frac{-Z^2}{n^2} (13.6 \text{ eV})$$

$$\Rightarrow \frac{1}{\lambda} = Z^2 \left(\frac{13.6 \text{ eV}}{hc} \right) \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right)$$

$$= (13.4)(3)^2 \left[1 - \frac{1}{16} \right] \text{ eV}$$

$$\Rightarrow \lambda = \frac{1242 \times 16}{(13.4) \times (9)(15)} \text{ nm} \approx 10.8 \text{ nm}$$

15. (b) $\frac{1}{\lambda_1} = -R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = \frac{5R}{36}$

$$\frac{1}{\lambda_2} = R \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = \frac{3R}{16}$$

$$\therefore \frac{\lambda_2}{\lambda_1} = \frac{80}{108}$$

$$\lambda_2 = \frac{80}{108} \lambda_1 = \frac{80}{108} \times 660 = 488.9 \text{ nm.}$$

16. (c) $E_n = -13.6 \frac{Z^2}{n^2}$

For He^+ , $E_2 = \frac{-13.6(2)^2}{2^2} = -13.60 \text{ eV}$

Ionization energy = $0 - E_2 = 13.60 \text{ eV}$

17. (d) Energy released by hydrogen atom for transition $n = 2$ to $n = 1$

$$\therefore \Delta E_1 = 13.6 \times \left(\frac{1}{1^2} - \frac{1}{2^2} \right) = \frac{3}{4} \times 13.6 \text{ eV}$$

$$= 10.2 \text{ eV}$$

This energy is absorbed by He^+ ion in transition from $n = 2$ to $n = n_1$ (say)

$$\therefore \Delta E_2 = 13.6 \times 4 \times \left(\frac{1}{4} - \frac{1}{n_1^2} \right) = 10.2 \text{ eV}$$

$$\Rightarrow n_1 = 4$$

So, possible transition is $n = 2 \rightarrow n = 4$

18. (3) Energy of photon = $\frac{hc}{\lambda} = \frac{12500}{980} = 12.75 \text{ eV}$

Energy of electron in n^{th} orbit is given by

$$E_n = \frac{-13.6}{n^2} \Rightarrow E_n - E_1 = -13.6 \left[\frac{1}{n^2} - \frac{1}{1^2} \right]$$

$$\Rightarrow 12.75 = 13.6 \left[\frac{1}{1^2} - \frac{1}{n^2} \right] \Rightarrow n = 4$$

$\therefore$ Electron will excite to $n = 4$

We know that ' R ' $\propto n^2$

$\therefore$ Radius of atom will be $16a_0$

19. (d) When electron jumps from $M \rightarrow L$ shell

$$\frac{1}{\lambda} = K \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = \frac{K \times 5}{36} \quad \dots (i)$$

When electron jumps from $N \rightarrow L$ shell

$$\frac{1}{\lambda'} = K \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = \frac{K \times 3}{16} \quad \dots (ii)$$

solving equation (i) and (ii) we get

$$\lambda' = \frac{20}{27} \lambda$$

20. (a) Wavelength of emitted photon from n^{th} state to the ground state,

$$\frac{1}{\Lambda_n} = RZ^2 \left(\frac{1}{1^2} - \frac{1}{n^2} \right)$$

$$\Lambda_n = \frac{1}{RZ^2 \left(1 - \frac{1}{n^2} \right)^{-1}}$$

Since n is very large, using binomial theorem

$$\Lambda_n = \frac{1}{RZ^2} \left(1 + \frac{1}{n^2} \right)$$

$$\Lambda_n = \frac{1}{RZ^2} + \frac{1}{RZ^2} \left(\frac{1}{n^2} \right)$$

As we know, $\lambda_n = \frac{2\pi r}{n} = 2\pi \left(\frac{n^2 h^2}{4\pi^2 m Z e^2} \right) \frac{1}{n} \propto n$

$$\Lambda_n \approx A + \frac{B}{\lambda_n^2}$$

21. (d) $h\nu_L = E_\infty - E_1 \quad \dots (i)$

$$h\nu_f = E_\infty - E_5 \quad \dots (ii)$$

$$E \propto \frac{Z^2}{n^2} \Rightarrow \frac{E_5}{E_1} = \left(\frac{1}{5} \right)^2 = \frac{1}{25}$$

$$\text{Eqn (i) / (ii)} \Rightarrow \frac{h\nu_L}{h\nu_f} = \frac{E_1}{E_5}$$

$$\Rightarrow \frac{\nu_L}{\nu_f} = \frac{25}{1} \Rightarrow \nu_f = \frac{\nu_L}{25}$$

22. (d) de-Broglie wavelength, $\lambda = \frac{h}{p}$

$$\frac{\lambda_B}{\lambda_G} = \frac{P_a}{P_B} = \frac{mv_G}{mv_B}$$

$$\text{Speed of electron } v \propto \frac{Z}{n}$$

$$\text{so } \frac{\lambda_B}{\lambda_G} = \frac{n_B}{n_G} = \frac{3}{1} \Rightarrow \lambda_B = 3\lambda_G$$

23. (b) Energy required to remove e^- from singly ionized

$$\text{helium atom} = \frac{(13.6)Z^2}{1^2} = 54.4 \text{ eV} \quad (\because Z=2)$$

Energy required to remove e^- from helium atom = $x \text{ eV}$

According to question, $54.4 \text{ eV} = 2.2x \Rightarrow x = 24.73 \text{ eV}$

Therefore, energy required to ionize helium atom

$$= (54.4 + 24.73) \text{ eV} = 79.12 \text{ eV}$$

24. (d) (A) Radius of muon = $\frac{\text{Radius of hydrogen}}{200}$

$$\text{Radius of H atom} = r = \frac{\epsilon_0 n^2 h^2}{\pi m e^2}$$

$$\text{Radius of muon} = r_\mu = \frac{\epsilon_0 n^2 h^2}{\pi \times 200 m e^2}$$

$$r_\mu = \frac{r}{200}$$

(B) Velocity relation given is wrong

(C) Ionization energy in e^- H atom

$$E = \frac{+me^4}{8\epsilon_0^2 n^2 h^2}$$

$$E_\mu = \frac{200me^4}{8\epsilon_0^2 n^2 h^2} = 200E$$

(D) Momentum of H-atom

$$mvr = \frac{nh}{2\pi}$$

Hence (A), (C), (D) are correct.

25. (b) From energy level diagram, using $\Delta E = \frac{hc}{\lambda}$

$$\text{For wavelength } \lambda_1, \Delta E = -E - (-2E) = \frac{hc}{\lambda_1}$$

$$\therefore \lambda_1 = \frac{hc}{E}$$

$$\text{For wavelength } \lambda_2, \Delta E = -E - \left(-\frac{4E}{3}\right) = \frac{hc}{\lambda_2}$$

$$\therefore \lambda_2 = \frac{hc}{\left(\frac{E}{3}\right)} \quad \therefore r = \frac{\lambda_1}{\lambda_2} = \frac{1}{3}$$

26. (c) Speed of electron in first orbit ($n = 1$) of hydrogen atom ($z = 1$),

$$v = \frac{e^2}{2\epsilon_0 h}$$

radius of Bohr's first orbit,

$$r = \frac{h^2 \epsilon_0}{\pi m e^2} \Rightarrow \epsilon_0 = \frac{r \pi m e^2}{h^2} \quad \dots\dots (i)$$

Acceleration of electron,

$$\frac{v^2}{r} = \frac{e^4}{4\epsilon_0^2 h^2} \times \frac{\pi m e^2}{h^2 \epsilon_0}$$

$$= \frac{e^4 \times \pi m e^2}{4h^4 \epsilon_0^3} \quad \dots\dots (ii)$$

eliminating ϵ_0 from eq (ii),

$$= \frac{e^4 \pi m e^2 h^6}{4h^4 r^3 \pi^3 m^3 e^6} \quad \text{from eq}^n (i)$$

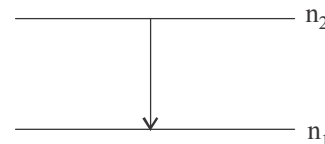
$$= \frac{h^2}{4\pi^2 m^2 r^3}$$

27. (d) Magnetic field at the centre of nucleus of H-atom,

$$B = \frac{\mu_0 I}{2r} \quad \dots\dots (i)$$

According to Bohr's model, radius of orbit $r \propto n^2$ from eq. (i) we can also write as $B \propto n^{-2}$

28. (b) A hydrogen atom makes a transition from $n = 2$ to $n = 1$



$$\text{Then wavelength} = Rc z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] = Rc(1)^2 \left[1 - \frac{1}{4} \right]$$

$$\lambda = Rc \left[\frac{3}{4} \right] \quad \dots(1)$$

For ionized lithium

$$\lambda = Rc(3)^2 \left[\frac{1}{n^2} \right] = Rc9 \left[\frac{1}{n^2} \right] \quad \dots(2)$$

$$Rc \left[\frac{3}{4} \right] = Rc9 \left[\frac{1}{n^2} \right]$$

$$\Rightarrow \frac{3}{4} = \frac{9}{n^2} \Rightarrow n = \sqrt{12} = 2\sqrt{3}$$

$\therefore$ The least quantum number must be 4.

29. (c) Kinetic energy of electron is

$$\text{K.E.} \propto \left(\frac{Z}{N} \right)^2$$

When the electron makes transition from excited state to ground state, then n increases and kinetic energy increases.

Total energy = - KE

$\therefore$ Total energy also decreases.

Potential energy is lowest for ground state.

30. (b) De-Broglie wavelength of electron $\lambda = \frac{h}{mV}$

$$\text{As we know, } V \propto \frac{1}{n}$$

So, $\lambda \propto n$

$$\lambda_4 = 4\lambda_1$$

λ_1 is the de-Broglie wavelength of the electron in the ground state.

31. (c) $qVB = \frac{mv^2}{r} \quad \dots\dots (i)$

$$\frac{nh}{2\pi} = mvr \quad \dots\dots (ii)$$

Multiplying equation (i) and (ii),

$$\frac{qBnh}{2\pi} = m^2 v^2$$

Now multiplying both sides by $\frac{1}{2m}$,

$$n \frac{qBh}{4\pi m} = \frac{1}{2} mv^2$$

$$\text{i.e. KE} = n \left[\frac{qBh}{4\pi m} \right]$$

32. (b) Radius of circular path followed by electron is given by,

$$r = \frac{mv}{qB} = \frac{\sqrt{2meV}}{eB} = \frac{1}{B} \sqrt{\frac{2m}{e}} V$$

$$\Rightarrow V = \frac{B^2 r^2 e}{2m} = 0.8V$$

For transition between 3 to 2.

$$E = 13.6 \left(\frac{1}{4} - \frac{1}{9} \right) = \frac{13.6 \times 5}{36} = 1.88 \text{ eV}$$

$$\text{Work function} = 1.88 \text{ eV} - 0.8 \text{ eV} = 1.08 \text{ eV} \approx 1.1 \text{ eV}$$

33. (c) Wave number $\frac{1}{\lambda} = RZ^2 \left[\frac{1}{n_1^2} - \frac{1}{n^2} \right]$

$$\Rightarrow \lambda \propto \frac{1}{Z^2}$$

$$\therefore \lambda Z^2 = \text{constant}$$

By question $n = 1$ and $n_1 = 2$

$$\text{Then, } \lambda_1 = \lambda_2 = 4\lambda_3 = 9\lambda_4$$

34. (b)
- (1) Davisson and Gemmer experiment-wave nature of electrons.
 - (2) Millikan's oil drop experiment - charge of an electron.
 - (3) Rutherford experiment - Existence of nucleus.
 - (4) Frank-Hertz experiment - Quantisation of energy levels.

35. (b) For first excited state, $n = 2$ and for Li^{++} $Z = 3$

$$E_n = \frac{13.6}{n^2} \times Z^2 = \frac{13.6}{4} \times 9 = 30.6 \text{ eV}$$

36. (d) $\Delta E = h\nu$

$$\nu = \frac{\Delta E}{h} = k \left[\frac{1}{(n-1)^2} - \frac{1}{n^2} \right] = \frac{k(2n-1)}{n^2(n-1)^2}$$

$$\approx \frac{2k}{n^3} \quad \text{or} \quad \nu \propto \frac{1}{n^3}$$

37. (a) $E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{E} = \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{12.5 \times 1.6 \times 10^{-19}} = 993 \text{ \AA}$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

(where Rydberg constant, $R = 1.097 \times 10^7$)

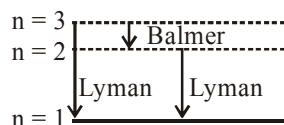
$$\text{or, } \frac{1}{993 \times 10^{-10}} = 1.097 \times 10^7 \left(\frac{1}{1^2} - \frac{1}{n_2^2} \right)$$

Solving we get $n_2 = 3$

Spectral lines

Total number of spectral lines = 3

Two lines in Lyman series for $n_1 = 1, n_2 = 2$ and $n_1 = 1, n_2 = 3$ and one in Balmer series for $n_1 = 2, n_2 = 3$



38. (c) As $F = \frac{mv^2}{r} = \frac{e^2}{4\pi\epsilon_0} \left(\frac{1}{r^2} + \frac{\beta}{r^3} \right)$

$$\text{and } mvr = \frac{nh}{2\pi} \Rightarrow v = \frac{nh}{2\pi mr}$$

$$\therefore m \left(\frac{nh}{2\pi mr} \right)^2 \times \frac{1}{r} = \frac{e^2}{4\pi\epsilon_0} \left(\frac{1}{r^2} + \frac{\beta}{r^3} \right)$$

$$\text{or, } \frac{1}{r^2} + \frac{\beta}{r^3} = \frac{mn^2 h^2 4\pi\epsilon_0}{4\pi^2 m^2 e^2 r^3}$$

$$\text{or, } \frac{a_0 n^2}{r^3} = \frac{1}{r^2} + \frac{\beta}{r^3} \quad \left(\because a_0 = \frac{\epsilon_0 h^2}{m\pi e^2} \text{ Given} \right)$$

For n^{th} atom

$$\therefore r_n = a_0 n^2 - \beta$$

39. (c) According to the question,

$$2\pi r = n\lambda = \frac{nh}{p} = \frac{nh}{mv}$$

$$\text{or } mvr = \frac{nh}{2\pi} \quad \text{or } mv = \frac{nh}{2\pi r}$$

$$F = qv_B = \frac{mv^2}{r} \quad \text{or, } q_B = \frac{mv}{r} = \frac{nh}{2\pi r}$$

$$\text{or, } r^2 = \frac{nh}{2\pi q_B} \quad \text{or, } r = \sqrt{\frac{nh}{2\pi q_B}}$$

$$\text{i.e., } r \propto n^{1/2}$$

40. (c) Magnetic moment of the hydrogen atom, when the electron is in n^{th} excited state, i.e., $n' = (n+1)$

$$\text{As magnetic moment } M_n = I_n A = i_n (\pi r_n^2)$$

$$i_n = eV_n = \frac{mz^2e^5}{4\epsilon_0 n^3 h^3}$$

$$r_n = \frac{n^2 h^2}{4\pi^2 k m e^2} \left(k = \frac{1}{4\pi\epsilon_0} \right)$$

Solving we get magnetic moment of the hydrogen atom for n^{th} excited state

$$M_n = \left(\frac{e}{2m} \right) \frac{nh}{2\pi}$$

41. (d) For ground state, the principal quantum no. (n) = 1. Principal quantum number 4 belongs to 3rd excited state. The possible number of the spectral lines from a state n to ground state is

$$= \frac{n(n-1)}{2} = \frac{4(4-1)}{2} = 6$$

42. (d) The energy of the system of two atoms of diatomic

$$\text{molecule } E = \frac{1}{2} I \omega^2$$

where I = moment of inertia

$$\omega = \text{Angular velocity} = \frac{L}{I},$$

L = Angular momentum

$$I = \frac{1}{2} (m_1 r_1^2 + m_2 r_2^2)$$

$$\text{Thus, } E = \frac{1}{2} (m_1 r_1^2 + m_2 r_2^2) \omega^2 \quad \dots (i)$$

$$E = \frac{1}{2} (m_1 r_1^2 + m_2 r_2^2) \frac{L^2}{I^2}$$

$$L = n \hbar$$

(According to Bohr's Hypothesis)

$$E = \frac{1}{2} (m_1 r_1^2 + m_2 r_2^2) \frac{L^2}{(m_1 r_1^2 + m_2 r_2^2)^2}$$

$$E = \frac{1}{2} \frac{L^2}{(m_1 r_1^2 + m_2 r_2^2)} = \frac{n^2 h^2}{8\pi^2 (m_1 r_1^2 + m_2 r_2^2)}$$

$$E = \frac{(m_1 + m_2) n^2 h^2}{8\pi^2 r^2 m_1 m_2}$$

43. (a) Velocity of electron in n^{th} orbit of hydrogen atom is given by :

$$V_n = \frac{2\pi K Z e^2}{nh}$$

Substituting the values we get,

$$V_n = \frac{2.2 \times 10^6}{n} \text{ m/s} \quad \text{or} \quad V_n \propto \frac{1}{n}$$

As principal quantum number increases, velocity decreases.

44. (c) $\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$ where R = Rydberg constant

$$\frac{1}{\lambda_{32}} = \left(\frac{1}{4} - \frac{1}{9} \right) = \frac{5}{36} \quad \Rightarrow \quad \lambda_{32} = \frac{36}{5}$$

Similarly solving for λ_{31} and λ_{21}

$$\lambda_{31} = \frac{9}{8} \quad \text{and} \quad \lambda_{21} = \frac{4}{3}$$

$$\therefore \frac{\lambda_{32}}{\lambda_{31}} = 6.4 \quad \text{and} \quad \frac{\lambda_{21}}{\lambda_{31}} \approx 1.2$$

45. (b) $E = \frac{hc}{\lambda}$

46. (b) If $n_1 = n$ and $n_2 = n + 1$

$$\text{Maximum wavelength } \lambda_{\text{max}} = \frac{n^2 (n+1)^2}{(2n+1) R}$$

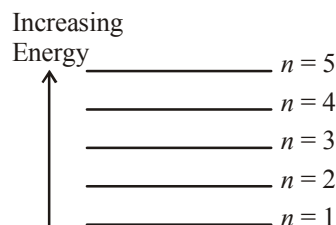
Therefore, for large n , $\lambda_{\text{max}} \propto n^3$

47. (b) Energy of excitation (ΔE) is

$$\Delta E = 13.6 Z^2 \left(\frac{1}{n_1} - \frac{1}{n_2} \right) eV$$

$$\Rightarrow \Delta E = 13.6 (3)^2 \left(\frac{1}{1^2} - \frac{1}{3^2} \right) = 108.8 eV$$

48. (c) It is given that transition from the state $n = 4$ to $n = 3$ in a hydrogen like atom result in ultraviolet radiation. For infrared radiation $\left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$ should be less. The only option is $5 \rightarrow 4$.



49. (b) Given,

$$\text{Centripetal force} = \frac{k}{r}$$

Then

$$\frac{k}{r} = \frac{mv^2}{r}$$

$$\Rightarrow k = mv^2 \quad \Rightarrow Tn = \frac{1}{2}mv^2 = \frac{1}{2}k$$

Tn is independent of n

Also,

$$\text{Angular momentum, } L = \frac{nh}{2\pi}$$

$$\Rightarrow mvr_n = \frac{nh}{2\pi} \quad (\because L = mvr)$$

$$\Rightarrow r_n = \frac{nh}{2\pi\sqrt{km}} \quad \left(\because m^2v^2 = km \right) \\ \left(\text{or } mv = \sqrt{km} \right)$$

Clearly, $r_n \propto n$

50. (d) We have to find the frequency of emitted photons. For emission of photons electron should make a transition from higher energy level to lower energy level. so, option (a) and (b) are incorrect.

Frequency of emitted photon is given by

$$h\nu = -13.6 \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right)$$

For transition from $n = 6$ to $n = 2$,

$$\nu_1 = \frac{-13.6}{h} \left(\frac{1}{6^2} - \frac{1}{2^2} \right) = \frac{2}{9} \times \left(\frac{13.6}{h} \right)$$

For transition from $n = 2$ to $n = 1$,

$$\nu_2 = \frac{-13.6}{h} \left(\frac{1}{2^2} - \frac{1}{1^2} \right) = \frac{3}{4} \times \left(\frac{13.6}{h} \right)$$

$$\therefore \nu_1 < \nu_2$$

51. (b) Energy of radiation that corresponds to energy difference between two energy levels n_1 and n_2 is given as

$$E = Rhc \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \quad \therefore E \propto \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$E \text{ will be maximum for the transition for which } \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

is maximum. Here n_2 is the higher energy level.

$$\text{Clearly, } \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \text{ is maximum for the third transition,}$$

i.e., $2 \rightarrow 1$. I transition is showing the absorption of energy.

52. (c) The wavelength of spectrum is given by

$$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\text{where } R = \frac{1.097 \times 10^7}{1 + \frac{m}{M}}$$

where m = mass of electron

M = mass of nucleus.

Thus, wavelength involved in the spectrum of hydrogen like atom depends upon masses of nucleus. The mass number of hydrogen and deuterium is 1 and 2 respectively, so spectrum of deuterium will be different from hydrogen.

53. (c) The energy required to remove the electron from the n^{th} orbit of hydrogen is given by

$$E_n = \frac{13.6}{n^2} \text{ eV/atom}$$

$$\text{For } n = 2, E_n = \frac{13.6}{4} = 3.4 \text{ eV}$$

Therefore the energy required to remove electron from $n = 2$ is + 3.4 eV.

Nuclei


TOPIC 1 Composition and Size of the Nuclei


- The radius R of a nucleus of mass number A can be estimated by the formula $R = (1.3 \times 10^{-15}) A^{1/3}$ m. It follows that the mass density of a nucleus is of the order of :
 $(M_{\text{prot.}} \cong M_{\text{neut.}} \approx 1.67 \times 10^{-27} \text{ kg})$ [Sep. 03, 2020 (II)]
 (a) 10^3 kg m^{-3} (b) $10^{10} \text{ kg m}^{-3}$
 (c) $10^{24} \text{ kg m}^{-3}$ (d) $10^{17} \text{ kg m}^{-3}$
- The ratio of the mass densities of nuclei of ^{40}Ca and ^{16}O is close to : [8 April 2019 II]
 (a) 1 (b) 0.1 (c) 5 (d) 2
- An unstable heavy nucleus at rest breaks into two nuclei which move away with velocities in the ratio of 8:27. The ratio of the radii of the nuclei (assumed to be spherical) is: [Online April 15, 2018]
 (a) 8:27 (b) 2:3 (c) 3:2 (d) 4:9
- Which of the following are the constituents of the nucleus? [2007]
 (a) Electrons and protons (b) Neutrons and protons
 (c) Electrons and neutrons (d) Neutrons and positrons
- If radius of the $^{27}_{13}\text{Al}$ nucleus is estimated to be 3.6 fermi then the radius of $^{125}_{52}\text{Te}$ nucleus be nearly [2005]
 (a) 8 fermi (b) 6 fermi
 (c) 5 fermi (d) 4 fermi

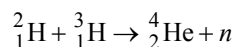
TOPIC 2 Mass-Energy Equivalence and Nuclear Reactions


- 4.5×10^5
 - 8×10^6
 - 6.82×10^5
 - 1.33×10^6
- Given the masses of various atomic particles $m_p = 1.0072 \text{ u}$, $m_n = 1.0087 \text{ u}$, $m_e = 0.000548 \text{ u}$, $m_{\bar{\nu}} = 0$, $m_d = 2.0141 \text{ u}$, where $p \equiv$ proton, $n \equiv$ neutron, $e \equiv$ electron, $\bar{\nu} \equiv$ antineutrino and $d \equiv$ deuteron. Which of the following process is allowed by momentum and energy conservation? [Sep. 06, 2020 (II)]
 - $n + n \rightarrow$ deuterium atom (electron bound to the nucleus)
 - $p \rightarrow n + e^+ + \bar{\nu}$
 - $n + p \rightarrow d + \gamma$
 - $e^+ + e^- \rightarrow \gamma$
- Find the Binding energy per nucleon for $^{120}_{50}\text{Sn}$. Mass of proton $m_p = 1.00783 \text{ U}$, mass of neutron $m_n = 1.00867 \text{ U}$ and mass of tin nucleus $m_{\text{Sn}} = 119.902199 \text{ U}$. (take $1\text{U} = 931 \text{ MeV}$) [Sep. 04, 2020 (II)]
 - 7.5 MeV
 - 9.0 MeV
 - 8.0 MeV
 - 8.5 MeV
- In a reactor, 2 kg of $^{235}_{92}\text{U}$ fuel is fully used up in 30 days. The energy released per fission is 200 MeV. Given that the Avogadro number, $N = 6.023 \times 10^{26}$ per kilo mole and $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$. The power output of the reactor is close to: [Sep. 02, 2020 (I)]
 - 35 MW
 - 60 MW
 - 125 MW
 - 54 MW
- Consider the nuclear fission
 $\text{Ne}^{20} \rightarrow 2\text{He}^4 + \text{C}^{12}$
 Given that the binding energy/nucleon of Ne^{20} , He^4 and C^{12} are, respectively, 8.03 MeV, 7.07 MeV and 7.86 MeV, identify the correct statement: [10 Jan. 2019 II]
 - energy of 12.4 MeV will be supplied
 - 8.3 MeV energy will be released
 - energy of 3.6 MeV will be released
 - energy of 11.9 MeV has to be supplied
- You are given that mass of $^7_3\text{Li} = 7.0160 \text{ u}$,
 Mass of $^4_2\text{He} = 4.0026 \text{ u}$
 and Mass of $^1_1\text{H} = 1.0079 \text{ u}$.
 When 20 g of ^7_3Li is converted into ^4_2He by proton capture, the energy liberated, (in kWh), is :
 [Mass of nucleon = $1 \text{ GeV}/c^2$] [Sep. 06, 2020 (I)]

11. Imagine that a reactor converts all given mass into energy and that it operates at a power level of 10^9 watt. The mass of the fuel consumed per hour in the reactor will be : (velocity of light, c is 3×10^8 m/s) [Online April 9, 2017]
 (a) 0.96 gm (b) 0.8 gm
 (c) 4×10^{-2} gm (d) 6.6×10^{-5} gm
12. Two deuterons undergo nuclear fusion to form a Helium nucleus. Energy released in this process is : (given binding energy per nucleon for deuteron = 1.1 MeV and for helium = 7.0 MeV) [Online April 8, 2017]
 (a) 30.2 MeV (b) 32.4 MeV
 (c) 23.6 MeV (d) 25.8 MeV
13. When Uranium is bombarded with neutrons, it undergoes fission. The fission reaction can be written as :
 ${}_{92}\text{U}^{235} + {}_0n^1 \rightarrow {}_{56}\text{Ba}^{141} + {}_{36}\text{Kr}^{92} + 3x + Q(\text{energy})$
 where three particles named x are produced and energy Q is released. What is the name of the particle x ? [Online April 9, 2013]
 (a) electron (b) α -particle
 (c) neutron (d) neutrino
14. Assume that a neutron breaks into a proton and an electron. The energy released during this process is : (mass of neutron = 1.6725×10^{-27} kg, mass of proton = 1.6725×10^{-27} kg, mass of electron = 9×10^{-31} kg). [2012]
 (a) 0.51 MeV (b) 7.10 MeV
 (c) 6.30 MeV (d) 5.4 MeV
15. Ionisation energy of Li (Lithium) atom in ground state is 5.4 eV. Binding energy of an electron in Li^+ ion in ground state is 75.6 eV. Energy required to remove all three electrons of Lithium (Li) atom is [Online May 19, 2012]
 (a) 81.0 eV (b) 135.4 eV
 (c) 203.4 eV (d) 156.6 eV
16. After absorbing a slowly moving neutron of mass m_N (momentum ≈ 0) a nucleus of mass M breaks into two nuclei of masses m_1 and $5m_1$ ($6m_1 = M + m_N$) respectively. If the de Broglie wavelength of the nucleus with mass m_1 is λ , the de Broglie wavelength of the nucleus will be [2011]
 (a) 5λ (b) $\lambda/5$ (c) λ (d) 25λ
- DIRECTIONS:** Questions number 17-18 are based on the following paragraph.
 A nucleus of mass $M + \Delta m$ is at rest and decays into two daughter nuclei of equal mass $\frac{M}{2}$ each. Speed of light is c . [2010]
17. The binding energy per nucleon for the parent nucleus is E_1 and that for the daughter nuclei is E_2 . Then
 (a) $E_2 = 2E_1$ (b) $E_1 > E_2$
 (c) $E_2 > E_1$ (d) $E_1 = 2E_2$
18. The speed of daughter nuclei is
 (a) $c \frac{\Delta m}{M + \Delta m}$ (b) $c \sqrt{\frac{2\Delta m}{M}}$
 (c) $c \sqrt{\frac{\Delta m}{M}}$ (d) $c \sqrt{\frac{\Delta m}{M + \Delta m}}$
19. **Statement-1:** Energy is released when heavy nuclei undergo fission or light nuclei undergo fusion and
Statement-2 : For heavy nuclei, binding energy per nucleon increases with increasing Z while for light nuclei it decreases with increasing Z . [2008]
 (a) Statement-1 is false, Statement-2 is true
 (b) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1
 (c) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1
 (d) Statement-1 is true, Statement-2 is false
20. If M_O is the mass of an oxygen isotope ${}^8\text{O}^{17}$, M_P and M_N are the masses of a proton and a neutron respectively, the nuclear binding energy of the isotope is [2007]
 (a) $(M_O - 17M_N)c^2$ (b) $(M_O - 8M_P)c^2$
 (c) $(M_O - 8M_P - 9M_N)c^2$ (d) M_Oc^2
21. When ${}^7_3\text{Li}$ nuclei are bombarded by protons, and the resultant nuclei are ${}^8_4\text{Be}$, the emitted particles will be [2006]
 (a) alpha particles (b) beta particles
 (c) gamma photons (d) neutrons
22. If the binding energy per nucleon in ${}^7_3\text{Li}$ and ${}^4_2\text{He}$ nuclei are 5.60 MeV and 7.06 MeV respectively, then in the reaction

$$p + {}^7_3\text{Li} \longrightarrow 2 {}^4_2\text{He}$$
 energy of proton must be [2006]
 (a) 28.24 MeV (b) 17.28 MeV
 (c) 1.46 MeV (d) 39.2 MeV
23. A nuclear transformation is denoted by $X(n, \alpha) {}^7_3\text{Li}$. Which of the following is the nucleus of element X ? [2005]
 (a) ${}^{10}_5\text{B}$ (b) ${}^{12}_6\text{C}$ (c) ${}^{11}_4\text{Be}$ (d) ${}^9_5\text{B}$
24. A nucleus disintegrated into two nuclear parts which have their velocities in the ratio of 2 : 1. The ratio of their nuclear sizes will be [2004]
 (a) $3^{1/2} : 1$ (b) $1 : 2^{1/3}$ (c) $2^{1/3} : 1$ (d) $1 : 3^{1/2}$
25. The binding energy per nucleon of deuteron (${}^2_1\text{H}$) and helium nucleus (${}^4_2\text{He}$) is 1.1 MeV and 7 MeV respectively. If two deuteron nuclei react to form a single helium nucleus, then the energy released is [2004]
 (a) 23.6 MeV (b) 26.9 MeV
 (c) 13.9 MeV (d) 19.2 MeV
26. When a ${}^{238}\text{U}$ nucleus originally at rest, decays by emitting an alpha particle having a speed ' u ', the recoil speed of the residual nucleus is [2003]
 (a) $\frac{4u}{238}$ (b) $-\frac{4u}{234}$
 (c) $\frac{4u}{234}$ (d) $-\frac{4u}{238}$

27. In the nuclear fusion reaction



given that the repulsive potential energy between the two nuclei is $\sim 7.7 \times 10^{-14} \text{ J}$, the temperature at which the gases must be heated to initiate the reaction is nearly

[Boltzmann's Constant $k = 1.38 \times 10^{-23} \text{ J/K}$] [2003]

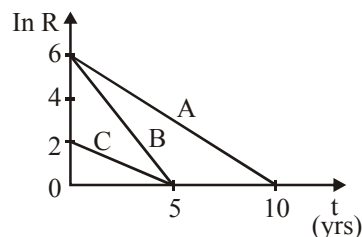
- (a) 10^7 K (b) 10^5 K (c) 10^3 K (d) 10^9 K

TOPIC 3 Radioactivity



28. Activities of three radioactive substances A, B and C are represented by the curves A, B and C, in the figure. Then their half-lives $T_1(A) : T_1(B) : T_1(C)$ are in the ratio :

$\frac{2}{2} : \frac{2}{2} : \frac{2}{2}$ [Sep. 05, 2020 (I)]



- (a) 2 : 1 : 1 (b) 3 : 2 : 1
(c) 2 : 1 : 3 (d) 4 : 3 : 1
29. A radioactive nucleus decays by two different processes. The half life for the first process is 10 s and that for the second is 100 s. The effective half life of the nucleus is close to : [Sep. 05, 2020 (II)]
(a) 9 sec. (b) 6 sec.
(c) 55 sec. (d) 12 sec.
30. In a radioactive material, fraction of active material remaining after time t is $9/16$. The fraction that was remaining after $t/2$ is : [Sep. 03, 2020 (I)]
(a) $\frac{4}{5}$ (b) $\frac{3}{5}$ (c) $\frac{3}{4}$ (d) $\frac{7}{8}$
31. The activity of a radioactive sample falls from 700 s^{-1} to 500 s^{-1} in 30 minutes. Its half life is close to : [7 Jan. 2020, II]
(a) 72 min (b) 62 min
(c) 66 min (d) 52 min
32. Two radioactive materials A and B have decay constants 10λ and λ , respectively. If initially they have the same number of nuclei, then the ratio of the number of nuclei of A to that of B will be $1/e$ after a time : [10 April 2019, I]

- (a) $\frac{1}{9\lambda}$ (b) $\frac{1}{11\lambda}$ (c) $\frac{11}{10\lambda}$ (d) $\frac{1}{10\lambda}$

33. Two radioactive substances A and B have decay constants 5λ and λ respectively. At $t = 0$, a sample has the same number of the two nuclei. The time taken for the

ratio of the number of nuclei to become $\left(\frac{1}{e}\right)^2$ will be :

[10 April 2019, II]

- (a) $1/2\lambda$ (b) $1/4\lambda$ (c) $1/\lambda$ (d) $2/\lambda$
34. In a radioactive decay chain, the initial nucleus is ${}^{232}_{90}\text{Th}$. At the end there are 6 α -particles and 4 β -particles which are emitted. If the end nucleus is ${}^A_Z\text{X}$, A and Z are given by :

[12 Jan. 2019, II]

- (a) $A = 208 ; Z = 80$ (b) $A = 202 ; Z = 80$
(c) $A = 208 ; Z = 82$ (d) $A = 200 ; Z = 81$
35. Using a nuclear counter the count rate of emitted particles from a radioactive source is measured. At $t = 0$ it was 1600 counts per second and $t = 8$ seconds it was 100 counts per second. The count rate observed, as counts per second, at $t = 6$ seconds is close to:

[10 Jan. 2019 I]

- (a) 200 (b) 150 (c) 400 (d) 360
36. A sample of radioactive material A, that has an activity of 10 mCi ($1 \text{ Ci} = 3.7 \times 10^{10} \text{ decays/s}$), has twice the number of nuclei as another sample of a different radioactive material B which has an activity of 20 mCi. The correct choices for half-lives of A and B would then be respectively: [9 Jan. 2019 I]
(a) 5 days and 10 days (b) 10 days and 40 days
(c) 20 days and 5 days (d) 20 days and 10 days
37. At a given instant, say $t = 0$, two radioactive substances

A and B have equal activities. The ratio $\frac{R_B}{R_A}$ of their activities after time t itself decays with time t as e^{-3t} . If the half-life of A is $\ln 2$, the half-life of B is:

[9 Jan. 2019, II]

- (a) $4\ln 2$ (b) $\frac{\ln 2}{2}$ (c) $\frac{\ln 2}{4}$ (d) $2\ln 2$
38. At some instant, a radioactive sample S_1 having an activity $5\mu\text{Ci}$ has twice the number of nuclei as another sample S_2 which has an activity of $10\mu\text{Ci}$. The half lives of S_1 and S_2 are [Online April 16, 2018]
(a) 10 years and 20 years, respectively
(b) 5 years and 20 years, respectively
(c) 20 years and 10 years, respectively
(d) 20 years and 5 years, respectively
39. A solution containing active cobalt ${}^{60}_{27}\text{Co}$ having activity of $0.8 \mu\text{Ci}$ and decay constant λ is injected in an animal's body. If 1 cm^3 of blood is drawn from the animal's body after 10 hrs of injection, the activity found was 300 decays per minute. What is the volume of blood that is flowing in the body? ($1 \text{ Ci} = 3.7 \times 10^{10} \text{ decay per second}$ and at $t = 10 \text{ hrs } e^{-\lambda t} = 0.84$) [Online April 15, 2018]
(a) 6 litres (b) 7 litres (c) 4 litres (d) 5 litres

40. A radioactive nucleus A with a half life T , decays into a nucleus B. At $t = 0$, there is no nucleus B. At sometime t , the ratio of the number of B to that of A is 0.3. Then, t is given by [2017]
- (a) $t = T \log(1.3)$ (b) $t = \frac{T}{\log(1.3)}$
 (c) $t = T \frac{\log 2}{\log 1.3}$ (d) $t = \frac{\log 1.3}{\log 2}$
41. Half-lives of two radioactive elements A and B are 20 minutes and 40 minutes, respectively. Initially, the samples have equal number of nuclei. After 80 minutes, the ratio of decayed number of A and B nuclei will be : [2016]
 (a) 1 : 4 (b) 5 : 4 (c) 1 : 16 (d) 4 : 1
42. Let N_β be the number of β particles emitted by 1 gram of Na^{24} radioactive nuclei (half life = 15 hrs) in 7.5 hours, N_β is close to (Avogadro number = 6.023×10^{23} /g. mole): [Online April 11, 2015]
 (a) 6.2×10^{21} (b) 7.5×10^{21}
 (c) 1.25×10^{22} (d) 1.75×10^{22}
43. A piece of wood from a recently cut tree shows 20 decays per minute. A wooden piece of same size placed in a museum (obtained from a tree cut many years back) shows 2 decays per minute. If half life of C^{14} is 5730 years, then age of the wooden piece placed in the museum is approximately: [Online April 19, 2014]
 (a) 10439 years (b) 13094 years
 (c) 19039 years (d) 39049 years
44. A piece of bone of an animal from a ruin is found to have ^{14}C activity of 12 disintegrations per minute per gm of its carbon content. The ^{14}C activity of a living animal is 16 disintegrations per minute per gm. How long ago nearly did the animal die? (Given half life of ^{14}C is $t_{1/2} = 5760$ years) [Online April 12, 2014]
 (a) 1672 years (b) 2391 years
 (c) 3291 years (d) 4453 years
45. A radioactive nuclei with decay constant $0.5/\text{s}$ is being produced at a constant rate of 100 nuclei/s. If at $t = 0$ there were no nuclei, the time when there are 50 nuclei is: [Online April 11, 2014]
 (a) 1s (b) $2\ln\left(\frac{4}{3}\right)\text{s}$
 (c) $\ln 2$ s (d) $\ln\left(\frac{4}{3}\right)\text{s}$
46. The half-life of a radioactive element A is the same as the mean-life of another radioactive element B. Initially both substances have the same number of atoms, then : [Online April 22, 2013]
 (a) A and B decay at the same rate always.
 (b) A and B decay at the same rate initially.
 (c) A will decay at a faster rate than B.
 (d) B will decay at a faster rate than A.
47. The counting rate observed from a radioactive source at $t = 0$ was 1600 counts s^{-1} , and $t = 8$ s, it was 100 counts s^{-1} . The counting rate observed as counts s^{-1} at $t = 6$ s will be [Online May 26, 2012]
 (a) 250 (b) 400 (c) 300 (d) 200
48. The decay constants of a radioactive substance for α and β emission are λ_α and λ_β respectively. If the substance emits α and β simultaneously, then the average half life of the material will be [Online May 19, 2012]
 (a) $\frac{2T_\alpha T_\beta}{T_\alpha + T_\beta}$ (b) $T_\alpha + T_\beta$
 (c) $\frac{T_\alpha T_\beta}{T_\alpha + T_\beta}$ (d) $\frac{1}{2}(T_\alpha + T_\beta)$
49. Which of the following Statements is correct? [Online May 12, 2012]
 (a) The rate of radioactive decay cannot be controlled but that of nuclear fission can be controlled.
 (b) Nuclear forces are short range, attractive and charge dependent.
 (c) Nuclei of atoms having same number of neutrons are known as isobars.
 (d) Wavelength of matter waves is given by de Broglie formula but that of photons is not given by the same formula
50. A sample originally contained 10^{20} radioactive atoms, which emit α -particles. The ratio of α -particles emitted in the third year to that emitted during the second year is 0.3. How many α -particles were emitted in the first year? [Online May 7, 2012]
 (a) 3×10^{18} (b) 3×10^{19}
 (c) 5×10^{18} (d) 7×10^{19}
51. The half life of a radioactive substance is 20 minutes. The approximate time interval $(t_2 - t_1)$ between the time t_2 when $\frac{2}{3}$ of it had decayed and time t_1 when $\frac{1}{3}$ of it had decayed is : [2011]
 (a) 14min (b) 20min (c) 28min (d) 7min
52. **Statement - 1 :** A nucleus having energy E_1 decays by β^- emission to daughter nucleus having energy E_2 , but the β^- rays are emitted with a continuous energy spectrum having end point energy $E_1 - E_2$.
Statement - 2 : To conserve energy and momentum in β^- decay at least three particles must take part in the transformation. [2011 RS]
 (a) Statement-1 is correct but statement-2 is not correct.
 (b) Statement-1 and statement-2 both are correct and statement-2 is the correct explanation of statement-1.
 (c) Statement-1 is correct, statement-2 is correct and statement-2 is not the correct explanation of Statement-1
 (d) Statement-1 is incorrect, statement-2 is correct.

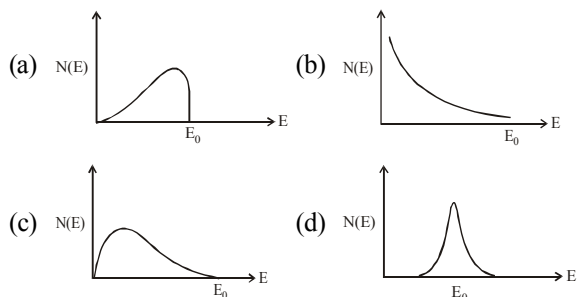
53. A radioactive nucleus (initial mass number A and atomic number Z) emits 3 α -particles and 2 positrons. The ratio of number of neutrons to that of protons in the final nucleus will be [2010]

(a) $\frac{A-Z-8}{Z-4}$ (b) $\frac{A-Z-4}{Z-8}$
 (c) $\frac{A-Z-12}{Z-4}$ (d) $\frac{A-Z-4}{Z-2}$

54. The half-life period of a radio-active element X is same as the mean life time of another radio-active element Y . Initially they have the same number of atoms. Then [2007]

- (a) X and Y decay at same rate always
 (b) X will decay faster than Y
 (c) Y will decay faster than X
 (d) X and Y have same decay rate initially

55. The energy spectrum of β -particles [number $N(E)$ as a function of β -energy E] emitted from a radioactive source is [2006]



56. Starting with a sample of pure ^{66}Cu , $\frac{7}{8}$ of it decays into Zn in 15 minutes. The corresponding half life is [2005]

- (a) 15 minutes (b) 10 minutes
 (c) $7\frac{1}{2}$ minutes (d) 5 minutes

57. The intensity of gamma radiation from a given source is I . On passing through 36 mm of lead, it is reduced to $\frac{I}{8}$. The thickness of lead which will reduce the intensity to $\frac{I}{2}$ will be [2005]

- (a) 9mm (b) 6mm
 (c) 12mm (d) 18mm

58. Which of the following **cannot** be emitted by radioactive substances during their decay? [2003]

- (a) Protons (b) Neutrinos
 (c) Helium nuclei (d) Electrons

59. A nucleus with $Z=92$ emits the following in a sequence:

$\alpha, \beta^-, \beta^-, \alpha, \alpha, \alpha, \alpha, \beta^-, \beta^-, \alpha, \beta^+, \beta^+, \alpha$

Then Z of the resulting nucleus is [2003]

- (a) 76 (b) 78 (c) 82 (d) 74

60. A radioactive sample at any instant has its disintegration rate 5000 disintegrations per minute. After 5 minutes, the rate is 1250 disintegrations per minute. Then, the decay constant (per minute) is [2003]

- (a) $0.4 \ln 2$ (b) $0.2 \ln 2$
 (c) $0.1 \ln 2$ (d) $0.8 \ln 2$

61. At a specific instant emission of radioactive compound is deflected in a magnetic field. The compound can emit

- (i) electrons (ii) protons
 (iii) He^{2+} (iv) neutrons

The emission at instant can be [2002]

- (a) i, ii, iii (b) i, ii, iii, iv
 (c) iv (d) ii, iii

62. If N_0 is the original mass of the substance of half-life period $t_{1/2} = 5$ years, then the amount of substance left after 15 years is [2002]

- (a) $N_0/8$ (b) $N_0/16$ (c) $N_0/2$ (d) $N_0/4$



Hints & Solutions



1. (d) Density of nucleus, $\rho = \frac{\text{Mass}}{\text{Volume}} = \frac{mA}{\frac{4}{3}\pi R^3}$

$$\Rightarrow \rho = \frac{mA}{\frac{4}{3}\pi(R_0 A^{1/3})^3} \quad (\because R = R_0 A^{1/3})$$

Here m = mass of a nucleon

$$\therefore \rho = \frac{3 \times 1.67 \times 10^{-27}}{4 \times 3.14 \times (1.3 \times 10^{-15})^3} \quad (\text{Given, } R_0 = 1.3 \times 10^{-15})$$

$$\Rightarrow \rho = 2.38 \times 10^{17} \text{ kg/m}^3$$

2. (a) Nuclear density is independent of atomic number.

3. (c) Let heavy nucleus breaks into two nuclei of mass m_1 and m_2 and move away with velocities V_1 and V_2 respectively.

According to question, $\frac{V_1}{V_2} = \frac{8}{27}$

$m_1 V_1 = m_2 V_2$ (Law of momentum conservation)

$$\Rightarrow \frac{m_1}{m_2} = \frac{V_2}{V_1} = \frac{27}{8}$$

$$\frac{\rho \times \frac{4}{3}\pi R_1^3}{\rho \times \frac{4}{3}\pi R_2^3} \quad \left(\because \text{density } \rho = \frac{\text{mass}}{\text{volume}} \right)$$

$$\Rightarrow \left(\frac{R_1}{R_2} \right) = \left(\frac{27}{8} \right)^{\frac{1}{3}} = \left(\frac{3}{2} \right)^{3 \times \frac{1}{3}} \quad \therefore \frac{R_1}{R_2} = \frac{3}{2}$$

4. (b)

5. (b) Radius of a nucleus,

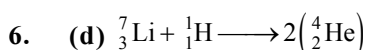
$$R = R_0(A)^{1/3}$$

Here, R_0 is a constant

A = atomic mass number

$$\therefore \frac{R_1}{R_2} = \left(\frac{A_1}{A_2} \right)^{1/3} = \left(\frac{27}{125} \right)^{1/3} = \frac{3}{5}$$

$$\Rightarrow R_2 = \frac{5}{3} \times 3.6 = 6 \text{ fermi}$$



$$\Delta m \rightarrow [m_{\text{Li}} + m_{\text{H}}] - 2[M_{\text{He}}]$$

$$\text{Energy released} = \Delta mc^2$$

$$\text{In use of 1 g Li energy released} = \frac{\Delta mc^2}{m_{\text{Li}}}$$

$$\text{In use of 20 g energy released} = \frac{\Delta mc^2}{m_{\text{Li}}} \times 20 \text{ g}$$

$$= \frac{[(7.016 + 1.0079) - 2 \times 4.0026]u \times c^2}{7.016 \times 1.6 \times 10^{-24}} \times 20 \text{ g}$$

$$= \left(\frac{0.0187 \times 1.6 \times 10^{-19} \times 10^9}{7.016 \times 1.6 \times 10^{-24}} \times 20 \right) = 480 \times 10^{10} \text{ J}$$

$$\therefore 1 \text{ J} = 2.778 \times 10^{-7} \text{ kWh}$$

$$\therefore \text{Energy released} = 480 \times 10^{10} \times 2.778 \times 10^{-7} = 1.33 \times 10^6 \text{ kWh}$$

7. (c) For the momentum and energy conservation, mass defect (Δm) should be positive. Since some energy is lost in every process.

$$(m_p + m_n) > m_d$$

8. (d) Mass defect,

$$\Delta m = (50m_p + 70m_n) - (m_{\text{sn}})$$

$$= (50 \times 1.00783 + 70 \times 1.008) - (119.902199) = 1.096$$

$$\text{Binding energy} = (\Delta m)C^2 = (\Delta m) \times 931 = 1020.56$$

$$\frac{\text{Binding energy}}{\text{Nucleon}} = \frac{1020.5631}{120} = 8.5 \text{ MeV}$$

9. (b) Power output of the reactor,

$$P = \frac{\text{energy}}{\text{time}}$$

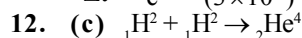
$$= \frac{2}{235} \times \frac{6.023 \times 10^{26} \times 200 \times 1.6 \times 10^{-19}}{30 \times 24 \times 60 \times 60} \approx 60 \text{ MW}$$

10. (d)

11. (c) Power level of reactor, $P = \frac{E}{\Delta t} = \frac{\Delta mc^2}{\Delta t}$

mass of the fuel consumed per hour in the reactor,

$$\frac{\Delta m}{\Delta t} = \frac{P}{c^2} = \frac{10^9}{(3 \times 10^8)^2} = 4 \times 10^{-2} \text{ gm}$$

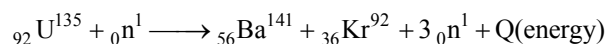


$$\text{Total binding energy of two deuterium nuclei} = 1.1 \times 4 = 4.4 \text{ MeV}$$

$$\text{Binding energy of a } ({}_2\text{He}^4) \text{ nuclei} = 4 \times 7 = 28 \text{ MeV}$$

$$\text{Energy released in this process} = 28 - 4.4 = 23.6 \text{ MeV}$$

13. (c) Nuclear fission equation



Hence particle x is neutron.

14. (a) ${}_0^1\text{n} \longrightarrow {}_1^1\text{H} + {}_{-1}^0\text{e} + \bar{\nu} + Q$

The mass defect during the process

$$\Delta m = m_n - m_H - m_e = 1.6725 \times 10^{-27} - (1.6725 \times 10^{-27} + 9 \times 10^{-31} \text{ kg})$$

$$= -9 \times 10^{-31} \text{ kg}$$

The energy released during the process

$$E = \Delta mc^2$$

$$E = 9 \times 10^{-31} \times 9 \times 10^{16} = 81 \times 10^{-15} \text{ Joules}$$

$$E = \frac{81 \times 10^{-15}}{1.6 \times 10^{-19}} = 0.511 \text{ MeV}$$

15. (d)

16. (c) Initial momentum of system, $p_i = 0$

Let p_1 and p_2 be the momentum of broken nuclei of masses m_1 and $5m_1$ respectively.

$$p_f = p_1 + p_2$$

From the conservation of momentum

$$p_i = p_f$$

$$0 = p_1 + p_2$$

$$p_1 = -p_2$$

From de Broglie relation, wavelength

$$\lambda_1 = \frac{h}{p_1} \text{ and } \lambda_2 = \frac{h}{p_2}$$

$$|\lambda_1| = |\lambda_2|$$

$$\lambda_1 = \lambda_2 = \lambda.$$

17. (c) In nuclear fission, the binding energy per nucleon of daughter nuclei is always greater than the parent nucleus.

18. (b) Mass defect, $\Delta M = \left[(M + \Delta m) - \left(\frac{M}{2} + \frac{M}{2} \right) \right]$

$$= [M + \Delta m - M] = \Delta m$$

$$\text{Energy released, } Q = \Delta Mc^2 = \Delta mc^2 \quad \dots(i)$$

From the law of conservation of momentum

$$(M + \Delta m) \times 0 = \frac{M}{2} v_1 - \frac{M}{2} \times v_2$$

$$\Rightarrow v_1 = v_2$$

$$\text{Now, } Q = \frac{1}{2} \left(\frac{M}{2} \right) v_1^2 + \frac{1}{2} \left(\frac{M}{2} \right) v_2^2 - \frac{1}{2}$$

$$(M + \Delta m) \times (0)^2$$

$$= \frac{M}{2} v_1^2 (\because v_1 = v_2) \quad \dots(ii)$$

From equation (i) and (ii), we get

$$\left(\frac{M}{2} \right) v_1^2 = \Delta mc^2$$

$$\Rightarrow v_1^2 = \frac{2\Delta mc^2}{M} \quad \Rightarrow v_1 = \sqrt{\frac{2\Delta m}{M}}$$

19. (d) We know that energy is released when heavy nuclei undergo fission or light nuclei undergo fusion. Therefore statement (1) is correct.

The second statement is false because for heavy nuclei the binding energy per nucleon decreases with increasing Z and for light nuclei, B.E/nucleon increases with increasing Z .

20. (c) Number of protons in oxygen isotope, $Z = 8$

$$\text{Number of neutrons} = 17 - 8 = 9$$

Binding energy

$$= [ZM_p + (A - Z)M_n - M]c^2$$

$$= [8M_p + (17 - 8)M_n - M]c^2$$

$$= [8M_p + 9M_n - M]c^2$$

$$= [8M_p + 9M_n - M_o]c^2$$

21. (c) ${}_3^7\text{Li} + {}_1^1\text{p} \longrightarrow {}_4^8\text{Be} + {}_0^0\gamma$

We see that both proton number and mass number are equal in both sides, so emitted particle should be massless gamma photons.

22. (b) Given,

$$\text{Binding energy per nucleon of } {}_3^7\text{Li} = 5.60 \text{ MeV}$$

$$\text{Binding energy per nucleon of } {}_2^4\text{He} = 7.06 \text{ MeV}$$

Let E be the energy of proton, then

$$E + 7 \times 5.6 = 2 \times [4 \times 7.06]$$

$$\Rightarrow E = 56.48 - 39.2 = 17.28 \text{ MeV}$$

23. (a) ${}_Z\text{X}^A + {}_0^1\text{n} \longrightarrow {}_3\text{Li}^7 + {}_2\text{He}^4$

Using conservation of mass number

$$A + 1 = 4 + 7$$

$$\Rightarrow A = 10$$

Using conservation of charge number

$$Z + 0 = 2 + 3 \quad \Rightarrow Z = 5$$

It is boron ${}_5\text{B}^{10}$

24. (b) Given :

$$\frac{v_1}{v_2} = \frac{2}{1}$$

From conservation of momentum $m_1 v_1 = m_2 v_2$

$$\Rightarrow \left(\frac{m_1}{m_2} \right) = \left(\frac{v_2}{v_1} \right) = \frac{1}{2}$$

We know that mass of nucleus, $m \propto A$

Nuclear size $R \propto A^{1/3} \propto m^{1/3}$

$$\frac{R_1}{R_2} = \left(\frac{m_1}{m_2} \right)^{1/3} \Rightarrow \frac{R_1^3}{R_2^3} = \frac{1}{2} \Rightarrow \left(\frac{R_1}{R_2} \right) = \left(\frac{1}{2} \right)^{1/3}$$

25. (a) The chemical reaction of process is $2^2_1\text{H} \rightarrow ^4_2\text{He}$

Binding energy of two deuterons,

$$4 \times 1.1 = 4.4 \text{ MeV}$$

Binding energy of helium nucleus = $4 \times 7 = 28 \text{ MeV}$

$$\text{Energy released} = 28 - 4.4 = 23.6 \text{ MeV}$$

26. (c) Mass of α particle, $m_\alpha = 4u$

Mass of nucleus after fission, $m_n = 234u$ From conservation of linear momentum we have

$$238 \times 0 = 4u + 234v$$

$$\therefore v = -\frac{4}{234}u$$

$$\therefore \text{Speed} = |\vec{v}| = \frac{4}{234}u$$

27. (d) The average kinetic energy per molecule at temperature T is

$$= \frac{3}{2}kT$$

Where k = Boltzmann's constant

This kinetic energy should be able to provide the repulsive potential energy

$$\therefore \frac{3}{2}kT = 7.7 \times 10^{-14}$$

$$\Rightarrow T = \frac{2 \times 7.7 \times 10^{-14}}{3 \times 1.38 \times 10^{-23}} = 3.7 \times 10^9 \text{ K}$$

28. (c) Since, $R = R_0 e^{-\lambda t}$

$$\ln R = \ln R_0 + (-\lambda \ln t)$$

$$\lambda = \frac{\ln 2}{t_{1/2}} = \text{Slope}$$

$$\lambda_A = \frac{6}{10} \Rightarrow T_A = \frac{10}{6} \ln 2$$

$$\lambda_B = \frac{6}{5} \Rightarrow T_B = \frac{5 \ln 2}{6}$$

$$\lambda_C = \frac{2}{5} \Rightarrow T_C = \frac{5 \ln 2}{6}$$

$$\therefore T_{1/2}^A : T_{1/2}^B : T_{1/2}^C = \frac{10}{6} : \frac{5}{6} : \frac{15}{6} = 2 : 1 : 3$$

29. (a) Let λ_1 and λ_2 be the decay constants of two process. N be the number of nuclei left undecayed after two process. From the law of radioactive decay we have

$$-\frac{dN}{dt} = \lambda_1 N + \lambda_2 N \quad \left[\because -\frac{dN}{dt} = \lambda N \right]$$

$$\Rightarrow -\frac{dN}{dt} = (\lambda_1 + \lambda_2)N$$

$$\Rightarrow \lambda_{\text{eq.}} = (\lambda_1 + \lambda_2)$$

$$\Rightarrow \frac{\ln 2}{T} = \frac{\ln 2}{T_1} + \frac{\ln 2}{T_2} \quad \left(\because \lambda = \frac{\ln 2}{T} \right)$$

$$\Rightarrow \frac{1}{T} = \frac{1}{T_1} + \frac{1}{T_2}$$

$$\Rightarrow \frac{1}{T} = \frac{1}{10} + \frac{1}{100} = \frac{11}{100} \quad [\text{Given: } T_1 = 10 \text{ s \& } T_2 = 100 \text{ s}]$$

$$\Rightarrow T = \frac{100}{11} = 9 \text{ sec.}$$

30. (c) As we know, for first order decay, $N(t) = N_0 e^{-\lambda t}$

According to question,

$$\frac{N(t)}{N_0} = \frac{9}{16} = e^{-\lambda t}$$

After time, $t/2$;

$$N(t/2) = N_0 e^{-\lambda(t/2)}$$

$$\frac{N(t/2)}{N_0} = \sqrt{e^{-\lambda t}} = \sqrt{\frac{9}{16}}$$

$$\therefore N(t/2) = \frac{3}{4} N_0$$

31. (b) We know that

$$\text{Activity, } A = A_0 e^{-\lambda t}$$

$$A = A_0 e^{-t \ln 2 / T_{1/2}} \quad \left(\because \lambda = \frac{\ln 2}{T_{1/2}} \right)$$

$$\Rightarrow 500 = 700 e^{-t \ln 2 / T_{1/2}}$$

$$\Rightarrow \ln \frac{5}{7} = \frac{30 \ln 2}{T_{1/2}} \quad (\because t = 30 \text{ minute})$$

$$\Rightarrow T_{1/2} = 30 \frac{\ln 2}{\ln 1.4} = 61.8 \text{ minute}$$

$$(\because \ln 2 = 0.693 \text{ and } \ln 1.4 = 0.336)$$

$$\Rightarrow T_{1/2} \approx 62 \text{ minute}$$

32. (a) As, $N = N_0 e^{-\lambda t}$

$$\text{so, } \frac{N_A}{N_B} = e^{(\lambda_B - \lambda_A)t} = \frac{1}{e} \Rightarrow (\lambda_B - \lambda_A)t = -1$$

$$\Rightarrow (\lambda_A - \lambda_B) \cdot t = 1$$

$$\Rightarrow t = \frac{1}{(\lambda_B - \lambda_A)} \quad t = \frac{1}{10\lambda - \lambda} = \frac{1}{9\lambda}$$

33. (a) Let N_1 and N_2 be the number of radioactive nuclei of substance at anytime t .

$$N_1(\text{at } t) = N_0 e^{-5\lambda t} \quad (\text{i})$$

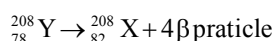
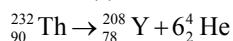
$$N_2(\text{at } t) = N_0 e^{-\lambda t} \quad (\text{ii})$$

Dividing equation (i) by (ii), we get

$$\frac{N_1}{N_2} = \frac{1}{e^2} = e^{-4\lambda t} \Rightarrow 4\lambda t = 2$$

$$\Rightarrow t = \frac{2}{4\lambda} = \left(\frac{1}{2\lambda}\right)$$

34. (c) When one α -particle emitted then daughter nuclei has 4 unit less mass number (A) and 2 unit less atomic (z) number (Z).



35. (a) According to question, at $t = 0$, $A_0 = \frac{dN}{dt} = 1600 \text{ C/s}$

and at $t = 8\text{s}$, $A = 100 \text{ C/s}$

$$\therefore \frac{A}{A_0} = \frac{1}{16} \text{ in } 8\text{s}$$

Therefore half life period, $t_{1/2} = 2\text{s}$

$$\therefore \text{Activity at } t = 6\text{s} = 1600 \left(\frac{1}{2}\right)^3 = 200 \text{ C/s}$$

36. (c) Activity $A = \lambda N$

For material, A $10 = (2N_0)\lambda_A$

For material, B $20 = N_0\lambda_B$

$$\Rightarrow \lambda_B = 4\lambda_A \therefore T_{1/2A} = 4T_{1/2B} \left[\because T_{1/2} = \frac{0.693}{\lambda} \right]$$

i.e. 20 days half-lives for A and 5 days $\left(T_{1/2}\right)_B$ For material B.

37. (c) Half life of A = $\ell \text{ n}2$

$$(t_{1/2})_A = \frac{\ell \text{ n}2}{\lambda}$$

$$\therefore \lambda_A = 1$$

at $t = 0$ $R_A = R_B$

$$N_A e^{-\lambda_A t} = N_B e^{-\lambda_B t}$$

$$N_A = N_B \text{ at } t = 0$$

$$\text{At } t = t \quad \frac{R_B}{R_A} = \frac{N_0 e^{-\lambda_B t}}{N_0 e^{-\lambda_A t}}$$

$$e^{-(\lambda_B - \lambda_A)t} = e^{-3t}$$

$$\Rightarrow \lambda_B - \lambda_A = 3$$

$$\lambda_B = 3 + \lambda_A = 4$$

$$(t_{1/2})_B = \frac{\ell \text{ n}2}{\lambda_B} = \frac{\ell \text{ n}2}{4}$$

38. (b) Given : $N_1 = 2N_2$

Activity of radioactive substance = λN

$$\text{Half life period } t = \frac{\ln 2}{\lambda} \text{ or, } \lambda = \frac{\ln 2}{T}$$

$$\lambda_1 N_1 = \frac{\ln 2}{t_1} \times N_1 = 5 \mu\text{Ci} \quad \dots (i)$$

$$\lambda_2 N_2 = \frac{\ln 2}{t_2} \times N_2 = 10 \mu\text{Ci} \quad \dots (ii)$$

Dividing equation (ii) by (i)

$$\frac{t_2}{t_1} \times \frac{N_1}{N_2} = \frac{1}{2}$$

$$\frac{t_2}{t_1} = \frac{1}{4} \Rightarrow t_1 = 4t_2$$

i.e., Half life of S_1 is four times of sample S_2 .
Hence 5 years and 20 years.

39. (d) Let initial activity = $N_0 = 0.8 \mu\text{Ci}$

$$0.8 \times 3.7 \times 10^4 \text{ dps}$$

Activity in 1 cm^3 of blood at $t = 10 \text{ hr}$,

$$n = \frac{300}{60} \text{ dps} = 5 \text{ dps}$$

$N =$ Activity of whole blood at time $t = 10 \text{ hr}$.

Total volume of the blood in the person, $V = \frac{N}{n}$

$$= \frac{N_0 e^{-\lambda t}}{n} = \frac{0.8 \times 3.7 \times 10^4 \times 0.7927}{5} \cong 5 \text{ litres}$$

40. (d) Let initially there are total N_0 number of nuclei
At time t

$$\frac{N_B}{N_A} = 0.3 (\text{given})$$

$$\Rightarrow N_B = 0.3N_A$$

$$N_0 = N_A + N_B = N_A + 0.3N_A$$

$$\therefore N_A = \frac{N_0}{1.3}$$

As we know $N_t = N_0 e^{-\lambda t}$

$$\text{or, } \frac{N_0}{1.3} = N_0 e^{-\lambda t}$$

$$\frac{1}{1.3} = e^{-\lambda t} \Rightarrow \ln(1.3) = \lambda t$$

$$\text{or, } t = \frac{\ln(1.3)}{\lambda} \Rightarrow t = \frac{\ln(1.3)}{\frac{\ln(2)}{T}} = \frac{\ln(1.3)}{\ln(2)} T$$

41. (b) For $A_{t/2} = 20 \text{ min}$, $t = 80 \text{ min}$, number of half lives $n = 4$

$\therefore$ Nuclei remaining = $\frac{N_0}{2^4}$. Therefore nuclei decayed

$$= N_0 - \frac{N_0}{2^4}$$

For $B_{t/2} = 40 \text{ min}$, $t = 80 \text{ min}$, number of half lives $n = 2$

$\therefore$ Nuclei remaining = $\frac{N_0}{2^2}$. Therefore nuclei decayed

$$= N_0 - \frac{N_0}{2^2}$$

$$\therefore \text{Required ratio} = \frac{N_0 - \frac{N_0}{2^2}}{N_0 - \frac{N_0}{2^2}} = \frac{1 - \frac{1}{16}}{1 - \frac{1}{4}} = \frac{15}{16} \times \frac{4}{3} = \frac{5}{4}$$

42. (b) We know that $N_\beta = N_0 (1 - e^{-\lambda t})$

$$N_\beta = \frac{6.023 \times 10^{23}}{24} \left[1 - e^{-\frac{\ell \ln 2}{15} \times 7.5} \right]$$

on solving we get,

$$N_\beta = 7.4 \times 10^{21}$$

43. (c) Given: $\frac{dN_0}{dt} = 20$ decays/min

$$\frac{dN}{dt} = 2 \text{ decays/min}$$

$$T_{1/2} = 5730 \text{ years}$$

As we know,

$$N = N_0 e^{-\lambda t}$$

$$\text{Log} \frac{N_0}{N} = \lambda t$$

$$\therefore t = \frac{1}{\lambda} \text{Log} \frac{N_0}{N}$$

$$= \frac{2.303 \times T_{1/2}}{0.693} \times \text{Log}_{10} \frac{N_0}{N}$$

$$\text{But } \frac{\frac{dN_0}{dt}}{\frac{dN}{dt}} = \frac{N_0}{N} = \frac{20}{2} = 10$$

$$\therefore t = \frac{2.303 \times 5730}{0.693} \times 1$$

$$= 19039 \text{ years}$$

44. (b) Given, for ^{14}C
 $A_0 = 16 \text{ dis min}^{-1} \text{ g}^{-1}$
 $A = 12 \text{ dis min}^{-1} \text{ g}^{-1}$
 $t_{1/2} = 5760 \text{ years}$

$$\text{Now, } \lambda = \frac{0.693}{t_{1/2}}$$

$$\lambda = \frac{0.693}{5760} \text{ per year}$$

$$\text{Then, from, } t = \frac{2.303}{\lambda} \log_{10} \frac{A_0}{A}$$

$$= \frac{2.303 \times 5760}{0.693} \log_{10} \frac{16}{12}$$

$$= \frac{2.303 \times 5760}{0.693} \log_{10} 1.333$$

$$= \frac{2.303 \times 5760 \times 0.1249}{0.693} = 2390.81 \approx 2391 \text{ years.}$$

45. (b) Let N be the number of nuclei at any time t then,

$$\frac{dN}{dt} = 100 - \lambda N \quad \text{or} \quad \int_0^N \frac{dN}{(100 - \lambda N)} = \int_0^t dt$$

$$-\frac{1}{\lambda} [\log (100 - \lambda N)]_0^N = t$$

$$\log (100 - \lambda N) - \log 100 = -\lambda t$$

$$\log \frac{100 - \lambda N}{100} = -\lambda t$$

$$\frac{100 - \lambda N}{100} = e^{-\lambda t} \quad 1 - \frac{\lambda N}{100} = e^{-\lambda t}$$

$$N = \frac{100}{\lambda} (1 - e^{-\lambda t})$$

As, $N = 50$ and $\lambda = 0.5/\text{sec}$

$$\therefore 50 = \frac{100}{0.5} (1 - e^{-0.5t})$$

Solving we get,

$$t = 2 \ln \left(\frac{4}{3} \right) \text{ sec}$$

46. (d) $(T_{1/2})_A = (t_{\text{mean}})_B$

$$\Rightarrow \frac{0.693}{\lambda_A} = \frac{1}{\lambda_B} \Rightarrow \lambda_A = 0.693 \lambda_B$$

$$\text{or } \lambda_A < \lambda_B$$

Also rate of decay = λN

Initially number of atoms (N) of both are equal but since

$\lambda_B > \lambda_A$, therefore B will decay at a faster rate than A

47. (d) As we know,

$$\left[\frac{N}{N_0} \right] = \left[\frac{1}{2} \right]^n \quad \dots(i)$$

n = no. of half life

N - no. of atoms left

N_0 - initial no. of atoms

By radioactive decay law,

$$\frac{dN}{dt} = kN$$

k - disintegration constant

$$\therefore \frac{\frac{dN}{dt}}{\frac{dN_0}{dt}} = \frac{N}{N_0} \quad \dots(ii)$$

From (i) and (ii) we get

$$\frac{\frac{dN}{dt}}{\frac{dN_0}{dt}} = \left[\frac{1}{2} \right]^n$$

$$\text{or, } \left[\frac{100}{1600} \right] = \left[\frac{1}{2} \right]^n \Rightarrow \left[\frac{1}{2} \right]^4 = \left[\frac{1}{2} \right]^n$$

$\therefore n = 4$, Therefore, in 8 seconds 4 half life had occurred in which counting rate reduces to 100 counts s^{-1} .

$$\therefore \text{Half life, } \frac{T_1}{2} = 2 \text{ sec}$$

In 6 sec, 3 half life will occur

$$\therefore \left[\frac{\frac{dN}{dt}}{1600} \right] = \left[\frac{1}{2} \right]^3 \Rightarrow \frac{dN}{dt} = 200 \text{ counts } s^{-1}$$

$$48. (c) T_{av} = \frac{T_\alpha T_\beta}{T_\alpha + T_\beta}$$

If α and β are emitted simultaneously.

49. (a) Radioactive decay is a continuous process. Rate of radioactive decay cannot be controlled. Nuclear fission can be controlled but not of nuclear fusion.

50. (b)

51. (b) Number of undecayed atom after time t_2 ;

$$\frac{N_0}{3} = N_0 e^{-\lambda t_2} \quad \dots(i)$$

Number of undecayed atom after time t_1 ;

$$\frac{2N_0}{3} = N_0 e^{-\lambda t_1} \quad \dots(ii)$$

Dividing (ii) by (i), we get

$$\begin{aligned} 2 &= e^{\lambda(t_2 - t_1)} \\ \Rightarrow \ln 2 &= \lambda(t_2 - t_1) \\ \Rightarrow t_2 - t_1 &= \ln 2 / \lambda \end{aligned}$$

52. (b) **Statement-1:** A nucleus having energy E_1 decays by β - emission to daughter nucleus having energy E_2 then β - rays are emitted with continuous energy spectrum with energy $E_1 - E_2$.

Statement-2: For energy conservation and momentum conservation at least three particles, daughter nucleus, β particle and antineutrino are required.

53. (b) When a radioactive nucleus emits 1 α -particle, the mass number decreases by 4 units and atomic number decreases by 2 units. When a radioactive nucleus emits 1 positron the atomic number decreases by 1 unit but mass number remains constant.

$$\therefore \text{Mass number of final nucleus} = A - 12$$

$$\text{Atomic number of final nucleus} = Z - 8$$

$$\therefore \text{Number of neutrons, } N_n = (A - 12) - (Z - 8) = A - Z - 4$$

$$\text{Number of protons, } N_p = Z - 8$$

$$\therefore \text{Required ratio} = \frac{N_n}{N_p} = \frac{A - Z - 4}{Z - 8}$$

54. (c) Let λ_X and λ_Y be the decay constant of X and Y .

Half life of X , = average life of Y

$$T_{1/2} = T_{av}$$

$$\Rightarrow \frac{0.693}{\lambda_X} = \frac{1}{\lambda_Y}$$

$$\Rightarrow \lambda_X = (0.693) \cdot \lambda_Y$$

$$\therefore \lambda_X < \lambda_Y.$$

Now, the rate of decay is given by

$$-\left(\frac{dN}{dt}\right)_x = \lambda_X N_0$$

$$-\left(\frac{dN}{dt}\right)_y = \lambda_Y N_0$$

As the rate of decay is directly proportional to decay constant, Y will decay faster than X .

55. (c) The range of energy of β -particles is from zero to some maximum value.

56. (d) It is given that

$$\frac{7}{8} \text{ of Cu decays in 15 minutes.}$$

$\therefore$ Cu left undecayed is

$$N = 1 - \frac{7}{8} = \frac{1}{8} = \left(\frac{1}{2}\right)^3$$

$\therefore$ No. of half lifes = 3

$$n = \frac{t}{T} \Rightarrow 3 = \frac{15}{T}$$

$$\Rightarrow T = \text{half life period} = \frac{15}{3} = 5 \text{ minutes}$$

57. (c) Let intensity of gamma radiation from source be I_0 .

$$\text{Intensity } I = I_0 \cdot e^{-\mu d},$$

Where d is the thickness of lead.

Applying logarithm on both sides,

$$-\mu d = \log \left(\frac{I}{I_0} \right)$$

$$\text{For } d = 36 \text{ mm, intensity} = \frac{I}{8}$$

$$-\mu \times 36 = \log\left(\frac{I/8}{I}\right) \dots\dots\dots(i)$$

For intensity $I/2$, thickness = d

$$-\mu \times d = \log\left(\frac{I/2}{I}\right) \dots\dots\dots(ii)$$

Dividing (i) by (ii),

$$\frac{36}{d} = \frac{\log\left(\frac{1}{8}\right)}{\log\left(\frac{1}{2}\right)} = \frac{3\log\left(\frac{1}{2}\right)}{\log\left(\frac{1}{2}\right)} = 3 \quad \text{or} \quad d = \frac{36}{3} = 12 \text{ mm}$$

58. (a) The radioactive substances emit α -particles (Helium nucleus), β -particles (electrons) and neutrinos. Protons cannot be emitted by radioactive substances during their decay.

59. (b) The number of α -particles released = 8

Decrease in atomic number = $8 \times 2 = 16$

The number of β^- -particles released = 4

Increase in atomic number = $4 \times 1 = 4$

Also the number of β^+ particles released is 2, which should decrease the atomic number by 2.

Therefore the final atomic number of resulting nucleus

$$= Z - 16 + 4 - 2 = Z - 14$$

$$= 92 - 14 = 78$$

60. (a) Initial activity, $A_o = 5000$ disintegration per minute
Activity after 5 min, $A = 1250$ disintegration per minute

$$A = A_o e^{-\lambda t}$$

$$\Rightarrow e^{-\lambda t} = \frac{A_o}{A}$$

$$\Rightarrow \lambda = \frac{1}{t} \log_e \frac{A_o}{A} = \frac{1}{5} \log_e \frac{5000}{1250}$$

$$= \frac{2}{5} \log_e 2 = 0.4 \log_e 2$$

61. (a) Charged particles are deflected in magnetic field. Electrons, protons and He^{2+} all are charged species. Hence, correct option is (a).

62. (a) After every half-life, the mass of the substance reduces to half its initial value.

$$N_0 \xrightarrow{5 \text{ years}} \frac{N_0}{2} \xrightarrow{5 \text{ years}} \frac{N_0/2}{2}$$

$$= \frac{N_0}{4} \xrightarrow{5 \text{ years}} \frac{N_0/4}{2} = \frac{N_0}{8}$$

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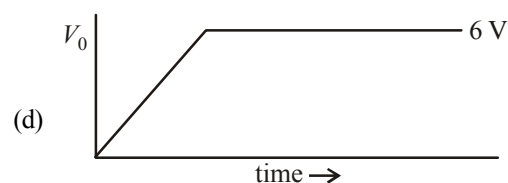
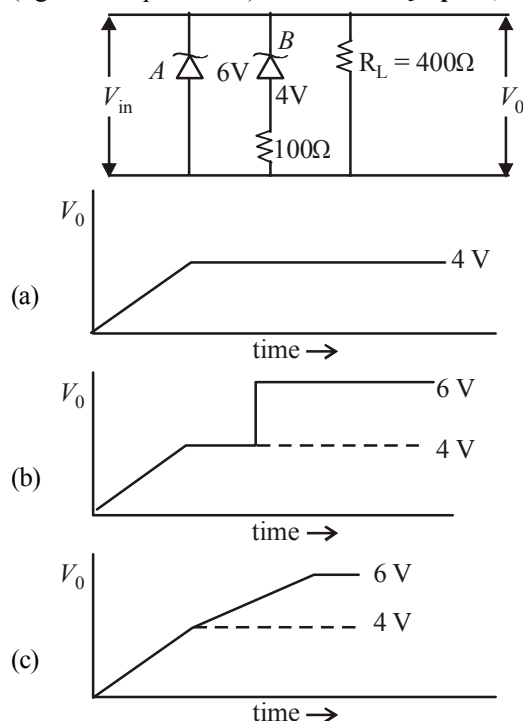
Semiconductor Electronics : Materials, Devices and Simple Circuits



TOPIC 1 Solids, Semiconductors and P-N Junction Diode

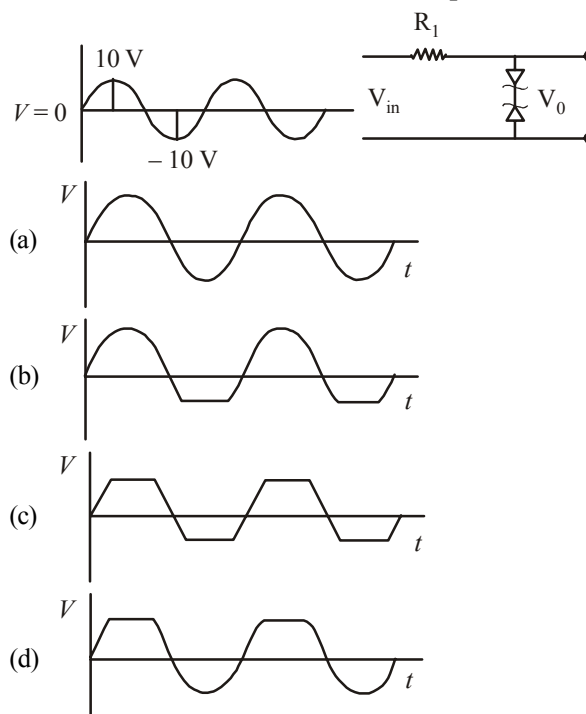


- With increasing biasing voltage of a photodiode, the photocurrent magnitude : [Sep. 05, 2020 (I)]
 - remains constant
 - increases initially and after attaining certain value, it decreases
 - Increases linearly
 - increases initially and saturates finally
- Two Zener diodes (*A* and *B*) having breakdown voltages of 6 V and 4 V respectively, are connected as shown in the circuit below. The output voltage V_0 variation with input voltage linearly increasing with time, is given by : [Sep. 05, 2020 (II)]
($V_{\text{input}} = 0$ V at $t = 0$)
(figures are qualitative)



- Take the breakdown voltage of the zener diode used in the given circuit as 6V. For the input voltage shown in figure below, the time variation of the output voltage is :
(Graphs drawn are schematic and not to scale)

[Sep. 04, 2020 (I)]



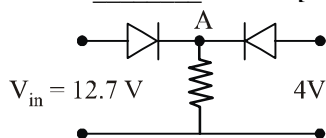
- When a diode is forward biased, it has a voltage drop of 0.5 V. The safe limit of current through the diode is 10 mA. If a battery of emf 1.5 V is used in the circuit, the value of minimum resistance to be connected in series with the diode so that the current does not exceed the safe limit is :

[Sep. 03, 2020 (I)]

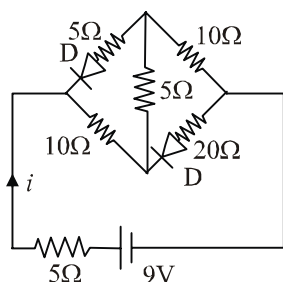
- 300 Ω
- 50 Ω
- 100 Ω
- 200 Ω

5. If a semiconductor photodiode can detect a photon with a maximum wavelength of 400 nm, then its band gap energy is: Planck's constant, $h = 6.63 \times 10^{-34}$ J.s. Speed of light, $c = 3 \times 10^8$ m/s [Sep. 03, 2020 (II)]
- (a) 1.1 eV (b) 2.0 eV
(c) 1.5 eV (d) 3.1 eV

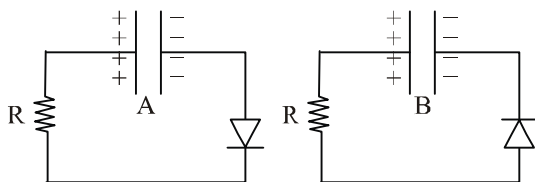
6. Both the diodes used in the circuit shown are assumed to be ideal and have negligible resistance when these are forward biased. Built in potential in each diode is 0.7 V. For the input voltages shown in the figure, the voltage (in Volts) at point A is _____. [NA 9 Jan. 2020 I]



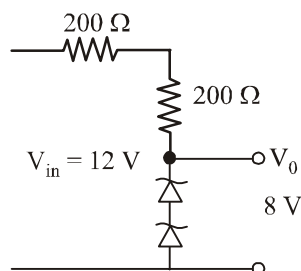
7. The current i in the network is: [9 Jan. 2020 II]



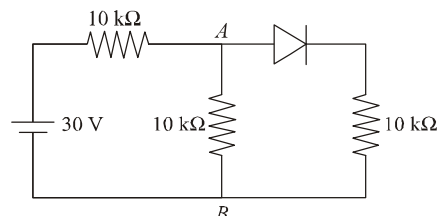
- (a) 0.2 A (b) 0.6 A (c) 0.3 A (d) 0 A
8. Two identical capacitors A and B, charged to the same potential 5V are connected in two different circuits as shown below at time $t = 0$. If the charge on capacitors A and B at time $t = CR$ is Q_A and Q_B respectively, then (Here e is the base of natural logarithm) [9 Jan. 2020 II]



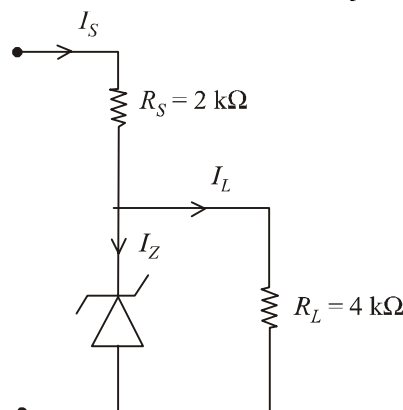
- (a) $Q_A = \frac{VC}{e}, Q_B = \frac{CV}{2}$ (b) $Q_A = VC, Q_B = CV$
(c) $Q_A = VC, Q_B = \frac{VC}{e}$ (d) $Q_A = \frac{CV}{2}, Q_B = \frac{VC}{e}$
9. The circuit shown below is working as a 8 V dc regulated voltage source. When 12 V is used as input, the power dissipated (in mW) in each diode is; (considering both zener diodes are identical) _____. [NA 9 Jan. 2020 II]



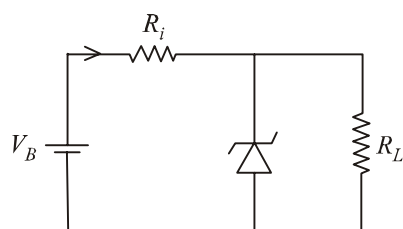
10. In the figure, potential difference between A and B is: [7 Jan. 2020 II]



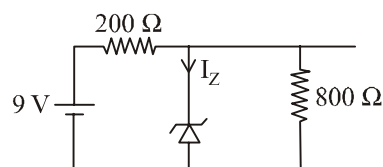
- (a) 10V (b) 5V (c) 15V (d) zero
11. Figure shows a DC voltage regulator circuit, with a Zener diode of breakdown voltage = 6V. If the unregulated input voltage varies between 10 V to 16 V, then what is the maximum Zener current? [12 Apr. 2019 II]



- (a) 2.5 mA (b) 1.5 mA (c) 7.5 mA (d) 3.5 mA
12. The figure represents a voltage regulator circuit using a Zener diode. The breakdown voltage of the Zener diode is 6 V and the load resistance is $R_L = 4$ k. The series resistance of the circuit is $R_i = 1$ k. If the battery voltage V_B varies from 8 V to 16 V, what are the minimum and maximum values of the current through Zener diode? [10 Apr. 2019 II]

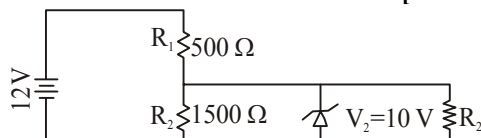


- (a) 0.5 mA; 6 mA (b) 1 mA; 8.5 mA
(c) 0.5 mA; 8.5 mA (d) 1.5 mA; 8.5 mA
13. The reverse breakdown voltage of a Zener diode is 5.6 V in the given circuit.

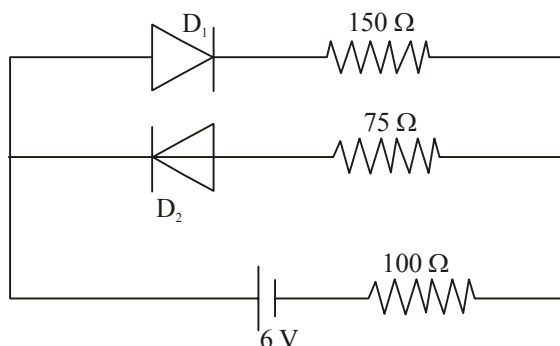


- The current I_z through the Zener is: [8 April 2019 I]
- (a) 10 mA (b) 17 mA
(c) 15 mA (d) 7 mA

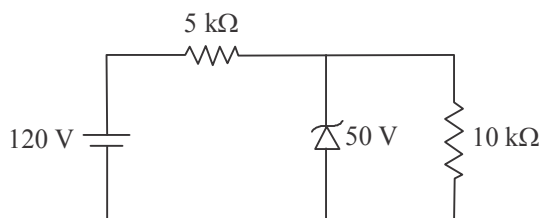
14. In the given circuit the current through Zener Diode is close to : [11 Jan. 2019 I]



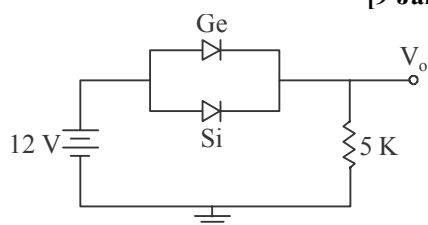
- (a) 0.0 mA (b) 6.7 mA
(c) 4.0 mA (d) 6.0 mA
15. The circuit shown below contains two ideal diodes, each with a forward resistance of $50\ \Omega$. If the battery voltage is 6V, the current through the $100\ \Omega$ resistance (in Amperes) is : [11 Jan. 2019 II]



- (a) 0.036 (b) 0.020
(c) 0.027 (d) 0.030
16. For the circuit shown below, the current through the Zener diode is: [10 Jan. 2019 II]

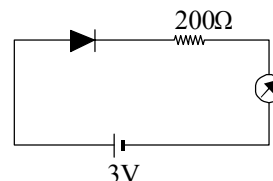


- (a) 9 mA (b) 5 mA
(c) Zero (d) 14 mA
17. Mobility of electrons in a semiconductor is defined as the ratio of their drift velocity to the applied electric field. If, for an n-type semiconductor, the density of electrons is $10^{19}\ \text{m}^{-3}$ and their mobility is $1.6\ \text{m}^2/(\text{V}\cdot\text{s})$ then the resistivity of the semiconductor (since it is an n-type semiconductor contribution of holes is ignored) is close to: [9 Jan. 2019 I]
- (a) $2\ \Omega\text{m}$ (b) $4\ \Omega\text{m}$ (c) $0.4\ \Omega\text{m}$ (d) $0.2\ \Omega\text{m}$
18. Ge and Si diodes start conducting at 0.3 V and 0.7 V respectively. In the following figure if Ge diode connection are reversed, the value of V_o changes by : (assume that the Ge diode has large breakdown voltage) [9 Jan. 2019 II]

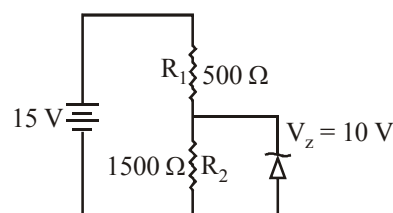


- (a) 0.8 V (b) 0.6 V
(c) 0.2 V (d) 0.4 V

19. The reading of the ammeter for a silicon diode in the given circuit is : [2018]



- (a) 0 (b) 15 mA
(c) 11.5 mA (d) 13.5 mA
20. In the given circuit, the current through zener diode is: [Online April 16, 2018]

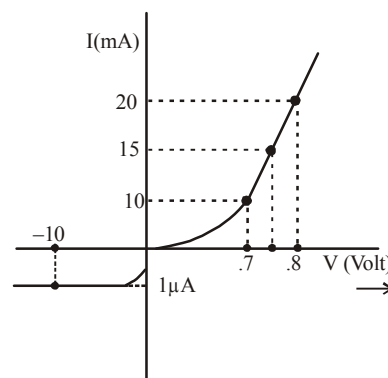


- (a) 2.5mA (b) 3.3mA
(c) 5.5mA (d) 6.7mA
21. What is the conductivity of a semiconductor sample having electron concentration of $5 \times 10^{18}\ \text{m}^{-3}$, hole concentration of $5 \times 10^{19}\ \text{m}^{-3}$, electron mobility of $2.0\ \text{m}^2\ \text{V}^{-1}\ \text{s}^{-1}$ and hole mobility of $0.01\ \text{m}^2\ \text{V}^{-1}\ \text{s}^{-1}$?

[Online April 8, 2017]

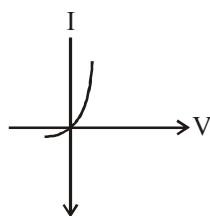
(Take charge of electron as $1.6 \times 10^{-19}\ \text{C}$)

- (a) $1.68\ (\Omega\text{-m})^{-1}$ (b) $1.83\ (\Omega\text{-m})^{-1}$
(c) $0.59\ (\Omega\text{-m})^{-1}$ (d) $1.20\ (\Omega\text{-m})^{-1}$
22. The V-I characteristic of a diode is shown in the figure. The ratio of forward to reverse bias resistance is : [Online April 8, 2017]

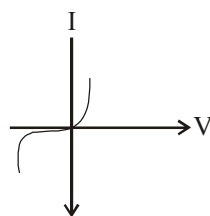


- (a) 10 (b) 10^{-6}
(c) 10^6 (d) 100

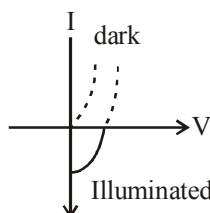
23. Identify the semiconductor devices whose characteristics are given below, in the order (i), (ii), (iii), (iv) : [2016]



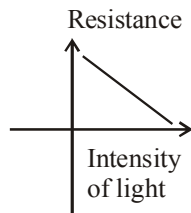
(i)



(ii)

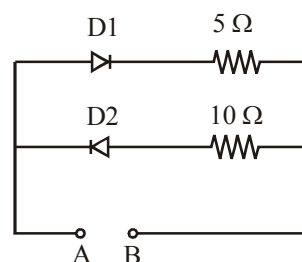


(iii)

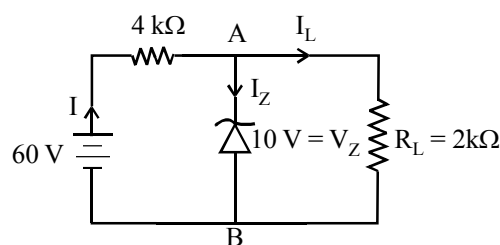


(iv)

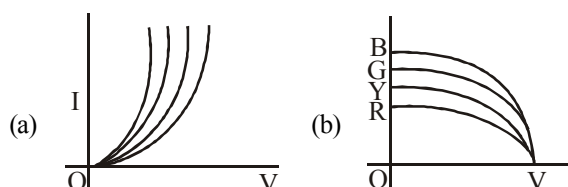
- (a) Solar cell, Light dependent resistance, Zener diode, simple diode
 (b) Zener diode, Solar cell, simple diode, Light dependent resistance
 (c) Simple diode, Zener diode, Solar cell, Light dependent resistance
 (d) Zener diode, Simple diode, Light dependent resistance, Solar cell
24. The temperature dependence of resistances of Cu and undoped Si in the temperature range 300-400 K, is best described by : [2016]
 (a) Linear increase for Cu, exponential decrease of Si.
 (b) Linear decrease for Cu, linear decrease for Si.
 (c) Linear increase for Cu, linear increase for Si.
 (d) Linear increase for Cu, exponential increase for Si.
25. An experiment is performed to determine the 1-V characteristics of a Zener diode, which has a protective resistance of $R = 100\ \Omega$, and a maximum power of dissipation rating of 1 W. The minimum voltage range of the DC source in the circuit is : [Online April 9, 2016]
 (a) 0–5 V (b) 0–24 V
 (c) 0–12 V (d) 0–8 V
26. A red LED emits light at 0.1 watt uniformly around it. The amplitude of the electric field of the light at a distance of 1 m from the diode is : [2015]
 (a) 5.48 V/m (b) 7.75 V/m
 (c) 1.73 V/m (d) 2.45 V/m
27. A 2V battery is connected across AB as shown in the figure. The value of the current supplied by the battery when in one case battery's positive terminal is connected to A and in other case when positive terminal of battery is connected to B will respectively be: [Online April 11, 2015]

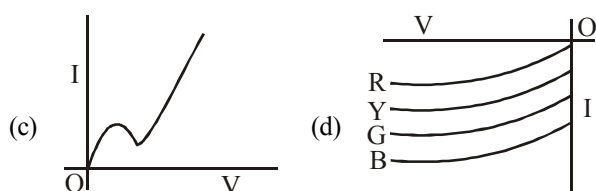


- (a) 0.4 A and 0.2 A (b) 0.2 A and 0.4 A
 (c) 0.1 A and 0.2 A (d) 0.2 A and 0.1 A
28. In an unbiased n-p junction electrons diffuse from n-region to p-region because : [Online April 10, 2015]
 (a) holes in p-region attract them
 (b) electrons travel across the junction due to potential difference
 (c) only electrons move from n to p region and not the vice-versa
 (d) electron concentration in n-region is more compared to that in p-region
29. The forward biased diode connection is: [2014]
 (a) (b)
 (c) (d)
30. For LED's to emit light in visible region of electromagnetic light, it should have energy band gap in the range of: [Online April 12, 2014]
 (a) 0.1 eV to 0.4 eV (b) 0.5 eV to 0.8 eV
 (c) 0.9 eV to 1.6 eV (d) 1.7 eV to 3.0 eV
31. A Zener diode is connected to a battery and a load as show below: [Online April 11, 2014]



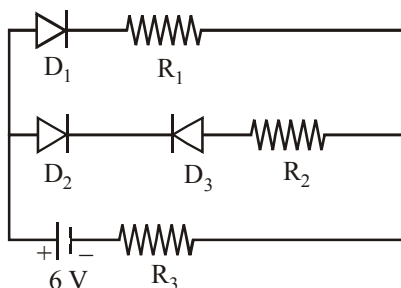
- The currents, I , I_Z and I_L are respectively.
 (a) 15 mA, 5 mA, 10 mA
 (b) 15 mA, 7.5 mA, 7.5 mA
 (c) 12.5 mA, 5 mA, 7.5 mA
 (d) 12.5 mA, 7.5 mA, 5 mA
32. The I-V characteristic of an LED is [2013]



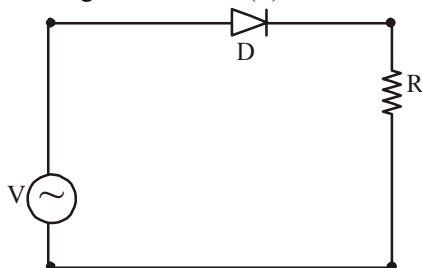


33. Figure shows a circuit in which three identical diodes are used. Each diode has forward resistance of $20\ \Omega$ and infinite backward resistance. Resistors $R_1 = R_2 = R_3 = 50\ \Omega$. Battery voltage is 6 V . The current through R_3 is :

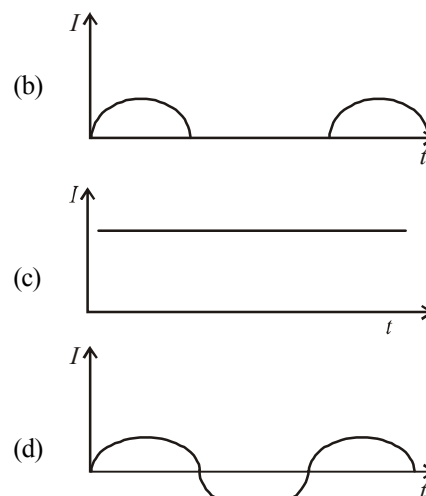
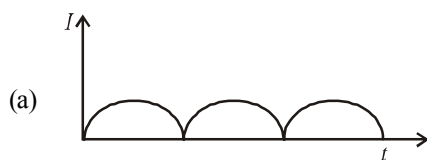
[Online April 22, 2013]



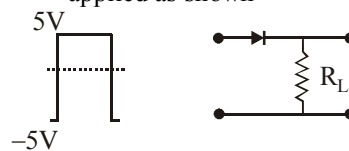
- (a) 50 mA (b) 100 mA
(c) 60 mA (d) 25 mA
34. This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
- Statement 1:** A pure semiconductor has negative temperature coefficient of resistance.
- Statement 2:** On raising the temperature, more charge carriers are released into the conduction band.
- [Online May 12, 2012]
- (a) Statement 1 is false, Statement 2 is true.
(b) Statement 1 is true, Statement 2 is false.
(c) Statement 1 is true, Statement 2 is true, Statement 2 is not a correct explanation of Statement 1.
(d) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.
35. A p - n junction (D) shown in the figure can act as a rectifier. An alternating current source (V) is connected in the circuit.



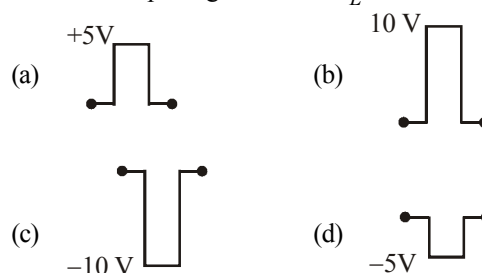
The current (I) in the resistor (R) can be shown by : [2009]



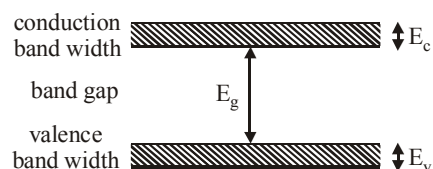
36. If in a p - n junction diode, a square input signal of 10 V is applied as shown [2007]



Then the output signal across R_L will be



37. Carbon, silicon and germanium have four valence electrons each. At room temperature which one of the following statements is most appropriate? [2007]
- (a) The number of free electrons for conduction is significant only in Si and Ge but small in C.
(b) The number of free conduction electrons is significant in C but small in Si and Ge.
(c) The number of free conduction electrons is negligibly small in all the three.
(d) The number of free electrons for conduction is significant in all the three.
38. If the lattice constant of this semiconductor is decreased, then which of the following is correct? [2006]



- (a) All E_c, E_g, E_v increase
(b) E_c and E_v increase, but E_g decreases
(c) E_c and E_v decrease, but E_g increases
(d) All E_c, E_g, E_v decrease

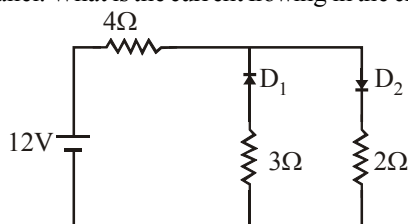
39. A solid which is not transparent to visible light and whose conductivity increases with temperature is formed by [2006]

(a) Ionic bonding
(b) Covalent bonding
(c) Vander Waals bonding
(d) Metallic bonding

40. If the ratio of the concentration of electrons to that of holes in a semiconductor is $\frac{7}{5}$ and the ratio of currents is $\frac{7}{4}$, then what is the ratio of their drift velocities? [2006]

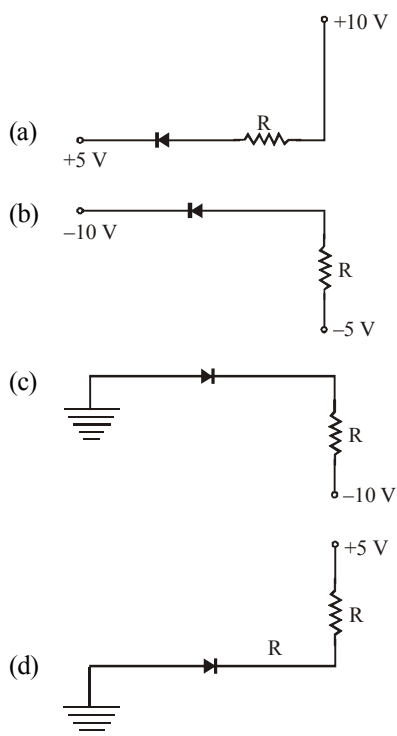
(a) $\frac{5}{8}$ (b) $\frac{4}{5}$ (c) $\frac{5}{4}$ (d) $\frac{4}{7}$

41. The circuit has two oppositely connected ideal diodes in parallel. What is the current flowing in the circuit? [2006]



(a) 1.71 A (b) 2.00 A (c) 2.31 A (d) 1.33 A

42. In the following, which one of the diodes reverse biased? [2006]



43. The electrical conductivity of a semiconductor increases when electromagnetic radiation of wavelength shorter than 2480 nm is incident on it. The band gap in (eV) for the semiconductor is [2005]
- (a) 2.5 eV (b) 1.1 eV (c) 0.7 eV (d) 0.5 eV

44. When p-n junction diode is forward biased then [2004]

(a) both the depletion region and barrier height are reduced
(b) the depletion region is widened and barrier height is reduced
(c) the depletion region is reduced and barrier height is increased
(d) Both the depletion region and barrier height are increased

45. A strip of copper and another of germanium are cooled from room temperature to 80K. The resistance of [2003]

(a) each of these decreases
(b) copper strip increases and that of germanium decreases
(c) copper strip decreases and that of germanium increases
(d) each of these increases

46. The difference in the variation of resistance with temperature in a metal and a semiconductor arises essentially due to the difference in the [2003]

(a) crystal structure
(b) variation of the number of charge carriers with temperature
(c) type of bonding
(d) variation of scattering mechanism with temperature

47. In the middle of the depletion layer of a reverse-biased p-n junction, the [2003]

(a) electric field is zero
(b) potential is maximum
(c) electric field is maximum
(d) potential is zero

48. At absolute zero, Si acts as [2002]

(a) non-metal (b) metal
(c) insulator (d) none of these

49. By increasing the temperature, the specific resistance of a conductor and a semiconductor [2002]

(a) increases for both (b) decreases for both
(c) increases, decreases (d) decreases, increases

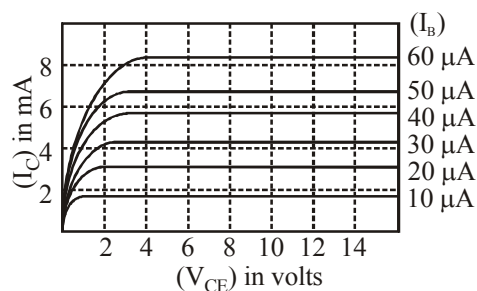
50. The energy band gap is maximum in [2002]

(a) metals (b) superconductors
(c) insulators (d) semiconductors.

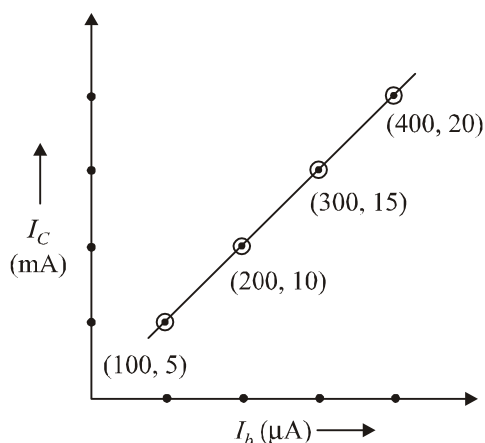
TOPIC 2 Junction Transistor



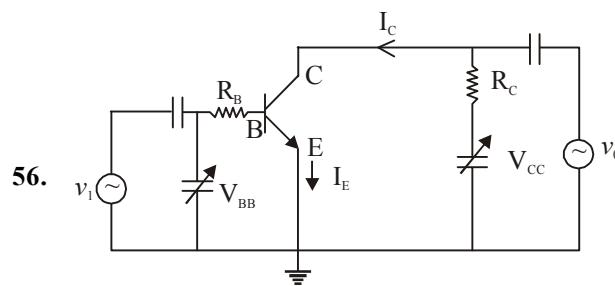
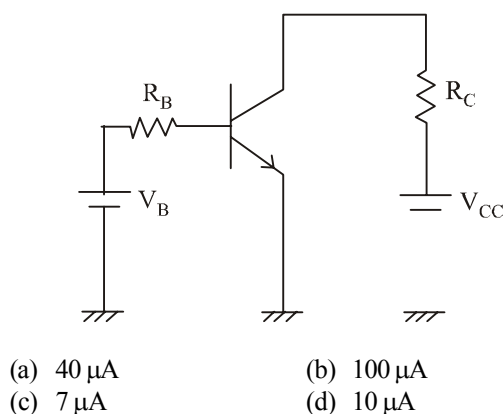
51. The output characteristics of a transistor is shown in the figure. When V_{CE} is 10 V and $I_C = 4.0$ mA, then value of β_{ac} is _____. [NA Sep. 06, 2020 (II)]



52. The transfer characteristic curve of a transistor, having input and output resistance $100\ \Omega$ and $100\ \text{k}\Omega$ respectively, is shown in the figure. The Voltage and Power gain, are respectively : [12 Apr. 2019 I]



- (a) $2.5 \times 10^4, 2.5 \times 10^6$ (b) $5 \times 10^4, 5 \times 10^6$
 (c) $5 \times 10^4, 5 \times 10^5$ (d) $5 \times 10^4, 2.5 \times 10^6$
53. An npn transistor operates as a common emitter amplifier, with a power gain of 60 dB. The input circuit resistance is $100\ \Omega$ and the output load resistance is $10\ \text{k}\Omega$. The common emitter current gain β is : [10 Apr. 2019 I]
 (a) 10^2 (b) 60 (c) 6×10^2 (d) 10^4
54. An NPN transistor is used in common emitter configuration as an amplifier with $1\ \text{k}\Omega$ load resistance. Signal voltage of $10\ \text{mV}$ is applied across the base-emitter. This produces a $3\ \text{mA}$ change in the collector current and $15\ \mu\text{A}$ change in the base current of the amplifier. The input resistance and voltage gain are: [9 April 2019 I]
 (a) $0.33\ \text{k}\Omega, 1.5$ (b) $0.67\ \text{k}\Omega, 300$
 (c) $0.67\ \text{k}\Omega, 200$ (d) $0.33\ \text{k}\Omega, 300$
55. A common emitter amplifier circuit, built using an npn transistor, is shown in the figure. Its dc current gain is 250, $R_C = 1\ \text{k}\Omega$ and $V_{CC} = 10\ \text{V}$. What is the minimum base current for V_{CE} to reach saturation ? [8 Apr. 2019 II]



56.

In the figure, given that V_{BB} supply can vary from 0 to 5.0 V, $V_{CC} = 5\ \text{V}$, $\beta_{dc} = 200$, $R_B = 100\ \text{k}\Omega$, $R_C = 1\ \text{k}\Omega$ and $V_{BE} = 1.0\ \text{V}$. The minimum base current and the input voltage at which the transistor will go to saturation, will be, respectively : [12 Jan. 2019 II]

- (a) $25\ \mu\text{A}$ and $3.5\ \text{V}$ (b) $20\ \mu\text{A}$ and $3.5\ \text{V}$
 (c) $25\ \mu\text{A}$ and $2.8\ \text{V}$ (d) $20\ \mu\text{A}$ and $2.8\ \text{V}$
57. In a common emitter configuration with suitable bias, it is given that R_L is the load resistance and R_{BE} is small signal dynamic resistance (input side). Then, voltage gain, current gain and power gain are given, respectively, by: (β is current gain, I_B, I_C, I_E are respectively base, collector and emitter currents:) [Online April 15, 2018]

- (a) $\beta \frac{R_L}{R_{BE}}, \frac{\Delta I_E}{\Delta I_B}, \beta^2 \frac{R_L}{R_{BE}}$
 (b) $\beta^2 \frac{R_L}{R_{BE}}, \frac{\Delta I_C}{\Delta I_B}, \beta \frac{R_L}{R_{BE}}$
 (c) $\beta^2 \frac{R_L}{R_{BE}}, \frac{\Delta I_C}{\Delta I_E}, \beta^2 \frac{R_L}{R_{BE}}$
 (d) $\beta \frac{R_L}{R_{BE}}, \frac{\Delta I_C}{\Delta I_B}, \beta^2 \frac{R_L}{R_{BE}}$

58. The current gain of a common emitter amplifier is 69. If the emitter current is $7.0\ \text{mA}$, collector current is : [Online April 9, 2017]

- (a) $9.6\ \text{mA}$ (b) $6.9\ \text{mA}$
 (c) $0.69\ \text{mA}$ (d) $69\ \text{mA}$

59. In a common emitter amplifier circuit using an n-p-n transistor, the phase difference between the input and the output voltages will be : [Online April 2, 2017]

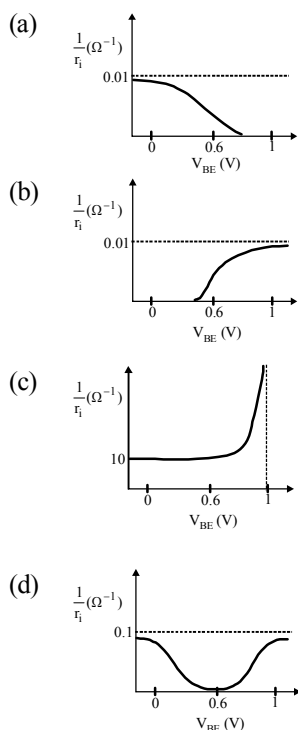
- (a) 135° (b) 180°
 (c) 45° (d) 90°

60. For a common emitter configuration, if α and β have their usual meanings, the incorrect relationship between α and β is : [2016]

- (a) $a = \frac{b}{1+b}$ (b) $a = \frac{b^2}{1+b^2}$
 (c) $\frac{1}{a} = \frac{1}{b} + 1$ (d) None of these

61. A realistic graph depicting the variation of the reciprocal of input resistance in an input characteristics measurement in a common emitter transistor configuration is :

[Online April 10, 2016]



62. The ratio (R) of output resistance r_o , and the input resistance r_i in measurements of input and output characteristics of a transistor is typically in the range :

[Online April 10, 2016]

- (a) $R \sim 10^2 - 10^3$ (b) $R \sim 1 - 10$
 (c) $R \sim 0.1 - 1.0$ (d) $R \sim 0.1 - 0.01$
63. An unknown transistor needs to be identified as a npn or pnp type. A multimeter, with +ve and -ve terminals, is used to measure resistance between different terminals of transistor. If terminal 2 is the base of the transistor then which of the following is correct for a pnp transistor?

[Online April 9, 2016]

- (a) +ve terminal 2, -ve terminal 3, resistance low
 (b) +ve terminal 2, -ve terminal 1, resistance high
 (c) +ve terminal 1, -ve terminal 2, resistance high
 (d) +ve terminal 3, -ve terminal 2, resistance high
64. An n-p-n transistor has three leads A, B and C. Connecting B and C by moist fingers, A to the positive lead of an ammeter, and C to the negative lead of the ammeter, one finds large deflection. Then, A, B and C refer respectively to:

[Online April 9, 2014]

- (a) Emitter, base and collector
 (b) Base, emitter and collector
 (c) Base, collector and emitter
 (d) Collector, emitter and base.
65. A working transistor with its three legs marked P , Q and R is tested using a multimeter. No conduction is found

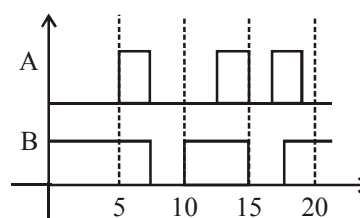
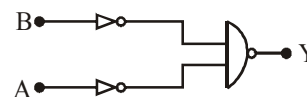
between P and Q . By connecting the common (negative) terminal of the multimeter to R and the other (positive) terminal to P or Q , some resistance is seen on the multimeter. Which of the following is true for the transistor? [2008]

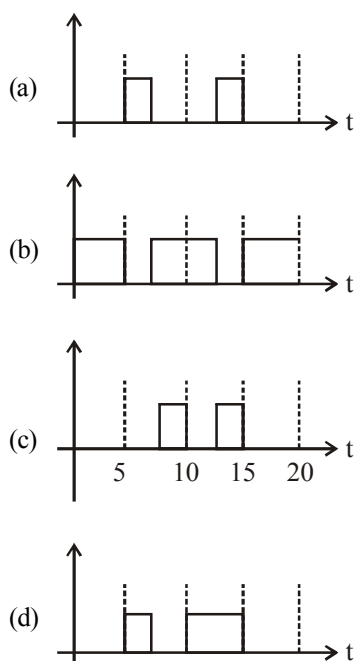
- (a) It is an npn transistor with R as base
 (b) It is a pnp transistor with R as base
 (c) It is a pnp transistor with R as emitter
 (d) It is an npn transistor with R as collector
66. In a common base mode of a transistor, the collector current is 5.488 mA for an emitter current of 5.60 mA. The value of the base current amplification factor (β) will be [2006]
- (a) 49 (b) 50
 (c) 51 (d) 48
67. In a common base amplifier, the phase difference between the input signal voltage and output voltage is [2005]
- (a) π (b) $\frac{\pi}{4}$
 (c) $\frac{\pi}{2}$ (d) 0
68. When npn transistor is used as an amplifier [2004]
- (a) electrons move from collector to base
 (b) holes move from emitter to base
 (c) electrons move from base to collector
 (d) holes move from base to emitter
69. For a transistor amplifier in common emitter configuration for load impedance of $1k\Omega$ ($h_{fe} = 50$ and $h_{oe} = 25$) the current gain is [2004]
- (a) -24.8 (b) -15.7
 (c) -5.2 (d) -48.78
70. The part of a transistor which is most heavily doped to produce large number of majority carriers is [2002]
- (a) emitter
 (b) base
 (c) collector
 (d) can be any of the above three.

TOPIC 3 Digital Electronics and Logic Gates

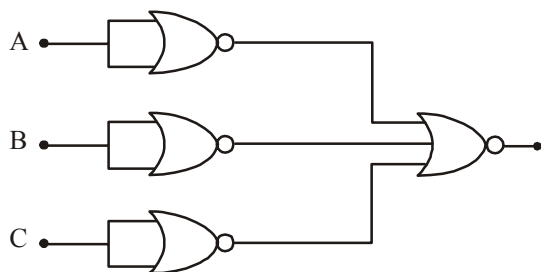


71. Identify the correct output signal Y in the given combination of gates (as shown) for the given inputs A and B . [Sep. 06, 2020 (I)]

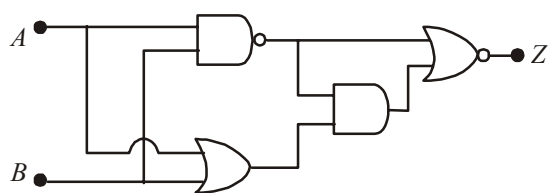




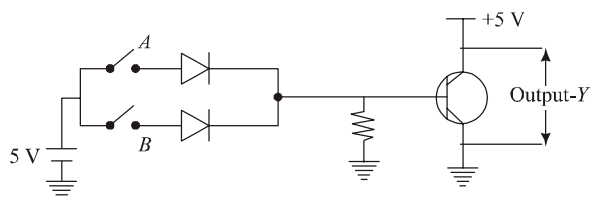
72. Identify the operation performed by the circuit given below:
[Sep. 04, 2020 (II)]



- (a) NAND (b) OR
(c) AND (d) NOT
73. In the following digital circuit, what will be the output at 'Z', when the input (A, B) are (1, 0), (0, 0), (1, 1), (0, 1):
[Sep. 02, 2020 (II)]

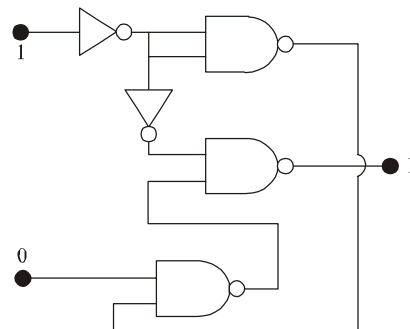


- (a) 0, 0, 1, 0 (b) 1, 0, 1, 1
(c) 1, 1, 0, 1 (d) 0, 1, 0, 0
74. Boolean relation at the output stage-Y for the following circuit is:
[8 Jan. 2020 I]

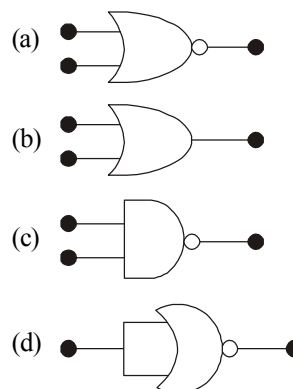


- (a) $\bar{A} + \bar{B}$ (b) $A + B$
(c) $A.B$ (d) $\bar{A}.\bar{B}$

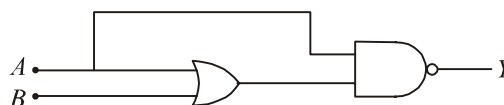
75. In the given circuit, value of Y is: [8 Jan. 2020 II]



- (a) 0
(b) toggles between 0 and 1
(c) will not execute
(d) 1
76. Which of the following gives a reversible operation?
[7 Jan. 2020 I]



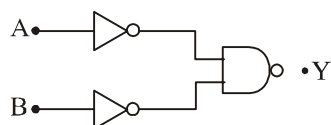
77. The truth table for the circuit given in the fig. is :
[9 April 2019 I]



- (a) $\begin{bmatrix} A & B & Y \\ 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} A & B & Y \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$
- (c) $\begin{bmatrix} A & B & Y \\ 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} A & B & Y \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

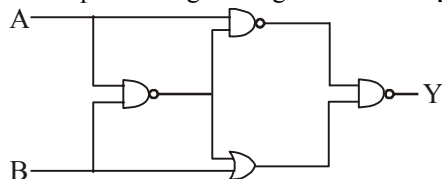
78. The logic gate equivalent to the given logic circuit is:

[9 Apr. 2019 II]



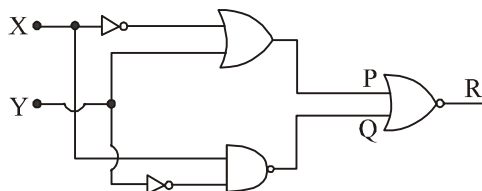
- (a) NAND (b) OR (c) NOR (d) AND

79. The output of the given logic circuit is: [12 Jan. 2019 I]



- (a) $A\bar{B} + \bar{A}B$ (b) $AB + \bar{A}\bar{B}$
(c) $A\bar{B}$ (d) $\bar{A}B$

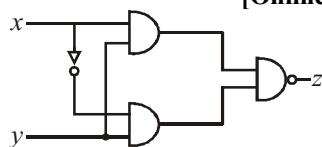
80. To get output '1' at R, for the given logic gate circuit the input values must be: [10 Jan. 2019 I]



- (a) $X = 0, Y = 1$ (b) $X = 1, Y = 1$
(c) $X = 1, Y = 0$ (d) $X = 0, Y = 0$

81. Truth table for the given circuit will be

[Online April 15, 2018]



(a)

x	y	z
0	0	1
0	1	1
1	0	1
1	1	0

(b)

x	y	z
0	0	0
0	1	0
1	0	0
1	1	1

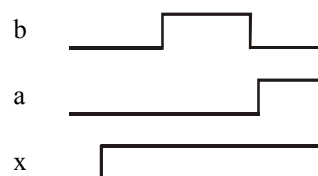
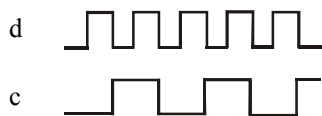
(c)

x	y	z
0	0	1
0	1	1
1	0	1
1	1	1

(d)

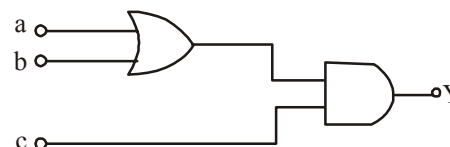
x	y	z
0	0	0
0	1	1
1	0	1
1	1	1

82. If a, b, c, d are inputs to a gate and x is its output, then, as per the following time graph, the gate is: [2016]



- (a) OR (b) NAND
(c) NOT (d) AND

83. To get an output of 1 from the circuit shown in figure the input must be: [Online April 10, 2016]



- (a) $a = 0, b = 0, c = 1$ (b) $a = 1, b = 0, c = 0$
(c) $a = 1, b = 0, c = 1$ (d) $a = 0, b = 1, c = 0$

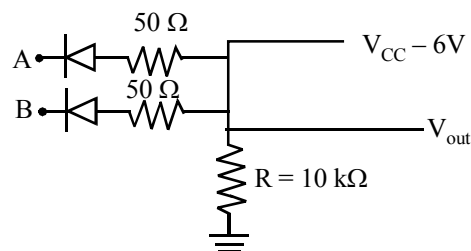
84. The truth table given in fig. represents:

[Online April 9, 2016]

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

- (a) OR-Gate (b) NAND-Gate
(c) AND-Gate (d) NOR-Gate

85.



Given: A and B are input terminals.

Logic 1 $\Rightarrow$ 5 V

Logic 0 $\Rightarrow$ < 1 V

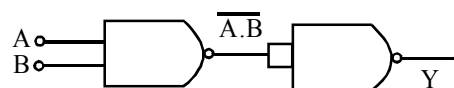
Which logic gate operation, the above circuit does?

[Online April 19, 2014]

- (a) AND Gate (b) OR Gate
(c) XOR Gate (d) NOR Gate

86. Identify the gate and match A, B, Y in bracket to check.

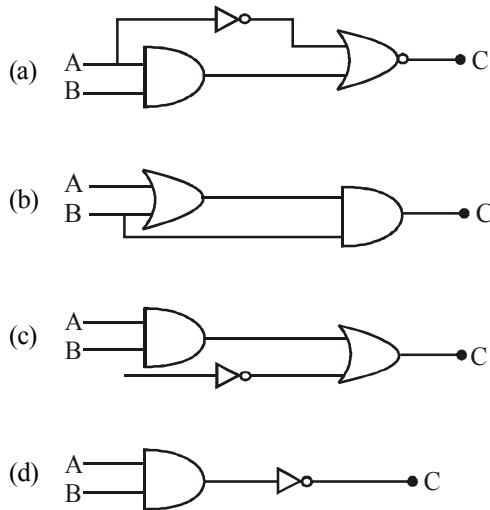
[Online April 9, 2014]



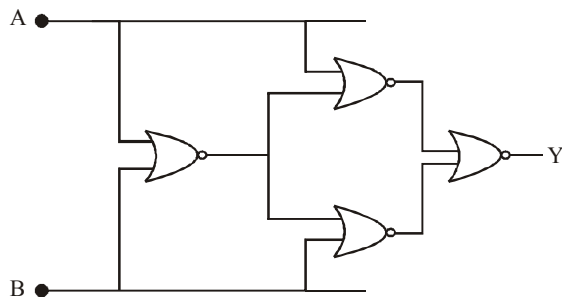
- (a) AND ($A = 1, B = 1, Y = 1$)
(b) OR ($A = 1, B = 1, Y = 0$)
(c) NOT ($A = 1, B = 1, Y = 1$)
(d) XOR ($A = 0, B = 0, Y = 0$)

87. Which of the following circuits correctly represents the following truth table ? [Online April 25, 2013]

A	B	C
0	0	0
0	1	0
1	0	1
1	1	0



88. A system of four gates is set up as shown. The 'truth table' corresponding to this system is : [Online April 23, 2013]



(a)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

(b)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	0

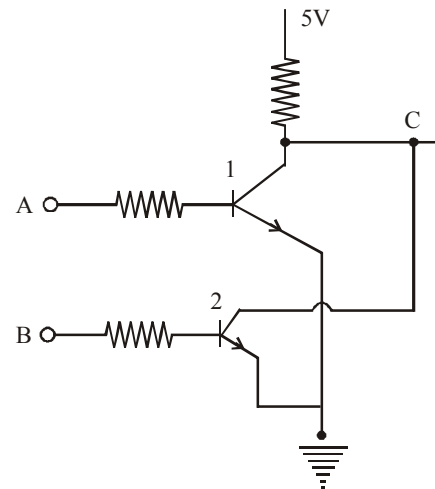
(c)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

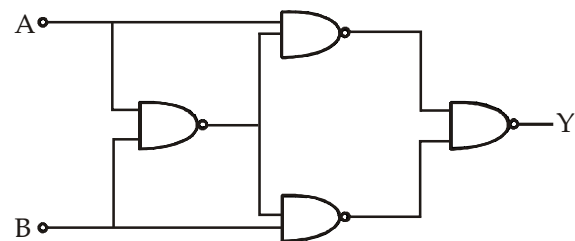
(d)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

89. Consider two *npn* transistors as shown in figure. If 0 Volts corresponds to false and 5 Volts correspond to true then the output at C corresponds to : [Online April 9, 2013]



- (a) A NAND B (b) A OR B
(c) A AND B (d) A NOR B
90. Truth table for system of four NAND gates as shown in figure is : [2012]



(a)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(b)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

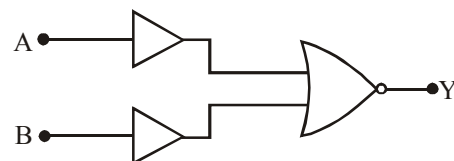
(c)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

(d)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	1

91. The figure shows a combination of two NOT gates and a NOR gate. [Online May 26, 2012]

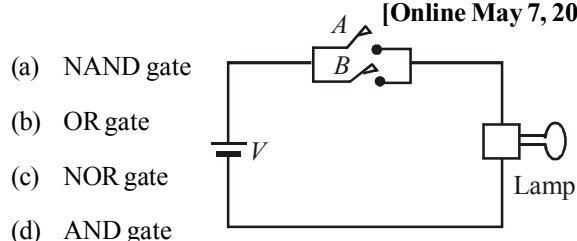


- The combination is equivalent to a
(a) NAND gate (b) NOR gate
(c) AND gate (d) OR gate

92. Which one of the following is the Boolean expression for NOR gate? [Online May 19, 2012]

(a) $Y = \overline{A+B}$ (b) $Y = \overline{A.B}$
 (c) $Y = A.B$ (d) $Y = \overline{A}$

93. Which logic gate with inputs A and B performs the same operation as that performed by the following circuit? [Online May 7, 2012]

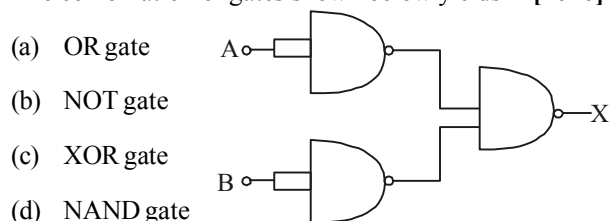


94. The output of an OR gate is connected to both the inputs of a NAND gate. The combination will serve as a:

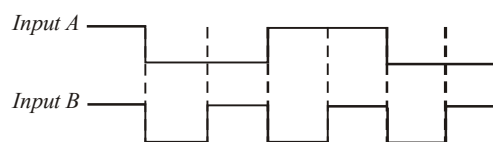
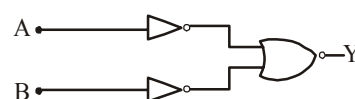
[2011 RS]

(a) NOT gate (b) NOR gate
 (c) AND gate (d) OR gate

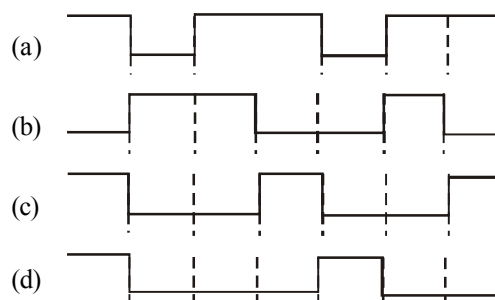
95. The combination of gates shown below yields [2010]



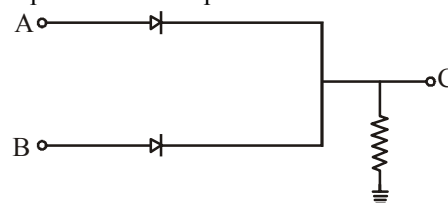
96. The logic circuit shown below has the input waveforms 'A' and 'B' as shown. Pick out the correct output waveform. [2009]



Output is



97. In the circuit below, A and B represent two inputs and C represents the output. [2008]



The circuit represents

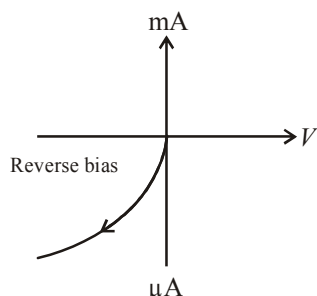
(a) NOR gate (b) AND gate
 (c) NAND gate (d) OR gate



Hints & Solutions



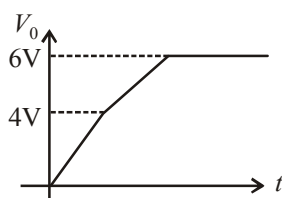
1. (d) I-V characteristic of a photodiode is as follows :



On increasing the biasing voltage of a photodiode, the magnitude of photocurrent first increases and then attains a saturation.

2. (c) Till input voltage reaches 4 V. No zener is in breakdown region so $V_0 = V_i$. Then now when V_i changes between 4 V to 6 V one zener with 4 V will breakdown and P.D. across this zener will become constant and remaining potential will drop across resistance in series with 4 V zener.

Now current in circuit increases abruptly and source must have an internal resistance due to which some potential will get drop across the source also so correct graph between V_0 and t will be



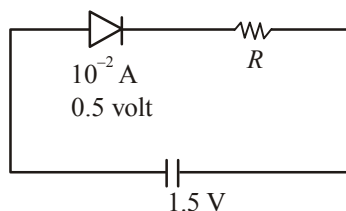
3. (c) Here two zener diodes are in reverse polarity so if one is in forward bias the other will be in reverse bias and above 6V the reverse bias will too be in conduction mode. Hence when $V > 6V$ the output will be constant. And when $V < 6V$ it will follow the input voltage.

4. (c) According to question, when diode is forward biased,

$$V_{\text{diode}} = 0.5 \text{ V}$$

$$\text{Safe limit of current, } I = 10 \text{ mA} = 10^{-2} \text{ A}$$

$$R_{\text{min}} = ?$$



Voltage through resistance

$$V_R = 1.5 - 0.5 = 1 \text{ volt}$$

$$iR = 1 (=V_R)$$

$$\therefore R_{\text{min}} = \frac{1}{i} = \frac{1}{10^{-2}} = 100 \Omega$$

5. (d) Given,

$$\text{Wavelength of photon, } \lambda = 400 \text{ nm}$$

A photodiode can detect a wavelength corresponding to the energy of band gap. If the signal is having wavelength greater than this value, photodiode cannot detect it.

$$\therefore \text{Band gap } E_g = \frac{hc}{\lambda} = \frac{1237.5}{400} = 3.09 \text{ eV}$$

6. (12) Right hand diode is reversed biased and left hand diode is forward biased.

Hence Voltage at 'A'

$$V_A = 12.7 - 0.7 = 12 \text{ volt}$$

7. (c) Both the diodes are reverse biased, so, there is no flow of current through 5Ω and 20Ω resistances.

Now, two resistors of 10Ω and two resistors of 5Ω are in series.

$$\text{Hence current } I \text{ through the network} = 0.3 \text{ A}$$

8. (c) In case I diode is reverse biased, so no current flows

$$\therefore Q_A = CV$$

In case II, current will flow as diode is forward biased. So, it offers negligible resistance to the flow of current and thus be replaced by short circuit. Now, the charge of capacitor will leak through the resistance and decay exponentially with time.

During discharging of capacitor

Potential difference across the capacitor at any instant

$$V' = Ve^{-\frac{t}{CR}}$$

$$\text{But } t = CR$$

$$V' = Ve^{-1} = \frac{V}{e}$$

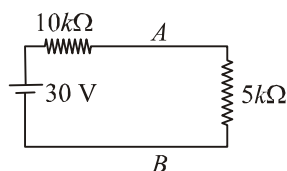
$$\therefore \text{Charge } Q_B = CV' = \frac{CV}{e}$$

9. (40) Current in the circuit, $I = \frac{12-8}{400} = 10^{-2} \text{ A}$

$$\text{Power dissipated in each diode, } P = VI$$

$$\Rightarrow P = 4 \times 10^{-2} = 40 \text{ mW}$$

10. (a) The given circuit has two $10k\Omega$ resistances in parallel, so we can reduce this parallel combination to a single equivalent resistance of $5k\Omega$.



Diode is in forward bias. So it will behave like a conducting wire.

$$V_A - V_B = \frac{30}{5+10} \times 5 = 10 \text{ V}$$

11. (d) Current in load resistance,

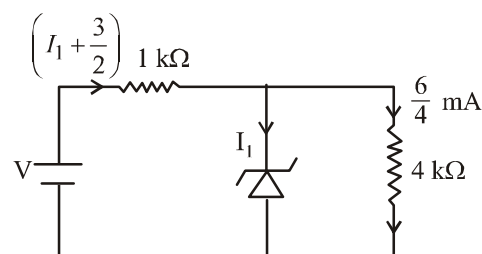
$$i_1 = \frac{6}{4 \times 10^3} = 1.5 \times 10^{-3} \text{ A} = 1.5 \text{ mA}$$

For $V = 16$ volt,

$$i_s = \frac{(16-6)}{2 \times 10^3} = 5 \text{ mA}$$

$$\therefore i_2 = i_s - i_1 = 5 - 1.5 = 3.5 \text{ mA}$$

12. (c)



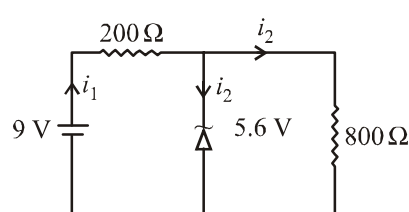
For voltage, $V = 8 \text{ V}$

$$\text{Current, } I_1 = \left(8 - 6 - \frac{3}{2}\right) = \frac{1}{2} = 0.5 \text{ mA}$$

For voltage, $V = 16 \text{ V}$

$$\text{Current, } I_2 = \left(16 - 6 - \frac{3}{2}\right) = 8.5 \text{ mA}$$

13. (a)



P.D. across 800Ω resistors = 5.6 V

$$\text{so, } I_{800\Omega} = \frac{5.6}{800} \text{ A} = 7 \text{ mA}$$

Now, P.D. across 200Ω resistors = $9 - 5.6 \text{ V} = 3.4 \text{ V}$

$$\text{so, } I_{200\Omega} = \frac{9-5.6}{200} = 17 \text{ mA}$$

so, current through zener diode = $I_2 = 17 - 7 = 10 \text{ mA}$

14. (a) Since voltage across zener diode does not reach to breakdown voltage therefore its resistance will be infinite & current through it is 0.

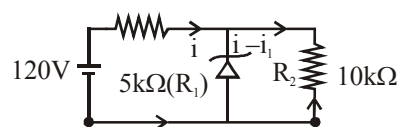
15. (b) As D_2 is reversed biased, so no current through 75Ω resistor.

$$\begin{aligned} \text{now } R_{eq} &= 150 + 50 + 100 \\ &= 300 \Omega \end{aligned}$$

$$\text{So, required current } I = \frac{\text{Battery Voltage}}{300}$$

$$I = \frac{6}{300} = 0.02$$

16. (a) The voltage across zener diode is constant



$$i_{(R_2)} = \frac{V}{R} = \frac{50}{10 \times 10^3} = 5 \times 10^{-3} \text{ A}$$

$$i_{(R_1)} = \frac{V}{R} = \frac{120-50}{5 \times 10^3} = \frac{70}{5 \times 10^3} = 14 \times 10^{-3} \text{ A}$$

$$\therefore i_{\text{zener diode}} = 14 \times 10^{-3} - 5 \times 10^{-3} = 9 \times 10^{-3} \text{ A} = 9 \text{ mA}$$

17. (c) As we know, current density,

$$j = \sigma E = ne v_d$$

$$\sigma = ne \frac{v_d}{E} = ne \mu$$

$$\frac{1}{\sigma} = \rho = \frac{1}{ne \mu} = \text{Resistivity}$$

$$= \frac{1}{10^{19} \times 1.6 \times 10^{19} - 19 \times 1.6}$$

$$\text{or } \rho = 0.4 \Omega \text{ m}$$

18. (d) Initially Ge and Si are both forward biased so current will effectively pass through Ge diode $\therefore V_o = 12 - 0.3 = 11.7 \text{ V}$

And if "Ge" is reversed then current will flow through "Si" diode

$$\therefore V_o = 12 - 0.7 = 11.3 \text{ V}$$

Clearly, V_o changes by $11.7 - 11.3 = 0.4 \text{ V}$

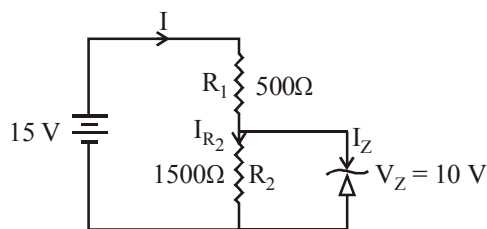
19. (c) Clearly from fig. given in question, Silicon diode is in forward bias.

$\therefore$ Potential barrier across diode

$$\Delta V = 0.7 \text{ volts}$$

$$\text{Current, } I = \frac{V - \Delta V}{R} = \frac{3 - 0.7}{200} = \frac{2.3}{200} = 11.5 \text{ mA}$$

20. (b)



The voltage drop across R_2 is $V_{R_2} = V_Z = 10\text{ V}$

The current through R_2 is

$$I_{R_2} = \frac{V_{R_2}}{R_2} = \frac{10\text{ V}}{1500\Omega} = 0.667 \times 10^{-2}\text{ A}$$

$$= 6.67 \times 10^{-3}\text{ A} = 6.67\text{ mA}$$

The voltage drop across R_1 is

$$V_{R_1} = 15\text{ V} - V_{R_2} = 15\text{ V} - 10\text{ V} = 5\text{ V}$$

The current through R_1 is

$$I_{R_1} = \frac{V_{R_1}}{R_1} = \frac{5\text{ V}}{500\Omega} = 10^{-2}\text{ A} = 10 \times 10^{-3}\text{ A} = 10\text{ mA}$$

The current through the zener diode is

$$I_Z = I_{R_1} - I_{R_2} = (10 - 6.67)\text{ mA} = 3.3\text{ mA}$$

21. (a) The conductivity of semiconductor

$$\sigma = e(\eta_e \mu_e + \eta_h \mu_h)$$

$$= 1.6 \times 10^{-19} (5 \times 10^{18} \times 2 + 5 \times 10^{19} \times 0.01)$$

$$= 1.6 \times 1.05 = 1.68$$

22. (b) Forward bias resistance $= \frac{\Delta V}{\Delta I} = \frac{0.1}{10 \times 10^{-3}} = 10\ \Omega$

$$\text{Reverse bias resistance} = \frac{10}{10^{-6}} = 10^7\ \Omega$$

$$\text{Ratio of resistances} = \frac{\text{Forward bias resistance}}{\text{Reverse bias resistance}} = 10^{-6}$$

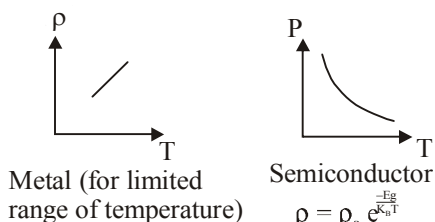
23. (c) Graph (p) is for a simple diode.

Graph (q) is showing the V Break down used for zener diode.

Graph (r) is for solar cell which shows cut-off voltage and open circuit current.

Graph (s) shows the variation of resistance h and hence current with intensity of light.

24. (a)



25. (c) The minimum voltage range of DC source is given by

$$V^2 = PR \quad \therefore P = 1\text{ watt}, R = 100\Omega$$

$$= 1 \times 100$$

$$\therefore V = 10\text{ volt.}$$

26. (d) Using $U_{av} = \frac{1}{2} \epsilon_0 E_0^2$

$$\text{But } U_{av} = \frac{P}{4\pi r^2 \times c}$$

$$\therefore \frac{P}{4\pi r^2} = \frac{1}{2} \epsilon_0 E_0^2 \times c$$

$$E_0^2 = \frac{2P}{4\pi r^2 \epsilon_0 c} = \frac{2 \times 0.1 \times 9 \times 10^9}{1 \times 3 \times 10^8}$$

$$\therefore E_0 = \sqrt{6} = 2.45\text{ V/m}$$

27. (a) When positive terminal connected to A then diode

$$D_1 \text{ is forward biased, current, } I = \frac{2}{5} = 0.4\text{ A}$$

When positive terminal connected to B then diode D_2

$$\text{is forward biased, current, } I = \frac{2}{10} = 0.2\text{ A}$$

$$E_0^2 = \frac{2P}{4\pi r^2 \epsilon_0 c} = \frac{2 \times 0.1 \times 9 \times 10^9}{1 \times 3 \times 10^8}$$

$$\therefore E_0 = \sqrt{6} = 2.45\text{ V/m}$$

28. (d) Electrons in an unbiased p - n junction, diffuse from n -region i.e. higher electron concentration to p -region i.e. low electron concentration region.29. (a) $\overrightarrow{P} \rightarrow \overrightarrow{n}$

For forward bias, p -side must be at higher potential than n -side. $\Delta V = (+)Ve$

30. (d) Energy band gap range is given by,

$$E_g = \frac{hc}{\lambda}$$

$$\text{For visible region } \lambda = (4 \times 10^{-7} \sim 7 \times 10^{-7})\text{ m}$$

$$E_g = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{7 \times 10^{-7}}$$

$$= \frac{19.8 \times 10^{-26}}{7 \times 10^{-7}}$$

$$= \frac{2.8 \times 10^{-19}}{1.6 \times 10^{-19}}$$

$$E_g = 1.75\text{ eV}$$

31. (d) Here, $R = 4 \text{ k}\Omega = 4 \times 10^3 \Omega$

$$V_i = 60 \text{ V}$$

$$\text{Zener voltage } V_z = 10 \text{ V}$$

$$R_L = 2 \text{ k}\Omega = 2 \times 10^3 \Omega$$

$$\text{Load current, } I_L = \frac{V_z}{R_L} = \frac{10}{2 \times 10^3} = 5 \text{ mA}$$

$$\text{Current through } R, I = \frac{V_i - V_z}{R}$$

$$= \frac{60 - 10}{4 \times 10^3} = \frac{50}{4 \times 10^3} = 12.5 \text{ mA}$$

From circuit diagram,

$$I = I_z + I_L$$

$$\Rightarrow 12.5 = I_z + 5$$

$$\Rightarrow I_z = 12.5 - 5 = 7.5 \text{ mA}$$

32. (a) For same value of current higher value of voltage is required for higher frequency hence (a) is correct answer.

33. (a) Here, diodes D_1 and D_2 are forward biased and D_3 is reverse biased.

Therefore current through R_3

$$i = \frac{V}{R'} = \frac{6}{120} = \frac{1}{20} \text{ A} = 50 \text{ mA}$$

34. (d) Temperature coefficient of resistance is negative for pure semiconductor. And no. of charge carriers in conduction band increases with increase in temperature.

35. (b) The given circuit will work as half wave rectifier as it conducts during the positive half cycle of input AC.

Forward biased in one half cycle and reverse biased in the other half cycle].

36. (a) The current will flow through R_L when the diode is forward biased.

37. (a) Si and Ge are semiconductors but C is an insulator. In Si and Ge at room temperature, the energy band gap is low due to which electrons in the covalent bonds gain kinetic energy and break the bond and move to conduction band. As a result, hole is created in valence band. So, the number of free electrons is significant in Si and Ge.

38. (c) A crystal structure is made up of a unit cell arranged in a particular way; which is periodically repeated in three dimensions on a lattice. The spacing between unit cells in various directions is called its lattice constants. As lattice constants increase the band-gap (E_g), also increases which means more energy would be required by electrons to reach the conduction band from the valence band. Automatically E_c and E_v decreases.

39. (b) Van der Waal's bonding is attributed to the attractive forces between molecules of a liquid. The conductivity of

semiconductors (covalent bonding) and insulators (ionic bonding) increases with increase in temperature.

Solid which is formed by covalent bond is not transparent to visible light and its conductivity increases with temperature.

40. (c) Relation between drift velocity and current is

$$I = nAeV_d$$

$$\frac{I_e}{I_h} = \frac{n_e e A v_e}{n_h e A v_h}$$

$$\Rightarrow \frac{7}{4} = \frac{7}{5} \times \frac{v_e}{v_h}$$

$$\Rightarrow \frac{v_e}{v_h} = \frac{5}{4}$$

41. (b) D_2 is forward biased.

D_1 is reversed biased. So, it will act like an open circuit.

So effective resistance of the circuit

$$R = 4 + 2 = 6 \Omega \quad \therefore i = \frac{E}{R} = \frac{12}{6} = 2 \text{ A}$$

42. (d) p -side connected to low potential and n -side is connected to high potential.

43. (d) Band gap = energy of photon of wavelength 2480 nm. So,

$$\text{Band gap, } E_g = \frac{hc}{\lambda}$$

$$= \left(\frac{6.63 \times 10^{-34} \times 3 \times 10^8}{2480 \times 10^{-9}} \right) \times \frac{1}{1.6 \times 10^{-19}} \text{ eV}$$

$$= 0.5 \text{ eV}$$

44. (a) In forward biasing, the p type is connected to positive terminal and n type is connected with negative terminal. So holes from p region and electron from n region are pushed towards the Junction which reduces the width of depletion layer. Also, distance between diffused holes and electrons decrease, which decreases electric field hence barrier potential.

45. (c) Copper is a conductor and in conductor resistance decreases with decrease in temperature. Germanium is a semiconductor. In semiconductor resistance increases with decrease in temperature.

46. (b) When the temperature increases, certain bounded electrons become free which tend to promote conductivity. Simultaneously number of collisions between electrons and positive kernels increases which decrease the relaxation time.

47. (a) In reverse biasing the width of depletion region increases, and current flowing through diode is zero. Thus, electric field is zero at middle of depletion region.

48. (c) Pure silicon, at OK, will contain all the electrons in bounded state. The conduction band will be empty. So there will be no free electrons (in conduction band) and holes (in valence band). Therefore no electrons from valence band are able to shift to conduction band due to thermal agitation. Pure silicon will act as insulator.

49. (c) Specific resistance (resistivity) is given by

$$\rho = \frac{m}{ne^2\tau}$$

where n = no. of free electrons per unit volume
and τ = average relaxation time

For a conductor with rise in temperature n increases. Increase in temperature results increase in number of collision between free electrons due to which relaxation time T decreases. But the decrease in τ is more dominant than increase in n resulting an increase in the value of ρ .

For a semiconductor with rise in temperature, n increases and τ decreases. But the increase in n is more dominant than decrease in τ resulting in a decrease in the value of ρ .

✚ ALTERNATE SOLUTION

$$\rho_2 = \rho_1(1 + \alpha\Delta T)$$

For conductor α is positive

$\therefore \rho_2 > \rho_1$ for ΔT positive i.e., increase in temperature.

For semi conductor α is negative

$\therefore \rho_2 < \rho_1$ for ΔT positive.

50. (c) In insulators, valence band is completely filled while conduction band is empty. The energy band gap is maximum in insulators.

51. (150)

At $V_{CE} = 10$ V and $I_C = 4$ mA

Change in base current, $\Delta I_B = (30 - 20) = 10$ μ A

Change in collector current, $\Delta I_C = (4.5 - 3) = 1.5$ mA

$$\beta = \left(\frac{\Delta I_C}{\Delta I_B} \right) = \frac{1.5 \text{ mA}}{10 \mu\text{A}} = 150$$

52. (Bonus) $\beta = \frac{\Delta i_c}{\Delta i_b} = \frac{200 - 100}{10 - 5} = 20$

$$\text{Voltage gain} = \beta \frac{R_2}{R_1} = \frac{20 \times 100 \times 10^3}{100} = 20 \times 10^3$$

$$\text{Power gain} = \beta^2 \frac{R_2}{R_1} = 20^2 \left(\frac{100 \times 10^3}{100} \right) = 4 \times 10^5$$

53. (a) Power gain = 60 = $10 \log \left(\frac{P_0}{P_i} \right)$

$$\Rightarrow 6 = \log \left(\frac{P_0}{P_i} \right)$$

$$\Rightarrow \frac{P_0}{P_i} = 10^6$$

$$= \beta^2 \left(\frac{R_{\text{out}}}{R_{\text{in}}} \right)$$

$$\Rightarrow 10^6 = \beta^2 \left(\frac{10000}{100} \right) \quad [\text{as } R_{\text{out}} = 10,000 \Omega, R_{\text{in}} = 100 \Omega]$$

$$\Rightarrow \beta = 100$$

54. (b) $\beta = \frac{\Delta I_C}{\Delta I_B} = \frac{3 \times 10^{-3}}{15 \times 10^{-6}} = 200$

$$\text{We have } \frac{V_0}{V_i} = \beta \frac{R^2}{R_1}$$

$$\text{or } \frac{V_0}{V_i} = 200 \left(\frac{1000}{R_1} \right)$$

$$\text{If } R_1 = 0.67 \text{ k}\Omega \Rightarrow \frac{V_0}{V_i} = 300$$

55. (a) Given, $\beta = 250$

$$\text{Voltage gain, } \frac{V_{CC}}{V_B} = \beta \frac{R_C}{R_B}$$

$$\frac{10}{V_B} = 250 \times \frac{10^3}{R_B}$$

$$\therefore \frac{V_B}{R_B} = \frac{1}{25 \times 10^3} = 40 \mu\text{A}$$

56. (a) At saturation, $V_{CE} = 0$

$$V_{CE} = V_{CC} - I_C R_C$$

$$\Rightarrow I_C = \frac{V_{CC}}{R_C} = 5 \times 10^{-3} \text{ A}$$

Current gain,

$$\beta_{dc} = \frac{I_C}{I_B}$$

$$I_B = \frac{5 \times 10^{-3}}{200} = 25 \mu\text{A}$$

At input side

$$V_{BB} = I_B R_B + V_{BE} \\ = (25 \text{ mA})(100 \text{ k}\Omega) + 1 \text{ V}$$

$$V_{BB} = 3.5 \text{ V}$$

57. (d) Current gain $\beta = \frac{\Delta I_C}{I_B}$

$$\text{Voltage gain } A_v = \text{Current gain} \times \text{Resistance gain} = \beta \frac{R_L}{R_{BE}}$$

$$\text{Power gain } A_p = (\text{Current gain})^2 \times \text{Resistance gain}$$

$$= \beta^2 \frac{R_L}{R_{BE}}$$

58. (b) Given, current gain of CE amplifier $\beta = 69$, $I_E = 7 \text{ mA}$

$$\text{or } \frac{I_C}{I_B} = 69$$

$$\text{We know that, } \alpha = \frac{\beta}{1+\beta} = \frac{69}{70} = \frac{I_C}{I_E}$$

$$I_C = I_E \times \frac{69}{70} = \frac{69}{70} \times 7$$

$$\text{Collector current, } I_C = 6.9 \text{ mA}$$

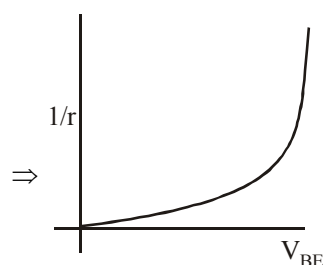
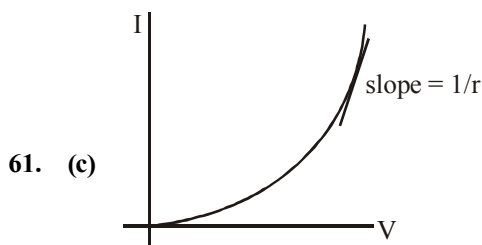
59. (b) In common emitter configuration for $n-p-n$ transistor input and output signals are 180° out of phase *i.e.*, phase difference between output and input voltage is 180° .

60. (b,d) We know that $\alpha = \frac{I_c}{I_e}$ and $\beta = \frac{I_c}{I_b}$

$$\text{Also } I_e = I_b + I_c$$

$$\therefore \alpha = \frac{I_c}{I_b + I_c} = \frac{\frac{I_c}{I_b}}{1 + \frac{I_c}{I_b}} = \frac{\beta}{1+\beta}$$

Option (b) and (d) are therefore incorrect.

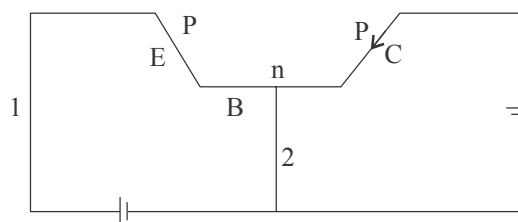


62. (c) For C.B. configuration $\frac{r_i}{r_o} \cong 0.1\Omega$

For CE and CC-configuration

$$\frac{r_i}{r_o} \approx 1\Omega$$

63. (c) Connecting circuit according to question, it is clear



+ve terminal 1, -ve terminal 2, resistance high.

64. (c) In the given question, A, B and C refer base, collector and emitter respectively.

65. (b) It is a $p-n-p$ transistor with B as base.

66. (a) Collector current, $I_C = 5.488 \text{ mA}$,
Emitter current $I_e = 5.6 \text{ mA}$

$$\alpha = \frac{I_c}{I_e} = \frac{5.488}{5.6},$$

$$\beta = \frac{\alpha}{1-\alpha} = 49$$

67. (d) In common base amplifier circuit, input and output voltage are in the same phase. So, the phase difference between input voltage signal and output voltage signal is zero.

68. (c) In npn transistor, electrons moves from emitter to base.

69. (d) In common emitter configuration for transistor amplifier current gain

$$A_i = \frac{-h_{fe}}{1 + h_{oe} R_L}$$

Where h_{fe} and h_{oe} are hybrid parameters.

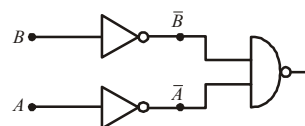
$$\therefore A_i = \frac{-50}{1 + 25 \times 10^{-6} \times 1 \times 10^3}$$

$$= -48.78$$

70. (a) Emitter main function is to supply the majority charge carriers towards the collector. Therefore emitter is most heavily doped.

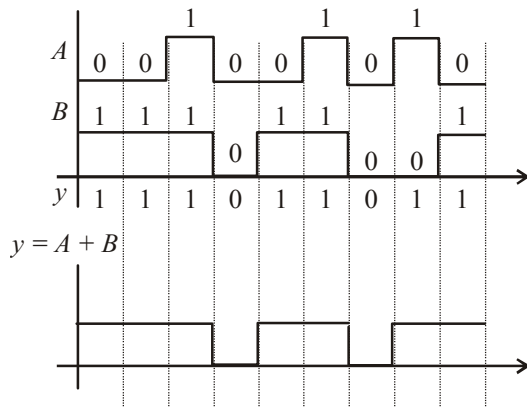
71. (a) Boolean expression,

$$y = \overline{\overline{A} \cdot \overline{B}} = \overline{\overline{A}} + \overline{\overline{B}} = A + B$$



Truth table :

A	B	Y
0	1	1
1	0	1
0	0	0
1	1	1



72. (c) When two inputs of NAND gate is shorted, it behaves like a NOT gate so boolean equation will be

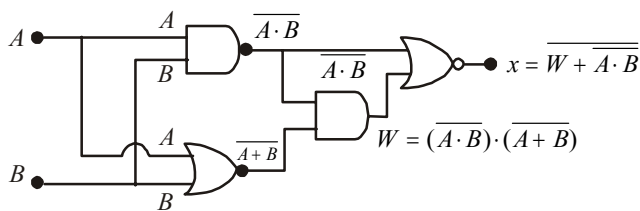
$$y = \overline{A + B + C}$$

$$y = A \cdot B \cdot C$$

A	B	C	
0	0	0	0
1	0	0	0
0	1	0	0
0	0	1	0
1	1	0	0
1	0	1	0
0	1	1	0
1	1	1	1

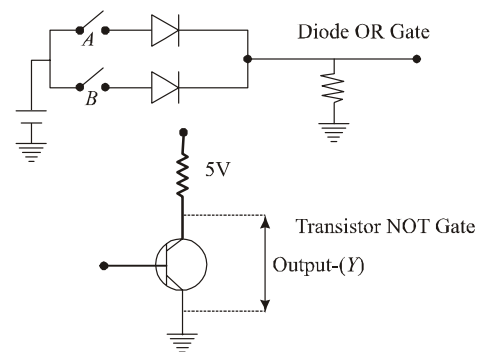
Thus, whole arrangement behaves like a AND gate.

73. (a)



A	B	$\overline{A \cdot B}$	$\overline{A + B}$	$W = (A \cdot B) \cdot (A + B)$	$Q = W + A \cdot B$	$\overline{Q} = x$
1	0	1	0	0	1	0
0	1	1	0	0	1	0
1	1	0	0	0	0	1
0	0	1	1	1	0	0

74. (d)

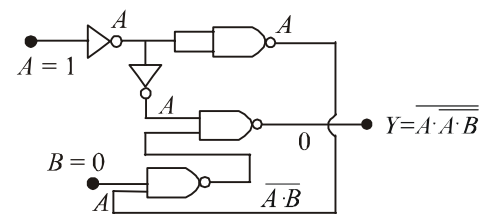


OR + NOT $\rightarrow$ NOR Gate

Hence Boolean relation at the output stage – Y for the circuit,

$$Y = \overline{A + B} = \overline{A} \cdot \overline{B}$$

75. (a)



$$Y = \overline{AB} \cdot A = \overline{AB} + \overline{A} = AB + \overline{A}$$

For $A = 1, B = 0$

$$Y = (1) \times 0 + 0$$

$$\Rightarrow Y = 0 + 0 = 0$$

76. (d) A logic gate is reversible if we can recover input data from the output. Hence NOT gate.

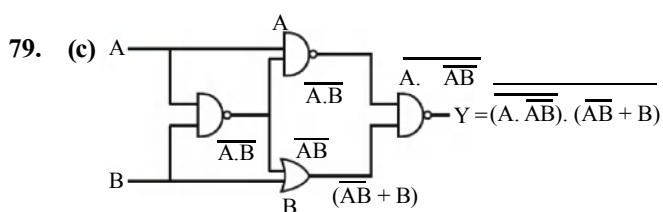
77. (c)

A	B	$(A + B)$	$(A + B) \cdot A$	$\overline{(A + B) \cdot A}$
0	0	0	0	1
0	1	1	0	1
1	0	1	1	0
1	1	1	1	0

78. (b) Truth table $\rightarrow$

The output is of OR-gate

A	B	$\overline{A}$	$\overline{B}$	$\overline{A \cdot B}$
0	0	1	1	0
0	1	1	0	1
1	0	0	1	1
1	1	0	0	1



$$Y = (\overline{A \cdot AB}) + (\overline{AB + B})$$

$$= A \cdot \overline{AB} + AB \cdot \overline{B}$$

$$= A \cdot (\overline{A} + \overline{B}) + AB \cdot \overline{B}$$

$$= A\overline{B}$$

80. (c) From the given logic circuit,

$$p = \overline{x} + y$$

$$Q = \overline{\overline{y}} \cdot x = y + \overline{x}$$

$$\text{Output, } R = \overline{P + Q}$$

To make output 1

$P + Q$ must be '0'

So, $x = 1, y = 0$

81. (c) Truth table of the circuit is as follows

x	y	$\overline{x}$	$a = x \cdot y$	$b = \overline{x} \cdot y$	$z = \overline{a \cdot b}$
0	0	1	0	0	1
0	1	1	0	1	1
1	0	0	0	0	1
1	1	0	1	0	1

82. (a) In case of an 'OR' gate the input is zero when all inputs are zero. If any one input is '1', then the output is '1'.

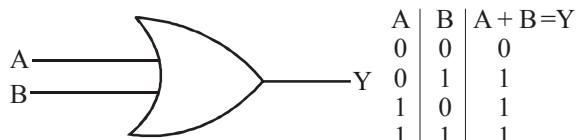
83. (c) Truth table for given logical circuit

a	b	$(a + b)$	c	$Y = (a + b) \cdot c$
0	0	0	0	0
0	1	1	1	1
1	0	1	1	1
1	1	1	0	0

Output of OR gate must be 1 and $c = 1$

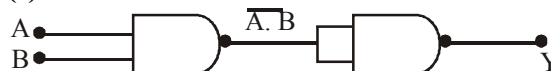
So, $a = 1, b = 0$ or $a = 0, b = 1$.

84. (a) It represents OR-Gate.



85. (a) AND Gate

86. (a)



$$Y = \overline{AB} \cdot \overline{AB} = \overline{AB} + \overline{AB} = AB + AB = AB$$

In this case output Y is equivalent to AND gate.

87. (a) For circuit 1

$$A \cdot B = \overline{Y + \overline{A}} = C$$

A	B	$\overline{Y + \overline{A}} = C$
0	0	0
0	1	0
1	0	0
1	1	0

88. (a) In the given system all four gate is NOR gate

Truth Table

A	B	$(y' = \overline{A + B})$	$y'' = \overline{(\overline{A + y'})}$	$y''' = \overline{(A + y'')}$	$y = \overline{y'' + y''}$
0	0	1	0	0	1
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

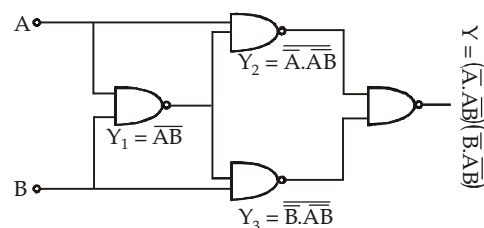
i.e.,

A	B	y
0	0	1
0	1	0
1	0	0
1	1	1

89. (a) The output at C corresponds to A NAND B or

$$\overline{A \cdot B} = C$$

90. (a)



By expanding this Boolean expression

$$Y = A \cdot \overline{B} + B \cdot \overline{A}$$

Thus the truth table for this expression should be (a).

91. (c) Truth table is as shown :

A	B	$\overline{A}$	$\overline{B}$	$\overline{A + B}$	$\overline{\overline{A + B}}$
0	0	1	1	1	0
0	1	1	0	1	0
1	0	0	1	1	0
1	1	0	0	0	1

Thus the combination of two NOT gates and one NOR gate is equivalent to a AND gate.

92. (a) NOR gate is the combination of NOT and OR gate.

Boolean expression for NOR gate is

$$Y = \overline{A + B}$$

93. (b) When either of A or B is 1 i.e. closed then lamp will glow.

In this case, Truth table

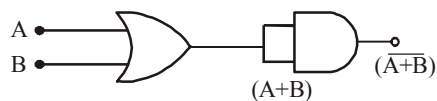
Inputs		Output
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

This represents OR gate.

94. (b) When both inputs of NAND gate are jointed to form a single input, it behaves as NOT gate

OR + NOT = NOR.

$$(\overline{A + B}) = \text{NOR gate}$$



95. (a) The final boolean expression of these gates is,

$$X = (\overline{A \cdot B}) = \overline{\overline{A + B}} = A + B \Rightarrow \text{OR gate}$$

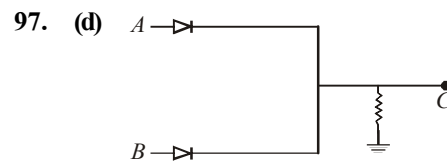
It means OR gate is formed.

96. (d) The final boolean expression

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

$$Y = (\overline{A + B}) = \overline{\overline{A \cdot B}} = A \cdot B.$$

Thus, it is an AND gate for which truth table is



The truth table for the above circuit is :

A	B	C
1	1	1
1	0	1
0	1	1
0	0	0

when either A or B conducts, the gate conducts. It means $C = A + B$ which is for OR gate.

Communication Systems



TOPIC 1 Communication Systems



- An amplitude modulated wave is represented by the expression $v_m = 5(1 + 0.6 \cos 6280t) \sin(211 \times 10^4 t)$ volts. The minimum and maximum amplitudes of the amplitude modulated wave are, respectively : **[Sep. 02, 2020 (I)]**
 - $\frac{3}{2} \text{ V}, 5 \text{ V}$
 - $\frac{5}{2} \text{ V}, 8 \text{ V}$
 - $5 \text{ V}, 8 \text{ V}$
 - $3 \text{ V}, 5 \text{ V}$
- In an amplitude modulator circuit, the carrier wave is given by, $C(t) = 4 \sin(20000 \pi t)$ while modulating signal is given by, $m(t) = 2 \sin(2000 \pi t)$. The values of modulation index and lower side band frequency are : **[12 April 2019 II]**
 - 0.5 and 10 kHz
 - 0.4 and 10 kHz
 - 0.3 and 9 kHz
 - 0.5 and 9 kHz
- A message signal of frequency 100 MHz and peak voltage 100 V is used to execute amplitude modulation on a carrier wave of frequency 300 GHz and peak voltage 400 V. The modulation index and difference between the two side band frequencies are : **[10 April 2019 II]**
 - $4; 1 \times 10^8 \text{ Hz}$
 - $4; 2 \times 10^8 \text{ Hz}$
 - $0.25; 2 \times 10^8 \text{ Hz}$
 - $0.25; 1 \times 10^8 \text{ Hz}$
- A signal $A \cos \omega t$ is transmitted using $v_0 \sin \omega_0 t$ as carrier wave. The correct amplitude modulated (AM) signal is: **[9 April 2019 I]**
 - $v_0 \sin \omega_0 t + \frac{A}{2} \sin(\omega_0 - \omega)t + \frac{A}{2} \sin(\omega_0 + \omega)t$
 - $v_0 \sin[\omega_0(1 + 0.01 A \sin \omega t)t]$
 - $v_0 \sin \omega_0 t + A \cos \omega t$
 - $(v_0 + A) \cos \omega t \sin \omega_0 t$
- The physical sizes of the transmitter and receiver antenna in a communication system are: **[9 April 2019 II]**
 - independent of both carrier and modulation frequency
 - inversely proportional to carrier frequency
 - inversely proportional to modulation frequency
 - proportional to carrier frequency
- The wavelength of the carrier waves in a modern optical fiber communication network is close to : **[8 April 2019 I]**
 - 2400 nm
 - 1500 nm
 - 600 nm
 - 900 nm
- In a line of sight radio communication, a distance of about 50 km is kept between the transmitting and receiving antennas. If the height of the receiving antenna is 70 m, then the minimum height of the transmitting antenna should be : **[8 April 2019 II]**
(Radius of the Earth = $6.4 \times 10^6 \text{ m}$).
 - 20 m
 - 51 m
 - 32 m
 - 40 m
- A 100 V carrier wave is made to vary between 160 V and 40 V by a modulating signal. What is the modulation index? **[12 Jan. 2019 I]**
 - 0.3
 - 0.5
 - 0.6
 - 0.4
- To double the covering range of a TV transmission tower, its height should be multiplied by: **[12 Jan 2019 II]**
 - $\frac{1}{\sqrt{2}}$
 - 2
 - 4
 - $\sqrt{2}$
- An amplitude modulated signal is given by $V(t) = 10[1 + 0.3 \cos(2.2 \times 10^4 t)] \sin(5.5 \times 10^5 t)$. Here t is in seconds. The sideband frequencies (in kHz) are, [Given $\pi = 22/7$] **[11 Jan 2019 II]**
 - 1785 and 1715
 - 178.5 and 171.5
 - 89.25 and 85.75
 - 892.5 and 857.5
- An amplitude modulated signal is plotted below :

 Which one of the following best describes the above signal? **[11 Jan. 2019 II]**
 - $(9 + \sin(2.5 \pi \times 10^5 t)) \sin(2\pi \times 10^4 t) \text{ V}$
 - $(1 + 9 \sin(2\pi \times 10^4 t)) \sin(2.5 \pi \times 10^5 t) \text{ V}$

- (c) $(9 + \sin(2\pi \times 10^4 t)) \sin(2.5\pi \times 10^5 t)$ V
 (d) $(9 + \sin(4\pi \times 10^4 t)) \sin(5\pi \times 10^5 t)$ V
12. A TV transmission tower has a height of 140 m and the height of the receiving antenna is 40 m. What is the maximum distance upto which signals can be broadcasted from this tower in LOS (Line of Sight) mode? (Given : radius of earth = 6.4×10^6 m). **[10 Jan. 2019 I]**
 (a) 65 km (b) 48 km (c) 80 km (d) 40 km
13. The modulation frequency of an AM radio station is 250 kHz, which is 10% of the carrier wave. If another AM station approaches you for license what broadcast frequency will you allot? **[10 Jan. 2019 I]**
 (a) 2750 kHz (b) 2900 kHz
 (c) 2250 kHz (d) 2000 kHz
14. In a communication system operating at wavelength 800 nm, only one percent of source frequency is available as signal bandwidth. The number of channels accommodated for transmitting TV signals of band width 6 MHz are (Take velocity of light $c = 3 \times 10^8$ m/s, $h = 6.6 \times 10^{-34}$ J-s) **[9 Jan. 2019 II]**
 (a) 3.75×10^6 (b) 3.86×10^6
 (c) 6.25×10^5 (d) 4.87×10^5
15. A telephonic communication service is working at carrier frequency of 10 GHz. Only 10% of it is utilized for transmission. How many telephonic channels can be transmitted simultaneously if each channel requires a bandwidth of 5 kHz? **[2018]**
 (a) 2×10^3 (b) 2×10^4 (c) 2×10^5 (d) 2×10^6
16. A carrier wave of peak voltage 14 V is used for transmitting a message signal. The peak voltage of modulating signal given to achieve a modulation index of 80% will be: **[2018]**
 (a) 11.2 V (b) 7 V (c) 22.4 V (d) 28 V
17. The number of amplitude modulated broadcast stations that can be accommodated in a 300 kHz band width for the highest modulating frequency 15 kHz will be: **[Online April 15, 2018]**
 (a) 20 (b) 10 (c) 8 (d) 15
18. The carrier frequency of a transmitter is provided by a tank circuit of a coil of inductance $49 \mu\text{H}$ and a capacitance of 2.5 nF . It is modulated by an audio signal of 12 kHz. The frequency range occupied by the side bands is: **[Online April 15, 2018]**
 (a) 18 kHz–30 kHz (b) 63 kHz–75 kHz
 (c) 442 kHz–466 kHz (d) 13482 kHz–13494 kHz
19. In amplitude modulation, sinusoidal carrier frequency used is denoted by ω_c and the signal frequency is denoted by ω_m . The bandwidth ($\Delta\omega_m$) of the signal is such that $\Delta\omega_m < \omega_c$. Which of the following frequencies is not contained in the modulated wave? **[2017]**
 (a) $\omega_m + \omega_c$ (b) $\omega_c - \omega_m$
 (c) ω_m (d) ω_c
20. A signal is to be transmitted through a wave of wavelength λ , using a linear antenna. The length l of the antenna and effective power radiated P_{eff} will be given respectively as :
 (K is a constant of proportionality) **[Online April 9, 2017]**
 (a) $\lambda, P_{\text{eff}} = K \left(\frac{1}{\lambda}\right)^2$ (b) $\frac{\lambda}{8}, P_{\text{eff}} = K \left(\frac{1}{\lambda}\right)$
 (c) $\frac{\lambda}{16}, P_{\text{eff}} = K \left(\frac{1}{\lambda}\right)^3$ (d) $\frac{\lambda}{5}, P_{\text{eff}} = K \left(\frac{1}{\lambda}\right)^{\frac{1}{2}}$
21. A signal of frequency 20 kHz and peak voltage of 5 Volt is used to modulate a carrier wave of frequency 1.2 MHz and peak voltage 25 Volts. Choose the correct statement. **[Online April 8, 2017]**
 (a) Modulation index = 5, side frequency bands are at 1400 kHz and 1000 kHz
 (b) Modulation index = 5, side frequency bands are at 21.2 kHz and 18.8 kHz
 (c) Modulation index = 0.8, side frequency bands are at 1180 kHz and 1220 kHz
 (d) Modulation index = 0.2, side frequency bands are at 1220 kHz and 1180 kHz
22. Choose the correct statement : **[2016]**
 (a) In frequency modulation the amplitude of the high frequency carrier wave is made to vary in proportion to the amplitude of the audio signal.
 (b) In frequency modulation the amplitude of the high frequency carrier wave is made to vary in proportion to the frequency of the audio signal.
 (c) In amplitude modulation the amplitude of the high frequency carrier wave is made to vary in proportion to the amplitude of the audio signal.
 (d) In amplitude modulation the frequency of the high frequency carrier wave is made to vary in proportion to the amplitude of the audio signal.
23. A modulated signal $C_m(t)$ has the form $C_m(t) = 30 \sin 300\pi t + 10 (\cos 200\pi t - \cos 400\pi t)$. The carrier frequency f_c , the modulating frequency (message frequency) f_m and the modulation index μ are respectively given by : **[Online April 10, 2016]**
 (a) $f_c = 200 \text{ Hz}; f_m = 50 \text{ Hz}; \mu = \frac{1}{2}$
 (b) $f_c = 150 \text{ Hz}; f_m = 50 \text{ Hz}; \mu = \frac{2}{3}$
 (c) $f_c = 150 \text{ Hz}; f_m = 30 \text{ Hz}; \mu = \frac{1}{3}$
 (d) $f_c = 200 \text{ Hz}; f_m = 30 \text{ Hz}; \mu = \frac{1}{2}$

24. An audio signal consists of two distinct sounds: one a human speech signal in the frequency band of 200 Hz to 2700 Hz, while the other is a high frequency music signal in the frequency band of 10200 Hz to 15200 Hz. The ratio of the AM signal bandwidth required to send both the signals together to the AM signal bandwidth required to send just the human speech is : **[Online April 9, 2016]**

(a) 2 (b) 5 (c) 6 (d) 3

25. A signal of 5 kHz frequency is amplitude modulated on a carrier wave of frequency 2 MHz. The frequencies of the resultant signal is/are : **[2015]**

(a) 2005 kHz, 2000 kHz and 1995 kHz
(b) 2000 kHz and 1995 kHz
(c) 2 MHz only
(d) 2005 kHz and 1995 kHz

26. Long range radio transmission is possible when the radio waves are reflected from the ionosphere. For this to happen the frequency of the radio waves must be in the range:

[Online April 19, 2014]

(a) 80 - 150 MHz (b) 8 - 25 MHz
(c) 1 - 3 MHz (d) 150 - 1500 kHz

27. For sky wave propagation, the radio waves must have a frequency range in between: **[Online April 12, 2014]**

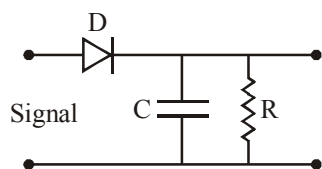
(a) 1 MHz to 2 MHz (b) 5 MHz to 25 MHz
(c) 35 MHz to 40 MHz (d) 45 MHz to 50 MHz

28. A transmitting antenna at the top of a tower has height 32 m and height of the receiving antenna is 50 m. What is the maximum distance between them for satisfactory communication in line of sight (LOS) mode?

[Online April 9, 2014]

(a) 55.4 km (b) 45.5 km (c) 54.5 km (d) 455 km

29. A diode detector is used to detect an amplitude modulated wave of 60% modulation by using a condenser of capacity 250 picofarad in parallel with a load resistance 100 kilo ohm. Find the maximum modulated frequency which could be detected by it. **[2013]**



(a) 10.62 MHz (b) 10.62 kHz
(c) 5.31 MHz (d) 5.31 kHz

30. Which of the following modulated signal has the best noise-tolerance ? **[Online April 25, 2013]**

(a) Long-wave (b) Short-wave
(c) Medium-wave (d) Amplitude-modulated

31. Which of the following statement is NOT correct?

[Online April 23, 2013]

(a) Ground wave signals are more stable than the sky wave signals.

- (b) The critical frequency of an ionospheric layer is the highest frequency that will be reflected back by the layer when it is vertically incident.
(c) Electromagnetic waves of frequencies higher than about 30 MHz cannot penetrate the ionosphere.
(d) Sky wave signals in the broadcast frequency range are stronger at night than in the day time.

32. This question has Statement-1 and Statement-2. Of the four choices given after the Statements, choose the one that best describes the two Statements.

Statement-1: Short wave transmission is achieved due to the total internal reflection of the e-m wave from an appropriate height in the ionosphere.

Statement-2: Refractive index of a plasma is independent of the frequency of e-m waves. **[Online April 22, 2013]**

- (a) Statement-1 is true, Statement-2 is false.
(b) Statement-1 is false, Statement-2 is true.
(c) Statement-1 is true, Statement-2 is true but Statement -2 is **not** the correct explanation of statement-1.
(d) Statement-1 is true, Statement-2 is true and Statement -2 is the correct explanation of Statement-1.

33. If a carrier wave $c(t) = A \sin \omega_c t$ is amplitude modulated by a modulator signal $m(t) = A \sin \omega_m t$ then the equation of modulated signal $[C_m(t)]$ and its modulation index are respectively **[Online April 9, 2013]**

(a) $C_m(t) = A(1 + \sin \omega_m t) \sin \omega_c t$ and 2
(b) $C_m(t) = A(1 + \sin \omega_m t) \sin \omega_m t$ and 1
(c) $C_m(t) = A(1 + \sin \omega_m t) \sin \omega_c t$ and 1
(d) $C_m(t) = A(1 + \sin \omega_c t) \sin \omega_m t$ and 2

34. A radar has a power of 1 kW and is operating at a frequency of 10 GHz. It is located on a mountain top of height 500 m. The maximum distance upto which it can detect object located on the surface of the earth

(Radius of earth = 6.4×10^6 m) is : **[2012]**

(a) 80 km (b) 16 km (c) 40 km (d) 64 km

35. A radio transmitter transmits at 830 kHz. At a certain distance from the transmitter magnetic field has amplitude 4.82×10^{-11} T. The electric field and the wavelength are respectively **[Online May 26, 2012]**

(a) 0.014 N/C, 36 m (b) 0.14 N/C, 36 m
(c) 0.14 N/C, 360 m (d) 0.014 N/C, 360 m

36. Given the electric field of a complete amplitude modulated wave as

$$\vec{E} = \hat{i} E_c \left(1 + \frac{E_m}{E_c} \cos \omega_m t \right) \cos \omega_c t.$$

Where the subscript c stands for the carrier wave and m for the modulating signal. The frequencies present in the modulated wave are **[Online May 19, 2012]**

(a) ω_c and $\sqrt{\omega_c^2 + \omega_m^2}$
(b) $\omega_c, \omega_c + \omega_m$ and $\omega_c - \omega_m$

- (c) ω_c and ω_m
(d) ω_c and $\sqrt{\omega_c \omega_m}$
37. A 10 kW transmitter emits radio waves of wavelength 500 m. The number of photons emitted per second by the transmitter is of the order of **[Online May 12, 2012]**
(a) 10^{37} (b) 10^{31} (c) 10^{25} (d) 10^{43}
38. This question has Statement – 1 and Statement – 2. Of the four choices given after the statements, choose the one that best describes the two statements. **[2011]**
Statement – 1 : Sky wave signals are used for long distance radio communication. These signals are in general, less stable than ground wave signals.
Statement – 2 : The state of ionosphere varies from hour to hour, day to day and season to season.
- (a) Statement–1 is true, Statement–2 is true, Statement–2 is the correct explanation of Statement–1.
(b) Statement–1 is true, Statement–2 is true, Statement–2 is not the correct explanation of Statement – 1.
(c) Statement – 1 is false, Statement – 2 is true.
(d) Statement – 1 is true, Statement – 2 is false.
39. Which of the following four alternatives is not correct ? We need modulation : **[2011 RS]**
(a) to reduce the time lag between transmission and reception of the information signal
(b) to reduce the size of antenna
(c) to reduce the fractional band width, that is the ratio of the signal band width to the centre frequency
(d) to increase the selectivity



Hints & Solutions



1. (b) From the given expression,
 $V_m = 5(1 + 0.6 \cos 6280t) \sin(211 \times 10^4 t)$
 Modulation index, $\mu = 0.6$

$$\therefore A_m = \mu A_c$$

$$\frac{A_{\max} + A_{\min}}{2} = A_c = 5 \quad \dots(i)$$

$$\frac{A_{\max} - A_{\min}}{2} = A_m = 3 \quad \dots(ii)$$

From equation (i) + (ii),

Maximum amplitude, $A_{\max} = 8$.

From equation (i) - (ii),

Minimum amplitude $A_{\min} = 2$.

2. (d) Modulation index, $\mu = \frac{A_m}{A_c} = \frac{2}{4} = 0.5$

$$\text{Given, } f_c = \frac{20000\pi}{2\pi} = 10000 \text{ Hz.}$$

$$\text{and } f_m = \frac{2000\pi}{2\pi} = 1000 \text{ Hz.}$$

$$\therefore \text{LSB} = f_c - f_m = 10000 - 1000 = 9000 \text{ Hz.}$$

3. (c) Range of frequency = $(f_c - f_m)$ to $(f_c + f_m)$
 $\therefore$ Band width = $2f_m = 2 \times 100 \times 10^6 \text{ Hz}$
 $= 2 \times 10^8 \text{ Hz}$

$$\text{and Modulation index} = \frac{A_m}{A_c} = \frac{100}{400} = 0.25$$

4. (a)

5. (b) Size of antenna,

$$l = \frac{\lambda}{4}. \text{ As } \lambda = \frac{c}{f} \therefore l \propto \frac{1}{f}$$

6. (b) Carrier waves of wavelength 1500 nm is used in modern optical fiber communication.

7. (c) $\text{LOS} = \sqrt{2h_T R} + \sqrt{2h_R R}$

$$\text{or } 50 \times 10^3 = \sqrt{2h_T \times 6.4 \times 10^6} + \sqrt{2 \times 70 \times 6.4 \times 10^6}$$

On solving, $h_T = 32 \text{ m}$

8. (c) Maximum amplitude = $E_m + E_c = 160$

$$E_m + 100 = 160$$

$$E_m = 160 - 100 = 60$$

Modulation index,

$$\mu = \frac{E_m}{E_c} = \frac{60}{100}$$

$$\mu = 0.6$$

9. (c) As we know, Range = $\sqrt{2hR}$

therefore to double the range height 'h' should be 4 times.

10. (c) Equation given

$$V(t) = 10[1 + 0.3 \cos(2.2 \times 10^4 t)]$$

$$\sin(5.5 \times 10^5 t)$$

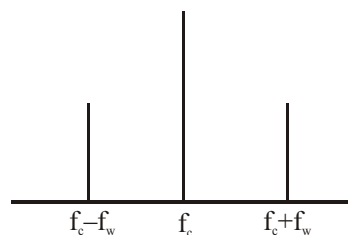
$$= 10 + 1.5[\sin(57.2 \times 10^4 t) + \sin(52.8 \times 10^4 t)]$$

$$\omega_c + \omega_w = 57.2 \times 10^4 = 2\pi f_1$$

$$f_1 = \frac{57.2 \times 10^4}{2 \times \left(\frac{22}{7}\right)} = 9.1 \times 10^4 \approx 91 \text{ KHz}$$

$$\omega_c - \omega_w = 52.8 \times 10^4$$

$$f_2 = \frac{52.8 \times 10^4}{2 \times \left(\frac{22}{7}\right)} \approx 84 \text{ KHz}$$



Upper side band frequency (f_1) is

$$f_1 = f_c - f_w = \frac{52.8 \times 10^4}{2\pi} \approx 85.00 \text{ kHz}$$

Lower side band frequency (f_2) is

$$f_2 = f_c + f_w = \frac{57.2 \times 10^4}{2\pi} \approx 90.00 \text{ kHz}$$

11. (c) After analysing the graph we may conclude that

(i) Amplitude varies as $8-10 \text{ V}$ or 9 ± 1

(ii) Two time period as

$100 \mu\text{s}$ (signal wave) & $8 \mu\text{s}$ (carrier wave)

So, equation of AM signal is

$$\left[9 \pm 1 \sin\left(\frac{2\pi t}{T_1}\right) \sin\left(\frac{2\pi t}{T_2}\right) \right]$$

$$= [9 \pm \sin(2\pi \times 10^4 t)] \sin(2.5\pi \times 10^5 t) \text{ V}$$

12. (a) Maximum distance upto which signal can be broadcasted

$$d_{\max} = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

where h_T and h_R are heights of transmission tower and receiving antenna respectively.

Putting values R, hT and hR

$$d_{\max} = \sqrt{2 \times 6.4 \times 106} \left[\sqrt{140} + \sqrt{40} \right]$$

$$\text{or, } d_{\max} \approx 65 \text{ km}$$

13. (d) According to question, modulation frequency, 250 Hz is 10% of carrier wave

$$f_{\text{carrier}} = \frac{250}{0.1} = 2500 \text{ KHZ}$$

$\therefore$ Range of signal $2500 \pm 250 \text{ KHZ} = 2250 \text{ Hz to } 2750 \text{ Hz}$ For 2000 KHZ

$$f_{\text{mod}} = 200 \text{ Hz}$$

$\therefore$ Range = 1800 KHZ to 2200 KHZ

14. (c) Frequency, $f = \frac{V}{\lambda} = \frac{3 \times 10^8}{8 \times 10^{-7}} = \frac{30}{8} \times 10^{14} \text{ Hz}$

$$= 3.75 \times 10^{14} \text{ Hz}$$

$$1\% \text{ of } f = 0.0375 \times 10^{14} \text{ Hz}$$

$$= 3.75 \times 10^{12} \text{ Hz} = 3.75 \times 10^6 \text{ MHz}$$

As we know, number of channels accommodated for transmission =

$$\frac{\text{total bandwidth of Channel}}{\text{bandwidth needed per channel}} = \frac{3.75 \times 10^6}{6} = 6.25 \times 10^5$$

15. (c) If n = no. of channels

$$10\% \text{ of } 10 \text{ GHz} = n \times 5 \text{ KHz or,}$$

$$\Rightarrow n = 2 \times 10^5$$

16. (a) Given : modulation index $m = 80\% = 0.8$

$$E_c = 14 \text{ V, } E_m = ?$$

$$\text{using, } m = \frac{E_m}{E_c} \Rightarrow E_m = m \times E_c = 0.8 \times 14 = 11.2 \text{ V}$$

17. (b) Given, modulating frequency $f_m = 15 \text{ KHz}$

$$\therefore \text{ Bandwidth of one channel} = 2f_m = 30 \text{ KHz}$$

$$\therefore \text{ No of channels accommodate} = \frac{300 \text{ KHz}}{30 \text{ KHz}} = 10$$

18. (c) Given : Inductance, $L = 49 \mu\text{H} = 49 \times 10^{-6} \text{ H}$, capacitance $C = 2.5 \text{ nF} = 2.5 \times 10^{-9} \text{ F}$

$$\text{Using } \omega = \frac{1}{\sqrt{LC}}$$

$$= \frac{1}{\sqrt{49 \times 10^{-6} \times \frac{2.5}{10} \times 10^{-9}}} = \frac{1}{7 \times 5 \times 10^{-8}} = \frac{10^8}{7 \times 5}$$

$$\text{or, } \frac{10^8}{7 \times 5} = 2\pi \times f = 2 \times \frac{22}{7} \times f \quad (\because \omega = 2\pi f)$$

$$\text{or, } f = \frac{10^7}{22} = \frac{10^4}{22} \text{ kHz} = 454.54 \text{ kHz}$$

Therefore frequency range $454.54 \pm 12 \text{ kHz}$

i.e., 442 kHz – 466 kHz

19. (c) Modulated carrier wave contains frequency ω_c and $\omega_c \pm \omega_m$

20. (a) Length of antenna = comparable to λ

Power radiated by linear antenna inversely depends on the square of wavelength and directly on the length of the antenna. Hence,

$$\text{Power } P = \mu \left(\frac{1}{\lambda} \right)^2$$

here $\mu = K$

21. (d) Modulation index (m) = $\frac{V_m}{V_0} = \frac{5}{25} = 0.2$

Given, frequency of carrier wave (f_c) = $1.2 \times 10^6 \text{ Hz}$
= 1200 kHz.

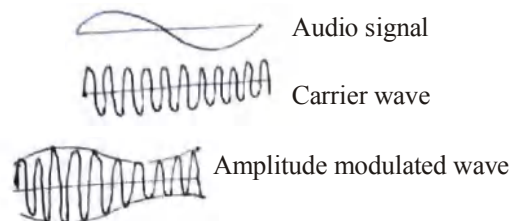
Frequency of signal (f_0) = 20 kHz.

Side frequency bands = $f_c \pm f_0$

$$f_1 = 1200 - 20 = 1180 \text{ kHz}$$

$$f_2 = 1200 + 20 = 1220 \text{ kHz}$$

22. (c) In amplitude modulation, the amplitude of the high frequency carrier wave made to vary in proportional to the amplitude of audio signal.



23. (b) Comparing the given equation with standard modulated signal wave equation, $m = A_c \sin \omega_c t + \frac{\mu A_c}{2}$

$$\cos(\omega_c - \omega_s)t - \frac{\mu A_c}{2} \cos(\omega_c + \omega_s)t$$

$$\mu \frac{A_c}{2} = 10 \Rightarrow \mu = \frac{2}{3} \text{ (modulation index)}$$

$$A_c = 30$$

$$\omega_c - \omega_s = 200\pi$$

$$\omega_c + \omega_s = 400\pi$$

$$\Rightarrow f_c = 150, f_s = 50 \text{ Hz.}$$

24. (c) Ratio of AM signal Bandwidths

$$= \frac{15200 - 200}{2700 - 200} = \frac{15000}{2500} = 6.$$

25. (a) Amplitude modulated wave consists of three frequencies are $\omega_c + \omega_m$, ω_c , $\omega_c - \omega_m$

i.e. 2005 kHz, 2000 kHz, 1995 kHz

26. (b) Frequency of radio waves for sky wave propagation is 2 MHz to 30 MHz.

27. (b) Sky wave propagation is suitable for frequency range 5 MHz to 25 MHz.

28. (b) Given : $h_R = 32$ m

$$h_T = 50 \text{ m}$$

Maximum distance, $d_M = ?$

$$\text{Applying, } d_M = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

$$= \sqrt{2 \times 6.4 \times 10^6 \times 50} + \sqrt{2 \times 6.4 \times 10^6 \times 32} = 45.5 \text{ km}$$

29. (b) **Given :** Resistance $R = 100$ kilo ohm

$$= 100 \times 10^3 \Omega$$

Capacitance $C = 250$ picofarad

$$= 250 \times 10^{-12} \text{ F}$$

$$\tau = RC = 100 \times 10^3 \times 250 \times 10^{-12} \text{ sec}$$

$$= 2.5 \times 10^7 \times 10^{-12} \text{ sec}$$

$$= 2.5 \times 10^{-5} \text{ sec}$$

The higher frequency which can be detected with tolerable distortion is

$$f = \frac{1}{2\pi m_d RC} = \frac{1}{2\pi \times 0.6 \times 2.5 \times 10^{-5}} \text{ Hz}$$

$$= \frac{100 \times 10^4}{25 \times 1.2\pi} \text{ Hz} = \frac{4}{1.2\pi} \times 10^4 \text{ Hz}$$

$$= 10.61 \text{ KHz}$$

This condition is obtained by applying the condition that rate of decay of capacitor voltage must be equal or less than the rate of decay modulated signal voltage for proper detection of modulated signal.

30. (b) Short-wave has the best noise tolerance.

31. (c) Above critical frequency (f_c), an electromagnetic wave penetrates the ionosphere and is not reflected by it.

32. (a) Effective refractive index of the ionosphere

$$n_{\text{eff}} = n_0 \left[1 - \frac{80.5N}{f^2} \right]^{1/2}$$

where f is the frequency of em waves

33. (c) Modulation index

$$m_a = \frac{E_m}{E_c} = \frac{A}{A} = 1$$

Equation of modulated signal $[C_m(t)]$

$$= E_{(c)} + m_a E_{(c)} \sin \omega_m t$$

$$= A (1 + \sin \omega_c t) \sin \omega_m t$$

$$(\text{As } E_{(c)} = A \sin \omega_c t)$$

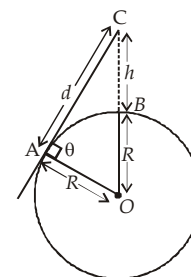
34. (a) Let d is the maximum distance, upto which it can detect the objects

From ΔAOC

$$OC^2 = AC^2 + AO^2$$

$$(h+R)^2 = d^2 + R^2$$

$$\Rightarrow d^2 = (h+R)^2 - R^2$$



$$d = \sqrt{(h+R)^2 - R^2} ; d = \sqrt{h^2 + 2hR}$$

$$d = \sqrt{500^2 + 2 \times 6.4 \times 10^6} = 80 \text{ km}$$

35. (d) Frequency of EM wave $\nu = 830$ KHz

$$= 830 \times 10^3 \text{ Hz.}$$

$$\text{Magnetic field, } B = 4.82 \times 10^{-11} \text{ T}$$

As we know, frequency, $\nu = \frac{c}{\lambda}$

$$\text{or } \lambda = \frac{c}{\nu} = \frac{3 \times 10^8}{830 \times 10^3}$$

$$\lambda \approx 360 \text{ m}$$

$$\text{And, } E = BC = 4.82 \times 10^{-11} \times 3 \times 10^8$$

$$= 0.014 \text{ N/C}$$

36. (b) The frequencies present in amplitude modulated wave are :

$$\text{Carrier frequency} = \omega_c$$

$$\text{Upper side band frequency} = \omega_c + \omega_m$$

$$\text{Lower side band frequency} = \omega_c - \omega_m$$

37. (b) Power = $\frac{nhc}{\lambda}$

(where, n = no. of photons per second)

$$\Rightarrow n = \frac{10 \times 10^3 \times 500}{6.6 \times 10^{-34} \times 3 \times 10^8} \approx 10^{31}$$

38. (b) For long distance communication, sky wave signals are used.

Also, the state of ionosphere varies every time.

So, both statements are correct.

39. (a) Low frequencies cannot be transmitted to long distances. Therefore, they are super imposed on a high frequency carrier signal by a process known as modulation. Speed of electro-magnetic waves will not change due to modulation. So there will be time lag between transmission and reception of the information signal.

MOCK TEST

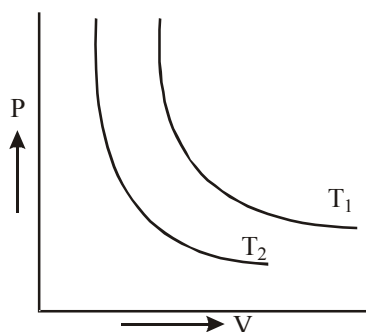
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PART-I (Multiple Choice Questions)

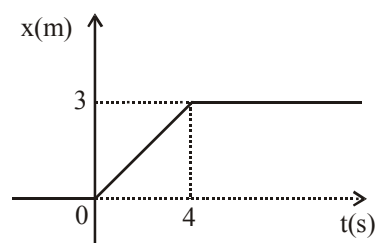
1. A d.c. supply of 120V is connected to a large resistance X. A voltmeter of resistance $10\text{k}\Omega$ placed in series in the circuit reads 4V. What is the value of X?
(a) $190\text{k}\Omega$ (b) $90\text{k}\Omega$
(c) $290\text{k}\Omega$ (d) $390\text{k}\Omega$
2. The period of the satellite of the earth orbiting very near to the surface of the earth is T_0 . What is the period of the geostationary satellite in terms of T_0 ?
(a) $\frac{T_0}{\sqrt{7}}$ (b) $\sqrt{7}T_0$
(c) $7T_0$ (d) $7\sqrt{7}T_0$
3. In "Al" and "Si". If temperature is changed from normal temperature to 70 K then
(a) The resistance of Al will increase and that of Si will decrease
(b) The resistance of Al will decrease and that of Si will increase
(c) Resistance of both decrease
(d) Resistance of both increase
4. When white light passes through a dispersive medium, it breaks up into various colours. Which of the following statements is true?
(a) Velocity of light for violet is greater than the velocity of light for red colour.
(b) Velocity of light for violet is less than the velocity of light for red.
(c) Velocity of light is the same for all colours
(d) Velocity of light for different colours has nothing to do with the phenomenon of dispersion
5. An engine has an efficiency of $\frac{1}{6}$. When the temperature of sink is reduced by 62°C , its efficiency is doubled. Temperatures of source and sink are
(a) $99^\circ\text{C}, 37^\circ\text{C}$ (b) $124^\circ\text{C}, 62^\circ\text{C}$
(c) $37^\circ\text{C}, 99^\circ\text{C}$ (d) $62^\circ\text{C}, 124^\circ\text{C}$

 Space for Rough Work 

6. Two isothermals are shown in figure at temperatures T_1 and T_2 . Which of the following relations is correct?

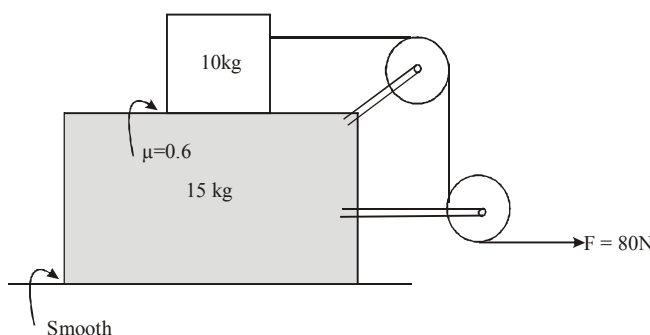


- (a) $T_1 > T_2$ (b) $T_1 < T_2$
 (c) $T_1 = T_2$ (d) $T_1 = \frac{1}{2} T_2$
7. During the experiment to determine the resistivity of a wire by a metre bridge, the jockey is moved gently along the wire from left to right to
- (a) find a deflection in the galvanometer towards left
 (b) find a deflection in the galvanometer towards right
 (c) find no deflection in the galvanometer
 (d) find a maximum deflection in the galvanometer
8. The circular head of a screw gauge is divided into 200 divisions and move 1 mm ahead in one revolution. If the same instrument has a zero error of -0.05 mm and the reading on the main scale in measuring diameter of a wire is 6 mm and that on circular scale is 45. The diameter of the wire is
- (a) 6.275 mm (b) 6.375 mm (c) 5.75 mm (d) 5.50 mm
9. The adjacent figure shows the position graph of one dimensional motion of a particle of mass 4 kg. The impulse at $t=0$ s and $t=4$ is given respectively as:



- (a) 0,0 (b) $0, -3 \text{ kg ms}^{-1}$
 (c) $+3 \text{ kg ms}^{-1}, 0$ (d) $+3 \text{ kg ms}^{-1}, -3 \text{ kg ms}^{-1}$
10. In case of a p-n junction diode at high value of reverse bias, the current rises sharply. The value of reverse bias is known as
- (a) cut off voltage (b) zener voltage
 (c) inverse voltage (d) critical voltage
11. A photon materializes into an electron-positron pair. The kinetic energy of the electron is found to be 0.19 MeV. What was the energy of the photon?
- (a) 0.38 MeV (b) 0.70 MeV
 (c) 1.40 MeV (d) None
12. If 10% of a radioactive material decays in 5 days, then the amount of the original material left after 20 days is approximately
- (a) 60% (b) 65%
 (c) 70% (d) 75%
13. Lights of two different frequencies, whose photons have energies 1 eV and 2.5 eV respectively, successively illuminate a metal whose work function is 0.5 eV. The ratio of the maximum speeds of the emitted electrons will be
- (a) 1:5 (b) 1:4
 (c) 1:2 (d) 1:1

14. A block of mass 15 kg is placed over a frictionless horizontal surface. Another block of mass 10 kg is placed over it, that is connected with a light string passing over two pulleys fastened to the 15 kg block. A force $F = 80\text{ N}$ is applied horizontally to the free end of the string. Friction coefficient between two blocks is 0.6. The portion of the string between 10 kg block and the upper pulley is horizontal. Pulley, string and connecting rods are massless. (Take $g = 10\text{ m/s}^2$)



The magnitude of acceleration of the 10 kg block is

- (a) 3.2 m/s^2 (b) 2.0 m/s^2 (c) 1.6 m/s^2 (d) 0.8 m/s^2

15. A body of mass 5 kg under the action of constant force

$\vec{F} = F_x \hat{i} + F_y \hat{j}$ has velocity at $t = 0\text{ s}$ as $\vec{v} = (6\hat{i} - 2\hat{j})\text{ m/s}$

and at $t = 10\text{ s}$ as $\vec{v} = +6\hat{j}\text{ m/s}$. The force $\vec{F}$ is:

(a) $(-3\hat{j} + 4\hat{j})\text{ N}$ (b) $\left(-\frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}\right)\text{ N}$

(c) $(3\hat{i} - 4\hat{j})\text{ N}$ (d) $\left(\frac{3}{5}\hat{i} - \frac{4}{5}\hat{j}\right)\text{ N}$

16. A plate of mass (M) is placed on a horizontal frictionless surface and a body of mass (m) is placed on this plate. The coefficient of dynamic friction between this body and the plate is μ . If a force $3\mu mg$ is applied to the body of mass (m) along the horizontal, the acceleration of the plate will be

(a) $\frac{\mu m}{M}g$ (b) $\frac{\mu mg}{M+m}$ (c) $\frac{3\mu mg}{M}$ (d) $\frac{2\mu mg}{M+m}$

17. Two rods of length d_1 and d_2 and coefficients of thermal conductivities K_1 and K_2 are kept touching each other. Both have the same area of cross-section the equivalent of thermal conductivity is

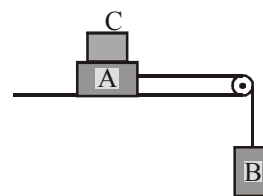
(a) $K_1 + K_2$ (b) $K_1 d_1 + K_2 d_2$
(c) $\frac{d_1 K_1 + d_2 K_2}{d_1 + d_2}$ (d) $\frac{d_1 + d_2}{(d_1 / K_1 + d_2 / K_2)}$

18. A particle starts S.H.M. from the mean position. Its amplitude is a and total energy E . At one instant its kinetic energy is $3E/4$, its displacement at this instant is

(a) $y = \frac{a}{\sqrt{2}}$ (b) $y = \frac{a}{2}$

(c) $y = \frac{a}{\sqrt{3/2}}$ (d) $y = a$

19. Two long parallel wires P and Q are both perpendicular to the plane of the paper with distance of 5 m between them. If P and Q carry current of 2.5 amp and 5 amp respectively in the same direction, then the magnetic field at a point half-way between the wires is
- (a) $\frac{3\mu_0}{2\pi}$ (b) $\frac{\mu_0}{\pi}$
- (c) $\frac{\sqrt{3}\mu_0}{2\pi}$ (d) $\frac{\mu_0}{2\pi}$
20. In a circuit L, C and R are connected in series with an alternating voltage source of frequency f. The current leads the voltage by 45° . The value of C is
- (a) $\frac{1}{\pi f(2\pi fL - R)}$ (b) $\frac{1}{2\pi f(2\pi fL - R)}$
- (c) $\frac{1}{\pi f(2\pi fL + R)}$ (d) $\frac{1}{2\pi f(2\pi fL + R)}$
22. The radius of curvature of a thin plano-convex lens is 10 cm (of curved surface) and the refractive index is 1.5. If the plane surface is silvered, then it behaves like a concave mirror of focal length (in cm)
23. Two masses A and B of 10 kg and 5 kg respectively are connected with a string passing over a frictionless pulley fixed at the corner of a table (as shown in figure). The coefficient of friction between the table and the block is 0.2. The minimum mass (in kg) of C that may be placed on A to prevent it from moving is equal to



24. Ammeter and voltmeter readings were recorded as 0.25 A and 0.5 V during the experiment to determine the resistance of a given wire using Ohm's law. What is the correct value of the resistance (in ohm)?
25. A wheel is rotating at 900 r.p.m. about its axis. When power is cut off it comes to rest in 1 minute. The angular retardation in (rad/s^2) is :

PART-II (Numerical Answer Questions)

21. An insect trapped in a circular groove of radius 12 cm. moves along the groove steadily and completes 7 revolutions in 100 sec. What is the linear speed (in m/s) of the motion ?

RESPONSE SHEET

- | | | | | |
|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1. (a) (b) (c) (d) | 2. (a) (b) (c) (d) | 3. (a) (b) (c) (d) | 4. (a) (b) (c) (d) | 5. (a) (b) (c) (d) |
| 6. (a) (b) (c) (d) | 7. (a) (b) (c) (d) | 8. (a) (b) (c) (d) | 9. (a) (b) (c) (d) | 10. (a) (b) (c) (d) |
| 11. (a) (b) (c) (d) | 12. (a) (b) (c) (d) | 13. (a) (b) (c) (d) | 14. (a) (b) (c) (d) | 15. (a) (b) (c) (d) |
| 16. (a) (b) (c) (d) | 17. (a) (b) (c) (d) | 18. (a) (b) (c) (d) | 19. (a) (b) (c) (d) | 20. (a) (b) (c) (d) |
| 21. <input type="text"/> | 22. <input type="text"/> | 23. <input type="text"/> | 24. <input type="text"/> | 25. <input type="text"/> |

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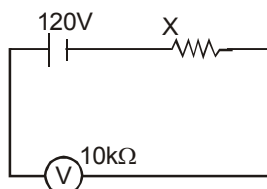
HINTS & SOLUTIONS

MOCK TEST-1

1. (c) Given $(10 \text{ k}\Omega)I = 4\text{V}$ OR $10 \text{ k}\Omega \cdot \frac{120\text{V}}{X+10\text{k}\Omega} = 4\text{V}$ OR

$$300 \text{ k}\Omega = X + 10 \text{ k}\Omega;$$

$$X = 290 \text{ k}\Omega.$$



2. (d) $T^2 \propto r^3$ law; $T_0^2 \propto R^3$ & $T_s^2 \propto (> 7R)^3$;

$$\left(\frac{T_s}{T_0}\right)^2 = 7^3; \quad T_s = 7\sqrt{7}T_0$$

3. (b) $T \downarrow$ (300K to 70K)

$$T \downarrow \quad R_{\text{metal}} \downarrow \quad R \uparrow \quad \text{semi-conductor} \\ (\text{Al}) \quad (\text{Si})$$

4. (b) As $\mu_v > \mu_r$, therefore, $v_v < v_r$.

5. (a) From $\eta = 1 - \frac{T_2}{T_1}$, $\frac{T_2}{T_1} = 1 - \eta = 1 - \frac{1}{6} = \frac{5}{6}$... (i)

$$\text{In 2nd case: } \frac{T_2 - 62}{T_1} = 1 - \eta' = 1 - \frac{2}{6} = \frac{2}{3} \dots \dots \text{(ii)}$$

$$\text{Using (i), } T_2 - 62 = \frac{2}{3}T_1 = \frac{2}{3} \times \frac{6}{5}T_2 = \frac{4}{5}T_2$$

$$\text{or } \frac{1}{5}T_2 = 62, \quad T_2 = 310 \text{ K} = 310 - 273 = 37^\circ\text{C}$$

$$T_1 = \frac{6}{5}T_2 = \frac{6}{5} \times 310 = 372 \text{ K} = 372 - 273 = 99^\circ\text{C}$$

6. (a) As isothermal at T_1 is farther from the origin than the isothermal at T_2 , therefore, $T_1 > T_2$

7. (c) The jockey is moved so as to get the null point (i.e., no deflection in the galvanometer).

8. (a) Pitch = 1 mm

Number of divisions on circular scale = 200

$$\text{L.C} = \frac{\text{Pitch}}{\text{Number of divisions on circular scale}}$$

$$= \frac{1 \text{ mm}}{200} = 0.005 \text{ mm} = 0.0005 \text{ cm}$$

Diameter of the wire = (Main scale reading + Circular scale reading $\times$ L.C.) - zero error

$$= 6 \text{ mm} + 45 \times 0.005 - (-0.05)$$

$$= 6 \text{ mm} + 0.225 \text{ mm} + 0.05 \text{ mm} = 6.275 \text{ mm}$$

9. (d) $t < 0$, $v_i = 0$ and $t > 0$, $v_f = \frac{3}{4} \text{ m/s}$

$$\therefore \text{Impulse} = m(v_f - v_i) = 4 \left(\frac{3}{4} - 0 \right)$$

$$= 3 \text{ kg ms}^{-1}$$

Little before 4 second $v = \frac{3}{4} \text{ m/s}$, little after 4

second, velocity becomes zero

Therefore, Impulse

$$= m(v_f - v_i) = 4 \left(0 - \frac{3}{4} \right) = -3 \text{ kg ms}^{-1}$$

10. (b) In reverse bias on p-n junction when high voltage is applied, electric break down of junction takes place, resulting large increase in reverse current. This high voltage applied is called zener voltage.

11. (c) Conservation of Energy,

Photon $\rightarrow e^+ + e^- + \text{energy}$

$$E_p = E_{re^+} + E_{re^-} + \text{K.E. of both } e^- + e^+$$

We know that rest mass energy of $e^+ + e^-$ is .5 MeV

$$\text{Hence } E_p = .5 + .5 + .19 \times 2 = 1.40 \text{ MeV}$$

12. (b) Let initial amount be 100 gm.

disintegrated Left

$$100 \text{ gm} \xrightarrow{5 \text{ days}} \frac{100 \times 10}{100} \quad 10 \quad 90$$

$$90 \xrightarrow{\text{Next 5 days}} \frac{90 \times 10}{100} \quad 9 \quad 81$$

$$81 \xrightarrow{\text{Next 5 days}} \frac{81 \times 10}{100} \quad 8.1 \quad \approx 73$$

$$73 \xrightarrow{\text{Next 5 days}} \frac{73 \times 10}{100} \quad 7.3 \quad \approx 65$$

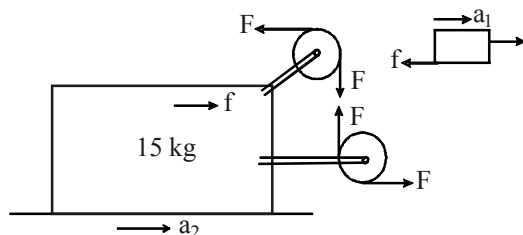
13. (c) Einstein equation $KE_{\max} = E - \text{Work function};$

$$\frac{1}{2}mv^2 = E - W$$

Using this concept,

$$\frac{\frac{1}{2}mV_1^2 \max}{\frac{1}{2}V_2^2 \max} = \frac{1 - .5}{2.5 - .5} = \frac{1}{4} \text{ or } \frac{V_1 \max}{V_2 \max} = \frac{1}{2}$$

14. (a) First, let us check upto what value of F , both blocks move together. Till friction becomes limiting, they will be moving together. Using the FBDs



10 kg block will not slip over the 15 kg block till acceleration of 15 kg block becomes maximum as it is created only by friction force exerted by 10 kg block on it.

$$a_1 > a_{2(\max)}$$

$$\frac{F - f}{10} = \frac{f}{15} \text{ for limiting condition as } f_{\max} \text{ is } 60\text{N.}$$

$$F = 100\text{ N}$$

Therefore, for $F = 80\text{N}$, both will move together.

Their combined acceleration, by applying NLM using both as system, $F = 25a$

$$a = \frac{80}{25} = 3.2 \text{ m/s}^2$$

15. (a) From question,

Mass of body, $m = 5 \text{ kg}$

Velocity at $t = 0$,

$$u = (6\hat{i} - 2\hat{j}) \text{ m/s}$$

Velocity at $t = 10\text{s}$,

$$v = +6\hat{j} \text{ m/s}$$

Force, $F = ?$

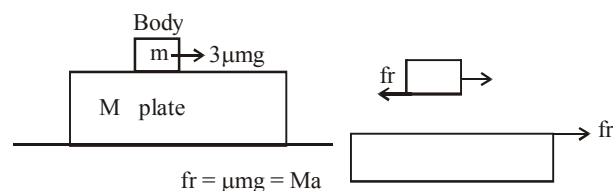
$$\text{Acceleration, } a = \frac{v - u}{t}$$

$$= \frac{6\hat{j} - (6\hat{i} - 2\hat{j})}{10} = \frac{-3\hat{i} + 4\hat{j}}{5} \text{ m/s}^2$$

Force, $F = ma$

$$= 5 \times \frac{(-3\hat{i} + 4\hat{j})}{5} = (-3\hat{i} + 4\hat{j})\text{N}$$

16. (a) $a = \frac{\mu mg}{M}$



17. (d) When two rods are connected in series

$$Q = \frac{A(T_1 - T_2)t}{\frac{d_1}{K_1} + \frac{d_2}{K_1}} = \frac{A(T_1 - T_2)t}{(d_1 + d_2)/K}$$

$$\therefore \frac{d_1 + d_2}{K} = \frac{d_1}{K_1} + \frac{d_2}{K_2}; \therefore K = \frac{(d_1 + d_2)}{\frac{d_1}{K_1} + \frac{d_2}{K_2}}$$

18. (b) Total energy, $E = \frac{1}{2}m\omega^2 a^2;$

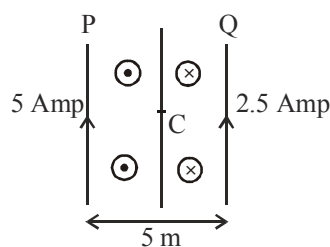
$$\text{K.E.} = \frac{3E}{4} = \frac{1}{2}m\omega^2 (a^2 - y^2).$$

$$\text{So, } \frac{3}{4} = \frac{a^2 - y^2}{a^2} \text{ or } y^2 = \frac{a^2}{4} \text{ or } y = \frac{a}{2}.$$

19. (a) When current flow in both wire in same direction then magnetic field at half way due to P wire.

$$\vec{B}_P = \frac{\mu_0 I_1}{2\pi \frac{5}{2}} = \frac{\mu_0 I_1}{\pi \cdot 5} = \frac{\mu_0}{\pi} \quad (\text{where } I_1 = 5\text{Amp})$$

The direction of B_P is downward $\odot$



Magnetic field at half way due to Q wire

$$\vec{B}_Q = \frac{\mu_0 I_2}{2\pi \frac{5}{2}} = \frac{\mu_0}{2\pi} [\text{upward } \odot]$$

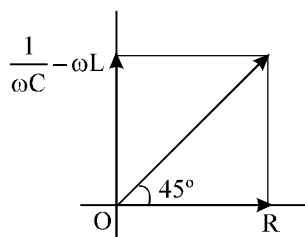
[where $I_2 = 2.5 \text{ Amp.}$]

Net magnetic field at half way

$$\vec{B} = \vec{B}_P + \vec{B}_Q = \frac{\mu_0}{\pi} + \frac{\mu_0}{2\pi} = \frac{3\mu_0}{2\pi} \text{ (downward } \odot)$$

Hence net magnetic field at midpoint = $\frac{3\mu_0}{2\pi}$

20. (d) From figure,



$$\tan 45^\circ = \frac{\frac{1}{\omega C} - \omega L}{R}$$

$$\Rightarrow \frac{1}{\omega C} - \omega L = R \Rightarrow \frac{1}{\omega C} = R + \omega L$$

$$C = \frac{1}{\omega(R + \omega L)} = \frac{1}{2\pi f(R + 2\pi fL)}$$

21. (5.3) This is an example of uniform circular motion.

$$\omega = \frac{2\pi}{T} = 2\pi n = 2\pi \times \frac{7}{100} = 0.44 \text{ rad/sec. ;}$$

$$V = R\omega = 0.44 \times 12 = 5.3 \text{ cm/sec.}$$

22. (20) The silvered plano convex lens behaves as a concave mirror; whose focal length is given by

$$\frac{1}{F} = \frac{2}{f_1} + \frac{1}{f_m}$$



If plane surface is silvered

$$f_m = \frac{R_2}{2} = \frac{\infty}{2} = \infty$$

$$\therefore \frac{1}{f_1} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = (\mu - 1) \left(\frac{1}{R} - \frac{1}{\infty} \right) = \frac{\mu - 1}{R}$$

$$\therefore \frac{1}{F} = \frac{2(\mu - 1)}{R} + \frac{1}{\infty} = \frac{2(\mu - 1)}{R} \Rightarrow F = \frac{R}{2(\mu - 1)}$$

Here $R = 20 \text{ cm}$, $\mu = 1.5$

$$\therefore F = \frac{20}{2(1.5 - 1)} = 20 \text{ cm}$$

23. (15) Let T be the tension in the string; f = frictional force between block A and table; m' = minimum mass of C.

For the just motion of block A on table

$$T = f = \mu R = \mu(m + m')g = 0.2(10 + m')g \quad \dots(i)$$

For the just motion of block B, $T = 5g \quad \dots(ii)$

From (i) and (ii), $5g = 0.2(10 + m')g$

$$\text{or } 5 = 2 + 0.2m' \text{ or } m' = \frac{5-2}{0.2} = 15 \text{ kg}$$

$$24. (2) R = \frac{V}{I} = \frac{0.5}{0.25} = 2 \Omega$$

25. ($\pi/2$) Angular retardation,

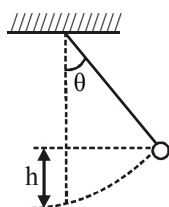
$$\alpha = \frac{\omega_2 - \omega_1}{t} = \frac{2\pi(n_2 - n_1)}{t} = \frac{2\pi(0 - 900/60)}{60} = \frac{\pi}{2} \text{ rad/s}^2.$$

MOCK TEST

2

PART-I (Multiple Choice Questions)

1. The horizontal component of impulse provided by string tension in moving from extreme point to lower most point –

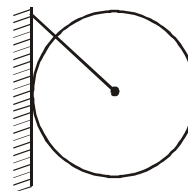


- (a) $m\sqrt{2gh}$
(b) $m\sqrt{2gh} (1 - \cos \theta)$
(c) $m\sqrt{2gh} (\cos \theta)$
(d) can't be found without knowing tension as a function of time
2. N divisions on the main scale of a vernier callipers coincide with N + 1 divisions on the vernier scale. If each division on the main scale is of a units, determine the least count of the instrument.
- (a) $\frac{a}{N+1}$ (b) $\frac{a}{N-1}$ (c) $\frac{2a}{N-1}$ (d) $\frac{2a}{N+1}$

3. A current of 4A produces a deflection of 30° in the galvanometer. The figure of merit is

- (a) 6.5 A/rad (b) 7.6 A/rad
(c) 7.5 A/rad (d) 8.0 A/rad

4. A uniform disc of radius R and mass m is connected to a wall by string of length 2R. The normal reaction of wall is –



- (a) mg (b) $mg/2$ (c) $mg/\sqrt{3}$ (d) 2mg
5. For the determination of the focal length of a convex mirror, a convex lens is required because
- (a) it is not possible to obtain the image produced by a convex mirror on the screen
(b) a convex lens has high resolving power so it helps to measure the focal length correctly
(c) a convex mirror always forms a real image which is diminished by the convex lens
(d) none of these

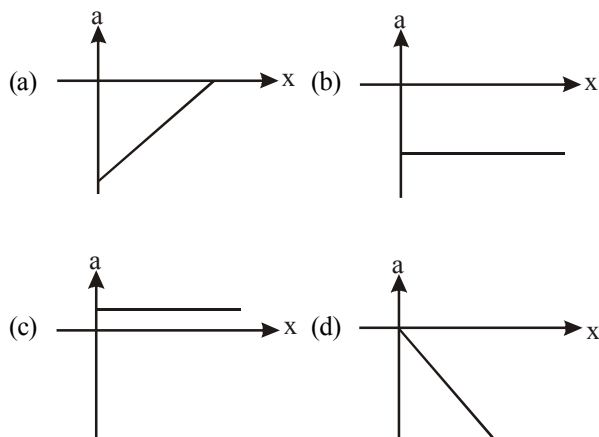
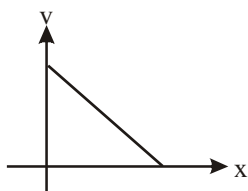


Space for Rough Work



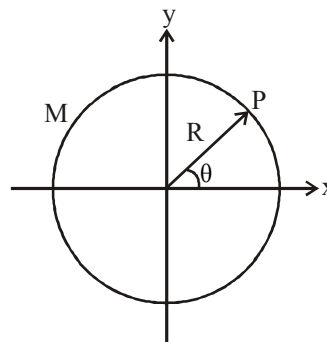
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6. A particle moves along x-axis with initial position $x = 0$. Its velocity varies with x-coordinate as shown in graph. The acceleration 'a' of this particle varies with x as –



7. Choose the wrong statement regarding the experiment to identify a diode, an LED, a resistor and a capacitor
- when pointer moves in one way when voltage is applied and doesn't move when reversed and there is light emission, the item is LED
 - when pointer moves in one way when voltage is applied and doesn't move when reversed and there is no emission of light, the item is a diode
 - when pointer moves in one way when voltage is applied and not when reversed, the item is a resistor
 - none of these

8. A ring of mass M and radius R lies in x-y plane with its centre at origin as shown. The mass distribution of ring is non-uniform such that at any point P on the ring, the mass per unit length is given by $\lambda = \lambda_0 \cos^2 \theta$ (where λ_0 is a positive constant). Then the moment of inertia of the ring about z-axis is –

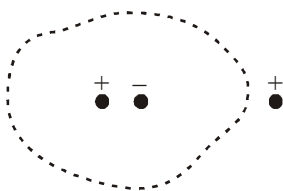


- (a) MR^2 (b) $\frac{1}{2}MR^2$ (c) $\frac{1}{2} \frac{M}{\lambda_0} R$ (d) $\frac{1}{\pi \lambda_0} R$
9. What happens when we do not stir the mixture continuously in the experiment to determine the specific heat capacity of a given solid ?
- Temperature of mixture is increased
 - Temperature of mixture is not constant
 - Temperature of mixture is constant
 - None of these
10. An ideal monatomic gas with pressure P , volume V and temperature T is expanded isothermally to a volume $2V$ and a final pressure P_i . If the same gas is expanded adiabatically to a volume $2V$, the final pressure is P_a . The ratio $\frac{P_a}{P_i}$ is
- $2^{-1/3}$ (b) $2^{1/3}$ (c) $2^{2/3}$ (d) $2^{-2/3}$

11. In a Young's double slit experiment with light of wavelength λ , fringe pattern on the screen has fringe width β . When two thin transparent glass (refractive index μ) plates of thickness t_1 and t_2 ($t_1 > t_2$) are placed in the path of the two beams respectively, the fringe pattern will shift by a distance

(a) $\frac{\beta(\mu-1)}{\lambda} \left(\frac{t_1}{t_2} \right)$ (b) $\frac{\mu\beta}{\lambda} \frac{t_1}{t_2}$
 (c) $\frac{\beta(\mu-1)}{\lambda} (t_1 - t_2)$ (d) $(\mu-1) \frac{\lambda}{\beta} (t_1 + t_2)$

12. Consider Gauss's law $\oint \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}$. Which of the following is true?



- (a) $\vec{E}$ must be the electric field due to the enclosed charge
 (b) If net charge inside the Gaussian surface = 0, then $\vec{E}$ must be zero everywhere over the Gaussian surface.
 (c) If the only charge inside the Gaussian surface is an electric dipole, then the integral is zero.
 (d) $\vec{E}$ is parallel to $d\vec{A}$ everywhere over the Gaussian surface.
13. A dipole consisting of two charges $+q$ and $-q$ joined by a massless rod of length ℓ , is seen oscillating with a small amplitude in a uniform electric field of magnitude E . The period of oscillation
- (a) is proportional to E
 (b) is proportional to $1/E$
 (c) is $\pi \sqrt{\frac{m\ell}{3qE}}$

(d) is proportional to $\frac{1}{\sqrt{E}}$ but $\neq \pi \sqrt{\frac{m\ell}{3qE}}$

14. A sinusoidal wave is propagating in negative x-direction in a string stretched along x-axis. A particle of string at $x = 2\text{m}$ is found at its mean position and it is moving in positive y-direction at $t = 1$ sec. The amplitude of the wave, the wavelength and the angular frequency of the wave are 0.1 meter, $\pi/2$ meter and 2π rad/sec respectively.

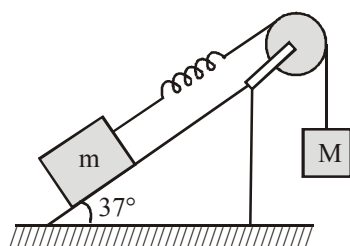
The equation of the wave is

- (a) $y = 0.1 \sin(4\pi(t-1) + 8(x-2))$
 (b) $y = 0.1 \sin((t-1) - (x-2))$
 (c) $y = 0.1 \sin(2\pi(t-1) + 4(x-2))$
 (d) None of these
15. In an experiment, a small steel ball falls through a liquid at a constant speed of 10 cm/s. If the steel ball is pulled upward with a force equal to twice its effective weight, how fast will it move upward?
- (a) 5 cm/s (b) Zero (c) 10 cm/s (d) 20 cm/s
16. A ball of mass 160 g is thrown up at an angle of 60° to the horizontal at a speed of 10 ms^{-1} . The angular momentum of the ball at the highest point of the trajectory with respect to the point from which the ball is thrown is nearly ($g = 10 \text{ ms}^{-2}$)
- (a) $1.73 \text{ kg m}^2/\text{s}$ (b) $3.0 \text{ kg m}^2/\text{s}$
 (c) $3.46 \text{ kg m}^2/\text{s}$ (d) $6.0 \text{ kg m}^2/\text{s}$
17. A block of mass m is attached with massless spring of force constant k . The block is placed over a fixed rough inclined surface for which the coefficient of friction is $\mu = 3/4$. The block of mass m is initially at rest. The block of mass M is released from rest with spring in unstretched state. The minimum value of M required to move the block up the plane is (neglect mass of string and pulley and friction in pulley.)



Space for Rough Work





- (a) $\frac{3}{5}m$ (b) $\frac{4}{5}m$ (c) $\frac{6}{5}m$ (d) $\frac{3}{2}m$

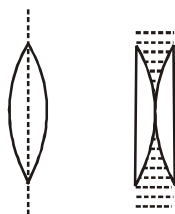
18. A charged particle of mass m and having a charge Q is placed in an electric field E which varies with time as $E = E_0 \sin \omega t$. What is the amplitude of the S.H.M. executed by the particle?

- (a) $\frac{QE_0}{m\omega^2}$ (b) $\frac{1}{2} \frac{QE_0}{m\omega^2}$
(c) $\frac{2QE_0}{m\omega^2}$ (d) None of these

19. In a photoelectric experiment, with light of wavelength λ , the fastest electron has speed v . If the exciting wavelength is changed to $3\lambda/4$, the speed of the fastest emitted electron will become –

- (a) $v\sqrt{\frac{3}{4}}$ (b) $v\sqrt{\frac{4}{3}}$
(c) less than $v\sqrt{\frac{4}{3}}$ (d) greater than $v\sqrt{\frac{4}{3}}$

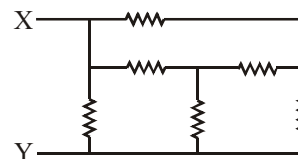
20. A thin convex lens of focal length 10cm and refractive index 1.5 is cut vertically into two equal pieces. They are placed as shown with a liquid of refractive index 3 between them. What is the focal length of the combination ?



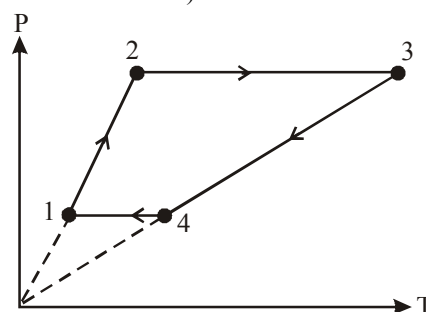
- (a) -10 cm. (b) $-10/4$ cm. (c) $-10/3$ cm. (d) None

PART-II (Numerical Answer Questions)

21. The activity of a radioactive substance drops to $1/32$ of its initial value in 7.5 h. Find the half life (in hours).
22. Six resistors of 10Ω each are connected as shown. The equivalent resistance (in ohms) between points X and Y is –



23. The electric potential V is given as a function of distance x by $V = (5x^2 + 10x - 9)$ volt. Value of electric field (in v/m) at $x = 1$ m is
24. Three moles of an ideal monoatomic gas perform a cycle shown in figure. The gas temperatures in different states are $T_1 = 200$ K, $T_2 = 400$ K, $T_3 = 1600$ K, and $T_4 = 800$ K. The work done (in kJ) by the gas during the cycle is (Take $R = 25/3$ J/mol-K)



25. A body is executing simple harmonic motion. At a displacement x from mean position, its potential energy is $E_1 = 2$ J and at a displacement y from mean position, its potential energy is $E_2 = 8$ J. The potential energy E (in Joule) at a displacement $(x + y)$ from mean position is

HINTS & SOLUTIONS

MOCK TEST-2

1. (a) $\Delta p = mv = m\sqrt{2gh}$

2. (a) Least count of vernier callipers
= value of one division of main scale – value of one division of vernier scale

Now, $N \times a = (N + 1) a'$

(a' = value of one division of vernier scale)

$$a' = \frac{N}{N+1} a$$

$$\therefore \text{Least count} = a - a' = \frac{a}{N+1}$$

3. (b) Here $I = 4A$

$$\theta = 30^\circ = \left(\frac{30 \times \pi}{180} \right)^c = \frac{\pi^c}{6}$$

$$\text{Now, } k = \frac{I}{\theta} = \frac{4}{\left(\frac{\pi}{6} \right)} = \frac{4 \times 6 \times 7}{22} = \frac{2 \times 6 \times 7}{11}$$

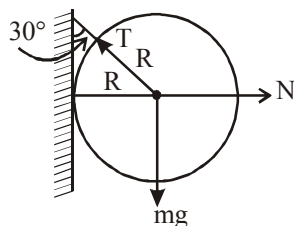
$$= \frac{84}{11} = 7.6 \text{ A/rad}$$

4. (c)

$$mg = T \cos 30^\circ$$

$$N = T \sin 30^\circ$$

$$\Rightarrow N = \frac{mg}{\sqrt{3}}$$

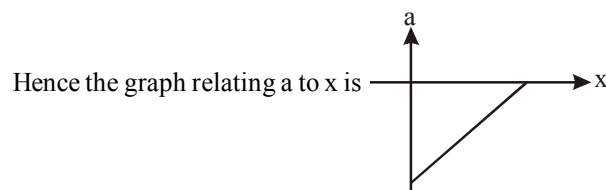


5. (a) A convex mirror always forms a virtual image which can't be cast on a screen. Hence, a convex lens is used to find the focal length of a convex mirror.

6. (a) The linear relationship between v and x is
 $v = -mx + C$ where m and C are positive constants.

$$\therefore \text{Acceleration, } a = v \frac{dv}{dx} = -m(-mx + C)$$

$$\therefore a = m^2x - mC$$



7. (c) For a resistor the pointer moves in both ways when voltage is applied.

8. (a) Divide the ring into infinitely small lengths of mass dm_1 . Even though mass distribution is non-uniform, each mass dm_1 is at same distance R from origin.

$\therefore$ MI of ring about z -axis is

$$= dm_1 R^2 + dm_2 R^2 + \dots + dm_n R^2 = MR^2$$

9. (b) If temp of mixture (T) is not constant then according to calculation we find wrong value of specific heat. If we stir continuously then temperature of whole mixture becomes same.

10. (d) For isothermal process :

$$PV = P_i 2V$$

$$P = 2P_i \quad \dots(i)$$

For adiabatic process

$$PV^\gamma = P_a (2V)^\gamma$$

($\because$ for monatomic gas $\gamma = 5/3$)

$$\text{or, } 2P_i V^{\frac{5}{3}} = P_a (2V)^{\frac{5}{3}} \quad [\text{From (i)}]$$

$$\Rightarrow \frac{P_a}{P_i} = \frac{2}{2^{\frac{5}{3}}} \Rightarrow \frac{P_a}{P_i} = 2^{-\frac{2}{3}}$$

11. (c) Shift = $\frac{\beta(\mu-1)}{\lambda} t_1 - \frac{\beta(\mu-1)}{\lambda} t_2$

$$= \frac{\beta(\mu-1)}{\lambda} (t_1 - t_2)$$

12. (c) $\phi = \frac{Q_{\text{net}}}{\epsilon_0}$ For dipole $Q_{\text{net}} = 0 \Rightarrow \phi = 0$

13. (c) $I = \frac{m\ell^2}{12};$

Restoring torque $= qE\ell \sin \theta \approx qE\ell \theta = \frac{d^2\theta}{dt^2}$

$\therefore \omega = \sqrt{\frac{qE\ell}{I}} = \sqrt{\frac{qE\ell \times 12}{m\ell^2}} = \frac{2\pi}{T}$

$\therefore T = \pi \sqrt{\frac{m\ell}{3qE}}$

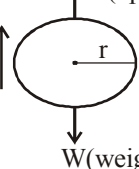
14. (c) The equation of wave moving in negative x-direction, assuming origin of position at $x = 2$ and origin of time (i.e. initial time) at $t = 1$ sec.

$y = 0.1 \sin(2\pi t + 4x)$

Shifting the origin of position to left by 2m, that is, to $x = 0$. Also shifting the origin of time backwards by 1 sec, that is to $t = 0$ sec.

$y = 0.1 \sin(2\pi(t-1) + 4(x-2))$

15. (c) Weight of the body

$W = mg = \frac{4}{3}\pi r^3 \rho g$ (viscous force) 

$T = \frac{4}{3}\pi r^3 \sigma g$

and $F = 6\pi\eta vr$

When the body attains terminal velocity net force acting on the body is zero. i.e.,

$W - T - F = 0$

And terminal velocity $v = \frac{2}{9} \frac{r^2(\rho - \sigma)g}{\eta}$

As in case of upward motion upward force is twice its effective weight, therefore, it will move with same speed 10 cm/s

16. (c) Given : $m = 0.160$ kg

$\theta = 60^\circ$

$v = 10$ m/s

Angular momentum $L = \vec{r} \times m \vec{v}$

$= H mv \cos \theta$

$= \frac{v^2 \sin^2 \theta}{2g} \cos \theta \quad \left[H = \frac{v^2 \sin^2 \theta}{2g} \right]$

$= \frac{10^2 \times \sin^2 60^\circ \times \cos 60^\circ}{2 \times 10}$

$= 3.46$ kg m²/s

17. (a) As long as the block of mass m remains stationary, the block of mass M released from rest comes down by $\frac{2Mg}{K}$ (before coming to rest momentarily again).

Thus the maximum extension in the spring is

$x = \frac{2Mg}{k}$ (1)

For block of mass m to just move up the incline

$kx = mg \sin \theta + \mu mg \cos \theta$ (2)

$2Mg = mg \times \frac{3}{5} + \frac{3}{4} mg \times \frac{4}{5}$ or $M = \frac{3}{5} m$

18. (a) For a particle undergoing SHM with an amplitude A and angular frequency ω , the maximum acceleration $= \omega^2 A$

Here the maximum force on the particle $= QE_0$

$\therefore$ maximum acceleration $= \frac{QE_0}{m} = \omega^2 A$

$\therefore A = \frac{QE_0}{m\omega^2}$

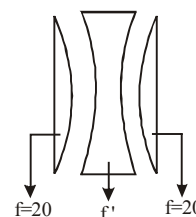
19. (d) $\frac{1}{2}mv^2 = \frac{hc}{\lambda} - \phi$

$\frac{1}{2}mv'^2 = \frac{hc}{(3\lambda/4)} - \phi = \frac{4hc}{3\lambda} - \phi$. Clearly, $v' > \sqrt{\frac{4}{3}}v$

20. (c) $\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

$R = 10$ cm.

$f' = (3 - 1) \left(\frac{1}{-10} - \frac{1}{10} \right) = -\frac{10}{4}$



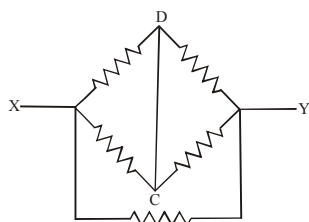
$$\frac{1}{f_{eq}} = \frac{1}{20} - \frac{4}{10} + \frac{1}{20} = \frac{2}{20} - \frac{4}{10}$$

$$\Rightarrow f_{eq} = -\frac{10}{3} \text{ cm}$$

21. (1.5) Using $\frac{A}{A_0} = \left(\frac{1}{2}\right)^{t/T}$ or $\frac{1}{32} = \left(\frac{1}{2}\right)^{7.5/T}$

or $\left(\frac{1}{2}\right)^5 = \left(\frac{1}{2}\right)^{7.5/T}$ or $5 = \frac{7.5}{T}$ i.e., $T = 1.5$ hours

22. (5)



Equivalent resistance = 5Ω

23. (-20) $E = \frac{-dV}{dx}$ $V = 5x^2 + 10x - 9$,

$$E = \frac{-d}{dx} (5x^2 + 10x - 9) = -(10x + 10)$$

On putting value of x in it we get,

$$E = -(10 \times 1 + 10) = -20 \text{ V/m}$$

24. (15) Process_{1→2} and Process_{3→4} are isochoric processes.

$$W_{12} = 0, W_{34} = 0, W_{23} = nR (T_3 - T_2) = 3R (1600 - 400)$$

$$= 3600 R$$

$$W_{41} = nR (T_1 - T_4) = 3R (200 - 800) = -1800 R$$

$$W = (3600 - 1800) R = 1800 R = 15 \text{ kJ}$$

25. (18) $E_1 = \frac{1}{2} kx^2$, $E_2 = \frac{1}{2} ky^2$,

$$E = \frac{1}{2} k(x+y)^2 = E_1 + E_2 + 2\sqrt{E_1 E_2}$$

$$= 2 + 8 + 2\sqrt{16} = 18 \text{ J}$$